RPMGLOBAL

AMT
User Documentation

Version 8.26

© 2024 RPMGLOBAL | All Rights Reserved

RPMGLOBAL

Contents

AMT overview 31
Registration and Activation 32
Software configuration 33
System Administration support 33
AMT navigation 34
My Functions 34
Information Panel 35
Equipment Navigator 35
My Favourites 36
Main Menu toolbar 36
Using filters 38
Copying data 38
AMT Home Screen 39
Dashboard 39
Asset Performance KPIs in the Dashboard 39
Dashboard audit trail 40
Quick Watch 41
Enabling Quick Watch 42
Setting the parameters 42
Quick buttons 43
Trigger Quick Watch 43
Notifications 44
Filter explanation 45
Reports overview 48
Report Icons 48
Report Parameters 49
Running Reports 50
Equipment Evaluation 52
Reports 52
A 52
B 63

C 70
D 95
E 99
F 108
G 121
H 122
I 127
J 130
K 131
L 132
M 139
O 150
P 151
R 173
S 178
T 186
U 190
V 195
W 196
Key Concepts 207
Asset hierarchy 207
Set up the asset hierarchy 207
Asset types 208
Life cycle costing 209
Usage profile 211
Projections 212
Projection types 212
Strategy task 213
Strategy Task Structure 215
Strategy task modes 215
Component code hierarchy 216
Unassigned 217
Cost bearer 217

Cost escalations ... 218
Current projections - actual costs ... 218
Current projections - future costs ... 218
Budget projections - costs to date ... 219
Budget projections - future costs ... 219
Alternate projection - actual costs and future costs ... 219
Archive projection ... 220
Centreline projections ... 220
Modelling Estimates (current and drafts) ... 220
Time zone support ... 220
Configuration required ... 220
Reference time zones ... 220
Actual Costs ... 223
Work order and work order settlement ... 223
Multi-Currency ... 224
Risk Rating ... 224
Changing the risk rating ... 225
Strategy task external identifier ... 225
What If Analysis ... 226
Work orders ... 226
Setup & Admin ... 228
Equipment ... 228
Fleet Administration ... 228
Equipment Administration ... 229
Multi-Editor Asset Details ... 237
Multi-Editor Plant Hire ... 237
Multi-Editor Asset Register ... 237
Multi-Editor MARC ... 238
Multi-Editor Quick Watch ... 240
Multi-Editor Asset Configuration ... 240
Equipment Editor ... 240
Copy and Create Equipment ... 248
(Rpt_675) Equipment Expiration ... 253

(Rpt_650) Equipment Audit 254
(Rpt_808) Asset Geolocation History 254
Projections 254
Projection types 255
Strategy Task Manager 256
Editable Strategy Task Manager 287
(Rpt_223) Projection Summary 287
(Rpt_222) Projection Details 289
(Rpt_440) Projection and Interval Analysis 289
(Rpt_425) Part Number Reference 290
Add/Copy/Delete projection 290
Export Projection Details (Old) 291
Export Projection 291
Strategy Task Import Wizard 291
Projections - Multi-editors 294
Multi-Task Scope Editor 294
Copy Strategy Tasks 294
Multi-Task Editor 296
Multi-Task % Cost Adjuster 296
Multi-Task Labour Editor 296
Multi-Equipment Delete Task 296
Forecast cost adjuster 296
Usage profile 297
Set up Usage Profile 298
Usage Profile Manager 298
Usage profile 301
(Rpt_248) Usage Profile Details 303
Monthly Usage Profile 303
Asset Ledger 304
Capital Details 304
Asset Ledger 305
Plant Asset Journals 310
CBD Analysis 312

Asset types 314
Configuration items 314
Capital Asset Recognition 316
Asset capability 316
AMT Asset Journals 317
Overheads 319
Overhead Navigator 319
Components 321
Serialised Component Manager 321
Machine Assembly 321
Create new Machine Assembly Workflow 321
View/Edit Machine Assembly Workflow 321
Depreciable Asset Recognition 322
Database configuration requirements 322
System administration requirements 322
AMT Depreciable Asset Process 323
Straight line depreciation 323
Units of activity depreciation 324
User Defined Financial Periods 324
User defined financial period depreciation status 325
Configuration of Entry Distribution Codes 326
Capital Asset Depreciation 329
Depreciable Asset Editor 330
Depreciation Strategy Task 333
Depreciable Asset Manager 339
Quick Links 341
Depreciable Asset Monthly Calculation 356
(Rpt_826) Depreciable Asset Register Report 357
Depreciable asset transactions 357
Asset Register Report 359
Functional Location 360
Functional hierarchy overview 360
Functional Location System Tables 361

Functional Location Manager 361
Set up the functional hierarchy 361
(Rpt_781) Functional Location Budget Analysis 362
Initial parameters 362
Report parameters 363
Additional filters 363
(Rpt_782) Functional Location Budget v Actual Analysis 363
Initial parameters 363
Report parameters 363
Additional filters 363
(Rpt_783) Functional Location Cost Analysis 363
Initial parameters 363
Report parameters 363
Additional filters 363
(Rpt_784) Functional Location ERP Master Data Analysis 364
Initial parameters 364
Report Parameters 364
Additional filters 364
(Rpt_785) Functional Location Forecasting Maturity 364
Initial parameters 364
Default parameters 364
Additional filters 364
(Rpt_786) Functional Location Cost Details 364
Initial parameters 364
Report parameters 365
Additional filters 365
Strategy Optimisation 366
Strategy definition 366
(Rpt_399) Maintenance Strategy Summary 367
(Rpt_400) Failure Mode Details 369
Failure Mode Assessment (FMA) 369
Decide on the maintenance strategy 370
Determine task scheduling 370

Last performed unit of measure or date 374
Last scheduled unit of measure or date 374
Last performed scheduling 374
Measuring performance 375
Overdue strategy tasks 376
Monitoring process 376
Maintenance strategy base operating parameters 378
Non-Planning task 378
Unassigned tasks 379
Life cycle analysis 379
(Rpt_217) PCV Chart 380
(Rpt_188) Life Cycle Cost Summary 380
(Rpt_216) PCV Analysis 381
(Rpt_251) Variance Analysis Worksheet 382
(Rpt_135) Budget vs Current Projection Details 382
(Rpt_150) Projection Comparison 383
(Rpt_815) Life Cycle Cost Summary - @Cost 383
Health - actual costs to date 383
Health - projected costs 386
Health assessment process 388
Cost Analyses 388
(Rpt_146) Cost Analysis - "Life Cycle" 388
(Rpt_147) Cost Analysis - "To Date" 389
(Rpt_243) Unassigned Cost Analysis 390
Analysis of Projection Assumptions 390
Strategy task analyser 392
Utilisation Analyser 393
(Rpt_171) Fleet Component Performance Summary 393
(Rpt_323) Component Variance Analysis Report 394
(Rpt_324) Component Variance Summary 394
Labour 395
Strategy optimisation - Labour analysis 395
(Rpt_182) Labour Hours Analysis 396

(Rpt_185) Labour Hrs Variance Analysis 396
(Rpt_832) Workload Optimisation 396
Component Risk 397
Risks and opportunities 397
(Rpt_177) High Opportunity 397
(Rpt_178) High Risk 398
Component Reliability Analysis 398
(Rpt_238) Technical Performance Analysis 398
(Rpt_239) Technical Performance Analysis Chart 399
(Rpt_240) Technical Performance Details 399
(Rpt_142) Component Life Distribution 400
(Rpt_389) Component History 400
(Rpt_393) Failure Indicators Analysis 401
(Rpt_391) Trending 401
Parts Usage Analysis 402
(Rpt_208) Parts Usage Analysis - Qty 402
(Rpt_210) Parts Usage Analysis - Value 402
(Rpt_212) Parts Usage Details 403
(Rpt_214) Parts Usage Interrogation 403
Potential Parts Warranty 403
Global Parts Usage Analysis 405
What If Analysis 407
Strategy optimisation - What If Analysis 407
Assess the What If Strategy 408
Add/Copy/Delete projection 409
(Rpt_254) What-if - Life Cycle Analysis 409
(Rpt_255) What-if - Period Analysis 409
(Rpt_252) What-if - Cost Analysis 409
(Rpt_253) What-if - Cost per UOM 410
(Rpt_251) Variance Analysis Worksheet 411
Strategy Admin KPIs 411
(Rpt_233) Standard Job Coverage 411
(Rpt_234) Standard Job Interrogation 411

(Rpt_259) WorkScope Coverage 412
Archiving 412
Budget and Forecast Manager 412
(Rpt_134) Archive Cost Worksheet 412
(Rpt_133) Archive Cost Analysis 413
(Rpt_119) Actual vs Archive Analysis 413
(Rpt _120) Actual vs Archive Period Analysis 414
(Rpt_251) Variance Analysis Worksheet 415
(Rpt_136) Archive - Costs/UOM 415
Long-Term Planning 416
Long-term planning screen 417
Reviewing new tasks 420
Group tasks 420
Bow Wave identification 421
Long-Term Planning - Advanced 421
STO Occurrence Movement Logic 422
Standard vs Advanced planning 423
Component changeout grid 424
Next Occ Export 425
Next Occ Import 429
Last Occ Export 429
Last Occ Import 429
(Rpt_633) Long-Term Planning Log 429
Initial parameters 430
Report parameters 430
Additional filters 430
(Rpt_632) Hours To PM 430
Initial parameters 430
Additional filters 430
Fleet condition 430
(Rpt_810) Job Status Audit Log 431
Initial parameters 431
Report parameters 432

Additional filters 432
Availability 432
(Rpt_172) Forecast Availability 432
Validate availability forecast 432
Costs 433
(Rpt_173) Forecast Cost Analysis 433
(Rpt_174) Forecast Worksheet 434
(Rpt_778) Forecast Worksheet (Serial Number) 434
(Rpt_816) Forecast Worksheet - @Cost 435
(Rpt_818) Forecast Worksheet (Dealer Adjusted) 435
Components 436
Long-term planning component reports 436
Component capacity overview 436
(Rpt_143) Component Status 437
(Rpt_145) Contribution Summary 438
(Rpt_204) Overdue Strategy Tasks 438
(Rpt_141) Changeout Schedule 439
(Rpt_140) Changeout Analysis 439
(Rpt_325) Changeout Analysis - Detailed 440
(Rpt_426) Part Number Reference - Modelling 440
Marketing and Customer Support Editor 441
(Rpt_453) Parts Demand Analysis 441
(Rpt_562) Forecast Rotables Stock 441
(Rpt_563) Rebuild Centre Forecast Workload 443
(Rpt_564) Changeout Forecast Worksheet 443
Component Forecast Management 443
Review component forecasts 445
Component Changeouts 445
Communicate rebuild forecast 446
Set rebuild strategy 446
Validate parts and components 446
Validate rebuild centre capacity 448
Consumables 449

(Rpt_443) Consumables Forecast 449
Labour 449
Validate labour resources 449
(Rpt_183) Labour Forecast - Hours 450
(Rpt_184) Labour Forecast - Persons 450
(Rpt_832) Workload Optimisation 451
Parts 451
Parts Forecast 451
Parts Forecast Manager 452
Export Parts Forecast 453
Create and import parts 454
Add parts using Parts Interface 454
Health - Parts forecasting 454
Review superseded parts 454
Shutdown, Turnaround, Outage 455
Audit Log – Requires STO Flag 455
STO Manager 455
Create an STO 455
STO Dates and Status 456
STO Category 457
Product Support Marketing 458
PSM Administration 458
PSM Manager 458
Product Support Marketing process - Central Admin 459
Product Support Marketing key principles - Central Admin 461
Product Support Marketing process - Marketing Director 461
Product Support Marketing key principles - Marketing Director 462
Product Support Marketing process - Sales Responsibility 463
Product Support Marketing key principles - Sales Responsibility 464
Set up Product Support Marketing customers 465
Product Support Marketing analysis 466
Product Support Marketing base potential adjustments 467
Product Support Marketing data validation 468

Product Support Marketing projections 468
Product Support Sales Responsibility Multi Editor 469
Sales Forecast Adjustment Factors 469
Gross margin factors 470
Product Support Marketing Multi-Equipment Editor 471
Is the equipment currently in AMT? 471
Marketing and Customer Support Editor 476
Multi-Customer Editor 477
Product Support Marketing annual utilisation 478
Market influences 479
PSM Reports 479
(Rpt_657) Forecast Market Analysis 479
(Rpt_658) Forecast Market Share Analysis 480
(Rpt_659) Actual Sales Analysis 480
(Rpt_660) Historical Market Share Analysis 481
(Rpt_634) Customer Audit 481
(Rpt_635) Customer Sales Activity 481
(Rpt_653) Sales Status and Strategy Forecast Health 482
(Rpt_654) Strategy Task Audit 482
(Rpt_663) Product Support Component Status 483
(Rpt_814) Labour Forecast (Dealer Adjusted) 483
(Rpt_818) Forecast Worksheet (Dealer Adjusted) 484
Daily Planning 485
Planning 485
Planning grid 486
Grid fields (standard display fields) 487
Business Rules for Operational Criticality, Risk Override and Risk Rating (optional fields) 489
Standard vs Advanced planning 489
Scheduling Gantt chart and grid descriptions 490
Scheduling 496
Search for Inventory Parts 496
Shift Handover Comments 497
PM Schedule 497

Work order authorisation 502
Activity Dashboard 503
Event Merging Wizard 511
Daily Planning Reports 511
Employee Availability and Allocation 516
Employee Setup 517
(Rpt_546) Workorder Authorisation List 519
Initial parameters 519
Report parameters 519
Additional filters 519
Statutory Compliance 519
Job ready planning 520
Job ready requirements 520
Job readiness indicator 520
Job documentation 520
Set up labour availability 521
Per day instead of per hour 521
Using the Resource Availability and Fill - Work group report 522
Labour Activity 522
Associate resources to a work order 522
Create a shutdown 523
Parts 523
Attach work order instructions 528
Labour 528
Miscellaneous 528
Planning a Work Order 529
Operations 529
Schedule a work order 529
User Work Order Status 530
Prepare Job Folders 530
User Work Order Status 531
Daily Operations 532
Equipment status 532

Health - Daily Operations 532
Events 534
External Downtime Wizard 535
Shift log 537
Work orders 538
Linking work order settlements 539
Backlogs 540
Equipment Down 542
Quick Backlog Capture 542
Product incident report capture 543
Product incident report listing 544
Submit warranty claims 545
Product incident report listing 546
Product incident report capture 546
View Workorder Status 547
Timesheet 548
Timesheet App Login 548
Timesheet confirmation - Supervisor 549
(Rpt_681) Timesheet Cost Analysis 550
(Rpt_679) Timesheet Journal 551
Work Order Settlement Closure 551
Notifications 552
Raise a new notification 552
Assign and action a notification 553
Audit a notification 553
Inspections 553
Daily Reports 554
Availability Reports 554
(Rpt_157) Daily Downtime Log 556
(Rpt_523) Shift Downtime Log 556
(Rpt_524) Shift Downtime Summary 556
(Rpt_158) Daily Log - Detailed 557
(Rpt_159) Daily Log - Summary 557

(Rpt_166) Equipment Down 558
(Rpt_160) Daily Machine Performance Check Sheet 558
(Rpt_231) Snapshot - The Worm 558
(Rpt_232) Snapshot Summary 559
(Rpt_355) Equipment Operational Performance Summary 559
(Rpt_198) Monthly Downtime Log 560
(Rpt_237) Workorder Details 561
(Rpt_518) Event Completion Performance 561
(Rpt_519) Event Completion Performance SCREEN 562
(Rpt_488) Workorder Labour Analysis 562
(Rpt_506) Equipment Down Alert 562
(Rpt_646) Availability Snapshot Worm 563
(Rpt_665) Workorder Duration Analysis 563
(Rpt_764) Inspection Completion Time 564
(Rpt_779) Planned Workorder Details 564
(Rpt_780) Extended Planned Workorder Details 564
(Rpt_830) User Work Order Status Audit Report 565
Timesheet Reports 565
(Rpt_679) Timesheet Journal 565
(Rpt_680) Employee Time Validator 566
(Rpt_681) Timesheet Cost Analysis 566
Usages & Measurements 567
Usages, measurements and production statistics 567
Record actual usages 567
Utilisation Analyser 570
(Rpt_413) Fleet Statistics 571
(Rpt_390) Equipment Statistics 571
(Rpt_118) Actual Usage Details 572
(Rpt_391) Trending 572
Record Actual Measurement Readings 572
Measurements dialog box overview 573
Analyse actual measurements 574
Usage validation 574

Production statistics data capture 574
Workorder Cost Analysis 577
Workorder Settlement Manager 577
(Rpt_257) Workorder Settlement Interrogation 588
(Rpt_321) Workorder Settlement List 588
(Rpt_322) Average Workorder Costs 589
Cost accrual 589
Warranty 589
Warranty approval 589
(Rpt_350) Warranty Claim 590
(Rpt_351) Warranty Analysis 590
(Rpt_528) Monthly Warranty Status 591
AMT User Reports 591
Core Tracker 592
Create a Reman process workflow 592
Reman process workflow 593
Reman Overview 594
Reman scheduling logic 595
View / Edit Reman process workflow 596
Reporting & Analyses 597
Summary 597
(Rpt_167) Equipment Performance Summary 597
(Rpt_382) Change Analysis 597
(Rpt_439) Key Stats Change Trending 598
(Rpt_541) Budget Equipment Cost Analysis 598
Equipment Performance 598
Availability 598
Reliability 602
Definitions 603
Stoppages and downtime 603
Utilisation 603
Consumption 605
Production Analysis 606

Costs 609
Costs 609
Total Costs (including Overheads) 618
Unassigned Costs 620
Maintenance Efficiency 622
Planning 622
Daily Operations 627
Equipment Evaluation 631
Summary Evaluation 631
Cost per UOM 632
Residual Value Analyses 633
Web Dashboard 639
System configuration and settings 639
KPI Calculation 639
Scheduled hours 639
Multiple units of measure 639
Dashboard structure 639
Dashboard navigation 640
Dashboard calculations 643
Dashboard recalculation 645
Dashboard targets 645
Printing the Dashboard 646
AMT IC 647
Intelligence Centre - Report Writing module overview 647
How does it work? 647
AMT data models 647
Cubes vs views 648
Data model reference 648
Accessing AMT data models 649
Administration 649
IC Equipment Manager 649
IC System Task Manager 649
Report Manager 649

Report Templates ... 658
Budgeting and forecasting ... 659
Budgeting ... 659
Budget and Forecast Manager ... 659
Changing budget status ... 667
Reports ... 667
(Rpt_494) Budget Compliance ... 672
(Rpt_540) Budget Forecast Performance ... 673
(Rpt_541) Budget Equipment Cost Analysis ... 673
(Rpt_543) Budget Component Performance ... 674
(Rpt_837) Budget Comments Report ... 674
Live Forecasting ... 674
(Rpt_173) Forecast Cost Analysis ... 674
(Rpt_174) Forecast Worksheet ... 675
(Rpt_818) Forecast Worksheet (Dealer Adjusted) ... 675
(Rpt_778) Forecast Worksheet (Serial Number) ... 676
(Rpt_816) Forecast Worksheet - @Cost ... 676
Knowledge Database ... 677
Electronic Workscope Manager ... 677
Electronic Workscope dialog box overview ... 677
Inspection configuration ... 679
Inspection display modes ... 679
Inspection item node ... 679
Inspection Type ... 680
Inspection steps ... 680
Electronic workscope ... 681
Library ... 681
Library jobs ... 682
Standard Job Manager ... 682
Calculation of standard job totals ... 683
Standard job overview ... 683
Standard job identification and version control ... 684
Standard job principles ... 684

Standard jobs and escalation 685
Standard job margins 685
Technical advisor tab 685
Create a standard job from Standard Job Manager 686
Create strategy tasks from standard jobs 690
Create operations 691
Define Labour and Misc requirements 692
Export standard jobs 693
Linking to standard jobs 695
Linking standard jobs to strategy tasks 695
Populate centreline maintenance strategy 695
Standard job security 696
Standard jobs and workscopes 696
(Rpt_398) Parts Interrogation 696
Initial parameters 696
Report parameters 696
Additional filters 696
(Rpt_529) Strategy Task - Standard Job Details 696
Tips 696
Initial parameters 696
Report parameters 697
Additional filters 697
(Rpt_833) Standard Job Version History 697
Initial parameters 697
Report parameters 697
Additional filters 697
EM Issues 697
EM Issues process 697
EMI technical analysis 699
EMI investigation 699
EMI evaluation 699
EMI authorisation 699
EMI monitor and resolve 699

EMI corrective action 700
Linking EMI to equipment, costs and availability 700
Application and operation metrics 700
Equipment design 700
Strategy optimisation - Investigation 701
(Rpt_163) EM Issue List 701
Condition Based Maintenance 702
Program Structure 702
Set up condition based maintenance message groups 702
Component structure 703
Component structure dialog box overview 704
Measurements 706
Measurement Translation Rules 706
Recording measurements against failure modes 709
MARC 710
MARC Financial 710
Characteristics of a MARC 710
Summary 711
Cash Flow 717
Costs / Contribution 718
Maintenance Accrual 722
Forecasting 726
Intercompany Accuracy 727
Contract tendering 731
Update current margins 731
Gross margin analysis 731
MARC Admin 732
Workflow management 732
Contract Invoicing 732
Contract Administration 733
Billings 736
Equipment billings - Audit 746
Equipment billings 746

Apply billing escalations 747
Overheads 747
(Rpt_241) Termination Summary 751
Contract termination 752
Plant Hire 758
Plant hire budgeting 758
Financial Performance 759
Plant Hire Performance Reporting 759
Budget Reporting 759
(Rpt_521) Site P&L 761
(Rpt_522) Site Performance 761
(Rpt_318) Fleet Cost and Recovery Schedule 762
(Rpt_148) Equipment Cost and Recovery Analysis 762
(Rpt_409) Equipment Cost and Recovery Schedule 763
(Rpt_376) Plant Hire Under/Over Recovery 763
Asset Validation 764
(Rpt_613) Asset Valuation 764
Plant Invoicing 764
(Rpt_375) Plant Hire Invoice Summary 764
(Rpt_831) Plant Hire Rates Audit Report 765
Plant Hire accounting process 765
Allocate invoices to strategy tasks 766
Cost Analysis 766
(Rpt_122) Asset Cost Worksheet 766
Setup 767
Configuring Plant Hire 767
Set/Edit Plant Hire Rate 768
Create a new Plant Hire project 768
Calculation logic 769
(Rpt_220) Plant Rate Analysis 771
Process Plant Hire Rates 771
Review and Edit Processed Hire Charges 772
Post Plant Hire Rates 774

Project Allocations Gantt 774
(Rpt_565) Plant Hire Rates Schedules 775
Centrelines 776
Reports 776
Administration 776
Branch Administration 776
Site Administration 776
Fleet Administration 777
Equipment Administration 778
Multi-Editor Asset Details 779
Equipment Editor 779
Create blank equipment - Centreline 781
Register modelling equipment 783
Copy and Create Equipment 784
Usage Profile Manager 785
(Rpt_311) Usage Profile Details - Centreline 786
Define application severity 786
Strategy Task Manager 786
Editable Strategy Task Manager 790
(Rpt_299) Projection Details - Centreline 791
(Rpt_442) Projection and Interval Analysis - Centrelines 791
(Rpt_301) Projection Summary - Centreline 791
(Rpt_484) Projection Summary - Customer - Centreline 792
(Rpt_427) Part Number Reference - Centreline 792
Copy Strategy Tasks 793
Multi-Task % Cost Adjuster 793
Multi-Task Editor 793
Multi-Task Labour Editor 793
Multi-Task Scope Editor 793
Add/Copy/Delete projection 793
Centreline Analyses 794
(Rpt_286) Cost Analysis - "Life Cycle" - Centreline 794
(Rpt_278) Cost per Interval - Centerline 794

(Rpt_282) Cost per UOM - by Usage - Centreline 794
(Rpt_284) Cost per UOM - Comparison - Centreline 795
(Rpt_280) Cost per UOM - by Date - Centreline 795
(Rpt_431) Cost per UOM per Interval - Centrelines 795
(Rpt_290) High Risk - Centreline 796
(Rpt_288) High Opportunity - Centreline 796
(Rpt_303) RVIC - Chart - Centreline 797
(Rpt_305) RVIC Analysis - Centerline 797
(Rpt_292) Labour Forecast - Hours - Centreline 798
(Rpt_294) Labour Forecast - Persons - Centreline 798
(Rpt_435) Labour Hours per Interval - Centrelines 798
(Rpt_296) Maintenance Ratio - Centreline 799
(Rpt_296) Maintenance Ratio - Centreline 799
(Rpt_438) Duration Per Interval - Centrelines 799
(Rpt_445) Consumables Forecast - Centrelines 800
(Rpt_531) Strategy Task - Standard Job Details - Centreline 800
Modelling 801
Administration 801
Equipment Hierarchy 801
Projections 811
Usage Profile 820
Asset Ledger 823
Plant Hire Administration 824
MARC Administration 825
Overheads 826
Equipment Evaluation 828
(Rpt_307) Summary Analysis - Modelling 828
(Rpt_281) Cost per UOM - by Date - Modelling 828
(Rpt_283) Cost per UOM - by Usage - Modelling 828
(Rpt_285) Cost per UOM - Comparison - Modelling 829
(Rpt_430) Cost per UOM per Interval - Modelling 829
(Rpt_289) High Opportunity - Modelling 829
(Rpt_291) High Risk - Modelling 830

(Rpt_306) RVIC Analysis - Modelling 830
(Rpt_304) RVIC - Chart - Modelling 831
Cost Analysis 831
(Rpt_277) Cost Analysis - "Life Cycle" - Modelling 831
(Rpt_279) Cost per Interval - Modelling 832
(Rpt_797) Asset Cost Worksheet - Modelling 832
(Rpt_342) Total Cost Worksheet - Modelling 833
(Rpt_383) Financial Summary - Modelling 833
(Rpt_388) Forecast Worksheet - Modelling 833
(Rpt_817) Forecast Worksheet - @Cost - Modelling 834
(Rpt_821) Forecast Worksheet (Dealer Adjusted) - Modelling 834
(Rpt_496) Budget Compliance - Modelling 834
(Rpt_497) Cost Trending - Modelling 835
Availability 835
(Rpt_ 287) Forecast Availability - Modelling 835
(Rpt_437) Duration Per Interval - Modelling 836
Labour Analysis 836
(Rpt_293) Labour Forecast - Hours - Modelling 836
(Rpt_295) Labour Forecast - Persons - Modelling 837
(Rpt_434) Labour Hours per Interval - Modelling 837
(Rpt_297) Maintenance Ratio - Modelling 837
What-if 838
(Rpt_316) What-if - Life Cycle Analysis - Modelling 838
(Rpt_317) What-if - Period Analysis - Modelling 838
(Rpt_314) What-if - Cost Analysis - Modelling 838
(Rpt_315) What-if - Cost per UOM - Modelling 839
(Rpt_313) Variance Analysis Worksheet - Modelling 839
Components 839
(Rpt_386) Changeout Schedule - Modelling 839
(Rpt_385) Changeout Analysis - Modelling 840
(Rpt_387) Changeout Analysis - Detailed - Modelling 840
Marketing and Customer Support Editor 840
(Rpt_454) Parts Demand Analysis - Modelling 841

Consumables 841
(Rpt_444) Consumables Forecast - Modelling 841
MARC Analyses 842
MARC Summary 842
MARC Cash Flow 842
MARC Cost at Cost 843
MARC Customer Reports 844
Quoter 847
Contract Tendering 852
Set Contract Tendering System Tables 852
View / Edit Contract Tendering Workflow 853
System Administration 854
AMT System Administration Responsibilities 854
System Administration Procedural Guidelines 854
User Profiles 854
User Accounts 854
Group Administration 855
Change Password 855
Message Group Administration 855
System Tables 856
Organisation 856
Financial 860
Current projections - actual costs 865
Current projections - future costs 865
Budget projections - costs to date 865
Budget projections - future costs 866
Alternate projection - actual costs and future costs 866
Archive projection 866
Centreline projections 866
Modelling Estimates (current and drafts) 867
Cost Reports 867
Maintenance 871
Parts & Standard Jobs 880

Equipment 886
Functional Location 888
Usages & Measurements 889
Other 892
Multi tenancy system tables 894
Data Translation and Validation 896
Work order settlement code translation 896
Set GL reconciling items 898
(Rpt_186) Ledger Missing Equipment 898
(Rpt_396) AMT to GL Reconciliation 899
Reconciling work order settlements 899
(Rpt_224) Reconciling Open Workorder Settlements 905
(Rpt_532) Workorder Settlement Transactions 905
Monthly contract equipment reconciliation 906
Configuration 906
Quote Documentation 907
Group Permissions 907
Cost Allocation Interface Details 908
External Downtime Equipment 909
External Downtime Site 909
AMT Utilities 910
Refresh Reference Data 910
User Group Function Manager 910
Recalculation Manager 910
Interface Administration 910
(Rpt_414) Interface Log 913
(Rpt_455) Generic Interface Log 914
Export Administration 914
Language Type 915
Language Editor 915
System Task Manager 915
Create Component Changeout Journal 916
Log Entries 916

Process Work Order Settlements ... 916
Audit Log ... 917
AMT Audit Log Report ... 918
Generic Value Pairs ... 918
Terminology ... 919
AMT Mobile overview ... 949
AMT Mobile - Login ... 949
Navigation - Mobile module ... 949
Customers app ... 950
View Market Analysis ... 951
Definitions ... 951
Items on the screen ... 951
View and update Alert Opportunities ... 952
View the Component Forecast ... 952
View and update Overdue Opportunities ... 953
Timesheet App Login ... 954
Time allocation ... 956
Clock out ... 957
Supervisor App ... 957
Timesheet confirmation - Supervisor ... 957
Workbench app ... 958
Select an Asset and start an Inspection ... 959
Statutory Compliance - Mobile ... 960
Inspections App Login ... 960
Select an Asset and start an Inspection ... 961
Work Orders - Mobile ... 961
Perform an Inspection ... 962
Quoter ... 963
Configurations ... 963
Quoter setup ... 963
Quote Report structure ... 964
Quoter user security ... 967
Available agreements ... 968

Quoter Manager - Add Quote Jobs 973
Quoter Manager- Approve and Authorise Quote 973
Quoter Manager - View and Edit Quote Jobs 974
Quoter Manager - Print Quote 974
Quoter centrelines 974
Administrator App 977
Maintenance App 977
Measurements 978
Defects 978
AMT Mobile Usages 978
Statutory Compliance 979
Offline Options 979
Content feedback 980
About 981

AMT overview

AMT Asset Maintenance Software is the only in-shift maintenance solution that dynamically links to lifecycle costing and strategy to provide an accurate lifecycle position and optimal maintenance strategies.

Below are high-level descriptions of functionality available in AMT. Some functions are only available if certain modules of AMT have been purchased.

Modelling	
Centrelines	Maintain centreline (baseline) life-cycle templates for assets.  Centrelines are not only used in the Modelling module, but are used throughout AMT. Centrelines are like templates for assets, for example, when you purchase new assets you can copy that asset (and its existing projections) from an existing centreline.
Modelling	Model fleet scenarios. Analysing and compare different asset purchase options - costs and productivity.
Reporting	
Dashboard	Report management KPIs in real time, graphically.
Financial	Real time forecasting, Life-cycle cost analyses, Asset Evaluation (Cost Per Tonne, Discounted Cash Flows), Benchmarking and determining optimal asset replacement points (CAPEX forecast).
Equipment Performance	Availability, reliability and cost performance reporting and analysis
Maintenance Efficiency	Efficiency measures for Strategy, Planning, Daily Operations (Execution) – that is, scheduling accuracy, backlog ratios and so on.
Specialised Financials	
Budgeting	Streamline creating and maintaining accounting budgets and forecasts. You can maintain multiple versions of budgets, export to corporate budgeting systems and perform analyses against actual performance.
Asset Register	Maintain asset register, calculate depreciation and other asset charges.
Plant Hire	Calculate plant hire rates, process monthly plant hire invoices and management reports on recovery of costs and maintain condition-based depreciation accounts.
MARC Financials	A specific set of reports designed to report on the current and forecast profitability of maintenance and repair contracts, cost cap contracts and so on.

Strategy	
Strategy Optimisation	Routines to define and optimise maintenance strategies – identify risk areas, play what-if scenarios, view fleet performance, understand life-cycle impact.
Component Management	Maintain technical information on component performance, analyse component performance (that is, bell-curve, analysis of failure modes and so on), and maintain component changeout schedule, optimise inventory, gain visibility, over rebuild centre forecasts.
Long Term Resource Planning	Forecast components and labour requirements, optimise scheduling efficiency, manage resources, maintain asset availability and coordinate maintenance activities.
Rotables Management	Component tracking and rotable inventory management – forecast rotable inventory levels to identify 'spikes', decision support around component strategy (exchange/reman/new/rebuild).
EM Issues	Log of asset issues, quantify their impact (costs, downtime), and the steps to resolve them. Provides a knowledge database to prevent others re-inventing the wheel.
Parts Usage	Analyse parts usage and identify potential warranty claims.
Planning	
Standard Jobs	Standard jobs with bill of materials, labour times and so on.
Library	Store all workscopes (job instructions, JSAs, inspection sheets), training materials, legal contracts and best practice information.
Planning	Manage the backlog, daily planning, raise purchase orders and full level resource.
Work Order Settlements	Capture asset costs (or import from ERP).
Parts Forecasting	Forecast the parts required for planned jobs.
Daily Operations	
Equipment Status/Shift Log	Service tool – electronic shift log, access to asset history and component status, capture downtime and can create work order settlements in ERP system.

Registration and Activation

The registration details of AMT are updated using the Registration/Activation function. For example, AMT may have been registered to Joe Blogs when AMT was first installed. This person may have left the company, so the registration details will need to be updated.

Changing registration details causes AMT to request a new product key from RPM. Therefore, only modify this information when absolutely necessary, and when the new product key has been issued, so that it can be entered immediately.

Software configuration

The software configuration settings fall into the following general categories.

- **Integration**: Interfaces enabled, file paths, file types and configuration.
- **Modules**: Features enabled and configuration.

System configuration settings are not editable. The configuration settings are reviewed if:

- AMT is upgraded and the new version has new configuration options.
- Additional interfaces are deployed.
- Additional modules are deployed.

System Administration support

All user enhancement requests, for example, new report or new functionality requests, and all possible system errors should be reported to the system administrator, who then records them in a log. The system administrator will then contact the AMT vendor.

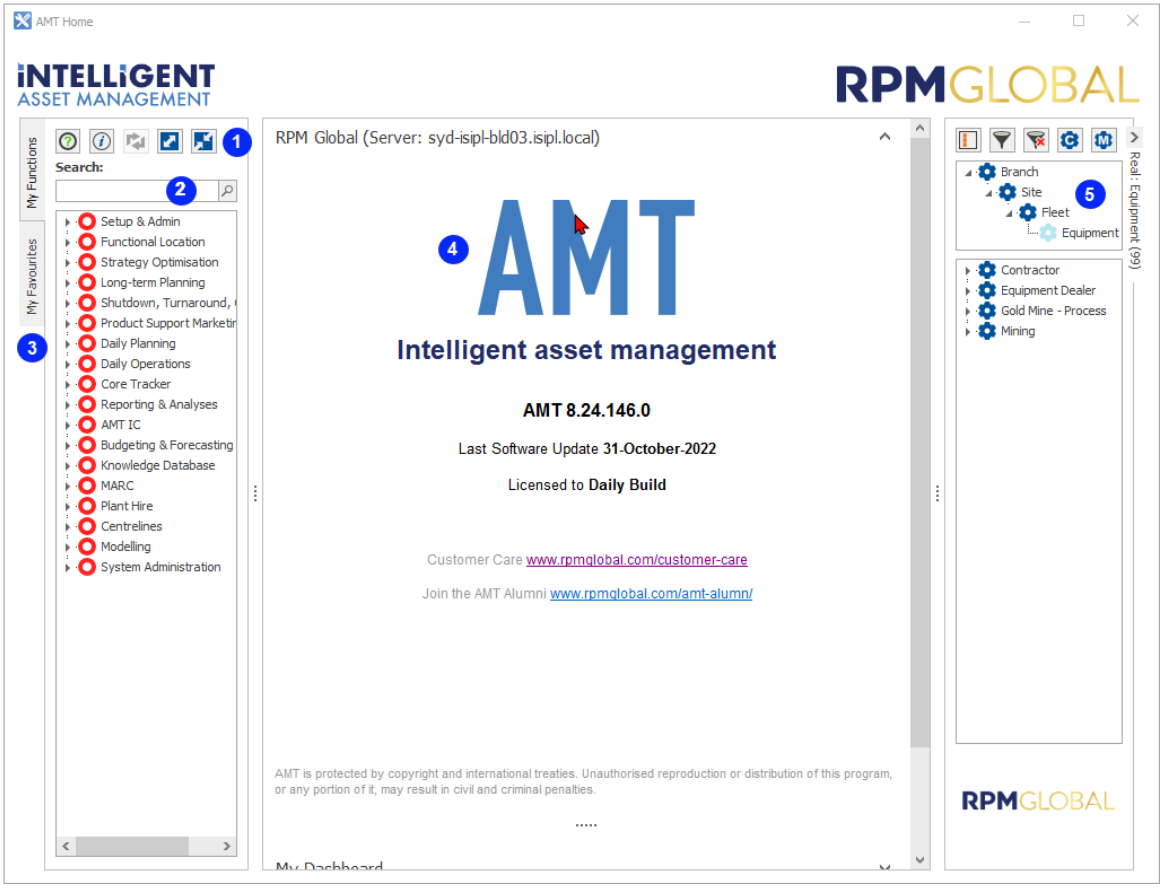
If you require assistance in using the system, you should consult:

- The AMT literature that is stored in the Library.
- The AMT champion either on site or in head office.
- The system administrator.

We recommend that users do not contact the AMT vendor directly. All communication should be directed to the system administrator, or a similarly designated person or group within the company, who would then contact the AMT vendor with the request.

AMT navigation

The AMT Home screen is divided into the following areas.



Number	Description
1	My Functions
2	Search Bar
3	My Favourites
4	Information Panel
5	Equipment Navigator

My Functions

My Functions gives you all functionality as per your access level.

The **My Functions** tab is where you can launch all functions and reports. Find the function you want by drilling down into the branch structure.

Alternatively, you can navigate to functions and reports using the **Search Bar**. Enter the name, or part of the name, of the function or report you are searching for.

The **My Favourites** tab is where you can see your favourite reports and functions, and preselected functionality. See My Favourites for more information.

Information Panel

The Information Panel provides you with information about the AMT version that is currently installed. You can also view the **My Dashboard**, **Quick Watch** and Notifications functions by clicking the relevant arrow to the right.

Equipment Navigator

The Equipment Navigator displays a list of all assets according to a pre-populated hierarchy. You can display the asset information in two ways.

1. Grouped by various categories: Fleet, Model, Site and so on. Filters are then available to limit the results:
   - Group by - Refers to how the data is grouped.
   - Equipment Hierarchy - These filters give you the flexibility to limit, very specifically, the data that displays in the reports.

   Click Filter  at the top of the panel. The **Equipment Editor** displays.

   The default filter is **Group By Branch, Site, Fleet** . You can clear that filter, then click the **Search** icon to select the items you want to filter on.

2. Listing Standard/Child Assets.
   i. Click **Change to siblings/Change to standard**  to toggle between Standard and Child Assets.
   ii. Click **Filter** at the top of the panel. The **Equipment Selector** displays. You can select the parent equipment and associated filters.

To set up the Equipment Navigator follow these links:

Branch

Site

Fleet

Below are the **icons** used in My Functions, My Favourites and Equipment Listing areas of AMT's home screen.

Item	Description
	Function List - Menu Header
	Function List – Sub-Menu Header
	Function List – Working Screen

Item	Description
	Function List – Report
	Equipment List – Group By
	Equipment List – Analyse By

My Favourites

The system administrator can create a custom list of reports and functions per user (which a subset of the complete list of functions and report available to that user). The custom list is called **My Favourites** and is set up from the AMT homepage.

Click on the **My Favourites** tab on the left side of AMT's main menu and then click **Edit**  . The Function

Hierarchy screen displays. You can create your own tree structure on the right-hand side, and then add into each branch the relevant reports/functions.

For example:

- My Monthly Reports
  - Cost Analysis – To Date
  - Cost Analysis – Life Cycle
  - Availability
  - High Risk
  - High Opportunity
- My Forecasting Reports
  - Changeout Analysis
  - Labour Forecast – Hours
  - Forecast Cost Worksheet

To save changes click **Save**. Close the **Function Hierarchy** screen, and click **Refresh** on the **My Favourites** tab to see the new structure. Now, every time you log into AMT, you will see your favourites first.

Click **My Functions** at any time to see the original list of all functions and reports you have access to.

Main Menu toolbar

Standard AMT buttons

Most AMT functions have the following standard buttons

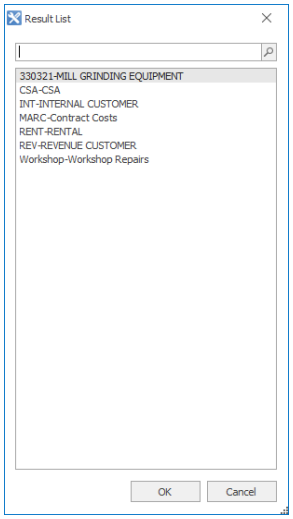
Button	Action
	Add
	Edit
	Delete. You can delete levels in the AMT Explorer or delete records from specific forms.
	Refresh
	Undo
	Apply Filters
	Cancel Filters
	Expand
	Collapse
	Preview Report (using default filters)
	Help
	System Administration
	Save
	Close

Using filters

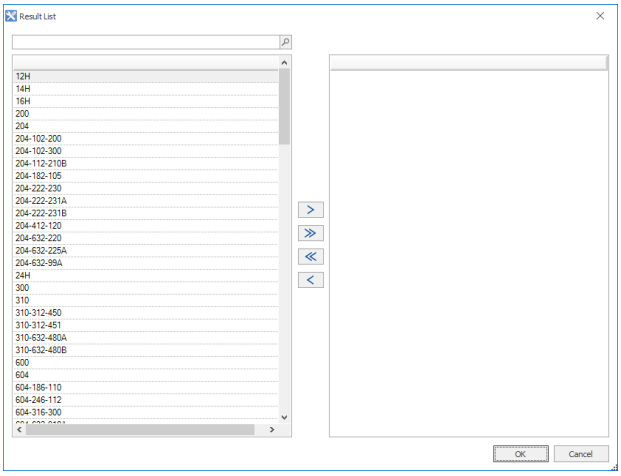
Filters are used to restrict data selections. On some filters you can select only one record, on other multi-select filters you can select multiple records.

- On a screen, click **Search** to open the filter results list.
- If the Results list is a single-select box, make a selection from the list and click **OK**.
- If the Results list is a multi-select box, move your selection to the right-side panel using the arrows, or double-click your selection and click **OK**.

Single-select box



Multi-select box



Move Field	Remove All Fields
Move All Fields	Remove Field

To change your selection, clear the current selection by clicking Clear Filters before you can open the results list.

Copying data

You can copy the data on some grids and export it to other applications such as Excel.

Right-click anywhere on the grid and select **Copy Table**, then paste the data into the third-party application.

You can paste data directly into some grids. Grids with this capability have a **Paste** option on the right-click menu.

AMT Home Screen

Dashboard

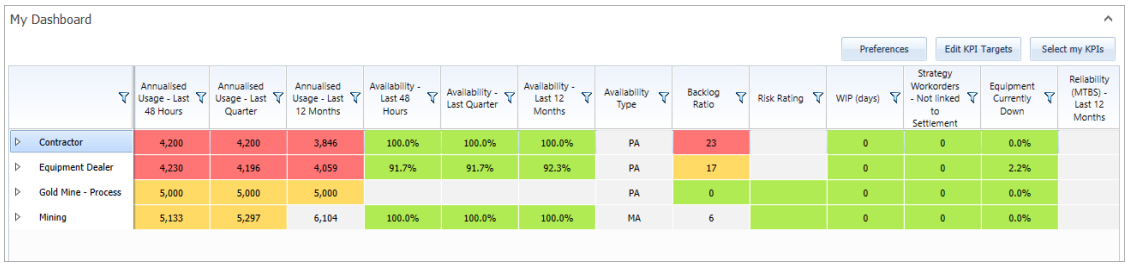
Asset Performance KPIs in the Dashboard

The following asset performance KPIs are pre-calculated and display in the AMT dashboard. The dashboard displays on your AMT home screen and shows data grouped by branch, site and fleet.

- Annualised Usage – Last 48 hours
- Annualised Usage – Last quarter
- Annualised Usage – Last 12 months
- Availability – Last 48 hours
- Availability – Last quarter
- Availability – Last 12 months
- Availability type (set at the fleet)
- Backlog ratio
- Assets currently down
- Reliability (MTBS) – Last 12 months
- Reliability (MTBS) – Last quarter

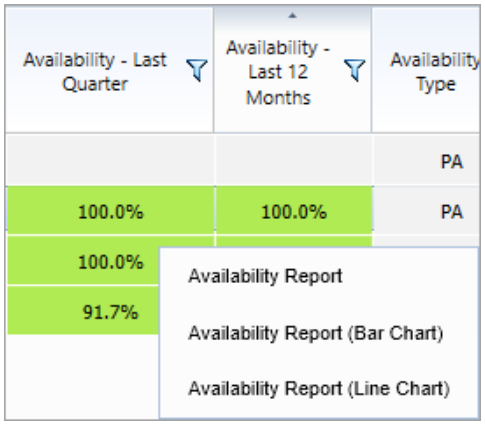
An example screenshot of the dashboard is shown below.

The dashboard shows many KPIs in addition to the above list. You can change the list by clicking any of the three tabs above the grid and changing the settings.



	Annualised Usage - Last 48 Hours	Annualised Usage - Last Quarter	Annualised Usage - Last 12 Months	Availability - Last 48 Hours	Availability - Last Quarter	Availability - Last 12 Months	Availability Type	Backlog Ratio	Risk Rating	WIP (days)	Strategy Workorders - Not linked to Settlement	Equipment Currently Down	Reliability (MTBS) - Last 12 Months
Contractor	4,200	4,200	3,846	100.0%	100.0%	100.0%	PA	23		0	0	0.0%	
Equipment Dealer	4,230	4,196	4,059	91.7%	91.7%	92.3%	PA	17		0	0	2.2%	
Gold Mine - Process	5,000	5,000	5,000				PA	0		0	0	0.0%	
Mining	5,133	5,297	6,104	100.0%	100.0%	100.0%	MA	6		0	0	0.0%	

Right-click any cell to see a list of available reports.



Dashboard audit trail

Measurement	Source	Type
Availability	Availability (Mechanical) or Availability (Physical) - Responsibility = <default Responsibility>.	Report
Backlog Ratio	Equipment Backlog Listing - Strategy Tasks = Yes.	Report
Breakdown %	Breakdown Downtime Events % - Responsibility = <default Responsibility>.	Report
Budget Margin %	GM Comparison	Report
Budget Variance	Equipment Plan Editor - Configuration - MARC	Screen
Cash Flow to Date	Cash Flow Statement.	Report
Equipment Currently Down	EQS - Equipment Down.	Screen
Maintenance Ratio	Maintenance Ratio - Cost Bearer = <default Cost Bearer>.	Report
MTFS After PM	MTFS After PM - Responsibility = <default Responsibility>. - PM Task Type = PM. - FS Task Type = All. - Planned = All. - Breakdown = All	Report
MTTR	MTTR - Responsibility = <default Responsibility>.	Report
PCV%	PCV Chart.	Report
Planned DT% (DT = Downtime)	Planned Downtime Hours % - Responsibility = <default Responsibility>.	Report
Projected Billings	MARC PCV Chart.	Report
Projected Margin %	GM Comparison.	Report
Reliability (MTBS)	MTBS - Responsibility = <default Responsibility>.	Report
Net Opportunity Risk	High Risk High Opportunity	Report

Measurement	Source	Type
	- Default filters.	
Risk: % of Projected Life-Cycle Costs	Dashboard calculation.	
Risk: Adjustment Margin %	Dashboard calculation.	
Risk: Avg Margin %	Dashboard calculation.	
Service Accuracy	Service Accuracy.	Report
Strategy Tasks - Action Required	Strategy Confirmation.	Screen
Strategy Tasks - Not linked to WOs	EQS - Tasks - Admin Alerts.	Screen
Strategy Tasks - Overdue	Overdue Strategy Tasks.	Report
Utilisation	Usage Analysis Chart - Gross Usages As = Annual - Avg Per Eqp Plan = Yes.	Report
WIP (days)	Work orders imported from the ERP system.	Interface

For those measurements which are based on a date range greater than one month, the date range used includes prior months plus month to date. For example, the last quarter is defined as the last three months as well as this month to date. Therefore, if the current date is 12-Jun-19 the last quarter is: 1-Mar-19 to 12-Jun-19 (March-April-May PLUS June to date).

Quick Watch

Quick Watch provides a real-time view of performance indicators for easy identification of areas needing attention. You can configure Quick Watch items to continually monitor important KPIs.

To view Quick Watch, in the **middle panel** of the **Home Page**, click the drop-down arrow beside **Quick Watch**.



An asset only becomes visible in Quick Watch once it has been enabled to do so and parameters have been set up. The next two sections describe these steps.

Enabling Quick Watch

To make an asset available for Quick Watch:

1. **Edit** the asset.
2. Click **Configuration**.
3. **Enable the Quick Watch KPIs** relevant to that asset.

Setting the parameters

Once the Quick Watch items have been enabled for the asset, additional parameters may be added using Quick Watch on the AMT home page. The parameters are described below.

Parameter	Description
Work order Cost Variance#	Compares actual cost to estimated cost for strategy tasks only, based on a percentage or cost (only examines negative variances). **Sort Order**: % and Value.
High Cost#	Non-Strategy tasks with an actual cost greater than a defined cost amount. **Sort Order**: Cost and Date.
PCV Change^	Movement in Projected Cost Variance (PCV) for an asset in the last five days. **Sort Order**: Absolute value of PCV movement.
Annual Cost Variance	Movement in financial year cost in the last five days. Requires an accounting budget for that site in order to make a comparison.  Accounting budgets are set up in the Budget and Forecast module. **Sort Order**: Absolute value of PCV movement.
WIP@	Work order settlements which have been opened for greater than 30 days from today's date. **Sort Order**: WIP days, date.
Strategy Breakdowns#	Completed and In Progress Breakdowns for strategy tasks only, based on minimum hours. **Sort Order**: Down Hours.
First Stoppage#	Highlights first stoppage after project maintenance if within a parameter. **Sort Order**: Hours since previous PM.
Overdue Component	Overdue component changeouts by value. **Sort Order**: Absolute value of PCV movement.

Parameter	Description
Financial Yr Cost Change^	Examining movement in financial year costs based on a minimum value. **Sort Order**: Absolute value of PCV movement.
Unscheduled Breakdowns#	Completed and in progress breakdowns for non-strategy tasks only, based on minimum hours. **Sort Order**: Down Hours.
No Recent Usage Readings	Show no usage reading within a defined timeframe. **Sort Order**: Days of Difference.

^ - Quick Watch only looks at snapshot data five days earlier from today.

- Quick Watch only looks at data in the last 14 days.

@ - Quick Watch only looks at data in the last 30 days.

A number of the KPIs reference a parameter, for example, define sort order, cost limit, within a parameter.

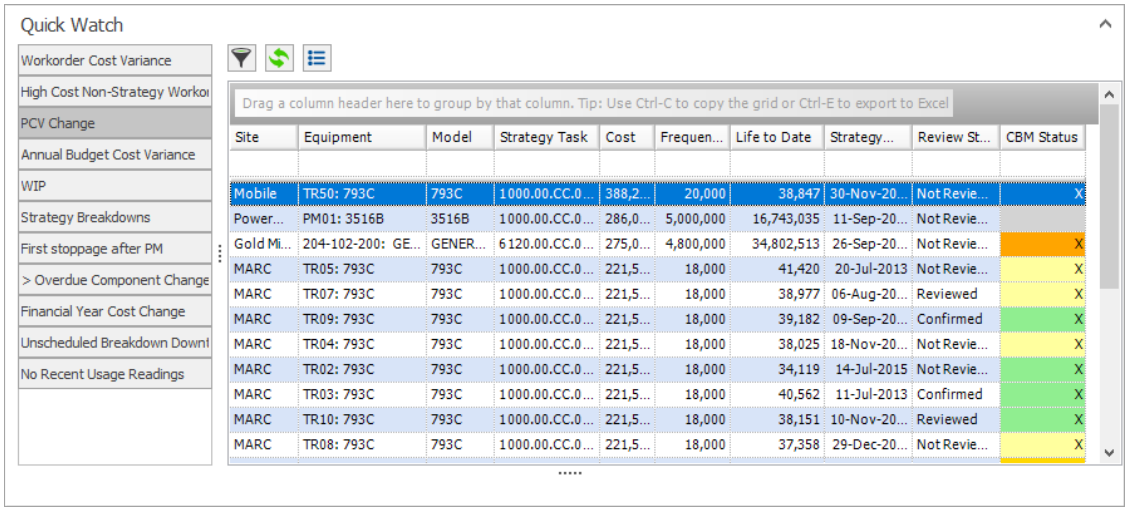
By default, the Quick Watch KPIs extract the Top 20 results. This can be changed by clicking **Apply Filter**  .

Quick buttons

Button	Description
	Apply Filters. Opens a new dialog box depending on the menu item on the left that you have selected. For example, opening WIP then clicking this button displays the WIP - Filters dialog box.
	Refreshes the current selection.
	Provides quick links to the work order settlement and work order dialog boxes.

Trigger Quick Watch

Once you have enabled the Quick Watch KPIs for the relevant assets, and the parameters have been set, Quick Watch is ready. Click on the relevant Quick Watch menu item to the left to see if, based on the parameters established, there is any data.



- You can sort and group columns.
- The data in the grid is not calculated, the data is live; that is, a KPIs data is compiled directly out of the AMT database each time a specific Quick Watch KPI item is clicked.
- **No Data** means that no data in the system met the parameters entered or there is no data to report (for example, if there are not any components overdue, AMT will say No Data).

Notifications

If AMT Mobile and/or Health functionality has been deployed and configured to create and manage Notifications, you are able to create, view and update Notifications in this window. In earlier versions of AMT, this window was labelled and contained Messages.

Notifications can be generated by AMT or imported from an external system. An example of a Notification may be a result of a condition monitoring measurement triggering follow-up action.

The Default fields displayed are:

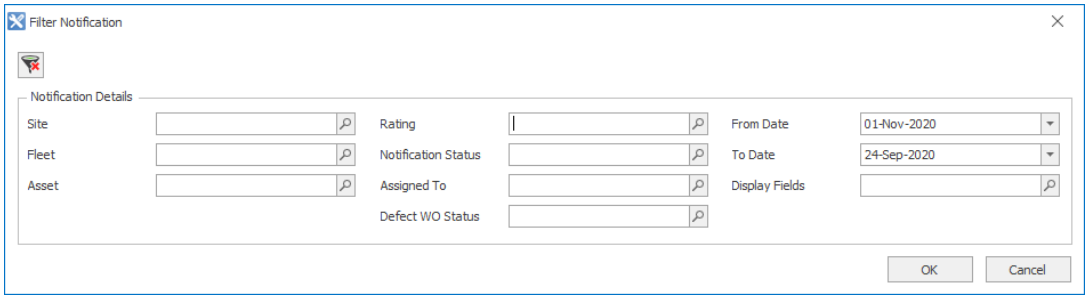
- Asset
- Title
- Date Raised
- Rating
- Counter (which identifies to you there have been multiple updates to the notification)
- Assigned To
- Status

Notification buttons

Icon	Description
	Filter notifications.
	Clear Filters.
	Refreshes the current selection.
	The count of notifications (1) with warning level 4.
	The count of notifications (3) with warning level 3.
	Add a new manual notification.  Adding a new notification takes you to the Notification's web screen.
	Edit an existing notification.  Editing an existing notification takes you to the Notification's web screen.

Filter explanation

Notification filter options operate as follows, and note, these differ in the date area from the filters when accessing Notifications on the web interface.



How filter notification works with the AMT Client window

1. Filters can be set in line with the standard AMT paradigm with the exception of **From Date** and **To Date**. It is a software design decision to ensure any saved date filters do not preclude you from seeing new notifications when the screen is opened.
2. **Date** filters can be changed from the default values and will apply for the time the screen is active. If you close the screen or navigate away from the screen, the date filters will return to the default.

3. Default Date filters will display:
    a. **Date From** = (midnight/00:00:00) Today's date - 1 day.
    b. **Date To** = (23:59:59) Today's date.
    c. If you run this screen at 2.15 in the afternoon on September 1, 2020, the results will display all notifications that have a date greater than 31-Aug-2020 00:00:00 and 01-Sep-2020 14:15:00.
4. The **Clear Filter** icon resets last used filters (excluding dates which are not saved).
5. Notifications display in a grid with the software design standard columns (Asset, Title, Date Raised, Rating, Counter, Assigned To, Status). Additional columns can be added to this grid based on one or more of the Display Fields selected.

    Display field options are:

    - Defect Workorder Status
    - Serial Number
    - Site
    - Customer
    - Inspection Type
    - Operational Impact
    - Source Workorder
    - External ID
    - Source
    - Occupational Health and Safety
    - Notification Number.

You can right-click on a Notification and select an action. An explanation of the actions are in the table below.

Selection	Description
Add Notification	Add a new manual notification.  Adding a new notification takes you to the notification's web screen.
Edit Notification	Edit a notification.  Editing an existing notification takes you to the notification's web screen.
Add Backlog Task	Create a backlog work order in the AMT Planning module.
Edit Failure Mode Assessment	If the notification was generated by AMT's Condition Based Maintenance (CBM) module, you can view and update CBM's Failure Mode Assessment.
Edit Strategy Task	If the notification was generated by AMT's CBM module, you can view the strategy task the notification is linked to. The notification is linked to a strategy task via a CBM component structure.

Selection	Description
Edit Equipment	View the equipment the notification is linked to.
Edit Measurements	If the notification was generated by an AMT Condition Monitoring Measurement rule, you can view and edit the measurement that triggered that rule.
Long-term Planning	Launch the Long-term Planning screen for the highlighted equipment.

Reports overview

The type of report produced by AMT depends on the module you are using. Some modules run within the AMT Client, producing Crystal reports.

AMT includes a small number of reports that are launched as PDF files, and therefore require Adobe Acrobat Reader to be installed on the client computer. These PDF reports are part of the following AMT modules:

- Parts Forecasting
- Rotables Manager
- EM Issues
- Sales Tool.

You can save or print both types of reports.

To see descriptions of all AMT reports and their parameters, click here.

Report Icons

Icon	Action
	Report.
	Preview.
	Select Filter Options.
	Clear Filters selected.
	Print.
	Export Report.
	Zoom in or out.
	Move highlighted fields to the selection window.
	Move all fields to the selection window.

Icon	Action
	Remove all fields from the selection window.
	Remove highlighted fields from the selection window.

Report Parameters

Default report parameters are set for all reports and charts. These are parameters display whenever a report or chart is selected.

If other parameters are available, you can view them by opening the report in **My Functions** or **My Favourites**.

AMT's report parameters have five categories, depending on what the report is for.

1. **Report Type**

   Choose how the information on the report displays, for example as text, a bar chart, a line chart or a pie chart.

2. **Date Selector**

   Narrows the data in the report to a specific point or period of time, for example Plan to Current Month, where Plan is the start date of the equipment plan.

3. **Analyse By**

   These options, which may also include Group By or Sort By, determine how the report data is to be shaped, for example, if running a cost report, do you want to see the costs per month? quarter? parts? fleet? component?

   Analyse By does not limit the set of data used in the report; the combination of **Date Selector**, **Equipment Hierarchy** and **Additional Filters** does this.

4. **Equipment Hierarchy**

   Select the level of the equipment hierarchy to which you want to report.

   - Use the **Set Selections** button  to open the **Results List** screen.
   - Highlight the required system in the left panel **Available** list.
   - Click  to move the system to the right panel (**Filter** list).
   - To move all systems, click  .
   - To move several non-continuous Systems, hold the Ctrl key and one by one highlight the systems you want, then press the  button.
   - To move several continuous systems, highlight the first record, hold the Shift key, and highlight the last record you want. Your selection is highlighted. Click **>** to move the records.
   - If records are moved to the **Filter** area (right panel) in error, click  or  to return the records to the **Results** list side.

5. **Additional Filters**

   You can select a combination of filters to limit the data in the report, to refine the focus and drill down to the root cause.

   Each report has a different set of available additional filters.

AMT applies a set of default filters to reports for two reasons. Firstly, some reports are run in a particular way every time, with default filters, so you do not need to continuously select filter options each time. You can always override the defaults. Secondly, it provides a shortcut because you can click **Preview** on the toolbar to run the report without having to open the report parameters. Mandatory selections to run a particular report are marked with an asterisk *****.

Running Reports

You can use reports to assess many aspects of your business and maintenance strategy, produce operational and financial forecasts and review KPIs.

Reports display on the My Functions tab with this icon . Double-click a report icon to open a **Report Parameters** dialog box with many filter options. Make sure you select the filters carefully to ensure the report returns the specific data required.

Highlighting a fleet, model or equipment plan in the **Equipment Navigator** before double-clicking the report auto-populates the filters with the highlighted selection.

Key concepts

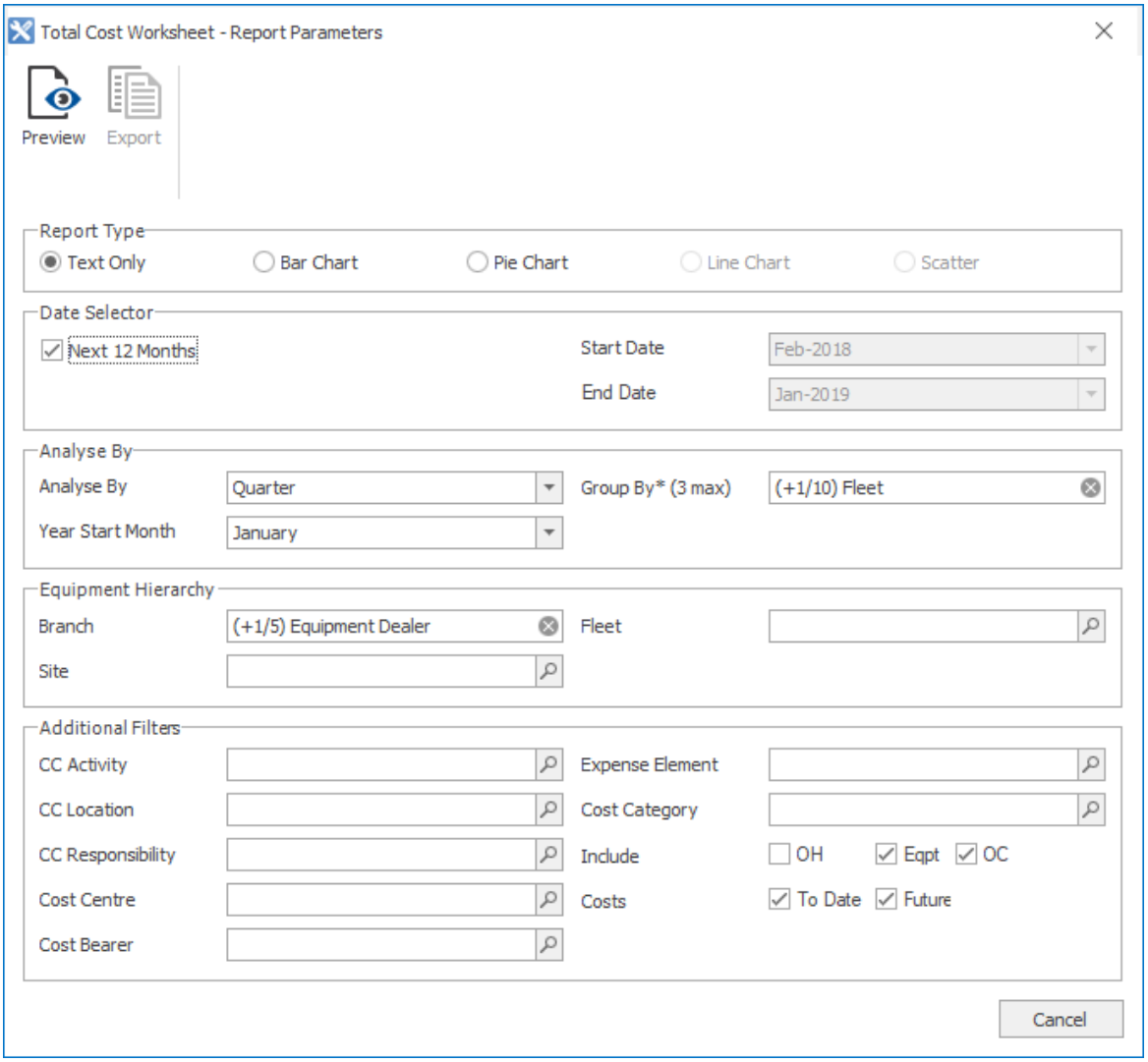
- **Analyse by**: Defines the categories or classifications in which the data is to be displayed, for example equipment class, equipment group.
- **Group by**: How the data is grouped, for example branch, site, fleet.
- **Filter**: The filters give you the flexibility to limit the data that displays in the reports.

Example

In the example screenshot below, we want to determine the quarterly equipment costs for a fleet for the next 12 months. Note the filter selections:

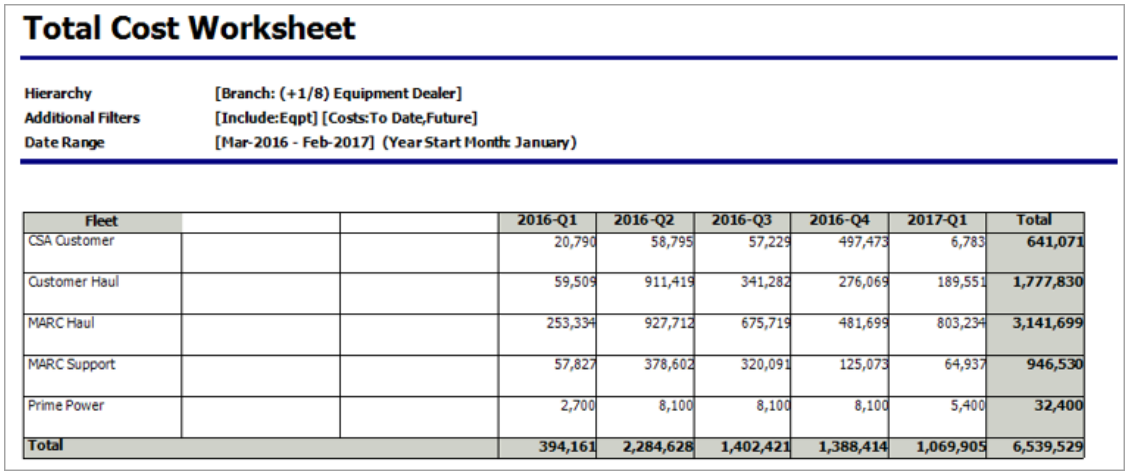
- Analyse by: Quarter
- Group by : Fleet
- Branch: User Specified Branch

- Include: Eqpt (Equipment)



Note that OH (overheads) and OC (other costs) are not included in this report query.

Click **Preview** at the top left of the screen to generate a Total Cost report similar to the screenshot below.



Total Cost Worksheet

Hierarchy [Branch: (+1/8) Equipment Dealer]
Additional Filters [Include:Eqpt] [Costs:To Date,Future]
Date Range [Mar-2016 - Feb-2017] (Year Start Month: January)

Fleet			2016-Q1	2016-Q2	2016-Q3	2016-Q4	2017-Q1	Total
CSA Customer			20,790	58,795	57,229	497,473	6,783	**641,071**
Customer Haul			59,509	911,419	341,282	276,069	189,551	**1,777,830**
MARC Haul			253,334	927,712	675,719	481,699	803,234	**3,141,699**
MARC Support			57,827	378,602	320,091	125,073	64,937	**946,530**
Prime Power			2,700	8,100	8,100	8,100	5,400	**32,400**
Total			**394,161**	**2,284,628**	**1,402,421**	**1,388,414**	**1,069,905**	**6,539,529**

Equipment Evaluation

Equipment can be evaluated in a number of ways – by how much it costs per operating hour, mean time between stoppages in the last 12 months and so on. A quick overview of some of the reports useful in evaluating equipment is outlined below.

Equipment Evaluation Summary Analysis

Navigate to **Reporting & Analyses** > **Equipment Evaluation** > **Summary Evaluation** > Summary Analysis.

Represents the cost benefit based on the revenue from the equipment. The revenue amount in this report is expressed in AMT's primary currency (it is not site or contract specific).

Cost Analysis – "Life Cycle"



There are numerous ways to navigate to this report, only one is listed below.

Navigate to **Reporting & Analyses** > **Costs** > **Costs** > Cost Analysis - "Life Cycle".

The Cost Analysis – "Life Cycle" report dissects costs (historic and future projected) in order to identify spikes, trends and perhaps the time when it would be appropriate to discontinue the use of the equipment.

RVIC Reports

Residual Value in Components Reports

Equipment Performance Summary

Navigate to **Reporting & Analyses** > **Summary** > **Equipment Performance Summary**.

The Equipment Performance Summary report shows the historic, current and expected future performance of equipment. Performance is represented by utilisation, costs, availability and reliability.

Reports

AMT reports are described alphabetically. Click the links on following pages to go to the relevant report lists.



The number in brackets in the report name is the report ID, for example (Rpt_118). This report ID is an AMT internal number. It can assist when making an enquiry using the portal.

You may not see all reports, depending on your AMT configurtaion.

A

(Rpt_118) Actual Usage Details

(Rpt_119) Actual vs Archive Analysis

(Rpt_120) Actual vs Archive Period Analysis

(Rpt_121) Agreement Summarye

(Rpt_122) Asset Cost Worksheet

(Rpt_124) Availability - Mechanical

(Rpt_125) Availability - Physical

(Rpt_133) Archive Cost Analysis

(Rpt_134) Archive Cost Worksheet

(Rpt_136) Archive - Costs UOM

(Rpt_274) Asset Cost Worksheet - Modelling

(Rpt_396) AMT to GL Reconciliation

(Rpt_410) Asset Utilisation

(Rpt_450) Agreement Summary 2 - Modelling

(Rpt_492) Availability to Target Summary

(Rpt_613) Asset Valuation

(Rpt_646) Availability Snapshot Worm

(Rpt_659) Actual Sales Analysis

(Rpt_808) Asset Geolocation History

"(Rpt_842) AMT Audit Log Report" on page 918

(Rpt_118) Actual Usage Details

Details the units of measure entered for a particular equipment, site, branch and so on. Shows both actual and cumulative readings, for example, engine hours, fuel litres, travel hours. **Reporting & Analyses** > **Equipment Performance** > **Utilisation** or **Daily Operations** > **Usages & Measurements**.

Initial parameters

|ControlType|, |IDlist|, |TextList|, |ProjectionTypes|, |UserID|, |AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|, |DateChecked|,|StartDate|,|EndDate|,|AnalyseBy|

Report parameters

Default.

Additional filters

None.

(Rpt_119) Actual vs Archive Analysis

This report is similar to the Actual vs. Budget Period Analysis report, but lets you expand on the information required for analysis. Dates can be set according to user requirement as can the "Analyse By" parameters.

Tips

It is critical that the Equipment Management group understand, report on, and develop continuous improvement initiatives based on current performance variances to Annual Budget expectations.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, Cost Centre, System, Responsibility, Sub System, Expense Element, Component Code, Labour Activity, Strategy Task, Parts/Labour/Misc, Task Type, Currency, Maintenance Type

(Rpt _120) Actual vs Archive Period Analysis

This report highlights the variances between an Annual Budget Projection and the Current Projection at the end of the financial year, or any specified period of time.

Tips

It is critical that the Equipment Management group understand, report on, and develop Continuous Improvement initiatives based on current performance variances to Annual Budget expectations. Data in this report can be analysed by month, quarter, year.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, Cost Centre, System, Responsibility, Sub System, Expense Element, Component Code, Labour Activity, Strategy Task, Parts/Labour/Misc, Task Type, Currency, Maintenance Type

(Rpt_121) Agreement Summary

This report provides a summary of the equipment life cycle costs updated from the details entered into the scoping area of the Sales Tool and Agreement Manager.

Initial parameters

SalesToolReport,|UserID|

Report parameters

Default.

Additional filters

Equipment Class, Eqp Type, Region, Equipment Group, Eqp Criticality, Division, Manufacturer, Eqp Location, Branch, Model, Eqp Category, Site, Fleet, Contract, Equipment Year, Serial Number

(Rpt_122) Asset Cost Worksheet

Flexible cost analysis report that combines the strategy task costs set up in a projection with the other operating costs set up for that equipment.

Tips

Lets you to evaluate and forecast all equipment related costs - repair and maintenance costs, insurance, fuel, depreciation and so on.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Project, Cost Bearer, Region, Cost Centre, Division, Expense Element, Industry Group, System, Cost Responsibility, Sub System, Component Code, Cost Type, Equipment Class, Eqp Criticality, Equipment Group, Manufacturer, Eqp Category

(Rpt_124) Availability - Mechanical

Shows the mechanical availability for the designated period of time. The downtime used in the calculation is dependant on the "responsibility" nominated by the user.

To Trend, "Analyse By" Month to compare equipment with fleets, analyse by equipment and fleet.

Tips

Mechanical Availability = Operating Hours / (Operating Hours + Downtime).



Operating Hours = SMU.

Initial parameters

|ControlType|,|IDlist|, |TextList|,|ProjectionTypes|,|UserID|,|LogoLocation|,|AutoLaunch|, |BranchID|,|SiteID|,FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|,|DateChecked|, |StartDate|,|EndDate|,|AnalyseBy|,|Responsibility|,|ResponsibilityID|

Report parameters

Default.

Additional filters

Responsibility, System, Task Type, Maint. Order Type, Planned, Planning Group, Breakdown,

(Rpt_125) Availability - Physical

Shows the physical availability for the designated period of time. The downtime used in the calculation is dependant on the "responsibility" nominated by the user.

To Trend, "Analyse By" Month. To compare equipment with fleets, analyse by equipment and fleet.

Tips

Physical Availability = (Scheduled Hours - Downtime) / Scheduled Hours.

Scheduled Hours are defined using the Scheduled Hours function, and typically represent the calendar hours of a month.

|ControlType|, |IDlist|, |TextList|, |ProjectionTypes|, |UserID|, |AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|, |Fleet|,|EqpPlan|,|DateChecked|,|StartDate|,|EndDate|,|AnalyseBy|, |Responsibility|, |ResponsibilityID|

Report parameters

Default.

Additional filters

Responsibility, System, Task Type, Maint. Order Type, Planned, Planning Group, Breakdown,

(Rpt_133) Archive Cost Analysis

Provides an analysis of costs.

Tips

Lets you to compare two projections in the report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, Cost Responsibility, System, Cost Centre, Sub System, Expense Element, Strategy Task, Maintenance Type, Task Type, Modifier Code, Component Code, Division, Region, Equipment Class, Eqp Criticality, Equipment Location, Equipment Group, Parts/Labour/Misc, Eqp Category, Currency, Cost Bearer

(Rpt_134) Archive Cost Worksheet

Used for forecasting costs by expense group. It is used primarily for budgeting costs. It shows historical and future costs at sell, categorised by up to five different areas.

Tips

Similar to the Forecast Worksheet report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, Maintenance Type, System, Cost Centre, Sub System, Responsibility, Component Code, Expense Element, Strategy Task, Labour Activity, Task Type, Region, Division, Eqp Criticality, Equipment Class, Equipment Group, Equipment Location, Parts/Labour/Misc, Division, Custody

(Rpt_136) Archive - Costs/UOM

Report lets users to analyse between projections and shows data by unit of measure for either month, quarter or year. You have the option to analyse by two user selections.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, Sub System, Cost Bearer, Component Code, Cost Type, Modifier Code, System, Task Type, Costs

(Rpt_274) Asset Cost Worksheet - Modelling

Flexible cost analysis report that the Strategy Task costs set up in a Projection with the Other Operating costs set up for that equipment.

Tips

Lets you evaluate and forecast all equipment related costs - repair and maintenance costs, insurance, fuel, depreciation and so on.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Project, Region, Division, Industry Group, Cost Responsibility, Component Code, Equipment Class, Equipment Group, Eqp Category, Cost Bearer, Cost Centre, Expense Element, System, Sub System, Cost Type, Eqp Criticality, Manufacturer

(Rpt_322) Average Workorder Costs

This report shows the average parts, labour and miscellaneous costs per projected task based on work order history.

Tips

This report is useful to compare against what a projected task is expected to cost in the future. If there is a significant difference, there could be a strong argument to change the costs contained in the projected task.

Initial parameters

|AutoLaunch|,|Site|,|Fleet|,|FleetID|,|Model|,|ModelID|,|EqpPlan|,|EqpPlanID|,
|System|,|SystemID|,|SubSystem|,|SubSystemID|,|ProjTaskID|,|ProjTaskID2|,
|UserID|TaskHeaderId|

(Rpt_396) AMT to GL Reconciliation

Summarises the balance between the work order costs in AMT and the costs in the General Ledger for maintenance contract equipment. The report also reconciles contract billings.

Tips

This report does not reconcile Overheads, Plant Hire or other Operating Costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Contract, AMT Equipment, Currency, Cost Bearer, Parts/Labour/Misc

Click here to go to a description of the report output.

Reconciling work order settlements

A reconciliation process between AMT and the work order settlement system must be performed at least once a month, regardless of whether there is an interface between the two systems or not.

If AMT does not balance to the work order settlement system, management will not have confidence in the financial reporting produced by AMT.

The items that **must** reconcile between the two systems are the costs that have been incurred by each asset registered in AMT. The type of costs which have to be reconciled is dependent on the nature of the assets

registered in AMT, for example, are the assets part of a maintenance contract, does the asset only incur direct maintenance costs and so on.



Types of costs which have to be reconciled based on the nature of the asset should be documented in a procedure prior to the installation of the software.

In certain implementations of AMT there will be a general ledger interface which provide you with the ability to perform a reconciliation within AMT and identify the items which do not balance.

For implementations which do not have this interface in place, the organisation will have to develop a procedure that ensures:

1. Reconciliations are performed on a routine basis.
2. Items which do not reconcile are investigated and addressed properly.

(Rpt_410) Asset Utilisation

Compares the actual utilisation with the budget utilisation. It also shows the breakdown and scheduled down time.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Asset Util UOM, Analyse Downtime By

(Rpt_450) Agreement Summary 2 - Modelling

Provides a summary of the equipment life cycle costs updated from the details entered into the scoping area of the Sales Tool/Agreement Manager.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Cost Bearer, Escalation, Availability Table, Labour Hours Table, Labour Rate Table, Lube Cost Table, Lube Quantity Table, Projection Details, Costs Equal Bilings, Comp-Job-Mod Desc, Cost Bearer for Lube only

(Rpt_492) Availability to Target Summary

The logic in this report is very similar to the Snapshot - The Worm report. In this report there is a measure of the number of hours downtime variance between "actual downtime hours" and " downtime hours to achieve the availability target".

Tips

Definition of "Daily Actual: This is the **previous** shift's availability. So, if currently 10:00 am, this is from 8:00 pm the previous night to 8:00 am this morning.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|YearStart|

Report parameters

Default.

Additional filters

Responsibility, Planned, Task Type, Breakdown, System, Maint. Order Type, Availability, Planning Group

(Rpt_613) Asset Valuation

Shows a listing of assets or equipment at a point in time and their evaluations. This incorporates the Capital Cost, normal Depreciation and the Component Based Depreciation.

Tips

Shows the real book value of an asset that should align to market value by adjusting for the remaining life on components.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|,|ProjectionTypes|

Report parameters

Default.

Additional filters

Region, Industry Group, Cost Centre, Equipment Group, Eqp. Criticality, Manufacturer, Project Customer Terminated, Cost Responsibility, Equipment Class, Eqp. Category, Project Name, Hire Status

(Rpt_646) Availability Snapshot Worm

Provides a combined graphical representation of actual downtime versus target and actual availability versus target. It illustrates the impact of downtime on availability, and highlights when each is trending towards falling short of targets, so to trigger preventative, proactive actions.

Tips

Similar in logic to the Snapshot - The Worm report but adds availability actuals versus target to the analysis.

If actual downtime exceeds, or is trending towards, alert (that is, allowable downtime while still achieving availability target), you need to consider what measures can be put in place to reduce future downtime so the alert is not exceeded.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Availability Type, Task Type, Responsibility

(Rpt_659) Actual Sales Analysis

Provides flexible reporting analysis of actual product support sales.

Tips

Analyses current data (updated nightly) and applies to future data, not historical data. You can dynamically drill down into selected areas.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report Parameters

Report Type, Date Selector

Additional filters

X-Axis, Y-Axis, Cost Or Sell, Data, Industry, Sales Zone L1, Classification, Account Manager, PLM, Major Comm Code, Minor Comm Code, Bec Code, Version

(Rpt_797) Asset Cost Worksheet - Modelling

Flexible cost analysis report that combines the Strategy Tasks costs set up in a Projection with the Other Operating Costs set up for that equipment.

Tips

Lets you evaluate and forecast all equipment-related costs - repair and maintenance costs, insurance, fuel, depreciation and so on.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Project, Region, Division, industry Group, Cost Responsibility, Component Cable, Equipment Class, Equipment Group, Eqp Category, Cost Bearer, Cost Centre, Expense Element, System, Sub System, Cost Type, Eqp Criticality, Manufacturer

(Rpt_808) Asset Geolocation History

This report provides the geolocation history for selection of assets.

Tips

Click on the GPS coordinates to view the location in your browser.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|, |ReportStyleKey|

Report parameters

Default.

Additional filters

None

(Rpt_842) AMT Audit Log Report

Initial parameters

|ControlType|,|Idlist|,|TextList|,|LogoLocation|,|UserID|,|LanguageKey|,
|ReportStyleKey|

Tips

The report can be run for all changes across all entities (equipment, functional location, strategy task or standard job), or by selecting one or more entities. Changes can be reported "from a date" to the current date and time by leaving the To Date and Time blank.

Report parameters

Default.

Additional filters

Entity Type, Entity, User, Action

B

(Rpt_126) Backlog Ageing

(Rpt_127) Billings Interrogation

(Rpt_128) Billings Alignment

(Rpt_129) Billings Profit Balance Chart

(Rpt_130) Billings Reconciliation

(Rpt_131) Breakdown Downtime Events %

(Rpt_132) Breakdown Downtime Hours %

(Rpt_135) Budget vs Current Projection Details

(Rpt_357) Backlog Work Analysis

(Rpt_494) Budget Compliance

(Rpt_496) Budget Compliance - Modelling

(Rpt_540) Budget Forecast Performance

(Rpt_541) Budget Equipment Cost Analysis

(Rpt_543) Budget Component Performance

(Rpt_837) Budget Comments Report

(Rpt_126) Backlog Ageing

This report displays two bar graphs showing the average days and the average number of backlogs. You are able to select the different task types or priority to assist with the analysis. Navigate to **Reporting & Analyses** > **Maintenance Efficiency** > **Planning**.

High backlogs are one of the key maintenance "lead indicators" and can indicate a number of potential issues.

1. Increase in maintenance workload from:
   - a change in the application/operating practices.
   - product problems.
   - quality of maintenance work.
2. Shortage of maintenance labour.
3. Shortage of maintenance parts.

As backlogs increase, the risk of failures increases. It also highlights a potential future drop in availability. The age of the backlog is a good way of determining the effectiveness of planning, scheduling and execution.

High levels of recent defect backlogs indicate maintenance execution is identifying new work during inspections. Low levels of older work orders indicate that backlogs are being executed and closed out in a timely manner.

Tips

Good practice requires the planner to manage all aspects of the known workload by having an up-to-date backlog listing, correctly coded to facilitate good planning, and ensure efficient and effective planning processes

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Priority, Task Type

(Rpt_127) Billings Interrogation

A text report that calculates the % of future billings that need to go to the company so that its billings are in line with its costs.

Tips

The report is used to practically adjust future MARC invoicing so that there is alignment between costs and billings within the companies. An example would be, if 65% of the costs are projected to be incurred by Company A then if only 40% of the billings to date.

Report parameters

Default.

Additional filters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

(Rpt_128) Billings Alignment

A text report that calculates the percentage of future Billings that need to go to the first Company so that its billings are inline with its costs. It is used to practically adjust future MARC invoicing so that there is alignment between costs and billings within the companies. An example would be, if 65% of the costs are projected to be incurred by Company A, if only 40% of the billings to date has gone to Company A, maybe 80% of future billings need to go to Company A in order for the total MARC billings to average 65% to Company A.

Report parameters

Default.

Additional filters

None

(Rpt_129) Billings Profit Balance Chart

A bar chart that shows the Billings, Costs and Margin for the group and each company individually.

Tips

This report is used to ensure there is a reasonable alignment of margin against each Company. For example, you do not want one company making a 30% margin while the other makes a 20% loss.

Report parameters

Default.

Additional filters

None.

(Rpt_130) Billings Reconciliation

Highlights if there is any variance between the actual billings to the customer and the expected billings from AMT. This needs to be reconciled to ensure that accurate data is in AMT and that the correct billings are being passed onto the client.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Use Actual UOM Reading, Overheads/Exp

See Reconcile Billings for more information.

(Rpt_131) Breakdown Downtime Events %

Measures the overall quality of the work performed. Like Mean Time Between Stoppages, it is also a measure of the reliability delivered to the customer. High levels of breakdowns can indicate a number of issues. Use the different filters and analyse by options to trend, then drill down to identify the cause. When analysing, the breakdowns focus on the components and cause of failures and how the failures could be prevented.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,|UserID|,|AutoLaunch|,
|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|,
|AnalyseBy|,|ResponsibilityID|

Report parameters

Default.

Additional filters

Cost Type, Responsibility, System, Task Type, Sub System, Strategy Task, Component Code

(Rpt_132) Breakdown Downtime Hours %

Measures the overall quality of the work performed. Like Mean Time Between Stoppages, it is also a measure of the reliability delivered to the customer. High levels of breakdowns can indicate a number of issues. Use the different filters and analyse by options to trend, then drill down to identify the cause. When analysing the breakdowns focus on the components and cause of failures and how the failures could be prevented.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,
|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,
|Branch|,|Site|,|Fleet|,|EqpPlan|,|AnalyseBy|,|ResponsibilityID|

Report parameters

Default.

Additional filters

Cost Type, Responsibility, System, Task Type, Sub System, Strategy Task, Component Code

(Rpt_135) Budget vs Current Projection Details

Lets the comparison of actual performance to the original budget.

Tips

Actual and projected costs are shown at cost at sell.

Report parameters

Default.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Additional filters

Cost Type, Component Code, System, Task Type, Sub System, Strategy Task, Cost Selector

(Rpt_357) Backlog Work Analysis

This report shows the average planned labour hours and average planned duration hours.

High backlogs are one of the key maintenance "lead indicators" and can indicate a number of potential issues.

1. Increase in maintenance workload from:
   - a change in the application/operating practices.
   - product problems.
   - quality of maintenance work.
2. Shortage of maintenance labour.
3. Shortage of maintenance parts.

As backlogs increase, the risk of failures increases. It also highlights a potential future drop in availability. The age of the backlog is a good way of determining the effectiveness of planning, scheduling and execution.

High levels of recent defect backlogs indicate maintenance execution is identifying new work during inspections. Low levels of older work orders indicate that backlogs are being executed and closed out in a timely manner.

Tips

This report only displays backlog tasks.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Priority, Strategy Task, Task Type

(Rpt_494) Budget Compliance

Compares actual costs for the current month, financial year to date and projected for the current financial year to a budget.

Tips

Only budgets created in AMT's Budgeting module can be used in this report for comparison.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Task Counter, Work Group

(Rpt_496) Budget Compliance - Modelling

Compares actual costs for the current month, financial year to date and projected for the current financial year to a budget.

Tips

Applicable to Modelling equipment only. Only budgets created in AMT's Budgeting module can be used in this report for comparison.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Task Counter, Work Group

(Rpt_540) Budget Forecast Performance

This report is based on Budgeting module data that allows comparison of a Budget, a Forecast and live AMT data.

Tips

"Forecast" is a nominated Budget.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|PeriodStart|,|PeriodEnd|,|UserID|

Report parameters

Default.

Additional filters

Cost Centre, CC Activity, CC Location, CC Responsibility, Cost Expense, Cost Category, CC Code, Cost Bearer, Include

(Rpt_541) Budget Equipment Cost Analysis

This report is based on Budgeting module data that allows comparison of a Budget, a Forecast and live AMT data.

Tips

This report can be run at the Equipment level only.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Task Counter, Work Group, Cost Bearer, Cost Centre, Cost Category, Expense Element, CC Activity, CC Location, CC Responsibility, Parts/Labour/Misc

(Rpt_543) Budget Component Performance

Based on Budgeting module data that allows comparison of component costs and occurrences in a Budget to live AMT.

Tips

Shows costs and occurrences for the Total Budget Period vs Actuals to Date vs Projected Future vs Total Projected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|,|MinCompCost|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Task Counter, Work Group, Cost Bearer, Cost Centre, Cost Category, Expense Element, CC Activity, CC Location, CC

Responsibility, Parts/Labour/Misc, Identified Components, Budget or Actual Annual Cost >= Nominated Amount

(Rpt_837) Budget Comments Report

Report displaying Comments of the selected Budget. You can select Latest comments only, All valid comments, All comments including deleted comments or Deleted comments. Strikethrough on the report indicates deleted comments.

Initial parameters

|ControlType|,|Idlist|,|TextList|,|LogoLocation|,|UserID|,|LanguageKey|,
|ReportStyleKey|

Report parameters

Default.

Additional filters

Employee, Branch, site, Fleet, Equipment, Cost Centre, Budget Status, Comment Level, Labour Activity, Show

C

(Rpt_137) Cash Flow Chart

(Rpt_138) Cash Flow Statement

(Rpt_139) CBD Analysis

(Rpt_140) Changeout Analysis

(Rpt_141) Changeout Schedule

(Rpt_142) Component Life Distribution

(Rpt_143) Component Status

(Rpt_144) Contract Invoice Details

(Rpt_145) Contribution Summary

(Rpt_146) Cost Analysis - "Life Cycle"

(Rpt_147) Cost Analysis - "To Date"

(Rpt_148) Equipment Cost and Recovery Analysis

(Rpt_149) Cost Cap Financial Summary

(Rpt_151) Cost Per Interval

(Rpt_152) Cost per UOM - by Date

(Rpt_153) Cost per UOM - by Usage

(Rpt_156) Costs

(Rpt_154) Cost per UOM - Comparison

(Rpt_155) Overhead Cost Variance Analysis

(Rpt_277) Cost Analysis - "Life Cycle" - Modelling

(Rpt_278) Cost per Interval - Centreline

(Rpt_280) Cost per UOM - by Date - Centreline

(Rpt_281) Cost per UOM - by Date - Modelling

(Rpt_282) Cost per UOM - by Usage - Centreline

(Rpt_283) Cost per UOM - by Usage - Modelling

(Rpt_284) Cost per UOM - Comparison - Centreline

(Rpt_285) Cost per UOM - Comparison - Modelling

(Rpt_286) Cost Analysis - "Life Cycle" - Centreline

(Rpt_333) Cash Flow Chart - Modelling

(Rpt_334) Cash Flow Statement - Modelling

(Rpt_336) Costs at Cost - Modelling

(Rpt_382) Change Analysis

(Rpt_385) Change Analysis - Modelling

(Rpt_386) Changeout Schedule - Modelling

(Rpt_387) Changeout Analysis - Detailed - Modelling

(Rpt_388) Forecast Worksheet - Modelling

(Rpt_389) Component History

(Rpt_392) Consumption per UOM

(Rpt_394) Consumption per Month

(Rpt_415) Component Status (Suppression Impact)

(Rpt_419) Contract Reconciliation

(Rpt_443) Consumables Forecast

(Rpt_444) Consumables Forecast - Modelling

(Rpt_445) Consumables Forecast - Centrelines

(Rpt_488) Contract Billings vs Cost Reconciliation

(Rpt_495) Cost Trending

(Rpt_497) Cost Trending - Modelling

(Rpt_535) Cost Per Interval - Customer

(Rpt_564) Changeout Forecast Workload

(Rpt_635) Customer Audit

(Rpt_635) Customer Sales Activity

(Rpt_651) Cost Per Interval Comparison

(Rpt_798) Changeout Analysis - Detailed - Modelling - Multiple Asset Quantity

(Rpt_800) Cost Analysis - "Life Cycle" - Modelling - Multiple Asset Quantity

(Rpt_811) Contract Revenue Recognition - @Cost

(Rpt_137) Cash Flow Chart

Bar chart displaying Cash Inflow and Outflows with the resulting Period (Net) Cash Flows.

- Use this report to identify periods of negative and low position cash flow.
- Cash inflows are based on the equipment's billing schedule.
- Cash outflows are based on the estimate projection's costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|AutoLaunch|,|FleetID|, |EqpPlanID|,|Fleet|,|EqpPlan|,|AnalyseBy|,|ProjectionTypeID|,|ProjectionType|

Report parameters

Default.

Additional filters

None.

(Rpt_138) Cash Flow Statement

Provides the ability to project future cash flows to highlight any potential problems so that management can start planning for issues now. Cash in and cash out are significantly different for contracts.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|AutoLaunch|,|BranchID|,|SiteID|, |FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|,|DateChecked|,|StartDate|,|EndDate|, |AnalyseBy|

Report parameters

Default.

Additional filters

None.

(Rpt_139) CBD Analysis

Displays a breakdown of each strategy task and the base details of the task. AMT calculates the actual CBD provision for each strategy task. It also calculates the actual CBD hire rate credits from the plant hire invoices and indicates the over or under price.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|LastMonth|,|ProjectionTypes|, |UserID|,|AutoLaunch|,|EqpPlanId|,|CurrencyId|,|Currency|,|EqpPlan|,

|ModelId|,|Model|,|SiteId|,|Site|,|AnalyseBy|

Report parameters

Default.

Additional filters

Comp Code, Expense Element, Task Type, Currency, Strategy Task, CBD Tasks Only

(Rpt_140) Changeout Analysis

Provides a graphical view of the number of components to be charged in the time period required. You can navigate to this report in the following ways:
Long-term Planning > **Components**.
Modelling > **Components**.

Tips

Date selector defaults to entire plan term but can be modified to suit user requirements.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Feature_32|

Report parameters

Default.

Additional filters

System, Component Code, Strategy Task, Part Classification, Modifier Code, Part Rating, Occurrence Type, Primary Part No, Cost Type, Rebuild Location, Sub System, Review Status, Task Type, Sales Status

(Rpt_141) Changeout Schedule

Provides details of next changeout scheduled to allow for parts supply, labour resources and production schedules to be managed effectively

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Task Type, Part Classification, System, Part Rating, Component Code, Primary Part No, Cost Type, Rebuild Location, Sub System, Review Status, Task Counter, Sales Status, Strategy Task

(Rpt_142) Component Life Distribution

Reports allows analysis in a graphical view of the component lives achieved and quick identification of problem areas.

Tips

Review major components using the strategy task filter.

Compare current expectation versus the trend shown in the report. Inconsistencies suggest a change in current projection frequency parameterAdditional.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|AutoLaunch|,
|QUOMID|,|QUOM|,|SiteID|,
|FleetID|,|ModelID|,|CompCodeID|,|TaskTypeID|,|ModifierID|,|AppCodeID|,|Site|,
|Fleet|,|Model|,|CompCode|,|TaskType|,|Modifier|,|AppCode|,
|EqpPlanID|,|EqpPlan|,|UserID|

Report parameters

Default.

Additional filters

+Comp Code, Original Part Type, Modifier Code, Warranty, Occurrence Type, Redo, +Strategy Task, Life Confirmed, Task Type, Rating Average, Task Counter, UOM

+ indicates that you must select a Strategy Task or Component Code.

(Rpt_143) Component Status

Shows a component overview of the machine including overdue components, next changeouts and historical workorders.

Tips

Run the report across a fleet of equipment, analysing by Component Code to compare between equipment Last Change Out Details (Life Used,Cost), and Next Changeout Details (Life Used).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|EqpPlanID|,|EqpPlan|,
|ProjectionTypes|,|LogoLocation|,|CostBearer|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Component Code, System, +Cost Bearer, Sub System, Task Type

 **+** indicates that Cost Bearer filters costs only (not Strategy Task list).

(Rpt_144) Contract Invoice Details

Provides an equipment billings summary for approval by the client. Ensures there is a consistent record of invoicing and that billings are managed correctly.

Tips

The report can include test time. Set this functionality in the **Equipment Editor** > **Configuration** tab > **Marc** quick button - **Record Test Time**.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Site, Currency, Contract, Use Actual UOM Reading, Schedule

See Contract Invoice Details for more information.

(Rpt_145) Contribution Summary

Shows graphically the projected margin versus the budget margin over time.

Tips

Costs are shown as Cost at Cost.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Additional filters

None.

(Rpt_146) Cost Analysis - "Life Cycle"

Shows graphically, or in a text report, the major cost areas of the equipment for the entire plan term or to the month to date.

Run the report for the life-cycle or a shorter period (for example, a year). Running as a Pie Chart, "Analyse By" Component shows the top 20 component cost areas.

Tips

Can be run in Text, Bar or Pie chart format. Bar Chart shows the top 10 costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|CostBearerID|,|ProjectionTypes|,|EquipmentTypes|,
|ProjHeaderID|,|ProjHeader|,|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,
|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|,
|AnalyseBy|,|CurrencyID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, Work Group, System, Cost Centre, Sub System, Cost Category, Strategy Task, Expense Element, Strategy Task Desc, Maintenance Type, Task Type, Task Mode, Component Code, Costs, Modifier Code, Parts/Labour/Misc, Cost Bearer, Currency, Task Counter

(Rpt_147) Cost Analysis - "To Date"

Lets you list and analyse cost values in descending value based on a wide variety of filters. This report is for analysis of costs to date, not for future projected costs to date, not for future projected costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Contract, Cost Type, Strategy Task, System, Occurrence Type, Sub System, Modifier Code, Component Code, Task Counter, Task Type, Cost Bearer, WO Status, P/L/M

(Rpt_149) Cost Cap Financial Summary

This report is specific to the Sales Tool module - Cost Cap agreement type. The report provides information about the Billings and Costs at Cost.

Tips

Billings amount = Calculated Cost +/- discounts +/- Risk Premium. Cost at Cost is calculated taking into account the financial margins in the system tables.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Manufacturer, Model, Equipment Group, Equipment Class, Region, Division, Equipment Location, Eqp Criticality, Eqp Category, Industry Group, Cost Responsibility, Cost Centre, CC Activity, CC Location, CC Responsibility, Expense Element, Cost Bearer, Cost Type, System, Sub System, Strategy Task, Component Code, Task Type, Modifier Code

(Rpt_151) Cost Per Interval

Cost per interval is a Standard contract report showing cost at sell per interval.

The report lets you nominate a period of hours to analyse what equipment costs will fall in that timeframe. Can be analysed using either the Budget or Current projection and by numerous other filters.

Tips

This report can assist management in understanding what costs have occurred during what period of time in the equipment life cycle and can assist in estimating other machine costs of a similar nature.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,|CostBearer|,
|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre. Expense Element, Maintenance Type, Item, Item

(Rpt_152) Cost Per UOM - by Date

Compares visually the expected hourly costs by machine. It shows the cumulative cost per UOM for the period of time selected.

"Analyse By" Model/System/Sub System for a date range. Lets you see the trending of unassigned costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_153) Cost per UOM - by Usage

Depicts the costs at intervals of the equipment usage (engine hours) throughout the term of its use.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task

(Rpt_156) Costs

This report provides the ability for analysing historic and Forecast costs using a variety of report filters.

Tips

Initially analyse costs by System. Then drill down into subsystems and components to identify the major contributors to high costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,
|Enable_Cost_At_Cost|,|CostBearer|,|ProjectionTypes|,|EquipmentTypes|,
|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Region, Eqp Criticality, Equipment Group, Eqp Category, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Modifier Code, Division, Equipment Class, Equipment Location, Parts/Labour/Misc,

Currency

(Rpt_154) Cost per UOM - Comparison

Compares visually the expected hourly costs by machine. The report shows the cumulative cost per UOM for the period of time selected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_277) Cost Analysis - "Life Cycle" - Modelling

Shows graphically, or in text, the costs for the entire plan term or to the month to date.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,|CostBearer|,|CostBearerID|,
|ProjectionTypes|,|EquipmentTypes|,|ProjHeaderID|,|ProjHeader|,
|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,
|Branch|,|Site|,|Fleet|,|EqpPlan|,|AnalyseBy|,|CurrencyID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Task Mode, Costs, Parts/Labour/Misc, Currency

(Rpt_278) Cost per Interval - Centerline

Lets you nominate a period of hours to analyse what equipment costs will fall in that timeframe. Can be analysed using either the Budget or Current projection and by numerous other filters.

Tips

This report can assist management in understanding what costs have occurred during what period of time in the equipment life cycle. It can also assist in estimating other machine costs of a similar nature.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modify Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Item, Item

(Rpt_279) Cost per Interval - Modelling

Lets you nominate a period of hours to analyse what equipment costs will fall in that timeframe. Can be analysed using either the Budget or Current projection and by numerous other filters.

Tips

This report can assist management in understanding what costs have occurred during what period of time in the equipment life cycle. It can also assist in estimating other machine costs of a similar nature.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Item, Item

(Rpt_280) Cost per UOM - by Date - Centreline

Visually compares the expected hourly costs by machine. It shows the cumulative cost per UOM for the period of time selected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_281) Cost per UOM - by Date - Modelling

Visually compares the expected hourly costs by machine. It shows the cumulative cost per UOM for the period of time selected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_282) Cost per UOM - by Usage - Centreline

Depicts the costs at intervals of the equipment usage (engine hours) throughout the term of its use.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_283) Cost per UOM - by Usage - Modelling

Depicts the costs at intervals of the equipment usage (engine hours) throughout the term of its use.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_284) Cost per UOM - Comparison - Centreline

Compares visually the expected hourly costs by machine. It shows the cumulative cost per UOM for the period of time selected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_285) Cost per UOM - Comparison - Modelling

Visually compares the expected hourly costs by machine. It shows the cumulative cost per UOM for the period of time selected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_286) Cost Analysis - "Life Cycle" - Centreline

Provides users the ability to analyse the overhead costs of a project against a projection.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|CostBearerID|,|ProjectionTypes|,|EquipmentTypes|,
|ProjHeaderID|,|ProjHeader|,|UserID|,|AutoLaunch|,|BranchID|,
|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,
|Fleet|,|EqpPlan|,|AnalyseBy|,|CurrencyID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Task Mode, Costs, Parts/Labour/Misc, Currency

(Rpt_323) Component Variance Analysis Report

This report has two sections:

1. The top section of the report provides a summary of the projected task's performance.
2. The bottom section of the report provides an analysis of the projected task's performance per equipment within the selected fleet.

Tip

This report helps to identify equipment which are experiencing a greater number of occurrences (such as component changeouts) compared to other equipment in the fleet. In other words, it is easy to identify "problem equipment".

Initial parameters

|AutoLaunch|,|Branch|,|BranchID|,|Site|,|SiteID|,|Fleet|,|FleetID|,|Model|,
|ModelID|,|EqpPlan|,|EqpPlanID|,|Component|,|ComponentCodeID|,|
ModifierIDNN|,|TaskTypeIDNN|,|ApplicationCodeIDNN|,|UserID|

(Rpt_324) Component Variance Summary

For the selected fleet and or projected tasks, this report shows Costs To Date (Budget vs. Actual, and Variance) and Projected Costs (Budget vs. Projected, and Variance).

Tip

This report is a simpler version of the Life Cycle Cost Summary report ("Future" columns in the Life Cycle Cost Summary report are excluded), and helps the user quickly identify Top 10 variances.

Initial parameters

|AutoLaunch|,|Branch|,|BranchID|,|Site|,|SiteID|,|Fleet|,|FleetID|,|Model|,
|ModelID|,|EqpPlan|,|EqpPlanID|,|CostType|,|CostTypeID|,|System|,|SystemID|,
|SubSystem|,|SubSystemID|,|Top|,|UnassignedVal|,|UnassignedDesc|,|
PlannedVal|,|PlannedDesc|,|SortVal|,|SortDesc|,|UserID|

Report parameters

Default.

(Rpt_333) Cash Flow Chart - Modelling

Bar chart displaying Cash Inflow and Outflows with the resulting Period (Net) Cash Flows.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,
|AutoLaunch|,|FleetID|,|EqpPlanID|,|Fleet|,|EqpPlan|,
|AnalyseBy|,|ProjectionTypeID|,|ProjectionType|

Report parameters

Default.

Additional information

None.

(Rpt_334) Cash Flow Statement - Modelling

Provides the ability to project forward future cash flows to highlight any potential problems so that management can start planning for issues now. Cash in and cash out are significantly different for contracts.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|
AutoLaunch|,|FleetID|,|EqpPlanID|,|Fleet|,|EqpPlan|,|AnalyseBy|,
|ProjectionTypeID|,|ProjectionType|,|GroupBy|

Report parameters

Default.

Additional filters

Business Group, Contract, Equipment, Projection Type, Contract Type, Fleet, Model

(Rpt_336) Costs at Cost - Modelling

Provides the ability for sharing historic and forecast costs using a variety of report filters.

Tips

Initially analyse costs by System, then drill down into sub systems and components to identify the major contributors to high costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,|CostBearer|,
|ProjectionTypes|,|EquipmentTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Region, Eqp Criticality, Equipment Group, Eqp Category, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Modifier Code, Divison, Equipment Class, Equipment Location, Parts/Labour/Misc, Currency

(Rpt_382) Change Analysis

Shows the change in key parameters over time, and lets you drill down into the reasons for the changes.

Tips

Costs shown are for the default Cost Bearer, and are shown as Cost at Sell.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Snapshots|,|UserID|

Report parameters

Default.

Additional filters

System, Strategy Task, Modifier Code, Occurrence Type, Cost Type, Sub System, Task Type, Component Code, Part Classification, Part Rating, Primary Part No, Rebuild Location, Review Status, Sales Status

(Rpt_385) Changeout Analysis - Modelling

Provides a graphical view of the number of components to be changed in the time period required.

Tips

Date selector defaults to entire plan term, but can be modified to suit user requirements.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Feature_32|

Report parameters

Default.

Additional filters

System, Strategy Task, Modifier Code, Occurrence Type, Cost Type, Sub System, Task Type, Component Code, Part Classification, Part Rating, Primary Part No, Rebuild Location, Review Status, Sales Status.

(Rpt_386) Changeout Schedule - Modelling

Provides details of next changeout scheduled to allow for parts supply, labour resources and production schedules to be managed effectively.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Task Type, System, Component Code, Cost Type, Sub System, Task Counter, Strategy Task, Part Classification, Part Rating, Primary Part No, Rebuild Location, Review Status, Sales Status

(Rpt_387) Changeout Analysis - Detailed - Modelling

Provides a graphical view of the number of components to be changed in the time period required.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|UserID|,
|Feature_32|

Report parameters

Default.

Additional filters

System, Strategy Task, Strategy Task Desc, Modifier Code, Occurrence Type, Cost Type, Sub System, Task Type, Component Type, Region, Division, Cost Responsibility, Equipment Location, Part Classification, Part Rating, Primary Part No, Rebuild Location, Review Status, Sales Status

(Rpt_389) Component History

Shows the technical details for component changeouts. Also shows the specific work order history and Partial Life and Swing components.

Tips

This report can be used for simple tracking component history where the rotables module is not being used. Also useful for looking at the history where swing and partial life parameters have been used.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|EquipmentTypes|,|AutoLaunch|,
|SerialNumber|,|IsExactMatch|,|EqpPlan|,|EqpPlanID|,|ProjTaskID|,
|ProjTask|,|SerailNumberDesr|,|UserID|

Report parameters

Default.

Additional filters

Component Serial No, Component Code, Strategy Task, Modifier Code, Task Type, CCC, Show IN USE Comp, Show Serialised

(Rpt_392) Consumption per UOM

Shows the consumption details - per month and per usage, for example, oil consumption is 0.1 litres per engine hour, or petrol per kilometre.

Tips

You are not able to select a forward date, the default for this report is the past six months - including this month.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Element, UOM

(Rpt_394) Consumption per Month

Shows the consumption details - per month and per usage, for example, oil consumption is 0.1 litres per engine hour, or petrol per kilometre.

Tips

You are not able to select a forward date, the default for this report is the past six months - including this month.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Element

(Rpt_415) Component Status (Suppression Impact)

Similar to the standard Component Status report except that the Last Occurrence details of strategy task, which is part of a suppression group, will reflect when the parent task was performed.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|EqpPlanID|,|EqpPlan|,|ProjectionTypes|,
|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Task Type, Component Cable, +Cost Bearer

+ indicates filters Costs only, (not Strategy Task List).

(Rpt_419) Contract Reconciliation

Displays To Date, Future and Total Projected Billings and Costs (at Sell).

Tips

This report only considers Contract Costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,
|EqpPlanID|,|EqpPlan|,|ProjectionTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

None

(Rpt_429) Cost per UOM per Interval

Similar to the Cost Per Interval report except that this report shows costs as a cost per UOM (for example, engine hours) rather than a total cost.

Tips

This report does not include actuals for Current and Alternate projections.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Labour Activity, Job Code, Costs

(Rpt_430) Cost per UOM per Interval - Modelling

Similar to the Cost Per Interval report except that this report shows costs as a cost per UOM (e.g. engine hours) rather than a total cost.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Labour Activity, Job Code, Costs

(Rpt_431) Cost per UOM per Interval - Centrelines

Report which lists the Primary Part Number, Group Number and Component Serial Number per strategy task for the selected equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Labour Activity, Job Code, Costs

(Rpt_443) Consumables Forecast

Shows Forecast Consumable Usage.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Part Type, Job Code, Show

(Rpt_444) Consumables Forecast - Modelling

Shows Forecast Consumable Usage.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Part Type, Job Code, Show

(Rpt_445) Consumables Forecast - Centrelines

Shows Forecast Consumable Usage.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Part Type, Job Code, Show

(Rpt_478) Contract Billings vs Cost Reconciliation

Lets you compare separate billing projections to costs.

Tips

An example: A user has created a billing projection for GET. This report lets you compare the GET billing projection (that is, GET revenue) versus GET costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Task Type, Cost Bearer, Cost Centre, Expense Element, Parts/Labour/Misc

(Rpt_486) Contract Header

This report generates the Contract Header for a Modelling equipment, where the Modelling equipment, where the Modelling equipment represents a contract developed in AMT by an OEM or Dealer.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

.

(Rpt_495) Cost Trending

Trends actual and projected costs of an equipment over time.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,
|EquipmentTypes|,|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Parts/Labour/Misc, Display, Currency

(Rpt_497) Cost Trending - Modelling

Trends actual and projected costs of an equipment over time.

Tips

Applicable to modelling equipment only.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,
|EquipmentTypes|,|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Parts/Labour/Misc, Display, Currency

(Rpt_535) Cost Per Interval - Customer

This is the standard Cost Per Interval report, but in addition it contains logic which enables it to balance to provisions made to an estimate created using the Sales Tool.

Tips

This report is only relevant where the Sales Tool module in AMT is used.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Labour Activity, Job Code

(Rpt_564) Changeout Forecast Worksheet

Combines current component life and cost forecast for comparison to original budget projection assumptions. Navigate to **Long-term Planning** > **Components**.

Tips

The cost per month is based on the number of occurrences of the strategy in the month multiplied by the current cost of the strategy task.

Life to date of the component in the equipment now, and life achieved at last changeout, recognise the impact of suppression.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|LogoLocation|,|UserID|,|Feature_32|

Additional filters

Task Type, System, Component Code, Cost Type, Sub System, Task Counter, Strategy Task

(Rpt_634) Customer Audit

Displays Customers of AMT.

Tips

Use the Blank Fields filter to identify Customers with missing classifications.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Report Type, Equipment Hierarchy

Additional filters

Display Columns, Blank Fields, Industry, PSS Responsibility

(Rpt_635) Customer Sales Activity

Displays Customer sales activity.

Tips

Use the Blank Fields filter to identify records with missing fields

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Display Fields, Blank Fields, Sales Activity

(Rpt_651) Cost Per Interval Comparison

Based on the standard Cost Per Interval report, this report lets you compare the Cost Per Interval of two different projections side by side.

Tips

Useful for validating centrelines against real equipment, and comparing different modelling equipment to identify variances which may not have been intended.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|UserID|,|LogoLocation|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Labour Activity, Job Code, Costs

(Rpt_798) Changeout Analysis - Detailed - Modelling

Provides a graphical view of the number of components to be changed in the time period required.

You can navigate to this report in the following ways:
Long-term Planning > **Components**.
Modelling > **Components**.

Tips

Date selector defaults to the entire plan term, but can be modified to suit requirements.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

System, Strategy Task, Strategy Task Desc, Modifier Code, Occurrence Type, Cost Type, Sub System, Task Type, Component Code, Region, Division, Cost Responsibility, Equipment Location, Part Classification, Part Rating, Primary Part No, Rebuild Location, Review Status, Sales Status, Project, Def. Maint. Period

(Rpt_800) Cost Analysis - "Life Cycle" - Modelling - Multiple Asset Quantity

Displays the costs for the entire plan term or to the month to date.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,|
CostBearer|,|CostBearerID|,|ProjectionTypes|,|EquipmentTypes|,|ProjHeaderID|,|ProjHeader|,
|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,
|EqpPlan|,|AnalyseBy|,|CurrencyID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub-System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Task Mode, Costs, Parts/Labour/Misc, Currency

(Rpt_811) Contract Revenue Recognition - @Cost

Displays To Date, Future and Total Projected Billings and Costs (at Cost).

Tips

The report only considers Contract Costs.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,
|UserID|,|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

None

(Rpt_325) Changeout Analysis - Detailed

Provides a graphical view of the number of components to be changed in the time period required.

Tips

Date selector defaults to entire plan term, but can be modified to suit user requirements.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|UserID|,
|Feature_32|

Report parameters

Default.

System, Strategy Task, Strategy Task Desc, Modifier Code, Occurrence Type, Cost Type, Sub System, Task Type, Component Type, Region, Division, Cost Responsibility, Equipment Location, Part Classification, Part Rating, Primary Part No, Rebuild Location, Review Status, Sales Status

D

(Rpt_157) Daily Downtime Lag

(Rpt_158) Daily Log - Detailed

(Rpt_159) Daily Log - Summary

(Rpt_160) Daily Machine Performance Check Sheet

(Rpt_161) Daily/Weekly Work Plan

(Rpt_162) Downtime Analysis

(Rpt_436) Duration Per Interval

(Rpt_437) Duration Per Interval - Modelling

(Rpt_438) Duration Per Interval - Centrelines

(Rpt_826) Depreciable Asset Register Report

(Rpt_157) Daily Downtime Log

Details the event details of the equipment and the downtime allocation information.

Tips

This report **does** use the site's shift start time for the Date Range set by the user. The Additional Filters only filter the data displayed on the right side of the report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|UserID|,|LogoLocation|

Report parameters

Default.

Additional filters

Task Type, EM Issue, +Responsibility

+ indicates that the Responsibility filter defines the data on the right side of the report.

(Rpt_158) Daily Log - Detailed

Lets you check the daily events and tasks that have occurred and shows the downtime responsibility.

Tips

This report **does not** use the site's shift start time for the Date Range you set.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, EM Status

(Rpt_159) Daily Log - Summary

Lets you check the daily events and tasks that have occurred and shows the downtime responsibility.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, EM Issue

(Rpt_160) Daily Machine Performance Check Sheet

This report identifies the maintenance KPIs for the machine throughout a given month and provides the ability to proactively plan to reach targets.

Tips

This report provides a good reconciliation point between the client and the maintenance provider. It highlights the amount of available downtime left in the given period. This lets planners to manage the downtime.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

DTResponsibility+, Task Type, System, Availability Type

+ indicates the Downtime Responsibility filter.

(Rpt_161) Daily/Weekly Work Plan

Provides a maintenance schedule to the site foreman and supervisors as notification of future work planned. It is event based with all work orders associated with the event. Allows for better allocation of resources and work flow.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|SiteID|,|Site|,
|FleetID|,|Fleet|,|EqpPlanID|,|EqpPlan|,|UserID|

Report parameters

Default.

Additional filters

Task Type, Work Group, Include Overdue, Work Location,Priority, Responsibility, Maint. Order Type, Planning Group, STO

(Rpt_162) Downtime Analysis

Provides a maintenance schedule to the site foreman and supervisors as notification of future work planned. It is an event-based report, with all work orders associated with the event. The report analyses different categories of downtime. Options include items such as responsibility, system, subsystem, and task type.

"Analyse By" component code to identify high downtime areas.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|SiteID|,|Site|,
|FleetID|,|Fleet|,|EqpPlanID|,|EqpPlan|,|UserID|

Report parameters

Default.

Additional filters

Responsibility, Cost Type, Reason, System, Activity, Sub System, Variance Cause, Component Code, EM Issue, Task Type, Preventable, Source, Planned, Maintenance Order Type, Redo, Planning Group. Breakdown

(Rpt_436) Duration Per Interval

Shows duration estimates per interval.

Tips

This report includes actuals for Current and Alternate projections.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Labour Activity, Job Code, Labour Type

(Rpt_437) Duration Per Interval - Modelling

Shows duration estimates per interval based on the duration estimates within strategy tasks.

Tips

This report is used to assist in calculating availability Forecasts.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Labour Activity, Job Code, Labour Types

(Rpt_438) Duration Per Interval - Centrelines

Shows duration estimates per interval based on the duration estimates within strategy tasks.

Tips

This report is used to assist in calculating availability forecasetss.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Labour Activity, Job Code, Labour Type

(Rpt_826) Depreciable Asset Register Report

Displays Asset Cost, Accumulated Depreciation Written Down Value, %WDV and Residual Value for each depreciable asset.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Region, Industry Group, Cost Centre, Equipment Group, Manufacturer, Project Customer, Project Name, Division, Cost Responsibility, Equipment Class, Equipment Category, Asset Status, Hire Status, Incl Terminated

E

(Rpt_148) Equipment Cost and Recovery Analysis

(Rpt_163) EM Issue List

(Rpt_164) Equipment Backlog Listing

(Rpt_165) Equipment Cash Flow Chart

(Rpt_166) Equipment Down

(Rpt_167) Equipment Performance Summary

(Rpt_168) Event Analysis

(Rpt_226) Employee Allocation

(Rpt_335) Equipment Cash Flow Chart - Modelling

(Rpt_341) Employee Working Status Summary

(Rpt_355) Equipment Operational Performance Summary

(Rpt_390) Equipment Statistics

(Rpt_409) Equipment Cost and Recovery Schedule

(Rpt_506) Equipment Down Alert

(Rpt_518) Event Completion Performance

(Rpt_519) Event Completion Performance SCREEN

(Rpt_542) Equipment KPI

(Rpt_650) Equipment Audit

(Rpt_675) Equipment Expiration

(Rpt_680) Employee Time Validator

(Rpt_780) Extended Planned Workorder Details

(Rpt_148) Equipment Cost and Recovery Analysis

Displays the productivity, costs, revenue and gross over/under recovery results.

Tips

Provides an overall summary of the equipment cost and recovery and shows the balance in the CBD provision. The report can be run in Text, Bar or Pie Chart formats. Bar Chart shows the top 10 costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|LastMonth|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Project, Currency

(Rpt_163) EM Issue List

Defines the equipment management issues that have been registered in the EM Issues module. Details the issue type, date of registration, status, actual downtime hours and so on. Navigate to **Knowledge Database** > **EM Issues** > **EM Issues List**).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report Parameters

Report Type, Equipment Hierarchy

Additional filters

Cost Type, Issue Type, System, Priority, Sub System, EM Status, Component Code, EM Stage, Sort First, Sort Second, Sort Third, No. To Show

(Rpt_164) Equipment Backlog Listing

Designed to manage the existing backlogs against each piece of equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|EqpPlanID|,
|EqpPlan|,|ProjectionTypes|,|UserID|,|BranchID|,
|SiteID|,|FleetID|,|Branch|,|Site|,|Fleet|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Perform At, Priority, OHS, Source, Raised By, Maint. Order Type, Planning Group, STO, Backlog Age, Strategy Tasks

(Rpt_165) Equipment Cashflow Chart

Graphical display of gross margin. Refer to the GM Comparison report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_166) Equipment Down

Shows the pieces of equipment that are currently under maintenance or are down for a particular reason. This data relays directly from the equipment status events and will only show accurately if this data is up to date.

Tips

Ensure that all Equipment Status events are kept up-to-date and that the event status is modified when an event is "In Progress" or "Completed".

Initial parameters

|Con trolType|,|IDist|,|TextList|,|Today|,|UserID|,|AutoLaunch|,
|BranchID|,|SiteID|,|FleetID|, |EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|

Report parameters

Report Type, Equipment Hierarchy.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, EM Issue

(Rpt_167) Equipment Performance Summary

Summaries a number of the key equipment performance indicators and gives the ability to analyse by different levels.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Industry Group

(Rpt_168) Event Analysis

Allows analysis of the actual event data from the equipment status model of the program.

Navigate to **Reporting & Analyses** > **Equipment Performance** > **Stoppages and Downtime** > Event Analysis.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|UserID|

Report parameters

Default.

Additional filters

Responsibility, System, Reason, Sub System, Activity, Component Code, Variance Cause, Task Type, EM Issue, Planned, Preventable, Redo, Breakdown, Maint. Order Type, Cost Type, Planning Group

(Rpt_226) Employee Allocation

Compares the number of hours each work group has been allocated versus the available man hours per work group for the week.

Tips

Date range can be set according to analysis requirements.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|siteID|,|site|,|WorkGroupID|,
|WorkGroup|,|LabourActivity|,|LabourActivityID|,|Today|,|48HoursFromNow|,|UserID|

Report parameters

Default.

Additional filters

Employee, Work Group, Labour Activity, Working Status

(Rpt_335) Equipment Cash Flow Chart - Modelling

Graphical display of cash flow. Refer to the Cash Flow Statement report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Business Group, Contract, Equipment, Contract Type, Fleet, Projection Type

(Rpt_341) Employee Working Status Summary

Lists employees and the tasks to which they have been assigned.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|siteID|,|site|,
|WorkGroupID|,|WorkGroup|,|Today|,|14DaysFromNow|,
|EmployeeWorkingStatus|,|UserID|

Report parameters

Default.

Additional filters

Employee, Work Group, Labour Activity, Working Status, Shift Start Time

(Rpt_355) Equipment Operational Performance Summary

This report contains Availability, MTBS, MTTR, and MTBF statistics. These KPIs are summarised at group level, and details shown at the analysis level. The responsibility filter determines which events to include in the report, and therefore include in MTBS.

Tips

The report begins at the shift start time on the Start Date selected, and if the End Date is in the past, the report calculates to the end of the End Date's shift. If the End Date selected is today, the report uses the current system time.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,
|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

DTResponsibility +

+ indicates the field is the Downtime Responsibility filter.

(Rpt_390) Equipment Statistics

A summary report on the measurements and usages set up against a piece of equipment from the Condition Based Maintenance module.

Tips

This report is only applicable to Real equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|AutoLaunch|,|EqpPlan|,|EqpPlanID|,|UserID|

Report parameters

Default.

Additional filters

Branch, Site, Fleet, Model, Equipment

(Rpt_409) Equipment Cost and Recovery Schedule

Displays the variance between the equipment costs and the revenue collected. The costs and revenue are shown by expense element. The report also displays the Component Based Depreciation by expense element.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Region, Division, Industry Group, Eqp Criticality, Equipment Class, Equipment Group, Eqp Category, Manufacturer, Cost Responsibility, Eqp Cost Centre, Eqp Customer, Location, Currency

(Rpt_506) Equipment Down Alert

Contains information about all equipment which are down.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, Responsibility, Equipment Location

(Rpt_518) Event Completion Performance

Highlights events which have taken longer than expected to complete by comparing Planned Downtime to Actual Downtime (per event).

Tips

The Main Downtime Variance reason is sourced from the event's downtime allocation records. If there are multiple downtime allocations, the report shows the Variance Cause from the record which is not the primary, and has the largest actual downtime.

Initial parameters

|ControlType|,|Idlist|,|TextList|,|AutoLaunch|,|Today|,|48HoursFromNow|,|UserID|

Report parameters

Default.

Additional filters

Component Code, Task Type, Raised By, Priority, Strategy Task, Event Status, Planned, Strategy Tasks, Include Default Variance Cause, 1 hour ago, Exceptions Only

(Rpt_519) Event Completion Performance SCREEN

Highlights events which have taken longer than expected to complete by comparing Planned Downtime to Actual Downtime (per event).

Tips

The Main Downtime Variance reason is sourced from the event's downtime allocation records. If there are multiple downtime allocations, the report shows the Variance Cause from the record which is not the primary, and has the largest actual downtime.

Initial parameters

|ControlType|,|Idlist|,|TextList|,|AutoLaunch|,|Today|,|48HoursFromNow|,|UserID|

Report parameters

Default.

Additional filters

Component Code, Task Type, Raised By, Priority, Strategy Task, Event Status, Planned, Strategy Tasks, Include Default Variance Cause, 1 hour ago, Exceptions Only

(Rpt_542) Equipment KPI

Details key Equipment KPIs for a specific time period.

Tips

Shows Hours Worked, Scheduled Hours, Downtime Hours (Planned vs Breakdown vs Other) and Stoppages (Planned vs Breakdown vs Other).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|,|Responsibility|,|LogoLocation|

Report parameters

Default.

Additional filters

Responsibility

(Rpt_650) Equipment Audit

Use this report to see what the configurations for the equipment are; you can also use the Blank Fields filter to identify Equipment with missing classifications.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Display Columns, Blank Fields, Product Type Code, Product Type Desc, Application Code, App Code Desc, Application Severity, Sale Indicator, Territory Indicator, Engine Make, Engine Serial No, Excl From Sales

(Rpt_675) Equipment Expiration

Useful to identify when equipment are due to complete their life cycle within the date range specified. This report only runs for REAL equipment.

Tips

RVIC % is calculated following the same logic used in the RVIC Analysis report, except that it runs for the End Date per equipment, and only includes strategy tasks which have the Task Type flagged in the Task Type system table as Component Changeout.

The report will highlight in Blue either the End Usage or End Date depending on which parameter the user has selected as the end point of the equipment.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Branch, Site, Fleet, Model, Equipment, Region, Division, Parent Equipment, Eqp Location, Eqp Category, Eqp Cost Centre, Cost Responsibility, Equipment Class, Equipment Group, Customer, Contract, Manufacturer, Industry Group, Eqp Criticality, Sales Resp.

(Rpt_680) Employee Time Validator

Useful to validate the amount of time an employee has worked over a period of time.

Tips

This report returns results for non-terminated employees only.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

None

(Rpt_780) Extended Planned Workorder Details

Lets you select areas of the downtime tasks that you wish to analyse with numerous display and sort fields to select from. A very open report, which lets you customise your required data.

Tips

Date range can be set according to analysis requirements.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Serial Number, Modifier, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

F

(Rpt_169) Financial Risk Summary

(Rpt_170) Financial Summary

(Rpt_171) Fleet Component Performance Summary

(Rpt_172) Forecast Availability

(Rpt_173) Forecast Cost Analysis

(Rpt_174) Forecast Worksheet

(Rpt_287) Forecast Availability - Modelling

(Rpt_318) Fleet Cost and Recovery Schedule

(Rpt_383) Financial Summary - Modelling

(Rpt_287) Forecast Availability - Modelling

(Rpt_393) Failure Indicators Analysis
(Rpt_400) Failure Mode Details
(Rpt_413) Fleet Statistics
(Rpt_562) Forecast Rotables Stock
(Rpt_657) Forecast Market Analysis
(Rpt_658) Forecast Market Share Analysis
(Rpt_778) Forecast Worksheet (Serial Number)
(Rpt_781) Functional Location Budget Analysis
(Rpt_782) Functional Location Budget v Actual Analysis
(Rpt_783) Functional Location Cost Analysis
(Rpt_784) Functional Location ERP Master Data Analysis
(Rpt_785) Functional Location Forecasting Maturity
(Rpt_786) Functional Location Cost Details
(Rpt_816) Forecast Worksheet - @Cost
(Rpt_817) Forecast Worksheet - @Cost - Modelling
(Rpt_818) Forecast Worksheet (Dealer Adjusted)
(Rpt_821) Forecast Worksheet (Dealer Adjusted) - Modelling

(Rpt_169) Financial Risk Summary

Summarises MARC contract financials and risk, and aims to provide a management control report for MARC agreements.

Tips

This report shows the contract financials and risks, and reports the variance between the budget and projected financial contribution. It also projects future cash flows.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_170) Financial Summary

Shows the base financial details of the equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Customer, Industry Group, Overheads/Exp

(Rpt_171) Fleet Component Performance Summary

Assists users in forecasting the correct projection by reviewing the strategy tasks with this historical data.

Tips

Compares averages to current strategy parameters. Significant differences suggest the current strategy parameter should be changed.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|AutoLaunch|,
|ModelID|,|Model|,|SiteID|,|Site|,|FleetID|,|Fleet|,|SytemID|,|System|,
|BranchId|,|Branch|,|EqpPlanId|,|EqpPlan|,|SubSytemId|,
|SubSystem|,|CompCodeId|,|CompCode|,|TaskTypeId|,|TaskType|,
|Warranty|,|Redo|,|LifeConfirmed|,|UserID|

Report parameters

Report Type, Equipment Hierarchy

Additional filters

Cost Type, Cost Selector, Sub System, Warranty, Task Type, Redo, System, Life Confirmed, Component Code, Suppression Impact

(Rpt_172) Forecast Availability

This report is used to determine whether the maintenance schedule meets the customer's availability expectations.

Relates to the Budgeting Module in AMT, and lets you define, set and edit availability parameters to forecast the availability of the fleet over a period of time. The report reports on the base downtime, monthly downtime and availability.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_173) Forecast Cost Analysis

Shows the strategy task costs at sell.

Tips

Significant differences suggest the current strategy parameter should be changed.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|LogoLocation|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Region, Eqp Criticality, Equipment Group, Exp Category, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Modifier Code, Division, Equipment Class, Equipment Location, Project, Def. Maint. Period, Parts/Labour/Misc, Currency

(Rpt_174) Forecast Worksheet

Used for forecasting costs by expense group. The report shows future and historical costs at sell, categorised by up to five different areas.

- You can navigate to this report in the following ways:
  Long-term Planning > **Costs**.
  Budgeting & Forecasting > **Live Forecasting**.
  Modelling > **Cost Analysis**.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,
|EquipmentTypes|,|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|,
|Feature_32|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

(Rpt_ 287) Forecast Availability - Modelling

This report relates to the Budgeting Module, and lets you define and set or edit availability parameters for Forecast the availability of the fleet over a period of time. It reports on the base downtime, monthly downtime and availability.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_318) Fleet Cost and Recovery Schedule

A high level Plant Hire report showing the revenues versus costs for equipment or groups of equipment. The report also shows the performance breakdown between Ownership, Operating and Major Components.

Tips

Used for high level performance plant hire performance analyses, by Region, Model, Contract and so on.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Region, Division, Industry Group, Cost Responsibility, Eqp Criticality, Equipment Date, Equipment Group, Eqp Category, Manufacturer, Cost Bearer, Cost Centre, Expense Element, Currency

(Rpt_383) Financial Summary - Modelling

Shows the base financial details of the equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Branch, Site, Fleet, Equipment, Business Group, Contract Type, Contract, Model

(Rpt_388) Forecast Worksheet - Modelling

Used for forecasting costs by expense group. It shows future and historical costs at sell, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|EquipmentTypes|,
|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|,|Feature_32|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

(Rpt_393) Failure Indicators Analysis

Shows the trending of CM elements up to the point of component failure.

Initial parameters

|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Task Occurrence, Element Type, Element UOM, Lead Time

(Rpt_400) Failure Mode Details

Shows the failure mode details set up in a Component Structure, and how the actual performance compares with the Component Structure assumptions.

Initial parameters

|AutoLaunch|,|EquimentTypes|,|EqpPlan|,|EqpPlanID|,|TaskHeader|,
|TaskHeaderId|,|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Task Type, Strategy Task, Warranty, Redo, Life Confirmed

(Rpt_413) Fleet Statistics

Shows the latest actual usage readings for equipment.

Tips

This report is only applicable to Real equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM

(Rpt_562) Forecast Rotables Stock

Displays, per month, the quantity of rotable components available for use. A positive number indicates excess stock on hand, a negative number indicates shortage and zero indicates supply equalled demand in the month.

The report also contains four KPIs: Total Period Demand, Maximum Stock Level, Minimum Stock Level and Annualised Stock Turn.

Data for this report is sourced from the Inventory Parts Table, Parts Table and Strategy Tasks (Next Due and Projected).

Tips

Use this report to validate whether there are sufficient rotables to meet future demand.

If shortages exist, look to "move" stock excesses to the month of shortage through adjusting Next Due dates and usage of Strategy Tasks, optimising turn time (update Total Budget Turn Time in Parts Manager), or purchasing new rotable parts into stock (increasing parts on order in the Inventory Parts table.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

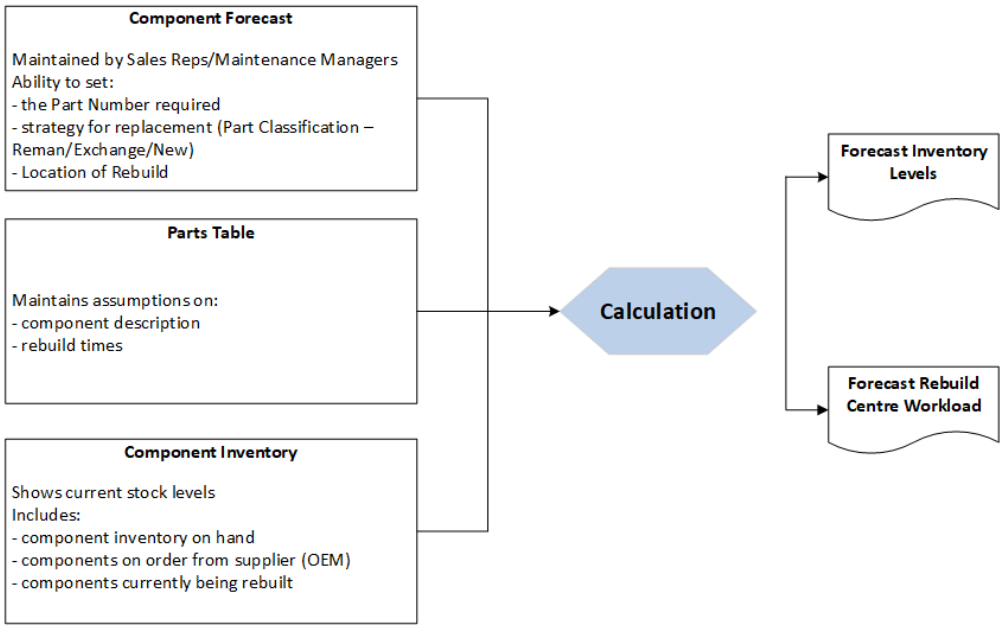
Default.

Additional filters

Part Number, Part Rating, Part Type, Component Code, Sales Status, Apply Minimum stock level <=, Minimum stock level <=, Apply Total Period Demand <=, Total Period Demand <=, Apply Average Stock Level <=, Average Stock Level <=, Apply Annualised Stock Turn <=, Annualised Stock Turn <=, Apply Annualised Stock Turn >=, Annualised Stock Turn >=

How component capacity works

- The life-cycle costing in AMT maintains a live component forecast.
- The strategy task contains some of the key parameters that drive the forecast.
- The strategy task is also used to set the strategy for the component replacement.
    - What type of component will be used (part classification)?
    - If it is to be rebuilt, then where will it be rebuilt?
- The strategy task includes:
    - Default assumptions for the component replacement strategy.
    - The specific plan for the next changeout.
- The parts table in AMT contains assumptions about the rebuild times for rotable components.
- The actual component inventory is imported into AMT and compared with the forecast to map out future projected inventory levels and identify expected shortfalls and excess inventory.



(Rpt_657) Forecast Market Analysis

Provides flexible reporting analyses of the calculated market potential, forecast and opportunity.

Tips

Analyses current data (updated nightly) and applies to future data, not historical. You can dynamically drill down into selected areas.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report Parameters

Report Type, Date Selector

Additional filters

X-Axis, Y-Axis, CostOrSell, Data, Industry, Sales Zone L1, Classification, Account Manager, PLM, Major Comm Code, Bec Code, Version, Minor Comms Code

(Rpt_658) Forecast Market Share Analysis

Provides flexible reporting analyses of the forecast market share.

Tips

Analyses current data (updated nightly) and applies to future data, not historical data. You can dynamically drill down into selected areas.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report Parameters

Report Type, Date Selector

Additional filters

X-Axis, Y-Axis, Industry, Sales Zone L1, Classification, Account Manager, PLM, Major Comm Code, Bec Code, Version, Minor Comm Code

(Rpt_778) Forecast Worksheet (Serial Number)

Used for forecasting costs by expense group. The report shows future and historical costs at sell, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|EquipmentTypes|, |ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Serial Number, Modifier, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

(Rpt_781) Functional Location Budget Analysis

Provides a comparison of the budget figures for the selected financial year and the actual results of the past 12 months.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,|LanguageKey|, |ReportStyleKey|,|FunctionalLocationName_Name|,|FunctionalLocationName_Value|

Report parameters

Default.

Additional filters

Currency, Cost Centre, Expense Element, Work Group, Cost Bearer, Parts, Labour, Misc, Toggle

(Rpt_782) Functional Location Budget v Actual Analysis

This report provides a dynamic report for comparison between budget and actual figures. You have the choice of multiple columns for analysis purposes.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,
|UserID|,|LanguageKey|,|ReportStyleKey|,|FunctionalLocationName_Name|,
|FunctionalLocationName_Value|

Report parameters

Default.

Additional filters

Currency, Cost Centre, Expense Element, Work Group, Cost Bearer, Parts Labour Misc, Toggle, Display Columns, UOM

(Rpt_783) Functional Location Cost Analysis

Provides a dynamic report for comparison between budget and actual figures. You have the choice of what you want to do.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,
|UserID|,|LanguageKey|,|ReportStyleKey|,
|FunctionalLocationName_Name|,|FunctionalLocationName_Value|

Report parameters

Default.

Additional filters

Currency, Cost Centre, Expense Element, Work Group, Cost Bearer, Parts Labour Misc, Toggle

(Rpt_784) Functional Location ERP Master Data Analysis

Provides analysis of the Functional Locations in AMT, linking to ERP Maintenance Plans. This helps you to see what level of the forecast is being driven by the ERP Maintenance Plan.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|,|FunctionalLocationName_Name|,
|FunctionalLocationName_Value|

Report Parameters

Report Type, Equipment Hierarchy

Additional filters

Currency, Cost Centre, Expense Element, Work Group, Cost Bearer, Parts Labour Misc, Toggle

(Rpt_785) Functional Location Forecasting Maturity

Provides an analysis of the maturity of the forecasts in AMT associated with Functional Locations. This is independent of the ERP linkage.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|,|FunctionalLocationName_Name|,
|FunctionalLocationName_Value|

Default parameters

Report Type, Equipment Hierarchy

Additional filters

Currency, Cost Centre, Expense Element, Cost Bearer, Parts Labour Misc, Toggle

(Rpt_786) Functional Location Cost Details

A flexible report with multiple columns to choose from to asses functional location costs.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,
|UserID|,|LanguageKey|,|ReportStyleKey|,
|FunctionalLocationName_Name|,|FunctionalLocationName_Value|

Report parameters

Default.

Additional filters

Currency, Cost Centre, Expense Element, Work Group, Cost Bearer, System, Sub System, Task Type, Component Code, Modifier Code, Task Counter, Cost Type, CC Activity, CC Location, Parks Labour Misc, Toggle, Display Columns, UOM

(Rpt_816) Forecast Worksheet - @Cost

Used for Forecasting costs by expense group. Shows future and historical costs at cost, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,
|EquipmentTypes|,|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|,
|Feature_32|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

(Rpt_817) Forecast Worksheet - @Cost - Modelling

Used for Forecasting costs by expense group. Shows future and historical costs at cost, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|EquipmentTypes|,|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|,|Feature_32|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

(Rpt_818) Forecast Worksheet (Dealer Adjusted)

Used for forecasting costs by expense group and optional dealer market share. Shows future and historical costs at sell, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Planned, Workorder, Currency

(Rpt_821) Forecast Worksheet (Dealer Adjusted)

- Modelling

Used for forecasting costs by expense group and optional dealer market share. Shows future and historical costs at sell, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Planned, Workorder, Currency

G

(Rpt_175) GM Comparison

(Rpt_176) GM Comparison Chart

(Rpt_455) Generic Interface Log

(Rpt_175) GM Comparison

Shows the projected gross margin versus the budgeted gross margin. It identifies at a gross margin level the expected equipment performance.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_176) GM Comparison Chart

Graphical display of the gross margin. Refer to the GM Comparison report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_455) Generic Interface Log

Reviews error messages related to importing data for from external systems.

Report Parameters

Report Type, Data Selector

Additional filters

Error Type, Import Reference

H

(Rpt_177) High Opportunity

(Rpt_178) High Risk

(Rpt_288) High Opportunity - Centreline

(Rpt_289) High Opportunity - Modelling

(Rpt_290) High Risk - Centreline

(Rpt_291) High Risk - Modelling

(Rpt_632) Hours TO PM

(Rpt_660) Historical Market Share Analysis

(Rpt_177) High Opportunity

Identifies components due for change just short of the contract end, and calculate the new frequency required to ensure the opportunity can be realised.

Identifies the quantity and value (at selling price) of components which are due for replacement within the contract period, but if managed properly, their life could be extended. Costs are saved by not having to replace the component.

Tips

Displays the last occurrence prior to equipment termination, the component life required and the cost benefit.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|MaxCompLife|,|MinCompCost|, |LogoLocation|,|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|, |Branch|,|Site|,|Fleet|,|EqpPlan|,|AnalyseBy|,|ProjectionHeaderID|, |ProjectionHeader|,|Currency|

Report parameters

Default.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, Min Comp. Cost, Max Life Used %, Min Comp Cur

(Rpt_178) High Risk

Identifies the risk components and calculates the life buffer available to prevent an additional component changeout prior to the equipment termination.

Identifies the quantity and value (at selling price) of components that are due to be replaced just after the contract concludes.

High risk contracts should include additional maintenance provisions and have the contract end date renegotiated.



This is an important report, and it should be approved by management before the contract is signed.

Tips

Shows the next projected occurrence after the equipment termination, the component life to avoid and the cost risk.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|MaxCompLife|,|MinCompCost|,
|LogoLocation|,|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,
|Branch|,|Site|,|Fleet|,|EqpPlan|,|AnalyseBy|,|ProjectionHeaderID|,
|ProjectionHeader|,|Currency|

Report parameters

Default.

Additional filters

Task type, Cost Type, System, Sub System, Component Code, Min. Comp Cost, Max Life Left %, Min Comp. Cur

(Rpt_288) High Opportunity - Centreline

Identifies components due for change just short of the contract end, and calculates the new frequency required to ensure the opportunity can be realised.

Tips

Shows the last occurrence prior to equipment termination, the component life required and the cost benefit.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|MaxCompLife|,
|MinCompCost|,|LogoLocation|,|UserID|,|AutoLaunch|,|BranchID|,
|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,
|Fleet|,|EqpPlan|,|AnalyseBy|,|ProjectionHeaderID|,|ProjectionHeader|,|Currency

Report parameters

Default.

Additional filters

Task Type, System, Component Code, Max Life Used %, Cost Type, Sub System, Min.Comp.Cost, Min. Comp. Cur.

(Rpt_289) High Opportunity - Modelling

Identifies components due for change just short of the contract end, and calculates the new frequency required to ensure the opportunity can be realised.

Tips

Shows the last occurrence prior to equipment termination, the component life required and the cost benefit.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|MaxCompLife|,
|MinCompCost|,|LogoLocation|,|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,
|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|,|AnalyseBy|,|
ProjectionHeaderID|,|ProjectionHeader|,|Currency|

Report parameters

Default.

Additional filters

Task Type, System, Component Code, Max Life Used %, Cost Type, Sub System, Min.Comp.Cost, Min. Comp. Cur.

(Rpt_290) High Risk - Centreline

Identifies the risk components and calculates the life buffer available to prevent an additional component changeout prior to the equipment termination.

Tips

Shows the next projected occurrence after the equipment termination, the component life to avoid and the cost risk.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|MaxCompLife|,
|MinCompCost|,|LogoLocation|,|UserID|,|AutoLaunch|,|BranchID|,
|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|,
|AnalyseBy|,|ProjectionHeaderID|,|ProjectionHeader|,|Currency|

Report parameters

Default.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, Min. Comp. Cost, Max Life Left %, Min. Comp.Cur.

(Rpt_291) High Risk - Modelling

Identifies the risk components and calculates the life buffer available to prevent an additional component changeout prior to the equipment termination.

Tips

Shows the next projected occurrence after the equipment termination, the component life to avoid and the cost risk.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|MaxCompLife|,
|MinCompCost|,|LogoLocation|,|UserID|,|AutoLaunch|,|BranchID|,
|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,
|EqpPlan|,|AnalyseBy|,|ProjectionHeaderID|,|ProjectionHeader|,|Currency|

Report parameters

Default.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, Min. Comp. Cost, Max Life Left %, Min. Comp.Cur.

(Rpt_632) Hours To PM

Gives a quick view of hours to the next Preventive Maintenance (PM) for selected equipment.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Report Type, Equipment Hierarchy

Additional filters

UOM

(Rpt_660) Historical Market Share Analysis

Provides flexible reporting analyses of the actual market share.

Tips

Analyses current data (updated nightly) and applies to future data, not historical data. You can dynamically drill down into selected areas.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report Parameters

Report Type, Date Selector

Additional filters

X-Axis, Y-Axis, Industry, Sales Zone L1, Classification, Account Manager, PLM, Major Comm Code, Bec Code, Version, Minor Comm Code

I

(Rpt_179) Inclusions and Exclusions

(Rpt_180) Intercompany Cost Analysis

(Rpt_181) Internal Summary

(Rpt_414) Interface Logs

(Rpt_764) Inspection Completion Time

(Rpt_179) Inclusions and Exclusions

Provides a listing of included and excluded tasks in any given agreement - relates to the Sales Tool module.

Initial parameters

SalesToolReport,|UserID|

Report parameters

Equipment

Additional filters

Equipment Class, Equipment Group, Manufacturer, Model, Fleet, Equipment Year, Serial Number, Eqp Type, Eqp Criticality, Eqp Location, Eqp Category, Contact, Region, Division, Branch, Site

(Rpt_180) Intercompany Cost Analysis

A text report that allows users to analyse the costs specifically for a Company. It also shows these costs against the Group costs and shows the % of the Group costs by Parts, Labour and Miscellaneous.

Tips

This is the only report that lets you analyse the costs within a Company (where there has been a split between Companies). The report also provides a validation against the percentage cost split figures entered in section D of the financial reporting parameters.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|Currency|,|ProjectionTypes|,|UserI

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Task Counter, Job Code, Cost Bearer, WO Status, Cost Selector

(Rpt_181) Internal Summary

List of reports that have been attached to a particular agreement type that need to be printed out with an estimation generated from the Sales Tool module.

Tips

Internal

Initial parameters

SalesToolReport,|UserID|

Report parameters

Equipment

Additional filters

Equipment Class, Equipment Group, Manufacturer, Model, Fleet, Equipment Year, Serial Number, Eqp type, Eqp Criticality, Eqp Location, Eqp Category, Contract, Region, Division, Branch, Site

(Rpt_414) Interface Log

Contains errors found by AMT when processing certain files.

Tips

Interface Files relevant to this report are:

- ImportEquipment
- ImportProjectedTask

Initial parameters

None

Default Parameters

Report Type, Date Selector

Additional filters

File Name

(Rpt_764) Inspection Completion Time

Shows details about the last inspection done on an equipment,

Tips

Displays the of the inspection, inspection type, as well as the employee responsible for the inspection.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Employee

J

(Rpt_417) Job Folders - Bulk Printing

(Rpt_810) Job Status Audit Log

(Rpt_417) Job Folders - Bulk Printing

Lets you print Eventscopes, Workscopes or Library documents for multiple tasks at once.

Tips

Choose filters carefully in order to limit the amount of printing. Choosing too few filters could result in printing too many pages.

Initial parameters

ExternalReport|,|UserID|

Report Parameters

Items to Include, Printer, Date Selector, Sort By, Equipment Hierarchy

Additional filters

Workorder Branch, Workorder Site, Work Group, Employee Resp., Labour Activity, Responsibility, Planning Group, Task Type, Work Location, Status, Priority, Raised By, Maint. Order Type, STO

(Rpt_810) Job Status Audit Log

Shows changes made by users to the job status of work orders.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,
|UserID|,|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Cost Type, Component Code, System, Sub System, Task Type, Strategy Task, Strategy Task Desc, User Name, Job Status

K

(Rpt_439) Key Stats Change Trending

(Rpt_439) Key Stats Change Trending

Lets you trend the movement in certain key stats over a period of time. For example, the ability to trend the movement in PCV or projected costs for this financial year.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Snapshots|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Sub System, Strategy Task, Industry Group, Contract Type, Customer, Cost Centre, System, Component Code, Task Type, Business Group, Contract, Maintenance Type, Responsibility

L

(Rpt_182) Labour Hours Analysis

(Rpt_183) Labour Forecast - Hours

(Rpt_184) Labour Forecast - Persons

(Rpt_185) Labour Hrs Variance Analysis

(Rpt_186) Ledger Missing Equipment

(Rpt_292) Labour Forecast - Hours - Centreline

(Rpt_293) Labour Forecast - Hours - Modelling

(Rpt_294) Labour Forecast Hours - Persons - Centreline

(Rpt_295) Labour Forecast Hours - Persons - Modelling

(Rpt_432) Labour Hours per Interval

(Rpt_434) Labour Hours per Interval - Modelling

(Rpt_435) Labour Hours per Interval - Centrelines

(Rpt_509) Labour Resource Scheduling & Utilisation

(Rpt_799) Labour Forecast - Hours - Modelling - Multiple Asset Quantity

(Rpt_814) Labour Forecast (Dealer Adjusted)

(Rpt_815) Life Cycle Cost Summary - @Cost

(Rpt_182) Labour Hours Analysis

Analysis of where time is being spent (particularly of breakdown labour) to identify product problems.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Contact, Strategy Task, Occurrence Type, Modifier Code, Task Counter, Cost Bearer, Cost Type, System, Sub System, Component Code, Task Type, WO Status

(Rpt_183) Labour Forecast - Hours

This report provides you with an understanding of how much labour is required for the contract.

The key drivers for this report are Labour Hours and Labour Costs set up in the strategy tasks within the estimate projection.

If labour is costed in the estimate projection, this report shows labour costs.

If labour costs are in overheads, labour costs will not be seen in this report. The Total Cost Worksheet report would have to be used to see this information.



Estimates can be benchmarked against each other, and against other equipment or centrelines in a number of reports by selecting **Equipment** as the **Analyse By** parameter in the report and including other equipment in the report's filters.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity

(Rpt_184) Labour Forecast - Persons

Lets you analyse maintenance work by persons required.

This report provides you with an understanding of how much labour is required for the contract.

The key drivers for this report are Labour Hours and Labour Costs set up in the strategy tasks within the estimate projection.

If labour is costed in the estimate projection, this report shows labour costs.

If labour costs are in overheads, labour costs will not be seen in this report. The Total Cost Worksheet report would have to be used to see this information.

Estimates can be benchmarked against each other, and against other equipment or centrelines in a number of reports by selecting **Equipment** as the **Analyse By** parameter in the report and including other equipment in the report's filters.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Cost Responsibility, Expense Element, Labour Activity, Productive Hrs+

+ indicates the average productive hours per person per month.

(Rpt_185) Labour Hrs Variance Analysis

Lets you analyse the actual labour hours required to complete strategy tasks compared to the target labour hours which have been set by the strategist or planner.

Shows actual average labour hours against the target. Identifies:

- incorrect targets (such as in Standard Jobs).
- inefficiencies due to low skill levels or poor task instructions.

Navigate to **Strategy Optimisation** > **Labour**.

Tips

The report can be run for all tasks or only those with set labour hours linked to work orders.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Work Group, Planned, Breakdown, Strategy Task, Component Code, Task Type, Modifier Code, Employee, Show

(Rpt_186) Ledger Missing Equipment

This report displays transaction records imported through the General Ledger interface that do not link to AMT equipment, and shows technical details for component changeouts. Can be extremely critical in highlighting data problems.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

GL Account, Invoice, Workorder, Serial Number

(Rpt_188) Life Cycle Cost Summary

Compares Cost to Date and Future Costs by Task and Sub System between Budget and Current projections. The report identifies cost variances by Task, and is used primarily to identify significant issues.

Applicable for those assets with Budget Protection. The PCV highlights projected cost over-runs from original/baseline assumptions.

Tips

Check for large variances between the Budget and Actual in the **To Date** columns. Large variances should be investigated further and the information qualified to assist decision making for the future costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Task Type, Strategy Task, UOM, Cost Bearer

(Rpt_292) Labour Forecast - Hours - Centreline

Lets you analyse maintenance work by hours required. The report can be grouped by a maximum of five categories for more specific analysis.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity

(Rpt_293) Labour Forecast - Hours - Modelling

Lets you analyse maintenance work by hours required. The report can be grouped by a maximum of five categories for more specific analysis.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity

(Rpt_294) Labour Forecast - Persons - Centreline

Lets you analyse maintenance work by persons required. The report can be grouped by a maximum of five categories for more specific analysis.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity, Productive Hours



+ indicates Average Productive Hours per Person per Month.

(Rpt_295) Labour Forecast - Persons - Modelling

Lets you analyse maintenance work by persons required. The report can be grouped by a maximum of five categories for more specific analysis.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity, Productive Hours



+ indicates Average Productive Hours per Person per Month.

(Rpt_432) Labour Hours per Interval

Shows labour hour requirements per interval.

Tips

This report includes actuals for Current and Alternate projections.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Item, Item, Labour Type

(Rpt_434) Labour Hours per Interval - Modelling

Shows labour hour requirements per interval based on the labour hour estimates within strategy tasks.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Item, Item, Labour Type

(Rpt_435) Labour Hours per Interval - Centrelines

Shows labour hour requirements per interval based on the labour hour estimates within strategy tasks.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Item, Item, Labour Type

(Rpt_509) Labour Resource Scheduling & Utilisation

Lets you understand the volume of work to be done in the next month, and how much is scheduled.

Tips

This report utilises snapshots of the seven day weekly work plan created in AMT. The day on which the work plan is saved is site specific and, therefore set up using site administration.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Report Type, Date Selector

Additional filters

Work Group

(Rpt_633) Long-Term Planning Log

Displays the long-term planning log.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Component Code, System, SubSystem, Task Type, Strategy Task, Strategy Task Desc, Reason, User Name

(Rpt 799) Labour Forecast - Hours - Modelling

Lets you analyse maintenance work by hours required.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity

(Rpt_814) Labour Forecast (Dealer Adjusted)

Displays the Dealer's market share.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,
|UserID|,|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity, Planned

(Rpt_815) Life Cycle Cost Summary - @Cost

Compares Cost to Date and Future Costs by Task and Sub-system between Budget and Current projections. This report identifies cost variances by Task and is used primarily to identify significant issues.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

System, Sub System, Task Type Strategy Task, UOM, Cost Bearer

M

(Rpt_189) Maintenance Accrual

(Rpt_190) Maintenance Accrual Details

(Rpt_191) Maintenance Ratio

(Rpt_192) MARC Analysis Of Actuals

(Rpt_193) MARC Financial Analysis

(Rpt_196) MARC PCV Chart

(Rpt_197) MARC Reconciliation Worksheet

(Rpt_198) Monthly Downtime Log

(Rpt_199) MTBF

(Rpt_200) MTBS

(Rpt_201) MTBUPS

(Rpt_202) MTFS_After_PM

(Rpt_203) MTTR

(Rpt_296) Maintenance Ratio - Centreline

(Rpt_297) Maintenance Ratio - Modelling

(Rpt_300) Projection Details - Modelling

(Rpt_332) MARC Financial Summary - Modelling

(Rpt_399) Maintenance Strategy Summary

(Rpt_493) Maintenance KPI Summary

(Rpt_507) Maintenance Compliance

(Rpt_528) Monthly Warranty Status

(Rpt_560) MARC Component Reconciliation

(Rpt_809) MARC PCV Chart - @Cost

(Rpt_189) Maintenance Accrual

Shows where revenue received (billings) have gone - to pay for the cost of sales or taken to profit (MARC GM), or if booked to the accrual account (Current MA). The report also shows the expected future sales (billings) and expected future costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Sub System

(Rpt_190) Maintenance Accrual Details

Calculates and identifies the accrual values required to the end of contract by taking into account the costs and billings to date and the percentage of contract completion.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Currency, Sub System

(Rpt_191) Maintenance Ratio

Measures the man hours to operating hours. If it is high then it indicates either reliability issues or low skill levels. A useful analysis is to trend this report and exclude component changeouts (remove the big spikes). If the trend is up this indicates issues.

Navigate to **Reporting & Analyses** > **Maintenance Efficiency** > **Daily Operations**.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|CostBearerID|,
|AutoLaunch|,|EqpPlanID|,|EqpPlan|,
|DateChecked|,|StartDate|,|EndDate|,|LabourType|,|ActualProjected|,
|AnalyseBy|,|ProjectionTypes|,|UserID|,|BranchID|,|SiteID|,|FleetID|,|Branch|,
|Site|,|Fleet|,|ProjectionHeaderID|,|ProjectionHeader|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Cost Bearer, Task Type, Strategy Task, Job Code, Labour, Show, Shop Job Codes

(Rpt_192) MARC Analysis Of Actuals

A text report that shows a summary of the actuals to date of costs (by parts, labour and miscellaneous), margin and billings. It also shows % billings complete. The report can be run for the Group or individually by Company.

Tips

This report provides an audit train for the Billing Profit Balance Chart report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Additional filters

Company

(Rpt_193) MARC Financial Analysis

This report contains the following summaries:

- MARC Budget
- Actuals to Date
- Future Projected
- Total Contract Protected

Each summary shows costs (by parts, labour and miscellaneous), margin, margin % and billings.

Tips

Provides a high-level analysis of the MARC financial status by Company.

Report parameters

Default.

Additional filters

Company

(Rpt_194) MARC Financial Summary

Shows the base financial details of the equipment, including percentage of contract completion, historical and projected costs and billings.

Costs in this report are cost at cost (except the net risk/opportunity values which are cost at sell).

To see overheads in this report, run it at the level of the overhead; that is, contract, branch, site or fleet.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|EquipmentTypes|,
|UserID|,|Currency|

Report parameters

Default.

Additional filters

Overheads/Eqp, Currency

(Rpt_196) MARC PCV Chart

Compares cumulative Actual and Projected Contract Billings, and Budget and Current Contract costs in order to identify differences between the two projections.

Tips

All costs are shown as Cost at Sell. Basis of Calculation = Budget & Current Projections only and Cost Bearer equal to Contract only.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Task Type, Strategy Task

(Rpt_197) MARC Reconciliation Worksheet

A detailed text report showing the calculations to support the specific methodologies defined in the Financial Report Parameters.

- %Completion (SMU and Billings).
- Projected MARC performance (Billings and Margins).
- Projected impact if the defined cost over-run occurs.
- MARC position to date (Billings, Costs, Margins).
- Calculation of the required Maintenance Accrual, Provision for cost over-run.
- Calculation of the Margin that can be recognised vs deferred.

Tips

This report is only applicable if the organisation has adopted the methodologies supported in the Financial Reporting Parameters form.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Today|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Company

(Rpt_198) Monthly Downtime Log

Lets you check the daily events and tasks, and to analyse the downtime allocations throughout the required period of time.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|UserID|,|LogoLocation|

Report parameters

Default.

Additional filters

Task Type, System, Event Comp Code, Responsibility, Variance Cause, Planned, Cost Type, Sub System, Activity, Cause, EM Issue

(Rpt_199) MTBF

Analyses and reports on the maintenance reliability of equipment by using historical event information.

To Trend, "Analyse By" month. To compare equipment with fleets, "Analyse By" Equipment and Fleet.

Tips

MTBF = Mean Time Between Failures.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Responsibility, System, Task Type, Sub System, Breakdown, Planned, Maint. Order Type, Enable Zero Value, Planning Group, Component Code

(Rpt_200) MTBS

Analyses and reports on the maintenance reliability of equipment by using historical event information.

Navigate to **Reporting & Analyses** > **Equipment Performance** > **Reliability** and run the MTBS report.

Tips

MTBS = Mean Time Between Stoppages. Basis of calculation - Usage / Number of stoppages. By default, all stoppages are taken into consideration - planned, unplanned, breakdown.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ResponsibilityID|,
|ProjectionTypes|,|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,
|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|,|DateChecked|,
|StartDate|,|EndDate|,|AnalyseBy|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Planned, Component Code, Responsibility, Task Type, Breakdown

(Rpt_201) MTBUPS

Analyses the mean time between unplanned stoppages.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Responsibility, System, Task Type, Sub System, Breakdown, Planned, Maint. Order Type, Component Code, Planning Group

(Rpt_202) MTFS_After_PM

This report analyses the hours between a PM service and the first stoppage that occurs after that PM service. This assists with understanding the reliability of the PM services being undertaken by the technicians.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,|LogoLocation|,
|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,
|Fleet|,|EqpPlan|,|AnalyseBy|,|ResponsibilityID|,|PMTaskTypeID|,|PMTaskType|,|FSTaskTypeID|,|FSTaskType|,|DateChecked|,|StartDate|,|EndDate|

Report parameters

Default.

Additional filters

PM Task Type*, +FS Task Type*, +FS Breakdown, PM Released By,
+FS Responsibility

* These filters define the attributes of the First Stoppage (FS).

(Rpt_203) MTTR

Analyses the time taken to complete a maintenance event. This may assist management in understanding the technical level of personnel, show faults in repair materials, and so on.

By itself, the MTTR (Mean Time To Repair) report does not mean a lot. An increasing MTTR could indicate many things - better planning (fewer and longer stoppages) or poor skill levels (tasks take longer). Therefore, it is important to view this report in a context, with the MTBS (Mean Time Between Stoppages) or MTBF (Mean Time Between Failures) report.

If MTBS is increasing then you would expect MTTR to increase. If, however, MTBS is constant and MTTR increases, then this indicates issues (poor skill levels).

Use the filters to capture only Breakdowns and avoid component changeouts (remove the spikes). This indicates the average time to repair the smaller general breakdowns and is a good indicator of response times (availability of people) and skill levels.

If Advanced Planning is active, use the Event Analysis report and review in conjunction with Event Delay Causes and Variances. This assists in identifying other causes that are non-skills related

Tips

You can run this report can be run by System or Sub system to assist with understanding the level of technical expertise on a site or shift.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,|UserID|,
|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,
|EqpPlan|,|AnalyseBy|,|ResponsibilityID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Planned, Responsibility, Task Taype, Component Code, Breakdown

(Rpt_296) Maintenance Ratio - Centreline

High level measure of efficiency of the direct Labour hours. The report only looks at productive labour hour, so needs to be viewed with the Labour Productivity report. It is calculated as Direct Labour Hours/Engine Hours.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|CostBearerID|,|AutoLaunch|,
|EqpPlanID|,|EqpPlan|,|DateChecked|,|StartDate|,|EndDate|,|LabourType|,
|ActualProjected|,|AnalyseByProjectionTypes|,|UserID|,|BranchID|,|SiteID|,
|FleetID|,|Branch|,|Site|,|Fleet|,|ProjectionHeaderID|,|ProjectionHeader|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Cost Bearer, Task Type, Strategy Task, Job Code, Labour, Show, Shop Job Codes

(Rpt_297) Maintenance Ratio - Modelling

High level measure of efficiency of the direct Labour hours. The report only looks at productive labour hour, so needs to be viewed with the Labour Productivity report. It is calculated as Direct Labour Hours/Engine Hours.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|CostBearerID|,|AutoLaunch|,
|EqpPlanID|,|EqpPlan|, |DateChecked|,|StartDate|,|EndDate|,
|LabourType|,|ActualProjected|,|AnalyseByProjectionTypes|,|UserID|,|BranchID|,
|SiteID|,|FleetID|,|Branch|,|Site|,|Fleet|,|ProjectionHeaderID|,|ProjectionHeader|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Cost Bearer, Task Type, Strategy Task, Job Code, Labour, Show, Shop Job Codes

(Rpt_332) MARC Financial Summary - Modelling

Shows the base financial details of the equipment, including percentage of contract completion, historical and projected costs and billings.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|EquipmentTypes|,
|UserID|,|Currency

Report parameters

Default.

Additional filters

Overheads/Exp, Currency

(Rpt_389) Component History

Shows the technical details for component changeouts. Also shows the specific work order history and Partial Life and Swing components.

Tips

This report can be used for simple tracking component history where the rotables module is not being used. Also useful for looking at the history where swing and partial life parameters have been used.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|EquipmentTypes|,|AutoLaunch|,
|SerialNumber|,|IsExactMatch|,|EqpPlan|,|EqpPlanID|,|ProjTaskID|,
|ProjTask|,|SerailNumberDesr|,|UserID|

Report parameters

Default.

Additional filters

Component Serial No, Component Code, Strategy Task, Modifier Code, Task Type, CCC, Show IN USE Comp, Show Serialised

(Rpt_399) Maintenance Strategy Summary

Documents the maintenance strategy for an equipment or selected strategy tasks. The report encompasses a number of subsidiary reports showing the failure mode assumptions.

Initial parameters

|AutoLaunch|,|EquipmentTypes|,|EqpPlan|,|EqpPlanID|,|TaskHeader|,|TaskHeaderId|, |ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Task Type, Strategy Task, Warranty, Redo, Life Confirmed

(Rpt_493) Maintenance KPI Summary

This is a downtime text report to show key parameters and target values.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Responsibility, Availability

(Rpt_507) Maintenance Compliance

Displays how much of the scheduled work was completed in a week.

Tips

This report utilises snapshots of the seven day weekly work plan created in AMT. The day on which the work plan is saved is site specific and, therefore set up using site administration.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Task Type, Work Group, System, Subsystem

(Rpt_528) Monthly Warranty Status

Lists work orders which have been flagged as warranty for claiming and auditing processes.

Tips

A work order is shown in this report if it has been flagged as "Warranty Claimable".

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Component Code, Task Type, System, Sub System, Claim Status, Warranty Claimable

(Rpt_560) MARC Component Reconciliation

Specific to reconciling components for maintenance contracts. The report reconciles components at the end of the equipment's life (RVIC) based on the actual last occurrence of each component and the frequency of the component from the budget projection.

Tips

Before running the report, make sure the equipment's life, as defined by its end usage or date is accurate.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,
|CostBearer|,|UserID|

Report parameters

Default.

Additional filters

System, Proj Task, Modifier Code, Occurrence Type, Cost Type, Sub System, Escalation, Task Type, Component Code, Region, Division, Cost Responsibility, Equipment Location, Cost Bearer

(Rpt_809) MARC PCV Chart - @Cost

Compares cumulative Actual and Projected Contract Billings, and Budget and Current Contract Projected Costs in order to identify differences between the two projections.

Tips

All costs are shown as Cost at Sell. Basis of Calculation = Budget & Current Projections only and Cost Bearer equal to Contract only.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|, |LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Schedule, System, Sub System, Task Type, Strategy Task

O

(Rpt_155) Overhead Cost Variance Analysis

(Rpt_204) Overdue Strategy Tasks

(Rpt_155) Overhead Cost Variance Analysis

Provides you with the ability to analyse the overhead costs of a project against a projection.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Overhead Category, Expense Element, Cost Centre, Overhead Code.

(Rpt_204) Overdue Strategy Tasks

Reviews the components that are overdue. These components need to be rescheduled and a decision made as to when these are expected to be done.

Tips

Utilise this report as part of ongoing data health checks.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|AutoLaunch|,
|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|

Report Parameters

Report Type. Hierarchy

Additional filters

Cost Type, System, Sub System, Component Code, Maint. Order Type, Planning Group, STO, Regulatory Code

P

(Rpt_150) Project Comparison

(Rpt_195) P&L - Summary

(Rpt_205) P&L - Detailed

(Rpt_206) P&L - Forecast

(Rpt_207) P&L - Interrogation

(Rpt_208) Parts Usage Analysis - Qty

(Rpt_209) Parts Usage Analysis - Qty - Global

(Rpt_210) Parts Usage Analysis - Value

(Rpt_211) Parts Usage Analysis - Value - Global

(Rpt_212) Parts Usage Details

(Rpt_213) Parts Usage Details - Global

(Rpt_214) Parts Usage Interrogation

(Rpt_215) Parts Usage Interrogation - Global

(Rpt_216) PCV Analysis

(Rpt_217) PCV Chart

(Rpt_218) Planned Downtime Events %

(Rpt_219) Planned Downtime Hours %

(Rpt_220) Plant Rate Analysis

(Rpt_221) PM Standard Jobs

(Rpt_222) Projection Details

(Rpt_223) Projection Summary

(Rpt_298) Plant Rate Analysis - Modelling

(Rpt_299) Projection Details - Centreline

(Rpt_300) Projection Details - Modelling

(Rpt_301) Projection Summary - Centreline

(Rpt_302) Projection Summary - Modelling

(Rpt_375) Plant Hire Invoice Summary

(Rpt_376) Plant Hire Under/Over Recovery

(Rpt_398) Parts Interrogation

(Rpt_425) Part Number Reference

(Rpt_426) Part Number Reference - Modelling

(Rpt_427) Part Number Reference - Centreline

(Rpt_440) Projection and Interval Analysis

(Rpt_441) Projection and Interval Analysis - Modelling

(Rpt_442) Projection and Interval Analysis - Centrelines

(Rpt_448) Production Statistics Analysis

(Rpt_449) Production Statistics Chart

(Rpt_453) Parts Demand Analysis

(Rpt_454) Parts Demand Analysis - Modelling

(Rpt_548) Planning Status

(Rpt_483) Projection Summary - Customer - Modelling

(Rpt_484) Projection Summary - Customer - Centreline

(Rpt_525) Plant Performance KPI
(Rpt_556) Projection Summary and Cost per Interval - Modelling
(Rpt_565) Plant Hire Rates Schedules
(Rpt_779) Planned Workorder Details
(Rpt_831) Plant Hire Rates Audit Report
"(Rpt_838) Parts Forecast" on page 452

(Rpt_150) Projection Comparison

Provides you with the ability to analyse equipment costs between the budget and current projections using numerous filters; a maximum of three filters in the Analyse by field. The report lets management to understand where cost overruns are occurring and to use other reports.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Cost Bearer, Costs, Component Code, Modifier Code, Task Type, Task Counter.

(Rpt_195) P&L - Summary

This report is formatted similar to a standard Profit and Loss statement (sometimes called an Operating Statement or an Income Statement).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|FinPeriod|,|ProjectionTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_205) P&L - Detailed

Shows the financial summary of life to date costs (budget, actual and variance) - broken into Sales Cost of Sales, Expenses and Net Margin. The report also projects the budget, actual and variance to the end of term.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|FinPeriod|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_206) P&L - Forecast

Shows a forecast of costs by either month, quarter, or year - broken into Sales, Cost of Sales, Expenses and Net Margin.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|FiTipsnPeriod|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_207) P&L - Interrogation

Lets you interrogate the life to date costs by analysing external work orders at cost or sell, contract, direct or indirect expenses at cost.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|FinPeriod|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Branch, Site, Fleet, Equipment, Model, Business Group, Contract Type, Contact, Industry group, Customer

(Rpt_208) Parts Usage Analysis - Qty

This report does not provide specific job or work order details, rather it sums up the parts usage by period (month/quarter/year). Flexible filters allow users to focus on specific parts and models, type of work and so on.

Tips

Lets you quickly identify the high usage parts, particularly for breakdowns.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, System, Modifier Code, Source of Supply, Occurrence Type, Ops Comp Code, Ops Job Code, Part Number, Symptom, Failure, Show, Breakdown, Parts Return, Sub System, Task Type, Cause, Operator, Comp Code, Part Type, Transaction Code, Repair, Sort

(Rpt_209) Parts Usage Analysis - Qty - Global

This report does not provide specific job or work order details, it sums up the parts usage by period (monthly, quarterly or yearly) using the Global Job Codes. See System Table - Job Code. Flexible filters allow you to focus on specific parts and models, type of work and so on, enabling you to quickly identify the high usage parts (particularly for breakdowns)

Tips

Lets you quickly identify the high turnover parts, for example, for breakdowns.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, System, Modifier Code, Source of Supply, Occurrence Type, Ops Comp Code, Ops Job Code, Part Number, Symptom, Failure, Show, Breakdown, Parts Return, Sub System, Task Type, Cause, Operator, Comp Code, Part Type, Transaction Code, Repair, Sort.

(Rpt_210) Parts Usage Analysis - Value

This report does not provide specific job or work order details, rather it sums up the parts usage by period (month/quarter/year). Flexible filters allow users to focus on specific parts and models, types of work and so on.

Analysis is based on value, not quantity, which allows users to focus on the high-cost areas.

Tips

The report lets you quickly identify the high value usage parts, particularly for breakdowns.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, Source of Supply, Occurrence Type, Ops Como Code, Ops Job Code, Part Number, Symptom, Failure, Show, Breakdown, Parts Return, Sub System, Task Type, Cause, Operation, Comp Code,

Part Type, Transaction Code, Repair, Sort.

(Rpt_211) Parts Usage Analysis - Value - Global

This report does not provide specific job or work order details, it sums up the parts usage by period (monthly, quarterly or yearly) using the Global Job Codes. See System Table - Job Code. Flexible filters allow you to focus on specific parts and models, type of work and so on, enabling you to quickly identify the high usage parts (particularly for breakdowns)

Analysis is based on part value, not quantity, which allows you to focus on high-cost areas.

Tips

Lets users quickly identify the high turnover parts (such as for breakdowns).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, System, Modifier Code, Source of Supply, Occurrence Type, Ops Comp Code, Ops Job Code, Part Number, Symptom, Failure, Show, Breakdown, Parts Return, Sub System, Task Type, Cause, Operator, Comp Code, Part Type, Transaction Code, Repair, Sort.

(Rpt_212) Parts Usage Details

Detailed listing of parts usage by work order number. Flexible reports parameters allow you to filter for specific models, type of work, component specific part number and so on.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, System, Comp Code, Task Type, Cause, Operation, Ops Job Code, Part Number, Symptom, Failure, Breakdown, Parts Return, Sub System, Modifier Code, Source of Supply, Occurrence Type, Ops Comp Code, Part Type, Transaction Code, Repair, Sort.

(Rpt_213) Parts Usage Details Global

Detailed listing of parts usage by work order number using the Global Codes. See System Table - Job Code. Flexible reporting parameters allow you to filter for specific models, types of work, components, specific part numbers and so on.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, System, Comp Code, Task Type, Cause, Operation, Ops Job Code, Part Number, Symptom, Failure, Breakdown, Parts Return, Sub System, Modifier Code, Source of Supply, Occurrence Type, Ops Comp Code, Part Type, Transaction Code, Repair, Sort.

(Rpt_214) Parts Usage Interrogation

The focus of this report is on what parts are needed for a certain type of job, and where a part has been used. The report provides a flexible interrogation of parts usage.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, System, Comp Code, Task Type, Cause, Operation, Ops Job Code, Part Number, Symptom, Failure, Breakdown, Parts Return, Sub System, Modifier Code, Source of Supply, Occurrence Type, Ops Comp Code, Part Type, Transaction Code, Repair, Sort.

(Rpt_215) Parts Usage Interrogation - Global

Detailed listing of parts usage by work order number using the Global Codes. See System Table - Job Code. The focus of this report is on what parts are needed for a certain type of job, or where a part has been used.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, System, Comp Code, Task Type, Cause, Operation, Ops Job Code, Part Number, Symptom, Failure, Breakdown, Parts Return, Sub System, Modifier Code, Source of Supply, Occurrence Type, Ops Comp Code, Part Type, Transaction Code, Repair, Sort.

(Rpt_216) PCV Analysis

Lists the projected cost variances based on the comparison of two projections selected by the user.

Tips

Run the report as a bar chart as it provides a Pareto analysis of the projected cost variances, that is, it will identify the top 10 projected variances, starting with cost overruns.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Sub System, Strategy Task, Industry Group, Contract Type, Customer, Cost Centre, Expense Element, Cost Selector, System, Component Code, Task Type, Business Group, Contract, Maintenance Type, Responsibility, Cost Bearer, Costs.

(Rpt_217) PCV Chart

Lets you tailor a report which compares the current and projected costs of different projections. The report also has the flexibility to compare costs for different Cost Bearers within the selected properties.

Tips

You must choose a filter from the Equipment Hierarchy area (Branch, Site and so on) prior to selecting two projections to Analyse By.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,
|UserID|,|AutoLaunch|,|BranchID||SiteID|,|FleetID|,|EqpPlanID|,
|Branch|,|Site|,|Fleet|,|EqpPlan|,|Projection1ID|,|Projection1|,|Projection2ID|,
|Projection2|

Report parameters

Default.

Additional filters

Cost Type, System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Sheet, Activity, Costs

(Rpt_218) Planned Downtime Events %

Ascertains the accuracy of the planned work being performed and analyses the planned downtime.

For both the Events and Hours, this report indicates the level of "control" with the maintenance organisation. If the trend is down, it indicates either reliability problems or planning inefficiencies.
An analysis of "non-planned" and "non-breakdowns" is a good way to identify planning inefficiencies. Planning inefficiencies lead to increased stoppages and downtime, that is, work that could have been done in a PM gets done as a separate stoppage.

Tips

This report allows analysis between the usage per annum and the plant rate by expense element.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|SiteID|,|Site|,|FleetID|,|Fleet|,
|EqpPlanID|,|EqpPlan|,|AnalyseBy|,|Responsibility|,|ResponsibilityID|,
|ProjectionTypes|,|UserID|,|BranchID|,|Branch|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Responsibility, Task Type, Strategy Task

(Rpt_219) Planned Downtime Hours %

Ascertains the accuracy of the planned work being performed and analyses the planned downtime. This report also measures the percentage of downtime hours that were flagged as planned. When analysing the downtime, analyse by activity, looking for planning-related downtime (that is, time spent waiting for parts).

For both the Events and Hours, this report indicates the level of "control" with the maintenance organisation. If the trend is down, it indicates either reliability problems or planning inefficiencies.
An analysis of "non-planned" and "non-breakdowns" is a good way to identify planning inefficiencies. Planning inefficiencies lead to increased stoppages and downtime, that is, work that could have been done in a PM gets done as a separate stoppage.

Navigate to **Reporting & Analyses** > **Maintenance Efficiency** > **Planning**.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|SiteID|,|Site|,|FleetID|,|Fleet|,
|EqpPlanID|,|EqpPlan|,|AnalyseBy|,|Responsibility|,|ResponsibilityID|,
|ProjectionTypes|,|UserID|,|BranchID|,|Branch|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Responsibility, Task Type, Strategy Task

(Rpt_220) Plant Rate Analysis

A breakdown of the expense elements and the associated costs. Displays three different scenarios based on the simulated hours you have entered into the report parameters.

Tips

This report allows analysis between the usage per annum and the plant rate by expense element.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Today|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Simulation Usage 2, Simulation Usage 3, Currency



Simulation usage is the annual usage using the primary UOM.

(Rpt_221) PM Standard Jobs

Provides details of costs and parts for a preventive maintenance service linked to a standard job.

Initial parameters

PMStdJobReport,|UserID|

Report parameters

Equipment

Additional filters

Equipment Class, Equipment Group, Manufacturer, Model, Fleet, Equipment Year, Serial Number, Eqp Type, Eqp Criticality, Eqp Location, Eqp Category, Contract, Region, Division, Branch, Site

(Rpt_222) Projection Details

Shows the base details of the projection including pricing, and highlights which tasks have associated documents in the library or has a standard job attached. Navigate to
Setup & Admin > **Projections** > Projection Details.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Bearer, Escalation

(Rpt_223) Projection Summary

Provides an overview of the costs for the equipment by projected tasks and sub system. Navigate to **Setup & Admin** > **Projections** > Projection Summary

Tips

Historical costs for the current projection are actual costs, historical costs for the budget projections are expected costs. Strategy tasks which do not have a projected cost will not appear in this report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|EqpPlanID|,
|EqpPlan|,|ProjHeaderID|,|ProjHeader|,|IsEscalated|,|ProjectionTypes|,
|LogoLocation|,|UserID|,|Currency|,|CurrencyID|

Report parameters

Default.

Additional filters

Cost Bearer, Currency, Escalation

(Rpt_298) Plant Rate Analysis - Modelling

A breakdown of the expense elements and the associated costs. The report displays three different scenarios based on the simulated hours you have entered into the report parameters.

Tips

This report allows analysis between the usage per annum and the plat rate by expense element.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Today|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Simulation Usage 2, Simulation Usage 3, Currency



Simulation usage is the annual usage using the primary UOM.

(Rpt_299) Projection Details - Centreline

Shows the base details of the projection, including pricing, and highlights which tasks have associated documents in the library, or has a standard job attached.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Bearer, Escalation

(Rpt_300) Projection Details - Modelling

Shows the base details of the projection, including pricing, and highlights which tasks have associated documents in the library, or has a standard job attached.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Bearer, Escalation

(Rpt_301) Projection Summary - Centreline

Gives an overview of the costs for the equipment by projected task and component sub system. Provides details of each projected task.

Tips

All costs are shown as cost at sell.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|EqpPlanID|,|EqpPlan|,
|ProjHeaderID|,|ProjHeader|,|ProjHeader|,|IsEscalated|,|ProjectionTypes|,
|LogoLocation|,|UserID|, |Currency|,|CurrencyID|

Report parameters

Default.

Additional filters

Cost Bearer, Escalation, Currency

(Rpt_302) Projection Summary - Modelling

Gives an overview of the costs for the equipment by projected task and component sub system. Provides details of each projected task.

Tips

All costs are shown as cost at sell.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|EqpPlanID|,|EqpPlan|,
|ProjHeaderID|,|ProjHeader|,|ProjHeader|,|IsEscalated|,|ProjectionTypes|,
|LogoLocation|,|UserID|, |Currency|,|CurrencyID|

Report parameters

Default.

Additional filters

Cost Bearer, Escalation, Currency

(Rpt_375) Plant Hire Invoice Summary

This report intends to provide a summary of the plant hire invoices for a month. It can, however, summarise the invoicing for any data range in months. It shows the utilisation for the period, the effective plant hire rates and a break down of the invoicing.

Tips

The utillisation shown is the primary UOM for that equipment. The effective plant hire rate is the sum of the fixed and variable portions of the invoice divided by the utilisation.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Additional filters

Region, Division, Industry Group, Eqp Criticality, Equipment Class, Equipment Group, Eqp Category, Terminated, Manufacturer, Cost Responsibility, Cost Centre. Hire Status, Project Name, Project Customer, Currency

(Rpt_376) Plant Hire Under/Over Recovery

Provides a flexible analysis of plant hire revenues against plant costs. The analysis is by the expense element but any grouping of equipment and date range can be selected.

Tips

The categorisations of ownership, operating and CBD costs are made in the expense element table.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Region, Division, Industry Group, Cost Responsibility, Eqp Criticality, Equipment Class, Equipment Group, Eqp Category, Manufacturer, Cost Centre, Project Name, Project Customer, Hire Status, Currency, Terminated

 **+** indicates Current Equipment Situation.

(Rpt_398) Parts Interrogation

Outlines the details from the Parts Manager on part number, source of supply, standard job reference, probability and so on.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

None

(Rpt_425) Part Number Reference

Report which lists the Primary Part Number, Group Number and Component Serial Number per strategy task for the selected equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report Parameters

Report Type, Equipment Hierarchy

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Modifier Code, Strategy Task, Region, Division, Cost Responsibility, Eqp Location

(Rpt_426) Part Number Reference - Modelling

Report which lists the Primary Part Number, Group Number and Component Serial Number per strategy task for the selected equipment.

Tips

Component Serial Number will always be blank for Modeling equipment

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Report Type, Equipment Hierarchy

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Modifier Code, Strategy Task, Region, Division, Cost Responsibility, Eqp Location

(Rpt_427) Part Number Reference - Centreline

Report which lists the Primary Part Number, Group Number and Component Serial Number per strategy task for the selected equipment.

Tips

Component Serial Number will always be blank for Centreline equipment

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Report Type, Equipment Hierarchy

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Modifier Code, Strategy Task, Region, Division, Cost Responsibility, Eqp Location

(Rpt_440) Projection and Interval Analysis

This report shows a strategy task's estimated cost, and then how much has actually been spent per interval and what is Forecasted to be spent per interval.

Tips

Actuals are not included if the report is based on a budget projection.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,

|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Costs, Escalation

(Rpt_441) Projection and Interval Analysis - Modelling

This report shows a strategy task's estimated cost, and then how much is Forecasted to be spent per interval.

Tips

Timing of expenditure is based on the scheduling parameters of each strategy task.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Costs, Escalation

(Rpt_442) Projection and Interval Analysis - Centrelines

Tips

Timing of expenditure is based on the scheduling parameters of each strategy task.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Costs, Escalation

(Rpt_448) Production Statistics Analysis

Analyses production statistics.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Prod Stat Code

(Rpt_449) Production Statistics Chart

Analyses production statistics and display them in chart form.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Prod Stat Code

(Rpt_453) Parts Demand Analysis

Forecasts expected parts demand based on probability of sale. You can navigate to this report in the following ways:
Long-term Planning > **Components**.
Modelling > **Components**.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|EquipmentTypes|,|ProjHeaderID|, |ProjHeader|,|LogoLocation|,|UserID|,|Feature_32|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Cod, Cost Bearer, Part Type, Part Manufacturer, Primary Part No, Sales Resp., Part Classification, Part Rating, Rebuild Location, Review Status, Sales Status, Task Mode

(Rpt_454) Parts Demand Analysis - Modelling

Forecasts expected parts demand based on probability of sale.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|EquipmentTypes|,|ProjHeaderID|, |ProjHeader|,|LogoLocation|,|UserID|,|Feature_32|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Cod, Cost Bearer, Part Type, Part Manufacturer, Primary Part No, Sales Resp., Part Classification, Part Rating, Rebuild Location, Review Status, Sales Status, Task Mode

(Rpt_458) Planning Status

Details the planning status of work orders.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Workorder Branch, Workorder Site, Status, Job Ready Status, Strategy Task, Component Code, Modifier Code, Task Type, Task Counter, Source, Raised By, Authorised By, Priority, Work Group, Employee Resp. Misc Category, Source of Supply, Supplier, Parts Plan. Status, Target Date (up to), Scheduled Date (up to), Exp. Delivery Date, WO #, Part #, PO #, Stock Code, Maint. Order Type, Planning Group, STO, Strategy Task, Reserved, Parts, Misc, Only show if Exp. Deliv. Date after Target Date, Only show if Exp. Deliv. Date after Date after Scheduled Data.

(Rpt_483) Projection Summary - Customer - Modelling

Gives an overview of the costs for the equipment by projected task and component sub system. Provides details of each projected task. Costs are adjusted by customer risk premium and discounts.

Tips

All costs are shown as cost at sell, but adjusted by customer risk premium and discounts.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,
|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Cost Bearer, Escalation

(Rpt_484) Projection Summary - Customer - Centreline

Delete this text and replace it with your own content. Gives an overview of the costs for the equipment by projected task and component sub system. Provides details of each projected task. Costs are adjusted by customer risk premium and discounts.

Tips

All costs are shown as cost at sell, but adjusted by customer risk premium and discounts.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,
|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Cost Bearer, Escalation

(Rpt_525) Plant Performance KPI

Provides a summary report of key equipment operational performance data to validate operational KPI calculations.

Tips

This report has a date selector to enable reporting for any nominated period.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|,|Responsibility|,|LogoLocation|

Report parameters

Default.

Additional filters

DT Resp. (Show), DT Res. (Calc), Availability Type, Prod Stat Code

(Rpt_556) Projection Summary and Cost per Interval - Modelling

Provides a summary of Equipment life cycle costs along with an analysis of costs per interval.

Tips

The interval is provided by the user when the report is run.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Costs, Escalation, Comp-Job-Mod Desc

(Rpt_565) Plant Hire Rates Schedules

Outputs Plant Hire Rate Schedules into a format which can be easily reviewed.

Tips

This report is useful to validate consistency of rates across equipment and rate periods set up in AMT are correct.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|LogoLocation|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Centre, Expense Element, UOM, Currency

(Rpt_663) Product Support Component Status

Shows the health of components, with the Frequency % statistic indicating whether the component is potentially being extended too far beyond its economic life.

Tips

The Current Component Condition field will only provide an indication of component health if AMTs Condition Based Maintenance module has been deployed to import and interpret condition monitoring data.

Current Component Usage takes into consideration the execution of parent components if the strategy task is part of a suppression group.

Next Due Target Life = Strategy Usage from the Long-term Planning > Component Changeout grid, where the value can be system calculated or manually adjusted by a user.

Next Due% = Current Component Usage (life) / Next Due Target Life.

Frequency% = Current Component Usage (life) / Frequency.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report Parameters

Report Type, Equipment Hierarchy

Additional filters

Cost Type, System, Component Code, Task Type, Strategy Task, Sales Status, Review Status, Frequency %

(Rpt_779) Planned Workorder Details

Lets you select areas of the downtime tasks that you wish to analyse with numerous display and sort fields to select from. A very open report, which lets you customise your required data.

Tips

Date range can be set according to analysis requirements.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|QUOMID|,|QUOM|,|
UserID|,|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,
|Site|,|Fleet|,|EqpPlan|,|DateChecked|,|StartDate|,|EndDate|,
|ISNOTAsDefault_StrategyTasksOnly|,|DisplayFields|,|DisplayFieldsLabels|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Serial Number, Modifier, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

(Rpt_831) Plant Hire Rates Audit Report

Identifies the history of plant hire rate changes over the life of a project to date or for a defined period.

Tips

This is an audit report.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Project Name

(Rpt_838) Parts Forecast

This allows the user to manually generate a specific Part Forecast.

Initial parameters

|ControlType|,|Idlist|,|TextList|,|LogoLocation|,|UserID|,|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Part Source, Part Number, Part Type, Show 100% parts only

R

(Rpt_224) Reconciling Open Workorder Settlements

(Rpt_226) RVIC - Chart

(Rpt_228) RVIC Analysis

(Rpt_303) RVIC - Chart - Centreline

(Rpt_304) RVIC - Chart - Modelling

(Rpt_305) RVIC Analysis - Centreline

(Rpt_306) RVIC Analysis - Modelling

(Rpt_326) Redo Analysis

(Rpt_411) Resource Availability and Fill - Work Group

(Rpt_563) Rebuild Centre Forecast Workload

(Rpt_224) Reconciling Open Workorder Settlements

Assists users to evaluate and action any open work orders in the data validation/ledger reconciliation area of the program. Report can be run to filter for work orders at cost or at sell and defaults to current month, but can be modified to suit the users. This report only displays data if AMT is configured in WIP mode (this mode is normally used only by OEMs).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Contract

(Rpt_225) Resource Availability and Fill - Employee

Displays the number of man hours allocated to each work group for the selected period. This assists with the management of the labour hours required to perform maintenance tasks effectively.

Basis of calculation: detail provided in the Planning details tab of the task referring to the employee.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Tips

Report parameters

Default.

Additional filters

Employee, Work Group, Labour Activity, Shift Start Time

(Rpt_227) RVIC - Chart

Displays the movement of Residual Life over the period of the chart dates.

This report charts the residual value in the components of the equipment selected. As components wear out, the value in the component is lost; as components are replaced, the value increases. Therefore, this report determines when the logical end date of the contract should be.

Tips

The chart does not display consolidated Totals for Fleet, only individual Equipment. This means that selections should consider the amount of data to be displayed in the chart. It is recommended that no more than 6 - 10 items of equipment be displayed.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Sub System, Task Type, System, Component Code, Cost Bearer

(Rpt_228) RVIC Analysis

Shows defined periods. Components can be used as a basis for contract variation or termination, and can also be used for assessing contingent liabilities.

Based on the same logic as the RVIC Chart report, but with the added flexibility to select a hypothetical contract end date and see the value each component would have at that date.

This report is important if the contract states that the customer will receive their equipment with a certain minimum value at the end of the contract. Use this report to determine whether the contract you have set up in AMT is designed to meet the customer's minimum value expectation.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Component Code, Task Type, Cost Bearer

(Rpt_303) RVIC - Chart - Centreline

Displays the movement of Residual Life over the period of the chart dates.

Tips

The chart does not display consolidated Totals for Fleet, only individual Equipment. This means that selections should consider the amount of data to be displayed in the chart. It is recommended that no more than 6-10 items of equipment be displayed.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Sub System, Task Type, System, Component Code, Cost Bearer

(Rpt_304) RVIC - Chart - Modelling

Displays the movement of Residual Life over the period of the chart dates.

Tips

The chart does not display consolidated Totals for Fleet, only individual Equipment. This means that selections should consider the amount of data to be displayed in the chart. It is recommended that no more than 6-10 items of equipment be displayed.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Sub System, Task Type, System, Component Code, Cost Bearer

(Rpt_305) RVIC Analysis - Centerline

Shows defined periods and components can be used as a basis for contract variation or termination, and can also be used for assessing contingent liabilities.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Component Code, Task Type, Cost Bearer

(Rpt_306) RVIC Analysis - Modelling

Shows defined periods and components can be used as a basis for contract variation or termination, and can also be used for assessing contingent liabilities.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Component Code, Task Type, Cost Bearer

(Rpt_326) Redo Analysis

Provides information on tasks marked as Redo, also known as rework, with the new costs and the costs of the task to be redone.

An analysis of any work that is being completed a second time, generally as a result of poor workmanship or lack of skill set.

This is not a KPI report, but a good report to view if any of the other reports are indicating potential work quality issues.



The Supervisor or Planner marks the work order as redo to track this work.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Sub system, Component Code, Cause, Part Number

(Rpt_411) Resource Availability and Fill - Work Group

Displays the number of man hours allocated to each work group for the selected period. This assists with the management of the labour hours required to perform maintenance tasks effectively.

Tips

Basis of Calculation: Detail provided in the planning details tab of the task referring to the employee.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|siteID|,|site|,|WorkGroupID|,|WorkGroup|,
|Today|,|14DaysFromNow|,|Mode|,|UserID|

Report parameters

Default.

Additional filters

Work Group, Shift Start Time

(Rpt_563) Rebuild Centre Forecast Workload

This report is useful for rebuild centres to forecast their future workload. The report also takes into consideration rebuild work currently in progress If the workload is too high in certain periods, contact those responsible for the component forecast to assess whether a rebuild can be pushed out or brought forward, or transferring the rebuild to a different location.

Tips

Use Review and Sales status filters to understand actual versus potential workload.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Part Classification, Part Rating, Part Number, Part Type, Source of Supply, Component Code, Review Status, Sales Status

S

(Rpt_229) Service Accuracy

(Rpt_230) Service Accuracy Variance

(Rpt_231) Snapshot - The Worm

(Rpt_232) Snapshot Summary

(Rpt_233) Standard Job Coverage

(Rpt_234) Standard Job Interrogation

(Rpt_235) Strategy Task Margin Analysis

(Rpt_236) Summary Analysis

(Rpt_307) Summary Analysis - Modelling

(Rpt_521) Site P&L

(Rpt_522) Site Performance

(Rpt_523) Shift Downtime Log

(Rpt_524) Shift Downtime Summary

(Rpt_531) Strategy Task - Standard Job Details - Centreline

(Rpt_653) Sales Status and Strategy Forecast Health

(Rpt_654) Strategy Task Audit

(Rpt_833) Standard Job Version History

(Rpt_229) Service Accuracy

Measures the % of preventive maintenance services performed within a nominated tolerance (normally 10%) of the defined interval. To edit tolerances, right-click on the Fleet in the Explorer.

Regular, timely and quality servicing forms the backbone of good maintenance practices. In addition, it is often a warranty requirement. The Service Accuracy report measures the efficiency of the planning process to plan and schedule services in line with the maintenance strategy. Not performing regular services can lead to breakdowns and unplanned stoppages, and also have a long-term impact on component lives. The Services Accuracy Variance report shows more detailed service information than the Service Accuracy report, and more filters to analyse Service Accuracy information.



Incorrect or infrequent capture of utilisation (usages) can also lead to poor service accuracy.

Navigate to **Reporting & Analyses** > **Maintenance Efficiency** > **Planning**.

Tips

Provides basic information in either text, line or bar graph.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|LogoLocation|,
|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,
|EqpPlan|,|AnalyseBy|

Report parameters

Default.

Additional filters

None.

(Rpt_230) Service Accuracy Variance

Measures the % of preventive maintenance services performed within a nominated tolerance (normally 10%) of the defined interval. To edit tolerances, right-click on the Fleet in the Explorer.

This report shows more detailed service information than the Service Accuracy report, and more filters to analyse Service Accuracy information.



Incorrect or infrequent capture of utilisation (usages) can also lead to poor service accuracy.

Navigate to **Reporting & Analyses** > **Maintenance Efficiency** > **Planning**.

Tips

Date range can be set according to analysis requirements.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,
|LogoLocation|,|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,

|Site|,|Fleet|,|EqpPlan|,|AnalyseBy

Report parameters

Default.

Additional filters

None.

(Rpt_231) Snapshot - The Worm

Provides a graphical representation of actual downtime versus target for a specific time period.

Tips

Run this report for a Month and analyse by Day prior to month end in order to monitor availability targets.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Responsibility, Task Type

(Rpt_232) Snapshot Summary

Summarises Availability, Utilisation and MTBS KPIs into a single report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Responsibility, Task Type

(Rpt_233) Standard Job Coverage

The preparation and usage of standard jobs leads to improved efficiency, standardisation of work performed and improved parts forecasting.

The report shows the percentage of the value of tasks that have standard jobs associated to them in the Standard Jobs Manager.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Task Type

(Rpt_234) Standard Job Interrogation

Details, either by equipment or strategy task what standard jobs are linked and whether a Rotable Repair Strategy is in place.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, Strategy Task, Rebuild Location, Source of Supply, Linked to Std Job, Strategy

(Rpt_235) Strategy Task Margin Analysis

Reports actual margins achieved on Parts, Labour and Misc costs by analysing work order history.

Tips

Compare the actual margins in the report (grouping by Fleet and Subsystem)to those in the Set Margin financial table to determine whether forecast margins should be adjusted to reflect actuals.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Task Counter, Job Code, Cost Bearer, WO Status, Company, Cost Selector

(Rpt_236) Summary Analysis

Provides summary details of the cost per UOM to run the equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, Cost Type, Contract, Costs, Cost Bearer, Discount Factor %, Revenue

+ indicates that the UOM selected is used as a basis for the equipment evaluation.

(Rpt_307) Summary Analysis - Modelling

Provides summary details of the cost per UOM to run the equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, Cost Type, Contract, Costs, Cost Bearer, Discount Factor %, Revenue

+ indicates that the UOM selected is used as a basis for the equipment evaluation.

(Rpt_521) Site P&L

Shows a Profit and Loss report format the difference between Plant Hire revenue and Plant Hire costs. The CBD Adjustment area draws on Plant Hire Journals to show CBD transactions (for example, component disposal, capitalisation, depreciation).

Tips

Only applicable for equipment in AMT configured to sue the Plant Hire module.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Region, Division, Industry Group, Eqp Criticality, Equipment Class, Equipment Group, Eqp Category, Manufacturer, Cost Responsibility, Project - Equipment, Cost Centre, CC Activity, CC Location, CC Responsibility, Expense Element, Currency, Terminated

(Rpt_522) Site Performance

Shows the profit and loss per expense element for the equipment selected, and then summarises the profit, or loss, across all expense elements.

Tips

Only applicable for equipment in AMT configured to use the Plant Hire module.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Region, Division, Industry Group, Eqp Criticality, Equipment Class, Equipment Group, Eqp Category, Manufacturer, Cost Responsibility, Project - Equipment, Cost Centre, CC Activity, CC Location, CC Responsibility, Expense Element, Currency, Terminated

(Rpt_523) Shift Downtime Log

Contains the event details of the equipment and the downtime allocation information.

Tips

This report is the same as the Daily Downtime Log report, except this report has the ability to select Time and Date, not just Date.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|UserID|,|LogoLocation|

Report parameters

Default.

Additional filters

Task Type, EM Issue, +Responsibility

+ indicates that the Responsibility filter defines the data on the right side of the report.

(Rpt_524) Shift Downtime Summary

Designed to show the primary work order executed on an equipment in a given shift, based on an event. The work order reported will always be the primary work order of the event.

Tips

The report only considers IP and C events which occur within or overlap the selected date range.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Task Type, EM Issue, Cost Type, System, Sub System, Comp Code

(Rpt_529) Strategy Task - Standard Job Details

Lists, for strategy tasks linked to standard jobs, the details of the parts from the standard job.

Tips

Labour and Misc data from the standard jobs are excluded from this report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Task Type, Cost Bearer

(Rpt_531) Strategy Task - Standard Job Details - Centreline

Lists, for strategy tasks linked to standard jobs, the details of the parts from the standard job.

Tips

Labour and Misc data from the standard jobs are excluded from this report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Task Type, Cost Bearer

(Rpt_653) Sales Status and Strategy Forecast Health

This report has two sections - one section focusing on the status of current sales opportunities (that is, the next occurrence of each component), and the second focusing on the health of the component forecast (for example, overdue components). The data for this report is primarily sourced from Long-term Planning and Component Changeout Grid functions.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Parts/Labour/Misc, Currency, Strategy Task, Strategy Task Desc, Cost Type, System, Sub System, Component Code, Cost Bearer, Task Type

(Rpt_654) Strategy Task Audit

Use the Blank Fields filter to identify Strategy Tasks to identify Strategy Tasks with missing classifications.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Display Columns, Blank Fields, Cost Type, System, Sub System, Component Code, Modifier Code, Task Type, Task Counter, Strategy Task, Com. Code - Major, Com.Code - Minor, BEC Code, Part Type, Marketing Grouping, Parts Prob.%, Labour Prob. %, Misc Prob. %, New Parts Purc. %

(Rpt_833) Standard Job Version History

Compares different versions of Standard Jobs and displays differences and change-related information.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Report Type, Date Selector

Additional filters

Modified by User, Branch, Price Group, Std Job Currency, Model, Standard Job Number, Component Code, Modifier Code, Task Type, Job Code, Source, Part Number

T

(Rpt_238) Technical Performance Analysis

(Rpt_239) Technical Performance Analysis Chart

(Rpt_240) Technical Performance Details

(Rpt_241) Termination Summary

(Rpt_242) Total Cost Worksheet

(Rpt_342) Total Cost Worksheet - Modelling

(Rpt_391) Trending

(Rpt_679) Timesheet Journal

(Rpt_681) Timesheet Cost Analysis

(Rpt_238) Technical Performance Analysis

Lets you analyse the specific areas of the downtime relating to parts, occurrence types and so on, and shows the related costs, duration and lives achieved from components.

Run for each major component in a Fleet. "Analyse By" Occurrence Type (Failure Mode). Shows the average life/cost for each Failure Mode.

Tips

This detailed information can be used by product and parts support groups and factories to understand the causes of failure and the corresponding parts. The parts information is derived from the EQS/Task/Strategy level of the program.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Component Code, Modifier Code, Occurrence Type, Task Type, Task Counter, Original Part Type, Warranty, Redo, Life Confirmed

(Rpt_239) Technical Performance Analysis Chart

Provides a graphical view of specific technical areas of the downtime - occurrence type, cause, symptom and repair codes.

"Analyse By" original part type to identify components with poor life expectancy and then review the associated rebuild centres.

Tips

Allows quick analysis of performance in the downtime area. An example could be: is maintenance work being performed before or after failure on a regular basis and are users being specific when entering downtime data to assist with later analysis.

Requires the Part Type system table to be set up correctly with the associated rebuild centre.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Component Code, Modifier Code, Occurrence Type, Task Type, Task Counter, Original Part Type, Warranty, Redo, Life Confirmed

(Rpt_240) Technical Performance Details

Similar to the Technical Performance Report, but has set parameters. The report provides you with information relating to parts, occurrence types, costs and durations.

Tips

This report shows the work order number relevant to the downtime task. The report always shows the total cost of the Job at the Job (work order) Date, even though some reports may split these costs over different transactional months (for the Transaction Date reporting).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Additional filters

Component Code, Modifier Code, Occurrence Type, Task Type, Task Counter, Original Part Type, Warranty, Redo, Life Confirmed

(Rpt_241) Termination Summary

Highlights the potential financial adjustments that may be associated with an early termination of the contract.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Component Code, Task Type

(Rpt_242) Total Cost Worksheet

Lets you analyse costs by several different categories within the budget period. You are able to group the report by three categories for viewing particular cost information.

This report can be run to include overheads in the total cost.

Tips

This report will return results for both real and modelling if there is a mixture of equipment in the fleets selected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

CC Activity, CC Location, CC Responsibility, Cost Centre, Cost Bearer, Expense Element, Cost Category, Include, Costs

(Rpt_342) Total Cost Worksheet - Modelling

Lets you analyse costs by several different categories within the budget period. You are able to group the report by three categories for viewing particular cost information.

Tips

This report will return results for both real and modelling, if there is a mixture of equipment in the fleet selected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

CC Activity, CC Location, CC Responsibility, Cost Centre, Cost Bearer, Expense Element, Cost Category, Include, Costs

(Rpt_391) Trending

Report shows the element trending leading up to a component failure.

Tips

This report will return results for Real, non-terminated equipment only.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Branch, Site, Fleet, Model, Equipment, Division, Parent Equipment, Eqp Location, Eqp Category, Eqp Cost Centre, Cost Responsibility, Element Type, Element UOM

(Rpt_679) Timesheet Journal

Analyses the timesheet entries for specific equipment and cost centres.

Tips

This report will return results for real equipment only.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Cost Centre, Workgroup, Employee

(Rpt_681) Timesheet Cost Analysis

Analyses individual timesheet entries and their associated work orders and strategy tasks.

Tips

This report returns results for real, non-terminated equipment only.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Strategy Task, Workorder, Employee, Supervisor Confirmed

U

(Rpt_243) Unassigned Cost Analysis

(Rpt_2444) Unassigned Cost per UOM - by Date

(Rpt_245) Unassigned Cost per UOM - by Usage

(Rpt_246) Unassigned Cost per UOM - Comparison

(Rpt_247) Usage Analysis Chart

(Rpt_248) Usage Profile Details

(Rpt_249) Usage vs Profile Snapshot

(Rpt_250) Utilisation Rate

(Rpt_311) Usage Profile Details - Centreline

(Rpt_312) Usage Profile Details - Modelling

(Rpt_505) Usage Analysis Worksheet

(Rpt_830) User Work Order Status Audit Report

(Rpt_243) Unassigned Cost Analysis

Allows the unassigned task costs in the current projections to be analysed in either a text or a graphical view. The "Report On" parameter provides an option to use the ERP codes on work order settlements or the AMT codes the work order settlement is matched with. AMT codes exist for component code, modifier code and task type, as well as cost type, system and sub system, as these are determined from the component hierarchy.

"Analyse By" Component Code identifies the components that have the highest unassigned costs.

Tips

The filters allow a thorough dissection of the costs by different cost bearers and by parts, labour and misc tasks.

This report always shows the total cost of the Job at the Job (Work order) Date, even though some reports may split these costs over different transactions.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Contract, Cost Type, Component Code, Modifier Code, Task Type, Report On, System, Sub System, Cost Bearer, WO Status, Task Counter, P/L/M

(Rpt_244) Unassigned Cost per UOM - by Date

Shows the unassigned costs per unit of measure for the entire equipment term to date.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_245) Unassigned Cost per UOM - by Usage

Shows the unassigned costs per unit of measure by elapsed machine usage for the entire equipment term to date.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_246) Unchanged Cost per UOM - Comparison

Shows the unassigned costs per unit of measure by elapsed machine usage for the entire equipment term to date.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_247) Usage Analysis Chart

Displays the machine usage. Enables users to analyse by equipment plan, branch, month and so on. The target usage profile is shown to gauge how close the actual is to the profile. **Reporting & Analyses** > **Equipment Performance** > **Utilisation**.

Tips

If the actual usage has a large variance form the target profile, you should review the usage of the machine and make the necessary changes to the usage profile.

Initial parameters

|ControlType|,|Idlist|,|TextList|,|UsageRateReport|,|ProjTaskID|,|RateAgainstQuomID|,
|IsNotShowAsPerc|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, Gross Usage As, Avg Per Eqp Plan

(Rpt_248) Usage Profile Details

Displays the usages pertaining to an equipment, with both actual and cumulative values showing. This report can be generated for a specific month period, but defaults are set to the last twelve months. than zero.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,1,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_249) Usage v Profile Snapshot

Compares the usage of the equipment (based on the UOM) to the target. This could highlight issues with the application or the client machine usage. **Reporting & Analyses** > **Equipment Performance** > **Utilisation**.

Tips

If the actual usage has a large variance from the target profile, users should review the usage of the machine and make the necessary changes to the usage profile.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|QUOM|,|ProjectionTypes|,|UserID|Additional filters

Report parameters

Default.

Additional filters

UOM, Avg Per Eqp Plan

(Rpt_250) Utilisation Rate

Displays the utilisation of the machine and the target to be achieved. **Reporting & Analyses** > **Equipment Performance** > **Utilisation**.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, Show Usage at UOM

(Rpt_311) Usage Profile Details - Centreline

Displays the different usage profiles set up with both actual and cumulative values showing. The report can be generated for a specific month period, but defaults are set to the last twelve months.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,4,|UserID|

Report parameters

Default.

Additional filters

Branch, Site, Fleet, Equipment

(Rpt_312) Usage Profile Details - Modelling

Displays the different usage profiles set up with both actual and cumulative values showing. The report can be generated for a specific month period, but defaults are set to the last twelve months.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,5,|UserID|

Report parameters

Default.

Additional filters

Branch, Site, Fleet, Equipment

(Rpt_505) Usage Analysis WorkSheet

Contains cumulative daily usage and average daily usage. **Reporting & Analyses** > **Equipment Performance** > **Utilisation**.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|QUOM|

Report parameters

Default.

Additional filters

Contract, UOM, Use Actual UOM Reading

(Rpt_830) User Work Order Status Audit Report

Provides a history of the user work order status for a work order. The report shows who changed what (Login Name of user who made the change), and when, giving a view of the lifecycle of the user-defined status of the work order.

Initial parameters

|ControlType|,|Idlist|,|TextList|,|LogoLocation|,|UserID|,|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

User, User WO Status

V

(Rpt_251) Variance Analysis Worksheet

(Rpt_313) Variance Analysis Worksheet - Modelling

(Rpt_251) Variance Analysis Worksheet

Lets you analyse between projections and group by a maximum of five categories to show variances in a period of time - default date is last month but this can be modified as required.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Strategy Task, Task Type, Maintenance Type, Cost Bearer, Cost Centre, Responsibility, Expense Element, Labour Activity, Parts/Labour/Misc, Currency

(Rpt_313) Variance Analysis Worksheet - Modelling

Lets you analyse between projections and group by a maximum of five categories to show variances in a period of time - default date is last month but this can be modified as required.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Strategy Task, Task Type, Maintenance Type, Cost Bearer, Cost Centre, Responsibility, Expense Element, Labour Activity, Parts/Labour/Misc, Currency

W

(Rpt_237) Workorder Details

(Rpt_252) What-if - Cost Analysis

(Rpt_253) What-if - Cost per UOM

(Rpt_254) What-if - Life Cycle Analysis

(Rpt_255) What-if - Period Analysis

(Rpt_256) Work Location Allocation

(Rpt_257) Workorder Settlement Interrogation

(Rpt_258) Workorder Settlement Interrogation (Cost)

(Rpt_259) WorkScope Coverage

(Rpt_314) What-if- Cost analysis - Modelling

(Rpt_315) What-if - Cost per UOM - Modelling
(Rpt_316) What-if - Life Cycle Analysis - Modelling
(Rpt_317) What-if - Period Analysis - Modelling
(Rpt_321) Workorder Settlement List
(Rpt_350) Warranty Claim
(Rpt_351) Warranty Analysis
(Rpt_352) Workplan - Workorder Summary
(Rpt_416) WorkPlan 2
(Rpt_488) Workorder Labour Analysis
(Rpt_508) Work Loading - Labour
(Rpt_532) Workorder Settlement Transactions
(Rpt_546) Workorder Authorisation List
(Rpt_665) Workorder Duration Analysis
(Rpt_832) Workload Optimisation

(Rpt_237) Workorder Details

Lets you select areas of the downtime tasks that they wish to analyse with numerous display and sort fields to select from. A very open report to allow you to customise your required data.

Tips

Date range can be set according to analysis requirements.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|AutoLaunch|,
|BranchID|,|SiteID|,|FleetID|,|EqpPlanIDZ|,|Branch|,|Site|,Fleet|,|EqpPlan|,
||DateChecked|,|StartDate|,|EndDate|,|ISNOTAsDefault|_StrategyTakskOnly|,|
DisplayFields|,|DisplayFieldsLabels|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Modifier Code, Task Type, Task Counter, Status, OHS, Perform At, Raised By, Part Number, Customer Reference, Workorder #, YTS/Backlog, Strategy Tasks, Suppression Impact, Primary WO Only, Durability Index, User WO Status

(Rpt_252) What-if - Cost Analysis

Compares the cost between two projections for a nominated time period.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Strategy Task, Task Type, Maintenance Type, Cost Centre, Responsibility, Expense Element, Labour Activity, Parts/Labour/Misc, Currency

(Rpt_253) What-if - Cost per UOM

Compares the Cost per UOM between two projections selected by the user.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Fleet, Equipment, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_254) What-if - Life Cycle Analysis

Allows analysis of costs between a budget or current projection and one created within the scenario manager.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Cost Selector, Component Code, Task Type, Strategy Task

(Rpt_255) What-if - Period Analysis

Compares the Cost between two projections for a nominated time period.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Strategy Task, Task Type, Maintenance Type, Cost Centre, Responsibility, Expense Element, Labour Activity, Parts/Labour/Misc, Currency

(Rpt_256) Work Location Allocation

Represents the current work plan by location with over and under allocation to particular locations highlighted.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|SiteID|,|Site|,|AnalyseBy|,
|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Work Location Site, Work Location, Event Status

(Rpt_257) Workorder Settlement Interrogation

Analyse the work order costs. Gives users the ability to monitor spending throughout a period, and for reconciliation of costs against the client enterprise resource planning system.

Tips

Always shows the total cost of the Job a the Job (Work order) Dare, even though some reports may split these costs over different transactional months (for Transaction Date Reporting).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|AutoLaunch|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Sub System, Task Type, Modifier Code, Job Code, WO Status, System, Strategy Task Desc, Strategy Task, Component Code, Task Counter, Cost Bearer, Cost Selector

(Rpt_258) Workorder Settlement Interrogation (Cost)

Analyses the work order costs at cost. Gives users the ability to monitor spending throughout a period and for reconciliation of costs against the client enterprise resource planning.

Tips

Always shows the total cost of the Job a the Job (Work order) Dare, even though some reports may split these costs over different transactional months (for Transaction Date Reporting).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Sub System, Task Type, Modifier Code, Job Code, WO Status, Cost Selector, System, Strategy Task, Component Code, Task Counter, Cost Bearer, Company

(Rpt_259) WorkScope Coverage

Shows the amount of workscope coverage relating to each specific model and the related tasks for the model. A workscope is a set of model-specific instructions on how tasks should be performed, for example, safety process, tooling requirements.

The preparation and usage of work scopes leads to improved quality of work performed

Tips

Ensuring that relevant documentation relating to work processes are held and maintained in AMT assets on-site users and also enables new contract managers and greenfield sites to utilise the specific documentation in their mobilisation.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID

Report parameters

Default.

System, Sub System, Component Code Task Type

(Rpt_314) What-if - Cost Analysis - Modelling

Compares the cost between two projections for a nominated time period.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Strategy Task, Task Type, Maintenance Type, Cost Centre, Responsibility, Expense Element, Labour Activity, Parts/Labour/Misc, Currency

(Rpt_315) What-if - Cost per UOM - Modelling

Compares the Cost per UOM between two projections selected by the user.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Fleet, Equipment, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_316) What-if - Life Cycle Analysis - Modelling

Allows analysis of costs between a budget or current projection and one created within the scenario manager.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Cost Selector, Component Code, Task Type, Strategy Task

(Rpt_317) What-if - Period Analysis - Modelling

Compares the Cost between two projections for a nominated time period.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Strategy Task, Task Type, Maintenance Type, Cost Centre, Responsibility, Expense Element, Labour Activity, Parts/Labour/Misc, Currency

(Rpt_321) Workorder Settlement List

Lets you validate work orders more effectively. Data is shown relating to the strategy task being linked, costs and usages of work order.

Tips

Shows the total cost of the Job at the Job (Work Order) Date, even though some reports may split these costs over different transactional months (for Transaction Data Reporting).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|BranchID|,|Branch|,
|SiteID|,|Site|,|FleetID|,|Fleet|,EqpPlanID|,|EqpPlan|,|SubSystemID|,
|SubSystem|,||TaskTypeID|,|TaskType|,|CompCodeID|,|CompCode|,
|ModCodeID|,|ModCode|,|JobStatusID|,|JobStatus|,|Checked|,|Strategy|,
|WONumber|,|ISNOTFutureWO|,|LastMonthDate|,|Today|,|AnalyseBy|,
|ProjectionTypes|,|IsLinkedToEquipment|,|UserID|,|StartDate|,|EndDate|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Modifier Code, Task Type, Task Counter, Strategy Task, Strategy, Job Status, Workorder Number, Internal WO Ref, WIP, Link to Workorder, Before Equip Start, AMT WO.

(Rpt_350) Warranty Claim

Report for suitable for providing to a technical services or warranty divisions as a summary to the warranty claim. You can group the report by three categories for viewing particular cost information.

Tips

Use of the Warranty Claim number will assist with running this report. If filters are left at a branch or site level, the information will generate for every warranty claim registered at that level.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Warranty Claim

(Rpt_351) Warranty Analysis

Provides users with the ability to analyse the warranty claims registered or pending in the system.

Tips

The ability to Analyse By provides you with the option to choose two fields to group the report by for ease of use and selection of specific warranty data.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Component Code, Task Type, System, Sub System, Claim Status, Potential Claim

(Rpt_352) Workplan - Workorder Summary

Provides a summary based on work orders that are Yet to Start, In Progress or Completed as a result of planning. The report lets the site foreman and supervisors plan the resources required for the upcoming outstanding work orders.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|GroupByID|,|GroupBy|,|SiteID|,
|Site|,|FleetID|,|Fleet|,|EqpPlanID|,|EqpPlan|,|UserID|

Report parameters

Default.

Additional filters

Workorder Branch, Workorder Site, Owner, Work Group, Employee Resp., Task Type, Work Location, Status, Priority, Maint. Order Type, Planning Group

(Rpt_416) WorkPlan 2

Provides a high level summary based on work orders that are Yet to Start, In Progress or Completed as a result of Planning.

Tips

The filter Status defaults to Yet to Start and In Progress to allow quick reporting of outstanding work orders only.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Workorder Branch, Workorder Site, Work Group, Employee Resp., Labour Activity, Task Type, Work Location, Status, Priority, Raised By, Maint. Order Type, Planning Group, STO

(Rpt_488) Workorder Labour Analysis

Contains Labour data.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Labour Activity, Work Group, WO-Operation, Cost Bearer, Cost Centre, Expense Element, Employee, Week Start

(Rpt_508) Work Loading - Labour

Compares available labour to that which was planned, and to that which was actually used.

Tips

This report utilises snapshots of the seven day weekly work plan created in AMT. The day on which the work plan is saved is site specific and, therefore set up using site administration.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Subsystem, Component, Task Type, Job Ready Status, Work Group, Priority

(Rpt_532) Workorder Settlement Transactions

Displays all transactions of work order settlements, including child work order settlements which are linked to a parent. The transaction details include splits between cost centres, cost bearers and work groups.

Tips

This report is useful for reconciling AMT's work orders to external systems.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Cost Bearer, Work Group, Cost Centre, Cost Category, Expense Element, Contract, Parts/Labour/Misc, WIP

(Rpt_546) Workorder Authorisation List

Audit report detailing who authorised which work orders to what value and how the authorised value compares to the costs which were ultimately charged to the work order.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Task Type, Status, Job Ready Status, Source, Raised By, Priority, Work Group, Employee Resp., Backlog Age, Strategy, Requested, Authorised, WO Number, Internal WO Ref, Authorisation Amt >=, Actual Amt >=, Variance Amt >=, Maint. Order Type, Planning Group, STO

(Rpt_665) Workorder Duration Analysis

Displays the planned and actual duration of work orders for the selected date range and indicates their Planned vs Actual Variance.

Tips

Sort by Workorder Duration Variance to identify the work orders with the highest or lowest variances.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Component Code, System, Sub System, Task Type, Sort By

(Rpt_832) Workload Optimisation

Compares Labour Hours by Month, Quarter or Year for two or more projections.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

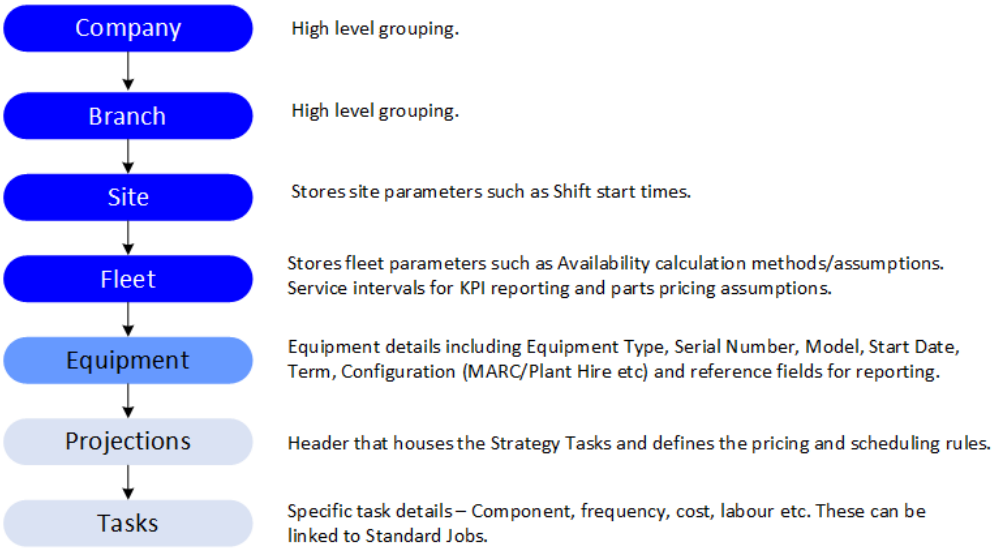
Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity

Key Concepts

Asset hierarchy

The diagram below shows the AMT asset hierarchy.



Set up the asset hierarchy

Every asset in AMT must belong to a fleet, which belongs to a site, which belongs to a branch. This is known as the asset hierarchy. See Asset Hierarchy for more information. Before you can register an asset in AMT, the fleet it belongs to must be set up - also the site and branch will need to be set up if they do not exist.

The table below shows the branch, site and fleet functions and the fields that need to be set up before registering the asset.

AMT Function	Key Fields	Descriptions
Branch Administration	Branch Name	The name of the branch (must be unique).
	Branch Code	The ERP code of the branch (if applicable) where costs for assets are captured.
Site Administration	Site Name	The name of the site (must be unique).
	Site Code	The ERP code of the site (if applicable).
	Shift start time	The start time of the first shift on site (24 hour).
Fleet Administration	Pricing Group	Typically, this is the default pricing group.

AMT Function	Key Fields	Descriptions
		However, if the site's parts are priced at a special rate (that is, includes a discount), then a specific pricing group can be selected. Pricing groups are set up in the system administration area of AMT, and require special parts pricing to be imported or entered into AMT.
	Service Interval	The service interval is only used for the Service Accuracy report. The value correlates to the primary Unit Of Measure (UOM) for that asset. For example, enter 250 if the PM service interval is 250 engine hours. If it is not applicable, enter 1.
	Service Tolerance	The service tolerance is only used for the Service Accuracy report. The value correlates to the primary Unit Of Measure (UOM) for that asset. For example, if the service interval is 250 and a 10% service tolerance is entered, preventative maintenance services performed within 225 – 275 of the previous service will be deemed okay, while those outside that range will be out of tolerance on the report. Enter 250 if the preventative maintenance service interval is 250 engine hours.
	Availability Type	Default availability calculation setting (physical or mechanical).
	Availability Calc Parameters	The parameters applicable to availability forecast reports (if such forecasting is required.

Asset types

There are three asset types in AMT:

1. **Centrelines**: Centrelines are generic assets that are used to store standard projection templates for a model. It is typically the starting point for registering real assets or performing modelling or feasibility

studies.

2. **Modelling**: These are dummy assets that have been set up to prepare tenders (OEMs), perform feasibility studies or benchmark different assets. A number of assets can be evaluated in different modelling scenarios.
3. **Real**: This is a real asset that is operating. Only real assets are enabled for planning and generating transactional data.

Life cycle costing

Life cycle costing is the process of projecting the costs and productivity for assets over their life time. For existing assets this includes the actual costs to date, and the forecast costs to the estimated termination date.

How does AMT do life cycle costing?
In AMT Projections you can set up the strategy tasks that contain the assumptions that are used to forecast costs. This is also known as the Maintenance Strategy.

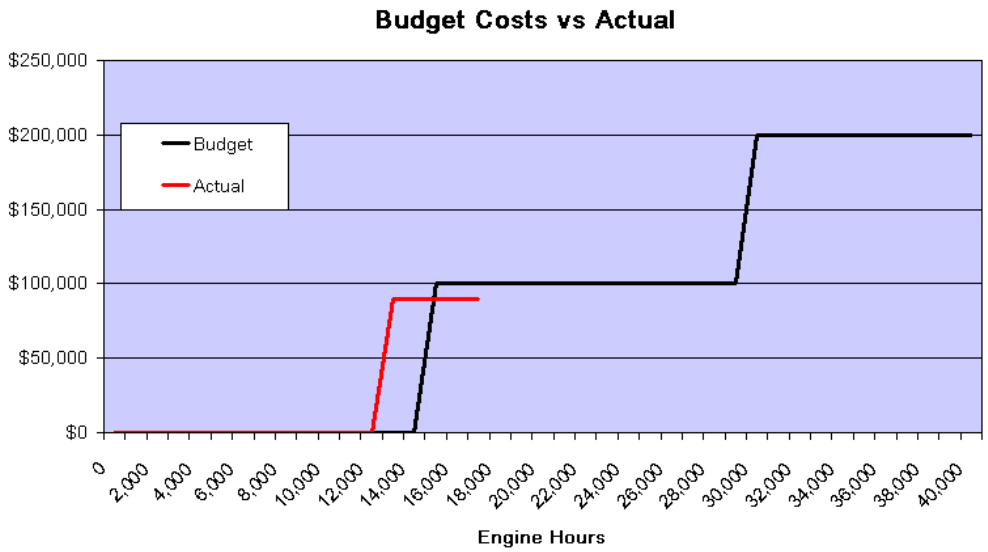
The strategy tasks can include any type of task such as:

1. Replacing major components.
2. Replacing minor components.
3. General repairs.
4. Servicing.
5. Consumables.

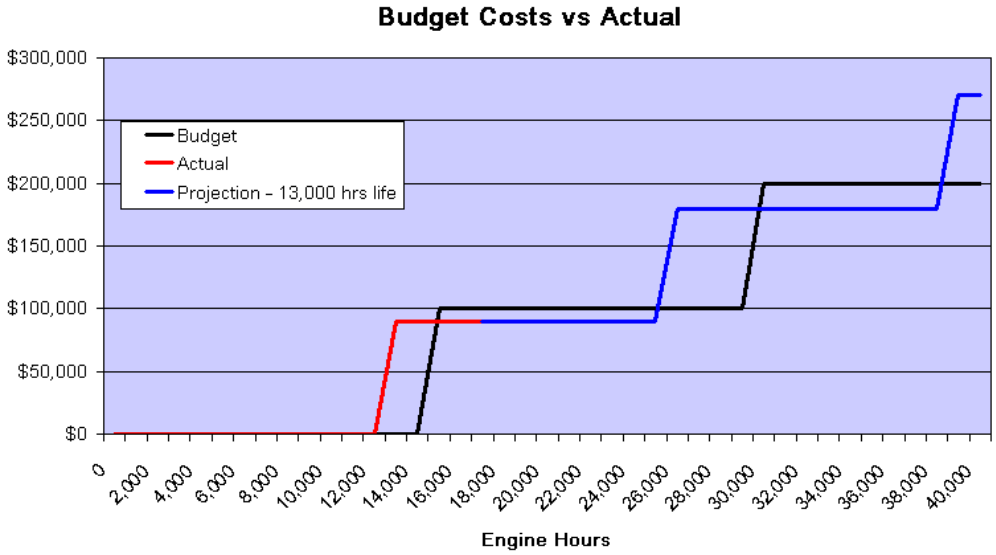
Fuel, tyres, ownership and operator costs can also be included.

Example

Assume that three years ago you purchased an asset (say, a truck), that consists of one component (an engine) that was budgeted to last 15,000 engine hours and costs $100,000 to replace. The life of the asset is 40,000 engine hours and you therefore budgeted to change out two engines and have a life cycle cost of $200,000 ($5.00 per hour). You are now at 18,000 engine hours and you see that you are $10,000 ahead of budget, but there was an engine replaced at 13,000 hours and cost $90,000. In the chart below the black line represents your initial assumptions (budget) and the red line your actual cumulative costs to date.

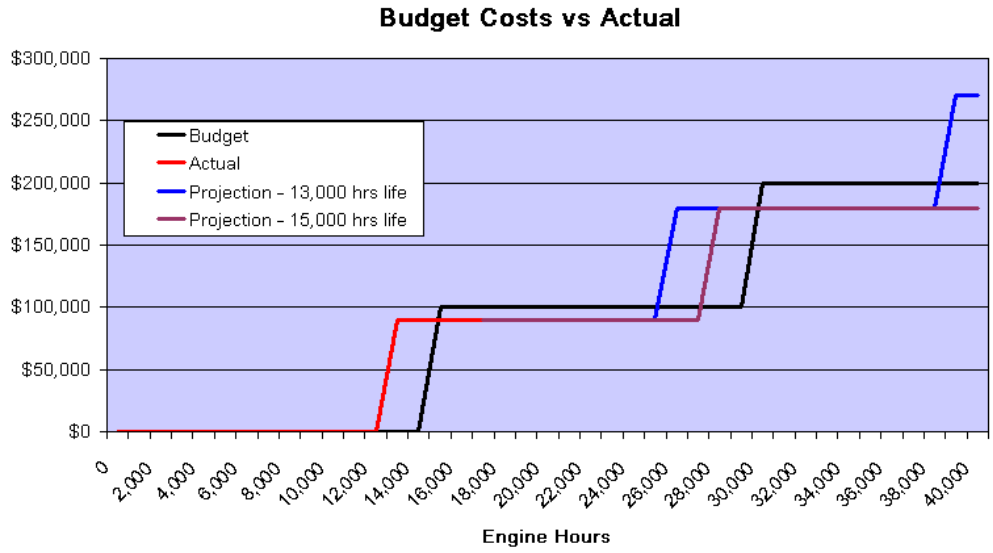


1. At 14,000 hrs you were $90,000 behind budget while at 15,000 hrs you are $10,000 ahead of budget. Nothing actually happened in this period, so it is misleading that there is such a big variance.
   If you assume that future engines will only last 13,000 hrs and cost $90,000, you will need two more engines and your life cycle cost will be $270,000 (the last engine will be required at 39,000 hrs). The blue line shows this forecast.



2. The variance between your budgeted life cycle cost of $200,000 and your current expected life-cycle cost of $270,000. The difference of -$70,000 is known as the Projected Cost Variance (PCV). This is a key measure in AMT that is calculated for all components (strategy tasks) and can be analysed at the component, asset, fleet and model level.

   If, however, you assume that future engines will last 15,000 hrs and cost $90,000 then you will need one more engine and your life cycle cost will be $180,000 (the last engine will be required at 28,000 hrs). The dark red line shows this forecast.



3. Each line in the chart represents a projection in AMT. An asset can have many projections. This lets you evaluate the life cycle of different strategies (or assumptions).

Usage profile

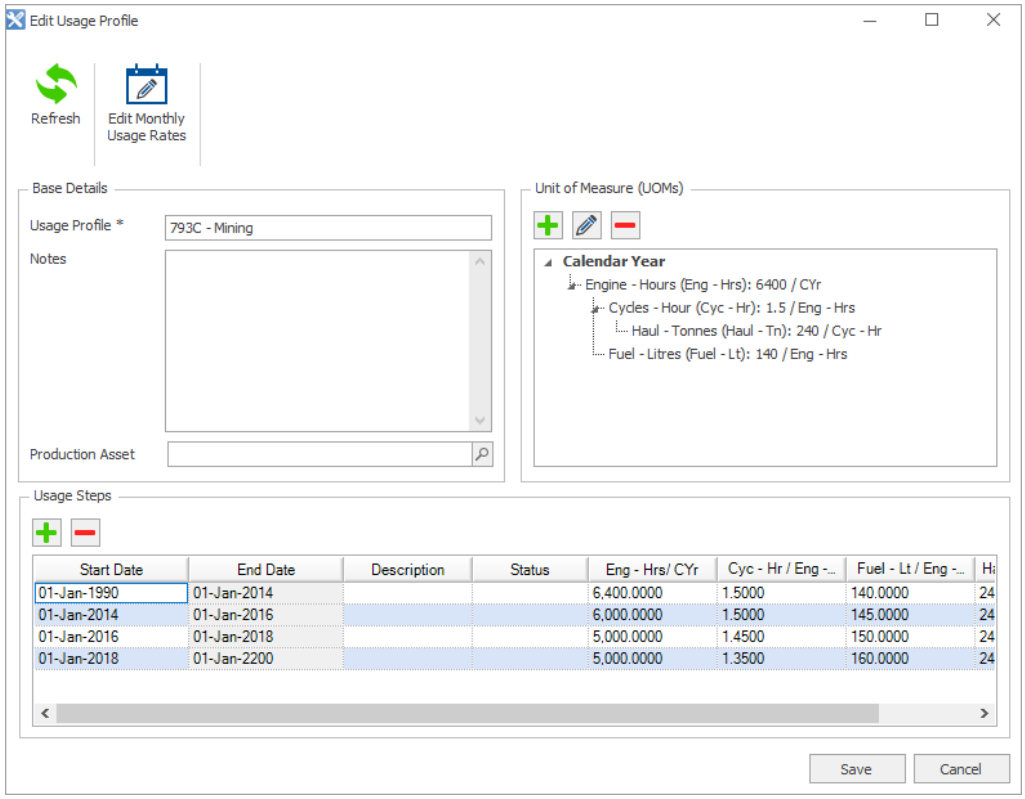
Unit of measure (UOM) refers to any cumulative usage related to an asset, such as machine hours, fuel litres, kilowatt hours and tonnes. They are also referred to as **Usages** in AMT. Usages are used to:

1. **Drive maintenance schedules**
   For example, an engine is scheduled based on fuel litres. In the strategy task you can select the UOM appropriate for that asset.
2. **Forecast productivity**
   For example, tonnes moved can be set up as a UOM. Tonnes moved can then be used to forecast production.

Usage Profiles enable you to:

- Set up a profile for the usage rates or utilisation for groups of assets.
- Set up relationships between the usages (for example, tonnes is a dependency of engine hours).
- Vary the usage rates over time to reflect changing operating conditions (for example, deeper pit, rainy season, winter and other conditions).

  The example screenshot below shows a usage profile with a hierarchy of usages.



- Usage profiles allow you to set up the severity of the situation. For example, if the pit in a mine becomes deeper, the fuel burn rate increases and the engine life in hours will decrease.
- Actual usages can be captured in AMT and plotted against the usage profile.

- Usage profiles are optional for assets. You can elect to either have no usage measure for an asset (all strategy tasks are then date-based) or apply a simple utilisation where one UOM is selected and one utilisation rate set (for example, 4,000 engine hours per annum).



The **Status** field within a usage profile step is only applicable for the tendering and forecasting of a specific type of equipment which is configured for this functionality, and therefore, in most cases, does not impact the usage forecast by AMT.

Projections

Projections house strategy tasks that are used to forecast costs and drive the planning process. Every real asset has at least one projection, which is the current projection. Additional projections can be created if the multi-projection option is set in the asset. For assets under a maintenance contract (MARC), a budget projection is also created, which contains maintenance and costs agreed under the maintenance contract.

Refer to projection types for the real asset projections.

All of the topics in this user guide under strategy definition are focused on **current projections**.

Projection types

A projection in AMT represents the life cycle costing for an asset, which can be broken down to the component level. An asset can have a number of projections, including current budget and estimate-draft. Projections house the strategy tasks that are used to forecast costs and drive the planning process.

For an asset owned by the business there are a number of different types of projections depending on the asset type. While the structure of each projection is the same, each type of projection behaves slightly differently.

Centreline asset projections

1. Centreline projection.
    - Centreline.
    - Always assumed to be current.
    - Cannot have a start date in the past - AMT updates the start date at the start of each month, so scheduled tasks can never become overdue.
    - Strategy tasks can be linked to standard jobs.

Modelling asset projections

1. Estimate-Current projection.
    - Modelling assets can only have one Estimate-Current projection.
    - Represents the latest assumptions.
    - Can link to standard jobs but pricing is not automatically updated if the standard job is changed (so that history is always maintained).
    - Enter the expected asset start dates (future or historical) and terms.
    - There is an option to escalate costs to today's values or show at original values.

2. Estimate-Draft projection.
    - Modelling assets can have many Estimate-Draft projections.
    - Provides full version control.
    - Behaves the same way as Estimate-Current.

Real asset projections

1. Current projection.
    - The live assumptions for an asset.
    - Drives the planning and execution of work.
    - Real-time updating for actual maintenance activities (work orders).
    - Reflects the real-time life cycle costing for an asset.
    - Leaves the link to standard jobs.
    - All strategy task costs are escalated to today's values.
2. Alternate projection.
    - Used for what-if analyses.
    - You can use a copy of the current projection in a test environment.
    - Behaves like a current projection.
3. Archive projection.
    - Save a copy of a projection.
    - Contains all the details of a projection, but does not change for anything in the future.
    - Used to provide an audit trail or reference point.
4. Budget projection.
    - For maintenance contracts, the budget projection usually reflects the commercial assumptions made to determine contractual billing rates.
    - Represents original assumptions or company benchmark performance.
    - Can link to standard jobs, but pricing is not automatically updated if a standard job is changed (so that history is always maintained).
    - Will schedule historical tasks to when they were expected to occur (so they do not become overdue). Therefore, the projection flexes with utilisation (it is not a data-based accounting budget).
    - Will escalate historical costs to the date they were expected to occur - called escalating costs over time.
    - Not impacted by actual occurrences, only utilisation.

Strategy task

A projection contains the schedule of work required to manage the asset through its life cycle, and the costs associated with that work. AMT breaks down an asset’s cost projection to the component level. The type of work performed on a component and its cost are defined in a strategy task.

You can create a strategy task by navigating to

- **Setup & Admin** > **Projections** > **Strategy Task Manager** , applying the relevant filters then clicking the **Add** icon.
- **Setup & Admin** > **Projections - Multi-editors** > **Copy Strategy Tasks**.

You can also get to these windows by navigating to

- **Centrelines** > **Administration**.
- **Modelling** > **Administration** > **Projections**.

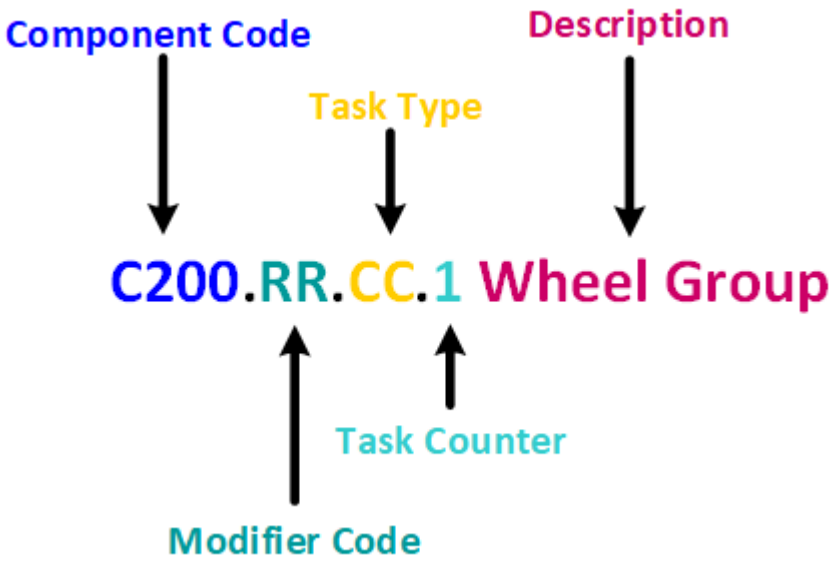
An AMT strategy task is a specific maintenance task that is forecast to occur periodically over the life cycle of the asset. At this level you are able to identify:

- costs to perform work on components, services and so on.
- type and frequency of work that will be required.
- parts and labour required.

Both strategy tasks and work orders are uniquely defined by a combination of four codes:

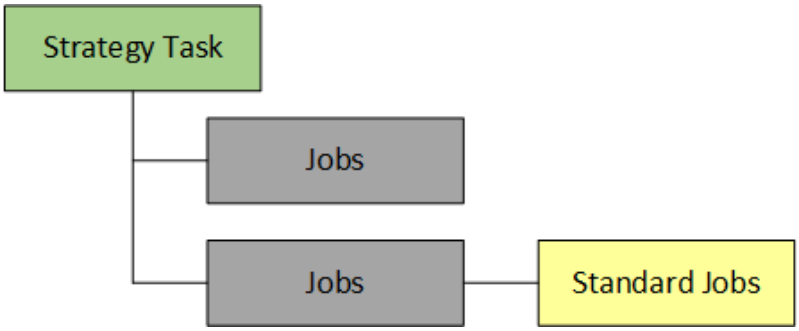
- Component Code (CC)
  What is being worked on or forecast (for example, HCYL – Hoist Cylinder).
- Modifier Code (MC)
  Used to provide an additional identification to the component code (such as the component's location on the machine, for example, right, left, upper and so on.).
- Task Type (TT)
  A code and description combination that identifies what is to happen at an occurrence of the task (for example, CC – component change out, GR – general repair).
- Task Counter (TC)
  Defines the sequence by which a task is performed, for example:
  - C220.RR.CC.1 (First/Freq 15,000 until 30,000hrs) reflects a remanufacturer strategy.
  - C220.RR.CC.2 (First 45,000hrs Freq. 15,000) reflects a rebuild strategy.

    Most Tasks do not use the Task Counter. It will be set to zero, for example, C200.RR.CC.0.



Strategy Task Structure

The structure of a strategy task in a projection is pictured below.



Each strategy task can have one or more **strategy task jobs** and each job has a cost, which is split into **parts**, **labour** and **miscellaneous**. You can manually enter the costs into each of these categories, or you can link the job to a standard job.

Strategy tasks can represent a variety of activities:

- Component changeouts
- PM services
- Inspections
- General repairs
- Unscheduled/miscellaneous work (that is, provision for costs)

Strategy tasks that are contained in the current projection are repetitive strategic maintenance activities, which typically require

1. Budgeting
2. A maintenance strategy
3. Standard jobs
4. Part, rotable component and costs to be forecast.
5. Long-term planning (for example, in order to consider lead times)
6. Short-term planning (for example, resources, location, support assets)

Strategy task modes

You can create a task in normal or unassigned mode. Below are examples of normal and unassigned task mode structures. Unassigned tasks have the same structure as simple mode tasks but have some limitations. Below is a summary of the characteristics of each task mode:

Normal

- Can be a planning task or a non-planning task.
- Can use advanced scheduling logic.
- Can be associated with a rotable part.
- Can have a rotable repair strategy.
- Can have jobs and therefore multiple currencies in cost build up.
- Can link to multiple standard jobs (if the task has multiple jobs).

Standard jobs can only be linked if the standard job module of AMT has been implemented. You should refer to the Standard Jobs Overview for more information.

You must consider the above characteristic before creating a task in the projection. It is not possible to swap or change between task modes after the task has been created.

Unassigned

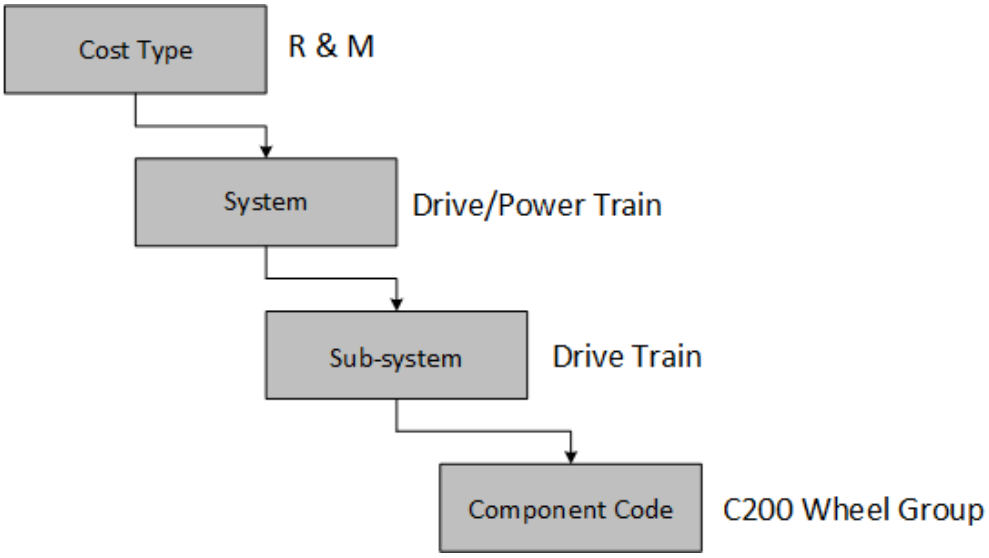
- Can only be a non-planning task.
- Cannot use advanced scheduling logic.
- Cannot be associated with a rotable part.
- Cannot have a rotable repair strategy.
- Cannot be linked to a standard job.

Unassigned is a strategy task mode to:

1. Budget for general repair costs at the subsystem level, to complete a full lifecycle costing model.
2. Group actual costs for non-strategy tasks into subsystems for high-level cost reporting.

Component code hierarchy

To provide greater flexibility in reporting and analysis, AMT has a four-tiered component code hierarchy. The hierarchy is set up in the system tables and are automatically applied when you select a component code.



1. A subsystem in the component code hierarchy is flagged as the default; any new component codes received by AMT through other software, such as SAP, are automatically added to the default subsystem.
2. Editors in AMT allow you to manipulate the component code hierarchy, such as adding or moving systems, subsystems and component codes.

Unassigned

To create a full life cycle cost budget or model, you need to consider not only the large expenses (such as major component changeouts, services) but also the smaller, usually more frequent, general repairs. It is impractical to budget for every possible type of repair; you don't want to create a strategy task for every type of repair. It would be preferable to budget for general repairs on electrics, general repairs on hydraulics, general repairs on the body and frame and so on.

In the back ground of each projection, AMT creates unassigned budget lines for every subsystem that exists in the Component Code Hierarchy system table. These lines are not used until one of the following:

- You elect to include one in a forecast by assigning it a frequency and cost estimate.
- An actual cost is incurred for a non-strategy task (see the next topics for explanations of actual costs and unassigned).

An unassigned task in a projection has the following codes:

- **Component Code**: Zxxx - Subsystem - Unassigned (for example, Z011 - Wheel and Tyres - Unassigned).
- **Modifier Code**: default modifier code (for example, 0).
- **Task Type**: UN - Unassigned.
- **Task Counter**: default task counter (for example, 0).

Z codes are managed dynamically by AMT. For example, if you add a subsystem to the Component Code Hierarchy, AMT creates an unassigned task with a Z code for it in the background.

Cost bearer

The cost bearer is the part of your business responsible for the costs incurred when performing work on the asset.

Reporting in AMT is primarily associated with maintenance costs. When work is being performed on the asset, the cost can be categorised by the responsibility for the cost. Not all work performed on the asset is categorised as a maintenance cost.

For example, a warranty repair. There may be a cost associated with the repair, but warranty is responsible for the cost, not the maintenance department.

All costs in AMT need to be categorised by the cost bearer.

Typical cost bearers include:

- CC - Contract or Maintenance (which is the default AMT cost bearer)
- W - Warranty cost.
- P - Policy.
- O - Operational cost.
- E - Customer (external) cost.

Cost bearer is defined per equipment and strategy task, and actual cost transactions. If a strategy task does not have a specific cost bearer assigned to it, the strategy task inherits the cost bearer from the equipment it is linked to.

In some cost and financial reports, AMT only reports on the cost bearer which is flagged in the system as the default. In these cases, the report's launch menu will not have a Cost Bearer filter.

All contract costs, defined as strategy tasks, must have the cost bearer set to contract as it is only these costs that are included in AMT's contract tendering reports.

Cost escalations

Cost escalation is an increase or decrease in the cost of parts, labour and miscellaneous costs over time.

Cost escalations are entered into the Escalation system table. Escalation indexes are created, and there is a default index for each of parts, labour and misc. These indexes can be applied to parts, labour or misc. at the strategy task level. The default parts, labour and misc. indexes are used where no index is applied.

Escalation percentage can be entered for any date for these Escalation Indexes.

Escalations are applied to all asset types. How escalations are applied in AMT depends on the projection and the analyses being performed.

Current projections - actual costs

As a general rule, for all reports in AMT, actual costs are never escalated. This includes all financial reports.

However, there are exceptions where the actual costs are escalated to today's value. This occurs when average cost analyses and benchmarking are performed. The forms and reports where this occurs are:

- Average work order costs.
- Fleet component performance summary.
- Strategy task analyser.
- Technical performance analysis.
- Technical performance analysis chart.
- Technical performance details.

In some other forms, you can view actual costs as escalated (to today's values), or unescalated.

Current projections - future costs

All future costs on the current projection use costs escalated to today's values. These costs are entered into strategy tasks and have a pricing date nominated in the strategy task. These costs are escalated from this date to today.

Example

On 1 January 2019, you enter a cost of $1,000 for a strategy task. If there is a 5% escalation on 1 March 2019, and you run a forecast report on 1 June 2019, the report uses $1,050 for the forecast cost for that strategy task.

Where a strategy task is linked to a standard job, AMT assumes the standard job pricing is always up-to-date and no escalation is applied.

All future costs are forecast at today's values. Do not enter future escalations into AMT.

Budget projections - costs to date

The costs entered into the budget projection have a pricing date that sets the point from which they are escalated. For the budget projection, the pricing date is set in the **Projection Header** form and applies to all strategy tasks.

The general principle is that budget costs are escalated over time. This means that the costs are escalated to the point that they were budgeted to occur.

Example

For the engine on truck 1 assume:

- Budget pricing date = 1 January 2016.
- Term = 25,000 hours.
- Engine strategy task: cost = $100,000 at a frequency of 10,000 hours.
- The truck reached 10,000 hours on 27 October 2017.
- Escalation history.
    - 5% on 1 December 2016.
    - 3% on 1 July 2017.
    - 4% on 1 Jan 2018.

If you run the life-cycle cost report on 1 May 2018, it will show:

- Budget costs to date: $108,15 (5% + 3% cumulative)
- Budget future costs: $112,476 (5% + 3% + 4% cumulative)

However, there are exceptions where the actual costs are escalated to today's value - when average cost analyses and benchmarking are performed. The forms and reports where this occurs are:

- Strategy Task Analyser
- Fleet Component Performance Summary

In some other forms you have the option to view actual costs as escalated (to today's values) or unescalated.

Budget projections - future costs

All future costs from the budget projection use costs escalated to today's values. These costs are entered into strategy tasks and have a pricing date set in the projection header.

See example above for Budget Project – costs to date.

A strategy task in the budget may reference a standard job, but it is not linked to the standard job. Therefore, the standard job will not update the cost in the strategy task. This ensures the budget never changes.

All future costs are forecast at today's values. Do not enter future escalations into AMT.

Alternate projection - actual costs and future costs

The same rules apply as for the current projection.

Archive projection

Archive projections are created from current, budget or alternate projections. Once created, the archive projection never changes. Therefore, it reflects the rules that apply to their source projection when it was created.

Centreline projections

Centreline projections only forecast future events and costs. There is no historical element.

All costs in the centreline projection use costs escalated to today's values. These costs are entered into strategy tasks and have a pricing date nominated in the strategy task. These costs are escalated from the pricing date to today.

Example

On 1 January 2018, you enter a cost of $1,000 for a strategy task. If there is a 5% escalation on 1 March 2018, and you run a forecast report on 1 June 2018, $1,050 is the forecast cost for that strategy task.

Where a strategy task is linked to a standard job, the program assumes the standard job pricing is always up-to-date and no escalation is applied.

All future costs are forecast at today's values. Do not enter future escalations into AMT.

Modelling Estimates (current and drafts)

The same rules apply as for budget projections.

Time zone support

Time zone support has been added to several key functions. Using the time zone support, you are able to view and work with dates and times relative to your own time zone, or the time zone where the asset is physically located.

Time zone support AMT works by converting dates and times between two reference time zones:

- The time zone of the AMT user.
- The time zone of the site where the physical asset is located.

Configuration required

A new AMT_TYPED_VARIABLE value, **SQLServerTimezonename** is required to be configured as part of the installation or upgrade.

Reference time zones

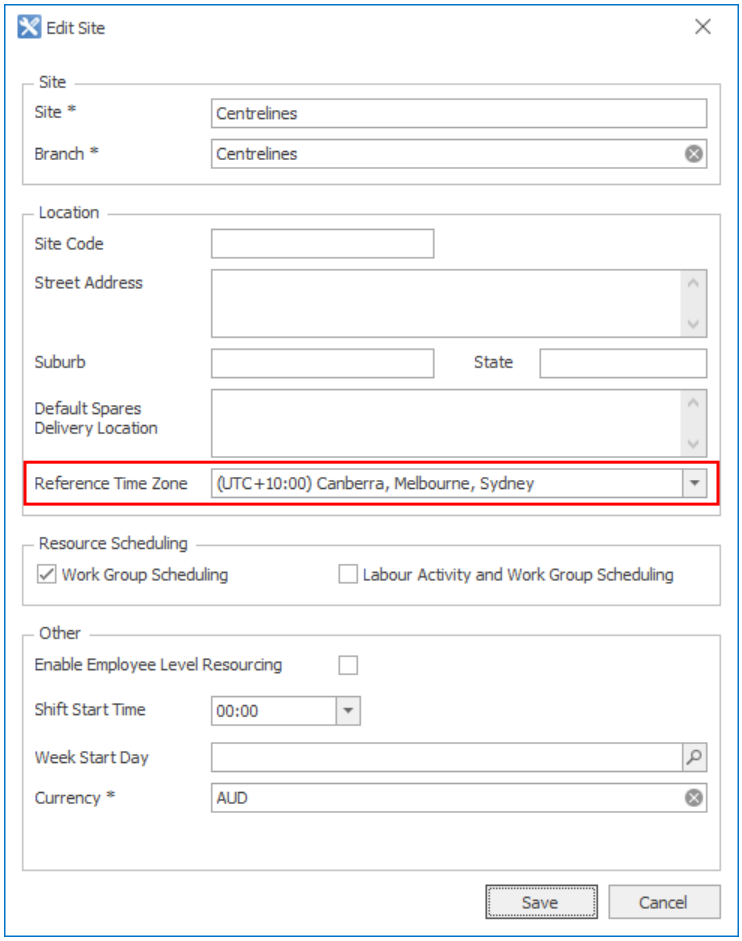
For the time zone feature to work, it is necessary to first specify the time zone of each AMT user and the time zone of each site in AMT.

User time zone

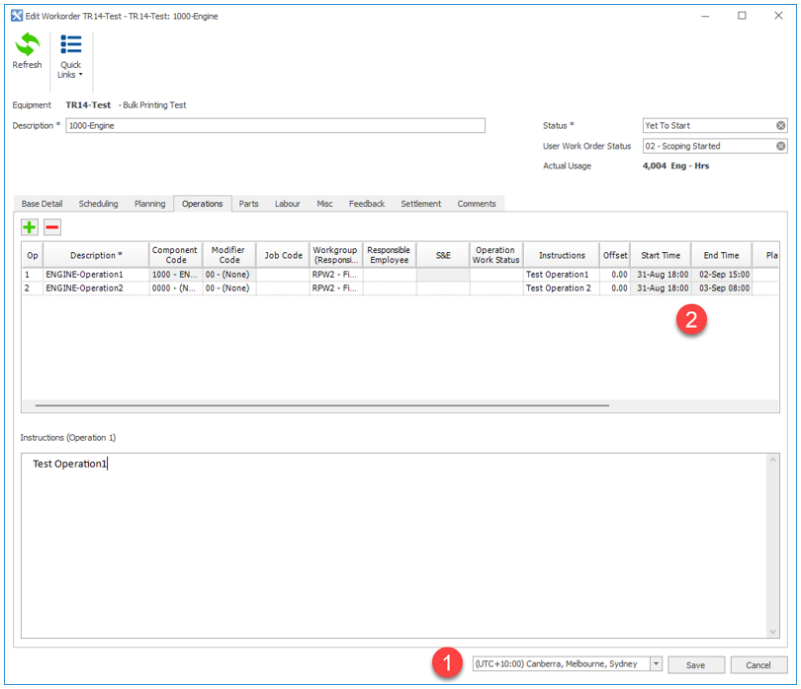
The time zone of an AMT user is controlled by the time zone of the computer the AMT client is installed on. This is automatic.

Site time zone

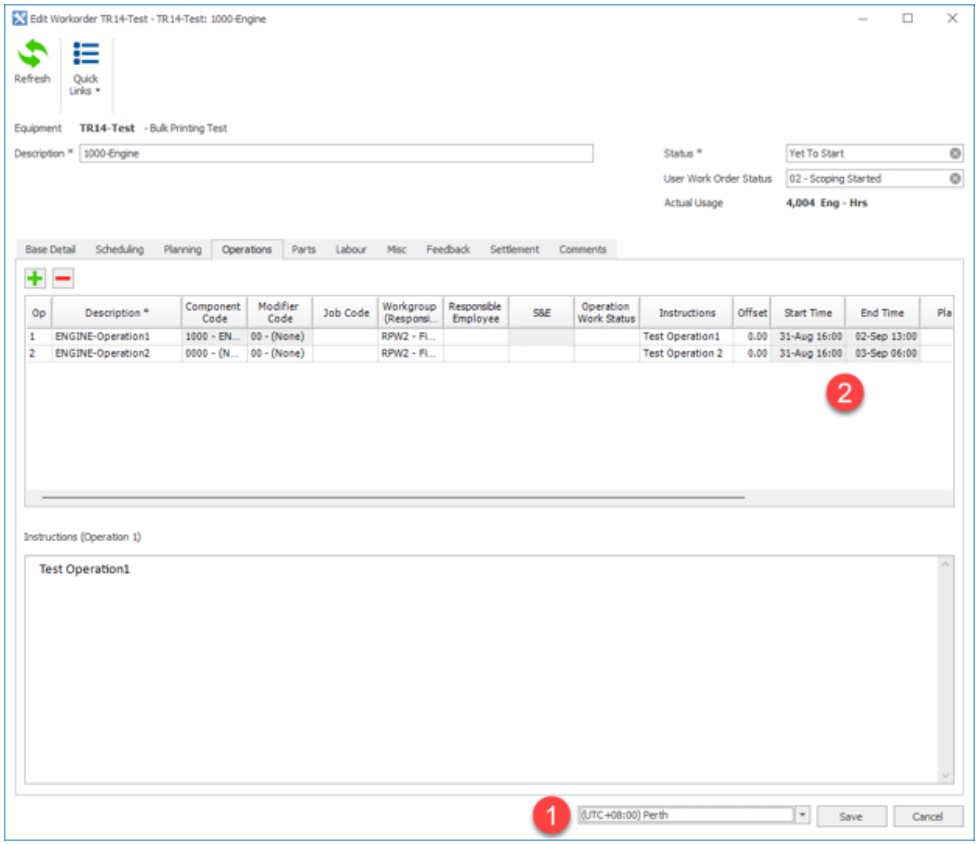
The site reference time zone needs to be specified for each site in AMT. This can be done via the **Edit Site** screen.



Once configured, AMT features where time zone support has been enabled display a control displaying the time zone the dates and times are relative to. For example, the image below shows the work order operations tab in AMT 8.24. Your time zone is shown in the drop-down list at the bottom of the form (**1**). The dates and times shown in the form (**2**) are relative to your time zone.



You can change the reference time zone by selecting a different time zone from the drop-down list (**1**). This will cause the dates and times to be updated to be relative to the newly selected time zone. For example, if you changed the reference time zone to Perth (**1**), the times at (**2**) are recalculated to be relative to Perth, which is two hours behind Sydney (see the screenshot below).



Actual Costs

Work order and work order settlement

Any time an asset stops for maintenance, there is a work order, regardless of whether costs were incurred.

Example

An operator stops the truck because of low power; the service supervisor instructs them to keep operating as the truck is coming in for a service the next day. Even though no job was performed, there is a record of the stoppage in the work order.

The work order settlement, which is linked to a work order, captures the accounting costs of that work order. However, not all work orders have a work order settlement (as with the example above).

Item	Work Order	Work Order Settlement
Engine Changeout	Yes	Specific (1 to 1)
Miscellaneous Repair	Yes	Monthly work order settlement
Maintenance Stoppage - No Cost	Yes	No work order settlement

The work order settlement updates the AMT strategy task, not the work order.

If a work order settlement is linked to a work order, the work order's details (codes and job date) override the work order settlement's details.

If AMT has a work order settlement where the component codes on the settlement do not match a strategy task (for the same asset), AMT has to decide where to allocate this work order cost. It can not be assigned to a strategy task, so where else can it go?

In this case, AMT looks at the component code of the work order settlement, and then it searches through the component code hierarchy to identify to which subsystem the component code belongs. If the component code is found in the component code hierarchy, AMT links the work order the parent subsystem.

The subsystem in a projection has the following codes;

- Component Code: Zxxx - Subsystem - Unassigned (for example, Z011 - Wheel and Tyres - Unassigned).
- Modifier Code: default modifier code (for example, 0).
- Task Type: UN - Unassigned.
- Task Counter: default task counter (for example, 0).

If AMT receives a work order settlement, where the component code on the settlement does not exist in the component code hierarchy, AMT inserts the new component code into the default subsystem in the component code hierarchy. The work order settlement will be linked to the default subsystem.

Even though a work order is linked to an unassigned task mode, the work order still maintains its original coding and keeps its original identity (the work order will still be seen in the Work Order Settlement Manager, Shift log, Planning and so on, with its original codes). It will not have a Z component code nor an unassigned task type. In high-level cost reports, which only show strategy tasks and unassigned strategy tasks, the work orders costs will be included in the unassigned strategy tasks.

Work orders should not be coded to Z unassigned component codes.

The task type UN should only be used for **unassigned** tasks in a projection; it should not be used when creating work orders.

When AMT reports the component life of an unassigned work order, the life is calculated from the last time a work order settlement was linked to the same Unassigned Task (for example, Z011 – Wheel & Tyres – Unassigned). In other words, just as a strategy task's life is calculated from the last work order settlement to link to the same strategy task, an unassigned work order's life calculation follows the same principle. An unassigned work order's life is not calculated from the last time there was an unassigned work order with the same codes.

Multi-Currency

AMT supports multiple currencies. Many reports can show financial amounts in any currency, and transactions (for example, work orders, billings) can be imported into AMT in any currency. If external systems provide the exchange rate to AMT it will be used, otherwise AMT uses a table of historical exchange rates.

The reporting currency is nominated at the site level. Any reports run for assets within that site show the site currency. Where assets from multiple sites are selected and there are financial values in the report, the report only runs if the site currencies are the same.

The exception to this is if the report allows for a currency to be selected. In this case, the report converts the data to the nominated currency. The following data can be maintained in multiple currencies:

- Standard jobs.
- Strategy task jobs (can also be linked to a standard job in different currencies).
- Projected billings.
- Costs (work order settlements).
- Billings.

You maintain the exchange rates from all currencies to the Default Currency in the System Tables (on a monthly basis enter the latest exchange rates).

For actual costs and billings transactions, the exchange rate at the time of the transaction is recorded (usually imported from the ERP as part of the standard interface routine). That transaction-specific exchange rate is what is used for converting amounts between currencies.

Risk Rating

AMT has the ability to automatically calculate a raw strategy task risk rating value by multiplying three rating attributes:

- Equipment > Asset Condition.
- Equipment > Equipment Criticality.
- Strategy Task > Operational Criticality.

This Strategy Task Risk Rating value is translated into a Calculated Risk Factor when AMT performs its Calculated Risk Assessment. The Calculated Risk Assessment process, in addition to calculating the risk factor

(a number) will translate this into customer-defined risk percentile bands. The percentile bands are defined in the Risk Grouping system table.

In the example below, the three system tables, Asset Condition, Equipment Criticality and Operational Criticality, have user-defined risk levels. The Risk Grouping field contains the risk percentile bands.

Equipment Criticality

Code	Desc	Rating
C	Critical	4
H	High	3
M	Medium	2
L	Low	1

Asset Condition

Code	Desc	Rating
New	New	1
Good	Good	2
OK	Medium	3
Fair	Fair	4
Poor	Poor	5

Operational Criticality

Code	Desc	Rating
C	Critical	5
U	Urgent	4
H	High	3
M	Medium	2
L	Low	1

Risk Grouping

Desc	Min%	Max%
Low	0	25
Medium	26	50
High	51	75
Critical	76	100

Example	Equipment Criticality	Asset Condition	Operational Criticality	Risk Calculation	Risk Calc as %	Calculated Risk Assessment
1	Critical = 4	New = 1	Critical = 5	20	20	Low
2	Critical = 4	Poor = 5	Urgent = 4	80	80	Critical
3	Critical = 4	Medium = 3	Critical = 5	60	60	High
4	Low = 1	Poor = 5	Critical = 5	25	25	Low
5	Medium = 2	Medium = 3	High = 3	18	18	Low

AMT will not allow a rating value to equal zero in either of the Equipment Criticality, Asset Condition or Operational Criticality system tables. Also, one code in each table must be flagged as the default.

The Calculated Risk Assessment value is visible on the Long-term Planning screen and in Budget and Forecast Manager.

Changing the risk rating

While AMT calculates a risk rating, you are still able to record your personal assessment of risk against a strategy task. This can be done in Long-term Planning and Budget and Forecast manager in the **Risk Override** field.

Strategy task external identifier

For some installations of AMT, the current maintenance strategy (represented in AMT as strategy tasks linked to an equipment current projection), is housed in an external system, such as SAP. An automated interface is used to send maintenance plans and job costs from the external system to AMT, typically on a daily basis.

In this scenario, the maintenance strategy is defined and maintained in the external system and AMT is used to forecast and analyse the current maintenance strategy. Changes to the maintenance strategy are then made in the external system, which then interfaces the updates back to AMT.

The link between the strategy task in AMT and the equivalent strategy task in the external system is defined in the **External Identifier** field in each AMT strategy task. The value in the field is typically formatted to be a concatenation of
[EquipmentID + ComponentCode + ModifierCode + ExternalSystemStrategyTaskID], which guarantees uniqueness.

What If Analysis

The What If functionality in AMT allows you to take copies of, modify and assess current strategies. What if projections, which are alternate projections, are used to assess different strategies, are typically based on the current projection of a piece of equipment. The current projection reflects the existing strategy for the equipment. The what if projection reflects what you are considering to implement as a new strategy, but do not want to commit the changes until they have been properly assessed.

It is typical to create a what if projection across an entire fleet, as maintenance strategies are usually developed and applied at the fleet level for a particular model, rather than at the equipment level. The following example of creating a what if projection will be based on these parameters. The same steps can be applied to create what if projections for individual pieces of equipment.

Work orders

You can access and search equipment history in Equipment Status on the Work Orders tab.

Grid fields (Key fields)

Field	Description
SM	Standard planning mode equipment (light grey in colour). Advanced planning mode equipment (mid-grey in colour).
Admin Alert	**!** displays in this column if: • The work order is strategy task, but you have not linked it to a work order settlement. • The work order is associated with a rotable item but you have not completed the details of the rotable going into or coming out.  Admin alerts only display if AMT is not being administered properly.
S	This column is shaded **green** if the backlog work order is a strategy task.
Status	Work orders of all statuses display unless filtered.
Linked	Indicates if the backlog is linked to a work order settlement. No colour indicates a work order settlement has not been linked to the backlog work order, neither has a request from AMT for a work order settlement been initiated. **Light blue** indicates a request to open a work order settlement has been made, but AMT has not received confirmation. **Dark blue** indicates a work order settlement has been opened (AMT has received confirmation).

Field	Description
Work Order	Actual number of the work order settlement linked to the backlog work order. <DUE> indicates that AMT has requested a work order settlement number but a confirmation has not been received.
Redo	If you have manually flagged the work order as a job that needs to be redone, a tick (✓) displays.
Cause	Enter a reason why the work order was executed.  For strategy tasks, cause defaults to strategy task but you can change this.
EM Issue No.	If the backlog work order is an equipment management issue, the identification number of the issue displays in this field.
Perform At	The operational stoppage at which the work order was planned to be performed.

Setup & Admin

Equipment

Fleet Administration

Fleets are the levels below Site in the asset hierarchy. Navigate to **Setup & Admin** > **Equipment** to use the **Fleet Administration** screen to add, edit and delete fleets in AMT.

Field	Description
Fleet	Name of the fleet.
Site	Site to which the fleet belongs.
Pricing Group	Parts can be grouped by price. The pricing group is usually the default pricing group. However, if the site's parts are priced at a special rate, that is, include a discount, then a specific pricing group can be selected.  Pricing groups are set up in the System Administration area of AMT, and require special parts pricing to be imported or entered into AMT.
Default	Select for the fleet to be the default fleet.
Service Interval	Standard interval between services, for example 250 hours. The service interval is used in the Service Accuracy report.
Tolerance%:	A KPI benchmark used in the Service Accuracy report to determine which services were performed within an acceptable tolerance of the standard service interval, for example 10%.
Availability Formula	Currently there are two options to work out asset availability. • Physical availability (based on scheduled/calendar hours). • Mechanical availability (based on operating hours).
Set Availability Calculation Parameters	Opens a special function designed to enable Availability Forecasting for the fleet.
Standard Job Parts Pricing Adjustment Factors	Use this function to manually adjust the pricing of parts for specific part types for this fleet. For example, if fluid was a specific part type, then you could increase the price of all fluid parts for this fleet by 10% (enter into AMT 0.1) using this function.

Availability Forecasting

Availability forecasting is enabled at the fleet level using the Fleet Administration function, (**Setup & Admin > Equipment**), by clicking **Availability Calculation**.

Field	Description
Enable Forecasting Availability	Check this box to enable availability forecasting for this fleet.
Annual Scheduled Hours	Average annual scheduled hours for the equipment (for example 365 days x 24 hours = 8760 hours)
Scheduling Efficiency	A measurement of how efficient the planning process is on site. • 100% means that if a number of strategy tasks fall in the same month, all tasks will be performed to occur at the same time. Therefore, there will be only one downtime event, the duration of which is the length of the strategy task with the longest duration. • 0% means that the strategy tasks will be performed when they are next scheduled to be performed. No optimisation of downtime is assumed. A realistic percentage should be entered.
Annual Base Downtime (Availability%)	The assumed rate of equipment availability per annum based on what is considered to be achievable targets for a standard maintenance regime.
Task Specific Downtime	Select the task types that should be included in the availability forecast calculation in order to see their impact on standard maintenance practice.

After the settings have been saved, you can view the availability forecast by running the Forecast Availability report.

Equipment Administration

You can view existing equipment (real, modelling and centrelines), as well as add and register new equipment in the system using the Equipment Administration function.

The equipment registration process consists of a number of steps that support the ongoing management and control of an operational asset. This process begins with defining a method to easily and uniquely identify an asset (depending on the asset type, this can be a difficult task). After the identification method is decided, you need to gather additional information about the asset to help determine classification and data collection requirements.

Before registering the asset in AMT, a number of questions need to be answered. For example:

- At which branch, site and fleet will the asset be located?
- Is this a complex asset requiring:
    - Sophisticated maintenance planning?
    - The ability to track a component's life using different units of measure (for example, machine hours, engine hours, fuel burn, travel time)?

  If so, a Usage Profile will have to be created.
- Is this a simple, low value asset requiring basic routine maintenance?
- What is the forecasted life cycle for the asset (for example 90,000 hours)?
- Will there be a rotable component strategy for certain components on the asset?
- What is the applicable cost centre or project that the asset will be assigned to?
- Is the asset is under a maintenance contract?
    - What are the contract details? (Contract number, contract name, contract account number, customer name and contract type).
    - If the asset is under contract, the contract needs to be set up in AMT using Contract Administration.

This kind of information determines how the asset will be registered in AMT.

To register equipment in AMT, either click **Add** in the Equipment Administration screen (which opens the Equipment Plan Editor screen), or run the Equipment Plan Editor function.

Asset Ledger overview

The Asset Ledger module of AMT helps you to register all the details related to capital expenditure for an item of equipment. These details include ownership and depreciation.

AMT has two forms of asset registration - maintenance and repair contracts (MARC) and standard. Maintenance and repair contracts are also known as customer support agreements, repair and maintenance projects, cost cap agreements and service agreements.

As much of the information is the same for both forms of asset registration, it has been combined where practical, and is identified as MARC or standard when the information is unique.

The menu for Asset Registration can be found under the Equipment Administration menu item **Setup & Admin > Equipment > Equipment Administration**.

At a high level the asset registration process consists of:

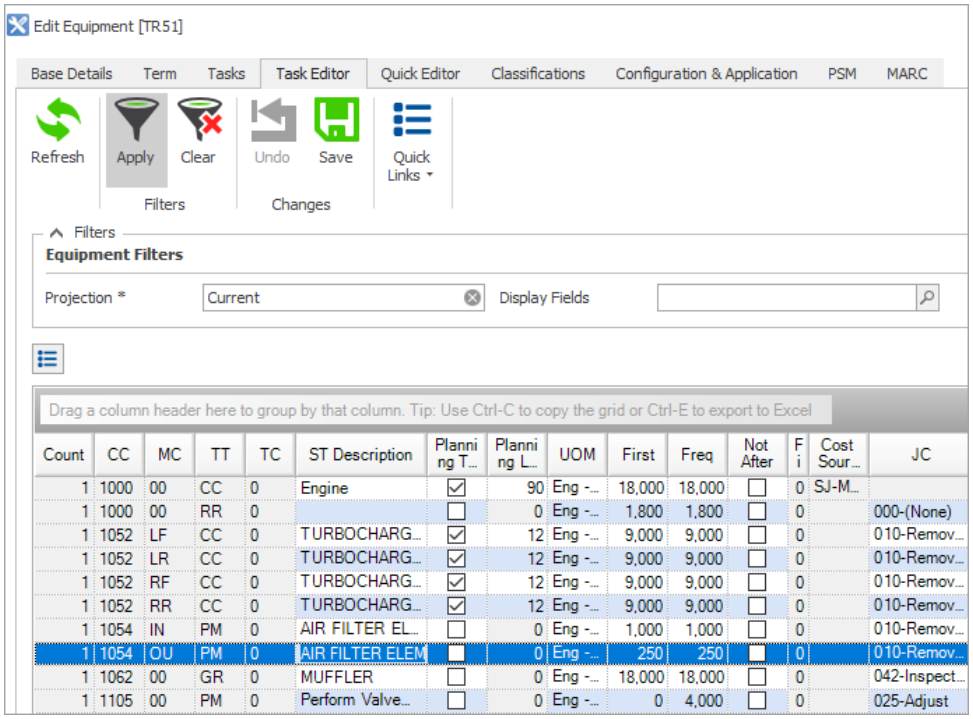
1. Enable the asset register
    - Configure the Asset Register for an equipment.
    - Use the Create an Asset function. Also refer to Entering Asset Details.
2. Add capital details
    - Equipment replacement and disposal values.
    - Depreciation cost codes and schedules.
    - Depreciation rates and projected termination.
    - Current position.
    - Use the Capital Details function.

3. Set up asset reports
    - Reports can be produced to indicate the current position. The reports available are:
        - Asset Valuation.
        - Equipment Cost and Recovery Analysis,

Editing strategy tasks

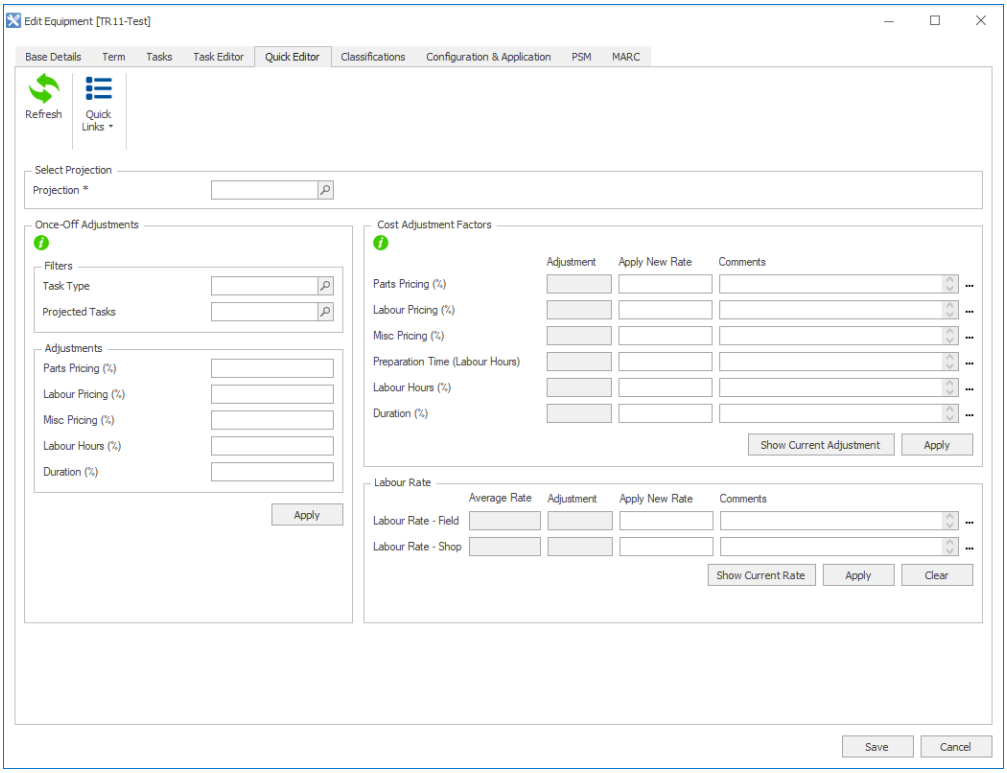
To open and edit a strategy task directly:

1. Navigate to **Setup & Admin** > **Equipment** > **Equipment Administration**.
2. Select **Eqp Type** and **Display Fields** then click the **Filter** icon.
3. Select an equipment and click **Edit**. The Edit Equipment dialog box displays.
4. Click the **Task Editor** tab.



You can edit directly any fields not shaded grey in the grid. The **Display Fields** filter allows you select additional fields for editing.

- On the same form, click the **Quick Editor** tab.

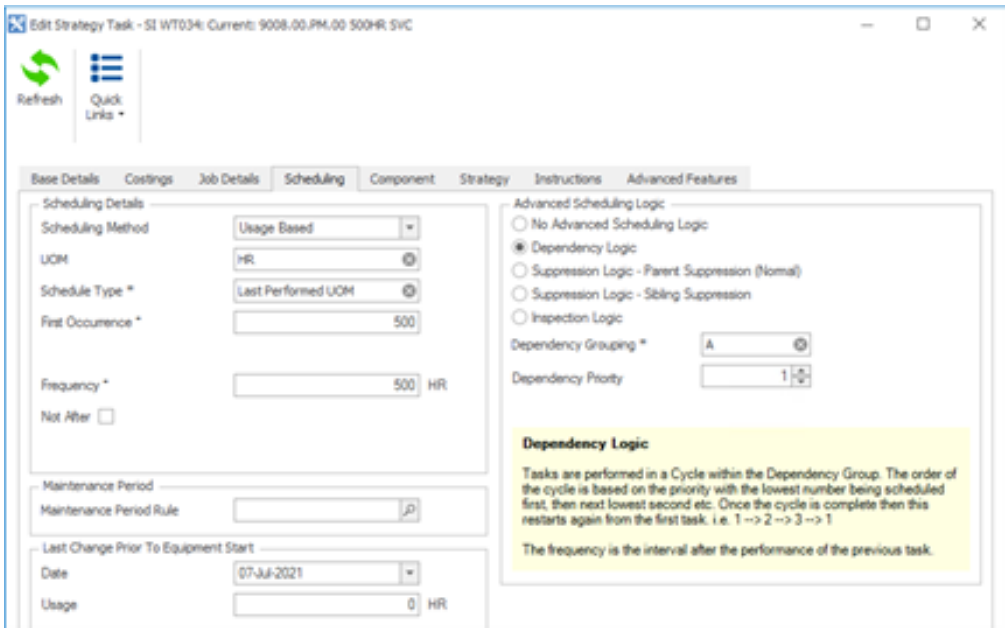


Panel	Description
Once-Off Adjustments	This makes direct adjustments to the amounts in the strategy task jobs. It does not leave an audit trail. It will only adjust strategy task jobs in Manual mode, therefore strategy task jobs linked to standard jobs will not be adjusted.
Cost Adjustment Factors	These do not adjust the strategy task jobs directly, rather, it applies the factors when calculating the total cost of the strategy task (much like escalation). In the strategy task (**Setup & Admin > Projections** > **Strategy Task Manager** > **Edit Strategy Task**, these factors can be viewed by going to the **Quick Links** > **Audit Trail** form.
Labour Rate	This applies to the labour rates entered to the labour cost of the strategy task job. It multiplies the labour hours in the strategy task job by the rates entered to obtain the labour cost. It will only adjust strategy task jobs in Manual mode, therefore strategy task jobs linked to standard jobs will not be adjusted.

Set up MARC parameters

You need to set up special parameters for MARC tenders.

On the **Edit Equipment** form, enable the **MARC** and **Multi Projection** check boxes so that the MARC tab displays.



Refer to Create an Equipment Billing Projection for more information.

Maintenance Contract (MARC) budget margins

Budget margins are set at the equipment plan level in AMT. Budget margins are only applied to the budget projection. Margins that are applied to the current projection are set up in the Set Margin system table .

Set the Budget Margins for an Equipment Plan

Use the Equipment Administration function to edit an equipment.

1. Click on the **Configuration** tab.
2. Click on the **MARC** button.
3. Enter the **Budget Parts Margin %**.
4. Enter the **Budget Labour Margin %**.
5. Enter the **Budget Miscellaneous Margin %**.
6. Click the **Save** button.

Validate billings

The main items you need to validate are:

1. **Billings Variance** - ensure the **Budget Billings** = **Budget Projected costs** (displayed in the **Edit Equipment** screen > **MARC** tab).
2. **Billings Variance** - on day one only - **Current Billings** = **Current Projected costs** (displayed in the **MARC** tab).
3. **Budget Margins** - **Budget Projection** margins are realistic and accurate (on the **MARC** tab).
4. **Current Projection Margins** - **Current Projection Margins** are realistic and accurate (in the **Set Margin** system table).

Set/Edit Other Operating Costs

This function allows you to set up the various costs that the organisation intends to recover from the equipment. Add other operating costs is achieved by clicking the add button on the menu and selecting the various fields that relate to the type of cost you want to attribute to the equipment.

You can select for the entire equipment period or for any given period of the equipment life. There are various calculation methods to choose from which include

- Annual % of book value (this method is only applicable if the AMT Asset Register is also being used as it requires the book value of an asset to be calculated each month based on its current depreciation).
- Fixed monthly amount.
- Rate per UOM.

Once the other cost is set up you can save the record by clicking the tick button. Then, to add other types of other operating costs, continue in the same method until all the costs are set up for processing to the equipment for the periods.

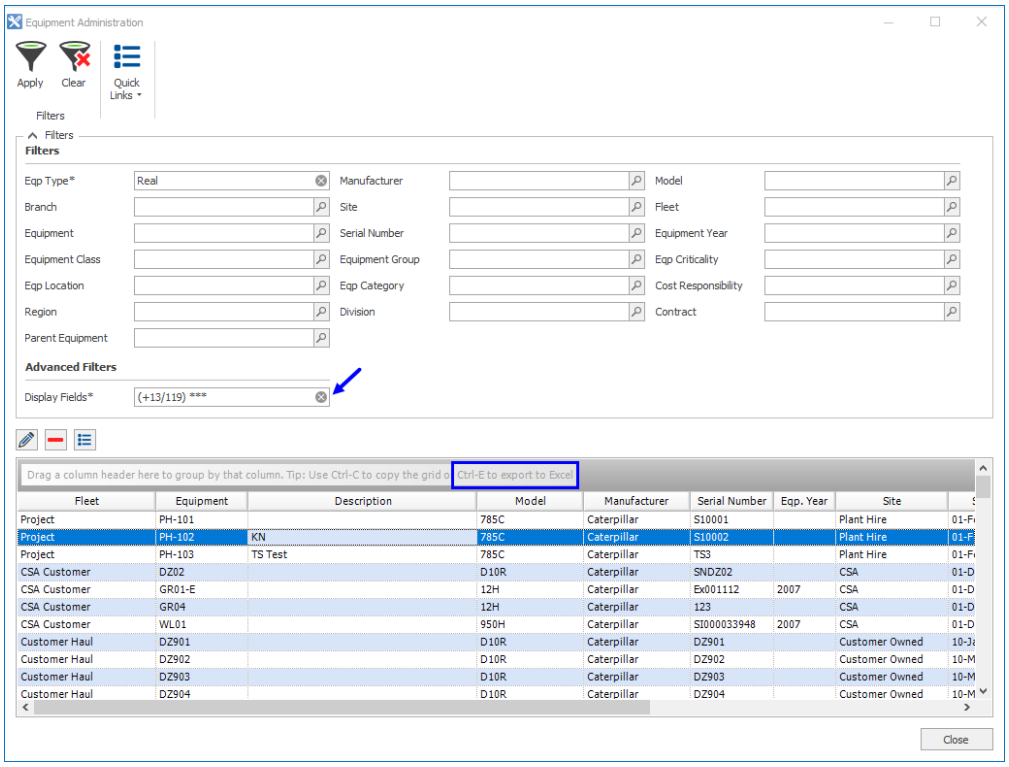
If there are specific Other Operating Costs that can be imported from the ERP then these would be setup as Strategy Tasks and imported through the standard Work Order Settlement interface.

Export asset data

You can export asset data to Excel.

Navigate to **Setup & Admin** > **Equipment** > **Equipment Administration**.

After you have filtered for the relevant asset and selected the display fields (the fields you want to see in the grid and export), press Ctrl-E to export the grid to Excel.



Modelling equipment

Characteristics of modelling equipment include:

- It can represent a tender, which can be for contract equipment or non-contract equipment, or
- It can be used to model different fleet scenarios in certain applications.

Modelling allows you to:

- Model different equipment or fleet scenarios made up of different types of models with different terms, start dates, utilisation profiles.
- Link tasks in the modelling equipment's projection to standard jobs.
- Apply pricing adjustments to one or more estimates.
- Have the option to save the previous version as an estimate-draft (and maintaining an audit trail), every time an estimate is changed.
- Convert an approved estimate to live equipment.

Create modelling equipment

Modelling equipment is created following the same process as registering a real equipment - using the **Equipment Administration** function. The only difference is that the equipment type for the equipment has to be set to **Modelling** instead of **Real**.

By choosing Modelling, the equipment plan is automatically allocated an empty estimate-current projection; estimate-current is a type of projection. Modelling projections (current or draft) can be based on:

- Centrelines.
- Existing equipment registered in AMT.
- New parameters you enter.

You can perform comparisons between new, existing and draft estimates, and once an estimate or model has been confirmed, it can be treated as live equipment.

Reports

Only reports which have the suffix **– Modelling** can be run for modelling equipment. All other reports will not function for this type of equipment.

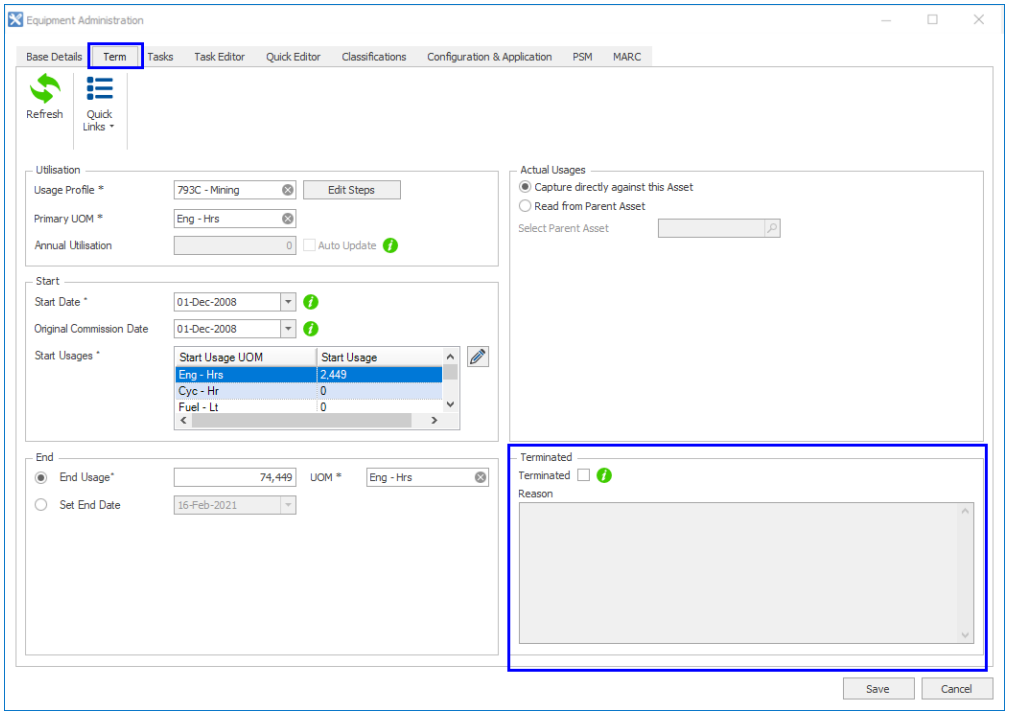
Linking modelling equipments to standard jobs

- Tasks in modelling equipment projections can be linked to standard jobs.
- If an estimate projection is sourced from a centreline, and that centreline is linked to one or more standard jobs, the links will be preserved in the estimate projection.
- Links to standard jobs can be removed at any time.
- If standard jobs are updated – in terms of labour activities or master parts prices or bill of materials for example – costs will **not** be automatically updated in estimate projections.
- Estimates act like budget and archive projections.
    - The initial link to a standard job results in updating the estimate's price.
    - From that point forward, the standard job is simply a reference document to the original assumptions.

Terminate an asset

Terminating an asset means calculating the financial consequences of terminating a contract before the agreed end date or end hours.

In the **Edit Equipment** screen, on the **Term** tab, select the **Terminated** check box.



When an asset is terminated you must also set the **Actual Usage Reading** to greater than, or equal to, the **End Usage**. For date-based equipment, the **End Date** must be set in the past. Even if an asset is flagged as terminated, if the end date is in the future, AMT continues to forecast costs to that point. Based on the end usage or end date entered, AMT recalculates the projections of the terminated equipment.

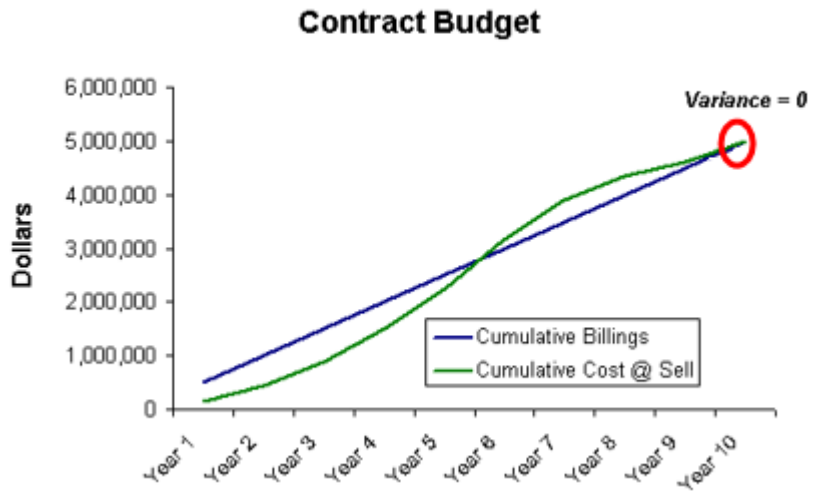
The asset's registration counter is appended automatically to the asset's equipment name.

You can no longer use Long-Term Planning functionality. However, Backlogs and Events will remain visible in Planning and Advanced Planning screens. All maintenance history is still accessible.

To fully terminate asset you must delete any overdue backlogs.

Validate budget variance

For each piece of equipment registered in AMT, it initially has a budget projection, and then copied from that is the current projection. Therefore, on day one, the budget and current projections will be identical. It is important to ensure that the budget was set up correctly because it is the baseline for future analysis.



The circled area on the image above is the amount the customer will be invoiced for the contract balances to the cost of work to be performed.

The cumulative budget billings total should equal the cumulative budget cost at sell total (that is, budget variance = 0).

The only time there is an exception to this rule is if there has been a discount applied to revenue but not costs (resulting in a negative variance) in order to win the contract, or the capturing of overhead costs and revenue is mixed (that is, overhead revenue might be included in the equipment billing rate but the overhead costs are not).

Multi-Editor Asset Details

Use the Multi-Editor Asset Details function to change Classification and/or Life Cycle Term parameters across many equipment.

Multi-Editor Plant Hire

Use the Multi-Editor Plant Hire function to change Classification, Life Cycle Term and/or Default Plant Hire Project and Probability parameters across many equipment.

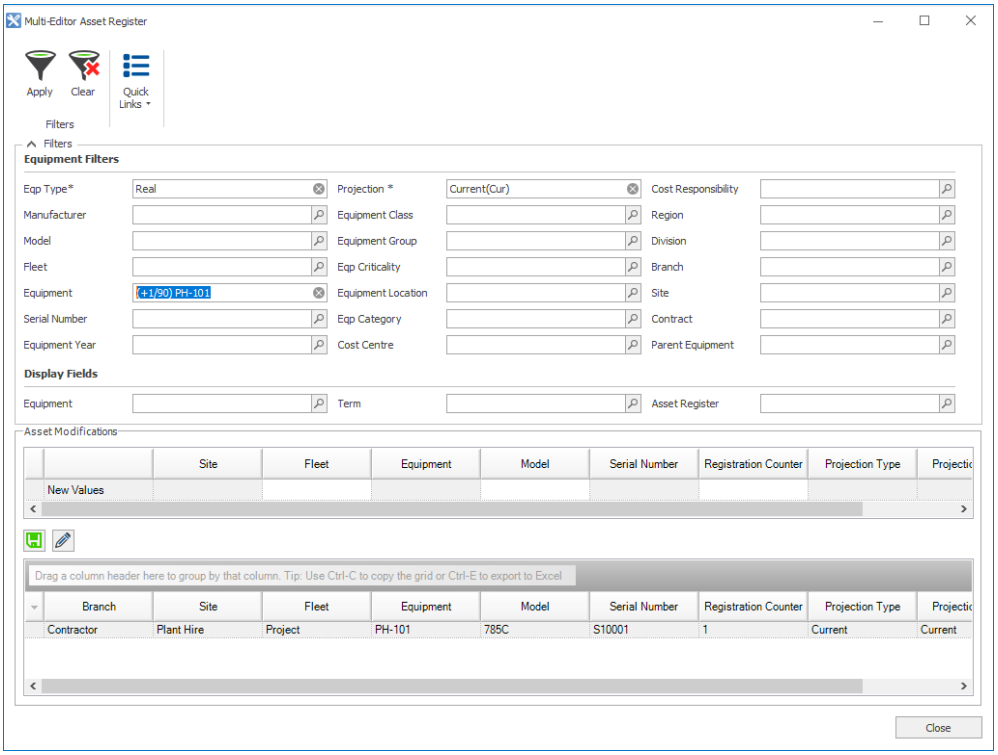
Multi-Editor Asset Register

Use the Multi-Editor Asset Register function to change Classification, Life Cycle Term and/or Asset Register parameters across many equipment.

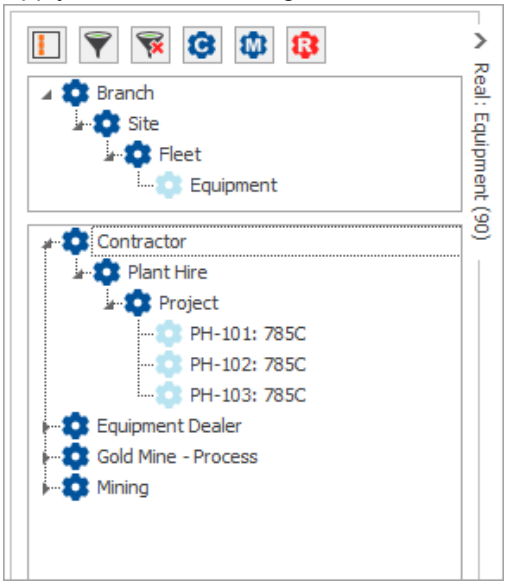
Validate asset details

There are two ways to validate asset details.

1. Navigate to **Setup & Admin** > **Equipment** > **Multi-Editor Asset Register**.



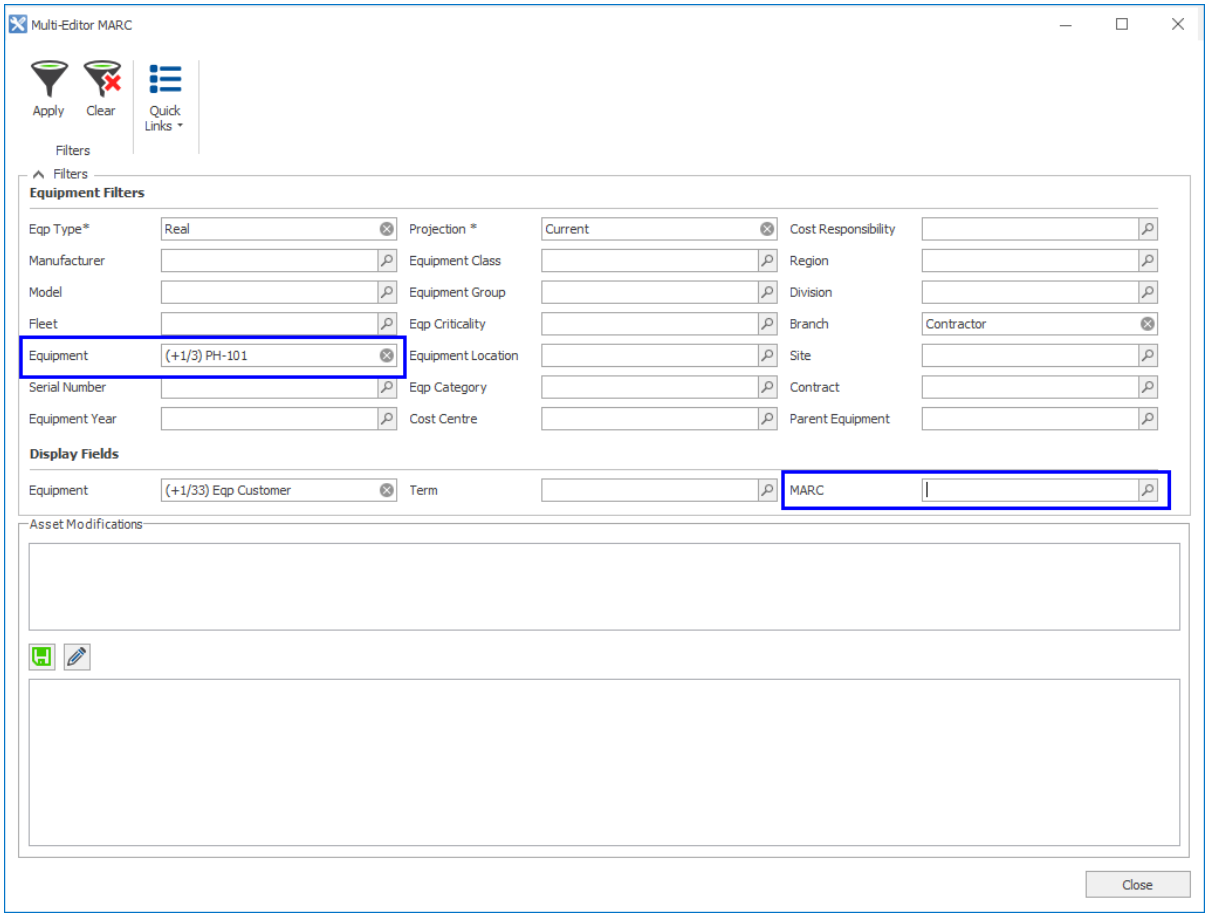
2. Apply the filters on the right side of the AMT home screen to display the asset in the explorer.



Multi-Editor MARC

Use the **Multi-Equipment Editor** to apply the MARC configuration base parameters to the equipment.

Navigate to **Setup & Admin** > **Equipment** > **Multi-Editor MARC**.



Filter for the relevant **Equipment**, then click the **Search** icon at the right of the **MARC** field, as shown in the screenshot above.

Click the double arrow to move all the MARC display fields to the right panel. The fields then display in the grid on the Multi-Editor form.

Field	Description
Billing Split%	Unless the billings revenue for this asset is split between two separate companies, Billings Split% should always equal 100%.
Budget Labour Margin	Estimated labour margins for the budget projection. These values should not change over time unless there is a contract renegotiation.
Budget Misc Margin	Estimated miscellaneous margins for the budget projection. These values should not change over time unless there is a contract renegotiation.
Budget Parts Margin	Estimated parts margins for the budget projection. These values should not change over time unless there is a contract renegotiation.
Contract Name	The Contract Name created using the Contract Administration function.

Field	Description
Include Esc in Pro Forma	Shows current billing rates in the Contract Invoice Details report for overhead billings. The ability to report such billings depends on the overheads configuration of the contract.
MARC	Enables the creation of MARC billing projections and the application of the parameters below. This flag is set in Asset Registration.
Record Test Time	Select if the contract requires the deduction of test hours from billable hours. The hours deducted must then be recorded in the event header.

Multi-Editor Quick Watch

Use the Multi-Editor Quick Watch function to change Classification and/or Quick Watch parameters across many equipment.

Multi-Editor Asset Configuration

Complete documentation for this topic will be available in a future release.

Equipment Editor

The Equipment Editor function is used to create new equipment in AMT.

The nature of the information required when registering an equipment depends on the type of equipment being registered. Valid equipment types are Real, Centreline, Modelling. Adding new equipment to AMT requires the completion of the following.

Base Details

- Equipment Type, Model, Serial Number and Registration Counter are important fields in this section.
- Check with your System Administrator before entering the serial number. External systems, which may be linked to AMT via interfaces, may require the serial number to be entered in a specific format. For example, some systems require the serial number to actually be “MODEL:SERIAL_NUMBER” because serial numbers are not unique unless prefixed by the Model.
    - **System Configuration - Utilisation**
        - No utilisation: select when only date based scheduling and reporting is required.
        - Simple: select when only a single unit of measure (for example, engine hours) is required for scheduling and reporting.
        - Advanced: select when multiple units of measure are required for scheduling and reporting (this requires a usage profile).
    - **System Configuration - Planning**
        - None: select when AMT will not be required to do daily planning for this asset.
        - Standard: refer to Standard vs. Advanced Planning.
        - Advanced: refer to Standard vs. Advanced.

- **System Configuration - Enabled**

  Some of the special setting options may be disabled if certain modules of AMT have not been purchased.

  - Asset Register: select this option to use AMT's asset register.
  - Other Operating Costs: select this option to use AMT's operating costs for the asset .
  - Plant Hire: select this option if the asset is Plant Hire.
  - MARC: select this option if the asset is under a maintenance contract
  - Multi-projection: select this option if more than one projection is required for the asset (maintenance contract equipment require this setting to be enabled)
- **Forecast Calculations**
  - The unassigned ratio is the ratio of the final hour/months unassigned cost to the first hour/month. The logic is that the average unassigned costs are still based on the value entered.
- **Equipment Interfaces**
  - Allow Equipment Detail Updates: select this option if the equipment import interface should be allowed to update asset details.
- **Quick Watch**

Term

- **Utilisation**: options presented depend on the type of utilisation selected in Configuration.
  - Usage Profile can only be selected with Advanced utilisation.
- **Start Date**: date when the asset began operation.
- **Original Commission Date**: date when the asset was commissioned.
- **Start**: for each usage listed, enter the starting usage. Typically it will be set to zero for new assets, but "second-hand" asset's will have a usage greater than zero.
- **End**: set either an End Usage or End Date for the asset. This then becomes the termination point for forecasting.

Tasks

On this tab you can view, add, edit and delete tasks in selected projections. Refer to About Strategy Tasks.

Quick Editors

On this tab you are able to make multiple changes to strategy tasks for this specific equipment. This feature incorporates many of the Multi Editor functions which already exist in AMT, but in this case, the changes are limited to the specific equipment being edited.

- **Multi-Task Cost Adjustor**

  Apply a Parts/Labour/Misc cost adjustments, or Labour Hours or Duration (hours) adjustments across multiple tasks (for example, for selected tasks, increase Parts cost by 5%).

- **Labour Rates**

  Apply new Labour Rates for Shop and Field for selected projected tasks.

- **Travel**

  Ability to change Travel costs (labour and misc) for selected projected tasks.

Classifications

- Refer to Terminology for a description of each of the classification options
  Cost Details.
- **Cost Bearer**: select the default cost bearer. If unsure, contract the AMT Administrator.

Configuration & Application

The Registration and Configuration Details available for selection depend on which options were enabled for the Model of equipment you are registering.

- **Registration Details**

  Information only fields documenting the registration information of an equipment.

- **Configuration Details**

  Information only fields capturing equipment configuration.

- **Application Details**

  Information only fields capturing details of the application in which the equipment is currently operating.

- **Other Details**

  Documents can be attached to the equipment. The process will copy the document from the specified location to the AMT in order that it can be accessed by AMT users.

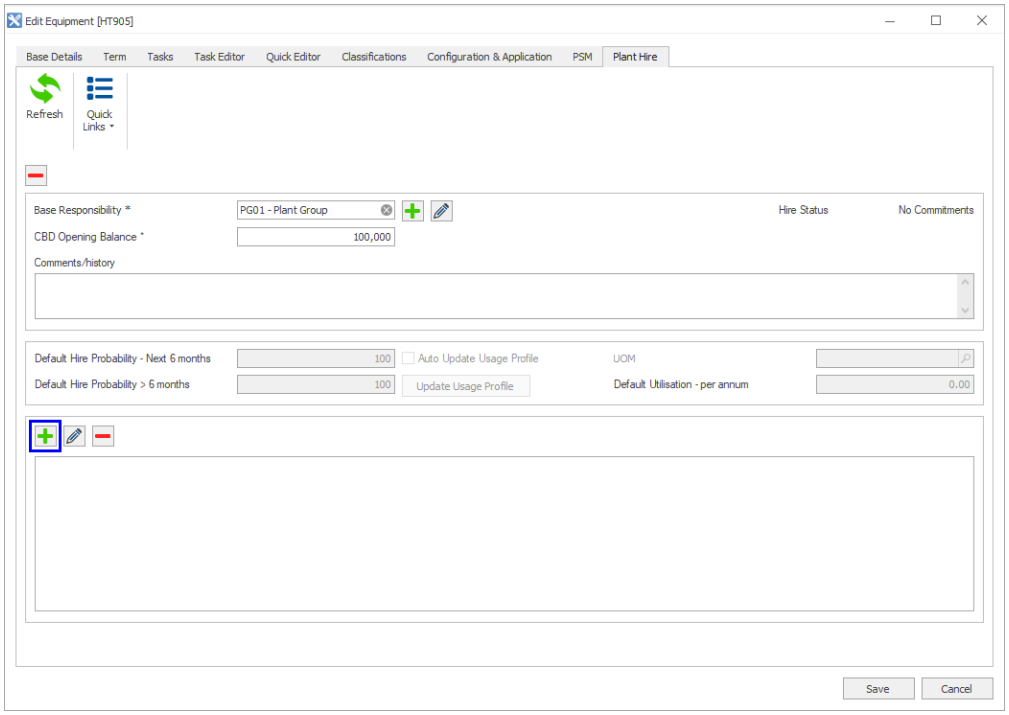
- **Notes:**

  Add any notes required.

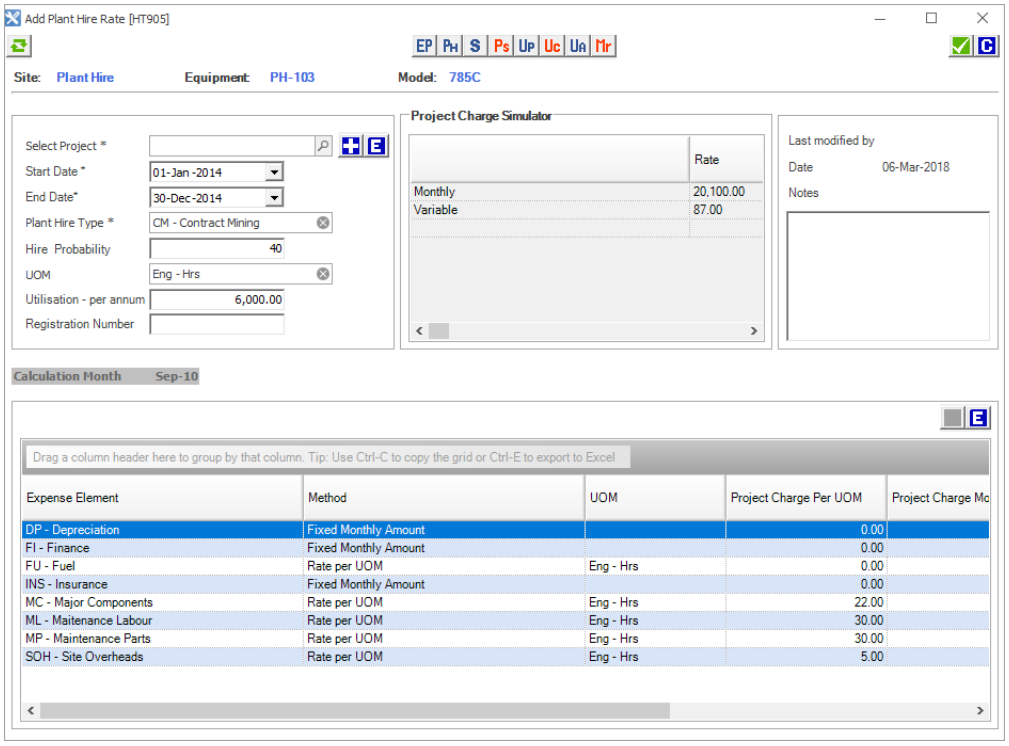
Add Project Rate Schedules

Project Rate Schedules contain the rates specific to an equipment and project for a defined period.

On the **Edit Equipment** > **Plant Hire** tab, click the **Add** icon in the lower half of the form (click **Edit** if you are updating a rate schedule).



The Add Plant Hire Rate form displays.



Key fields and key grid fields are described in the table below:

Field	Description
Project	Select a project, or click  to add a new one. Different project rate schedules can be set up for the same equipment and project to cater for escalation of rates or any other changes in the terms.
Start and End Dates	The plant hire period cannot start prior to the start date of the equipment in AMT. Project rate schedules date ranges cannot overlap.
Plant Hire Type	This field is for information purposes and does not impact any calculations.
Hire Probability UOM Utilisation	These fields are used for forecasting and budgeting future revenue. These fields are ignored for current project rate schedules where the probability is assumed to be 100% and the current usage profile is referenced. Current project rate schedules are where the Start date is in the past and the End date is in the future.
Project Charge Simulator	This is for information purposes. You can enter a simulated or expected monthly equipment usage, based on which AMT calculates an expected plant hire monthly charge. The charge is split into two categories: the monthly fixed charged and the variable units of measure charge, to give a total expected charge. The completion of the expected monthly usage is important as it is used in the plant rate analysis report and also to give an indication of the expected changes.
Grid fields	
Expense Element	Defines the plant rate categories. Select either a fixed monthly charge or unit of measure-based charge.
CC	Cost Centre. Defines the cost centre allocated to the invoices.
EDC	Entry Distribution Code. Is only applicable if the plant hire invoices are being exported to an external system. The Entry Distribution Code system table defines the accounting entries to be assigned.

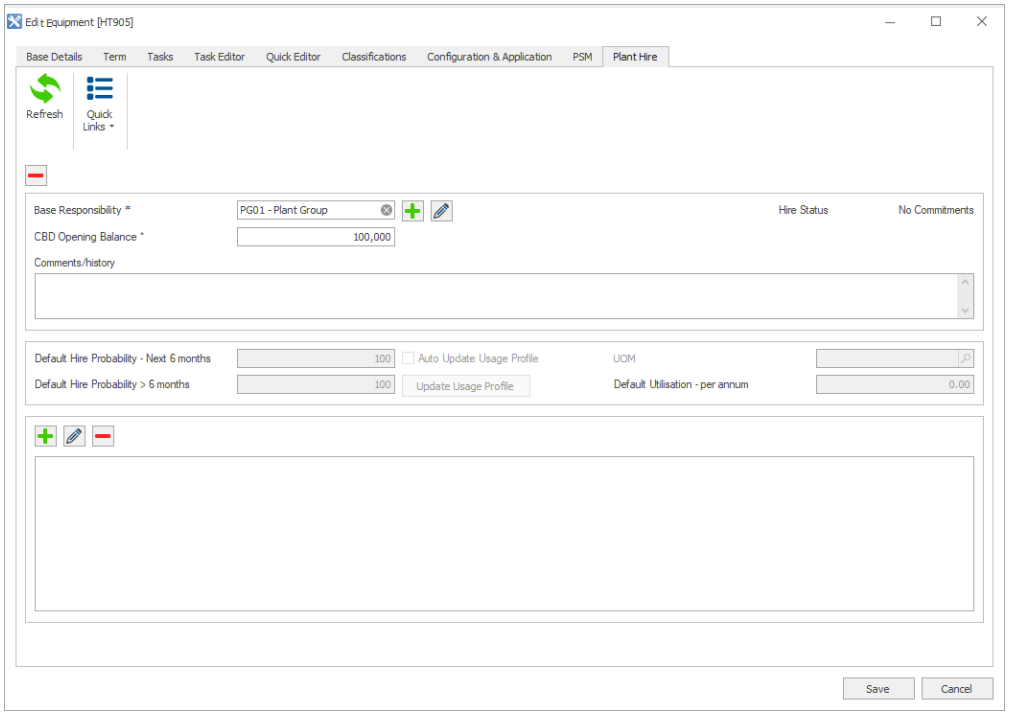
The buttons at the top middle of the screen are quick links to the following forms:

Button	Form
	Edit Equipment.
	Edit Projection.
	Strategy Task Editor.
	Projection Summary Report.
	Edit Usage Profile.
	Usage Analysis Chart.
	Actual Usage.
	Maintenance Ratio Report.

Set up the Plant Hire base responsibility

You need to enable the Plant Hire functionality before you can set up plant hire projects and rates.

1. Navigate to **Setup & Admin** > **Equipment** > **Equipment Editor**. Select the Equipment and click **OK**.
2. Select the **Plant Hire** tab and click **Refresh**. The Plant Hire tab displays.
3. Click **Show Details** to display the fields on the form.



The fields on this tab are described in the table below:

Field	Description
Base Responsibility	The default project that AMT assumes the equipment will belong to if it is not assigned to a specific Project Rate Schedule. This is relevant for assigning costs that are charged while the equipment is idle (for example, financing costs and minimum depreciation). Click  to register a new project, or  to edit an existing project.
CBD Opening Balance	The opening balance for component based depreciation. This field is relevant if an existing equipment is registered in AMT, and there is a pre-existing component based depreciation balance in the accounting system which AMT requires in order to reflect the current actual component based depreciation balance.
Default Hire Section	These fields set the assumptions for forecasting revenue when the equipment is not assigned to a specific project rate schedule.

It is important that you save the changes to this screen after creating the base responsibility project. You will not be able to add a plant hire rate until this information is saved.

Create an equipment billing projection

A budget billings schedule (part of the budget projection) reflects the amounts agreed to with the customer at the commencement of the contract.

A current billings schedule (part of the current projection) is the same as the budget billing schedule at the commencement of the contract. However, it must be updated whenever a renegotiation of the billings with the customer is agreed, which usually occurs due to an escalation of costs.

The current billings schedule reflects actual billing rates in the past, and expected billing rates in the future.

Create a billing projection by navigating to the **Equipment Editor** for a **MARC Equipment**, and clicking the **MARC** tab.

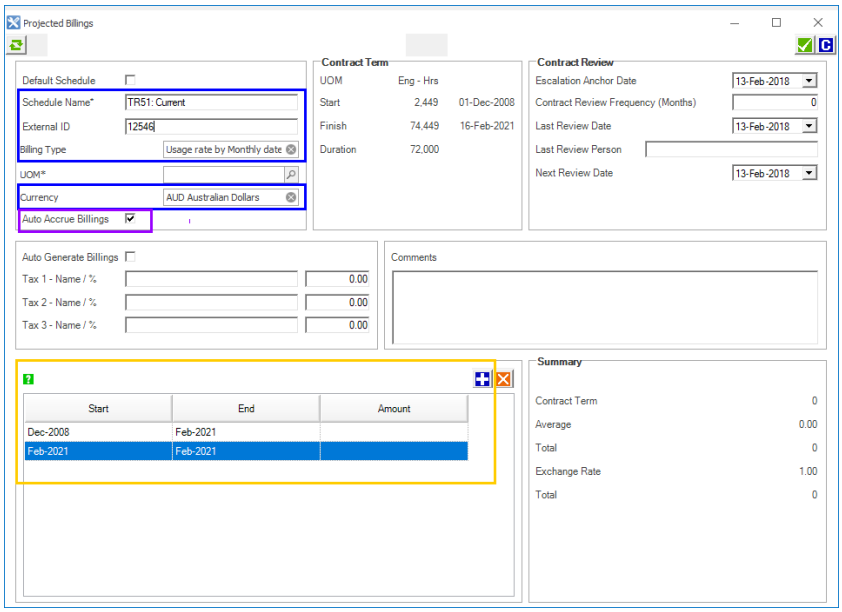
To add a new child billing projection, highlight the projection on the **Billings and Costs Summary** grid, and click **Add**.

To edit an existing child billing projection, highlight the parent billing projection on the **Billings and Cost Summary** grid, and click **Edit**.

Notes

- You can not edit a parent billing projection because it is the summary of its child billing projection.
- It is possible in AMT to have more than one current billing projection. For example, there can be separate current billing projections for engine, hydraulics, drivetrain, powertrain.
- There can be a billing projection per currency.
- From a reporting point of view, all current billing projections are totalled together to produce a total current billing.

Projected billings form



On the **Projected Billings** screen, complete at least the following fields.

Field	Description
Schedule Name	The name of the asset the billing schedule is for.

Field	Description
External ID	Used to link billing records imported into AMT to a specific billing projection (for example, using this ID AMT can distinguish whether an actual billing record links to the engine billing projection or the hydraulic billing projection). If such a distinction is not made, all actual billing records imported into AMT will link to the **default** billing projection (which should not be the budget billing projection).
Billing Type	There are four different billing types: • Usage rate by monthly date range • Usage rate by usage steps (traditional) • Monthly rate by monthly date range • Daily rate by date range Current billing projection is normally set as the **Default**. The **Billing Type** you select must align with the billing method contained in the Maintenance Contract document. Once you have selected **Billing Type**, you can add billing steps to the projection (highlighted in gold on the screenshot above).
Currency	This is the main currency in which financial data are expressed.

If you select the **Auto Accrue Billings** check box (highlighted in purple on the screenshot above), at the end of the contract period, AMT will accrue for billing excess (negative accrual) or shortfall (positive accrual).

Copy and Create Equipment

Create and copy new assets

The term asset is used interchangeably with equipment.

You can create a new blank asset in the following ways:

- **Setup & Admin** > **Equipment** > **Create Blank Equipment - Real**.
- **Centrelines** > **Administration** > **Create Blank Equipment - Centreline**.
- **Modelling** > **Administration** > **Equipment Hierarchy** > **Create Blank Equipment - Modelling**.

You can also create an asset by copying an existing asset. There are two ways to do this.

- **Modelling** > **Administration** > **Equipment Hierarchy** > **Create Real Equipment From Modelling**.
- **Setup & Admin** > **Equipment** > **Copy and Create Equipment**:

You can also copy and create equipment under the **Centreline** and **Modelling** menus.

Refer to Entering Asset Details for information on the fields on the form.

Entering asset details

Depending on how you create an asset, some or all of the fields on the screen will be populated. The screen has four tabs.

- Base Details
- Term
- Classifications
- PSM

The fields on each tab are described below. Click the links to expand them for more information. The fields are all on real, centreline and modelling asset screens unless otherwise specified.



The term asset is used interchangeably with equipment.

Base Details tab

Field	Description
Equipment Details panel	
Asset Name	**Required field** Name of the asset, generally its operational name. The name must be unique.
Manufacturer	**Required field**. The manufacturer of the asset.
Equipment Type	**Required field**. The type of asset, for example, Real.
Fleet	**Required field**. The group the asset belongs to, for example, MARC Haul.
Serial Number	**Required field**. **Real assets only**. A unique alphanumeric combination, which identifies a single, real asset.
Model	**Required field**. A specific type of asset, with similar construction, features and operation (generally a manufacturer model).
Model Serial Number RegistrationCounter	**Required fields**. For real assets only, the combination of these three fields must be **unique**. This is because these three fields are often the unique identifiers used in APIs.
Registration Counter	**Required field**. **Real assets only**. 1. Enables the same asset to be registered multiple times. For example, if a haul truck is converted to a water truck, or you want to re-register it after a certified rebuild or it moves location. This field should always be set to

Field	Description
	one unless it is being registered for a second time. 2. If the asset is registered multiple times, the following rules apply: i. It must have a new asset name. ii. Only one asset can be "live" at any point in time; therefore, all assets except one must be terminated.
Parent Asset	This field is used to create a functional hierarchy of assets. The hierarchy is primarily used in the following ways: 1. An alternative display of asset hierarchy on the AMT home screen. 2. A filter for forms and reports.
Rotable	Select the check box if the asset is rotable.
System Configuration	
Utilisation	**No utilisation**: Select this option **only when** date-based scheduling and reporting is required. **Simple**: Select this option when only a single unit of measure, for example engine hours, is required for scheduling and reporting. **Advanced**: Select this option when you have multiple units of measure you want to track against an asset, and you want to vary the utilisation rate, for example wet season versus dry season utilisation rate. If advanced utilisation is applicable for the asset, a Usage Profile is required.
Planning	**None**: Select this option when daily planning and scheduling is not required. **Simple**: Select this option when simple planning is required. Simple planning excludes the ability to schedule down events and capture downtime information. **Advanced**: Select this option to used the advanced planning functionality. This enables the scheduling of down Events and capture downtime information.
Enabled	**Real and modelling assets only** **MARC**:Select this option if the asset is under a maintenance contract, that is, suitable for dealers or suppliers. **Multi-projection**: Select this option if more than one projection is required for the asset. **Asset Register**: Clear this check box for MARC equipment. **Plant Hire**:Select if the Asset is Plant Hire. Clear this check box for MARC equipment.
Quoter - Type Enabled	**Centreline assets only**

Field	Description
	Select the type of quotation. • Life cycle costing • Maintenance agreement • Service agreement

Term tab

Field	Description
Utilisation	The options displayed depend on the utilisation mode you selected on the **Base Details** tab. **No Utilisation**: All fields are displayed. **Simple Utilisation**: Select the primary UOM and enter the annual forecast utilisation. **Advanced Utilisation**: Select the usage profile which has the UOMs and utilisation assumptions.
Start	**Start Date**: Defines the starting point for an asset, and all scheduling commences on that date. Transactions, for example, work orders, usages and measurements, are ignored. **Commissioning Date**: The date when the asset was first purchased or commissioned. This is primarily a reference field, and is only used to calculate the asset's life. **Start Usages**: The usage readings at the Start Date, the first valid usage readings for an asset in AMT.
End	Select either End Usage or End Date. AMT calculates the opposite entry. End Date represents the point that AMT will stop life-cycle costing, for example, scheduling.
Actual Usages	**Real assets only** Select either Capture directly against the Asset or Read from Parent Asset. If you select Read from Parent Asset, select the Parent Asset.
Terminated	To Terminate an asset, you need to do two things. 1. Select the Terminated flag. 2. Set an end date in the past, or a usage reading that is in the past.

Classifications tab

Field	Description
Location Criticality Category Responsibility Region Division Industry Group	These are flexible classifications used for reporting. The fields and name of the classification can be changed by customers during the implementation.
Owner	Enter the owner of the asset.
Customer	Select from the list. This field primarily applies to dealers; if the asset is classified as being under a contract (MARC), this field is disabled, as it takes the customer from the contract that the asset is linked to.
Status	Select the status of the asset from the list. Asset status is set up in **System Administration** > **System Tables** > **Equipment**.
Default Cost Centre	The default cost centre that is applied to new strategy tasks and work orders that are created or imported without a cost centre.
Default Cost Bearer	The default cost bearer that is applied to new strategy tasks and work orders that are created or imported without a cost bearer.
Address	Address of who the asset is assigned to.
WBS	Free text classifications that usually align to ERP Work Breakdown Structure (WBS) Elements. Not used in any AMT reports.

PSM tab

Field	Description
OEM Details **Code** fields are used for BI Cube reporting. You can enter anything in these fields.	
Equipment Type	Select either E (equipment) or M (machine).
Equipment Division Code	Enter the division code.
Equipment Division Description	Enter a description of the division.
Equipment Industry Code	Enter the industry code.
Equipment Industry Description	Enter a description of the industry.

Field	Description
Product Type Code	Enter the product type code.
Product Type Description	Enter a description of the product type.
Engine	
Engine Make	Enter the make of the engine.
Engine Model	Enter the model of the engine.
Engine Serial Number.	Enter the serial number of the engine.
Application	
Application Code	Enter the application code.
Application Code Description	Enter a description of the application code.
Application Severity	Select the severity of the application.
Other	Enter or select information in relevant fields.

(Rpt_675) Equipment Expiration

Useful to identify when equipment are due to complete their life cycle within the date range specified. This report only runs for REAL equipment.

Tips

RVIC % is calculated following the same logic used in the RVIC Analysis report, except that it runs for the End Date per equipment, and only includes strategy tasks which have the Task Type flagged in the Task Type system table as Component Changeout.

The report will highlight in Blue either the End Usage or End Date depending on which parameter the user has selected as the end point of the equipment.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Branch, Site, Fleet, Model, Equipment, Region, Division, Parent Equipment, Eqp Location, Eqp Category, Eqp Cost Centre, Cost Responsibility, Equipment Class, Equipment Group, Customer, Contract, Manufacturer,

Industry Group, Eqp Criticality, Sales Resp.

(Rpt_650) Equipment Audit

Use this report to see what the configurations for the equipment are; you can also use the Blank Fields filter to identify Equipment with missing classifications.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Display Columns, Blank Fields, Product Type Code, Product Type Desc, Application Code, App Code Desc, Application Severity, Sale Indicator, Territory Indicator, Engine Make, Engine Serial No, Excl From Sales

(Rpt_808) Asset Geolocation History

This report provides the geolocation history for selection of assets.

Tips

Click on the GPS coordinates to view the location in your browser.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|, |ReportStyleKey|

Report parameters

Default.

Additional filters

None

Projections

Projections house strategy tasks that are used to forecast costs and drive the planning process. Every real asset has at least one projection, which is the current projection. Additional projections can be created if the multi-projection option is set in the asset. For assets under a maintenance contract (MARC), a budget projection is also created, which contains maintenance and costs agreed under the maintenance contract.

Refer to projection types for the real asset projections.

All of the topics in this user guide under strategy definition are focused on **current projections**.

Projection types

A projection in AMT represents the life cycle costing for an asset, which can be broken down to the component level. An asset can have a number of projections, including current budget and estimate-draft. Projections house the strategy tasks that are used to forecast costs and drive the planning process.

For an asset owned by the business there are a number of different types of projections depending on the asset type. While the structure of each projection is the same, each type of projection behaves slightly differently.

Centreline asset projections

1. Centreline projection.
    - Centreline.
    - Always assumed to be current.
    - Cannot have a start date in the past - AMT updates the start date at the start of each month, so scheduled tasks can never become overdue.
    - Strategy tasks can be linked to standard jobs.

Modelling asset projections

1. Estimate-Current projection.
    - Modelling assets can only have one Estimate-Current projection.
    - Represents the latest assumptions.
    - Can link to standard jobs but pricing is not automatically updated if the standard job is changed (so that history is always maintained).
    - Enter the expected asset start dates (future or historical) and terms.
    - There is an option to escalate costs to today's values or show at original values.
2. Estimate-Draft projection.
    - Modelling assets can have many Estimate-Draft projections.
    - Provides full version control.
    - Behaves the same way as Estimate-Current.

Real asset projections

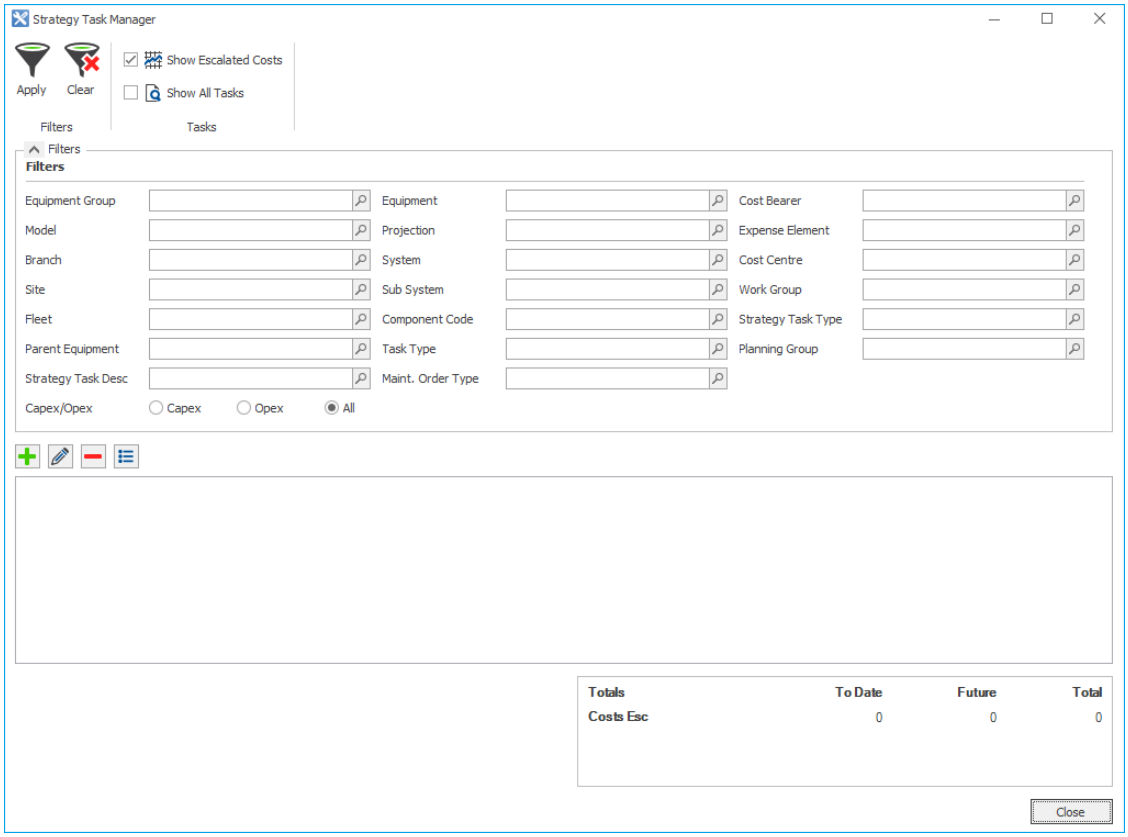
1. Current projection.
    - The live assumptions for an asset.
    - Drives the planning and execution of work.
    - Real-time updating for actual maintenance activities (work orders).
    - Reflects the real-time life cycle costing for an asset.
    - Leaves the link to standard jobs.
    - All strategy task costs are escalated to today's values.
2. Alternate projection.
    - Used for what-if analyses.
    - You can use a copy of the current projection in a test environment.
    - Behaves like a current projection.

3. Archive projection.
    - Save a copy of a projection.
    - Contains all the details of a projection, but does not change for anything in the future.
    - Used to provide an audit trail or reference point.
4. Budget projection.
    - For maintenance contracts, the budget projection usually reflects the commercial assumptions made to determine contractual billing rates.
    - Represents original assumptions or company benchmark performance.
    - Can link to standard jobs, but pricing is not automatically updated if a standard job is changed (so that history is always maintained).
    - Will schedule historical tasks to when they were expected to occur (so they do not become overdue). Therefore, the projection flexes with utilisation (it is not a data-based accounting budget).
    - Will escalate historical costs to the date they were expected to occur - called escalating costs over time.
    - Not impacted by actual occurrences, only utilisation.

Strategy Task Manager

You can navigate to strategy task manager in the following ways:

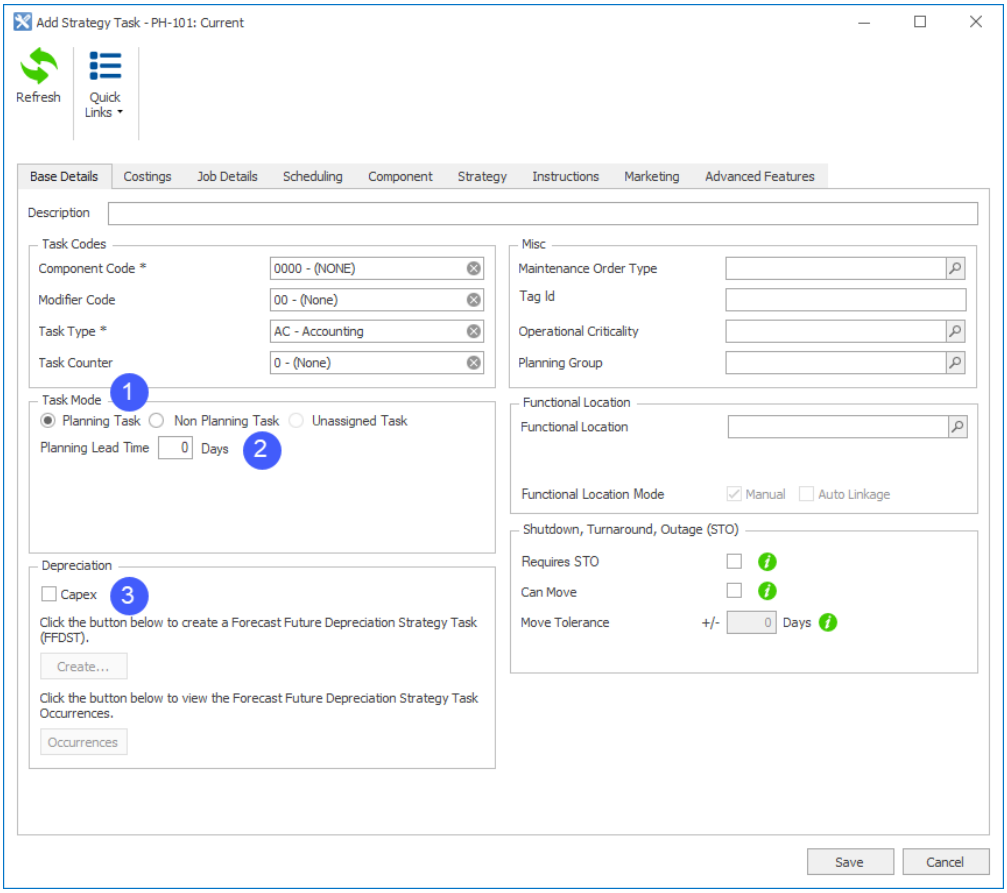
- Navigate to **Setup & Admin** > **Projections** > **Strategy Task Manager**.
- Navigate to **Centrelines** > **Administration** > **Strategy Task Manager**.
- On the right panel of the **Home** screen, select **Real Equipment**  . Use the filters to select the relevant equipment, right-click the equipment and select **Current Projection**.



Double-click a strategy task to open the **Edit Strategy Task** dialog box.

Key fields on Edit Strategy Task

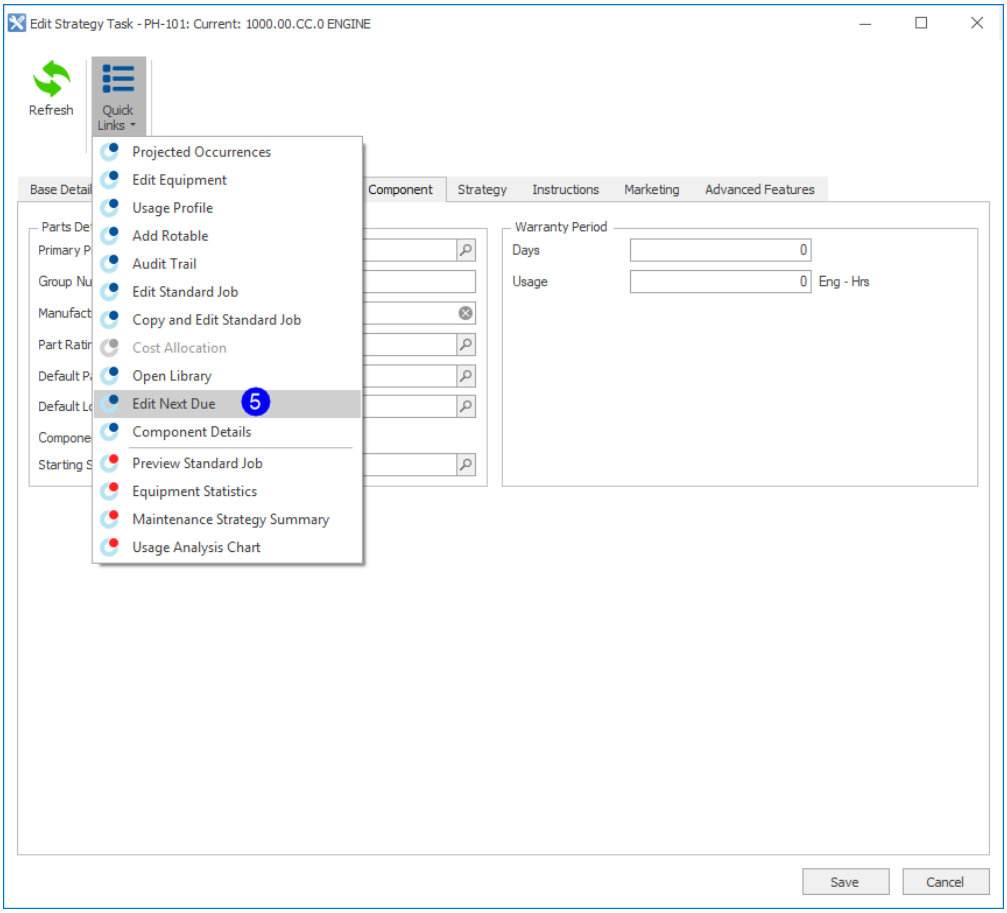
1. The Task Mode must be **Planning Task** for the component to display in the long term planning forms.
2. The **Planning Lead Time** defines when the strategy task will display in the long term planning forms.



3. The **Capex** check box, when selected, identifies the strategy task as a Capital expense. If the check box is not selected, the strategy task is an Operating expense (Opex). If a linked FFDST exists, the Capex check box is selected and cannot be changed. If a linked FFDST does not exist, the Capex check box is not selected by default, but is selectable.
4. For Dealers, **Auto generate Workorder History** should be selected for non-contract equipment. For contract equipment, it should be cleared as the dealer will be creating contract work orders for the work. These fields are on the **Advanced Features** tab.



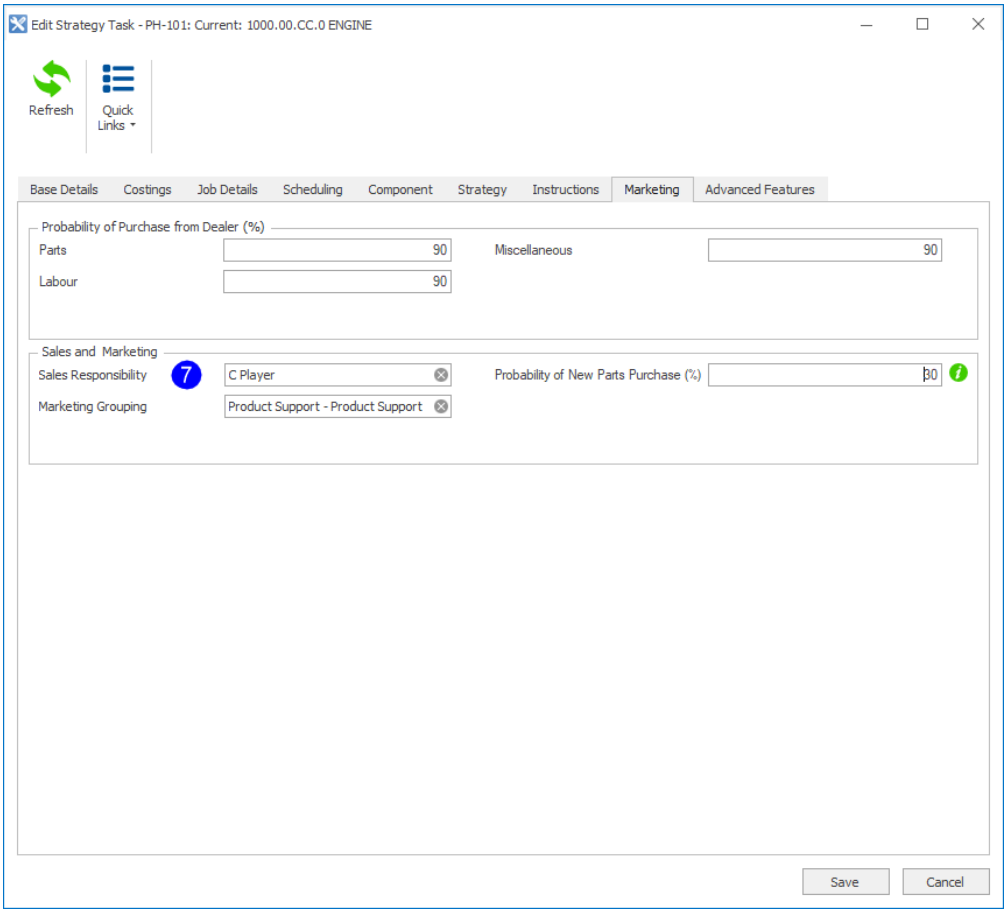
5. The **Parts Details** section displays the default settings for that strategy task. Click **Quick Links** at the top of the screen and then click **Edit Next Due**.



6. The three **Component Capacity Management** fields initially default to the settings in the **Base Details** tab, but you can edit the values.

   The defaults are only applied when a changeout takes place, so you have to set them manually initially.

7. Sales responsibility should be the sales rep, prime product sales person or account manager. This field is on the **Marketing** tab.



STO business process rules

Can Move check box

The following rules apply, depending on whether or not you have selected the **STO** and **Can Move** check boxes.

1. **STO Required is selected (that is, Y), Can Move is not selected (that is, N)**

   This means that the occurrence needs to be completed in an STO and the dates of the occurrence cannot move (unless there is manual intervention).

2. **STO Required is selected (that is, Y), Can Move is selected (that is, Y) and Tolerance = 0 days**

   This means there is no limit to the number of days that an occurrence can move, but the date movement is constrained by the STO dates (for example, must be within the STO dates).

3. **STO Required is selected (that is, Y), Can Move is selected (that is, Y) and Tolerance => 0 days (1, 2 ...)**

   This means that the occurrence move is limited by the number of days in the tolerance and constrained by the linked STO dates (for example, must be within the STO dates).

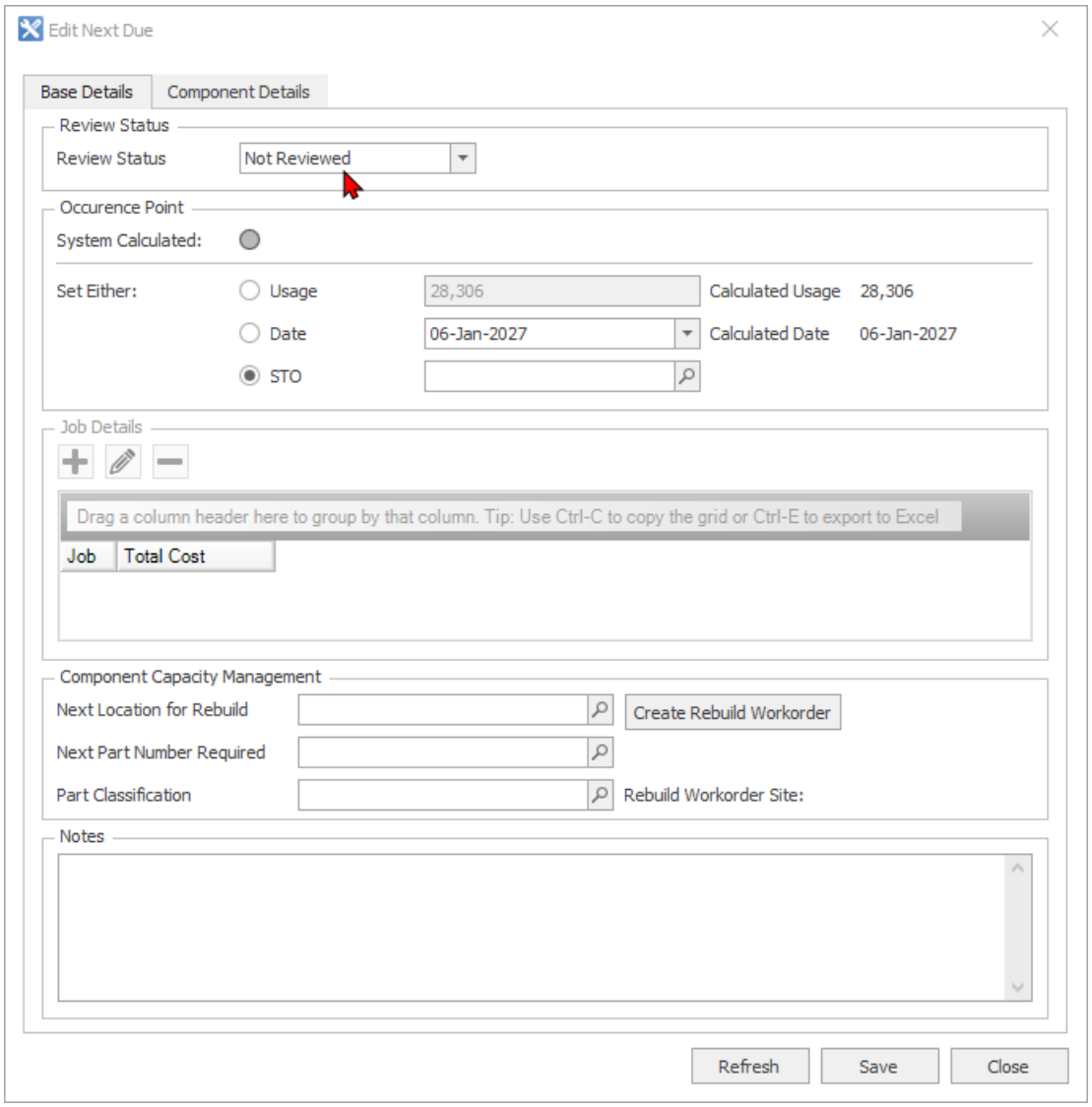
Adjust the scope of a next due job

There will be times where the scope of the next occurrence does not necessarily match the standard scope for a job. An example of this is when due to timing, an overhaul job will be a minor overhaul rather than a major

overhaul. For transmission type components this could be a “re-race & reseal” as opposed to a full rebuild.

For the next occurrence, AMT lets you adjust the cost information to whatever suits the actual plan. This can be done using a standard job or some other cost source.

Navigate to **Edit Next Due** by opening **Strategy Task Manager**, click  to display the **Strategy Tasks**, then double-click one of the tasks. Click **Quick Links** and select **Edit Next Due**.



By updating the details for the next due, you are able to more accurately forecast the expected costs based on a modified job scope.

The **Strategy Task Manager** shows the new value for future costs once you have updated the next due. The updated cost is also shown in cost forecast reports such as the **Forecast Worksheet**.

Multi-task editors

A number of editing tools have been developed to make the job of adjusting strategy task details easier, see summary below.

Editor	Description
Strategy Task Manager	Allows you to view tasks across different equipment in order to identify differences in strategy and make adjustments.
Multi-Task % Cost Adjuster	Apply a parts, labour, or misc cost adjustment across multiple tasks (for example, for selected tasks, increase the parts cost by 5%).
Multi-Task Editor	Ability to change the characteristics of a strategy task across multiple equipment (for example, for all engine changeout tasks, change labour hours from 60 to 85).
Multi-Task Labour Editor	Ability to change the standard and overtime rates for selected strategy tasks.
Multi-Task Scope Editor	Ability to change travel, mileage, freight, and crane costs for selected strategy tasks.
Add / Copy / Delete Projection.	This function is used to copy a projection from one equipment to many equipment.
Copy Projected Task	Ability to copy a task from one projection to another.
Multiple Equipment Delete Task	Ability to delete a task across multiple equipment.

Assign maintenance period rule to strategy tasks

The Maintenance Period Rule (MPR) is used by AMT to identify which tasks should be performed during a shutdown, a standby period or during a production period. This is assigned at the **Strategy Task** level and allows a single asset to have tasks to be performed at different periods. An example could be a monthly service and a major overhaul. The service can be done while the asset is in production, and the overhaul is performed at a shutdown. If no Maintenance Period Rule is chosen for a task, AMT will not show the task in Long Term Planning – Shutdown Planning.

Definitions of different Maintenance Period Rules:

- **SD**: Shut Down. This type of event is when the asset is going to be down for a significant period of time. This is usually accompanied by a reduction in utilisation for the period. For example, the process plant is down so various maintenance tasks can be performed.
- **ST**: Standby. This type of event is a standby or turnaround event. The asset or group of assets is operating, but there is a period of time required to overhaul removed components. For example, a gearbox is removed from a conveyor and replaced with a spare. The Standby period is used to repair or refurbish the component to return it to a condition where it can be used as a spare.
- **IP**: In Production. This is a strategy task job that can happen during production times. This means that there is no defined maintenance period while this work occurs. The asset will most likely be offline, but not in a shutdown type situation. For example, a monthly inspection or replacing a hydraulic cylinder is expected to only take a few hours.

Assign tasks to Defined Maintenance Periods

Long-term Planning – Shutdown Planning is the function used by AMT to assign the future occurrences of the maintenance tasks to the Defined Maintenance Periods. AMT projects the expected future occurrences of the tasks and then lets you assign them to a particular defined maintenance period. AMT then adjusts the forecast based on the decisions you make. It is an iterative process and must be completed in order.

The **Freq. Optimised** column displays a green smiley face when the following conditions are met:

- The task has been assigned to a defined maintenance period.
- The task has been assigned to a defined maintenance period that has the greatest scheduled frequency <110% For example, for an unassigned task, if the **Before Freq** % is 73% and the **After Freq** % is 110.5%, then the 73% option will show a green smiley face.



The 110% cutoff is hard-coded into AMT and cannot be altered.

The table below describes each of the columns on the grid in the **Long-term Planning - Shutdown Planning** form.

Column	Description
Asset	The equipment name.
RA	Rotable Asset. Displays a ✓ if the asset has been set up as a rotable asset.
UOM	Unit of measure used for the task.
Component	The component code for the task and strategy task description.
Frequency	The frequency of the strategy task. Sometimes called the interval.
Cost	The total cost of parts, labour and misc. for the strategy task.
MPR	The Maintenance Period Rule for the Strategy Task.
Tolerance %	A hard-coded 10% tolerance factor used by AMT for the **Frequency Optimised** field.
Next Occ	The number of the next occurrence. 1 means the next occurrence. 2 is the one after that and so on.
Scheduled Date	The currently scheduled date for the task. This can be allocated in **Long-term Planning - Shutdown Planning**, **Long-term Planning**, or by editing the **Next Due** in a strategy task.
Scheduled Frequency	The current frequency of the task if it occurs at the scheduled date. This is the projected life for the component for that occurrence. If the task is system calculated then this value will be equal to frequency.

Column	Description
System Calc	A flag to indicate if the scheduling of the task is system calculated or if a user has allocated a particular date, usage or defined maintenance period.
MP	The assigned Maintenance Period.
MPC	The Maintenance Period Category for the assigned maintenance period.
Assign MPR	The Maintenance Period Rule of the assigned maintenance period.
MPR Conflict	Displays a **!** if the assigned maintenance period rule is different to the strategy task maintenance period rule.
APA Conflict	Asset Production Assignment conflict. For an asset flagged as a rotable asset, this displays a conflict when a task with the **MPR ST – Standby** is scheduled to occur at a date when the asset is assigned to a usage profile step. To remove the conflict, you need to either schedule the task for a period where the asset is not being utilised, or change the maintenance period rule. There is a conflict when the **!** is displayed. This means that an **ST – Standby Task** can be assigned to a shutdown type defined maintenance period when that defined maintenance period occurs at a point in time where the usage profile has been deselected for the asset.
Freq. Optimised	Displays a green smile face when the following conditions are met: • The task has been assigned to a defined maintenance period. • The task has been assigned to a defined maintenance period that has the greatest scheduled frequency <100%.
Scheduled Frequency %	This is the percentage of the strategy task frequency that will be achieved based on the scheduled date. It is essentially the percentage of component life at the scheduled date.
C.V.V	Component Value Variance. The extra value expected (positive) or forgone based on the scheduled frequency. It is used to make a decision on whether the scheduled date for a component has a large cost impact on the business.
MP Before	The name of the defined maintenance period before the current scheduled date.
Before Freq %	The percentage of scheduled frequency that will be achieved if the task is assigned to occur at the start of **MP Before**.
MP After	The name of the defined maintenance period after the current scheduled date.
After Freq %	The percentage of scheduled frequency that will be achieved if the task is assigned to occur at the start of **MP After**.

Column	Description
Project Code	The project code is setup in the **Project Code** system table and can be allocated to a task. This is used as a financial grouping structure to allow you to group multiple defined maintenance periods. For reporting purposes, you can more easily group the costs from the shutdown and standby (turnaround) period following the shutdown.

Completing Strategy Tasks

Depending on how assets were created, some or all of the settings will be populated on the Edit Strategy Task window.

Review and validate the strategy

Once you have created the projections, you need to validate the costs before the equipment, plant or asset manager signs off. To do this, navigate to **Setup & Admin** > **Projections** and open the Projection Summary report.

Standard job interrogation

Navigate to **Strategy Optimisation** > **Strategy Admin KPIs** > Standard Job Interrogation.



You should only run this report if strategy tasks were linked to standard jobs and were linked to primary part numbers.

Additional reports

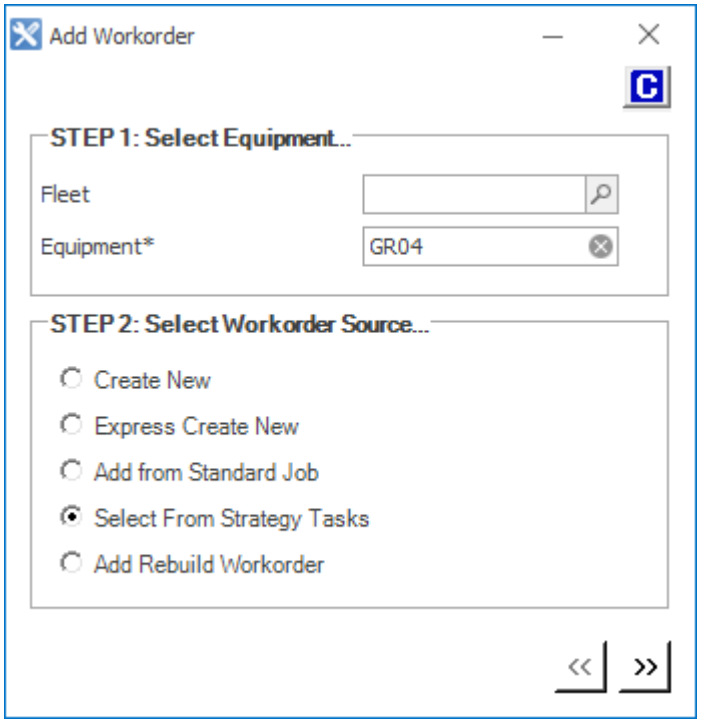
The reports listed below are useful for reviewing and validating the strategy.

Reports	Reviewing	Validating Report
Productivity	Forecast availability. Forecast utilisation.	**Forecast Availability.** **Usage Analysis Chart/Details.**
Costs	Total costs. Cost per unit of measure.	**Cost Analysis - "Life-cycle".** **Cost per UOM.** **Cost Analysis - "Life-cycle" (Task Type).**
Labour	Total labour hours by skill type. Labour hours to operating hours.	**Labour Hours Forecast.** **Maintenance Ratio.**
Resources	Parts/component inventory required.	**Component Forecast.**
Consumables	Consumable usage.	Consumables Forecast**.**
Economic Life	Analyse value in components over time.	**RVIC Chart.**

The Maintenance Strategy Summary report also provides a useful record of the strategy.

Filtering for Strategy Tasks

When creating a work order from a strategy task, you can now filter for strategy tasks versus having to scroll through a long list of strategy tasks.



Filters provided are:

- Component Code
- Modifier Code
- Task Type
- Description

The **Component Code**, **Modifier Code** and **Task Type** are multi-select filters. The **Description** will search the Strategy Task Description for any letters that match the search criteria.



The Description is the description that is entered on the Strategy Task. This is not a mandatory field, and is left blank in some customer's AMT environments.

The **Add Work Order** > **Select from Strategy Task** is available in the following areas:

- Daily Planning > Planning > Edit > Add Work Order
- Daily Planning > Planning > Filter on Work Orders > Right mouse click in Results Grid > Add Work Order
- Daily Planning > Backlog > Edit > Add Work Order
- Daily Planning > Events > Edit > Add Event
- Daily Planning > Events > Open Event > Work Orders > Add Work Order
- Daily Planning > Equipment Down > Search > Select and Edit Event > Add Work Order
- Daily Planning > Shift Log > Search > Select and Edit Event > Add Work Order

Define repair strategy and resources

For some components there may be a number of replacement options to consider. These are described in the table below.

Repair Strategy	Description	Pros and Cons
Repair	Involves repairing the component and placing it back in the same equipment.	Often the lowest repair cost but high downtime.
Rebuild	Involves replacing the component with an exchange item and rebuilding the component at an in-house facility.	Downtime is optimised; the company has control over the rebuild and can optimise rebuild cost but there is no warranty. However, does require the company to maintain their own exchange inventory which can be costly.
Replace with Reman	Involves replacing the component with an OEM Reman item and returning the old component to the OEM in return for a core credit.	Downtime and inventory are optimised, locked into fixed price but will receive some warranty.
Replace new	Involves replacing the component with a new component and scrapping the old one.	This is generally the most costly option and will only be chosen if it is justified through the increased quality (for example, a critical component), or if other options are not available.

When selecting the appropriate option you will need to consider the impact on safety, environmental, production and costs.

Influencing these are items such as the physical location of the site, availability of reman or new components from the OEM and capacity and quality of rebuild facilities.

In the AMT strategy task, you can record this information by selecting the appropriate **Part Type** (New, Reman, Exchange and so on) in the **Job Details** and entering the **Primary Part Number** in the **Base Details** tab and **Notes** in the **Strategy** tab.

You need to decide resources, costs and time required to perform the task.

1. Parts = BOM or parts list.
2. Labour - hours by skill type.
3. Outwork or subcontract.
4. Special tooling or facilities required.

The aim is that when this task is scheduled in short term planning, the Planner knows exactly what needs to be arranged. This particularly important for regularly performed tasks such as preventive maintenance services. For certain items like preventive maintenance services, the OEM recommendations are a good reference point.

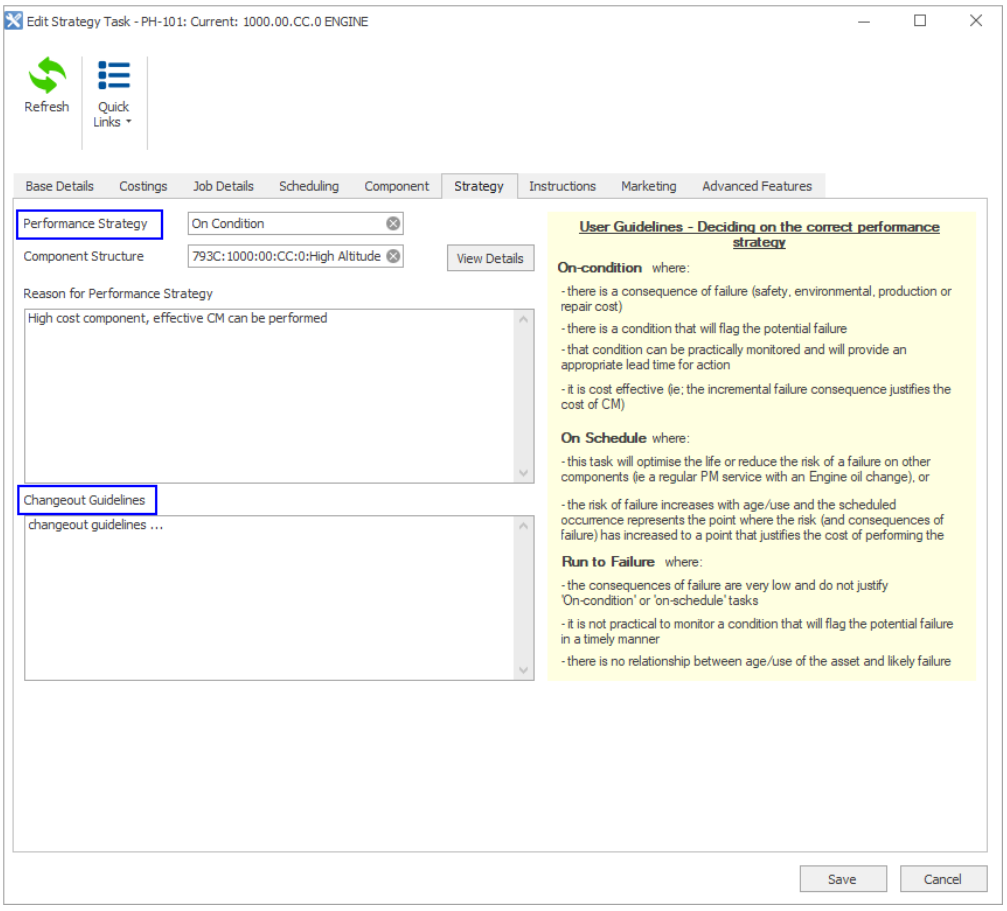
This detail can be recorded in Standard Jobs that can be referenced from the strategy task.

Document strategy changes

Best practice demands that any change to the strategy - especially long-term adjustments - are documented and filed in a central location which can be easily accessed by personnel across the company.

Documentation is required for auditing purposes and because other sites in the organisation could learn from strategies developed elsewhere, without having to reinvent the knowledge. Also, new staff need to understand the reasons behind the current strategy and why it was modified over time.

Strategy changes for a strategy task can be documented within each task on the **Edit Strategy Task** form either in the **Reason for Performance Strategy** field or the **Changeout Guidelines** text box.



Even if the client has a management of change process for managing changes within the organisation, it is still recommended that you, or the company have a standard format for requesting and documenting strategy changes, particularly long term adjustments, and that these documents be uploaded to the AMT Library.

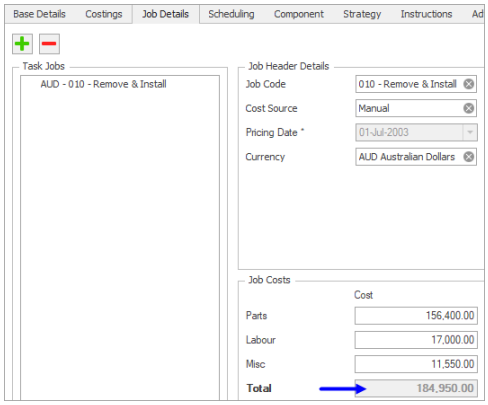
Alternatively, you can upload a scanned copy of the management of change authorised approval in place of a standard format, for auditing purposes.

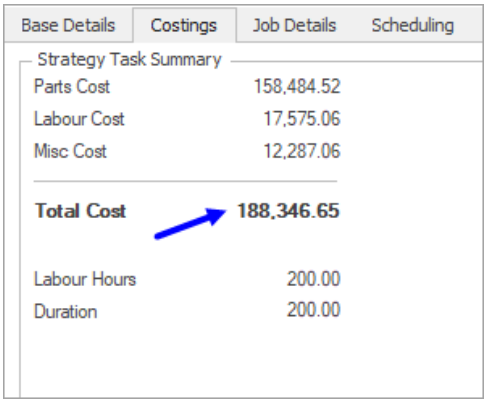
Create a special Document Type in the AMT Document Library, for example Strategy Reviews, to assist in finding these documents easily in the future.

For more information, see Library.

Strategy task adjustment factors

You can apply a number of pricing adjustments when translating the strategy task job cost to the strategy task cost. See the example Edit Strategy Task **Job Details** and **Costings** tabs as shown in the example screenshots below.



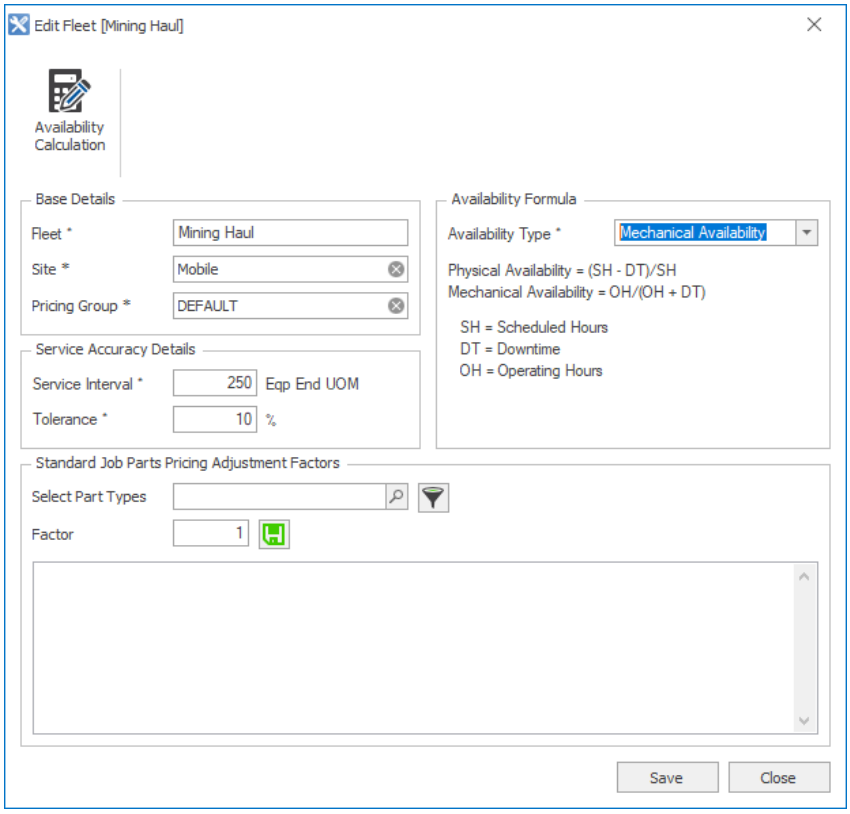


The Audit Trail form, **Edit Strategy Task** > **Quick Links** > **Audit Trail** provides the audit trail for the calculations.

The adjustments you can apply are:

Escalation - Escalation factors entered in the system table and applied from the pricing date in the Strategy Task Job.

Standard Job Pricing Adjustment Factors - These are set in the fleet administration system table editor, and applies adjustments to parts pricing only escalation.



Cost Adjustment Factors - covers a range of items which you can enter in the Audit Trail form or the Multi-Task Scope Editor.

- Pats Cost.
- Labour Split % (will only cost the percentage of labour entered).
- Labour Cost.
- Labour Hours.
- Labour Preparation Time (will add the time at the start of every job).
- Duration.
- Labour Rate.
- Freight.
- Misc pricing.

Strategy optimisation - Action

As with the strategy optimisation investigation, the Action steps vary depending on the issue and action steps required. If the action is related to an asset then the first step is to update AMT with the new strategy. This involves:

- Updating the strategy tasks (frequencies, costs, standard jobs, work scopes).
- Documenting the reasons for the change in strategy (see the **Edit Strategy Task** > **Strategy** tab).
- If applicable, updating the component structure (failure modes, probabilities).

The Maintenance Strategy Summary report (**Strategy Optimisation** > **Strategy Definition**) provides a detailed summary of the maintenance strategy for an asset.

If one-off corrective tasks are required, you can create them directly in **Daily Planning**. Alternatively, you can create them automatically from EM Issues.

The corrective action will need to be monitored to ensure it is carried out and achieves the desired result.

Planning vs Non-Planning tasks

It is a deliberate decision to make a strategy task planning or non-planning. A strategy task should not be set as non-planning if you want to pro-actively manage the task and receive warnings about when it is due.

In other words, the planning task setting should be dictated by the maintenance strategy for the particular task and not what is more convenient for you.

The decision to not manage the task actively in AMT, that is, setting it as **non-planning** should be dictated by the prevailing maintenance strategy which states that the task is not to be managed actively in the field. If used properly, the advantage of the non-planning setting is that you do not have to manually reschedule the strategy task when it falls due. Rather, the system reforecasts the component automatically in 33% increments until the task is executed.

- This logic follows RCM principles and results in a smoothing of costs over the life cycle of the equipment; thereby minimising the potential bow wave effect in financial reports.
- The Rescheduling Percentage (in the above case is 33%) is a setting within the AMT database and applies to all non-planning tasks. The default setting is 33%. Check with the AMT Administrator to confirm the setting for the existing AMT installation.

Tasks flagged as planning tasks display a **Next Due** tab. AMT calculates and shows when the component is next due. You can change the usage or date when the component is next due. AMT also displays the forecast risk or opportunity of the component. The screen is similar to the Long-term Planning screen.

Repair Reserve tasks

The concept of repair reserve (RR) reflects the weighted average of cost and frequency of the component structure linked to the strategy task.

If you want AMT to include the possibility of the different failures of this component in the overall cost forecast for this equipment, create a repair reserve task for this component.

You can create a repair reserve task manually on the **Add Strategy Task** dialog box by selecting the task type **RR**; however, a manual repair reserve task cannot be linked to another strategy task. If a repair reserve task is created it will display in the projection and in reports.

Standard scheduling

Standard scheduling means that the strategy task relies on the **First**, **Frequency** and **Not After** parameters entered when you edit a strategy task.

Therefore, the forecasting of future occurrences of the strategy task is not influenced by the execution of other strategy tasks (unlike suppression and dependency logics). AMT treats the strategy task in isolation.

Typically most strategy tasks are set up with this standard form of scheduling.

Scheduling Methods

Depending on the nature of scheduling required by site for particular strategy tasks (for example, PM services), the tasks could be set up using any of the following logics available in AMT:

- Standard.
- Dependency.
- Suppression (Parent or Sibling).
- Inspection.

As well as basing the strategy tasks on one of the above logics, you can also choose whether the next occurrence of the task is based on either:

- Last Performed UOM/Date.
- Last Scheduled UOM/Date.

Strategy task

A projection contains the schedule of work required to manage the asset through its life cycle, and the costs associated with that work. AMT breaks down an asset's cost projection to the component level. The type of work performed on a component and its cost are defined in a strategy task.

You can create a strategy task by navigating to

- **Setup & Admin** > **Projections** > **Strategy Task Manager** , applying the relevant filters then clicking the **Add** icon.
- **Setup & Admin** > **Projections - Multi-editors** > **Copy Strategy Tasks**.

You can also get to these windows by navigating to

- **Centrelines** > **Administration**.
- **Modelling** > **Administration** > **Projections**.

An AMT strategy task is a specific maintenance task that is forecast to occur periodically over the life cycle of the asset. At this level you are able to identify:

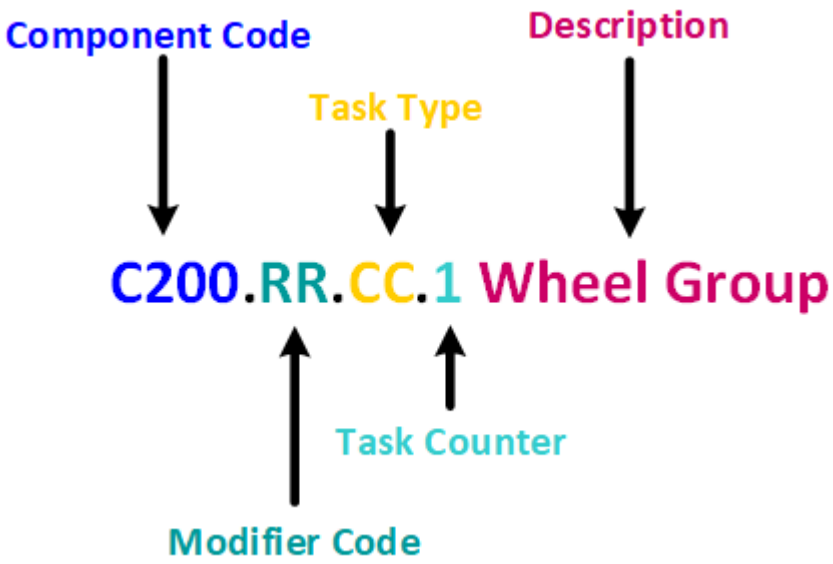
- costs to perform work on components, services and so on.
- type and frequency of work that will be required.
- parts and labour required.

Both strategy tasks and work orders are uniquely defined by a combination of four codes:

- Component Code (CC)
  What is being worked on or forecast (for example, HCYL – Hoist Cylinder).
- Modifier Code (MC)
  Used to provide an additional identification to the component code (such as the component's location on the machine, for example, right, left, upper and so on.).
- Task Type (TT)
  A code and description combination that identifies what is to happen at an occurrence of the task (for example, CC – component change out, GR – general repair).
- Task Counter (TC)
  Defines the sequence by which a task is performed, for example:
  - C220.RR.CC.1 (First/Freq 15,000 until 30,000hrs) reflects a remanufacturer strategy.
  - C220.RR.CC.2 (First 45,000hrs Freq. 15,000) reflects a rebuild strategy.

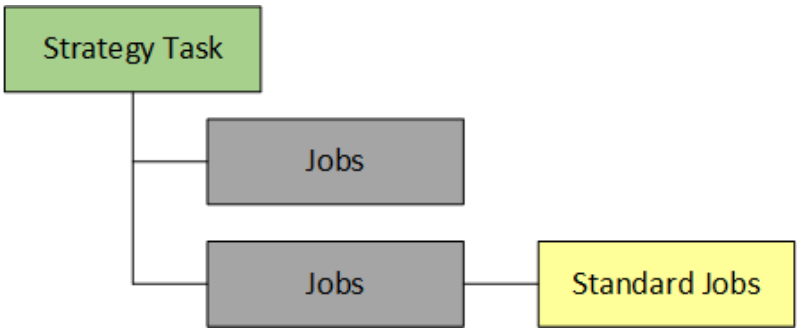
    

    Most Tasks do not use the Task Counter. It will be set to zero, for example, C200.RR.CC.0.



Strategy Task Structure

The structure of a strategy task in a projection is pictured below.



Each strategy task can have one or more **strategy task jobs** and each job has a cost, which is split into **parts**, **labour** and **miscellaneous**. You can manually enter the costs into each of these categories, or you can link the job to a standard job.

Strategy tasks can represent a variety of activities:

- Component changeouts
- PM services
- Inspections
- General repairs
- Unscheduled/miscellaneous work (that is, provision for costs)

Strategy tasks that are contained in the current projection are repetitive strategic maintenance activities, which typically require

1. Budgeting
2. A maintenance strategy
3. Standard jobs
4. Part, rotable component and costs to be forecast.

5. Long-term planning (for example, in order to consider lead times)
6. Short-term planning (for example, resources, location, support assets)

Strategy task modes

You can create a task in normal or unassigned mode. Below are examples of normal and unassigned task mode structures. Unassigned tasks have the same structure as simple mode tasks but have some limitations. Below is a summary of the characteristics of each task mode:

Normal

- Can be a planning task or a non-planning task.
- Can use advanced scheduling logic.
- Can be associated with a rotable part.
- Can have a rotable repair strategy.
- Can have jobs and therefore multiple currencies in cost build up.
- Can link to multiple standard jobs (if the task has multiple jobs).

Standard jobs can only be linked if the standard job module of AMT has been implemented. You should refer to the Standard Jobs Overview for more information.

You must consider the above characteristic before creating a task in the projection. It is not possible to swap or change between task modes after the task has been created.

Unassigned

- Can only be a non-planning task.
- Cannot use advanced scheduling logic.
- Cannot be associated with a rotable part.
- Cannot have a rotable repair strategy.
- Cannot be linked to a standard job.

Unassigned is a strategy task mode to:

1. Budget for general repair costs at the subsystem level, to complete a full lifecycle costing model.
2. Group actual costs for non-strategy tasks into subsystems for high-level cost reporting.

Edit strategy task details



The table below describes the various fields within a strategy task.

Base Details

Field/Panel	Description
Description	What will display as the description on the work order and some forms and reports. If it is not entered, the description in those areas default to a concatenation of the four strategy task codes, for example, 1000:00:CC.0 - Engine.
Task Codes	The combination of component code, modifier code, task type and task counter must be unique for a projection.
Misc.	**Tag Id** is a field that displays on work scopes and work orders. It generally applies to fixed plant assets to help trades identify the correct asset when performing maintenance. Operational criticality is one of the fields used by AMT to automatically calculate a strategy task's Risk Rating. The other fields included in the calculation are Equipment Criticality and Asset Condition; both are defined at the Equipment level.
Task Mode	**Planning task** - AMT automatically creates a work order when the planning lead time is reached (that is, when it is 33 days from the scheduled date). Planning tasks also display in long-term planning and will become overdue if not performed by the scheduled date (AMT life-cycle costing assumes overdue components will be done today). **Non-planning task** - Items you do not want to schedule, but would rather manually create work orders when needed. If a non-planning task becomes overdue, AMT automatically reschedules it for a percentage of its frequency (Frequency multiplier - this percentage is defined during AMT configuration and applied to all non-planning tasks in the system). For example, if a strategy task with a frequency of 10,000 Eng-Hrs becomes overdue and the frequency multiplier is 30% it ill be reschedules for a further 3,000 Eng-Hrs. **Unassigned tasks** - These are system created and cannot be edited. AMT forecasts a consistent daily cost for these items.
Functional Location	The place at which a maintenance task is performed.
CAPEX	Flag which identifies whether the strategy task is a CAPEX activity or not. This field is used in reporting.
Shutdown, Turnaround, Outage	Identifies that the type of work, when performed, occurs during a defined shutdown period.d shutdown period.

Costings

Field/Panel	Description
Indexation	Select the applicable escalation category. See Cost Escalations.
Strategy Task Summary	This totals the costs and hours for all jobs. The job costs are adjusted for escalation and pricing adjustment factors.

Job Details

Field/Panel	Description
Job Header	**Job Code** - Identifies the type of job. **Cost Source** - Identifies the source for the costs of the job. There are four sources. • Manual - costs are entered manually. • Standard Job - select from the list of AMT jobs. Depending on the configuration of AMT, selecting a standard job may automatically create a linked Standard Job Cost Allocation for AMT to forecast multiple expense elements and labour activities contained in a single standard job. • Cost Allocation - lets you create multiple cost records fro a single job. This is used in two situations: where multiple expense elements and cost centres apply, or when interfacing with ERP maintenance plans and standard jobs. • Consumables - lets you enter the consumable quantities and unit prices to calculate the parts cost. Note the type of consumables is identified by the job code. **Pricing Date** - Date that the costs are applicable. For manually entered costs, escalation is applied from the pricing date. The escalation schedule is defined in the Base Details tab under Indexation. Note that for current, alternate and centreline projections the pricing date has no impact on jobs that are linked to standard jobs as AMT assumes the pricing of the standard jobs is always up-to-date. **Currency** - Mandatory field. Currency to be used. **Cost Centre** - Optional. If you leave this field blank, when the work order is created it will default to the cost centre in the asset classifications if there is one. **Work Group** - Optional. Allows you to select a work group. **OH&S** - Optional. One of Environmental, Safety, Statutory. **Part Type** - Optional. Select where the part type is from. **Commodity Code** - Optional. Only used in the AMT PSM module. **Source of Supply** - Optional. Used by some inventory systems. **BEC Code** - Optional. Only used in the AMT PSM module.
Job Costs	Lets you add or change parts, labour and miscellaneous costs.
Expense Element	An optional financial code, particularly used for financial budgeting and cost analysis.

Field/Panel	Description
Labour/Duration	**Labour Type** - This only impacts the forecast maintenance ratio report which only includes the field labour hours in the calculation. **Labour Activity** - Reflects the labour skill type. **Labour Hours** - Direct labour hours to perform the job. **Duration** - The elapsed time to perform the job.

Scheduling

Field/Panel	Description
Scheduling Details	**Scheduling Method** - Usage or Date. Date means scheduling in days. **UOM** - If the Date was not selected, then any UOM linked to that asset in a usage profile or its primary UOM can be selected as a scheduling method. **Schedule Type** - See Last Performed UOM/Date or Last Scheduled UOM/Date. **Not After** - Selecting the check box sets the end point at which scheduling will stop. If not selected, the scheduling continues to the end of the asset's term.
Last Change Prior To Equipment Start	These fields have **no impact** on scheduling. They are only used to determine the life a strategy task achieved at its first actual occurrence. It is only applicable if used assets are registered in AMT.
External Linkage	For some installations of AMT, the current maintenance strategy (represented in AMT as strategy tasks linked to an equipment current projection), is housed in an external system, such as SAP. An automated interface is used to send maintenance plans and job costs from the external system to AMT, typically on a daily basis. In this scenario, the maintenance strategy is defined and maintained in the external system, AMT is used to forecast and analyse the current maintenance strategy. Changes to the maintenance strategy are then made in the external system, which then interfaces the updates back to AMT. The link between the strategy task in AMT and the equivalent strategy task in the external system is defined in the **External Identifier** field in each AMT strategy task. The value in the field is determined per AMT installation and should not be altered. To link to a maintenance plan from an external ERP system, select **ERP Maintenance Plan**, and select from the list.  ERP Maintenance should not be changed unless first consulting the AMT System Administrator.
Advanced Scheduling Logic	If the task is a Planning task, Advanced Scheduling Logic options display. If the task is not flagged as a Planning task, you will not see this section.

Advanced Scheduling Logic

Dependency

Suppression - Parent

Suppression - Sibling

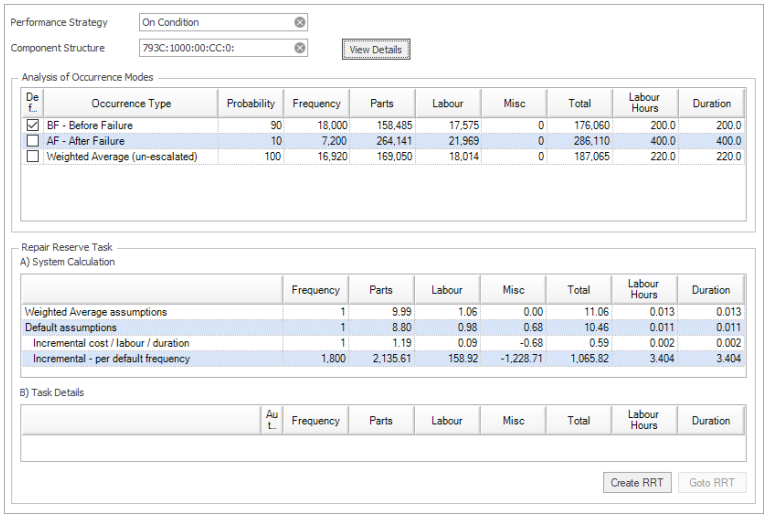
Inspection Logic

Component

Field/Panel	Description
Parts Details	**Primary Part Number** - Enables component forecasting by part number. **Group Number** - A secondary parts reference number that enables searching for a group of parts in parts books. **Part Rating** - Used to define the size and capacity of a component, for example, 175kw for pumps. Only used for reporting and analytics. **Default Part Classification** - Used to define the type of component replacement strategy (Repair, Replace with new, Exchange, Reman). This field is primarily used for the Component Capacity Management module. **Default Location For Rebuild** - Used to define the default rebuild location of the component if it is to be rebuilt. This field is primarily used for forecasting the component repairs by the receiving rebuild location. **Serial Number** - Used in the Component Tracking module.
Warranty Period	This is an input into an alert that displays in the work order **Feedback** tab. If this strategy task is performed for a second time within the tolerances set then a red flag displays on the work order highlighting that it is a potential warranty claim.

Strategy

Field/Panel	Description
Performance Strategy	There are three performance strategies available. • **On Condition** • **On Schedule** • **Run To Failure** If you select **On Condition** you must select the **Component Structure**. The component structure is like a standard job for reliability rules and assumptions. In AMT it can be used simply to factor in contingency costs and document the maintenance strategy around failure modes. Or, it can be used in the Condition Based Monitoring module to house the condition monitoring interpretation rules. See Component Structures for more information.

Field/Panel	Description
View Details	If **On Condition** and a **Component Structure** are selected, and the **View Details** button is clicked, the display of the form changes. An example screenshot is below.  This form shows: • **Analysis of Occurrence Modes** - this is the interpreted component structure. • **Repair Reserve Task A) System Calculation** - shows the calculations for the recommended repair reserve task for that risk profile. • **Repair Reserve Task B) Task details** - shows the repair reserve task that has been created, if there is one. • **Create RRT** button - enables you to create a repair reserve task if one has not been created. A repair reserve task is a type of strategy task specifically for factoring contingency costs.

Instructions

Field/Panel	Description
Electronic Workscopes	Lets you link the strategy task to electronic work scopes with are used in AMT Field or AMT Mobile.
Linked Library Documents	Lets you link the strategy task to any documents (PDF, Word) in either the AMT Library or any external document management system.

Advanced Features

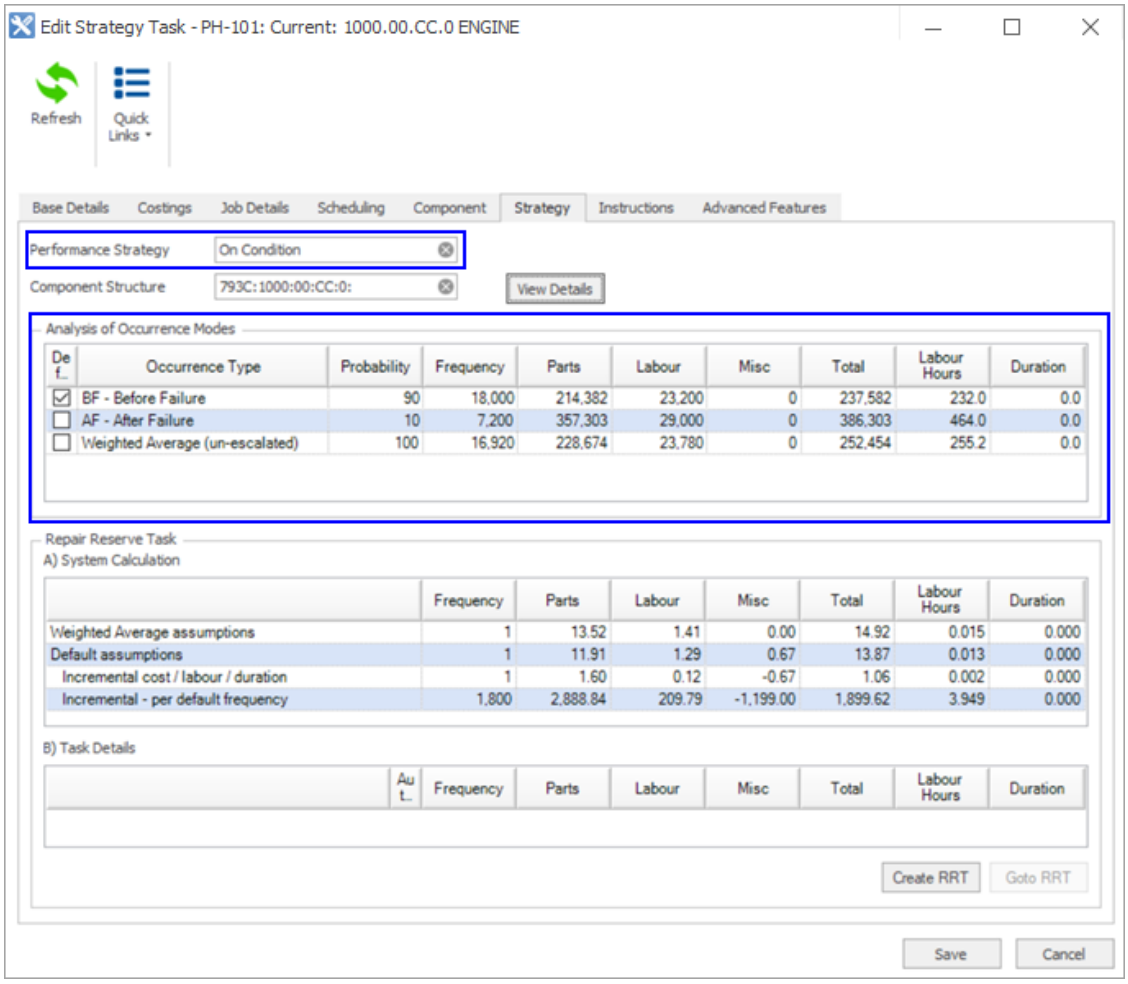
Field/Panel	Description
Automated Actions	Lets you generate and create a work order history, external work orders and planned events or work orders. If you select Work Order History, AMT will automatically create completed work order settlements for the strategy task amount.
Manual Cost Source	If you select this check box, cost allocations are not added or updated from an external source. This field is only applicable when interfacing AMT with an enterprise resource planning system (this would be the cost allocation cost source)

Assign CBM rules to maintenance task

A component structure can be linked to a strategy task by setting the **Performance Strategy** parameter of the strategy task equal to **On Condition**.

As a result, the default option of the component structure inherits the scheduling (frequency) and job details (costs) of the strategy task (not vice-versa); which AMT uses for forecasting (not the other way around).

The other occurrence types of the component structure being recalculated proportionally (see example screenshot below).

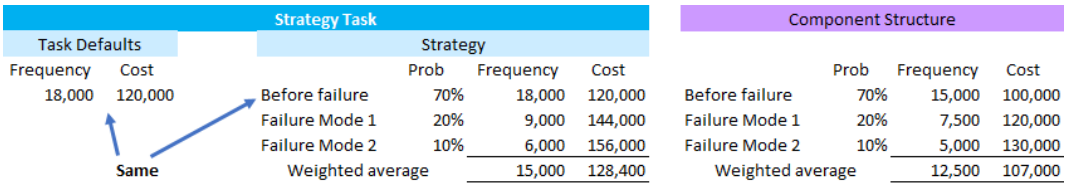


What does linking a component structure to a strategy task do?

- It does NOT change projected costs (the costs from the jobs within the strategy task are still used).
- References maintenance strategy.
- Displays repair reserve requirement.
- Enables component based maintenance.

Interpreted Component Structure

- When a task is linked to a component structure it does not copy the information directly from the component structure, rather it interprets or converts the information.
- The calculation method uses the current pricing, lives in the task and uses the ratios in the component structure to create the failure modes. See an example below.



In the above example:

- The **Before failure** is the same as the basic data set up in the strategy task, that is, the cost could come from a standard job.
- 9,000 Frequency in the strategy task's Failure Mode 1 is calculated as 7,500/15,000 x 18,000 (component structure failure mode 1 frequency/component structure before failure frequency x strategy task before failure frequency).
- $156,000 in the strategy task's Failure Mode 2 is calculated as 120,000/100,000 x 120,000 (component structure failure mode 1 cost/component structure before failure cost x strategy task before failure cost).

Why calculate it this way?

- It means one component structure can support equipment on different sites with different application severities and costing.
- When there are escalations you do not have to change the component structure.

Instructions

Field/Panel	Description
Electronic Workscopes	Lets you link the strategy task to electronic workscopes which are used in the AMT Client or AMT Mobile.
Linked Library Documents	Lets you link the strategy tasks to any documents (PDF, Word) in either the AMT Library or any external document management system.

Advanced features

Field/Panel	Description
Automated Actions	Lets you generate and create a work order history, external work orders and planned events or work orders. If you select Work Order History, AMT automatically creates completed work order settlements for the strategy task amount.
Manual Cost Source	If you select this check box, cost allocations are not added or updated from an external source. This field is only applicable when interfacing AMT with an enterprise resource planning system (this would be the cost allocation cost source).

System configuration - cost bearer code

The cost bearer code is a general code that determines which entity bears a cost. All contract costs, defined as strategy tasks, must have the cost bearer set to contract as it is only these costs that are included in AMT's contract tendering reports.

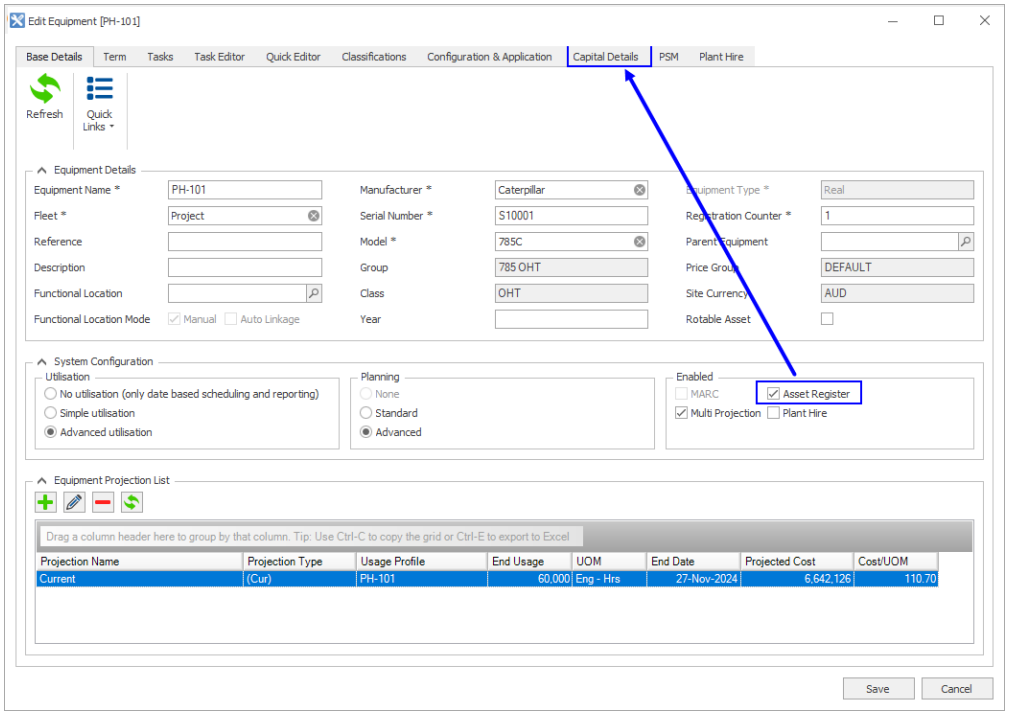
Configuration items

Ensure the following items have been configured.

Equipment configuration

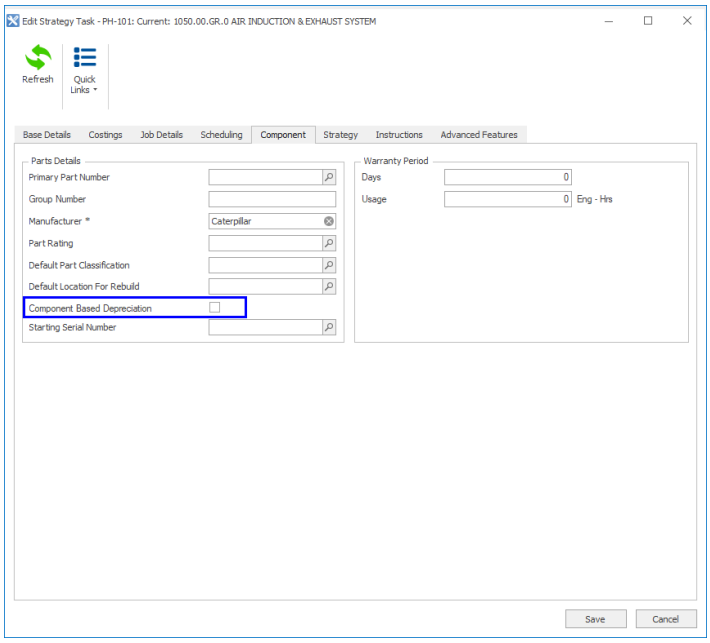
Navigate to the **Equipment Editor**, select the equipment to edit, and click **OK** .

Select the **Asset Register** check box and click **Refresh**. The Asset Ledger functionality is enabled and the Capital Details tab displays at the top of the screen.



Component Based Depreciation (CBD) Flag

Navigate to the **Strategy Task Editor > Component** tab, and select the **Component Based Depreciation** check box.

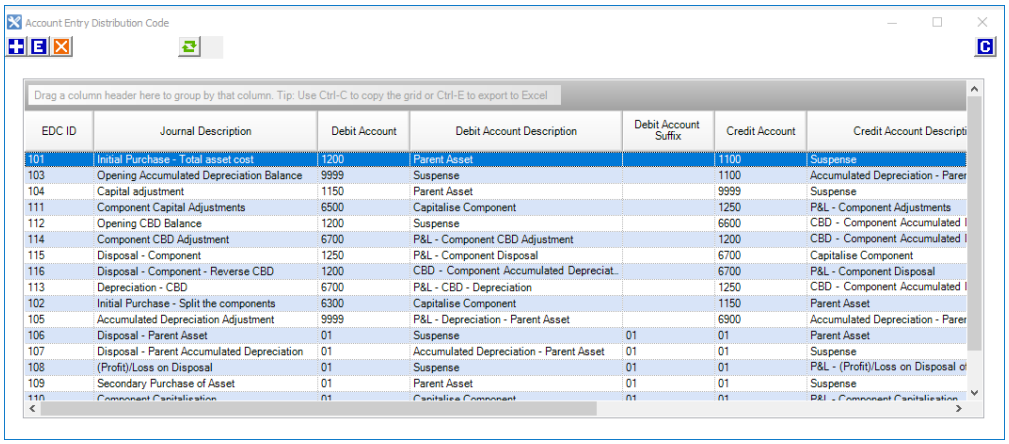


Entry Distribution Codes (EDC)

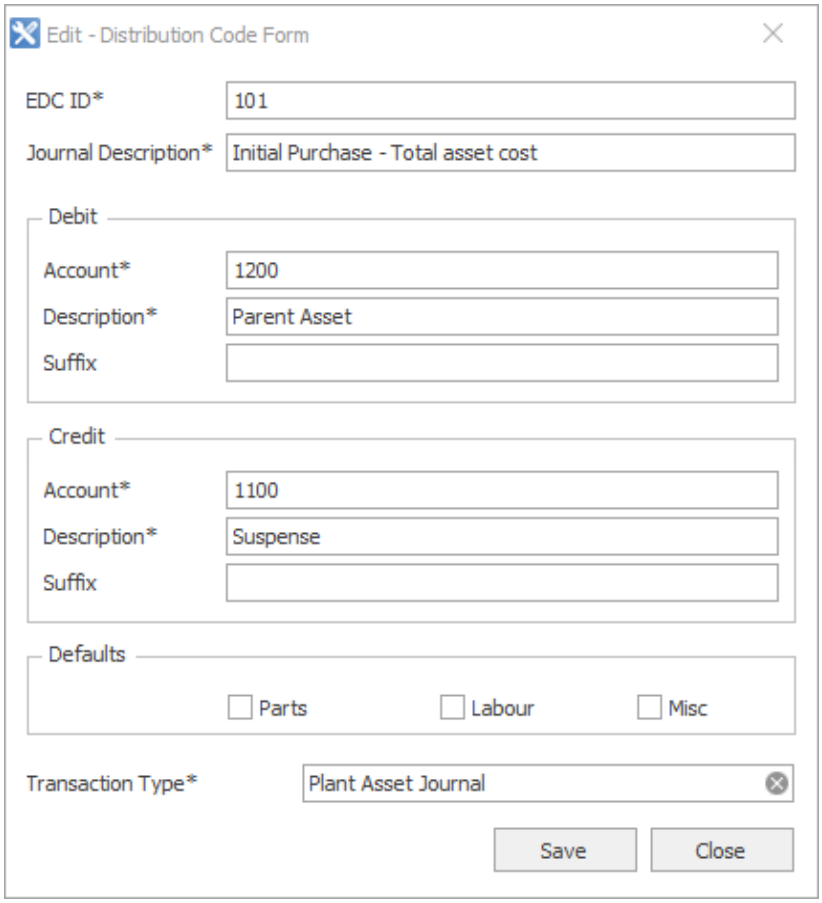
The Entry Distribution Codes are very important to both the Asset Ledger and Plant Hire functionality as they define the calculation rules. The EDC is a System Table that is set up during implementation.

Entry distribution codes are used to identify the balanced transactional journal that is created by AMT when an asset process is executed. The account codes associated with the debit and credit detail will align to the ERP financial ledger codes (representing balance sheet and P&L accounts) and support any interfaces developed to post transactions to the ERP general ledger.

To view and edit the EDC table, navigate to **System Administration** > **System Tables** > **Financial** > **Entry Distribution Code**. Click  to Edit the table.



- EDCs 101-117 are hard coded in AMT and should not be deleted or have their meaning changed.
- EDCs 200-227 are Depreciation specific. Refer to Configuration of Entry Distribution Codes for more information.
- The EDCs identify all the types of transactions that you need to create.
- You can change the **Journal Description** as long as the intention or use of the EDC is not changed. For example, you can change *Depreciation -CBD* to *Component Depreciation*.
- The **Debit Account**, **Debit Account Description**, **Debit Account Suffix**, **Credit Account**, **Credit Account Description** and **Credit Account Suffix** columns and fields are for information purposes. They do not affect the calculations in AMT. They should be set to correlate to the terminology and accounts for your company, but it is not important if they are not.



Capital Asset Recognition

Recognising assets once AMT is configured requires the recognition of the entire asset value at the parent level, and the recognition of the individual child assets.

A parent capital asset will have one or more child capital assets; an asset must have **at least one** child asset for depreciation purposes.

Asset capability

From an engineering perspective it is important to understand the capability of the asset.

1. Is the asset capable of meeting the required level of performance?
2. What are the application and operating factors that can influence the performance of the asset.

For example, it may be important to consider the impact of poor haul roads, overloading, or very steep haul roads on the performance of a haul truck, and specifically which components are impacted by which factors.

AMT Asset Journals

Entry distribution codes are used to identify the balanced transactional journal that is created by AMT when an asset process is executed. The account codes associated with the debit and credit detail will align to the ERP financial ledger codes (representing balance sheet and P&L accounts) and support any interfaces developed to post transactions to the ERP general ledger.

The table below identifies the asset journals and transaction IDs created by the AMT asset processes in depreciable asset recognition.

Process	Transaction ID	Entry Distribution Code	Debit Account	Credit Account
Capitalisation (Parent)	20	200	Capital	Capital Clearing
		222	Capital Clearing	Accumulated Depreciation
		223[2]	Residual Value	Residual Value
Capitalisation (Child/Component)	21	201	Capital	Capital Clearing
	22	202	Capital Clearing	Accumulated Depreciation
	24	204[2]	Residual Value	Residual Value
Monthly Depreciation	23	203	Depreciation (P&L)	Accumulated Depreciation
Additional Capital	25	205	Capital	Capital Clearing
	34	214	Capital Clearing	Accumulated Depreciation
	24	204[2]	Residual Value	Residual Value
Split In (Transfer In)[1]	44	224	Capital - Split Value (IN)	Capital - Split Value (IN)

Process	Transaction ID	Entry Distribution Code	Debit Account	Credit Account
	45	225	Accumulated Depreciation - Split Value (IN)	Accumulated Depreciation - Split Value (IN)
	24	204[2]	Residual Value (IN)	Residual Value (IN)
Split Out (Transfer Out)[1]	46	226	Capital - Split Value (OUT)	Capital - Split Value (OUT)
	47	227	Accumulated Depreciation - Split Value (OUT)	Accumulated Depreciation - Split Value (OUT)
	24	204[2]	Residual Value (OUT)	Residual Value (OUT)
Transfer In (Foreign Exchange)	40	220	Captial	Intercompany
	41	221	Intercompany	Accumulated Depreciation
	24	204[2]	Residual Value (IN)	Residual Vale (IN)
Transfer Out (Foreign Exchange)	28	208	Intercompany	Capital
	29	209	Accumulated Depreciation	Intercompany
	24	204[2]	Residual Value (OUT)	Residual Value (OUT)
Disposal (Gain)	35	215	Disposal Clearing	Capital
	36	216	Accumulated Depreciation	Disposal Clearing
	27	207	Cash	Disposal Clearing
	26	206	Disposal Clearing	Profit on Disposal (P&L)

Process	Transaction ID	Entry Distribution Code	Debit Account	Credit Account
Disposal (Loss)	35	215	Disposal Clearing	Capital
	36	216	Accumulated Depreciation	Disposal Clearing
	27	207	Cash	Disposal Clearing
	32	212	Loss on Disposal (P&L)	Disposal Clearing
Impairment	30	210	Impairment Loss (P&L)	Accumulated Impairment Losses
Revaluation	33	213	Capital	Revaluation Surplus

[1] Split in and split out are used for transactions where transfers occur with no change in currency.

[2] Residual value is being recognised by a journal transaction to support calculation of depreciable asset value.

Except where stated, AMT calculates the journals and the journal values in the background based on the information entered on the screen.

Editable Strategy Task Manager

This editor is specially designed to look and feel like a spreadsheet, where you can make changes to multiple strategy tasks across multiple equipment.

Each line in the grid shows a Strategy Task (combination of **CC**, **MC**, **TT**, **TC** and **UOM**). The count of equipment that have this Strategy Task is shown in the **Count** column. One line in the grid could represent the same Strategy Task for many equipment, therefore, when that line is updated, many Strategy Tasks are being updated at the same time. Strategy Task Job parameters can be updated in Strategy Tasks that have exactly one Projected Job. Linked Standard Jobs or Cost Allocations also restrict the Job Parameters that can be updated.

Cells which are shaded **grey** cannot be edited.

Click **Undo** to reverse changes up to the point at which the data on the form was last saved or displayed.

(Rpt_223) Projection Summary

Provides an overview of the costs for the equipment by projected tasks and sub system. Navigate to **Setup & Admin** > **Projections** > Projection Summary

Tips

Historical costs for the current projection are actual costs, historical costs for the budget projections are expected costs. Strategy tasks which do not have a projected cost will not appear in this report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|EqpPlanID|,
|EqpPlan|,|ProjHeaderID|,|ProjHeader|,|IsEscalated|,|ProjectionTypes|,
|LogoLocation|,|UserID|,|Currency|,|CurrencyID|

Report parameters

Default.

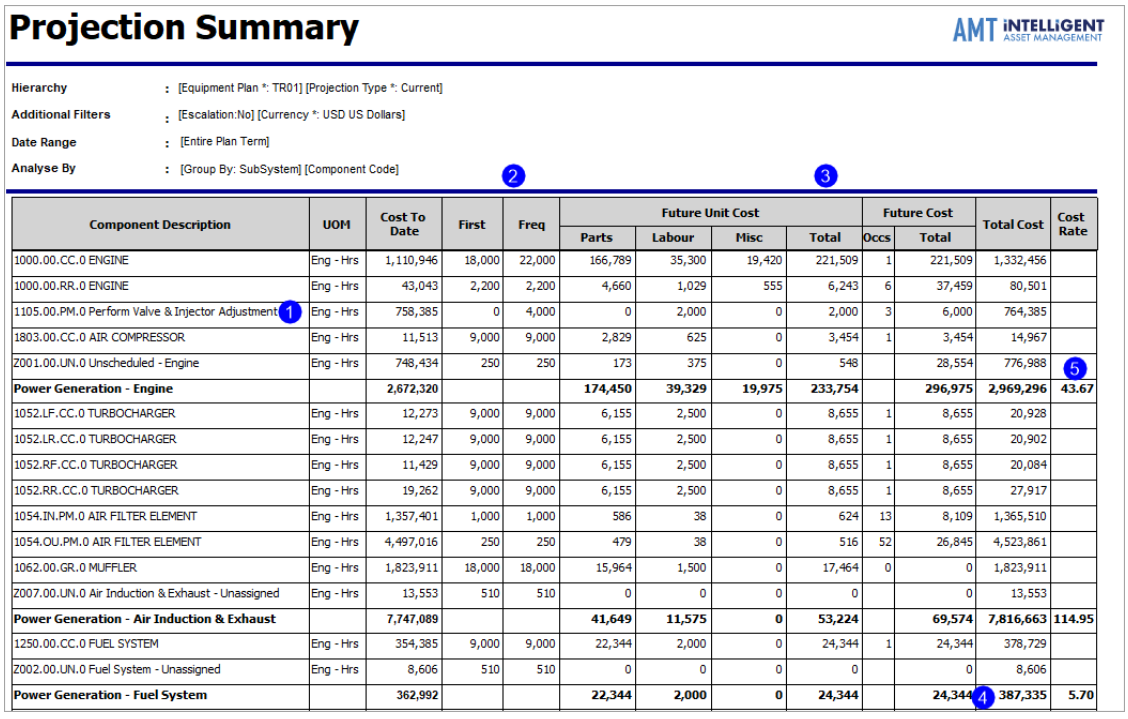
Additional filters

Cost Bearer, Currency, Escalation

Projection Summary Report

Projection Summary report

Select the **Equipment Plan**, **Projection Type** and set the **Escalation** to **No**.



Projection Summary

AMT INTELLIGENT ASSET MANAGEMENT

Hierarchy : [Equipment Plan *: TR01] [Projection Type *: Current]
Additional Filters : [Escalation:No] [Currency *: USD US Dollars]
Date Range : [Entire Plan Term]
Analyse By : [Group By: SubSystem] [Component Code]

Component Description	UOM	Cost To Date	First	Freq	Future Unit Cost				Future Cost		Total Cost	Cost Rate
					Parts	Labour	Misc	Total	Occs	Total		
1000.00.CC.0 ENGINE	Eng - Hrs	1,110,946	18,000	22,000	166,789	35,300	19,420	221,509	1	221,509	1,332,456	
1000.00.RR.0 ENGINE	Eng - Hrs	43,043	2,200	2,200	4,660	1,029	555	6,243	6	37,459	80,501	
1105.00.PM.0 Perform Valve & Injector Adjustment	Eng - Hrs	758,385	0	4,000	0	2,000	0	2,000	3	6,000	764,385	
1803.00.CC.0 AIR COMPRESSOR	Eng - Hrs	11,513	9,000	9,000	2,829	625	0	3,454	1	3,454	14,967	
Z001.00.UN.0 Unscheduled - Engine	Eng - Hrs	748,434	250	250	173	375	0	548		28,554	776,988	
Power Generation - Engine		**2,672,320**			**174,450**	**39,329**	**19,975**	**233,754**		**296,975**	**2,969,296**	**43.67**
1052.LF.CC.0 TURBOCHARGER	Eng - Hrs	12,273	9,000	9,000	6,155	2,500	0	8,655	1	8,655	20,928	
1052.LR.CC.0 TURBOCHARGER	Eng - Hrs	12,247	9,000	9,000	6,155	2,500	0	8,655	1	8,655	20,902	
1052.RF.CC.0 TURBOCHARGER	Eng - Hrs	11,429	9,000	9,000	6,155	2,500	0	8,655	1	8,655	20,084	
1052.RR.CC.0 TURBOCHARGER	Eng - Hrs	19,262	9,000	9,000	6,155	2,500	0	8,655	1	8,655	27,917	
1054.IN.PM.0 AIR FILTER ELEMENT	Eng - Hrs	1,357,401	1,000	1,000	586	38	0	624	13	8,109	1,365,510	
1054.OU.PM.0 AIR FILTER ELEMENT	Eng - Hrs	4,497,016	250	250	479	38	0	516	52	26,845	4,523,861	
1062.00.GR.0 MUFFLER	Eng - Hrs	1,823,911	18,000	18,000	15,964	1,500	0	17,464	0	0	1,823,911	
Z007.00.UN.0 Air Induction & Exhaust - Unassigned	Eng - Hrs	13,553	510	510	0	0	0	0		0	13,553	
Power Generation - Air Induction & Exhaust		**7,747,089**			**41,649**	**11,575**	**0**	**53,224**		**69,574**	**7,816,663**	**114.95**
1250.00.CC.0 FUEL SYSTEM	Eng - Hrs	354,385	9,000	9,000	22,344	2,000	0	24,344	1	24,344	378,729	
Z002.00.UN.0 Fuel System - Unassigned	Eng - Hrs	8,606	510	510	0	0	0	0		0	8,606	
Power Generation - Fuel System		**362,992**			**22,344**	**2,000**	**0**	**24,344**		**24,344**	**387,335**	**5.70**

Item	Description
1	Strategy task.

Item	Description
2	Component life estimates.
3	Total cost per strategy task occurrence.
4	Projected life cycle cost.
5	Cost per hour.

Validation

- Are the strategy tasks set up with the correct codes?
- Are the estimated component lives accurate based on the operating environment, historic performance and other factors?
- Has sufficient provision been built into the life cycle for minor repairs, that is, have unassigned tasks been created in the relevant subsystems with budgets?
- Is the cost per hour reasonable?

You can see additional details in the following reports.

(Rpt_222) Projection Details

Shows the base details of the projection including pricing, and highlights which tasks have associated documents in the library or has a standard job attached. Navigate to **Setup & Admin** > **Projections** > Projection Details.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Bearer, Escalation

Projection Details Report

Navigate to **Setup & Admin** > **Projections** > Projection Details.

This report shows additional details of each strategy task for the selected projection.

(Rpt_440) Projection and Interval Analysis

This report shows a strategy task's estimated cost, and then how much has actually been spent per interval and what is Forecasted to be spent per interval.

Tips

Actuals are not included if the report is based on a budget projection.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,

|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Costs, Escalation

(Rpt_425) Part Number Reference

Report which lists the Primary Part Number, Group Number and Component Serial Number per strategy task for the selected equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report Parameters

Report Type, Equipment Hierarchy

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Modifier Code, Strategy Task, Region, Division, Cost Responsibility, Eqp Location

Add/Copy/Delete projection

You can create, copy, delete or import projections of the following types:

- Budget
- Current
- Alternate. Alternate projections are useful for What-If analysis.
- Archive. Archive projections are useful for annual budgets.

Copy projected task

This function is used primarily for copying strategy tasks. If the strategy task

- Exists: then the function overwrites the old with the new.
- Does not exist: the procedure creates a new task.

The update of some of the strategy task's details can be customised using the editor. This is especially important for setting the first occurrence of the strategy task for used equipment.

Export Projection Details (Old)

Function superseded by the Export Projection function.

Export Projection

Enables you to export one or more projections, and optionally, any standard jobs linked to the strategy tasks within the selected projections.

Parameter	Description
Export to today's pricing	Escalate or de-escalate the strategy task's costs to reflect today's pricing and change the pricing date of a strategy task to today's pricing.  If the strategy task is reimported and linked to a standard job, the strategy task will inherit the pricing of the standard job.
Export Standard Job Details	Export standard jobs linked to strategy tasks in a separate file. The standard job reference identifier is contained in the strategy task data so that the standard jobs can be re-linked to the strategy tasks if required.
Convert to Global Codes	Convert the codes of strategy tasks to the codes in the corresponding AMT system tables, Global Component Code and Global Job Code.
Convert to Primary Currency	Convert costs to the Primary Currency of the AMT database, which is defined in the Currency system table, using rates in the Exchange Rate system table.
Nominate Projection File Name	The file is saved on the AMT server. Contact the System Administrator for the file location details. The file is .csv format.
Nominate Standard Job File Name	Only applicable if the **Export Standard Job Details** check box is selected. The file is saved on the AMT server. Contact the System Administration for file location details. The file is .csv format.

Strategy Task Import Wizard

 This function is only relevant to specific AMT solution configurations.

When importing Strategy Tasks from an External ERP (for example, SAP, Oracle, Ellipse) there is often a requirement to customise exactly how specific tasks are updated from the external ERP. The Strategy Task Import Wizard enables these key decisions to be customised by specific Strategy Task.

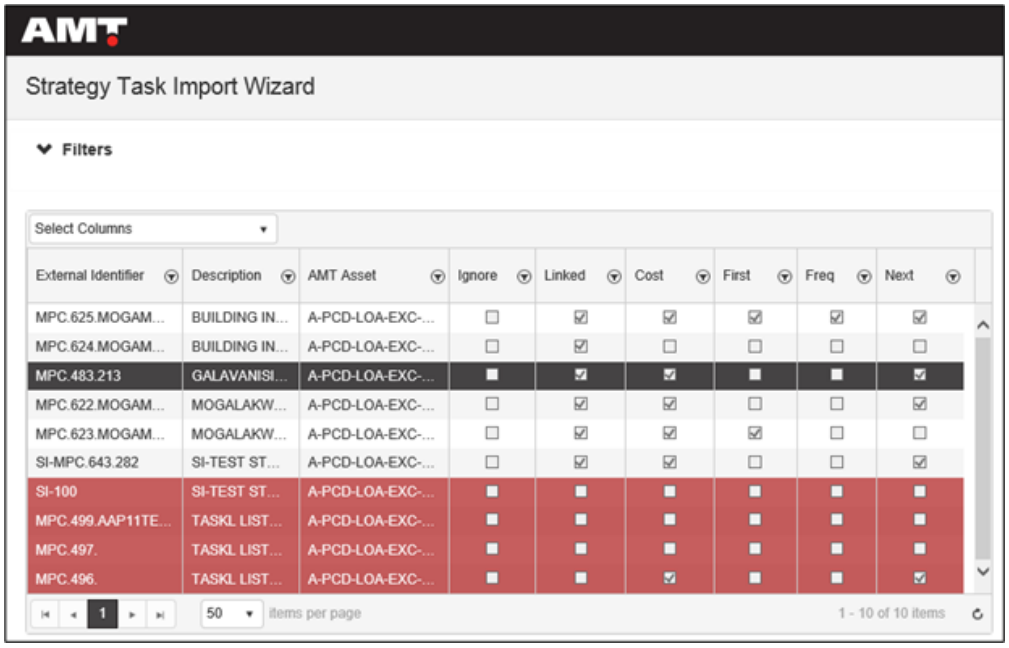
A configuration enables the Strategy Task Import Wizard. If not enabled, tasks load directly into AMT without going through the Strategy Task Import Wizard.

The defining link between the AMT Strategy Task and the ERP Strategy Task is the **External Identifier**. This is a key field in the Strategy Task import interface file. The first time the task is imported using the Strategy Task Import Wizard, if the task already exists in AMT with the External Identifier nominated, the task links automatically to AMT and will overwrite details from the External System.

An AMT system-level configuration also enables that the four key codes (Component Code, Modifier Code, Task Type, Task Counter) can be set in AMT with the linkage being the External Identifier. Workorder Settlements with the External Identifier then link based on the External Identifier.

The Import Wizard process is a combination of specific actions to be taken for new strategy tasks from the ERP, and then a decision on how the ERP updates the Strategy Task in AMT.

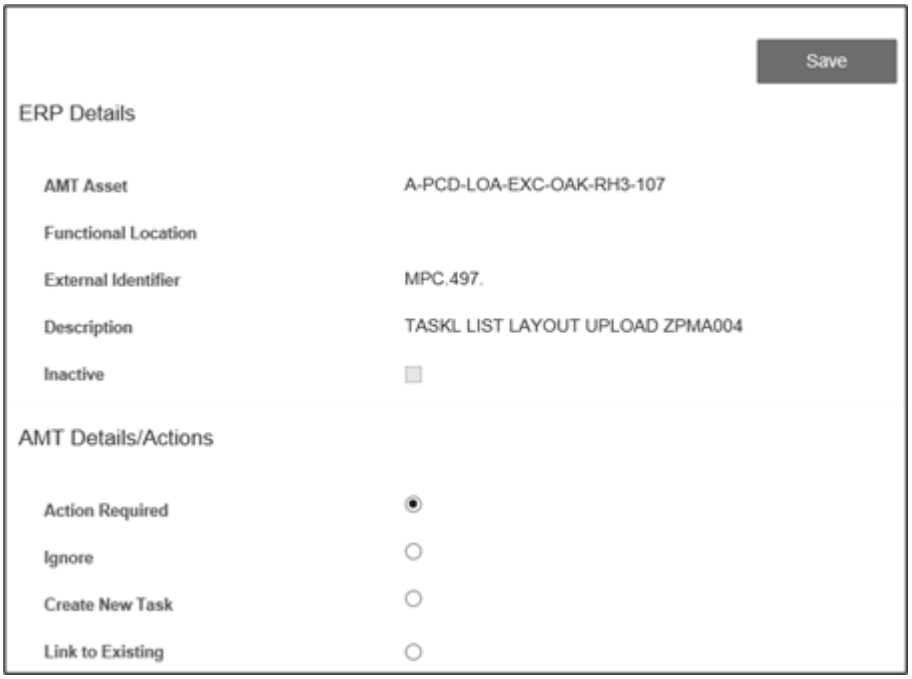
The Import Wizard shows a list of all strategy tasks that have come from the ERP. This lists all tasks with the External Identifier populated. Filters can limit which tasks to show. Tasks highlighted in **red** are tasks which have newly come from the ERP and an initial action is required. Columns also show the decisions made as to how the ERP task impacts AMT.



When a task first comes into AMT there are initial actions required.

- **Action Required** - No decision has yet been made on this strategy task and will remain highlighted **red** until a decision is made.
- **Ignore the task** - The external task is ignored and does not update into the AMT strategy task.
- **Create a new task** - A new strategy task is created in AMT.
- **Link to existing** - Nominate an existing AMT task to link to. The External Identifier in the AMT strategy

task is updated.

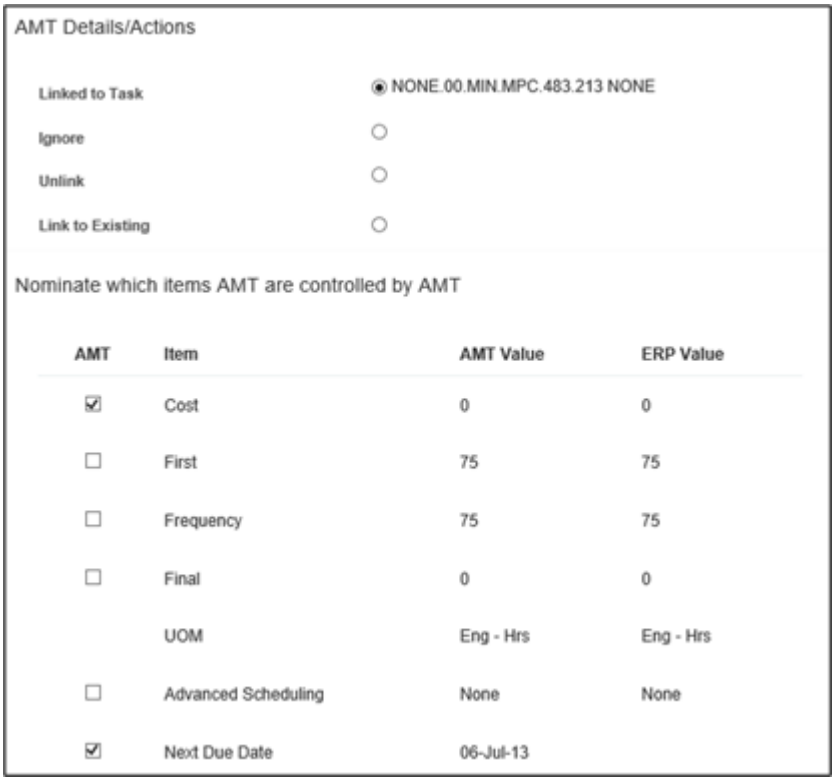


Where there is a linkage between the ERP task and the AMT strategy task, you can nominate specific interface decisions for the linkage between the ERP and the AMT strategy task.

When an ERP Task links to an AMT strategy task, specific decisions can be made about the relationship between AMT and the ERP for that specific task.

- Does cost come from AMT or the ERP? This is very useful where there may not be accurate parts costs in the ERP for large components.
- Do you set First or Frequency, both or none in AMT? First is useful when loading equipment into AMT part-way through the life.
- Ability to set Final in AMT. This is used when the ERP task is no longer active, but you do not want to delete the AMT strategy task as you want to keep a linkage to the task work order analysis.
- Ability to set advanced scheduling in AMT.
- Ability to have AMT control the next occurrence date. This is used when AMT controls the long-term plan

for specific strategy tasks.



Projections - Multi-editors

Multi-Task Scope Editor

The Multi-Task Scope Editor lets you edit the cost adjustment factors across multiple assets in the same fleet. In the strategy task these factors can be viewed by going to the Audit Trail form **Edit Strategy Task** > **Quick Links** > **Audit Trail** .

Copy Strategy Tasks

Copy Strategy Tasks lets you copy and overwrite strategy tasks from one projection to another.

Multi-editing strategy tasks

There are a number of administration tools in AMT to help you add, modify, copy and delete strategy tasks to reflect the requirements of the contract tender. The tools are described in the table below.

Most of these tools are under **Setup & Admin** > **Projections - Multi-editors**, **Modelling** > **Administration** > **Projections** and **Centrelines** > **Administration**.

Editor	Description
Multi-Task Scope Editor	Lets you edit the cost adjustment factors across multiple assets in the same fleet. In the strategy task these factors can be viewed by going to the Audit Trail form **Edit Strategy Task** > **Quick Links** > **Audit Trail** .
Copy Strategy Tasks	Lets you copy and overwrite strategy tasks from one projection to another.
Multi-Task % Cost Adjuster	Makes direct adjustments to the amounts in the strategy task.  It does not leave an audit trail. It will only adjust strategy task jobs in Manual mode, therefore strategy task jobs linked to standard jobs will not be adjusted.
Multi-Task Editor	This is the most flexible multi-editor, and enables you to edit both strategy tasks and strategy task job fields. The Job Filters section applies to strategy task job fields.  There can be many strategy task jobs in one strategy task.
Multi-Task Labour Editor	This applies the labour rates entered to the labour cost of the strategy task job. It multiplies the labour hours in the strategy task job by the rates entered to obtain the labour cost. It will only adjust strategy task jobs in Manual mode, therefore strategy task jobs linked to standard jobs will not be adjusted.
Multi-Equipment Delete Task	Enables you to delete multiple strategy tasks.
Forecast Cost Adjuster	Enables you to update projected costs based on actual cost history. This allows more accurate forecasting of costs, facilitating increased budget accuracy. This process was principally developed for process plant budgeting where you may have 10,000 assets registered and you want to quickly create strategy tasks for all of them based on their cost history.  This process should only be run with **extreme care**.
Copy in Projection	Copy a projection from one asset to one or more assets.
Copy a Projected Task	Copy a task from one projection to many projections. If the task is linked to a standard job, it will retain the link.

Multi-Task Editor

This is the most flexible multi-editor, and enables you to edit both strategy tasks and strategy task job fields. The Job Filters section applies to strategy task job fields.



There can be many strategy task jobs in one strategy task.

Multi-Task % Cost Adjuster

The Multi-Task Percent Cost Adjuster makes direct adjustments to the amounts in the strategy task.



It does not leave an audit trail.

It will only adjust strategy task jobs in Manual mode, therefore strategy task jobs linked to standard jobs will not be adjusted.

Multi-Task Labour Editor

This applies the labour rates entered to the labour cost of the strategy task job. It multiplies the labour hours in the strategy task job by the rates entered to obtain the labour cost.

It will only adjust strategy task jobs in Manual mode, therefore strategy task jobs linked to standard jobs will not be adjusted.

Multi-Equipment Delete Task

Enables you to delete multiple strategy tasks.

Forecast cost adjuster

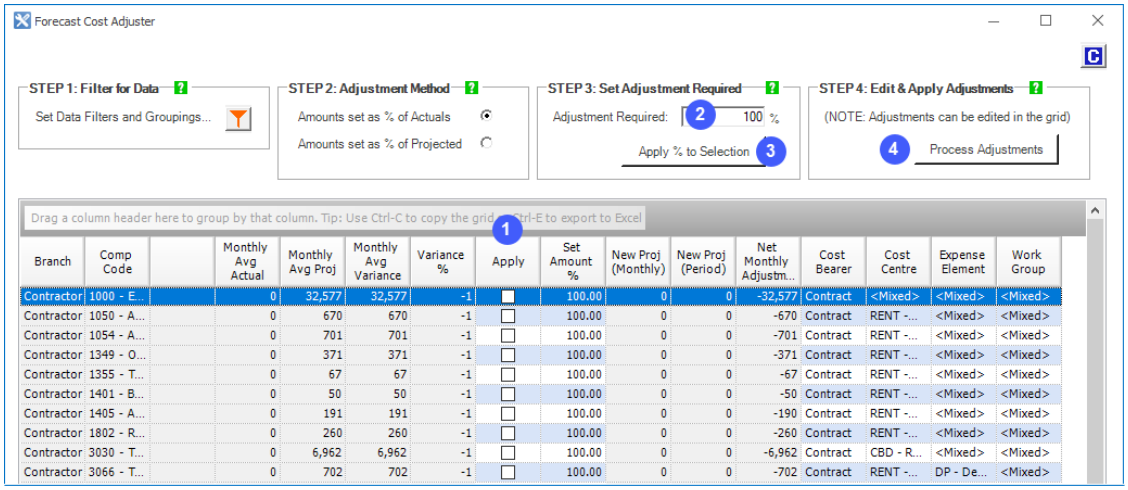
You can update projected costs based on actual cost history using forecast cost adjuster. This provides more accurate forecasting of costs, facilitating increased budget accuracy.

Costs can be adjusted using one of two methods:

1. Adjustment amount set as a percentage of Actual costs.
2. Adjustment amount set as a percentage of Projected costs.

Using the forecast cost adjuster

Navigate to **Setup & Admin** > **Projections - Multi-editors** > **Forecast Cost Adjuster**.



1. Select the check boxes of all costs that are to be adjusted.
2. Enter a multiplier in the **Adjustment Required** field.
3. Click **Apply % to Selection** to apply the adjustment. You can repeat this process until the desired costs are achieved.
4. Click **Process Adjustments** to save the cost adjustments.

Cost adjustments are recorded in a special task in the projection with the component code Z029 (Cost Adjuster – Unassigned).

Usage profile

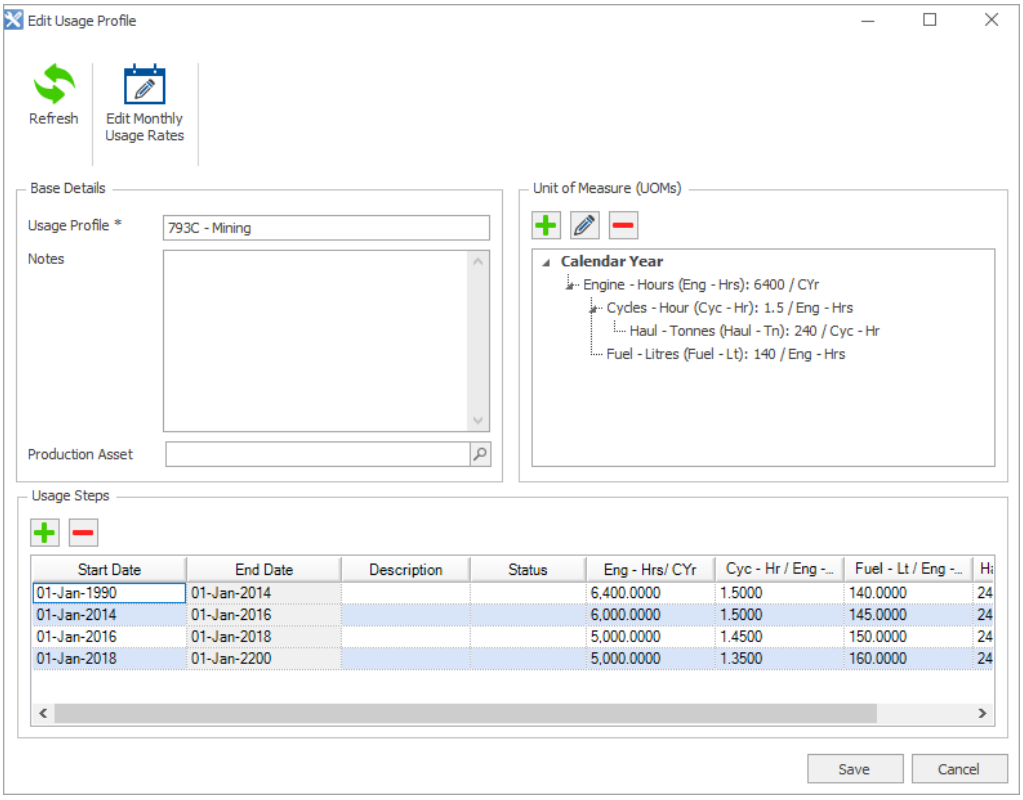
Unit of measure (UOM) refers to any cumulative usage related to an asset, such as machine hours, fuel litres, kilowatt hours and tonnes. They are also referred to as **Usages** in AMT. Usages are used to:

1. **Drive maintenance schedules**
   For example, an engine is scheduled based on fuel litres. In the strategy task you can select the UOM appropriate for that asset.
2. **Forecast productivity**
   For example, tonnes moved can be set up as a UOM. Tonnes moved can then be used to forecast production.

Usage Profiles enable you to:

- Set up a profile for the usage rates or utilisation for groups of assets.
- Set up relationships between the usages (for example, tonnes is a dependency of engine hours).
- Vary the usage rates over time to reflect changing operating conditions (for example, deeper pit, rainy season, winter and other conditions).

  The example screenshot below shows a usage profile with a hierarchy of usages.



- Usage profiles allow you to set up the severity of the situation. For example, if the pit in a mine becomes deeper, the fuel burn rate increases and the engine life in hours will decrease.
- Actual usages can be captured in AMT and plotted against the usage profile.
- Usage profiles are optional for assets. You can elect to either have no usage measure for an asset (all strategy tasks are then date-based) or apply a simple utilisation where one UOM is selected and one utilisation rate set (for example, 4,000 engine hours per annum).

The **Status** field within a usage profile step is only applicable for the tendering and forecasting of a specific type of equipment which is configured for this functionality, and therefore, in most cases, does not impact the usage forecast by AMT.

Set up Usage Profile

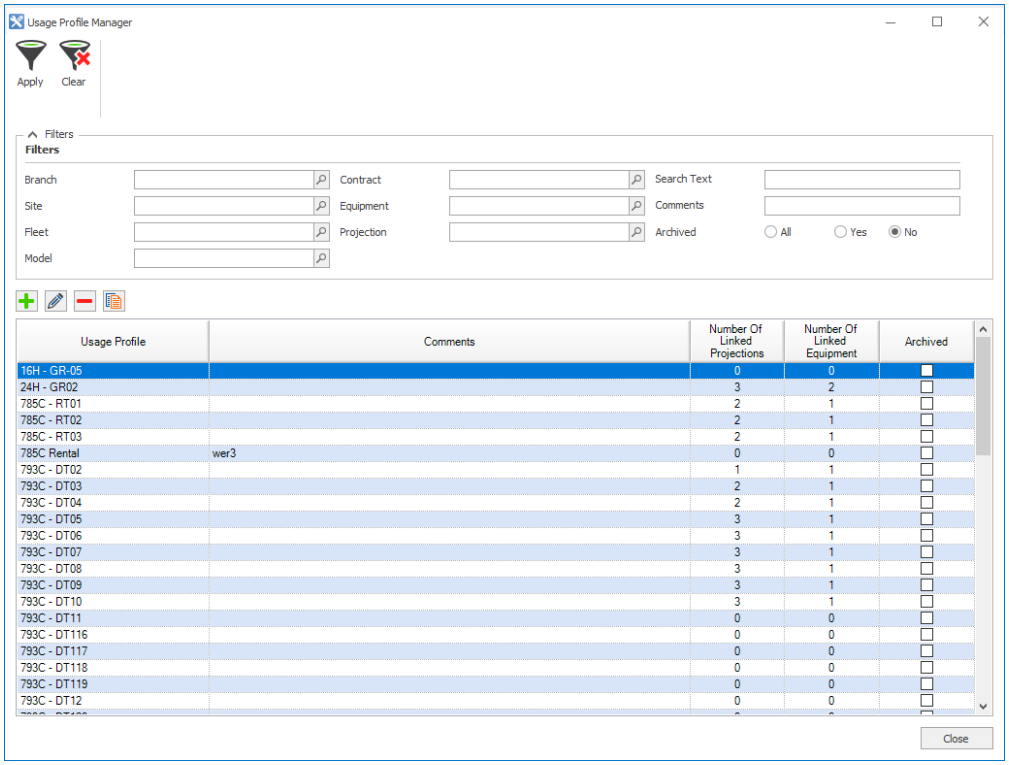
Setting up a usage profile for contract tendering is **optional**; it depends on the level of detail and complexity required for the asset. It is needed if either:

- More than one Unit of Measure is needed, for example, engine hours and travel time; or
- Utilisation rates vary over the asset's life, for example, due to weather conditions, different applications in which the asset will operate.

Navigate to **Centrelines** > **Administration** > **Usage Profile Manager**.

Usage Profile Manager

You can open the Usage Profile Manager by navigating to **Setup & Admin** > **Usage Profiles** > **Usage Profile Manager** or **Centrelines** > **Administration** > **Usage Profile Manager**.



Use the Usage Profile Manager function in AMT to add, edit and delete usage profiles.

To copy a usage profile click  .

Usage is the consumption of a resource by an asset. This consumption can vary over time due to changes in the working application and in working patterns. For example, consumption may be altered by changes in weather patterns; high rainfall may slow production, dry fine weather may increase production, or by physical changes in loads, haulage road conditions, holidays, and so on.

The usage profile maps the changes in consumption over time and allows the consumption to be measured by different units of measure. Usage profiles can have multiple steps and multiple branches in the consumption tree.

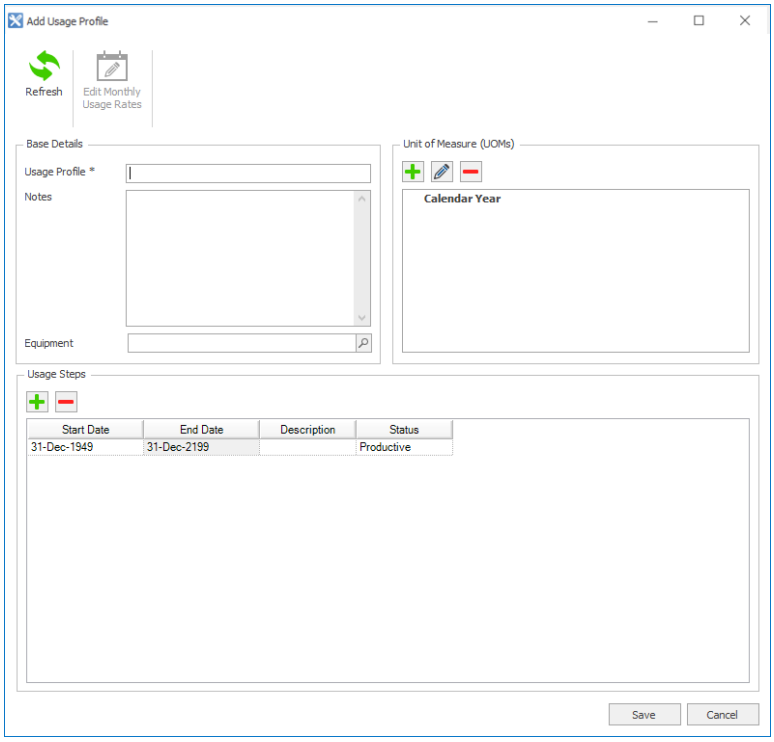
All usages are expressed in qualified units of measure (UOM) such as engine hours or litres of fuel. The most common unit of measure is engine hours. A usage profile is a named pattern for the rate of consumption of UOMs by an asset. Usage profiles can be linked to a single asset in a single projection or a range of assets of the same model across a fleet or site.

Notes

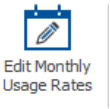
- A range of assets with different models can use the same usage profile, and different projections can use different usage profiles to model application, utilisation, or other what-if scenarios.
- The final usage step within a usage profile cannot have a usage value of zero.

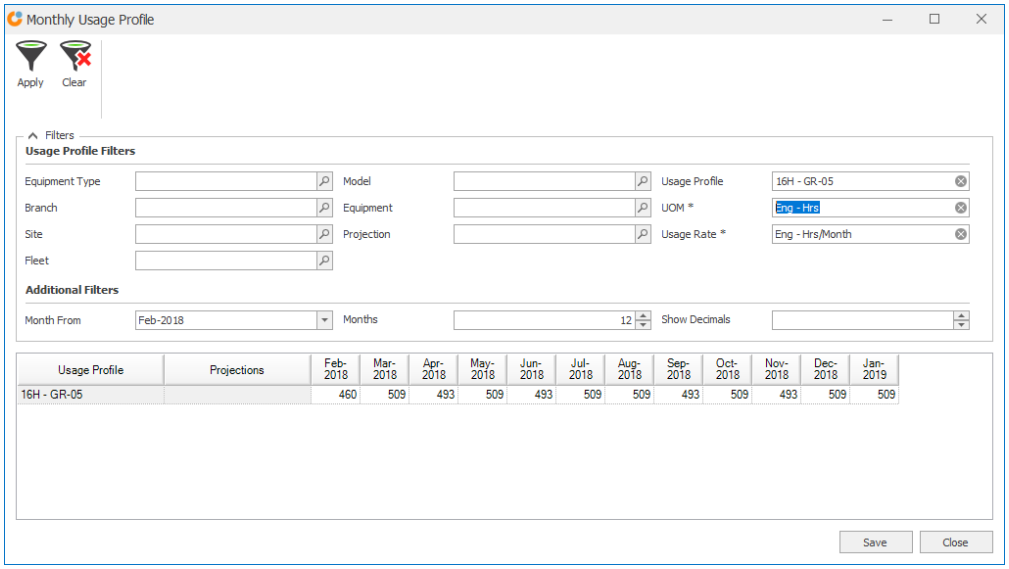
Add Usage Profile

Click  on the Usage Profile Manager.



Field	Description
Usage Profile	Enter a name for the profile.
Notes	Add notes about the profile.
Equipment	Click **Search** and select the relevant asset.
Unit of Measure	Click **Add** to add a Unit of Measure to the parent UOM - **Calendar Year**. You can have multiple parent UOMs within a profile.
Usage Steps	Click the **Add** icon to add steps. You can configure each usage step to have different usage rates. This feature is valuable if you want to create a cyclical usage profile or a profile that changes over time. The length of each cycle is defined by selecting the relevant **Start Date** of each step. You can use different usages for each step.

To edit the monthly usage rates click 



You can double-click on a cell in the grid to manually adjust the data. Right-click on the grid to see a variety of copy and paste options.

If you are pasting tables from an external spreadsheet back into the grid, the number of cells copied from the external spreadsheet must be the same as the number of cells in the grid.

Set up a Shutdown Planning Usage Profile

When setting up a **Shutdown Planning Usage Profile**, use **Shutdown Profile** as the **Usage Profile** name.

The **Start Date** and **End Dates** are system defaults and are not editable.

Assign usage profiles to assets

The usage profile in AMT is used to forecast expected utilisation of the asset. With the **Long-Term Planning – Shutdown Planning** module, you can select steps in the usage profile where the asset will be utilised. This functionality is used to more accurately forecast utilisation of assets where they will only be used at certain times or periods in the productive life of an operation.

Usage profile

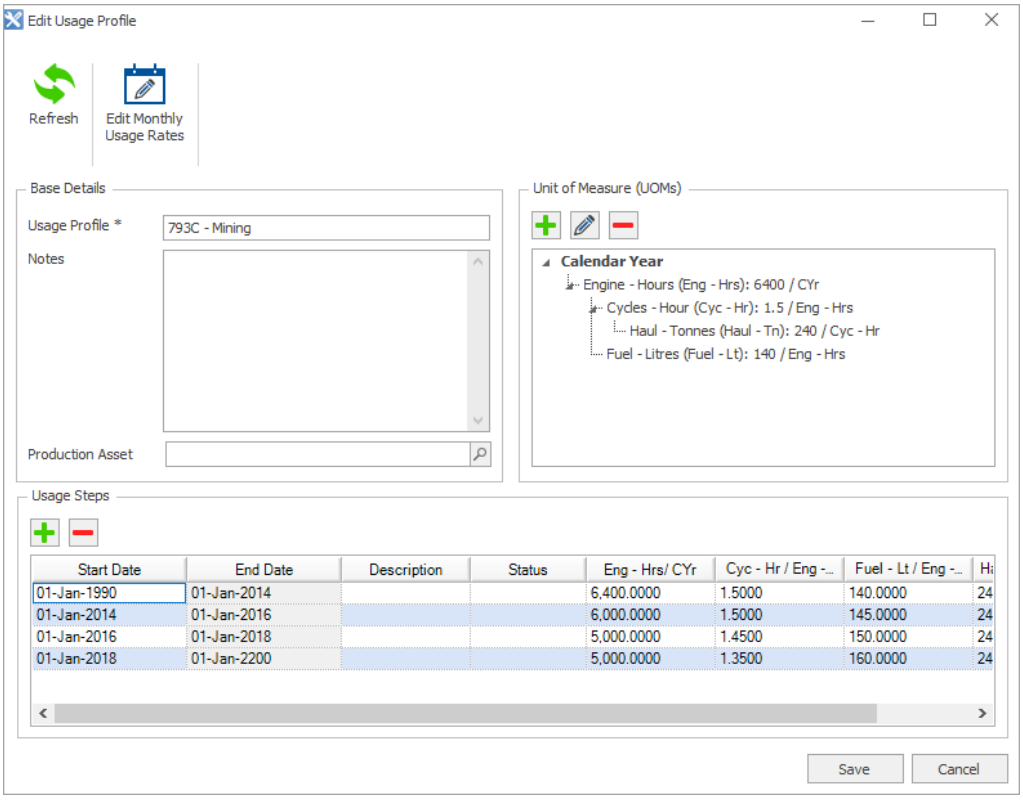
Unit of measure (UOM) refers to any cumulative usage related to an asset, such as machine hours, fuel litres, kilowatt hours and tonnes. They are also referred to as **Usages** in AMT. Usages are used to:

1. **Drive maintenance schedules**
   For example, an engine is scheduled based on fuel litres. In the strategy task you can select the UOM appropriate for that asset.
2. **Forecast productivity**
   For example, tonnes moved can be set up as a UOM. Tonnes moved can then be used to forecast production.

Usage Profiles enable you to:

- Set up a profile for the usage rates or utilisation for groups of assets.
- Set up relationships between the usages (for example, tonnes is a dependency of engine hours).
- Vary the usage rates over time to reflect changing operating conditions (for example, deeper pit, rainy season, winter and other conditions).

  The example screenshot below shows a usage profile with a hierarchy of usages.



- Usage profiles allow you to set up the severity of the situation. For example, if the pit in a mine becomes deeper, the fuel burn rate increases and the engine life in hours will decrease.
- Actual usages can be captured in AMT and plotted against the usage profile.
- Usage profiles are optional for assets. You can elect to either have no usage measure for an asset (all strategy tasks are then date-based) or apply a simple utilisation where one UOM is selected and one utilisation rate set (for example, 4,000 engine hours per annum).

The **Status** field within a usage profile step is only applicable for the tendering and forecasting of a specific type of equipment which is configured for this functionality, and therefore, in most cases, does not impact the usage forecast by AMT.

Utilisation and Usage Profiles

There are three utilisation configuration options for an asset:

- **No Utilisation**
  No utilisation is tracked against the asset and tasks are scheduled based on date. This generally applies to fixed plant.

- **Simple Utilisation**
  One unit of measure (UOM) is set as well as one consistent utilisation rate. For example, a car might have 20,000km per annum set as its utilisation assumption.
- **Advanced Utilisation**
  When you have multiple UOMs you want to track against an asset or you want to vary the utilisation rate. For example, a wet season versus a dry season utilisation rate.

If advanced utilisation is applicable for the asset, you need to set up a usage profile.

Define application severity

Application severity covers the physical environment and operating practices. In AMT, application severity is represented by Usage Profiles.

(Rpt_248) Usage Profile Details

Displays the usages pertaining to an equipment, with both actual and cumulative values showing. This report can be generated for a specific month period, but defaults are set to the last twelve months. than zero.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,1,|UserID|

Report parameters

Default.

Additional filters

None.

Monthly Usage Profile

This function allows you to modify the projected usage per month of one or more usage profiles.

Right-click editing options on the grid are as follows.

- **Copy Table** - copy and paste into a spreadsheet program like Microsoft Excel.
- **Paste** - pastes data copied in the spreadsheet program into the grid.
- **Copy Across** - copy the usage value in the selected cell across all months in the same row.
- **Copy Down** - copy the usage value in the selected cell down, that is, make the usage the same value for all usage profiles for the same month.
- **Copy Daily Rate Access** - AMT will divide the usage value by the number of days in the selected month to determine the daily usage rate. AMT then applies that rate to all months in the same row (that is, daily rate multiplied by the number of days in the respective months).

Any values you change are displayed in **red** until saved.

Asset Ledger

Capital Details

You can open the Capital Details screen in the following ways:

- **Setup & Admin** > **Asset Ledger**. Select an equipment and click OK. The **Capital Details** tab displays.
- **Modelling** > **Administration** > **Asset Ledger** > **Capital Details** tab. Select an equipment and click OK. The **Capital Details** tab displays.
- On the right of the home screen, in the **Equipment Navigator** on the right of the screen, drill-down into **Contractor** > **Plant Hire** > **Project** and double-click an equipment. Click **Show Details** to display the input fields.

The capital detail function in AMT has a number of input fields. First nominate if the equipment is owned or leased. Select the check box to indicate if the equipment will be replaced. If this check box is selected the estimated replacement cost field can be updated with the cost. You can also enter the estimated disposal price at this point.

Other fields to complete for depreciation are:

- Component code.
- Cost bearer, for example project or plant group.
- Expense element.
- Journal applicable. Select this box if transactions will be posted to the general ledger using an interface.
- Entry distribution code (general ledger codes for posting via an interface).

The current depreciation position is displayed on the screen to indicate the gross cost, accumulated depreciation, book value and CBD provision at the end of the month. The depreciation rate is also shown here, whether on a fixed basis or UOM. On the lower half of the screen you can set up capital schedules and add or adjust the capital details.

After the capital details are set up, you must save the records before the acquisition details or schedules can be set up.

Acquisition/Disposal Details

- Enter the purchase price and any accumulated depreciation.
- These details set the book value to be depreciated in the schedules. The purchase order, delivery and warranty details of the equipment are also stored on this screen for reference by AMT.

Technical Details

- Specifications and general information for an item of equipment.
- Record information such as registration, chassis ID, wheelbase and tare weights. The current driver or operator of a vehicle can also tracked in this screen.

Capital Schedule

- After the base details are complete, you can create a capital schedule.
- The capital schedule is added from the capital detail screen by clicking **Add**.
- The screen displays reference data and contains fields to nominate various parameters for the capital schedule.

- The starting period defaults to the current month but should be changed to the equipment start month if no processing has been performed previously.
    - The capital adjustment/addition section allows any purchases or adjustments to the book value to be modified. There may be an additional capital item purchased for the equipment throughout its life and this adjusts the value to be depreciated.
    - The depreciation schedule can be given a nominated residual value to remain at the end of the term. The schedule has the option of monthly or by a UOM. The end point will change depending on whether the schedule is by date or UOM.
- The capital schedule grid displays a summary section to indicate the current position.



It is important for you to refresh the screens after changes are made so that the summary displays the modified information.

Asset Ledger

Using the Asset Ledger module of AMT you can record all financial transactions against assets and equipment ledger accounts (cost and accumulated depreciation). You can also capitalise and depreciate components.

This functionality is in addition to the standard information that can be set up when registering assets in AMT.

This module in AMT is closely associated with the **Plant Hire** module when some of the component journals impact some Plant Hire reports.

Asset Ledger terminology

Asset Register vs Asset Ledger

Typical registration of assets in AMT is performed with the *Asset Register*. Recording of the capital cost and depreciation is performed with the *Asset Ledger*.

Parent Assets

The main capital asset, such as a truck.

Child Assets

Specific components on a parent asset that have been capitalised, for example engine.

Component Based Depreciation (CBD)

Component Based Depreciation Strategy Tasks that relate to child assets (components that have been capitalised in the asset ledger). Depreciation at a component level, rather that a parent asset level.

Entry Distribution Code (EDC)

A transaction type for journals.

Journals

Journals are the transactions that record the entries in the Asset Ledge. Work orders generally record maintenance costs, journals record financial transactions against the Asset Ledger. In other modules of AMT, Journals can also be used for other financial transactions.

Plant Rates

How an asset's hire rate is broken down.

Plant Invoices

The monthly invoices for the hire of the asset. The invoices are often based on the utilisation of the asset in that month.

Project

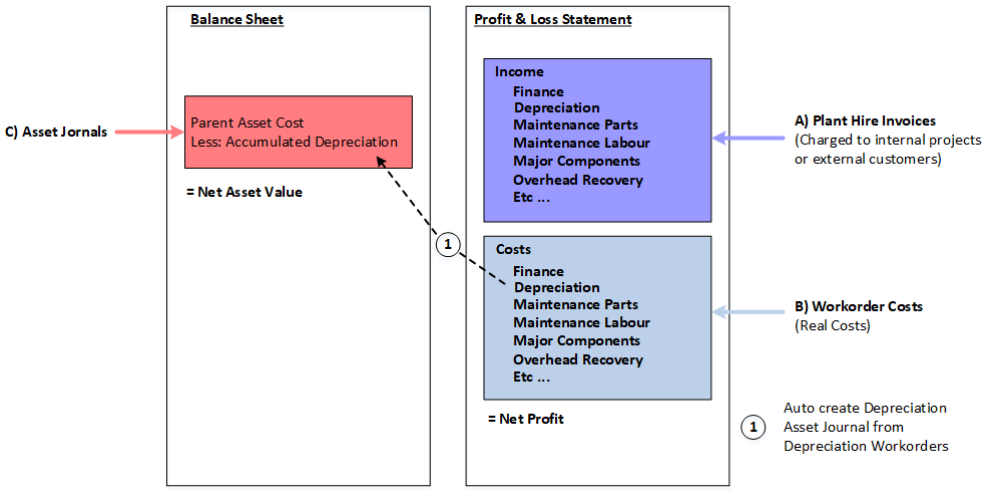
Specific hire projects or contracts for an asset. An asset can have many hire projects in its life, but can only ever belong to one project at a time.

Accounting process

The accounting process is best explained in two steps, **Single Asset** and **Capital Components**.

1. Single Asset

An asset with no capitalised component, that is, a single asset that is being depreciated.



The asset ledger maintains the asset balance sheet accounts, which are comprised of:

- Parent Asset Cost - for Haul Truck.
- Parent Asset Accumulated Depreciation.

All transactions impacting the asset ledger are recorded in Asset Journals.

Depreciation on the parent asset must be recorded on a work order (for the profit & loss statement) and a journal (for the asset ledger). Therefore, in the AMT database it is recorded twice. See Automated Journals for more information.

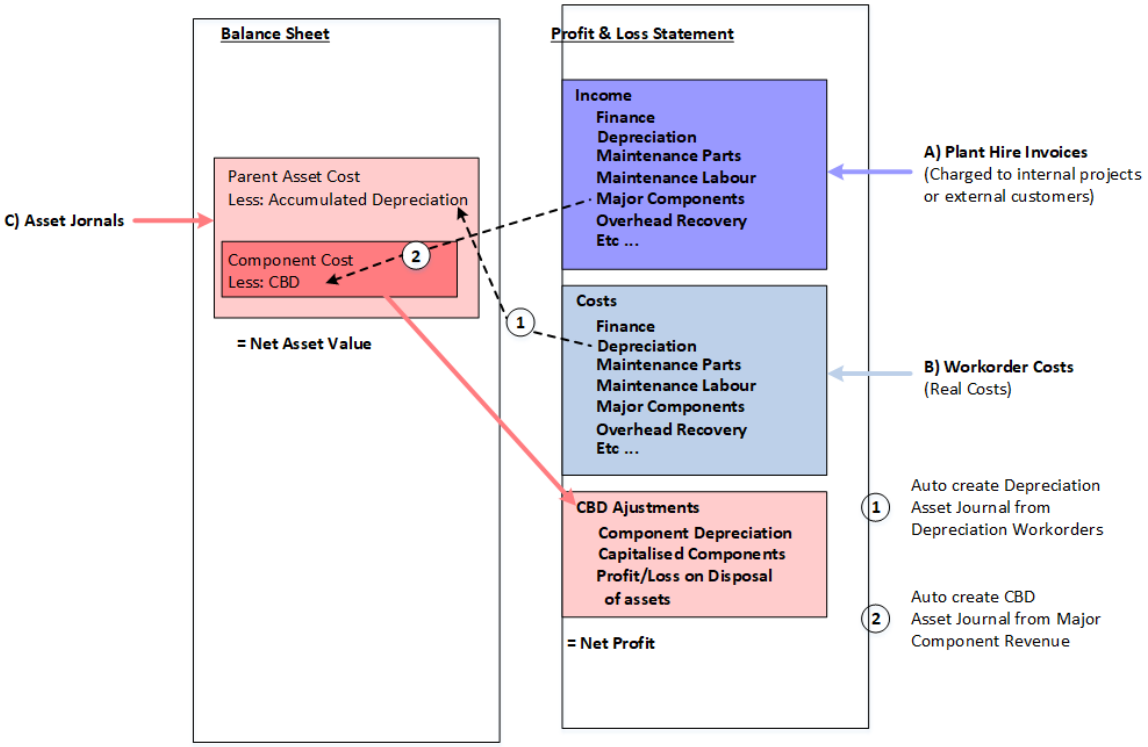
2. Capitalised Components

Specific components within an asset can be capitalised and depreciated.

There are two additional asset balance sheet accounts:

- Component Capitalised Cost (also referred to as the child asset, for example, Engine).
- Component Based Depreciation (CBD).

All transactions affecting the asset ledger are recorded in asset journals.



Internal Asset Transfers are the journal entries for the asset ledger that also affect the profit & loss (P&L). The entries are for items such as depreciation of components, disposal of components and capitalisation of component changeouts. These journals should NOT be captured on work orders, but should be entered in the asset journals.

Define asset structure

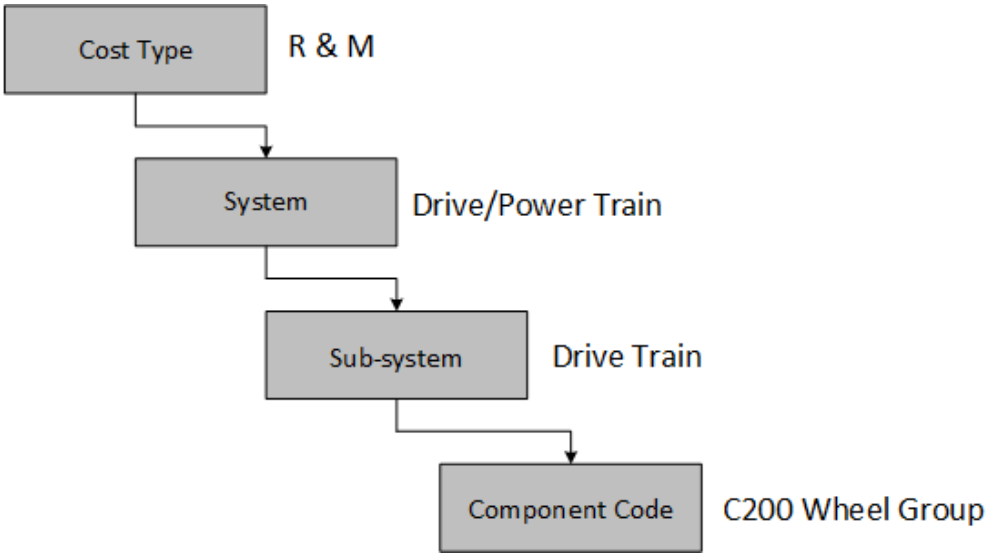
Defining asset structure in a projection is an iterative exercise. You need to define the asset in term of maintainable components or cost buckets.

The level of detail is determined by base parameters, such as criticality, and cost. There are diminishing returns from too much detail. For example, a more detailed asset structure is required for a prime loader vs support vehicles.

Items to consider when defining asset structure include:

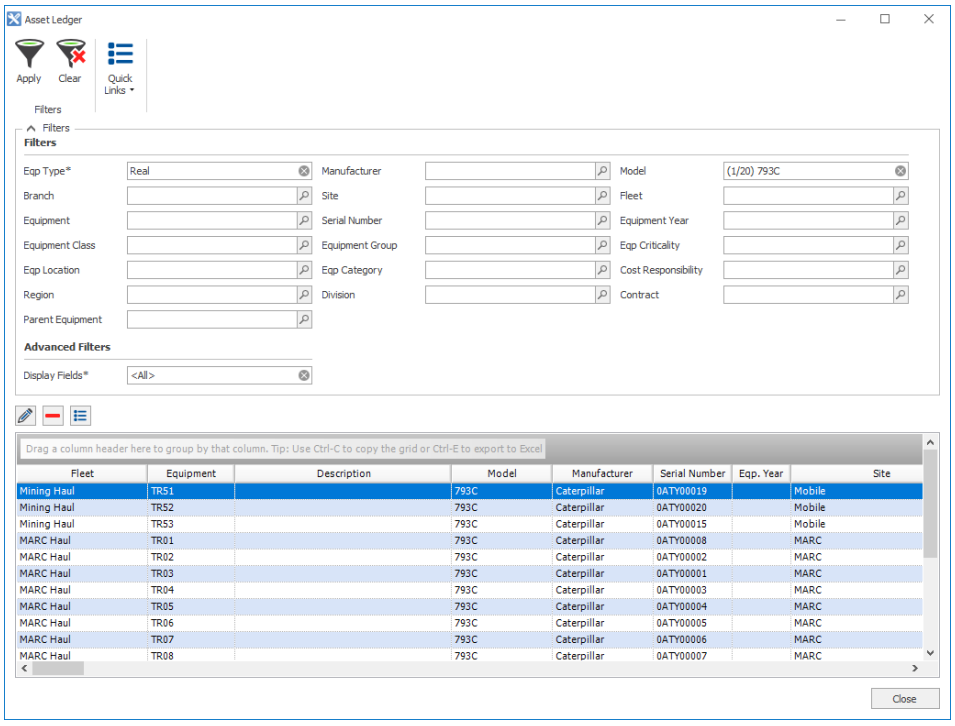
1. **High cost areas**. Generally the Pareto principle applies and a few major components comprise most of the costs.
2. **High downtime areas**. As above in respect to downtime.
3. **System areas**. These are key areas of the asset such as electrical systems.
4. **PM Services**. Preventive maintenance actions.
5. **Unscheduled maintenance**. This is a provision for cost and downtime to cover miscellaneous and unscheduled works.
6. **Safety**. It is important to validate that no safety or environmental areas have been missed.

In order to allow greater flexibility in reporting and analysis AMT has a four-tiered component code hierarchy. These are set up in the system tables and are automatically applied when you select a component code.



Viewing the Asset Ledger

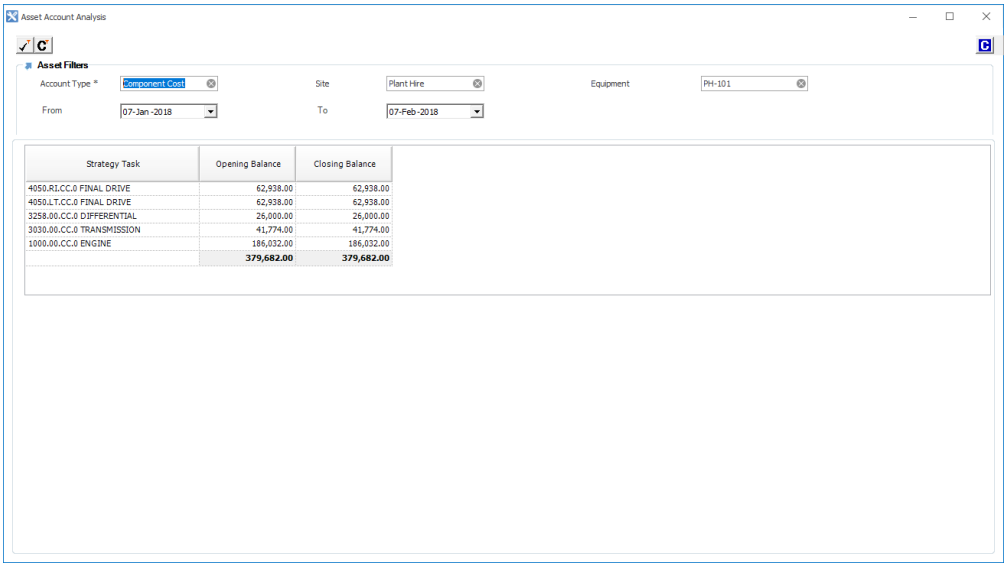
You can access the asset ledger by navigating to **Setup & Admin** > **Asset Ledger**.



Features of the screen include:

- You can export to Excel.
- Each row in the table displays the breakdown of the parent and child assets and the total asset details.
- Click the **Quick Links** button above the grid to access the asset account analysis, displaying the full

transaction history for any date range for the four asset account types.



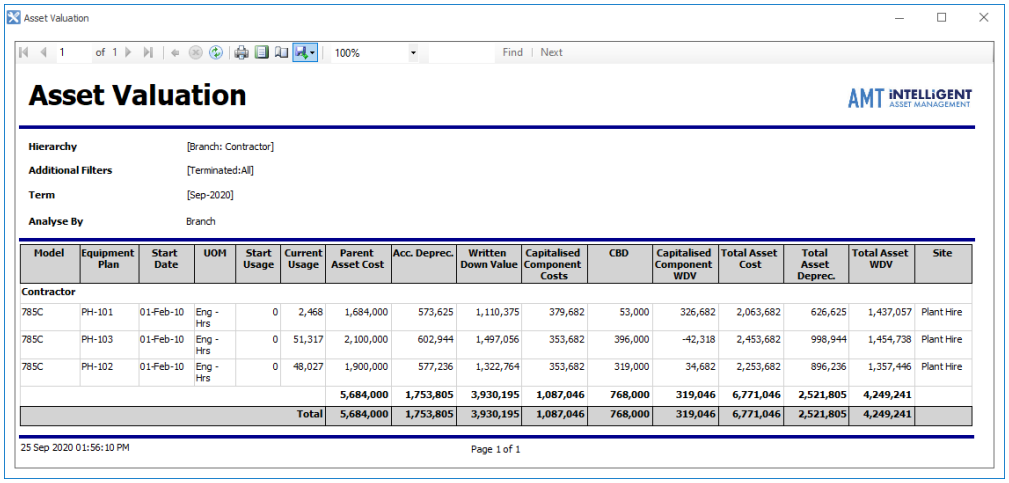
Asset Ledger Reports

There are two Asset Ledger Reports; Asset Valuation and CBD analysis.

Asset Valuation

Plant Hire > **Asset Valuation**.

Displays similar information to the Asset Register form. This report shows the real book value of an asset that should align to market value by adjusting for the remaining life on components.

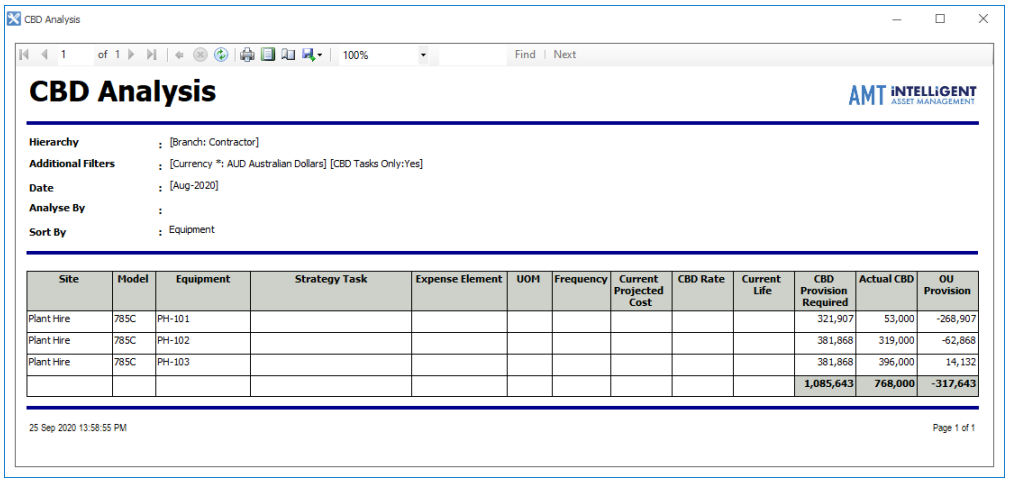


CBD Analysis

Setup & Admin > **Asset Ledger** > **CBD Analysis**.

This report compares a generic required component based depreciation provision with the actual component based depreciation provision by strategy task.

The generic required component based depreciation looks at the current Strategy Task value and the current life on the component to calculate the required provision. It is an estimate, as the strategy task value might not equal exactly the previous capitalised component value. However, the report is designed to provide a reasonability check, not an exact figure.


CBD Analysis

CBD Analysis

AMT INTELLIGENT ASSET MANAGEMENT

Hierarchy : [Branch: Contractor]
Additional Filters : [Currency *: AUD Australian Dollars] [CBD Tasks Only:Yes]
Date : [Aug-2020]
Analyse By :
Sort By : Equipment

Site	Model	Equipment	Strategy Task	Expense Element	UOM	Frequency	Current Projected Cost	CBD Rate	Current Life	CBD Provision Required	Actual CBD	OU Provision
Plant Hire	785C	PH-101								321,907	53,000	-268,907
Plant Hire	785C	PH-102								381,868	319,000	-62,868
Plant Hire	785C	PH-103								381,868	396,000	14,132
										1,085,643	768,000	-317,643

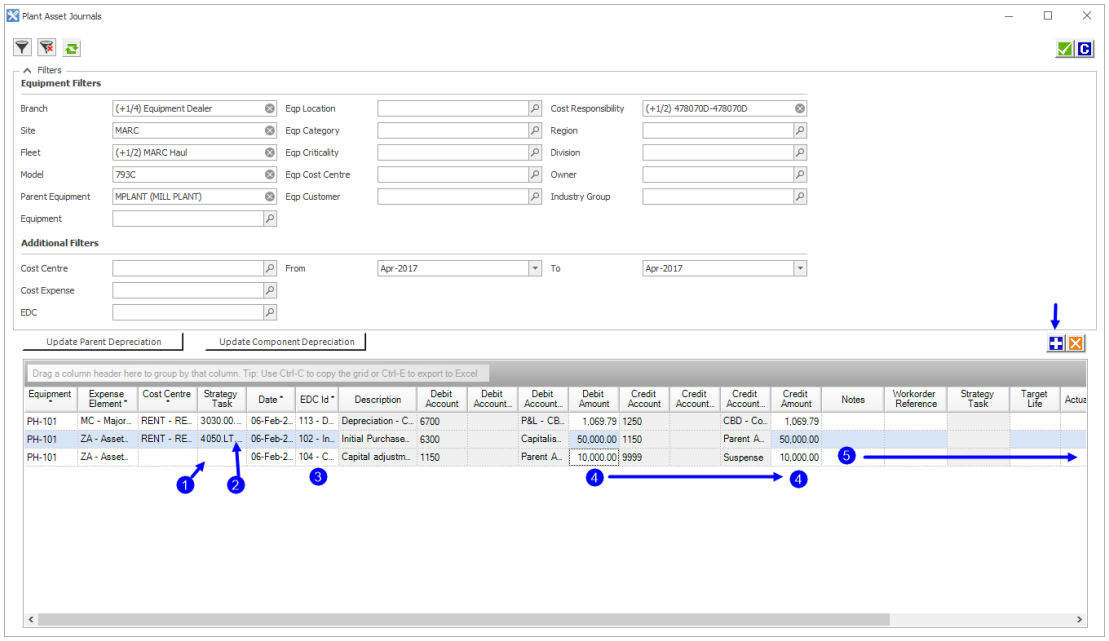
25 Sep 2020 13:58:55 PM Page 1 of 1

Plant Asset Journals

Plant asset journals are recorded on the **Plant Asset Journal** screen, which you can access by navigating to **Setup & Admin** > **Asset Ledger** > **Plant Asset Journals**.

There are three ways for journals to be created:

- Manually – You can manually create a journal by clicking  .
- Imported, for example, from Excel.
- Automatically created – See Automated Journals for more information.



Screenshot legend

Refer to the screenshot above.

1. If the journal relates to the **Parent Asset**, the **Strategy Task** should be blank.
2. If the journal relates to a **Component** it is important to select the **Strategy Task** that is the component identifier. If the **Strategy Task** is left blank, the journal will have no impact.
3. The **EDC ID** determines the type of journal and will automatically define the accounts to be debited and credited. Therefore, instead of trying to determine what account to debit or credit, you identify the transaction type and AMT populates the correct codes.
4. As a general rule, the **Debit Amount** and **Credit Amount** entered should be the same.
5. All the fields to the right of **Credit Amount**, that is, from the **Notes** field, are purely for information purposes and have no impact on any reports.

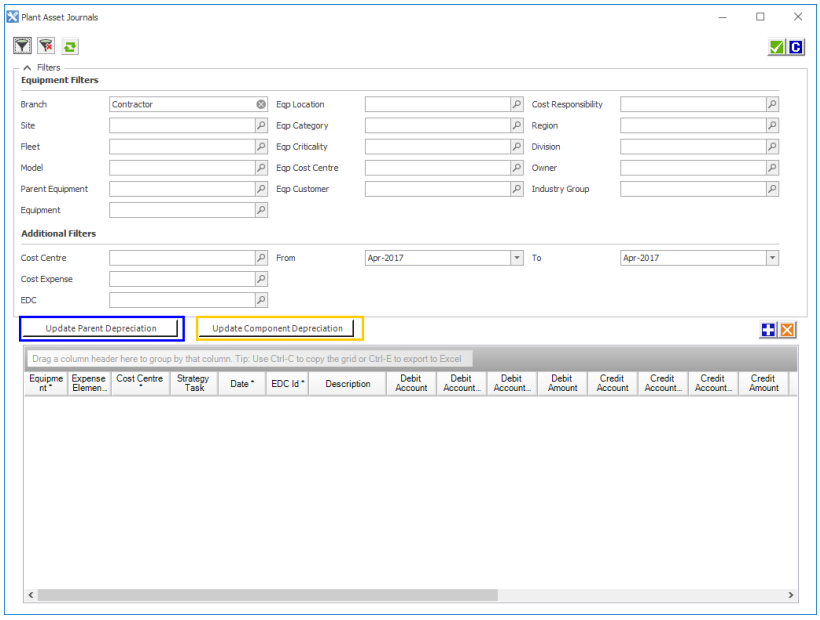
Automated journals

To streamline the journal process, you can configure AMT to automatically create certain journals.

Parent Asset Depreciation

You can automatically set AMT to create a journal (EDC 105) for depreciation cost on work orders.

Alternatively, you can manually initiate this routine by, on the **Plant Assets Journal** form, click **Update Parent Depreciation** (highlighted in the screenshot below). All depreciation on work orders is updated.



Component Depreciation

You can automatically set AMT to create a journal (EDC 113) for component depreciation from the Component Revenue (Plant Hire charge) screen, that is, the transfer of revenue directly to the component based depreciation account.

This routine looks at the revenue for component based depreciation flagged expense elements that has been apportioned to Strategy Tasks and creates corresponding journals for them.

Alternatively, you can initiate this routine by clicking on **Update Component Depreciation** on the Plant Asset Journals screen (right button highlighted in the screenshot above).

In the Invoice Details form you can edit:

- Start and end usage.
- The revenue per expense element.

If you make manual adjustments, the reasons for the adjustment should be noted in the **Comments** section.

CBD Analysis

(Rpt_139) CBD Analysis

Displays a breakdown of each strategy task and the base details of the task. AMT calculates the actual CBD provision for each strategy task. It also calculates the actual CBD hire rate credits from the plant hire invoices and indicates the over or under price.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|LastMonth|,|ProjectionTypes|,
|UserID|,|AutoLaunch|,|EqpPlanId|,|CurrencyId|,|Currency|,|EqpPlan|,
|ModelId|,|Model|,|SiteId|,|Site|,|AnalyseBy|

Report parameters

Default.

Additional filters

Comp Code, Expense Element, Task Type, Currency, Strategy Task, CBD Tasks Only

Component Based Depreciation

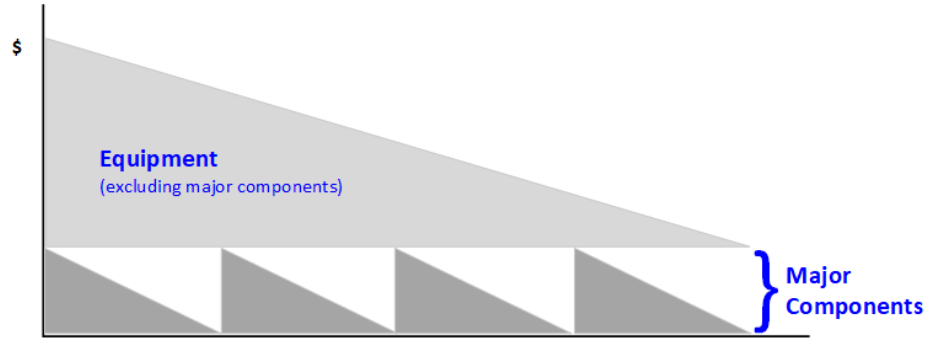
Component based depreciation (CBD) provides a greater level of accuracy for depreciating an asset and reflecting its true asset value. This is achieved by extracting the major components from an equipment and capitalising and depreciating them separately.

For example, a haul truck with a purchase price of $3,000,000 could be split into the following components:

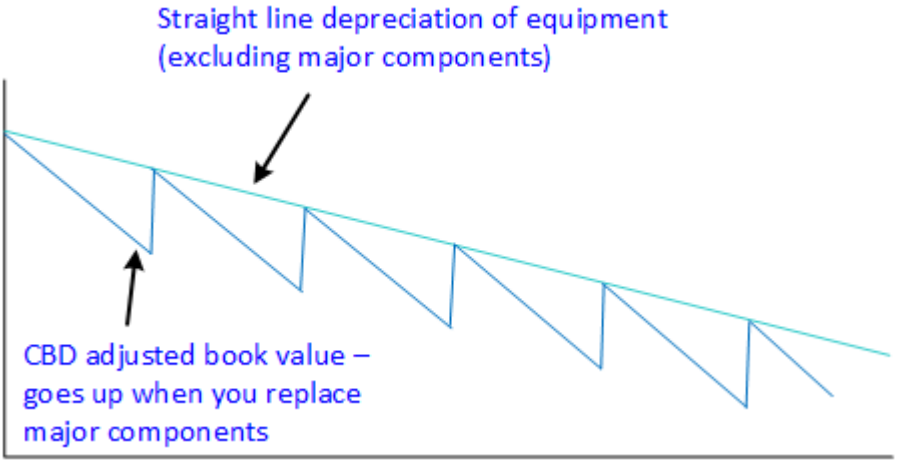
Item	Price
Base Haul Truck	2,580,000
Engine	200,000
Transmission	60,000
Final Drive - Left	80,000
Final Drive - Right	80,000
Total	3,000,000

This approach recognises that the value of equipment is determined by its age and condition of its components. The major components are depreciated over their specific lives which creates accumulated depreciation (known as the CBD account).

Equipment depreciation is split into two elements: equipment and major components.



Impact on asset value.



This approach

- smooths operating costs, which helps to avoid incurring large costs. Depreciation is a more consistent expense than having to expense the costs of component rebuilds.
- prevents the business from taking profit early from money that should be set aside for future maintenance.
- aligns book value to market value.

CBD Summary

This function displays the current CBD status of an equipment.

- **CBD Provision Required** - sum of the expected CBD balance of all strategy task costs linked to the CBD expense element, recognising the current life of each component,

- **Actual CBD** - the amount of actual plant hire revenue received for the expense element flagged in the Expense Element system table as CBD.
- **OU Provision** - (O) Over or (U) Under Provision is calculated as Actual CBD less CBD Provision required. Under Provision is a negative amount.

To drill down to the source data, highlight a row and select the CBD Analysis report from the **Reports** menu.

Asset types

There are three asset types in AMT:

1. **Centrelines**: Centrelines are generic assets that are used to store standard projection templates for a model. It is typically the starting point for registering real assets or performing modelling or feasibility studies.
2. **Modelling**: These are dummy assets that have been set up to prepare tenders (OEMs), perform feasibility studies or benchmark different assets. A number of assets can be evaluated in different modelling scenarios.
3. **Real**: This is a real asset that is operating. Only real assets are enabled for planning and generating transactional data.

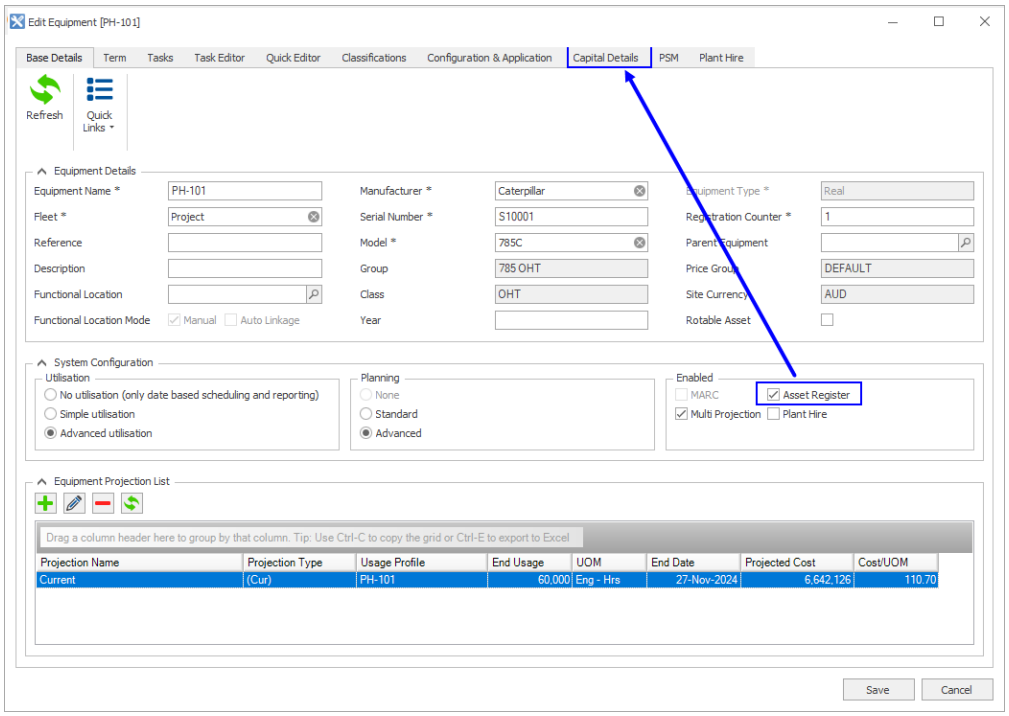
Configuration items

Ensure the following items have been configured.

Equipment configuration

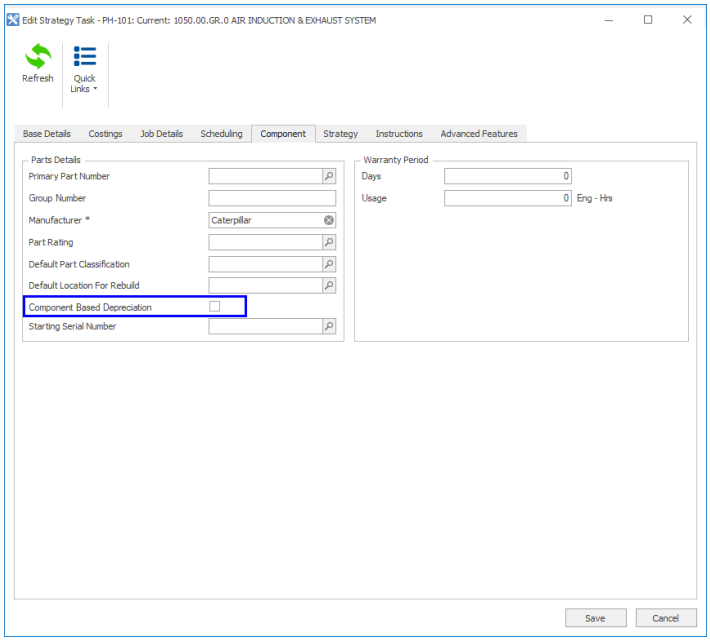
Navigate to the **Equipment Editor**, select the equipment to edit, and click **OK** .

Select the **Asset Register** check box and click **Refresh**. The Asset Ledger functionality is enabled and the Capital Details tab displays at the top of the screen.



Component Based Depreciation (CBD) Flag

Navigate to the **Strategy Task Editor > Component** tab, and select the **Component Based Depreciation** check box.

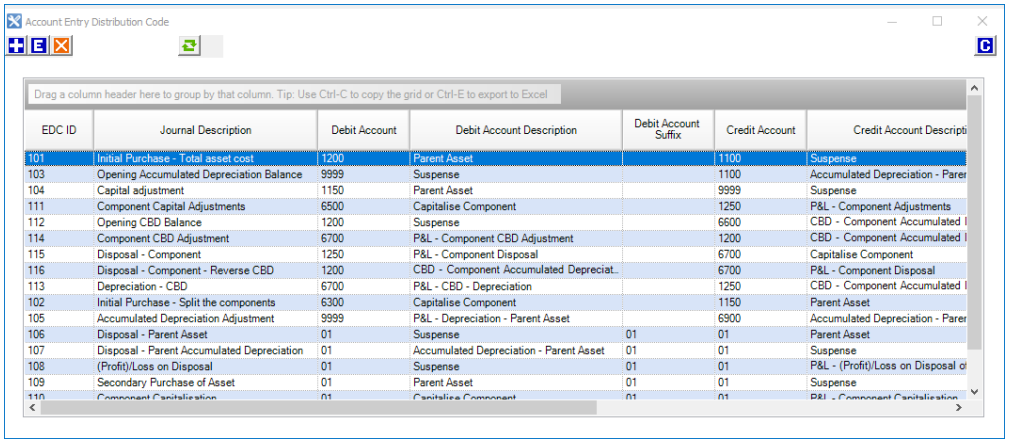


Entry Distribution Codes (EDC)

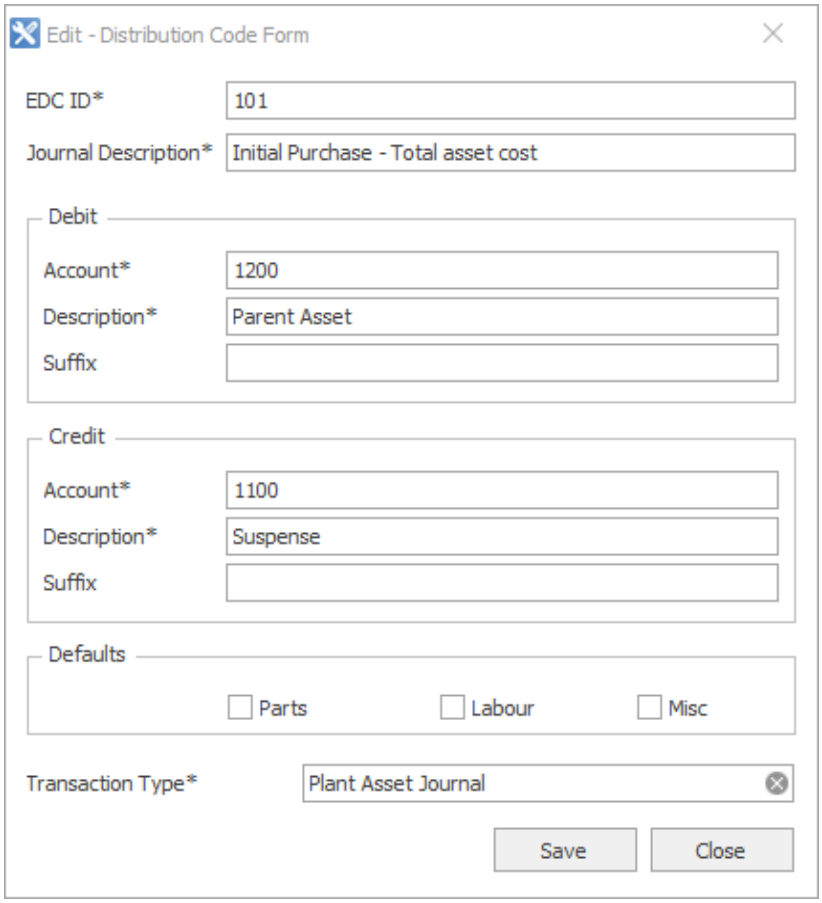
The Entry Distribution Codes are very important to both the Asset Ledger and Plant Hire functionality as they define the calculation rules. The EDC is a System Table that is set up during implementation.

Entry distribution codes are used to identify the balanced transactional journal that is created by AMT when an asset process is executed. The account codes associated with the debit and credit detail will align to the ERP financial ledger codes (representing balance sheet and P&L accounts) and support any interfaces developed to post transactions to the ERP general ledger.

To view and edit the EDC table, navigate to **System Administration** > **System Tables** > **Financial** > **Entry Distribution Code**. Click  to Edit the table.



- EDCs 101-117 are hard coded in AMT and should not be deleted or have their meaning changed.
- EDCs 200-227 are Depreciation specific. Refer to Configuration of Entry Distribution Codes for more information.
- The EDCs identify all the types of transactions that you need to create.
- You can change the **Journal Description** as long as the intention or use of the EDC is not changed. For example, you can change *Depreciation -CBD* to *Component Depreciation*.
- The **Debit Account**, **Debit Account Description**, **Debit Account Suffix**, **Credit Account**, **Credit Account Description** and **Credit Account Suffix** columns and fields are for information purposes. They do not affect the calculations in AMT. They should be set to correlate to the terminology and accounts for your company, but it is not important if they are not.



Capital Asset Recognition

Recognising assets once AMT is configured requires the recognition of the entire asset value at the parent level, and the recognition of the individual child assets.

A parent capital asset will have one or more child capital assets; an asset must have **at least one** child asset for depreciation purposes.

Asset capability

From an engineering perspective it is important to understand the capability of the asset.

1. Is the asset capable of meeting the required level of performance?
2. What are the application and operating factors that can influence the performance of the asset.

For example, it may be important to consider the impact of poor haul roads, overloading, or very steep haul roads on the performance of a haul truck, and specifically which components are impacted by which factors.

AMT Asset Journals

Entry distribution codes are used to identify the balanced transactional journal that is created by AMT when an asset process is executed. The account codes associated with the debit and credit detail will align to the ERP financial ledger codes (representing balance sheet and P&L accounts) and support any interfaces developed to post transactions to the ERP general ledger.

The table below identifies the asset journals and transaction IDs created by the AMT asset processes in depreciable asset recognition.

Process	Transaction ID	Entry Distribution Code	Debit Account	Credit Account
Capitalisation (Parent)	20	200	Capital	Capital Clearing
		222	Capital Clearing	Accumulated Depreciation
		223[2]	Residual Value	Residual Value
Capitalisation (Child/Component)	21	201	Capital	Capital Clearing
	22	202	Capital Clearing	Accumulated Depreciation
	24	204[2]	Residual Value	Residual Value
Monthly Depreciation	23	203	Depreciation (P&L)	Accumulated Depreciation
Additional Capital	25	205	Capital	Capital Clearing
	34	214	Capital Clearing	Accumulated Depreciation
	24	204[2]	Residual Value	Residual Value
Split In (Transfer In)[1]	44	224	Capital - Split Value (IN)	Capital - Split Value (IN)

Process	Transaction ID	Entry Distribution Code	Debit Account	Credit Account
	45	225	Accumulated Depreciation - Split Value (IN)	Accumulated Depreciation - Split Value (IN)
	24	204[2]	Residual Value (IN)	Residual Value (IN)
Split Out (Transfer Out)[1]	46	226	Capital - Split Value (OUT)	Capital - Split Value (OUT)
	47	227	Accumulated Depreciation - Split Value (OUT)	Accumulated Depreciation - Split Value (OUT)
	24	204[2]	Residual Value (OUT)	Residual Value (OUT)
Transfer In (Foreign Exchange)	40	220	Captial	Intercompany
	41	221	Intercompany	Accumulated Depreciation
	24	204[2]	Residual Value (IN)	Residual Vale (IN)
Transfer Out (Foreign Exchange)	28	208	Intercompany	Capital
	29	209	Accumulated Depreciation	Intercompany
	24	204[2]	Residual Value (OUT)	Residual Value (OUT)
Disposal (Gain)	35	215	Disposal Clearing	Capital
	36	216	Accumulated Depreciation	Disposal Clearing
	27	207	Cash	Disposal Clearing
	26	206	Disposal Clearing	Profit on Disposal (P&L)

Process	Transaction ID	Entry Distribution Code	Debit Account	Credit Account
Disposal (Loss)	35	215	Disposal Clearing	Capital
	36	216	Accumulated Depreciation	Disposal Clearing
	27	207	Cash	Disposal Clearing
	32	212	Loss on Disposal (P&L)	Disposal Clearing
Impairment	30	210	Impairment Loss (P&L)	Accumulated Impairment Losses
Revaluation	33	213	Capital	Revaluation Surplus

[1] Split in and split out are used for transactions where transfers occur with no change in currency.

[2] Residual value is being recognised by a journal transaction to support calculation of depreciable asset value.

Except where stated, AMT calculates the journals and the journal values in the background based on the information entered on the screen.

Overheads

Overhead Navigator

Some maintenance contracts include overheads costs as part of the direct equipment maintenance costs and billings calculations. This is captured in the estimate-current projection's costs and billing schedule.

Other types of maintenance contracts classify contract overhead costs and revenue separately. The overhead costs are factored into the direct equipment maintenance costs, but the revenue is captured separately or vice-versa. The **Overheads** module of AMT handles this case; this module is a separate area where overhead costs and revenue can be estimated and managed for contracts



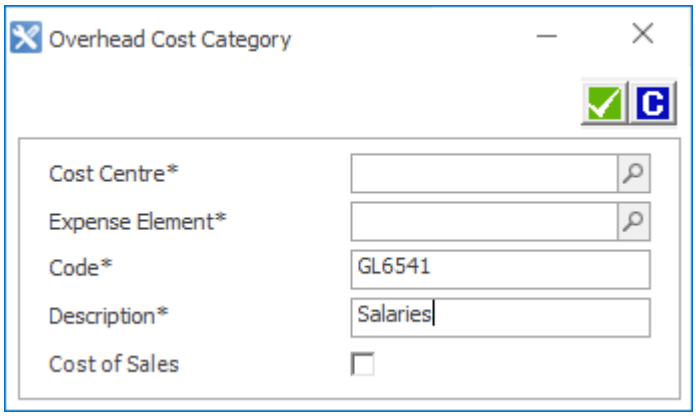
The Overheads module can also be used for non-contract overheads.

Before setting up overheads, it is important to validate:

- How are overheads costed in the contract; are they part of the direct maintenance costs or separate?
- How will overhead costs will be recovered - as a separate fixed monthly fee or calculated into the utilisation billing rate?
- How are these costs and revenue being captured in the ERP system?

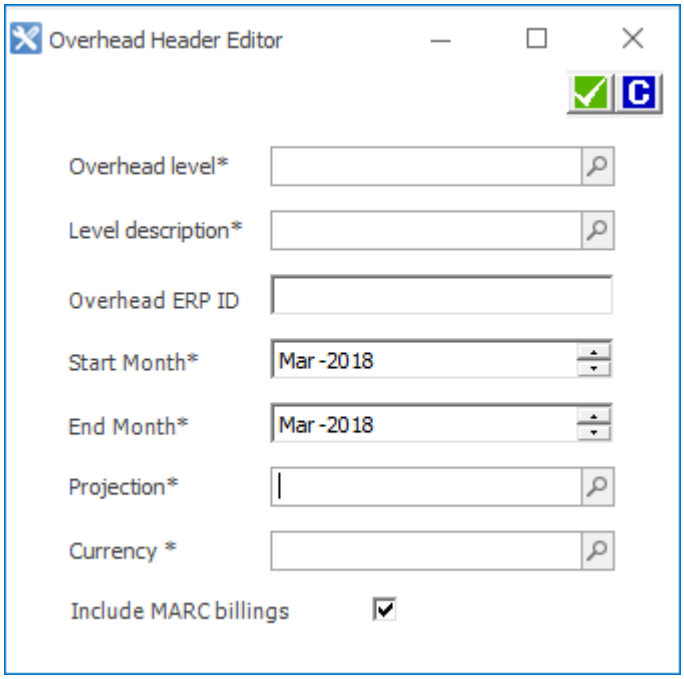
- How are these costs and revenue captured if AMT manages the equipment or contract when they become live?

Set up overheads by navigating to **System Administration** > **System Tables** > **Financial** > **Overhead Cost Category**, then clicking . The Overhead Cost Category dialog box displays.



The **Code** field is the overhead cost category definition. Select the **Cost of Sales** check box means to report overhead costs as cost of sales rather than expenses in some profit & loss reports.

To set up overhead costs and billings for a contract tender, use the **Overhead Header Editor** function. Navigate to **Setup & Admin** > **Overheads** > **Overhead Navigator**, then click .



Field	Description
Overhead level	The overhead level at which the overheads are costed and the overhead billing (management fee) is charged. For example, does the management fee cover all

Field	Description
	equipment under contract? Does it only apply to specific fleets or a specific branch or site?
Level description	Describes the overhead level chosen.
Overhead ERP ID	Not relevant for modelling overheads (only real overheads).
Start and End Month	Set to the period of time during which the overheads will be incurred.
Projection	Select the projection type.
Currency	Select the currency of the overheads.
Include MARC Billings	If a non-utilisation fee is charged to the customer to recover overhead costs, select this check box. If the check box is not selected, you can set up a schedule of overhead costs, but not management fees.

Populate the overhead's schedule with costs and billings on the **Overhead Header Editor**. You can either enter values directly on the grid, or copy and paste them from Excel.

Components

Serialised Component Manager

To perform Component Tracking in AMT, component serial numbers need to be registered in Serialised Component Manager. You can manually create component serial number details, upload them via a .csv file or they can be interfaced into AMT.

The only mandatory field is **Component Serial Number**. The other fields are for information only.

Machine Assembly

Create new Machine Assembly Workflow

Complete documentation for this topic will be available in a future release.

View/Edit Machine Assembly Workflow

Complete documentation for this topic will be available in a future release.

Depreciable Asset Recognition

Database configuration requirements

There are configuration requirements for AMT Depreciable Asset capability.

A setting in the AMT database determines whether certain menu options and their underlying processes are visible to you.



If you can't see **Depreciable Asset Recognition** under **Setup & Admin**, raise a ticket with RPMGlobal Customer Support to ask for the feature to be enabled.

System administration requirements

A number of system tables need to be configured before you use AMT Depreciable Assets.

On a monthly basis, a system administration user needs to manage the current period.

There can only be one user-defined financial period with a status of current.

Journals post to the current period. Journals cannot post to closed or future periods.

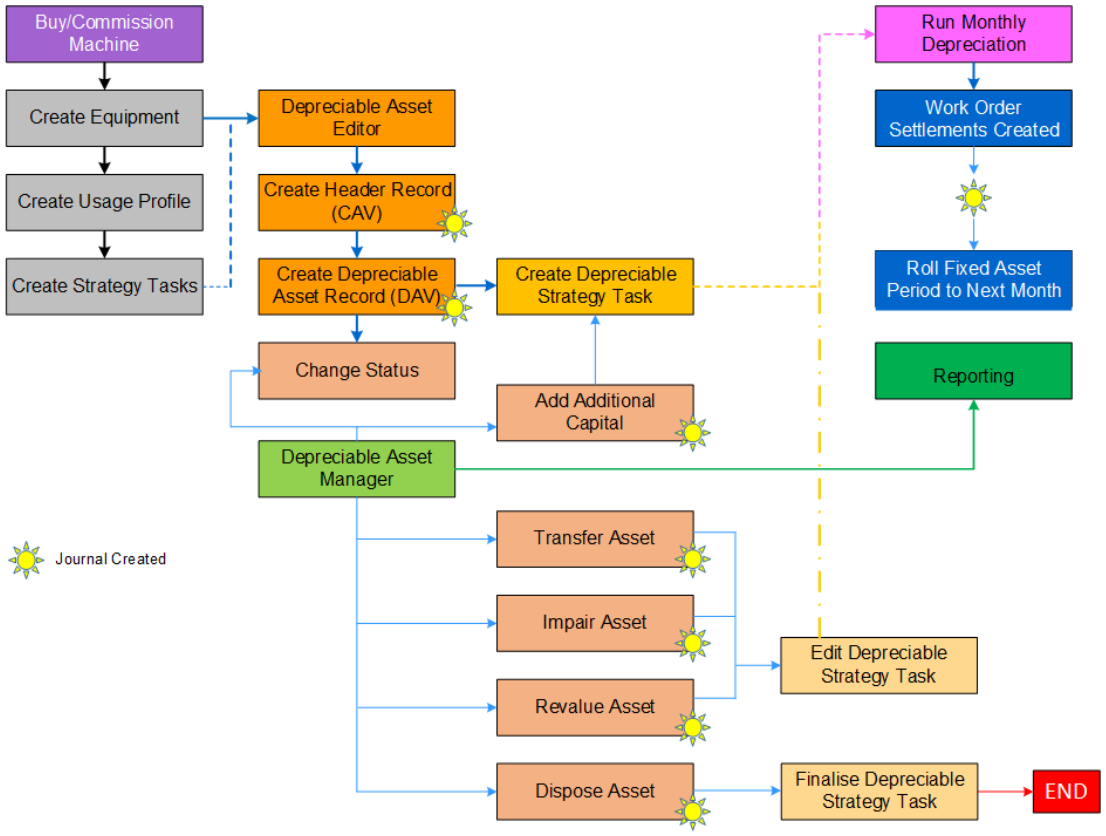
Generally, user-defined financial periods have a status of either closed, current or future. A period only has a status of open if a previously closed period is re-opened to post adjusting entries for depreciation purposes. Close an open period immediately after any adjusting entries are processed.

Depreciation journals can be corrected in a previous period only if the previous period's status is changed from closed to open. Consult with your financial team before undertaking this process, to ensure you do not violate rules for profit and loss recognition.

The ability to change a status to open is subject to a user's AMT security.

The ability to modify and adjust user defined financial periods must be controlled and contained to personnel who understand the impact of any changes they make to the period status.

AMT Depreciable Asset Process



Straight line depreciation

$$\text{Depreciation per annum} = \frac{(\text{Cost} - \text{Residual Value})}{\text{Useful Life}}$$

$$\text{Depreciation per annum} = (\text{Cost} - \text{Residual Value}) \times \text{Rate of Depreciation}$$

Cost	Initial acquisition or construction costs related to the asset, as well as any subsequent capital expenditure.
Residual Value	Residual value is the estimated proceeds from the disposal of an asset at the end of its useful life. Residual value is also known as its scrap, or salvage value, The portion of an asset's cost equal to the residual value is not depreciated because it is expected to be recovered at the end of an asset's useful life.
Useful Life	The estimated time period that the asset is expected to be used, starting from the date it is available for use up until the date of its disposal or termination of use. Useful life is normally expressed in years or months.

Rate of Depreciation	The percentage of useful life that is consumed in a single accounting period. The rate of depreciation is calculated as $\text{Rate of Depreciation} = \frac{1}{\text{Useful Life}} \times 100\%$ The depreciation expense needs to be adjusted downwards when an asset is acquired or disposed of during the accounting period to avoid distributing depreciation for the time when the asset was not available for use.

Advantages

- Easy to calculate.
- Useful when the pattern of economic benefits are hard to determine with precision.
- Suitable for depreciating assets that provide similar levels of economic benefits throughout their useful life (for example, buildings).

Disadvantages

- May not reflect the true pattern of an asset's economic benefits.

Units of activity depreciation

$$\text{Depreciation per annum} = \frac{(\text{Cost} - \text{Residual Value})}{\text{Useful Life}}$$

$$\text{Stage of Completion \%} = \frac{\text{Value of Work Certified as Complete}}{\text{Total Expected Production or Usage}} \times 100$$

This depreciation method is also known as Units of Production or Units of Usage depreciation, and calculates depreciation on the basis of expected output or usage.

Advantages

- Most accurately reflects the pattern of consumption of economic benefits.
- Suitable in the case of fixed assets that depreciate in proportion of units of activity rather than just the passage of time.

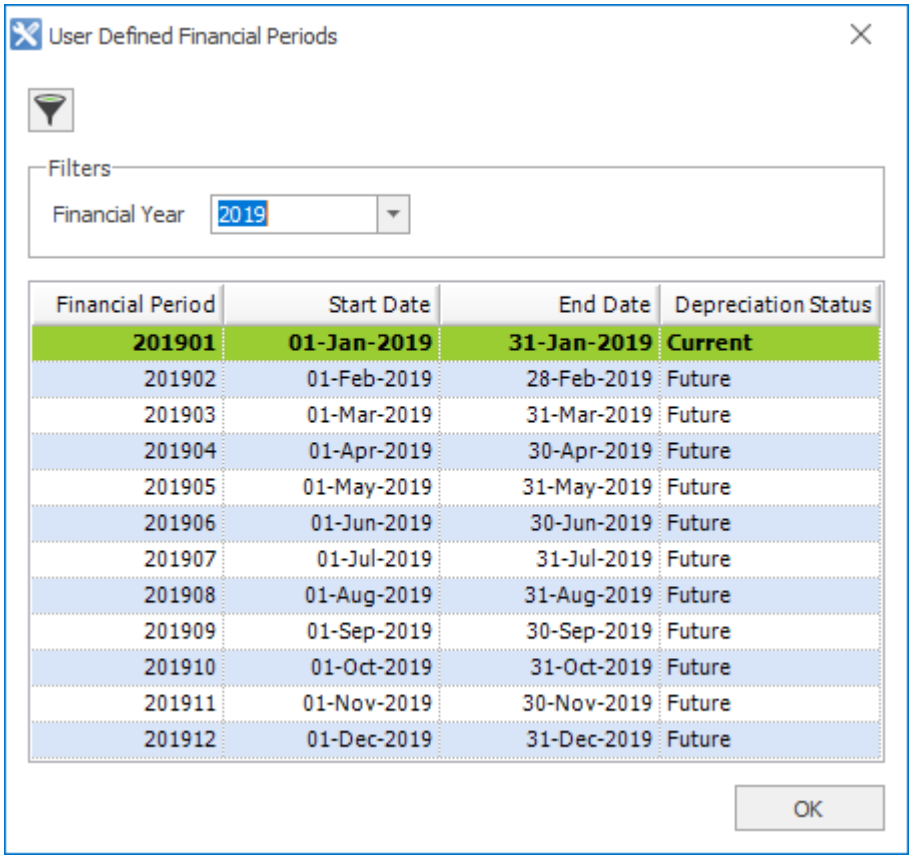
Disadvantages

- Difficult to determine and measure a reasonable basis of activity.

User Defined Financial Periods

Use the User Defined Financial Periods system table in AMT to define the financial periods that do not align with the first and last days of a calendar month. You can define financial periods that conform with any external ERP.

Navigate to **System Administration** > **System Tables** > **Financial**.



Financial Period	Start Date	End Date	Depreciation Status
201901	**01-Jan-2019**	**31-Jan-2019**	**Current**
201902	01-Feb-2019	28-Feb-2019	Future
201903	01-Mar-2019	31-Mar-2019	Future
201904	01-Apr-2019	30-Apr-2019	Future
201905	01-May-2019	31-May-2019	Future
201906	01-Jun-2019	30-Jun-2019	Future
201907	01-Jul-2019	31-Jul-2019	Future
201908	01-Aug-2019	31-Aug-2019	Future
201909	01-Sep-2019	30-Sep-2019	Future
201910	01-Oct-2019	31-Oct-2019	Future
201911	01-Nov-2019	30-Nov-2019	Future
201912	01-Dec-2019	31-Dec-2019	Future

Initial configuration requires the generation of at least one year's financial periods. Financial periods are created that align to AMT's existing financial periods.

The difference between the User Defined Financial Periods system table and the existing Financial Periods system table are:

1. It allows AMT to be configured for monthly depreciation processes (under Depreciable Asset Recognition).
2. Flexible period start and end date definitions can be developed in the future.

User defined financial period depreciation status

Each user defined financial period has an associated status which determines whether the period is:

- Current - the period where the asset-related journals are posted.
- Closed - the previous period (with respect to the current period) that does not accept asset-related journals.
- Future - the future period (with respect to the current period) that does not accept asset-related journals.
- Open[1] - the previous closed period (relative to the current period) that has been re-opened to accept asset-related journals.



This is subject to the user's AMT security.

There can only be one user defined financial period with a status of current.

Journals will post to the current period. Journals cannot be posted to closed or future periods.

Generally, user defined financial periods have a status of either closed, current or future. A period only has a status of open, if a previously closed period is re-opened to post adjusting entries for depreciation purposes. An open period should be closed immediately after the adjusting entries have been processed.

Depreciation journals can be corrected in a previous period only if the previous period's status is changed from closed to open. Consult with your financial team if you are thinking about undertaking this process to ensure rules for profit and loss recognition are not violated.

Journals cannot be deleted after they are created. Transactions need to be reversed.

Configuration of Entry Distribution Codes

Extra entry distribution codes (EDCs) have been added to the depreciable asset recognition processes for the following reasons.

1. They are required to support all the processes supported by depreciable asset recognition.
2. You may not have transitioned from component based depreciation to depreciable asset recognition when upgrading to release 8.17 or later of AMT.

   If you have previously used component based depreciation in releases earlier than 8.17, it is important that you keep the different processes distinct and not have a confusing set of transactions in the database.

3. It provides a distinction between the two functional solutions, which will help if you need to contact customer support.

Entry distribution codes are used to identify the balanced transactional journal that is created by AMT when an asset process is executed. The account codes associated with the debit and credit detail will align to the ERP financial ledger codes (representing balance sheet and P&L accounts) and support any interfaces developed to post transactions to the ERP general ledger.

The entry distribution codes need to be populated with account code details that correlate with the financial ERP general ledger and P&L accounts. This will align journal entries created by AMT with those recognised in the ERP, or will support extracting the AMT journal information to import into the ERP solution.

The table below provides an example of account codes and the description information expected.

AMT Entry Distribution Codes with accounting details

EDC ID	Journal Description	Debit Acct	Debit Journal Description	Credit Acct	Credit Journal Description
200	Capital Asset Value (control account) - parent	10000	Capital - Parent Control	10000	Capital - Parent Control
201	Capital - Initial Purchase - component	10000	Capital - Initial Purchase	99999	Work in Progress Clearing
202	Accumulated	99999	Work In	10001	Accumulated

EDC ID	Journal Description	Debit Acct	Debit Journal Description	Credit Acct	Credit Journal Description
	Depreciation (Initial Purchase)		Progress Clearing		Depreciation
203	Depreciation	64904650	Depreciation	10001	Accumulated Depreciation
204	Salvage Value - component	99999	Work In Progress Clearing	99999	Work In Progress Clearing
205	Additional Capital Cost	10000	Additional Capital Purchase	99999	Work In Progress Clearing
206	Profit on Disposal (P&L account)	99998	Disposal Clearing	6902000	Profit - Asset Disposal
207	Cash Received on Disposal	11100	Cash Received - Asset Disposal	99998	Disposal Clearing
208	Transfer (OUT) - Capital	80001	Intercompany	100000	Capital Cost - Asset Transfer
209	Transfer (OUT) - Accumulated Depreciation	10001	Accumulated Depreciation - Asset Transfer	80001	Intercompany
210	Impairment Loss (P&L account)	6904400	Asset Impairment Loss	10012	Accumulated Impairment Losses
211	Accumulated Impairment Losses (Balance Sheet)	10012	Accumulated Impairment Losses	99997	Impairment Clearing
212	Loss on Disposal (P&L account)	6904201	Disposal Asset Loss	99998	Disposal Clearing
213	Revaluation Reserve (Balance Sheet account)	10000	Capital - Asset Revaluation	70000	Revaluation Surplus
214	Accumulated Depreciation	99999	Work In Progress	10001	Accumulated Depreciation

EDC ID	Journal Description	Debit Acct	Debit Journal Description	Credit Acct	Credit Journal Description
	(Additional Capital)				
215	Capital Disposed	99998	Disposal Clearing	10000	Capital
216	Accumulated Depreciation Disposed	10001	Accumulated Depreciation	99998	Disposal Clearing
217	Impairment Reversal (Capital)	10000	Capital	99997	Impairment Clearing
218	Gain In Value of Asset	99997	Impairment Clearing	10011	Gain In Asset Value (Impairment)
219	Revaluation Surplus	99997	Impairment Clearing	70000	Revaluation Surplus
220	Transfer (IN) - Capital	10000	Capital	80000	Intercompany Account
221	Transfer (IN) - Accumulated Depreciation	80000	Intercompany Account	10001	Accumulated Depreciation
222	Accumulated Depreciation (control account) - parent	10001	Accumulated Depreciation	10001	Accumulated Depreciation
223	Salvage Value (control account) - parent	99999	Work In Progress Clearing	99999	Work In Progress Clearing
224	Asset Capital Split (IN)	10000	Capital	99999	Work In Progress
225	Accumulated Depreciation Split (IN)	99999	Work In Progress	10001	Accumulated Depreciation
226	Asset Capital Split (OUT)	99999	Work In Progress	10000	Capital

EDC ID	Journal Description	Debit Acct	Debit Journal Description	Credit Acct	Credit Journal Description
227	Accumulated Depreciation Split (OUT)	10001	Accumulated Depreciation	99999	Work In Progress

Capital Asset Depreciation

Depreciation is when you distribute the cost of a fixed asset over its useful life. Fixed assets are tangible (physical) assets held by an entity for the production or supply of goods and services, for rental to others, or for administrative purposes.

AMT supports the following types of depreciation:

- Straight Line

  The same depreciation is distributed over the entire useful life of the asset.

- Units of Activity

  The depreciation amount varies for each period in proportion to the change in the level of activity. Units of Activity depreciation calculates depreciation on the basis of expected output or usage.

Recognising Depreciable Assets

All depreciation occurs at the child level, which is a fundamental difference from AMT's component-based depreciation.

After child assets (depreciable assets) have been recognised and the associated journals created, a depreciation strategy task must be created for each depreciable asset. The depreciation strategy task supports the calculation of monthly depreciation.

There is no particular right or wrong way to recognise depreciable assets, however the following rules need to be observed during the capital expenditure (capex) recognition process.

Creating depreciable assets

Rule 1

An asset's capital value must be greater than or equal to zero.

Rule 2

Any accumulated depreciation initially recognised, or residual value, must be greater than or equal to zero.

Rule 3

If an asset's capital value is zero, it cannot have accumulated depreciation or a residual value greater than zero.



Do not recognise an asset in AMT if its initial capital cost is zero.

Rule 4

Accumulated depreciation initially recognised cannot be greater than the asset's capital value less the residual value.

Rule 5

The residual value of an asset cannot be greater than the capital value less any accumulated depreciation initially recognised.

Rule 6

The parent (or header) capital asset details (capital, accumulated depreciation and residual value) must balance to the sum of the individual component (child) depreciable asset records created.



This rule applies when creating a new asset and before any depreciation in AMT has been applied.

Once depreciation has been generated for the asset, this rule is not enforced due to the processes of additional capital, transfers, disposals and so on.

Depreciable Asset Editor

You can create the capital asset header and depreciable asset child (components) records using the Depreciable Asset Editor.

You can create capital asset header records and leave the process without creating any depreciable asset (child) records. If you leave the process at this step, no depreciation is calculated. Calculation of depreciation only occurs when a depreciable asset is created with an associated depreciation strategy task.

If both capital asset headers and depreciable asset records are created at the same time, AMT expects the depreciable asset records to balance with the values of the header record on clicking **Save**. This means the record needs to balance on capital value, any accumulated depreciation, and residual value. If you click **Cancel** during the create depreciable asset process, no depreciable asset records are saved and no journals created; the capital asset header will remain.

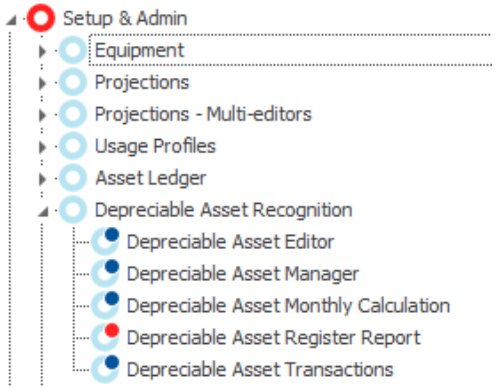
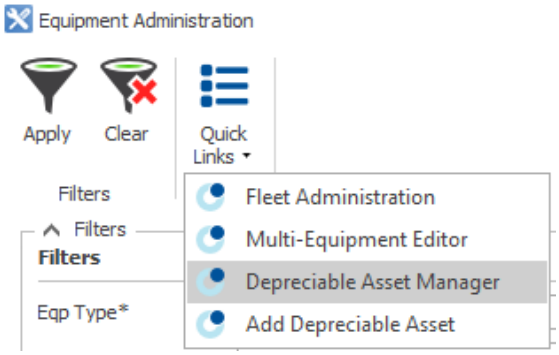
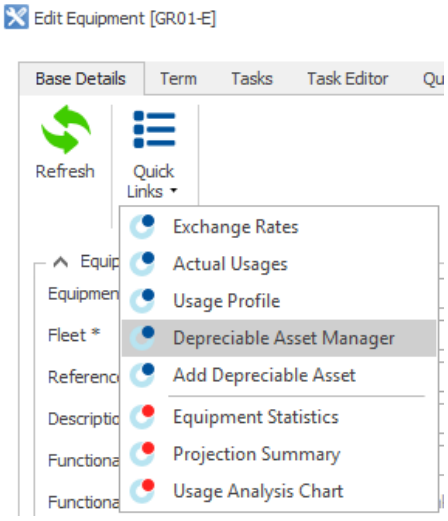
When creating assets, all values must be zero or greater; values cannot be negative.

If you launch the Depreciable Asset Editor from the menu without having an equipment selected, you need to select an equipment item from the **Equipment** list.



If only real equipment type is entered, the list contains all the equipment available in the database. Entering additional criteria filters the list to a manageable size.

You can launch the Depreciable Asset Editor in the following ways:

1	Setup & Admin > Depreciable Asset Recognition	
2	Setup & Admin > Equipment Administration - Quick Links menu	
3	Edit Equipment - Quick Links menu	

When editing an asset, some fields are not editable because there is an existing strategy task linked to that depreciable asset.

The following fields display on the **Depreciable Asset Value** dialog box.

Column	Description
Asset Identifier	The name or serial number of the equipment selected on the initial screen.
External Asset ID	The external ERP asset ID or specific identifier for the depreciable asset.
Description	Asset description. If it is a long description, click the ellipsis to see the full description.
Serial Number	The serial number associated with this depreciable asset.
Capital Paid	The capital cost associated with the child or component asset.
Accumulated Depreciation	The total depreciation that has been calculated in the previous accounting periods since capitalising the asset.
Residual Value	The estimated proceeds from the disposal of a child or component asset at the end of its useful life.
Capitalised Date	The date from which the asset was valued. The default is the current date. You cannot enter a date in the future.
Supplier	The supplier from which the child or component was purchased.
Purchase Order	The purchase order associated with the child or component purchase.
Currency	Defaults to the site's currency.
Authorised By	Enter the name of the person who approved the capitalisation. This field is a list of AMT employees. If the approver is not an AMT employee, you can enter their name in the field.
Classification 1	General ledger classification.
Classification 2	General ledger classification.
Entered By	The default is the logged in user.
Status	The default is Active.

Also in the **Asset Details** panel you can do the following tasks:

	Add a depreciable asset.
	Edit a depreciable asset.
	Remove a depreciable asset.
	Add a depreciation strategy task.

Depreciation Strategy Task

The depreciation strategy task supports calculating the depreciation for an asset by either straight line or units of activity.

Each depreciable asset must have a strategy task that is configured to support the calculation of depreciation.

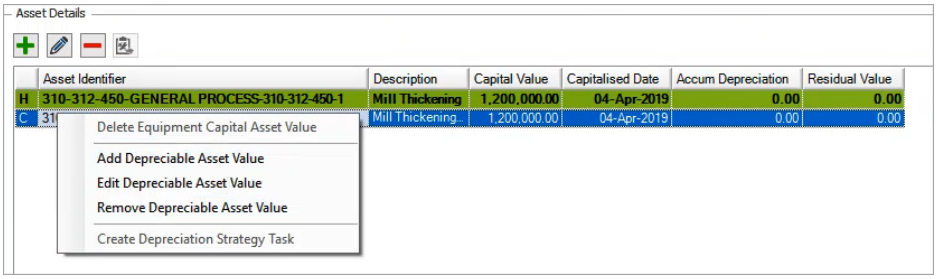
If there is no depreciation strategy task, there is no monthly depreciation.

The strategy task creates both a forecast depreciation schedule for cost forecasting purposes, as well as actual depreciation values for the current depreciation. Once the strategy task information has been created, and depreciation run, it cannot be deleted.

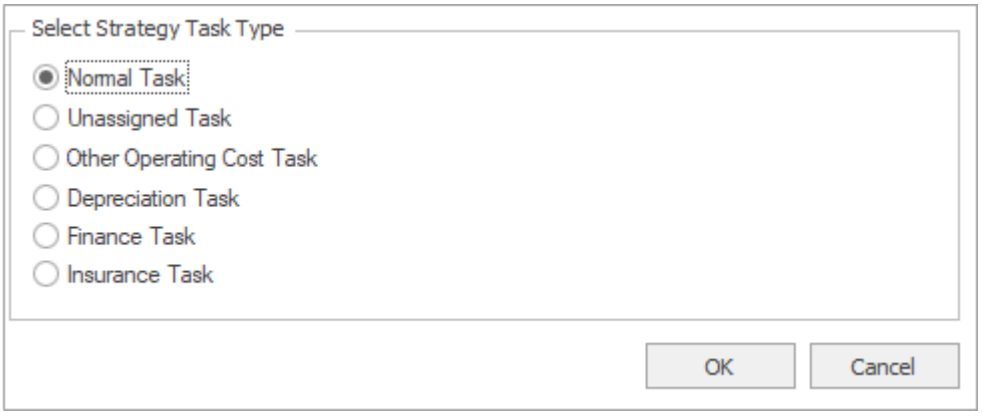
You can add a **Depreciable Asset Strategy Task** in the following ways:

1. Navigate to **Setup & Admin** > **Depreciable Asset Recognition** > **Depreciable Asset Editor**. Select the equipment, and on the **Equipment Depreciable Editor** dialog box, click **Action** .

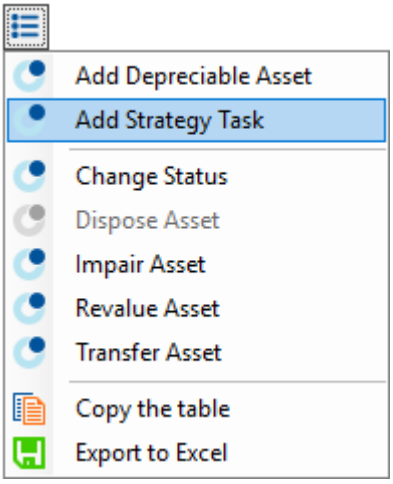
2. In the **Depreciable Asset Editor**, right-click on the depreciable asset.



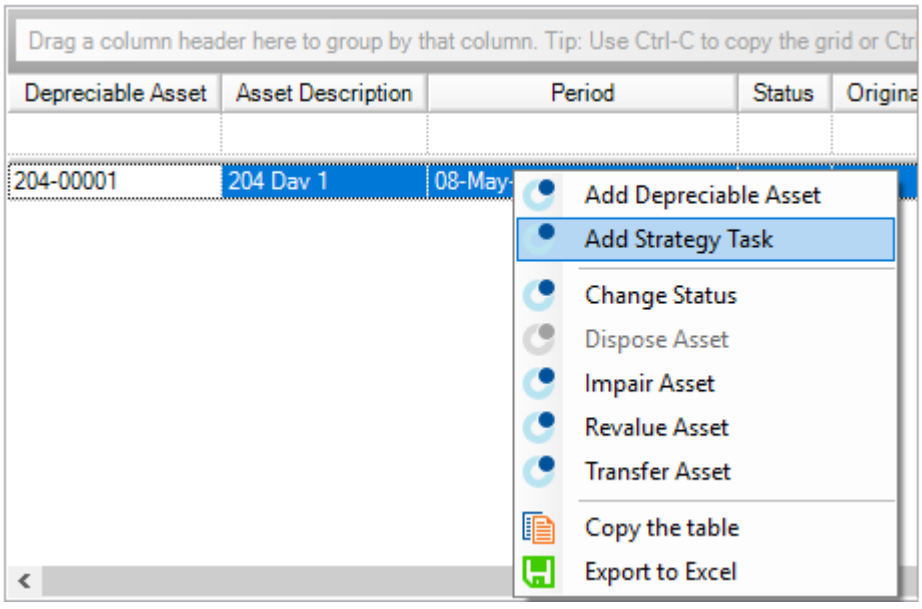
3. In the **Edit Equipment** dialog box, click **Add**.



4. In the **Depreciable Asset Manager**,click the **Quick Links** menu button and click **Add Strategy Task**.



5. In the **Depreciable Asset Manager**, right-click on a depreciable asset and click **Add Strategy Task**.



The **Edit Depreciation Strategy Task** dialog box has the following panels:

- Costs
- Life of Depreciable Asset
- Scheduling Details
- Equipment Details

The following editable fields display on the **Edit Depreciation Strategy Task**.

Field	Description
Costs	
Depreciable Asset	This field is populated with the asset to be depreciated which was selected when the strategy task was created. The format of the depreciable asset information is: Equipment name . Depreciable asset number - Equipment description.
Depreciable Capital Cost	This field is populated from the depreciable asset. It is either the initial capital cost less residual value or the initial capital cost less residual value less accumulated depreciation (at the date the strategy task is created). You need to adjust the depreciable capital cost during the life of an asset when the life has been extended or reduced but with no change to the written down value (WDV). In this case, the depreciable capital cost is set to the opening WDV of the asset in the period during which the life is changed.  During the life of the asset, changing this value and the total usage will result in a new cost per UOM rate for future depreciation calculations.
Depreciation Type	Either U-usage or S-straight line. The type changes depending on the **Life of Depreciable Asset** selection (**Total Usage** or **Straight Line**).
Residual Value	This field is defaults to the residual value entered on the **Depreciable Asset Record** when an asset is initially created. The estimated proceeds from the sale or disposal of the asset at the end of its useful life. You are able to modify the value over the asset's life.  This figure can be manually overwritten if the residual value is adjusted later in an asset's life due to a change in circumstance; for example, the price of scrap metal or other market expectation that can impact the residual value received. A modification to the residual value would usually be offset by a change in the depreciable capital cost.
Cost per Month	The monthly depreciation cost of the asset. Displayed information depends on the information entered in the **Life of Depreciable Asset** section and the **Depreciable Capital Cost**.
Manual Override	Select to enable a manual monthly depreciation amount to be directly entered, which displays next to this check box when selected. This amount overwrites any information in the **Life of Depreciable Assets** and

Field	Description
	Scheduling Details sections for Straight Line or Usage. The amount is a manually applied amount each month until the method is changed or the WDV is zero.
Life of Depreciable Asset	
Select either **Total Usage** or **Straight Line**. Relevant options are then enabled.	
Total Usage	
UOM	Defined unit of measure, based on the UOM of end life.  If you change the UOM, and select **Use Usage Projection**, the Av Monthly Usage Rate is calculated.
Avg Monthly Usage Rate	Enter an average monthly usage rate. This usage rate overwrites the rate calculated from the forecast usage projection. The **Cost per Month** is calculated as the forecast usage per month from the projection multiplied by this rate.
Use Usage Projection	Select this option to use the forecast projection details to calculate the depreciation rate. The rate is the depreciation capital cost divided by the forecast projection usage details and displays in the **Avg Monthly Usage Rate**. Clear the check box to overwrite any calculated figure in the Avg **Monthly Usage Rate**. If the check box is selected, the field is protected. Used to calculate a deprecation forecast for the life of the asset.
Straight Line Select to enable straight line depreciation. Changes the **Depreciation Type** in the **Costs** section to **S – Straight Line**. **Straight Line Method** option. Select the time period for the straight-line depreciation: 1. **Years** 2. **% per Year** 3. **Months** **Duration or %** option **Years** – the number of years over which to calculate depreciation. Years are multiplied by 12 to calculate the months; the depreciable capital cost is then divided by this number to calculate the monthly depreciation amount. **% per Year** – a number that represents the percentage of the depreciable capital cost to be recognised in	

Field	Description
the year. The percentage multiplied by the depreciable capital cost is divided by 12 to calculate the monthly depreciation amount. **Months** – the number of months by which the depreciable capital cost is divided to calculate the monthly depreciation amount.	
Scheduling Details	
Usage Based a. **Entire Asset Period** Select to ensure that the period for depreciation reflect the life on the equipment record. This field should not be selected when an asset already has usage readings. If you select **Entire Asset Period**, you cannot enter values in the **Usage From** and **Usage To** fields or the **Month From** and **Month To** fields. b. **Usage From** If **Entire Asset Period** is not selected, enter the usage from which depreciation should start to calculate. This field is greater than or equal to zero. Enter the usage value up to two decimal places. c. **Usage To** If **Entire Asset Period** is not selected, enter the usage when depreciation should end (the life of the depreciable asset). The usage will generally be equal to, or less than, the equipment's total life usage; however, it may be greater than the equipment's total life usage if it is a rotable asset. This field is greater than zero and greater than the Usage From value. For a previously depreciated asset, that is one with an opening accumulated depreciation greater than zero, this figure will be **Usage From** + **Total Usage** (from the **Life of Depreciable Asset** section). Enter the usage value up to two decimal places. **Date Based (Monthly)** a. **Entire Asset Period** Select to ensure the period for depreciation reflects the start and end dates on the equipment record. This field should not be selected when an asset already has usage readings. If you select **Entire Asset Period**, you cannot enter values in the **Usage From** and **Usage To** fields or the **Month From** and **Month To** fields. b. **Month From** If **Enter Asset Period** is not selected, enter the date when depreciation starts. c. **Month To** If **Enter Asset Period** is not selected, enter the date when depreciation finishes. This date range corresponds to the duration information.	
Equipment Details	

Field	Description
All of the fields in this panel should be pre-populated.	
Description	Strategy task description that identifies the monthly asset depreciation.
Component Code	The component code to which depreciation is allocated.
Modifier Code	An additional reference to the component code, typically the component's location on the asset. For example, RF - Right Front, TOP – Top.
Expense Element	The depreciation expense code, which exists in the ERP accounting system.
Task Type	Code and description combination identifying what is to happen at an occurrence of a strategy task.
Escalation Index	Depreciation doesn't escalate. This field is standard across AMT strategy tasks. Create and enter a zero rated escalation code in this field.
Task Counter	Enter the task counter (which may be None) to represent the incremental changeout number for components that have been changed out during the life of the total asset. This field identifies the number of times a component has been changed out.
Cost Bearer	Used to distinguish between maintenance and other costs.
Enable User WO Settlement Creation	Select to ensure AMT creates a work order settlement for the monthly depreciation calculated. Do not select this check box if AMT will not calculate depreciation.
Cost Centre	The organisational or departmental entity of a company to which the depreciation is allocated. The cost exists in the ERP accounting system.
Pricing Date	Pricing Date is a common field across AMT strategy tasks. It does not have an impact on depreciation.

Depreciable Asset Manager

Use the Depreciable Asset Manager to review the depreciable assets you create in the Depreciable Asset Editor. It also supports additional processes that can occur on assets, for example disposal, transfer, additional capital. Navigate to **Setup & Admin** > **Depreciable Asset Recognition**.

You can set **Equipment**, **Depreciable Asset** and **Additional** filters, or click **Apply** to display all the depreciable assets already completed. The results display in the grid at the bottom of the dialog box.

The fields on the dialog box are described in the table below.

Field	Description
Branch	Select a branch.
Site	Select a site.
Fleet	Select a fleet.
Model	Select a model.
Capital Asset	Parent capital identifier.
Depreciable Asset	Depreciable capital identifier.
Status	The status of the asset - active, idle (Inactive), transferred, disposed.
Transaction Type	Process performed on the asset.
All Periods	When selected, the life to death position displays. The other date fields are unavailable when **All Periods** is selected.
Period From Period To	Enter if results for a particular period are required.
Display Fields	Optional fields to include in the display grid.

The following columns display in the grid by default. You can filter on other columns by selecting them in the **Display Fields** field in the **Additional Filters** panel.

Column	Description
Depreciable Asset ID	The name or serial number of the equipment
Period	The accounting period.
Status	Whether the asset is active, disposed or idle.
Original Asset Cost	The total price associated with the purchase of an asset.
Capital Value	The value of the asset.
Opening WDV	Opening Written Down Value (WDV) The opening value of the asset.
Depreciation for Period	Depreciation value over the accounting period.

Column	Description
Closing WDV	Closing Written Down Value (WDV) The closing value of the asset.
Depreciation Type (S/U)	Type of depreciation used, either straight line or units of activity.
Asset Classification 1 Asset Classification 2	The type of asset, for example, equipment.

You can also copy the grid, and export the grid to Excel.

You can drag the columns in the grid to group them differently. Clicking the cell below the column name gives you search criteria. Depending on the column type they include the following:

- Clear
- Equal
- Not Equal
- Begins With
- Ends With
- Contains
- Empty
- Not Empty
- Greater Than
- Greater Than or Equal
- Less Than
- Less Than or Equal

Quick Links

Click **Quick Links**  , or right-click an asset to quick the following processes.

Process	Description
Add Depreciable Asset	Opens the **Depreciable Asset Editor**. If no depreciation has been calculated for the asset, you can adjust the parent capital asset's values. Then either add a new depreciable asset or adjust a previously created depreciable asset to ensure the parent values balance to the sum of the depreciable totals. If depreciation has been calculated for the asset, add a new depreciable asset using the **Depreciable Asset Editor** screen. With asset processes run for existing assets, AMT will not validate the parent totals with the child totals when saved.

Process	Description
Add Strategy Task	Opens the Depreciation Strategy Task Editor.
Change Status	Allows the depreciable asset's status to be modified from active to inactive, or inactive to active.
Dispose Asset	Disposing a depreciable asset and its associated transactions.
Impair Asset	Impairing a depreciable asset and its associated transactions.
Revalue Asset	Revaluing a depreciable asset and its associated transactions.
Transfer Asset	Transfer an asset from one site or fleet to another.

Add a Depreciable Asset

Adding a depreciable asset occurs after depreciation has been run for that asset. An asset can have additional capital added by right-clicking the asset in the Asset Details results grid or clicking the Quick Links button above the grid. This opens the Depreciable Asset Editor with the capital header and current depreciable assets created for the selected asset. The Depreciable Asset Editor opens with the next sequential depreciable asset number populated.

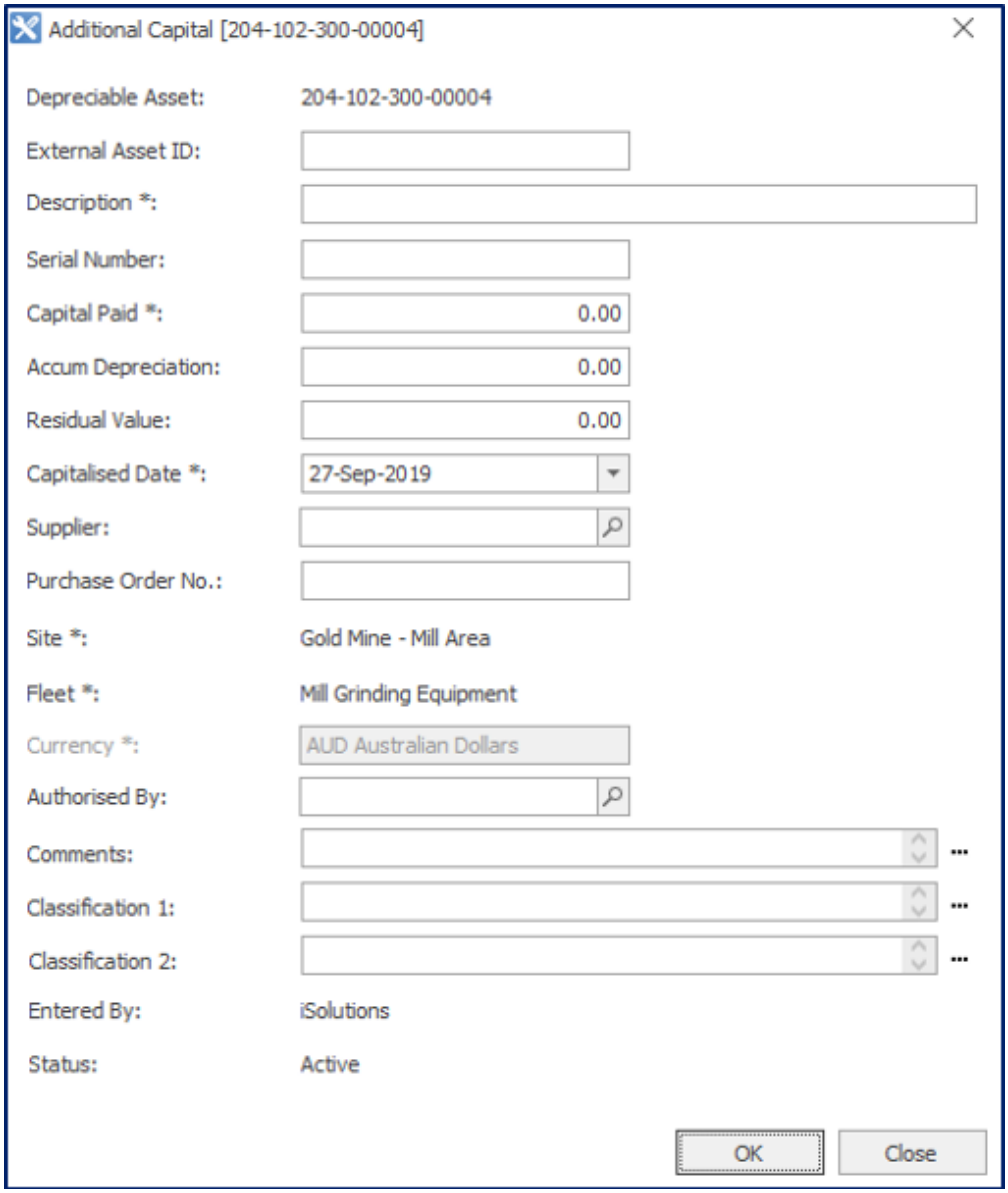
AMT no longer validates that the parent header detail values equal the sum of the underlying child or component values. These values will no longer be valid with the addition of Additional Capital or other processes of Impairment, Disposal, Revaluation and Transfer that can occur.

Add a Depreciable Asset to an existing Asset

This process adds a new depreciable asset in sequence to an existing asset when the asset has already been depreciated, the depreciation agreed, and period closed financially. If Finance support are re-opening the month and updating the depreciation recognition, AMT can support this process.

If a new asset needs to be created and depreciation has been run, but the period is still the current period, it is better to open the **Depreciable Asset Editor** for this asset. Adjust the header values for the new depreciable asset values and create a new, or adjust an existing, depreciable asset. Then re-run depreciation for this asset.

Navigate to the **Additional Capital** dialog box by navigating to **Setup & Admin** > **Depreciable Asset Recognition** > **Depreciable Asset Manager**, then clicking the link on the **Quick Links** menu, or right-click on the grid.



AMT automatically increments the asset number to the next sequential number.

Enter data in the following fields:

Field	Description
External Asset ID	The external ERP asset identifier or specific identifier by which the depreciable child asset is known.
Description	The description of the asset.
Serial Number	The specific serial number associated with this depreciable child asset.

Field	Description
Capital Paid	The original capital cost associated with the child asset.
Accumulated Depreciation	The accumulated depreciation associated with the child asset, if this asset has been previously depreciated.
Residual Value	The residual value of the child asset.
Capitalised Date	This field defaults to the date entered at the header level. The date cannot be in the future.
Supplier	The supplier from whom the child asset was purchased.
Purchase Order Number	The purchase order associated with the above supplier.
Authorised By	Select the name of the AMT employee who approved the asset captitalisation. If the approver is not an AMT employee, enter their name manually into the field.
Comments	Enter any relevant comments about the asset.
Classification 1 and 2	General Ledger classifications.

The transaction information created is referenced with EDC codes 205 and 214 and the associated account codes.

Balancing capital header and child assets is not required to recognise assets during the Add Depreciable Asset process. Balancing only occurs during the initial capital recognition process for the equipment.

Change a Depreciable Asset's status

You can change the status of a depreciable asset in the **Depreciation Asset Manager**, by either clicking **Quick Links** or right-clicking the asset in the results grid and clicking **Change Status**.

Completing the form modifies the state of the asset from active (depreciating) to inactive (not depreciating) and writes an audit record.

The fields on the screen are described in the table below.

Field	Description
Date	The date the status changed. Defaults to today's date. The date must be in the current period, and cannot be in the future or in a previous period.
Status	Select either active, idle (inactive), or transferred (disposed). Transferred or disposed

Field	Description
	cannot be modified by this process.
Usage	Enter the life to date usage of the status change.
UOM	Enter a Unit of Measure associated with the usage.
Reason	Enter the reason for the asset's status change.

Disposal of assets

Sometimes, the parent asset and all child assets (components) need to be disposed. This involves five sets of journals:

- Disposal of Parent asset - EDC - writes off the capitalised cost. Navigate to the **Asset Ledger** screen.
- Reverse Accumulated depreciation on parent asset - EDC. Navigate to the **Plant Asset Journals** screen.
- Record profit or loss on disposal of a Parent Asset. Navigate to the **Plant Asset Journals** screen.
- Disposal of component - EDC 115 - writes off the capitalised cost of the component.
- Reverse CBD - EDC 116 - reverses the accumulated depreciation on the disposed component.

Record profit or loss on disposal does not impact the Asset Ledger. It is only needed if:

- The Plant Hire module is being used.
- The Profit/Loss on disposal is not captured through the normal cost interfaces.

Dispose an Asset

An asset may be disposed of at any time after its first depreciation run. If an asset has been used and depreciation recognised in AMT, the depreciable asset must be disposed. Disposal means the asset is scrapped or sold.

If an asset is disposed, it cannot be reinstated with the same asset identifier.

If no depreciation has occurred or needs to be recognised, open the Depreciable Asset Editor to adjust the initial capital header amount and the subsequent depreciable assets. If you are disposing of the entire machine, all records need to be deleted.

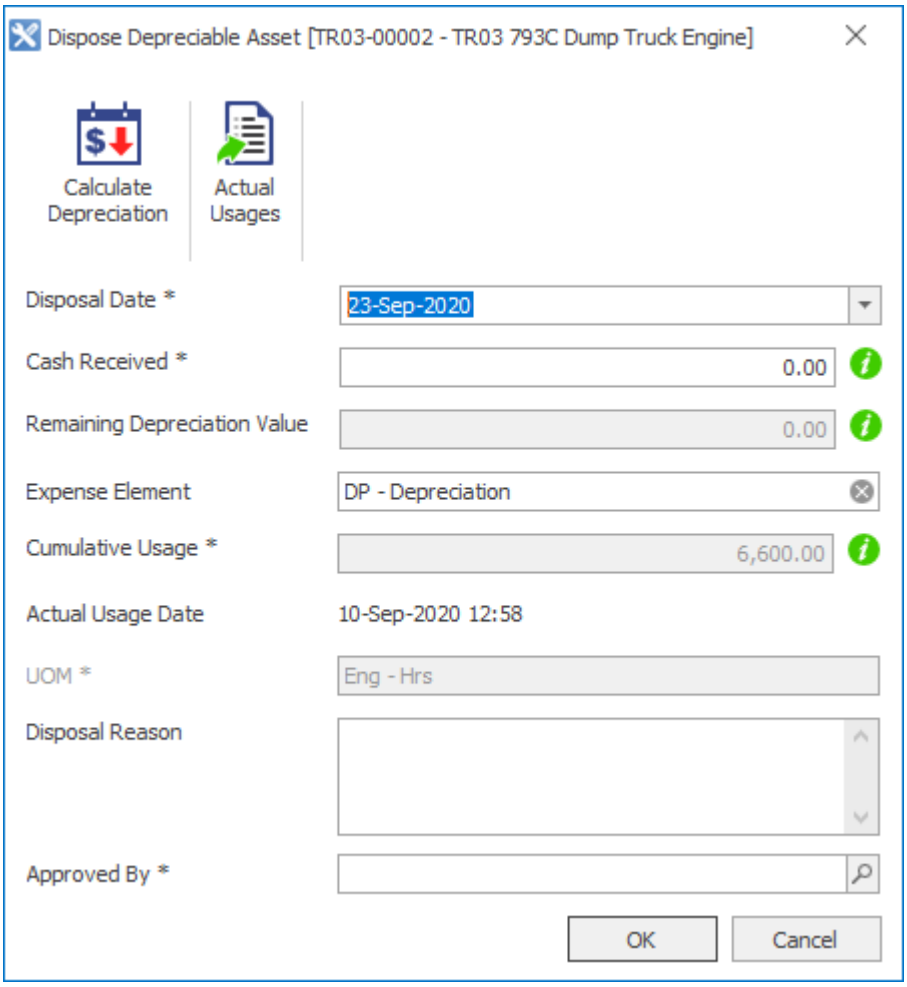
AMT is not designed to recognise fully depreciated assets. These assets can be recognised as equipment, managed from a maintenance perspective and have no associated depreciation.

It would be unusual for an asset to be capitalised in the same month as it was disposed without at least some depreciation being run. If this is the case and no depreciation has occurred or needs to be recognised, open the **Depreciable Asset Editor** and adjust the initial capital header amount and the subsequent depreciable assets. If you're disposing of the entire machine, all records need to be deleted.

You can dispose a depreciable asset using the **Depreciation Asset Manager**, by either clicking **Quick Links** or right-clicking the asset in the results grid and clicking **Dispose Asset**.

Completing the fields on the screen changes the status of the asset to be disposed, writes the journal transactions and records an audit record.

1. In the **Depreciation Asset Manager**, either click **Quick Links** > **Dispose Asset**, or right-click the depreciable asset in the grid.



The fields on the form are described in the table below.

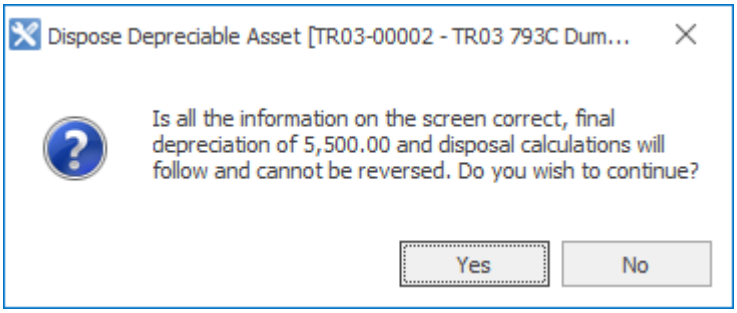
Field	Description
Disposal Date	Defaults to today's date. The date must be in the current period, and cannot be in the future or in a previous period.
Cash Received	Enter the cash received for the disposal asset. This value can be zero.
Remaining Depreciation Value	The remaining depreciation value will be calculated by AMT based on the usage information recorded in AMT. Depreciation will be the number of units used or consumed at the disposal date less the last depreciation calculation usage multiplied by the current unit rate for the asset.

Field	Description
	Based on the date of disposal, AMT presents the latest cumulative information for that date in the Cumulative Usage field. If this field is correct, then you do not need to make any adjustments. If the Cumulative Usage field is not correct, you need to adjust the Actual Usages information by selecting the Actual Usages icon and entering or adjusting the actuals information in this screen. When complete they will return to the Disposal screen. Calculate Depreciation process will calculate and present the depreciation for the month up until the disposal date for this asset. Perform a visual assessment of this information to confirm its accuracy.
Expense Element	Defaults to the expense element on the Depreciation Strategy Task. It can be overwritten to another expense element configured for depreciation if needed.
Cumulative Usage	The Cumulative Usage defaults to the latest cumulative usage held in AMT for the disposal date. You need to check this value for its accuracy with the cumulative usage reading provided to you during the disposal process. For example, if the cumulative usages are only entered twice per day at 0600 and 1800 and the part failed at midday, there could easily be hours that need to be adjusted depending on the date when the depreciable asset is disposed. If this cumulative usage information is not correct, click on the Actual Usages icon. The Actual Usages screen will open and can be completed as necessary to record the cumulative usage at the disposal date and time of the depreciable asset.
UOM	Enter a unit of measure for the usage. If the asset is depreciated by usage, the UOM will default.
Approved By	Select the name from the list of AMT users, or, If the approver is not an AMT employee, you can enter their name.

2. When the usage information is correct, depreciation calculated and all mandatory information has been added, click **OK** to finalise the disposal calculation process.

   You will be presented with a warning ‘confirmation’ message to acknowledge as the disposal calculations

cannot be reversed.

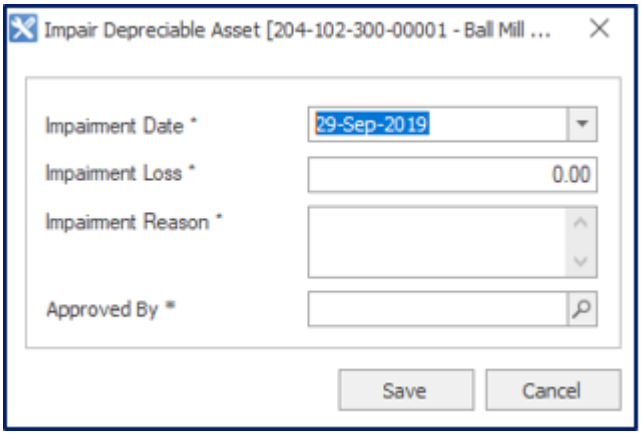


After clicking **OK**, the message above displays asking if you want continue.
If you want to modify the depreciation strategy task now, it opens.
If you do not want to modify the depreciation strategy task now, you will need to modify it at a later time.
If you do not modify it, and forget about it, depreciation calculations and associated journal transactions may not occur as expected.

Impair an Asset

To impair an asset, do the following.

1. Click **Quick Links** > **Impair Asset**, or right-click on the depreciable asset in the grid.



The fields on the screen are described in the table below.

Field	Description
Impairment Date	Defaults to today's date. The date must be in the current period, and cannot be in the future or in a previous period.
Impairment Loss	Enter the value of the impairment loss to be recognised. The amount must be positive.
Impairment Reason	Enter a reason for impairing of the asset.

Field	Description
Approved By	Select an employee name from a list of valid AMT users, or enter the name manually if the approver is not an AMT user.

2. Click **Save**.

After clicking **Save**, a message displays asking if you want to modify the depreciation strategy task now.

If you want to modify the depreciation strategy task now, it opens.

If you do not want to modify the depreciation strategy task now, you will need to modify it at a later time.

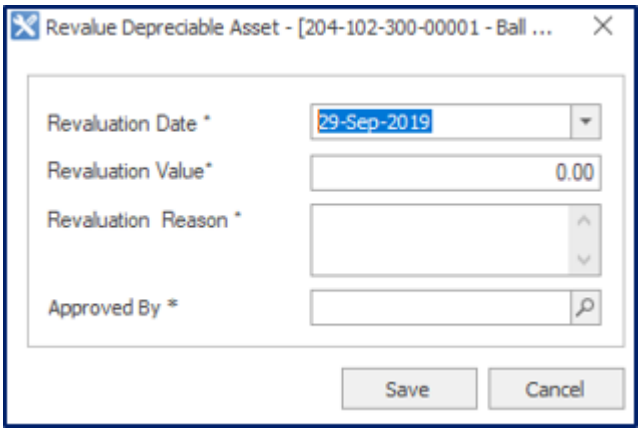
If you do not modify it, and forget about it, depreciation calculations and associated journal transactions may not occur as expected.

AMT will support both a negative and positive impairment transaction. However, the negative impairment should only be used where it is necessary to adjust take-on balances and any transaction should always include appropriate commentary for audit purposes.

Revalue Asset

Revalue an asset by doing the following.

1. Click **Quick Links** > **Revalue Asset**, or right-click the asset in the results grid and click **Revalue Asset**.



The fields on the screen are described in the table below.

Field	Description
Revaluation Date	Defaults to today's date. The date must be in the current period, and cannot be in the future or in a previous period.

Field	Description
Revaluation Value	Enter the revalued amount of the asset. This is the amount by which the asset value has increased.
Revaluation Reason	Enter a reason for revaluing of the asset.
Approved By	Select the name from the list of AMT users, or, If the approver is not an AMT employee, you can enter their name.

2. Click **Save**.



After clicking **Save**, a message displays asking if you want to modify the depreciation strategy task now.

If you want to modify the depreciation strategy task now, it opens.

If you do not want to modify the depreciation strategy task now, you will need to modify it at a later time.

If you do not modify it, and forget about it, depreciation calculations and associated journal transactions may not occur as expected.

AMT will support both a negative and positive revaluation transaction. However, a negative revaluation should only be used where it is necessary to adjust take-on balances and any transaction should always include appropriate commentary for audit purposes.

Split an Asset

You can split or transfer an asset using the **Split/Transfer** screen. Splitting an asset creates a new asset in the original asset sequence.

A depreciable asset can be split by a percentage or an amount. An asset may need to be split to support:

- Disposing of part of the asset.
- Transferring part of the asset to another asset.
- A change in depreciation associated with part of the asset.

The split process occurs in the same underlying currency as the original asset.

After splitting an asset in AMT, you have to perform other tasks to complete the business process; for example

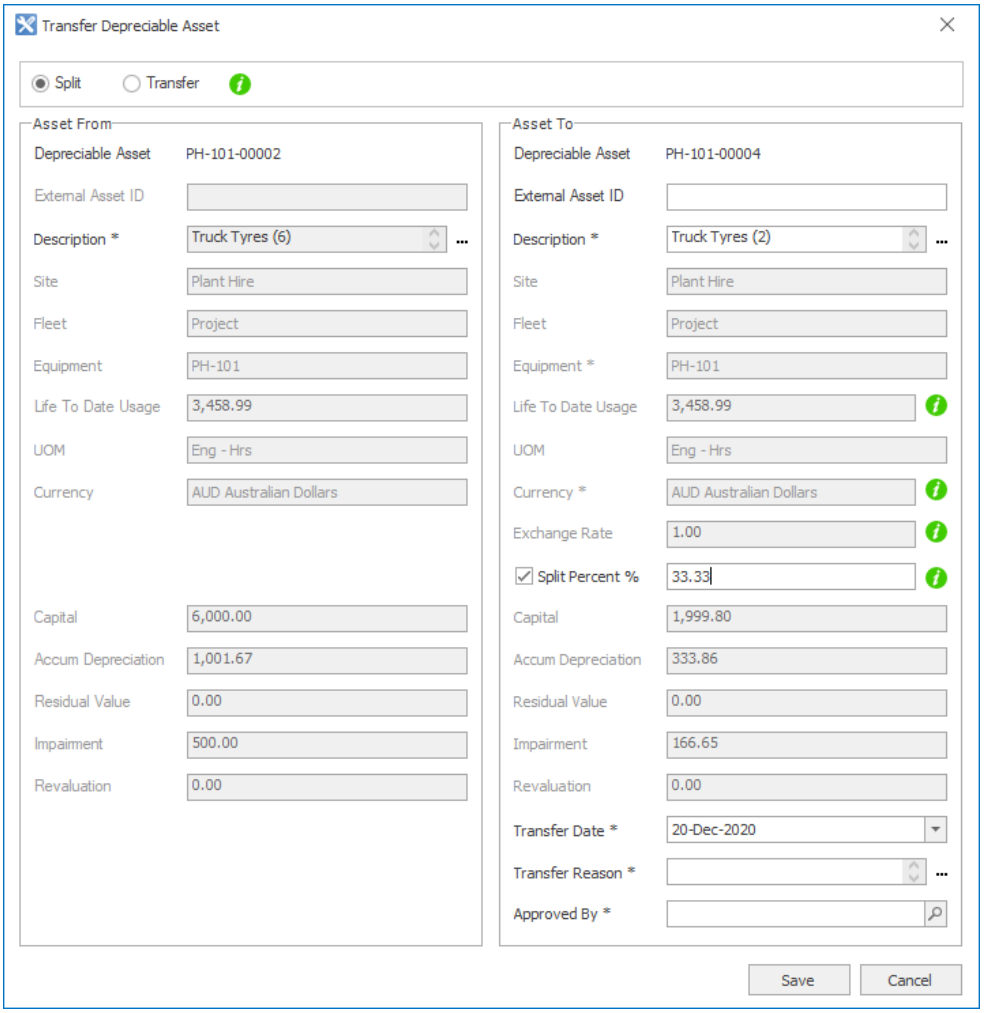
- For a depreciation split:
  - Create a new depreciation strategy task for the asset value that was split out.
  - Adjust the existing depreciation strategy task for the remaining asset.
- For a disposal:
  - Dispose of the split asset.
  - Adjust the existing depreciation strategy task for the remaining asset.

- For a transfer:
  - Transfer the split asset.
  - Create a new depreciation strategy task for the split asset.
  - Adjust the existing depreciation strategy task for the remaining asset.

Any impairment or revaluation values associated with the asset can be split with the asset depending on an organisation's requirements. AMT will not allow the Written Down Values of either asset created to breach accounting rules; this means that AMT will not allow the process to create negative written down values for either asset.

To split an asset, do the following.

1. Click **Quick Links** > **Split Asset**, or right-click the asset in the results grid and click **Split Asset**.
2. Check that the correct **From** depreciable asset details are visible in the **Asset From** section before completing the details in the **Asset To** section. The "To Asset" is a new depreciable asset with the next number in the depreciable asset sequence.



The fields on the screen are described in the table below.

Field	Description
External Asset ID	The external ERP Asset ID or specific identifier by which that asset is known.
Split Percent %	Select to split the **From** and **To** assets based on a percentage of the original asset value. If you do not want to sprit by percentage, enter the To asset values manually. AMT applies this percentage to the depreciable asset and calculates the capital, accumulated depreciation and residual values.
Capital	Enter the capital value to be split from the original asset. The capital is calculated from the percentage split value entered, or you can enter a value. It cannot be greater than the **From** capital value.
Accumulated Depreciation	Enter the accumulated depreciation value to be split from the originating asset. The accumulated depreciation is calculated from the percentage split value entered, or you can enter a value. The value cannot be greater than the **From** accumulated depreciation.
Residual Value	The residual value is calculated from the percentage split value entered, or you can enter a value. The value cannot be greater than the **From** accumulated depreciation.
Transfer Date	Enter the date when the split was performed. The date cannot be in the future. AMT defaults to the current date.
Transfer Reason	Enter a reason for the split.
Approved By	Select the name from the list of AMT users, or, If the approver is not an AMT employee, you can enter their name.

3. Click **Save**.

   If values are entered manually, AMT will not allow any negative written down values to result from the split process.

Transfer an Asset

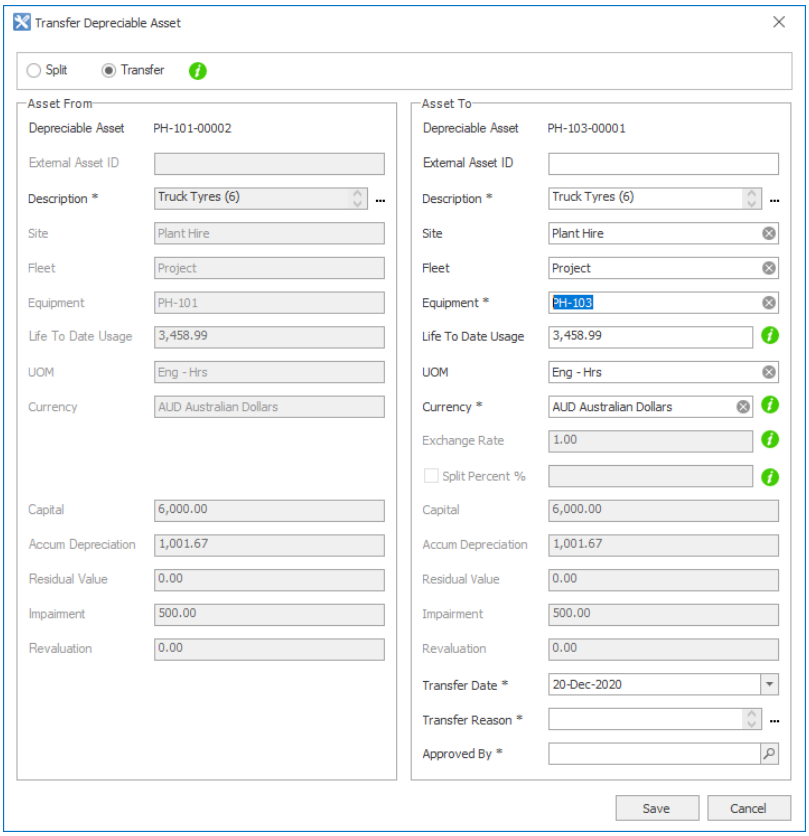
The transfer process ensures 100% of the **From** asset is transferred to the new **To** asset. An asset transfer can occur in the same currency or different currencies. The depreciation strategy task associated with the From asset should be adjusted to ensure no further depreciation can occur that relates to it.

A new depreciation strategy task must be created for the new **To** asset created in this process.

To transfer an asset, do the following.

1. Click **Quick Links** > **Transfer Asset**, or right click the depreciable asset in the results grid and click **Transfer Asset**.

2. Check that the correct **From** depreciable asset details are visible in the **Asset From** section. before completing the details in the **Asset To** section. The **To** asset will be a new depreciable asset with the next number in the depreciable asset sequence.



The fields on the screen are described in the table below.

Field	Description
External Asset ID	The external ERP Asset ID or specific identifier by which that asset is known.
Description	Defaults to the same description as the **From** asset. You can overwrite it.
Site	Select the site if it is different from the **From** asset. The site determines the currency of the asset.
Fleet	Select the fleet if it is different from the **From** asset.
Equipment	Select the equipment to which the asset is being transferred.
Life to Date Usage	Enter the cumulative life to date usage at the date of transfer.
UOM	Enter the unit of measure associated with the life to date usage.

Field	Description
Currency	Enter the currency if it is different from the **From** asset.
Exchange Rate	AMT defaults to the last exchange rate. You can override it by entering another rate.
Split Percent %	You cannot use a split percent in the transfer process, you can only transfer 100% of the asset. If only part of an asset is being transferred, split the asset first, then transfer the new split asset.
Capital	Calculated from the exchange rate and the **From** capital value.
Accumulated Depreciation	Calculated from the exchange rate and the **From** accumulated depreciation value.
Residual Value	Calculated from the exchange rate and the **From** residual value.
Impairment Value	Calculated from the exchange rate and the **From** impairment value.
Revaluation Value	Calculated from the exchange rate and the **From** revaluation value.
Transfer Date	Select the transfer date. The date can be today or a date in the past; it cannot be in the future.
Transfer Reason	Enter a reason for the transfer.
Approved By	Enter the name of the person who approved the transfer. Select the name from a list of AMT employees or enter the name if the approver is not registered as an AMT employee.

3. **Click OK**. An audit record is written.

How to use the Transfer Asset screen

Action	Description
Split one asset into two assets by percentage.	Use the **Split** option and enter the percentage.
Split one asset into two assets by value.	Use the **Split** option and enter **Accumulated Depreciation** and **Salvage Value**.
Transfer an asset from one site or fleet to another with the same currency.	Use the **Split** option with 100% or full accumulated value details.
Transfer an asset from one site or fleet to another with the different currency.	Use the **Transfer Asset** option.

Transfer to a different site or fleet with the same currency

The fields on the **Transfer Asset** screen are described in the table below.

Field	Description
External Asset ID	Enter the external ERP asset ID or specific identifier by which that asset is known.
Site	Select a site.
Fleet	Select a fleet.
Equipment	Select equipment associated with the selected site and fleet.
Life to Date Usage	Enter life to date usage at the time of transfer.
UOM	Enter the unit of measure associated with the usage. Will be set by default if the asset is depreciated by units of activity.
Currency	Set by default from the site currency and the equipment currency selected.
Exchange Rate	The exchange rate defaults to the latest AMT exchange rate held for the two currencies; this should be 1.0 when the currencies are the same.
Capital	Calculated by AMT based on the exchange rate. Should be equal to the **Transfer From Capital**.
Accumulated Depreciation	Calculated by AMT based on the exchange rate. Should be equal to the **Transfer From Accumulated Depreciation**.
Residual Value	Calculated by AMT based on the exchange rate. Should be equal to the **Transfer from Residual Value**.
Transfer Date	Enter the date when the transfer was performed. AMT defaults to the current date.
Transfer Reason	Enter a reason for the transfer.
Approved By	Select the name from the list of AMT users, or, If the approver is not an AMT employee, you can enter their name.

Transfer to a different site or fleet with different currency

The difference between transferring to a different site or fleet with the same currency and different currency is:

Field	Description
Exchange Rate	The exchange rate defaults to the latest AMT exchange rate held for the two currencies; this can be overwritten.

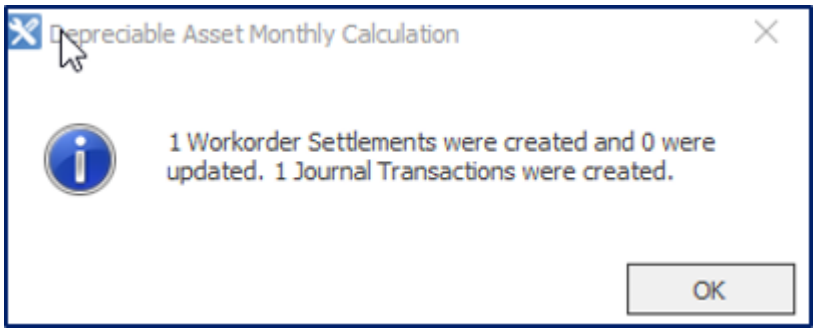
Depreciable Asset Monthly Calculation

Depreciation calculations can be triggered in two ways:

- Manually by selecting the menu option. Navigate to **Setup & Admin** > **Depreciable Asset Recognition** > **Depreciable Asset Monthly Calculation**.
- Automated with the AMT system task.

We recommend that you manually calculate monthly depreciation until you're satisfied with the calculation and process. Calculating depreciation manually supports re-running the process and limiting the calculations by various filters, for example, by site, by equipment and/or specific asset.

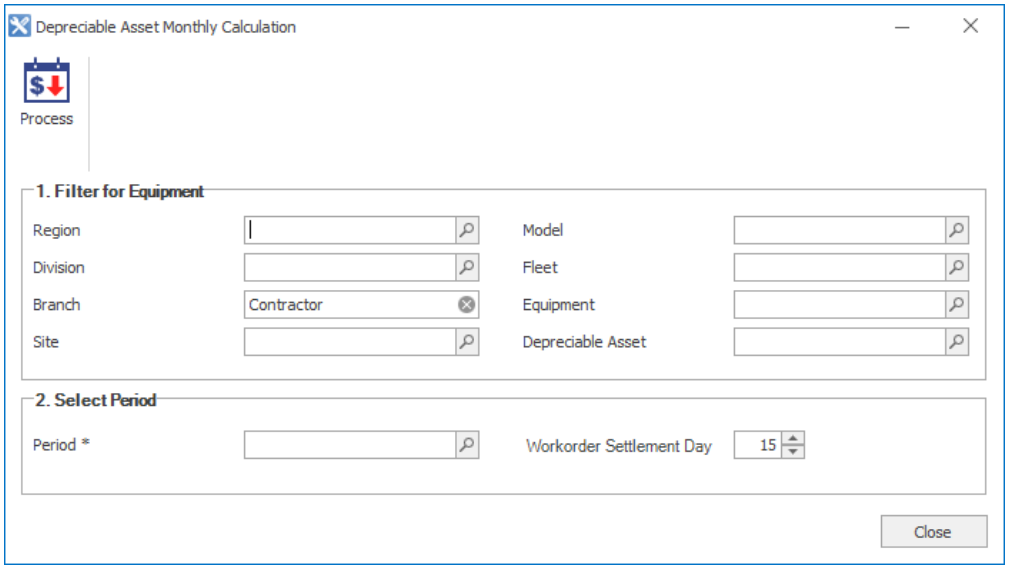
When the manual process has completed, AMT notifies you of the number of work order settlements created and updated, and the number of journal transactions created.



If depreciation is being run for the first time in a month for the selected assets, there should be no work order settlements updated. Updates only occur after depreciation has been run once for the selected asset within that period.

Running monthly depreciation manually

Navigate to **Setup & Admin** > **Depreciable Asset Recognition** > **Depreciable Asset Monthly Calculation**.



You can filter for specific depreciable assets. The filters operate the same way as other filters in AMT; you can make multiple selections where appropriate.

If there is only one current period and no open periods, the current period defaults into the **Period** field. If there is one or more open periods, you must select a period.

Work Order Settlement Day is included as the date recorded on the work order settlement for the depreciation. It is not the day to which depreciation is charged.

Running monthly depreciation with a system task

Depreciation calculations on a monthly basis can be automated with an AMT system task. The system task, **Monthly Depreciation Calculation**, runs all depreciation calculations for all active assets with a current depreciation strategy task.

Set the system task with a recurring frequency of once per month and run or trigger it after actual monthly usages are finalised.

(Rpt_826) Depreciable Asset Register Report

Displays Asset Cost, Accumulated Depreciation Written Down Value, %WDV and Residual Value for each depreciable asset.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Region, Industry Group, Cost Centre, Equipment Group, Manufacturer, Project Customer, Project Name, Division, Cost Responsibility, Equipment Class, Equipment Category, Asset Status, Hire Status, Incl Terminated

Depreciable asset transactions

The recognition of assets and subsequent asset processes for depreciation, transfers, splits, impairments, revaluations and disposals all create transactions in AMT. These transactions will be balanced (that is, debits equal credits), and the accounts associated with the transactions are recorded against entry distribution codes 200 to 227,

You can review the fixed asset transactions you have created. There are no mandatory fields, so you can either select particular fields to filter on, or click **Apply** to display all of the transactions.

The **Additional Filters** panel is an important area you can use to focus the results to specific transaction types, cost centres, data ranges, and so on.

In the **Equipment Filters** section, in addition to the standard AMT equipment search options, you can make multiple selections in the Capital Asset and Depreciable Asset fields.

Results grid

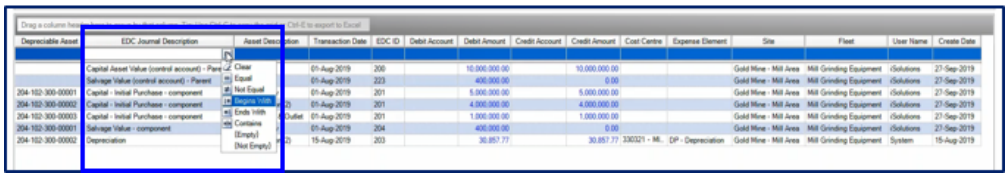
The following columns display in the results grid by default. You can add more columns to the grid by selecting them in the **Display Fields** field in the **Additional Filters** panel.

Column	Description
DAV ID	Depreciated Asset Value ID or serial number
DAV Description	Description of the depreciated asset value.
Transaction Date	Date the asset was created.
EDC ID	Entry distribution code ID.
EDC Journal Description	Description of the entry distribution code journal transaction.
Debit Account	Account debited.
Debit Amount	Amount debited.
Credit Account	Account credited.
Credit Amount	Amount credited.
Cost Centre	The codes that costs are charged against.
Expense Element	The expense element category.
Site	The name of the site that the asset belongs to.
Fleet	The fleet that the asset belongs to.
User Name	The name of the user of the asset.
Create Date	The date the transaction was created.

You can drag one or more column headers to the grid header to group transactions.

You can export results to Excel using **Ctrl-E**.

You can filter column results in the grid. Select the icon in the field under the column heading and choose a filter option. This capability is useful when you're looking for a particular transaction amount or transactions greater than or less than an specific amount.



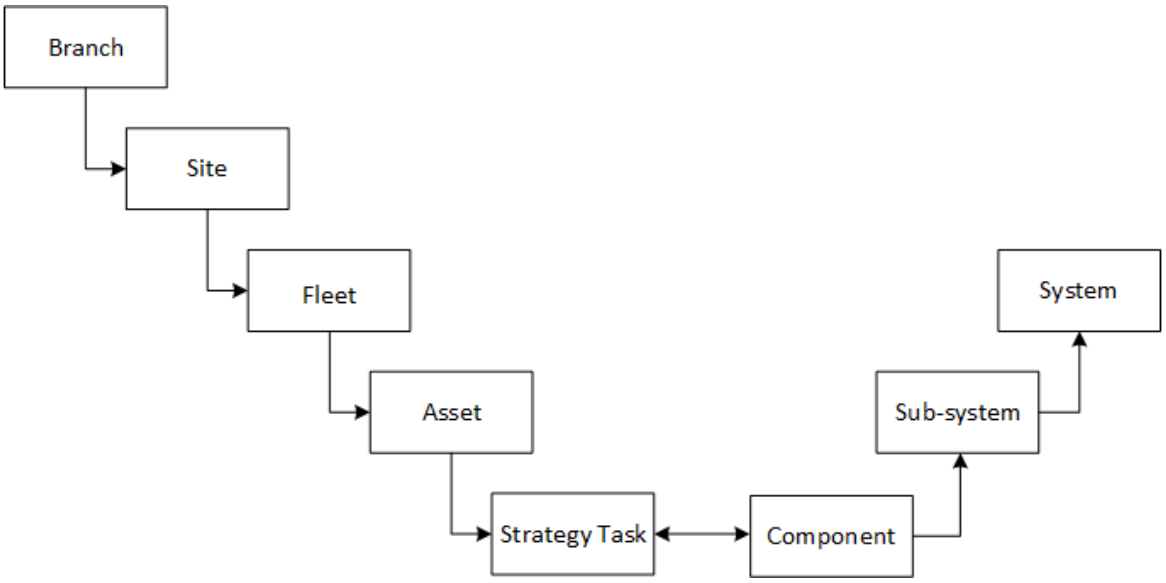
Asset Register Report

Review the depreciable assets that have been created. Navigate to **Setup & Admin** > **Depreciable Asset Recognition** > **Depreciable Asset Register Report**.

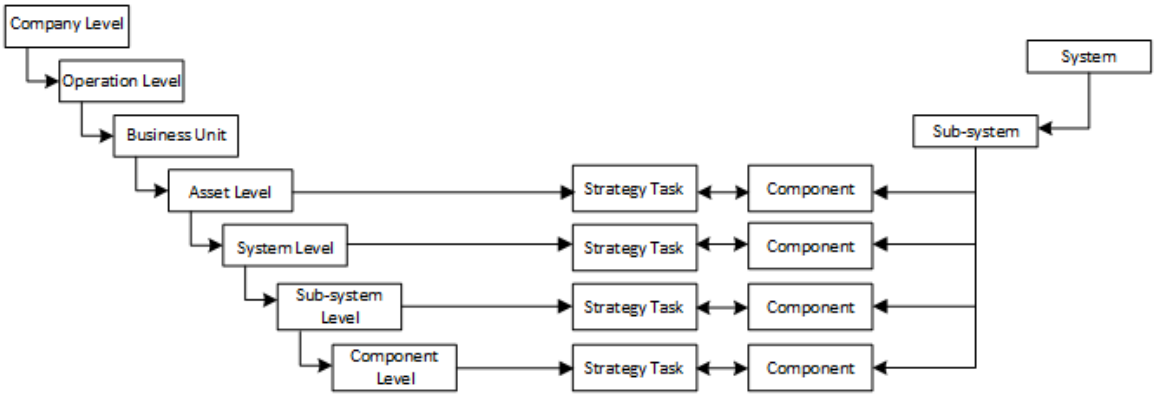
Functional Location

Functional hierarchy overview

AMT traditionally has been designed to operate on a hierarchy of a branch, site, fleet and asset. Each asset has strategy tasks and components. The components themselves form part of a hierarchy of systems and subsystems. Using multiple hierarchies working together gives a significant amount of flexibility in reporting.



The functional hierarchy operates differently to the above structure in that an asset is assigned to a functional location which does not have to follow the branch-site-fleet hierarchy. It lets you set up a structure to represent the existing structure in your ERP system, or to set up a new structure that represents the operational structure of your site or plant. Below is an example showing a possible structure. You can assign a strategy task to any level, although this would usually be done at the asset level or below.



In the background, AMT continues to operate mostly the same, the functional hierarchy structure is used to visually represent what is happening and to allow reporting based on that structure. When creating the assets,

they will still be assigned to a branch, site and fleet, however these will not be used for functional hierarchy reporting.

The functional hierarchy structure in AMT is designed to:

- Represent your assets and components based on a functional location and hierarchy
- Report based on the structure of your plant or operation
- Work with ERP systems to automatically create the same structure

Prior to starting the functional hierarchy process it is recommended you develop an understanding of how you want to set up your structure, to reduce rework if you want to change something.

Functional Location System Tables

There are three system tables you should review before creating a functional location hierarchy.

- Functional Location Category
- Functional Location Level
- Functional Location Attribute

Navigate to **System Administration** > **System Tables** > **Functional Location**.

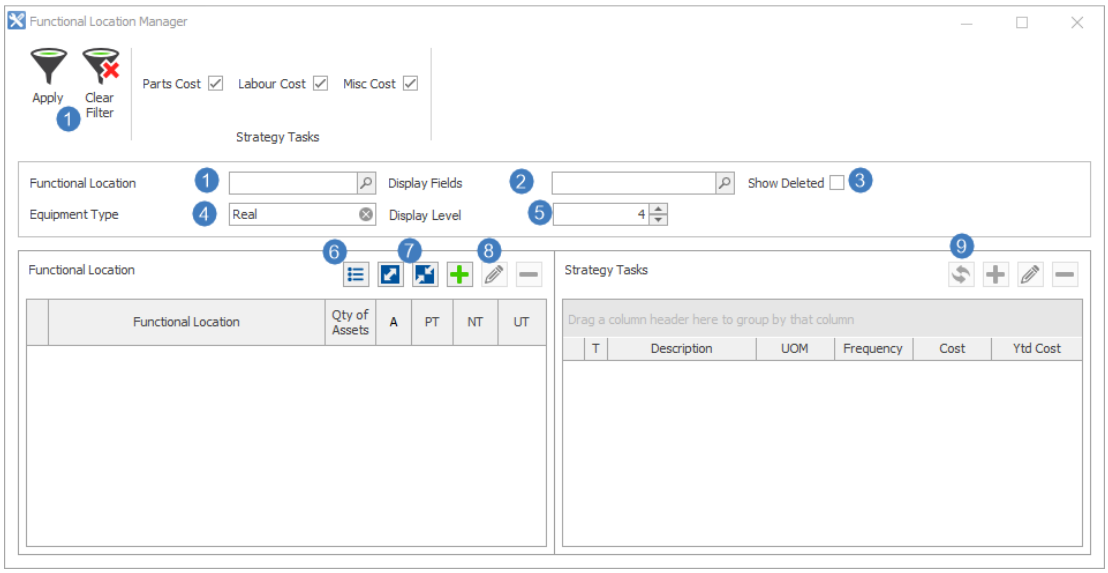
Functional Location Manager

Set up the functional hierarchy

You can set up a functional hierarchy in two ways in AMT. You can manually create one in the Functional Location Manager, or you can create an interface between AMT and your ERP system which will allow functional locations to be created automatically in AMT. Using an interface with an ERP system is the preferred method as it will allow your ERP system and AMT to be reporting based on the same structure.

The interface will be set up by an RPM Interface Consultant due to the complex nature of the system.

Navigate to **Functional Location** > **Functional Location Manager**.



Number	Description
1	Filter by **Functional Location**. If you enter a few characters of the asset name before clicking the Search icon the list is reduced to results only containing those characters.
2	Select which **Attributes** to display.
3	Display deleted records.
4	Select the type of **Equipment** to display.
5	Select how many **Levels** to display.
6	Quick Links - **View Asset**, **Export data**, **Copy grid**.
7	**Expand** or **Collapse** the functional hierarchy tree.
8	**Add**, **Edit** or **Delete** a functional location.
9	**Refresh**, **Add**, **Edit** or **Delete** a strategy task for the selected asset.

(Rpt_781) Functional Location Budget Analysis

Provides a comparison of the budget figures for the selected financial year and the actual results of the past 12 months.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,|LanguageKey|, |ReportStyleKey|,|FunctionalLocationName_Name|,|FunctionalLocationName_Value|

Report parameters

Default.

Additional filters

Currency, Cost Centre, Expense Element, Work Group, Cost Bearer, Parts, Labour, Misc, Toggle

(Rpt_782) Functional Location Budget v Actual Analysis

This report provides a dynamic report for comparison between budget and actual figures. You have the choice of multiple columns for analysis purposes.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,
|UserID|,|LanguageKey|,|ReportStyleKey|,|FunctionalLocationName_Name|,
|FunctionalLocationName_Value|

Report parameters

Default.

Additional filters

Currency, Cost Centre, Expense Element, Work Group, Cost Bearer, Parts Labour Misc, Toggle, Display Columns, UOM

(Rpt_783) Functional Location Cost Analysis

Provides a dynamic report for comparison between budget and actual figures. You have the choice of what you want to do.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,
|UserID|,|LanguageKey|,|ReportStyleKey|,
|FunctionalLocationName_Name|,|FunctionalLocationName_Value|

Report parameters

Default.

Additional filters

Currency, Cost Centre, Expense Element, Work Group, Cost Bearer, Parts Labour Misc, Toggle

(Rpt_784) Functional Location ERP Master Data Analysis

Provides analysis of the Functional Locations in AMT, linking to ERP Maintenance Plans. This helps you to see what level of the forecast is being driven by the ERP Maintenance Plan.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|,|FunctionalLocationName_Name|,
|FunctionalLocationName_Value|

Report Parameters

Report Type, Equipment Hierarchy

Additional filters

Currency, Cost Centre, Expense Element, Work Group, Cost Bearer, Parts Labour Misc, Toggle

(Rpt_785) Functional Location Forecasting Maturity

Provides an analysis of the maturity of the forecasts in AMT associated with Functional Locations. This is independent of the ERP linkage.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|,|FunctionalLocationName_Name|,
|FunctionalLocationName_Value|

Default parameters

Report Type, Equipment Hierarchy

Additional filters

Currency, Cost Centre, Expense Element, Cost Bearer, Parts Labour Misc, Toggle

(Rpt_786) Functional Location Cost Details

A flexible report with multiple columns to choose from to asses functional location costs.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,
|UserID|,|LanguageKey|,|ReportStyleKey|,
|FunctionalLocationName_Name|,|FunctionalLocationName_Value|

Report parameters

Default.

Additional filters

Currency, Cost Centre, Expense Element, Work Group, Cost Bearer, System, Sub System, Task Type, Component Code, Modifier Code, Task Counter, Cost Type, CC Activity, CC Location, Parks Labour Misc, Toggle, Display Columns, UOM

Strategy Optimisation

The maintenance strategy drives life-cycle costs, component schedules, parts forecasts, labour forecasts, maintenance plans and component risks and opportunities. In AMT, maintenance strategies are housed in Projections.

The strategy process consists of a number of related and parallel activities that support the maintenance effort. The core strategy process covers

- Setting up the original maintenance strategy.
- Periodic strategy reviews, and validating data.
- Identifying opportunities.
- Investigating.
- Testing alternate strategic solutions.
- Actioning and adjusting the short or long term strategy.
- Documenting changes and reporting.

The strategy functions in AMT are designed to support this process, and a key outcome of this process more accurate forecast of financials, components, availability, parts, labour, rotables and so on.

Strategy management and optimisation is the most difficult area of maintenance to identify specific and easily measurable KPIs that gauge performance. Overall, the asset management group is responsible for monitoring equipment performance and managing strategy to optimise asset group performance; so the real measure of the asset management group performance is the equipment performance. Unfortunately, there are many other factors that also impact equipment performance, so these cannot be viewed as directly under their control.

- Projected Cost Variance (PCV)
- Performance vs Budget
- Availability
- Reliability
- % Changeouts before failure
    - A key measure of the effectiveness of the equipment health activities. This needs to be viewed in conjunction with the average component life against budget - it is a fine balance between being too conservative and too aggressive.
- Standard Job Coverage
    - The preparation and usage of standard jobs leads to improved planning efficiency, standardisation of work performed and improved parts forecasting. At this point there is no specific KPI measurement in AMT for standard job coverage but it can be manually calculated relatively easily.
- Work Scope Coverage
    - The preparation and usage of work scopes leads to improved quality of work performed. At this point there is no specific KPI measurement in AMT for standard job coverage but it can be manually calculated relatively easily.

Strategy definition

The strategy for an asset is defined in the current projection. The projection contains

- High cost or high risk items which need to be forecast and tracked. These are defined in AMT as strategy tasks.

- The life expectancy of these strategy tasks.
- The cost to replace or repair these strategy tasks.
- The expected utilisation of the asset.

The output of the process is a full life-cycle cost projection for the asset. An example is shown below.

Projection Summary

AMT INTELLIGENT ASSET MANAGEMENT

Hierarchy : [Equipment Plan *: TR01] [Projection Type *: Current]
Additional Filters : [Escalation:Normal] [Currency *: USD US Dollars]
Date Range : [Entire Plan Term]
Analyse By : [Component Code]

Component Description	UOM	Cost To Date	First	Freq	Future Unit Cost				Future Cost		Total Cost	Cost Rate
					Parts	Labour	Misc	Total	Occs	Total		
1000.00.CC.0 ENGINE	Eng - Hrs	1,110,946	18,000	22,000	166,789	35,300	19,420	221,509	1	221,509	1,332,456	
1000.00.RR.0 ENGINE	Eng - Hrs	43,043	2,200	2,200	4,660	1,029	555	6,243	6	37,459	80,501	
1052.LF.CC.0 TURBOCHARGER	Eng - Hrs	12,273	9,000	9,000	6,237	2,585	0	8,822	1	8,822	21,094	
1052.LR.CC.0 TURBOCHARGER	Eng - Hrs	12,247	9,000	9,000	6,237	2,585	0	8,822	1	8,822	21,068	
1052.RF.CC.0 TURBOCHARGER	Eng - Hrs	11,429	9,000	9,000	6,237	2,585	0	8,822	1	8,822	20,250	
1052.RR.CC.0 TURBOCHARGER	Eng - Hrs	19,262	9,000	9,000	6,237	2,585	0	8,822	1	8,822	28,083	
1054.IN.PM.0 AIR FILTER ELEMENT	Eng - Hrs	1,357,401	1,000	1,000	587	38	0	624	13	8,116	1,365,517	
1054.OU.PM.0 AIR FILTER ELEMENT	Eng - Hrs	4,497,016	250	250	479	38	0	517	51	26,353	4,523,369	
1062.00.GR.0 MUFFLER	Eng - Hrs	1,823,911	18,000	18,000	16,177	1,551	0	17,727	0	0	1,823,911	
1105.00.PM.0 Perform Valve & Injector Adjustment	Eng - Hrs	758,385	0	4,000	0	2,000	0	2,000	3	6,000	764,385	
1250.00.CC.0 FUEL SYSTEM	Eng - Hrs	354,385	9,000	9,000	22,642	2,068	0	24,709	1	24,709	379,095	
1353.00.CC.0 RADIATOR	Eng - Hrs	22,831	18,000	18,000	15,300	4,135	0	19,435	1	19,435	42,266	
1355.00.CC.0 Thermostats	Eng - Hrs	461	6,000	6,000	274	500	0	774	3	2,321	2,782	

(Rpt_399) Maintenance Strategy Summary

Documents the maintenance strategy for an equipment or selected strategy tasks. The report encompasses a number of subsidiary reports showing the failure mode assumptions.

Initial parameters

|AutoLaunch|,|EquipmentTypes|,|EqpPlan|,|EqpPlanID|,|TaskHeader|,|TaskHeaderId|, |ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Task Type, Strategy Task, Warranty, Redo, Life Confirmed

Maintenance strategy

Maintenance strategy is an input to the standard job process. This process drives life-cycle costs, component schedules, parts forecasts, labour forecasts, maintenance plans and component risks and opportunities. The standard job, which defines the resources, costs and time required to perform the task, is then associated to the strategy task, giving it a much greater level of detail.

Maintenance strategy adjustment

Examples of strategy changes that can impact AMTs life cycle forecasting include the decision to:

- extend the engine life from 14,000 to 16,000 hours for all future engine changeouts.
- undertake 500 hour preventive maintenance services (as opposed to 250 hour).
- use rebuilt boom cylinders rather than purchase new.
- use a subcontractor to overhaul buckets, blades and beds.
- change rebuild centres.
- use cheap tracks vs tracks that can be overhauled (such as pin turn and rebush).

These are parameters that not only affect the next changeout, but every changeout until the end of the asset's life cycle.

Changes to the long-term strategy are made to the asset's current **projection**.

For Maintenance Contract equipment, the budget projection is only changed if:

- the asset is under a maintenance contract and the contract has been renegotiated.
- the baseline life cycle parameters for the asset have changed.

Changes to the budget projection usually require authorisation from a contract or central plant manager, or, depending on the structure of the organisation the Asset Management Group, after going through the change management process.

Optimising the long-term strategy in AMT is done by changing the asset's current projection parameters, which include:

- Planning vs Non-Planning Task and Lead Days.
- Scheduling Parameters and Logic, for example, First, Frequency and Suppression.
- Jobs to be performed and job costs.
- Labour Hours required to perform the Task and Duration.
- Probability of Occurrences and Failure modes.

You can make changes to the strategy tasks directly in the strategy task, or other special purpose editors as shown in the table below.

Type of Edit Required	AMT Function
Change to specific selected tasks.	Strategy Task Manager.
Change costs by a nominated factor across multiple tasks.	Multi-Task % Cost Adjuster.
Adjust parts prices on linked standard jobs.	Find the relevant Fleet using the Fleet Administration function under which the asset is located, edit and apply the price factors.
Change costs across multiple equipment in the same fleet.	Multi-Task Editor.
Change the characteristics of a strategy task across multiple assets.	Multi-Task Editor. Multi-Task Labour Editor. Multi-Task Scope Editor.

Type of Edit Required	AMT Function
Copy a strategy task to another projection.	Copy New Projected Task.
Delete a task from multiple equipment.	Multiple Equip Delete Task.
Update forecasted costs based on actual cost history.	Forecast Cost Adjuster.

(Rpt_400) Failure Mode Details

Shows the failure mode details set up in a Component Structure, and how the actual performance compares with the Component Structure assumptions.

Initial parameters

|AutoLaunch|,|EquimentTypes|,|EqpPlan|,|EqpPlanID|,|TaskHeader|, |TaskHeaderId|,|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

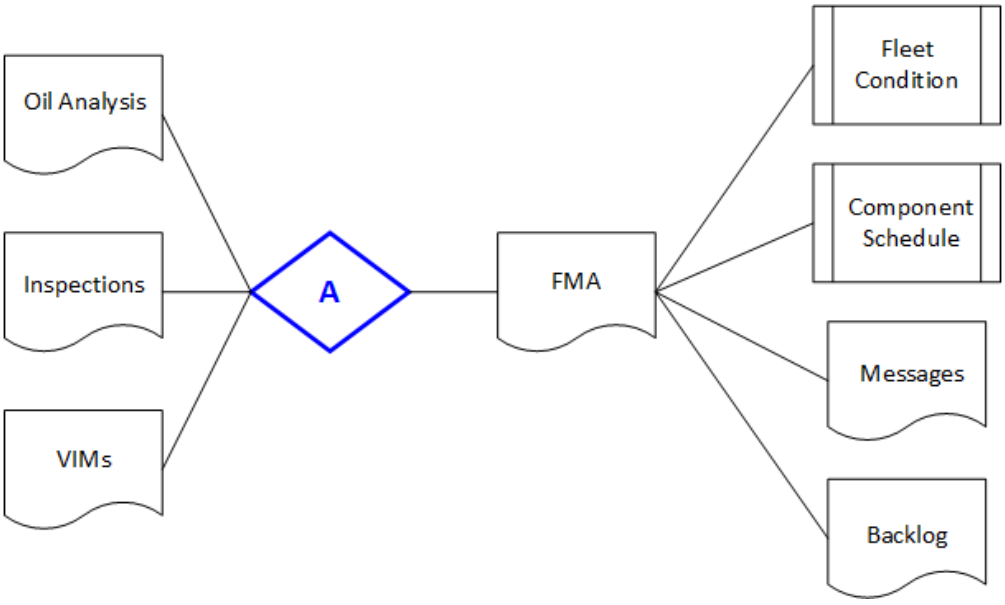
Default.

Additional filters

System, Sub System, Component Code, Task Type, Strategy Task, Warranty, Redo, Life Confirmed

Failure Mode Assessment (FMA)

Actions resulting from the FMA.



The failure mode assessment shows the details of the interpretation of the measurements (there could be many measurements for that failure mode assessment). You can update the review status (Not Reviewed, Reviewed, Closed) and enter review comments.

You can access the failure mode assessment from the AMT home screen by doing the following.

1. In the middle panel, click the arrow beside Notifications.
2. Right-click on the grid and select **Edit Failure Mode Assessment**.

You can also access the failure mode assessment from the **Long-Term Planning** function. The long-term planning function shows the status of the component and the trending of the failure mode assessment and measurements.

The Fleet Condition function presents a graphical view of the status of the fleet and components.

Decide on the maintenance strategy

For items identified in the asset structure, you need to determine and record the maintenance strategy for each item. The strategy generally falls into one of three categories.

1. On Condition.
    - There is a consequence of failure (safety, environment, production or repair cost).
    - There is a condition that will flag the potential failure, that condition can be practically monitored and will provide an appropriate lead time for action.
    - It is cost-effective - the failure consequence justifies the cost of condition monitoring.
2. On Schedule.
    - The task will optimise the life or reduce the risk of failure on other components, for example, a regular PM service with an engine oil change).

      OR
    - The risk of failure increases with age/use, and the scheduled occurrence represents the point where the risk (and consequences of failure) has increased to a point that justifies the cost of performing the task.
3. Run to Failure.
    - The consequences of failure are very low and do not justify On Condition or On Schedule tasks.
    - It is not practical to monitor a condition that will flag the potential failure.
    - There is no relationship between the age/use of the asset and probability of failure.

You can record the maintenance strategy for a component, and the reasons for selecting it, in the **Strategy** tab of the strategy task.

Determine task scheduling

Scheduling records when you plan to perform the strategy task. Considerations include the following.

1. Is the task date-based or usage-based? For example, certain safety inspections will be date-based, while a preventive maintenance service is generally usage-based.
2. If usage-based, what is the appropriate usage measure? For example, should an engine changeout be based on engine hours or the quantity of fuel consumed?

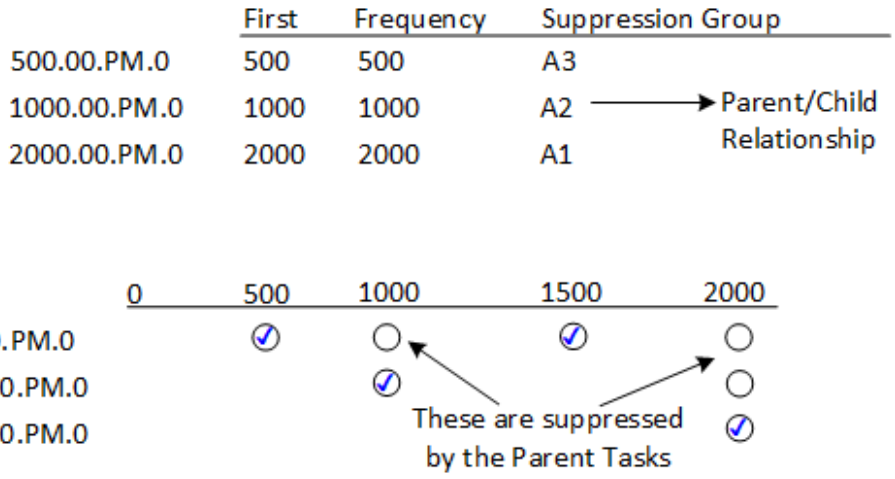
3. What is the appropriate scheduling method? In AMT there are a number of methods.
    - Normal.
    - Parent suppression.
    - Sibling suppression.
    - Dependency.
    - Inspection.

Scheduling parameters - Parent suppression

If you have tasks which are associated with each other, for example, preventive maintenance services or child components of a major component where the major component is the engine, and the minor components are the turbos and radiator, a form of suppression logic may be appropriate.

Parent suppression logic is based on the performance of the PARENT task. Parent tasks suppress any children tasks if they fall within a certain tolerance of the parent task. The performance of the parent task implies that the child task has been performed as well, and therefore the component life of the child task is reset along with the parent's.

With suppression, strategy tasks are grouped in a hierarchy so that A1 is the parent of A2 is the parent of A3 is the parent of A4 and so on. So when A1 occurs then this resets the life for all of its children. See the example below.



In this example, the three tasks are set up as preventive maintenance services, and are all within the suppression grouping "A". The 2000 hour service is the highest member of the group (sequence number 1). The 1000 hour service is the child of the 2000 hour service, and the 500 hour service is the child of the 1000 hour service.

This example illustrates that child tasks of a parent task are suppressed when AMT schedules them to occur at the same time. Consequently, (in this example) at the 1000 life-cycle point, the 1000.00.PM.0 task is scheduled to occur. Therefore, the 1000.00.PM.0 task is sent through to AMT's Planning module, and the 500.00.PM.0 is rescheduled based on that task's Frequency parameter.

Scheduling parameters - Sibling suppression

If you have tasks which are associated with each other, for example, preventive maintenance services or child components of a major component, a form of suppression logic may be appropriate.

Sibling suppression logic is where scheduled occurrences of future tasks are set so that they fall at the scheduled interval (or multiples thereof) after the most recent last occurrence of any projected task in the suppression grouping. This is used primarily for preventive maintenance scheduling where there is a requirement for maintenance to occur at a set scheduling interval. For example:

Task	First	Frequency	Last Performed	Future Occurrences		
500.00.PM.0	500	500	480		1480	
1000.00.PM.0	1000	1000		980		
2000.00.PM.0	2000	2000				1980

Scheduling Interval	500

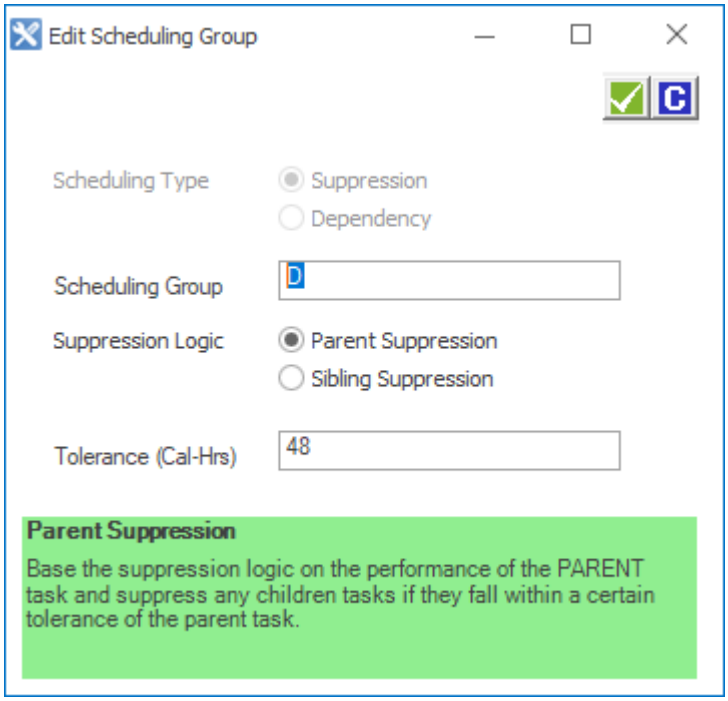
In the above example, future occurrences of tasks within the sibling group are based on the latest last performed task within the group PLUS the scheduling interval (which in this example is set to 500 hours). Calculation summary in order of preventive maintenance:

- PM 500: 480 (actual).
- PM 1000: 480 + 500%= 980.
- PM 500: 980 + 500% = 1480.
- PM 2000: 1480 + 500% = 1980.

% Scheduling Interval = 500

The performance of a parent task resets the life of any children tasks.

Scheduling Intervals for each suppression grouping are maintained in the system table **Scheduling Group**. Navigate to **System Administration** > **System Tables** > **Maintenance**



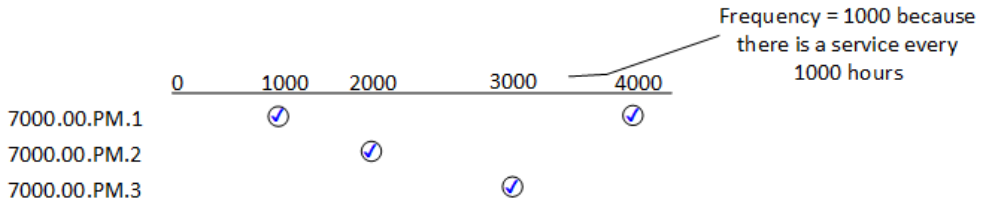
Scheduling - Dependency

If you have tasks which are always executed in a sequence, (for example, preventive maintenance service #1 followed by preventive maintenance service #2 followed by preventive maintenance service #3 and then start again at #1), dependency logic is appropriate.

Dependency is where tasks follow each other in a sequence where the lowest number is scheduled to occur first, followed by the second lowest. Once the cycle is complete, then this restarts from the lowest number.

The frequency parameter of the strategy task should be set to the interval after the performance of the preceding task.

	First	Frequency	Dependency Group
7000.00.PM.1	1000	1000	B1
7000.00.PM.2	2000	1000	B2
7000.00.PM.3	3000	1000	B3



In the above example, there are three preventive maintenance tasks which occur starting at 1000 hours, 2000 hours and 3000 hours. Each has been set up in the Dependency grouping “B” and the first task to be executed in the sequence is B1 (because it has the sequence number 1) followed by B2 and B3.

Once the initial round as been completed, that is, after B3 (the last in the sequence) is executed, AMT goes back to B1 and starts again, determined using the Frequency parameters.

AMT still requires a work order to be linked to the strategy task which was executed otherwise the strategy task will be **overdue** and AMT will not forecast the next occurrence (regardless of its scheduling logic).

Scheduling - Inspection

The next occurrence of this projected task will be scheduled at the nominated offset prior to the earliest next occurrence of the “related projected tasks”. The related projected tasks are the planning tasks for the nominated task type.

The inspection logic is used primarily for pre-preventive maintenance inspections where the pre-preventive maintenance inspection is scheduled at some offset prior to a preventive maintenance service.

AMT still requires a work order to be linked to the strategy task which was executed otherwise the strategy task will be **overdue** and AMT will not forecast the next occurrence (regardless of its scheduling logic).

Last performed unit of measure or date

Last performed unit of measure or date means that the forecast next occurrence of the strategy task is based on the last time the strategy task was actually executed.

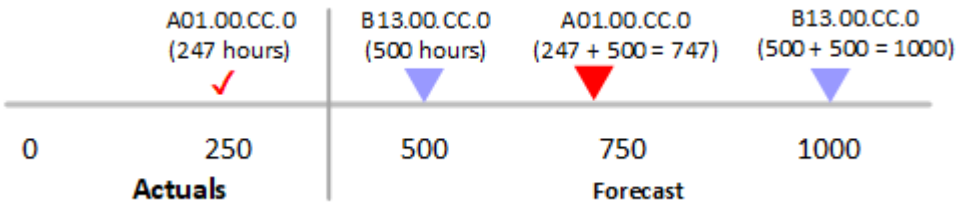
Last scheduled unit of measure or date

Last scheduled unit of measure or date means that the forecast next occurrence of the strategy task is based on when the strategy task was scheduled to occur. It ignores the usage or date when the job was actually performed.

Last performed scheduling

Last performed means that the forecast next occurrence of the strategy task is based on the last time the strategy task was actually executed. Below is a simplified example.

Example	First	Frequency	Scheduling Logic
A01.00.CC.0	250	500	Last Performed
B13.00.CC.0	500	500	Last Performed



In the above example, the projection is set up with the A1.00.CC.0 task forecast to be first performed at 250 hours and the next 500 hours thereafter. Therefore, the second occurrence of the A1.00.CC.0 task is forecast to occur at 750 hours.

However, A1.00.CC.0 is then actually performed on the equipment at 247 hours (3 hours earlier). Therefore, because the strategy task is based on the Last Performed Unit Of Measure, the next occurrence is forecast by AMT to occur at 747 hours (247 + 500). In other words, AMT takes into consideration when the task was actually performed when forecasting the next occurrence.



AMT still requires a work order to be linked to the strategy task executed, otherwise the strategy task will be overdue and AMT will not forecast the next occurrence (regardless of whether Last Performed or Last Scheduled logic is used).

Measuring performance

On an ongoing basis, for example, monthly or quarterly, you can make comparisons between actual performance and the expected position based on the strategy review. You can also make comparisons against the original budget as well, as this is just another type of projection.

The reports below can be used for comparison.

AMT Report	Description
Budget vs Current Projection Details Strategy Optimisation > Life Cycle Analysis	Even though the report title is Budget vs Current, you can compare any two projections to see differences in strategy task parameters (cost and frequency) and forecast cost.
Actual vs Archive Period Analysis Strategy Optimisation > Archiving	Even though the report title is Actual vs Archive, you can compare any two projections to see the differences in costs per period (month, quarter, year).
What-if - Cost Analysis Strategy Optimisation > What-if Analysis	This text report is similar to the Actual vs Archive report. However, instead of breaking down costs into blocks of time (for example, month, quarter, year) it breaks down costs into various cost categories (for example, equipment, component, system) over a user-specified range.
Variance Analysis Worksheet Strategy Optimisation > What-if Analysis	This text report allows you to analyse the causes of cost variances between the what-if projection and a second projection.
PCV Chart Strategy Optimisation > Life Cycle Analysis	Provides a high-level graphical snapshot of actual performance to date vs budget to date and projected.

Overdue strategy tasks

Overdue strategy tasks can be caused by a number of factors, for example a strategy task:

- Has been completed but AMT has not received any work order settlements from the enterprise resource planning system.
    - Use the Work Order Settlement Manager function in AMT to check for missing work order settlement.
- Has been completed, AMT has received work order settlements from the enterprise resource planning system, but the work order settlement did not have the correct coding to indicate new life.
    - Use the Work Order Settlement Manager function in AMT to review and change a work order settlement's codes.
- Has been completed but has not been linked to a work order settlement received from the enterprise resource planning system (this negates the issue of the work order settlement having incorrect coding in the previous point).
- Has not been completed by its strategy or target date and therefore is genuinely overdue.
    - In this case the work order for this strategy task needs to be rescheduled.

The frequency set for the strategy task in the current projection is unrealistic based on the existing maintenance strategy in place on site.

The remedy for these types of overdue strategy tasks may be to re-set the frequencies. Useful functions in AMT for this purpose are:

- Multi-Task Editor.
- Strategy Task Manager.
- Equipment Plan Editor (for adjusting the frequencies of a single equipment).

The effects of overdue strategy tasks are significant, for example, AMT will:

- Continue to flag the work order as yet to be performed (even though it might have been done), which results in a double-up of costs, and if unrecognised, may result in an additional, unnecessary component change out.
- Project the next occurrence of the strategy task inaccurately.
- Produce inaccurate forecasts (component, parts, rotable and financial).

If all processes are running smoothly, there should never be any overdue components.

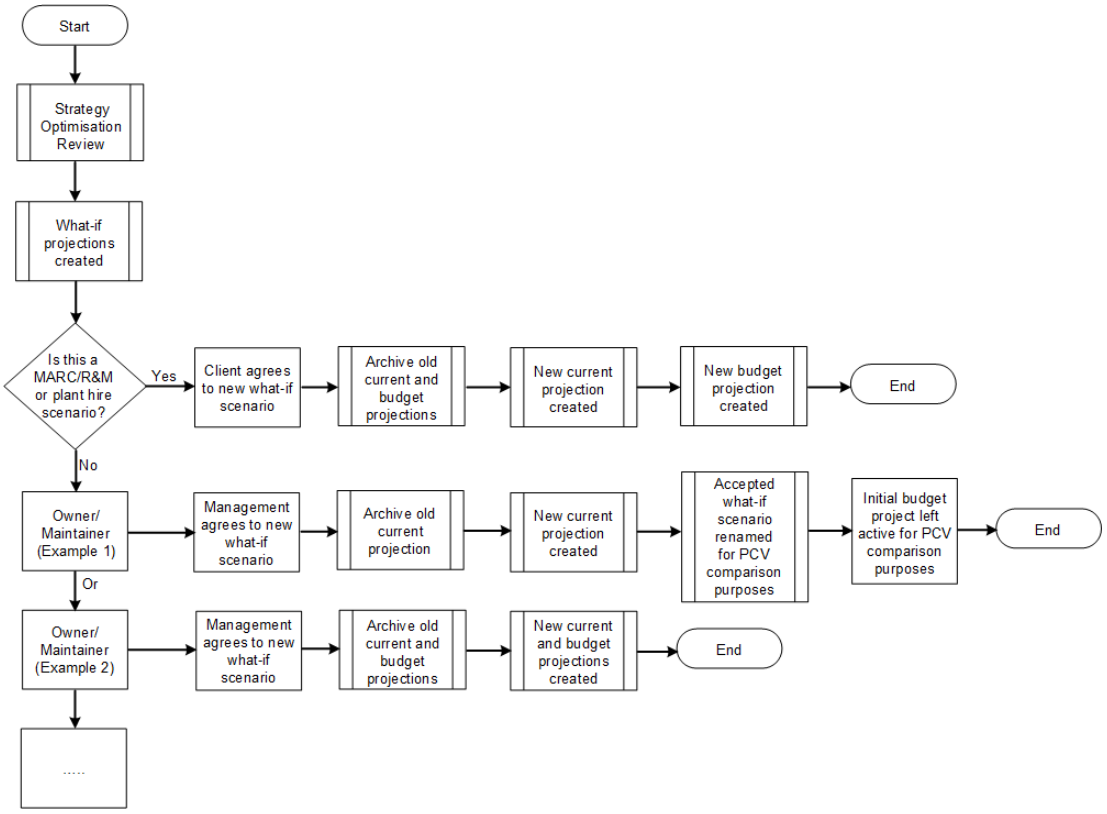
Monitoring process

Whenever you make significant changes to the strategy, make a baseline copy of the current projection so that you can measure future performance against the current projection.

There are different methods for managing these baseline projections, and each client needs to develop a business process that best suits them to manage these projections. This can be done by creating a new budget projection to compare against, and archiving the previous budget projection. Some potential overview processes are listed in the diagram below.

To assist in managing projections, incorporate a date and time stamp into the projection name to avoid any future confusion; for example, Truck 1 – Budget 20170925, Truck 1 – Alt2 20180912, and so on.

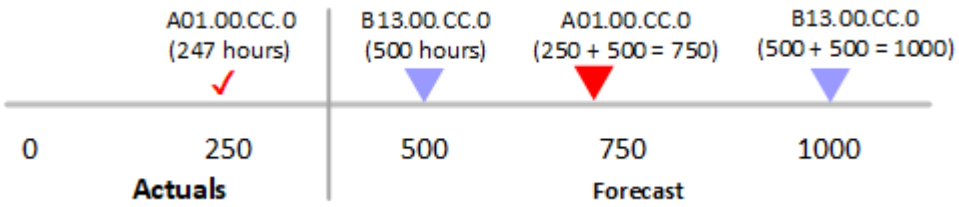
When to use budget projections, and the projection naming convention used is a business decision; personnel should conform to internal policy in relation to managing projections.



Last scheduled scheduling

Last scheduled means that the forecast next occurrence of the strategy task is based on when the strategy task was scheduled to occur. It ignores the usage or date when the actual job was performed. See the example below.

Example	First	Frequency	Scheduling Logic
A01.00.CC.0	250	500	Last Scheduled
B13.00.CC.0	500	500	Last Scheduled



In the above example, the projection is set up with the A1.00.CC.0 task forecast to be first performed at 250 hours and the next 500 hours thereafter. Therefore, the second occurrence of the A1.00.CC.0 task is forecast to occur at 750 hours.

The A1.00.CC.0 task is then actually performed on the equipment at 247 hours (3 hours earlier). However, because the strategy task is based on the Last Scheduled Unit Of Measure, the next occurrence is forecast by AMT to occur at 750 hours (250 + 500). In other words, AMT ignores when the task actually occurred when forecasting the next occurrence.

AMT still requires a work order to be linked to the strategy task executed, otherwise the strategy task will be overdue and AMT will not forecast the next occurrence (regardless of whether Last Performed or Last Scheduled logic is used).

Maintenance strategy base operating parameters

You should develop the maintenance strategy to deliver a defined level of performance for the asset (equipment). For example, a higher level of performance will be expected for a pump that is used continuously as compared to a standby pump.

The base operating parameters you need to understand are listed in the table below.

Parameter	Description
Target life	If it is a short period then you may make different component repair decisions as opposed to if it is a long period.
Criticality	What is the impact of a breakdown on safety, production and costs.
Required performance levels	What is the required availability, reliability and productivity of the asset.
Incremental productivity	Are there any benefits to exceeding the performance levels.

Non-Planning task

These are strategy tasks which do not require forward planning (for example, run to failure components such as lights and mirrors) or are tasks that represent provisions for general maintenance (for example, electrical general repairs, hydraulic system general repairs).

Such strategy tasks do not need to appear on the planner's screen (because they do not need preparation and there is no specific due date), so are forecasted in the background.

Nevertheless, these tasks have scheduling parameters like any other strategy task, and therefore become overdue. Yet the planner will not be aware when this happens because the tasks are hidden. Therefore, AMT implements an automatic reforecasting logic to "push out" these tasks. An example of the logic employed is explained below.

Assume there is a strategy task with a frequency of 2000 hours:

- If a non-planning task becomes overdue for the first time, the following formula is applied
  - Next Occurrence = (33% * Task's Frequency) + Task's Frequency, for example, (33% * 2000) + 2000 = 2666 (Next Occ)

If the same non-planning tasks becomes overdue again, then

- Next Occurrence = (33% * Task's Frequency) + Last Forecasted Occurrence,
  for example, (33% * 2000) + 2666 = 3332 (Next Occ)

If a non-planning task is performed, then the standard formula is applied

- Next Occurrence = Last Occurrence + Task's Frequency,
  for example, performed at 3000, therefore 3000 + 2000 = 5000 (Next Occ)

If the next occurrence then becomes overdue, AMT reschedules the next occurrence using the following formula

- Next Occurrence = (33% * Task's Frequency) + Last Occurrence + Task's Frequency,
  for example, (33% * 2000) + 5000 + 2000 = 7666 (Next Occ)

Unassigned tasks

The purposes of unassigned tasks are to forecast costs for general work on subsystems, and to capture actual costs of non-projected/strategy tasks into appropriate subsystems.

Unassigned tasks can only be non-planning tasks.

They cannot

- use advanced scheduling logic.
- be associated with a rotable part.
- have a rotable repair strategy.

You must consider these characteristics before creating a task in the projection. It is not possible to swap/change between normal and unassigned task modes once you have created the task.

Unassigned tasks utilise the unassigned ratio value (set at the equipment level) which determines for the unassigned calculation whether the unassigned costs are expected to occur consistently over the life cycle or whether the costs should be skewed toward the end of the equipment's life.

Life cycle analysis

Life cycle analysis examines the performance of equipment to date, in the future, and for the entire life of the equipment in order to identify the existence and cause of any significant cost variances from the budget, so that:

- Work order settlement mistakes can be identified and rectified.
- Budget structures can be tested and verified.
- Projection Cost Variances (PCV) can be identified and strategies developed to rectify.
- Necessary changes to future projections can be identified (that is, implement the strategy).

There are four main types of variance:

- Negative cost variances from the budget.
- Positive cost variances from the budget.
- Variances where actual costs exist with zero budget costs.
- Variances where budget costs exist with zero actual costs.

Each variance has to be examined to identify the cause, and only if applicable, the remedy.

To identify significant cost variances use the following functions in AMT.

- Life Cycle Cost Summary.
- PCV Analysis.
- Costs Analysis to Date.

- Strategy Task Analyser.
- Work Order Settlement Interrogation.

A number of reports in the Strategy module are designed to analyse historical and projected costs based on a single projection. Other reports are designed to compare the difference between two projections. In order to have confidence in both types of analysis, it is critical that the health of AMTs data is at an acceptable level. The primary health driver in this context is the quality and accuracy of transactional data (that is, work order related data) and projection assumptions (estimated cost and estimated life). Therefore, in order to understand the current position as reported by AMT, it is necessary to validate data in AMT. The AMT Health Assessment is the process through which the quality of the data in AMT can be assessed.

Refer to Health Assessment Process for more information.

(Rpt_217) PCV Chart

Lets you tailor a report which compares the current and projected costs of different projections. The report also has the flexibility to compare costs for different Cost Bearers within the selected properties.

Tips

You must choose a filter from the Equipment Hierarchy area (Branch, Site and so on) prior to selecting two projections to Analyse By.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,
|UserID|,|AutoLaunch|,|BranchID||SiteID|,|FleetID|,|EqpPlanID|,
|Branch|,|Site|,|Fleet|,|EqpPlan|,|Projection1ID|,|Projection1|,|Projection2ID|,
|Projection2|

Report parameters

Default.

Additional filters

Cost Type, System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Sheet, Activity, Costs

(Rpt_188) Life Cycle Cost Summary

Compares Cost to Date and Future Costs by Task and Sub System between Budget and Current projections. The report identifies cost variances by Task, and is used primarily to identify significant issues.

Applicable for those assets with Budget Protection. The PCV highlights projected cost over-runs from original/baseline assumptions.

Tips

Check for large variances between the Budget and Actual in the **To Date** columns. Large variances should be investigated further and the information qualified to assist decision making for the future costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Task Type, Strategy Task, UOM, Cost Bearer

(Rpt_216) PCV Analysis

Lists the projected cost variances based on the comparison of two projections selected by the user.

Tips

Run the report as a bar chart as it provides a Pareto analysis of the projected cost variances, that is, it will identify the top 10 projected variances, starting with cost overruns.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Sub System, Strategy Task, Industry Group, Contract Type, Customer, Cost Centre, Expense Element, Cost Selector, System, Component Code, Task Type, Business Group, Contract, Maintenance Type, Responsibility, Cost Bearer, Costs.

(Rpt_251) Variance Analysis Worksheet

Lets you analyse between projections and group by a maximum of five categories to show variances in a period of time - default date is last month but this can be modified as required.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Strategy Task, Task Type, Maintenance Type, Cost Bearer, Cost Centre, Responsibility, Expense Element, Labour Activity, Parts/Labour/Misc, Currency

(Rpt_135) Budget vs Current Projection Details

Lets the comparison of actual performance to the original budget.

Tips

Actual and projected costs are shown at cost at sell.

Report parameters

Default.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Additional filters

Cost Type, Component Code, System, Task Type, Sub System, Strategy Task, Cost Selector

(Rpt_150) Projection Comparison

Provides you with the ability to analyse equipment costs between the budget and current projections using numerous filters; a maximum of three filters in the Analyse by field. The report lets management to understand where cost overruns are occurring and to use other reports.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Cost Bearer, Costs, Component Code, Modifier Code, Task Type, Task Counter.

(Rpt_815) Life Cycle Cost Summary - @Cost

Compares Cost to Date and Future Costs by Task and Sub-system between Budget and Current projections. This report identifies cost variances by Task and is used primarily to identify significant issues.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

System, Sub System, Task Type Strategy Task, UOM, Cost Bearer

Health - actual costs to date

There are two areas that need to be reviewed when assessing the health of actual costs to date:

- Cost reconciliation to accounting system.
- Strategy tasks correctly coded.

Costs reconcile to accounting system

It is important to ensure that the actual costs in AMT balance to the costs recorded in the accounting system. The accounting reconciliation is typically done on a monthly basis. Site management should ensure that this reconciliation is being performed, and any reconciling items are corrected.

Strategy tasks correctly coded

If a strategy task is incorrectly coded, AMT does not recognise that it has been performed and will allocate the costs to "unassigned". This will both distort the analyses of the actual cost variance and cause the future projections to be inaccurate.

There are three key indicators that highlight incorrect coding:

- High actual cost variance in unassigned.
- % of totals costs in unassigned.
- Overdue components.

Each of these are explained below.

High actual cost variance in unassigned

The example report below highlights costs that should have been allocated to a strategy task that have in fact gone to unassigned.

Life Cycle Cost Summary

AMT INTELLIGENT ASSET MANAGEMENT

Hierarchy : [Equipment: TR01: 793C]
Date Range : [Entire Plan Term]
Additional Filters : [UOM *: Eng - Hrs] [Currency: AUD (Australian Dollars)]

Component Description	To Date			Future			Totals		
	Budget	Actual	Variance	Budget	Proj (Cur)	Variance	Budget	Proj (Cur)	Variance
4250.00.GR.0 BRAKING SYSTEM	50,841	1,158	49,683	10,348	11,523	-1,175	61,190	12,681	48,509
4251.LF.CC.0 BRAKE GRP	46,273	0	46,273	15,424	16,523	-1,099	61,697	16,523	45,174
4251.LR.CC.0 BRAKE GRP	55,961	0	55,961	18,654	0	18,654	74,615	0	74,615
4251.RF.CC.0 BRAKE GRP	46,273	0	46,273	15,424	16,523	-1,099	61,697	16,523	45,174
4251.RR.CC.0 BRAKE GRP	55,961	0	55,961	18,654	0	18,654	74,615	0	74,615
4268.00.CC.0 BRAKE PUMP	0	0	0	0	11,696	-11,696	0	11,696	-11,696
4268.MO.CC.0 BRAKE PUMP	38,163	0	38,163	9,541	0	9,541	47,704	0	47,704
Z024.00.UN.0 Brake System - Unassigned	0	25,789	-25,789	0	0	0	0	25,789	-25,789
Brake System	**293,473**	**26,948**	**266,525**	**88,045**	**56,265**	**31,780**	**381,518**	**83,213**	**298,305**
4300.00.GR.0 STEERING SYSTEM	38,704	1,572	37,132	7,878	8,251	-374	46,581	9,824	36,758
4303.00.GR.0 STEERING CYLD	5,140	0	5,140	0	5,909	-5,909	5,140	5,909	-768
4303.LT.CC.0 STEERING CYLD	18,235	0	18,235	6,078	2,713	3,366	24,314	2,713	21,601
4303.RI.CC.0 STEERING CYLD	18,235	0	18,235	6,078	2,713	3,366	24,314	2,713	21,601
4305.00.CC.0 STEERING LINKAGE	25,238	0	25,238	8,413	9,985	-1,573	33,650	9,985	23,665
4306.00.CC.0 STEERING PUMP	11,626	3,684	7,942	3,875	4,733	-858	15,501	8,417	7,084
Z025.00.UN.0 Steering System - Unassigned	0	4,350	-4,350	0	0	0	0	4,350	-4,350
Steering System	**117,178**	**9,606**	**107,572**	**32,322**	**34,303**	**-1,981**	**149,500**	**43,909**	**105,591**
R&M - Braking/Steering System	**410,651**	**36,553**	**374,097**	**120,368**	**90,569**	**29,799**	**531,018**	**127,122**	**403,896**
Z010.00.UN.0 Cooling System (Drive/Power Train) - Una	0	2,633	-2,633	0	0	0	0	2,633	-2,633
Cooling System (Drive/Power Train)	**0**	**2,633**	**-2,633**	**0**	**0**	**0**	**0**	**2,633**	**-2,633**
4050.LT.CC.0 FINAL DRIVE	231,930	129,984	101,946	77,310	129,396	-52,086	309,240	259,380	49,860
4050.RI.CC.0 FINAL DRIVE	231,930	120,916	111,014	77,310	129,396	-52,086	309,240	250,312	58,928
Z008.00.UN.0 Drive Train - Unassigned	0	41,223	-41,223	0	0	0	0	41,223	-41,223

% of Total costs in Unassigned

The target is for not more than 15% of total costs to be put in unassigned. It is very poor if more than 25% of total costs are in unassigned.

To calculate the percentage of total costs in unassigned you need to run the **Cost Analysis - To date** report and analyse by task type. Do this for each fleet.

Overdue components

The overdue components have in fact been performed but not recognised by AMT.

The overdue percentage is calculated as the number of overdue tasks versus all strategy items which have a planning task type.

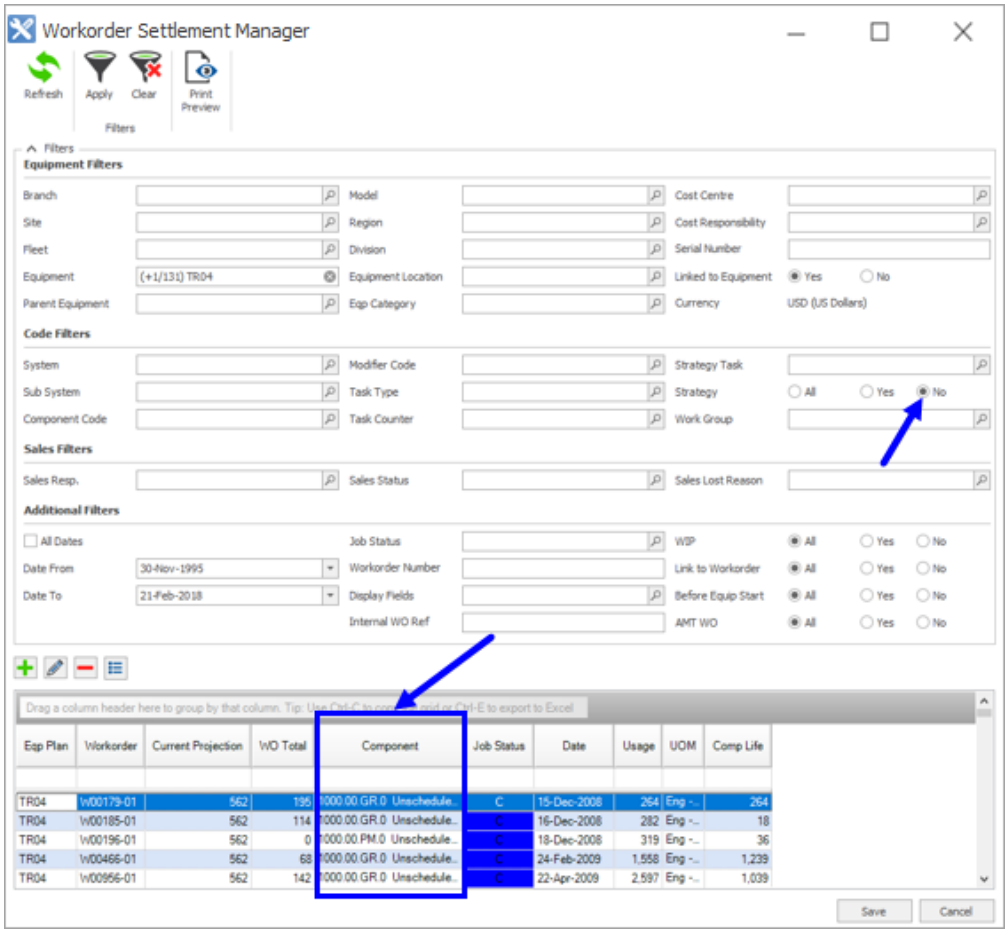
Overdue Task

AMT INTELLIGENT ASSET MANAGEMENT

Hierarchy : [Dealer: Company A] [Branch: Central Branch] [Site: North Site] [Fleet: Wheel Loaders]
Additional Filters :

			Last Projected Occurrence				
Fleet - Equipment	Component	Occurence Type	Plan Usage	QUOM	Plan Date	Planned Status	Cost
Wheel Loaders - LD0	4267.00.CO.0 PARKING BRAKE	On Condition	29,205	Eng-Hr	12-Aug-03	✔	3,932
	4268.US.CC.0 BRAKE PUMP	Before Failure	10,000	Eng-Hr	24-Jun-00	✔	980
	4312.00.CC.0 HMU	Before Failure	20,000	Eng-Hr	21-Feb-02	✔	952
	5073.PD.CC.0 GEAR PUMP	(NONE)	10,000	Eng-Hr	24-Jun-00	✔	830
	7524.00.PM.0 250 SVC HOUR MAINTENANCE	Service	30,997	Eng-Hr	25-Dec-03	✔	718
	7525.00.PM.0 500 SVC HOUR MAINTENANCE	Service	30,811	Eng-Hr	14-Dec-03	✔	999
	7542.00.PPI.0 SCHEDULED OIL SAMPLING(SOS)	Service	30,561	Eng-Hr	29-Nov-03	✔	0
Total - Wheel Loaders - LD001							**8,411**
Wheel Loaders - LD0	1358.00.CO.0 BELT TIGHTENER	(NONE)	14,574	Eng-Hr	23-Mar-01	✔	892
	4263.00.CC.0 BRAKE ACCUMULATOR	(NONE)	10,000	Eng-Hr	17-Jun-00	✔	1,828
	4268.US.CC.0 BRAKE PUMP	Before Failure	10,000	Eng-Hr	17-Jun-00	✔	980
	4307.00.CO.0 STEERING VALVE	(NONE)	10,000	Eng-Hr	17-Jun-00	✔	2,290
	4307.SE.CO.0 STEERING VALVE	(NONE)	10,000	Eng-Hr	17-Jun-00	✔	2,293
	4312.00.CC.0 HMU	Before Failure	20,000	Eng-Hr	16-Jan-02	✔	952
	5073.PD.CC.0 GEAR PUMP	(NONE)	10,000	Eng-Hr	17-Jun-00	✔	830
	5084.CE.CC.0 HOIST PUMP	Before Failure	30,781	Eng-Hr	09-Nov-03	✔	10,250
	7524.00.PM.0 250 SVC HOUR MAINTENANCE	Service	31,603	Eng-Hr	27-Dec-03	✔	718
→	7542.00.PPI.0 SCHEDULED OIL SAMPLING(SOS)	Service	31,353	Eng-Hr	13-Dec-03	✔	0
Total - Wheel Loaders - LD002							**21,033**

To correct incorrect coding, open the Work Order Settlement Manager, and use the filters to identify the specific work orders. Double-click on any specific work order to edit the coding. You can also use Work Order Settlement Manager to identify incorrect coding. Filter for high value non-strategy items and vice versa.



In the example screenshot above, many of the work orders listed are probably strategy tasks that had incorrect modifier codes.

Health - projected costs

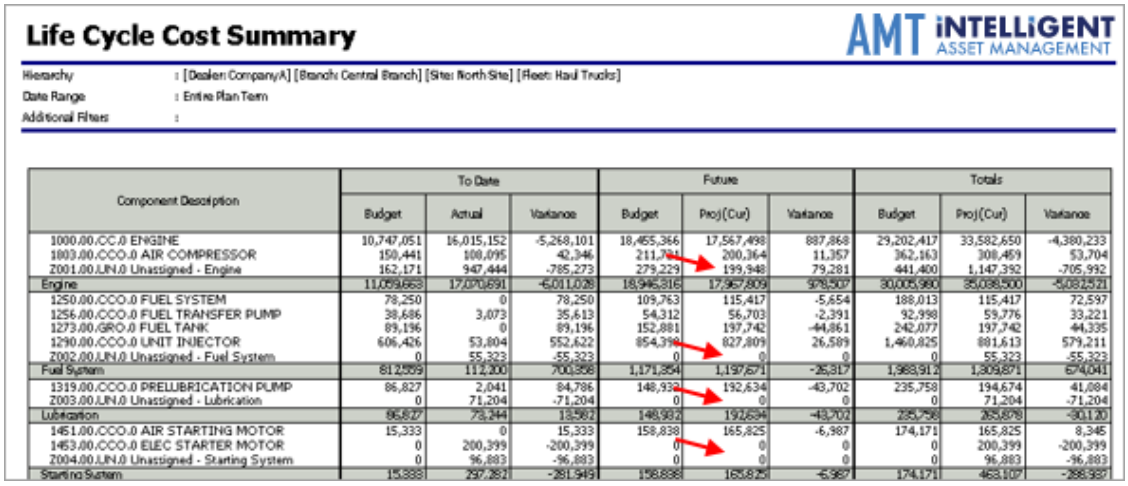
Projection parameters are realistic (expected component lives and costs). Indicators of problems with projected cost can include:

- Inconsistencies between actual cost variance and projected cost variance.
- Variances between actual average lives and costs and projected lives and costs.
- Strategy tasks correctly coded.
- Short-term strategy is not being maintained.
- Overdue Items.

Each of these are explained in more detail below. Click the links to expand them.

Inconsistencies between actual cost variance and projected cost variance

At a high level, if actual costs to date are 20% over budget you would expect future costs to be about 20% over budget (unless there are some significant changes or extenuating circumstances). Review the Life Cycle Cost Summary report looking for inconsistencies between the actual cost variance and projected cost variance.



Life Cycle Cost Summary

AMT INTELLIGENT ASSET MANAGEMENT

Hierarchy : [Dealer:Company A] [Branch: Central Branch] [Site: North Site] [Fleet: Haul Trucks]
Date Range : Entire Plan Term
Additional Filters :

Component Description	To Date			Future			Totals		
	Budget	Actual	Variance	Budget	Proj(Cur)	Variance	Budget	Proj(Cur)	Variance
1000.00.CC.0 ENGINE	10,747,051	16,015,152	-5,268,101	18,455,366	17,567,498	887,868	29,202,417	33,582,650	-4,380,233
1003.00.CCO.0 AIR COMPRESSOR	150,441	100,095	42,346	211,263	200,364	11,357	362,163	308,459	53,704
Z001.00.UN.0 Unassigned - Engine	162,171	947,444	-785,273	279,229	199,948	79,281	441,400	1,147,392	-705,992
Engine	11,059,663	17,070,691	-6,011,028	18,945,816	17,967,809	978,507	30,005,980	35,038,500	-5,032,521
1250.00.CCO.0 FUEL SYSTEM	78,250	0	78,250	109,763	115,417	-5,654	188,013	115,417	72,597
1256.00.CCO.0 FUEL TRANSFER PUMP	38,686	3,073	35,613	54,312	56,703	-2,391	92,998	59,776	33,221
1273.00.GRO.0 FUEL TANK	89,196	0	89,196	152,881	197,742	-44,861	242,077	197,742	44,335
1290.00.CCO.0 UNIT INJECTOR	606,426	53,804	552,622	854,396	827,809	26,589	1,460,825	881,613	579,211
Z002.00.UN.0 Unassigned - Fuel System	0	55,323	-55,323	0	0	0	0	55,323	-55,323
Fuel System	812,559	112,200	700,358	1,171,354	1,197,671	-26,317	1,983,912	1,309,871	674,041
1319.00.CCO.0 PRELUBRICATION PUMP	86,827	2,041	84,786	148,932	192,634	-43,702	235,758	194,674	41,084
Z003.00.UN.0 Unassigned - Lubrication	0	71,204	-71,204	0	0	0	0	71,204	-71,204
Lubrication	86,827	73,244	13,582	148,932	192,634	-43,702	235,758	265,878	-30,120
1451.00.CCO.0 AIR STARTING MOTOR	15,333	0	15,333	158,838	165,825	-6,987	174,171	165,825	8,345
1453.00.CCO.0 ELEC STARTER MOTOR	0	200,399	-200,399	0	0	0	0	200,399	-200,399
Z004.00.UN.0 Unassigned - Starting System	0	96,883	-96,883	0	0	0	0	96,883	-96,883
Starting System	15,333	297,282	-281,949	158,838	165,825	-6,987	174,171	463,107	-288,937

See the items highlighted. The projected future costs appear to be unrealistically low when recognising their actual costs to date.

Variances between actual average lives and costs and projected lives and costs

The fleet projection screen in AMT should be reviewed, identifying where average occurrences and costs differ significantly from the budget for major components (as they usually make up 60% of the overall budget cost). The projection parameters for the current projection should then be altered to account for what is actually happening. Double-click on any specific work order to edit the coding.

Strategy tasks correctly coded

If a strategy task is incorrectly coded then AMT does not recognise that it has been performed and will allocate the costs to 'unassigned'. This will both distort the analyses of the ACV and will cause the future projections to be inaccurate See Validating actual costs.

Short-term strategy is not being maintained

It is important that the short-term strategy in AMT reflects the actual work plan. As specific components are moved in and out, this needs to be reflected in AMT. Therefore, items within the planning window that have not been reviewed are not included.

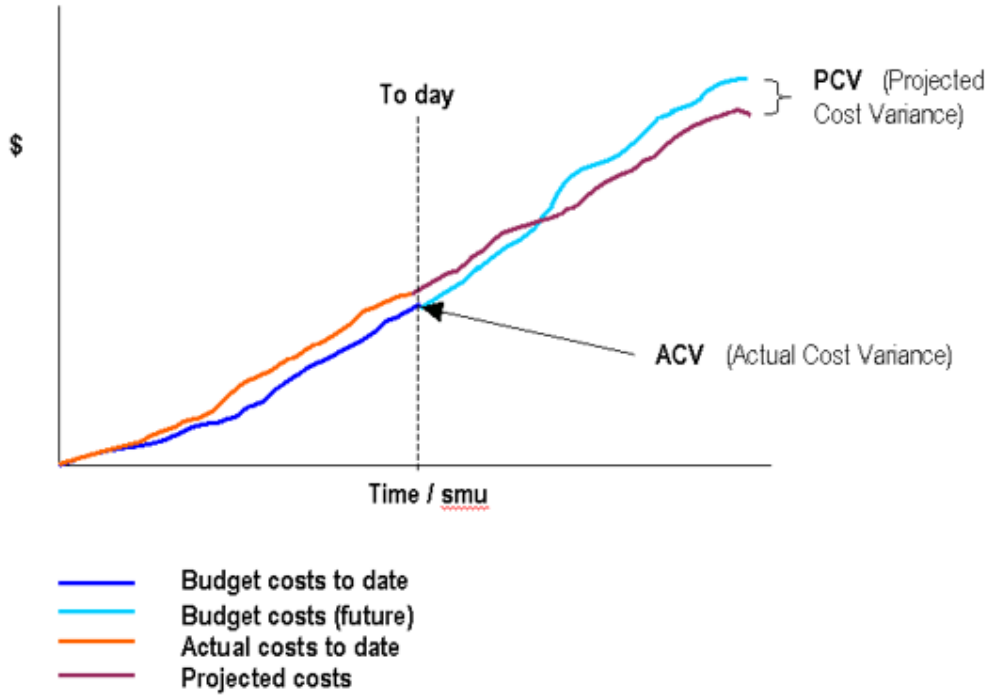
The Action Required % is calculated as the number of action required tasks or overdue tasks versus all strategy items with a Planning task type.

Overdue items

Overdue items are either items that have been done but incorrectly coded so AMT has not recognised them (see Validating actual costs), or items that are showing as overdue because AMT has not been updated with the planned perform date.

The Overdue % is calculated as the number of overdue tasks versus all strategy items with a Planning task type.

Health assessment process



The chart above summarises the base data in AMT primarily used by maintenance and finance groups:

- Budget (to date and future) – this is the original budget costs adjusted by internal parts and labour escalations (budget projection in AMT).
- Actual costs to date – work order costs imported from the accounting system.
- Projected costs – Projected future costs (current projection in AMT).

For the reporting and analyses in AMT to be meaningful, the above sources of data need to be accurate. It is critical that this data is accurate because it not only impacts the maintenance strategy, it also impacts financial reporting, budgeting and forecasting.

Cost Analyses

(Rpt_146) Cost Analysis - "Life Cycle"

Shows graphically, or in a text report, the major cost areas of the equipment for the entire plan term or to the month to date.

Run the report for the life-cycle or a shorter period (for example, a year). Running as a Pie Chart, "Analyse By" Component shows the top 20 component cost areas.

Tips

Can be run in Text, Bar or Pie chart format. Bar Chart shows the top 10 costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|CostBearerID|,|ProjectionTypes|,|EquipmentTypes|,
|ProjHeaderID|,|ProjHeader|,|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,
|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|,
|AnalyseBy|,|CurrencyID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, Work Group, System, Cost Centre, Sub System, Cost Category, Strategy Task, Expense Element, Strategy Task Desc, Maintenance Type, Task Type, Task Mode, Component Code, Costs, Modifier Code, Parts/Labour/Misc, Cost Bearer, Currency, Task Counter

(Rpt_147) Cost Analysis - "To Date"

Lets you list and analyse cost values in descending value based on a wide variety of filters. This report is for analysis of costs to date, not for future projected costs to date, not for future projected costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Contract, Cost Type, Strategy Task, System, Occurrence Type, Sub System, Modifier Code, Component Code, Task Counter, Task Type, Cost Bearer, WO Status, P/L/M

(Rpt_243) Unassigned Cost Analysis

Allows the unassigned task costs in the current projections to be analysed in either a text or a graphical view. The "Report On" parameter provides an option to use the ERP codes on work order settlements or the AMT codes the work order settlement is matched with. AMT codes exist for component code, modifier code and task type, as well as cost type, system and sub system, as these are determined from the component hierarchy.

"Analyse By" Component Code identifies the components that have the highest unassigned costs.

Tips

The filters allow a thorough dissection of the costs by different cost bearers and by parts, labour and misc tasks.

This report always shows the total cost of the Job at the Job (Work order) Date, even though some reports may split these costs over different transactions.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Contract, Cost Type, Component Code, Modifier Code, Task Type, Report On, System, Sub System, Cost Bearer, WO Status, Task Counter, P/L/M

Analysis of Projection Assumptions

In simple terms, Projection Assumptions are the parameters which determine at what points in time a strategy task is forecast, and how much it is estimated to cost per occurrence. These parameters are stored in the **Scheduling** and **Job Details** tabs when editing strategy tasks.

The Strategy module contains functionality which helps you validate the accuracy of these parameters, by checking whether the assumptions and estimations are reasonable. **Strategy Task Analyser** is the key function in this area (**Strategy Optimisation** > **Analysis of Projection Assumptions**.

Using **Strategy Task Analyser** you can evaluate the strategy task assumptions against actual history. An important acronym to understand on the **Strategy Task Analyser**report is **ACI**, which means Annualised Cost Impact. Annualised Cost Impact quantifies how the strategy task assumptions differ from actual historical performance. This is used to prioritise the review of the strategy task assumptions.

Negative ACI numbers reflect a predicted negative impact on future costs, due to disconnect between actual performances to data based on actual work orders, and estimated performance in the future, based on strategy tasks projection assumptions. In other words, if current performance to date does not align to the expected performance (based on strategy tasks parameters) then there is a disconnection. ACI extrapolates that disconnect in the future, and is reflected by negative costs.

For example, a engine's life is estimated to be 15,000 hours and costs $100,000 to replace. History shows the engine has an average life of 8,000 hours and costs $150,000 to replace, so you are replacing the engine more

regularly and at a higher cost than originally estimated. This scenario would be reflected with a negative ACI amount.

Columns on the Strategy Task Analyser report are described below.

ACI Field	Description
UOM	Unit of measure of the strategy task.
Actual Occs	Number of actual occurrences of the strategy task based on the work order settlement history linked to the strategy task (it does not include occurrences of a Parent strategy task if the strategy task is part of a suppression or suppression advanced scheduling group).
Capex	Identifies the strategy task as a Capital (Capex) or Operating (Opex) expense.
SJ	Indicates whether the strategy task is linked to a standard job.
SD	Scheduling Details. Indicates whether the strategy task has advanced scheduling applied, and if so, is the strategy task a Parent or Child in the scheduling group. This is an indicator only. Calculations within the grid (such as average actual frequency, average actual cost and so on) are not impacted by this relationship, which is standard for AMT reports. M = Multiple combinations of Suppression or Dependency in use. Sp = Suppression - Parent. Sc = Suppression - Child. D = Dependency.
ACI Frequency	The additional cost (if a negative number) over and above the current forecasted cost of the strategy task **if** the Actual Average Frequency were to be the basis for the strategy task's estimated life in the future. In other words, if you changed the Frequency parameter of the strategy task to equal the Actual Average Frequency, then the projected future cost of the strategy task would increase by the ACI Frequency value.  This statistic is based on the work order settlement history linked to the strategy task.
ACI Cost	The additional cost (if a negative number) over and above the current forecasted cost of the strategy task IF Actual Average Cost were to be the basis for strategy task's estimated job cost in the future.
ACI Total	Combined impact of ACI Cost and ACI Frequency (that is, ACI Cost + ACI Frequency = ACI Total) on future costs.
ALI Labour	The additional labour hours (if a negative number) over and above the current forecasted labour hours of the strategy task IF the Actual Average Labour Hours

ACI Field	Description
	were to be the basis for the strategy task's estimated labour hours in the future.
ALI Total	Combined impact of Frequency and Labour (that is amount of additional labour hours in the future based on Actual Average Frequency and Actual Average Labour Hours).
ALI Duration	The additional duration (if a negative number) over and above the current forecasted duration of the strategy task IF the Actual Average Duration were to be the basis for the strategy task's estimated duration in the future.
ADI Total	Combined impact of Frequency and Duration (that is, the amount of additional labour hours in the future based on Actual Average Frequency and Actual Average Duration).
Proj vs Actual / UOM Variance	Variance per UOM of the existing Project Amount vs the hypothetical Projected Amount based on Actual Averages.

Strategy task analyser



Strategy Task Analyser replaces Multi-Equipment Projection Editor and Component Variance Analysis.

Navigate to **Strategy Optimisation** > **Analysis of Projection Assumptions** > **Strategy Task Analyser**.

Annualised Cost Impact (ACI) quantifies how the strategy task assumptions differ from the actual historical performance. Use this number to prioritise the review of strategy task assumptions.

Calculations

ACI Frequency = Current Proj Cost / Current Proj Frequency * Annual Utilisation – Current Proj Cost / Actual Avg Frequency * Annual Utilisation

ACI Cost = Current Proj Cost / Actual Avg Frequency * Annual Utilisation – Actual Avg Cost / Actual Avg Frequency * Annual Utilisation

ACI Total = Current Proj Cost / Current Proj Frequency * Annual Utilisation – Actual Avg Cost / Actual Avg Frequency * Annual Utilisation

ALI Labour = Current Proj Labour Hours / Current Proj Frequency * Annual Utilisation – Current Proj Labour Hours / Actual Avg Frequency * Annual Utilisation

ALI Total = Current Proj Labour Hours / Current Proj Frequency * Annual Utilisation – Actual Avg Labour Hours / Actual Avg Frequency * Annual Utilisation

ADI Duration = Current Proj Duration / Current Proj Frequency * Annual Utilisation – Current Proj Duration / Actual Avg Frequency * Annual Utilisation

ADI Total = Current Proj Duration / Current Proj Frequency * Annual Utilisation – Actual Avg Duration / Actual Avg Frequency * Annual Utilisation

Utilisation Analyser

This function allows you to compare Actual Utilisation with Forecast Utilisation and quickly make adjustments to Forecast Utilisation.

If an equipment is not utilised at the same rate as the Annual Utilisation (Simple Util Eqp) or the Usage Profile (Advanced Util Eqp), the Strategy Task Target Date may not be an accurate indication of when the Strategy Task should be performed. Utilisation Analyser is designed to identify equipment where Actual Utilisation differs from the Forecast Utilisation and allows you to quickly update Forecast Utilisation.

You can analyse and adjust equipment configured for Simple Utilisation, which means changes are applied to selected, individual equipment.

You can adjust Usage Profiles, which are linked to equipment configured for Advanced Utilisation. This means changes made will update the selected usage profiles, which in turn update the usage forecast of equipment linked to those usage profiles.



When changes are made to Usage Profiles, the change is only applied to the current usage profile step only, not future steps.

Utilisation filters

- **Days to Analyse Actuals** - how, back in history from today, should actual usages records be included in the analysis.
- **Variance Greater Than** - only identify variances of actual usage history versus future usage forecast greater than a specific usage amount.
- **Variance% Greater Than** - only identify variances of actual usage history vs future usage forecast greater than a specific percentage.

The calculation AMT performs is similar for **Simple Utilisation** and **Advanced Utilisation**. AMT first calculates the Daily Average Actual Usage based on the **Days to Analyse Actuals** filter. This daily rate is used to calculate the Annual Average Actual Usage. The Annual Average Actual Usage rate is then compared to the equipment's current Annual Forecast Usage (Simple Utilisation) or usage profile's Current Annual Forecast Usage (Advanced utilisation). The usage variance amount is displayed in the Annual Variance column. This is where the **Variance Greater Than** parameter filters results. The usage variance percentage is displayed in the Annual Variance% column. This is where the **Variance% Greater Than** parameter filters results.

The **Simple Utilisation** > **Set Annual Utilisation To**, or the **Advanced Utilisation** > **Set Annual Forecast Usage To** fields are automatically populated with the value from the **Annual Average Actual Usage** field, implying that you may want to change the forecast usage rate to align to the historical annual usage pattern. Of course, you may not wish to make a change at all, or set a different value. Either way, if a change it to be made, select the **Select** check box, adjust the **Set Annual Forecast Usage/Utilisation To** value as needed, then click **Apply Changes** to update the Equipment or Usages Profiles.



To see which equipment are linked to a usage profile prior to making a change, on the Advanced Utilisation tab, highlight the Usage Profile row, click **Actions** and select **Equipment List**.

(Rpt_171) Fleet Component Performance Summary

Assists users in forecasting the correct projection by reviewing the strategy tasks with this historical data.

Tips

Compares averages to current strategy parameters. Significant differences suggest the current strategy parameter should be changed.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|AutoLaunch|,
|ModelID|,|Model|,|SiteID|,|Site|,|FleetID|,|Fleet|,|SytemID|,|System|,
|BranchId|,|Branch|,|EqpPlanId|,|EqpPlan|,|SubSytemId|,
|SubSystem|,|CompCodeId|,|CompCode|,|TaskTypeId|,|TaskType|,
|Warranty|,|Redo|,|LifeConfirmed|,|UserID|

Report parameters

Report Type, Equipment Hierarchy

Additional filters

Cost Type, Cost Selector, Sub System, Warranty, Task Type, Redo, System, Life Confirmed, Component Code, Suppression Impact

(Rpt_323) Component Variance Analysis Report

This report has two sections:

1. The top section of the report provides a summary of the projected task's performance.
2. The bottom section of the report provides an analysis of the projected task's performance per equipment within the selected fleet.

Tip

This report helps to identify equipment which are experiencing a greater number of occurrences (such as component changeouts) compared to other equipment in the fleet. In other words, it is easy to identify "problem equipment".

Initial parameters

|AutoLaunch|,|Branch|,|BranchID|,|Site|,|SiteID|,|Fleet|,|FleetID|,|Model|,
|ModelID|,|EqpPlan|,|EqpPlanID|,|Component|,|ComponentCodeID|,|
ModifierIDNN|,|TaskTypeIDNN|,|ApplicationCodeIDNN|,|UserID|

(Rpt_324) Component Variance Summary

For the selected fleet and or projected tasks, this report shows Costs To Date (Budget vs. Actual, and Variance) and Projected Costs (Budget vs. Projected, and Variance).

Tip

This report is a simpler version of the Life Cycle Cost Summary report ("Future" columns in the Life Cycle Cost Summary report are excluded), and helps the user quickly identify Top 10 variances.

Initial parameters

|AutoLaunch|,|Branch|,|BranchID|,|Site|,|SiteID|,|Fleet|,|FleetID|,|Model|,
|ModelID|,|EqpPlan|,|EqpPlanID|,|CostType|,|CostTypeID|,|System|,|SystemID|,
|SubSystem|,|SubSystemID|,|Top|,|UnassignedVal|,|UnassignedDesc|,|
PlannedVal|,|PlannedDesc|,|SortVal|,|SortDesc|,|UserID|

Report parameters

Default.

Labour

Strategy optimisation - Labour analysis

The Strategy module enables you to forecast expected labour requirements based on labour data in the strategy task, and compare estimated to actual, where actual is captured on work orders.

The labour elements stored in strategy tasks are:

- Labour Hours.
- Labour Activity.
- Labour Type.
- Labour Split.
- Labour Travel Cost (OEM and Plant Hire modules).

(Rpt_182) Labour Hours Analysis

Analysis of where time is being spent (particularly of breakdown labour) to identify product problems.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Contact, Strategy Task, Occurrence Type, Modifier Code, Task Counter, Cost Bearer, Cost Type, System, Sub System, Component Code, Task Type, WO Status

(Rpt_185) Labour Hrs Variance Analysis

Lets you analyse the actual labour hours required to complete strategy tasks compared to the target labour hours which have been set by the strategist or planner.

Shows actual average labour hours against the target. Identifies:

- incorrect targets (such as in Standard Jobs).
- inefficiencies due to low skill levels or poor task instructions.

Navigate to **Strategy Optimisation** > **Labour**.

Tips

The report can be run for all tasks or only those with set labour hours linked to work orders.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Work Group, Planned, Breakdown, Strategy Task, Component Code, Task Type, Modifier Code, Employee, Show

(Rpt_832) Workload Optimisation

Compares Labour Hours by Month, Quarter or Year for two or more projections.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity

Component Risk

Risks and opportunities

Variations in the strategy of a piece of equipment, or the timing of actual or future projected strategy task occurrences, could result in a number of potential risks and opportunities. These risks and opportunities need to be identified, quantified, and used as inputs into further strategy adjustments in order to limit risks and maximise opportunities.

- Risks are defined as the costs of strategy tasks, which are currently scheduled to occur just outside the end date of an equipment plan, but may, if the current strategy frequency is not achieved, fall inside the equipment plan period.
    - Run the High Risk report in AMT to see risk components.
- Opportunities are defined as the costs of strategy tasks, which are currently scheduled to occur just inside the end date of an equipment plan, but could, if the strategy frequency is extended, fall outside the equipment plan period.
    - Run the High Opportunity report in AMT to see the opportunities.



The realisation of opportunities and the avoidance of risks are sometimes conditional on the contractual obligations between maintenance supplier and equipment owner. Before assessing either opportunity or risk it is recommended that contractual arrangements be assessed, particularly in relation to residual value in components clauses and “Pain and Gain” clauses.

(Rpt_177) High Opportunity

Identifies components due for change just short of the contract end, and calculate the new frequency required to ensure the opportunity can be realised.

Identifies the quantity and value (at selling price) of components which are due for replacement within the contract period, but if managed properly, their life could be extended. Costs are saved by not having to replace the component.

Tips

Displays the last occurrence prior to equipment termination, the component life required and the cost benefit.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|MaxCompLife|,|MinCompCost|,
|LogoLocation|,|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,
|Branch|,|Site|,|Fleet|,|EqpPlan|,|AnalyseBy|,|ProjectionHeaderID|,
|ProjectionHeader|,|Currency|

Report parameters

Default.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, Min Comp. Cost, Max Life Used %, Min Comp Cur

(Rpt_178) High Risk

Identifies the risk components and calculates the life buffer available to prevent an additional component changeout prior to the equipment termination.

Identifies the quantity and value (at selling price) of components that are due to be replaced just after the contract concludes.

High risk contracts should include additional maintenance provisions and have the contract end date renegotiated.



This is an important report, and it should be approved by management before the contract is signed.

Tips

Shows the next projected occurrence after the equipment termination, the component life to avoid and the cost risk.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|MaxCompLife|,|MinCompCost|,
|LogoLocation|,|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,
|Branch|,|Site|,|Fleet|,|EqpPlan|,|AnalyseBy|,|ProjectionHeaderID|,
|ProjectionHeader|,|Currency|

Report parameters

Default.

Additional filters

Task type, Cost Type, System, Sub System, Component Code, Min. Comp Cost, Max Life Left %, Min Comp. Cur

Component Reliability Analysis

(Rpt_238) Technical Performance Analysis

Lets you analyse the specific areas of the downtime relating to parts, occurrence types and so on, and shows the related costs, duration and lives achieved from components.

Run for each major component in a Fleet. "Analyse By" Occurrence Type (Failure Mode). Shows the average life/cost for each Failure Mode.

Tips

This detailed information can be used by product and parts support groups and factories to understand the causes of failure and the corresponding parts. The parts information is derived from the EQS/Task/Strategy level of the program.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Component Code, Modifier Code, Occurrence Type, Task Type, Task Counter, Original Part Type, Warranty, Redo, Life Confirmed

(Rpt_239) Technical Performance Analysis Chart

Provides a graphical view of specific technical areas of the downtime - occurrence type, cause, symptom and repair codes.

"Analyse By" original part type to identify components with poor life expectancy and then review the associated rebuild centres.

Tips

Allows quick analysis of performance in the downtime area. An example could be: is maintenance work being performed before or after failure on a regular basis and are users being specific when entering downtime data to assist with later analysis.

Requires the Part Type system table to be set up correctly with the associated rebuild centre.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Component Code, Modifier Code, Occurrence Type, Task Type, Task Counter, Original Part Type, Warranty, Redo, Life Confirmed

(Rpt_240) Technical Performance Details

Similar to the Technical Performance Report, but has set parameters. The report provides you with information relating to parts, occurrence types, costs and durations.

Tips

This report shows the work order number relevant to the downtime task. The report always shows the total cost of the Job at the Job (work order) Date, even though some reports may split these costs over different transactional months (for the Transaction Date reporting).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Additional filters

Component Code, Modifier Code, Occurrence Type, Task Type, Task Counter, Original Part Type, Warranty, Redo, Life Confirmed

(Rpt_142) Component Life Distribution

Reports allows analysis in a graphical view of the component lives achieved and quick identification of problem areas.

Tips

Review major components using the strategy task filter.

Compare current expectation versus the trend shown in the report. Inconsistencies suggest a change in current projection frequency parameterAdditional.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|AutoLaunch|,
|QUOMID|,|QUOM|,|SiteID|,
|FleetID|,|ModelID|,|CompCodeID|,|TaskTypeID|,|ModifierID|,|AppCodeID|,|Site|,
|Fleet|,|Model|,|CompCode|,|TaskType|,|Modifier|,|AppCode|,
|EqpPlanID|,|EqpPlan|,|UserID|

Report parameters

Default.

Additional filters

+Comp Code, Original Part Type, Modifier Code, Warranty, Occurrence Type, Redo, +Strategy Task, Life Confirmed, Task Type, Rating Average, Task Counter, UOM

+ indicates that you must select a Strategy Task or Component Code.

(Rpt_389) Component History

Shows the technical details for component changeouts. Also shows the specific work order history and Partial Life and Swing components.

Tips

This report can be used for simple tracking component history where the rotables module is not being used. Also useful for looking at the history where swing and partial life parameters have been used.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|EquipmentTypes|,|AutoLaunch|,
|SerialNumber|,|IsExactMatch|,|EqpPlan|,|EqpPlanID|,|ProjTaskID|,
|ProjTask|,|SerailNumberDesr|,|UserID|

Report parameters

Default.

Additional filters

Component Serial No, Component Code, Strategy Task, Modifier Code, Task Type, CCC, Show IN USE Comp, Show Serialised

(Rpt_393) Failure Indicators Analysis

Shows the trending of CM elements up to the point of component failure.

Initial parameters

|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Task Occurrence, Element Type, Element UOM, Lead Time

(Rpt_391) Trending

Report shows the element trending leading up to a component failure.

Tips

This report will return results for Real, non-terminated equipment only.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Branch, Site, Fleet, Model, Equipment, Division, Parent Equipment, Eqp Location, Eqp Category, Eqp Cost Centre, Cost Responsibility, Element Type, Element UOM

Parts Usage Analysis

(Rpt_208) Parts Usage Analysis - Qty

This report does not provide specific job or work order details, rather it sums up the parts usage by period (month/quarter/year). Flexible filters allow users to focus on specific parts and models, type of work and so on.

Tips

Lets you quickly identify the high usage parts, particularly for breakdowns.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, System, Modifier Code, Source of Supply, Occurrence Type, Ops Comp Code, Ops Job Code, Part Number, Symptom, Failure, Show, Breakdown, Parts Return, Sub System, Task Type, Cause, Operator, Comp Code, Part Type, Transaction Code, Repair, Sort

(Rpt_210) Parts Usage Analysis - Value

This report does not provide specific job or work order details, rather it sums up the parts usage by period (month/quarter/year). Flexible filters allow users to focus on specific parts and models, types of work and so on.

Analysis is based on value, not quantity, which allows users to focus on the high-cost areas.

Tips

The report lets you quickly identify the high value usage parts, particularly for breakdowns.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, Source of Supply, Occurrence Type, Ops Como Code, Ops Job Code, Part Number, Symptom, Failure, Show, Breakdown, Parts Return, Sub System, Task Type, Cause, Operation, Comp Code, Part Type, Transaction Code, Repair, Sort.

(Rpt_212) Parts Usage Details

Detailed listing of parts usage by work order number. Flexible reports parameters allow you to filter for specific models, type of work, component specific part number and so on.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, System, Comp Code, Task Type, Cause, Operation, Ops Job Code, Part Number, Symptom, Failure, Breakdown, Parts Return, Sub System, Modifier Code, Source of Supply, Occurrence Type, Ops Comp Code, Part Type, Transaction Code, Repair, Sort.

(Rpt_214) Parts Usage Interrogation

The focus of this report is on what parts are needed for a certain type of job, and where a part has been used. The report provides a flexible interrogation of parts usage.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, System, Comp Code, Task Type, Cause, Operation, Ops Job Code, Part Number, Symptom, Failure, Breakdown, Parts Return, Sub System, Modifier Code, Source of Supply, Occurrence Type, Ops Comp Code, Part Type, Transaction Code, Repair, Sort.

Potential Parts Warranty

This function displays the parts that AMT has identified for a potential warranty claim and the action taken.

The table below shows the special columns in the Potential Parts Warranty grid.

Field	Description
Workorder & Segment	Work order and its segment as it already exists in AMT or, if it does not exist, it is the work order and its segment from the Parts Usage interface file.
S	Will be **green** if the work order is linked to a strategy task.
Qty	Part quantity used on a particular Work Order-Segment.
Value	Part unit sell cost multiplied by quantity.
W	Charged to Warranty (based on the work order's cost bearer).
Warr Term	Warranty Term of the part's associated Warranty Group.
Claimable	Indicates whether AMT has deduced that the part could be warranty (using the above logic). Can be overwritten by a user.
Reviewed By	Person who has validated whether or not the part is warranty.
Claim Reference	Identification number of the warranty claim submitted.
Notes	Any notes relating to the claim or the manual entries made by a user.

Notes:

- You can edit **blue** cells directly in the grid.
- If AMT automatically identified a part as being Potential Warranty (such as Claimable) it can be overwritten by the user and vice versa.
- PU = Parts Usage - values direct from the interface file.

For a part to be considered potential warranty, AMT applies a specific set of logic, starting with the Parts Warranty Group system table.

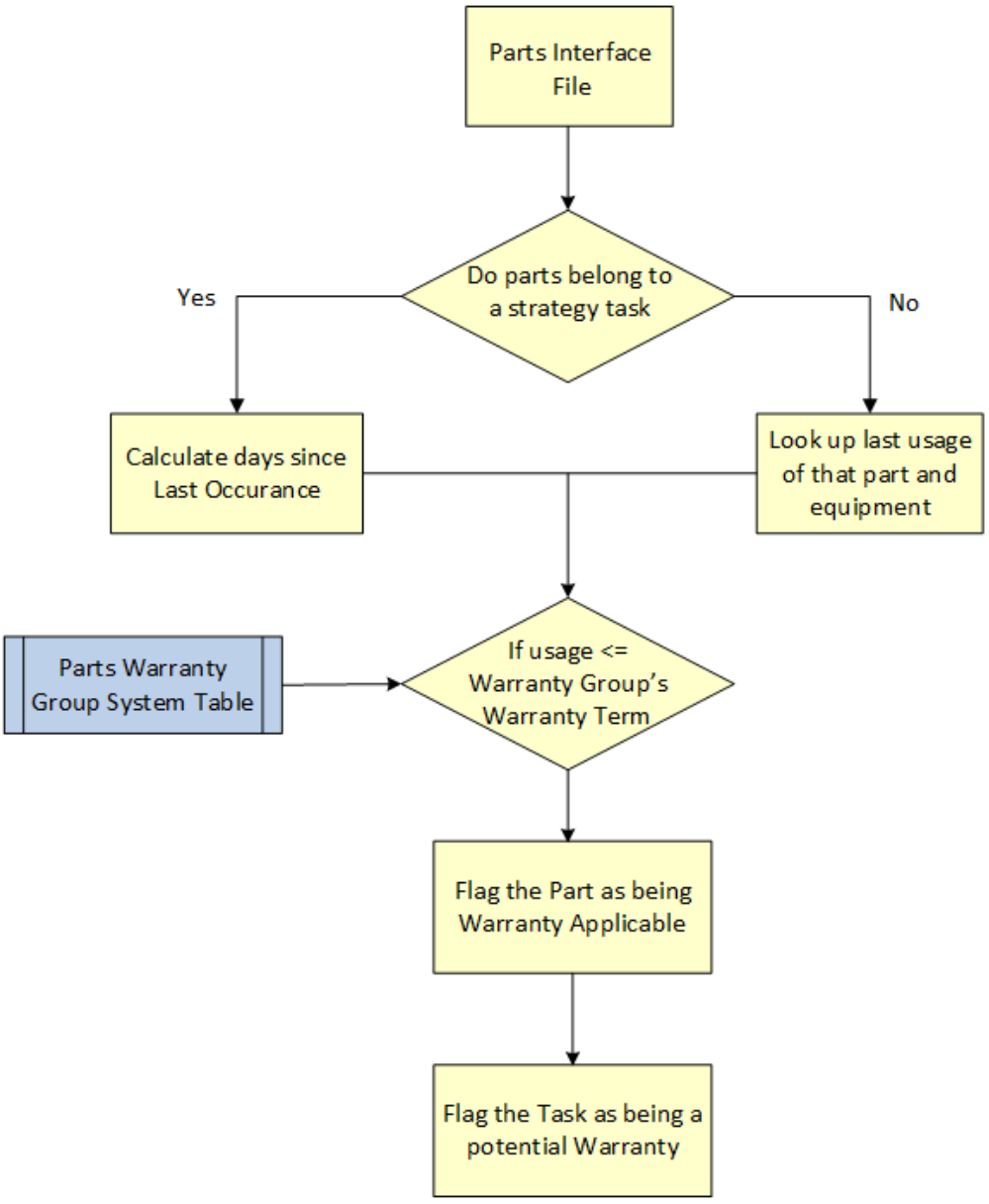
The purpose of this table is to list the different groupings of parts with their respective warranty periods. For example, the warranty period for an Engine part may be different to Tyres or Undercarriage.

This system table is only required if AMT is going to be used to identify potential parts warranty. It is not required for standard parts usage analysis.

Accordingly, this system table contains

- Warranty Group Code
- Warranty Group Description
- Warranty Term (days)

When a new part is registered in AMT the warranty group to which the part belongs is identified. Upon receiving actual parts usage data, AMT compares when the part was actually used to that of the warranty period for parts in the same Warranty Group in order to flag whether the part could be warranty applicable. The logic is outlined below.



Parts Interface File is a generic term representing actual part usage data, which is either recorded on an AMT work order or imported into AMT using the parts usage interface. System-level configuration determines which approach has been configured.

The lookup for non-strategy tasks interrogates actual parts usage history to find the most recent date when that part was used. AMT then uses the Job Date and Usage from that record to compare the Parts Warranty Group parameters (stored in the Parts Warranty table) to determine whether the part used is potentially under warranty.

Global Parts Usage Analysis

(Rpt_209) Parts Usage Analysis - Qty - Global

This report does not provide specific job or work order details, it sums up the parts usage by period (monthly, quarterly or yearly) using the Global Job Codes. See System Table - Job Code. Flexible filters allow you to focus

on specific parts and models, type of work and so on, enabling you to quickly identify the high usage parts (particularly for breakdowns)

Tips

Lets you quickly identify the high turnover parts, for example, for breakdowns.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, System, Modifier Code, Source of Supply, Occurrence Type, Ops Comp Code, Ops Job Code, Part Number, Symptom, Failure, Show, Breakdown, Parts Return, Sub System, Task Type, Cause, Operator, Comp Code, Part Type, Transaction Code, Repair, Sort.

(Rpt_211) Parts Usage Analysis - Value - Global

This report does not provide specific job or work order details, it sums up the parts usage by period (monthly, quarterly or yearly) using the Global Job Codes. See System Table - Job Code. Flexible filters allow you to focus on specific parts and models, type of work and so on, enabling you to quickly identify the high usage parts (particularly for breakdowns)

Analysis is based on part value, not quantity, which allows you to focus on high-cost areas.

Tips

Lets users quickly identify the high turnover parts (such as for breakdowns).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, System, Modifier Code, Source of Supply, Occurrence Type, Ops Comp Code, Ops Job Code, Part Number, Symptom, Failure, Show, Breakdown, Parts Return, Sub System, Task Type, Cause, Operator, Comp Code, Part Type, Transaction Code, Repair, Sort.

(Rpt_213) Parts Usage Details Global

Detailed listing of parts usage by work order number using the Global Codes. See System Table - Job Code. Flexible reporting parameters allow you to filter for specific models, types of work, components, specific part numbers and so on.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, System, Comp Code, Task Type, Cause, Operation, Ops Job Code, Part Number, Symptom, Failure, Breakdown, Parts Return, Sub System, Modifier Code, Source of Supply, Occurrence Type, Ops Comp Code, Part Type, Transaction Code, Repair, Sort.

(Rpt_215) Parts Usage Interrogation - Global

Detailed listing of parts usage by work order number using the Global Codes. See System Table - Job Code. The focus of this report is on what parts are needed for a certain type of job, or where a part has been used.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Planned, Strategy Tasks, System, Comp Code, Task Type, Cause, Operation, Ops Job Code, Part Number, Symptom, Failure, Breakdown, Parts Return, Sub System, Modifier Code, Source of Supply, Occurrence Type, Ops Comp Code, Part Type, Transaction Code, Repair, Sort.

What If Analysis

The What If functionality in AMT allows you to take copies of, modify and assess current strategies. What if projections, which are alternate projections, are used to assess different strategies, are typically based on the current projection of a piece of equipment. The current projection reflects the existing strategy for the equipment. The what if projection reflects what you are considering to implement as a new strategy, but do not want to commit the changes until they have been properly assessed.

It is typical to create a what if projection across an entire fleet, as maintenance strategies are usually developed and applied at the fleet level for a particular model, rather than at the equipment level. The following example of creating a what if projection will be based on these parameters. The same steps can be applied to create what if projections for individual pieces of equipment.

Strategy optimisation - What If Analysis

Once the analysis of components and labour has been performed, you can test alternate solutions before implementing changes. By doing this you:

- Do not impact the "live" strategy of the asset, because the current projection is not changed yet.
- Are able to quantify the impact of the proposed solution, which may be required as part of company policy.

Not all solutions require testing. If this is the case, you are able to make changes directly to the "live" strategy, that is, the current projection.

The "What If" functionality in AMT allows you to take copies of the current projection, make changes and run reports to see the impact, without impacting the asset or the live strategy.

"What If" projections (alternate projections) are typically based on the current projection of a piece of equipment.

Once you have decided which changes are to be made to the live strategy, you can then update the relevant parameters in the current projection.

Maintenance organisations should have a clear policy in place defining the circumstances in which authorisation to change the strategy is required vs authorisation is not required.

Assess the What If Strategy

A number of reports have been designed to assist you in assessing the What If projection (strategy) they have created and manipulated. Typically, you will compare the What If projection to the Current projection (because the Current projection reflects the current “living” strategy for a piece of equipment or fleet). A brief description of each report follows.

Add/Copy/Delete Projection

Use this tool to copy a projection from one equipment to another equipment to create Estimate-Draft projections.

What If – Life Cycle Analysis

This text report is based on the comparison of the What If projection to a second projection (typically the Current projection). It lists the Frequency and Cost of each strategy task occurrence for both projections, and calculates the resultant Total Projected Life Cycle Cost for each strategy task based on the Equipment’s Life Cycle. Variances between projections are highlighted. Variances are the result of strategy task parameters in the What If projection equating to cost savings or overruns compared to the second projection.

What-If – Period Analysis

This report is based on the comparison of the What If projection to a second projection (typically the Current projection) and can be produced in text or as a bar chart. Depending on the Analyse By and Filter options selected, the report breaks down the costs of each projection over a user-specified date range into blocks (for example, Month, Quarter, Year). Variances between projections are highlighted.

What If – Cost Analysis

This text report is similar to the What If – Period Analysis report. However, instead of breaking down costs into blocks of time (for example, Month, Quarter, Year) it breaks down costs into various cost categories (for example, Equipment, Component, System and so on) over a user-specified date range.

What-If – Cost Per UOM

This report, produced as a line chart, illustrates the cumulative Cost per UOM (for example, Engine Hours) for each projection (for example, the What-If projection and the Current projection). The UOM selected must be common to both projections. If the What-If projection is a copy of the projection it is to be compared against, there should be consistency unless the you changed the UOM in a strategy task in either projection after the initial copy was created.

Various Analysis Worksheet

This text report lets you analyse the causes of cost variances between the What-If projection and a second projection. You can break down the Cost Variances (that is, the Cost differences between the two projections) into various categories, into specified blocks of time, over a specified date range.

Add/Copy/Delete projection

You can create, copy, delete or import projections of the following types:

- Budget
- Current
- Alternate. Alternate projections are useful for What-If analysis.
- Archive. Archive projections are useful for annual budgets.

(Rpt_254) What-if - Life Cycle Analysis

Allows analysis of costs between a budget or current projection and one created within the scenario manager.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Cost Selector, Component Code, Task Type, Strategy Task

(Rpt_255) What-if - Period Analysis

Compares the Cost between two projections for a nominated time period.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Strategy Task, Task Type, Maintenance Type, Cost Centre, Responsibility, Expense Element, Labour Activity, Parts/Labour/Misc, Currency

(Rpt_252) What-if - Cost Analysis

Compares the cost between two projections for a nominated time period.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Strategy Task, Task Type, Maintenance Type, Cost Centre, Responsibility, Expense Element, Labour Activity, Parts/Labour/Misc, Currency

(Rpt_253) What-if - Cost per UOM

Compares the Cost per UOM between two projections selected by the user.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Fleet, Equipment, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_251) Variance Analysis Worksheet

Lets you analyse between projections and group by a maximum of five categories to show variances in a period of time - default date is last month but this can be modified as required.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Strategy Task, Task Type, Maintenance Type, Cost Bearer, Cost Centre, Responsibility, Expense Element, Labour Activity, Parts/Labour/Misc, Currency

Strategy Admin KPIs

(Rpt_233) Standard Job Coverage

The preparation and usage of standard jobs leads to improved efficiency, standardisation of work performed and improved parts forecasting.

The report shows the percentage of the value of tasks that have standard jobs associated to them in the Standard Jobs Manager.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Task Type

(Rpt_234) Standard Job Interrogation

Details, either by equipment or strategy task what standard jobs are linked and whether a Rotable Repair Strategy is in place.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, Strategy Task, Rebuild Location, Source of Supply, Linked to Std Job, Strategy

(Rpt_259) WorkScope Coverage

Shows the amount of workscope coverage relating to each specific model and the related tasks for the model. A workscope is a set of model-specific instructions on how tasks should be performed, for example, safety process, tooling requirements.

The preparation and usage of work scopes leads to improved quality of work performed

Tips

Ensuring that relevant documentation relating to work processes are held and maintained in AMT assets on-site users and also enables new contract managers and greenfield sites to utilise the specific documentation in their mobilisation.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID

Report parameters

Default.

System, Sub System, Component Code Task Type

Archiving

Budget and Forecast Manager

For more information, click here.

(Rpt_134) Archive Cost Worksheet

Used for forecasting costs by expense group. It is used primarily for budgeting costs. It shows historical and future costs at sell, categorised by up to five different areas.

Tips

Similar to the Forecast Worksheet report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, Maintenance Type, System, Cost Centre, Sub System, Responsibility, Component Code, Expense Element, Strategy Task, Labour Activity, Task Type, Region, Division, Eqp Criticality, Equipment Class, Equipment Group, Equipment Location, Parts/Labour/Misc, Division, Custody

(Rpt_133) Archive Cost Analysis

Provides an analysis of costs.

Tips

Lets you to compare two projections in the report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, Cost Responsibility, System, Cost Centre, Sub System, Expense Element, Strategy Task, Maintenance Type, Task Type, Modifier Code, Component Code, Division, Region, Equipment Class, Eqp Criticality, Equipment Location, Equipment Group, Parts/Labour/Misc, Eqp Category, Currency, Cost Bearer

(Rpt_119) Actual vs Archive Analysis

This report is similar to the Actual vs. Budget Period Analysis report, but lets you expand on the information required for analysis. Dates can be set according to user requirement as can the "Analyse By" parameters.

Tips

It is critical that the Equipment Management group understand, report on, and develop continuous improvement initiatives based on current performance variances to Annual Budget expectations.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, Cost Centre, System, Responsibility, Sub System, Expense Element, Component Code, Labour Activity, Strategy Task, Parts/Labour/Misc, Task Type, Currency, Maintenance Type

(Rpt _120) Actual vs Archive Period Analysis

This report highlights the variances between an Annual Budget Projection and the Current Projection at the end of the financial year, or any specified period of time.

Tips

It is critical that the Equipment Management group understand, report on, and develop Continuous Improvement initiatives based on current performance variances to Annual Budget expectations. Data in this report can be analysed by month, quarter, year.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, Cost Centre, System, Responsibility, Sub System, Expense Element, Component Code, Labour Activity, Strategy Task, Parts/Labour/Misc, Task Type, Currency, Maintenance Type

(Rpt_251) Variance Analysis Worksheet

Lets you analyse between projections and group by a maximum of five categories to show variances in a period of time - default date is last month but this can be modified as required.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Strategy Task, Task Type, Maintenance Type, Cost Bearer, Cost Centre, Responsibility, Expense Element, Labour Activity, Parts/Labour/Misc, Currency

(Rpt_136) Archive - Costs/UOM

Report lets users to analyse between projections and shows data by unit of measure for either month, quarter or year. You have the option to analyse by two user selections.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, Sub System, Cost Bearer, Component Code, Cost Type, Modifier Code, System, Task Type, Costs

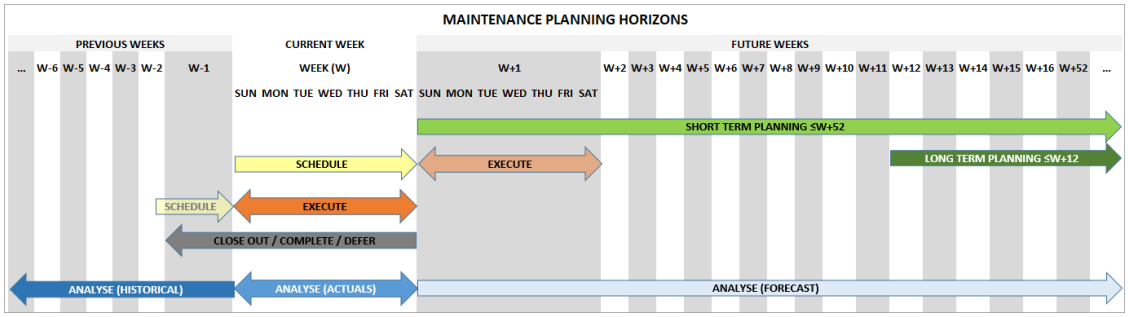
Long-Term Planning

Long term planning is based around the concept of ensuring that major maintenance (strategic) tasks have minimal impact on the production and operations targets. To expand on this, it is the formal identification of the major strategy activities and ensuring minimal impact on the production and cost targets by the optimisation of labour resources, parts and materials and tooling through accurate forecasting.

The Long Term Planning module is used to analyse and produce the following information:

- A W+12 forecast for the short term planning and W+1 scheduling cycles
- Provide multiple inputs for the annual budgeting and forecasting cycles
- Availability requirements forecast
- One year and beyond availability and utilisation projections
- Manage lead times on long lead time components
- Identify opportunities to optimise the Dynamic Life Cycle Costing
- Assist in identifying fleet replacement opportunities

Current best practice is to update and produce the W+12 (12 week/3 month) forecast on a weekly basis as an integral part of the short term process, that is, the W+1 maintenance schedule. The maintenance planning horizons are explained in the diagram below.



This is then reviewed with the operations team as part of the initial draft or first cut maintenance schedule and, along with the preventive maintenance services, forms the basis of next week's (W+1) schedule.

Strategy changes applicable only to the next occurrence of a strategy task are called long term planning adjustments. For example:

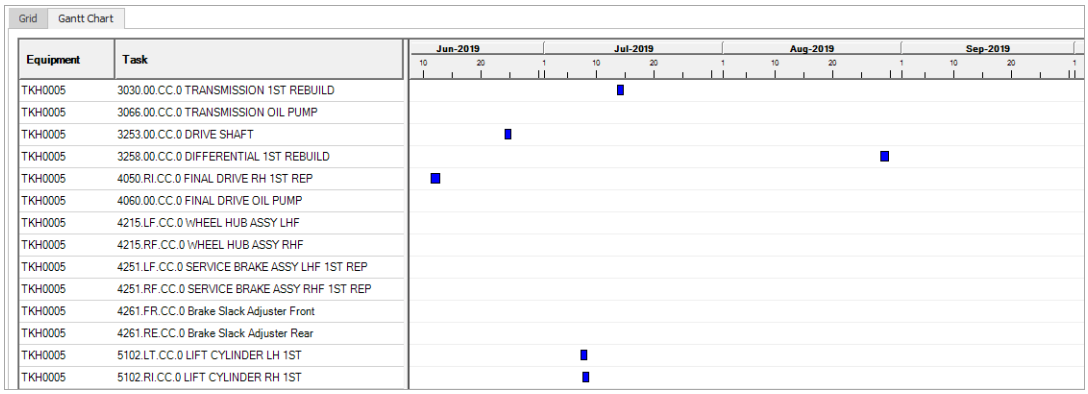
- For the transmission on a loader, based on condition monitoring analysis, the component may need to be replaced three months earlier than expected.
- For the suspension on a truck there may not be any rotable or service exchange components available when the suspension is due for replacement in three months, therefore new components will have to be used.

These are strategy changes which will only affect the next occurrence, that is, the strategy for the components which are currently in the machine. Changes to long term planning only affect the next occurrence, they do not alter the parameters of the overall maintenance strategy frequency or interval.

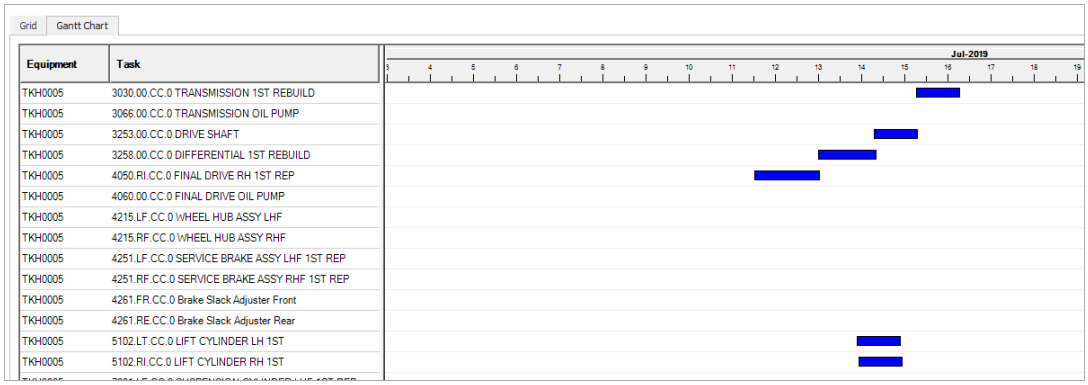
The Long-term Planning grid shows all planned strategy tasks for the equipment selected in the filters (non-planned strategy tasks are excluded).

Part of the long term planning process is the alignment or staging of these strategy tasks at a high level. For example, a truck may have the following components come due for replacement within three months of each other:

- Main drive shaft
- Transmission
- Right final drive
- Both lift (hoist) cylinders
- Differential.



Instead of executing each strategy task individually over a longer period, the long term planner would seek to align and stage these tasks to increase uptime by consolidating all these tasks into one major outage. There are many factors to consider. For example, operations may be at a particularly critical phase of the mine plan at this point, and a longer outage may not be feasible for compliance with the mine plan, despite providing better availability overall. The long term planner may have to stage these into two or more shorter events to ensure compliance, but effectively reducing maintenance efficiency in the process.



Once the long term planner has done some basic high-level staging of the strategy tasks to create a consolidation of complimentary tasks, the W+12 forecast is given to the short term planner who fine tunes the staging of the components, parts or services for anything that falls within the W+12 (12 week) forecast.

When these have been planned and the components or parts ordered, the scheduler takes control of the strategy tasks and includes them in the W+1 (weekly maintenance schedule). Click here to see the long term planning process.

Long-term planning screen

Filters

Filters allow you to view, group and sort the tasks to suit the analyses being performed.

Grid fields

Field	Description
Equip. Plan	Identification number of a piece of equipment.
Component	Component code combination of the strategy task followed by the description of the component code.
S	Suppression grouping.
D	Dependency grouping.
Job Cost	Total cost of all jobs associated with the selected strategy task for the next occurrence of the strategy task (expressed in the primary currency).
Duration	Total job duration of all jobs associated with the selected strategy task for the next occurrence of the strategy task.
Life To Date	Current life of the component (calculated as current equipment usage – usage of most recent work order settlement for this task).
Strategy Date	Date at which point the strategy task should be executed.
Strategy Usage	Usage at which point the strategy task should be executed.
Overdue	! Indicates that the task should have been completed but it has not (based on strategy date versus today's date).
Reviewed Status	**Not Reviewed**: task's strategy date or usage values have not been adjusted or validated. **Reviewed**: task's strategy date or usage values have been adjusted or validated. **Confirmed**: task's strategy date or usage has been locked in.
System Calculate	A tick will display if the strategy usage or date has been calculated by the system. Once you have determined the optimal life for the component, update the strategy usage or date for the strategy task. This results in the system calculated flag disappearing, and either the strategy date or strategy usage field displaying in light blue.
Message	Indicates whether an internal conditional based maintenance warning message has been generated.
PCV	Projected cost variance of the strategy task. (**Note**: The value will be zero if there is no matching strategy task in the equipment's budget projection).

Field	Description
RVIC	% life remaining on the strategy task at present.
Occurrence Type (FM)	Failure mode (only shown if the strategy task is linked to a component structure).
Measurement/Usage	Actual measurement or usage reading relevant to the occurrence type.
R1 - RD6	Last six actual usage or measurement readings.

Gantt chart legend

Bar	Description
	Overdue (next occurrence).
	System calculated and not overdue (next occurrence).
	Not system calculated and not overdue and not confirmed (next occurrence).
	Confirmed (next occurrence).
	Future occurrences (beyond the next).
	Future occurrence of a strategy task which belongs either to a dependency or suppression scheduling group.

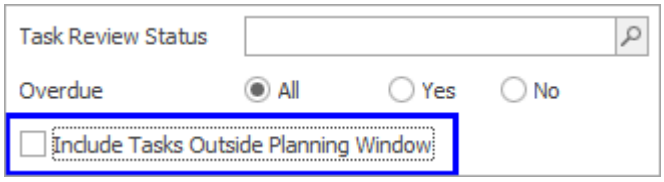
Quick Buttons

Button	Description
	Displays all future occurrences of the strategy task projected. The list is read only and is for information purposes only.
	Displays the strategy task from the current projection. Again this should only be modified after careful consideration as changes can have wide reaching effects on future projections.
	Opens the usage profile linked to the selected equipment plan. This should only be modified after careful consideration as changes can have wide reaching effects on future projections.
	Open the Record Actual Usages function.

Button	Description
W	Displays the work order's settlement history. If there are errors in any work order settlement, for example a new life event that has been incorrectly coded, then correct the codes of the work order settlement.
Cs	Component status report.
Mu	Displays current equipment usage readings and measurements.
Uc	Usage v Profile Snapshot report.
W+	Create a work order settlement for the selected work order.
FA	Edit failure mode assessment (CBM Module).
T+	Create a backlog work order in planning.
M	Opens the Record Actual Measurements function.
TR	Displays the Trending report.

Reviewing new tasks

When reviewing new tasks, it is best to clear the **Include Tasks Outside Planning Window** check box so that you can focus on the tasks that have entered the planning window, as they have the highest priority.



Group tasks

When refining the long-term plan you should look for synergies, such as:

- the timing of major shutdowns.
- tasks performed on the same type of component in different locations (for example, aligning left and right

cylinders to be performed at the same time).
- minor components with parent components (for example, brakes aligned with final drives).

On the Long-term Planning Gantt chart you can visually group tasks.

Bow Wave identification

The term Bow Wave refers to a very high spike of costs forecast to be incurred in the very short-term (AMT can also forecast Bow waves in the long-term). Ideally, costs should be spread out over time as this helps to streamline cash flow, resource planning, inventory management and so on. Bow waves can also undermine the accuracy of AMT's forecasts if they are monitored (see Forecasting Accuracy). Some of the causes of bow waves are a lack of smoothing in work plans and overdue tasks.

One way of identifying the possible occurrence of a bow wave in AMT is to run the Forecast Cost Analysis report.

Ideally, the level of costs will be fairly consistent over time. Periodically, there will be some spikes as major components changeouts fall due. However, large spikes (particularly ones forecast to occur in the current month or next month) should be investigated immediately. Unless major work is being done, the primary cause of large costs spikes forecast to be incurred in the current or next month are a result of overdue components (see Overdue Strategy Tasks). AMT assumes that any task which is overdue will be performed tomorrow, so AMT creates a bow wave. You need to reschedule overdue tasks in order to reduce the bow wave.

To drill down to the cause of the bow wave, refine filters and run the Forecast Cost Analysis report again or run the Forecast Worksheet report.

Long-Term Planning - Advanced

The Long-Term Planning - Advanced screen helps Planners to manage their work. If you have access to the original Long-Term Planning screen you will have automatic access to the Advanced screen if the Shutdown, Turnaround, Outage (STO) feature is enabled.

On this screen, Planners can do the following:

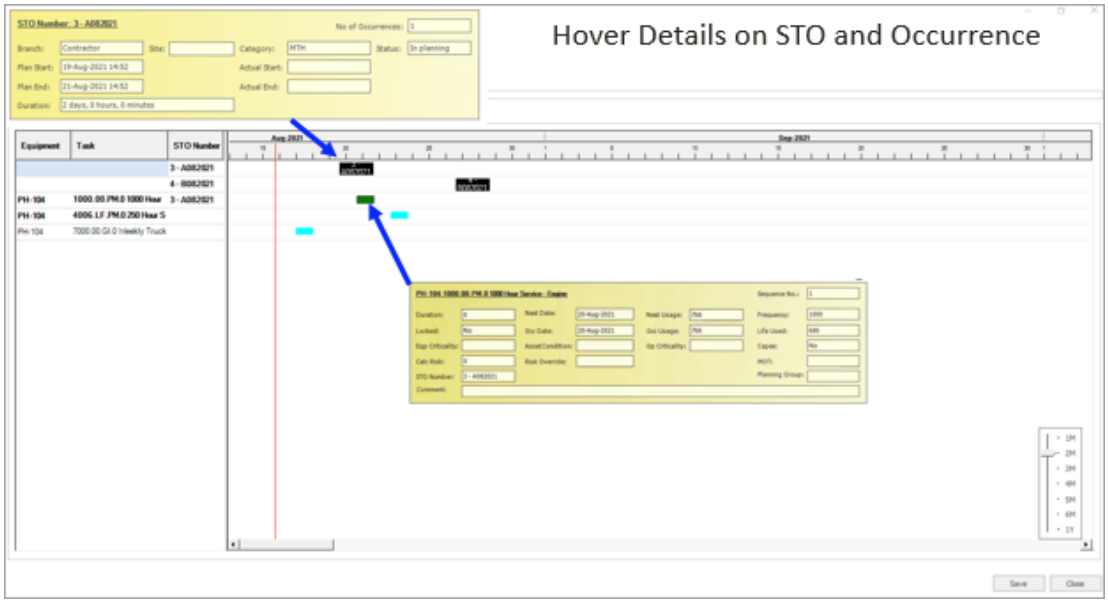
- Move and confirm strategy task occurrences.
- Visualise overdue occurrences.
- Visualise work that must be performed during a shutdown.
- See whether work that requires a shutdown has been assigned to a shutdown event.
- Add work to shutdowns.
- Manage and shift shutdowns and associated work subject to constraints and tolerances.

You can view your work in the grid or on the Gantt chart view. The grid view lists individual occurrences (sequence number = occurrence number). The Gantt chart shows the task as a single line and the occurrences within that line.

- **STO Required** on Strategy Task sets bold text,
- **STO Not Required** on Strategy Task sets normal text.

See Long-Term Planning for the meaning of the bars on the Gantt chart. Work that must be performed within a shutdown displays in bold text. Any work included in a shutdown has an allocated STO number. Work that is not required to be performed as part of a shutdown (based on asset or work configuration) but has, by the Planner's discretion, been included in the shutdown, will display with an allocated STO number but without bold text.

The screenshot below shows the details displayed when you hover over an STO and an Occurrence.



You can right-click an Occurrence and **Highlight Linked Occurrences**.

STO Occurrence Movement Logic

STO Start Date used for checking movement depends on STO Status.

- Status = In Planning, dates used are Planned Start and Planned End Dates.
- Status = In Progress, start date used is Actual Start Date.
    - Actual End Date = Actual Start Date + Planned Duration.

AMT will calculate the new occurrence date.

- New Occurrence Date = Previous Occurrence Date + (STO Previous Start Date – STO New Start Date).
- When tolerances are defined a further check is required.
    - New Occurrence Date must be between.
      (Previous Occurrence Date – Tolerance) and (Previous Occurrence Date + Tolerance).
        - If the new date is not within the tolerance range, the message given.

AMT checks if the new occurrence date is within the new STO Start and End Dates.

- If Yes, AMT will then check if the occurrence can be completed within the STO period.
    - If Yes, the occurrence will be moved with the STO.
    - If No, message received.
- If No, message received and one of three actions is taken

1. Remove occurrence from STO.
2. Adjust STO dates.
3. Manually update occurrence date to fall within STO period.

Standard vs Advanced planning

The Planning function in AMT is different to the Advanced Planning function. The fundamental difference between the type modes of planning is that advanced planning requires you to create events.

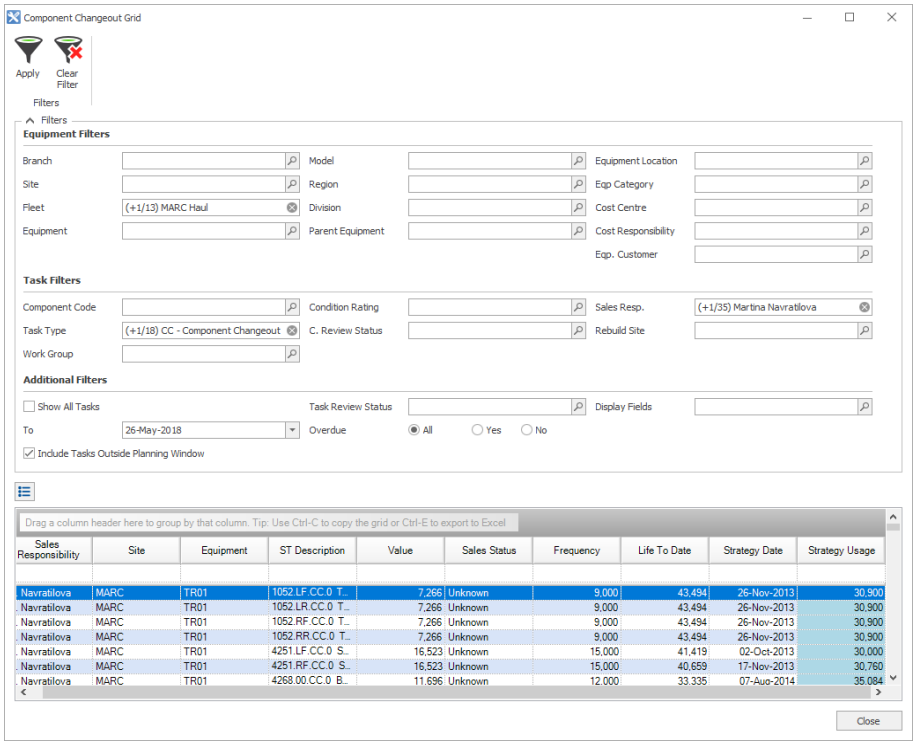
Characteristic	Planning	Advanced Planning
Event		
Ability to create an event.	N	Y
Ability to view and manipulate events on a Gantt chart.	N	Y
Ability to create one or more downtime allocations.	N	Y
Work order		
Ability to plan strategy tasks generated by projections.	Y	Y
Ability to create a non-strategy task.	Y	Y
Ability to set up start and end dates/times per work order.	Y	Y
Ability to view work orders on a Gantt chart.	Y	Y
Ability to have multiple operations.	Y	Y
Ability to add multiple parts.	Y	Y
Ability to create an internal work order settlement.	Y	Y
Ability to request an external work order settlement.	Y	Y
Ability to raise a purchase order.	Y	Y
Ability to record actual labour and durations at the work order level.	Y	Y
Ability to allocate labour resources.	Y	Y
Ability to capture actual maintenance performed.	Y	Y

Characteristic	Planning	Advanced Planning
Work order settlements		
Ability to request work order settlements from an external system.	Y	Y
Ability to self-generate work order settlements.	Y	Y
General planning and reporting		
Ability to generate a work plan.	Y	Y
Ability to review resource allocations.	Y	Y
Ability to view age of backlogs.	Y	Y
Ability to analyse downtime by Responsibility.	N	Y
Ability to perform Event Analysis	N	Y
Ability to perform Downtime Analysis	N	Y

Component changeout grid

This form is generally used by sales representatives, and displays sales-related fields.

Navigate to **Long-term Planning** > **Component Changeout Grid**.



Select the relevant filters and click  .

The blue-shaded fields in the grid indicate what has strategy dates influencing the schedule.

You can move columns around by clicking and dragging the column header.

Next Occ Export

On this screen you can create a .csv file which you can export the **Strategy Task Next Due** and **Sales Status**fields to Excel. You can export to an Internal file or a Customer Facing file. This makes it easy for you to bulk update and reimport the file using **Next Occ Import**.

The Import and Export functions share a common file format.

Field	Export/Import File Name	Editable	Internal File	Customer File	Req. For Import	Notes
Branch	Branch/Code		X	X		
Site	SiteCode		X	X		
Fleet	FleetCode		X	X		

Field	Export/Import File Name	Editable	Internal File	Customer File	Req. For Import	Notes
Customer	CustomerCode		X	X	X	
Model	ModelCode		X	X	X	
Equipment	AssetName		X	X		
Serial Number	SerialNumber		X	X	X	
Registration Counter	RegistrationCounter		X	X	X	
Component Code	ComponentCode		X	X	X	
Modifier Code	ModiferCode		X	X	X	
Task Type	TaskTypeCode		X	X	X	
Task Counter	TaskCounterCode		X	X	X	
ST Description	StrategyTaskDescription		X	X		
Frequency	FrequencyValue		X	X		
Life to Date	LifeToDateValue		X	X		
Primary Part Number	Primary Part Number		X	X		
Next Part Number	NextPartNumberCode	X	X			Part

Field	Export/Import File Name	Editable	Internal File	Customer File	Req. For Import	Notes
Required						Number in Next Occ. If Next Occ does not have a Part Number, the Primary Part Number from the Strategy Task Editor Component tab is displayed.
Source of Supply	SourceOfSupplyCode		X	X	X	Part is identified by part number, source of supply and organisation.
Strategy UOM	StrategyUOMCode		X	X	X	This field was added as it is required in import. It does not show in the form. If it is a date-based task the field will be empty.
Strategy Usage	StrategyUsageValue		X	X		
New Strategy	NewStrategyUsageValue	X	X	X		Can only

Field	Export/Import File Name	Editable	Internal File	Customer File	Req. For Import	Notes
Usage						specify EITHER Usage OR Date - if the user specifies both then this is an error.
Strategy Date	StrategyDate		X	X		
New Strategy Date	NewStrategyDate	X	X	X		Can only specify EITHER Usage OR Date - if the user specifies both then this is an error.
Sales Status	SalesStatusCode	X	X	X		
Review Status	ReviewStatusCode	X	X			
Part Classification	PartClassificationCode	X	X			Part Classification in Next Occ. If Next Occ does not have Part Classification then the default Part Classification from the

Field	Export/Import File Name	Editable	Internal File	Customer File	Req. For Import	Notes
						Strategy Class Editor Component tab is displayed.
Sales Lost Reason	SalesLostReasonCode	X	X			
Sales Status Comments	SalesStatusCommentsNote	X	X			
PO Number	PurchaseOrderNumber	X	X	X		

Next Occ Import

This screen lets you import updated .csv files containing **Strategy Task Next Due** and **Sales Status** fields which have been bulk-updated after being exported using Next Occ Export.

The Import and Export functions share a common file format. See **Next Occ Export**.

Last Occ Export

On this screen you can create a .csv file which you can export the last occurrence of the **Strategy Task Next Due** and **Sales Status**fields to Excel. You can export to an Internal file or a Customer Facing file. This makes it easy for you to bulk update and reimport the file using **Last Occ Import**.

Last Occ Import

This screen lets you import updated .csv files containing the last occurrence of the **Strategy Task Next Due** and **Sales Status** fields which have been bulk-updated after being exported using Last Occ Export.

(Rpt_633) Long-Term Planning Log

Displays the long-term planning log.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Component Code, System, SubSystem, Task Type, Strategy Task, Strategy Task Desc, Reason, User Name

(Rpt_632) Hours To PM

Gives a quick view of hours to the next Preventive Maintenance (PM) for selected equipment.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Report Type, Equipment Hierarchy

Additional filters

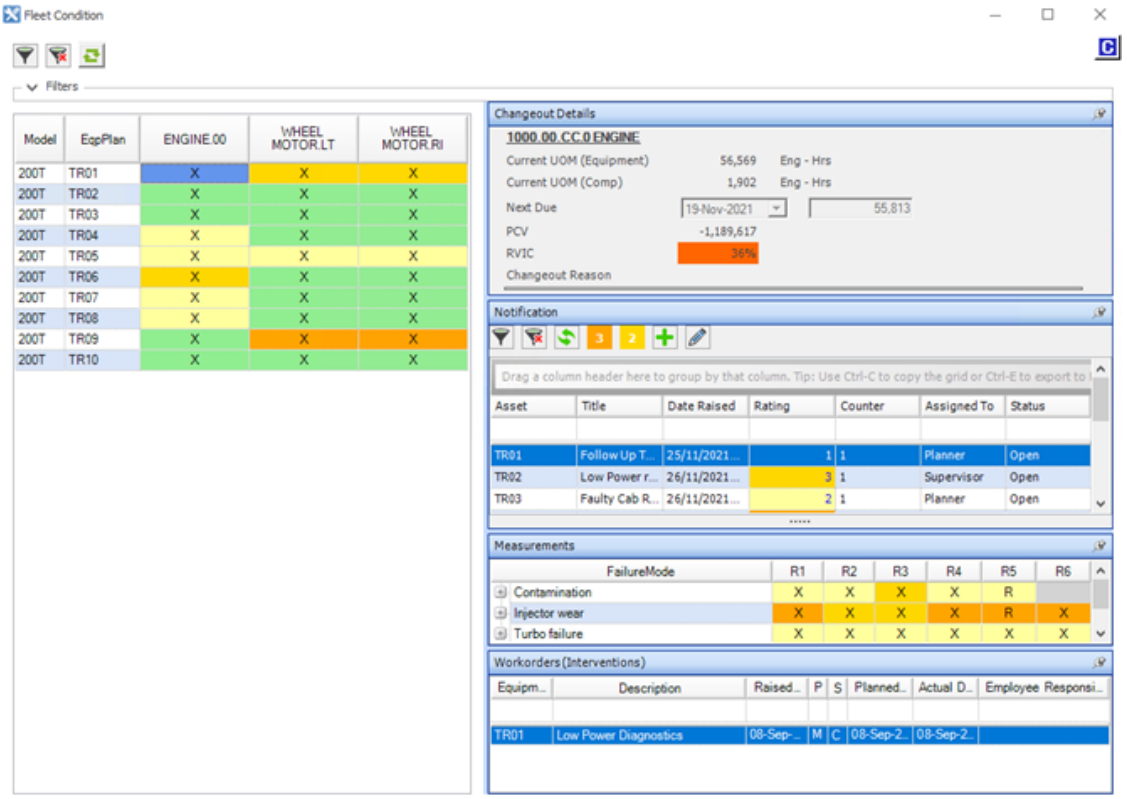
UOM

Fleet condition

The objective of monitoring the condition of equipment is so that the sales representative can monitor the status of customers' equipment as an input into working with the customer to optimise their component lives.

This feature is only available if the Condition Based Maintenance (CBM) module is being used. The Condition Based Maintenance module imports condition monitoring data and interprets it against components. It also automates the creation of emails, AMT messages and AMT work orders for alerts and exceptions.

Navigate to **Long-term Planning** > **Fleet Condition**.



On this dialog box, you can view the health of components in one list (the grid on the left), along with corresponding notifications, measurements and condition monitoring interventions on the right.

The colour of the cell in both grids indicates the status, from OK to High Alert. The letter in the cell indicates the status of the last failure mode assessment made by AMT.

- X = Not reviewed.
- R = Reviewed.
- C = Closed.

When you click on a cell in the grid on the left, these details display on the right:

- Changeout details: Next due details, Projected Cost Variance and Residual Value in Components.
- Notifications: Details of the last six readings.
- Messages: AMT messages (alerts) generated for that component.
- Work orders (interventions): The work orders performed on that component since its last changeout that have been flagged as interventions. An intervention means that the component has had something done to extend its life.

(Rpt_810) Job Status Audit Log

Shows changes made by users to the job status of work orders.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,
|UserID|,|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Cost Type, Component Code, System, Sub System, Task Type, Strategy Task, Strategy Task Desc, User Name, Job Status

Availability

(Rpt_172) Forecast Availability

This report is used to determine whether the maintenance schedule meets the customer's availability expectations.

Relates to the Budgeting Module in AMT, and lets you define, set and edit availability parameters to forecast the availability of the fleet over a period of time. The report reports on the base downtime, monthly downtime and availability.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

None.

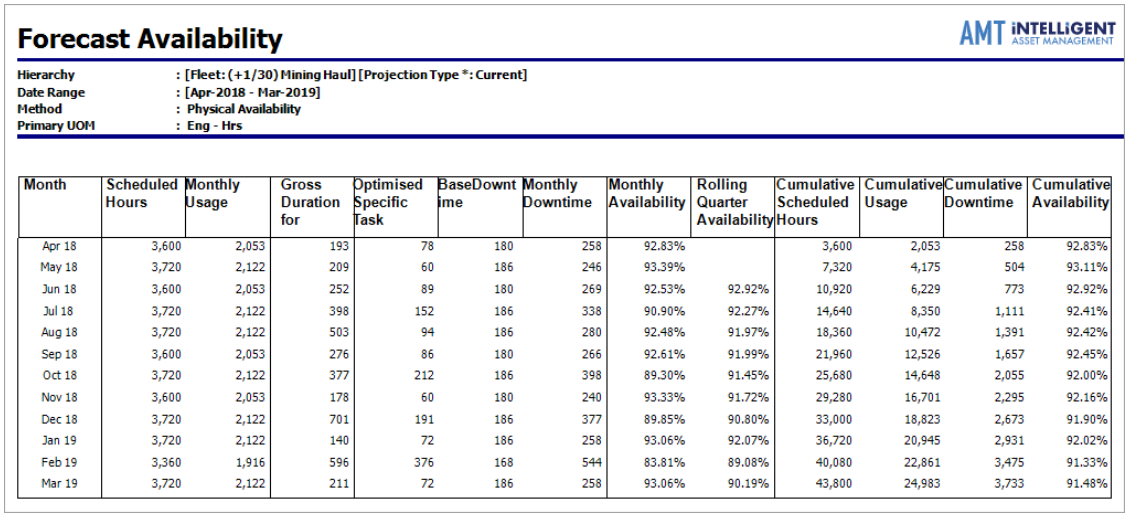
Validate availability forecast

It is important to validate the impact of the plan on availability.

The production department has an expectation that assets will be available for use a certain percentage of time. Sometimes these expectations are in the form of departmental or maintenance contract KPIs.

Navigate to **Long-term Planning** > **Availability** > Forecast Availability.

The planner should run an availability forecast for the next 12 months for the fleet, and then a forecast per asset.



Forecast Availability

AMT iNTELLiGENT ASSET MANAGEMENT

Hierarchy : [Fleet: (+1/30) Mining Haul] [Projection Type *: Current]
Date Range : [Apr-2018 - Mar-2019]
Method : Physical Availability
Primary UOM : Eng - Hrs

Month	Scheduled Hours	Monthly Usage	Gross Duration for	Optimised Specific Task	BaseDownt ime	Monthly Downtime	Monthly Availability	Rolling Quarter Availability	Cumulative Scheduled Hours	Cumulative Usage	Cumulative Downtime	Cumulative Availability
Apr 18	3,600	2,053	193	78	180	258	92.83%		3,600	2,053	258	92.83%
May 18	3,720	2,122	209	60	186	246	93.39%		7,320	4,175	504	93.11%
Jun 18	3,600	2,053	252	89	180	269	92.53%	92.92%	10,920	6,229	773	92.92%
Jul 18	3,720	2,122	398	152	186	338	90.90%	92.27%	14,640	8,350	1,111	92.41%
Aug 18	3,720	2,122	503	94	186	280	92.48%	91.97%	18,360	10,472	1,391	92.42%
Sep 18	3,600	2,053	276	86	180	266	92.61%	91.99%	21,960	12,526	1,657	92.45%
Oct 18	3,720	2,122	377	212	186	398	89.30%	91.45%	25,680	14,648	2,055	92.00%
Nov 18	3,600	2,053	178	60	180	240	93.33%	91.72%	29,280	16,701	2,295	92.16%
Dec 18	3,720	2,122	701	191	186	377	89.85%	90.80%	33,000	18,823	2,673	91.90%
Jan 19	3,720	2,122	140	72	186	258	93.06%	92.07%	36,720	20,945	2,931	92.02%
Feb 19	3,360	1,916	596	376	168	544	83.81%	89.08%	40,080	22,861	3,475	91.33%
Mar 19	3,720	2,122	211	72	186	258	93.06%	90.19%	43,800	24,983	3,733	91.48%

Column descriptions

- **Gross Duration for Specific Tasks**

  If **Task Types** were selected in the **Task Specific Downtime** section, this column reflects the sum of all durations for selected Task Types from the asset's current projection.

  If no Task Types were selected, the value is blank.

- **Optimised Specific Downtime**

  The **Planning Efficiency** factor is applied to **Gross Duration** to determine the number.

  If no Task Types were selected, the value is blank.

- **Base Downtime**

  Determined from the **Annual Base Downtime** parameters entered.

- **Monthly Downtime**

  Base Downtime + Optimised Specific Downtime.

- **Monthly Availability**

  The formula used depends on the **Availability Type** set, and the downtime used in the formula is derived from the **Monthly Downtime** column. Availability statistics are calculated using the **Availability Type** set at the **Fleet Level**, and are either **Physical** or **Mechanical**.

Costs

(Rpt_173) Forecast Cost Analysis

Shows the strategy task costs at sell.

Tips

Significant differences suggest the current strategy parameter should be changed.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|LogoLocation|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Region, Eqp Criticality, Equipment Group, Exp Category, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Modifier Code, Division, Equipment Class, Equipment Location, Project, Def. Maint. Period, Parts/Labour/Misc, Currency

(Rpt_174) Forecast Worksheet

Used for forecasting costs by expense group. The report shows future and historical costs at sell, categorised by up to five different areas.

- You can navigate to this report in the following ways:
  Long-term Planning > **Costs**.
  Budgeting & Forecasting > **Live Forecasting**.
  Modelling > **Cost Analysis**.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,
|EquipmentTypes|,|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|,
|Feature_32|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

(Rpt_778) Forecast Worksheet (Serial Number)

Used for forecasting costs by expense group. The report shows future and historical costs at sell, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|EquipmentTypes|,
|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Serial Number, Modifier, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

(Rpt_816) Forecast Worksheet - @Cost

Used for Forecasting costs by expense group. Shows future and historical costs at cost, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,
|EquipmentTypes|,|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|,
|Feature_32|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

(Rpt_818) Forecast Worksheet (Dealer Adjusted)

Used for forecasting costs by expense group and optional dealer market share. Shows future and historical costs at sell, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Planned, Workorder, Currency

Components

Long-term planning component reports

Navigate to **Long-term Planning** > **Components**.

There are a number of reports in Long-term Planning that can be used to view and validate the component changeout schedule. There are also reports to forecast costs and labour requirements.

The reports are:

- Component Status
- Overdue Strategy Tasks
- Changeout Schedule
- Changeout Analysis
- Changeout Analysis - Detailed
- Part Number Reference

Component capacity overview

Central inventory coordinators manage component inventory. Typically these are rotable components that are more complex to manage than normal parts because:

- They can be rebuilt.
- There are lead times to rebuild a component before it can be available in inventory again.
- There are different replacement options, including replace with new, replace with OEM remanufactured and replace with exchange.
- There are more capacity constraints for the rebuild centres.

Set Component Capacity Management System Tables

The system tables relevant to component capacity management are:

- Part Classification
- Part Type
- Part Rating

Part Classification

This table is used to define the type of part, particularly for rotable components, for example, OEM Reman, Exchange.

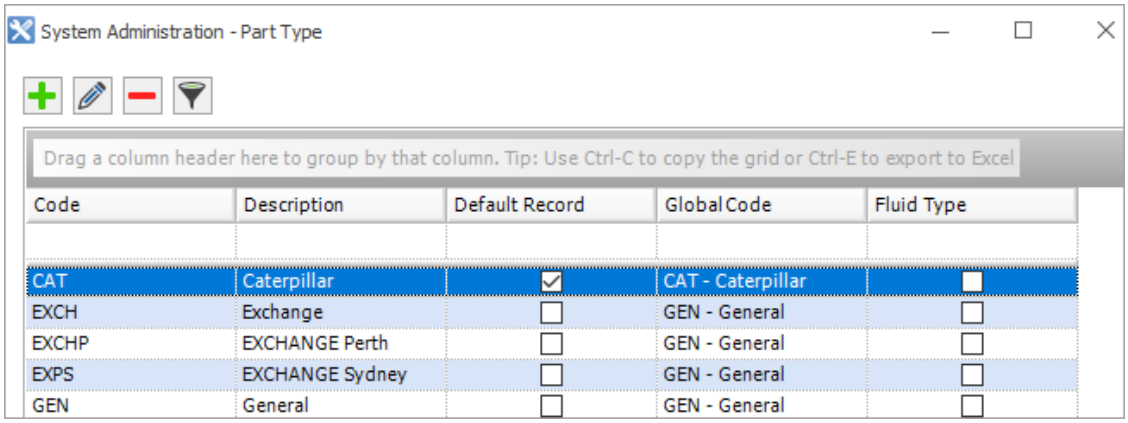
Core Retained means the component core will be rebuilt by the dealer. If it is not retained, it means the core (or part) is either disposed of or returned to the OEM to be rebuilt. AMT uses this flag to identify the components that will be rebuilt and returned to inventory.

Part Type

Part Type is very similar to Part Classification but it is not used in the component capacity management area as:

- You can only have one part type for a specific part number, but you can have different part classifications that apply to the same part number reflecting different the strategies.
- It generally comes from an external system so does not give you enough flexibility.

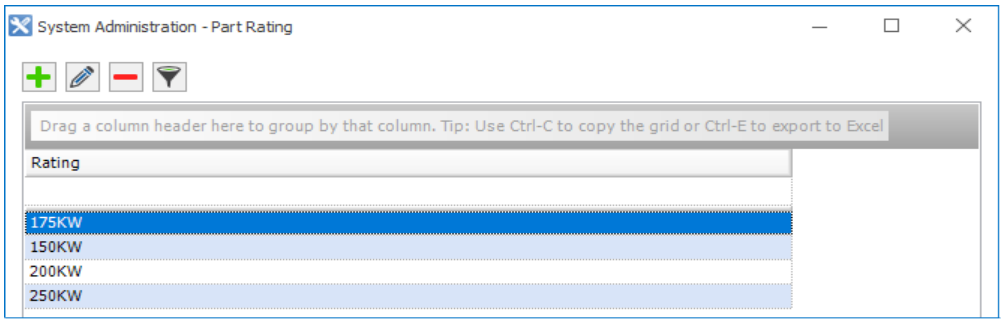
Part type is used to set up the basic parameters for parts and standard jobs.



Global Code and **Fluid Type** are not relevant for this area.

Part Rating

This table is used to define the rating or size of a part, for example, for engines, it can describe the size or capacity of the engine as 3600hp. This enables an additional breakdown of the component code, for example, all engines have the same component code, but using part rating you can group them by their capacity.



(Rpt_143) Component Status

Shows a component overview of the machine including overdue components, next changeouts and historical workorders.

Tips

Run the report across a fleet of equipment, analysing by Component Code to compare between equipment Last Change Out Details (Life Used,Cost), and Next Changeout Details (Life Used).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|EqpPlanID|,|EqpPlan|,
|ProjectionTypes|,|LogoLocation|,|CostBearer|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Component Code, System, +Cost Bearer, Sub System, Task Type

+ indicates that Cost Bearer filters costs only (not Strategy Task list).

(Rpt_415) Component Status (Suppression Impact)

Similar to the standard Component Status report except that the Last Occurrence details of strategy task, which is part of a suppression group, will reflect when the parent task was performed.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|EqpPlanID|,|EqpPlan|,|ProjectionTypes|, |LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Task Type, Component Cable, +Cost Bearer

+ indicates filters Costs only, (not Strategy Task List).

(Rpt_145) Contribution Summary

Shows graphically the projected margin versus the budget margin over time.

Tips

Costs are shown as Cost at Cost.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Additional filters

None.

(Rpt_204) Overdue Strategy Tasks

Reviews the components that are overdue. These components need to be rescheduled and a decision made as to when these are expected to be done.

Tips

Utilise this report as part of ongoing data health checks.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|AutoLaunch|,
|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|

Report Parameters

Report Type. Hierarchy

Additional filters

Cost Type, System, Sub System, Component Code, Maint. Order Type, Planning Group, STO, Regulatory Code

(Rpt_141) Changeout Schedule

Provides details of next changeout scheduled to allow for parts supply, labour resources and production schedules to be managed effectively

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Task Type, Part Classification, System, Part Rating, Component Code, Primary Part No, Cost Type, Rebuild Location, Sub System, Review Status, Task Counter, Sales Status, Strategy Task

(Rpt_140) Changeout Analysis

Provides a graphical view of the number of components to be charged in the time period required. You can navigate to this report in the following ways:
Long-term Planning > **Components**.
Modelling > **Components**.

Tips

Date selector defaults to entire plan term but can be modified to suit user requirements.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Feature_32|

Report parameters

Default.

Additional filters

System, Component Code, Strategy Task, Part Classification, Modifier Code, Part Rating, Occurrence Type, Primary Part No, Cost Type, Rebuild Location, Sub System, Review Status, Task Type, Sales Status

(Rpt_325) Changeout Analysis - Detailed

Provides a graphical view of the number of components to be changed in the time period required.

Tips

Date selector defaults to entire plan term, but can be modified to suit user requirements.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|UserID|,
|Feature_32|

Report parameters

Default.

System, Strategy Task, Strategy Task Desc, Modifier Code, Occurrence Type, Cost Type, Sub System, Task Type, Component Type, Region, Division, Cost Responsibility, Equipment Location, Part Classification, Part Rating, Primary Part No, Rebuild Location, Review Status, Sales Status

(Rpt_426) Part Number Reference - Modelling

Report which lists the Primary Part Number, Group Number and Component Serial Number per strategy task for the selected equipment.

Tips

Component Serial Number will always be blank for Modeling equipment

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Report Type, Equipment Hierarchy

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Modifier Code, Strategy Task, Region, Division, Cost Responsibility, Eqp Location

Marketing and Customer Support Editor

To make changes to any missing classifications:

- Navigate to **Product Support Marketing** > **PSM Administration** > **Marketing and Customer Support Editor**, filter for the required equipment and select the **Display Fields** filter.
- Double-click **Sales Responsibility**, **% of New Parts Purchase**, **Parts % of Purchase from Dealer**, **Labour % of Purchase from Dealer**, **Misc % of Purchase from Dealer**, **Marketing Group** and click **OK**.
- Once you have made all the selections, click **Apply Filters**, make the changes and click **Apply Changes**.

(Rpt_453) Parts Demand Analysis

Forecasts expected parts demand based on probability of sale. You can navigate to this report in the following ways:
Long-term Planning > **Components**.
Modelling > **Components**.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|EquipmentTypes|,|ProjHeaderID|, |ProjHeader|,|LogoLocation|,|UserID|,|Feature_32|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Cod, Cost Bearer, Part Type, Part Manufacturer, Primary Part No, Sales Resp., Part Classification, Part Rating, Rebuild Location, Review Status, Sales Status, Task Mode

(Rpt_562) Forecast Rotables Stock

Displays, per month, the quantity of rotable components available for use. A positive number indicates excess stock on hand, a negative number indicates shortage and zero indicates supply equalled demand in the month.

The report also contains four KPIs: Total Period Demand, Maximum Stock Level, Minimum Stock Level and Annualised Stock Turn.

Data for this report is sourced from the Inventory Parts Table, Parts Table and Strategy Tasks (Next Due and Projected).

Tips

Use this report to validate whether there are sufficient rotables to meet future demand.

If shortages exist, look to "move" stock excesses to the month of shortage through adjusting Next Due dates and usage of Strategy Tasks, optimising turn time (update Total Budget Turn Time in Parts Manager), or purchasing new rotable parts into stock (increasing parts on order in the Inventory Parts table.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

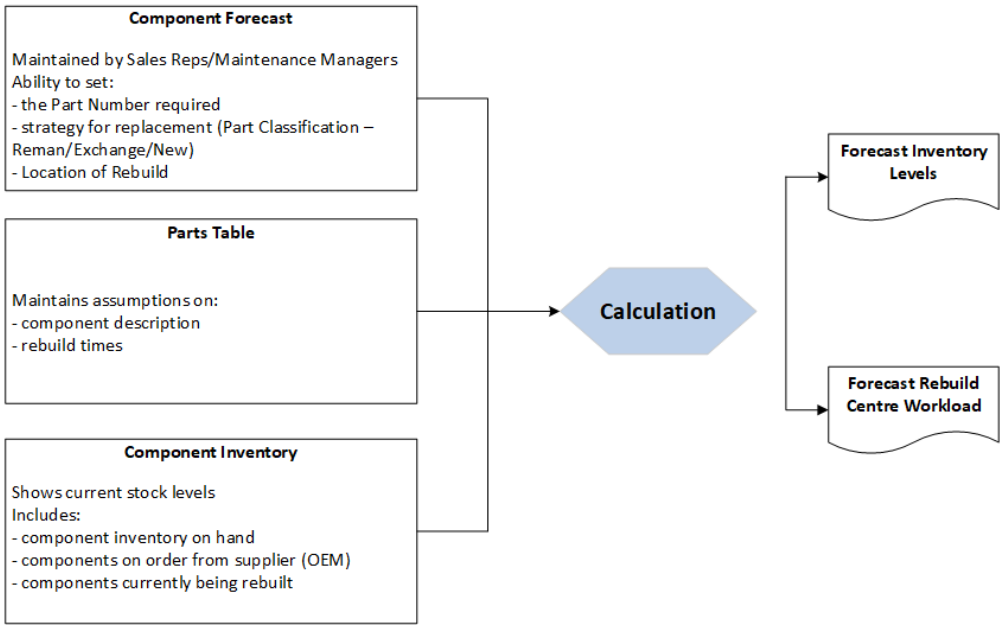
Default.

Additional filters

Part Number, Part Rating, Part Type, Component Code, Sales Status, Apply Minimum stock level <=, Minimum stock level <=, Apply Total Period Demand <=, Total Period Demand <=, Apply Average Stock Level <=, Average Stock Level <=, Apply Annualised Stock Turn <=, Annualised Stock Turn <=, Apply Annualised Stock Turn >=, Annualised Stock Turn >=

How component capacity works

- The life-cycle costing in AMT maintains a live component forecast.
- The strategy task contains some of the key parameters that drive the forecast.
- The strategy task is also used to set the strategy for the component replacement.
    - What type of component will be used (part classification)?
    - If it is to be rebuilt, then where will it be rebuilt?
- The strategy task includes:
    - Default assumptions for the component replacement strategy.
    - The specific plan for the next changeout.
- The parts table in AMT contains assumptions about the rebuild times for rotable components.
- The actual component inventory is imported into AMT and compared with the forecast to map out future projected inventory levels and identify expected shortfalls and excess inventory.



(Rpt_563) Rebuild Centre Forecast Workload

This report is useful for rebuild centres to forecast their future workload. The report also takes into consideration rebuild work currently in progress If the workload is too high in certain periods, contact those responsible for the component forecast to assess whether a rebuild can be pushed out or brought forward, or transferring the rebuild to a different location.

Tips

Use Review and Sales status filters to understand actual versus potential workload.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Part Classification, Part Rating, Part Number, Part Type, Source of Supply, Component Code, Review Status, Sales Status

(Rpt_564) Changeout Forecast Worksheet

Combines current component life and cost forecast for comparison to original budget projection assumptions. Navigate to **Long-term Planning** > **Components**.

Tips

The cost per month is based on the number of occurrences of the strategy in the month multiplied by the current cost of the strategy task.

Life to date of the component in the equipment now, and life achieved at last changeout, recognise the impact of suppression.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|LogoLocation|,|UserID|,|Feature_32|

Additional filters

Task Type, System, Component Code, Cost Type, Sub System, Task Counter, Strategy Task

Component Forecast Management

The topics and screens for Component Forecast Management are:

- Record Actual Usages.
- Monitor Equipment Condition.
- Update component forecasts using the Component Changeout form.
- Create a changeout record using the Long-term Planning form.

- Update historical changeout details using the Work Order Settlement Manager form.
- Validate the component forecast using Long-term Planning reports.

Component capacity forecast process validation

Process validation items include the following.

Number	Title	Description	Target
1	Strategy defined.	All major components should have a strategy defined.	All.
2	Capacity balanced.	Forecast rotable levels should be positive.	All.
3	Rebuild capacity achievable.	Rebuild centres have confirmed capacity to meet forecast workload.	All.

Component forecast management process validation

The validation items for this process include:

Process Number	Title	Description	Target
1	Overdue Tasks	There should be no tasks that are overdue.	None
2	Tasks Not Reviewed	All tasks should be reviewed if they are within a week of impacting the planning window.	<10%
3	Tasks Not Confirmed	All tasks within six weeks of the target date should be confirmed.	None
4	Sales Status	The sales status should be maintained with all tasks within six weeks of the target date having a sales status of confirmed.	
5	Component Forecast	The 12-month component forecast is optimised and realistic.	
6	Customer Communication	Reports sent to the various stakeholders.	Conducted Monthly

Review component forecasts

There are a number of screens and reports you can use to review the component forecasts:

- Long-Term Planning reports - Navigate to **Long-term Planning** > **Components**.
- Long-Term Planning form - Navigate to **Long-term Planning** > **Long-term Planning**.
- Component Changeout form - Navigate to **Long-term Planning** > **Component Changeout Grid**.

Component Changeouts

Component changeouts require three journals:

- Component disposal:
  - EDC 115 - Disposal - Component. Writes off the capitalised cost of the component. Navigate to the **Asset Ledger** screen.
  - EDC 116 - Disposal - Component - Reverse CBD. Reverses the accumulated depreciation on the disposed component. Navigate to the **Plant Asset Journals** screen.
- Capitalise component being installed:
  - EDC 110 - Component Capitalisation. Usually the work order cost of the component changeout. Navigate to the **Plant Asset Journals** screen.

Review last changeout

You need to review and update the last changeout details of the strategy tasks, both when the task occurred and the latest sales information.

Navigate to **Long-term Planning** > **Component Changeout Grid**. Filter for the relevant equipment and select the **Display Fields** filter. Double-click any of the relevant sales-related fields, and click **OK**. Once you have made all your selections select the **Show All Tasks** check box, then click **Apply Filter**.

You should be able to see the strategy task occurrences and adjust them accordingly.

Review last changeout (Sales Status)

Follow the instructions above. Once you have made your selections and selected **Show All Tasks**, right-click on the relevant **Strategy Task** and click **Edit Sales Status**.

The fields the Sales Responsibility user needs to complete are listed in the table below. Then specify the **Sale Details** and **Save**.

SALES STATUS DETAILS	
Field	**Description**
Sales Responsibility	This needs to be confirmed in order to understand what the Sales Responsibility's pipeline looks like and who is responsible for the sale related to the particular job.
Sales Status	This is from your perspective as to how likely is it that the dealer will get the sale. The options are: Confident, Lost, Sale Confirmed, Unknown and Unlikely.

SALES STATUS DETAILS	
Field	**Description**
PO Number	The purchase order number for the job.
Sales Lost Reason	If the dealer did not get the sale, what is the reason for losing the sale? The options are: Delivery Constraint. Other, Price, Quality, Unknown.
Customer Contacted	If the customer has been contacted the check box should be selected.
Customer Contacted Employee	If the customer contacted the dealership, which employee was contacted? This filter displays all the employees in the Employee system table.
Customer Contact Date	The date when the customer was contacted.
Comments	Any general comments.

Communicate rebuild forecast

Navigate to **Long-term Planning** > **Components**. The following reports can be used to view the forecast by rebuild centre.

- Changeout Schedule
- Changeout Analysis
- Changeout Analysis Detailed
- Rebuild Centre Forecast Workload

Set rebuild strategy

Setting the rebuild strategy involves refining the strategy for the component changeouts that are forecast in the next 3-6 months.

The assumption is that the forecast is accurate. The decisions to make when refining the strategy are:

- What component should be installed (new, remanufacture, exchange, rebuild)? The component installed dictates what will happen to the component being removed.
- If the component is to be rebuilt in the company, then where should it be rebuilt?

Validate parts and components

Planners need to validate that the required parts and components will be available on the dates planned, whether there any constraints and so on. This helps them understand whether the long term plan being developed is realistic (from a parts resourcing point of view).

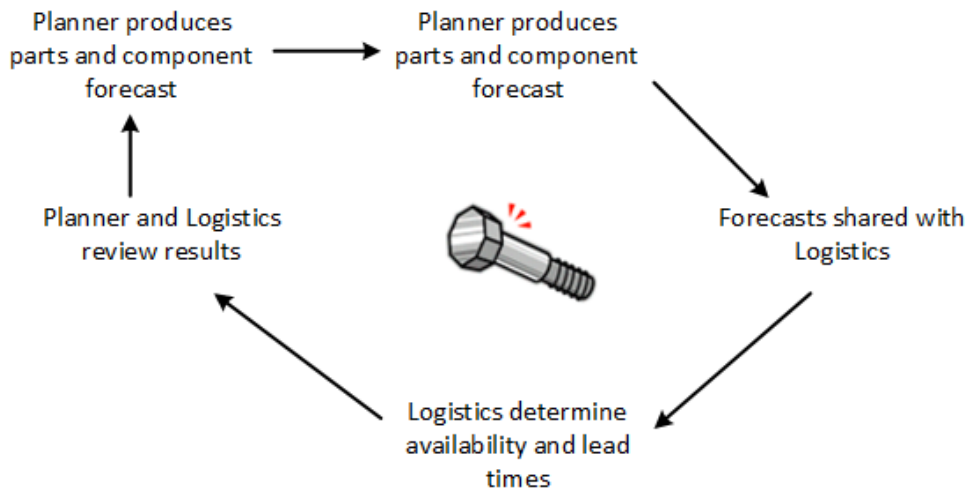
Navigate to **Long-term Planning** > **Components** and run the Changeout Analysis - Detailed report for analysis.

A useful way to run this report is to set the **Analyse By** option to include **Primary Part Number**. This requires the people responsible for maintaining strategy tasks to ensure that the **Primary Part Number** field on the **Strategy Task Editor > Component** tab of the strategy task is populated.

The long term planner then validates the demand forecast with supply, which means liaising with the inventory procurement department.

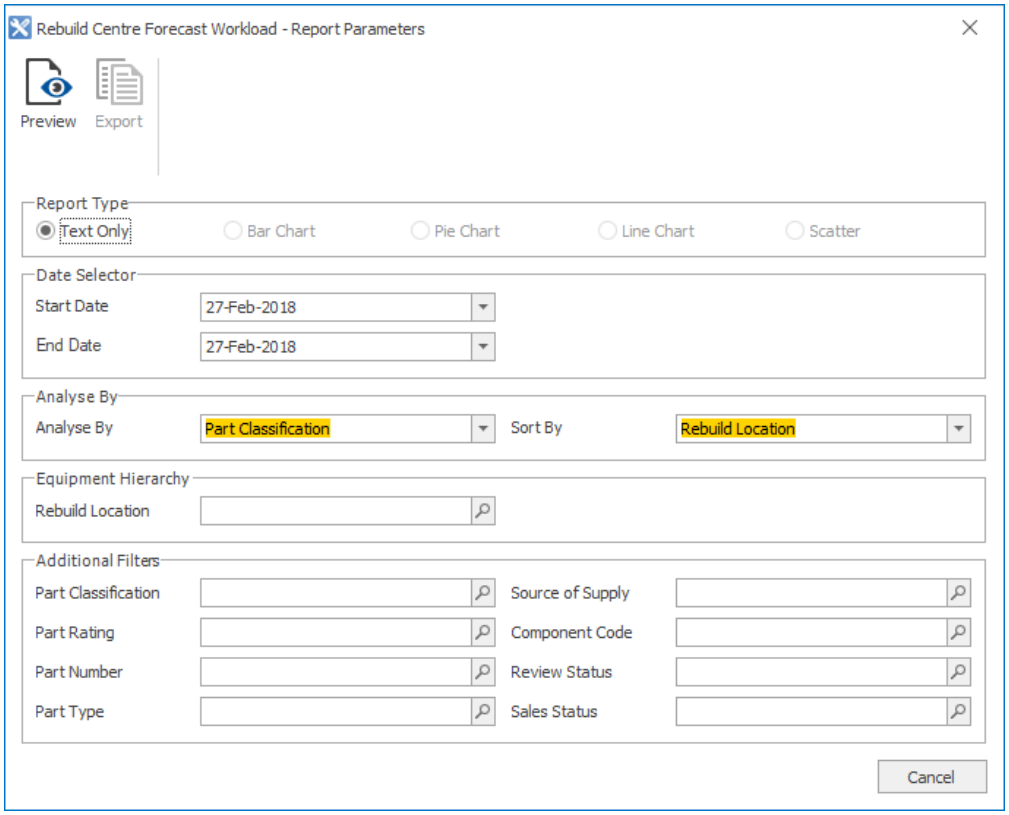
Responsibilities are shown in the table below.

Planning window responsibilities for ordering parts and components		
Role	**Window**	**Parts Ordering Scope**
Long Term Planner	Greater than W+4 to end life of mine	Parts or components • Greater than W+4 to life of mine forecast • With a lead time greater than six months • That require capital expenditure approval
Short Term Planner	Less than W+4	Parts or components • Less than W + 4 forecast • Lead time of less than six months • Falls within the annual capital expenditure scope
Scheduler	W+1	N/A



Validate rebuild centre capacity

You need to validate the rebuild centre capacity to ensure that it has the capacity to handle the component rebuilds being assigned to it. To do this, run the Rebuild Centre Forecast Workload report. Navigate to **Long-term Planning** > **Components** > **Rebuild Centre Forecast Workload**.

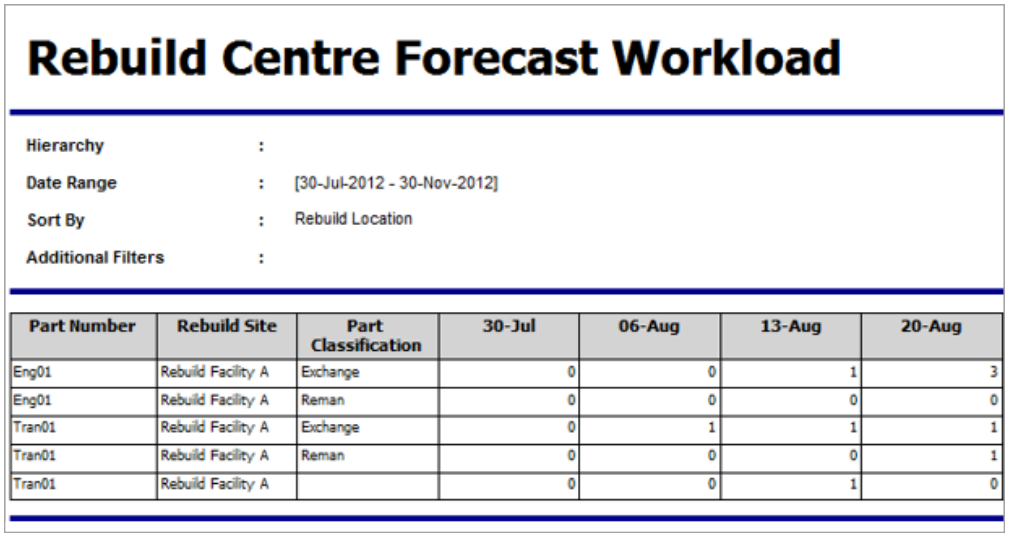


You can Analyse By **Part Classification**, **Part Type**, **Part Rating**, **Component Code** or **Part Description**. You can Sort By **Rebuild Location**, **Part Number** or whatever you selected in the **Analyse By** filter.

The report is useful for rebuild centres to forecast their future workload. The report also considers rebuild work currently in progress.

You can use the **Review Status** and **Sales Status** filters to understand actual vs potential workload.

An example of the report is shown below.



Rebuild Centre Forecast Workload

Hierarchy :
Date Range : [30-Jul-2012 - 30-Nov-2012]
Sort By : Rebuild Location
Additional Filters :

Part Number	Rebuild Site	Part Classification	30-Jul	06-Aug	13-Aug	20-Aug
Eng01	Rebuild Facility A	Exchange	0	0	1	3
Eng01	Rebuild Facility A	Reman	0	0	0	0
Tran01	Rebuild Facility A	Exchange	0	1	1	1
Tran01	Rebuild Facility A	Reman	0	0	0	1
Tran01	Rebuild Facility A		0	0	1	0

The report shows the average number of components under repair in the rebuild centre during the week, rounded up. Therefore, if a component is expected to take three weeks to build, it will display in three consecutive weeks.

Consumables

(Rpt_443) Consumables Forecast

Shows Forecast Consumable Usage.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Part Type, Job Code, Show

Labour

Validate labour resources

The planner validates labour in the following ways

- Demand versus supply.
- Skills required versus skills available.

Labour Activity defines skills types in AMT, and is set at the strategy task **Job** level.

Navigate to **Long-term Planning** > **Labour**, and use the Labour Forecast - Hours and Labour Forecast – Persons reports to view the labour demand and the types of skills required, and to view whether the plan is causing spikes in labour demand.

The planner must assess whether there are sufficient labour resources on site to meet the demand.

(Rpt_183) Labour Forecast - Hours

This report provides you with an understanding of how much labour is required for the contract.

The key drivers for this report are Labour Hours and Labour Costs set up in the strategy tasks within the estimate projection.

If labour is costed in the estimate projection, this report shows labour costs.

If labour costs are in overheads, labour costs will not be seen in this report. The Total Cost Worksheet report would have to be used to see this information.



Estimates can be benchmarked against each other, and against other equipment or centrelines in a number of reports by selecting **Equipment** as the **Analyse By** parameter in the report and including other equipment in the report's filters.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity

(Rpt_184) Labour Forecast - Persons

Lets you analyse maintenance work by persons required.

This report provides you with an understanding of how much labour is required for the contract.

The key drivers for this report are Labour Hours and Labour Costs set up in the strategy tasks within the estimate projection.

If labour is costed in the estimate projection, this report shows labour costs.

If labour costs are in overheads, labour costs will not be seen in this report. The Total Cost Worksheet report would have to be used to see this information.



Estimates can be benchmarked against each other, and against other equipment or centrelines in a number of reports by selecting **Equipment** as the **Analyse By** parameter in the report and including other equipment in the report's filters.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Cost Responsibility, Expense Element, Labour Activity, Productive Hrs+

+ indicates the average productive hours per person per month.

(Rpt_832) Workload Optimisation

Compares Labour Hours by Month, Quarter or Year for two or more projections.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity

Parts

Parts Forecast

This report is used to determine the quantity of parts to be ordered. The format of the report is the same as the Parts Usage Details report, however the calculation logic is different. For example, if a quantity of 3.6 is forecast should you order 3 or 4?

The forecast of the parts for a specific period is based on the multiplication and summation of fractions. The handling of the fractions for the forecasting is critical as items can only be forecast as integers. There is the ability to nominate the level to round up the forecast. For example, is a part forecast at 0.4, 0.5, 0.6 or 1? The default is 0.5.

Month	Parts Required	Cumulative Required	Rounded Required	Cumulative Previously Forecast	Parts Forecast
1	0.5	0.5	1	0	1
2	1.5	2.0	2	1	1
3	0.5	2.5	3	2	1
4	0.5	3.0	3	3	0
5	1.5	4.5	5	3	2
6	0.5	5.0	5	5	0
7	0.5	0.25	6	5	1

(Rpt_838) Parts Forecast

This allows the user to manually generate a specific Part Forecast.

Initial parameters

|ControlType|,|Idlist|,|TextList|,|LogoLocation|,|UserID|,|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Part Source, Part Number, Part Type, Show 100% parts only

Parts Forecast Manager

Parts Forecast Manager contains a listing of parts forecasts automatically generated by AMT. You can also manually generate a specific forecast. AMT can calculate and save one parts forecast per site per month. This is automatically calculated on the first day of the each month. You can, however, manually update the current months forecast at any time. Parts Forecast Manager's opening screen presents you with the option to view the existing parts forecasts generated by the system.

To trigger the calculation of a fresh parts forecast for a specific branch, click  next to the relevant branch.

AMT will change the calculating status of this forecast to **True**, and schedule the recalculation to occur overnight when the server's usage is low.



For a parts forecast to be calculated, strategy tasks of the equipment within the branch must be linked to standard jobs (where the standard jobs contain parts). The only exception relates to the Rotable Forecast report (see below).

Button	Action
	Apply filters
	Cancel filters
	Return to AMT Web main menu
	Save
	Close or Cancel
	Log out

The remaining options on the menu relate to specific parts forecast reports.

- **Parts Forecast**
  - This report is used to determine the quantity required to be ordered.
  - The forecast for each period (month) is based on the cumulative requirement to the end of the period less the total number of items forecast for previous periods.
- **Parts Usage**
  - This report shows the forecast demand for the next 12 months.
  - The parts required for each of the tasks are multiplied by the percentage usage of the individual parts (in the standard job).
- **Rotable Forecast**
  - This report has been superseded by the Forecast Rotables Stock report.

Export Parts Forecast

AMT generates a . csv flat file for the selected parts forecast. The fields in the export file are:

- Branch
- Source of Supply
- Part Number

- Period (Year-Month)
- Quantity

Create and import parts

A list of current parts needs to be available in order to create or modify a standard job. Part details are normally held within an enterprise resource planning system. These parts can be imported in AMT using the Part and Part Price Update interface or can be created manually in AMT using Parts Manager.

Add parts using Parts Interface

New parts are interfaced into AMT using the Add/Update Parts Pricing interface. Check with your system administrator in order to determine if there is an interface between AMT and your inventory system. This interface may not be automatic, it may require manual loading using AMTs Interface Administration function.

Health - Parts forecasting

The focus areas for parts forecasting that need to be checked are:

- Are parts forecasts being produced?
  - Check they are being produced on a monthly basis, checked and sent to the parts group.
- Are they accurate?
  - Their accuracy is dependent on the quality of the standard jobs and the quality of the current projection.
- Are they comprehensive?
  - Parts forecasts will only be produced where standard jobs are set up in AMT and linked to strategy tasks.

Review superseded parts

It is common for parts to be superseded over time by new and improved parts. Superseded parts are usually identified in the enterprise resource planning system, and they can also be identified in AMT.

You can superseded a part in AMT using the Part and Part Price Update interface process, in which there is a flag passed by the enterprise resource planning system to supersede it, or you can do it manually in AMT using Parts Manager.

It is also possible to import part supersessions via integration. The parts supersession interface will supersede parts on standard jobs, and primary part numbers on strategy tasks.

Shutdown, Turnaround, Outage

Shutdown, Turnaround and Outage (STO) are all standard industry terms. An STO can vary in duration from short (one hour or less due to an unexpected event on a fixed plant item) to many days or months. Therefore, an STO can be categorised into anticipated or opportunistic.

Anticipated: These are planned and scheduled into many months or years in advance.

Opportunistic: These are unexpected and therefore require agility to plan and schedule any work to be done.



Opportunistic STOs could be managed by existing event functionality except for the requirement that there may be multiple equipment involved and this isn't possible with existing event functionality.

When you add a strategy task you can add an STO task option if the task is typically performed during an STO. Selecting **Requires STO** automatically assigns the STO task to the closest STOs due date. If the due date is midway between two STOs, the system defaults to the earlier period. You can manually overwrite this at any point of the STO planning process.

You can see the STOs on the Gantt Chart in Long-term Planning - Advanced.

Audit Log – Requires STO Flag

This report is designed to show changes made by users to the **Requires STO** check box on Equipment, Functional Location, Strategy Task and Standard Job.

Entries are sorted in reverse chronological order (most recent changes display at the top of the report).

STO Manager

STO Manager lets you register, review and manage shutdown, turnaround or outage details. The STO can be created at the Branch level.

Clicking the **Apply Filter** button returns a list of STOs that match the filters you have entered.

You can edit the following columns in the grid:

- Planned Start Date
- Planned Finish Date
- Duration (Planned)
- Actual Start Date
- Actual Finish Date
- Status
- Planning Group

Create an STO

Click  above the grid to add a new STO. STOs are added with an **In Planning** status. When you enter an **Actual Start Date**, the status changes to **In Progress**. This means that the **Planned Start Date** cannot be

modified. When you enter an **Actual Finish Date**, the STO will have a **Completed** status.

Can Move check box

The following rules apply, depending on whether or not you have selected the **STO** and **Can Move** check boxes.

1. **STO Required is selected (that is, Y), Can Move is not selected (that is, N)**

   This means that the occurrence needs to be completed in an STO and the dates of the occurrence cannot move (unless there is manual intervention).

2. **STO Required is selected (that is, Y), Can Move is selected (that is, Y) and Tolerance = 0 days**

   This means there is no limit to the number of days that an occurrence can move, but the date movement is constrained by the STO dates (for example, must be within the STO dates).

3. **STO Required is selected (that is, Y), Can Move is selected (that is, Y) and Tolerance => 0 days (1, 2 ...)**

   This means that the occurrence move is limited by the number of days in the tolerance and constrained by the linked STO dates (for example, must be within the STO dates).

STO Dates and Status

Date filters

The various date filters return the following STOs to the grid.

- No dates entered - returns all.
- Start Date & Time - returns all from this day forward.
- End Date & Time (no start) details - returns all up to and including end date.
- End Date must always be greater than the Start Date.

Status

 Date = Date and Time

In Planning

- STO has Planned Start and End Date.

In Progress

- STO has Actual Start Date.
  - Changing status to In Progress requires an Actual Start Date to be added.
- Actual Start Date does not need to be the same as Planned Start Date.
  - Actual Start Date can be now or in the past.
- Planned Duration derives Actual End Date.

Completed

- STO has Actual End Date.
    - Changing status to Completed requires an Actual End Date to be added.
    - Date can be now or in the past; it cannot be in the future.
- Actual Duration will be calculated from Actual Start and End Dates.

Deferred & Abandoned

- Deferred - No work has been completed; no Actual Start Date.
- Abandoned – Work has been completed; Actual Start Date exists.

Archived

- Archived Flag is checked.

STO Category

This is a system table where you can set up STO categories. Examples of STO categories are Weekly, Monthly, Annually and Bi-annual.

Product Support Marketing

Product Support Marketing shows the overall sales potential for a specific dealer, in its territory. Product Support Marketing also reflects the market share that the dealer has within the industry based on its sales records, and highlights where to target future sales.

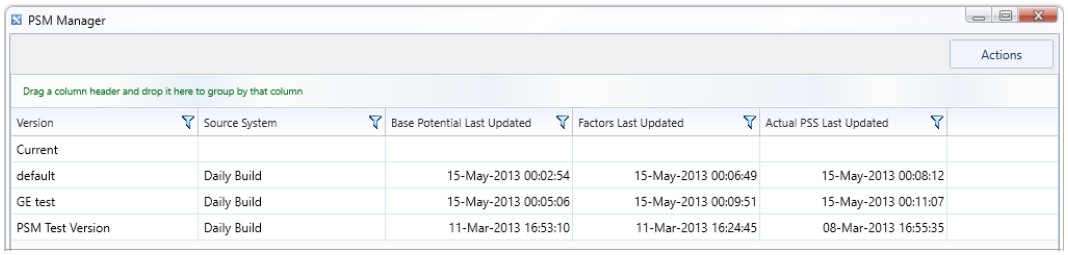
PSM Administration

PSM Manager

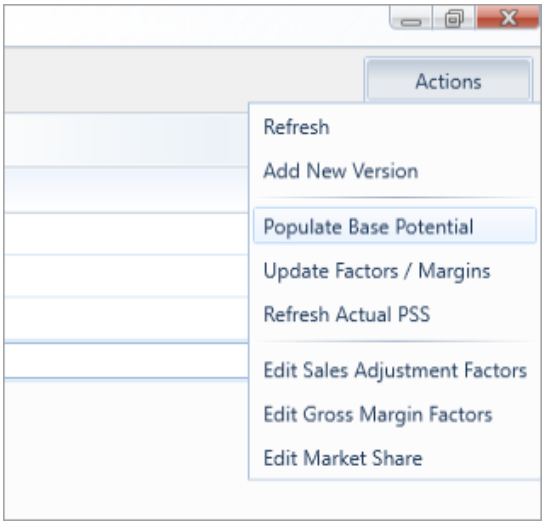
The PSM Manager is a screen for PSM administrators to manage Product Support Marketing cubes; this is where all of the market data is stored. From this screen PSM administrators can manage existing PSM cubes, create new PSM cube versions (for example, archives), populate Base Potential, edit and apply sales adjustment factors and gross margins, and refresh cubes with Actual Product Support Sales.

You can also create a Version to represent a Sales Budget, for example, Financial Year 2022 sales budget.

1. Under **Product Support Marketing** > **PSM Administration**, double-click **PSM Manager**.
2. Click **Actions**, then click **Add New Version**.



3. Highlight the new version, select **Actions**, then click **Populate Base Potential**.

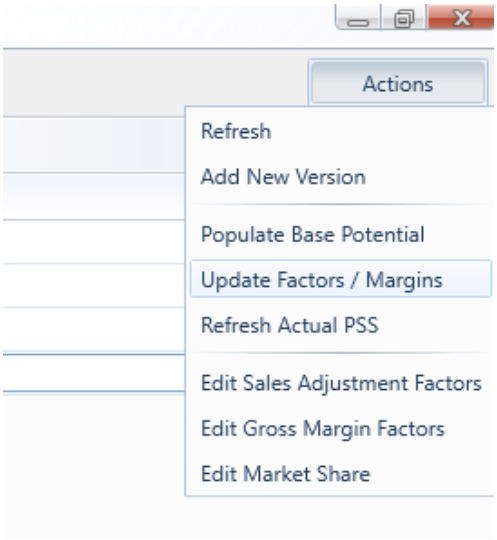


4. When you create a new version, the following information is used to populate the Base Potential:

- Equipment (entire life cycle).
- Strategy Tasks (all occurrences within the life cycle of the equipment) with relevant actual and projected costs.
- Sales Adjustment factors.
- Gross Margin Factors.

At this point, AMT populates the Base Potential within the PSM cube. You can review the various PSM reports and decide if adjustments need to be made.

1. On PSM Manager, highlight the version you want, click **Actions**, then click **Update Factors / Margins**.

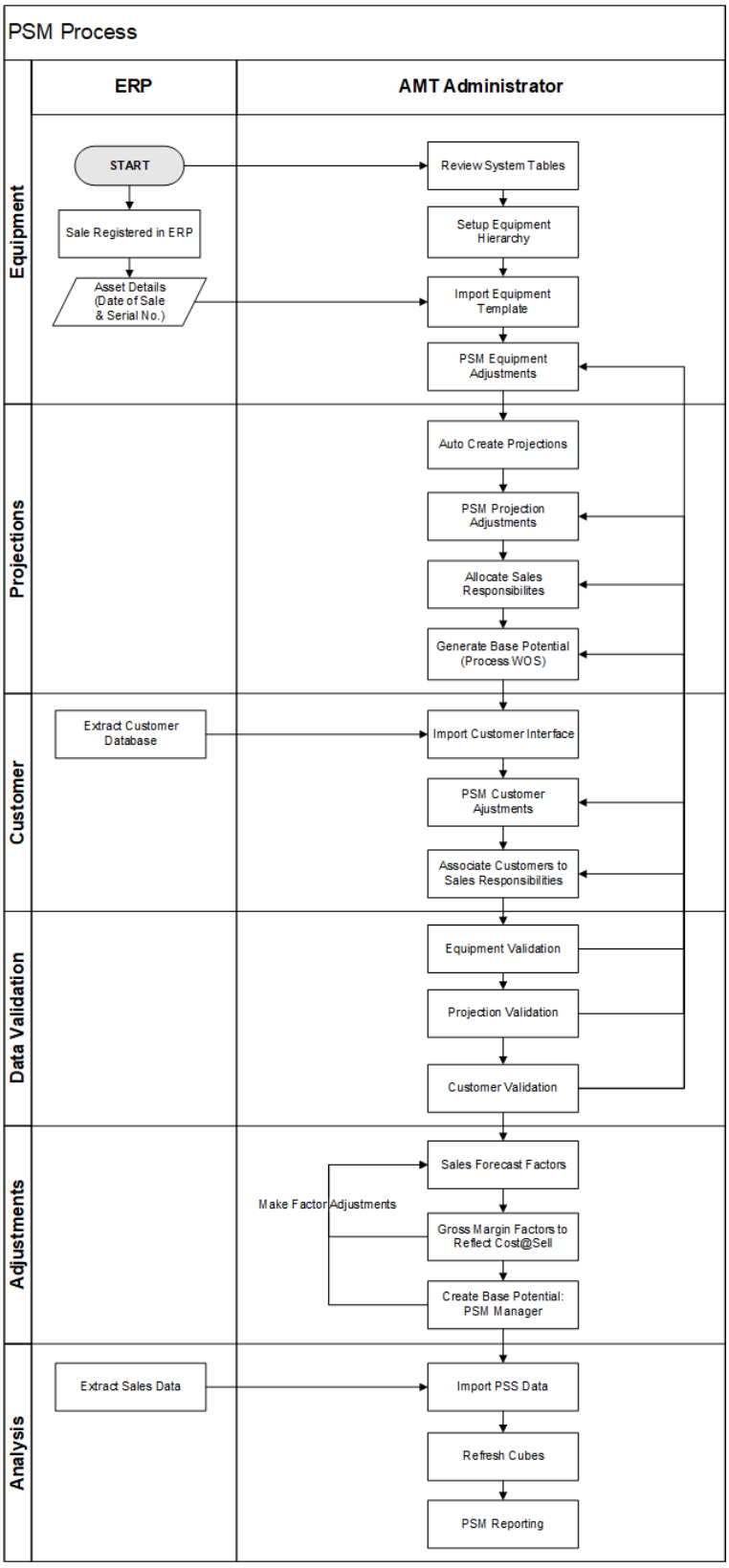


2. Highlight the version you want, click **Actions** and select the relevant quick links.
    - There are quick links to the Sales Adjustment Factor, Gross Margin Factors and Market Share (Marketing & Customer support editor) screens. This allows for quick and easy adjustments to be made.
3. Highlight the version you want, select **Actions**, then select **Edit Sales Adjustment Factors**.

    You have the option to manually update the Base Potential if changes are made to the Factors / Margins.

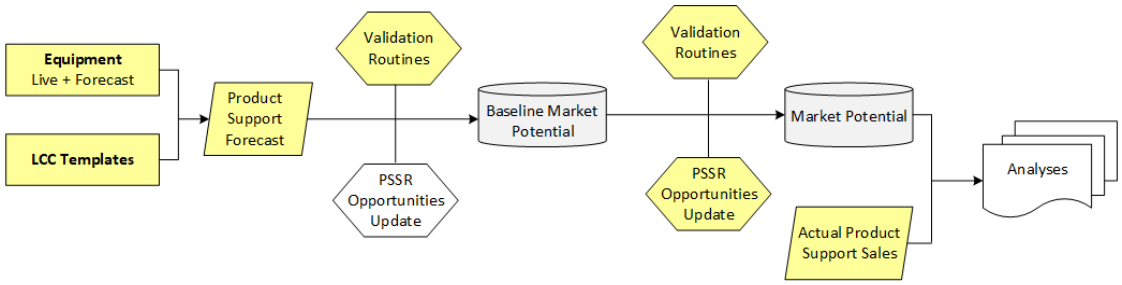
Product Support Marketing process - Central Admin

The flowchart below shows the whole Product Support Marketing process.


PSM Process
ERP
AMT Administrator
Equipment
START
Sale Registered in ERP
Asset Details (Date of Sale & Serial No.)
Review System Tables
Setup Equipment Hierarchy
Import Equipment Template
PSM Equipment Adjustments
Projections
Auto Create Projections
PSM Projection Adjustments
Allocate Sales Responsibilites
Generate Base Potential (Process WOS)
Customer
Extract Customer Database
Import Customer Interface
PSM Customer Ajustments
Associate Customers to Sales Responsibilities
Data Validation
Equipment Validation
Projection Validation
Customer Validation
Adjustments
Sales Forecast Factors
Make Factor Adjustments
Gross Margin Factors to Reflect Cost@Sell
Create Base Potential: PSM Manager
Analysis
Extract Sales Data
Import PSS Data
Refresh Cubes
PSM Reporting

Product Support Marketing key principles - Central Admin

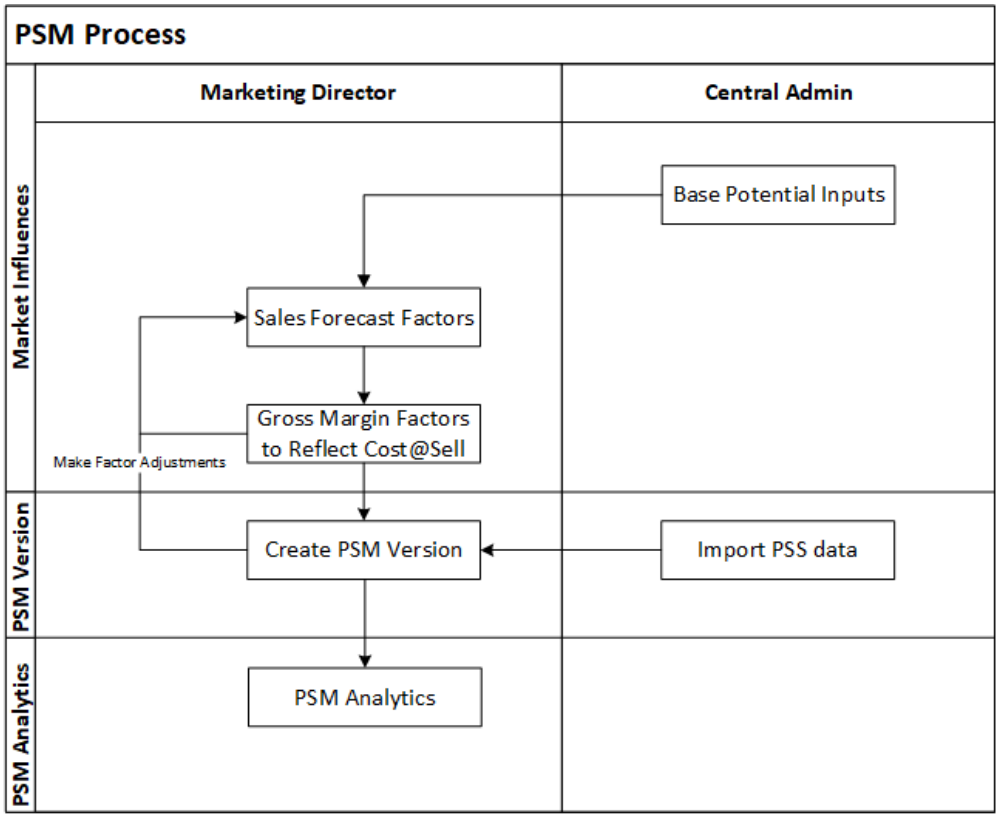
The flow chart below depicts the flow of the Product Support Marketing process. The coloured blocks highlight the areas where **Central Admin** users are actively involved.



- The AMT Product Support Marketing Module allows the dealer to understand the size of the market it is operating in and its market share.
- All of the known equipment, along with the relative life cycle costing, which the dealer has sold, need to be loaded into AMT whether they are managed by MARC or CSA contracts or not.
- A series of customised information fields are displayed at each step and are available for population; these include but are not limited to, start date, estimated life, and salesperson responsible for the deal.
- If the equipment is not actively managed, AMT assumes that all historical jobs have been performed according to the projection and marks them as being completed. AMT then forecasts all future jobs according to the projection, and together the two form what is termed the base potential, that is, all possible sales to be made within the dealer's region.
- The base potential can be adjusted to incorporate sales forecast adjustment factors, such as an expected decline in the mining industry, expected increase in demand for construction equipment or currency fluctuation.
- The dealers gross margin factors are applied to the base potential to reflect cost@sell prices.
- All of the dealer's recorded sales data is then imported into AMT; this information is read only in AMT.
- The difference between the base potential and the enterprise resource planning sales will reflect the missed opportunities to the dealer. The dealer can also deduct what percentage of the market it dominates and in which areas it can focus on in order to increase sales.

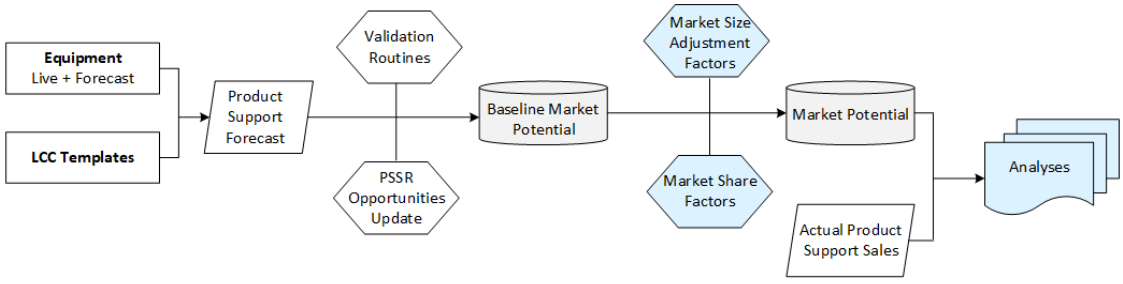
Product Support Marketing process - Marketing Director

The flowchart below shows the Product Support Marketing process for the **Marketing Director** user.



Product Support Marketing key principles - Marketing Director

The flow chart below depicts the flow of the Product Support Marketing Process. The coloured blocks highlight the areas where **Marketing Director** users are actively involved.

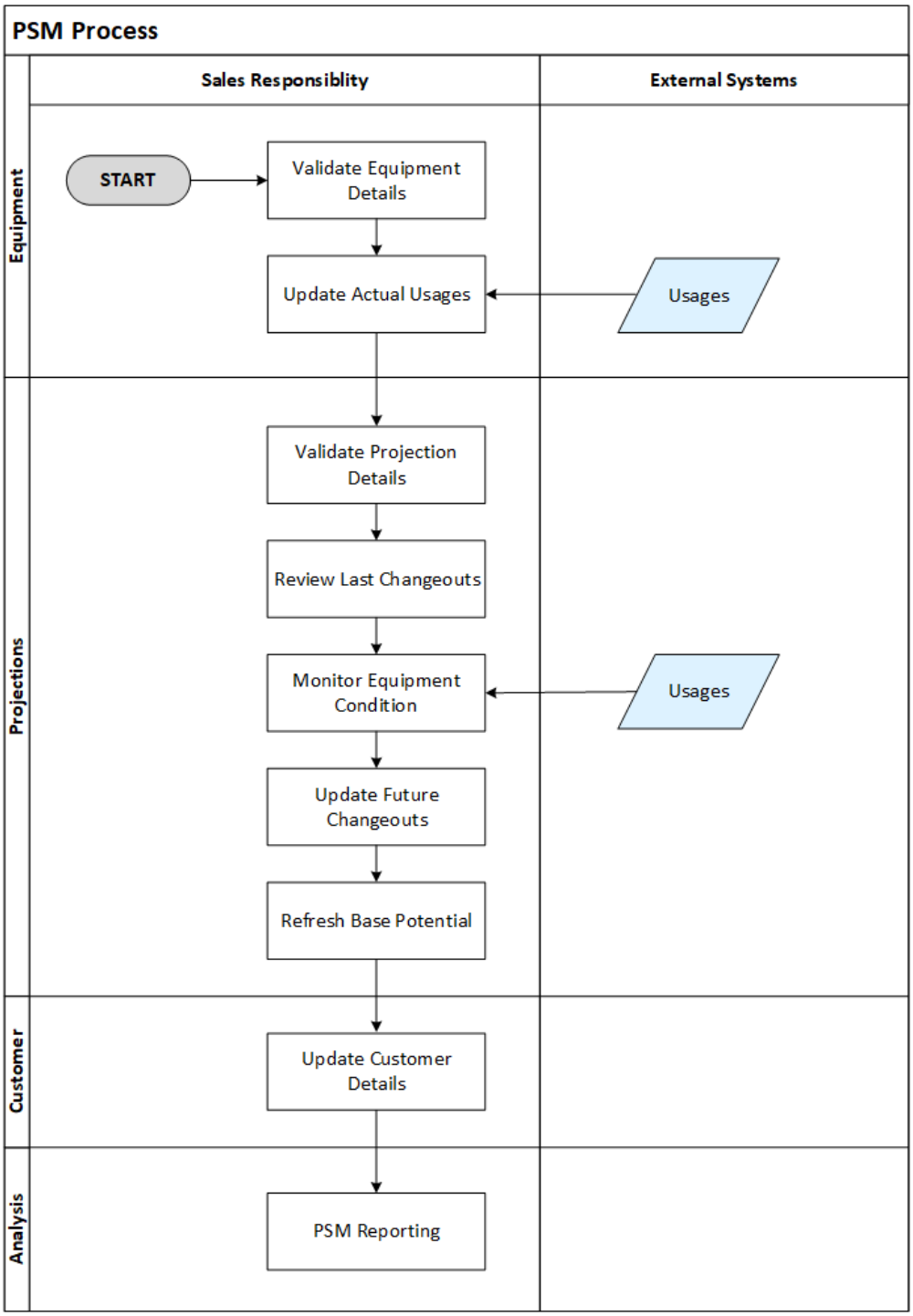


- The AMT Product Support Marketing Module allows the dealer to understand the size of the market it is operating in and its market share.
- All of the known equipment, along with the relative life cycle costing, which the dealer has sold, need to be loaded into AMT whether they are managed by MARC or CSA contracts or not.
- A series of customised information fields are displayed at each step and are available for population; these include but are not limited to, start date, estimated life, and salesperson responsible for the deal.
- If the equipment is not actively managed, AMT assumes that all historical jobs have been performed according to the projection and marks them as being completed. AMT then forecasts all future jobs according to the projection, and together the two form what is termed the base potential, that is, all possible sales to be made within the dealer's region.

- The base potential can be adjusted to incorporate sales forecast adjustment factors, such as an expected decline in the mining industry, expected increase in demand for construction equipment or currency fluctuation.
- The dealers gross margin factors are applied to the base potential to reflect cost@sell prices.
- All of the dealer's recorded sales data is then imported into AMT; this information is read only in AMT.
- The difference between the base potential and the enterprise resource planning sales will reflect the missed opportunities to the dealer. The dealer can also deduct what percentage of the market it dominates and in which areas it can focus on in order to increase sales.

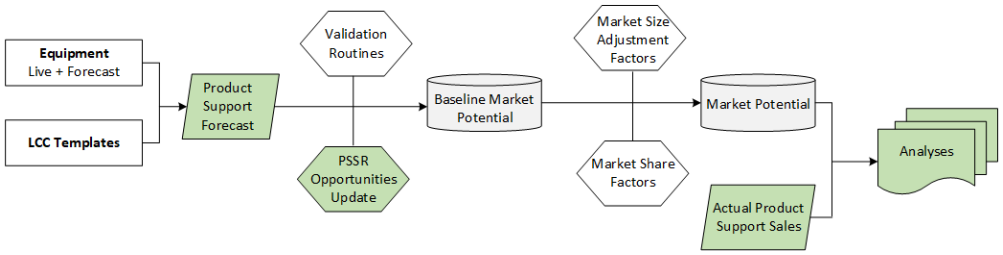
Product Support Marketing process - Sales Responsibility

The flowchart below shows the Product Support Marketing process for the **Sales Responsibility** user.



Product Support Marketing key principles - Sales Responsibility

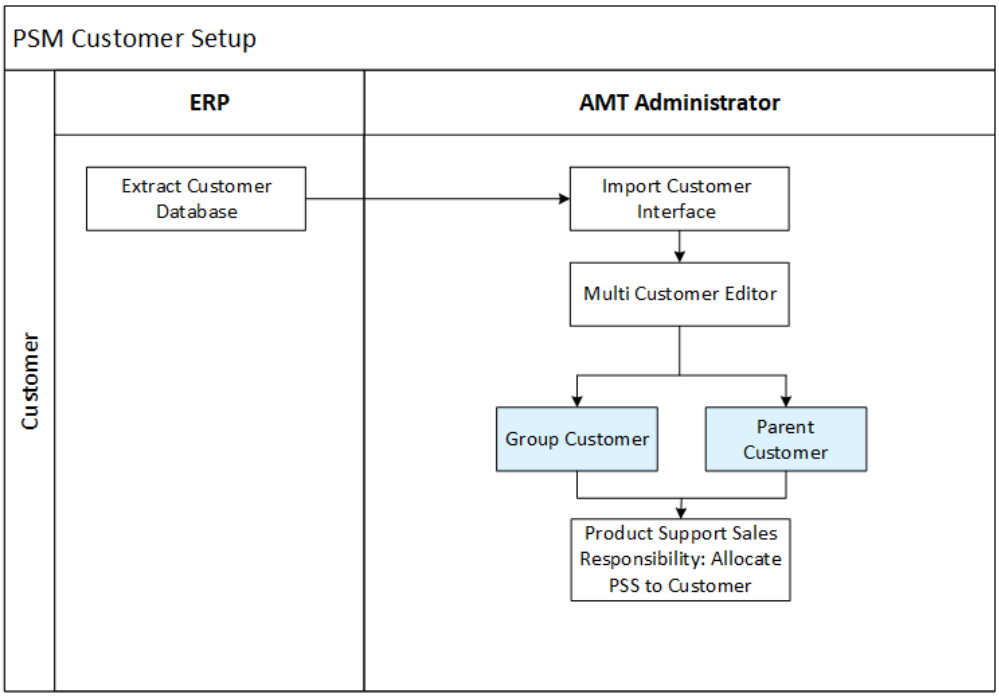
The flow chart below depicts the flow of the Product Support Marketing Process. The coloured blocks highlight the areas where **Sales Responsibility** users are actively involved.



- The AMT Product Support Marketing Module allows the dealer to understand the size of the market it is operating in and its market share.
- All of the known equipment, along with the relative life cycle costing, which the dealer has sold, need to be loaded into AMT whether they are managed by MARC or CSA contracts or not.
- A series of customised information fields are displayed at each step and are available for population; these include but are not limited to, start date, estimated life, and salesperson responsible for the deal.
- If the equipment is not actively managed, AMT assumes that all historical jobs have been performed according to the projection and marks them as being completed. AMT then forecasts all future jobs according to the projection, and together the two form what is termed the base potential, that is, all possible sales to be made within the dealer's region.
- The base potential can be adjusted to incorporate sales forecast adjustment factors, such as an expected decline in the mining industry, expected increase in demand for construction equipment or currency fluctuation.
- The dealers gross margin factors are applied to the base potential to reflect cost@sell prices.
- All of the dealer's recorded sales data is then imported into AMT; this information is read only in AMT.
- The difference between the base potential and the enterprise resource planning sales will reflect the missed opportunities to the dealer. The dealer can also deduct what percentage of the market it dominates and in which areas it can focus on in order to increase sales.

Set up Product Support Marketing customers

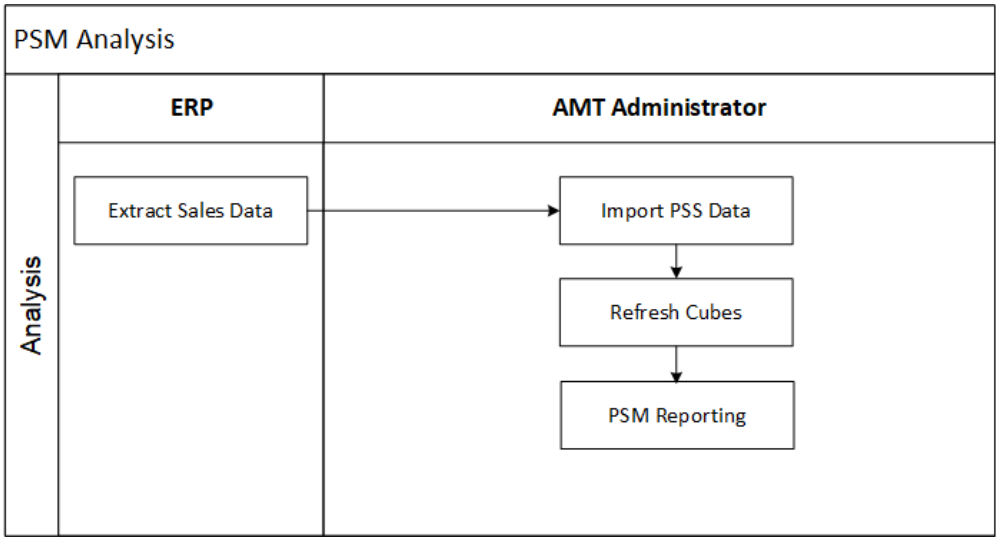
The flowchart below shows the process of setting up Product Support Marketing customers.



The enterprise resource planning system will contain a list of customers against which sales would have been recorded. These need to be imported into AMT.

Product Support Marketing analysis

The flowchart below shows how to get the sales information from the enterprise resource planning system into AMT, and the steps involved in the analysis.



Extract sales data: The sales data is all be recorded in the ERP system so it will need to be extracted in order to populate the Product Support Sale Data templates.

Import PSS data: It is likely that the initial load of product support sale data from the enterprise resource system will involve a large number of records so you need to import them from the enterprise resource system via the interface.

For details of the data required by the interface, and the format of the interface file, contact the AMT Administrator and request the interface specification document.

Important points to bear in mind are that all amounts are assumed to be in AMT's main currency and if the cost bearer does not already exist in AMT, then that record will be rejected.

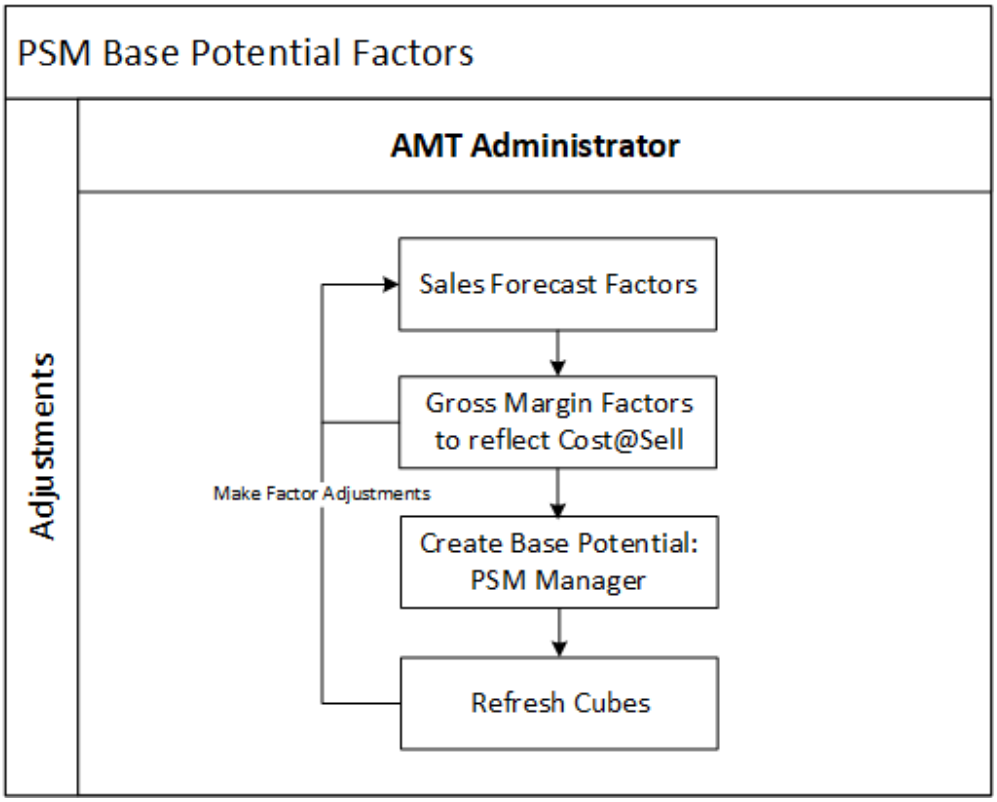
The product support sale data is read only and cannot be edited.

Refresh cubes: Update to the latest product support sale data by navigating to **Product Support Marketing** > **PSM Administration** > **PSM Manger**. Highlight the applicable version, click **Actions** at the top right of the screen and click **Refresh Actual PSS**.

PSM reporting: The Central Admin role is generally not the person running or interpreting the analytics. For information on the OLAP product support marketing cube refer to the PSM (Marketing Director) and PSM (Sales Rep) topics in this help.

Product Support Marketing base potential adjustments

The flowchart below shows the process for adjusting the base potential to cater for market factor and gross margins.



To adjust **Sales Forecast Factors**, navigate to **Product Support Marketing** > **PSM Administration** > **Sales Forecast Adjustment Factors**.

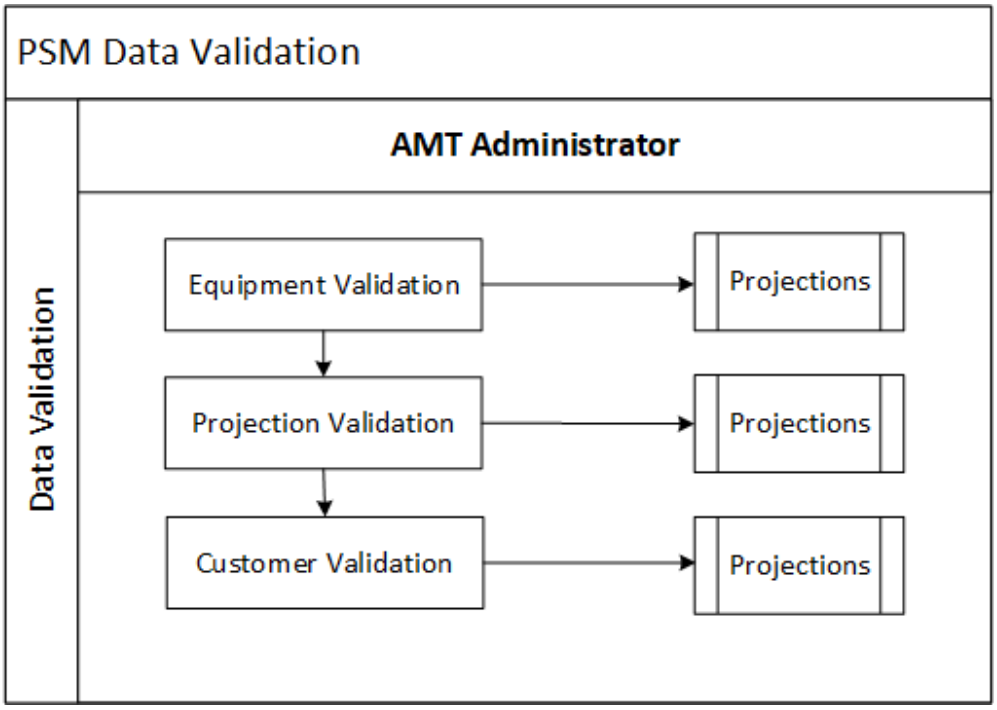
To adjust **Gross Margin Factors**, navigate to **Product Support Marketing** > **PSM Administration** > **Gross Margin Factors**.

To create **Base Potential**, navigate to **Product Support Marketing** > **PSM Administration** > **PSM Manager**. This is a screen for product support marketing Administrators to manage product support marketing cubes; this is where all the market data is stored.

To refresh cubes, navigate to **Product Support Marketing** > **PSM Administration** > **PSM Manager**. Make any changes then click **Actions** at the top right of the screen and click **Refresh**.

Product Support Marketing data validation

The flowchart below shows the process of validating all of the data required for determining the base potential has been populated.



Before you populate the product support marketing cubes you should confirm that all of the required information has been completed.

To validate **Equipment**, navigate to **Product Support Marketing** > **PSM Administration** > **Product Support Multi-Equipment Editor**.

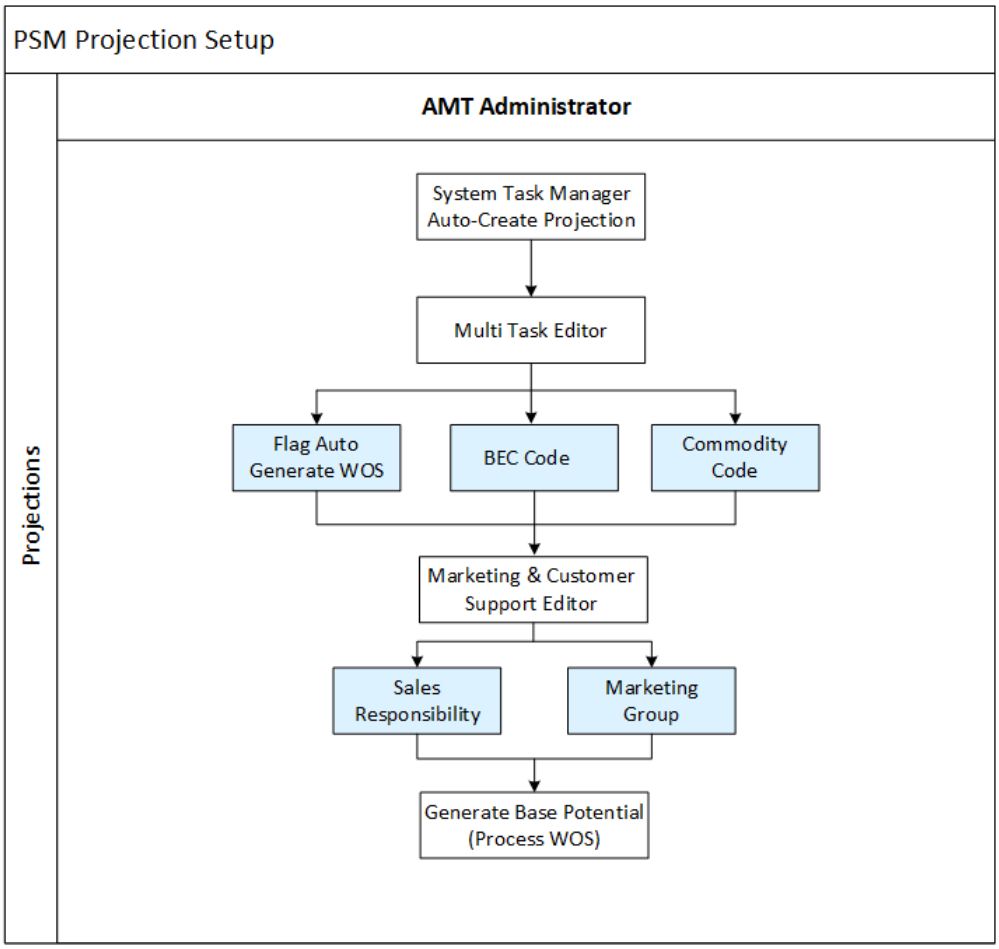
To validate **Projections** you can use the Strategy Task Audit report to see which strategy tasks have missing classifications. Navigate to **Product Support Marketing** > **PSM Reports** > **Strategy Task Audit**.

To validate **Customers**, navigate to either **Product Support Marketing** > **PSM Administration** > **Multi-Customer Editor**, or **System Administration** > **System Tables** > **Other** > **Multi-Customer Editor**.

Product Support Marketing projections

The flowchart below shows the process for setting up projections, which form part of the base potential. The base potential is the theoretical potential sales a dealer could have made for equipment.

The system defines base potential as the following:
If the average actual utilisation for the last six months varies more than 5% of the annual utilisation, AMT will update the annual utilisation to that of the average for the last six months.



Setting up the projections and base potential is a manual process, so every time you look at or update the base potential you should regenerate the work order history. It may be beneficial to perform this process once a month to avoid too much of a backlog. It is best if the AMT system administrator performs the task and the permissions are set accordingly.

It is difficult to undo this process when it has been run. It can generate a very large number of work order settlements, which you can only delete one by one using the **Work Order Settlement Manager**.

Product Support Sales Responsibility Multi Editor

Complete documentation for this topic will be available in a future release.

Sales Forecast Adjustment Factors

These factors are applied to the whole future market potential and are accumulated across all of the factors resulting in an adjustment in the base potential. These are best illustrated by examples, as shown below.

- Fleet of 4 x 777 trucks with strategy task: 1000.00.CC.0 engine changeout
  - Pricing date = 1-Jan-2019
  - Parts = $100,000
  - Labour = $20,000
  - Misc = $5,000
  - Total per engine = $125,000
- Forecast in 2020 = 4 engine changeouts for this 777 fleet
  - Base future potential = 4 engines x $125,000 = $500,000
- Sales forecast adjustments.
  - Expected changes in workplace safety add additional tasks to workscopes
    - +2% Labour cost
  - Parts prices expected to increase by 5% in the next 12 months
    - +5% Parts cost
  - Renegotiation with subcontractor expected to reduce outwork costs
    - -20% Misc cost
  - Customer to buy 3 new trucks in 2013, offered discount on overhauls of existing fleet
    - -5% Parts, labour and misc
- Adjusted sales forecast 2020 (4 x engines) = $491,720
  - Parts = $99,750
  - Labour = $19,380
  - Misc = $3,800
  - Total per engine = $122,930

The new sales adjusted forecast reflects the adjusted base future potential at cost.

You can create sales forecast adjustment factors by navigating to **Product Support Marketing** > **PSM Administration** > **Sales Forecast Adjustment Factors**. This is where you can specify rules for adjustment factors to be applied to sales forecasts; there is no limit, and all adjustment factor rules that match are applied (they are multiplied together to give a total adjustment factor).

Gross margin factors

These are factors which would bring about a change in the dealers standard gross margin. This is best illustrated by using the example from Sales Forecast Factors.

These are factors which would bring about a change in the dealers standard gross margin.

This is best illustrated by using the example in

- Adjusted forecast 2020 (4 x engines) = $491,720
  - Parts = $99,750
  - Labour = $19,380
  - Misc = $3,800
  - Total per engine = $122,930
- Expected gross margins in 2020
  - Parts margin is expected to grow from 28% to 30%
  - Labour margin is expected to tighten to 15% due to workplace health and safety changes
  - Misc margin is expected to remain unchanged at 10%

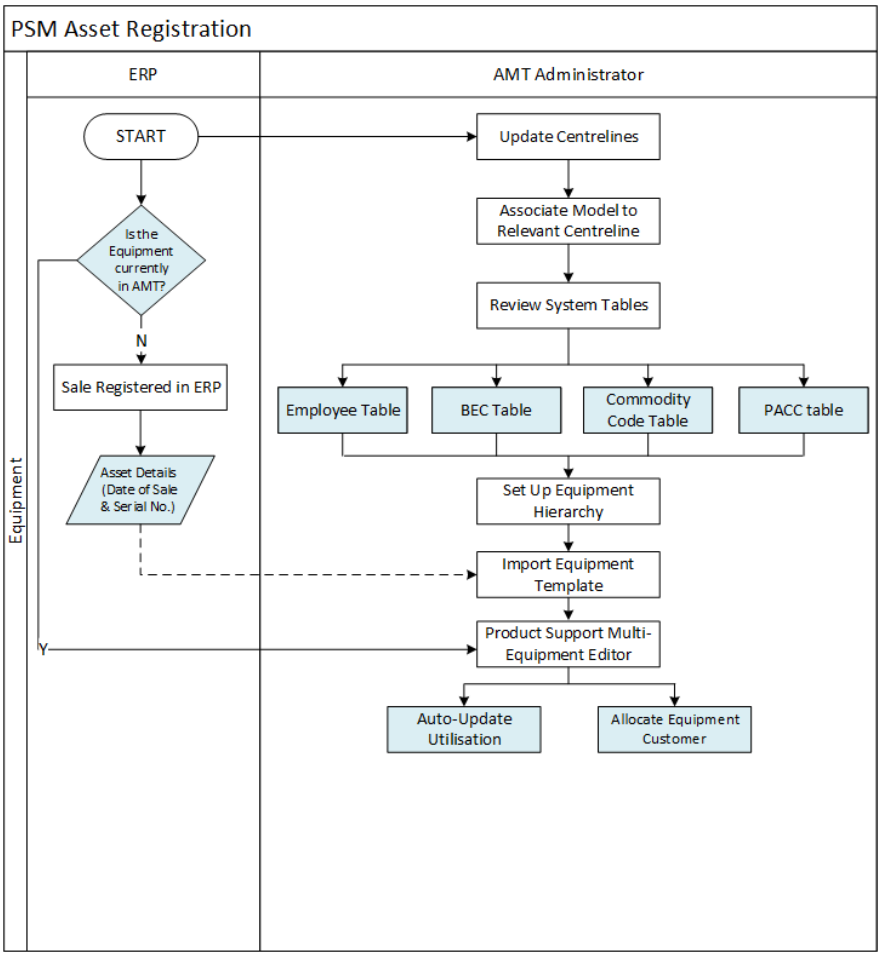
- Adjusted sales forecast at sell 2020 (4 x engines) = $814,888
  - Parts Sell = $142,500
  - Labour Sell = $57,000
  - Misc Sell = $4,222
  - Total Sell per engine = $203,722

The gross margin adjustments calculate the base future potential at sell (the price the customer pays).

You can create gross margin factors by navigating to **Product Support Marketing** > **PSM Administration** > **Gross Margin Factors**.

Product Support Marketing Multi-Equipment Editor

The flowchart below shows the process of loading equipment not already in AMT from the enterprise resource planning system using a template, and then adjusting all relevant equipment to determine the market potential.



Is the equipment currently in AMT?

It may be the case that some of the equipment currently resides in AMT as it is registered under a CSA or MARC contract, in which case only a few additional fields need to be edited. However, the rest of the machines which a

dealer has sold but is not managing, need to be recorded in AMT. The source of this information will be in the dealers enterprise resource planning system.



For equipment which already exists and is actively maintained in AMT it is imperative that parameters are not overwritten in the process, only additional information should be added.

There are two ways to deal with not updating the current settings of equipment already loaded in AMT:

- Exclude the equipment already loaded in AMT from the equipment template used to import recorded equipment sales from the enterprise resource planning system.
- Temporarily disable equipment updates. The AMT Administrator will need to do this.

There are a few built-in measures which will stop you from changing certain settings:

- If the Planning Mode is **Advanced** or **Simple**; you will not be able to activate the **Auto Update Usage** function.
- If the equipment already has projections they will not get replaced if **Auto Create Projection** is enabled.
- If the equipment is out of the dealer's territory the **Auto Create Projection** will not work.

If the equipment has been excluded from the sales forecast the **Auto Create Projection** will not work, nor will these equipment form part of the base potential.

To check if equipment is currently in AMT, navigate to **Product Support Marketing** > **PSM Administration** > **Product Support Multi-Equipment Editor**.

Create Product Support Marketing (PSM) versions

The Marketing Director may create several different PSM versions. They may have created a sales budget in the previous financial year and instead of refreshing the version it can be archived and a new version created for the following financial year.

The **PSM Manager** is a screen where you can manage product support marketing cubes; this is where all of the market data is stored. From this screen the you can manage existing product support marketing cubes, create new product support marketing cube versions (for example, archives), populate base potential, edit and apply sales adjustment factors and gross margins, and refresh particular product support marketing versions with actual product support sale.

You can create a version to represent a sales budget, for example, financial year 2020 sales budget, the full market potential over a long term horizon, what-if scenarios and so on.

To open the **PSM Manager**, navigate to **Product Support Marketing** > **PSM Administration** > **PSM Manager**.

Load equipment for Product Support Marketing

Once you have set up the system tables and equipment hierarchy, you can register the asset in AMT. It is likely that the initial load of equipment for Product Support Marketing purposes will be large. The quickest way to load the equipment is to import them from the enterprise resource planning system via the interface.

For details of the data required by the interface, and the format of the interface file, contact the **AMT Administrator** and request the interface specification document.

Equipment which is loaded for the purpose of Product Support Marketing need to be configured accordingly. The table below lists these configurations and should be noted in addition to those in the interface specification document.

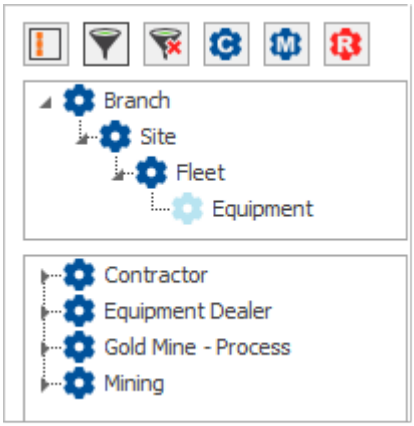
EQUIPMENT LOAD - PSM RELATED FIELDS		
Field	**Description**	**Example**
Planning_Type	0 = No Planning (Default – this is where planning is not performed at all in AMT). 1 = Advanced Planning (this is where downtime is captured in AMT). 2 = Simple Planning (this is where downtime is not captured in AMT). Set = 0 if not supplied. The significance of setting the field to 0 (No Planning) is that you will then be able to use the Auto Update Usage feature.	0
Estimated_Life_ Confirmed	Allowed values are 1 (Yes) and 0 (No).	1
Auto_Create_ Protection	Automatically create the equipment's current projection using the centreline linked to the model as the template. This only works for equipment: 1. with no tasks in the current projection. 2. not terminated. 3. not flagged to be excluded from sales forecast and 4. with **Territory Indicator** not set to "O" (Out of Territory). Default = 0. This ensures that you do not replace the current projections of the equipment which are already in AMT and are being actively managed.	0
OEM_Equipment_ Type	Allowed values are M (Machine) or E (Engine).	M
Sale_Indicator	S = Not sold by the dealer.	S
Exclude_from_Sales_ Forecast	Allowed values are 1 (Yes) and 0 (No). Default = 0. If the field is marked as 1 (Yes), then you will not be able to autocreate the projections for the equipment, nor will the equipment form part of the base potential.	1
Territory_Indicator	I - In Territory. O - Out of Territory. F - National Fleet Use. X - Unknown.	O

EQUIPMENT LOAD - PSM RELATED FIELDS		
Field	**Description**	**Example**
OEM_Equipment_ Division_Code	Specifies the division in which the machine operates. The code is defined by the dealership.	G
OEM_Equipment_ Division_Desc	Specifies the description of the division in which the machine operates.	General Line
Product_Type_Code	Specifies the product type. The code is specified by the dealer.	C
Product_Type_Desc	Specifies the product type description.	Hydraulic Shovel
Engine_Make	Specifies the make of the engine, which may be different to that of the machine.	Cummins
Engine_Model	Specifies the model of the machine.	QSK45 2000
Engine_Serial_ Number	Specifies the serial number of the machine.	6001
Equipment_Grouping_ A	Spare field.	
Equipment_Grouping_ B	Spare field.	
Equipment_Grouping_ C	Spare field.	
Application_Code	Specifies the application in which the machine operates. The code is defined by the dealership.	3
Application_Code_ Desc	Specifies a description of the application in which the machine operates.	PWC:570 Non-Metallic Mi
Application_Severity	Allowed values are High, Medium, Normal and Low.	High
OEM_Equipment_ Industry_Code	OEM Equipment Industry Code.	BR
OEM_Equipment_ Industry_Desc	Specifies the industry in which the machine operates.	Building - Residential
Sales_Responsibility_ Employee_Number	Unique identifier of an employee in AMT	260985

Validate equipment details

The steps below outline the process of validating equipment details and updating actual usages. The sales responsibility person is usually in daily contact with the customer, and confirms the details and updates them where relevant. This process is important because it aligns the base potential to what is really happening out in the market.

You need to validate the fields described in the table below. In the panel on the right of the **Home** screen, navigate to **Equipment**. Find the relevant Equipment and double-click it.



EQUIPMENT DETAILS VALIDATION	
Field	**Description**
Serial Number	The serial number links the sales data coming from the enterprise resource planning system to that of the equipment in AMT.
Customer	Crucial in order to get an understanding of the buying power of the customer and to understand who owns which equipment.
Sales Responsibility	This needs to be confirmed in order to understand what the sales responsibility's pipeline looks like, and which equipment or customer falls under their responsibility.
Start Usage	Indicates how much the machine had been used prior to the current application.
End Usage (if Terminated)	Indicates how much the machine had been used over its life.
Annual Utilisation	Affects the forecast projection.

You will have to validate fields such as **Serial Number**, **Start Usage** and **End Usage** individually, however, **Annual Utilisation**, **Customer** and **Sales Responsibility** can be edited through the **Product Support Multi-Equipment Editor**.

For equipment which is actively maintained in AMT (Marc or CSA Contracts) it is imperative to understand the impact of changing certain parameters such as **Start Usage** and **Annual Utilisation**. If these parameters are incorrect, changes should be made in conjunction with the relevant stakeholders and follow the correct business processes involved.

To validate the fields mentioned above:

1. On the **Base Details** tab, modify the **Serial Number**.
2. On the **Terms** tab, to the right of the **Start Usages** table, click  and update the **Start Usages**. If the machine has been Terminated you can update the **End Usage** as well.

To modify the **Sales Responsibility** and **Customer** fields, navigate to **Product Support Marketing** > **PSM Administration** > **Product Support Multi-Equipment Editor**.

Audit equipment

A good way for you to quickly review the equipment product support marketing configurations is to run the Equipment Audit report. Navigate to **Setup & Admin** > **Equipment** and double-click **Equipment Audit**.

You can use the report to see what the configurations for the equipment are; you can also see any missing fields which should have been populated. You can make changes by using the **Product Support Multi-Equipment Editor**. Navigate to **Product Support Marketing** > **PSM Administration** > **Product Support Multi-Equipment Editor**.

1. Filter for the relevant equipment and select the **Blank Fields** filter.
2. Select fields to populate and click **OK**.
3. Select the **PSM** filter, select the same fields and click **OK**.
4. On the **Product Support Multi-Equipment Editor** apply the filter, and enter the new values in the middle grid.

Marketing and Customer Support Editor

To make changes to any missing classifications:

- Navigate to **Product Support Marketing** > **PSM Administration** > **Marketing and Customer Support Editor**, filter for the required equipment and select the **Display Fields** filter.
- Double-click **Sales Responsibility**, **% of New Parts Purchase**, **Parts % of Purchase from Dealer**, **Labour % of Purchase from Dealer**, **Misc % of Purchase from Dealer**, **Marketing Group** and click **OK**.
- Once you have made all the selections, click **Apply Filters**, make the changes and click **Apply Changes**.

Validate projection details

You can validate equipment projection details by updating historic and forecast changeouts. The sales responsibility is usually in daily contact with the customer and confirms the details and updates them where relevant. This process is important because it aligns the base potential to what is really happening out in the market. It also aids the dealer in understanding the stock demands for the future and allow for on-time rebuilds and delivery.

You need to validate the fields in the table below. With the exception of **Job Costs** and **Scheduling Details**, you can adjust the details using the **Marketing and Customer Support Editor**.

PROJECTION DETAILS VALIDATION	
Field	**Description**
Serial Number	Needs to be confirmed as it links the sales data coming from the ERP system to that of the equipment in AMT.
Sales Responsibility	Needs to be confirmed in order to understand what the sales responsibility's pipeline looks like and which equipment or customer falls under their responsibility.
% of New Parts Purchase	Relates to rotable components. If the demand is going to be met by dealer exchange inventory, this will be 0%.
Parts - Probability of purchase from Dealer	The probability that the customer will purchase the parts to perform the job from the dealer.
Labour - Probability of purchase from Dealer	The probability that the customer will use labour supplied by the dealer to perform the job.
Misc. - Probability of purchase from Dealer	The probability that the customer will use miscellaneous items from the dealer to perform the job.
Marketing Group	An additional grouping for strategy tasks which may be used by the sales responsibility to simplify groupings.
Job Costs	Parts, labour and miscellaneous costs to perform the job.
Scheduling Details	The frequency at which the job occurs.

Job costs and scheduling details

You can review these on the **Strategy Task Analyser**. Navigate to **Strategy Optimisation** > **Analysis of Projection Assumptions** > **Strategy Task Analyser**.

Multi-Customer Editor

This screen lets you apply Customer-level changes to multiple customers at once.

Import customer listings for Product Support Marketing

It is likely that the initial load of customers for product support marketing purposes will be large, so the quickest way to import them is from the enterprise resource planning system via the interface.

For details of the data required by the interface, and the format of the interface file, contact your AMT Administrator and request the interface specification document.

Customers which are loaded for the purpose of product support marketing need to be configured accordingly. The table below lists some of these configurations and should be noted in addition to the interface specification document.

CUSTOMER LOAD - SPECIAL FIELDS TO NOTE		
Field	**Description**	**Setting**
Cost_Bearer	This code must already exist in AMT otherwise it will reject the customer line.	
Account_Manager	This is the employee number of the Account Manager for this customer, as per the Employee Table. This number must already exist in AMT otherwise it will reject the customer line.	
Sales_Responsibility	This is the employee number of the prime product sales responsibility for this customer, as per the Employee Table. This number must already exist in AMT otherwise it will reject the customer line.	
PSM_Include	These customers will be for product support marketing purposes. By selecting 1 (Yes) the Auto Exclude function is activated, and AMT only brings in active revenue customers.	1 - Yes.

Once the Customers have been imported you can view them in the **Customer Table**, and edit them in the **Multi-Customer Editor**. Navigate to either **Product Support Marketing** > **PSM Administration** > **Multi-Customer Editor**, or **System Administration** > **System Tables** > **Other** > **Multi-Customer Editor**.

Product Support Marketing annual utilisation

The **Annual Utilisation** field plays an important role in product support marketing, as it is one of the primary inputs for the forecast.

If the equipment is actively maintained then certain reports can be used to validate the accuracy of the current usage parameter before making a change. The annual utilisation should represent what is actually happening as closely as possible.

You can view the annual utilisation for the last 12 months, by going to **My Dashboard**, drilling down to the required equipment, and looking at the **Annualised Usage** column.



	Annualised Usage - Month To Date	Reliability (MTBS) - Last 12 Months	Annualised Usage - Last 12 Months	Availability - Last 12 Months	Availability Type
HT905			3,000	100.0%	MA
TR785-01			5,901	100.0%	MA
TR785-02			6,000	100.0%	MA
TR785-03			6,000	100.0%	MA
WL901			2,000	100.0%	MA
WL902			2,000	100.0%	MA
WL903	2,000		2,000	100.0%	MA
WL904	2,000		2,000	100.0%	MA
WL905	2,000		2,000	100.0%	MA
▷ MARC	5,504		5,496	92.2%	PA
▷ Power systems					MA
▷ Gold Mine - Process	5,000		5,000		PA
▷ Mining	5,133		5,358	100.0%	MA

There are several reports you can use to review the usage, for example **Usage vs. Profile Snapshot**, **Usage Profile Details**, **Utilisation Analyser**, **My Dashboard** and so on.

Market influences

Once all of the equipment, projections and customers have been set up in AMT by the central administration role, the sales responsibility users will update their respective history and sales forecasts; any assets which are not managed by the dealer will have been set up and the respective history created by the central administration role. All of this information results in the base potential, and it is the Marketing Director's responsibility to adjust the base potential to reflect the dealers sales forecast factors and gross sales margins.

The Marketing Director has a thorough understanding of the factors influencing the dealers' market. The factors relevant to product support marketing would be those influencing dealer sales and margins; these are defined as sales forecast factors and gross margin factors.

PSM Reports

(Rpt_657) Forecast Market Analysis

Provides flexible reporting analyses of the calculated market potential, forecast and opportunity.

Tips

Analyses current data (updated nightly) and applies to future data, not historical. You can dynamically drill down into selected areas.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report Parameters

Report Type, Date Selector

Additional filters

X-Axis, Y-Axis, CostOrSell, Data, Industry, Sales Zone L1, Classification, Account Manager, PLM, Major Comm Code, Bec Code, Version, Minor Comms Code

(Rpt_658) Forecast Market Share Analysis

Provides flexible reporting analyses of the forecast market share.

Tips

Analyses current data (updated nightly) and applies to future data, not historical data. You can dynamically drill down into selected areas.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report Parameters

Report Type, Date Selector

Additional filters

X-Axis, Y-Axis, Industry, Sales Zone L1, Classification, Account Manager, PLM, Major Comm Code, Bec Code, Version, Minor Comm Code

(Rpt_659) Actual Sales Analysis

Provides flexible reporting analysis of actual product support sales.

Tips

Analyses current data (updated nightly) and applies to future data, not historical data. You can dynamically drill down into selected areas.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report Parameters

Report Type, Date Selector

Additional filters

X-Axis, Y-Axis, Cost Or Sell, Data, Industry, Sales Zone L1, Classification, Account Manager, PLM, Major Comm Code, Minor Comm Code, Bec Code, Version

(Rpt_660) Historical Market Share Analysis

Provides flexible reporting analyses of the actual market share.

Tips

Analyses current data (updated nightly) and applies to future data, not historical data. You can dynamically drill down into selected areas.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report Parameters

Report Type, Date Selector

Additional filters

X-Axis, Y-Axis, Industry, Sales Zone L1, Classification, Account Manager, PLM, Major Comm Code, Bec Code, Version, Minor Comm Code

(Rpt_634) Customer Audit

Displays Customers of AMT.

Tips

Use the Blank Fields filter to identify Customers with missing classifications.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Report Type, Equipment Hierarchy

Additional filters

Display Columns, Blank Fields, Industry, PSS Responsibility

(Rpt_635) Customer Sales Activity

Displays Customer sales activity.

Tips

Use the Blank Fields filter to identify records with missing fields

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Display Fields, Blank Fields, Sales Activity

(Rpt_653) Sales Status and Strategy Forecast Health

This report has two sections - one section focusing on the status of current sales opportunities (that is, the next occurrence of each component), and the second focusing on the health of the component forecast (for example, overdue components). The data for this report is primarily sourced from Long-term Planning and Component Changeout Grid functions.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Parts/Labour/Misc, Currency, Strategy Task, Strategy Task Desc, Cost Type, System, Sub System, Component Code, Cost Bearer, Task Type

(Rpt_654) Strategy Task Audit

Use the Blank Fields filter to identify Strategy Tasks to identify Strategy Tasks with missing classifications.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Display Columns, Blank Fields, Cost Type, System, Sub System, Component Code, Modifier Code, Task Type, Task Counter, Strategy Task, Com. Code - Major, Com.Code - Minor, BEC Code, Part Type, Marketing Grouping, Parts Prob.%, Labour Prob. %, Misc Prob. %, New Parts Purc. %

(Rpt_663) Product Support Component Status

Shows the health of components, with the Frequency % statistic indicating whether the component is potentially being extended too far beyond its economic life.

Tips

The Current Component Condition field will only provide an indication of component health if AMTs Condition Based Maintenance module has been deployed to import and interpret condition monitoring data.

Current Component Usage takes into consideration the execution of parent components if the strategy task is part of a suppression group.

Next Due Target Life = Strategy Usage from the Long-term Planning > Component Changeout grid, where the value can be system calculated or manually adjusted by a user.

Next Due% = Current Component Usage (life) / Next Due Target Life.

Frequency% = Current Component Usage (life) / Frequency.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report Parameters

Report Type, Equipment Hierarchy

Additional filters

Cost Type, System, Component Code, Task Type, Strategy Task, Sales Status, Review Status, Frequency %

(Rpt_814) Labour Forecast (Dealer Adjusted)

Displays the Dealer's market share.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,
|UserID|,|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity, Planned

(Rpt_818) Forecast Worksheet (Dealer Adjusted)

Used for forecasting costs by expense group and optional dealer market share. Shows future and historical costs at sell, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

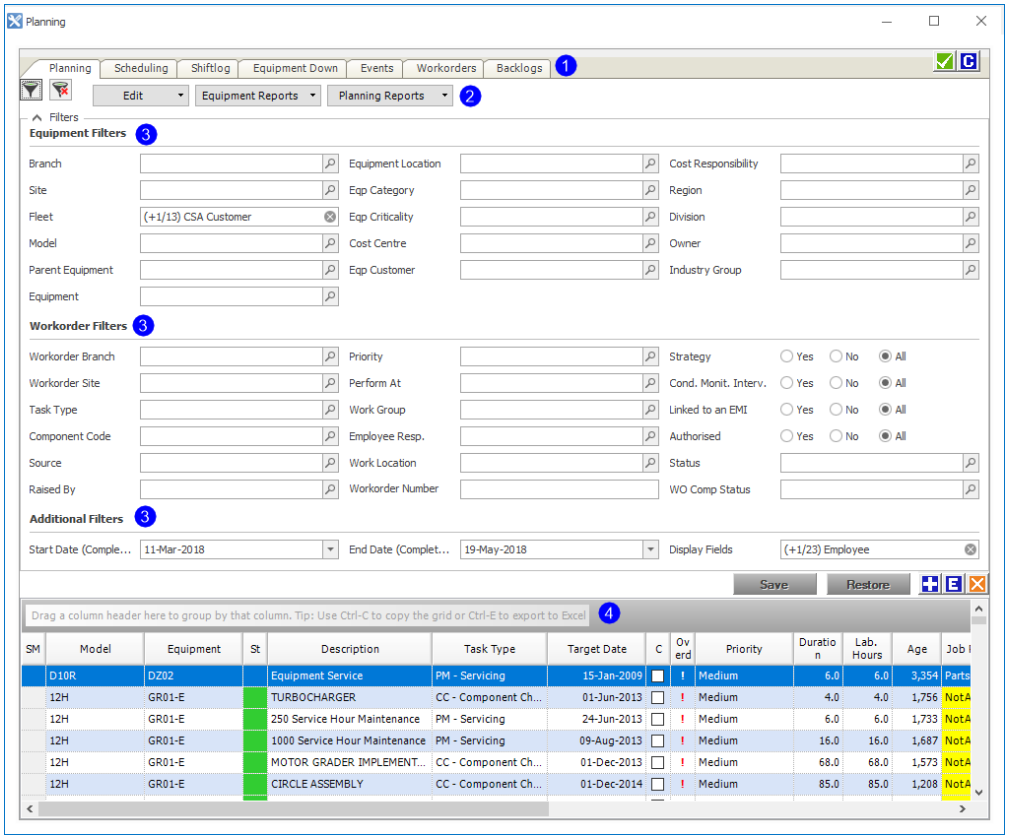
Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Planned, Workorder, Currency

Daily Planning

Planning

The Planning form is designed to enable various stakeholders in the planning process to perform all AMT-related planning tasks within this form.



The screen is divided into four sections.

1. **Functions**: Planning, Scheduled, Shiftlog, Equipment Down, Events, Work orders and Backlogs.
2. **Menu options**: Edit, Equipment Reports, Planning Reports.
3. **Filters**: Equipment, Work order and Additional.
4. **Grid**: contains the data you filtered for.

AMT has two Planning modes:

- **Standard** - suited for applications where downtime is not captured such as general line, PM contracts, process plants, workshop and so on.
- **Advanced** - mining focused and suited for downtime capture applications as it integrates with the **Equipment Status** module.

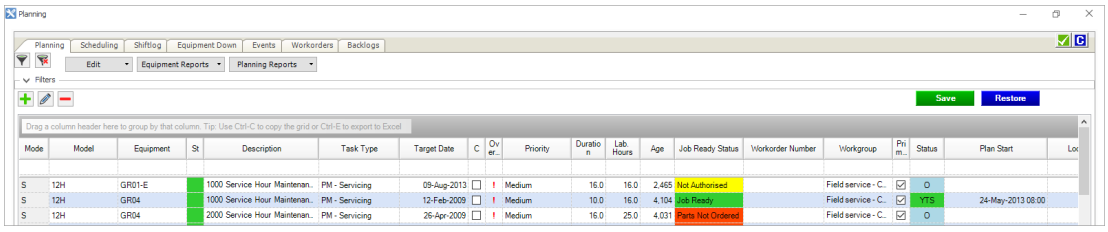
Navigate to **Daily Planning** > **Planning**.

Tab	Description
Planning	The Planning grid shows all planning work orders - both strategy work orders (components and PMs) and defects.
Scheduling	The Scheduling Gantt Chart graphically represents a plan and allows you to manipulate the plan, allocate additional backlogs, workgroups and employees to the plan using drag-and-drop.
Shiftlog	Shows events within a specified date range and is designed to quickly update the event status and create unplanned events with only the essential information.
Equipment Down	Graphically represents the live status of assets. Over a specified period of time all yet to start, in progress and completed events can be seen in a timeline.
Events	Enables you to create detailed events with the respective work order. When adding work orders to the Events tab, if they don't have the same shutdown, turnaround, outage (STO) as the Event STO there will be an error and the work orders will not be added.
Workorders	Lets you review all planning work orders with additional details such as admin alerts.
Backlogs	Lets you review all planning work orders with the ability to generate backlog-specific reports.

Planning grid

The grid has a set of default columns which are displayed, however, you can select which columns you would like to view. When you exit planning by clicking **Save**, the columns you selected are saved so that when you return, the same columns will be displayed automatically.

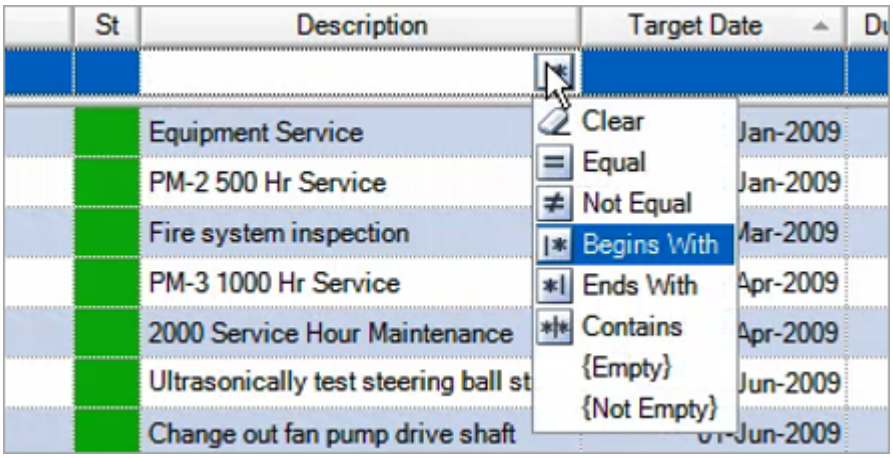
Navigate to **Daily Planning** > **Planning**.



Grids are searchable, allowing you to, depending on the column information, search based on:

- Equal
- Not Equal
- Begins With
- Ends With
- Contains
- Greater Than (for dates and numbers)

- Greater Than or Equal To (for dates and numbers)
- Less Than (for dates and numbers)
- Less Than or Equal To (for dates and numbers)
- (Empty)
- (Not Empty)
- Clear – option to remove the search criteria

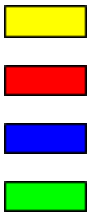
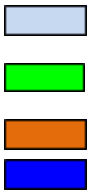


Searchable grids are available in:

- Daily Planning > Planning > Planning
- Daily Planning > Planning > Scheduling > Work Orders
- Daily Planning > Planning > Shift Log
- Daily Planning > Planning > Events
- Daily Planning > Planning > Work Orders

Grid fields (standard display fields)

Field	Key	Description
Mode	S A	Simple. Can't assign events to work order. Advanced. Can assign events to work order.
SM	 	Standard planning mode equipment. Advanced planning mode equipment.
Strategy	 	Strategy task. Condition monitoring intervention. Non-strategy and non-condition based maintenance task.

Field	Key	Description
Target Date		The Performed by Date which is set in the work order.
Confirmed		Strategy confirmed. This box is selected if the strategy date or usage was flagged as locked in the Long-Term Planning screen.
Overdue	!	Overdue.
Duration		Number of hours required to perform the work order (elapsed time).
Labour Hours		Number of labour hours required to perform the work order.
Age		Number of days the work order has been in the Planning screen.
Job Ready Status		Not Authorised. Authorised - Parts not ordered. Authorised - Not ready. Parts Ready - Authorised and ready.
Work Order Number		Work order settlement number.
Work Group		Group scheduled to perform the work order.
Primary		Primary task of event (if Advance Planning Mode).
Status		Outstanding = Yet to be planned. Yet to Start = Work order has been planned. In Progress = Work order is currently being performed. Completed = Work order is finished.
Plan Start		Planned date when the work order will be performed.
Location		Location where work order will be performed.

Notes

- Additional fields can be displayed as column headings adjusting Display Fields, which is an option in the **Additional Filters** section of the Grid.
- For **Advanced Planning equipment**, the **Event Start** field should be included in the Grid.

- Double-click a work order to open it. If a work order is part of a planned event (that is, it has been scheduled) then it will open the event first. The event will show all work orders and you can open the work order from there.

Business Rules for Operational Criticality, Risk Override and Risk Rating (optional fields)

- If a work order is created from a strategy task, the default value for **Operational Criticality** is inherited from the source strategy task. The inherited value is read-only until the work order is saved.
- If a work order is created from a strategy task, the default value for **Risk Override** is inherited from the source strategy task. The inherited value is read-only until the work order is saved.
- **Risk Override** is disabled for new work orders until the work order has been saved.
- **Operational Criticality** and **Risk Override** can only be updated for work orders with a status of **Outstanding**, **Yet to Start** or **Deferred**.

Standard vs Advanced planning

The Planning function in AMT is different to the Advanced Planning function. The fundamental difference between the type modes of planning is that advanced planning requires you to create events.

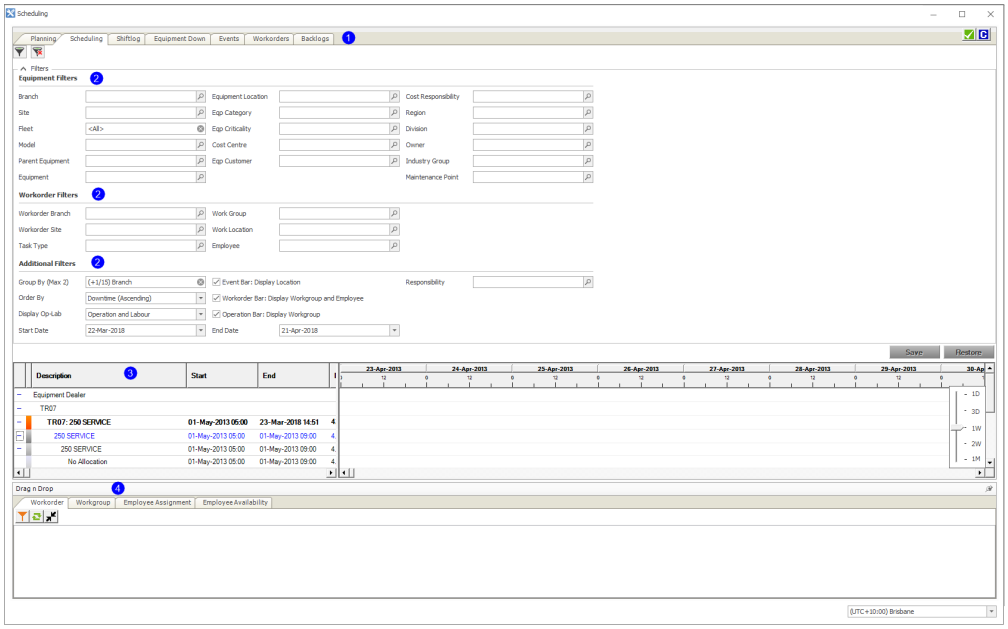
Characteristic	Planning	Advanced Planning
Event		
Ability to create an event.	N	Y
Ability to view and manipulate events on a Gantt chart.	N	Y
Ability to create one or more downtime allocations.	N	Y
Work order		
Ability to plan strategy tasks generated by projections.	Y	Y
Ability to create a non-strategy task.	Y	Y
Ability to set up start and end dates/times per work order.	Y	Y
Ability to view work orders on a Gantt chart.	Y	Y
Ability to have multiple operations.	Y	Y
Ability to add multiple parts.	Y	Y
Ability to create an internal work order settlement.	Y	Y

Characteristic	Planning	Advanced Planning
Ability to request an external work order settlement.	Y	Y
Ability to raise a purchase order.	Y	Y
Ability to record actual labour and durations at the work order level.	Y	Y
Ability to allocate labour resources.	Y	Y
Ability to capture actual maintenance performed.	Y	Y
Work order settlements		
Ability to request work order settlements from an external system.	Y	Y
Ability to self-generate work order settlements.	Y	Y
General planning and reporting		
Ability to generate a work plan.	Y	Y
Ability to review resource allocations.	Y	Y
Ability to view age of backlogs.	Y	Y
Ability to analyse downtime by Responsibility.	N	Y
Ability to perform Event Analysis	N	Y
Ability to perform Downtime Analysis	N	Y

Scheduling Gantt chart and grid descriptions

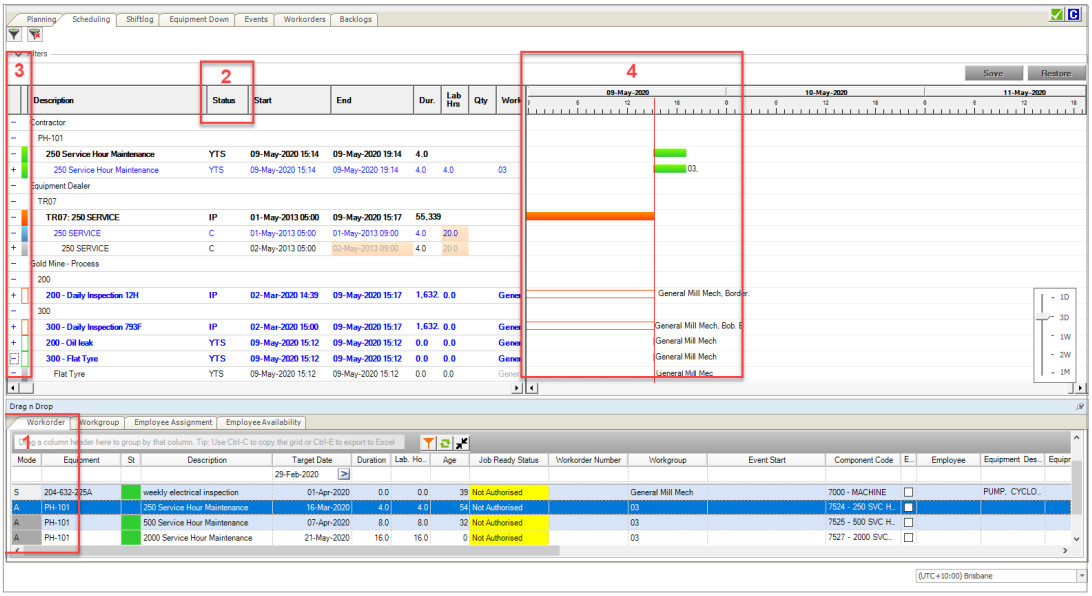
Navigate to **Daily Planning** > **Planning** > **Scheduling tab**.

The screen is divided into four sections.



1. **Functions**: Planning, Scheduled, Shiftlog, Equipment Down, Events , Workorders, Backlogs.
2. **Filters**: Equipment, Work Order and Additional.
3. **Grid**: Contains the data you filtered for.
4. **Drag-n-Drop**: Contains Work Order, Work Group, Employee Assignment and Employee Availability tabs which can be dragged and dropped onto the Gantt Chart to generate a schedule.

The screenshot below indicates aesthetic changes to assist you with understanding and visualisation of the work order statuses and the equipment planning mode.



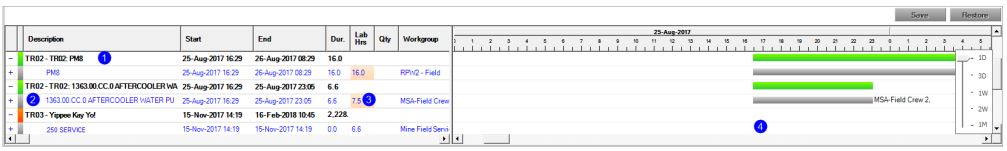
1. The first column in the Work Order results grid has a new heading, **Mode**. This column indicates the Equipment Planning Mode – Standard Planning (S) and Advanced Planning (A). Individual rows within the results grid will now display either an A or an S as well as the previously Dark Grey and Light Grey theme. The use of the A and S assists with the new grid filtering mentioned previously.

2. A new status column has been added to the Scheduling Grid which will display the status of the Event and Work Order.
3. Colour visualisation within the Scheduling Grid has been enhanced for work order status from previous AMT versions. The full colour explanation is included below:

	Colour	AMT Entity	Meaning
	Black	Shutdown	Shutdown
	Solid Green Bold Black Font	Event	Yet To Start Event
	Solid Orange Bold Black Font	Event	In Progress Event
	Solid Orange Blue Font	Advanced Planning Work Order	In Progress Work Order
	Solid Green Blue Font	Advanced Planning Work Order	Yet To Start Work Order
	Solid Teal Blue Font	Advanced Planning Work Order	Completed Work Order
	Gold Blue Font	Advanced Planning Work Order	Abandoned Work Order
	Moccasin Blue Font	Advanced Planning Work Order	Deferred Work Order
	Orange Outline Bold Blue Font	Standard Planning Work Order	In Progress Work Order
	Green Outline Bold Blue Font	Standard Planning Work Order	Yet To Start Work Order

4. Colour visualisation of the bars in the Gantt chart is enhanced and aligns with those in the Scheduling grid.
5. No changes have occurred to Operation and Labour Activity displays.

Gantt chart descriptions



Item	Description
1	Equipment and Event.
2	Work order.
3	If the downtime or uptime of a work order or operation does not fall within the planned downtime, orange shading displays.
4	In this example, the grey bar for the last labour resources does not display even though there are labour hours. This is because the operation's duration is 0.

Work order grid

The Work Order results grid, by default, displays the ability to group and search the grid. The grouping ability on the grid can be hidden by using the **Collapse** icon.



This will give some additional screen real estate if grouping is not required.

The limited space available on this screen creates some challenges for all the information needed. You will see that, at times, this icon set will move within the header area.

Work order grid column descriptions

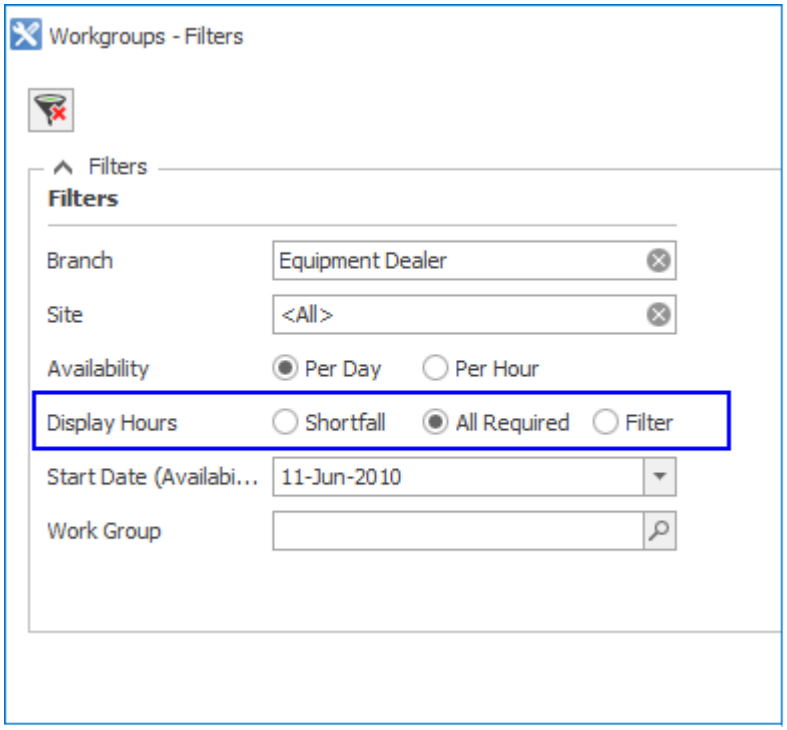
Field	Description
SM	Standard mode. Advanced mode.
Strategy	Strategy task. Condition monitoring intervention. Non-strategy and Non-condition based maintenance task.
Target Date	The performed by date which is set in the work order.
Duration	Number of hours required to perform the work order (elapsed time).
Labour Hours	Number of labour hours required to perform the work order.
Age	Number of days the work order has been in the Planning screen.
Job Ready Status	Not authorised.

Field	Description
	Authorised - Parts required. Authorised - Parts ordered. Ready - Authorised and parts ready.
Work Order Number	Work order settlement number.
Work Group	Group scheduled to perform the work order.

Work group grid column descriptions

Field	Description
Work Group	Work group name.
Code	Work group code
Default	This box is selected if the work group is the default work group for the site.
Target	Target allocation percentage (which is set per work group in the work group system table).
Date (daily)	< Target (each Work Group has a Target). OK (less than 100% but greater than or equal to Target) Over-Allocated. Status is calculated by comparing the work group's available man hours vs. time allocated to actual work orders.

Work group filter descriptions



Display Hours	**Shortfall** The remaining hours in the shift for that day. **All Required** The labour hours in the work order for an employee. **Filter**

Employee Assignment grid column descriptions

Field	Description
Work Group	Work group name.
Lab. Activity	Default labour activity assigned to the employee in the employee system table.
Employee	Employee name (Surname, Initial).

Employee availability grid column descriptions

Field	Description
Work Group	Work group name.
Lab. Activity	Default labour activity assigned to the employee in the employee system table.
Employee	Employee name (Surname, Initial).
Date/Time (hourly)	Non-work time. Not available time. Allocated. Partly available. Available. Conflict. Status is calculated by comparing the employee's available hours vs. time allocated to actual work orders.

Scheduling

Scheduling involves setting the **when**, **where** and **who** for work orders.

In **Advanced Planning** mode, a work order is scheduled in AMT if it is linked to a planned down event.

In **Standard Planning** mode, a work order is scheduled in AMT if it has status YTS.

In practical terms, this means that on the Planning grid, if the work order shows a plan down date it is scheduled.



In Advanced Planning mode if you open that work order it will first open the event.

Search for Inventory Parts

This is a function that can search for parts in an external inventory system, providing integration to such a system has been deployed as part of the AMT solution.

The search uses the combination of information in the search fields to return part numbers, part descriptions, stock code, part unit cost, quantities on hand at the **Site**, and quantities on hand across multiple warehouses **Total**. Integration logic determines the definition of Site and Total.

Shift Handover Comments

This function is designed to be used by technicians, supervisors and planners to communicate progress or issues between shifts.

Shift supervisors, for example, can create Shift Handover Comments at site or work order level. Delay code, delay reason and delay hours can (optionally) be recorded to capture the impact of an issue. If a comment is linked to a work order, use **Quick Links** to navigate to the work order to see its details.

PM Schedule

This function provides the same functionality as the Long-term Planning function, except that it is limited to the Task Types which are flagged in the Task Type table as **PM Planning** task types.

Example - Dependency with inspection logic

The image below shows how PM Services using dependency logic should be set up.

Inspection logic only works when a part of a dependency grouping.

Equipment	Projection	SS	Component Code	MC	TT	TC	Description	STT	P	Sup	Dep
SI WT...	Curre...	USAGE BASED	9005 - 250HR SVC	00	IN	00	250HR Replace Outer Air Filters	Normal	☑		
SI WT...	Curre...	USAGE BASED	9008 - 500HR SVC	00	PM	00	500HR PM Service	Normal	☑		A1
SI WT...	Curre...	USAGE BASED	9013 - 1000HR SVC	00	PM	00	1000HR PM Service	Normal	☑		A2
SI WT...	Curre...	USAGE BASED	9017 - 1500HR SVC	00	PM	00	1500HR SERVICE	Normal	☑		A3
SI WT...	Curre...	USAGE BASED	9018 - 2000HR SVC	00	PM	00	2000HR PM Service	Normal	☑		A4

The PM tasks follow each other in a sequence. A1, then A2 then A3 then A4 then back to A1. So whenever the last task was done, the next is scheduled based on the last performed.

Hypothetically, if PM services had been done in the past, we can identify that PM 1 was performed at 500 hours (which can be confirmed by looking at the work order history using Workorder Settlement Manager).

Consequently, AMT will calculate the other PM services in the dependency group to occur at:

- PM 2 at 1000 hours
- Followed by PM 3 at 1500 hours
- Followed by PM 4 at 2000 hours

Make sure you enter SMU readings regularly, and the first/frequency for each strategy task is set up correctly. Reports to use:

- Hours to PM
- Workorder Settlement Manager

Inputs

Input	Description
Base strategy data	Ensure you set up the strategy tasks with the correct logic and frequency.
Long-term planning	Use this to make sure servicing is aligned.

Input	Description
Workorders	The Planning tab is used to action each work order. Work order settlement Manager will keep the servicing aligned.

See the example below showing inspection logic with dependency service regime (as explained earlier).

- 250 hours on inspection logic.
- 500 hours on dependency A1
- 1000 hours dependency A2
- 1500 hours on dependency A3
- 2000 hours on dependency A4

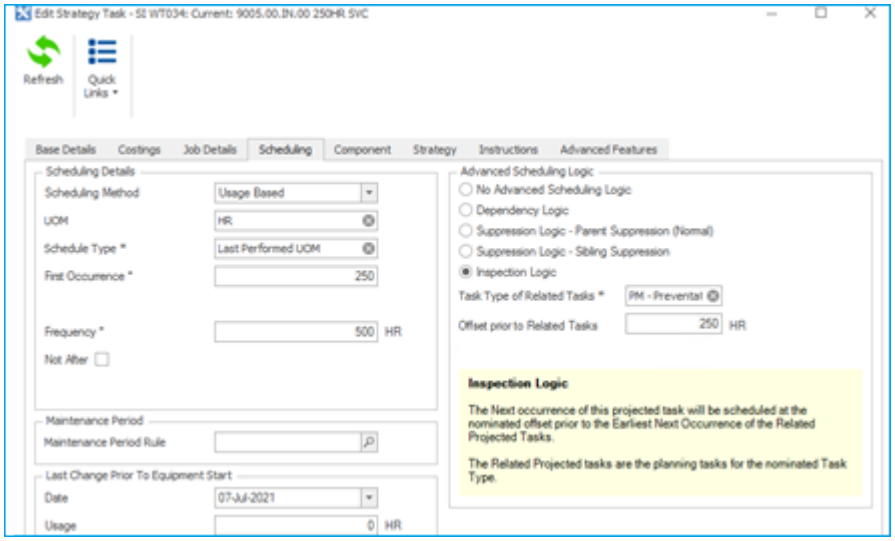
Equipment	Projection	SS	Component Code	MC	TT	TC	Description	STT	P	Sup	Dep
SI WT...	Curre...	USAGE BASED	9005 - 250HR SVC	00	IN	00	250HR Replace Outer Air Filters	Normal	☑		
SI WT...	Curre...	USAGE BASED	9008 - 500HR SVC	00	PM	00	500HR PM Service	Normal	☑		A1
SI WT...	Curre...	USAGE BASED	9013 - 1000HR SVC	00	PM	00	1000HR PM Service	Normal	☑		A2
SI WT...	Curre...	USAGE BASED	9017 - 1500HR SVC	00	PM	00	1500HR SERVICE	Normal	☑		A3
SI WT...	Curre...	USAGE BASED	9018 - 2000HR SVC	00	PM	00	2000HR PM Service	Normal	☑		A4

All on last performed UOM.

Create a strategy task

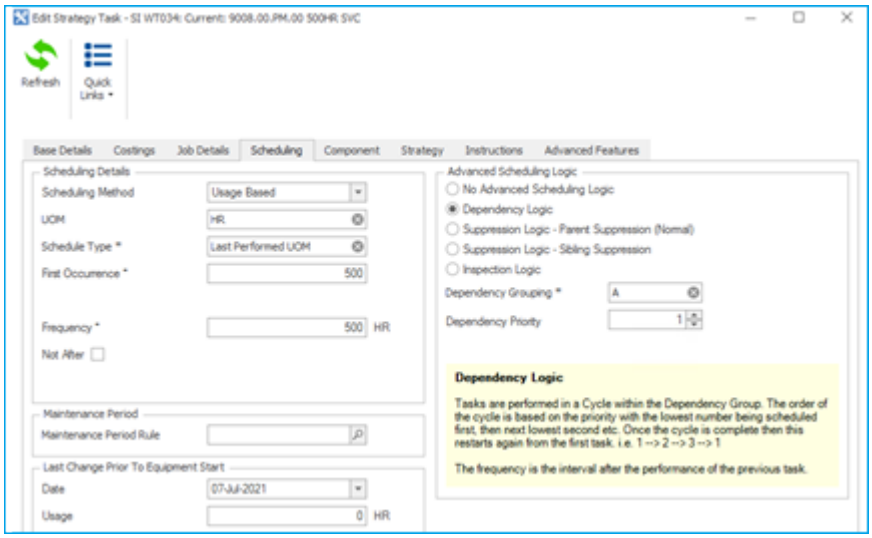
Inspection logic:

- Task Type - In
- Schedule Type - Last Performed UOM
- Inspection Logic selected with offset at the interval required.



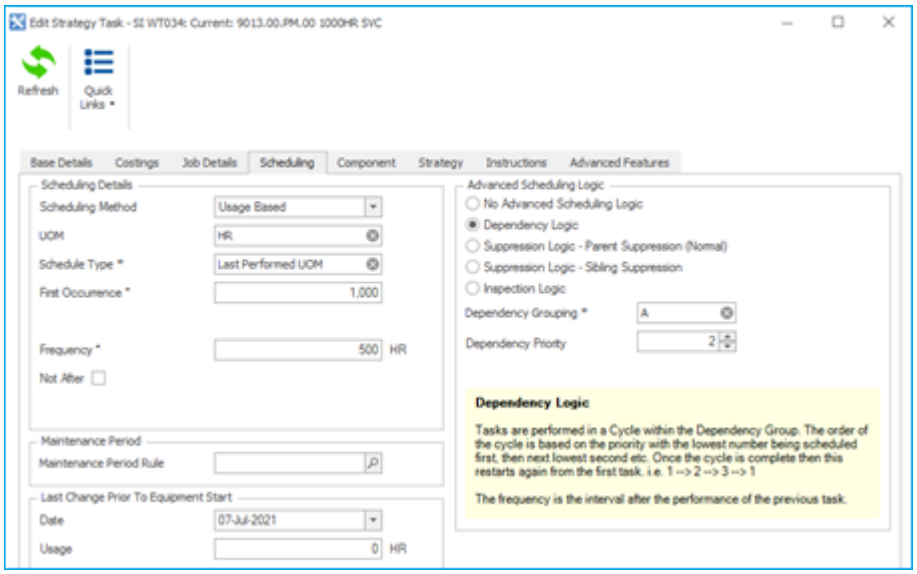
Dependency logic:

- Task Type - PM
- Dependency - A1 as first service to be performed in the regime.
- 500 hr frequency



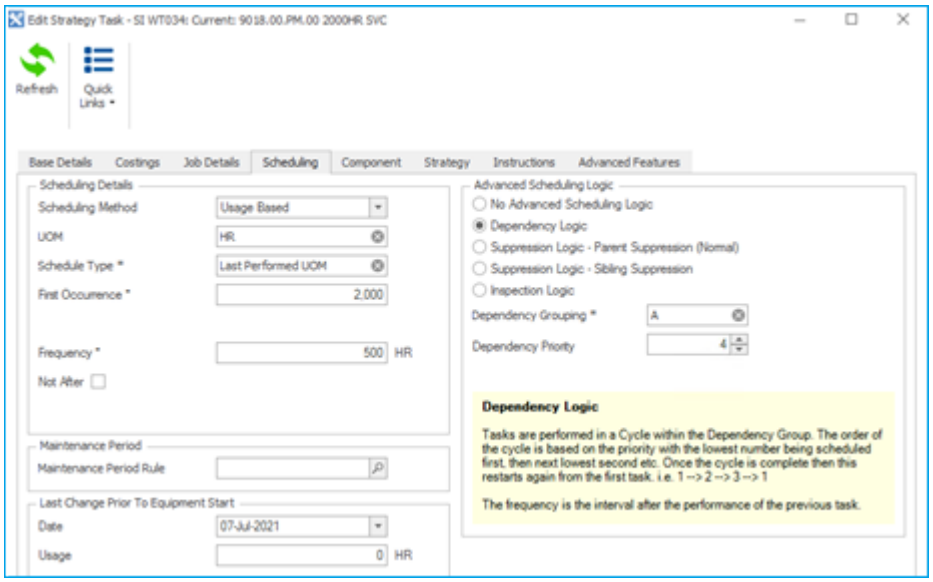
Dependency logic

- Task type - PM
- Dependency A2 as second service to be performed in the regime. 500 hour frequency, but the first occurrence is 100 hours



Dependency logic:

- Task type - PM
- Dependency - A3 as third service to be performed in the regime. 500 hour frequency, but the first occurrence is 1500 hours

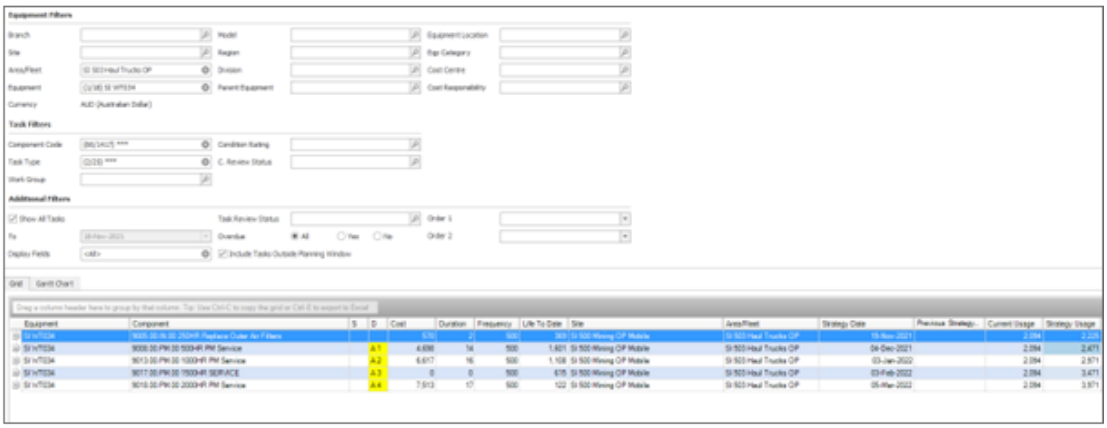


Dependency logic:

- Task type - PM
- Dependency - A4 and fourth service to be performed in the regime. 500 hour frequency, but first occurrence is 2000 hours.

The Long-term Planning screen is used to see all the strategy tasks.





Example - Parent Suppression for PMs

The images below show how PM services should be set up in AMT using suppression.

Task Summary

Drag a column header here to group by that column. Tip: Use Ctrl-C to copy the grid or Ctrl-E to export to Excel

Equipment	Projection	SS	Component Code	MC	TT	TC	ST Description	P	Sup	Dep	Cost Source	R	Part #	UOM	First	Frequency	Currency	Parts Esc	Labour Esc	Misc
TR02	Curre..	Engine	1000 - ENGI..	00	CC	0	Engine	☑	B1		SJ-Multi..	☐	E001	Eng -..	18,000	22,000	USD	166,789	35,300	
TR02	Curre..	Air Inducti..	1052 - TURB..	LF	CC	0	TURBOCH..	☑	B4		SJ-BUIL..	☐		Eng -..	9,000	9,000	USD	1,713	1,000	
TR02	Curre..	Air Inducti..	1052 - TURB..	LR	CC	0	TURBOCH..	☑	B4			☐		Eng -..	9,000	9,000	USD	6,237	2,585	
TR02	Curre..	Air Inducti..	1052 - TURB..	RF	CC	0	TURBOCH..	☑	B4			☐		Eng -..	9,000	9,000	USD	6,237	2,585	
TR02	Curre..	Air Inducti..	1052 - TURB..	RR	CC	0	TURBOCH..	☑	B4			☐		Eng -..	9,000	9,000	USD	6,237	2,585	
TR02	Curre..	Cooling S..	1355 - THER..	00	CC	0	Thermostats	☑	B4			☐		Eng -..	6,000	6,000	USD	274	500	
TR02	Curre..	Cooling S..	1359 - FAN..	00	CC	0	FAN DRIVE	☑	B4			☐		Eng -..	9,000	9,000	USD	1,469	1,034	
TR02	Curre..	Cooling S..	1361 - WAT..	00	CC	0	WATER P..	☑	B4			☐		Eng -..	9,000	9,000	USD	1,230	750	
TR02	Curre..	Cooling S..	1363 - AFTE..	00	CC	0		☑	B4			☐		Eng -..	9,000	9,000	USD	1,148	750	
TR02	Curre..	Starting S..	1451 - AIR S..	00	CC	0	AIR START..	☑	B4			☐		Eng -..	9,000	9,000	USD	3,967	646	
TR02	Curre..	Engine	1803 - AIR C..	00	CC	0	AIR COMP..	☑	B4			☐		Eng -..	9,000	9,000	USD	2,866	646	
TR02	Curre..	PM	7527 - 2000..	00	PM	0	PM8	☑	PM2501		SJ-BUIL..	☐		Eng -..	2,000	2,000	USD	1,692	1,600	
TR02	Curre..	PM	7526 - 1000..	00	PM	0	PM4	☑	PM2502		SJ-BUIL..	☐		Eng -..	1,000	1,000	USD	1,692	1,200	
TR02	Curre..	PM	7525 - 500 S..	00	PM	0	PM2	☑	PM2503		SJ-BUIL..	☐		Eng -..	500	500	USD	932	800	
TR02	Curre..	PM	7524 - 250 S..	00	PM	0	PM1	☑	PM2504		SJ-BUIL..	☐		Eng -..	0	250	USD	294	400	

The hierarchy is that B1 is the parent of B2 is the parent of B3 is the parent of B4 and so on. So when B1 occurs this then resets the life for all of its children. 2000 and 3000 hours services are at the same level = B2. They both suppress the 1000 service.

Resultant Forecast (Component Schedule Report)

Component	Part No	R	Rotable Serial No	Mod Cod	Task Type	Occ Type	Strategy Locked	O'due	Due Date	Usage	Life Used	UOM	Proj Interval	Proj Cost	Proj Lab Hrs	Proj Duration
Machine: RF_01																
7524 250 SVC HOUR MAINTEN				00	PM				31-Dec-06	250	26	Eng Hr	250	250	0	0
7525 500 SVC HOUR MAINTEN				00	PM				30-Jan-07	500	26	Eng Hr	500	500	0	0
7524 250 SVC HOUR MAINTEN				00	PM				02-Mar-07	750	26	Eng Hr	250	250	0	0
7526 1000 SVC HOUR MAINTE				00	PM				01-Apr-07	1,000	26	Eng Hr	1,000	1,000	0	0
7524 250 SVC HOUR MAINTEN				00	PM				02-May-07	1,250	26	Eng Hr	250	250	0	0
7525 500 SVC HOUR MAINTEN				00	PM				01-Jun-07	1,500	26	Eng Hr	500	500	0	0
7524 250 SVC HOUR MAINTEN				00	PM				02-Jul-07	1,750	26	Eng Hr	250	250	0	0
7527 2000 SVC HOUR MAINTE				00	PM				01-Aug-07	2,000	26	Eng Hr	2,000	2,000	0	0
7524 250 SVC HOUR MAINTEN				00	PM				31-Aug-07	2,250	26	Eng Hr	250	250	0	0
7525 500 SVC HOUR MAINTEN				00	PM				01-Oct-07	2,500	26	Eng Hr	500	500	0	0
7524 250 SVC HOUR MAINTEN				00	PM				31-Oct-07	2,750	26	Eng Hr	250	250	0	0
7545 3000 SVC HOUR MAINTE				00	PM				01-Dec-07	3,000	26	Eng Hr	3,000	3,000	0	0
7524 250 SVC HOUR MAINTEN				00	PM				31-Dec-07	3,250	26	Eng Hr	250	250	0	0
7525 500 SVC HOUR MAINTEN				00	PM				31-Jan-08	3,500	26	Eng Hr	500	500	0	0
7524 250 SVC HOUR MAINTEN				00	PM				01-Mar-08	3,750	26	Eng Hr	250	250	0	0
7527 2000 SVC HOUR MAINTE				00	PM				01-Apr-08	4,000	26	Eng Hr	2,000	2,000	0	0
7524 250 SVC HOUR MAINTEN				00	PM				01-May-08	4,250	26	Eng Hr	250	250	0	0
7525 500 SVC HOUR MAINTEN				00	PM				31-May-08	4,500	26	Eng Hr	500	500	0	0
7524 250 SVC HOUR MAINTEN				00	PM				01-Jul-08	4,750	26	Eng Hr	250	250	0	0
7526 1000 SVC HOUR MAINTE				00	PM				31-Jul-08	5,000	26	Eng Hr	1,000	1,000	0	0
7524 250 SVC HOUR MAINTEN				00	PM				31-Aug-08	5,250	26	Eng Hr	250	250	0	0
7525 500 SVC HOUR MAINTEN				00	PM				30-Sep-08	5,500	26	Eng Hr	500	500	0	0
7524 250 SVC HOUR MAINTEN				00	PM				31-Oct-08	5,750	26	Eng Hr	250	250	0	0
7543 6000 SVC HOUR MAINTE				00	PM				30-Nov-08	6,000	26	Eng Hr	6,000	6,000	0	0

Work order authorisation

These are the actions that need to be planned and actioned before the planner authorises the work order.

- **Work order authorised**
  Has the work order been authorised? Ideally the work order should be authorised before being captured into AMT. In some circumstances work order authorisation can be applied in bulk by a manager or supervisor, other modes include ‘automatic authorisation’ where work orders are automatically authorised based on the approval limit at the user level. Refer to your system administrator for more information.
- **Work order specifications**
  Are the planning requirements clearly understood? This covers the resources required (parts, labour, tooling) and the job instructions. This is really a validation of the earlier processes but is nevertheless an important checkpoint. If the planning requirements are unclear, the planner should return the defect to its source for clarification before proceeding with any planning activities.
- **Work order targets**
  Are there clear targets set for the duration and labour hours?

There are three authorisation modes that are used in AMT.

- **No authorisation**
  All work orders default to being authorised.
- **Simple authorisation**
  Work orders are to be checked as authorised manually (AMT will save the username and date of the authorisation, and the work order's planned total cost).
- **Advanced authorisation**
  Authorisation levels are set by the user and AMT validates the user has the required authorisation before

authorising a work order. In addition, a work order will become unauthorised if additional parts are added to the work order causing the authorised amount to be exceeded.

Work order authorisation wizard

To streamline the authorisation process, you can view and authorise many work orders at the same time using the work order authorisation wizard.

Navigate to **Daily Planning** > **Workorder Authorisation**.

If you do not have sufficient authority to authorise a work order, you can request it by clicking **Request Authorisation**.

Activity Dashboard

The Activity Dashboard function allows a planner to quickly review "health" indicators of the fleet, which is an important input for formulating a weekly plan that addresses the priorities.

The Activity Dashboard has four sections:

1. **Inspection Status**: reflects Workstep Ratings from AMT Mobile.
2. **Condition Monitoring Status**: measurements and severity.
3. **Messages Open**: number of open messages generated from condition monitoring translation rule triggers per alert level.
4. **Open Defect Workorders**: number of non-strategy task work orders by priority.

Activity Dashboard Functionality

- Clicking on a row drills down to the equipment level.
- The number in a cell represents the number of instances per equipment.
- Right-clicking on the instance takes you to the source of the data.
- Once an action it disappears from the grid.

Overview of Dashboards

The Dashboard is a management system that enables organisations to clarify their vision and strategy, and translate them into action. It keeps managers informed of equipment and operational performance. It provides feedback around both the internal business processes and external outcomes in order to continuously improve strategic performance and results.

The Dashboard provides a real-time graphical view of performance indicators for easy identification of areas needing attention. This includes both a short-term snapshot that focuses on the last 48 hours and month-to-date performance and a longer-term overview. Using Balance Scorecard principles, you have the ability to drill down both organisationally and functionally.

When fully deployed, the Dashboard places strategic planning at the centre of an enterprise. By establishing measurements covering multiple aspects of AMT operations, performance can be continuously monitored. The use of progressive alarm signals enables exception reporting to be clearly focused.

Toolbar buttons

Button	Action
	Targets/Benchmarks
	Home
	Save
	Close/Cancel
	Edit
	Drill Down
	Log out

Web Dashboard

System configuration and settings

KPI Calculation

A system task runs on the AMT server every hour to recalculate the statistics in the Dashboard. This process is automatically postponed to the next hour if the load on the AMT server is too high.

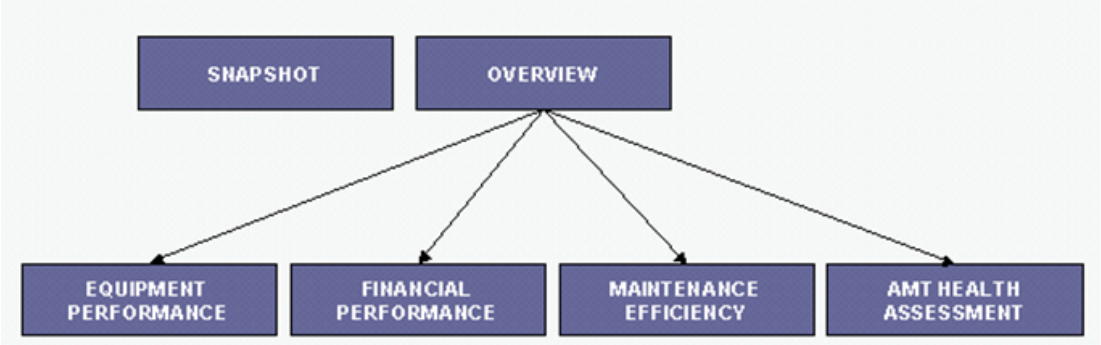
Scheduled hours

If scheduled hours have not been set up for an equipment, then the **Availability** statistic in the Dashboard will report **NaN** (no date). Create scheduled hours using the function **Scheduled Hours**.

Multiple units of measure

An asset can be linked to a usage profile, which can contain multiple units of measure. If more than one of these units of measure is flagged in the Units of Measure system table as being **Operating Hours**, this may cause usage-based statistics in the Dashboard to calculate inaccurately.

Dashboard structure



The Dashboard is not limited to Equipment Performance reporting; it can be used for a wide variety of reporting. Being web-based, it has the flexibility to be accessed by a wide range of users in an organisation.

The actual display is spreadsheet style with colour indicating levels of performance. Actual performance statistics are shown, and depending on the benchmarks set up in the Dashboard, shading indicates the result of the measurement as shown in the table below. This helps you to quickly see areas of poor performance which need follow-up.

Key	Description
	Good
	Normal
	Poor
	Alert
NaN	No value, that is, no data on which to calculate the measurement.

Dashboard navigation

Navigation within the Dashboard is done using your mouse and standard Internet browser buttons, Back, Forward, Print and so on.

The opening screen is the Snapshot. It displays the total number of machines registered and a summary of key measurements at the branch level, however, depending on how AMT was set up, the highest level may be the company. Drilling down into the measurements is possible, from the highest level to the lowest (individual equipment plans).

Navigation to dashboard measurements is shown in the table below.

Section	Group	Sub-Group	Measurement
Snapshot *Last 48 Hours*			Equipment Currently Down
			Availability
			Utilisation
			Strategy Tasks - Overdue
			Backlog Ratio
			WIP (days)
			Availability
			Reliability (MTBS)
			PCV%

Section	Group	Sub-Group	Measurement
	Financial Performance	Risk	Net Opportunity Risk
			% of Projected Life-Cycle Costs
		Financial Margin	Projected Billings
			Budget Margin %
			Projected Margin %
			Risk Adjustment Margin %
			Avg Margin %
		Cash Flow to Date	Cash Flow to Date
		AMT Health	AMT Health
	Maintenance Efficiency	Strategy	PCV%
			Strategy Tasks - Action Required
			Strategy Tasks - Overdue
		Planning	Backlog Ratio
			Planned DT % (DT = Downtime)#
			Service Accuracy#
		Daily Operations	Maintenance Rate#
			Breakdown (Events) %#
			MTFS after PM#
			MTTR#
			WIP (Days)@
		Customer Service	Budget Variance

Section	Group	Sub-Group	Measurement
	AMT Health		Budget Variance
			Strategy Tasks - Action Required
			Strategy Tasks - Overdue
			Strategy Tasks - Not linked to WOs



#000 Rolling 3 month KPI.
@ Rolling 12 month KPI.

Dashboard calculations

The table below contains an alphabetical listing of all measurements reported by the Dashboard, and a description of how the figure is calculated, regardless of date range or time period.

Date range and time period is dependent upon the section of the Dashboard in which a measurement is located. For example, measurements under Snapshot are calculated using data from the last 48 hours. Measurements calculated under Overview are month to date. The table below only describes the base calculation method, regardless of the time period.

Measurement	Calculation Method	OEM Specific
AMT Health	Average of the health indicators.	
Availability	Depends on the availability set up in AMT for each fleet. PA = Physical Availability. MA = Mechanical Availability.	
Backlog Ratio	Number of backlog tasks / Number of equipment	
Breakdown % #	Number of events where breakdown = True / Total number of events.	
Budget Margin %	(Projected billing for the budget projection - Projected cost@cost for the budget projection) / Projected billing for the budget projection.	Y
Budget Variance	Projected billing for the budget projection - Projected cost@sell not escalated for the budget projection.	Y
Cash Flow to Date	Actual billings to date - Work orders cost@cost where cost bearer = <default cost bearer>.	Y

Measurement	Calculation Method	OEM Specific
Equipment Currently Down	(Number of equipment currently down / Number of equipment) * 100.	
Maintenance Ratio#	Labour hours / Equipment usage.	
MTFS After PM#	Mean time to first stoppage after PM. Total of usage (in Eng-Hrs) from PM event to the next downtime event / Total number of PM events. A PM event is where there is a PM task in the event.	
MTTR#	Total downtime for the events / Total number of events.	
PCV%	(Projected total budget projection = Projected total current projection) / Projected budget projection.	
Planned DT% (DT = Downtime)#	Total downtime of events where planned = true / Total downtime of all events.	
Projected Billings	Total projected billing based on the billings profile for the current projection.	Y
Projected Margin %	(Projected billing for the current projection - Projected cost@cost for the current projection) / Projected billing for the current projection.	Y
Reliability (MTBS)	Total usage of the equipment / Total number of events.	
Risk: Net Opportunity Risk	The sum of the opportunity of risk components (life used <20% at the end of the equipment life) - the sum of the risk components (life remaining <20% at the end of the equipment life). A positive value is greater opportunity than risk.	
Risk: % of Projected Life-Cycle Costs	Net opportunity risk / Projected sell of current projection.	
Risk: Adjustment Margin %	(Projected billing for the current projection - Projected cost@cost for the current projection + Net opportunity risk components) / Projected billing for the current projection.	Y
Risk: Avg Margin %	([Projected margin %] + [Risk adjust margin %]) / 2	Y
Service Accuracy#	The number of services (PM events) within the allowed tolerance for a PM / Total number of PM events (tolerance is set using **Fleet Administration**).	

Measurement	Calculation Method	OEM Specific
Strategy Tasks - Action Required	The number of strategy tasks which are either overdue or (in the planning window and are system calculated and not reviewed) / Total number of strategy tasks. Only Strategy tasks which are flagged as planning tasks are considered in this statistic.	
Strategy Tasks - Not linked to WOs	The number of strategy tasks in completed or in progress events in the last 12 months that are not linked to a work order.	
Strategy Tasks - Overdue	(Number of strategy tasks overdue / Number of strategy tasks) * 100. Only strategy tasks which are flagged as planning tasks are considered in this statistic.	
Utilisation	Usage / Number of equipment.	
WIP (days)@	(Total WIP cost / Total work order cost last year) * 365.	Y

Rolling 3 month KPI.
@ Rolling 12 month KPI.

For those measurements which are based on a date range greater than one month, the date range used includes prior months plus month to date. For example, the last quarter is defined as the last three months as well as this month to date. Therefore, if the current date is 12-Jun-19, the last quarter is: 1-Mar-19 to 12-Jun-19 (March-April-May PLUS June to date).

Dashboard recalculation

You can customise the Dashboard to recalculate as often as practical given the nature of sites and equipment registered on the system and the AMT server. For example, if there are not many breakdowns, recalculation may only be required every four hours.

The System Administrator sets the frequency of the Dashboard's recalculation in the registry on the AMT server. The process takes several minutes.

You are not notified by the system when a recalculation is taking place.

Dashboard targets

For every measurement reported on the Dashboard there are four targets (benchmarks) that define the range of calculation results equating to Good, Normal, Poor, and Alert performance. Desired targets can be applied to all fleets across all branches (known as defaults), or specific targets can be set at the fleet level.

Access to editing the targets is typically granted to the AMT Administrator only.
It is recommended that the determination of target ranges however, should be a joint decision between site, branch and Head Office management

Event Merging Wizard

The Event Merging Wizard allows merging of events (driven by Planned Date Range) under a parent event. The Event Merging Wizard allows **Completed, Yet To Start** and **In Progress** events to be merged, however, some rules apply.

The Event Merging Wizard requires a Parent Event to be selected to enable this function to operate. You cannot have In Progress or Completed work orders merged to a Yet to Start Event - a message will display showing an error. However, all work order statuses can be merged to an In Progress or Completed Event.

If an External Downtime Event is selected, this must be assigned to the Parent Event.

Refer to the Event Merging Logic topic for information about the logic applied by AMT when merging events, whether they are internal or external events.

Daily Planning Reports

(Rpt_161) Daily/Weekly Work Plan

Provides a maintenance schedule to the site foreman and supervisors as notification of future work planned. It is event based with all work orders associated with the event. Allows for better allocation of resources and work flow.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|SiteID|,|Site|,
|FleetID|,|Fleet|,|EqpPlanID|,|EqpPlan|,|UserID|

Report parameters

Default.

Additional filters

Task Type, Work Group, Include Overdue, Work Location,Priority, Responsibility, Maint. Order Type, Planning Group, STO

(Rpt_164) Equipment Backlog Listing

Designed to manage the existing backlogs against each piece of equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|EqpPlanID|,
|EqpPlan|,|ProjectionTypes|,|UserID|,|BranchID|,
|SiteID|,|FleetID|,|Branch|,|Site|,|Fleet|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Perform At, Priority, OHS, Source, Raised By, Maint. Order Type, Planning Group, STO, Backlog Age, Strategy Tasks

(Rpt_632) Hours To PM

Gives a quick view of hours to the next Preventive Maintenance (PM) for selected equipment.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Report Type, Equipment Hierarchy

Additional filters

UOM

(Rpt_417) Job Folders - Bulk Printing

Lets you print Eventscopes, Workscopes or Library documents for multiple tasks at once.

Tips

Choose filters carefully in order to limit the amount of printing. Choosing too few filters could result in printing too many pages.

Initial parameters

ExternalReport|,|UserID|

Report Parameters

Items to Include, Printer, Date Selector, Sort By, Equipment Hierarchy

Additional filters

Workorder Branch, Workorder Site, Work Group, Employee Resp., Labour Activity, Responsibility, Planning Group, Task Type, Work Location, Status, Priority, Raised By, Maint. Order Type, STO

(Rpt_204) Overdue Strategy Tasks

Reviews the components that are overdue. These components need to be rescheduled and a decision made as to when these are expected to be done.

Tips

Utilise this report as part of ongoing data health checks.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|AutoLaunch|,
|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|

Report Parameters

Report Type. Hierarchy

Additional filters

Cost Type, System, Sub System, Component Code, Maint. Order Type, Planning Group, STO, Regulatory Code

(Rpt_458) Planning Status

Details the planning status of work orders.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Workorder Branch, Workorder Site, Status, Job Ready Status, Strategy Task, Component Code, Modifier Code, Task Type, Task Counter, Source, Raised By, Authorised By, Priority, Work Group, Employee Resp. Misc Category, Source of Supply, Supplier, Parts Plan. Status, Target Date (up to), Scheduled Date (up to), Exp. Delivery Date, WO #, Part #, PO #, Stock Code, Maint. Order Type, Planning Group, STO, Strategy Task, Reserved, Parts, Misc, Only show if Exp. Deliv. Date after Target Date, Only show if Exp. Deliv. Date after Date after Scheduled Data.

(Rpt_256) Work Location Allocation

Represents the current work plan by location with over and under allocation to particular locations highlighted.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|SiteID|,|Site|,|AnalyseBy|,
|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Work Location Site, Work Location, Event Status

(Rpt_352) Workplan - Workorder Summary

Provides a summary based on work orders that are Yet to Start, In Progress or Completed as a result of planning. The report lets the site foreman and supervisors plan the resources required for the upcoming outstanding work orders.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|GroupByID|,|GroupBy|,|SiteID|,
|Site|,|FleetID|,|Fleet|,|EqpPlanID|,|EqpPlan|,|UserID|

Report parameters

Default.

Additional filters

Workorder Branch, Workorder Site, Owner, Work Group, Employee Resp., Task Type, Work Location, Status, Priority, Maint. Order Type, Planning Group

(Rpt_416) WorkPlan 2

Provides a high level summary based on work orders that are Yet to Start, In Progress or Completed as a result of Planning.

Tips

The filter Status defaults to Yet to Start and In Progress to allow quick reporting of outstanding work orders only.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Workorder Branch, Workorder Site, Work Group, Employee Resp., Labour Activity, Task Type, Work Location, Status, Priority, Raised By, Maint. Order Type, Planning Group, STO

Workplan - Export to MS Project



This function requires Microsoft Project to be installed on the same computer as the AMT client application.

This function exports a workplan based on outstanding work orders from Standard or Advanced Planning to Microsoft Project. You can select whether to only export work order headers, or include work order operations and work order operations labour as well.

Group By and **Sort By** options determine how the work orders are presented within MS Project.

(Rpt_425) Part Number Reference

Report which lists the Primary Part Number, Group Number and Component Serial Number per strategy task for the selected equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report Parameters

Report Type, Equipment Hierarchy

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Modifier Code, Strategy Task, Region, Division, Cost Responsibility, Eqp Location

(Rpt_830) User Work Order Status Audit Report

Provides a history of the user work order status for a work order. The report shows who changed what (Login Name of user who made the change), and when, giving a view of the lifecycle of the user-defined status of the work order.

Initial parameters

|ControlType|,|Idlist|,|TextList|,|LogoLocation|,|UserID|,|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

User, User WO Status

Employee Availability and Allocation

(Rpt_225) Resource Availability and Fill - Employee

Displays the number of man hours allocated to each work group for the selected period. This assists with the management of the labour hours required to perform maintenance tasks effectively.

Basis of calculation: detail provided in the Planning details tab of the task referring to the employee.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Tips

Report parameters

Default.

Additional filters

Employee, Work Group, Labour Activity, Shift Start Time

(Rpt_411) Resource Availability and Fill - Work Group

Displays the number of man hours allocated to each work group for the selected period. This assists with the management of the labour hours required to perform maintenance tasks effectively.

Tips

Basis of Calculation: Detail provided in the planning details tab of the task referring to the employee.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|siteID|,|site|,|WorkGroupID|,|WorkGroup|,
|Today|,|14DaysFromNow|,|Mode|,|UserID|

Report parameters

Default.

Additional filters

Work Group, Shift Start Time

Employee availability

This function shows the work status of an employee over the select time period. An employee is considered to have allocated work when you assign the employee, or the employee's work group, to a work order in an event.

(Rpt_226) Employee Allocation

Compares the number of hours each work group has been allocated versus the available man hours per work group for the week.

Tips

Date range can be set according to analysis requirements.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|siteID|,|site|,|WorkGroupID|,
|WorkGroup|,|LabourActivity|,|LabourActivityID|,|Today|,|48HoursFromNow|,|UserID|

Report parameters

Default.

Additional filters

Employee, Work Group, Labour Activity, Working Status

Employee Setup

Set available man hours

This function is used for work group level resource levelling by allowing you to specify and smooth available resources, per hour per day, at the employee or work group level.

Navigate to **Daily Planning** > **Employee Setup** > **Set Available Man Hours.**



Work group resource levelling is set at the branch level (using the Branch Administration function).

The **Refresh Data From Employee** button populates the man hours table with working time set up for the individual employees on the selected site, from which you can make alternations.

Set employee working time

This function allows you to set up work time (rosters) for individual employees. The work time entered reflects a single shift, or in other words, a single period of work at the nominated status.

Field	Description
Start	Date and time when the working status period starts.
Finish	Date and time when the working status period ends.
Work Group	Work group the employee belongs to on site.
Labour Activity	Default labour activity the employee performs.
Working Status	Available (for work), Annual Leave, Sick Leave, Training.

For example, entering 1-Jul-18 07:00 to 1-Jul-18 16:00 (Status = Available) means that the person is available between those hours on that specific day.

- Be careful; entering 1-July-18 07:00 to 31-July-18 16:00 (Status = Available) means that the person is available every single hour from 1-July-18 07:00 to 31-July-18 16:00. This set up would be unrealistic.
- Non-work time (for example, annual leave) can be added as a work time allocation in order to provide a complete picture for the employee.

Once employees work time have been set up, the next step is to review the total available man hours on site, and if need be, refine to optimise those resources.

Notes

- AMT defaults all sites to use work group level resourcing.
- To use employee level resourcing, edit the site using Site Administration, and select the **Enable Employee Level Resourcing** check box.
- Work group and employee information is stored in separate AMT system tables.
    - Work groups and employees are related to a specific site.
    - Employees can still be added to a site, and used in planning, even if the site does not use the employee information for employee resource levelling.

Copy Employee Working Time

This function is used to copy an employee's working time, be it a roster employee or an individual employee, to one or more employees.

(Rpt_341) Employee Working Status Summary

Lists employees and the tasks to which they have been assigned.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|siteID|,|site|,
|WorkGroupID|,|WorkGroup|,|Today|,|14DaysFromNow|,
|EmployeeWorkingStatus|,|UserID|

Report parameters

Default.

Additional filters

Employee, Work Group, Labour Activity, Working Status, Shift Start Time

(Rpt_546) Workorder Authorisation List

Audit report detailing who authorised which work orders to what value and how the authorised value compares to the costs which were ultimately charged to the work order.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Task Type, Status, Job Ready Status, Source, Raised By, Priority, Work Group, Employee Resp., Backlog Age, Strategy, Requested, Authorised, WO Number, Internal WO Ref, Authorisation Amt >=, Actual Amt >=, Variance Amt >=, Maint. Order Type, Planning Group, STO

Statutory Compliance

This screen gives you the ability to review all work orders which have been nominated as statutory jobs. You are able to see the name of the asset, the relevant strategy task, when it is next due, when it was last performed and what the most recent work order number was. Navigate to **Daily Planning** > **Statutory Compliance**. The AMT Mobile screen opens in your web browser.

For each strategy task, the Statutory Compliance screen shows:

1. When the task was last performed.
2. The last work order number.
3. The next due date.

Colour coding is used to highlight the status:

Colour Coding	Description
	Not yet within 10% of frequency (system configuration variable).
	Within 10% of frequency (system configuration variable).
	Past compliance.

Job ready planning

The Job Ready step involves the planning required in order to be able to schedule the work order.

Job ready requirements

Work order instructions - are the appropriate work order instructions and check sheets available? These could either be generic documents attached or instructions detailed on the defect capture forms.

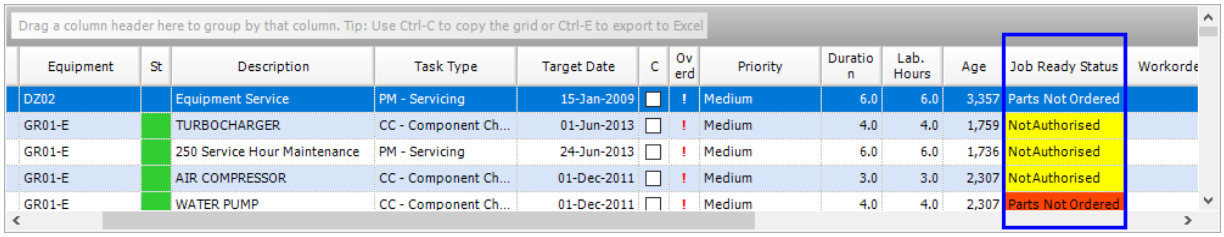
- **Parts available** - Are all the parts required in stock? Parts not in stock will need to be ordered and their lead times identified.
- **Specialised labour** - Is specialised labour required that may not be readily available? If so the lead time and availability dates will need to be ascertained.
- **Materials** - Are there any additional materials or consumables that need to be ordered to complete the work?
- **Outwork** - Is any outwork (sub-contract) required that may not be readily available? If so the lead time and availability dates will need to be ascertained.
- **Tooling** - Is any specialised tooling, facilities, vehicles or craneage required that may not be readily available? If so the lead time and availability dates will need to be ascertained.
- **Work Permits** - Are required permits for hot work, confined space work, working at heights completed and authorised for the work orders?
- **Workorder** - Has an enterprise resource planning work order been opened for the work order?



This is only applicable if AMT is interacting with another work order system.

Job readiness indicator

On the planning grid and work order this displays the job readiness status of the work order. Navigate to **Daily Planning** > **Planning**.



The job ready status is usually manually maintained on the form (in some installations AMT looks up the status of individual parts in the parts tab). The job readiness colour will only change once the job has been authorised.

Job documentation

AMT can automatically generate reports and house copies of documents designed to help in the practical execution of jobs and work orders.

Documents automatically generated by AMT are:

Eventscope Header

This report is generated for a selected event, and is designed to provide planned and capture actual downtime information for a planned event.

- Select an Event Log in Equipment Status, or a Work Order linked to an Event in Planning.
- Click the Eventscope button.

Workscope

This report is generated for a selected work order, and is designed to provide practical guidance to the person executing the work order. The report's data comes from where the work order was first identified and registered in AMT (for example, a Daily Inspector identifying a new defect, recording it on a form, and giving it to Admin to enter into AMT).

- Select a Work Order in Planning or within an Event in Equipment Status (via the Event Log).
- Click the Workscope button.

Job Folder

- This report is a combination of the Eventscope Header and Workscope documents, but also included details of each operation in the work order.
- Details items such as parts lists, list of allocate labour resources.



All of these reports can be generated in the Planning and Equipment Status modules of AMT. The quality of information in the Workscope and the Service report is directly dependent on the quality of information captured from the field or workshop and entered into AMT.

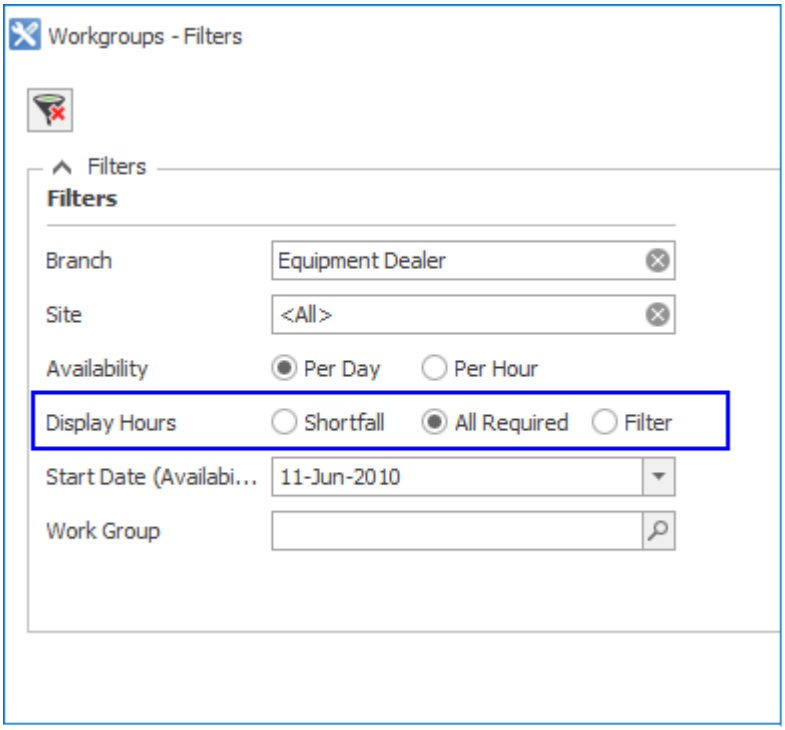
Set up labour availability

There are two ways you can set up labour availability as described below.

Per day instead of per hour

1. Navigate to **Daily Planning** > **Planning** > **Scheduling tab**.
2. In the **Drag n Drop** panel, click the **Workgroup** tab, then click the **Filter** icon.

3. Select **Per Day** as shown below.



Using the Resource Availability and Fill - Work group report

1. Navigate to **Daily Planning** > **Employee Availability and Allocation** > Resource Activity and Fill - Work Group.
2. Fill in the fields as required, selecting **Analyse By Site** and **Labour Activity**.
3. Click **Preview**.

Labour Activity

The Labour Activity field allows you to plan and schedule labour activity resources where you have multiple trades required per work order, for example Heavy Diesel Fitter, Boiler Maker, Tyre Fitter and so on.

The Labour Activity field is on

- **Daily Planning** > **Planning**
  - Planning, Scheduling tabs.
- **Daily Operations** > **Equipment Status**
  - Planning, Scheduling tabs.

Associate resources to a work order

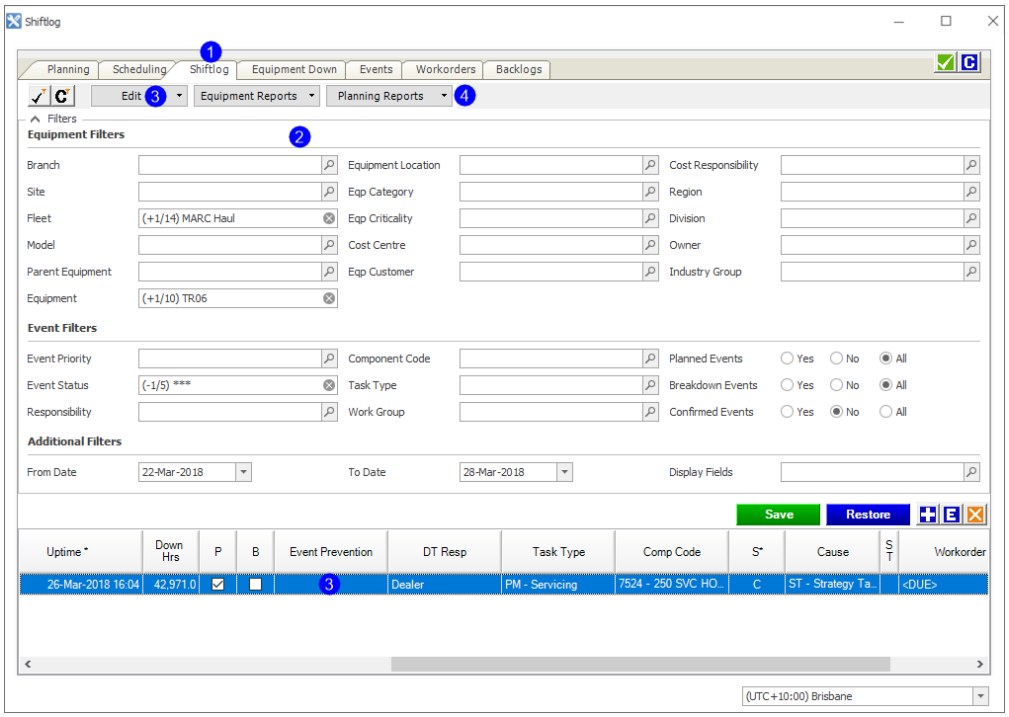
Resource assignment is determining where the work will be done and who will do it.

You can assign labour at the workgroup or employee level. This is a configuration setting at the site level.

Workgroup Level Resourcing does not take into consideration employee rosters. Rather, you can manually allocate the man hours available per day per hour to the work order.

Employee Level Resourcing lets you to specify the availability for individual employees.

Navigate to **Daily Planning** > **Planning**.



AMT assigns resources using the following steps.

1. Click the **Shiftlog** tab.
2. Select appropriate filters.
3. Either select the event in the grid, right-click it and click **Edit Event**, or click the **Edit** drop-down list and click **Edit Event**.
4. Select the relevant **Location** where the work will be carried out.
5. Click **Save**, right-click **Planning Reports** and click **Work Allocation Report** to see whether certain locations have been over or under booked.

Create a shutdown

You can create a shutdown which can be associated with work orders, from more than one asset, which are planned or are in progress.

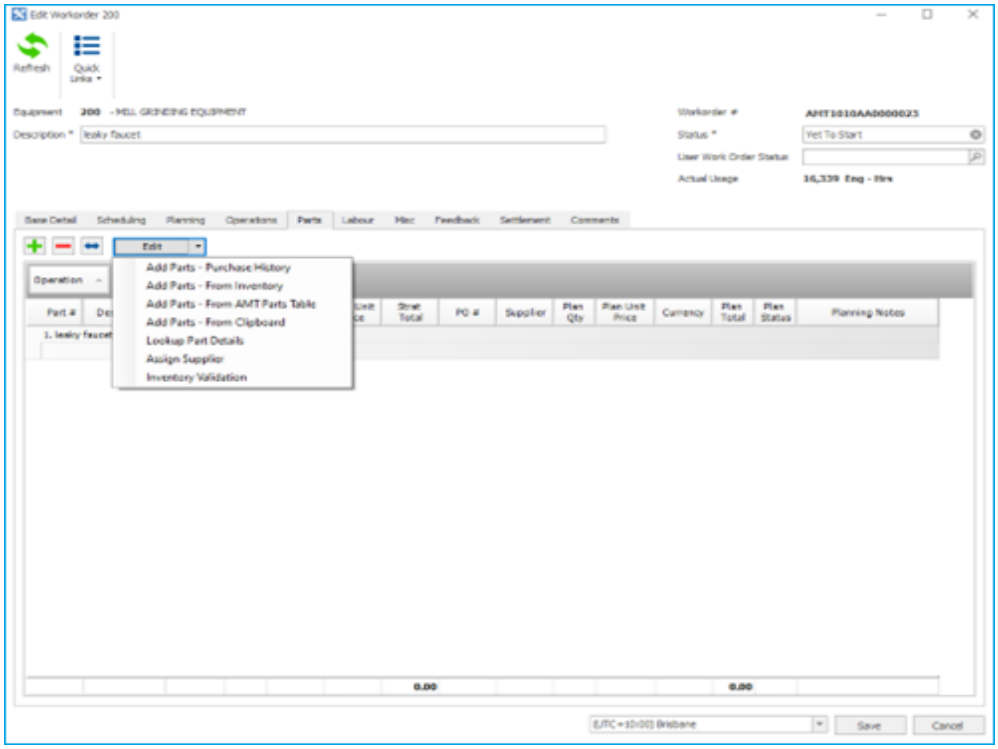
Navigate to **Daily Planning** > **Planning** and set the **Display Fields** filter to **Shutdown**.

Shutdowns do not affect the downtime or availability calculations of the assets to which they are associated.

Parts

Navigate to **Daily Planning** > **Planning** > **Workorders** tab. Double-click the work order and click the **Parts** tab. Click the links below to see more information.

Enter the required parts into the work order



There are a number of ways in which you can enter the parts required. Some are found on the **Edit** drop-down list.

Manually
Highlight the **Operation**, add a new row, then enter the parts details directly into the grid.

Manually with lookup
Use the **Lookup Parts Details** function. This will look up the part description and pricing from the AMT Parts Table. It is a useful way to validate the correct part number.

Automatically from strategy work order
If a strategy work order has a standard job linked, when AMT automatically creates the planning work order it will populate it with the standard job parts and labour details.

From standard job
Select and import parts from a standard job.

Search part table
Lets you drag and drop parts from the AMT Parts Table.

Add parts purchase history
Every time a work order is completed the planned parts for that work order are copied to a history table. The **Search Purchase History** function allows you to search the history.

Add Parts from Inventory
This requires a special interface. You can look up your internal stores to see stock levels and select parts to order.

Add Parts Part Book
For certain installations of AMT you can add parts from external parts book platforms supplied from OEMs.

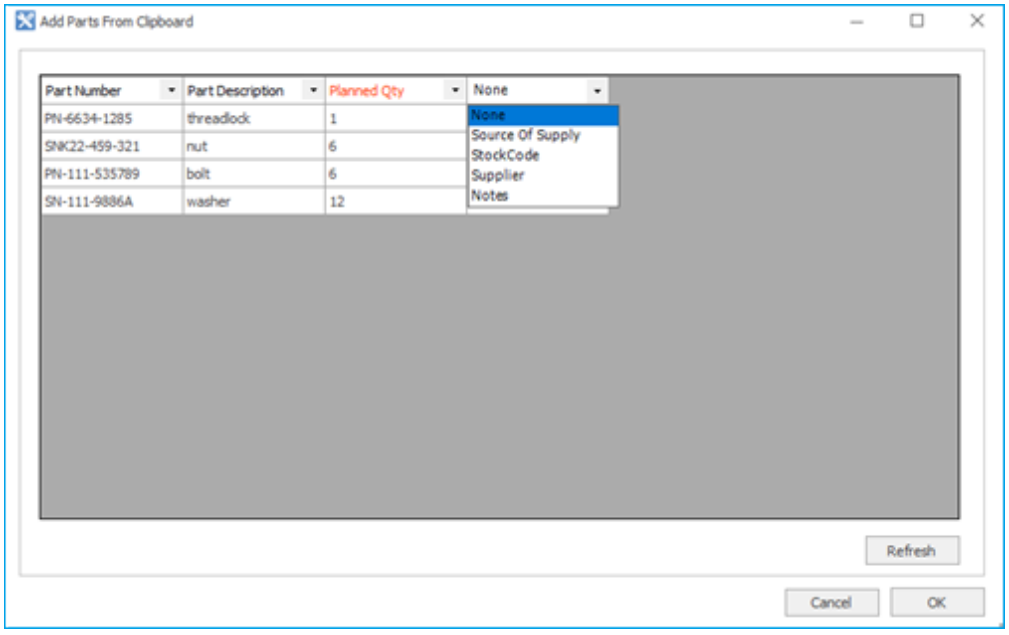
Add Parts from Clipboard

Enables you to paste a list of parts from Excel into an AMT work order. The functionality allows a list of parts to be copied to the clipboard from Excel, and the clipboard is displayed as a grid in AMT. The grid allows you to identify the type of AMT field in each column. The following fields are supported in the new functionality:

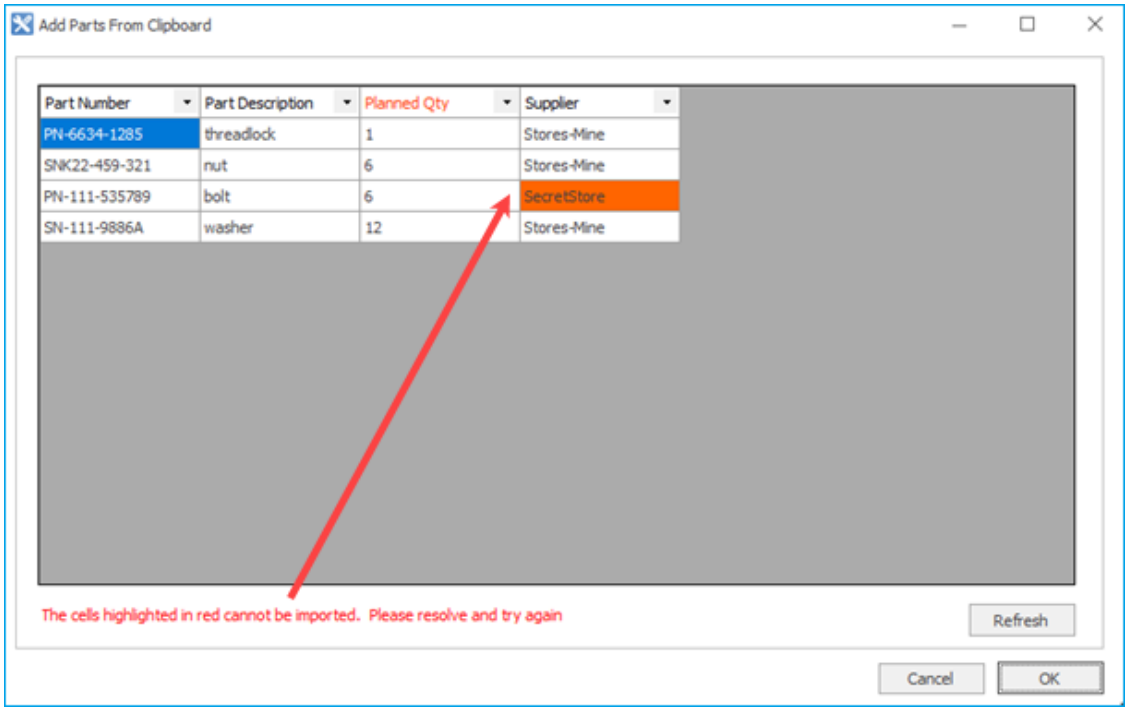
- Part Number
- Part Description
- Source of Supply
- Stock Code
- Supplier
- Planned Quantity
- Planning Notes

Not all fields need to be included in the data copied to the clipboard. The minimum fields required for the functionality to work are **Planned Quantity** with either **Part Number** or **Part Description**.

Upon selecting **Add Parts - From Clipboard** a grid displays with the contents of the clipboard and you can use the drop-down menus at the top of each column to specify the AMT fields in the column.

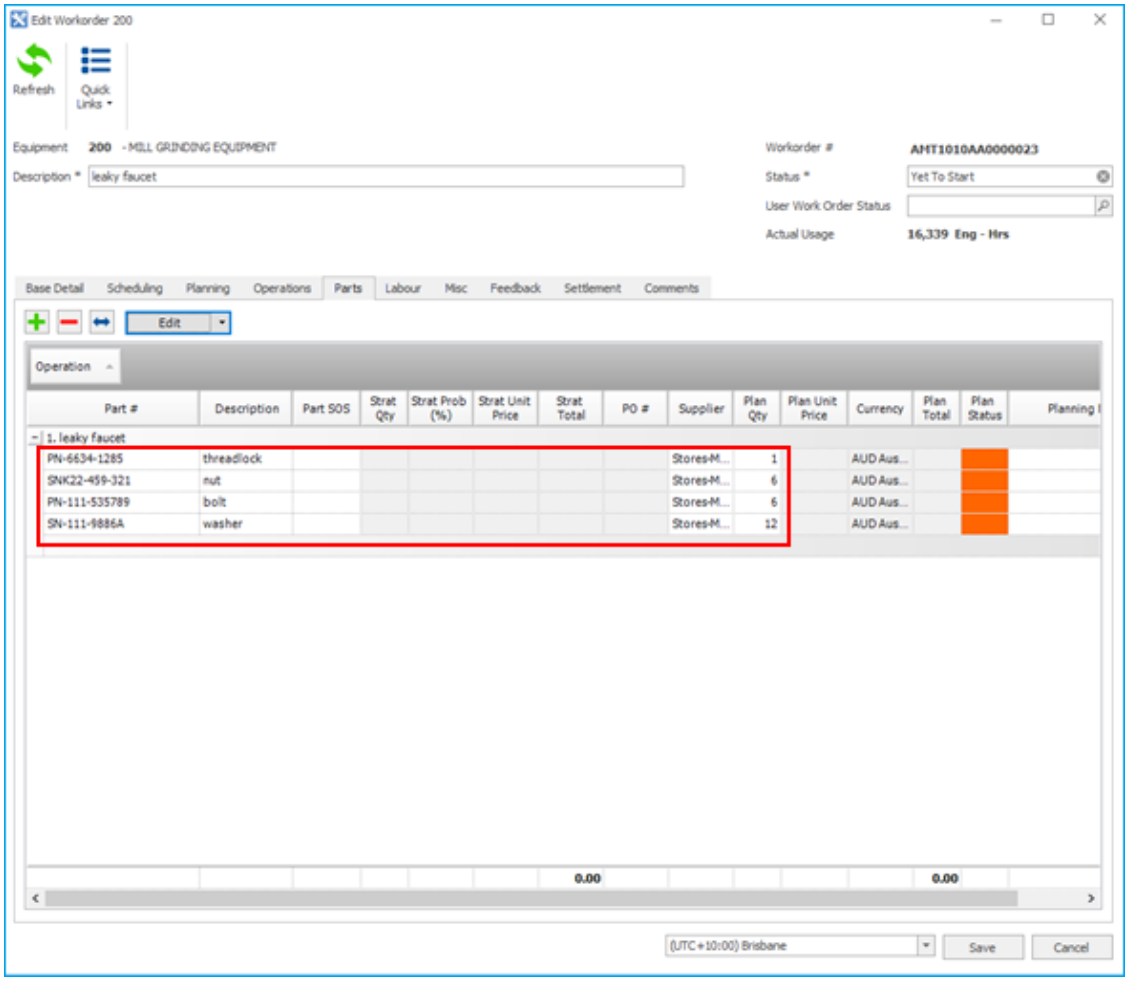


The data in the grid is read only. The column headings can be modified if data was accidentally mapped to the wrong column. If there are validation errors in the data from the clipboard, the cells in error are highlighted.The data will need to be fixed in the data source and copied to the clipboard again.



Click **Refresh** to reload the corrected data from the clipboard into the grid. Click **OK** to validate the refreshed data.

Once the parts data passes validation, it is appended to the work order parts grid as new parts on the pre-selected work order operation. The functionality operated on one operation at a time. If parts need to be added to different operations, the **Add Parts - Paste From Clipboard** process needs to be repeated with the parts list for each required operation. The functionality does not try to match existing parts and update them.



Standard AMT functionality is now being used to save the work order, validate the parts against inventory, assign supplier and so on.

Order Parts

1. Generate the purchase order or internal requisition number
2. Click the **Quick Links** button above the Purchase Order grid and click **Assign No** to automatically generate the purchase order number. You can manually override it with another reference number, for example, the PO Number from an external system.

Send off the purchase orders or parts requisitions

Depending on your AMT configuration you can do this in the following ways.

- Print and fax, post or manually deliver it.
- Email it.
- Send it electronically through an interface.

Monitor the planning status of parts

- Yellow = No purchase order created.
- Blue = Parts ordered but not yet available.
- Red = Parts not yet ordered.
- Green = Parts are available.

There are three configuration options for doing this in AMT.

Purchase Order Status
Update the **Expected Delivery Date** of the purchase order or internal requisition and that determines the status for all parts linked to that purchase order. **Orange** = Expected delivery date in the future. **Green** = Expected delivery date in the past.

Specific Parts Status (Manual)
You can edit the status for each part, the expected delivery date, and if the part has been received.

Specific Parts Status (Imported)
An interface from an external inventory system updates the status for each part, the expected delivery date and if it is reserved. You can right-click the parts line to see additional information about that part (such as quantity in stores).

Reporting the planning status of parts

You have the ability to run the Planning Status report, which is designed to assist maintenance planners fine tune planning schedules spanning the next day to any period in the planning horizon. This includes the ability to report on the planning parts status at the part level for respective work orders. Navigate to **Daily Planning** > **Daily Planning Reports** > **Planning Status.**

Attach work order instructions

You can enter work order instructions manually (see parts) and attach documents. Documents attached to strategy tasks will automatically be linked to work orders when the system creates the work order. You can manually add or remove attached documents.

Labour

Generally labour is only assigned once you know when the work order is scheduled. However, at this stage you can still enter the required labour (quantity and skill type).

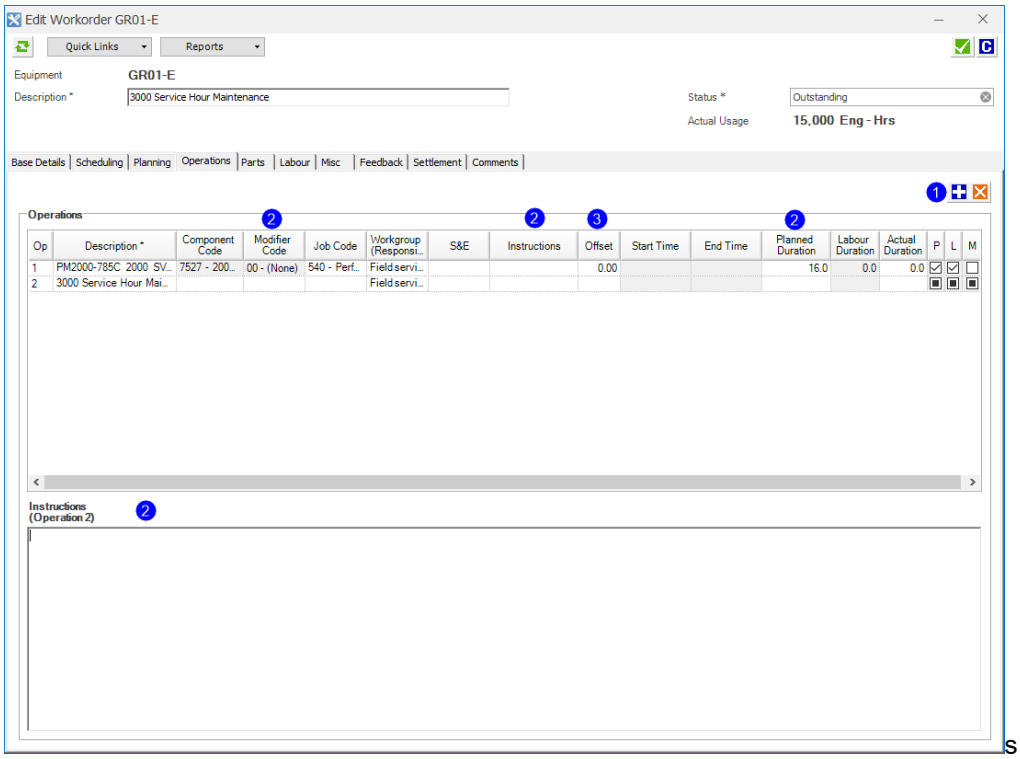
Miscellaneous

This covers any other planning requirement that is not parts or internal labour. Typically this would be items such as outwork, tooling, special permits or other materials. If a purchase order or internal requisition is required for any of these items then follow the same steps outlined for parts to assign a supplier and generate a purchase order or internal requisition.

Planning a Work Order

Operations

Navigate to **Daily Planning** > **Planning** > **Workorders** tab. Double-click the work order and click the **Operations** tab.



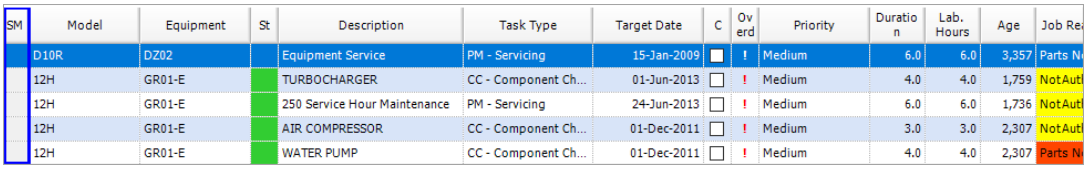
s

Item	Description
1	All work orders are created with a Single Operation by default. You can click **Add** to add multiple operations if required.
2	You can enter: • **Base Coding** - Component codes, modifier codes, job codes. • **Task Instructions** - Enter either in the field or the bigger text block below the grid. • **Planned Duration**.
3	Offset allows you to schedule the start of each operation. An offset of 2 will schedule the start of that operation in two hours after the scheduled start time of that work order.

Schedule a work order

For events, all work orders associated with the event will be rescheduled. To schedule a work order, you can either create a new planned down event for that work order or link the work order to an existing event.

Navigate to **Daily Planning** > **Planning**. Select the appropriate filters and set **Display Fields** to **Event Start** then click **Apply Filters**.



SM	Model	Equipment	St	Description	Task Type	Target Date	C	Overd	Priority	Duration	Lab. Hours	Age	Job Re
	D10R	DZ02		Equipment Service	PM - Servicing	15-Jan-2009	☐	!	Medium	6.0	6.0	3,357	Parts N
	12H	GR01-E		TURBOCHARGER	CC - Component Ch...	01-Jun-2013	☐	!	Medium	4.0	4.0	1,759	NotAutl
	12H	GR01-E		250 Service Hour Maintenance	PM - Servicing	24-Jun-2013	☐	!	Medium	6.0	6.0	1,736	NotAutl
	12H	GR01-E		AIR COMPRESSOR	CC - Component Ch...	01-Dec-2011	☐	!	Medium	3.0	3.0	2,307	NotAutl
	12H	GR01-E		WATER PUMP	CC - Component Ch...	01-Dec-2011	☐	!	Medium	4.0	4.0	2,307	Parts N

The **Event Start** is applicable to **Advanced Planning** assets only. The **SM column** on the far left will be coloured **Light Grey** if the asset uses **Standard Planning**, **Dark Grey** if the asset uses **Advanced Planning**.Create the event by editing the **Event Start** field for the desired work order.

User Work Order Status

User work order status codes reflect the state of a work order from a user perspective not covered by the system status.

These codes are available to assist managing the status (or state) of a work order from a user perspective during the planning and scheduling lifecycle of a work order. For example, you may wish to know work is pending production, pending scope, awaiting parts or awaiting external labour. Such states can be defined as a user status and applied to work orders. Work orders can be searched using this status as well. The user status cannot be changed on a work order with a system status of **In progress**, **Abandoned** or **Completed**.

The following screens have the **User WO Status** field:

- **Daily Planning** > **Planning**
  - Planning, Work Orders, Backlogs tabs.
- **Daily Operations** > **Equipment Status**
  - Planning, Work Orders, Backlogs tabs.
- **Daily Operations** > **Daily Reports** > Workorder Details

You can filter for the user work order status on these screens. The field has a one-to-one relationship with a work order. When you add a work order and set a user work order status you have the option of adding comments about the work order.

You set up the user work order status codes in **System Administration** > **System Tables** > **Maintenance** > **User Work Order Status**.

Prepare Job Folders

A Job Folder can consist of the following.

- **Eventscope** - details all work orders to be performed during an event (for example, asset stoppage).
- **Workscopes** - work order operations with parts and labour allocations.
- Attached documents such as:
  - Detailed work order instructions or references to product handbooks, check sheets (safety , environmental and statutory).
  - Permits and certificates.
  - Timesheets - to record the time spent on the job (this will vary based on the different payroll systems).

- **Bill of Materials** (included in the AMT Workscope report).
- Backlog listing.

You would collect the parts and specialised tooling required, and have them waiting at the workshop bay or other agreed location.

Navigate to **Daily Planning** > **Planning** and set the **Display Fields** filter to **Event Start**. Select the work order to be printed, then click the **Equipment Reports** drop-down list and click **Print Job Folder**.

Work plans and job folders are handed to the supervisor or foreman on site to perform the work. The mechanic completes the following documents.

- Eventscope - record the downtime details, update the work order status (if completed), and the duration of each work order.
- Work order - record the repair notes and sign-off details on the work scopes.
- Labour - depending on the process followed, the mechanics may complete timesheets or enter labour details directly onto the work scopes.
- Parts return advice - details of parts to be returned to stores.

Upon return to the workshop all documentation (including defect capture forms from above) is given to the supervisor.

User Work Order Status

User work order status codes reflect the state of a work order from a user perspective not covered by the system status.

These codes are available to assist managing the status (or state) of a work order from a user perspective during the planning and scheduling lifecycle of a work order. For example, you may wish to know work is pending production, pending scope, awaiting parts or awaiting external labour. Such states can be defined as a user status and applied to work orders. Work orders can be searched using this status as well. The user status cannot be changed on a work order with a system status of **In progress**, **Abandoned** or **Completed**.

The following screens have the **User WO Status** field:

- **Daily Planning** > **Planning**
  - Planning, Work Orders, Backlogs tabs.
- **Daily Operations** > **Equipment Status**
  - Planning, Work Orders, Backlogs tabs.
- **Daily Operations** > **Daily Reports** > Workorder Details

You can filter for the user work order status on these screens. The field has a one-to-one relationship with a work order. When you add a work order and set a user work order status you have the option of adding comments about the work order.

You set up the user work order status codes in **System Administration** > **System Tables** > **Maintenance** > **User Work Order Status**.

Daily Operations

Equipment status

Equipment status, in the **Daily Operations** module of AMT, provides operations with the practical tools they need for daily decision making and the ability to streamline the initiation of work orders, capture of technical data and update planning. Equipment status integrates planning, dispatch and work order settlement systems to provide a seamless flow of information and an integrated history database for analytics.

Specific functionality includes:

- Real time graphical display of equipment status and work plans.
- Single reference point for field service to access equipment history, component status, work plans, outstanding repairs and so on.
- Capture and analysis of downtime (integrated to dispatch).
- Daily reporting – Event log, monthly availability snapshot and so on.
- Creation of work order settlements.
- Integration of the production and cost data into a single structured database for analytics.
- Maintenance efficiency reporting.
- Equipment performance reporting.

Related Topics

Equipment Down

Events

Backlogs

Work Orders

Health - Daily Operations

Equipment Status is both a tool for daily operations personnel and the main point of capture of operational and technical data for equipment performance and maintenance efficiency reporting and analysis.

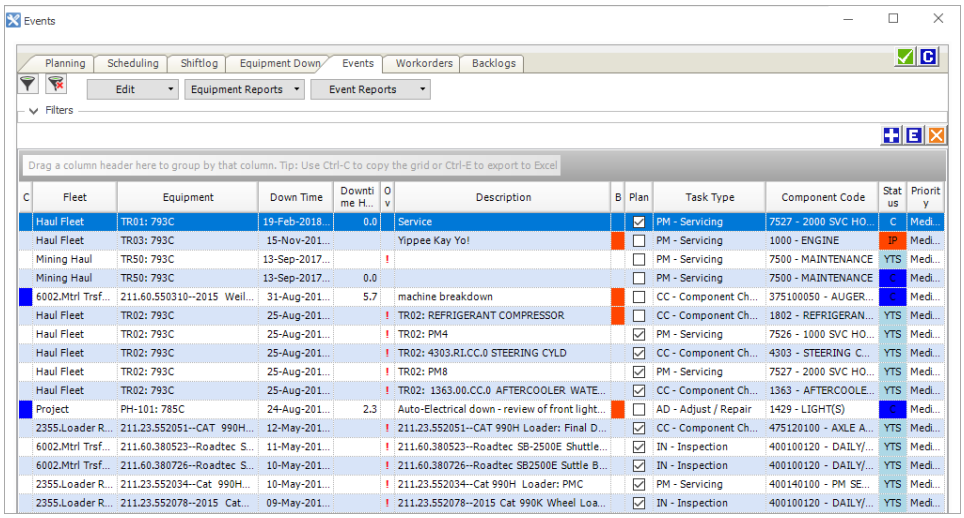
To ensure that Equipment Status is being maintained effectively, the key aspects to check are:

1. Downtime capture is up-to-date.
    a. As a minimum, the downtime events should be up-to-date and accurate. View **Daily Operations** > **Equipment Status** > **Equipment Down** tab and check if the machines currently down are being displayed accurately.
    b. Check that monthly downtime reconciliations are performed with the customer.

    If there is an interface to Dispatch and the interface goes down, then Equipment Status should continue to be maintained manually.

2. Event and Task technical details are accurately captured.
    a. All workscopes are completed and sent to the Maintenance Coordinator for updating AMT within 24 hours of the machines down event being completed.

b. View **Daily Operations** > **Equipment Status** > **Events** tab and check that all completed Events that are greater than 48 hours have been confirmed. See the example screenshot below.



The blue shading on the first column indicates that the Event has been confirmed. Note that in the filters you can only filter for the unconfirmed items.

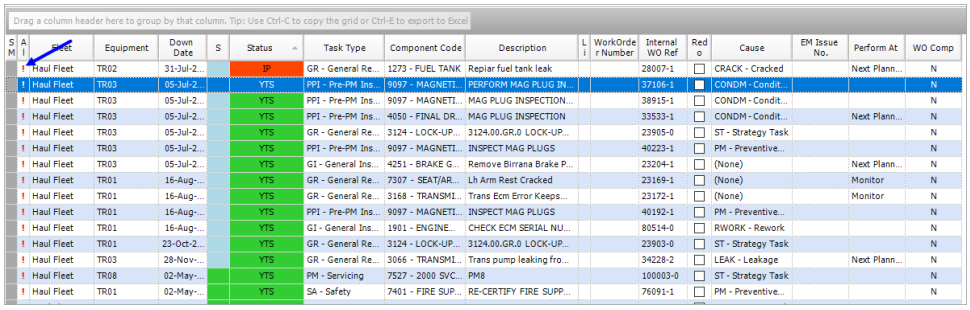
c. Analyse the Event details for the last six months for reasonableness of coding by running the Event Analysis report. Navigate to **Reporting & Analyses** > **Equipment Performance** > **Stoppages and Downtime** > **Event Analysis**. Run the **Bar Chart** report a number of times, analysing by Variance Cause, Activity, Breakdown, Planned and Component Code and check for reasonableness.

d. Analyse the Downtime details for the last six months for reasonableness of coding. Navigate to **Reporting & Analyses** > **Equipment Performance** > **Stoppages and Downtime** > **Downtime Analysis**. Run the **Bar Chart** report a number of times, analysing by **Activity**, **Responsibility**, **Variance Cause** and check for reasonableness.

3. Tasks linked to work orders.

This is especially critical for strategy tasks.

a. See **Dashboard** > **Overview** > **AMT Health** > **Not linked to WO**. This the number of strategy tasks that have been completed in EQS and are not lined to a work order. Also view the Task screen in Equipment Status and filter for Admin Alert items - these are strategy tasks that are either not linked to a work order or are linked to a rotable item that needs to be updated. See the red exclamation mark (**!**) in the example screenshot below.



4. Daily reports produced from AMT.
    a. Check that the Daily Log report is produced from AMT and used in customer morning meetings. Navigate to **Daily Operations** > **Daily Reports** > **Daily Log - Summary**.
    b. Check that the Snapshot - The Worm report is used to monitor monthly downtime against the target. Navigate to **Daily Operations** > **Daily Reports** > **Snapshot - The Worm**.

Events

The Events tab on the Equipment Status screen is designed to replace the manual shift log and the whiteboard. Events captures, through user input, all information regarding what is happening, and what happened, during a maintenance event, such as:

- Whether the event was planned, unplanned or a breakdown.
- Down and up times.
- Work orders performed, deferred or abandoned.
- Responsibility for user-nominated segments of the total downtime.
- Symptoms, causes and repairs.
- Parts which caused a breakdown.
- Which rotable component went into the machine, which came out and so on.

Grid Fields (Key Fields)

Field	Description
C	This column is shaded **blue** and shows that the AMT event has been confirmed. The purpose of confirming an event is to flag that you have entered the basic information pertaining to a downtime event such as downtime allocations, or the work order details have not been entered into AMT. Without the basic information, analyses of downtime, causes of failure and so on may be compromised
Down Time	• If the event's status is **YTS**, the date reflects the time planned for the event to occur. • If the event's status is **IP** or **C**, the date is the actual time the machine went down. • If the event's status is **A** or **D**, the date reflects the time the event should have occurred.
Down Hours	Total downtime hours for completed events
Overdue	**!** Indicates that the work order should have been completed but it has not (based on target date versus today's date).
Description	Description of the event (not the work order).
B	This column is shaded **red** indicating the event is a breakdown.
Plan	A tick (✓) displays if the event is planned.

Field	Description
Task Type	Task Type of the event's Primary Task.
Component Code	Component Code of the event's Primary work order.
Status	• **YTS** = Yet to Start. • IP = In Progress. • **C** = Completed. • **A** = Abandoned (the event will never take place). • **D** = Deferred (the event remains in the planning window).
Priority	Priority of the event.

External Downtime Wizard

AMT can be configured to import actual downtime events from an external system (for example, a Fleet Management system).

It is not uncommon for the external downtime system to use a different name, number or code to identify a Site and to identify an Equipment compared to how each is respectively defined in AMT. For example, the first truck in the fleet might be registered as T001 in the external downtime system, but in AMT it is registered as HT01 to integrate to other systems. Therefore, for AMT to recognise that T001 (external downtime system) equals HT01 (AMT) in the interface process, a mapping table must be created and maintained when new equipment are loaded into AMT. It is not something which the users need to maintain.

- To configure the mapping between external downtime equipment to that in AMT, use the **External Downtime Equipment** function.
- To configure the mapping between external downtime sites to that in AMT, use the **External Downtime Site Equipment** function.

Once all site and equipment mapping has been set up, AMT will recognise the downtime notifications provided by the external downtime system through the automated interface. The next step is for you to register the notifications imported into AMT through the interface as **Events**.

External downtime events can either be configured to automatically create an Event in AMT, or to be placed in a holding area for a user to review. This holding area is accessible using the External Downtime Wizard.

Using this Wizard, users will see all downtime notifications received by AMT, and then determine which of the notifications will be:

- ignored (this does not delete the notification).
- linked to an existing AMT downtime event (for example, a Planned Event), in which case you are able to search for and select the AMT event to link the notification to.
- used to create a new AMT event (for example, a Breakdown Event).

After accepting the external downtime event (that is, making it an AMT Event), the AMT Event needs to be reviewed. If AMT receives an Uptime notification from the external downtime system, all work orders in the AMT Event, and the Event itself will be set to **Completed**. Work orders not completed need to be deferred, deleted or

abandoned. You must supply missing work orders performed during the downtime or missing downtime allocation information as required.

Remember, the quality of reporting from AMT (and any system) is dependent on the quality of data in the system. Equipment Status is the cornerstone of the data integrity in AMT. This is the area where all the work gets done.

Additional notifications may be received by AMT for the same Event. The notifications typically fall into two categories:

1. Uptime (date:time notification of when the downtime was completed.
2. Shift changes (.ran uptime and downtime notification that represents the equipment still be down during the change of shift).

The database parameter is used to determine whether an additional notification is interpreted as the equipment has in fact gone back to work (which equates to an Event's Actual Uptime) or whether the notification signals a shift change (which is not relevant for AMT and is ignored). There may be occasions that a notification is not handled by the logic, in which case you will have to manage the notification using the External Downtime Wizard.

There are two ways in which an External Downtime Events can be removed:

1. Unlink the internal AMT Event from the External Downtime Event. This will disassociate the two events, however, the External Downtime Event will remain visible in Equipment Status.
2. Delete the Event, just as you can delete an internal AMT Event. However, once an External Downtime Event is deleted it will remain visible in the External Downtime Wizard. You can choose to ignore the Event so that another user does not manually try to recreate it.



If AMT is configured to auto-create External Downtime Events such as AMT Events, all External Downtime notifications will appear in AMT. You can then delete the External Event if it is not required. If the External Events represents planned work, merge the External Event with an existing AMT Planned Event. This is achieved by opening the new External Event, clicking the Merge Yet to Start Event function, and selecting the AMT Yet to Start Event. If the External Event represents breakdown work, update the internal Event's Downtime Allocation and work order details.

Event Merging Logic

AMT supports retaining manually entered planned downtime allocations and actual downtime allocations when merging planned AMT events and external events from the AMT EPF Asset Statistic Adapter. This in intended to support the following scenarios.

1. Planners create planned downtime allocations in planned events as part of the planning process. When work on the equipment commences, the external downtime notification is sent to AMT from the external downtime management system. The user in AMT links the planned event to the imported event from the external downtime management system. Previously, the planned downtime allocation records from the planning process were lost and replaced with any actual downtime allocation records from the external downtime system. Now, the planned downtime allocation records will be retained in the merge process so the user can see the planned downtime allocations from the planning process, and the actual downtime allocations (if any) from the external system.
2. An equipment breakdown results in the downtime management system creating a downtime event in AMT. At this time, the downtime event does not contain any downtime allocations. The planner decides to include an already planned PM service into the event. The PM service work order already existed in AMT

with a YTS Event with planning downtime allocations. When merging the planned PM event with the external downtime event, the planned downtime allocations should be retained. The status of the external event (In Progress) will not change, but the event type would be taken from the AMT planned event rather than using the event type from the external downtime event.

Users can add new downtime allocation records to an In Progress event which contain actual downtime hours. The new downtime allocation records added will not be deleted when the external downtime management system updates the AMT event.

In both cases, if AMT receives a downtime allocation record from the external system which aligns to an existing downtime allocation of the same event, then AMT will merge the rows. Matching is based on the combination of Activity, Variance Cause and Responsibility. Note that, if the match occurs on the primary downtime allocation record, a second downtime allocation record is created. This is because the primary downtime allocation record must continue to be the record used to balance to the event's total downtime. In the situation where actual downtime entered by the users in AMT is greater than the actual downtime from the external downtime management system, AMT will create a new (non-primary) downtime allocation record with a negative value to balance the downtime allocations in AMT with external downtime. This automatic downtime allocation record will use the Variance Cause defined as the Default UnallocatedDowntime, and AMT's default Downtime Responsibility.

Selecting one of the Variance Causes as the Default Unallocated Downtime is a pre-requistie for the mergic logic to work as described. Refer to the Variance Cause **system table for more information.**

Shift log

Only available for real equipment with Planning Mode = Advanced

The **Shift Log** tab on the **Equipment Status** screen is where you can capture downtime quickly. The event and its primary work order are created as a single line entry; to which a work order settlement can be linked (using drag and drop).

Additional work orders can be added to the event by either editing the event, or by clicking the **Add Work Order to Event** button.

Grid fields (Standard fields)

Field	Description
+	Multiple work orders linked to the event.
Eqp	Equipment name.
Model	Equipment model.
E	Flags if the event is generated from an external source via the Downtime module.
Description	Event description.

Field	Description
S	Event status (Yet to start, In progress, Completed).
Downtime	If Event status = Yet to start then downtime = Planned downtime date/time. If Event status = In progress or completed then downtime = Actual downtime date/time.
Uptime	If Event status = Yet to start then downtime = Planned uptime date/time. If Event status = In progress = Expected uptime date/time. If Event status = Completed then downtime = Actual uptime date/time.
Down Hrs	Total downtime hours for the event.
P	Planned.
B	Breakdown.
Event Prevention	(only applicable to breakdown events).
DT Resp	Primary downtime allocation responsibility.
Task Type	Task type code for the primary work order of the event.
Comp Code	Component code for the primary work order of the event.
S	Primary work order status (Yet to start, In progress, Completed).
Cause	Cause code for the primary work order.
ST	Shaded green if the primary work order is a strategy task.
Work Order#	Work order settlement linked to the work order.
Cost	Cost (at sell) of the work order settlement linked to the work order.
Labour Hrs	Actual labour hours.



Additional fields can be displayed as column headings by selecting them in **Display Fields**.

Work orders

You can access and search equipment history in Equipment Status on the Work Orders tab.

Grid fields (Key fields)

Field	Description
SM	Standard planning mode equipment (light grey in colour). Advanced planning mode equipment (mid-grey in colour).
Admin Alert	! displays in this column if: • The work order is strategy task, but you have not linked it to a work order settlement. • The work order is associated with a rotable item but you have not completed the details of the rotable going into or coming out. Admin alerts only display if AMT is not being administered properly.
S	This column is shaded **green** if the backlog work order is a strategy task.
Status	Work orders of all statuses display unless filtered.
Linked	Indicates if the backlog is linked to a work order settlement. No colour indicates a work order settlement has not been linked to the backlog work order, neither has a request from AMT for a work order settlement been initiated. **Light blue** indicates a request to open a work order settlement has been made, but AMT has not received confirmation. **Dark blue** indicates a work order settlement has been opened (AMT has received confirmation).
Work Order	Actual number of the work order settlement linked to the backlog work order. <DUE> indicates that AMT has requested a work order settlement number but a confirmation has not been received.
Redo	If you have manually flagged the work order as a job that needs to be redone, a tick (✓) displays.
Cause	Enter a reason why the work order was executed. For strategy tasks, cause defaults to strategy task but you can change this.
EM Issue No.	If the backlog work order is an equipment management issue, the identification number of the issue displays in this field.
Perform At	The operational stoppage at which the work order was planned to be performed.

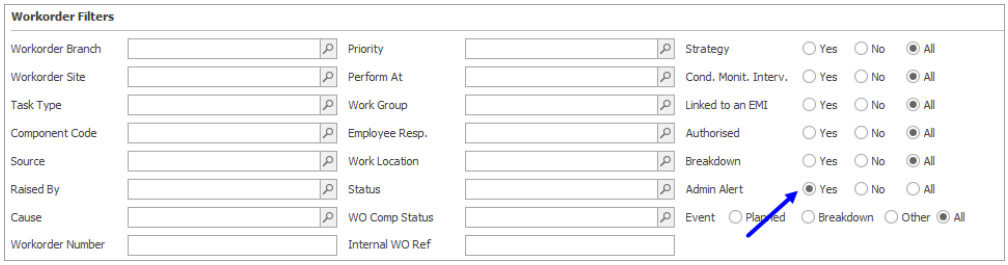
Linking work order settlements

Linking of work order settlements from external systems to strategy tasks and AMT work orders should happen automatically. If not, or the automatic linking has failed, you will have to manually link them.

Manual linking

If the work order settlement corresponds to a strategy task (that is, a task in the asset's current projection), the work order settlement should be linked to the strategy task in the Equipment Status (if you have this module installed). This ensures AMT correctly recognises the date when the strategy task occurred, and therefore will forecast from the correct date into the future. To link work order settlements to work orders, navigate to **Daily Operations** > **Equipment Status**, and click the **Workorders** tab.

Strategy work orders which have not been linked to work order settlements can be found by filtering for **Admin Alerts**.



After applying the filters, the list returned by AMT displays Admin Alerts, indicated by a red cross in the far left column of the grid. Admin Alerts can correspond to two administration problems:

1. The work order is a strategy task and is not linked to a work order settlement, in which case the work order settlement column will not have a work order settlement number, and
2. The work order is a strategy task which is linked to a rotable part but you have not updated the rotables management record in the work order.

If you need to find the work order that corresponds to the work order settlement created previously, then:

- Right-click the work order and select .
- Click the **Settlement** tab.
- Click **Link Workorder Settlement**.
- Search for the work order settlement using filters.

Once you have found the work order, select the work order settlement and save it, and the linking process is complete. AMT uses the actual down time of the event (in which the work order resides) as the strategy task's last occurrence, from which future occurrences are forecast.

Use the Component Status and Technical Performance Analysis reports to validate the linking.

Long-term Planning > **Components** > **Component Status**.

Strategy Optimisation > **Component Reliability Analysis** > **Technical Performance Analysis**.

The reports show the last occurrence of each strategy task in the asset's current projection and the linked work order settlement.

Backlogs

The Backlogs tab on the Equipment Status screen provides a simple way of accessing the list of outstanding work orders (that is, work orders which have yet to be executed) for an equipment, which is useful when windows of opportune maintenance are presented during any given day.

Grid fields (Key fields)

Field	Description
SM	Standard planning mode equipment (light grey in colour). Advanced planning mode equipment (mid-grey in colour).
Raised Date	Date when the backlog work order was first raised or identified.
Status	Current status of the backlog in the system. **A**: The backlog work order will not be executed as you have deemed it to be invalid (for example, an early engine failure resulting in an entire engine replacement negated the need to perform an outstanding minor repair to a turbo). It is suggested that backlogs are abandoned rather than deleted as this provides a more complete machine history. **O**: the backlog work order has not been executed yet and is valid. **YTS**: the backlog work order is planned.
Linked	Signifies whether the backlog has been linked to a work order settlement. No colour indicates a work order settlement has not been linked to the backlog work order nor has a request from AMT for a work order settlement been initiated. **LIGHT BLUE** indicates a request to open a work order settlement has been made, but AMT has yet to receive confirmation through the interface. **DARK BLUE** indicates a work order settlement has been opened (AMT has received confirmation).
Work Order Number	Actual number of the work order settlement linked to the backlog work order. <DUE> indicates that AMT has requested a work order settlement number but no confirmation has been received through the interface.
Exp. Duration	Expected time required to complete the backlog work order.
Job Readiness	**Not authorised**. **Authorised, Not Ready**. **Authorised, Parts not ordered**. **Job Ready, (authorised and parts ready)**.
S	This column is shaded **Green** if the backlog work order is a strategy task.
Strategy Date	If the backlog work order is a strategy task, this date is the forecast date for the work order to be performed.
Locked	If the user responsible for an equipment's strategy has through various analyses.

Field	Description
Overdue	! Indicates that the work order should have been completed but it has not (based on target date versus today's date).
EM Issue No.	If the backlog work order is an equipment management issue, the identification number of the equipment management issue displays in this field.

Equipment Down

The primary purpose of the Equipment Down display of Equipment Status is to provide a graphical display of the current status of the equipment – whether a machine is in operation, undergoing maintenance or maintenance has been completed. Through manipulation of filters, planned events in the future can be seen, as well as maintenance event history.

Gantt chart legend

Bar	Description
	Completed event. The event has an actual downtime value and an uptime value.
	In Progress event. The event has an actual downtime value not no uptime value.
	Yet to start event. The event does not have either an actual downtime or uptime value.
	Signifies the status of downtime notifications, which have not yet been linked to an AMT event. The downtime notification can include both down and up times. The presence of black coloured events greater than one day prior indicates that the downtime notifications from dispatch have not been linked to AMT events. It is recommended that a target be set so that at the end of each daily operations shift, all downtime notification events are linked so that there are no black events.

The vertical black line indicates the current date and time, and is automatically refreshed by AMT approximately every five minutes.

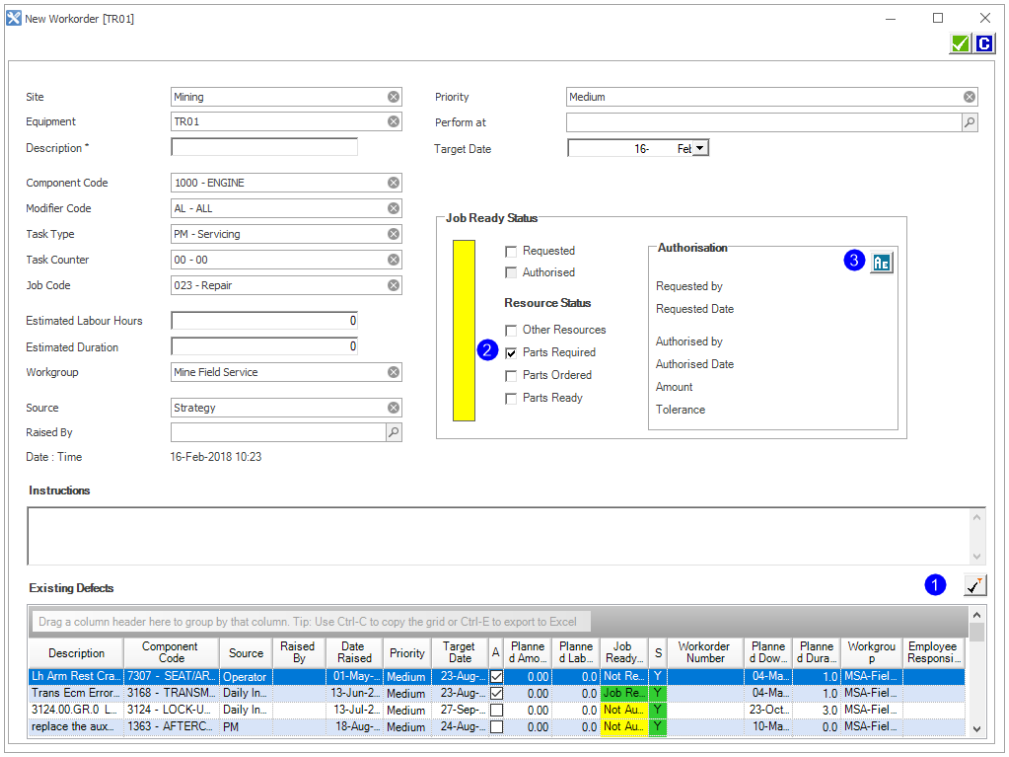
Quick Backlog Capture

There are two ways to manually create a new defect from the Planning Grid in AMT.

1. Add backlog using the full Work Order Editor.
2. Add backlog using the Quick Backlog Capture screen.

The full Work Order Editor should only be used where the work order has already been authorised and operations and associated parts, labour and misc. requirements are known.

Navigate to **Daily Operations** > **Quick Backlog Capture**.



Item	Description
1	Filter to check that backlog does not already exist. Backlog defects should not be entered twice.
2	Indicate status of parts.
3	Click to authorise the job to be performed.

The majority of backlogs will have Task Type = GR (General Repair).

PM services (Task Type = PM) and Component Changeouts (Task Type = CC) are usually strategy tasks and therefore do not need to be manually added as a backlog. If unsure, check with the person who is responsible for the asset's maintenance strategy

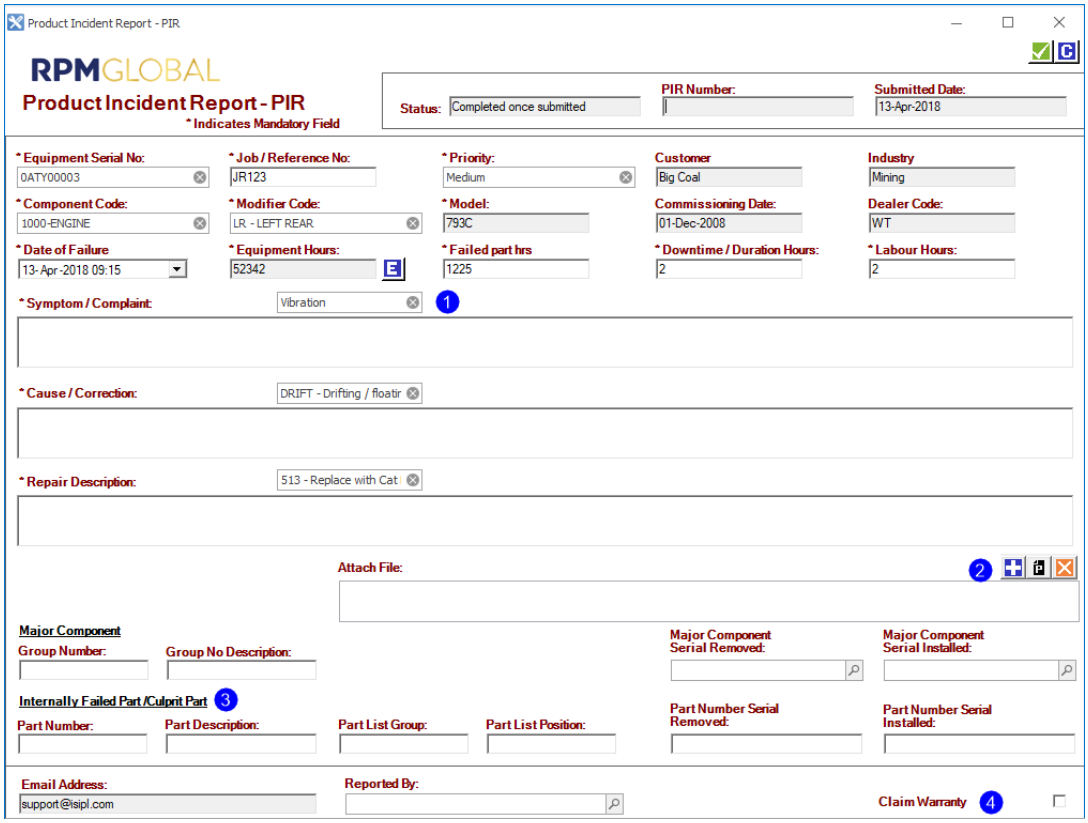
If you select the OH&S value (which is flagged in the OH&S system table as being **Auditable Record**) in the work order, certain fields become disabled. This feature is designed to be used to "lock" work orders. The fields can be re-enabled by selecting a different OH&S entry.

More information can be captured in AMT for a backlog; to do so, in Planning, search for the backlog task, open it and add the additional information.

Product incident report capture

AMT has a form which provides a structured process for registering and processing Product Incidents Reports and warranty claims.

Navigate to **Daily Operations** > **Product Incident Report Capture**.



Item	Description
1	**Symptom**, **Cause** and **Repair** are details on why the component failed. You can filter for the closed option, and enter comments.
2	Attach relevant documentation.
3	Enter extra failed part information.
4	Select the **Claim Warranty** check box if you want to claim the incident under warranty.

Product incident report listing

Typically, Product Incident Reports are reviewed in order to:

- identify reoccurring failures of specific components on a common model of asset.
- identify reoccurring failures of specific piece parts.
- view potential warranty claims, and create warranty claims for them.
    - The warranty feature of AMT, which is different to the Product Incident Report feature, can identify work orders which are potential warranty through parts usage analysis, which lets such work orders to be reviewed and flagged as warranty claims.

- assess whether a particular customer is experiencing a number of incidents on their asset which may be leading to dissatisfaction.

Navigate to **Daily Operations** > **Product Incident Report Listing**. Select and apply the relevant filters, then double-click a record in the grid to open the Product Incident Report Capture form.

Submit warranty claims



You can only submit warranty claims if you have the Warranty Interface installed.

If the Warranty Interface is in place you can automatically submit a warranty claim to a warranty system by clicking **Submit Warranty Claim** (on the Edit Work order screen), and select the warranty system to submit the claim to. AMT will then:

- Record the **Date**, **Time** and **Username** related to the request.
- Send a request to the selected warranty system to process the warranty claim. This claim is processed real-time in order to provide live feedback to you.

If the warranty claim is denied, the warranty system sends an error message back to AMT, and the **Claim Status** changes to **N/A**. The error message displays in a pop-up box. You should click the **Details** button to read why the claim was denied. If the warranty claim is approved, the **Claim Status** changes to **Claimed**.

Warranty approval

To assess and approve warranty claims, navigate to **Daily Operations** > **Warranty** > **Warranty Approval**. Filter for the relevant assets and set the **Warranty Claimable** = **Yes**. Review the details of the work order by clicking the **Work Order** in the grid and click **Edit**.

The Edit Work Order screen opens displaying the **Warranty** tab. On this tab you can enter additional details concerning the warranty claim. A work order can have the following claim statuses:

- **N/A**: If the work order is marked as not claimable.
- **Claim O/S**: If the work order is marked claimable but no claim has been submitted.
- **Claimed**: If either an amount is entered in the **Claimed** field (in the Cost Analysis section), or a claim has been submitted through the interface by clicking **Submit Warranty Claim**).
- **Closed**: If an amount is entered in the **Recovered** field (in the Cost Analysis section).



Flagging a work order as being Claimable is not limited to the product incident report process. A standard AMT user can flag a normal, everyday work order as being claimable.

Warranty claim

Warranty management is a key to providing an indicator of one of two main problems. High levels of warranty claims may indicate:

- Poor rebuild centre practices.
- Poor site practices.
    - Poor warehousing practices.
    - Poor maintenance execution practices.

A high number of successful warranty claims indicate that the rebuild centre may not be entirely at fault. Alternately, further investigation may find that the rebuild centre itself is not at fault, but the actual product itself, which may instigate a review of the product in use and determination of the economic implications of maintaining the existing fleet.

A low number of successful warranty claims may indicate that there are issues in how components are warehoused or transported; alternately it may indicate that there are issues with the maintenance execution practices, triggering a review of the existing practices and an improvement project at a site level.

Additionally, if there are multiple sites using similar asset makes and models, a comparison of warranty claims across the sites may also act as a review trigger for maintenance practices and result in an improvement project.

Warranty claim analysis should be carried out in conjunction with Equipment Management Issues.

Reports in AMT

Navigate to **Daily Operations** > **Warranty** > Warranty Claims.

This report is a summary of the warranty claim. It provides basic details of the component, relevant machine history and life achieved. It is suitable to give to a technical services or warranty division.

Product incident report listing

Typically, Product Incident Reports are reviewed in order to:

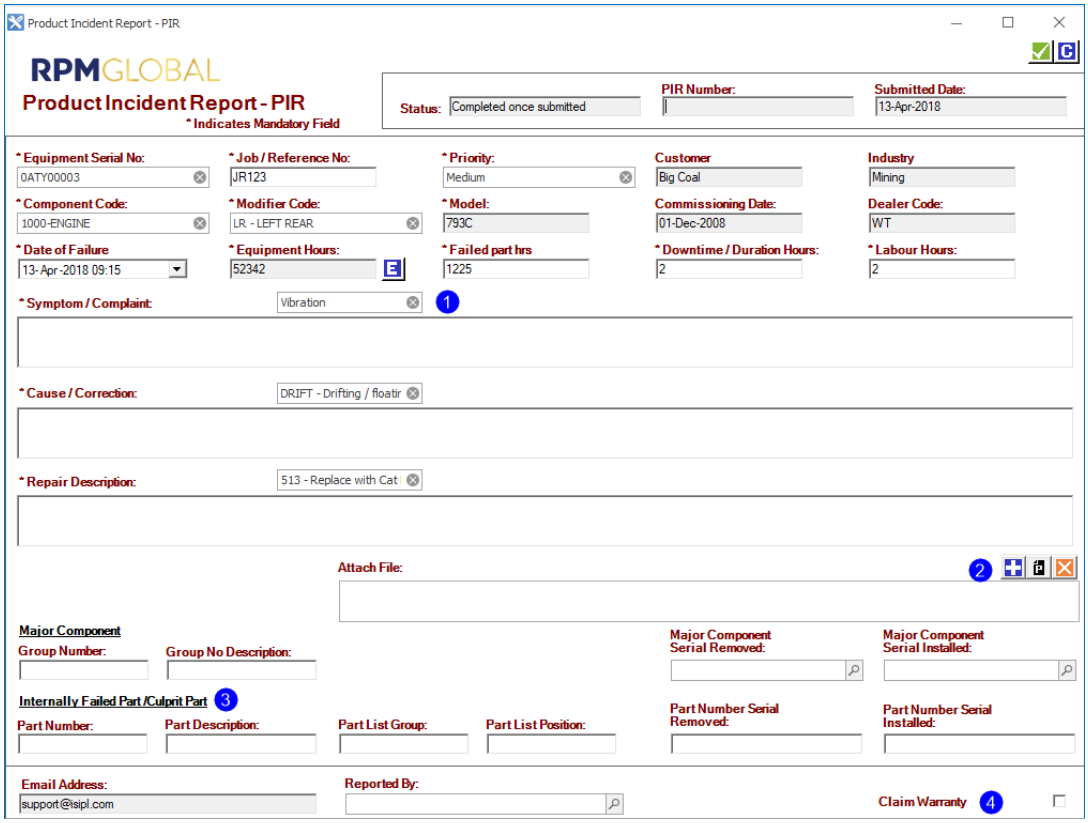
- identify reoccurring failures of specific components on a common model of asset.
- identify reoccurring failures of specific piece parts.
- view potential warranty claims, and create warranty claims for them.
    - The warranty feature of AMT, which is different to the Product Incident Report feature, can identify work orders which are potential warranty through parts usage analysis, which lets such work orders to be reviewed and flagged as warranty claims.
- assess whether a particular customer is experiencing a number of incidents on their asset which may be leading to dissatisfaction.

Navigate to **Daily Operations** > **Product Incident Report Listing**. Select and apply the relevant filters, then double-click a record in the grid to open the Product Incident Report Capture form.

Product incident report capture

AMT has a form which provides a structured process for registering and processing Product Incidents Reports and warranty claims.

Navigate to **Daily Operations** > **Product Incident Report Capture**.



Item	Description
1	**Symptom**, **Cause** and **Repair** are details on why the component failed. You can filter for the closed option, and enter comments.
2	Attach relevant documentation.
3	Enter extra failed part information.
4	Select the **Claim Warranty** check box if you want to claim the incident under warranty.

View Workorder Status

This function provides a view into work orders (current and historical), including:

- Work order status (for example, in progress, complete).
- Authorisation status (authorised or not authorised).
- Who raised the work order and when.
- Planned Downtime and Duration.
- Work Group Responsible.

- Employee Responsible.
- Planned Labour Hours.

The above details can be edited, along with work order Feedback details.

You can interrogate work order history. Right-click on the grid, click **Copy**, and paste into Excel for further analysis.

Timesheet

The timesheet function in AMT has been designed as a simple solution to review and capture the actual hours worked by employees. It is not intended to replace existing payroll or ERP systems.

There are two modes where the timesheet capture process will apply:

1. AMT is used as the work order system – actual hours are entered into the timesheet form. The actual hours will update the work order operation with the actual duration and the work order cost.
2. The timesheet function is used purely for data capture and reporting purposes – actual labour costs will not update the work order as these will come from the payroll or ERP system. The timesheet form will be used purely for entering and reporting on employee work time.

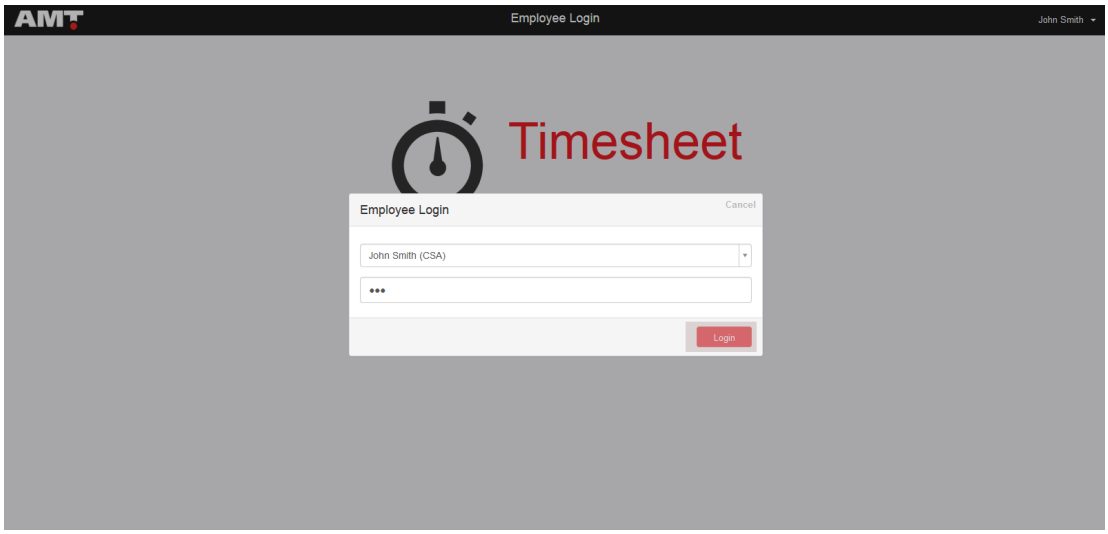
The capturing of employee timesheets is essential for some organisations. This information includes: the productivity, actual hours, and labour rate for each work order.

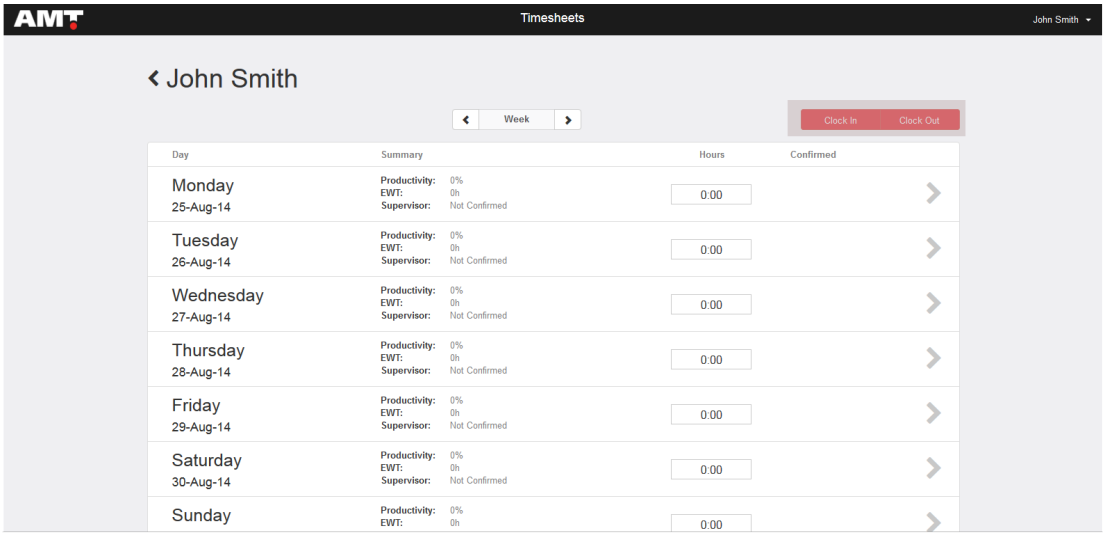
Entering timesheet data via the timesheet form is restricted to the work order operations, which are linked to equipment that you have access to, based on the Branch or Site restrictions of your User account.

Timesheet App Login

This app is designed to give you the ability to enter and review daily timesheets from anywhere.

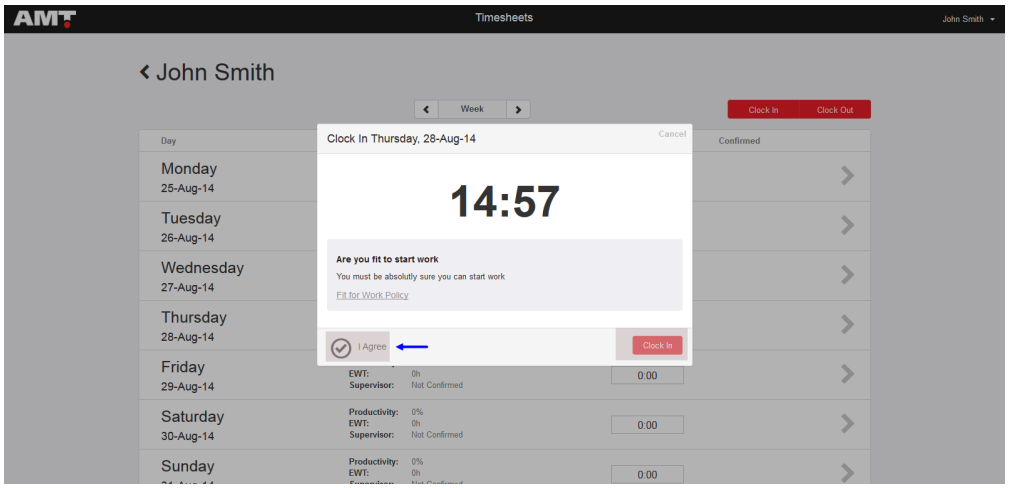
On the **Home Screen**, click the **Timesheet** app. The Employee Login screen displays. Each employee will have a login and password for the Timesheet app, and they must be entered prior to entering any time information. If you are unsure of this information, contact your AMT Administrator.





When an employee commences a shift it is a requirement that they **Clock In** and confirm they are fit for work. They do this by clicking **Clock In** at the top right of the screen.

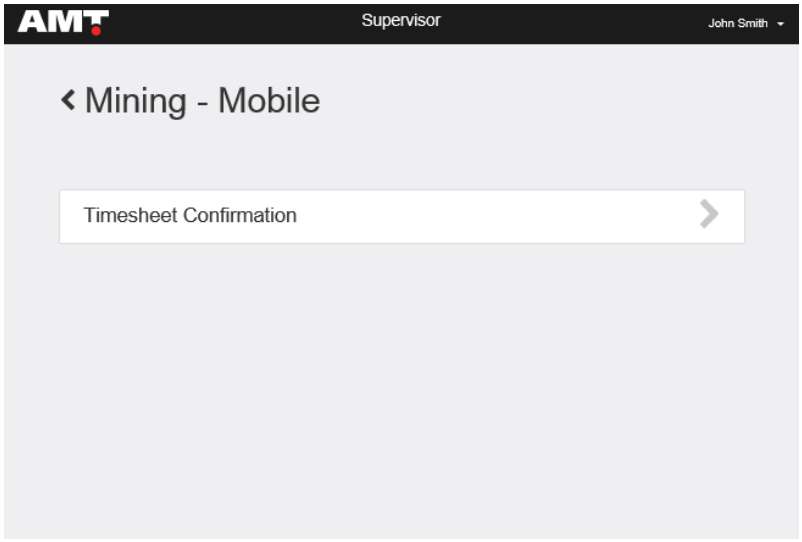
When they click Clock In, another screen displays. showing the Clock In time and a configurable message, which specifies any relevant company policies employees are required to adhere to. For example, in the example screenshot below the employee must click **I agree** to confirm they are fit for work and then click **Clock In**.



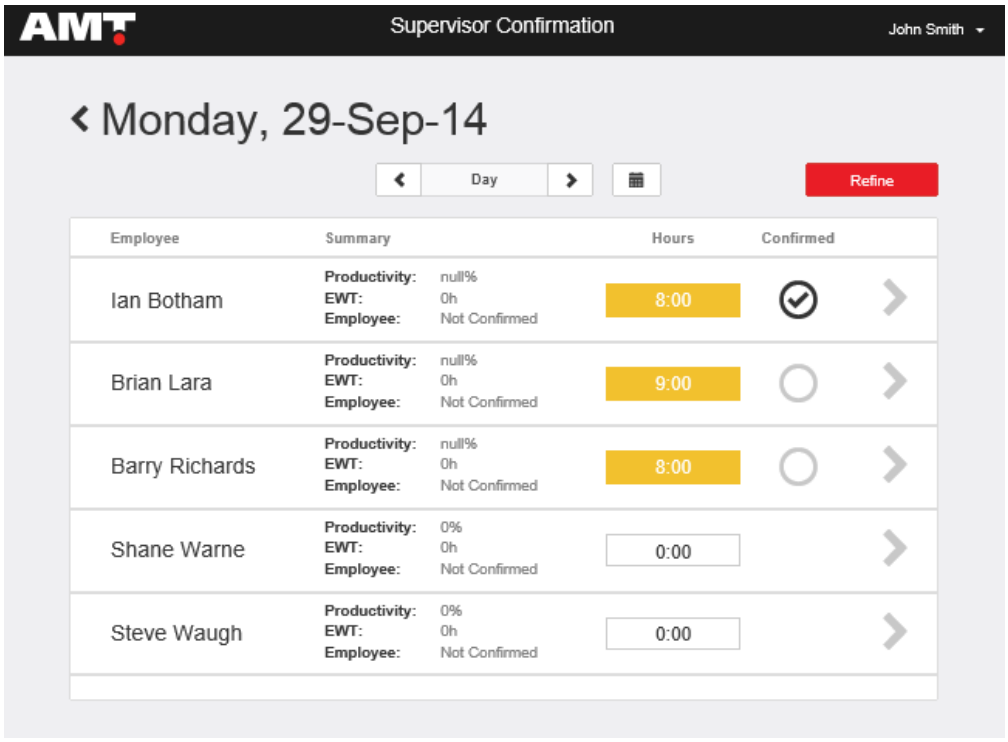
The configurable message is set up by the AMT Administrator.

Timesheet confirmation - Supervisor

Once you have **Selected a Site** to supervise, select **Timesheet Confirmation**.



The following screen displays



This screen lists the employee's time allocations by day for a particular site. This enables you to quickly scroll through the days and confirm employee timesheets by selecting the **Confirmed** circle.

You can search for specific employees by clicking **Refine** at the top right of the screen.

(Rpt_681) Timesheet Cost Analysis

Analyses individual timesheet entries and their associated work orders and strategy tasks.

Tips

This report returns results for real, non-terminated equipment only.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Strategy Task, Workorder, Employee, Supervisor Confirmed

(Rpt_679) Timesheet Journal

Analyses the timesheet entries for specific equipment and cost centres.

Tips

This report will return results for real equipment only.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

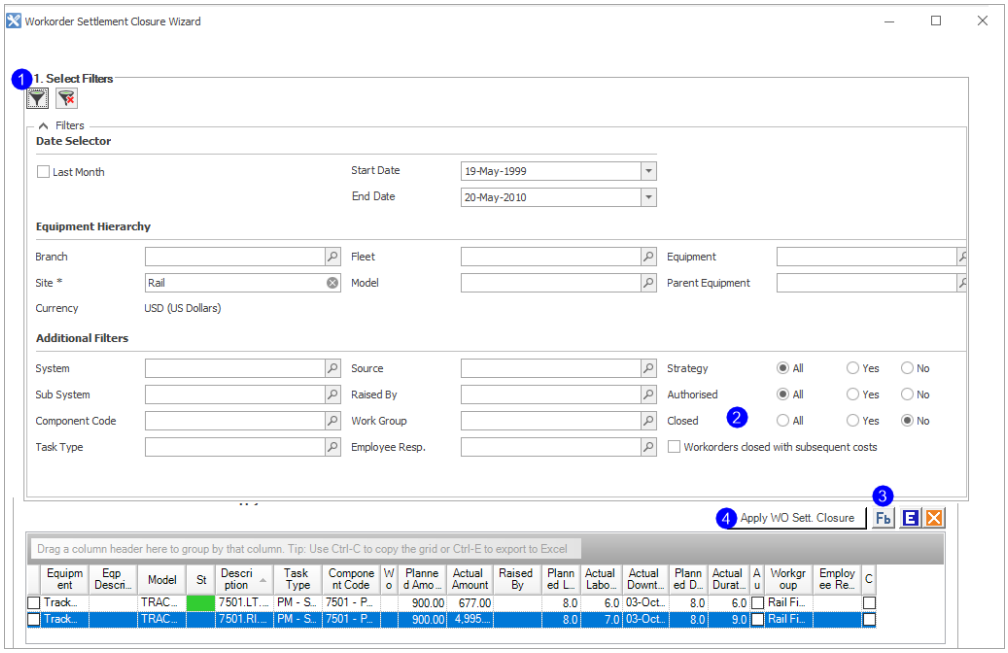
Additional filters

Cost Centre, Workgroup, Employee

Work Order Settlement Closure

If AMT does not interface with an external enterprise resource planning work order system, the supervisor must also close work order settlements.

Navigate to **Daily Operations** > **Workorder Settlement Closure**.



Item	Description
1	Select all required fields and click Apply Filters.
2	Set Closed to No.
3	Highlight work order and click the Feedback icon.
4	Apply the Work Order Settlement Closure.

When you click the **Feedback**, check the values on the **Parts**, **Labour** and **Misc** tabs are correct.

Back on the Work Order Settlement Closure Wizard dialog box, select the work orders to be closed and click **Apply WO Set Closure**.

Notifications

The Notification Control Board is used to filter, view and action notifications raised by AMT Mobile, condition monitoring translate rule triggers, and notifications created manually. You can use filters to view relevant notifications, and then edit or view to see more details of specific notifications. Notification Control Board supports a defined workflow process to manage notifications, which is described below.

Raise a new notification

Like a work order, a notification is assigned to an AMT equipment **Asset** and assigned relevant component, modifier, task type and task counter codes. The notification is given a **Title** to describe it, and can be linked to supporting documents (for example, photos, electronic documents). You can add **Comments**, and assign a

Rating (priority) of 1 to 4 and a status to the notification. Additional electronic documents (including photos) can be linked.

Once created, AMT automatically assigns the notification a unique identification number.

The Counter column indicates that there have been multiple updates to the notification.

Assign and action a notification

An existing notification can be assigned to a role within the organisation (**Assigned To**) or an employee (**Assigned to Person**).

Emails can be sent, showing as from the AMT User Account's email address. The email contains a hyperlink to the notification.

To action the notification, an AMT work order can be generated.

A condition monitoring translation rule can be defined to auto-create work orders. The work order will appear in AMT's Planning module.

Audit a notification

As the status of a notification changes, an audit record is created. The audit record contains the date of the status change, which AMT user made the change, current notification rating level, new notification status, and to which role the notification is currently assigned.

Inspections

The Inspection Control Board provides a place to view outcomes of all inspections performed using AMT Mobile. You are able to view

- the person who performed the inspection, and the date and time it was performed.
- default repair instructions captured by the person completing the inspection.
- attachments, for example, photos from the inspection.
- whether the inspection is a safety or environmental issue.
- The Operational Impact of the inspection item.

It is not the objective of this form to manage the actioning of issues identified by inspections. This is done in the Notification Control Board. All inspections with alerts automatically create Notifications. The following items determine if a notification is automatically created by AMT for an AMT Mobile inspection item.

- Alert level selected by the user on AMT mobile.
- Translation rule applied to an actual measurement capture on AMT Mobile.
- Operational Impact of the inspection item.

If you want to create a notification from an inspection item, it can be done manually by clicking the **Add Notification** button.

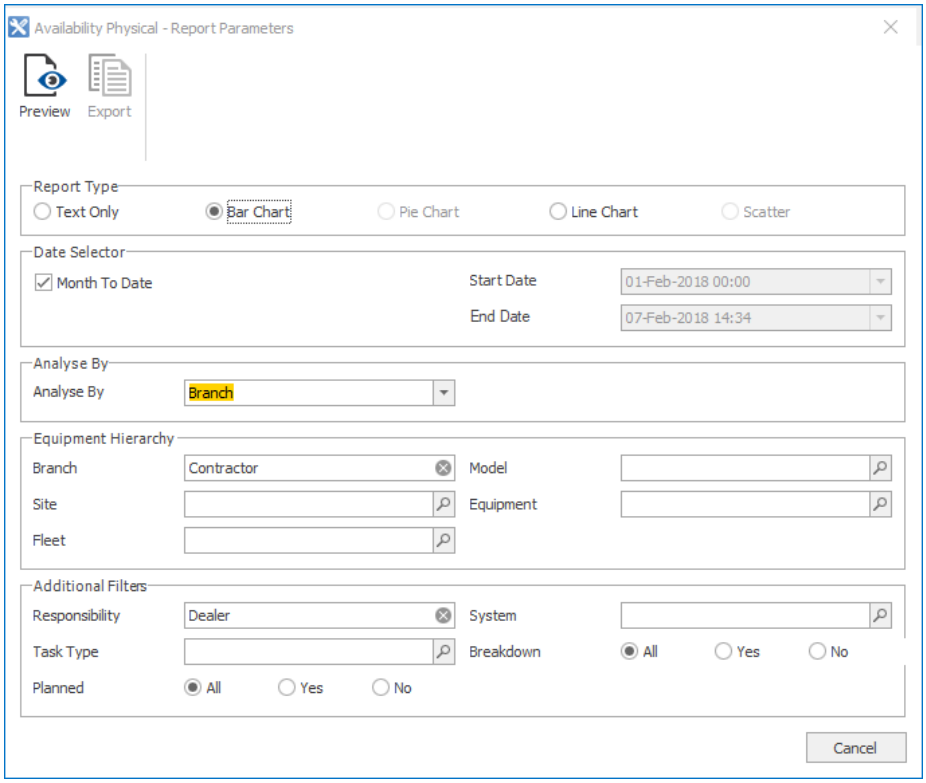
Daily Reports

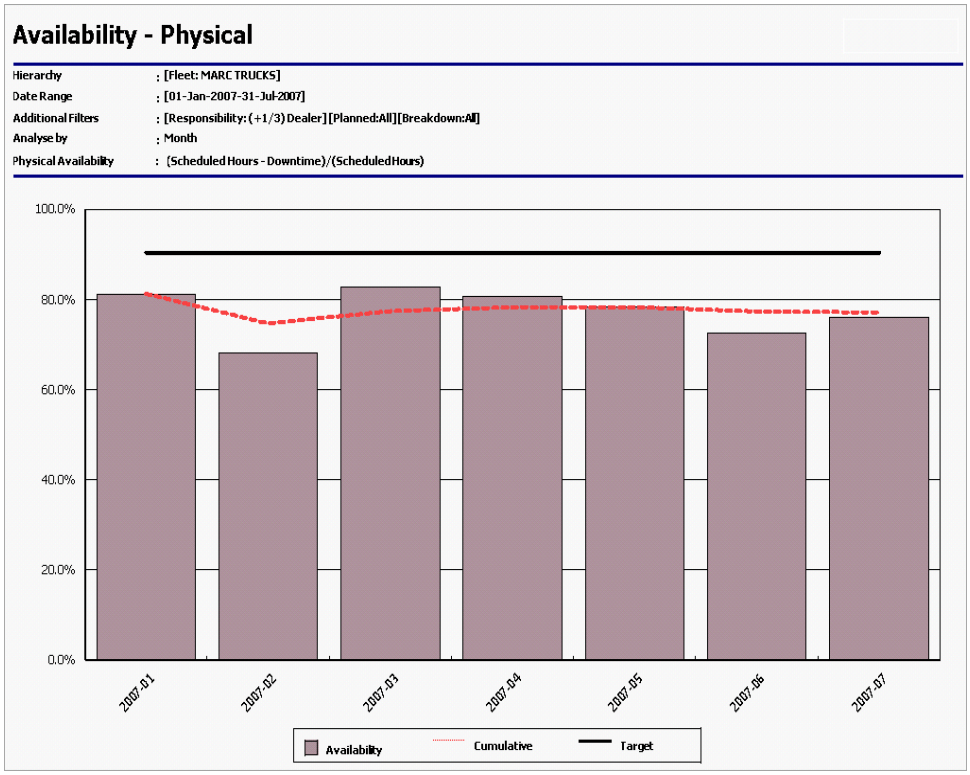
Availability Reports

The type of availability applicable to an asset is set at the fleet level. Fleets can be edited in the **Fleet Administration** screen. The availability formulas in AMT are:

- Physical Availability = *Scheduled Hours - Downtime* / *Scheduled Hours*.
- Mechanical Availability = *Operating Hours* / *Operating Hours + Downtime*.

Navigate to the two reports by clicking **Reporting & Analyses** > **Equipment Performance** > **Availability**. The screenshots below show an example of the **Report Parameters** screen and the report output for the Availability - Physical report. The Availability - Mechanical report screens have the same format.





Some additional reports that show availability in AMT are:

- Snapshot – The Worm: Navigate to **Daily Operations** > **Daily Reports** > **Snapshot - the Worm**.
- Availability Snapshot Worm: **Daily Operations** > **Daily Reports** > **Availability Snapshot Worm**.
- Availability to Target Summary: **Reporting & Analyses** > **Equipment Performance** > **Availability** > **Availability To Target Summary**.

(Rpt_157) Daily Downtime Log

Details the event details of the equipment and the downtime allocation information.

Tips

This report **does** use the site's shift start time for the Date Range set by the user. The Additional Filters only filter the data displayed on the right side of the report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|UserID|,|LogoLocation|

Report parameters

Default.

Additional filters

Task Type, EM Issue, +Responsibility



+ indicates that the Responsibility filter defines the data on the right side of the report.

(Rpt_523) Shift Downtime Log

Contains the event details of the equipment and the downtime allocation information.

Tips

This report is the same as the Daily Downtime Log report, except this report has the ability to select Time and Date, not just Date.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|UserID|,|LogoLocation|

Report parameters

Default.

Additional filters

Task Type, EM Issue, +Responsibility



+ indicates that the Responsibility filter defines the data on the right side of the report.

(Rpt_524) Shift Downtime Summary

Designed to show the primary work order executed on an equipment in a given shift, based on an event. The work order reported will always be the primary work order of the event.

Tips

The report only considers IP and C events which occur within or overlap the selected date range.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Task Type, EM Issue, Cost Type, System, Sub System, Comp Code

(Rpt_158) Daily Log - Detailed

Lets you check the daily events and tasks that have occurred and shows the downtime responsibility.

Tips

This report **does not** use the site's shift start time for the Date Range you set.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, EM Status

(Rpt_159) Daily Log - Summary

Lets you check the daily events and tasks that have occurred and shows the downtime responsibility.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, EM Issue

(Rpt_166) Equipment Down

Shows the pieces of equipment that are currently under maintenance or are down for a particular reason. This data relays directly from the equipment status events and will only show accurately if this data is up to date.

Tips

Ensure that all Equipment Status events are kept up-to-date and that the event status is modified when an event is "In Progress" or "Completed".

Initial parameters

|Con trolType|,|IDist|,|TextList|,|Today|,|UserID|,|AutoLaunch|,
|BranchID|,|SiteID|,|FleetID|, |EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|

Report parameters

Report Type, Equipment Hierarchy.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, EM Issue

(Rpt_160) Daily Machine Performance Check Sheet

This report identifies the maintenance KPIs for the machine throughout a given month and provides the ability to proactively plan to reach targets.

Tips

This report provides a good reconciliation point between the client and the maintenance provider. It highlights the amount of available downtime left in the given period. This lets planners to manage the downtime.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

DTResponsibility+, Task Type, System, Availability Type

+ indicates the Downtime Responsibility filter.

(Rpt_231) Snapshot - The Worm

Provides a graphical representation of actual downtime versus target for a specific time period.

Tips

Run this report for a Month and analyse by Day prior to month end in order to monitor availability targets.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Responsibility, Task Type

(Rpt_232) Snapshot Summary

Summarises Availability, Utilisation and MTBS KPIs into a single report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Responsibility, Task Type

(Rpt_355) Equipment Operational Performance Summary

This report contains Availability, MTBS, MTTR, and MTBF statistics. These KPIs are summarised at group level, and details shown at the analysis level. The responsibility filter determines which events to include in the report, and therefore include in MTBS.

Tips

The report begins at the shift start time on the Start Date selected, and if the End Date is in the past, the report calculates to the end of the End Date's shift. If the End Date selected is today, the report uses the current system time.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,
|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

DTResponsibility +

+ indicates the field is the Downtime Responsibility filter.

(Rpt_198) Monthly Downtime Log

Lets you check the daily events and tasks, and to analyse the downtime allocations throughout the required period of time.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|UserID|,|LogoLocation|

Report parameters

Default.

Additional filters

Task Type, System, Event Comp Code, Responsibility, Variance Cause, Planned, Cost Type, Sub System, Activity, Cause, EM Issue

(Rpt_237) Workorder Details

Lets you select areas of the downtime tasks that they wish to analyse with numerous display and sort fields to select from. A very open report to allow you to customise your required data.

Tips

Date range can be set according to analysis requirements.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|AutoLaunch|,
|BranchID|,|SiteID|,|FleetID|,|EqpPlanIDZ|,|Branch|,|Site|,Fleet|,|EqpPlan|,
||DateChecked|,|StartDate|,|EndDate|,|ISNOTAsDefault|_StrategyTakskOnly|,|
DisplayFields|,|DisplayFieldsLabels|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Modifier Code, Task Type, Task Counter, Status, OHS, Perform At, Raised By, Part Number, Customer Reference, Workorder #, YTS/Backlog, Strategy Tasks, Suppression Impact, Primary WO Only, Durability Index, User WO Status

(Rpt_518) Event Completion Performance

Highlights events which have taken longer than expected to complete by comparing Planned Downtime to Actual Downtime (per event).

Tips

The Main Downtime Variance reason is sourced from the event's downtime allocation records. If there are multiple downtime allocations, the report shows the Variance Cause from the record which is not the primary, and has the largest actual downtime.

Initial parameters

|ControlType|,|Idlist|,|TextList|,|AutoLaunch|,|Today|,|48HoursFromNow|,|UserID|

Report parameters

Default.

Additional filters

Component Code, Task Type, Raised By, Priority, Strategy Task, Event Status, Planned, Strategy Tasks, Include Default Variance Cause, 1 hour ago, Exceptions Only

(Rpt_519) Event Completion Performance SCREEN

Highlights events which have taken longer than expected to complete by comparing Planned Downtime to Actual Downtime (per event).

Tips

The Main Downtime Variance reason is sourced from the event's downtime allocation records. If there are multiple downtime allocations, the report shows the Variance Cause from the record which is not the primary, and has the largest actual downtime.

Initial parameters

|ControlType|,|Idlist|,|TextList|,|AutoLaunch|,|Today|,|48HoursFromNow|,|UserID|

Report parameters

Default.

Additional filters

Component Code, Task Type, Raised By, Priority, Strategy Task, Event Status, Planned, Strategy Tasks, Include Default Variance Cause, 1 hour ago, Exceptions Only

(Rpt_488) Workorder Labour Analysis

Contains Labour data.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Labour Activity, Work Group, WO-Operation, Cost Bearer, Cost Centre, Expense Element, Employee, Week Start

(Rpt_506) Equipment Down Alert

Contains information about all equipment which are down.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, Responsibility, Equipment Location

(Rpt_646) Availability Snapshot Worm

Provides a combined graphical representation of actual downtime versus target and actual availability versus target. It illustrates the impact of downtime on availability, and highlights when each is trending towards falling short of targets, so to trigger preventative, proactive actions.

Tips

Similar in logic to the Snapshot - The Worm report but adds availability actuals versus target to the analysis.

If actual downtime exceeds, or is trending towards, alert (that is, allowable downtime while still achieving availability target), you need to consider what measures can be put in place to reduce future downtime so the alert is not exceeded.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Availability Type, Task Type, Responsibility

(Rpt_665) Workorder Duration Analysis

Displays the planned and actual duration of work orders for the selected date range and indicates their Planned vs Actual Variance.

Tips

Sort by Workorder Duration Variance to identify the work orders with the highest or lowest variances.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Component Code, System, Sub System, Task Type, Sort By

(Rpt_764) Inspection Completion Time

Shows details about the last inspection done on an equipment,

Tips

Displays the of the inspection, inspection type, as well as the employee responsible for the inspection.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Employee

(Rpt_779) Planned Workorder Details

Lets you select areas of the downtime tasks that you wish to analyse with numerous display and sort fields to select from. A very open report, which lets you customise your required data.

Tips

Date range can be set according to analysis requirements.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|QUOMID|,|QUOM|,|
UserID|,|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,
|Site|,|Fleet|,|EqpPlan|,|DateChecked|,|StartDate|,|EndDate|,
|ISNOTAsDefault_StrategyTasksOnly|,|DisplayFields|,|DisplayFieldsLabels|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Serial Number, Modifier, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

(Rpt_780) Extended Planned Workorder Details

Lets you select areas of the downtime tasks that you wish to analyse with numerous display and sort fields to select from. A very open report, which lets you customise your required data.

Tips

Date range can be set according to analysis requirements.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Serial Number, Modifier, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

(Rpt_830) User Work Order Status Audit Report

Provides a history of the user work order status for a work order. The report shows who changed what (Login Name of user who made the change), and when, giving a view of the lifecycle of the user-defined status of the work order.

Initial parameters

|ControlType|,|Idlist|,|TextList|,|LogoLocation|,|UserID|,|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

User, User WO Status

Timesheet Reports

(Rpt_679) Timesheet Journal

Analyses the timesheet entries for specific equipment and cost centres.

Tips

This report will return results for real equipment only.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Cost Centre, Workgroup, Employee

(Rpt_680) Employee Time Validator

Useful to validate the amount of time an employee has worked over a period of time.

Tips

This report returns results for non-terminated employees only.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

None

(Rpt_681) Timesheet Cost Analysis

Analyses individual timesheet entries and their associated work orders and strategy tasks.

Tips

This report returns results for real, non-terminated equipment only.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Strategy Task, Workorder, Employee, Supervisor Confirmed

Usages & Measurements

Usages, measurements and production statistics

The capturing of vital asset information is a key component in the asset management process. This information can range from idle time, operating hours, to cycles used, litres consumed, kilometres travelled, tonnes moved. The collection of this data is not only important for an operational perspective in measuring performance but is equally critical in management of the asset strategy from a maintenance standpoint. There are three main areas where information can be maintained in AMT:

- Actual Usages - can be maintained manually or via an interface.
- Measurement - can be maintained manually or via an interface.
- Production Statistics.

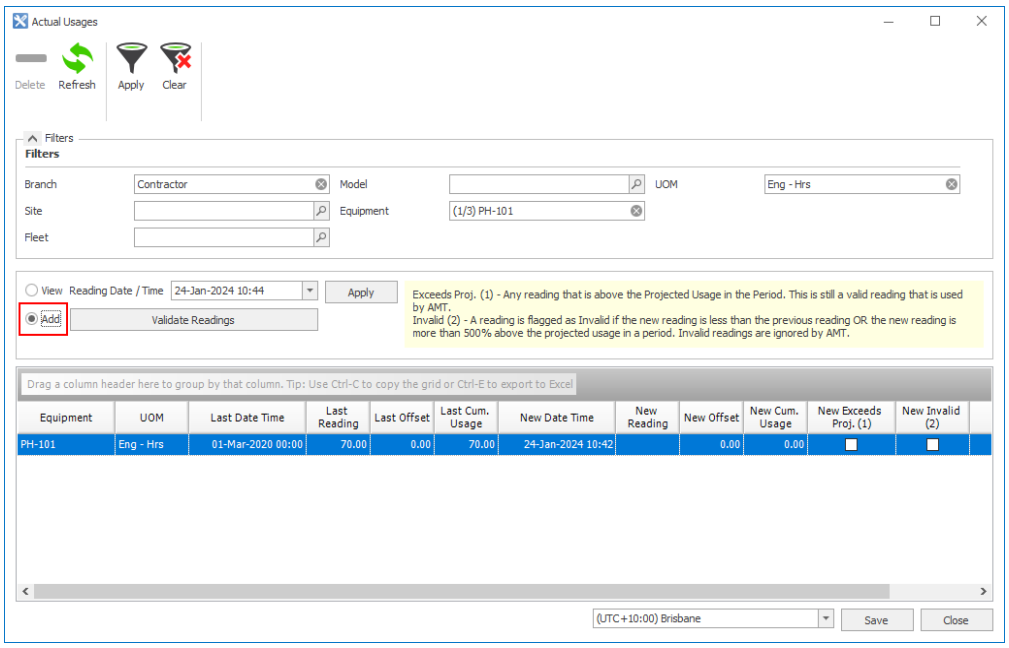
Record actual usages

Actual Usage readings, for example, engine hours, travel time, test time, can be received by AMT through an automatic interface such as Dispatch or manual data entry by a user. Usages entered into the system cannot be deleted. Usage readings are cumulative and any error in the past should be corrected by future readings. AMT has a number of validation steps that are applied before recognising whether the usage entered is legitimate.

AMT calculates usages for an equipment (based on the equipment's usage profile) until a valid usage is entered, at which point AMT will adjust usages. AMT calculates usages to keep planning schedules up to date. Any subsequent adjustment, after a valid usage has been entered, also adjusts schedules. This approach is less risky than relying solely on data input, which could be invalid or received sporadically, resulting in schedules being inaccurate relative to the equipment's life cycle, or changing dramatically.

Navigate to **Daily Operations** > **Usages & Measurements** > **Record Actual Usages**.

Select the filters and click **Add** to apply them.



Grid Fields

Field	Description
Equipment	The equipment name.
UOM	The unit of measure selected in the filters.
Date Time	The date and time of the last actual usage reading.
Reading	The last actual usage reading (entered manually or automatically using an interface).
Offset	The last actual offset reading entered manually into AMT.
Cumulative Usage	The last actual usage reading plus last offset.
Exceeds Proj (1)	The reading above the projected usage in the period to a maximum of 500%. This is still a valid reading used by AMT.
Invalid (2)	The reading is flagged as invalid if the new reading is less than the previous reading OR the new reading is more than 500% above the projected usage in a period. Invalid readings are ignored by AMT.
Manual Entry	The usage was entered manually, not imported.
Invalid Reason	The reason for an invalid record. See below for more information.

Invalid Reason column

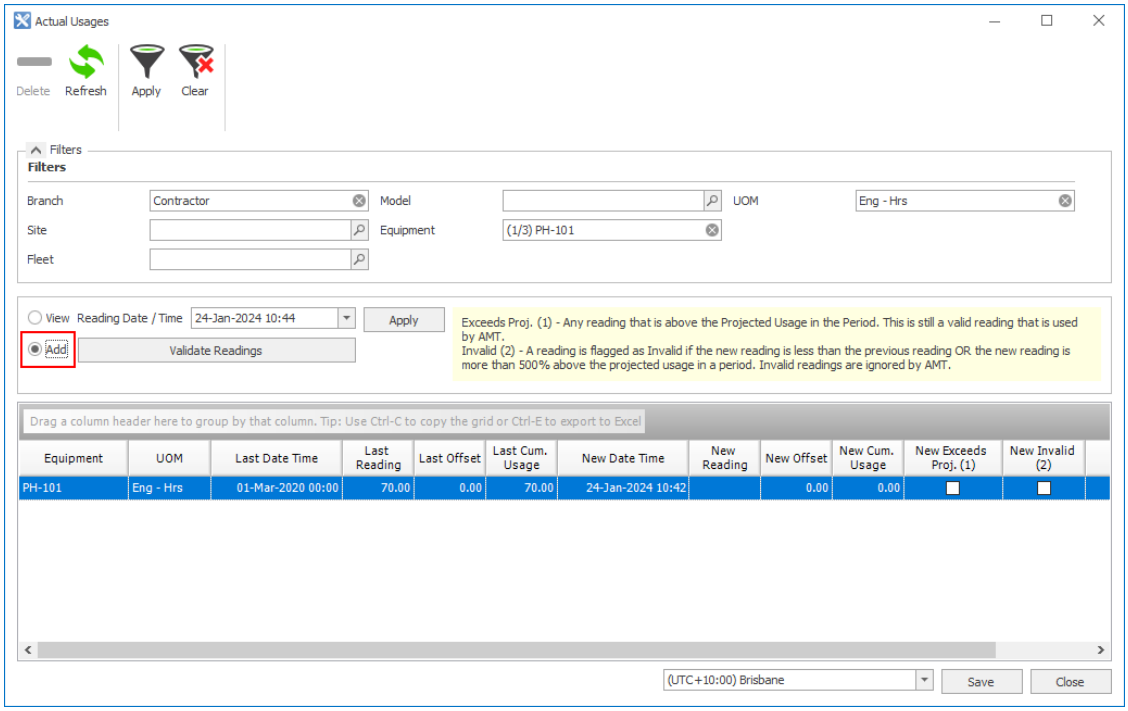
The Invalid Reason column displays a message explaining the reason for an Invalid Record.

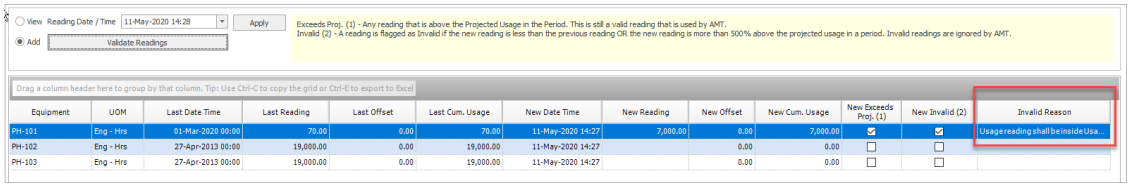
This gives you an opportunity to review the reason and make any necessary adjustments to correct if it results from a data entry issue.

The column will be populated on **Save** or **Validate** when manually creating records. The column will not populate as you manually update multiple records in the one screen, as the database record isn't saved and logic triggered until the screen information is saved.

When the AMT Client is not used to load records and the information is received from an external source, records will be processed through the database routine and therefore any invalid entry reasons are visible in **View** mode.

The new column exists at the end of the data entry and review screens.





Messages received for Invalid Records are as follows:

- Usage reading exceeds the maximum allowable growth tolerance (xxx%)
- Usage reading is less than the previous reading (Reading - Reading date)
- Usage reading exceeds the end usage (xxx)
- Usage reading is greater than the next existing usage reading (reading - reading date)
- Duplicate usage reading [Existing reading: Equipment = "xx", QUOM = "xx", ReadingDate = "xx"]

Actual usages

As part of the Work Finalisation process, typically the actual usages of an asset will be updated on the work scopes. Before recording actual usages you must ensure there is at least one Unit of Measure (UOM) marked as Operating Hours in the Units of Measure system table - **System Administration** > **Usages & Measurements** > **Units of Measure**.

To record actual usages, navigate to **Daily Operations** > **Usages & Measurements** > **Record Actual Usages**.

You can right-click on the grid and copy the data into a spreadsheet program. In the spreadsheet you can complete the **New Date Time**, **New Reading** and **New Offset** values then past the full grid back into AMT.

Record and validate usage readings

It is important that usages are current for assets so that they are as accurate as possible for the long-term planning review of tasks.

New usage readings meters can be managed in AMT by means of using the offset. Occasionally the meter reader is changed or damaged on an asset and the counter needs to be reset and started again. When this happens, you set the offset to be equal to the last reading prior to replacement/reset (for example, 34,238 engine hours). The new reading will then be zero.

Navigate to **Daily Operations** > **Usages & Measurements** > **Record Actual Usages**.

s

Update actual usage

The actual usages of equipment should be maintained and updated as often as possible, it is recommended at least once a month. By keeping the usage readings current, the forecast accuracy will be maintained. The actual usage can either be imported via an interface or manually updated within AMT.

For details of the data required by the interface, and the format of the interface file, contact the AMT Administrator and request the interface specification document.

To manually update the actual usages, navigate to **Daily Operations** > **Usages & Measurements** > **Record Actual Usages**.

Utilisation Analyser

This function allows you to compare Actual Utilisation with Forecast Utilisation and quickly make adjustments to Forecast Utilisation.

If an equipment is not utilised at the same rate as the Annual Utilisation (Simple Util Eqp) or the Usage Profile (Advanced Util Eqp), the Strategy Task Target Date may not be an accurate indication of when the Strategy Task should be performed. Utilisation Analyser is designed to identify equipment where Actual Utilisation differs from the Forecast Utilisation and allows you to quickly update Forecast Utilisation.

You can analyse and adjust equipment configured for Simple Utilisation, which means changes are applied to selected, individual equipment.

You can adjust Usage Profiles, which are linked to equipment configured for Advanced Utilisation. This means changes made will update the selected usage profiles, which in turn update the usage forecast of equipment linked to those usage profiles.



When changes are made to Usage Profiles, the change is only applied to the current usage profile step only, not future steps.

Utilisation filters

- **Days to Analyse Actuals** - how, back in history from today, should actual usages records be included in the analysis.
- **Variance Greater Than** - only identify variances of actual usage history versus future usage forecast greater than a specific usage amount.
- **Variance% Greater Than** - only identify variances of actual usage history vs future usage forecast greater than a specific percentage.

The calculation AMT performs is similar for **Simple Utilisation** and **Advanced Utilisation**. AMT first calculates the Daily Average Actual Usage based on the **Days to Analyse Actuals** filter. This daily rate is used to calculate the Annual Average Actual Usage. The Annual Average Actual Usage rate is then compared to the equipment's current Annual Forecast Usage (Simple Utilisation) or usage profile's Current Annual Forecast Usage (Advanced utilisation). The usage variance amount is displayed in the Annual Variance column. This is where the **Variance Greater Than** parameter filters results. The usage variance percentage is displayed in the Annual Variance% column. This is where the **Variance% Greater Than** parameter filters results.

The **Simple Utilisation** > **Set Annual Utilisation To**, or the **Advanced Utilisation** > **Set Annual Forecast Usage To** fields are automatically populated with the value from the **Annual Average Actual Usage** field, implying that you may want to change the forecast usage rate to align to the historical annual usage pattern. Of course, you may not wish to make a change at all, or set a different value. Either way, if a change it to be made, select the **Select** check box, adjust the **Set Annual Forecast Usage/Utilisation To** value as needed, then click **Apply Changes** to update the Equipment or Usages Profiles.



To see which equipment are linked to a usage profile prior to making a change, on the Advanced Utilisation tab, highlight the Usage Profile row, click **Actions** and select **Equipment List**.

(Rpt_413) Fleet Statistics

Shows the latest actual usage readings for equipment.

Tips

This report is only applicable to Real equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM

(Rpt_390) Equipment Statistics

A summary report on the measurements and usages set up against a piece of equipment from the Condition Based Maintenance module.

Tips

This report is only applicable to Real equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|AutoLaunch|,|EqpPlan|,|EqpPlanID|,|UserID|

Report parameters

Default.

Additional filters

Branch, Site, Fleet, Model, Equipment

(Rpt_118) Actual Usage Details

Details the units of measure entered for a particular equipment, site, branch and so on. Shows both actual and cumulative readings, for example, engine hours, fuel litres, travel hours. **Reporting & Analyses** > **Equipment Performance** > **Utilisation** or **Daily Operations** > **Usages & Measurements**.

Initial parameters

|ControlType|, |IDlist|, |TextList|, |ProjectionTypes|, |UserID|, |AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|, |DateChecked|,|StartDate|,|EndDate|,|AnalyseBy|

Report parameters

Default.

Additional filters

None.

(Rpt_391) Trending

Report shows the element trending leading up to a component failure.

Tips

This report will return results for Real, non-terminated equipment only.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

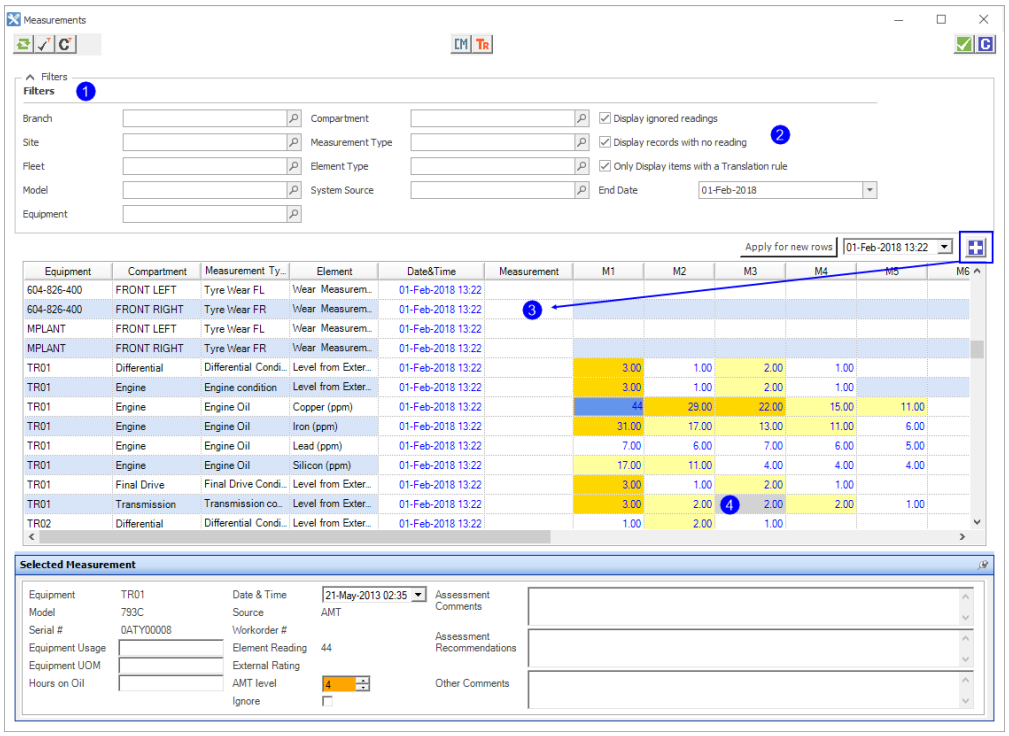
Additional filters

Branch, Site, Fleet, Model, Equipment, Division, Parent Equipment, Eqp Location, Eqp Category, Eqp Cost Centre, Cost Responsibility, Element Type, Element UOM

Record Actual Measurement Readings

Navigate to **Daily Operations** > **Usages & Measurements** > **Record Actual Measurement Readings**.

Measurements dialog box overview



Icons

Icon	Description
CM	Opens the CM Translation Rules dialog box.
TR	Opens the Standard Report - Trending.

Fields and Columns

Refer to the numbers on the screenshot above.

Number	Field	Description
1	Filters.	Select filters to apply on the appropriate fields.
2	Display ignored readings.	Ignored readings are coloured grey, indicating they are out of range or do not meet the CM translation rule.
	Display records with no reading.	Select if new measurement and has no previous readings.

Number	Field	Description
	Only display items with a translation rule.	Items with translation rules.
3	Measurement.	Click **Add** to add a new Measurement column.
4	Ignored measurement.	Cells are grey when the **Display ignored readings** check box is selected.

Analyse actual measurements

Having measurements in AMT gives you the ability to trend

- Consumption - per unit of measure.
- Consumption - per month.

and report the latest readings:

- Fleet statistics.
- Equipment statistics.

Usage validation

Validation is as follows:

- Invalid (Scenario 1): Check if the new reading is lower than the previous reading.

  AMT does not allow an entry of this type for manual readings, but this is possible if the readings are imported.

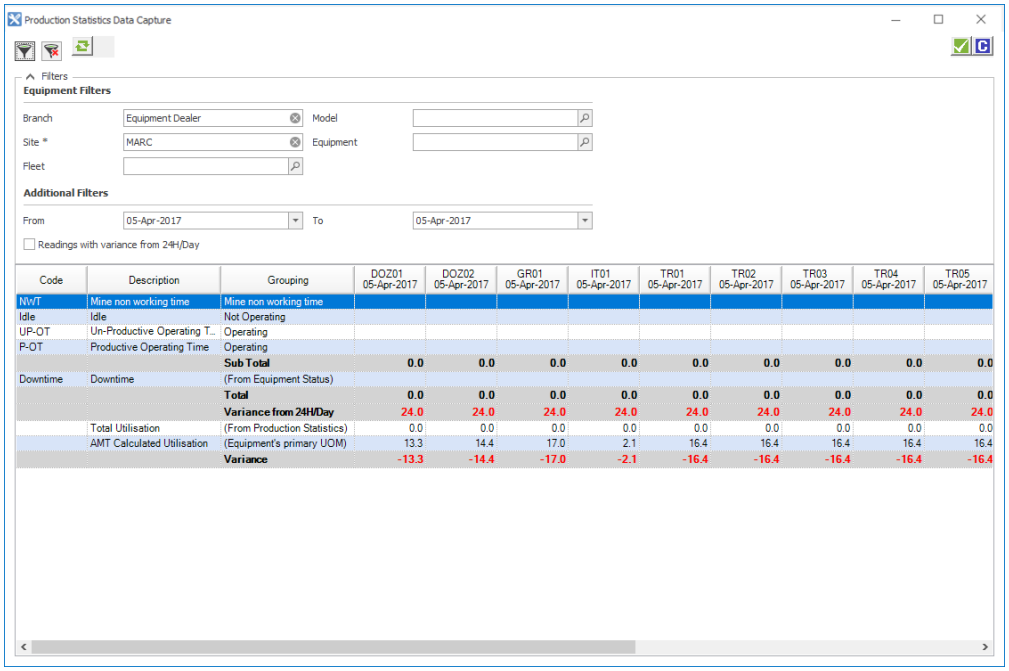
- Invalid (Scenario 2): Check if the new reading is more than the projected usage in the period, and exceeds the threshold by more than the 500% threshold.
- Exceeds Projected Usage: Check if the new reading is more than the projected usage in the period, but no more than the 500% threshold.

Occasionally the meter reader is changed or damaged on an asset and the counter needs to be reset and started again. When this happens, the offset is set to be equal to the last reading prior to replacement or reset. The new reading might then be equal to zero.

Production statistics data capture

Refer to Production Statistics for general information.

To record Production Statistics, navigate to **Daily Operations** > **Usages and Measurements** > **Production Statistics Data Capture**. Filter for the relevant Site and Assets, then click on a grouping to add a new record.



Code	Description	Grouping	DOZ01 05-Apr-2017	DOZ02 05-Apr-2017	GR01 05-Apr-2017	IT01 05-Apr-2017	TR01 05-Apr-2017	TR02 05-Apr-2017	TR03 05-Apr-2017	TR04 05-Apr-2017	TR05 05-Apr-2017
NWT	Mine non working time	Mine non working time									
Idle	Idle	Not Operating									
UP-OT	Un-Productive Operating T...	Operating									
P-OT	Productive Operating Time	Operating									
		Sub Total	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Downtime	Downtime	(From Equipment Status)									
		Total	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
		Variance from 24H/Day	**24.0**	**24.0**	**24.0**	**24.0**	**24.0**	**24.0**	**24.0**	**24.0**	**24.0**
	Total Utilisation	(From Production Statistics)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	AMT Calculated Utilisation	(Equipment's primary UOM)	13.3	14.4	17.0	2.1	16.4	16.4	16.4	16.4	16.4
		Variance	**-13.3**	**-14.4**	**-17.0**	**-2.1**	**-16.4**	**-16.4**	**-16.4**	**-16.4**	**-16.4**

The maintenance **Downtime** is showing null values, which indicates there is no event in AMT and no downtime has been captured for the completed work. This may occur on site if the production statistics data is captured on one form, and the downtime is captured on a shiftlog or breakdown form. It is also possible that the asset wasn't down for any scheduled or unscheduled maintenance during that period.

- The **Variance** shows the difference between the total productive and non-productive utilisation in 24 hours.
- **Total Utilisation (From production statistics)** shows productive or non-productive utilisation which is based on the grouping.
- **AMT Utilisation** is based on the asset's primary unit of measure, for example, engine hours.

Once the downtime record has been updated in AMT, this figure is reflected in production statistics. If a mistake is made, you can correct the entry be editing the hours with a zero value then saving.

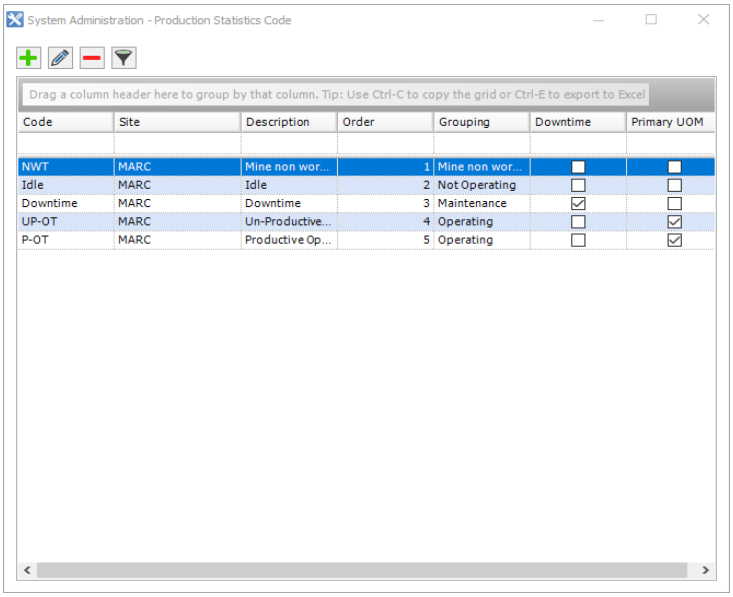
Data in the grid can be copied and pasted into a spreadsheet program, edited and pasted back into AMT.

Production Statistics Code

The Production Statistics function in AMT has been designed to capture productive and non-productive statistics for operational equipment.

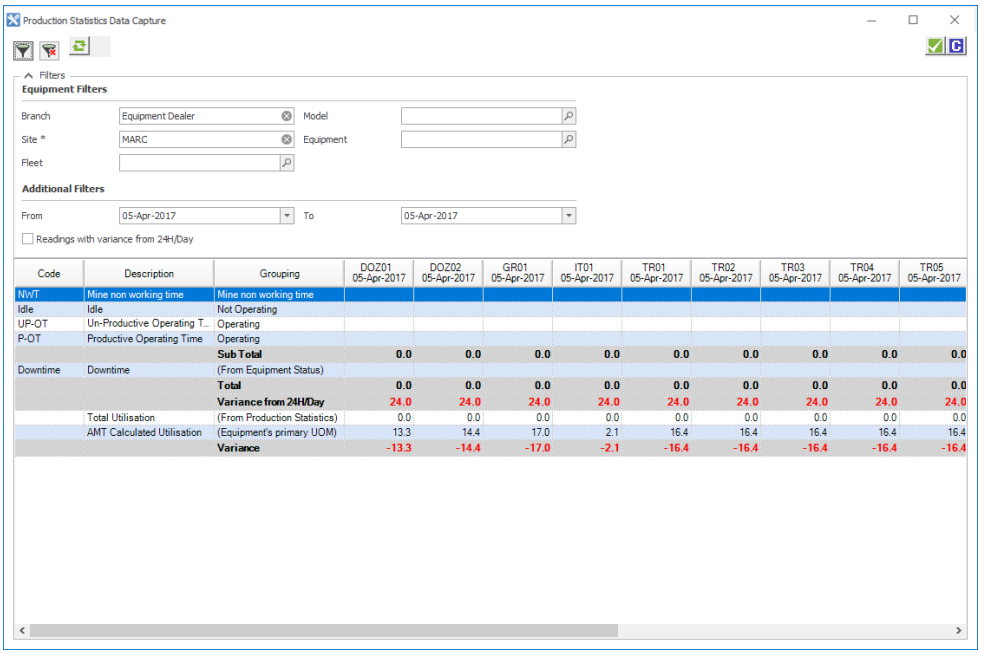
While the actual downtime for equipment is sourced from the Equipment Status or the Shiftlog, the intention is to capture the productivity detail for a machine to understand and report on the utilisation of equipment.

- Actual productive and non-productive statistics codes are entered into the Production Statistics Code system table by administration staff. Navigate to **System Administration** > **System Tables** > **Usages & Measurements**.



- A Production Downtime Code is added to the production statistics.
- The production statistics utilisation rate is compared to AMT's utilisation which is based on the primary UOM (for example, engine hrs).
- Reports are generated which enable a better understanding of asset utilisation. The reports assist management to understand problem areas and enable corrective steps.

- To capture production statistics data, navigate to **Daily Operations** > **Usages & Measurements** > **Production Statistics Data Capture**.



Code	Description	Grouping	DOZ01 05-Apr-2017	DOZ02 05-Apr-2017	GR01 05-Apr-2017	IT01 05-Apr-2017	TR01 05-Apr-2017	TR02 05-Apr-2017	TR03 05-Apr-2017	TR04 05-Apr-2017	TR05 05-Apr-2017
NWT	Mine non working time	Mine non working time									
Idle	Idle	Not Operating									
UP-OT	Un-Productive Operating T...	Operating									
P-OT	Productive Operating Time	Operating									
		Sub Total	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
Downtime	Downtime	(From Equipment Status)									
		Total	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
		Variance from 24H/Day	**24.0**	**24.0**	**24.0**	**24.0**	**24.0**	**24.0**	**24.0**	**24.0**	**24.0**
	Total Utilisation	(From Production Statistics)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	AMT Calculated Utilisation	(Equipment's primary UOM)	13.3	14.4	17.0	2.1	16.4	16.4	16.4	16.4	16.4
		Variance	**-13.3**	**-14.4**	**-17.0**	**-2.1**	**-16.4**	**-16.4**	**-16.4**	**-16.4**	**-16.4**

Production Statistics Reports

The core reports for production statistics are:

- Production Statistics Analysis
- Production Statistics Chart

These reports are located under **Reporting & Analyses** > **Equipment Performance** > **Production Analysis**.

The reports have a similar layout and fields.

When generating production statistics reports:

- Group 1 shows the production statistics grouping (for example, Idle).
- Group 2 shows the production statistics description (for example, Crib Break).
- Group 3 is only applicable for maintenance downtime, if there are no downtime events for the assets, no records will display.

  When a downtime event exists, the downtime activity code will be shown on the report.

The Production Statistics Chart helps management to better understand asset utilisation. The report displays the percentage of productive and non-productive utilisation, the downtime, and non-working time for assets.

Workorder Cost Analysis

Workorder Settlement Manager

This function is designed to allow work order settlements to be reviewed, validated, merged, and in some cases added and edited.

In Workorder Settlement Manager, you can

- Use a wide variety of filters to search for sets of work order settlements.
- Edit certain details of a work order settlement imported into AMT via an interface.
- Merge two or more work order settlements.
- Mark work order settlements as reviewed and checked.
- Add internal AMT manual work order settlements.

Special filters

Filter	Description
Linked to Equipment	Used to find work order settlements which have been imported into AMT from an external system, but the data may contain incorrect equipment information (for example, an incorrect serial number). Therefore, AMT will not be able to link the work order settlement record to an equipment.
Strategy	Include or exclude work order settlements that are linked to strategy tasks.
Job Status	Indicates whether the maintenance (the costs of which the work order settlement is capturing the costs of) is Yet to Start, In Progress or Completed.
Linked to work order	Include or exclude work order settlements linked to work orders in Equipment Status.
Before Equip Start	Include or exclude work order settlements which have a Job Date prior to the Start Date of the equipment.
Maintenance Order Type (MOT)	Select multiple values, for example Breakdown, Corrective Maintenance, Preventative Maintenance.
AMT WO	Include or exclude work order settlements manually created in AMT.

Grid fields

Field	Description
Checked	Indicates whether someone has edited any details of the work order settlement in AMT. If selected, the work order settlement's codes will not be changed when the work order settlement is next imported from the enterprise resource planning system.
Serial Number	Serial Number on the work order settlement.
Equipment	If the work order settlement is linked to an equipment, this column will show the Equipment Name, otherwise it will be blank.
Work Order	This is the combination of work order settlement Number + Task Number.

Field	Description
	Only parent work order settlements are shown in the work order settlement manager table.
WIP	If selected, this indicates the work order settlement is Work In Progress. If unselected, this means that the work order settlement is closed.
Current Projection	If the work order settlement is linked to a strategy task (the Strategy check box is selected), the estimated cost of the strategy task per occurrence, from the equipment's current projection, is shown. If the work order settlement is not linked to a strategy task (the Strategy check box is cleared), the estimated cost per occurrence of the unassigned subsystem (to which the work order settlement's component code is assigned is shown).
Contract Total	Work Order Settlement's total cost based on the equipment's default cost bearer. For equipment under maintenance contract, this cost is shown at Sell.
WO Total	Work Order Settlement's total cost (regardless of cost bearer). For equipment under maintenance contract, this cost is shown at Sell.
Component	Shows the Work Order Settlement's codes plus component code description.
Description	Description of the work order settlement from the interface file.
Occ Type	Occurrence Type.
Job Code	Job Code.
Job Status	Indicates whether the maintenance (the costs of which the work order settlement is capturing the costs of) is Yet to Start, In Progress or Completed.
CCC	Identifies whether the work order settlement is flagged as End Of Life, Partial Life, Swing In or Swing Out (these indicators are set on the work order in Equipment Status).
Life Conf	Life achieved by the component. Normally this is calculated automatically by AMT but you can overwrite it on the work order, which is linked to a work order settlement, in Equipment Status.
Date	This is the Job Date that represents the date the actual maintenance was performed.
Usage	Usage on the equipment at the Job Date

Field	Description
Comp Life	Usage difference between when the work order settlement was last performed and the usage when this current work order settlement was performed. Can be thought of as Component Life.
Strategy	This box is selected if the work order settlement's codes match that of a strategy task in the equipment's current projection.
Linked	This box is selected if the work order settlement is linked to a work order in the Equipment Status module of AMT.
MOT	If the work order settlement was created manually by the AMT user, the MOT code can be edited on the settlement detail screen. If the settlement is created from an external system, the MOT code will be read only. If the work order settlement is linked to an AMT planning work order, the MOT code will be read only and will come from the linked planning work order.

Work order settlements

The Work Order Settlement Interface process is critical to AMT. AMT requires work order settlements for:

1. recognising the occurrence of strategy tasks, for example, engine changeout, preventive maintenance services.
2. understanding what costs have been incurred by assets.
3. determining what has been charged to the contract.
4. analysing actual cost vs budget vs forecast.

Settlements can be manually created in AMT or sourced from an external system when the external system manages the "Work" process and collects costs.

The Work Order Settlement interface is used to automate the process of creating and updating AMT work order settlements with data from an external system.

Without reliable and accurate settlement data, AMT will not recognise the occurrence of the Strategy Task and will not allocate the costs to correct budget lines. This will in turn impact the reliability and accuracy of the component, parts, labour and financial forecasts produced by AMT.

If AMT does not receive work order settlements, or the work order settlement's codes do not match or translate to the codes of the strategy task in the current projection of the asset, AMT will not:

1. recognise the occurrence of the strategy task (that is, new life) and therefore it will be overdue.
2. project the next occurrence of the strategy task.
3. allocate the costs on the work order settlement to the correct budget line, instead the costs will be allocated into the "unassigned" bucket.

All of the above impact the reliability and accuracy of the component, parts, labour and financial forecast produced by AMT. Therefore, it is critical that a structured word order settlement data validation process is implemented, and is conducted once a week by someone responsible on site.

AMT maintains an equipment strategy that relies on a work order settlement system. Work order settlements are imported into AMT from a work order settlement system to provide:

1. the financial history of work performed on assets.
2. task occurrence data that is used to adjust asset strategies, and update projections and forecasts.

Manual work order settlement entry

Navigate to Work Order Settlement Manager by clicking either **Daily Operations** > **Workorder Cost Analysis** > **Workorder Settlement Manager** or **Reporting & Analyses** > **Costs** > **Costs** > **Workorder Settlement Manager**.

By clicking  at the bottom of the form, you can manually create a work order settlement. All work order settlements created in AMT are prefixed by **AMT**, and it is recommended that the work order settlement number you enter is traceable, so that it maintains the audit trail to the original work order settlement number. You must identify the:

1. **Equipment** to which the work order settlement will be charged.
2. **Date** the work order settlement was opened.
3. **Codes** for the work order settlement. If the work order settlement relates to a strategy task in the current projection, you should select the task from the list as this ensures the work order settlement links correctly.
4. **Cost Bearer**.
5. **Total Costs**, and if applicable **Sell**, for **Parts**, **Labour** and **Misc**.

After you save the work order settlement, AMT allocates the costs to the appropriate line (strategy task) in the current projection, and if applicable, forecasts the next occurrence.

Even though the costs are now in AMT, you should link the work order settlement to the work order in **Equipment Status** (**Daily Operations** > **Equipment Status**). This ensures AMT correctly forecasts the next occurrence from when the work order actually occurred (which is indicated by the downtime held in equipment status). Refer to the section on Linking Work Order Settlements.

Sources for work order settlements

AMT receives work order settlements from an external enterprise resource planning system. For data cleansing purposes, you can also manually create AMT work order settlements.

1. **Work order**: Is a specific maintenance job performed on an asset, which can be planned or executed as a result of a breakdown. It can represent a strategy task, backlog or defect. A work order settlement can link to a work order in order to provide the actual cost of performing the work order.
2. **Work order settlement**: In a transaction system, (for example, SAP, DBS) all actual costs (parts, labour, miscellaneous) are captured on a work order settlement.

Effects of poor work order settlement management in AMT

The effects of poor work order settlement management in AMT include the following.

1. Bow-wave of costs.
2. Unreliability of maintenance accrual and profit recognition reports.
3. Incorrect parts forecasts.
4. Incorrect components/rotables forecasts.
5. Incorrect component schedules.

6. Large buckets of unassigned costs which require data cleansing (which is labour intensive and time consuming).
7. Inability to analyse equipment and cost data meaningfully.

Finding Work Order Settlements

In Work Order Settlement Manager (**Daily Operations** > **Workorder Cost Analysis** > **Workorder Cost Analysis**, or **Reporting & Analyses** > **Costs** > **Costs**) you can:

1. Use a wide variety of filters to search for sets of work order settlements.
2. Edit a work order settlement imported into AMT from an external interface.
3. Merge two or more work order settlements to match AMT strategy tasks.
4. Flag work order settlements as reviewed.
5. Add manual work order settlements.

Special filters

Filter	Description
Linked to Equipment	Used to find work order settlements which have been imported into AMT from an external system, but the data may contain incorrect equipment information (for example, incorrect serial number). Therefore, AMT will not be able to link the work order settlement record to an equipment.
Strategy	Include or exclude work order settlements that are linked to strategy tasks.
Job Status	Indicates whether the maintenance (the costs of which the work order settlement is capturing the costs of) is Yet to Start, In Progress or Completed.
Link to work order	Include or exclude work order settlements linked to work orders in Equipment Status.
Before Equip Start	Include or exclude work order settlements which have a Job Date prior to the Start Date of the equipment.
AMT WO	Include or exclude work order settlements manually created in AMT.

Grid fields

Column	Description
Checked	Indicates whether someone has edited any details of the work order settlement in AMT. If selected, the work order settlement's codes will not be changed when the work order settlement is next imported from the ERP system.
Equipment	If the work order settlement is linked to an asset, this column will show the asset number, otherwise it will be blank.

Column	Description
WIP	If selected, this check box indicates the work order settlement is a Work In Progress. If cleared, this means that the work order settlement is closed.
Contract Total	Work order settlement's total cost based on the asset's default cost bearer.  For equipment under maintenance contract, this cost is shown at Sell.
WO Total	The work order settlement's total cost (regardless of cost bearer).  For equipment under maintenance contract, this cost is shown at Sell.
Component	Shows the work order settlement's codes plus component code description.
Occ Type	Occurrence Type.
Job Code	Displays the work order settlement job code and description.
Date	This is the date the work order settlement was opened, or if linked to a downtime event, it is the date when the work order was performed.
Usage	The asset's primary actual unit of measure reading at the time the work order settlement was begun.
Comp Life	Usage difference between when the work order settlement was last performed and the usage when this current work order settlement was performed. Can be thought of as Component Life.
Strategy	This check box is selected if the work order settlement's codes match that of a strategy task in the equipment's current projection.
Linked	This check box is selected if the work order settlement is linked to a work order in the Equipment Status module of AMT.

Merging work order settlements

It is sometimes required to merge two or more work order settlements that have not been merged during the enterprise resource planning-AMT interface process. For example, if work order settlements are designated in two currencies, one currency for imported parts, another for local labour, these two work order settlements should be combined in order to compare work order settlement costs with budget costs, which represent combined parts and labour costs.

Another example is where a work order settlement is opened for the remove and install job on site, but the recondition or rebuild of the component is done off-site and costed on a separate work order settlement. Both work order settlements need to be merged in order to reflect the total cost of the work, and a single new-life occurrence (not two).

Work order settlements are merged during an AMT process of work order settlement importation. Refer to the notes below for more information.

Notes

AMT cycles through the new or updated work order settlements, and work order settlements are merged if one of the following circumstances is true.

- If one or more work order settlements contain a duty free work order settlement ID, the work order settlements are merged; the work order settlement identified by the duty free work order settlement ID becomes the parent (or header) for the group of work order settlements.
- If more than work order settlement represents the same piece of work in the same maintenance plan (that is, the work order settlements share both maintenance plan number and maintenance plan call number), the work order settlements are merged and one work order settlement is arbitrarily chosen to be the parent (or header) for the group of work order settlements.

AMT collects a set of new and modified work order settlements from the enterprise resource planning system, then cycles through all new or modified work order settlements in the batch, and for each work order settlement, the following steps are performed.

- If the work order settlement already exists in AMT and there has been no modification, the work order settlement is ignored, otherwise, attributes of the work order settlements are determined as follows:
    - The set of codes that define a work order settlement (component code, modifier code, task counter, job code, task type).
    - If there is a standard job associated with the work order settlement, the codes of the work order settlement are set to the same values as the codes for the standard job or, if there is no standard job associated with the work order settlement, the codes of the work order settlement are set to the same values as the codes for the first operation in the work order settlement.
    - Task type and task counter are determined based on the work activity code for the first operation in the work order settlement.
    - Equipment plan is determined from a combination of the equipment serial number (and hence equipment ID) and the start date of the work order settlement.
    - Occurrence type is determined from the component code, modifier code and job code, where possible.
- Associated actual usages (an actual usage is a cumulative reading, such as 577 engine hours or 21000 km. The actual usage is the true usage reading at a particular point in time.) are added to the Actual Usages table, according to the following rules:
    - One and only one actual usage is added for all grouped work order settlements.
    - The unit of measure for the actual usage is either the unit of measure for the strategy task in the current projection that the work order settlement links to or, if there is no strategy task, then the unit of measure is the unit of measure in the projection that the work order settlement links to.
- If the imported work order settlement has no associated actual usages, an actual usage is calculated as follows:
    - The last actual usage and the last actual usage date are determined from the Actual Usages table (**Note**: this is always possible, since every equipment plan has a start usage).
    - The next actual usage and the next actual usage date are determined from the Actual Usages table (if a next actual usage exists).
    - If the next actual usage was determined, the actual usage for the work order settlement is calculated by extrapolation from the last actual usage, the next actual usage and their associated dates.

- The state of a work order settlement is used to determine whether the costs associated with the work order settlement should be included in actuals for the piece of equipment, as follows:
    - If the start date of the job is in the past - in other words, the job has occurred, the costs will always be included in actuals. The reason for this is that the projections that include the piece of work have been notified of the occurrence.
    - If the job is a change out and is linked to at least one strategy task, and hasn't yet started, the costs will never be included in actuals. This is because the projections that include the piece of work have not been notified of the occurrence, which means that the costs are already included in the projected occurrence.
    - Open work order settlements that do not fall into the category above, the costs will be included in actuals, because these work order settlements represent unassigned jobs, so the costs of these jobs have not been projected.

Work order settlement validation

There are many conditions that may indicate the presence of a work order settlement error, including but not limited to:

- The work order settlement total (cost) is significantly higher or lower than the current projection value.

  This may indicate that there may be an error in the component code combination. This is perhaps the most important test because errors may indicate the new life (change out) indicators may be wrong, which may impact on overdue components and on future projections
- The component code may be blank or display an unfamiliar number, perhaps indicating an error in the work order settlement set up in the enterprise resource planning system.
- The work order total may be zero.

If a potential error is found, highlight the work order settlement in the grid and double-click. The Edit Work Order Settlement form will open. At this point you may be required to choose one of the following:

- The error can be fixed in the enterprise resource planning system and subsequently the enterprise resource planning fix will be re-transmitted to AMT via the work order settlement interface, or
- If the error cannot be fixed in the enterprise resource planning system, the error can be fixed in AMT directly.
- It is recommended that wherever possible fixes are made to the enterprise resource planning system.

If the fix is to be made in AMT, then AMT allows you to change any of the following:

- Job date.
- Usage.
- Component code combination.
    - Component code.
    - Modifier code.
    - Task type.
    - Task counter.
- Occurrence type.
- Job code.

In addition, you are able to manually merge work order settlements. Refer to the Merge Work Order Settlements topic.

Validating work order settlements

As part of best practice and diligent cost and contract management, period reviews of contracts are performed. One of the most important areas reviewed and validated are work order settlements. From a contract point-of-view, poor work order settlement management results in

1. misallocated costs.
2. incorrect forecasts.
3. unreliable financial reports.

Golden rules

If AMT receives a work order settlement from the enterprise resource planning system, and that work order settlement does not have codes which align to a strategy task, the costs go into unassigned.

If AMT received a work order settlement from the enterprise resource planning system and that work order settlement does not have codes which align to a strategy task, AMT will not recognise the job was completed and therefore will

1. flag the strategy task as being overdue.
2. incorrectly forecast future occurrences.

Work order settlement KPIs

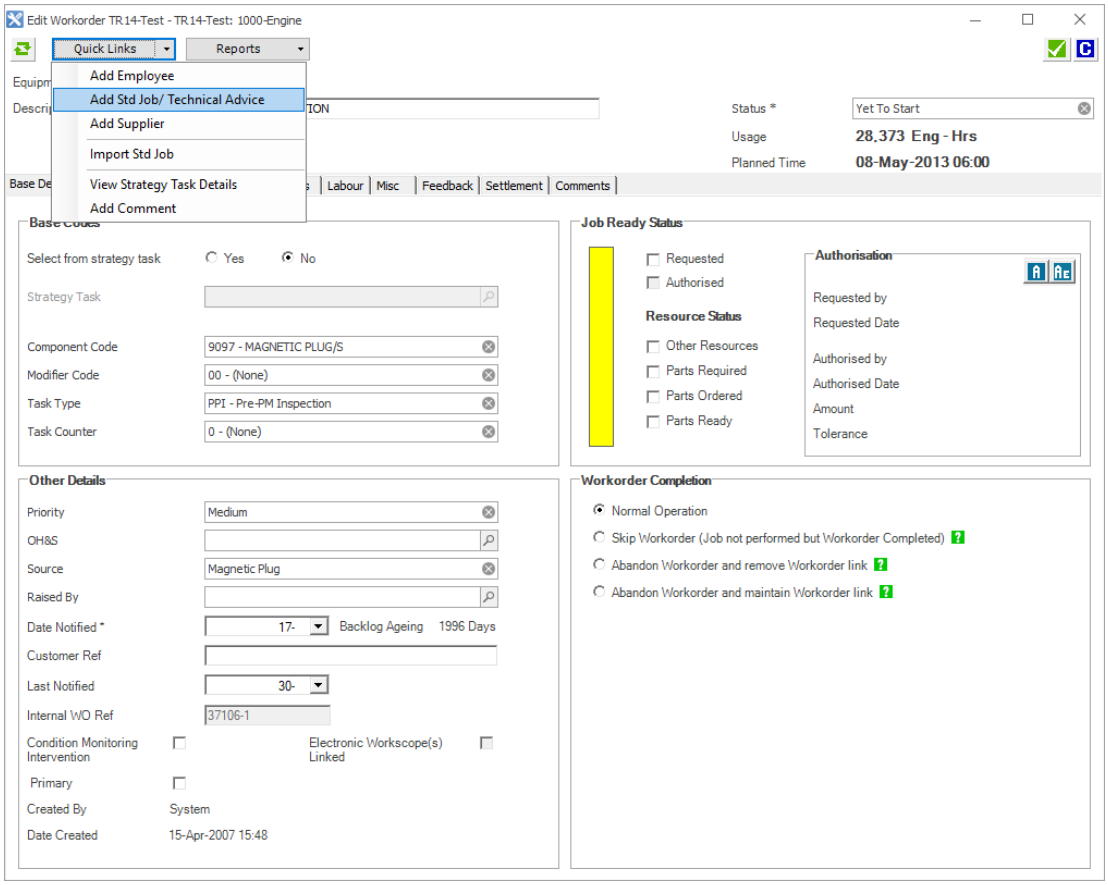
KPIs produced by AMT for assessing the quality of work order settlement management include:

1. Cost Analysis - "To Date".
   Strategy Optimisation > Cost Analyses.
   Reporting & Analyses > Costs > Costs.
   a. % Unassigned costs.
2. AMT Health.
   Dashboard > Overview > AMT Health.
   a. Strategy Tasks Overdue %.
   b. Number of Strategy Tasks not linked to work order settlements.
3. Work orders.
   Daily Operations > Equipment Status > Workorders tab.
   a. Admin Alerts.

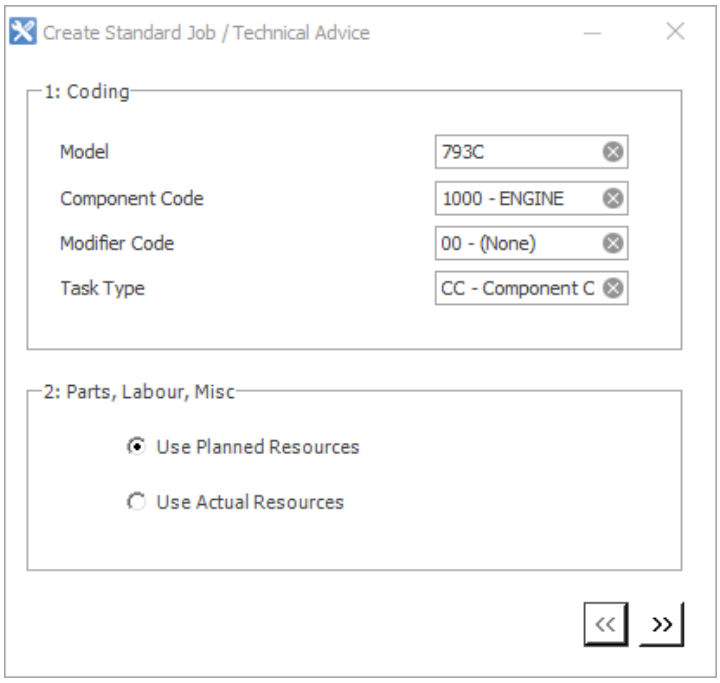
Create a standard job from a work order

If you execute a work order which is considered to be a "standard" for how this job should be done in future, you can trigger the Technical Advisor process.

Open the relevant work order. Click **Quick Links** then click **Add Std Job/Technical Advice** to trigger the creation of a standard job on the work order.



This opens the Create Standard Job /Technical Advice dialog box.



The **Coding** section lets you define standard job coding. It defaults to the work order codes.

The **Parts, Labour, Misc** section lets you flag whether planned or actual parts, labour or misc details should be the basis of the standard job.

Click the **right arrow** button to move forward. Check the standard job details and click **Save**.

(Rpt_257) Workorder Settlement Interrogation

Analyse the work order costs. Gives users the ability to monitor spending throughout a period, and for reconciliation of costs against the client enterprise resource planning system.

Tips

Always shows the total cost of the Job a the Job (Work order) Dare, even though some reports may split these costs over different transactional months (for Transaction Date Reporting).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|AutoLaunch|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Sub System, Task Type, Modifier Code, Job Code, WO Status, System, Strategy Task Desc, Strategy Task, Component Code, Task Counter, Cost Bearer, Cost Selector

(Rpt_321) Workorder Settlement List

Lets you validate work orders more effectively. Data is shown relating to the strategy task being linked, costs and usages of work order.

Tips

Shows the total cost of the Job at the Job (Work Order) Date, even though some reports may split these costs over different transactional months (for Transaction Data Reporting).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|BranchID|,|Branch|,
|SiteID|,|Site|,|FleetID|,|Fleet|,EqpPlanID|,|EqpPlan|,|SubSystemID|,
|SubSystem|,||TaskTypeID|,|TaskType|,|CompCodeID|,|CompCode|,
|ModCodeID|,|ModCode|,|JobStatusID|,|JobStatus|,|Checked|,|Strategy|,
|WONumber|,|ISNOTFutureWO|,|LastMonthDate|,|Today|,|AnalyseBy|,
|ProjectionTypes|,|IsLinkedToEquipment|,|UserID|,|StartDate|,|EndDate|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Modifier Code, Task Type, Task Counter, Strategy Task, Strategy, Job Status, Workorder Number, Internal WO Ref, WIP, Link to Workorder, Before Equip Start, AMT WO.

(Rpt_322) Average Workorder Costs

This report shows the average parts, labour and miscellaneous costs per projected task based on work order history.

Tips

This report is useful to compare against what a projected task is expected to cost in the future. If there is a significant difference, there could be a strong argument to change the costs contained in the projected task.

Initial parameters

|AutoLaunch|,|Site|,|Fleet|,|FleetID|,|Model|,|ModelID|,|EqpPlan|,|EqpPlanID|,
|System|,|SystemID|,|SubSystem|,|SubSystemID|,|ProjTaskID|,|ProjTaskID2|,
|UserID|TaskHeaderId|

Cost accrual

This is a system setting which enables AMT to accrue the cost for work order settlements which are classified as work in progress (WIP). Normally, the cost reported for work order settlements is that which has actually been costed to the work order settlement to date.

If a work order settlement is flagged as work in progress, and its cost to date is less than that of the strategy task to which is linked, AMT will accrue additional cost to the work order settlement based on a percentage

Example

- WO1234 for 1000.00.CC.0 Engine cost to date = $5000 (WIP).
- Strategy task estimated cost is $150,000.
- Accrual% = 100% (that is, accrue up to 100% of the strategy task's estimated value). Therefore WO1234 will be showing $150,000 in all reports until such time as either the WIP value is > $150,000 or the work order is closed (that is, no longer WIP).

The extent of the accrual can be customised (that is, will the work order settlement be accrued to 100% of the strategy task's cost or only 80%?)

Ask your AMT Administrator whether Cost Accrual Logic has been enabled, and if so, what parameters have been implemented to govern the calculation.

Warranty

Warranty approval

To assess and approve warranty claims, navigate to **Daily Operations** > **Warranty** > **Warranty Approval**. Filter for the relevant assets and set the **Warranty Claimable** = **Yes**. Review the details of the work order by clicking the **Work Order** in the grid and click **Edit**.

The Edit Work Order screen opens displaying the **Warranty** tab. On this tab you can enter additional details concerning the warranty claim. A work order can have the following claim statuses:

- **N/A**: If the work order is marked as not claimable.
- **Claim O/S**: If the work order is marked claimable but no claim has been submitted.
- **Claimed**: If either an amount is entered in the **Claimed** field (in the Cost Analysis section), or a claim has been submitted through the interface by clicking **Submit Warranty Claim**).
- **Closed**: If an amount is entered in the **Recovered** field (in the Cost Analysis section).



Flagging a work order as being Claimable is not limited to the product incident report process. A standard AMT user can flag a normal, everyday work order as being claimable.

(Rpt_350) Warranty Claim

Report for suitable for providing to a technical services or warranty divisions as a summary to the warranty claim. You can group the report by three categories for viewing particular cost information.

Tips

Use of the Warranty Claim number will assist with running this report. If filters are left at a branch or site level, the information will generate for every warranty claim registered at that level.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Warranty Claim

(Rpt_351) Warranty Analysis

Provides users with the ability to analyse the warranty claims registered or pending in the system.

Tips

The ability to Analyse By provides you with the option to choose two fields to group the report by for ease of use and selection of specific warranty data.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Component Code, Task Type, System, Sub System, Claim Status, Potential Claim

(Rpt_528) Monthly Warranty Status

Lists work orders which have been flagged as warranty for claiming and auditing processes.

Tips

A work order is shown in this report if it has been flagged as "Warranty Claimable".

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Component Code, Task Type, System, Sub System, Claim Status, Warranty Claimable

AMT User Reports

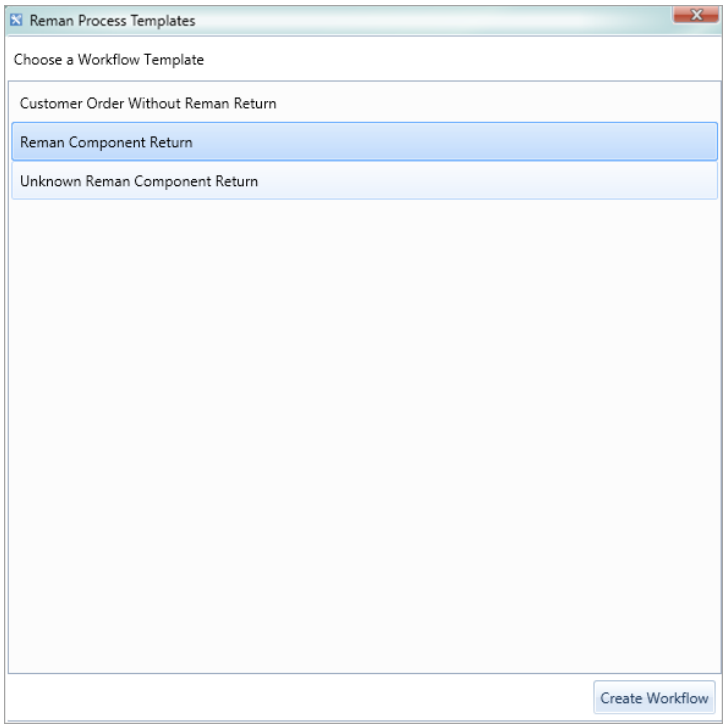
Published IC reports will appear under the **AMT User Reports** menu option for end users to access.

Core Tracker

Create a Reman process workflow

The Reman workflow creation step initiates the entire workflow process and therefore care should be taken in choosing the correct workflow template. In the workflow creation, data that applies to both the entire workflow and to the first workflow step is entered and the workflow and its current status will become available for you to view.

Navigate to **Core Tracker** > **Create new Reman Process Workflow**.



There are three Reman Process Templates you can select from, which cater to different scenarios.

1. **Customer Order Without Reman Return**.

   This is for a component issue and dispatch where a reman return component is not expected to be received in return.

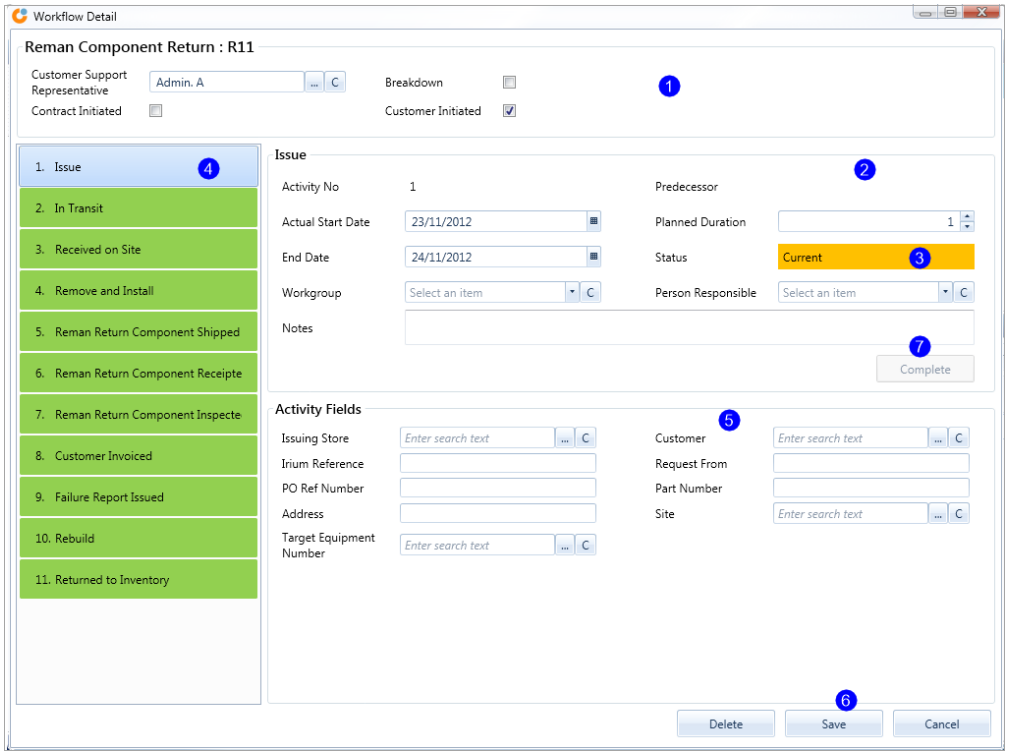
2. **Reman Component Return**.

   This is for a standard component issue where a corresponding component is returned for remanufacture.

3. **Unknown Reman Component Return**.

   This is for a component being unexpectedly returned for remanufacture without the associated issue of a component.

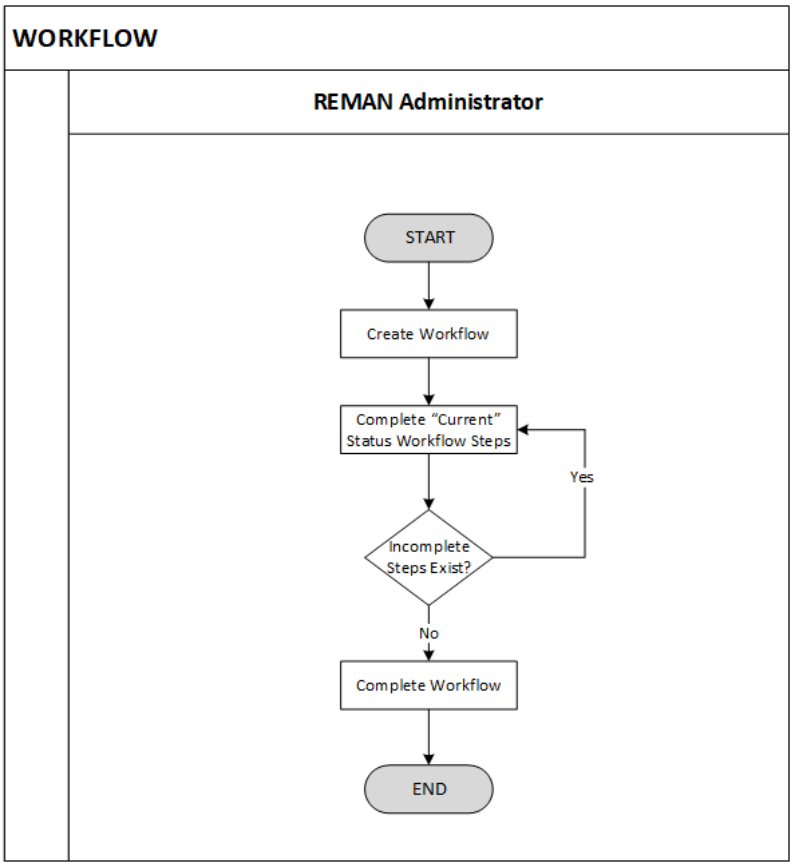
An example of the Workflow detail screen is below.



Item	Description
1	Workflow Header. These fields refer to the entire workflow process.
2	Step Header. The fields are relevant to the step you are on.
3	Status Indicator. Tells you what stage of the process this step is at.
4	Steps in the Process. The light blue indicates the step you are currently on.
5	Activity Fields. The fields are relevant to the step you are on.
6	Save the information entered. Delete will delete the entire workflow. Cancel reverts data back to the time of the last save.
7	Click Complete to complete the step.

Reman process workflow

The flowchart below illustrates the Reman process workflow.



How the Reman workflow works in AMT

The AMT Reman process workflow function provides the ability to record data in a step-by step layout, helping you to adhere to nominated business processes and to quickly determine the current progress of activities as they take place.

The workflow is represented by a series of linked steps which need to be completed in a predetermined sequence with data being entered at each step. The workflow records the start and completion dates of each step, enabling you to tell at a glance which step of the process you are up to.

You are able to monitor the progression of a workflow in real time, which enables you to more accurately predict process completion dates, and also provides a means to identify delays or hold-ups in a process.

All these things lead to maximising the efficiency of a process and the timely resolution of any issues that may arise during the process.

Reman Overview

Reman is an abbreviation of Remanufacture. It is referred to as Reman throughout this help file.

The AMT Reman Process Workflow function provides the ability to record data in a step-by step layout helping you to adhere to nominated business processes and to quickly determine the current progress of activities as they take place.

The workflow is represented by a series of linked steps which need to be completed in a predetermined sequence with data being entered at each step. The workflow records the start and completion dates of each step, enabling you to tell at a glance which step of the process you are up to.

You are able to monitor the progression of a workflow in real time, which enables you to more accurately predict process completion dates, and also provides a means to identify delays or holdups in a process.

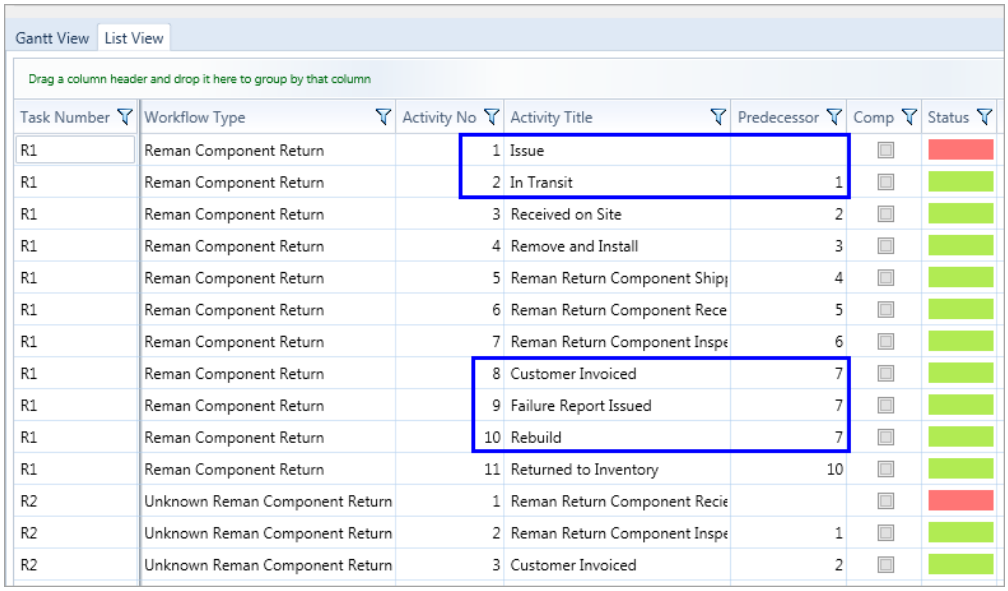
All these things lead to maximising the efficiency of a process and the timely resolution of any issues that may arise during the process.

Reman scheduling logic

Predecessors

Workflow processes are guided by specified relationships and rules. The most important concept is **predecessor**, which is a value originally defined when setting up the template. Predecessors determine which step must be completed prior to any of its dependent steps being completed.

In the example screenshot below, you can see that step 2, In Transit cannot be completed until step 1, Issue has been completed. In addition, steps 8, 9 and 10 cannot be completed until step 7 has been completed (in this case, steps 8, 9 and 10 are dependent only on the completion of step 7 and not on the completion of the previous sequential step).

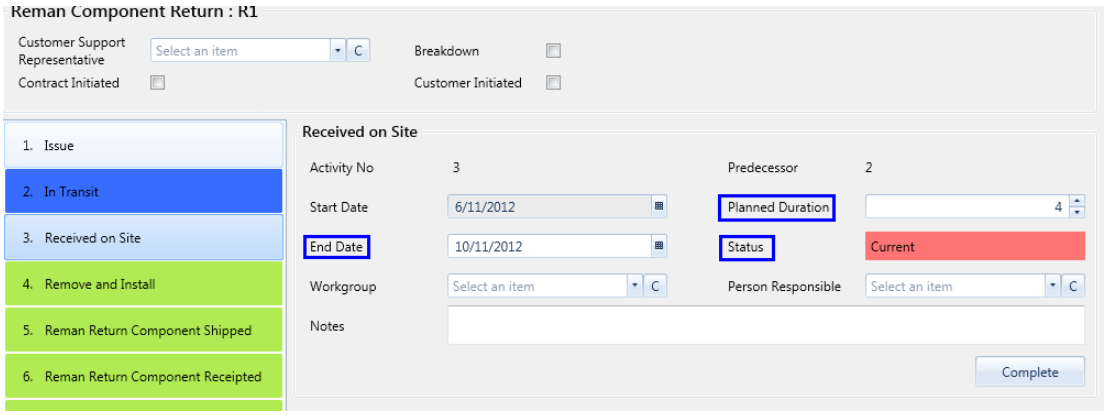

Gantt View | List View

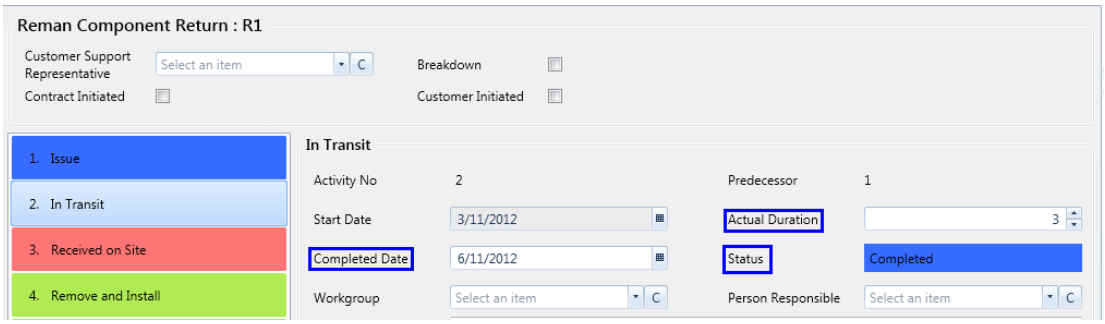
Drag a column header and drop it here to group by that column

Task Number	Workflow Type	Activity No	Activity Title	Predecessor	Comp	Status
R1	Reman Component Return	1	Issue			
R1	Reman Component Return	2	In Transit	1		
R1	Reman Component Return	3	Received on Site	2		
R1	Reman Component Return	4	Remove and Install	3		
R1	Reman Component Return	5	Reman Return Component Shipp	4		
R1	Reman Component Return	6	Reman Return Component Rece	5		
R1	Reman Component Return	7	Reman Return Component Inspe	6		
R1	Reman Component Return	8	Customer Invoiced	7		
R1	Reman Component Return	9	Failure Report Issued	7		
R1	Reman Component Return	10	Rebuild	7		
R1	Reman Component Return	11	Returned to Inventory	10		
R2	Unknown Reman Component Return	1	Reman Return Component Recie			
R2	Unknown Reman Component Return	2	Reman Return Component Inspe	1		
R2	Unknown Reman Component Return	3	Customer Invoiced	2		

Step end dates

When you complete a step, the field description changes from **Planned Duration** and **End Date** with the Status of **Current** to **Actual Duration**, **Completed Date** and Status of **Completed**, as shown in the example screenshot below.





When a step is completed any subsequent steps, based on the predecessor setting, inherits the **Completed Date** as a **Start Date**.

Any steps that were dependent on the newly completed step will change to a status of current and be available for completion. Any step that is not yet ready for completion will have a status of Future and will not be able to be completed until its predecessor has been completed.

View / Edit Reman process workflow

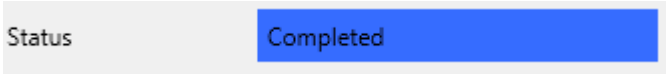
Navigate to **Core Tracker** > **View / Edit Reman Process Workflow**.

The Reman Workflow Manager screen enables you to retrieve, filter, sort and view existing workflow data. You can view the data in a Gantt chart and as a list. You can select optional filter fields to further refine the data result.

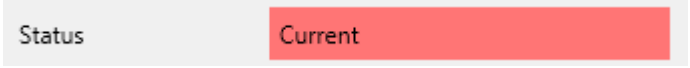
From the result screens you can open a relevant workflow step for more detailed examination or modification of a workflow step.

You can determine the status of the steps by their colour in both the Gantt chart and list view.

- **Blue** represents a **completed** step.



- **Red** represents the **current** step.



- **Green** denotes a **future** (uncompleted) step.



Reporting & Analyses

Summary

(Rpt_167) Equipment Performance Summary

Summaries a number of the key equipment performance indicators and gives the ability to analyse by different levels.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Industry Group

(Rpt_382) Change Analysis

Shows the change in key parameters over time, and lets you drill down into the reasons for the changes.

Tips

Costs shown are for the default Cost Bearer, and are shown as Cost at Sell.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Snapshots|,|UserID|

Report parameters

Default.

Additional filters

System, Strategy Task, Modifier Code, Occurrence Type, Cost Type, Sub System, Task Type, Component Code, Part Classification, Part Rating, Primary Part No, Rebuild Location, Review Status, Sales Status

(Rpt_439) Key Stats Change Trending

Lets you trend the movement in certain key stats over a period of time. For example, the ability to trend the movement in PCV or projected costs for this financial year.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Snapshots|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Sub System, Strategy Task, Industry Group, Contract Type, Customer, Cost Centre, System, Component Code, Task Type, Business Group, Contract, Maintenance Type, Responsibility

(Rpt_541) Budget Equipment Cost Analysis

This report is based on Budgeting module data that allows comparison of a Budget, a Forecast and live AMT data.

Tips

This report can be run at the Equipment level only.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Task Counter, Work Group, Cost Bearer, Cost Centre, Cost Category, Expense Element, CC Activity, CC Location, CC Responsibility, Parts/Labour/Misc

Equipment Performance

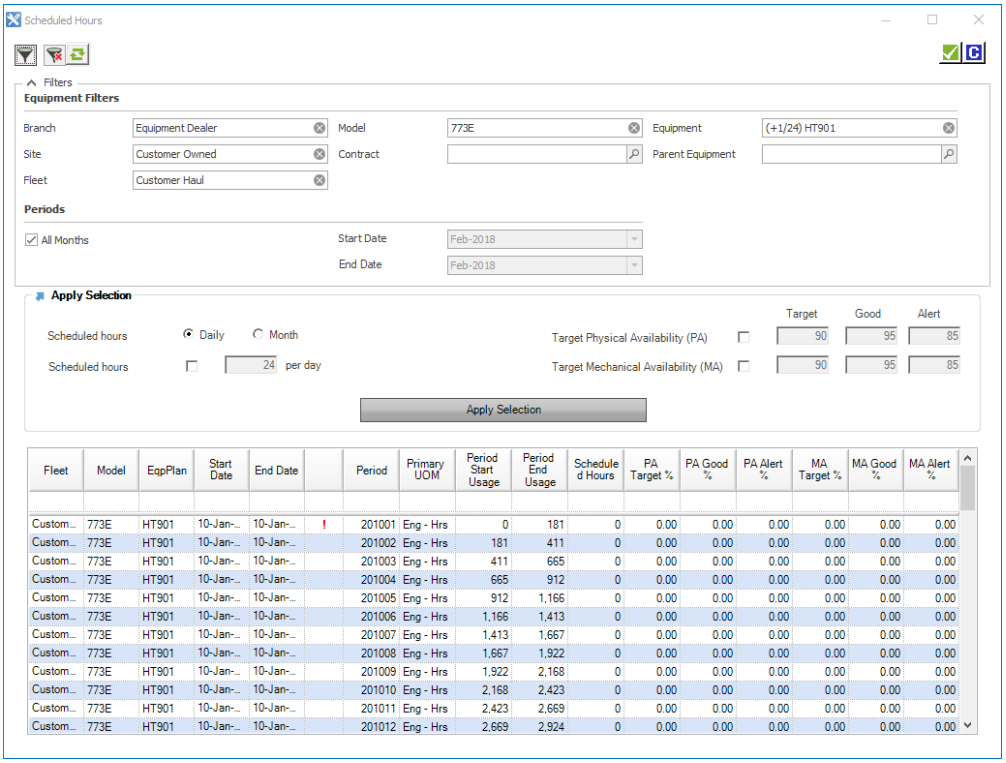
Availability

The reports for viewing availability (actuals) are:

- Availability - Physical
- Availability - Mechanical

The formula for physical availability is based on scheduled hours. Scheduled hours are set up using the **Scheduled Hours** screen, which typically represents calendar hours. Availability targets, both physical and mechanical, are also set up using the **Scheduled Hours** screen.

Navigate to **Scheduled Hours** by clicking **Reporting & Analyses** > **Equipment Performance** > **Availability**. Set the filters in the top panel to select the required assets and click  .



Fleet	Model	EqpPlan	Start Date	End Date		Period	Primary UOM	Period Start Usage	Period End Usage	Scheduled Hours	PA Target %	PA Good %	PA Alert %	MA Target %	MA Good %	MA Alert %
Custom...	773E	HT901	10-Jan-...	10-Jan-...	!	201001	Eng - Hrs	0	181	0	0.00	0.00	0.00	0.00	0.00	0.00
Custom...	773E	HT901	10-Jan-...	10-Jan-...		201002	Eng - Hrs	181	411	0	0.00	0.00	0.00	0.00	0.00	0.00
Custom...	773E	HT901	10-Jan-...	10-Jan-...		201003	Eng - Hrs	411	665	0	0.00	0.00	0.00	0.00	0.00	0.00
Custom...	773E	HT901	10-Jan-...	10-Jan-...		201004	Eng - Hrs	665	912	0	0.00	0.00	0.00	0.00	0.00	0.00
Custom...	773E	HT901	10-Jan-...	10-Jan-...		201005	Eng - Hrs	912	1,166	0	0.00	0.00	0.00	0.00	0.00	0.00
Custom...	773E	HT901	10-Jan-...	10-Jan-...		201006	Eng - Hrs	1,166	1,413	0	0.00	0.00	0.00	0.00	0.00	0.00
Custom...	773E	HT901	10-Jan-...	10-Jan-...		201007	Eng - Hrs	1,413	1,667	0	0.00	0.00	0.00	0.00	0.00	0.00
Custom...	773E	HT901	10-Jan-...	10-Jan-...		201008	Eng - Hrs	1,667	1,922	0	0.00	0.00	0.00	0.00	0.00	0.00
Custom...	773E	HT901	10-Jan-...	10-Jan-...		201009	Eng - Hrs	1,922	2,168	0	0.00	0.00	0.00	0.00	0.00	0.00
Custom...	773E	HT901	10-Jan-...	10-Jan-...		201010	Eng - Hrs	2,168	2,423	0	0.00	0.00	0.00	0.00	0.00	0.00
Custom...	773E	HT901	10-Jan-...	10-Jan-...		201011	Eng - Hrs	2,423	2,669	0	0.00	0.00	0.00	0.00	0.00	0.00
Custom...	773E	HT901	10-Jan-...	10-Jan-...		201012	Eng - Hrs	2,669	2,924	0	0.00	0.00	0.00	0.00	0.00	0.00

Scheduled hours can only be set for assets configured for advanced planning and the asset's primary UOM must be flagged as Operating Hours (in the UOM system table).

- A single row in the grid represents a calendar month during the asset's life cycle.
- The **!** in the grid is an alert, indicating that the assets life cycle starts or ends during the period selected, and therefore, the scheduled hours for this period may be less than other periods.
- Set the parameters in the **Apply Selection panel** and click **Apply Selection** to see the changes in the grid. The changes are highlighted in red.

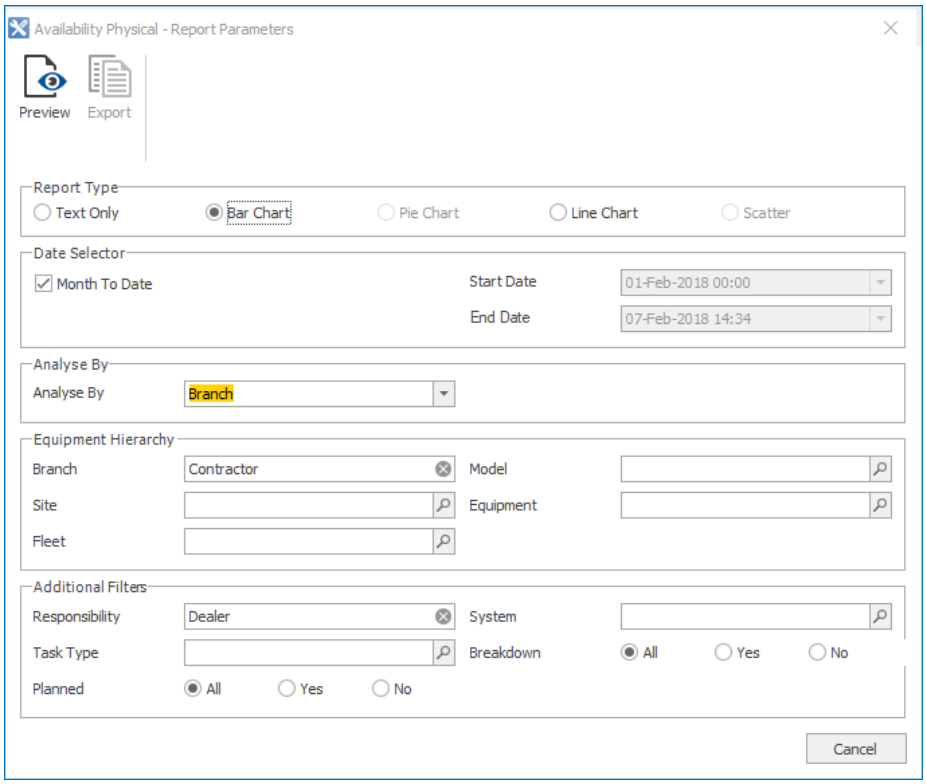
Fleet	Model	EqpPlan	Start Date	End Date		Period	Primary UOM	Period Start Usage	Period End Usage	Scheduled Hours	PA Target %	PA Good %	PA Alert %	MA Target %	MA Good %	MA Alert %
Custom...	773E	HT901	10-Jan-...	10-Jan-...	!	201001	Eng - Hrs	0	181	0	90.00	95.00	85.00	0.00	0.00	0.00
Custom...	773E	HT901	10-Jan-...	10-Jan-...		201002	Eng - Hrs	181	411	0	90.00	95.00	85.00	0.00	0.00	0.00

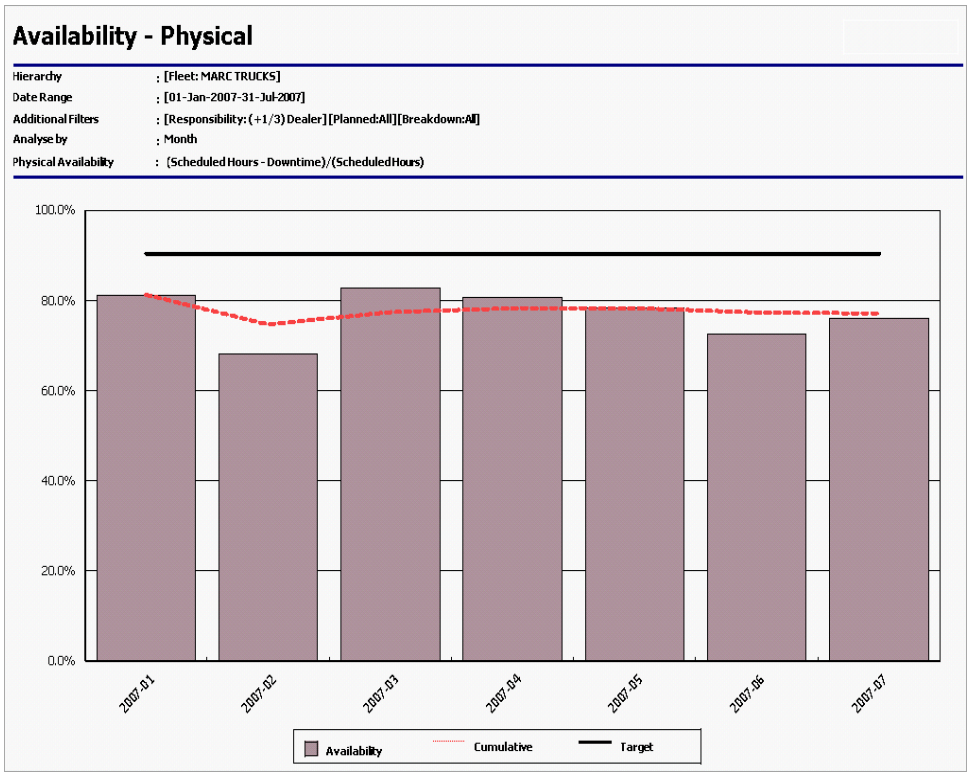
Availability Reports

The type of availability applicable to an asset is set at the fleet level. Fleets can be edited in the **Fleet Administration** screen. The availability formulas in AMT are:

- Physical Availability = *Scheduled Hours - Downtime* / *Scheduled Hours*.
- Mechanical Availability = *Operating Hours* / *Operating Hours + Downtime*.

Navigate to the two reports by clicking **Reporting & Analyses** > **Equipment Performance** > **Availability**. The screenshots below show an example of the **Report Parameters** screen and the report output for the Availability - Physical report. The Availability - Mechanical report screens have the same format.





Some additional reports that show availability in AMT are:

- Snapshot – The Worm: Navigate to **Daily Operations** > **Daily Reports** > **Snapshot - the Worm**.
- Availability Snapshot Worm: **Daily Operations** > **Daily Reports** > **Availability Snapshot Worm**.
- Availability to Target Summary: **Reporting & Analyses** > **Equipment Performance** > **Availability** > **Availability To Target Summary**.

(Rpt_124) Availability - Mechanical

Shows the mechanical availability for the designated period of time. The downtime used in the calculation is dependant on the "responsibility" nominated by the user.

To Trend, "Analyse By" Month to compare equipment with fleets, analyse by equipment and fleet.

Tips

Mechanical Availability = Operating Hours / (Operating Hours + Downtime).

Operating Hours = SMU.

Initial parameters

|ControlType|,|IDlist|, |TextList|,|ProjectionTypes|,|UserID|,|LogoLocation|,|AutoLaunch|, |BranchID|,|SiteID|,FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|,|DateChecked|, |StartDate|,|EndDate|,|AnalyseBy|,|Responsibility|,|ResponsibilityID|

Report parameters

Default.

Additional filters

Responsibility, System, Task Type, Maint. Order Type, Planned, Planning Group, Breakdown,

(Rpt_125) Availability - Physical

Shows the physical availability for the designated period of time. The downtime used in the calculation is dependant on the "responsibility" nominated by the user.

To Trend, "Analyse By" Month. To compare equipment with fleets, analyse by equipment and fleet.

Tips

Physical Availability = (Scheduled Hours - Downtime) / Scheduled Hours.

Scheduled Hours are defined using the Scheduled Hours function, and typically represent the calendar hours of a month.

|ControlType|, |IDlist|, |TextList|, |ProjectionTypes|, |UserID|, |AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|, |Fleet|,|EqpPlan|,|DateChecked|,|StartDate|,|EndDate|,|AnalyseBy|, |Responsibility|, |ResponsibilityID|

Report parameters

Default.

Additional filters

Responsibility, System, Task Type, Maint. Order Type, Planned, Planning Group, Breakdown,

Availability Target

A contractual level of productivity (for example, 95%) that has been agreed with a customer. There are usually defined penalties for failing to meet the agreed level.

Scheduled Hours

Navigate to **Reporting & Analyses** > **Equipment Performance** > **Availability** > **Scheduled Hours**.

Use this function to set up the scheduled hours for equipment per month, as well as mechanical and physical availability targets for the day or month. If you set the targets for a month, the scheduled hours figure calculated by AMT is the number of days in the chosen month multiplied by 24 hours in a day. For example, April has 30 days. Therefore scheduled hours = 30 x 24 = 720 hours. If the scheduled hours required are different to those calculated by AMT, enter the desired scheduled hours, and the appropriate availability targets and click **Apply Selection** to apply those parameters to the selected fleet or equipment.

Notes

- You do not need to select an equipment if the scheduled hours and availability targets are the same for all equipment in the same fleet.
- If you forget to set the scheduled hours and targets for a month, AMT uses the previous month's settings for the current month.
- Adjustments to scheduled hours figures will only show in reports after 24 hours, because the report tables in AMT are refreshed every night.

Reliability

Breakdowns and poor planning are the key drivers of poor reliability.

In AMT, every breakdown is flagged with how it could have been prevented – that is, what health activity could have prevented that breakdown. A useful analysis, therefore, is to analyse breakdown events by **Preventable**. This highlights which are the key processes to focus on to prevent breakdowns. To highlight poor planning, analyse non-breakdown events by **Task Type**. Ideally, all planned events should be either preventive maintenance services or component changeouts - all other tasks should be planned to be performed at those stoppages.

The core KPIs to measure the reliability of assets are:

- Mean Time Between Failure (MTBF) **Reporting & Analyses** > **Equipment Performance** > **Reliability** > **MTBF**
- Mean Time Between Stoppage (MTBS) **Reporting & Analyses** > **Equipment Performance** > **Reliability** > **MTBS**
- Mean Time Between Unplanned Stoppage (MTBUPS) **Reporting & Analyses** > **Equipment Performance** > **Reliability** > **MTBUPS**

Related reports include:

- Mean Time to First Stoppage after Preventive Maintenance (MTFS After PM). **Reporting & Analyses** > **Maintenance Efficiency** > **Daily Operations** > **MTFS After PM**
- **Reporting & Analyses** > **Maintenance Efficiency** > **Daily Operations** > **MTTR**

Definitions

MTBF = Usage / number of Failures
where a failure is a non-planned breakdown event.

MTBS = Usage / number of Stoppages
where a stoppage is an event.

MTBUPS = Usage / number of Unplanned Stoppages
where an unplanned stoppage is a non-planned event.

MTTR = Downtime / number of Stoppages
where a stoppage is an event.



Only events with their status as in progress or complete are included.

Stoppages and downtime

The core reports in AMT for stoppages and downtime are:

- Event Analysis **Reporting & Analyses** > **Equipment Performance** > **Stoppages and Downtime** > **Event Analysis**.
- Downtime Analysis **Reporting & Analyses** > **Equipment Performance** > **Stoppages and Downtime** > **Downtime Analysis**

In the Downtime Analysis report, you can find significant or exceptional downtime and then drill down to find the root cause.

Click the bar on the report that you want to drill down in to, then select the item from the list to drill down on.

Utilisation

(Rpt_118) Actual Usage Details

Details the units of measure entered for a particular equipment, site, branch and so on. Shows both actual and cumulative readings, for example, engine hours, fuel litres, travel hours. **Reporting & Analyses** > **Equipment Performance** > **Utilisation** or **Daily Operations** > **Usages & Measurements**.

Initial parameters

|ControlType|, |IDlist|, |TextList|, |ProjectionTypes|, |UserID|,
|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|,
|DateChecked|,|StartDate|,|EndDate|,|AnalyseBy|

Report parameters

Default.

Additional filters

None.

(Rpt_247) Usage Analysis Chart

Displays the machine usage. Enables users to analyse by equipment plan, branch, month and so on. The target usage profile is shown to gauge how close the actual is to the profile. **Reporting & Analyses** > **Equipment Performance** > **Utilisation**.

Tips

If the actual usage has a large variance form the target profile, you should review the usage of the machine and make the necessary changes to the usage profile.

Initial parameters

|ControlType|,|Idlist|,|TextList|,|UsageRateReport|,|ProjTaskID|,|RateAgainstQuomID|, |IsNotShowAsPerc|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, Gross Usage As, Avg Per Eqp Plan

(Rpt_249) Usage v Profile Snapshot

Compares the usage of the equipment (based on the UOM) to the target. This could highlight issues with the application or the client machine usage. **Reporting & Analyses** > **Equipment Performance** > **Utilisation**.

Tips

If the actual usage has a large variance from the target profile, users should review the usage of the machine and make the necessary changes to the usage profile.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|QUOM|,|ProjectionTypes|,|UserID|Additional filters

Report parameters

Default.

Additional filters

UOM, Avg Per Eqp Plan

(Rpt_250) Utilisation Rate

Displays the utilisation of the machine and the target to be achieved. **Reporting & Analyses** > **Equipment Performance** > **Utilisation**.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, Show Usage at UOM

(Rpt_505) Usage Analysis WorkSheet

Contains cumulative daily usage and average daily usage. **Reporting & Analyses** > **Equipment Performance** > **Utilisation**.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|QUOM|

Report parameters

Default.

Additional filters

Contract, UOM, Use Actual UOM Reading

Health - Utilisation

Check that usage readings are entered regularly (at least once a week). If usages are not entered into AMT, the program assumes utilisation is occurring in line with the usage profile, therefore there is not a high risk of the usage reading being incorrect.

If a machine is stopped for a number of days or months, you should still enter a usage reading into AMT, otherwise AMT assumes the machine is operating and therefore could schedule tasks to occur while the machine is parked.

Also review whether the actual usage pattern is reflective of the expected usage. If the actuals do not reflect the expected, AMT's forecasting in terms of the timing of strategy task occurrences could be either too early or too late. If there is a significant difference, consider changing the annual utilisation parameters set up in the equipment plan.

Consumption

The core reports for consumption are:

- Consumption per UOM
- Consumption per Month

These reports are located under **Reporting & Analyses** > **Equipment Performance** > **Consumption**.

Both of the reports have basically the same layout and fields. Click the links above for report information.

(Rpt_392) Consumption per UOM

Shows the consumption details - per month and per usage, for example, oil consumption is 0.1 litres per engine hour, or petrol per kilometre.

Tips

You are not able to select a forward date, the default for this report is the past six months - including this month.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Element, UOM

(Rpt_394) Consumption per Month

Shows the consumption details - per month and per usage, for example, oil consumption is 0.1 litres per engine hour, or petrol per kilometre.

Tips

You are not able to select a forward date, the default for this report is the past six months - including this month.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Element

Production Analysis

Analyse production statistics

There are two reports in AMT which provide accurate reporting for productive and non-productive work time which helps understand asset performance:

- Production Statistics Analysis.
- Production Statistics Chart.

Navigate to **Reporting & Analyses** > **Equipment Performance** > **Production Analysis** and double-click the relevant report.

Analyse usages

The Asset Performance reporting area of AMT allows the integration of a wide range of performance data to produce a variety of asset performance and asset productivity reports. For utilisation and usage analysis, navigate to **Reporting and Analyses** > **Equipment Performance** > **Utilisation**.

You can view, in report format:

1. Actual usages which have been recorded.
2. Expected usages, past and future based on the asset's usage profile.
3. Actual vs Expected usages.

(Rpt_410) Asset Utilisation

Compares the actual utilisation with the budget utilisation. It also shows the breakdown and scheduled down time.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Asset Util UOM, Analyse Downtime By

(Rpt_448) Production Statistics Analysis

Analyses production statistics.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Prod Stat Code

(Rpt_449) Production Statistics Chart

Analyses production statistics and display them in chart form.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Prod Stat Code

(Rpt_525) Plant Performance KPI

Provides a summary report of key equipment operational performance data to validate operational KPI calculations.

Tips

This report has a date selector to enable reporting for any nominated period.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|,|Responsibility|,|LogoLocation|

Report parameters

Default.

Additional filters

DT Resp. (Show), DT Res. (Calc), Availability Type, Prod Stat Code

(Rpt_542) Equipment KPI

Details key Equipment KPIs for a specific time period.

Tips

Shows Hours Worked, Scheduled Hours, Downtime Hours (Planned vs Breakdown vs Other) and Stoppages (Planned vs Breakdown vs Other).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|,|Responsibility|,|LogoLocation|

Report parameters

Default.

Additional filters

Responsibility

Costs

Costs

(Rpt_217) PCV Chart

Lets you tailor a report which compares the current and projected costs of different projections. The report also has the flexibility to compare costs for different Cost Bearers within the selected properties.

Tips

You must choose a filter from the Equipment Hierarchy area (Branch, Site and so on) prior to selecting two projections to Analyse By.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,
|UserID|,|AutoLaunch|,|BranchID||SiteID|,|FleetID|,|EqpPlanID|,
|Branch|,|Site|,|Fleet|,|EqpPlan|,|Projection1ID|,|Projection1|,|Projection2ID|,
|Projection2|

Report parameters

Default.

Additional filters

Cost Type, System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Sheet, Activity, Costs

(Rpt_188) Life Cycle Cost Summary

Compares Cost to Date and Future Costs by Task and Sub System between Budget and Current projections. The report identifies cost variances by Task, and is used primarily to identify significant issues.

Applicable for those assets with Budget Protection. The PCV highlights projected cost over-runs from original/baseline assumptions.

Tips

Check for large variances between the Budget and Actual in the **To Date** columns. Large variances should be investigated further and the information qualified to assist decision making for the future costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Task Type, Strategy Task, UOM, Cost Bearer

(Rpt_815) Life Cycle Cost Summary - @Cost

Compares Cost to Date and Future Costs by Task and Sub-system between Budget and Current projections. This report identifies cost variances by Task and is used primarily to identify significant issues.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

System, Sub System, Task Type Strategy Task, UOM, Cost Bearer

(Rpt_146) Cost Analysis - "Life Cycle"

Shows graphically, or in a text report, the major cost areas of the equipment for the entire plan term or to the month to date.

Run the report for the life-cycle or a shorter period (for example, a year). Running as a Pie Chart, "Analyse By" Component shows the top 20 component cost areas.

Tips

Can be run in Text, Bar or Pie chart format. Bar Chart shows the top 10 costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|CostBearerID|,|ProjectionTypes|,|EquipmentTypes|,
|ProjHeaderID|,|ProjHeader|,|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,
|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|,
|AnalyseBy|,|CurrencyID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, Work Group, System, Cost Centre, Sub System, Cost Category, Strategy Task, Expense Element, Strategy Task Desc, Maintenance Type, Task Type, Task Mode, Component Code, Costs, Modifier Code, Parts/Labour/Misc, Cost Bearer, Currency, Task Counter

(Rpt_147) Cost Analysis - "To Date"

Lets you list and analyse cost values in descending value based on a wide variety of filters. This report is for analysis of costs to date, not for future projected costs to date, not for future projected costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Contract, Cost Type, Strategy Task, System, Occurrence Type, Sub System, Modifier Code, Component Code, Task Counter, Task Type, Cost Bearer, WO Status, P/L/M

(Rpt_495) Cost Trending

Trends actual and projected costs of an equipment over time.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,
|EquipmentTypes|,|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Parts/Labour/Misc, Display, Currency

(Rpt_151) Cost Per Interval

Cost per interval is a Standard contract report showing cost at sell per interval.

The report lets you nominate a period of hours to analyse what equipment costs will fall in that timeframe. Can be analysed using either the Budget or Current projection and by numerous other filters.

Tips

This report can assist management in understanding what costs have occurred during what period of time in the equipment life cycle and can assist in estimating other machine costs of a similar nature.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,|CostBearer|,
|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre. Expense Element, Maintenance Type, Item, Item

(Rpt_651) Cost Per Interval Comparison

Based on the standard Cost Per Interval report, this report lets you compare the Cost Per Interval of two different projections side by side.

Tips

Useful for validating centrelines against real equipment, and comparing different modelling equipment to identify variances which may not have been intended.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|UserID|,|LogoLocation|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Labour Activity, Job Code, Costs

Workorder Settlement Manager

This function is designed to allow work order settlements to be reviewed, validated, merged, and in some cases added and edited.

In Workorder Settlement Manager, you can

- Use a wide variety of filters to search for sets of work order settlements.
- Edit certain details of a work order settlement imported into AMT via an interface.
- Merge two or more work order settlements.
- Mark work order settlements as reviewed and checked.
- Add internal AMT manual work order settlements.

Special filters

Filter	Description
Linked to Equipment	Used to find work order settlements which have been imported into AMT from an external system, but the data may contain incorrect equipment information (for

Filter	Description
	example, an incorrect serial number). Therefore, AMT will not be able to link the work order settlement record to an equipment.
Strategy	Include or exclude work order settlements that are linked to strategy tasks.
Job Status	Indicates whether the maintenance (the costs of which the work order settlement is capturing the costs of) is Yet to Start, In Progress or Completed.
Linked to work order	Include or exclude work order settlements linked to work orders in Equipment Status.
Before Equip Start	Include or exclude work order settlements which have a Job Date prior to the Start Date of the equipment.
Maintenance Order Type (MOT)	Select multiple values, for example Breakdown, Corrective Maintenance, Preventative Maintenance.
AMT WO	Include or exclude work order settlements manually created in AMT.

Grid fields

Field	Description
Checked	Indicates whether someone has edited any details of the work order settlement in AMT. If selected, the work order settlement's codes will not be changed when the work order settlement is next imported from the enterprise resource planning system.
Serial Number	Serial Number on the work order settlement.
Equipment	If the work order settlement is linked to an equipment, this column will show the Equipment Name, otherwise it will be blank.
Work Order	This is the combination of work order settlement Number + Task Number.  Only parent work order settlements are shown in the work order settlement manager table.
WIP	If selected, this indicates the work order settlement is Work In Progress. If unselected, this means that the work order settlement is closed.
Current Projection	If the work order settlement is linked to a strategy task (the Strategy check box is selected), the estimated cost of the strategy task per occurrence, from the equipment's current projection, is shown.

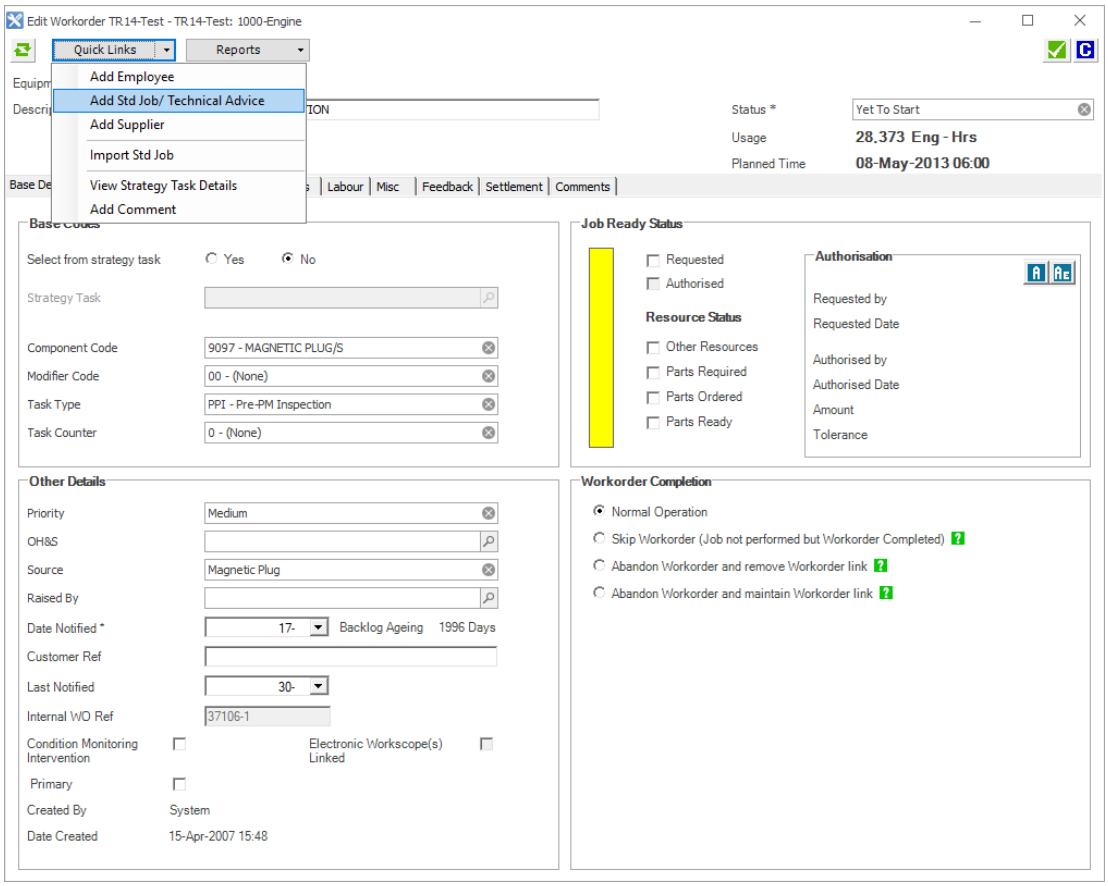
Field	Description
	If the work order settlement is not linked to a strategy task (the Strategy check box is cleared), the estimated cost per occurrence of the unassigned subsystem (to which the work order settlement's component code is assigned is shown).
Contract Total	Work Order Settlement's total cost based on the equipment's default cost bearer.  For equipment under maintenance contract, this cost is shown at Sell.
WO Total	Work Order Settlement's total cost (regardless of cost bearer).  For equipment under maintenance contract, this cost is shown at Sell.
Component	Shows the Work Order Settlement's codes plus component code description.
Description	Description of the work order settlement from the interface file.
Occ Type	Occurrence Type.
Job Code	Job Code.
Job Status	Indicates whether the maintenance (the costs of which the work order settlement is capturing the costs of) is Yet to Start, In Progress or Completed.
CCC	Identifies whether the work order settlement is flagged as End Of Life, Partial Life, Swing In or Swing Out (these indicators are set on the work order in Equipment Status).
Life Conf	Life achieved by the component. Normally this is calculated automatically by AMT but you can overwrite it on the work order, which is linked to a work order settlement, in Equipment Status.
Date	This is the Job Date that represents the date the actual maintenance was performed.
Usage	Usage on the equipment at the Job Date
Comp Life	Usage difference between when the work order settlement was last performed and the usage when this current work order settlement was performed. Can be thought of as Component Life.
Strategy	This box is selected if the work order settlement's codes match that of a strategy task in the equipment's current projection.
Linked	This box is selected if the work order settlement is linked to a work order in the Equipment Status module of AMT.

Field	Description
MOT	If the work order settlement was created manually by the AMT user, the MOT code can be edited on the settlement detail screen. If the settlement is created from an external system, the MOT code will be read only. If the work order settlement is linked to an AMT planning work order, the MOT code will be read only and will come from the linked planning work order.

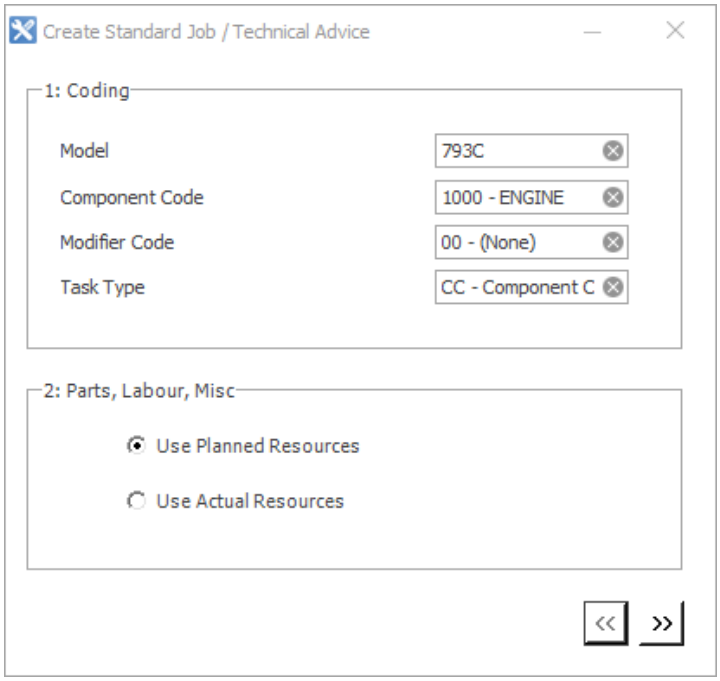
Create a standard job from a work order

If you execute a work order which is considered to be a "standard" for how this job should be done in future, you can trigger the Technical Advisor process.

Open the relevant work order. Click **Quick Links** then click **Add Std Job/Technical Advice** to trigger the creation of a standard job on the work order.



This opens the Create Standard Job /Technical Advice dialog box.



The **Coding** section lets you define standard job coding. It defaults to the work order codes.

The **Parts, Labour, Misc** section lets you flag whether planned or actual parts, labour or misc details should be the basis of the standard job.

Click the **right arrow** button to move forward. Check the standard job details and click **Save**.

Finding Work Order Settlements

In Work Order Settlement Manager (**Daily Operations** > **Workorder Cost Analysis** > **Workorder Cost Analysis**, or **Reporting & Analyses** > **Costs** > **Costs**) you can:

1. Use a wide variety of filters to search for sets of work order settlements.
2. Edit a work order settlement imported into AMT from an external interface.
3. Merge two or more work order settlements to match AMT strategy tasks.
4. Flag work order settlements as reviewed.
5. Add manual work order settlements.

Special filters

Filter	Description
Linked to Equipment	Used to find work order settlements which have been imported into AMT from an external system, but the data may contain incorrect equipment information (for example, incorrect serial number). Therefore, AMT will not be able to link the work order settlement record to an equipment.
Strategy	Include or exclude work order settlements that are linked to strategy tasks.

Filter	Description
Job Status	Indicates whether the maintenance (the costs of which the work order settlement is capturing the costs of) is Yet to Start, In Progress or Completed.
Link to work order	Include or exclude work order settlements linked to work orders in Equipment Status.
Before Equip Start	Include or exclude work order settlements which have a Job Date prior to the Start Date of the equipment.
AMT WO	Include or exclude work order settlements manually created in AMT.

Grid fields

Column	Description
Checked	Indicates whether someone has edited any details of the work order settlement in AMT. If selected, the work order settlement's codes will not be changed when the work order settlement is next imported from the ERP system.
Equipment	If the work order settlement is linked to an asset, this column will show the asset number, otherwise it will be blank.
WIP	If selected, this check box indicates the work order settlement is a Work In Progress. If cleared, this means that the work order settlement is closed.
Contract Total	Work order settlement's total cost based on the asset's default cost bearer.  For equipment under maintenance contract, this cost is shown at Sell.
WO Total	The work order settlement's total cost (regardless of cost bearer).  For equipment under maintenance contract, this cost is shown at Sell.
Component	Shows the work order settlement's codes plus component code description.
Occ Type	Occurrence Type.
Job Code	Displays the work order settlement job code and description.
Date	This is the date the work order settlement was opened, or if linked to a downtime event, it is the date when the work order was performed.
Usage	The asset's primary actual unit of measure reading at the time the work order settlement was begun.

Column	Description
Comp Life	Usage difference between when the work order settlement was last performed and the usage when this current work order settlement was performed. Can be thought of as Component Life.
Strategy	This check box is selected if the work order settlement's codes match that of a strategy task in the equipment's current projection.
Linked	This check box is selected if the work order settlement is linked to a work order in the Equipment Status module of AMT.

(Rpt_257) Workorder Settlement Interrogation

Analyse the work order costs. Gives users the ability to monitor spending throughout a period, and for reconciliation of costs against the client enterprise resource planning system.

Tips

Always shows the total cost of the Job a the Job (Work order) Dare, even though some reports may split these costs over different transactional months (for Transaction Date Reporting).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|AutoLaunch|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Sub System, Task Type, Modifier Code, Job Code, WO Status, System, Strategy Task Desc, Strategy Task, Component Code, Task Counter, Cost Bearer, Cost Selector

Total Costs (including Overheads)

(Rpt_242) Total Cost Worksheet

Lets you analyse costs by several different categories within the budget period. You are able to group the report by three categories for viewing particular cost information.

This report can be run to include overheads in the total cost.

Tips

This report will return results for both real and modelling if there is a mixture of equipment in the fleets selected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

CC Activity, CC Location, CC Responsibility, Cost Centre, Cost Bearer, Expense Element, Cost Category, Include, Costs

(Rpt_155) Overhead Cost Variance Analysis

Provides you with the ability to analyse the overhead costs of a project against a projection.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Overhead Category, Expense Element, Cost Centre, Overhead Code.

Unassigned Costs

(Rpt_243) Unassigned Cost Analysis

Allows the unassigned task costs in the current projections to be analysed in either a text or a graphical view. The "Report On" parameter provides an option to use the ERP codes on work order settlements or the AMT codes the work order settlement is matched with. AMT codes exist for component code, modifier code and task type, as well as cost type, system and sub system, as these are determined from the component hierarchy.

"Analyse By" Component Code identifies the components that have the highest unassigned costs.

Tips

The filters allow a thorough dissection of the costs by different cost bearers and by parts, labour and misc tasks.

This report always shows the total cost of the Job at the Job (Work order) Date, even though some reports may split these costs over different transactions.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Contract, Cost Type, Component Code, Modifier Code, Task Type, Report On, System, Sub System, Cost Bearer, WO Status, Task Counter, P/L/M

(Rpt_244) Unassigned Cost per UOM - by Date

Shows the unassigned costs per unit of measure for the entire equipment term to date.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_245) Unassigned Cost per UOM - by Usage

Shows the unassigned costs per unit of measure by elapsed machine usage for the entire equipment term to date.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_246) Unchanged Cost per UOM - Comparison

Shows the unassigned costs per unit of measure by elapsed machine usage for the entire equipment term to date.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

Maintenance Efficiency

Planning

(Rpt_126) Backlog Ageing

This report displays two bar graphs showing the average days and the average number of backlogs. You are able to select the different task types or priority to assist with the analysis. Navigate to **Reporting & Analyses** > **Maintenance Efficiency** > **Planning**.

High backlogs are one of the key maintenance "lead indicators" and can indicate a number of potential issues.

1. Increase in maintenance workload from:
   - a change in the application/operating practices.
   - product problems.
   - quality of maintenance work.
2. Shortage of maintenance labour.
3. Shortage of maintenance parts.

As backlogs increase, the risk of failures increases. It also highlights a potential future drop in availability. The age of the backlog is a good way of determining the effectiveness of planning, scheduling and execution.

High levels of recent defect backlogs indicate maintenance execution is identifying new work during inspections. Low levels of older work orders indicate that backlogs are being executed and closed out in a timely manner.

Tips

Good practice requires the planner to manage all aspects of the known workload by having an up-to-date backlog listing, correctly coded to facilitate good planning, and ensure efficient and effective planning processes

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Priority, Task Type

(Rpt_357) Backlog Work Analysis

This report shows the average planned labour hours and average planned duration hours.

High backlogs are one of the key maintenance "lead indicators" and can indicate a number of potential issues.

1. Increase in maintenance workload from:
   - a change in the application/operating practices.
   - product problems.
   - quality of maintenance work.
2. Shortage of maintenance labour.
3. Shortage of maintenance parts.

As backlogs increase, the risk of failures increases. It also highlights a potential future drop in availability. The age of the backlog is a good way of determining the effectiveness of planning, scheduling and execution.

High levels of recent defect backlogs indicate maintenance execution is identifying new work during inspections. Low levels of older work orders indicate that backlogs are being executed and closed out in a timely manner.

Tips

This report only displays backlog tasks.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Priority, Strategy Task, Task Type

(Rpt_229) Service Accuracy

Measures the % of preventive maintenance services performed within a nominated tolerance (normally 10%) of the defined interval. To edit tolerances, right-click on the Fleet in the Explorer.

Regular, timely and quality servicing forms the backbone of good maintenance practices. In addition, it is often a warranty requirement. The Service Accuracy report measures the efficiency of the planning process to plan and schedule services in line with the maintenance strategy. Not performing regular services can lead to breakdowns and unplanned stoppages, and also have a long-term impact on component lives. The Services Accuracy Variance report shows more detailed service information than the Service Accuracy report, and more filters to analyse Service Accuracy information.



Incorrect or infrequent capture of utilisation (usages) can also lead to poor service accuracy.

Navigate to **Reporting & Analyses** > **Maintenance Efficiency** > **Planning**.

Tips

Provides basic information in either text, line or bar graph.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|LogoLocation|,
|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,
|EqpPlan|,|AnalyseBy|

Report parameters

Default.

Additional filters

None.

(Rpt_230) Service Accuracy Variance

Measures the % of preventive maintenance services performed within a nominated tolerance (normally 10%) of the defined interval. To edit tolerances, right-click on the Fleet in the Explorer.

This report shows more detailed service information than the Service Accuracy report, and more filters to analyse Service Accuracy information.



Incorrect or infrequent capture of utilisation (usages) can also lead to poor service accuracy.

Navigate to **Reporting & Analyses** > **Maintenance Efficiency** > **Planning**.

Tips

Date range can be set according to analysis requirements.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,
|LogoLocation|,|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,

|Site|,|Fleet|,|EqpPlan|,|AnalyseBy

Report parameters

Default.

Additional filters

None.

(Rpt_219) Planned Downtime Hours %

Ascertains the accuracy of the planned work being performed and analyses the planned downtime. This report also measures the percentage of downtime hours that were flagged as planned. When analysing the downtime, analyse by activity, looking for planning-related downtime (that is, time spent waiting for parts).

For both the Events and Hours, this report indicates the level of "control" with the maintenance organisation. If the trend is down, it indicates either reliability problems or planning inefficiencies.
An analysis of "non-planned" and "non-breakdowns" is a good way to identify planning inefficiencies. Planning inefficiencies lead to increased stoppages and downtime, that is, work that could have been done in a PM gets done as a separate stoppage.

Navigate to **Reporting & Analyses** > **Maintenance Efficiency** > **Planning**.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|SiteID|,|Site|,|FleetID|,|Fleet|,
|EqpPlanID|,|EqpPlan|,|AnalyseBy|,|Responsibility|,|ResponsibilityID|,
|ProjectionTypes|,|UserID|,|BranchID|,|Branch|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Responsibility, Task Type, Strategy Task

(Rpt_493) Maintenance KPI Summary

This is a downtime text report to show key parameters and target values.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Responsibility, Availability

(Rpt_508) Work Loading - Labour

Compares available labour to that which was planned, and to that which was actually used.

Tips

This report utilises snapshots of the seven day weekly work plan created in AMT. The day on which the work plan is saved is site specific and, therefore set up using site administration.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Subsystem, Component, Task Type, Job Ready Status, Work Group, Priority

(Rpt_509) Labour Resource Scheduling & Utilisation

Lets you understand the volume of work to be done in the next month, and how much is scheduled.

Tips

This report utilises snapshots of the seven day weekly work plan created in AMT. The day on which the work plan is saved is site specific and, therefore set up using site administration.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Report Type, Date Selector

Additional filters

Work Group

Daily Operations

(Rpt_191) Maintenance Ratio

Measures the man hours to operating hours. If it is high then it indicates either reliability issues or low skill levels. A useful analysis is to trend this report and exclude component changeouts (remove the big spikes). If the trend is up this indicates issues.

Navigate to **Reporting & Analyses** > **Maintenance Efficiency** > **Daily Operations**.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|CostBearerID|,
|AutoLaunch|,|EqpPlanID|,|EqpPlan|,
|DateChecked|,|StartDate|,|EndDate|,|LabourType|,|ActualProjected|,
|AnalyseBy|,|ProjectionTypes|,|UserID|,|BranchID|,|SiteID|,|FleetID|,|Branch|,
|Site|,|Fleet|,|ProjectionHeaderID|,|ProjectionHeader|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Cost Bearer, Task Type, Strategy Task, Job Code, Labour, Show, Shop Job Codes

(Rpt_432) Labour Hours per Interval

Shows labour hour requirements per interval.

Tips

This report includes actuals for Current and Alternate projections.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Item, Item, Labour Type

Mean Time to First Session After PM

This report goes hand in hand with the Service Accuracy report, and measures the quality of PM Services. A down event after a PM indicates poor quality PM services, which directly impact costs and productivity. An extended interval until the first stoppage after PM indicates a quality PM service.

View this report in conjunction with the Service Accuracy, Event Analysis and Event Completion Performance reports, as well as Backlog age.



Event analysis is only available if Advanced Planning is active.

(Rpt_202) MTFS_After_PM

This report analyses the hours between a PM service and the first stoppage that occurs after that PM service. This assists with understanding the reliability of the PM services being undertaken by the technicians.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,|LogoLocation|,
|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,
|Fleet|,|EqpPlan|,|AnalyseBy|,|ResponsibilityID|,|PMTaskTypeID|,|PMTaskType|,|FSTaskTypeID|,|FSTaskTyp
e|,|DateChecked|,|StartDate|,|EndDate|

Report parameters

Default.

Additional filters

PM Task Type*, +FS Task Type*, +FS Breakdown, PM Released By,
+FS Responsibility

* These filters define the attributes of the First Stoppage (FS).

(Rpt_203) MTTR

Analyses the time taken to complete a maintenance event. This may assist management in understanding the technical level of personnel, show faults in repair materials, and so on.

By itself, the MTTR (Mean Time To Repair) report does not mean a lot. An increasing MTTR could indicate many things - better planning (fewer and longer stoppages) or poor skill levels (tasks take longer). Therefore, it is important to view this report in a context, with the MTBS (Mean Time Between Stoppages) or MTBF (Mean Time Between Failures) report.

If MTBS is increasing then you would expect MTTR to increase. If, however, MTBS is constant and MTTR increases, then this indicates issues (poor skill levels).

Use the filters to capture only Breakdowns and avoid component changeouts (remove the spikes). This indicates the average time to repair the smaller general breakdowns and is a good indicator of response times (availability of people) and skill levels.

If Advanced Planning is active, use the Event Analysis report and review in conjunction with Event Delay Causes and Variances. This assists in identifying other causes that are non-skills related

Tips

You can run this report can be run by System or Sub system to assist with understanding the level of technical expertise on a site or shift.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,|UserID|,
|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,
|EqpPlan|,|AnalyseBy|,|ResponsibilityID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Planned, Responsibility, Task Taype, Component Code, Breakdown

(Rpt_326) Redo Analysis

Provides information on tasks marked as Redo, also known as rework, with the new costs and the costs of the task to be redone.

An analysis of any work that is being completed a second time, generally as a result of poor workmanship or lack of skill set.

This is not a KPI report, but a good report to view if any of the other reports are indicating potential work quality issues.



The Supervisor or Planner marks the work order as redo to track this work.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Sub system, Component Code, Cause, Part Number

(Rpt_131) Breakdown Downtime Events %

Measures the overall quality of the work performed. Like Mean Time Between Stoppages, it is also a measure of the reliability delivered to the customer. High levels of breakdowns can indicate a number of issues. Use the different filters and analyse by options to trend, then drill down to identify the cause. When analysing, the breakdowns focus on the components and cause of failures and how the failures could be prevented.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,|UserID|,|AutoLaunch|,
|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|,
|AnalyseBy|,|ResponsibilityID|

Report parameters

Default.

Additional filters

Cost Type, Responsibility, System, Task Type, Sub System, Strategy Task, Component Code

(Rpt_132) Breakdown Downtime Hours %

Measures the overall quality of the work performed. Like Mean Time Between Stoppages, it is also a measure of the reliability delivered to the customer. High levels of breakdowns can indicate a number of issues. Use the different filters and analyse by options to trend, then drill down to identify the cause. When analysing the breakdowns focus on the components and cause of failures and how the failures could be prevented.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|ProjectionTypes|,
|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,
|Branch|,|Site|,|Fleet|,|EqpPlan|,|AnalyseBy|,|ResponsibilityID|

Report parameters

Default.

Additional filters

Cost Type, Responsibility, System, Task Type, Sub System, Strategy Task, Component Code

(Rpt_507) Maintenance Compliance

Displays how much of the scheduled work was completed in a week.

Tips

This report utilises snapshots of the seven day weekly work plan created in AMT. The day on which the work plan is saved is site specific and, therefore set up using site administration.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Task Type, Work Group, System, Subsystem

Equipment Evaluation

Summary Evaluation

(Rpt_236) Summary Analysis

Provides summary details of the cost per UOM to run the equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, Cost Type, Contract, Costs, Cost Bearer, Discount Factor %, Revenue

+ indicates that the UOM selected is used as a basis for the equipment evaluation.

Cost per UOM

(Rpt_152) Cost Per UOM - by Date

Compares visually the expected hourly costs by machine. It shows the cumulative cost per UOM for the period of time selected.

"Analyse By" Model/System/Sub System for a date range. Lets you see the trending of unassigned costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_153) Cost per UOM - by Usage

Depicts the costs at intervals of the equipment usage (engine hours) throughout the term of its use.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task

(Rpt_154) Cost per UOM - Comparison

Compares visually the expected hourly costs by machine. The report shows the cumulative cost per UOM for the period of time selected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_429) Cost per UOM per Interval

Similar to the Cost Per Interval report except that this report shows costs as a cost per UOM (for example, engine hours) rather than a total cost.

Tips

This report does not include actuals for Current and Alternate projections.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Labour Activity, Job Code, Costs

Residual Value Analyses

(Rpt_227) RVIC - Chart

Displays the movement of Residual Life over the period of the chart dates.

This report charts the residual value in the components of the equipment selected. As components wear out, the value in the component is lost; as components are replaced, the value increases. Therefore, this report determines when the logical end date of the contract should be.

Tips

The chart does not display consolidated Totals for Fleet, only individual Equipment. This means that selections should consider the amount of data to be displayed in the chart. It is recommended that no more than 6 - 10 items of equipment be displayed.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Sub System, Task Type, System, Component Code, Cost Bearer

(Rpt_228) RVIC Analysis

Shows defined periods. Components can be used as a basis for contract variation or termination, and can also be used for assessing contingent liabilities.

Based on the same logic as the RVIC Chart report, but with the added flexibility to select a hypothetical contract end date and see the value each component would have at that date.

This report is important if the contract states that the customer will receive their equipment with a certain minimum value at the end of the contract. Use this report to determine whether the contract you have set up in AMT is designed to meet the customer's minimum value expectation.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Component Code, Task Type, Cost Bearer

Residual value in components

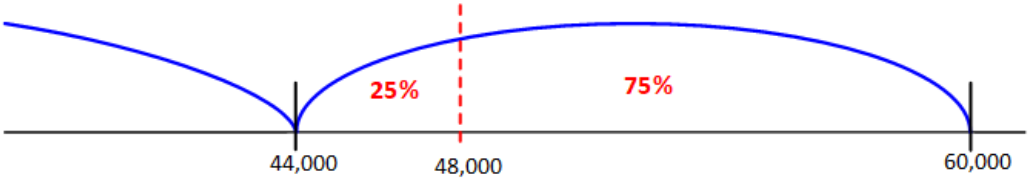
Residual value in components (RVIC) is a key concept that is used to identify risk and opportunities and forecast optimal asset replacement points.

Residual value in components (RVIC) represents how much value remains in a component at a specific point in the future. .

Example

In the example illustration below, a component is scheduled for changeout at 44,000hrs then 60,000hrs. If the asset is due to be scrapped at 48,000hrs, at that point there is 75% of life remaining in that component. If it is a

large component such as an engine, it is not a good time to scrap the asset. Alternatively, there is a large saving that can be achieved if the engine life can be extended to 48,000hrs.



RVIC is shown in the high opportunity and risk reports, component change out schedules and asset termination reports. It is also used in the MARC financial reports to predict contract margins.

Residual Value in Components reports

Residual value in components (RVIC) represents how much value remains in a component at a specific point in the future. This concept is typically used for:

- Varying maintenance contracts.
- Terminating maintenance contracts.
- Assessing contingent liabilities.
- Determining the economic life of an asset.

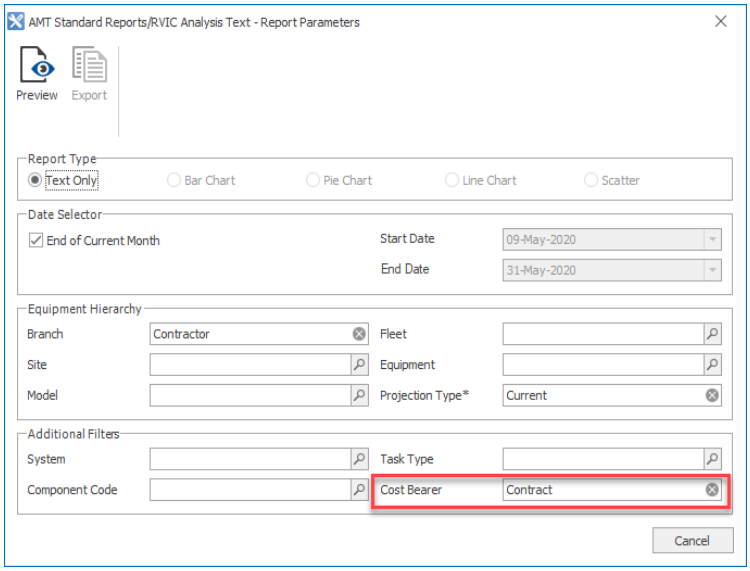
The RVIC calculation is based on:

- Specific end date.
- Current usage of the equipment.
- Cost of the strategy task (that is, the component).
- Percentage of life of the component remaining at the end date.
- Hence, Residual Value = Cost x Life Remaining.

Both the RVIC Analysis and RVIC Chart Report Parameters have been enhanced to include the Cost Bearer. The field supports the ability to multi-select one or more, or all cost bearers.

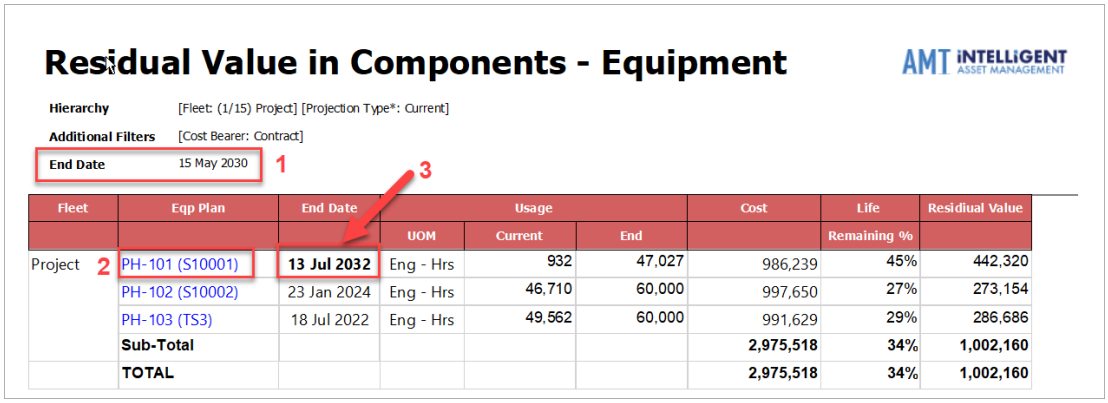
When launched, the report will place the default cost bearer into the Report Parameter field. This will allow the report to perform its calculation as in previous versions.

If you decide to modify the cost bearer, this selection will be remembered as per standard AMT functionality.



The menu option **RVIC Analysis** produces the **Residual Value in Components – Equipment** report. It is now written in SSRS (SQL Server Reporting Service).

The Residual Value in Component – Equipment report displays first; summarised by **Fleet** and **Equipment**.



Residual Value in Components - Equipment

AMT INTELLIGENT ASSET MANAGEMENT

Hierarchy [Fleet: (1/15) Project] [Projection Type*: Current]

Additional Filters [Cost Bearer: Contract]

End Date 15 May 2030 1

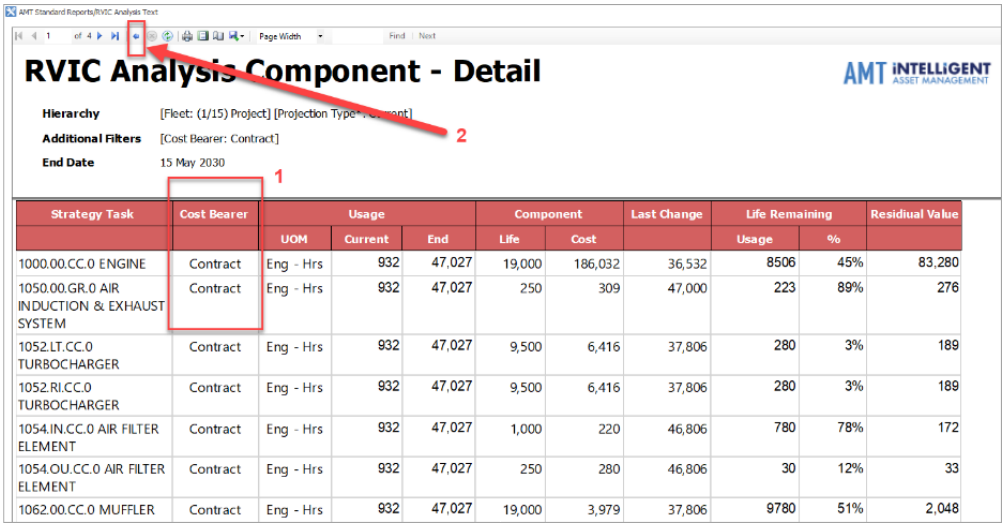
3

Fleet	Eqp Plan	End Date	Usage			Cost	Life	Residual Value
			UOM	Current	End		Remaining %	
Project 2	PH-101 (S10001)	**13 Jul 2032**	Eng - Hrs	932	47,027	986,239	45%	442,320
	PH-102 (S10002)	23 Jan 2024	Eng - Hrs	46,710	60,000	997,650	27%	273,154
	PH-103 (TS3)	18 Jul 2022	Eng - Hrs	49,562	60,000	991,629	29%	286,686
	Sub-Total					**2,975,518**	**34%**	**1,002,160**
	TOTAL					**2,975,518**	**34%**	**1,002,160**

1. End Date found in the Report Header displays the End Date input as a Report Parameter. Previously this was not displayed, but may have been displayed in the End Date column for an Equipment if the Equipment Projected End Date was after the report parameter end date.
2. The Residual Analysis Component - Detail report is available with the blue hyperlink.
3. End Date column in the report displays the Equipment's Projected End Date per the equipment record. If the Equipment's End Date is greater than the End Date for the report, the font will be bold.

When in the **RVIC Analysis Component - Detail** report (screenshot below):

1. New Cost Bearer column displays with information dependent on the report parameters. If no cost bearer selection is made, the report will display the equipment's Component RVIC details for the default cost bearer.
2. You can drill back to the summary report where you may choose to select another equipment to investigate in more detail.



Strategy Task	Cost Bearer	Usage			Component		Last Change	Life Remaining		Residual Value
		UOM	Current	End	Life	Cost		Usage	%	
1000.00.CC.0 ENGINE	Contract	Eng - Hrs	932	47,027	19,000	186,032	36,532	8506	45%	83,280
1050.00.GR.0 AIR INDUCTION & EXHAUST SYSTEM	Contract	Eng - Hrs	932	47,027	250	309	47,000	223	89%	276
1052.LT.CC.0 TURBOCHARGER	Contract	Eng - Hrs	932	47,027	9,500	6,416	37,806	280	3%	189
1052.RI.CC.0 TURBOCHARGER	Contract	Eng - Hrs	932	47,027	9,500	6,416	37,806	280	3%	189
1054.IN.CC.0 AIR FILTER ELEMENT	Contract	Eng - Hrs	932	47,027	1,000	220	46,806	780	78%	172
1054.OU.CC.0 AIR FILTER ELEMENT	Contract	Eng - Hrs	932	47,027	250	280	46,806	30	12%	33
1062.00.CC.0 MUFFLER	Contract	Eng - Hrs	932	47,027	19,000	3,979	37,806	9780	51%	2,048

The **RVIC Chart** report parameters are expanded in line with the **Detail Analysis** reports. The **RVIC Chart** report has not been upgraded to SSRS in this version.

These changes apply across all the **RVIC** reporting within **Real Equipment**, **Centrelines** and **Modelling Equipment**.

Monitor specific contract variance items

Typically there are clauses in full MARC regarding the reconciliation of specific maintenance items. For example:

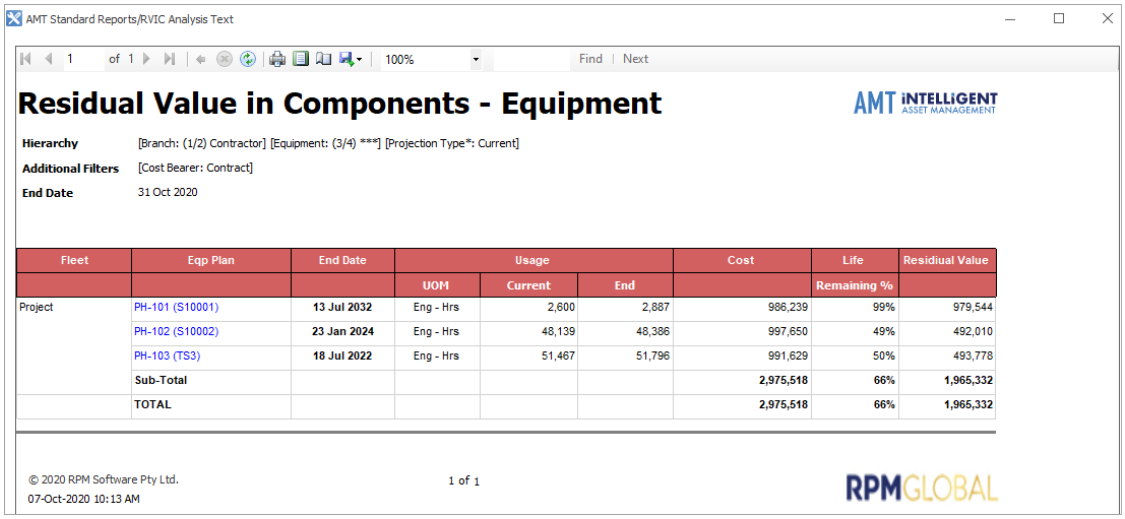
- Reconciling the condition of nominated components at the conclusion of the contract term.
- Reconciling maintenance cost vs. revenue for wear items.

Component condition

AMT's Residual Value in Component (RVIC) report forecasts the expected life remaining and value remaining in components at specific points in time (for example, contract completion).

The report in AMT is the RVIC Analysis report. Navigate to **Reporting and Analyses** > **Equipment Evaluation** > **Residual Value Analysis**.

An example of this report is below.


AMT Standard Reports/RVIC Analysis Text

1 of 1 100% Find | Next

Residual Value in Components - Equipment

AMT INTELLIGENT ASSET MANAGEMENT

Hierarchy [Branch: (1/2) Contractor] [Equipment: (3/4) ***] [Projection Type*: Current]

Additional Filters [Cost Bearer: Contract]

End Date 31 Oct 2020

Fleet	Eqp Plan	End Date	Usage			Cost	Life	Residual Value
			UOM	Current	End		Remaining %	
Project	PH-101 (S10001)	13 Jul 2032	Eng - Hrs	2,600	2,887	986,239	99%	979,544
	PH-102 (S10002)	23 Jan 2024	Eng - Hrs	48,139	48,386	997,650	49%	492,010
	PH-103 (TS3)	18 Jul 2022	Eng - Hrs	51,467	51,796	991,629	50%	493,778
	Sub-Total					**2,975,518**	**66%**	**1,965,332**
	TOTAL					**2,975,518**	**66%**	**1,965,332**

© 2020 RPM Software Pty Ltd.
07-Oct-2020 10:13 AM

1 of 1

RPMGLOBAL

Residual Value = 1 - ((End Usage - Last Change) / Life) x Cost.

Reconciling wear items

Some maintenance contracts have clauses concerning the performance of specific parts. That is, specific groups of parts have a specific budget cost and revenue stream which is constantly monitored and if significant variances occur, there can be a sharing of the costs. A common example are wear items. For such contracts, there should be multiple billing projections, such as a billing schedule for the wear items, or undercarriage, and a second billing schedule for the balance of the machine.

Separate actual billing records, for each billing schedule, are imported into AMT like any other billing.

The report in AMT is the Contract Billings vs Cost Reconciliation report. Navigate to **MARC** > **MARC Admin** > **Billings**. An example of the report is shown below.

Contract Billings vs Cost Reconciliation

Hierarchy : [Equipment: DOZ01: D10R]

Date Range : [May-2003 - May-2009]

Additional Filters : [System: (+1/15) Implements/GET/UC] [Sub System: (+1/3) Undercarriage] [Parts/Labour/Misc:Parts,Labour,Misc]

Analyse By : Sub System, Year

Equipment	Schedule	2003	2004	2005	2006	2007	2008	2009	Total
DOZ01	Implement/GET/UC	59,597	185,628	430,941	429,667	256,919	0	0	**1,362,752**
Total		**59,597**	**185,628**	**430,941**	**429,667**	**256,919**	**0**	**0**	**1,362,752**

Sub System	2003	2004	2005	2006	2007	2008	2009	Total
Undercarriage	46,128	69,340	127,346	158,169	322,725	0	0	**723,709**
Total	**46,128**	**69,340**	**127,346**	**158,169**	**322,725**	**0**	**0**	**723,709**
Variance	**13,469**	**116,288**	**303,595**	**271,498**	**-65,806**	**0**	**0**	**639,043**

Web Dashboard

System configuration and settings

KPI Calculation

A system task runs on the AMT server every hour to recalculate the statistics in the Dashboard. This process is automatically postponed to the next hour if the load on the AMT server is too high.

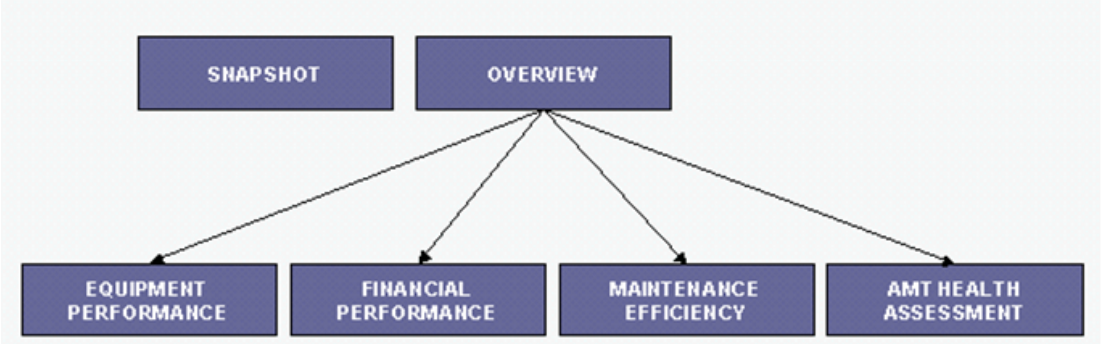
Scheduled hours

If scheduled hours have not been set up for an equipment, then the **Availability** statistic in the Dashboard will report **NaN** (no date). Create scheduled hours using the function **Scheduled Hours**.

Multiple units of measure

An asset can be linked to a usage profile, which can contain multiple units of measure. If more than one of these units of measure is flagged in the Units of Measure system table as being **Operating Hours**, this may cause usage-based statistics in the Dashboard to calculate inaccurately.

Dashboard structure



The Dashboard is not limited to Equipment Performance reporting; it can be used for a wide variety of reporting. Being web-based, it has the flexibility to be accessed by a wide range of users in an organisation.

The actual display is spreadsheet style with colour indicating levels of performance. Actual performance statistics are shown, and depending on the benchmarks set up in the Dashboard, shading indicates the result of the measurement as shown in the table below. This helps you to quickly see areas of poor performance which need follow-up.

Key	Description
	Good

Key	Description
	Normal
	Poor
	Alert
NaN	No value, that is, no data on which to calculate the measurement.

Dashboard navigation

Navigation within the Dashboard is done using your mouse and standard Internet browser buttons, Back, Forward, Print and so on.

The opening screen is the Snapshot. It displays the total number of machines registered and a summary of key measurements at the branch level, however, depending on how AMT was set up, the highest level may be the company. Drilling down into the measurements is possible, from the highest level to the lowest (individual equipment plans).

Navigation to dashboard measurements is shown in the table below.

Section	Group	Sub-Group	Measurement
Snapshot *Last 48 Hours*			Equipment Currently Down
			Availability
			Utilisation
			Strategy Tasks - Overdue
			Backlog Ratio
			WIP (days)
			Availability
			Reliability (MTBS)
			PCV%

Section	Group	Sub-Group	Measurement
	Financial Performance	Risk	Net Opportunity Risk
			% of Projected Life-Cycle Costs
		Financial Margin	Projected Billings
			Budget Margin %
			Projected Margin %
			Risk Adjustment Margin %
			Avg Margin %
		Cash Flow to Date	Cash Flow to Date
		AMT Health	AMT Health
	Maintenance Efficiency	Strategy	PCV%
			Strategy Tasks - Action Required
			Strategy Tasks - Overdue
		Planning	Backlog Ratio
			Planned DT % (DT = Downtime)#
			Service Accuracy#
		Daily Operations	Maintenance Rate#
			Breakdown (Events) %#
			MTFS after PM#
			MTTR#
			WIP (Days)@
		Customer Service	Budget Variance

Section	Group	Sub-Group	Measurement
	AMT Health		Budget Variance
			Strategy Tasks - Action Required
			Strategy Tasks - Overdue
			Strategy Tasks - Not linked to WOs



#000 Rolling 3 month KPI.
@ Rolling 12 month KPI.

Dashboard calculations

The table below contains an alphabetical listing of all measurements reported by the Dashboard, and a description of how the figure is calculated, regardless of date range or time period.

Date range and time period is dependent upon the section of the Dashboard in which a measurement is located. For example, measurements under Snapshot are calculated using data from the last 48 hours. Measurements calculated under Overview are month to date. The table below only describes the base calculation method, regardless of the time period.

Measurement	Calculation Method	OEM Specific
AMT Health	Average of the health indicators.	
Availability	Depends on the availability set up in AMT for each fleet. PA = Physical Availability. MA = Mechanical Availability.	
Backlog Ratio	Number of backlog tasks / Number of equipment	
Breakdown % #	Number of events where breakdown = True / Total number of events.	
Budget Margin %	(Projected billing for the budget projection - Projected cost@cost for the budget projection) / Projected billing for the budget projection.	Y
Budget Variance	Projected billing for the budget projection - Projected cost@sell not escalated for the budget projection.	Y
Cash Flow to Date	Actual billings to date - Work orders cost@cost where cost bearer = <default cost bearer>.	Y

Measurement	Calculation Method	OEM Specific
Equipment Currently Down	(Number of equipment currently down / Number of equipment) * 100.	
Maintenance Ratio#	Labour hours / Equipment usage.	
MTFS After PM#	Mean time to first stoppage after PM. Total of usage (in Eng-Hrs) from PM event to the next downtime event / Total number of PM events. A PM event is where there is a PM task in the event.	
MTTR#	Total downtime for the events / Total number of events.	
PCV%	(Projected total budget projection = Projected total current projection) / Projected budget projection.	
Planned DT% (DT = Downtime)#	Total downtime of events where planned = true / Total downtime of all events.	
Projected Billings	Total projected billing based on the billings profile for the current projection.	Y
Projected Margin %	(Projected billing for the current projection - Projected cost@cost for the current projection) / Projected billing for the current projection.	Y
Reliability (MTBS)	Total usage of the equipment / Total number of events.	
Risk: Net Opportunity Risk	The sum of the opportunity of risk components (life used <20% at the end of the equipment life) - the sum of the risk components (life remaining <20% at the end of the equipment life). A positive value is greater opportunity than risk.	
Risk: % of Projected Life-Cycle Costs	Net opportunity risk / Projected sell of current projection.	
Risk: Adjustment Margin %	(Projected billing for the current projection - Projected cost@cost for the current projection + Net opportunity risk components) / Projected billing for the current projection.	Y
Risk: Avg Margin %	([Projected margin %] + [Risk adjust margin %]) / 2	Y
Service Accuracy#	The number of services (PM events) within the allowed tolerance for a PM / Total number of PM events (tolerance is set using **Fleet Administration**).	

Measurement	Calculation Method	OEM Specific
Strategy Tasks - Action Required	The number of strategy tasks which are either overdue or (in the planning window and are system calculated and not reviewed) / Total number of strategy tasks. Only Strategy tasks which are flagged as planning tasks are considered in this statistic.	
Strategy Tasks - Not linked to WOs	The number of strategy tasks in completed or in progress events in the last 12 months that are not linked to a work order.	
Strategy Tasks - Overdue	(Number of strategy tasks overdue / Number of strategy tasks) * 100. Only strategy tasks which are flagged as planning tasks are considered in this statistic.	
Utilisation	Usage / Number of equipment.	
WIP (days)@	(Total WIP cost / Total work order cost last year) * 365.	Y

Rolling 3 month KPI.
@ Rolling 12 month KPI.

For those measurements which are based on a date range greater than one month, the date range used includes prior months plus month to date. For example, the last quarter is defined as the last three months as well as this month to date. Therefore, if the current date is 12-Jun-19, the last quarter is: 1-Mar-19 to 12-Jun-19 (March-April-May PLUS June to date).

Dashboard recalculation

You can customise the Dashboard to recalculate as often as practical given the nature of sites and equipment registered on the system and the AMT server. For example, if there are not many breakdowns, recalculation may only be required every four hours.

The System Administrator sets the frequency of the Dashboard's recalculation in the registry on the AMT server. The process takes several minutes.

You are not notified by the system when a recalculation is taking place.

Dashboard targets

For every measurement reported on the Dashboard there are four targets (benchmarks) that define the range of calculation results equating to Good, Normal, Poor, and Alert performance. Desired targets can be applied to all fleets across all branches (known as defaults), or specific targets can be set at the fleet level.

Access to editing the targets is typically granted to the AMT Administrator only.
It is recommended that the determination of target ranges however, should be a joint decision between site, branch and Head Office management

Printing the Dashboard

There are two ways to print the Dashboard:

Using your internet browser

If you want a hardcopy printout of the Dashboard itself , navigate within the Dashboard to the measurements you want to print. Make sure the desired measurements are displayed on the screen and then, using your internet browser, select **Print** from the browser's main menu or click the **Printer** button located on the browser's toolbar. What you see on your screen should be printed directly to the chosen printer.

Copy, paste or crop

Navigate to the particular measurements from the Dashboard which you want to print. Take a screen shot by pressing **Print Screen** on your keyboard. Open your word processing application and press **Ctrl-V**. The image should now be visible on the page. Then use a cropping tool to tighten the image as required.

Individual hard copy reports for each of the measurements can be generated from within AMT. This allows you greater flexibility in specifying report types, such as Pie Chart, Text and Line Chart, and parameters such as precise Date Ranges, Filters and Analyse By criteria.

When printing the Dashboard using your internet browser, you may have to change the page orientation from portrait to landscape.

RPMGLOBAL

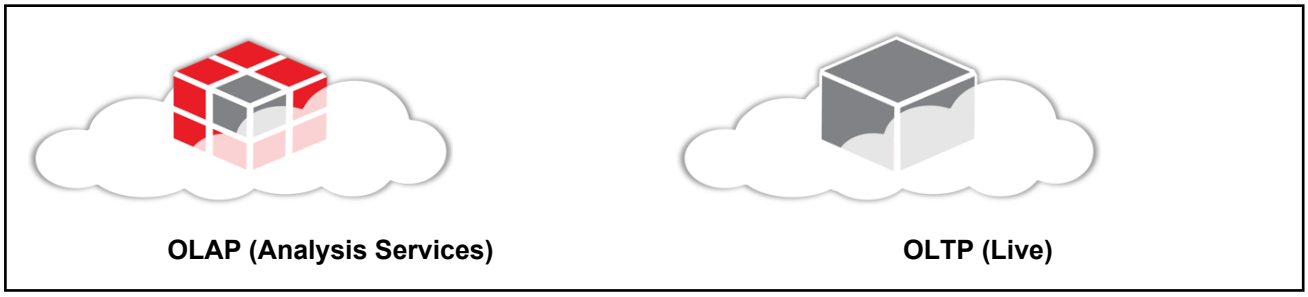
AMT IC

Intelligence Centre - Report Writing module overview

The AMT Report Writing module lets you easily write your own customised reports, tapping into AMT's existing Data Structures. Reports can be written so that they retain the look and feel of existing AMT reports, or can be customised in a variety of ways to meet specific needs.

Reports are written using Microsoft's Report Builder.

How does it work?



OLTP	Cubes	Availability and Scheduled Hours Budgeting Cost Stoppages and Downtime Strategy Task
	Views	Budgeting Equipment Statistics Utilisation Workflow Manager
OLAP	Cubes	Work Order Actual Downtime

AMT data models

AMT reports are written off two types of database; the live, On Line Transactional Processing database used by AMT, and a data warehouse On Line Analytical Processing database. It is important that the differences between the two are understood so that anyone writing reports can decide which is appropriate for any particular reporting requirement.

Live data models tap directly into AMT's transactional database and display the most updated information, whereas On Line Transactional Processing data models do not display live data (given that data needs to be extracted and consolidated), but they do allow customers to consolidate and query data from multiple AMT installations in a single place.

Consolidating data into an AMT data warehouse is an advanced feature used by organisations with multiple AMT installations (for example, a multinational with separate AMT installations in several different countries).



By default, the process of extracting and consolidating data occurs daily through a system task. There is some flexibility in terms of how often this task can be scheduled (depending on data volumes). Please contact your System Administrator to discuss the possibility of shortening the intervals.

Cubes vs views

While the approach to writing reports using either models or views is essentially the same, there are a few steps in the process of developing the individual datasets and filters that may differ within SQL Server's Report Builder. These differences, wherever they occur, are highlighted in various help topics.

Data model reference

A list of data models is below.

Available live data models

1. Availability and Scheduled Hours Cube.
    - Budgeting cube, budgeting views
    - Cost cube
    - Stoppages and downtime cube
    - Strategy task cube
    - Equipment statistics views
    - Utilisation views
    - Workflow manager views

In addition, these dimensions can also be used as data models, that is, these dimensions are useful in their own right for reporting.

1. Customer view.
2. Equipment details view.
3. System table view.

Available On Line Analytical Processing (consolidated) data models

1. Work orders.
2. Actual downtime.
3. PSM.

Each *View* data model has the following details:

1. List of views and tables exposed by the data model.
2. List of all attributes exposed by the views.

Each *Dimension* has the following details:

1. Description of attributes.
2. Structure of any hierarchies, for example, component code hierarchy, equipment hierarchy.

Accessing AMT data models

Available data models can be found under:

{server location}/ReportServer/Amt Data Models/Live Models/AMT Data Model – Live Cubes

{server location}/ReportServer/Amt Data Models/Live Models/AMT Data Model – Live Views

{server location}/ReportServer/Amt Data Models/OLAP Models/OLAP Cubes

Administration

IC Equipment Manager

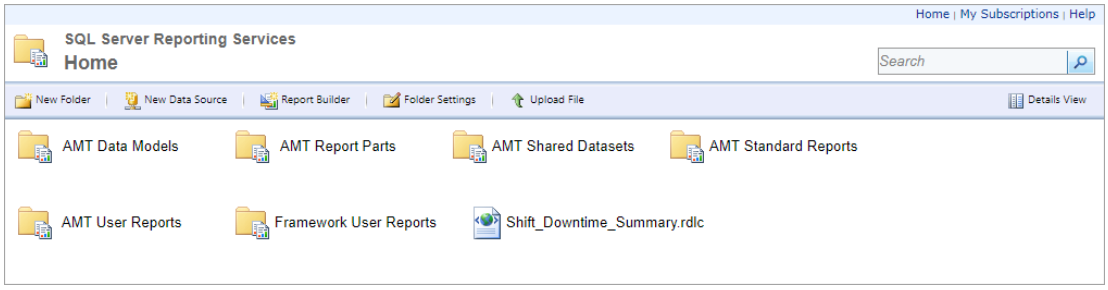
Complete documentation for this topic will be available in a future release.

IC System Task Manager

Complete documentation for this topic will be available in a future release.

Report Manager

The default configuration of SSRS Report Manager is shown in the example screenshot below.



AMT Data Models

Contains shared data sources that provide access to AMT data models. It is recommended that you use these shared data sources, rather than embedding data sources in specific reports – this ensures that appropriate credentials are used to connect to SQL Server, SSRS, and SSAS. The data sources that are available are as follows:

- Live Cubes.
- Live Views.
- OLAP Cubes.
- OLAP Views.

AMT Report Parts

Contains reusable items such as report templates (see Report Templates for more details on how to use report templates).

AMT Standard Reports

Contains reports based on the AMT Data Models (written and tested by RPM) that ship with every AMT installation.

AMT User Reports

Contains reports that you and other users write.

Launching Report Builder

You can launch Report Builder in the following ways.

From within AMT

1. Navigate to **AMT IC** > **Administration** > **Report Manager**.
2. From the Report Manager main screen, launch **Report Builder**.

From SQL Server Reporting Services (SSRS) Report Manager

1. In a web browser, enter the URL for your report server in the address bar. The default URL for Report Manager is http: //<servername>/Reports.
2. From the Report Manager main screen, launch **Report Builder**.

Using the ClickOnce URL

- http://<servername>/reportserver/reportbuilder/ReportBuilder_3_0_0_0.application

  This option assumes SSRS has been installed with default options.

Microsoft Report Builder

Report Builder is a report authoring tool written by Microsoft that ships with SQL Server Reporting Services (SSRS). Report Builder features a Microsoft Office-like authoring environment with enhanced data layout, data visualisation, richly formatted text, and on-demand rendering.

Report Builder is used to write reports that access AMT data. Report Builder version 3.0 was shipped for the first time with SQL Server 2008 R2 and the help topics related to Report Builder are based on Report Builder 3.0 functionality.

SSRS Fundamentals

This topic provides an overview of SQL Server Reporting Services (SSRS). Click the links below for more information.

Report development process

It is recommended that you follow the steps below to develop reports.

1. Select a tool.
2. Decide where to get the data (Data sources).
3. Decide what data to get (Datasets).
4. Design report layout (Report items).
5. Preview Report.
6. Deploy report to report server.
7. Publish report to AMT users.

Data sources

A data source is identified by a connection string which defines where to get the data from, and authentication details.

SSRS supports a broad range of data sources, for example, SQL Server, SQL Azure, Oracle, OLE DB, ODBC, Analysis Services, SAP NetWeaver BI, Hyperion, XML.

A data source can be shared by many reports or embedded in one report.

Datasets

A dataset defines a query for a data source and contains a collection of fields based on data columns. Both text and graphical query designers are provided in Report Builder 3.0.

A query can be one of several types:

- Text queries selectively query data.
- Table queries select all data from a table.
- Stored procedure queries retrieve data using stored procedures in the database.

A dataset has a time out value (in seconds). It is recommend that report queries are set to a reasonable timeout value (such as 30 seconds) – this minimises the impact on the database if queries are slow. A dataset consists of several collections:

Fields - Columns retrieved by the query or calculated based on an expression.

Filters - Used to exclude rows after every query execution.

Parameters - Used to modify the **where** clause of query text, or to provide input for a stored procedure.

From SQL Server 2008 R2 and later, shared datasets are supported. A shared dataset is actually an RSD file, which complies with a particular XML schema designed for datasets.

Report items

Report Items are disconnected from data.

Tablix Data Regions are bound to data and are used to display data in a similar format to how it is viewed in the query designer (with optional summarisation).

Graphical Data Regions use data visualisation techniques to summarise data for comparative purposes or for pattern discovery.

Type	Icon	Name	Description	Earliest Version Supported
Report Item		Textbox	Static text or dynamic text from a dataset or an expression	2000
		Line	Graphic element with defined endpoints, color, and width	2000
		Rectangle	Graphic element or container for free-form layout of multiple report items	2000
		Image	Binary image data embedded in report, stored externally, or retrieved by a dataset	2000
		Subreport	External report embedded in a parent report	2000
Tablix Data Region		Table	Structure of textboxes with fixed columns and dynamic, data-driven rows	2000
		Matrix	Structure of textboxes with dynamic, data-driven rows and columns	2000
		List	Free-form structure with data-driven repeating section	2000
Graphical Data Regions		Chart	Graphic display of data in standard chart form	2000
		Gauge	Graphic display of performance data in gauge form	2008
		Map	Graphic display of spatial data	2008 R2
		Data Bar	Graphic display of a single point as a bar or column	2008 R2
		Sparkline	Graphic display of a data series over a defined period of time	2008 R2
		Indicator	Graphic display of conditional data using icon sets	2008 R2

Tablix Data Regions

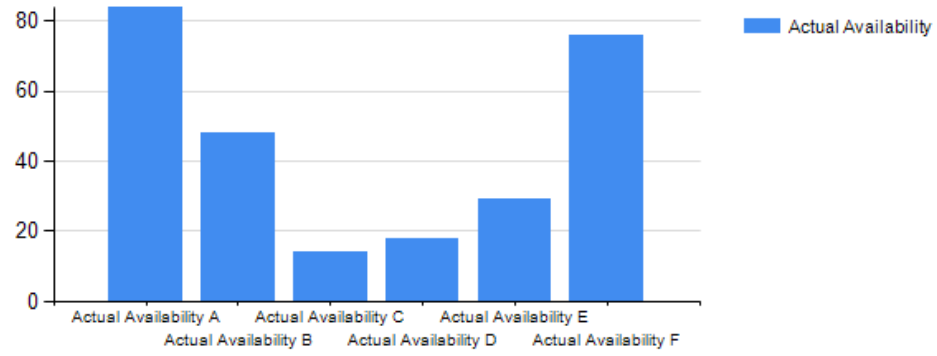
Table: Displays a fixed number of columns.

Year	Sales Amount
Calendar 2006	$30,674,773.17
Calendar 2007	$42,011,037.16
Calendar 2008	$25,828,762.10

Matrix: Displays a variable number of columns (*crosstab*).

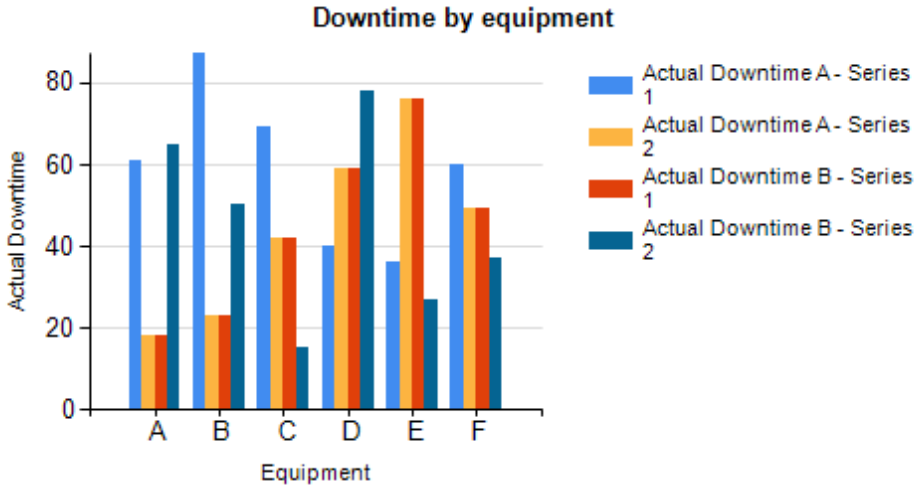
Category	Quarter 1, 2007	Quarter 2, 2007	Quarter 3, 2007	Quarter 4, 2007
Accessories	34..12%	33.24%	46.47%	52.52%
Bikes	3.89%	3.59%	1.40%	12.58%
Clothing	18.44%	18.52%	8.54%	15.28%
Components	5.14%	3.26%	5.05%	6.48%

List: A free-form area that repeats its contents depending on grouping.



Graphical data regions

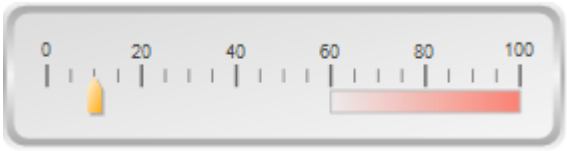
Chart



Data Bar

		Sales Amount
Accessories	2005	20,235
	2006	92,735
	2007	296,533
	2008	161,794
Bikes	2005	7,395,349
	2006	19,956,015
	2007	25,551,775
	2008	13,399,243
Clothing	2005	34,376
	2006	485,587
	2007	871,864
	2008	386,013
Components	2005	615,475
	2006	3,610,092
	2007	5,482,497
	2008	2,091,012

Data Bar (Gauge)



Indicator

Country	Prev Yr Avg Sales	
France	-2.35%	
Germany	3.28%	
United Kingdom	-0.80%	
Canada	-10.35%	
United States	2.72%	
Australia	-18.15%	

Sparkline

		Sales Amount	
Accessories	2005	20,235	
	2006	92,735	
	2007	296,533	
	2008	161,794	
Bikes	2005	7,395,349	
	2006	19,956,015	
	2007	25,551,775	
	2008	13,399,243	
Clothing	2005	34,376	
	2006	485,587	
	2007	871,864	
	2008	386,013	
Components	2005	615,475	
	2006	3,610,092	
	2007	5,482,497	
	2008	2,091,012	

Displaying reports in AMT

New reports, or changes to existing reports are discovered and handled by an AMT system task called **SSRS Manager**. This process is configured by default to run automatically every five minutes. However, you can run it manually using the **System Task Manager** function in AMT.

Save your reports under the following location so that AMT can discover them.
{serverlocation}/ReportServer/AMT User Reports/

Reports are added as a new AMT function. The function is automatically added to the function hierarchy of groups with **Automatically Add New IC Reports to Hierarchy** selected. You can edit this setting in the **Group Administration** function in AMT.

For users in groups that do not have this flag checked, AMT administrators can set report permissions separately and add the report to the group hierarchy in any appropriate location using **User Group Function Manager**.



It is necessary to restart the AMT client to see hierarchy changes.

Introduction to SSRS Parameters

Report parameters can be used to control report data, restrict data returned to the report, connect related reports together, and vary report presentation. It is important to understand how parameters and dataset queries are related and how to include parameters in expressions.

After you publish a report, you can manage parameters independently from the report definition (for example, when developing custom subscriptions).

When you view a report through the Report Manager URL, a toolbar displays each parameter so that you can specify values.

Parameters Pane

The report viewer toolbar displays a prompt and default value for each parameter. Parameter layout on the toolbar is formatted automatically. The order is determined by the order that parameters display in the Report Data pane. You can modify the order. The parameters, with the exception of @Hierarchy, are displayed as filters in AMT.

@FromDate, @Todate Parameters

These parameters have a **DateTime** data type and display a date picker control both in AMT and Report Manager.

@Site parameter

This parameter has a **Text** data type. @Site has an available values list, so valid values display in a list for you to choose values. @Site is multivalued, so a **Select All** option displays that enables you to clear all and select all values in the list.

@DateRange, @Hierarchy, @AddFilters

These parameters are only populated when the report is run from within AMT - you can ignore them otherwise.

Parameters Button

In the Report Builder preview, on the Ribbon, click **Parameters**.

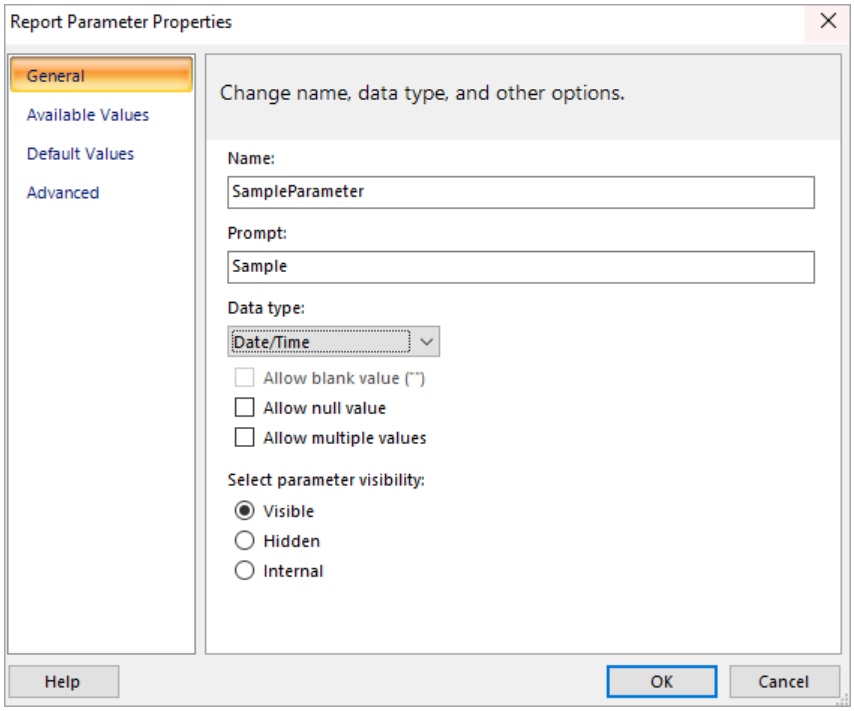
View Report Button

On the Report Viewer toolbar, click **View Report** to run the report after you enter parameter values. If all parameters have default values specified, the report runs automatically when viewed in Report Manager.

Rendering report parameters

When creating a parameter, there are a number of settings that define how the parameter will be displayed in both the SSRS URL and in AMT. To add a new parameter manually, right-click the **Parameters** folder on the left pane and click **Add Parameters**.

Below is a summary of important parameter settings and how they impact AMT.



General

Prompt: Name displayed to the report user.

Data Type: If you select Date/Time, AMT displays a date picker.

Allow Null Value and Allow Multiple Values: Applied when rendering a filter to make it required or multi-value.

Available Values

Select either **None**, **Specify Values**, which will be hard-coded values or **Get values from a query**, which populates values through an extra dataset.

Default Values

You can define default values for parameters. This is useful, for instance, to automatically chose a date range to cover the next six months without having to set it every time the report is run. Values can be hard-coded or read from a dataset.

Defining SSRS data sets to access AMT data models

A data set specifies the data that you want to use from a given data source. In Intelligence Centre, data sources are either a cube or a set of related views. The data set includes a query that specifies a set of fields. As you drag these fields to the design surface, you create expressions that select data when the report runs.

The procedures to create datasets from views and cubes is similar, but there are important differences to be aware of. Both procedures are described below.

Cubes

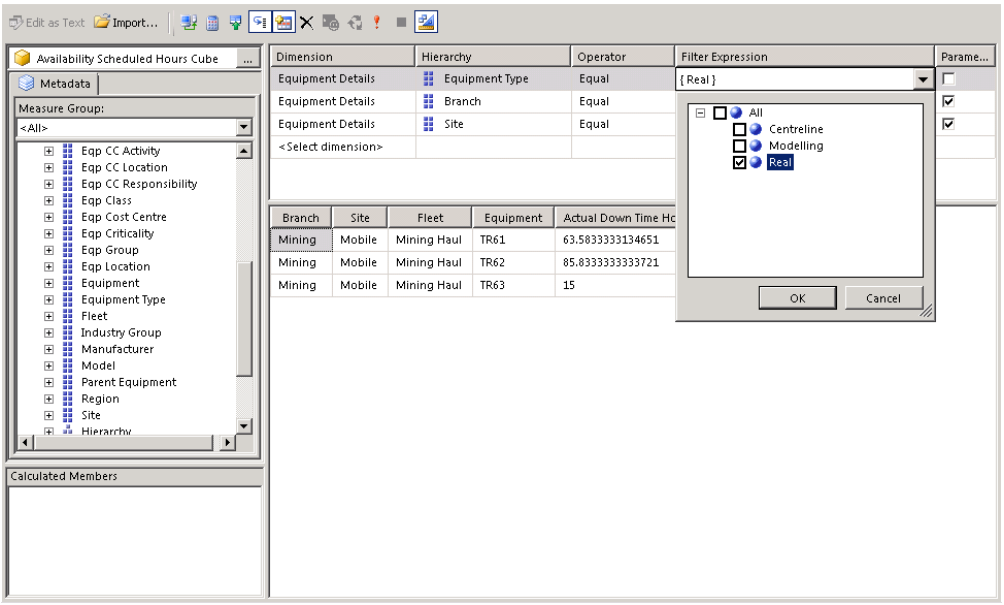
A cube is composed of dimensions, measures and KPIs. Each cube data set must have at least one measure and can have as many fields as required.

Fields, measures and KPIs are added to a query by dragging and dropping them into the bottom right pane.

Dimensions can be dragged into the top right pane and can be used as hard coded filters (where a filter expression is typed in or a list of values are selected) or parameters that allow you to select values at runtime (parameters can also be populated at runtime by automated subscriptions).

The screenshot below shows a query being built in Query Designer. This query selects Actual Down Time Hours from the Availability and Scheduled Hours cube. Dimensions included in the query are Branch, Site, Fleet and Equipment.

A filter expression has been defined to select only Real equipment.

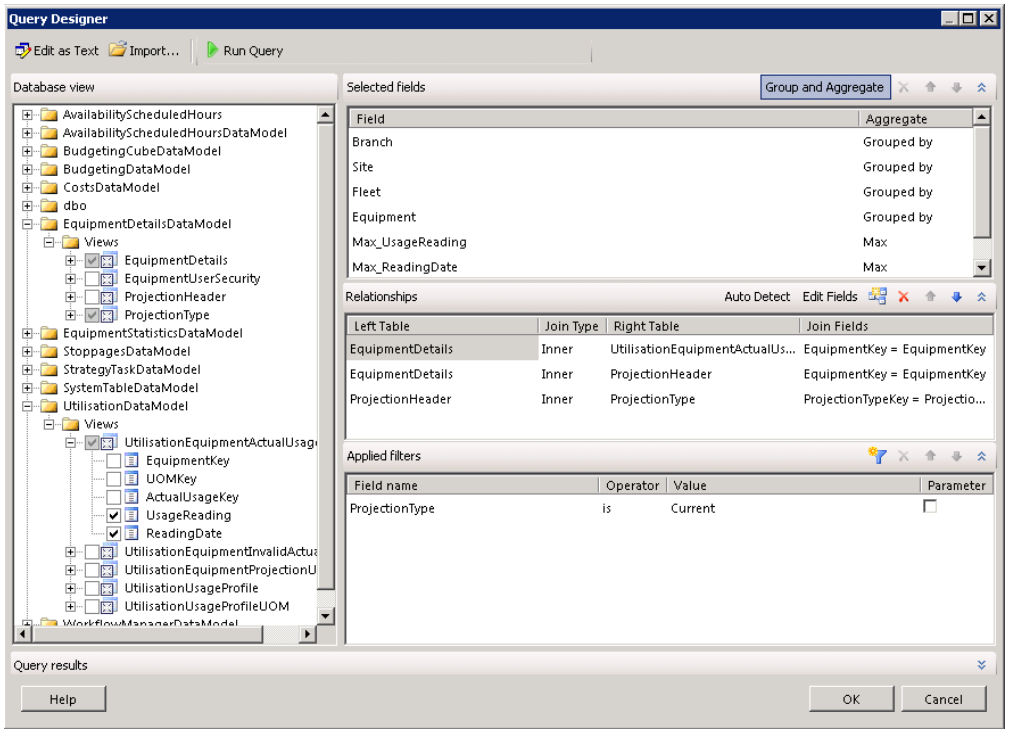


Each parameter created using a cube automatically generates a new data set used to populate it with available values (for example, a Branch parameter would show all available branches). Also, the order in which parameters are created also impacts the way into which parameter cascading takes place (for example, when you choose a certain branch, it is desirable that only sites belonging to that particular branch are selectable).

Views

Similar to Cubes, selecting the fields to use on the data set is just a matter of dragging and dropping them into the top right pane, while filters are added at the bottom.

When using Views, joins between different tables need to be specified manually. For example, to select the utilisation of an equipment, the following views need to be joined: EquipmentDetailsDataModel, ProjectionType, UtilisationEquipmentActualUsage. Relationships (joins) are specified in the middle section of the Query Designer.



For performance reasons, when defining queries the keys should be used, rather than a description field (for example, use EquipmentKey rather than Equipment). Filters set as parameters do not have a data set created automatically to populate it with available values; report writers must do this as a separate step.

Report Templates

To provide consistency with existing AMT reports, there are landscape and portrait templates provided in the report builder. The templates are under **AMT Report Parts**.

{server location}/ReportServer/AMT Report Parts/ReportTemplate-Landscape.rdl

{server location}/ReportServer/AMT Report Parts/ReportTemplate-Portrait.rdl

To use a report template, open it in Report Builder and save it as a different name under **AMT User Reports**.

{server location}/ReportServer/AMT User Reports/MyNewReport.rdl

The templates come with the following parameters.

- @AnalyseByName.
- @Hierarchy.
- @AddFilters.
- @ReportHeader.
- @DateRange.
- @Location.

@Hierarchy, @AddFilters and @DateRange are automatically populated when the report is run from within AMT, but are left blank when the report is run from SSRS Report Manager.

Budgeting and forecasting

AMT's Budgeting and Forecasting module provides the ability to create and save numerous accounting style budgets. These budgets can be based on existing equipment, future equipment (modelling equipment) and overhead projections.

The budgeting process depends on the initial strategy being set and maintained in AMT. Budgets are then created, maintained, finalised and exported to external systems, for example XERAS Enterprise, from the Budgeting and Forecasting module.

The budgeting process is different to creating archives or alternates in AMT. Key differences include:

- Budgets carry summary information (unlike the archives or alternates, which carry details of each strategy task).
- Budgets can be edited before they are finalised.
- Editing the budgets is flexible, and you can copy data to and from spreadsheets.
- Numerous versions can be created and used to compare to one another.
- The recalculation process is simplified within budgeting as all the main recalculations have already taken place within the strategy optimisation area of AMT.

The budgeting area of AMT can be accessed from Budget and Forecast Manager.

Before creating a budget, it is important to validate the source data to ensure the live AMT forecasts are accurate. To do this, perform the AMT Health Check process.

Budgeting

Budget and Forecast Manager

Navigate to **Budgeting & Forecasting** > **Budgeting** > **Budget & Forecast Manager**.

Most of the budgeting data displays in grids. You can copy between spreadsheets to easily edit data. There are some general rules for the grid operation.

- Data is not saved to the database until the grid is filtered you click **Save**.
- Data can be copied from the grid to a spreadsheet. Right-click on the grid, select **Copy the Table** and then paste into a spreadsheet.
- Actual data cannot be edited and is shown in **blue**.
- Editable records in the grid go into edit mode when you click the record. You can enter data directly into these editable fields or a range of data can be pasted from a spreadsheet. To paste a range of data, right-click the top left cell for the paste and select **Paste**.
- Some grids show edited records in **red**.
- Data in *italics* cannot be edited (determined by the Branch/Sites, and/or Cost Centres defined in your User Account).

Buttons above the grid

Budget System Task Status	Shows Budget System Task Statuses ordered by Requested Time, with the earliest at the top,
Calculate Budget	Recalculates the Budget in the background. Any other users who have made changes to this Budge will lose their changes.
Export Budget	Displays the Budget Export Wizard where you can create Export files.
Copy Budget	Copies the selected Budget and creates a new budget from it.
Add New Budget	Lets you create a new Budget.
Edit Budget	Opens the Budget Header Manager.
Delete Budget	Deletes the selected Budget.

Create a budget

The steps to creating the budget include:

- Set up all equipment (real and modelling) in AMT. Equipment is correctly set up in AMT when:
  - Equipment term and expected usages are confirmed.
  - Availability parameters are confirmed.
  - Work order settlements are imported and correctly linked and cleansed.
  - Strategy is maintained to show correct expected change outs.
  - All future (modelling) equipment is set up with expected purchase prices and purchase dates.
- Set up all overheads in AMT using **Overheads Navigator**.
- Set up a budget scenario in **Budget & Forecast Manager**.
- Import selected equipment and overheads into the **Budget & Forecast Manager**.

To create a budget, click **Add** then

1. Enter a **Scenario Name**.
2. Enter a **Version Number**.
3. Select a **Budget Type**.
4. Select a **Year**.
5. Select a **Start Month**.
   - The **End Month** will be automatically updated based on the **Period** associated to the **Budget Type** selected.
   - Budget Types and their associate budget periods are defined in the Budget Type system table.
6. Adjust the **End Month** (if required).
7. Select a **Currency**.

- All costs imported into the budget will be converted into the budget's currency.

8. Select a **Message Group**.
    - The AMT users associated to the Message Group will receive emails every time the budget's status is updated.
    - Message Groups are defined in the Message Group function.

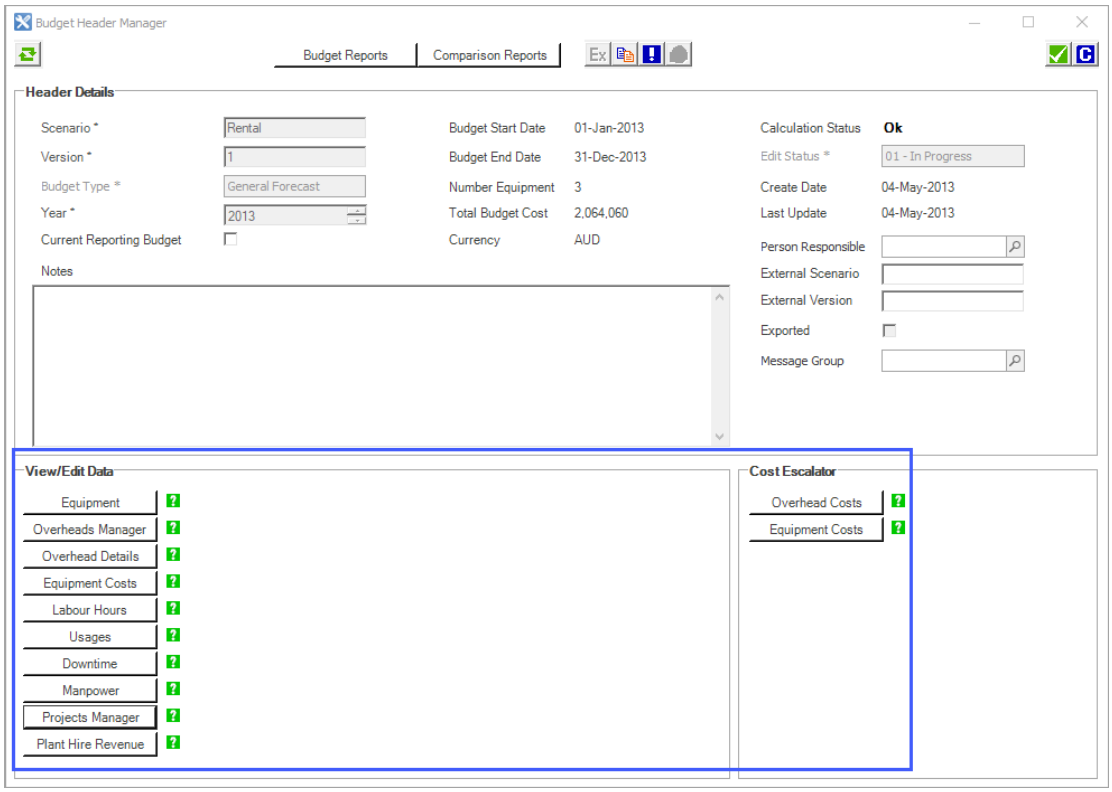
After clicking **Create New Budget**, AMT creates a system task to import the equipment into the budget and place it into a queue. To see your budget's place in the queue, navigate back to the **Budget & Forecast Manager** screen and click **Budget System Task System** .

A failed budget system task is automatically re-run up to three times. If it fails after three attempts the task is abandoned, and a record is written into the AMT BUDGET_ERROR table.

The data imported into a budget is a one-off snapshot of AMT's core data. Changes which occur in core AMT do not automatically update the budget. Changes in the core environment can be updated or reflected in the budget using the Advanced Update Routines.

The **My Equipment** filter and **My Cost Centre** filter, which appear on different budget module screens, are linked to your User Account. If you view equipment or cost centres which are not set up in your User Account, the data in the grid will show in italics and cannot be changed.

Once the budget is created, the following functions, which are at the bottom of the **Budget Header Manager** screen, can be used to view and modify the historical costs or changes to the budget strategy.



Equipment

Equipment enables you to add or remove Real and Modelling equipment from the budget.

When adding equipment, the **Equipment Filters** section determines which equipment can be added (based on the user's Branch/Site permissions set up in their User Account). The **Cost Filters** section, including cost centre activity, location, and responsibility, then refines which specific cost data, from the selected equipment, are imported into the budget. For example, for the equipment selected, "Asset#123", only import strategy task costs linked to expense element "Gaskets".



Eqp Cost Centre is the cost centre at the equipment level. **Cost Centre** is the cost centre at the strategy task level.

The equipment imported can be viewed in the Budget Manager grid.

- **Start Date** - equipment life cycle start date (which cannot be changed).
- **End Date** - initially the equipment's life cycle end date from the source projection but can be changed.
- **%RVIC** - RVIC of the equipment at the end of the budget period.
- **Opt. End Period** - AMT calculates the Optimal End Period (termination or sell date) if the RVIC is lower than the existing End Date's RVIC (+/- 12 months).
- **Total Costs** - equipment costs within the budget's start and end dates.

Once the equipment is imported into the budget, attributes of the equipment itself can be edited if required.

- **Model/Equipment** - these values should not be changed in order to maintain the link between the equipment in budgeting and the source equipment in core AMT. **Equipment Name** must be unique in the budget.
- **Replaces Equipment** - which equipment (if any) this specific equipment will replace.
- **Purchase Price** - can be added if the equipment is purchased during the budget period. This information will appear on the Budget Summary, Capital Forecast and Capital Purchases Forecast report.
- **New** - Modelling equipment added to a budget will be flagged automatically by AMT as **New**, but this can be toggled by the user for any equipment type.
- **Termination Date** - date on which the equipment will be terminated. If this date is extended, the End Date of the equipment must be extended in core AMT and the equipment re-imported into the budget.
- **Optimal Termination Period** - information provided automatically by AMT, where AMT automatically calculates the termination period.
- **Estimated Sell Price** - value used in the Capital Forecast and Budget Summary reports.
- **Scheduling Efficiency** - used to adjust Budgeting's Availability Forecast. Must be between 0-100%.
- **Cost Margins** – only relevant to equipment configured as MARC, and is used to calculate Cost at Cost for part, labour and misc. for costs manually added to the budget.
- **Ignore Budget Costs for the Equipment** - exclude the equipment in budget cost reports.

Overheads Data

Overheads Data allows you to import an overhead projection and actuals from the core of AMT into the budget. The Budget module is designed to import Modelling Overheads and Real Overheads.

To import an overhead projection, the overhead projection must exist in the core of AMT.

Multiple overhead projections can be imported into a single budget.

Overhead Details

Overhead Details enables you to view actual and future overhead costs and modify future overhead costs in an overhead projection.

Actual costs values are coloured **blue**. Values entered manually are coloured **red** until the data is saved, then they will return to normal. Data within the grid can be exported to an external spreadsheet program, modified, then copied and pasted back into AMT. Actual data will not be overwritten.



If you are having problems pasting data back into AMT, ensure you are pasting into a future month.

You can add additional overhead cost categories into the projection by clicking the **Add** button, as long as the overhead cost category already exists in the imported overhead cost projection. This inserts a new row into the overhead projection, which you can then edit.

As with all changes in budgeting, modifications made to forecasted values will not update the source overhead projection in the core of AMT.

Equipment Costs

Equipment Costs enables you to view actual and future equipment costs and modify future equipment costs and modify future equipment costs.

The grid groups costs are based on a combination of

- Equipment
- Component (Strategy Task)
- Work Group
- Expense Element
- Cost Bearer
- P (parts)/L (labour)/M (miscellaneous)

Use the **Display Fields** filter to add additional grouping fields.

Actual costs values are coloured **blue**. Values entered manually are be coloured **red**. Data within the grid can be exported to an external spreadsheet program, modified, then copied and pasted back into AMT. Actuals data will not be overwritten.



If you are having problems pasting data back into AMT, ensure you are pasting into a future month.

Use filters to narrow focus on strategy tasks to be changed, rather than showing everything.

A special editor is also available for changing the projected occurrences of a strategy task. To trigger this editor, right-click on the row of the strategy task and select **Edit Occurrences**.

Cost Occurrence Manager also provides a facility to add additional costs for a completely new type of cost, for example, creating a one-off refurbishment cost (basically any cost to the equipment that will occur in the budget period, which is not already costed in existing strategy tasks or as an overhead). The **Add** button on the screen allows you to add such a cost.



The Component (strategy task) options are limited to those available in the source projection of the equipment. Keep in mind that the way the cost is added and coded in the budget should ideally reflect

how the actual costs will be reflected in AMT when the actual cost is incurred. This will maintain an “apples with apples” comparison when running reports.

Comments can be recorded on a specific cell in the grid by right-clicking and selecting **Comments**. Along with the comment, AMT captures the date and time of the comment, and the user who entered the comment.

Risk Override values can be entered manually on tasks in the grid.

The **Risk Override** value entered applies to the entire task, not specific parts/labour/misc costs of the task or single occurrences of a task.

Manual changes made to costs do not update the budget's labour hours or downtime data.

The Cost Occurrence Manager also has a **Financial Code multi-editor grid**, allowing for batch modifications of financial codes of imported tasks, and simplifiying financial reassignments, improving the efficiency of budget adjustments.

You can easily adjust critical financial codes such as **Cost Centre**, **Cost Bearer**, and **Expense Element**.

As with all changes in budgeting, modifications made to forecasted values will not update the source projection in the core of AMT.

Labour Hours

Labour Hours enables you to view the labour hours associated with strategy tasks forecasted in the future.

Actual labour hours values are coloured blue. Values entered manually are coloured red. Data within the grid can be exported to an external spreadsheet program, modified, then copied and pasted back into AMT.

Manual changes made to labour hours do not update the budget's cost, actual occurrences or downtime data.

Changing the labour hours on a given task can automatically change the associated labour costs and vice versa, keeping labour rates consistent.

You can **Refresh Labour Rates**, which enables you to update labour rates that might have changed on AMT live, changing the associated costs for selected assets.

A warning message alerts you about the potential impact on labour costs before proceeding.

As with all changes in budgeting, modifications made to forecasted values will not update the source projection in the core of AMT.

Usages

Usages enables you to view actual and future equipment usages and modify future equipment usages.

For equipment configured for **Simple Utilisation**, projected usages per month shown in the grid is derived from the equipment’s **Annual Utilisation** assumption. For equipment configured for **Advanced Utilisation**, projected usages per month shown in the grid is derived from the **Usage Profile** linked to the equipment.

Actual usage values are coloured **blue**. Values entered manually are coloured red. Data within the grid can be exported to an external spreadsheet program, modified, then copied and pasted back into AMT. Actual data will not be overwritten.

If you are having problem pasting data back into AMT, ensure you are pasting into a future month.

As with all changes in budgeting, modifications made to forecasted values will not update the source projection in the core of AMT.



Usage Profile relationships imported for the live AMT environment are not maintained in Budgeting. Rather, AMT takes the forecast of usages per UOM from the Usage Profile, and replicates that forecast per UOM in Budgeting. If you change a Parent UOM in Budgeting, there is no cascade down to Child UOMs. You have to make the changes to other UOMs separately.



If you increase forecast usage rates significantly, this may cause future budget periods to forecast no cost. Remember, the budget module takes a snapshot from core AMT for the defined budget period when the budget is first created. The budget module does not take another snapshot to “back fill” future months due to increased utilisation. There are two reasons for this:

1. The budget module does not snapshot all the underlying core AMT parameters required to re-forecast future occurrences accurately (for example, strategy task frequencies, advanced scheduling logic, manual next occurrence changes, and so on.). Only the forecast produced by core AMT based on these parameters is snapshotted.

2. Mixing core future costs with a budget snapshot costs will result in an inaccurate budget. For example, an occurrence of a strategy task forecast originally in May (when the budget snapshot was created) might now in core be forecast in September. Back-filling September into a budget will result in the occurrence being budgeted twice: once in May and once in September.

If a significant change to an equipment’s forecast usage is required, this change should be made in core AMT. This allows AMT to recalculate (that is, re-forecast) future strategy task occurrences, which can then be reimported into a budget (by adding the equipment to the budget).

Downtime

Downtime enables you to view the downtime associated with strategy tasks forecasted in the future. This enables budgeting to produce **Availability Forecast** reports.

For Availability Forecasting, budgeting draws on the following data.

- Estimated durations per strategy task (source: projections imported into the budget).
- Scheduling Efficiency % (source: equipment details in the budget).
- Baseline Downtime (source: availability parameters per fleet set up using the Fleet Administration function).

Values entered manually are coloured **red**. Data within the grid can be exported to an external spreadsheet program, modified, then copied and pasted back into AMT.



Manual changes made to downtime do not update the budget’s costs or labour hours data.

As with all changes in budgeting, modifications made to forecasted values will not update the source projection in the core of AMT.



Actual durations (historic data) is not imported into budgeting.

Manpower

Manpower enables you to manually create a manpower (that is, headcount) forecast in AMT. AMT does not automatically calculate a manpower forecast from strategy tasks or any other source. It is entirely up to the user to create a manpower forecast here in budgeting if required, or to copy a manpower budget from another budget.

The manpower forecast is based on the following codes.

- Cost Centre
- Labour Activity
- Movement Type
    - BL: Baseline
    - NH: New Hire
    - PI: Promotion In
    - PO: Promotion Out
    - TR: Termination

A row in the grid is based on the combination of the above three codes. Values entered manually are coloured **red** until the data is saved, in which case it will return to normal. Data within the grid can be exported to an external spreadsheet program, modified, then copied and pasted back into AMT.

Multiple rows in the grid can be created at once by selecting multiple **Movement Types**



Manpower data does not relate to other budget data such as labour hours.

Projects Manager



Only relevant to equipment configured as Plant Hire.

Projects Manager enables the user to view Projects to which Plant Hire equipment are linked. Project linkages created in the core of AMT are visible here and cannot be changed.

The **Replace Equipment Plant Revenue Details** button triggers the revenue for listed projects to be recalculated based on usage adjustments made within the budget. This process does not draw on plant rate schedule data from core AMT. Plant rate schedule data is imported into budgeting when the equipment is first added to the budget.

If changes have been made to the plant rate scheduled in core AMT, and you want the budget to reflect these changes, the equipment must be deleted from the budget and re-added. The delete process deletes all data relating to the equipment in budgeting (that is costs, downtime, usages and so on).

Plant Hire Revenue



Only relevant to equipment configured as Plant Hire.

Plant Hire Revenue contains the forecast monthly revenue by equipment, plant rate schedule, cost centre and expense element.

Actual revenues are coloured blue. Values entered manually are coloured red. Data within the grid can be exported to an external spreadsheet program, modified, then copied and pasted back into AMT. Actual data will not be overwritten.

Overhead Costs/Equipment Costs

Escalations (cost adjustments) can be made to a budget. However, these adjustments only apply to the projected costs. Escalations to Equipment or Overheads can applied through the respective Cost Escalator functions, accessed on the **Budget Header Manager** screen.

Escalation rules

- Once applied, the escalation is not reversible unless you apply an appropriate percentage to achieve a reversal.
- Multiple escalations can be applied, so after applying one set of escalations, another can be applied at a different rate to other periods of time, resulting in a cumulative escalated result. Notes should be made to record the different adjustments made.
- Escalations apply to costs only.
- Escalations do not impact Plant Hire revenue data of equipment configured for Plant Hire.

Changing budget status

The status of a budget can be changed. However, budget status permissions, defined in a User Account determines whether a user can change a budget status, and which statuses can they change. For example, a user may be able to change an **In Progress** budget to **Under Review**, but cannot change the status to **Authorised**.

Budget status permissions are maintained in User Accounts. The Budget Status workflow is maintained in the Budget Status system table.

Reports

There are two main types of reports - budget reports and comparison reports.

Budget reports – reports on the nominated budget.

Budget Summary

- **Budget Summary**
  Combines equipment, overhead, CAPEX and manpower into one high-level report.
- **Total Cost Analysis**
  Analyses actual or projected equipment or overhead costs using common financial codes such as cost centre and expense element.
- **Total Cost per UOM**
  Calculates the cost per UOM (for example, cost per hour) for equipment and overheads over a selected time period based on actual or projected costs.
- **Headcount Forecast**
  Produces a headcount forecast based directly on the data entered into the **Manpower** screen. No calculations are performed.
- **Budget Change History**
  Lists the audit changes to the budget.

Equipment Details

- **Equipment Summary**.
  Presents a summary of equipment in the budget in terms of fleet population, total cost, average availability, total utilisation and net CAPEX.
- **Equipment Details**
  Lists each individual equipment included in the budget along with key attributes. Unlike other reports, this report calculates the RVIC% of the equipment at the end of the budget period, and calculates the Optimal End Date (termination or sell date). Both calculations are for information purposely only. AMT calculates an Optimal End Date if RVIC is lower than the existing End Date's RVIC (+/- 12 months).
- **Equipment Numbers**
  Line graph which illustrates the rise and fall in equipment numbers due to equipment terminations and acquisitions, over the budget period.
- **Capital Forecast**
  This report details the equipment which will be purchased as New in the budget period, and the equipment which will be sold. The report calculates total purchases less total equipment sales, as well as suggests the optimal sales point. AMT calculates the optimal sales point by looking at the lowest RVIC point +/- 12 months from the end date.
  For an equipment to be considered in this report as a New purchase, it must be flagged as new, have a start date in the Budget period, and have a purchase price. For an equipment to be considered as being sold, it must have an end date in the budget period and a sell price.
- **Capital Purchase Forecast**
  Bar chart report which shows the volume of New equipment purchased in terms of purchase value. For an equipment to be considered in this report, it must be flagged as new, have a start date in the budget period, and have a purchase price.

Usages and Availability

- **Equipment Usages Summary**
  Summarises per equipment per month utilisation, downtime hours, availability, RVIC and RVIC%.
- **Utilisation and Availability Forecast Details**
  Combination line/bar graph illustrating both utilisation and availability per month over a selected date range within the budget period.
- **Forecast Availability Analysis**
  Flexible report to analyse the availability forecast (for example, by equipment per month, by fleet per annum).
- **Fleet Usage and Availability Summary**
  Tabular report presenting forecast availability, forecast usage, equipment number and average equipment usage per month/quarter/year.

Equipment Cost Analysis

- **Equipment Cost Analysis**
  Same as the standard AMT **Cost Analysis – Life Cycle** report which can analyse actual and projected costs.
- **Changeout Analysis**
  Same as the standard AMT **Changeout Analysis** report which can contain actual and projected changeout quantities.

- **Component Changeout Analysis**
  Similar to the standard AMT Changeout Analysis – Detailed report, and is more flexible than the Changeout Analysis report

Plant Group

- **Plant Hire P&L**
  This report is only relevant for Plant Hire equipment. It is the same as the standard AMT Site P&L report for Plant Hire.

  The CBD calculation logic is relevant if you are using the Plant Hire CBD methodology and not the new Depreciable Assets methodology. The CBD adjustment area of the report is redundant if the Plant Hire methodology is not used.

  Some points on the calculation logic in the CBD Adjustment area:

  - **CBD Component Depreciation** – for future periods this defaults to the same as the Plant Revenue for CBD flagged Expense Elements.
  - **Capitalised Component** – for future periods this defaults to the same as the component changeout costs (that is Costs for CBD flagged Expense Elements).
  - **Forecast CBD Under/Over** – This is the difference between the Plant Hire Rate for CBD items and the Current Projection Rate.

    Example

    Assume an Equipment is forecast to do 5,000 hours per annum and its engine is setup with a frequency of 15,000 and a cost of $150,000. Its plant rate, therefore, is calculated and set at $10 per hour ($150,000 divided by 15,000 = $10/hour).

    If at some point in the future the engine strategy task cost is changed to $180,000 and the plant rate is not changed then AMT will calculate the CBD Under/Over amount as:

    - Current Projection average annualised cost ($180,000/15,000 x 5,000) $60,000
    - Plant Rate annualised revenue ($10 x 5,000) $50,000
    - Annual variance $10,000
    - Monthly Variance ($10,000 / 12) $ 833

    This follows the ACI logic in AMT, meaning the calculation works regardless of whether the strategy task is Unit of Measure-based or Days-based.

    A positive Forecast CBD Under/Over amount indicates CBD components are not being recovered fully by the prevailing CBD component rate. A negative amount indicates component recoveries are greater than component costs, and will reduce the overall plant result.

    This is the only report in Budgeting that references “live AMT” in its calculations. Therefore, this report can change from one day to the next, even though the Budget has not changed.
  - **Profit Loss on Component Disposal** – for future periods this is 0.
  - **Capitalised Component Adjustment** – for future periods this is 0.
  - **Component CBD Adjustment** – for future periods this is 0.

    - **Forecast CBD Over/Under** – a positive number indicates CBD components are not being recovered fully by the CBD component rate. A negative number indicates recovery > cost and will reduce the overall plant result (refer to the Plant Hire Handbook for more information).

- **Equipment Analysis**
  This report is only relevant for Plant Hire equipment.

  This is a flexible report for analysing the equipment numbers by different groupings. You can opt to focus on current equipment allocated to projects or to also include potential future equipment allocated to projects.

- **Revenue Analysis**

  This report is only relevant for Plant Hire equipment.

  This report allows you to analyse a Project's plant hire revenue per equipment. You can opt to focus on forecast revenue or potential revenue (which incorporates project probabilities).

Comparison reports – compare the nominated budget to other budgets

Budget Summary

- **Total Cost Analysis**
    - **Cost Analysis**
      Flexible report comparing high-level costs (for example, cost centre, expense elements, cost bearer) between two budgets for a specific time period. The report shows the cost per budget and calculates the variance. **OH** refers to Overheads.
    - **Cost Period Analysis**
      Similar to the Cost Analysis report but compares the difference between two budgets over time (that is, compares costs per month, quarter, year).
    - **Cost Variance Analysis**
      Similar to the Cost Period Analysis report but does not show the cost per budget, only the resultant variance per period (that is, month, quarter, year).
    - **YTS Comparison Report**
      Compares cost per budget in terms of Month To Date and Year To Date. Year to Date is determined by the Start Date of the report.

Equipment Cost Analysis

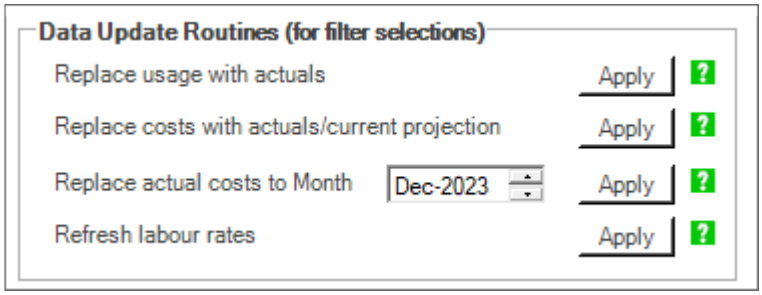
- **Cost Analysis**
  Flexible report comparing equipment costs between two budgets for a specific time period. The report shows the cost per budget and calculates the variance.
- **Cost Period Analysis**
  Similar to the Cost Analysis report but compares the difference between two budgets over time (that is, compares costs per month, quarter, year).
- **Cost Variance Analysis**
  Similar to the Cost Period Analysis report but does not show the cost per budget, only the resultant variance per period (that is, month, quarter, year).
- **Equipment Cost per UOM**
  Compares cost, usage and cost per UOM between equipment over a specific time period.
- **Strategy Task Variance Analysis**
  Analyses differences in strategy tasks between different budgets in terms of number of occurrences, cost

and utilisation. Therefore, if a strategy task will incur more cost in one budget compared to another, this report will identify whether it is driven by changes in utilisation (which may impact occurrences) or early failures or straight cost.

Advanced update routines

Update routines are located in the **Budget Equipment Manager** form .

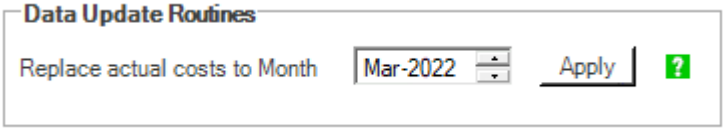
The update routine only applies to the asset filtered for in the form. The data is sourced for the updates from the respective projections of the equipment that appear in the grid.



Routine	Description
Replace usage with actuals	This is especially useful if you want to compare a budget created some time ago to the actual equipment usages achieved. The budget created was based on certain usages, but they can be quite different, and any variances need to be analysed based on the actual usages achieved. This routine replaces the usages with actual usages; future usages are not impacted. The budget calculation routine recalculates the budget using the updated usages but still using the strategy task assumptions from when the budget was originally created. This process impacts Plant Hire revenue for equipment configured for Plant Hire.
Replace costs with actuals/current projection	Replaces both the costs to date and future costs. Primarily used if you want to get the latest budget information on the equipment but want to keep the usage rates already set up in the budget. This process repopulates the costs tables in the budget from current projections, based on the budget's usage data. Manual changes made to future budget costs in Budgeting will be overwritten by this process. This process impacts Plant Hire revenue for equipment configured for Plant Hire.

Routine	Description
Replace actual costs to Month 'x'	Replaces the actual costs to a nominated month (up to the previous month). This replaces both actuals and projected in the nominated month. This function behaves in the same way as **Replace costs with actuals/current projection** but can be limited to a date range. You can end up with a mismatch, for example, you may have planned to do an engine in August, but it is done in July. Now you replace actuals to the month of July and the engine costs appear in both July (actuals) and August (projected). This scenario could happen in reverse, planned to do in July but doing in August.
Refresh Labour Rates	This will potentially impact the existing labour costs for the selected assets. Only forecast labour costs or rates are impacted.

Update routines are also located in the **Overheads Manager** form. The update routines are applied only to the overheads which appear in the grid.



Routine	Description
Replace actual costs to Month 'x'	This replaces both actuals and projected in the nominated month. Therefore, for the current month there would only be actual costs if you elected to replace these costs.

(Rpt_494) Budget Compliance

Compares actual costs for the current month, financial year to date and projected for the current financial year to a budget.

Tips

Only budgets created in AMT's Budgeting module can be used in this report for comparison.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Task Counter, Work Group

(Rpt_540) Budget Forecast Performance

This report is based on Budgeting module data that allows comparison of a Budget, a Forecast and live AMT data.

Tips

"Forecast" is a nominated Budget.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|PeriodStart|,|PeriodEnd|,|UserID|

Report parameters

Default.

Additional filters

Cost Centre, CC Activity, CC Location, CC Responsibility, Cost Expense, Cost Category, CC Code, Cost Bearer, Include

(Rpt_541) Budget Equipment Cost Analysis

This report is based on Budgeting module data that allows comparison of a Budget, a Forecast and live AMT data.

Tips

This report can be run at the Equipment level only.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Task Counter, Work Group, Cost Bearer, Cost Centre, Cost Category, Expense Element, CC Activity, CC Location, CC Responsibility, Parts/Labour/Misc

(Rpt_543) Budget Component Performance

Based on Budgeting module data that allows comparison of component costs and occurrences in a Budget to live AMT.

Tips

Shows costs and occurrences for the Total Budget Period vs Actuals to Date vs Projected Future vs Total Projected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|,|MinCompCost|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Task Counter, Work Group, Cost Bearer, Cost Centre, Cost Category, Expense Element, CC Activity, CC Location, CC Responsibility, Parts/Labour/Misc, Identified Components, Budget or Actual Annual Cost >= Nominated Amount

(Rpt_837) Budget Comments Report

Report displaying Comments of the selected Budget. You can select Latest comments only, All valid comments, All comments including deleted comments or Deleted comments. Strikethrough on the report indicates deleted comments.

Initial parameters

|ControlType|,|Idlist|,|TextList|,|LogoLocation|,|UserID|,|LanguageKey|,
|ReportStyleKey|

Report parameters

Default.

Additional filters

Employee, Branch, site, Fleet, Equipment, Cost Centre, Budget Status, Comment Level, Labour Activity, Show

Live Forecasting

(Rpt_173) Forecast Cost Analysis

Shows the strategy task costs at sell.

Tips

Significant differences suggest the current strategy parameter should be changed.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|LogoLocation|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Region, Eqp Criticality, Equipment Group, Exp Category, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Modifier Code, Division, Equipment Class, Equipment Location, Project, Def. Maint. Period, Parts/Labour/Misc, Currency

(Rpt_174) Forecast Worksheet

Used for forecasting costs by expense group. The report shows future and historical costs at sell, categorised by up to five different areas.

- You can navigate to this report in the following ways:
  Long-term Planning > **Costs**.
  Budgeting & Forecasting > **Live Forecasting**.
  Modelling > **Cost Analysis**.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,
|EquipmentTypes|,|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|,
|Feature_32|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

(Rpt_818) Forecast Worksheet (Dealer Adjusted)

Used for forecasting costs by expense group and optional dealer market share. Shows future and historical costs at sell, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Planned, Workorder, Currency

(Rpt_778) Forecast Worksheet (Serial Number)

Used for forecasting costs by expense group. The report shows future and historical costs at sell, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|EquipmentTypes|,
|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Serial Number, Modifier, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

(Rpt_816) Forecast Worksheet - @Cost

Used for Forecasting costs by expense group. Shows future and historical costs at cost, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,
|EquipmentTypes|,|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|,
|Feature_32|

Report parameters

Default.

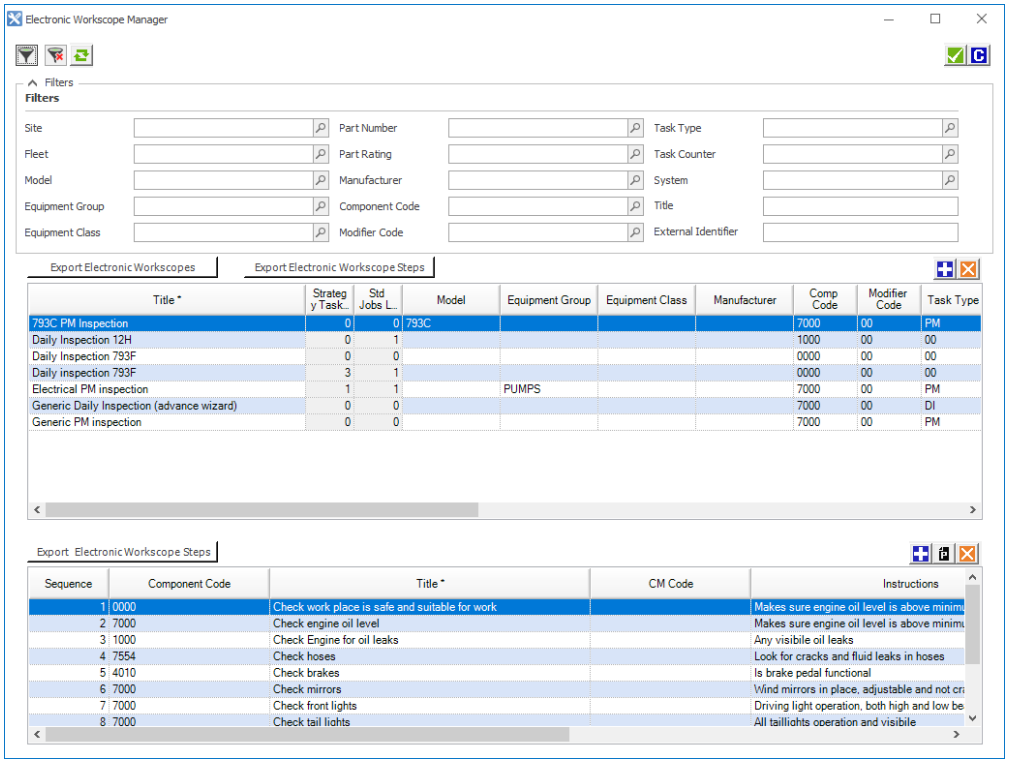
Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

Knowledge Database

Electronic Workscope Manager

Electronic Workscope Manager is used to define inspection templates. To access Electronic Workscope Manager, navigate to **Knowledge Database** > **Electronic Workscope Manager**.



Electronic Workscope dialog box overview

Fields and Columns for the top grid

For a description of the columns in the bottom grid, refer to Inspection Steps.

Column	Description	Example
Title	The title of the workscope.	Daily inspection 793F.
Strategy Task Links	The number of linked strategy tasks.	2
Std Jobs Linked	The number of linked standard jobs.	1
Model	The asset that the workscope is for.	773E

Column	Description	Example
Equipment Group	A grouping, or family, of model codes.	793, OHT.
Equipment Class	A grouping of the same class of equipment.	OHT, TTT, Rail.
Manufacturer	The manufacturer of the asset.	Caterpillar.
Comp Code	The code for a specific component.	C0723 Fuel Tank.
Modifier Code	A code that provides additional identification to the component code, typically the component's location on the asset.	RF - Right Front. TP - Top.
Task Type	Code and description combination identifying what is to happen at an occurrence of a strategy task.	CC - Component. Changeout. GI - General Inspection.
Task Counter	Identifies the sequence by which a strategy task is performed.	1st - First Overhaul. 365 - 365 Day Service.
Sub System	Part of the asset hierarchy.	Fuel System.
Primary Part Number	The part number that identifies the whole component. In some circumstances it is a Kit number. Primary part numbers are linked to strategy tasks.	
Site	The location of a fleet.	Gold Mine - Mill Area.
Fleet	A group of one or more assets used for a specific purpose.	Mining Haul, Process Plant.
Last Mod Date	The date the workscope was last modified.	22-May-2013.
Inspection Config	The inspection configuration. One of: Simple Wizard. Simple Checklist. Advanced Wizard. Advanced Checklist.	Simple Grid. See Inspection Display Modes.
Inspection Type	The type of inspection.	Maintenance.

Inspection configuration



There is no editor for inspection configurations; the configurations can be loaded into the AMT database by RPM. Inspection configurations can be reviewed and modified in a workshop with RPM when AMT Mobile is deployed. Changes or new configurations can only be performed by RPM. Please contact your RPM representative for more information.

The inspection configuration defines the nodes that make up an inspection, and the configuration of each node in that inspection. It is the *workflow* you must follow to complete the inspection. The possible nodes are:

- **Record Usages**. View and edit Usage Readings.
- **Record Measurements**. View and record Measurements. Measurements are interpreted (colour-coded ratings) in real time.
- **View Defects**. View and record Defects.
- **Perform Inspections**. Defined in the Electronic Workscope.

Inspection display modes

Depending on who is performing an inspection, different types of inspection require different levels of detail in the AMT Mobile user interface. This is achieved by four different display modes. Each inspection configuration is linked to one of the following inspection display modes:

1. Simple Wizard
2. Simple Checklist
3. Advanced Wizard
4. Advanced Checklist

For example, operators may only need to use a simple checklist, so they can quickly perform a pre-start inspection on an asset. However, Mechanics performing a detailed preventive maintenance inspection might want to note down some details and comments of each inspection task, or add photos and videos as attachments, so would use an advanced checklist.

Display Modes	Feature
Simple	The Inspector marks each inspection step as **OK** or **Defect**.
Advanced	The Inspector gives each inspection a rating from 1 to 4.
Wizard	Each inspection step is one page, making it easy to add inspection details for each step.
List	All inspection steps display on one page, in the style of a check list.

Inspection item node

During an inspection, the inspector can view and edit defects and record usages and measurements. There are four inspection item nodes in AMT.

Inspection item nodes can be set up to display in any order and repeat multiple times.

1. Example 1: View Defects – Record Measurements – Record Usages – Perform Inspections.
2. Example 2: Record Usages – Perform Inspections – Record Measurement – Record Usages – View Defects.

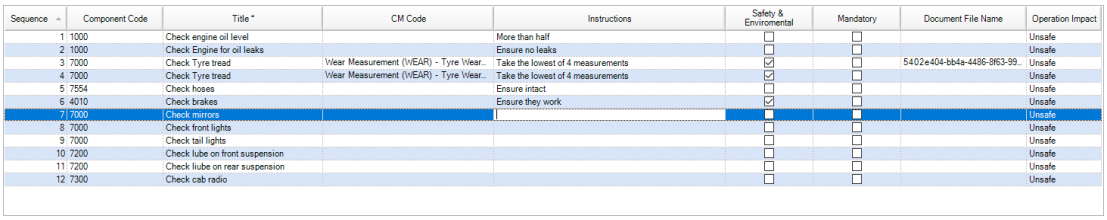
There is no editor for inspection item node configurations; the configurations can be loaded into the AMT database by RPM. Inspection item node configuration can be reviewed and modified in a workshop with RPM when AMT Mobile is deployed. Changes or new configurations can only be performed by RPM. Please contact your RPM rep for more information.

Inspection Type

Inspection type helps users, for example operators, supervisors, mechanics, and inspectors find relevant inspections. Inspections that you can select in the AMT Mobile Inspections module are filtered by inspection type.

Inspection steps

Each inspection is made up of a series of inspection steps. The steps are defined in the grid on the bottom half of the Electronic Workscope Manager screen.



Sequence	Component Code	Title *	CM Code	Instructions	Safety & Enviromental	Mandatory	Document File Name	Operation Impact
1	1000	Check engine oil level		More than half	☐	☐		Unsafe
2	1000	Check Engine for oil leaks		Ensure no leaks	☐	☐		Unsafe
3	7000	Check Tyre tread	Wear Measurement (WEAR) - Tyre Wear...	Take the lowest of 4 measurements	☑	☐	5402e404-bb4a-4486-8f63-99...	Unsafe
4	7000	Check Tyre tread	Wear Measurement (WEAR) - Tyre Wear...	Take the lowest of 4 measurements	☑	☐		Unsafe
5	7554	Check hoses		Ensure intact	☐	☐		Unsafe
6	4010	Check brakes		Ensure they work	☑	☐		Unsafe
7	7000	Check mirrors			☐	☐		Unsafe
8	7000	Check front lights			☐	☐		Unsafe
9	7000	Check tail lights			☐	☐		Unsafe
10	7200	Check lube on front suspension			☐	☐		Unsafe
11	7200	Check liube on rear suspension			☐	☐		Unsafe
12	7300	Check cab radio			☐	☐		Unsafe

Fields and Columns for the bottom grid

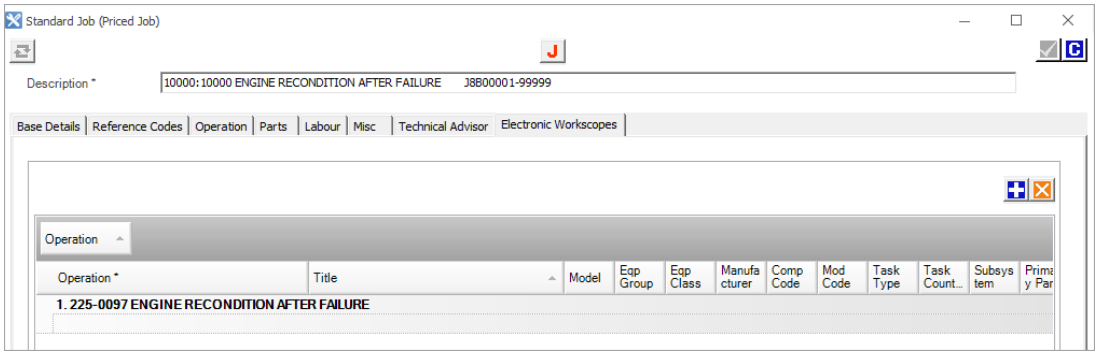
For a description of the columns in the top grid, refer to Electronic Workscope Manager.

Column	Description	Example
Sequence	The order in which you perform the Inspection Steps.	1, 2, 3,4 ... and so on.
Component Code	The code for a specific component.	1000 - Engine.
Title	The title of the inspection step.	Check engine oil level.
CM Code	The name of the CM (Configure Measurement) translation rule. The data in this field is from the list of measurement codes. You can not create a CM translation rule if the CM code does not exist in the list of measurement codes.	Copper (CU) - Engine Oil - Engine.
Instructions	Extra instructions relating to the step.	Take the lowest of four

Column	Description	Example
		measurements.
Safety & Environmental	Whether this inspection is Safety or Environmental related.	
Mandatory	Whether the step is mandatory.	
Document File Name	The file name for this inspection task.	File name.
Operation Impact	The severity of the impact if this asset fails this inspection task.	Unsafe, Potential Damage, Non Critical.

Electronic workscope

An electronic workscope can be linked to a standard job, which will flow through the strategy task and strategy task's work orders.



Library

Navigate to the library by clicking **Knowledge Database** > **Library**. The library opens in your web browser.

The library provides a single repository for a wide variety of equipment management related documents. You can add any electronic document to the AMT Library, for example, corporate guidelines, procedures, workscopes (equipment specific job instructions) and technical updates.

You have the ability to group documents for specific work orders and have these automatically linked to planned work. As the library is web-based, a wide range of users from different sites can access it. This allows organisations to leverage the efforts of individual sites and standardise processes. The results are improved efficiency and quality work performed.

You can link documents to SharePoint and other similar systems where separate credentials are required in AMT Standard Job Operations, Strategy Tasks and Work Orders. Both on-premises and cloud-hosted SharePoint deployments are supported.

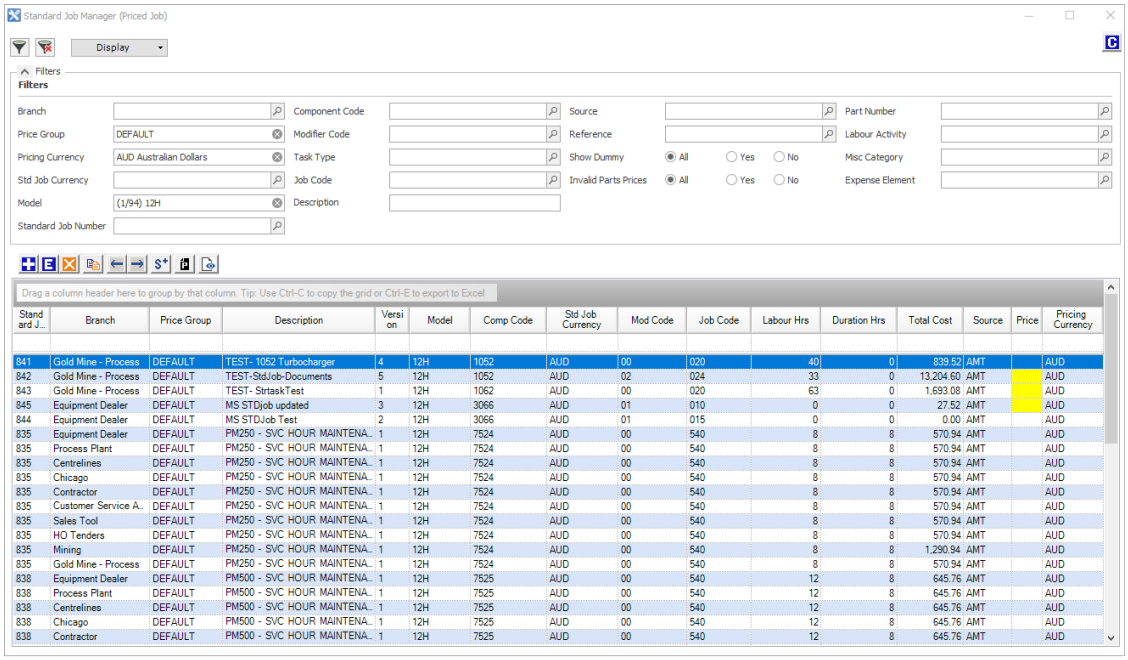
You will be prompted to supply the appropriate SharePoint credentials to access documents linked in SharePoint from AMT.

Library jobs

You can create jobs in the AMT Library. The jobs represent a specific maintenance task to be performed. By creating a job in the Library, you can link related documents to that job. If you want to search in the Library to find, for example, all jobs for a particular service on a particular model of equipment, you can search for a Job (which contains links to all the relevant documents) rather than searching document by document.

Standard Job Manager

In Standard Job Manager you can view standard jobs (those created in AMT and those imported from an external system), and add, delete and edit standard job attributes.



Button	Action
	Create a new standard job
	Edit the highlighted standard job
	Delete the highlighted standard job
	Copy the highlighted standard job

Button	Action
	Import standard jobs from external systems
	Export standard jobs
	Create strategy tasks from standard jobs
	Preview/Print
	Save
	Close/Cancel

Calculation of standard job totals

The method of calculation of standard job totals (that is, parts cost, labour cost, misc. cost, total cost, labour hours and duration hours) will be dependent on the value of the Use Summary Totals flag, as follows:

- If the Overwrite Cost box is selected, the Parts Cost, Labour Cost, Misc Cost, Labour Hours and Duration Hours amounts are manually entered by the user; Total Cost will be calculated as the sum of Parts Cost, Labour Cost and Misc Cost.
- If the Overwrite Cost box is not selected, the Labour Hours and Duration Hours amounts will be calculated as the sum of the respective values for all Standard Job Operations that make up the job. The Labour Cost, Parts Cost and Misc Cost amounts will be calculated as follows.
    - Labour Cost is calculated as the sum of the labour cost of all standard job operations that make up the job; the labour cost of each standard job operation is calculated as Labour Rate Per Hour * Labour Hours.
    - Parts Cost is calculated as the sum of the part cost of all standard job operation parts in all standard job operations that make up the job where the part is not miscellaneous; the part cost of each standard job operation part is calculated as Unit Price of Part * Quantity * Probability.
    - Misc Cost is calculated as the sum of the misc cost of all standard job operations.
- Total Cost is calculated as the sum of labour cost, parts cost and misc cost. For OEMs, the total cost showing in the grid is cost at sell.



The unit price of the part price is calculated as:
- If Credit = Partial, then Unit Price = Part Sell – Partial Credit Sell.
- If Credit = None, then Unit Price = Part Sell.

Standard job overview

A standard job is a repeatable task that can be carried out on a model of equipment (for example repair engine before failure). A standard job is represented as a financial breakdown of the work. A standard job can also be represented as a list of labour activities, each with an associated list of parts. When included in a standard job, a set of labour activities is referred to as a standard job operation (or simply operation). It is valid for a standard job

to contain no operations. In this case, the standard job is represented purely as a financial breakdown of a piece of work, with no list of operations.

A standard job has two aspects:

- A set of attributes that define the standard job.
- A set of pricing attributes for each branch. Whenever one or more of the attributes that define a standard job changes, each of the sets of pricing attributes will be affected. Conversely, whenever one or more of the pricing attributes (for a particular branch) changes, the definition of the standard job will be unaffected, and hence the pricing attributes for other branches will also be unaffected.

There are two sources for standard jobs:

- External systems, such as an enterprise resource system (for example, SAP, Ellipse, DBS) which are imported through a specific interface (contact your system administrator for more information).
- AMT (Standard jobs created in AMT).

Other points to note about standard jobs:

- A standard job can consist of a number of operations (steps).
- Each operation has a labour requirement and a bill of materials.
- You can nominate the probability of parts usage for a part entered into the bill of materials.

Standard job identification and version control

Standard jobs are uniquely identified by a **Standard Job Number**. The Standard Job Number is an integer value generated by AMT. The Standard Job Number cannot be changed.

Standard jobs also have a **version number**. The version number starts at 1 when a standard job is created, and is automatically incremented by AMT whenever the standard job is changed. Changes to standard jobs are logged, and can be audited via the Standard Job Version Details report.

The standard job number and version number are usually displayed side-by-side in AMT. If a version number is not shown, assume version 1.

Standard job principles

As part of the process of creating a standard job in AMT, a base priced job is created. Each base priced job has a nominated price group, branch and currency. The currency for the standard job is used to identify the currency for any miscellaneous costs, and labour rates added into the standard job.

In addition to the base priced job, AMT also automatically creates a priced job when the same standard job is used in another price group, branch or currency.

Standard jobs can be created for any currency, however, if the currency is not specified it will default to the currency in the price group.

Every standard job has a pricing group, for example, a pricing group may need to be set up for a particular customer because they receive a 5% discount on parts.

Each labour activity, such as a workshop, field, has a rate which is branch specific. You can change labour rates using the system table.

The projections of centrelines, estimates, real assets and so on can contain links to one or more standard jobs, so changes to a standard job can cause changes to the linked projections.

If a standard job is linked to a projected job (within a strategy task), then any changes to the standard job will automatically update the projected job with the new costs and the projected job's pricing date with the current date.

You can copy and change standard jobs.

Standard jobs and escalation

Standard job totals will always be expressed in current pricing. For standard jobs linked to an external system, escalation (the process by which prices are converted to current pricing) will be assumed to have been carried out by the external system. However, for standard jobs that are created in AMT, the standard job totals will be updated to current pricing in AMT whenever an escalation is applied in AMT.

Standard job margins

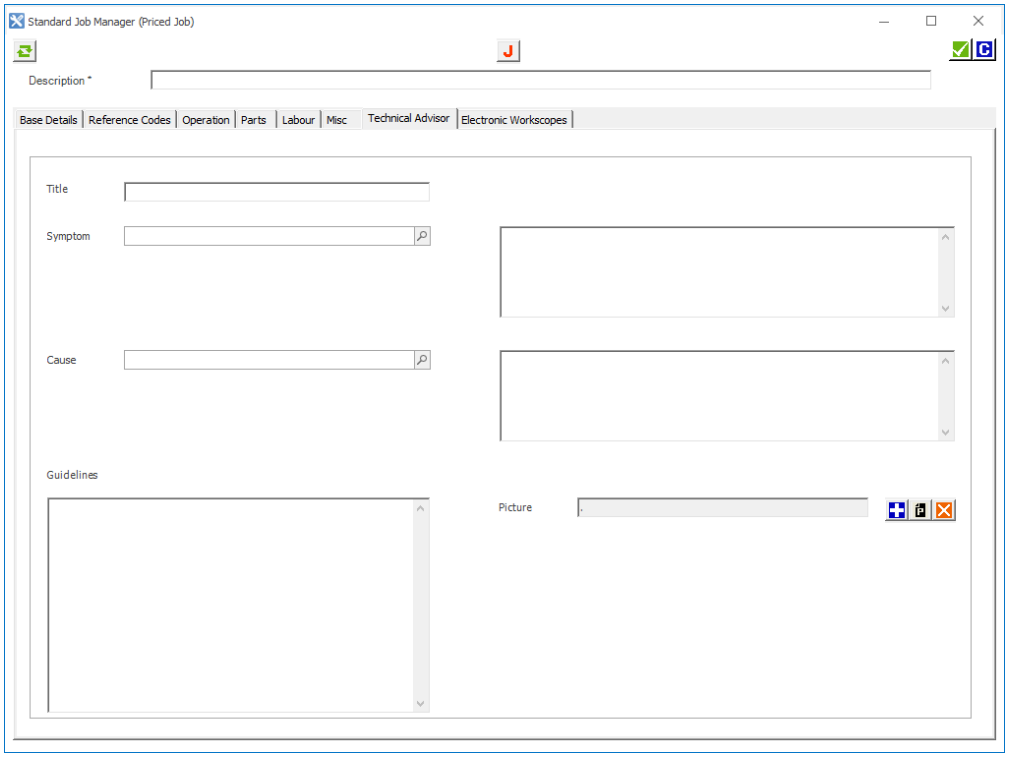
This system setting is only relevant to AMT installations at OEMs. The cost shown in the standard job for parts, labour and miscellaneous items is cost at sell. The cost at cost of the standard job is held in the background. By default, the cost at cost of the standard job is calculated from the margins contained in the Set Margin system table. However, there are two system configuration options that enable cost at cost for the standard job's parts to be based on the cost at cost of value each piece part within the standard job.



If the settings enable cost at cost to be calculated from the part rather than the set margin system table, but the total cost at cost of all parts in the standard job are not greater than zero, AMT will revert to calculating the cost at cost from the Set Margin system table. Otherwise AMT would be projecting a 100% margin for the parts.

Technical advisor tab

The Technical Advisor tab on the Standard Job screen contains information about the failure the standard job is designed to be used for to repair, and the repair guidelines.



Enter a **Description** of the standard job, a **Symptom** and a **Cause** with a code and description. You can enter extra information in the boxes to the right of the Symptom and Cause fields. Also enter Guidelines on how to perform the repair.

Create a standard job from Standard Job Manager

You can create standard jobs in AMT either using the Standard Job Import process or creating the standard jobs manually in AMT.

It is not always necessary to create a separate standard job for each position (such as modifier code) of a component on an asset. For example, if the parts to repair the right final drive are the same as the left, then only one standard job should be created.

Standard Jobs imported from an external system cannot be edited in AMT, they can only be deleted. If you want to modify an imported standard job, it should either be modified at the source and re-imported into AMT, or you should copy the external standard job in Standard Job Manager (using the copy function) and then modify that copied version in AMT.

The structure of a standard job in AMT is as follows.

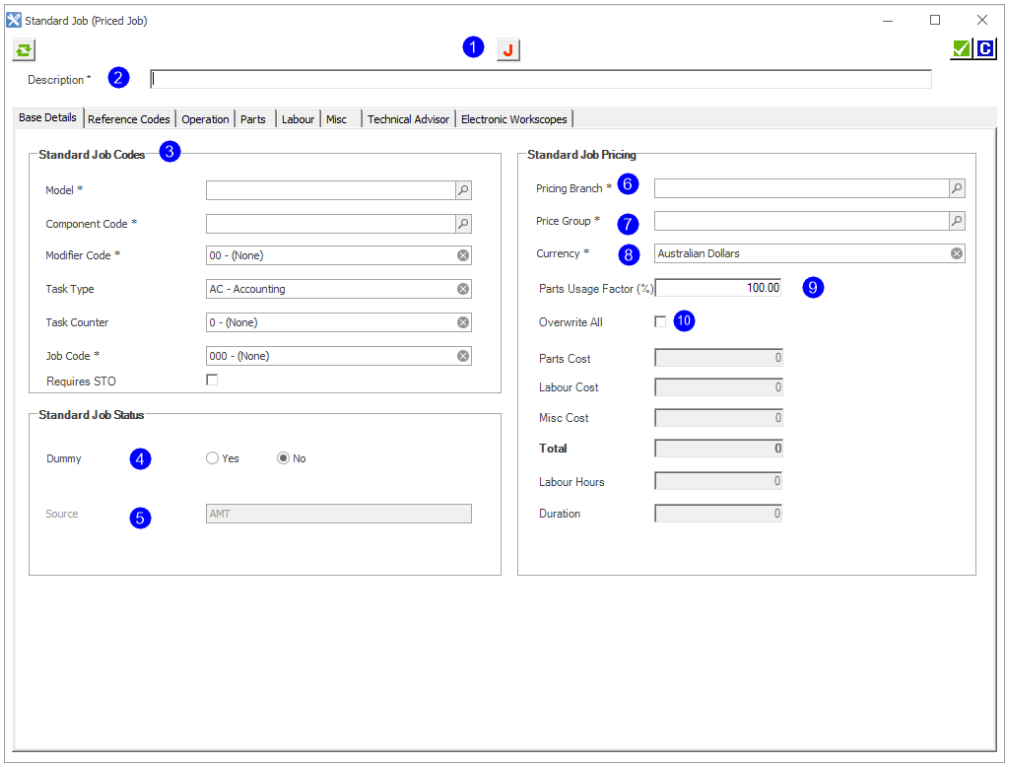
- **Standard Job Header**
  Contains description and coding to define the standard job.
- **Operation**
  Definable steps required to complete the standard job.
- **Parts**
  Bill of materials.

- **Labour**
  Labour jobs and costs.
- **Misc**
  Other costs.

A standard job cannot have parts, labour or misc. items without an operation. Most standard jobs only have one operation, which is given the same codes as the standard job header.

Navigate to **Knowledge Database** > Standard Job Manager. Click the **Add** icon above the grid.

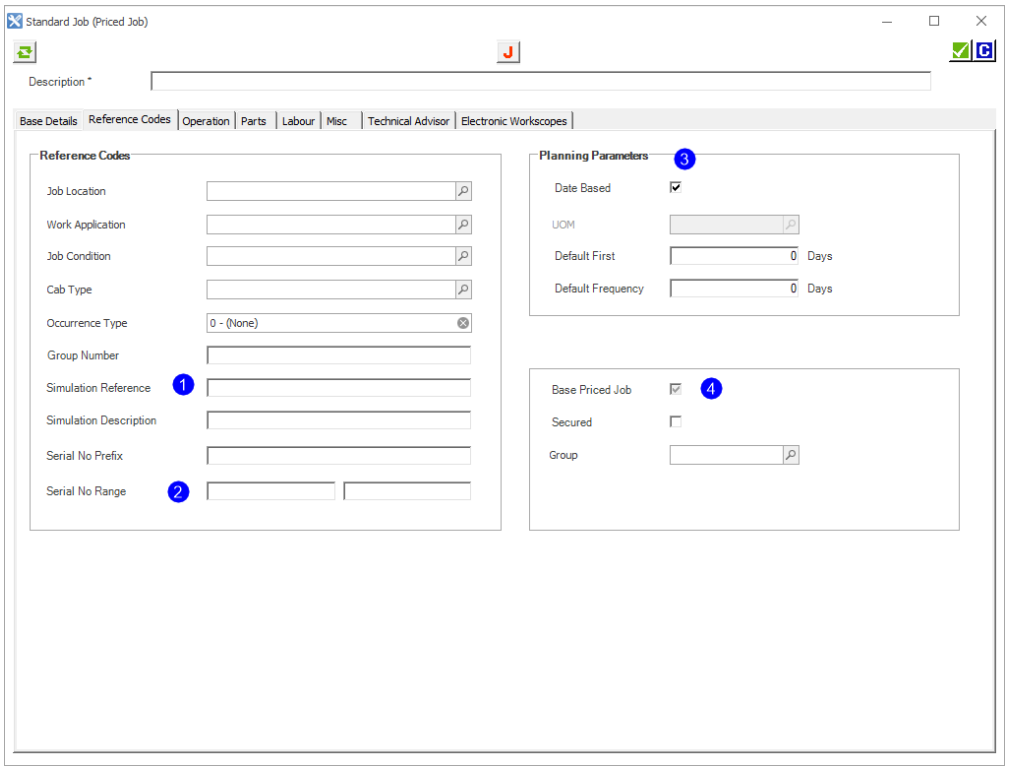
Base Details tab



Item	Description
1	Click to print the standard job's details.
2	Description of the job.
3	Configuration filters used to uniquely define the standard job.
4	The Dummy flag indicates that the standard job was created only to transfer a task and is not a real standard job. For example, if you have a task in AMT for electricals at a frequency of 1000 hours then this is more a provision of monies for general electrical work. It is not something that is scheduled to be performed at 1000 hours and will not have any parts list supporting the standard job. The default selection is **No**.
5	When a standard job is created in AMT this field is set to **AMT** automatically.

Item	Description
6	Determines the labour rate used to calculate labour costs in the standard job. Each branch in AMT can have its own labour rate.
7	Determines the cost of parts. Different pricing groups can have different costs for parts.
8	Currency used for pricing.
9	Has all prices as calculated fields.
10	Enables the fields below the check box and enables you to overwrite the Parts Usage Factor (9). If selected, you do not have to add specific parts, labour and misc items. See Define Labour and Misc Requirements.

Reference Codes tab



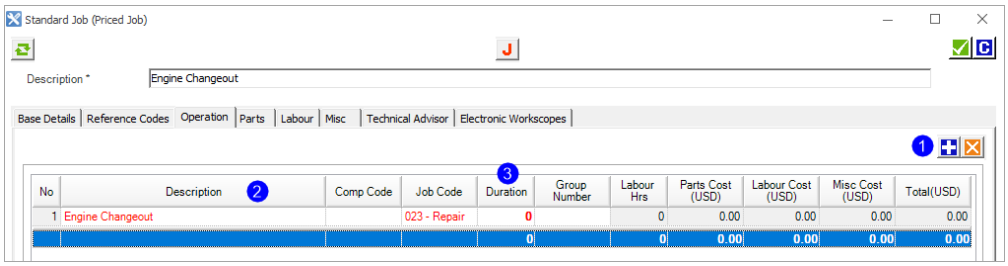
Item	Description
1	Not a mandatory field. Primarily relates to imported data from an external source that has a designated reference number. When manually creating a standard job header this field is still required. Try to keep the naming conventions consistent, for example: date of creation and model/component code - 20180115-XYZ-1250.

Item	Description
2	Provides extra information which can be recorded to define the standard job.
3	These fields are required if you want to be able to create a strategy task from the standard job.
4	Authority controls preventing unauthorised users from creating or updating standard jobs.

Operation tab

In order to allocate parts, labour or misc. cost items to the standard job, you must create an operation. An operation reflects a specific step required to execute a standard job. For example, for a Remove and Install standard job, the first operation may be Remove and the second operation Install. In the majority of cases, the standard job will only consist of one operation, and it has the same coding as the standard job header.

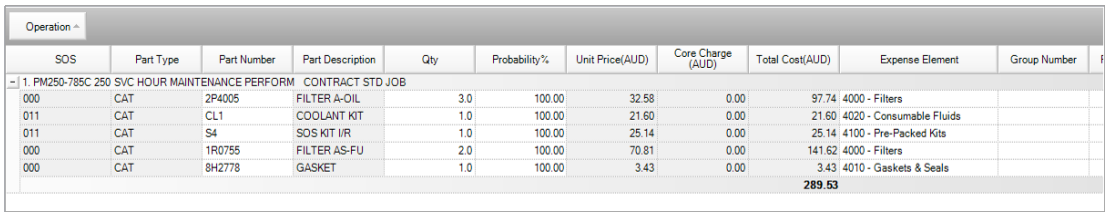
On the Standard Job Manager, click the **Operation** tab.



Item	Description
1	Click to add an operation.
2	Operation will by default inherit the codes from the header, but you are able to change them.
3	The time required to complete the operation.

Parts tab

For each operation you can add parts.



Item	Description
1	Add a part.
2	Add multiple parts. Highlight the operation, click the parts icon. A new window opens where you can select the parts then drag and drop them onto this window.
3	Highlight the operation before making changes.
4	Red text in this field indicates there has been a change.

You can view part sell, full credit sell and partial credit sell values for a part using Parts Manager.

Click the links below for more information.

Define Labour and Miscellaneous Requirements

Standard Job Technical Advisor

Standard Job Electronic Workscope

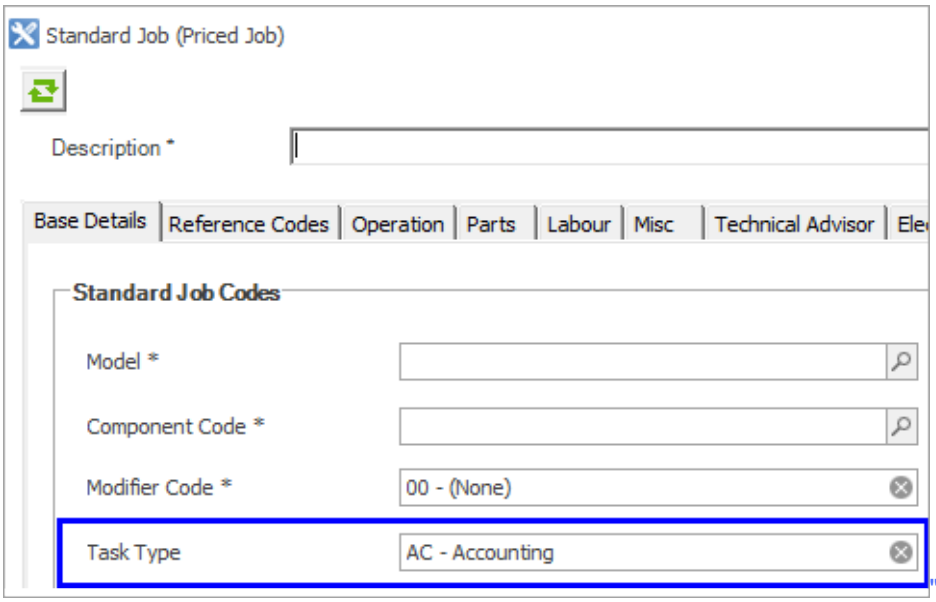
Standard Job Overview

Standard Job Manager

Create strategy tasks from standard jobs

AMT has the ability to create strategy tasks from standard jobs. However, there are some requirements which the standard job must adhere to.

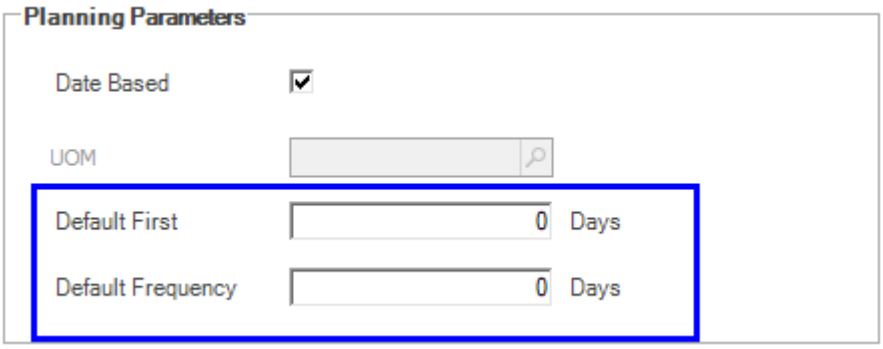
- On the **Base Details** tab, the **Task Type** must be selected.



"Create Work Order Settlement History" on page 916

On the **Reference Codes** tab:

- the **Default First** and **Default Frequency** fields must be greater than zero.



The standard job import function and standard integration functions let you set some extra parameters on the standard job for creating a strategy task from a standard job. These fields can only be set via the import or integration functions and are not visible in the AMT Client:

- **PlanningTask** is a flag. A value of 0 creates a non-planning strategy task and a value of 1 creates a planning strategy task.
- **LeadDays** defines the planning lead time. This is required to be non-zero if the **PlanningTask** flag is set to 1 for a planning strategy.
- **NotAfter** is a usage which defines that the strategy should not occur after this usage.

When the above fields are not set on the standard job via the import process or integration, AMT makes some assumptions when creating projected tasks from standard jobs, such as:

- not flagging the **Task** as a Planning Task.
- not setting any **Lead Days**.
- setting **Not After** to zero.

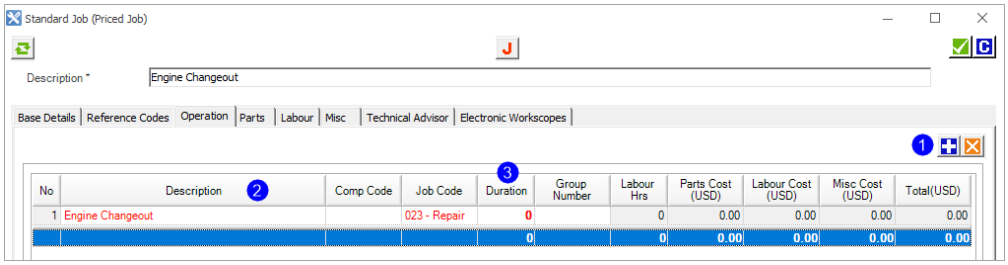
The **Cost Bearer** will be set to the default Cost Bearer and the strategy task **Currency** will default to that of the standard job, however it can be overwritten.

Therefore, additional configuration of the strategy task has to be done manually using **Strategy Task Manager**, **Editable Strategy Task Manager**, or using the multi-editors.

Create operations

In order to allocate parts, labour or misc. cost items to the standard job, you must create an operation. An operation reflects a specific step required to execute a standard job. For example, for a Remove and Install standard job, the first operation may be Remove and the second operation Install. In the majority of cases, the standard job will only consist of one operation, and it has the same coding as the standard job header.

On the Standard Job Manager, click the **Operation** tab.



Item	Description
1	Click to add an operation.
2	Operation will, by default, inherit the codes from the header, but you are able to change them.
3	The time required to complete the operation.

Define Labour and Misc requirements

The method of calculation of standard job totals (that is, parts cost, labour cost, misc. cost, total cost, labour hours and duration hours) will depend on whether the **Overwrite All** check box is selected on the standard jobs Base Details tab.

If Overwrite All is selected

If Overwrite All is selected in the base priced job, you will need to manually enter the total parts, labour and misc costs, total labour hours and total duration hours. Total cost is calculated as the sum of these manual cost entries.

If a new priced job is created, AMT gets the details from the base priced job. If the base priced job has the Overwrite All check box selected, then the new priced job is created as Overwrite All. The manual pricing is based on the exchange rate between the base priced job and the new priced job created.

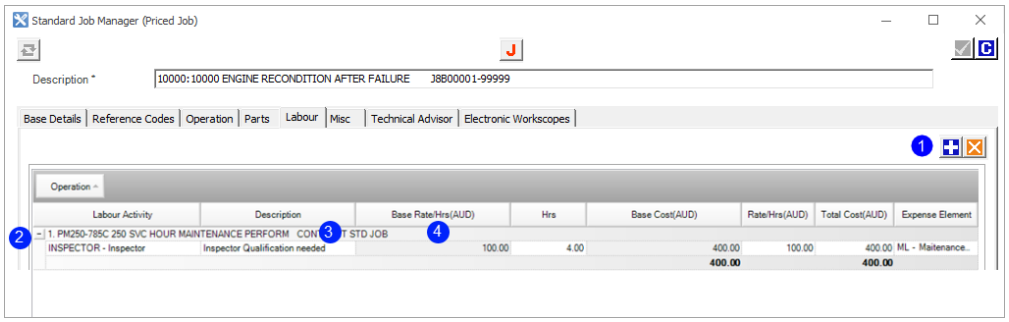
If Overwrite All is not selected

The labour hours and duration hour's amounts are calculated as the sum of the respective values for all standard job operations that make up the standard job.

The parts, labour and misc. cost amounts will be calculated as follows:

- Parts Cost is calculated as the sum of the part cost of all priced job operation parts in all priced job operations that make up the job; and are calculated as (Cost Per Part - Credit) x Quantity x Probability.
- Part prices may show as Invalid if there is no record of the required parts price in the Parts Manager for the selected price group.
- Parts Usage Factor percentage can be adjusted to increase or decrease the costing of parts only costs.
- Labour Cost is calculated as the sum of the miscellaneous costs of all prices job operation misc.
- Total Cost is calculated as the sum of parts, labour and misc. costs.

Creating a standard job in AMT does not add the job to the ERP system. The standard job is stored in AMT and is not automatically updated by price changes stored in the ERP system unless there is a parts interface in place.

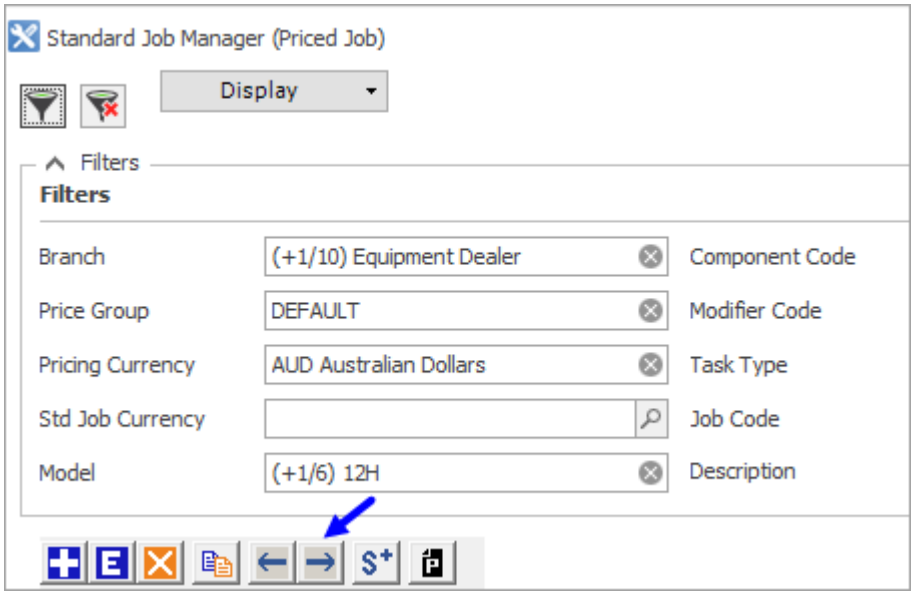


Item	Description
1	Click to Add Labour.
2	Highlight the operation first.
3	Labour items can be entered as free text.
4	Costs can be entered manually.

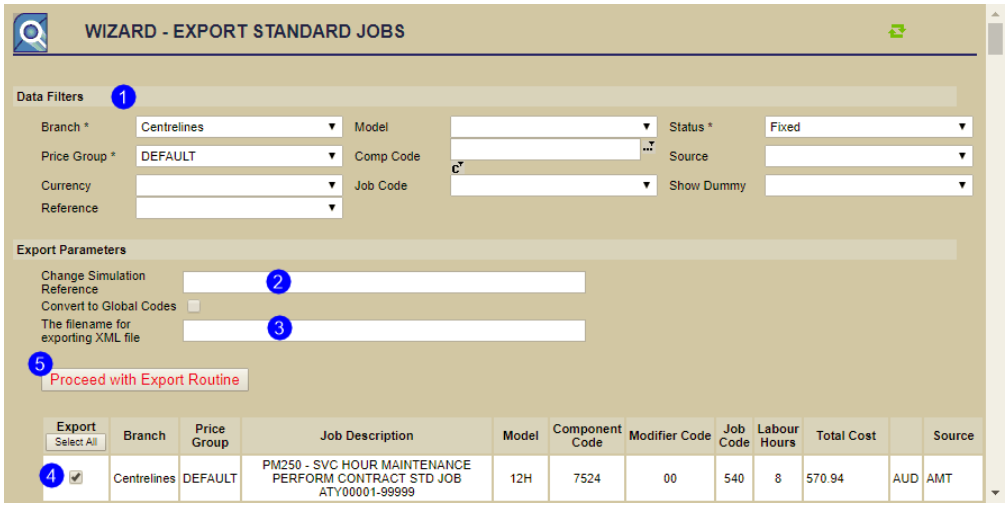
Export standard jobs

When exporting a standard job you have the option to create the export file using either normal AMT codes or using global codes if a conversion is necessary. Global codes are defined in system tables which can be accessed via the **System Administration** menu item.

To export a standard job, navigate to **Knowledge Database** > **Standard Job Manager** and click the **Export Standard Jobs** icon.



This will open the Export Standard Job Wizard in a new tab in your web browser.



Item	Description
1	Filter for the relevant standard jobs.
2	Enter a Simulation Reference if the standard jobs require a unique identifier, for example, the identifier used by the program into which the exported standard jobs will be imported.
3	Enter a unique file name.
4	Select which jobs to export.
5	Click to export the standard jobs into an XML file.

Linking to standard jobs

There are two types of links that can apply to standard jobs:

Link to External System

A standard job can be mapped to an external system, such as SAP, so that when there are changes in the external system, those changes are reflected in the standard job. This mapping triggers updates in one direction only; that is, a standard job created in AMT is never sent to an external system, and a standard job linked to an external system cannot be modified in AMT. However, a standard job can be duplicated to create a new, unlinked standard job in AMT.

Link to Projection

Projections can contain links to one or more standard job, so that changes to a standard job can cause changes to the linked projections.

- If a standard job is linked to a strategy task in a current, alternate or centreline projection then any changes to the standard job will automatically update the strategy task with the new costs.
- When an archived, estimate-current, estimate-draft or budget projection is linked to a standard job, changes to the standard job will no longer trigger updates to the projected job amounts.

Linking standard jobs to strategy tasks

You can link a standard job to a strategy task in a projection. Types of projections in AMT include centreline, estimate, budget, current, archive, alternate and so on.

In order to link standard jobs, certain codes in the strategy task must match the codes of the standard job. The codes are:

- Model of the asset (depending on system configuration).
- Component code.
- Job code.

Depending on the projection type, the standard job will impact the strategy task to which it is linked in the following ways.

- Centreline: The costs and pricing date of the projected job in the centreline will change if the costs in the standard job changes or if the standard job changes.
- Estimate: The costs of the projected job in the estimate will reflect the initial cost of the standard job, but will not be updated even if the standard job changes.
- Budget and Archive: Behave the same as the estimate.
- Current and Alternate: Behave the same as the centreline.

Populate centreline maintenance strategy

Populating the centreline's maintenance strategy involves selecting the AMT standard jobs that define the maintenance strategy.

AMT can create strategy tasks from standard jobs, however, the standard job must adhere to some requirements. The AMT standard job must have a:

- First parameter greater than zero.
- Frequency parameter greater than zero.
- Task type.

Standard job security

In addition to the standard security features in AMT (which applies to functions, reports, branches and sites), it is also possible to restrict the users who can edit specific standard jobs.

To secure a standard job you need to set the **Secured** flag in the **Reference Code** section of a standard job, and nominate the **Group** which can edit the standard. A Group is created by the AMT system administrators using the **Message Group Administration** function, and select users are linked to the group.



Standard job security can only be applied to standard jobs which are created in AMT (that is, the System Source = AMT).

Standard jobs and workscopes

Having standard jobs and workscopes for high value and regularly performed tasks is a key quality driver.

- Check that the strategist has a timetable or schedule for the preparation of standard jobs and workscopes for all high value and regularly tasks.
- View the listing of standard jobs for that branch.
- View the listing of workscopes for that branch.

(Rpt_398) Parts Interrogation

Outlines the details from the Parts Manager on part number, source of supply, standard job reference, probability and so on.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

None

(Rpt_529) Strategy Task - Standard Job Details

Lists, for strategy tasks linked to standard jobs, the details of the parts from the standard job.

Tips

Labour and Misc data from the standard jobs are excluded from this report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Task Type, Cost Bearer

(Rpt_833) Standard Job Version History

Compares different versions of Standard Jobs and displays differences and change-related information.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Report Type, Date Selector

Additional filters

Modified by User, Branch, Price Group, Std Job Currency, Model, Standard Job Number, Component Code, Modifier Code, Task Type, Job Code, Source, Part Number

EM Issues

Equipment Management Issues covers all issues directly related to an equipment type or model. Issues include product problems such as equipment design, equipment manufacture, OEM updates, continuous improvement initiatives, significant application and operation issues, poor maintenance practices (such as redo, inappropriate workscopes), and general performance items that need to be addressed.

EM Issues process

EM Issues provides a structured process for registering, investigating and resolving issues. Technical findings can be recorded, documents (such as photos) attached and correspondence logged. EM Issues is web-based, allowing access by a wide range of users from different sites. As the database grows, EM Issues will provide a practical means for leveraging the knowledge of the organisation. The lessons learned from one group are readily accessible to others, and with integration to the AMT suite, the impact of an EM Issue is quantified – costs, equipment breakdowns, downtime, labour hours and so on. In addition, the corrective actions determined can automatically create repair work orders for planners.

An EM Issue in AMT has a status. The EM Issue statuses are:

- Open
- Closed
- Abandoned

An Open Issue can go through a number of stages:

- Registration
- Technical Analysis
- Evaluation
- Authorisation
- Investigation
- Short Term Action
- Long Term Action
- Monitor and Resolution

You can close an EM Issue at any time in the process. However, the EM Issue should only be closed either:

- Following an Evaluation, whereby the EM Issue is deemed insignificant for it to proceed into an Investigation cycle (which can be a costly and time consuming exercise). In this case the EM Issue is Abandoned.
- After there is agreement that an Action and Follow-up are confirmed as successfully addressing the EM Issue – then the issue is completed. Company Procedures should be referenced for specific criteria as to when an EM Issue is opened and closed.

Navigate to **Knowledge Database** > **EM Issues** > **EM Issues**.

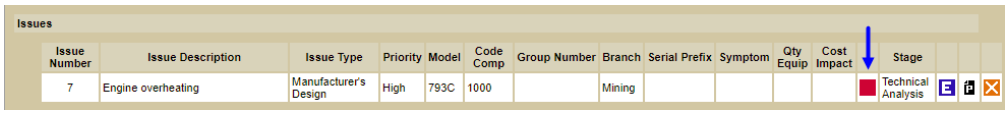
Toolbar buttons

Button	Description
Button	Description
	Add an EM Issue
	Apply Filters
	Cancel Filters
	Equipment Management Issue List report
	Save
	Jump back to AMT Web main menu
	Close EM Issues

Button	Description
	Log out

Click **Apply Filters**  to see all the issues in the database. The coloured boxes on the right of each issue indicate the following:

- Green = not overdue
- Red = overdue



EMI technical analysis

A Technical Analysis is performed to describe in detail the symptom and cause of the EM Issue. Documents which support the analysis, should be attached to the EM Issue.

EMI investigation

Once further investigation into an EM Issue has been authorised, you should log all communications, project plans, progress feedback and so on. Then, at any time, anyone in the company with access to AMT can see details on the investigation as well as estimated time to completion.

EMI evaluation

The evaluation of an EM Issue must be objective. The evaluation should be based on the EM Issues' impact on the key drivers of equipment management, such as equipment design, application and operation, and operational efficiency (strategy, planning, daily operations, administration and customer service). The impact is typically quantified in terms of the number of equipment effected by the EM Issue and resulting cost and availability side effects.

EMI authorisation

The outcomes of the evaluations of EM Issues potentially are a reprioritisation, a status alteration, and a possible recommendation for further investigation, which is submitted for authorisation.

EMI monitor and resolve

A follow-up usually occurs after a short-term or long-term action has been implemented. The follow-up is a check to ensure that the action has been implemented correctly (as per the specification) and that the implemented action has had the desired effect. It could be dangerous to assume that the action will work perfectly first time. Perform proactive checks and tests (whatever is necessary) to verify the quality of the

implemented action, and whether the action has indeed resolved the issue. Once all parties are satisfied that the EM Issue has been resolved, it should be closed.

EMI corrective action

EM Issues facilitates two kinds of corrective action:

Short-term action

Typically, a short-term intermediate solution is applied in the interim period, during which time a long-term solution is investigated. For example, an EM Issue is registered as a leakage of oil in the piston because of a mistake in the engineering specification of that component. A short-term action is to fix the oil leak on the piston by doing XYZ. This solution is applied until the OEM designs a new piston that will correct the leaks, and that component is installed in the equipment (the long-term action).

Long-term action

A long-term action represents a solution where no foreseeable, additional corrections are required for the EM Issue in order for it to be resolved. Another example of a long-term action is a decision by the OEM to develop a transmission especially for trucks in high-altitude applications, and a decision by the Dealer to install this new component once it becomes available.

Most short-term actions require some form of maintenance activity, therefore a backlog will be created and applied to the relevant pieces of equipment upon saving a short-term or long-term action, based on the data you enter.

Linking EMI to equipment, costs and availability

Equipment status contains the history of all maintenance performed on equipment registered in AMT – failure modes, costs of repair, downtime, components worked on and so on.

As an EM Issue is related to equipment, you can link an EM Issue to historical maintenance on one or more pieces of equipment in **Equipment Status**, which automatically quantifies the EM Issue in terms of quantity of equipment affected, cost impact and availability.

The results of the linking can be seen when you run the **Equipment Management Issue List** report.

Application and operation metrics

Standard application and operation metrics should be in place for all equipment. These should be directly related to the equipment high risk areas.

Where the product performance and operating practices are consistently breached, this should be logged and prioritised in EM Issues. As with equipment design it is important to evaluate the:

- Number of current design issues and their priorities.
- Overall impact design issues have had on availability, reliability and costs.
- Potential impact the current design issues will have if not addressed.
- Progress being made on the current issues.

Equipment design

All equipment design issues should be logged and prioritised in EM Issues. While there is no specific KPI report for equipment design it is important to evaluate the:

- Number of current design issues and their priorities.
- Overall impact design issues have had on availability, reliability and costs.
- Potential impact the current design issues will have if not addressed.
- Progress being made on the current issues.

Equipment management issues should be looked at by site level and across multiple sites; while there may be only one or two instances of an issue occurring at one site, across multiple sites it may become apparent that the issue is quite repetitive and warrant an improvement project.

Repetitive equipment management issues across multiple sites may indicate:

- Poor maintenance practices or skills deficiencies
- Product problems
- Rebuild centre problems
- Warehousing or transporting Issues

For a more complete picture, equipment management issues should be analysed in conjunction with the warranty claims history.

Strategy optimisation - Investigation

The investigation process varies depending on the issues identified. If the issue is specific to an equipment model, that is, a product problem, use the EM Issues module in AMT to document and manage the resolution process. Navigate to **Knowledge Database** > **EM Issues** > **EM Issues**.

If the issue is work management process or maintenance operating system (MOS) related, for example, there is a shortage of skilled mechanics, and not specific to an equipment model, this forms part of the broader business improvement (BI) initiatives and it should be registered in some other continuous improvement log.

The basic investigation approach is:

1. **Record**: Document the issue, describing the symptoms and their impact.
2. **Evaluate**: In order to prioritise the issue you need to evaluate the impact on the fleet, and wider implications for the business, paying particular attention to the estimated impact on production and costs.
3. **Authorise**: Based on the above analysis and depending on the internal processes, authorisation may be required before resources are assigned to execute a detailed excavation of the root cause.
4. **Investigate Root Cause**: The steps taken depend on the issue identified and effects on the business.
5. **Evaluate Strategies (Simple)**: Once the root cause is identified there may be a number of corrective options that need to be evaluated. These should all be evaluated in terms of impact on productivity, cost and resource requirements.
6. **Evaluate Strategies ("what-if")**: If a more detailed analysis of different strategies is required then the "what-if" function in AMT can be used.
7. **Determine Action**: Based on the above process, determine the optimal corrective action.

(Rpt_163) EM Issue List

Defines the equipment management issues that have been registered in the EM Issues module. Details the issue type, date of registration, status, actual downtime hours and so on. Navigate to **Knowledge Database** > **EM Issues** > **EM Issues List**).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report Parameters

Report Type, Equipment Hierarchy

Additional filters

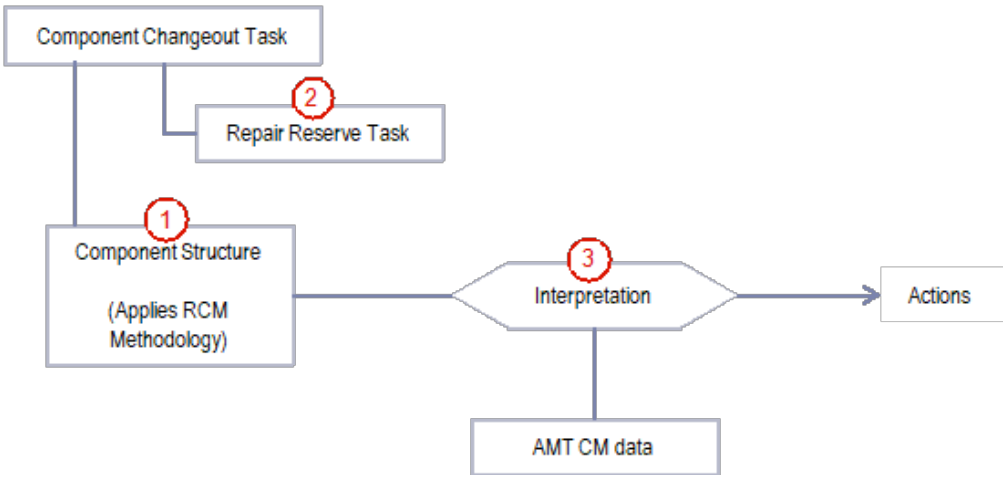
Cost Type, Issue Type, System, Priority, Sub System, EM Status, Component Code, EM Stage, Sort First, Sort Second, Sort Third, No. To Show

Condition Based Maintenance

Condition based maintenance covers three main objectives:

- Reliability centred maintenance (RCM) tool: Document a detailed maintenance strategy using the RCM methodology.
- Component contingency: Factor component contingency costs into projections.
- Condition Based Maintenance: Interpret condition monitoring data against RCM rules to optimise maintenance decisions and actions.

Program Structure



The Component Structure (1) contains the RCM assumptions. The structure provides a detailed documented maintenance strategy for a component.

The Repair Reserve Task (2) factors in the contingency cost for that component.

Condition based maintenance data imported into AMT is automatically interpreted (3) against the reliability centred maintenance rules in the component structure to influence actions.

Set up condition based maintenance message groups

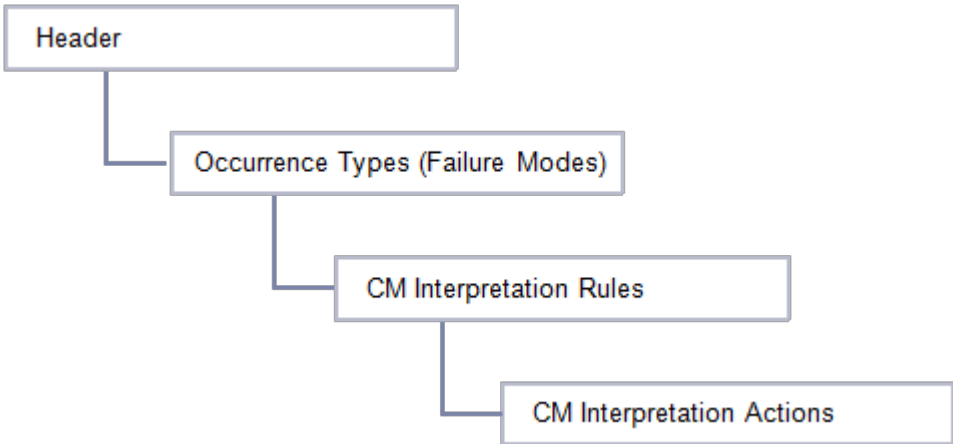
To configure who receives messages generated from condition based maintenance alerts, set up groups and assign users to the groups in **System Administration** > **User Profiles** > **Message Group Administration**.

Component structure

Measurement codes and translation rules are created and attached to specific failure modes so the measurements and any associated alert can be assigned to a possible component failure. Failure modes are assigned to components by setting up a component structure.

The component structure defines the failure modes for a particular component by model of equipment. This structure is later assigned to particular items of plant.

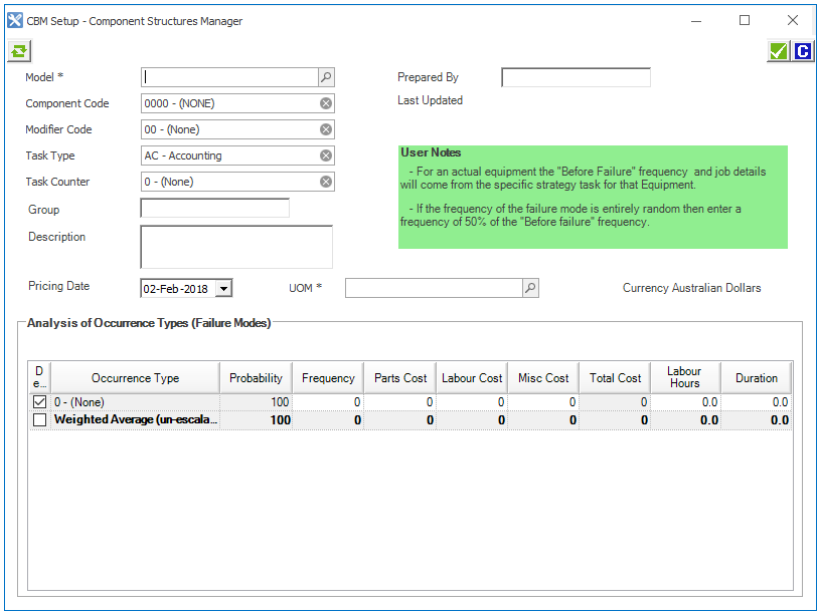
The component structure hierarchy comprises four levels:



The number of levels completed in the hierarchy and the detail entered depends on your objectives.

- For condition based maintenance all four levels must be completed.
- For reliability centred maintenance or component contingency, only the first two levels need to be completed.

To create a component structure, navigate to **Knowledge Database** > **Condition Based Maintenance** > **Component Structures**. The **Component Structure Manager** dialog box opens. Click **Add**  .



Component structure dialog box overview

Icons

Icon	Description
	Refresh.
	Save.
	Close the dialog box.

Component structure header (top section)

Contains header information about the Component Structure, particularly what model and component it relates to.

Fields and columns

Column	Description	Example
Model	The type of asset.	793C, WAR001.
Component Code	Indicates which component the measurement was taken from.	4000 for Engine.
Modifier Code	A code that provides additional identification to the component code, typically the component's location on the asset.	RF - Right Front. TP - Top.
Task Type	A code and description combination identifying what is to happen at an occurrence of a strategy task.	AC - Accounting.
Task Counter	Identifies the sequence in which a strategy task is performed.	1st - First Overhaul. 365 - 365 Day Service.
Group	Lets you set up multiple component structures for the same component and model, which may be required for different applications.	
Pricing Date	This field is for information purposes only, it indicates the pricing date for the costs set	

Column	Description	Example
	up in the occurrence types. It has no impact on financial forecasts.	
UOM (Unit of Measure)	A defined unit of measure.	ENG - Hrs Engine Hours Fuel - Lt Days

Analysis of occurrence types (failure modes)

Within a component structure there can be many occurrence types. These are displayed in the grid.

Column	Description	Example
Default	The generic 'before-failure' scenario (that is, if there is no failure what is the expected life of a component).	Select the check box.
Occurrence Type	The failure mode. Failure modes should reflect the component's failed state and why it failed.	Ceased – piston failure
Probability	The probability of all of the occurrence types. Must add up to 100%.	100%.
Frequency	The UOM between projected occurrences.	15,000 (engine hours between each engine replacement).
Parts Cost	The cost of Parts.	
Labour Cost	The cost of Labour.	
Misc Cost	Any Miscellaneous Costs.	
Total Cost	The total cost of the failure.	
Labour Hours	The number of labour hours.	
Duration	The duration of the failure.	

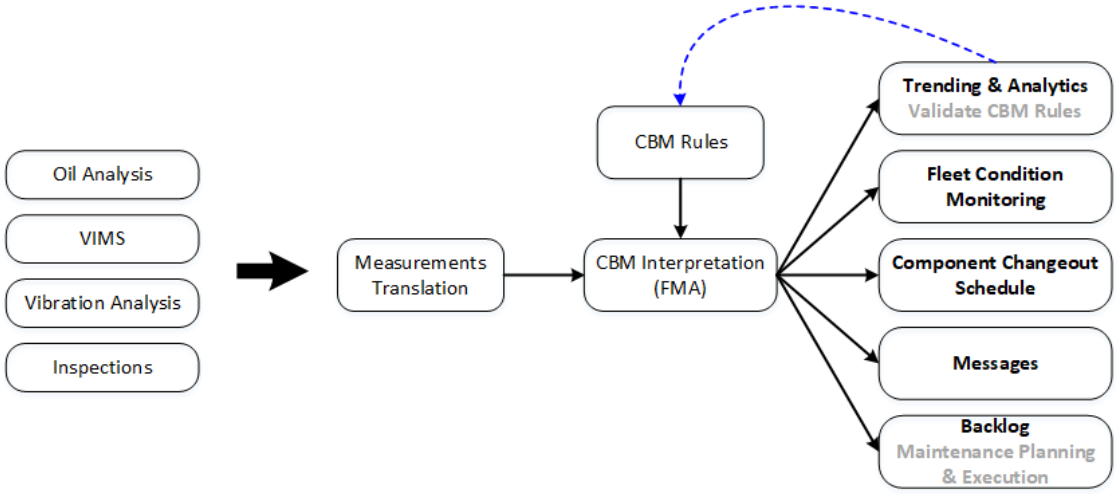
Condition Based Monitoring definition

The process to interpret condition monitoring data against reliability centred maintenance (RCM) rules set up in a component structure to optimise maintenance decisions and actions.

Measurements

Reliability centred maintenance

Reliability centred maintenance (RCM) is a process of interpreting condition monitoring data against reliability assumptions to optimise maintenance decisions. RCM is more complex than the condition monitoring features covered in this help, and should be used for high value components where the cost of failure is significant and justifies the additional complexity. The diagram below illustrates the components of live RCM at a high level.

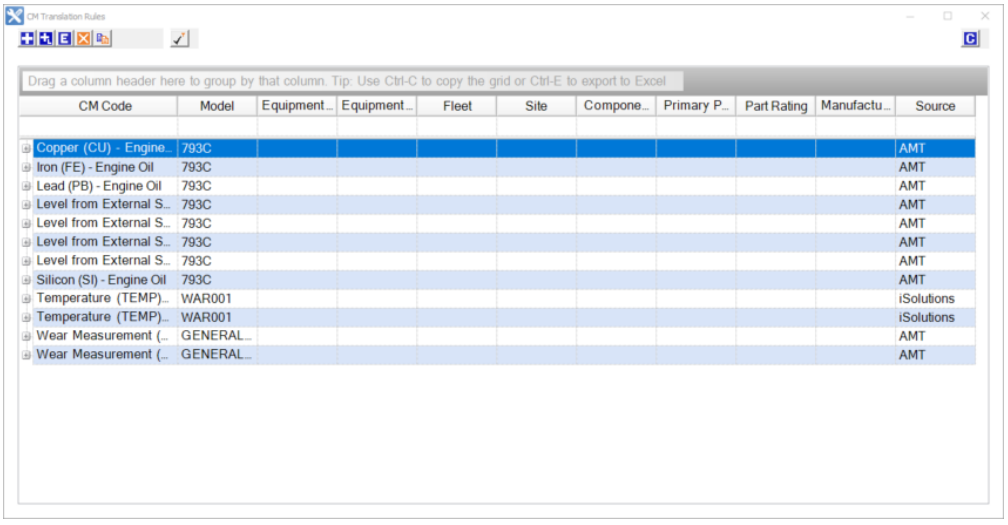


Measurement Translation Rules

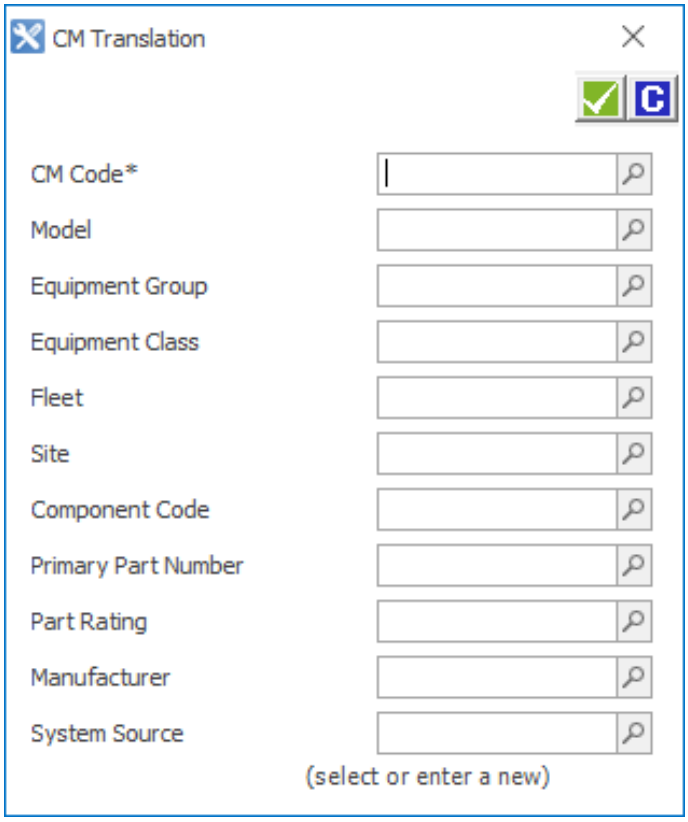
After you have configured measurement codes, you can configure the measurement translation rules. Translation rules have two main purposes:

- To prevent invalid measurements from being accepted by AMT.
- To establish ratings for which level of measurement it is, and whether it will generate a notification or work order.

To configure measurement translation rules, navigate to **Knowledge Database** > **Condition Based Maintenance** > **Measurement Translation Rule**.



The dialog box below displays when you click **Add**.



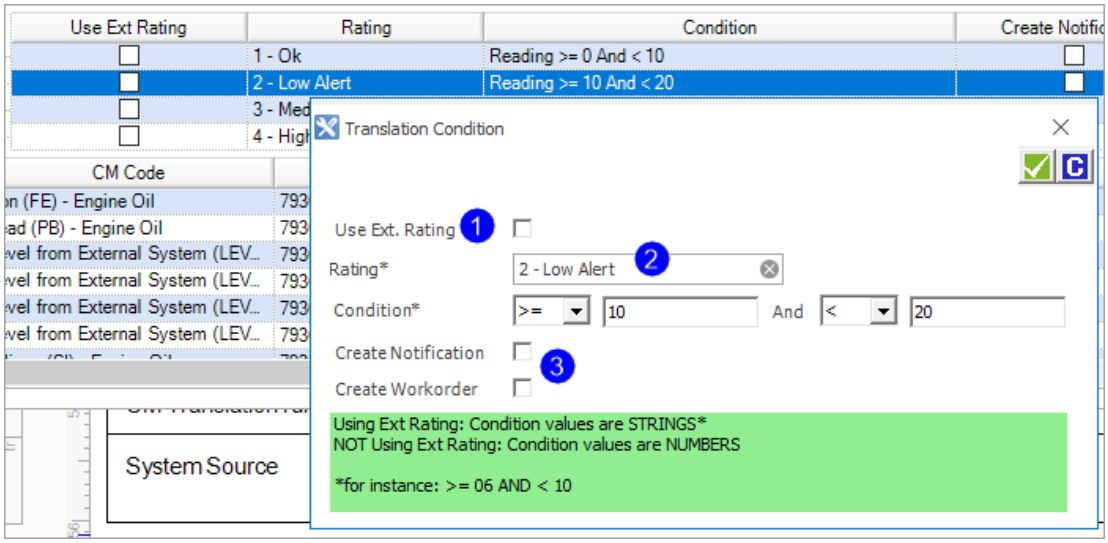
Fields and Columns

Column	Description	Example
CM Code	The only required field. Used for the name of the CM translation rule. This field data comes from the list of measurement codes. You can only create a CM Translation Rule if the CM code does exists in the list of measurement codes.	Copper (CU) - Engine Oil.
Model	The type of asset.	793C, WAR001.
Equipment Group	A grouping, or family, of model codes.	793, OHT.
Equipment Class	A grouping of the same class of equipment.	OHT, TTT, Rail.
Fleet	A grouping of one or more assets used for a specific purpose.	Mining Haul, Process Plant.
Site	The location of a fleet.	Gold Mine - Mill Area.

Column	Description	Example
Component Code	The code for a specific component.	C0723 Fuel Tank.
Primary Part Number	The part number that identifies the whole component, or in some circumstances, a kit number (for example, a PM service kit number).	0302135 - BOLT-LOCKING
Manufacturer	The manufacturer of the asset.	Caterpillar, Hyundai.
Model, Equipment Group, Equipment Class, Fleet, Site, Component Code, Primary Part Number, Part Rating and Manufacturer are not compulsory fields; they are used as filters. If you leave these fields blank, the CM Translation rule applies for all assets.		
System Source	Who or what identified the need to perform a certain work order on an asset.	AMT, SAP, Builder.

Clicking **+** to the left of the row expands the row.

Double-click a row to enter the translation condition.



1	Select the **Use Ext. Rating** check box if the external system sends a rating.
2	The AMT Low Alert rating when the condition is met.
3	Select these check boxes if you want the rating level to generate a notification or a work order.

Recording measurements against failure modes

When measurements or usages are entered or imported, AMT runs a routine to check if any criteria have been met. If criteria are met, a failure mode assessment (FMA) record is created and the actions for that Criterion are triggered.

Measurements can be imported from many sources such as oil analysis and inspections. When measurements are imported, AMT translates them to a rating of 1 (Ok) through to 4 (High Alert) using the measurement translation rules.

MARC

MARC Financial

MARC Financial is an area of AMT specifically designed for reporting the financial status of maintenance and repair contracts (MARC).

Each equipment supplier (OEM) will have their own corporate policies for profit recognition, allocation of profits between internal departments and calculating maintenance and provision requirements. They will also have their own accounting methods for implementing those policies. Therefore, the objective of MARC Financials is to provide some standard financial performance reporting of MARC agreements.

In some cases, these reports will not match the specific policies of the OEM, in which case the reports will be downloaded by the OEM to extract the data needed to feed into specific reports outside of AMT.

AMT provides specialised reporting for maintenance contracts.

- This reporting compares billings (revenue) to cost at cost.
- AMT can incorporate overheads if required.
- These financial reports are usually restricted to accountants and managers.

AMT also facilitates operational analysis of contract performance.

- This reporting compares actual and projected cost performance to the contract budget.
- This type of reporting in AMT uses Cost at Sell.
- Allows information-sharing with customers.
- Site management use this area of reporting.

Both sets of analysis are based on common sources of data.

- Actuals to date: billings and costs are provided to AMT by external systems.
- Projected: billings and costs are sourced from projections (current and budget).

If the data imported into, and maintained in AMT are not in a healthy state, the reliability of both financial and operational financial reporting is undermined. The AMT Health Assessment process reviews the health of AMT and identifies what areas need to be improved and how.

It is recommended that the Health Assessment process is conducted monthly, and the health status of AMT is reported monthly by site in their existing monthly reports.

Characteristics of a MARC

- Definition of what is to be maintained (for example, components, services, ground engaging tools, undercarriage).
- Inclusion or exclusion of labour, travel, oils.
- Provision for unscheduled or unplanned maintenance (for example, breakdown work).
- Definition of responsibilities (for example, planning, procurement, breakdowns, downtime, resourcing).
- Identification of requirements (for example, workshops, warehouse, vehicles, accommodation).
- Definition of maintenance costs excluded from the contract (for example, operator accident damage).
- Definition of how escalations will be calculated to cover rise and fall (for example, inflation).
- Definition of contract term (based on equipment hours, date or a combination of both).

- Fixed revenue rate calculated per equipment based on the agreed scope of maintenance.
- Equipment availability, KPI responsibility.
- Contract termination conditions (for example, component life expectations, billing adjustments).

It is important when setting up a MARC in AMT that the configuration of the system aligns to the requirements of the contract.

Summary

(Rpt_170) Financial Summary

Shows the base financial details of the equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Customer, Industry Group, Overheads/Exp

(Rpt_169) Financial Risk Summary

Summarises MARC contract financials and risk, and aims to provide a management control report for MARC agreements.

Tips

This report shows the contract financials and risks, and reports the variance between the budget and projected financial contribution. It also projects future cash flows.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Responsibility|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_194) MARC Financial Summary

Shows the base financial details of the equipment, including percentage of contract completion, historical and projected costs and billings.

Costs in this report are cost at cost (except the net risk/opportunity values which are cost at sell).

To see overheads in this report, run it at the level of the overhead; that is, contract, branch, site or fleet.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|EquipmentTypes|,
|UserID|,|Currency|

Report parameters

Default.

Additional filters

Overheads/Eqp, Currency

(Rpt_196) MARC PCV Chart

Compares cumulative Actual and Projected Contract Billings, and Budget and Current Contract costs in order to identify differences between the two projections.

Tips

All costs are shown as Cost at Sell. Basis of Calculation = Budget & Current Projections only and Cost Bearer equal to Contract only.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

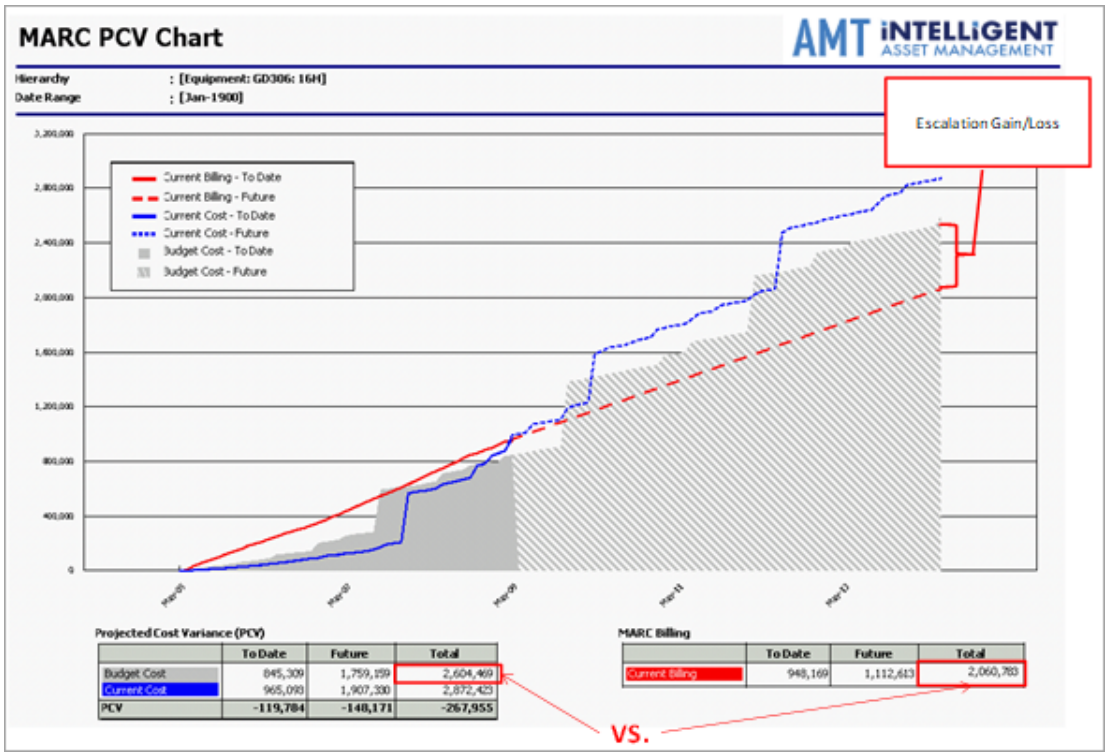
Additional filters

System, Sub System, Task Type, Strategy Task

Assess escalation gain and loss

Escalation gains and losses occur when internal costs escalate faster or slower than billing rates escalations.

Use the AMT report **MARC PCV Chart** to identify whether the contract is experiencing an escalation gain or loss. Navigate to **MARC** > **MARC Financial** > **Summary**. An example of the report is shown below.



Calculation

- Escalation Gain = Total Current Billings > Total Budget Cost Projection (escalated).
- Escalation Loss = Total Current Billings < Total Budget Cost Projection (escalated).

(Rpt_809) MARC PCV Chart - @Cost

Compares cumulative Actual and Projected Contract Billings, and Budget and Current Contract Projected Costs in order to identify differences between the two projections.

Tips

All costs are shown as Cost at Sell. Basis of Calculation = Budget & Current Projections only and Cost Bearer equal to Contract only.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Schedule, System, Sub System, Task Type, Strategy Task

(Rpt_149) Cost Cap Financial Summary

This report is specific to the Sales Tool module - Cost Cap agreement type. The report provides information about the Billings and Costs at Cost.

Tips

Billings amount = Calculated Cost +/- discounts +/- Risk Premium. Cost at Cost is calculated taking into account the financial margins in the system tables.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Manufacturer, Model, Equipment Group, Equipment Class, Region, Division, Equipment Location, Eqp Criticality, Eqp Category, Industry Group, Cost Responsibility, Cost Centre, CC Activity, CC Location, CC Responsibility, Expense Element, Cost Bearer, Cost Type, System, Sub System, Strategy Task, Component Code, Task Type, Modifier Code

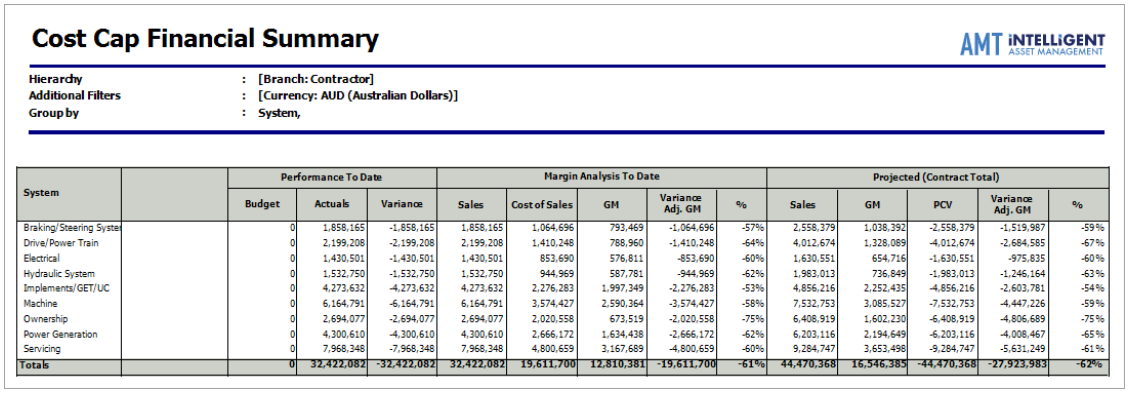
Assess cost cap contracts

Cost cap and component cost cap contracts are a special variety of contracts where the Dealer guarantees a maximum component replacement cost. Such contracts do not normally include services and defect repairs (like a standard MARC).

A component cost cap contract does not have an equipment billing projection in AMT (although it may have an overhead projection to cover management fees and on site overhead costs). Therefore, the analysis of this type of contract is based on comparing the sell of work performed (which reflects what the customer pays) versus the

cost of the work (internal dealer cost); and a cost overrun is reflected in a reduction in margin because the sell is fixed.

The report in AMT is called Cost Cap Financial Summary. Navigate to **MARC** > **MARC Financial** > **Summary**. An example of the report is shown below.



Cost Cap Financial Summary

AMT INTELLIGENT ASSET MANAGEMENT

Hierarchy : [Branch: Contractor]
Additional Filters : [Currency: AUD (Australian Dollars)]
Group by : System,

System		Performance To Date			Margin Analysis To Date					Projected (Contract Total)				
		Budget	Actuals	Variance	Sales	Cost of Sales	GM	Variance Adj. GM	%	Sales	GM	PCV	Variance Adj. GM	%
Braking/Steering Syste		0	1,858,165	-1,858,165	1,858,165	1,064,696	793,469	-1,064,696	-57%	2,558,379	1,038,392	-2,558,379	-1,519,987	-59%
Drive/Power Train		0	2,199,208	-2,199,208	2,199,208	1,410,248	788,960	-1,410,248	-64%	4,012,674	1,328,089	-4,012,674	-2,684,585	-67%
Electrical		0	1,430,501	-1,430,501	1,430,501	853,690	576,811	-853,690	-60%	1,630,551	654,716	-1,630,551	-975,835	-60%
Hydraulic System		0	1,532,750	-1,532,750	1,532,750	944,969	587,781	-944,969	-62%	1,983,013	736,849	-1,983,013	-1,246,164	-63%
Implements/GET/UC		0	4,273,632	-4,273,632	4,273,632	2,276,283	1,997,349	-2,276,283	-53%	4,856,216	2,252,435	-4,856,216	-2,603,781	-54%
Machine		0	6,164,791	-6,164,791	6,164,791	3,574,427	2,590,364	-3,574,427	-58%	7,532,753	3,085,527	-7,532,753	-4,447,226	-59%
Ownership		0	2,694,077	-2,694,077	2,694,077	2,020,558	673,519	-2,020,558	-75%	6,408,919	1,602,230	-6,408,919	-4,806,689	-75%
Power Generation		0	4,300,610	-4,300,610	4,300,610	2,666,172	1,634,438	-2,666,172	-62%	6,203,116	2,194,649	-6,203,116	-4,008,467	-65%
Servicing		0	7,968,348	-7,968,348	7,968,348	4,800,659	3,167,689	-4,800,659	-60%	9,284,747	3,653,498	-9,284,747	-5,631,249	-61%
Totals		**0**	**32,422,082**	**-32,422,082**	**32,422,082**	**19,611,700**	**12,810,381**	**-19,611,700**	**-61%**	**44,470,368**	**16,546,385**	**-44,470,368**	**-27,923,983**	**-62%**

Section	Fields
Performance To Date	• Budget: The capped cost to date at sell. This figure also represents the maximum possible sales to date under the cap. • Actual: Actual work done to date at sell. • Variance: Budget less actuals.
Analysis To Date	• Sales: Work order cost at sell to date. • Cost of Sales: Work order cost at cost to date. • Variance Adj. GM: ◦ If Performance To Date Variance < 0 then Variance Adj. GM = GM + Variance (reduce sales to the maximum allowed under the cap = budget). ◦ If Performance To Date Variance > 0 then Variance Adj. GM = GM (no change because actual costs are below the cap). • % = Variance Adj. GM / Sales.
Projected	• Sales and GM (balance to Maintenance Accrual report). • PCV = Cost at Sell (balances to MARC PCV Chart. • Variance Adj. GM ◦ If Performance To Date Variance < 0 then Variance Adj GM = GM + PCV. ◦ If Performance To Date Variance > 0 then Variance Adj. GM = GM. • % = Variance Adj. GM / Sales.

Suggested filters

- Group By: select the options which allow an easy comparison to the groupings in the original contract.
- Components: ensure the components covered by the cost cap contract are selected.
- Cost Bearer: ensure the appropriate cost bearer is selected for the types of work orders associated with this type of contract.

Interpretation guidelines

- With a cost cap, sales are capped. Therefore, if actual costs exceed capped price (capped sales), the margin is reduced.
- Sales in this report are based on work order cost at sell.
- Variance adjusted margin reflects that if costs have exceeded the budget for the period to date (that is, exceeded the maximum sales under the cap), then sales are reduced to the budgeted sales maximum (think of these as “allowable sales”) in order to report the actual margin that will be achieved.

(Rpt_155) Overhead Cost Variance Analysis

Provides you with the ability to analyse the overhead costs of a project against a projection.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Overhead Category, Expense Element, Cost Centre, Overhead Code.

(Rpt_560) MARC Component Reconciliation

Specific to reconciling components for maintenance contracts. The report reconciles components at the end of the equipment's life (RVIC) based on the actual last occurrence of each component and the frequency of the component from the budget projection.

Tips

Before running the report, make sure the equipment's life, as defined by its end usage or date is accurate.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,
|CostBearer|,|UserID|

Report parameters

Default.

Additional filters

System, Proj Task, Modifier Code, Occurrence Type, Cost Type, Sub System, Escalation, Task Type, Component Code, Region, Division, Cost Responsibility, Equipment Location, Cost Bearer

Cash Flow

(Rpt_137) Cash Flow Chart

Bar chart displaying Cash Inflow and Outflows with the resulting Period (Net) Cash Flows.

- Use this report to identify periods of negative and low position cash flow.
- Cash inflows are based on the equipment’s billing schedule.
- Cash outflows are based on the estimate projection’s costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|AutoLaunch|,|FleetID|,
|EqpPlanID|,|Fleet|,|EqpPlan|,|AnalyseBy|,|ProjectionTypeID|,|ProjectionType|

Report parameters

Default.

Additional filters

None.

(Rpt_138) Cash Flow Statement

Provides the ability to project future cash flows to highlight any potential problems so that management can start planning for issues now. Cash in and cash out are significantly different for contracts.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,
|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|,|DateChecked|,|StartDate|,|EndDate|,
|AnalyseBy|

Report parameters

Default.

Additional filters

None.

(Rpt_165) Equipment Cashflow Chart

Graphical display of gross margin. Refer to the GM Comparison report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

None.

Costs / Contribution

(Rpt_145) Contribution Summary

Shows graphically the projected margin versus the budget margin over time.

Tips

Costs are shown as Cost at Cost.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Additional filters

None.

(Rpt_195) P&L - Summary

This report is formatted similar to a standard Profit and Loss statement (sometimes called an Operating Statement or an Income Statement).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|FinPeriod|,|ProjectionTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_205) P&L - Detailed

Shows the financial summary of life to date costs (budget, actual and variance) - broken into Sales Cost of Sales, Expenses and Net Margin. The report also projects the budget, actual and variance to the end of term.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|FinPeriod|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_206) P&L - Forecast

Shows a forecast of costs by either month, quarter, or year - broken into Sales, Cost of Sales, Expenses and Net Margin.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|FiTipsnPeriod|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_207) P&L - Interrogation

Lets you interrogate the life to date costs by analysing external work orders at cost or sell, contract, direct or indirect expenses at cost.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|FinPeriod|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Branch, Site, Fleet, Equipment, Model, Business Group, Contract Type, Contact, Industry group, Customer

(Rpt_175) GM Comparison

Shows the projected gross margin versus the budgeted gross margin. It identifies at a gross margin level the expected equipment performance.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_176) GM Comparison Chart

Graphical display of the gross margin. Refer to the GM Comparison report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_235) Strategy Task Margin Analysis

Reports actual margins achieved on Parts, Labour and Misc costs by analysing work order history.

Tips

Compare the actual margins in the report (grouping by Fleet and Subsystem)to those in the Set Margin financial table to determine whether forecast margins should be adjusted to reflect actuals.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Task Counter, Job Code, Cost Bearer, WO Status, Company, Cost Selector

(Rpt_258) Workorder Settlement Interrogation (Cost)

Analyses the work order costs at cost. Gives users the ability to monitor spending throughout a period and for reconciliation of costs against the client enterprise resource planning.

Tips

Always shows the total cost of the Job a the Job (Work order) Dare, even though some reports may split these costs over different transactional months (for Transaction Date Reporting).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Sub System, Task Type, Modifier Code, Job Code, WO Status, Cost Selector, System, Strategy Task, Component Code, Task Counter, Cost Bearer, Company

Costs (at Sell and Cost)

At the start of a contract, the forecast of costs for equipment are contained within the equipment's current projection. For maintenance contracts, the costs in the current projection, for parts, labour and misc, are set up at sell (that is, the costs include a margin).

AMT then calculates the forecasted cost at cost by subtracting a set margin from the sell figure, Cost at Cost = Cost at Sell * (1 - Margin). The margins used by AMT for current projections are defined in a special AMT system table called Set Margin.

Actual costs incurred (at sell and at cost) are imported into AMT via the work order settlement interface or can be entered into AMT manually. Actual costs should always be reconciled monthly with the company's General Ledger. Reconciliation ensures confidence in AMT's financial reporting.

Actual costs are reported in AMT verbatim (the costs reported are that which were sent to AMT via an interface or entered manually into AMT by a user). AMT does not change actuals.

However, the forecast of cost at cost from today to the end of the equipment's life cycle, remains a calculated number

Forecasted Cost at Cost = Forecasted Cost at Sell * (1 – Margins).

Maintenance Accrual

(Rpt_189) Maintenance Accrual

Shows where revenue received (billings) have gone - to pay for the cost of sales or taken to profit (MARC GM), or if booked to the accrual account (Current MA). The report also shows the expected future sales (billings) and expected future costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Sub System

(Rpt_190) Maintenance Accrual Details

Calculates and identifies the accrual values required to the end of contract by taking into account the costs and billings to date and the percentage of contract completion.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Currency, Sub System

Assess maintenance accrual

To assist a Dealer's accounting or finance group determine maintenance accruals and profit recognition AMT has three specific reports. The reports are described below. They are all under **MARC** > **MARC Financial** > **Maintenance Accrual**.

Maintenance Accrual

- This report calculates what the Maintenance Accrual (MA) or Provision account should be (MA required) based on future revenues and costs.
- The report complies with US-GAP method for maintenance provisioning and is Sarbanes-Oxley (SOX) Compliant.

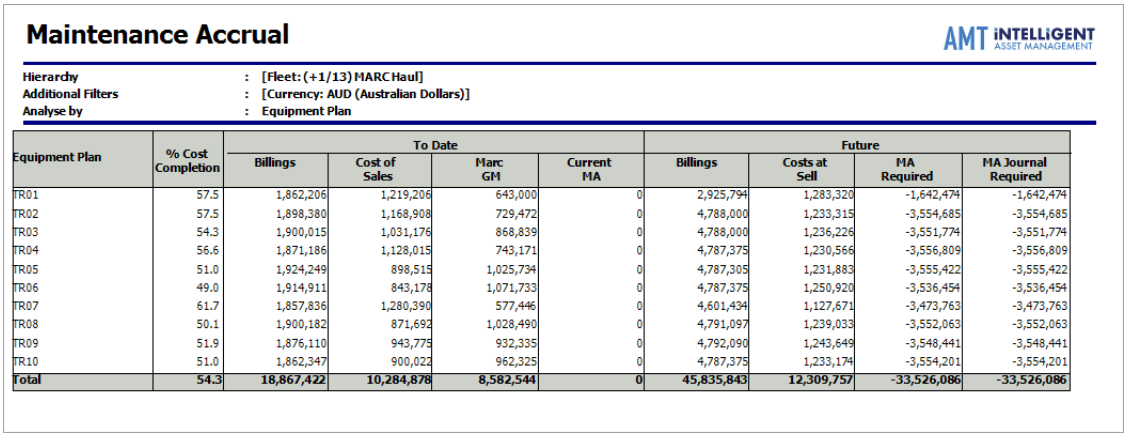
Maintenance Accrual Details

- Based on the Maintenance Accrual report, but includes details on margins and a projected contract position.
- The report complies with US-GAP method for maintenance provisioning and is Sarbanes-Oxley (SOX) Compliant.

Contract Reconciliation

- Report which includes To Date, Future and Total Projected Billings vs Costs at Sell.
- Does not include cost at cost figures.

The most common report used by the Dealer to initially assess the maintenance provisioning requirements is the **Maintenance Accrual** report explained in detail below.



Maintenance Accrual

AMT iNTELLiGENT ASSET MANAGEMENT

Hierarchy : [Fleet: (+1/13) MARC Haul]
Additional Filters : [Currency: AUD (Australian Dollars)]
Analyse by : Equipment Plan

Equipment Plan	% Cost Completion	To Date				Future			
		Billings	Cost of Sales	Marc GM	Current MA	Billings	Costs at Sell	MA Required	MA Journal Required
TR01	57.5	1,862,206	1,219,206	643,000	0	2,925,794	1,283,320	-1,642,474	-1,642,474
TR02	57.5	1,898,380	1,168,908	729,472	0	4,788,000	1,233,315	-3,554,685	-3,554,685
TR03	54.3	1,900,015	1,031,176	868,839	0	4,788,000	1,236,226	-3,551,774	-3,551,774
TR04	56.6	1,871,186	1,128,015	743,171	0	4,787,375	1,230,566	-3,556,809	-3,556,809
TR05	51.0	1,924,249	898,515	1,025,734	0	4,787,305	1,231,883	-3,555,422	-3,555,422
TR06	49.0	1,914,911	843,178	1,071,733	0	4,787,375	1,250,920	-3,536,454	-3,536,454
TR07	61.7	1,857,836	1,280,390	577,446	0	4,601,434	1,127,671	-3,473,763	-3,473,763
TR08	50.1	1,900,182	871,692	1,028,490	0	4,791,097	1,239,033	-3,552,063	-3,552,063
TR09	51.9	1,876,110	943,775	932,335	0	4,792,090	1,243,649	-3,548,441	-3,548,441
TR10	51.0	1,862,347	900,022	962,325	0	4,787,375	1,233,174	-3,554,201	-3,554,201
Total	**54.3**	**18,867,422**	**10,284,878**	**8,582,544**	**0**	**45,835,843**	**12,309,757**	**-33,526,086**	**-33,526,086**

Section	Fields
	• % Cost of Completion: based on cost forecast.
To Date	• Billings: Billings interface files from ERP. • Cost Of Sales: Contract cost at cost. Work order interface files from ERP. • MARC GM: Profit to date. • Current MA: Always equals zero unless the maintenance accrual interface is operational. Therefore, AMT assumes all billings and costs to date have been taken to profit, that is, the balance shown in the To Date - MARC GM column.
Future	• Billings: Current billings schedule per asset. Current billings projection total less billings to date. • Costs at Sell: Current projection per asset. Forecasted cost at cost based on current projection. • MA Required: Future Cost of sales less future billings. Provision balance AMT recommends. • MA Journal: MA required less current MA.

Other reports to analyse the data

Click each link to expand a brief explanation of other reports available in AMT to further analyse the data driving these reports.

MARC Financial Summary

This report shows projected margins based on actuals and forecast, and also quantifies the contract's risk profile.

MARC PCV Chart

This report shows graphically the status of the contract over its entire life cycle vs. budget vs. revenue. It will show bow-waves of overdue maintenance and spikes in maintenance (for example, overhaul periods).

The report also illustrates whether the contract is making an escalation gain or loss.

High Risk

This report identifies actual risk in the contract period at the component level.

A risk is defined as a component which falls due just outside the contract term, which if not managed closely, may fall into the contract term and result in the dealer having to perform extra unbudgeted work (dealer will not receive revenue for this work).

Cash Flow

This report shows historic and projected contract cash flows. Inflows are equipment billings and outflows are work order settlements and strategy task costs.

P&L Summary and P&L Forecast

This report shows the To Date and Projected financials of a contract in a profit and loss format based on the date range selected, and includes overhead billings and overheat costs if applicable.

(Rpt_419) Contract Reconciliation

Displays To Date, Future and Total Projected Billings and Costs (at Sell).

Tips

This report only considers Contract Costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,
|EqpPlanID|,|EqpPlan|,|ProjectionTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

None

(Rpt_811) Contract Revenue Recognition - @Cost

Displays To Date, Future and Total Projected Billings and Costs (at Cost).

Tips

The report only considers Contract Costs.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,
|UserID|,|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

None

Forecasting

(Rpt_156) Costs

This report provides the ability for analysing historic and Forecast costs using a variety of report filters.

Tips

Initially analyse costs by System. Then drill down into subsystems and components to identify the major contributors to high costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,
|Enable_Cost_At_Cost|,|CostBearer|,|ProjectionTypes|,|EquipmentTypes|,
|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Region, Eqp Criticality, Equipment Group, Eqp Category, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Modifier Code, Division, Equipment Class, Equipment Location, Parts/Labour/Misc,

Currency

Forecasted margin calculation

The standard AMT configuration is to have forecasted cost at cost calculated based on user-defined margins stored in the Set Margin system table for current projections and the Equipment system table for budget projections. However, it is possible to enable both budget and current margins to be calculated from standard jobs linked to strategy tasks; rather than relying on the default margins.

If this system setting has been enabled, the following rules apply

- Standard Job margins with be used for strategy tasks linked to standard jobs where the margin calculated in the standard job > 0%.
- If the calculated margin is <= 0%, or the strategy task is not linked to a standard job, the default margins from the Set Margin system table will be used.

Ask your AMT Administrator whether this special setting has been enabled.

While AMT can hold the cost at cost of a part, AMT does not show the cost at cost of a part; there are no reports or screens showing this value. If you want to see this data then you have to go to the source of the part's price in the external system from which it was sourced.

Forecasting accuracy

Depending on the modules of AMT which have been implemented, AMT produces a number of forecasts including:

- Component
- Parts
- Rotables
- Financials

The health of AMT, which also reflects the health of the forecasts produced by AMT, can be viewed in the Dashboard module of AMT. In this section of the Dashboard, AMT's health is rated and certain key health performance indicators are displayed.



Not all factors influencing the forecasting accuracy of AMT are shown in the Dashboard.

Additional factors affecting the accuracy of the forecasts produced by AMT include:

- Correct set up of an equipment's budget projection.
- Accurate current projection parameters (for example life, cost).
- Correct set up of an equipment's budget billings schedule (if under contract), refer to Billings Projections.
- Accurate current billings schedule (if under contract), refer to
    - Billings Projections
    - Billings Reconciliation
- Accurate component changeout history, refer to Work Order Settlement Validation.
- Bow wave of costs, refer to Bow Wave Identification.
- Overdue components, refer to Overdue Strategy Tasks.
- Accurate future expected margins.
- Up-to-date escalation history, refer to Escalation.

Intercompany Accuracy

Financial Report Parameters

The Intercompany MARC Financials area of AMT supports the process of reviewing MARC performance to determine profit recognition and intercompany accounting equity.

It provides very specific functionality that will only be applicable to a few organisations. It is applicable in the following circumstances:

- Dealer has multi-currency, and
- Dealer supplies the MARC services from two independent companies (that is, labour from one company and parts from another company), and
- Dealer recognises MARC profits based on % MARC Billings completion.



It is recommended that MARC profit recognition is based on % work or services completion not on billing completion.

The objectives of this area of AMT are to:

- calculate the profit that can be recognised and the MARC accrual requirements.
- ensure there is alignment between the MARC billings and costs within each company (that is, do not want only 30% of the billings going to one company when it is incurring 80% of the costs).
- evaluate the margins earned by each company.
- recognise one-off MARC adjustments that have not been captured elsewhere in AMT.

If organisations are utilising Intercompany Analysis, the Maintenance Accrual and Maintenance Accrual Detail reports in AMT will NOT be applicable.

The Financial Reporting Parameters form is used to set the parameters for the calculations.



Selecting a combination of Contract and Fleet which are not associated will result in a blank form. You must select a Contract and Fleet which are linked.

Financial reporting parameters

- % Completion at which margin recognition should begin:
  - Margin recognition is based on % Billings Completion. This alone could be too aggressive. This feature allows organisations to delay profit recongnition to a certain point.
  - For example, if 20% is entered, there will be no profit recognition until 20% of the total contract billings have been earned. Thereafter, the profit will be amortised evenly over the remaining 80% of the contract billings.
- % Cost over-run provision:
  - This allows organisations to ensure that the maintenance accrual always covers the potential of a cost over-run. The calculation uses cost@cost.
  - For example, if you enter 20% and future costs@cost are projected to be $1,000,000, the program will ensure that future billings + maintenance accrual > $1,200,000. If it is less it will reduce the profit recognition and increase the maintenance accrual required.
- MARC adjustments:
  - Allows users to enter any one-off adjustment by the company that is not otherwise being recognised by AMT.
  - For example, in calculating the profit recognition, the company may wish to recognise that there is a likely termination adjustment to the customer of $200,000.
- % Cost Split:
  - The % cost split between companies can be entered by labour and miscellaneous and for parts by strategy task.
  - This is used to calculate the future costs@cost by company for the purposes of projecting future margins.
  - The Billings split by Company is entered in the Equipment Plan.

(Rpt_128) Billings Alignment

A text report that calculates the percentage of future Billings that need to go to the first Company so that its billings are inline with its costs. It is used to practically adjust future MARC invoicing so that there is alignment

between costs and billings within the companies. An example would be, if 65% of the costs are projected to be incurred by Company A, if only 40% of the billings to date has gone to Company A, maybe 80% of future billings need to go to Company A in order for the total MARC billings to average 65% to Company A.

Report parameters

Default.

Additional filters

None

(Rpt_129) Billings Profit Balance Chart

A bar chart that shows the Billings, Costs and Margin for the group and each company individually.

Tips

This report is used to ensure there is a reasonable alignment of margin against each Company. For example, you do not want one company making a 30% margin while the other makes a 20% loss.

Report parameters

Default.

Additional filters

None.

(Rpt_180) Intercompany Cost Analysis

A text report that allows users to analyse the costs specifically for a Company. It also shows these costs against the Group costs and shows the % of the Group costs by Parts, Labour and Miscellaneous.

Tips

This is the only report that lets you analyse the costs within a Company (where there has been a split between Companies). The report also provides a validation against the percentage cost split figures entered in section D of the financial reporting parameters.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|Currency|,|ProjectionTypes|,|UserI

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Task Counter, Job Code, Cost Bearer, WO Status, Cost Selector

(Rpt_192) MARC Analysis Of Actuals

A text report that shows a summary of the actuals to date of costs (by parts, labour and miscellaneous), margin and billings. It also shows % billings complete. The report can be run for the Group or individually by Company.

Tips

This report provides an audit train for the Billing Profit Balance Chart report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Additional filters

Company

(Rpt_193) MARC Financial Analysis

This report contains the following summaries:

- MARC Budget
- Actuals to Date
- Future Projected
- Total Contract Protected

Each summary shows costs (by parts, labour and miscellaneous), margin, margin % and billings.

Tips

Provides a high-level analysis of the MARC financial status by Company.

Report parameters

Default.

Additional filters

Company

(Rpt_197) MARC Reconciliation Worksheet

A detailed text report showing the calculations to support the specific methodologies defined in the Financial Report Parameters.

- %Completion (SMU and Billings).
- Projected MARC performance (Billings and Margins).
- Projected impact if the defined cost over-run occurs.
- MARC position to date (Billings, Costs, Margins).
- Calculation of the required Maintenance Accrual, Provision for cost over-run.
- Calculation of the Margin that can be recognised vs deferred.

Tips

This report is only applicable if the organisation has adopted the methodologies supported in the Financial Reporting Parameters form.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Today|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Company

Contract tendering

Some of the topics and screens relating to contract tendering relate to more than one module in AMT.

- Overview of MARC Financials
- MARC Financials Principles
    - Equipment Billings
    - Equipment Costs
    - Overhead Billing and Costs
    - Escalation - Equipment Costs
    - Escalation - Equipment Billings
    - Escalation - Overheads Billings and Costs

Update current margins

The Contract Manager must assess the accuracy of the parts, labour and misc margins used in AMT to forecast cost at cost (actual cost at cost history is contained in work order settlements). The margin assumptions are stored in the AMT System Table - Set Margin.

Navigate to **System Administration** > **System Tables** > **Financial** > **Set Margin**.

Depending on the configuration of AMT, adjustments to margins made here may not impact margins calculated for strategy tasks linked to standard jobs.

Before updating margins, you first need to review what the historical margins have been.

Gross margin analysis

MARC Financial reports are the only reports in AMT that show

- Cost at Cost.
- Overheads (expenses).
- Gross and Net Margins.

High-level gross margin figures can be viewed in the Financial Performance section of the Dashboard (see Dashboard Overview and follow the link to Dashboard Structure). The Dashboard shows four different gross margins, each of which is calculated over the equipment's entire life-cycle:

Budget

Based on the budget projection (billings and costs) of a piece of equipment.

Projected

Based on the current projection (billings and costs) of a piece of equipment.

Risk Adjusted

There is a risk associated with the variability of component lives and the impact on the accuracy of projections. For example, if you enter 15,000 hours as projected life of an engine, AMT will assume all future engines will last exactly 15,000 hours. However, if you assume that 15,000 is only the average and that there will probably be some form of distribution around the average (a bell curve), what will the impact be on the financial projections? AMT assesses this risk by quantifying the high risk and high opportunity components and evaluating the impact on the projected margin – called risk adjusted margin.

Average

It is unlikely that all high risk and high opportunity items will occur so the average margin is the average of the risk adjusted margin and the projected margin. The average margin therefore is the most realistic estimate of projected margin.

MARC Admin

Workflow management

This is the process of managing the MARC equipment. It is covered in other topics. The sub-processes cover:

- Long-term Planning.
- Short-term Planning and Scheduling.
- Work Execution and Finalisation.
- Strategy Optimisation.
- Performance Reporting.
- Senior Management Reporting.

During this process work order costs are incurred by the contract and posted to the contract account in the ERP.

Contract Invoicing

Process contract invoices

The workflow for processing invoices is as follows.

The administrator receives the Contract Invoice Details report from the planner and processes the invoice in the enterprise resource planning system. This triggers the enterprise resource planning system to automatically upload the invoice into AMT as equipment billings.

View Contract Invoices

Complete documentation for this topic will be available in a future release.

Post Contract Invoices

Complete documentation for this topic will be available in a future release.

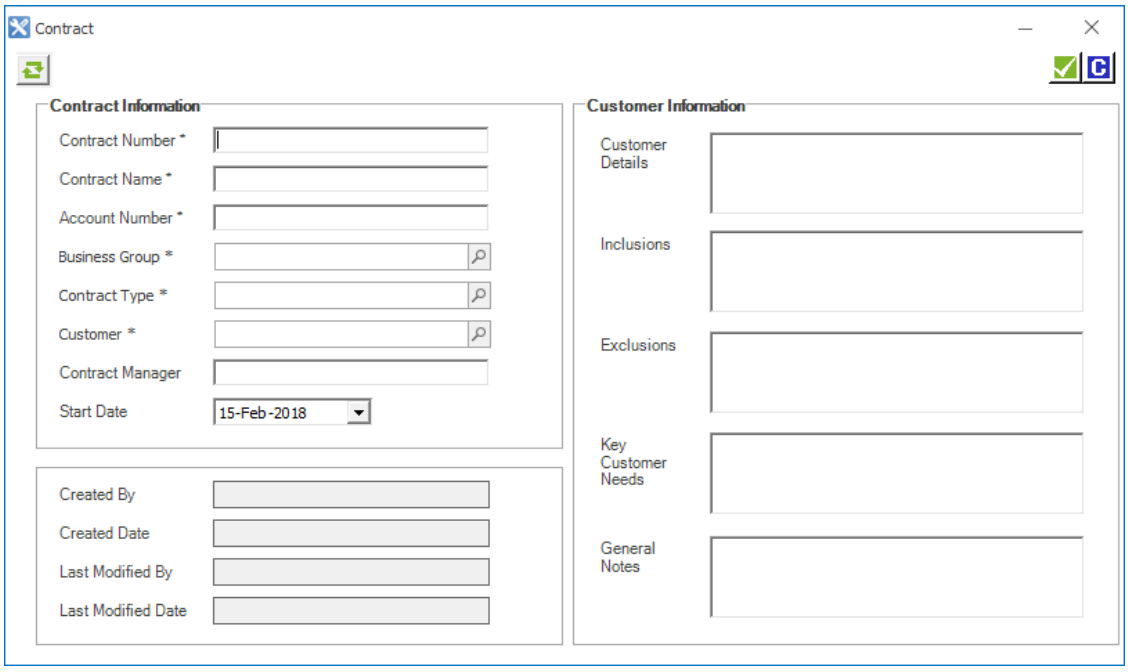
Finalise Contract Invoices

Complete documentation for this topic will be available in a future release.

Contract Administration

Navigate to **MARC** > **MARC Admin** > **Contract Administration** > **Contract Administration**.

Click the **Apply Filter** button to review whether the contract has already been set up. If not set up, click the **Add** button and fill in the form.



Field	Description
Contract Number	**Required Field** The contract's account-keeping number.
Contract Name	**Required Field** The title of the contract.
Account Number	**Required Field** The identification number of the contract in the accounting system.

Field	Description
Business Group	**Required Field** The business group of the contract.
Contract Type	**Required Field** The category of the contract, for example. MARC, Cost Cap, HME.
Customer	**Required Field** The name of the customer.
Contract Manager	The person responsible in the organisation for managing the contract.
Start Date	The date the contract commenced.
Customer Details	The contract details of the customer.
Inclusions	Contract inclusions, for example, engines, lube truck, labour.
Exclusions	Contract exclusions, for example, tyres, operators.
Key Customer Needs	Important needs such as providing the customer with monthly reports.
General Notes	Other relevant information.

Rollover

This topic only applies to MARC assets.

Using contract rollover functionality you can take the current projection from a terminated asset and apply it to:

- another asset.
- the new life cycle of the same fleet.

The termination and rollover process is done in the case where a contract reaches the end of its life, but gets renewed. Rollover is relevant when the new contract has different terms, rates and so on.

Neither current backlogs nor events are carried across to the new equipment. The history remains on the old asset and is not transferred. If the project is copied to the new life of the same equipment, the “last done” for the strategy tasks can be rolled over to the new equipment in order to maintain the forecast of next occurrences.

Equipment rollover

If an equipment was terminated under a maintenance contract but will still be managed in AMT as a non-contract equipment in order to track performance, forecast components (for example, using AMT’s Component Capacity Management module), forecast and capture product support parts and service sales (for example,

using AMT's Product Support Marketing module), then the equipment should be re-registered in AMT albeit as a non-contract.

To re-create the equipment you need to remember to:

- Set the Registration Counter appropriately (for example, customer number instead of contract number).
- Set the Start Usage to be the usage one hour after the terminate equipment's end usage.
- Not configure the equipment to be MARC as it is no longer under contract.
- Consider whether Standard or Advance Planning mode is applicable.

When it comes to creating the projections for a non-contract equipment, only a current projection is required because there is no contract budget and no budget or current billing schedule. The current projection does however need to recognise when each strategy task was last performed in order to forecast the next occurrence correctly and report component life achieved correctly.

Contract extensions

It is common for maintenance contracts of all types to be extended. Sometimes the extension is based on the original budget terms, the contract term is simply extended. Other changes to the original contract are required which results in a new contract budget and billing schedule.

Simple contract extension

A simple contract extension is one where the contract term is extended, the original budget does not change rather, the costs are extrapolated for another "x" months or years and additional steps are added to the existing billing schedule.

Questions to ask are:

- Is the scope of the contract the same? Are we doing the same work as before?
- Is the performance of the contract measured on the entire contract, including the additional period?
- Do we evaluate maintenance accruals based on the entire contract, including the additional period?
- Is the pricing from the original contract (factoring any escalation clause) still valid?
- Will the ERP use the same contract number for this extension?

If the answers to these questions is "Yes" you can apply a simple contract extension in AMT. The steps required in AMT are:

- Update the Contract Term to reflect the extended contract period.
- Add additional steps into both the budget and current billing projections based on the agreed rates with the customer.
- Check the recalculated budget variance on the Dashboard is within tolerance.

Normally there is no need to update the registration counter as a simple extension is a continuation of the same contract number in the accounting system.

Complex contract extension

A complex contract extension has a combination of the following characteristics:

- The scope of work and responsibilities to perform the work has changed.
- Pricing of repair options and types of repair options are renegotiated.
- Equipment operating conditions have changed substantially.

- Customer application management practices have changed significantly.
- Fleet composition is changed.
- On-site resource requirements have changed significantly.

Such a contract extension is practically a brand new contract albeit with an asset that already has history. Therefore, in AMT, a number of steps need to be performed in a specific sequence:

1. Terminate the existing asset and move it to a Terminated Fleet.
2. Create the asset again in AMT with the same Model and Serial number, but with a new Registration Counter to reflect the number contract.
3. Update the new assets start usage and end usage date.
4. Create a new budget projection, where the scheduling parameter “First“ of each strategy task reflects when each is next due under the new contract’s term.
5. Create a new billings projection to reflect the new contract, but starting at the equipment’s new contract start hours (for example, if the contract extension starts at 36,000 hours then this start usage for the first billing step).
6. Validate the budget variance and make adjustments are required.
7. Create a new current billing projection as a copy of the budget billings projection. Roll over the current projection.
8. Create a new current projection as a copy of the budget projection and update the “First” parameter to reflect when the each strategy task is next due for that specific equipment (based on its work order history).
9. Create a new overhead budget and current projection and populate with expected billings and costs (if applicable).

If a large number of assets are to undergo this process, and the start hours for each asset is the same (even though in reality some assets may not have reached these hours yet or have even gone over those hours) it is recommended that an asset per model is created in Modelling. Modelling lets you set up the projections listed above and then copy to multiple equipment.

In addition to using Modelling functionality to define the contract budget cost and billing projections, AMT’s Equipment Rollover logic can be used to quickly set up the equipment’s current projection.

Billings

MARC billings

MARC billings is the revenue for maintenance contracts (for example, MARC, CSA, TM&R). AMT projects the revenue for MARCs at the equipment level and sometimes at the overhead level (sometimes referred to as management fees).

Create budget and current billings



The **MARC** check box has to be selected on the **Base Details** tab to enable the **MARC** tab.

Refer to the MARC Billings topic for general information.

Escalation

Escalation refers to price changes over time and is sometimes referred to as Indexation. For example, if the initial projection is priced in year one, for a life cycle spanning several years, then actual costs could be higher

than the initial pricing, simply as a function of inflation over time. To harmonise comparisons between projection comparisons, escalation is applied to budget prices to adjust these values to current levels.

Future projections of both current and budget projections are made at today's escalated values.

Summary of Cost Escalation Rules In AMT

Projection Type	Actual Costs	Future Costs
Budget	Not escalated	Not escalated.
Current and Alternate	Not escalated	Escalated to today's value.
Archived	Not escalated	Escalated to the point of archiving.

Billings projection

This topic refers to MARC Asset Registration only.

The purpose of a billings projection is to detail the expected revenue to be received from customers for work performed under a maintenance contract.

The budget billing projection (part of the budget projection) reflects the amounts agreed to with the customer at the commencement of the contract. The current billing projection (part of the current projection) is the same as the budget billing projection at the commencement of the contract. However, it must be updated whenever a renegotiation of the billing rates with the customer is agreed, which usually occurs due to an escalation of costs. Consequently, the current billings projection will reflect actual billing rates in the past, and expected billing rates in the future.

Current and budget billing projections can have many child billing projections, but at least one billing projection is required for each budget and current projection. For each individual child billing projection, you can select from one of four different billing methods:

- Usage rate by usage steps (default)
- Usage rate by monthly date range
- Monthly rate by monthly date range
- Daily rate by date range

Future billings (that is, from today to the end of the equipment’s life cycle) can be calculated using one of two methods, detailed below.

Auto accrue billings (default method)

- AMT assumes an equipment will earn the total billings calculated in the billings schedule. Even if the company has not billed as much as it should have to date, AMT assumes they will.
- No matter what the actual billings are to date, AMT always assumes the equipment will not earn more or less than the current billing projection’s calculated total billing amount.
- Therefore, extraordinary revenue that the equipment has earned, or will not earn. **must** be incorporated into current billing projections. Otherwise, AMT will forecast the incorrect total billing revenue for the equipment.

Non-auto accrue billings

- AMT calculates the future billings from last invoice period and last unit of measure or month of the latest invoice.

Rules

- The desired future billing calculation method is configured within each child billing projection.
- There can only be one currency per billing projection.
- The billing projection flagged as the default projection will have actual billings (interface files) linked to it.
- Only equipment flagged as being MARC can have billing projections (this flag is set at the equipment level).
- Each projection can have a different billings projection.
- The billings projection of archived projection cannot be modified.

Equipment costs

A dealer essentially has two costs:

- **Cost@Sell**
  Otherwise known as **Sell**. This is the price paid by the end customer for goods and services, and includes a margin.
- **Cost@Cost**
  Otherwise known as **Cost**. This is the dealer's actual internal cost for producing the goods and services.

A MARC is quoted to the customer at **Sell**. The Sell for one customer may be different to the Sell for another customer because some, especially large, customers may receive discounts. The Sell price includes a margin (parts margin plus service margin). All MARC financial reports presented to the customer are based on Sell.

Cost is the dealer's internal cost and is commercially sensitive (only a few people in the company will ever see these numbers). The dealer's finance group reviews the performance of a contract based on Cost (Sell figures are irrelevant).

In order to quote and report on the performance of a MARC, you need the ability to report on both Cost and Sell. AMT has both values.

AMT imports actual Sell and Cost history figures as Work Order Settlements from the enterprise resource planning system, and forecasts Sell and Cost based on Strategy Tasks.

- For budget and modelling projections:
  - **Sell** = Strategy task job costs.
  - **Cost** = Strategy tasks job costs x (1 - budget margins at the equipment level).
- For current, centreline and archive projections:
  - **Sell** = Strategy task job costs.
  - **Cost** = Strategy task job costs x (1- set margins).

Equipment Contract Details



This function is only relevant to equipment configured as MARC, which must be selected from the Equipment Selector.

Upon running this function, you are presented with the MARC tab of the equipment. The screen is split into a number of sections, which are explained below.

Contract Parameters

- **Contract** is a look-up field which references a Contract using the Contract Administration function.
- **Record Test Time** should be selected if actual usages consumed during a downtime event are to be excluded from the equipment's billing forecast. Selecting this check box will result in a **Test Time** box appearing in every AMT Event for this equipment, where you can enter the number of hours to be dedicated from the equipment's expected contract billing for the month in which the Event is performed.
- **Billings Split %** is only used in select MARC Intercompany Analysis report.
- **Budget Margins** are used to calculate the parts, labour and miscellaneous Cost@Cost values for the linked budget projection.

Billings and Cost Summary

This section contains billing projections, and at least one linked billing schedule. A billing projection is a summary of all linked billing schedules. If a billing schedule's Start or End date do not align to the equipment's Start date/usage or End date/usage, the misaligned value will be displayed in **red** text.

The Cost@Sell and Cost@Cost values in the grid, for the Budget projection, are not escalated. Cost values for all other projection types are escalated.

The Re-link Actual Billings button triggers AMT to link existing actual billing records, which potentially are not linked to the equipment. Triggering this process is only required if chances to equipment model, serial number or registration counter were made, which may cause previously linked actual billings to be unlinked.

To create or edit a billing schedule, refer to Create an equipment billing projection.

Scope

Provides additional information related to the scope of the contract. The attributes listed are sourced from the Tech Fields defined against the Equipment Group to which the equipment is linked using the equipment's Model. The selections made are for information only.

(Rpt_144) Contract Invoice Details

Provides an equipment billings summary for approval by the client. Ensures there is a consistent record of invoicing and that billings are managed correctly.

Tips

The report can include test time. Set this functionality in the **Equipment Editor** > **Configuration** tab > **Marc** quick button - **Record Test Time**.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

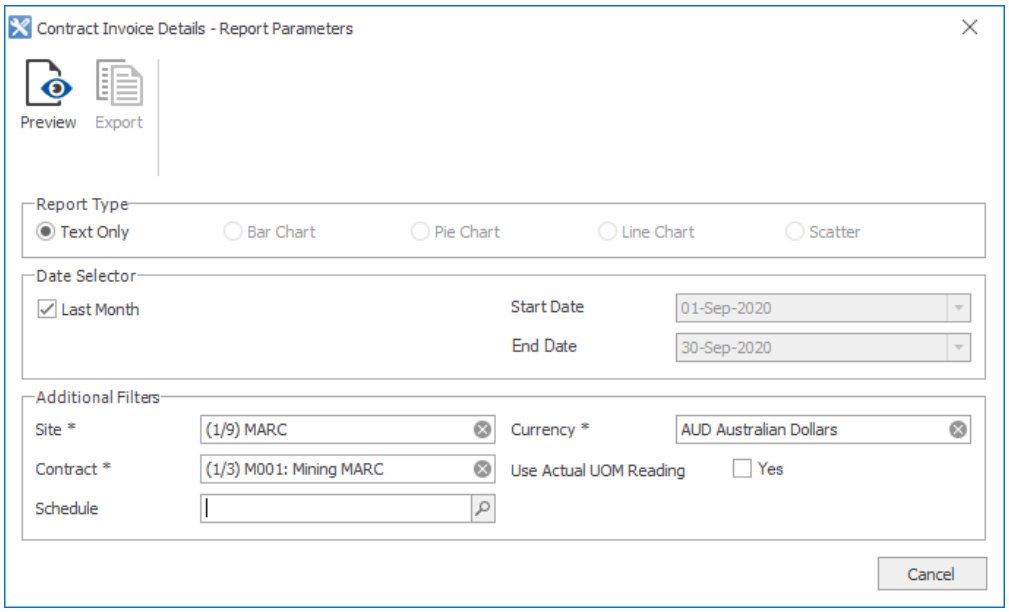
Site, Currency, Contract, Use Actual UOM Reading, Schedule

See Contract Invoice Details for more information.

Contract Invoice Details

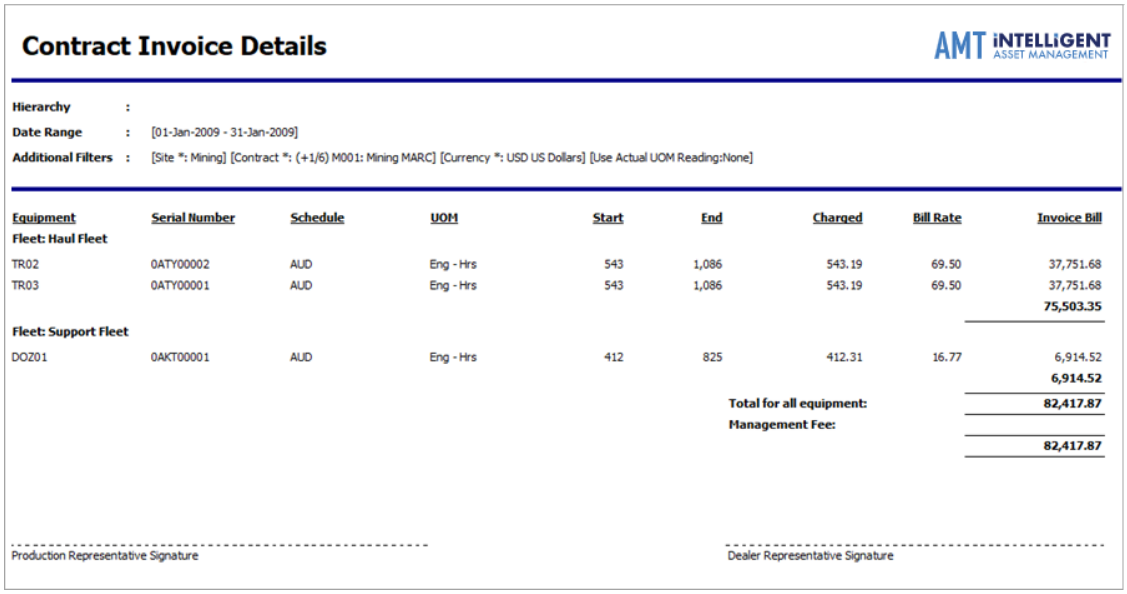
The planner runs the Contract Invoice Details report from AMT to validate charges to the customer for the current month for a specific contract. Contracts are set up using the **Contract Administration** function.

Navigate to **MARC** > **MARC Admin** > **Billings** > **Contract Invoice Details**.



Select the **Date Range**, and select the required filters for the required fields, **Site**, **Currency** and **Contract**. Also select the **Use Actual UOM Reading** check box.

An example of the report is shown in the screenshot below.



Contract Invoice Details

AMT INTELLIGENT ASSET MANAGEMENT

Hierarchy :
Date Range : [01-Jan-2009 - 31-Jan-2009]
Additional Filters : [Site *: Mining] [Contract *: (+1/6) M001: Mining MARC] [Currency *: USD US Dollars] [Use Actual UOM Reading:None]

Equipment	Serial Number	Schedule	UOM	Start	End	Charged	Bill Rate	Invoice Bill
Fleet: Haul Fleet								
TR02	0ATY00002	AUD	Eng - Hrs	543	1,086	543.19	69.50	37,751.68
TR03	0ATY00001	AUD	Eng - Hrs	543	1,086	543.19	69.50	37,751.68
								75,503.35
Fleet: Support Fleet								
DOZ01	0AKT00001	AUD	Eng - Hrs	412	825	412.31	16.77	6,914.52
								6,914.52
						Total for all equipment:		**82,417.87**
						Management Fee:		
								82,417.87

Production Representative Signature

Dealer Representative Signature

Start and **End** are the actual usage readings entered into AMT (as the **Actual UOM** check box was selected).

Bill Rate is the billing rate from the asset's current projection for the usage block the asset is in.

Invoice Bill = the charged usage multiplied by the bill rate.

Management fees are overhead billing (if applicable). These are included in the report if the

- site or contract has an overhead current projection.
- overhead current projection was set up to include MARC billings.

The report will also deduct test time from the charged usage if

- the individual equipment has the **Record Test Time** setting enabled.
- Actual test time has been recorded in events.

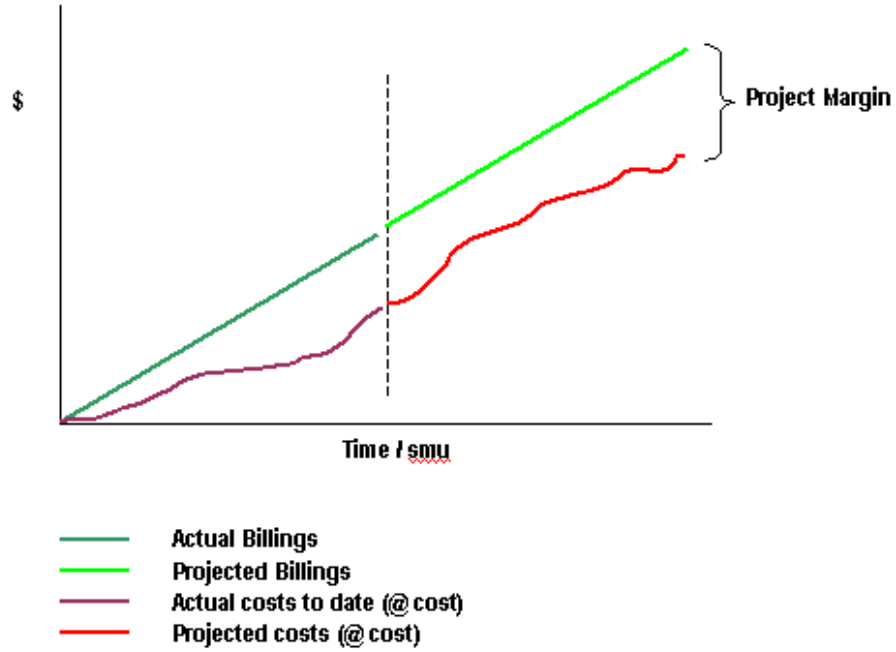
  Test time is those hours consumed during maintenance activities for which the dealer is unable to charge the customer. Test time is only applicable if it is specifically written into the maintenance contract.

Record test time

Test time is set up at the equipment level. Test time is not usually applicable unless stated in a maintenance contract. Any test time recorded is then deducted from AMT's **Contract Invoice Details** and **Billings Reconciliation** reports.

Health - Maintenance contract financials

The focus for customer service is on assessing the financial performance of the contract and ensuring compliance with the contract terms.



The financial performance is assessed as billings less cost at cost to calculate the contract margin. The margin is recognised over the contract based on the company's accounting policy.

For the financial reporting in AMT to be meaningful (and subsequent profit recognition) the base data needs to be accurate. The following key indicators should be checked.

Actual and projected billings

Check that the actual billings invoiced balances to the invoicing schedule in AMT. Variances will either be caused by incorrect billings to the customer or the billings schedule in AMT being incorrect. Both causes need to

be rectified for AMT projected billings to be accurate.

Run the report by fleet to identify any billings variances.



A monthly timing difference may occur, so check the reconciliation. If the actual billings schedule is correct it indicates the projected billings is likely to be correct.

Actual costs and projected costs

See Actual Costs and Projected Costs.

(Rpt_127) Billings Interrogation

A text report that calculates the % of future billings that need to go to the company so that its billings are in line with its costs.

Tips

The report is used to practically adjust future MARC invoicing so that there is alignment between costs and billings within the companies. An example would be, if 65% of the costs are projected to be incurred by Company A then if only 40% of the billings to date.

Report parameters

Default.

Additional filters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Actual Billings

This function allows you to view existing equipment billing records imported into AMT, add new actual billing records and edit actual billing records manually created by users.

Actual billing records imported from an external system cannot be edited. If a change is required, the change should be made in the source system's billing record, and reimported into AMT.

Overhead Billings, linked to an Overhead projection are not visible on this form.

(Rpt_130) Billings Reconciliation

Highlights if there is any variance between the actual billings to the customer and the expected billings from AMT. This needs to be reconciled to ensure that accurate data is in AMT and that the correct billings are being passed onto the client.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Use Actual UOM Reading, Overheads/Exp

See Reconcile Billings for more information.

Reconcile billings

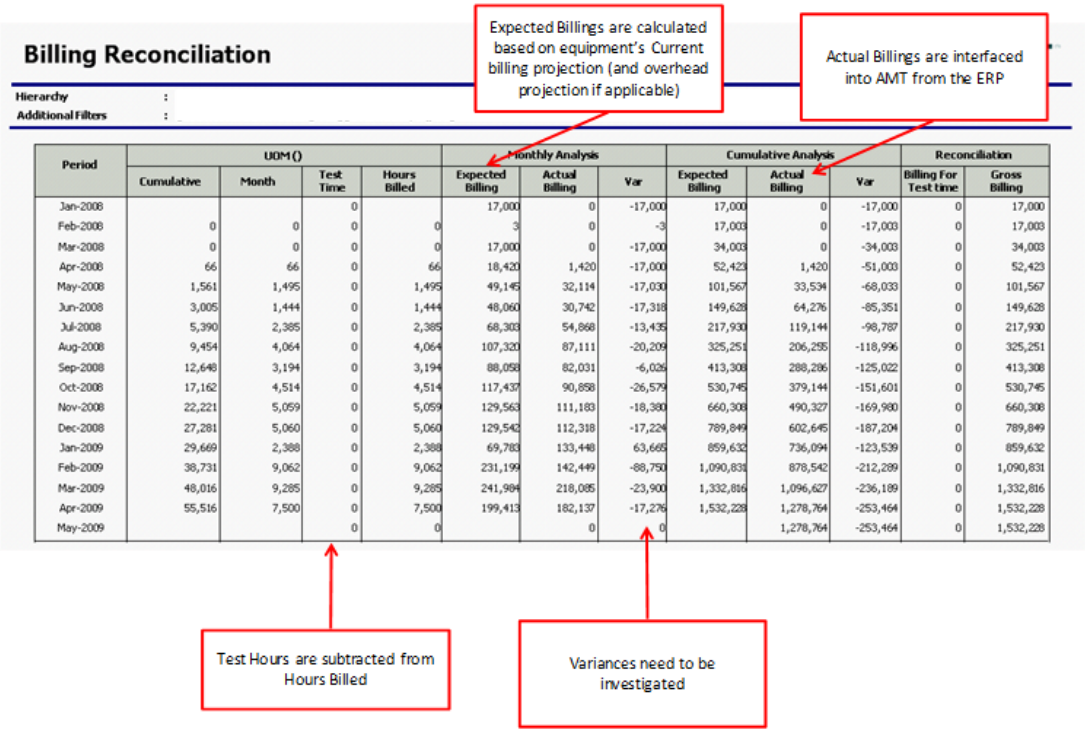
A billing reconciliation is a comparison between expected billings (based on a calculation) and actual billings in order to identify billing variances.

The **Billing Reconciliation** report calculates cumulative current billings based on the formula Actual Usage x Current Billing Rates. This is then compared with actual billings, and the billing variance is calculated.

Substantial billing variances should always be investigated.

Navigate to **MARC** > **MARC Admin** > **Billings** > **Billings Reconciliation**.

An example of the report is shown in the screenshot below.



Period	UOM ()				Monthly Analysis			Cumulative Analysis			Reconciliation	
	Cumulative	Month	Test Time	Hours Billed	Expected Billing	Actual Billing	Var	Expected Billing	Actual Billing	Var	Billing For Test time	Gross Billing
Jan-2008			0		17,000	0	-17,000	17,000	0	-17,000	0	17,000
Feb-2008	0	0	0	0	3	0	-3	17,003	0	-17,003	0	17,003
Mar-2008	0	0	0	0	17,000	0	-17,000	34,003	0	-34,003	0	34,003
Apr-2008	66	66	0	66	18,420	1,420	-17,000	52,423	1,420	-51,003	0	52,423
May-2008	1,561	1,495	0	1,495	49,145	32,114	-17,030	101,567	33,534	-68,033	0	101,567
Jun-2008	3,005	1,444	0	1,444	48,060	30,742	-17,318	149,628	64,276	-85,351	0	149,628
Jul-2008	5,390	2,385	0	2,385	68,303	54,868	-13,435	217,930	119,144	-98,787	0	217,930
Aug-2008	9,454	4,064	0	4,064	107,320	87,111	-20,209	325,251	206,295	-118,996	0	325,251
Sep-2008	12,648	3,194	0	3,194	88,058	82,031	-6,026	413,308	288,286	-125,022	0	413,308
Oct-2008	17,162	4,514	0	4,514	117,437	90,858	-26,579	530,745	379,144	-151,601	0	530,745
Nov-2008	22,221	5,059	0	5,059	129,563	111,183	-18,380	660,308	490,327	-169,980	0	660,308
Dec-2008	27,281	5,060	0	5,060	129,542	112,318	-17,224	789,849	602,645	-187,204	0	789,849
Jan-2009	29,669	2,388	0	2,388	69,783	133,448	63,665	859,632	736,094	-123,539	0	859,632
Feb-2009	38,731	9,062	0	9,062	231,199	142,449	-88,750	1,090,831	878,542	-212,289	0	1,090,831
Mar-2009	48,016	9,285	0	9,285	241,984	218,085	-23,900	1,332,816	1,096,627	-236,189	0	1,332,816
Apr-2009	55,516	7,500	0	7,500	199,413	182,137	-17,276	1,532,228	1,278,764	-253,464	0	1,532,228
May-2009			0	0		0	0		1,278,764	-253,464	0	1,532,228

- **Expected Billing** is the billing amount AMT calculated based on actual usages entered into AMT, and the current projection's billing schedule.
- **Actual Billings** are imported into AMT from the enterprise resource planning system.
- Obtain copies of actual customer invoices and reconcile with actual billings to verify data interfaces.
- Be sure that actual billing periods coincide with budget periods (accounting invoice time lag effect).
- Compare actual billing rates from invoices to current projection billing rates.
- Compare current projection billing steps with maintenance contract.
- Set the **Use Actual UOM** reading to base the report's calculations on usage readings entered into AMT (AMT automatically calculates expected usages based on an asset's usage profile until an actual usage is entered.

- If a variance exists, the administrator should ask the contract manger to investigate.
- Once the reconciliation is validated, the contract manager signs off on the reconciliation between AMT and the enterprise resource planning system.

(Rpt_478) Contract Billings vs Cost Reconciliation

Lets you compare separate billing projections to costs.

Tips

An example: A user has created a billing projection for GET. This report lets you compare the GET billing projection (that is, GET revenue) versus GET costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Sub System, Component Code, Task Type, Cost Bearer, Cost Centre, Expense Element, Parts/Labour/Misc

Submit reconciliation report

Once billings and work orders have been reconciled (and overhead billings and overhead costs if applicable), a summary report is compiled by the Contract Accountant and sent to the Contract Manager. The report should include:

- The current reconciliation status of each contract.
- A list of outstanding key issues.
- The person responsible for rectifying each issue.
- The target date for resolving each issue.
- An update of the issues from the previous month.
- A summary of adjustments made to existing MARC Accounting Procedures.
- A training plan for those that need to be trained on the updated MARC Accounting Procedures.

Based on the findings and corrections made during the reconciliation process, it may be necessary for the Contract Accountant to update the company's existing set of MARC accounting procedures.

Once updated, the relevant people who need to be trained on the new procedures should be identified, and a training plan organised. This information must be contained in the report submitted to the Contract Manager.

Billings escalation

To easily escalate billing rate per billing steps or ranges (including the subsequent billing rates per step/range) you can use the **Billings Escalation** function to apply a certain percentage across all of the subsequent billing steps or ranges.

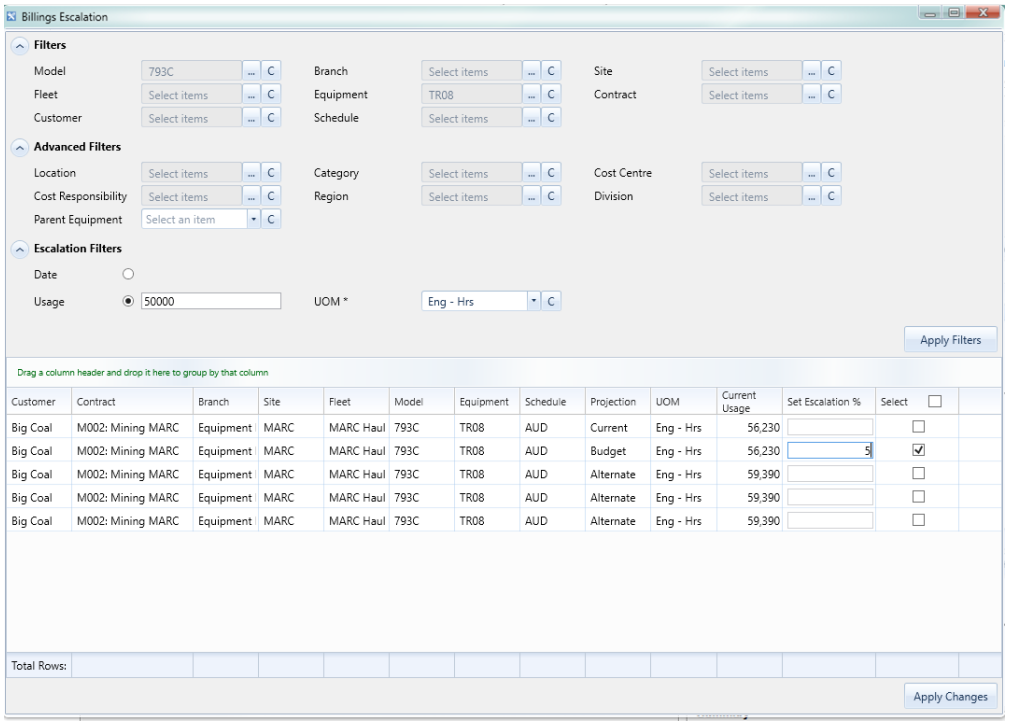


This function is mainly used by the AMT MARC administrator.

Once there has been a billing rate revision and the billing rates need to be escalated by a certain percentage, you can use the function to quickly escalate or de-escalate the future billing rate or amounts for many contracts at once, instead of manually go into each Billings Projection and editing the amounts.

Billings escalation can either be applied from a specific usage reading or a date.

Navigate to Billings Escalation by clicking **MARC >MARC Admin > Billings > Billings Escalation**.



Select the filters you want to apply by clicking ... , then click **Apply Filters**.

Click C to clear the filters.

You can escalate the future billing rates of multiple assets at the same time, however you should exercise some caution in doing this due to assets being in different stages of the billing cycle.

If you are doing an escalation by date, you need to make sure that an actual usage exists for the date from which the escalation is to be applied. If there is no reading, AMT calculates an estimated the reading at that date. It is advised that you enter a reading for that date if there is no reading.

Escalation - Equipment billings

It is typical for businesses to pass on price increases, such as internal cost escalations, to customers. In terms of maintenance contracts, the increases are passed onto customers through renegotiating the billing rates. However, it is not necessarily the case that an escalation of internal costs by 5% results in the customer agreeing to increase the billing rates by 5%. Clauses in the contract determine how escalations will be calculated, negotiated and applied. This is why billings are escalated independently of the costs, and hence why AMT does not automatically escalate the billings by the same percentage as the costs – new rates must be entered by users.

If new billing rates are not entered into AMT, AMT's billing forecast (revenue) will be either under or overstated. AMT will always forecast the total revenue that will be earned based on the billing schedule alone. When

entering new rates into AMT, the new rates must be entered per asset, because the usage at which the billing rate changes will be different for every asset.

Example

- On 1 July 2019, a customer agreed to a billings increase of 5%, resulting in the rate increasing to $27.50 per hour.
    - Truck 001 had 26,029 hours, and from that point $27.50 was applied.
    - Truck 002 had 25,892 hours, and from that point $27.50 was applied, and so on.

Refer to Apply Billing Escalations for more information.

Equipment billings - Audit

Billings are set up for each equipment as a billing projection. Actual billings are imported into AMT via an interface, and are reported verbatim.

The forecast of equipment billings from today to the end of the equipment's life cycle is derived from the equipment's current billing projection:

- AMT assumes an equipment will earn the total billings calculated in the current billings projection, not 1 cent more or less. Even if to date the company has not billed as much as it should have, AMT assumes there will be a catch up
- It is therefore critical that current billing projections in AMT always reflects what the company believes the specific equipment will earn in terms revenue

Billings projections must be adjusted every time billing rates are renegotiated. It is typical for businesses to pass on price increases such as internal cost escalations to customers. In terms of maintenance contracts, the increases are passed onto customers through renegotiating the billing rates.

However, it is not necessarily the case that an escalation of internal costs by 5% results in the customer agreeing to increasing the billing rates by 5%. Clauses in the contract determine how escalations will be calculated, negotiated and applied. This is why billings are escalated independently of the costs. Hence, AMT does not automatically escalate the billings by the same % as the costs – new rates must be entered by users.

If new billing rates are not entered into AMT, AMT's billing forecast (revenue) will be either under or overstated (remember AMT will always forecast the total revenue that will be earned based on the billing schedule alone). When entering new rates into AMT, the new rates must be entered per equipment because the usage at which the billing rate changes will be different for every equipment

Equipment billings

Billings is an AMT term referring to contract revenue. Billings can be generated at the equipment level or contract level (for example, a management fee). Billings are set up for each equipment as a billing projection.

For equipment billings, the Dealer calculates the total billings based on the work estimated to maintain the equipment.

- Actual equipment billing revenue generated under the contract can be either imported into AMT from an Enterprise Resource Planning (ERP) system, or manually created in AMT.
- Future billing revenue is forecast in AMT based on each asset's current billing projection, and whether the Auto Accrual Billings flag is switched on or off within each child billing projection.

Billings projections must be adjusted every time billing rates are renegotiated. It is typical for businesses to pass on price increases such as internal cost escalations to customers. In terms of maintenance contracts, the increases are passed onto customers through renegotiating the billing rates.

However, it is not necessarily the case that an escalation of internal costs by 5% results in the customer agreeing to increasing the billing rates by 5%. Clauses in the contract determine how escalations will be calculated, negotiated and applied. This is why billings are escalated independently of the costs. Hence, AMT does not automatically escalate the billings by the same % as the costs – new rates must be entered by users.

If new billing rates are not entered into AMT, AMT's billing forecast (revenue) will be either under or overstated (remember AMT will always forecast the total revenue that will be earned based on the billing schedule alone). When entering new rates into AMT, the new rates must be entered per equipment because the usage at which the billing rate changes will be different for every equipment

Apply billing escalations

Periodically, the billing rates currently charged to the customer under the maintenance contract will be reviewed. Typically, this occurs once or twice per annum after internal cost escalations have been officially announced and applied. The Contract Manager is usually responsible for negotiating billing rate changes with the customer under the conditions of the maintenance contract document. The output of these negotiations (that is, changes to billing rates) must be entered into the current billing projection for each equipment.

Overheads

Some maintenance contracts include overheads costs as part of the direct equipment maintenance costs and billings calculations. This is captured in the estimate-current projection's costs and billing schedule.

Other types of maintenance contracts classify contract overhead costs and revenue separately. The overhead costs are factored into the direct equipment maintenance costs, but the revenue is captured separately or vice-versa. The **Overheads** module of AMT handles this case; this module is a separate area where overhead costs and revenue can be estimated and managed for contracts

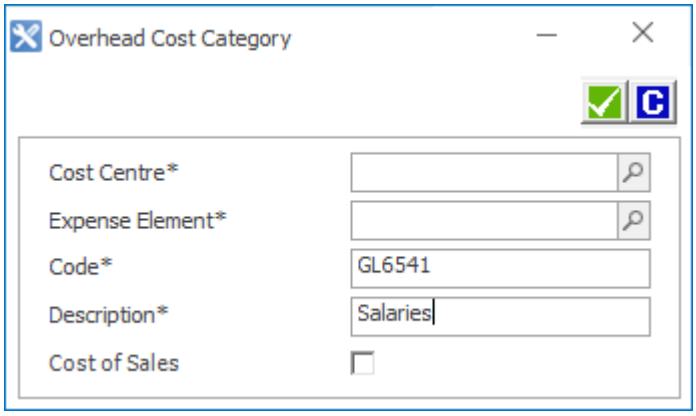


The Overheads module can also be used for non-contract overheads.

Before setting up overheads, it is important to validate:

- How are overheads costed in the contract; are they part of the direct maintenance costs or separate?
- How will overhead costs will be recovered - as a separate fixed monthly fee or calculated into the utilisation billing rate?
- How are these costs and revenue being captured in the ERP system?
- How are these costs and revenue captured if AMT manages the equipment or contract when they become live?

Set up overheads by navigating to **System Administration** > **System Tables** > **Financial** > **Overhead Cost Category**, then clicking  . The Overhead Cost Category dialog box displays.



The **Code** field is the overhead cost category definition. Select the **Cost of Sales** check box to report overhead costs as cost of sales rather than expenses in some profit & loss reports.

When a dealer defines and calculates a contract's overhead costs, there are numerous expense categories (accounts) considered, sometimes detailed such as printer consumables. AMT is not concerned with that degree of detail. Like equipment, AMT is concerned with tracking the significant cost items (for example, wages/salaries, accommodation, vehicles, travel) with minor expenses grouped into general categories or as Other Expenses.

Overhead Navigator

Some maintenance contracts include overheads costs as part of the direct equipment maintenance costs and billings calculations. This is captured in the estimate-current projection's costs and billing schedule.

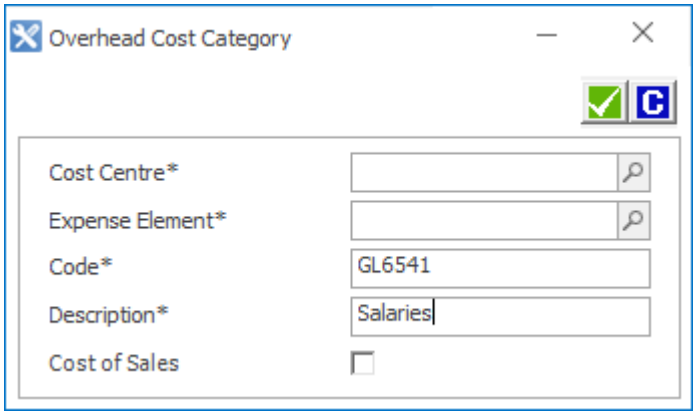
Other types of maintenance contracts classify contract overhead costs and revenue separately. The overhead costs are factored into the direct equipment maintenance costs, but the revenue is captured separately or vice-versa. The **Overheads** module of AMT handles this case; this module is a separate area where overhead costs and revenue can be estimated and managed for contracts

The Overheads module can also be used for non-contract overheads.

Before setting up overheads, it is important to validate:

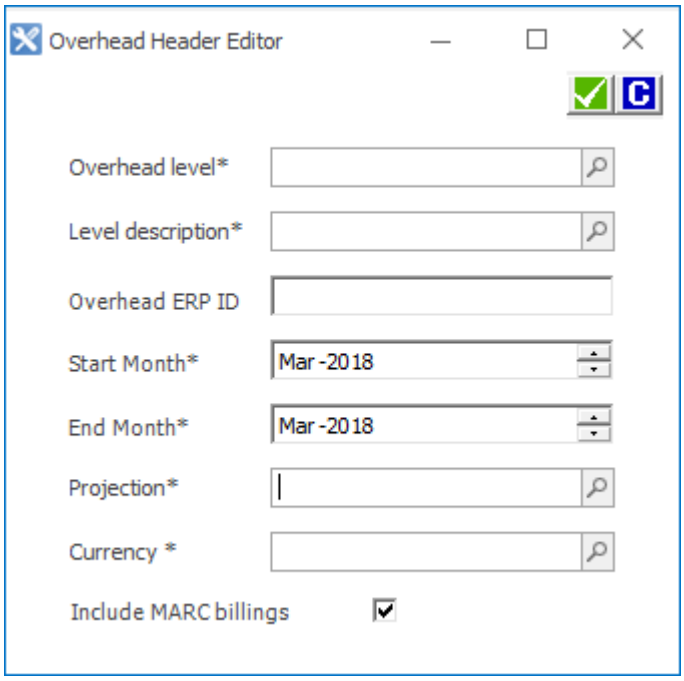
- How are overheads costed in the contract; are they part of the direct maintenance costs or separate?
- How will overhead costs will be recovered - as a separate fixed monthly fee or calculated into the utilisation billing rate?
- How are these costs and revenue being captured in the ERP system?
- How are these costs and revenue captured if AMT manages the equipment or contract when they become live?

Set up overheads by navigating to **System Administration** > **System Tables** > **Financial** > **Overhead Cost Category**, then clicking . The Overhead Cost Category dialog box displays.



The **Code** field is the overhead cost category definition. Select the **Cost of Sales** check box means to report overhead costs as cost of sales rather than expenses in some profit & loss reports.

To set up overhead costs and billings for a contract tender, use the **Overhead Header Editor** function. Navigate to **Setup & Admin** > **Overheads** > **Overhead Navigator**, then click .



Field	Description
Overhead level	The overhead level at which the overheads are costed and the overhead billing (management fee) is charged. For example, does the management fee cover all equipment under contract? Does it only apply to specific fleets or a specific branch or site?
Level description	Describes the overhead level chosen.

Field	Description
Overhead ERP ID	Not relevant for modelling overheads (only real overheads).
Start and End Month	Set to the period of time during which the overheads will be incurred.
Projection	Select the projection type.
Currency	Select the currency of the overheads.
Include MARC Billings	If a non-utilisation fee is charged to the customer to recover overhead costs, select this check box. If the check box is not selected, you can set up a schedule of overhead costs, but not management fees.

Populate the overhead's schedule with costs and billings on the **Overhead Header Editor**. You can either enter values directly on the grid, or copy and paste them from Excel.

Overhead billings and costs

The decision to incorporate overheads in AMTs financials depends on the nature of the maintenance contract. Examples of overhead billings are:

- An hourly billing rate to recover parts and miscellaneous costs (equipment level).
- A fixed monthly rate to recover labour and site overheads.

In the above case, the company may want to capture all costs and charges in AMT in order to:

- Report on the entire contract cost and revenue.
- Provide benchmarks for developing new contracts in terms of what is required to support a maintenance contract.

Typically, overheads costs are entered into AMT at cost (no mark-up) and overhead billings are entered into AMT as a fixed charge per month. Check with the AMT administrator to see if overheads have been set up.

The budget and current projections for overheads are set up in AMT at one of the following levels branch, site, fleet or contract. The level chosen will depend on what the fixed monthly charge is designed to recover, for example, is there a separate charge per fleet? The budget and current projections for overheads (which is treated as a separate entity in AMT) are set up using Overhead Navigator.

The overhead projections are the basis for overhead cost and revenue forecasting. Actual overhead costs and billings are imported into AMT via an interface and reported verbatim. Overhead costs are imported into AMT per equipment (that is, the overhead costs are apportioned per equipment). Overhead billings are not apportioned. The reason why overhead costs are apportioned is to allow you to assess the running and support costs of an equipment.

Notes

- Not all financial reports incorporate overheads.
- For those reports which include overheads, the report has to be run at the overhead level (branch, site, fleet or contract) in order to see the overhead cost and billing values (actuals and projected).

Overhead revenue is called overhead billings in AMT, sometimes referred to as a management fee. The management fee for a maintenance contract is typically calculated based on the overhead costs for which they are to recover.

Overhead costs are defined in AMT in the Overhead Cost Category system table.

When a dealer defines and calculates a contract's overhead costs, there are numerous expense categories (accounts) considered, sometimes very detailed categories such as printer consumables. AMT's MARC financials is not concerned with that degree of detail. Rather, like equipment, AMT is concerned with tracking the significant cost items (for example, wages and salaries, accommodation, vehicles, travel) with all other minor expenses grouped into general categories or simply other expenses.

Overhead Navigator is used to define overhead revenue and costs for a contract in AMT. Overheads are set up at one of the following levels: branch, site, fleet or contract.

To properly run MARC financial reports for a contract which includes overheads and see overhead revenue, you must know which level the overhead projection was defined at because you must run the report for that specific branch, site, fleet or contract. Running MARC financial reports at any other level will exclude overhead revenue (actuals and forecast).

To check the level of an overhead projection navigate to **MARC** > **MARC Admin** > **Overheads** > **Overhead Navigator**.

Escalation - Overhead billings and cost

Overheads escalations are independent of equipment cost and billing escalations.

- Budget overhead projections can be escalated by either applying an escalation factor in the projection for a month or manually changing the costs per month to reflect the new cost.
- Current overhead projections are escalated by manually changing the costs per month to reflect the new cost.

Both the budget and current overhead projections can be maintained in Excel then copied and pasted into the overhead schedule in AMT for ease of creating and updating costs and billings.

(Rpt_241) Termination Summary

Highlights the potential financial adjustments that may be associated with an early termination of the contract.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Component Code, Task Type

Contract termination

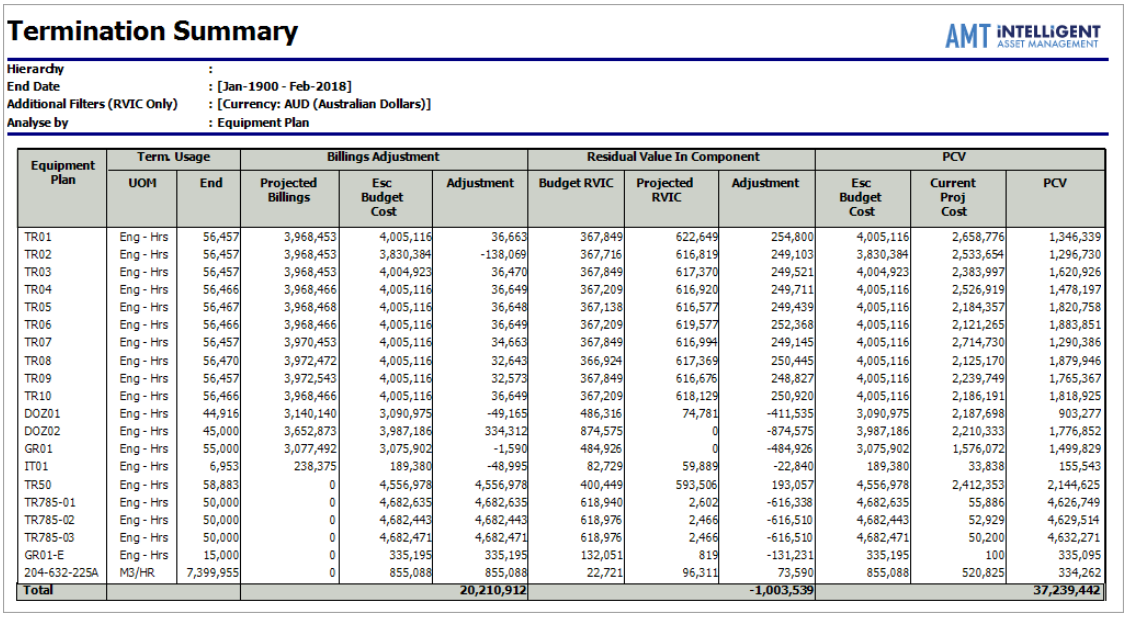
A contract can be terminated based on the contract term being reached or the contract being cancelled prematurely (for example, machine catches fire and burns down). AMT needs to be updated in both cases in order to stop maintenance being forecast, and for financial reports to be run accurately.

Termination summary report

If an asset is terminated the contract may require a reconciliation to take place. The objective of AMT's Termination Summary report is to provide:

- A summary of the termination adjustments that may apply for completed equipment; or
- An estimate of the termination adjustments that may apply at a future date.

Navigate to **MARC** > **MARC Admin** > **Termination Summary**. An example of the report is shown below.



Termination Summary

AMT iNTELLiGENT ASSET MANAGEMENT

Hierarchy :
End Date : [Jan-1900 - Feb-2018]
Additional Filters (RVIC Only) : [Currency: AUD (Australian Dollars)]
Analyse by : Equipment Plan

Equipment Plan	Term. Usage		Billings Adjustment			Residual Value In Component			PCV		
	UOM	End	Projected Billings	Esc Budget Cost	Adjustment	Budget RVIC	Projected RVIC	Adjustment	Esc Budget Cost	Current Proj Cost	PCV
TR01	Eng - Hrs	56,457	3,968,453	4,005,116	36,663	367,849	622,649	254,800	4,005,116	2,658,776	1,346,339
TR02	Eng - Hrs	56,457	3,968,453	3,830,384	-138,069	367,716	616,819	249,103	3,830,384	2,533,654	1,296,730
TR03	Eng - Hrs	56,457	3,968,453	4,004,923	36,470	367,849	617,370	249,521	4,004,923	2,383,997	1,620,926
TR04	Eng - Hrs	56,466	3,968,466	4,005,116	36,649	367,209	616,920	249,711	4,005,116	2,526,919	1,478,197
TR05	Eng - Hrs	56,467	3,968,468	4,005,116	36,648	367,138	616,577	249,439	4,005,116	2,184,357	1,820,758
TR06	Eng - Hrs	56,466	3,968,466	4,005,116	36,649	367,209	619,577	252,368	4,005,116	2,121,265	1,883,851
TR07	Eng - Hrs	56,457	3,970,453	4,005,116	34,663	367,849	616,994	249,145	4,005,116	2,714,730	1,290,386
TR08	Eng - Hrs	56,470	3,972,472	4,005,116	32,643	366,924	617,369	250,445	4,005,116	2,125,170	1,879,946
TR09	Eng - Hrs	56,457	3,972,543	4,005,116	32,573	367,849	616,676	248,827	4,005,116	2,239,749	1,765,367
TR10	Eng - Hrs	56,466	3,968,466	4,005,116	36,649	367,209	618,129	250,920	4,005,116	2,186,191	1,818,925
DOZ01	Eng - Hrs	44,916	3,140,140	3,090,975	-49,165	486,316	74,781	-411,535	3,090,975	2,187,698	903,277
DOZ02	Eng - Hrs	45,000	3,652,873	3,987,186	334,312	874,575	0	-874,575	3,987,186	2,210,333	1,776,852
GR01	Eng - Hrs	55,000	3,077,492	3,075,902	-1,590	484,926	0	-484,926	3,075,902	1,576,072	1,499,829
IT01	Eng - Hrs	6,953	238,375	189,380	-48,995	82,729	59,889	-22,840	189,380	33,838	155,543
TR50	Eng - Hrs	58,883	0	4,556,978	4,556,978	400,449	593,506	193,057	4,556,978	2,412,353	2,144,625
TR785-01	Eng - Hrs	50,000	0	4,682,635	4,682,635	618,940	2,602	-616,338	4,682,635	55,886	4,626,749
TR785-02	Eng - Hrs	50,000	0	4,682,443	4,682,443	618,976	2,466	-616,510	4,682,443	52,929	4,629,514
TR785-03	Eng - Hrs	50,000	0	4,682,471	4,682,471	618,976	2,466	-616,510	4,682,471	50,200	4,632,271
GR01-E	Eng - Hrs	15,000	0	335,195	335,195	132,051	819	-131,231	335,195	100	335,095
204-632-225A	M3/HR	7,399,955	0	855,088	855,088	22,721	96,311	73,590	855,088	520,825	334,262
Total					20,210,912			-1,003,539			37,239,442

The three potential adjustments that are calculated are:

- **Billing Adjustment**: to adjust the actual billings to equal the budget costs for the revised contract term.
- **Residual Value in Component (RVIC)**: to calculate the difference in the actual expected value in a component and that budgeted for.
- **Projected Cost Variance (PCV)**: the expected variance between actual costs and budget costs.

Refer to the specific contract to see if these adjustments are applicable.

Click the links to see a description of the report fields.

Termination Usage

Field	Description
UOM	Unit of Measure (for example, Eng - Hrs, Engine Hours).
End	Project usage at the End Date entered.

To validate this usage take the last actual usage reading or date from the Actual Usages Table, and add the expected usage from the usage profile to the end date.

If the equipment plan finishes before the end date then the term usage will be the end usage for the equipment plan.

Example:

Today is 1 January, 2017.

Truck 1 has a usage reading of 20,000 engine hours. Truck 1 is projected to run at 6,000 engine hours per annum.

If you enter 1 June, 2017 as the end date in the report parameters then the calculated engine hours at 1 June, 2017 will be 23,000 being calculated as 20,000 + 6,000 x 6/12. 23,000 will display as the end termination usage.

Billings Adjustment

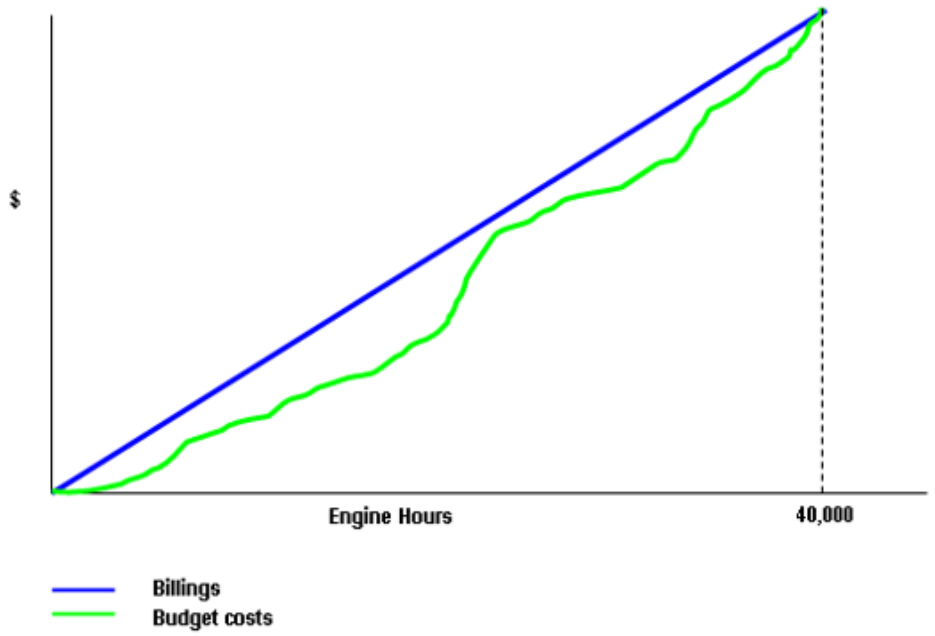
The purpose of the billings adjustment is to calculate the adjustment required to make the billings equal the budget costs at that point in the equipment life-cycle.



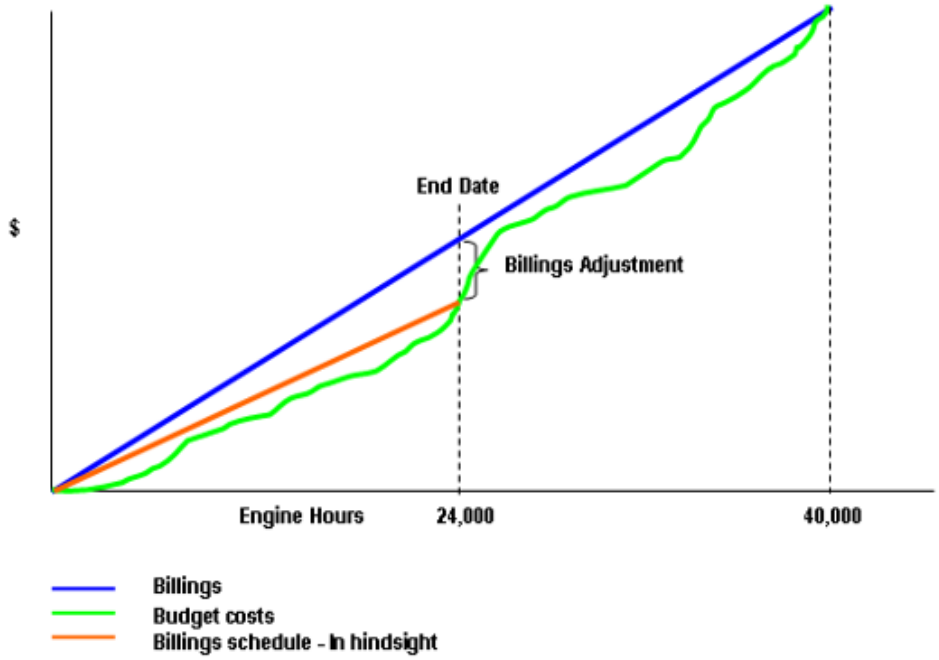
The actual costs incurred have **no impact** on this report.

Field	Description
Projected Billings	This is the projected billings to the end date using the current projection's billings schedule only. It represents the expected actual billings to the end date. This can be validated by calculating the theoretical billings from the current projection billings schedule.
Esc Budget Cost	This is the budget costs to the end date (escalated to when they were budgeted to occur). This can be validated and analysed using the Budget Cost Worksheet report and other cost analyses reports in the Equipment Financial menu. Ensure Budget Projection is selected.
Adjustment	Is calculated as Esc Budget costs – Projected Billings A positive figure means the customer owes the Dealer the amount. A negative figure means the Dealer owes the customer the amount. If you have tasks which are always executed in a sequence, (for example, PM service #1 followed by PM service #2 followed by PM service #3 and then start again at #1), dependency logic is appropriate.

At the start of a contract it is assumed that the billings schedule is set to balance to the budget costs over the term of the contract. However, during the contract timing differences case the billings not to balance to the costs.



Early Termination

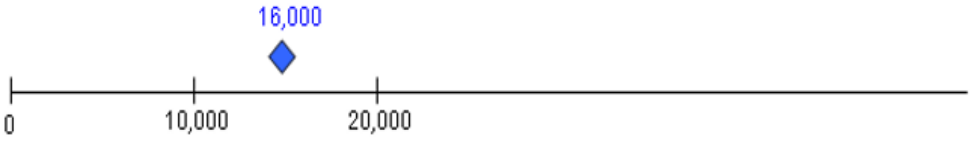


- At the point of termination, which you select through the end date, the billings adjustment is calculated as the difference between the projected billings to that point and the budget costs at that point.

- Explained differently, if at the start of the contract it was known that the contract term would be 24,000 not 40,000 then the billings schedule would have been set as the orange line. The Billings adjustment then is correcting the blue line to balance to the orange line.

Example:

Term is 20,000 hours. Engine is budgeted to occur at 16,000 hours and cost $100,000. MARC rate is $5.00 per hour ($100,000 / 20,000hrs).



If termination occurs at:

- 12,000 hours then Billing adjustment = Budget Cost – Projected Billings, for example, $0 – (12,000 x $5.00) = - $60,000 (Dealer pays Customer).
- 18,000 hours then Billing adjustment = Budget Cost – Projected Billings, for example, $100,000 – (18,000 x $5.00) = $10,000.

Residual Value in Components (RVIC)

The methodology behind the RVIC adjustment is that through the MARC the Dealer will deliver:

- a fixed maintenance cost.
- the asset in a defined condition at the end of the contract.

Field	Description
Budget RVIC	This is the Residual Value In Component (RVIC) calculated at the end date using the original budget parameters (Budget Projection). The budget costs for a task are escalated to today's values. This can be validated and analysed using the RVIC Analysis report in the Equipment Financial menu. **Note**: Select Budget Projection. This can be validated and analysed using the Budget Cost Worksheet report and other cost analyses reports in the Equipment Financial menu. **Note**: Select Budget Projection. Example: Budget life = 10,000 Eng Hrs. Budget Repair Cost = $20,000. If the End Date is 30 Nov, 2018 and the Term Usage is 26,000 Eng Hrs then the Budget RVIC (4,000 / 10,000 x $20,000) = $8,000
Projected RVIC	This is the projected actual (RVIC) calculated at the end date using the expected actual life remaining on components but calculating the value using the budget parameters for life and cost. The budget costs for a task are escalated to today's values.

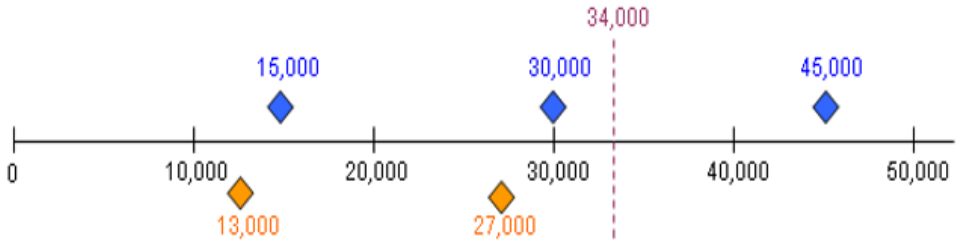
Field	Description
	These numbers are more difficult to validate. The best method is to run the Termination Summary but select one component in the filters. Then run the RVIC Analysis report in the Equipment Financial menu (**Note**: Select Current Projection). From this report determine the expected last change out point for the component. Then use the Budget Parameters to calculate the Projected RVIC for that component and reconcile to the Termination Summary. **Note**: Obtain the Budget Parameters from the RVIC report run using the Budget Parameters. Example: Using the example above, if the projected last change out is at 19,000 Eng Hrs, Projected RVIC (3,000 / 10,000 x $20,000) = $6,000.

The Billings Adjustment ensures the billings are in line with the budget maintenance costs. The RVIC adjustment compensates the Dealer or customer where the expected equipment is in a different condition to that defined in the contract.

The condition of the equipment is defined in the contract through the budget assumptions used to calculate the budget costs.

Example

- Original contract term = 50,000.
- Termination point = 34,000 (calculated Termination Usage for the End Point you enter).
- Budget parameters for an Engine (blue) – 15,000 engine hours and $120,000 repair cost.
- Actual Engine change outs (orange) - 13,000 and 27,000 engine hours.



At 34,000 engine hours:

- Budget RVIC (15,000 – 4,000) / 15,000 x $100,000 = $73,333.
- Projected RVIC (15,000 – 7,000) / 15,000 x $100,000 = $53,333.
- Adjustment ($53,333 - $73,333) = - $20,000 (Dealer owes the customer).

In this example, at termination the engine had 3,000 hours less life on it than budgeted which represent 20% of the budgeted life. Therefore the adjustment is 20% of $100,000.

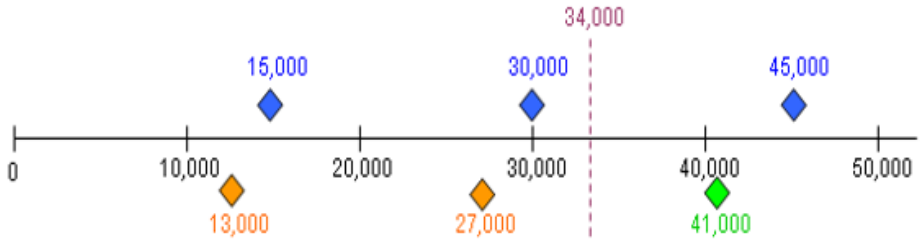
Projected Cost Variance

Projected Cost Variance (PCV) measures the difference between the original budgeted costs and the projected costs at a point in time. Usually the PCV reflects the expected variance at the end of the contract. However, in the Termination Summary report the PCV is calculated at the end date you select.

Field	Description
Esc Budget Cost	This is the budget costs to the end date (escalated to when they were budgeted to occur). This can be validated and analysed using the Budget Cost Worksheet report and other cost analyses reports in the Equipment Financial menu. **Note**: Ensure Budget Projection is selected.
Current Proj Costs	This is the actual date plus the projected costs to the end date using the **current** projection parameters.

Example:

- Original contract term = 50,000.
- Termination point = 34,000 (calculated Termination Usage for the End Point you enter).
- Budget parameters for Engine (blue).
  - 15,000 engine hours and $120,000 repair cost.
- Actual Engine change outs:
  - 13,000 engine hours and cost of $100,000.
  - 27,000 engine hours and cost of $115,000.
- Current projection parameters for engine (green).
  - 14,000 engine hours and $110,000 repair cost.



- PCV at end of contract (50,000 engine hours).
  - Budget Costs (3 x $120,000) = $360,000.
  - Projected Costs ($100,000 + $115,000 + $110,000) = $325,000.
  - Result ($360,000 – $325,000) = $ 35,000.
- PCV at termination (34,000 engine hours).
  - Budget Costs (2 x $120,000) = $240,000.
  - Projected Costs ($100,000 + $115,000) = $215,000.
  - Result ($240,000 - $215,000) = $ 25,000.

Plant Hire

The Plant Hire module provides functionality to support the process of hiring or renting assets.

The Plant Hire functionality includes setting up projects with defined hire rates, calculating and recording monthly revenue transactions, performance reporting, managing the future assignment of assets and forecasting the revenue and profitability of the plant group.

Plant Hire Rate calculations include all the related asset costs that may need to be recovered, including depreciation, finance, overheads, lubricants and maintenance. The rate can be set as a combination of fixed monthly charge, rate per unit of measure, such as Eng-Hrs, and % of book value. The automated monthly invoicing provides improved administration efficiencies while retaining flexibility for manual adjustments. Use the module's reporting to analyse plant hire invoices and measure the effectiveness of the recovery of costs and highlight non-performing assets.

This module is closely associated with the Asset Ledger module where depreciation and Component Based Depreciation is managed.

Plant hire budgeting

In addition to the standard cost budgeting functionality, you can also create future budgets of plant hire revenue.

The same process is followed to create a budget for plant revenue as for costs (see Budget and Forecast Manager).

When equipment is added to a budget, AMT automatically adds the plant revenue details for equipment that is flagged **Plant Hire** on the **Equipment Administration** screen. This can include modelling equipment.

Two Plant Hire tables are populated.

- **Projects Manager**

  This table has the header information for the equipment or plant rate schedule. You cannot edit data on this screen. If you have edited the **Usages** on the **Budget Header Manager** screen, click **Replace Equipment Plant Details** on the Projects Manager screen to recalculate the plant hire revenue based on the usages in budgeting.

- **Plant Hire Revenue**

  Contains the forecast monthly revenue by equipment, plant rate schedule, cost centre and expense element. You can manually edit any of the amounts on the grid. These are highlighted red to indicate they have been edited.



The Budget and Forecast Manager modules are the only areas where forecasts of plant revenue can be created. For more information, refer to Budgeting and Forecasting.

Financial Performance

Plant Hire Performance Reporting

The performance reports provide different formats for comparing the Plant Hire Revenue with equipment costs. On some reports the movement to and from the component based depreciation account (Asset Journals) is also shown. Navigate to **Plant Hire** > **Financial Performance**.

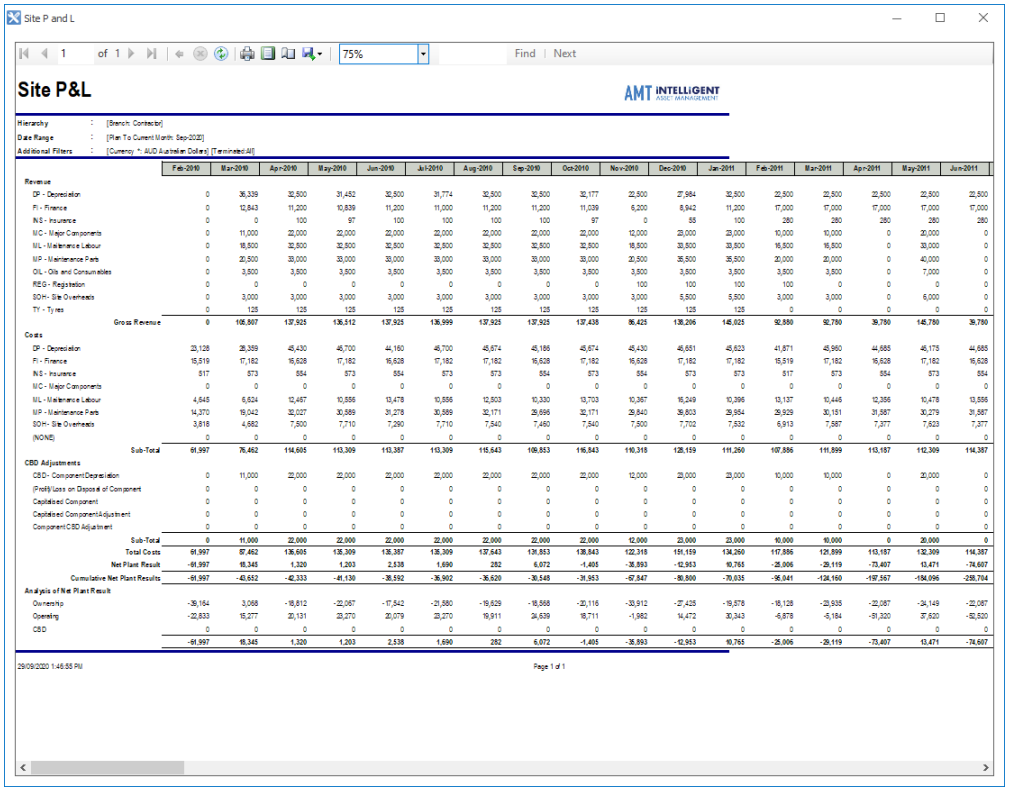
Report Name	Description
Site P&L	Shows revenue and costs analysed by Expense Element. The costs are also adjusted by the component based depreciation transfers.
Site Performance	Shows the same information as the site P&L report, analysed by asset.
Fleet Cost and Recovery Schedule	Summarises performance information into ownership, operating and component based depreciation categories.
Equipment Costs and Recovery Analysis	Shows detailed performance information for a single asset.
Equipment Cost and Recovery Schedule	Provides utilisation, rate and performance information by equipment or project.
Plant Hire Under/Over Recovery	Comparison of revenue and costs that can be run for groups of equipment.

Budget Reporting

The reports specific to Plant Hire Budgeting are described below.

Site P&L

Navigate to **Plant Hire** > **Financial Performance**. This report displays, in a profit and loss report format, the difference between plant hire revenue and plant hire costs. The **CBD Adjustments** area draws on plant hire journals to show component based depreciation transactions, for example, component disposal, capitalisation, depreciation.



Points on the CBD Adjustments calculation logic

- CBD Component Depreciation – for future periods this defaults to the same as the plant revenue for component based depreciation-flagged expense elements.
- Profit/Loss on Disposal of Component– for future periods this is 0.
- Capitalised Component – for future periods this defaults to the same as the component changeout costs, such as costs for component based depreciation-flagged expense elements.
- Capitalised Component Adjustment – for future periods this is 0.
- Component CBD Adjustment – for future periods this is 0.
- Forecast CBD Under/Over – the difference between the plant hire rate for component based depreciation items and the current projection rate – see the Note below.

Example: Forecast CBD Under/Over

Assume an equipment is forecast to do 5,000 hours per annum and its engine is set up with a frequency of 15,000 and a cost of $150,000. At that time its plant rate is calculated and set at $10.00 per hour.

If, at some point in the future, the engine strategy task cost is changed to $180,000 and the plant rate is not changed, AMT calculates the CBD Under/Over amount as:

Item	Cost
Current projection average annualised cost ($180,000/15,000x5,000)	$60,000

Item	Cost
Plant Rate annualised revenue ($10.00x5,000)	$50,000
Annual variance	$10,000
Monthly variance	$833

This follows the Annualised Cost Impact (ACI) logic in AMT, so the calculation works regardless of the unit of measure or if it is date-based tasks.

This is the only report in **Budgeting** that references live AMT data in its calculations. Therefore, this report can change from one day to the next, even though the Budget has not changed.

(Rpt_521) Site P&L

Shows a Profit and Loss report format the difference between Plant Hire revenue and Plant Hire costs. The CBD Adjustment area draws on Plant Hire Journals to show CBD transactions (for example, component disposal, capitalisation, depreciation).

Tips

Only applicable for equipment in AMT configured to sue the Plant Hire module.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Region, Division, Industry Group, Eqp Criticality, Equipment Class, Equipment Group, Eqp Category, Manufacturer, Cost Responsibility, Project - Equipment, Cost Centre, CC Activity, CC Location, CC Responsibility, Expense Element, Currency, Terminated

(Rpt_522) Site Performance

Shows the profit and loss per expense element for the equipment selected, and then summarises the profit, or loss, across all expense elements.

Tips

Only applicable for equipment in AMT configured to use the Plant Hire module.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Region, Division, Industry Group, Eqp Criticality, Equipment Class, Equipment Group, Eqp Category, Manufacturer, Cost Responsibility, Project - Equipment, Cost Centre, CC Activity, CC Location, CC Responsibility, Expense Element, Currency, Terminated

(Rpt_318) Fleet Cost and Recovery Schedule

A high level Plant Hire report showing the revenues versus costs for equipment or groups of equipment. The report also shows the performance breakdown between Ownership, Operating and Major Components.

Tips

Used for high level performance plant hire performance analyses, by Region, Model, Contract and so on.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Region, Division, Industry Group, Cost Responsibility, Eqp Criticality, Equipment Date, Equipment Group, Eqp Category, Manufacturer, Cost Bearer, Cost Centre, Expense Element, Currency

(Rpt_148) Equipment Cost and Recovery Analysis

Displays the productivity, costs, revenue and gross over/under recovery results.

Tips

Provides an overall summary of the equipment cost and recovery and shows the balance in the CBD provision. The report can be run in Text, Bar or Pie Chart formats. Bar Chart shows the top 10 costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|LastMonth|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Project, Currency

(Rpt_409) Equipment Cost and Recovery Schedule

Displays the variance between the equipment costs and the revenue collected. The costs and revenue are shown by expense element. The report also displays the Component Based Depreciation by expense element.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Region, Division, Industry Group, Eqp Criticality, Equipment Class, Equipment Group, Eqp Category, Manufacturer, Cost Responsibility, Eqp Cost Centre, Eqp Customer, Location, Currency

(Rpt_376) Plant Hire Under/Over Recovery

Provides a flexible analysis of plant hire revenues against plant costs. The analysis is by the expense element but any grouping of equipment and date range can be selected.

Tips

The categorisations of ownership, operating and CBD costs are made in the expense element table.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Region, Division, Industry Group, Cost Responsibility, Eqp Criticality, Equipment Class, Equipment Group, Eqp Category, Manufacturer, Cost Centre, Project Name, Project Customer, Hire Status, Currency, Terminated

 **+** indicates Current Equipment Situation.

Asset Validation

(Rpt_613) Asset Valuation

Shows a listing of assets or equipment at a point in time and their evaluations. This incorporates the Capital Cost, normal Depreciation and the Component Based Depreciation.

Tips

Shows the real book value of an asset that should align to market value by adjusting for the remaining life on components.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|,|ProjectionTypes|

Report parameters

Default.

Additional filters

Region, Industry Group, Cost Centre, Equipment Group, Eqp. Criticality, Manufacturer, Project Customer Terminated, Cost Responsibility, Equipment Class, Eqp. Category, Project Name, Hire Status

Plant Invoicing

(Rpt_375) Plant Hire Invoice Summary

This report intends to provide a summary of the plant hire invoices for a month. It can, however, summarise the invoicing for any data range in months. It shows the utilisation for the period, the effective plant hire rates and a break down of the invoicing.

Tips

The utillisation shown is the primary UOM for that equipment. The effective plant hire rate is the sum of the fixed and variable portions of the invoice divided by the utilisation.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Additional filters

Region, Division, Industry Group, Eqp Criticality, Equipment Class, Equipment Group, Eqp Category, Terminated, Manufacturer, Cost Responsibility, Cost Centre. Hire Status, Project Name, Project Customer, Currency

(Rpt_831) Plant Hire Rates Audit Report

Identifies the history of plant hire rate changes over the life of a project to date or for a defined period.

Tips

This is an audit report.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,
|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Project Name

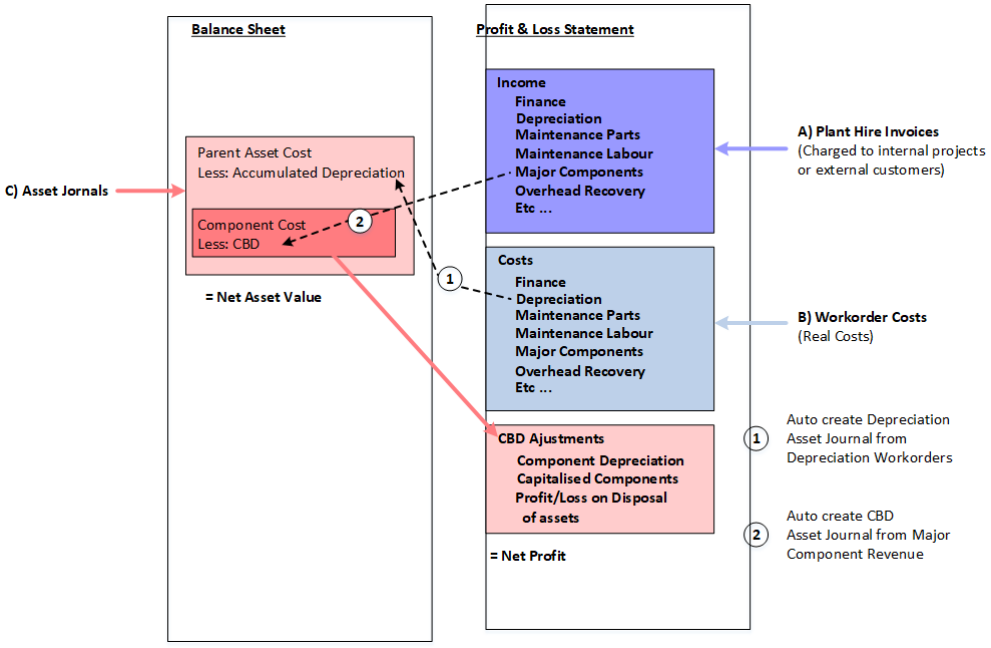
Plant Hire accounting process

There are three transaction data tables in AMT that are relevant to plant hire:

- **Plant hire invoices** – monthly invoices for the use of the asset. They can be generated by AMT or imported from external sources.
- **Real costs** – actual costs against the asset and include depreciation of the parent asset, finance, general maintenance costs, actual component changeout costs, recovery of plant group overheads. Real costs should be set up in strategy tasks, captured on work orders and display in standard AMT cost reporting.
- **Asset journals** – entries to the asset balance sheet accounts comprising:
    - Parent asset cost, such as a haul truck.
    - Parent asset accumulated depreciation.
    - Component capitalised cost (also referred to as child asset) – such as an engine.
    - Component cost accumulated depreciation (also referred to as Component Based Depreciation).

Internal asset transfers

Journal entries between the profit & loss account and the component asset accounts, such as depreciation of components, disposal of components and capitalisation of component changeouts. These journals should not be captured on work orders, rather they are entered in the asset journals.



Depreciation (of the parent asset) is the one item that is both a real cost and an asset journal and should populate both data tables. To streamline this process there is a configuration option in AMT to auto-generate asset journals for depreciation on parent assets from depreciation costs on work orders.

Organisations can elect to create asset journals for component based depreciation that match the revenue received for those components. Another way to view this is as the transfer of revenue directly to the component based depreciation account.

Allocate invoices to strategy tasks

When invoices are created or edited, AMT creates a supporting table where it splits the invoices amounts by strategy task, and applies the amounts proportionately based on the annualised cost of the strategy task. It will only do this for the component based depreciation expense elements and the strategy tasks flagged as component based depreciation.

If AMT is configured to auto generate asset journals for component depreciation (see Automated Journals), the depreciation amounts per strategy task are calculated.

Cost Analysis

(Rpt_122) Asset Cost Worksheet

Flexible cost analysis report that combines the strategy task costs set up in a projection with the other operating costs set up for that equipment.

Tips

Lets you to evaluate and forecast all equipment related costs - repair and maintenance costs, insurance, fuel, depreciation and so on.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Project, Cost Bearer, Region, Cost Centre, Division, Expense Element, Industry Group, System, Cost Responsibility, Sub System, Component Code, Cost Type, Equipment Class, Eqp Criticality, Equipment Group, Manufacturer, Eqp Category

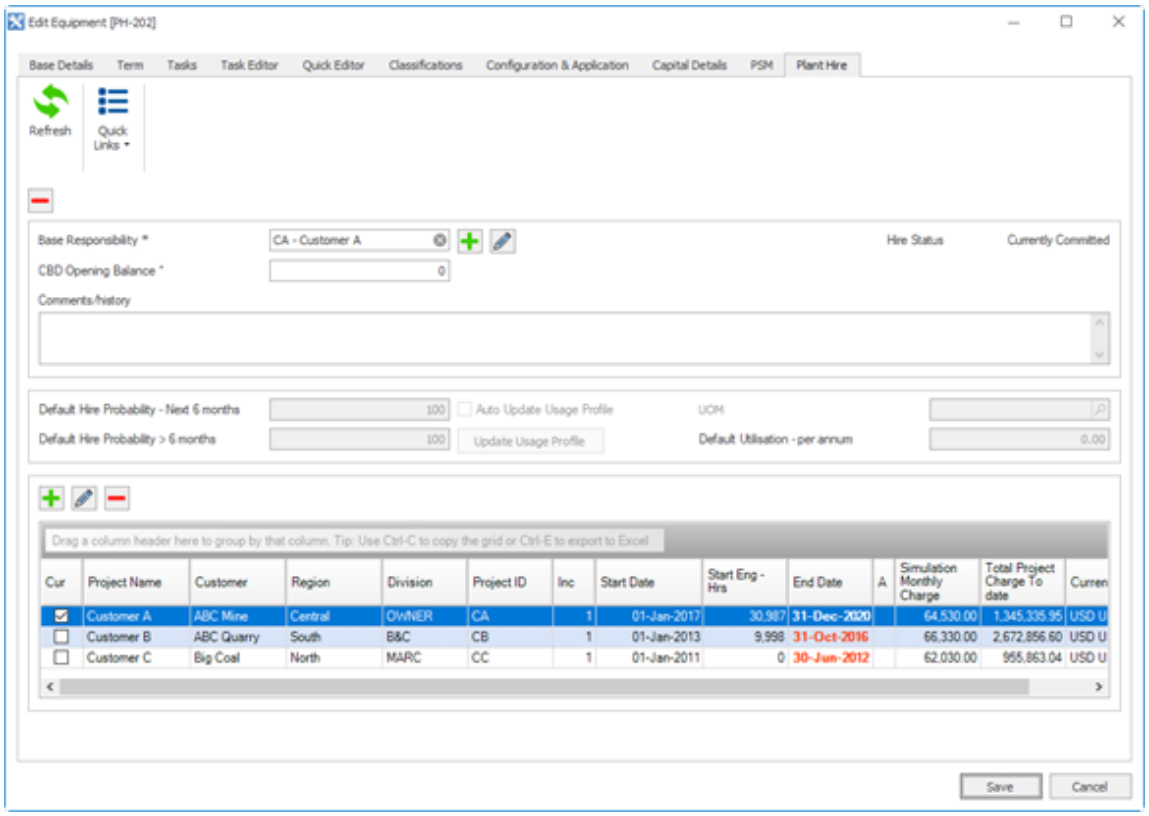
Setup

Configuring Plant Hire

Before setting up any equipment under plant hire it is important to note the following:

- Asset journals – transactions that record the entries against the asset ledger, such as the balance sheet.
- Equipment configuration – on the **Equipment Plan** > **Base Details** screen, select the **Plant Hire** check box to enable the plant hire functionality for that equipment. After you have selected the check box, refresh the screen and the **Plant Hire** tab displays.
- Expense Elements – Expense Element is the principle system table underlying the plant hire functionality, used for grouping categories for the plant rates, revenue transactions and costs. There needs to be consistency between how the costs are coded and how the rates are set up. In the Expense Element system table there are check boxes to group items into Ownership, Operating and Component Based Depreciation.
- CBD check box on strategy tasks – This check box enables you to nominate which are the component based depreciation components. It is only used to filter the strategy tasks shown on the component based depreciation. Typically it does not need to be selected, as it is the expense element that groups costs and revenue. It is important that there is consistency between the components that are flagged as component based depreciation and the expense elements assigned. Do not assign a component based depreciation expense element to a strategy task if the strategy task itself is not flagged for component based depreciation.
- Cost bearer - the code that indicates if the costs are for the Plant Group or Projects.
- One Equipment to one usage profile – to correctly enable the application of probabilities in the budgeting features you need to have one equipment to one usage profile. Therefore, you cannot have simple utilisation. The budgeting will still work for simple utilisation assets but all future projects will have a probability of 100%.

Set/Edit Plant Hire Rate



Base Responsibility represents a default project that AMT assumes the equipment will belong to, if it is not assigned to a specific project rate schedule. This is relevant for:

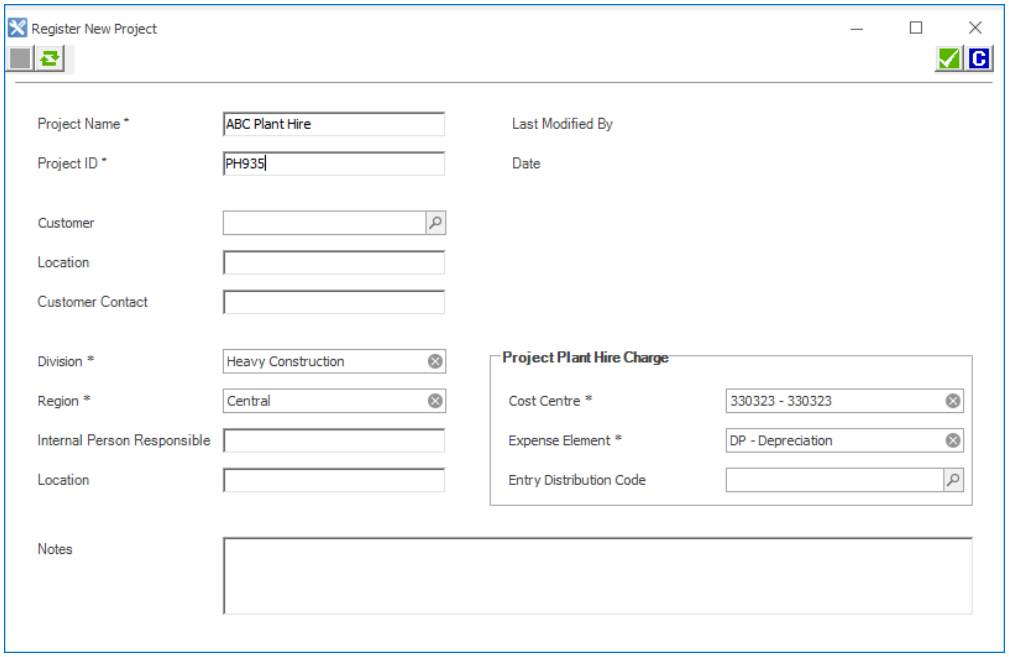
- assigning costs that are charged while the equipment is idle (such as financing costs and minimum depreciation.
- Forecasting revenue.

Default Hire section -sets the assumptions for forecasting revenue when the equipment is not assigned to a specific project rate schedule.

It is important to save your changes after creating the base responsibility project. You can't add a plant hire rate until the base responsibility information is saved.

Create a new Plant Hire project

To create a new project, click  beside the **Base Responsibility** field on the Plant Hire tab, then click



Key fields are described in the table below:

Field	Description
Project Name	The name of the plant hire project.
Project ID	Number to identify the project. This needs to match the Project ID used on the import files when importing Plant Hire Revenue transactions.
Division	The division to which the equipment belongs, for example mining, heavy construction.
Region	Where the equipment is located.
Cost Centre Expense Element Entry Distribution Code	These fields populate the export interface file for plant hire invoices. The codes are sent to the enterprise resource planning (ERP) system to determine the debit entry when creating the journal. These codes have no impact on any functionality within AMT, and are only relevant in cases where AMT is generating the plant hire revenue transactions and exporting them to the ERP.

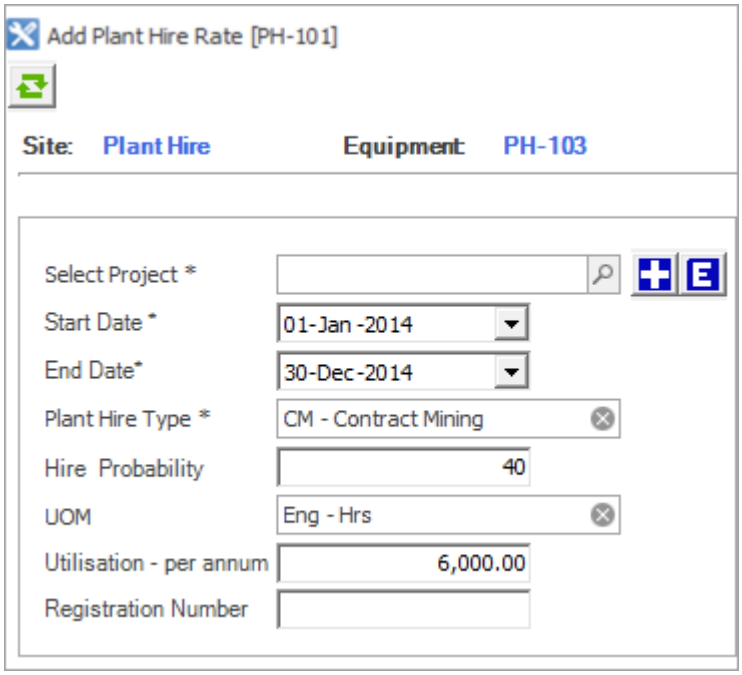
Calculation logic

To calculate future revenue AMT looks at the project rate schedules, or applies default assumptions for periods where there are no project rate schedules. AMT will, in some cases, also incorporate the probabilities and utilisation assumptions defined in the project rate schedules to the calculation of future revenue and future costs.

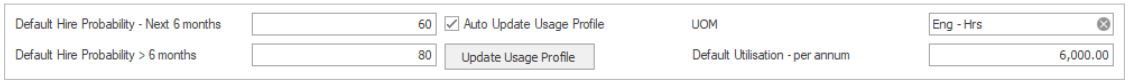
Click the headings to expand them and see more information.

Key inputs

On the **Add Plant Hire Rate** screen, you can enter a **Hire Probability** and **Annual Utilisation**.



On the **Edit Plant Hire** screen, you can enter a **Default Hire Probability** (next six months and > six months), and **Annual Utilisation**. You can set AMT to **Auto Update Usage Profile**, or click **Update Usage Profile** to update it manually.



Factoring in probabilities and utilisation assumptions

To ensure future revenue and future costs calculations are consistent, AMT adjusts the usage profile by product of the probability and utilisation assumptions. This adjusted usage profile then drives the calculation of future revenue and future costs.

Example: If a future project rate schedule has a probability of 60% at a utilisation of 5,000 Eng-hrs, the usage profile will be set to 3,000 Eng-Hrs per annum for the period of the project rate schedule. For this logic to apply:

1. Each asset must have its own usage profile.
2. The **UOM** nominated (see Key Inputs) must be the highest level in the usage Profile. It only adjusts that UOM, so there will be a cascade-down impact on the dependent UOMs in that Usage profile. It will not impact UOMs above it in the Usage Profile.
3. You must either enable **Auto Update Usage Profile** on the **Edit Plant** Hire form, or click **Update Usage Profile** to update it manually.

Specific calculation logic

The specific calculation logic for forecasting revenue depends on the hire status.

1. Current project.

   Applies the rates in the plant rate schedule to the utilisation assumptions (can be simple utilisation or a usage profile). The **Hire Probability** and **Utilisation Assumption** set in the project rates schedule have no impact.

2. Future project (Committed/Probable/Possible)

   Applies the rates in the plant rate schedule to the **Utilisation Assumption** (can be simple utilisation or a usage profile). The **Hire Probability** and **Utilisation Assumption** set in the project rates schedule updates the **Usage Profile** if the conditions outlined above are met.

3. No project for that period.

   If there is a future plant rate schedule it applies those rates, otherwise the rates are taken from the previous plant rate schedule. The **Probabilities** and **Utilisation Assumption** set in the **Equipment** > **Plant Hire** screen are applied to update the usage profile if the conditions outlined above are met.



There can be different probabilities for the next six months and later. Generally the probability in the next six months is lower, that is, there is no possible project identified for it now.

(Rpt_220) Plant Rate Analysis

A breakdown of the expense elements and the associated costs. Displays three different scenarios based on the simulated hours you have entered into the report parameters.

Tips

This report allows analysis between the usage per annum and the plant rate by expense element.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Today|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Simulation Usage 2, Simulation Usage 3, Currency

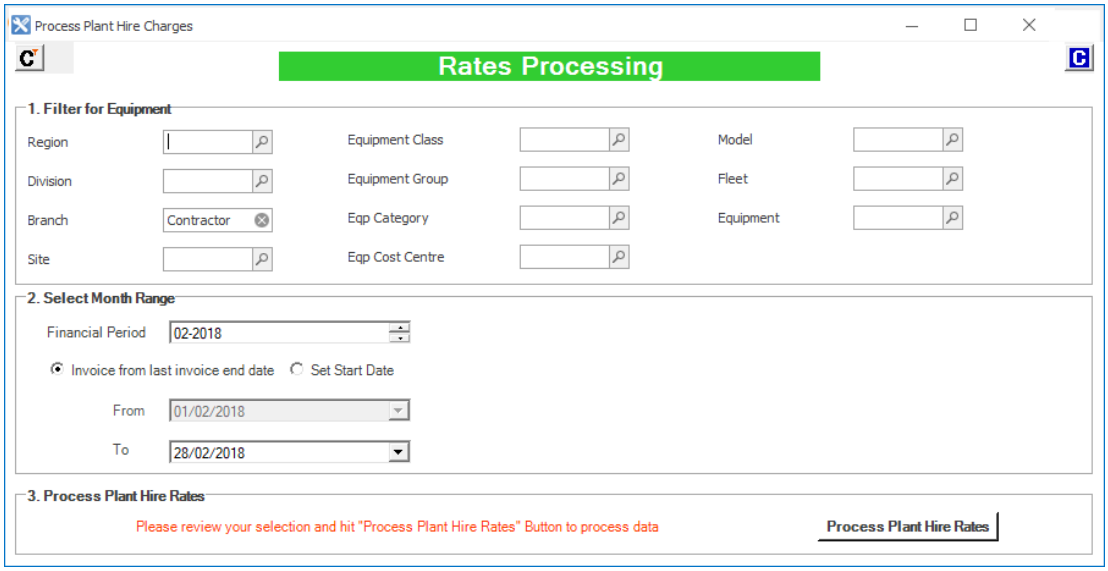


Simulation usage is the annual usage using the primary UOM.

Process Plant Hire Rates

Once the plant hire rates are approved, you can process the charges to the project. You can create plant hire invoices in AMT, or import them from an external system. If you import invoices from an external system, they are the same as if they were created by AMT, you can edit, delete and overwrite them.

Navigate to **Plant Hire** > **Setup** > **Process Plant Hire Rates**.

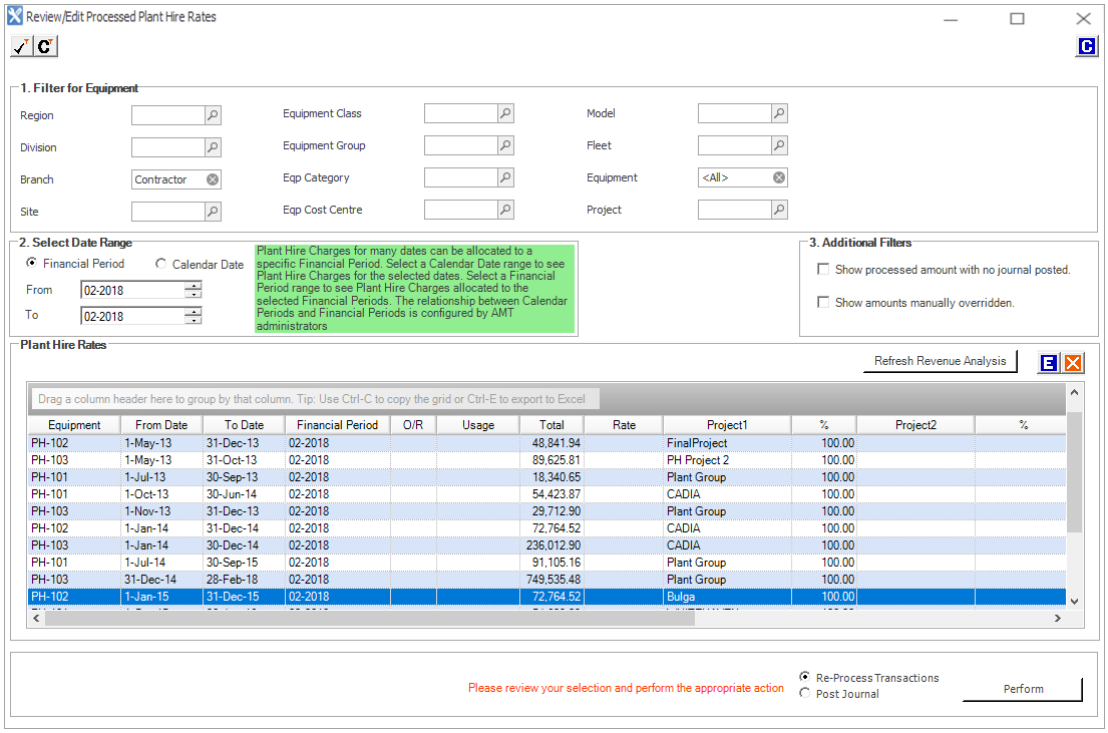


Select the parameters for processing, which can be from a regional level to an equipment level. The period is nominated in monthly increments. When all the parameters are selected, click **Process Plant Hire Rates**. This process generates revenue for the equipment to each project it is hired to.

The process overwrites any invoice for the same project rates schedule month unless that invoice has been posted.

Review and Edit Processed Hire Charges

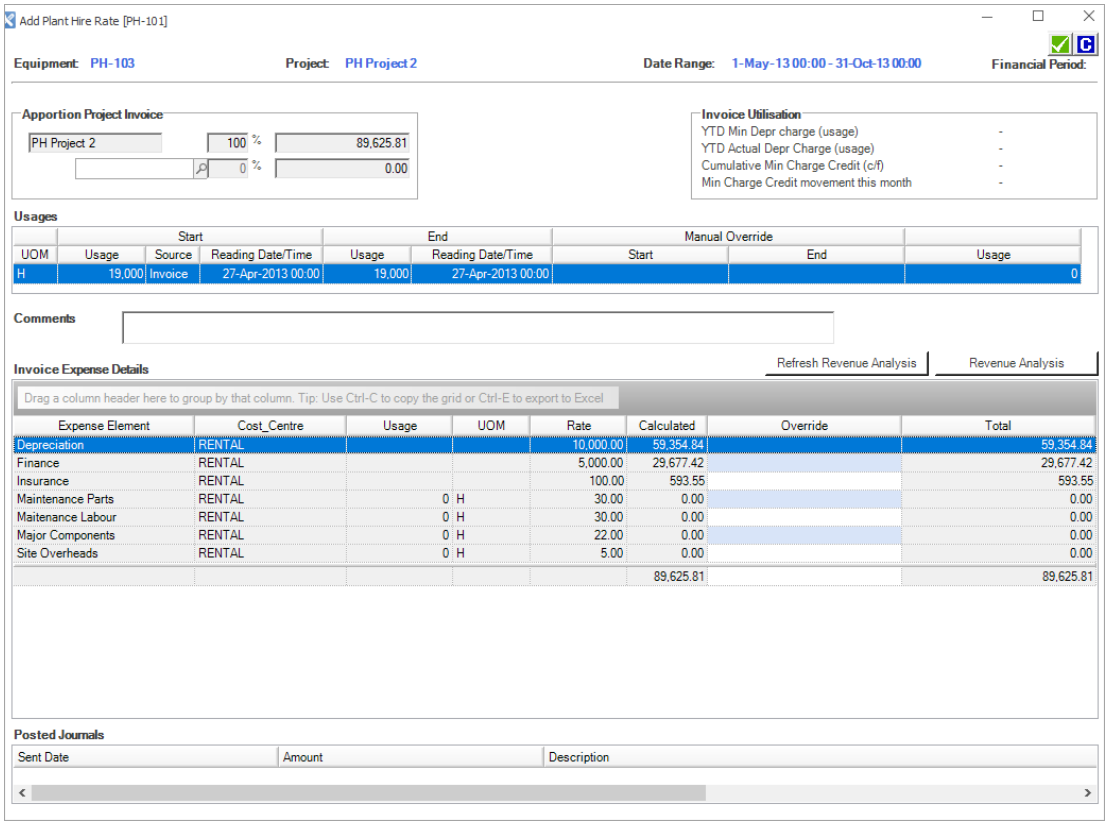
Navigate to **Plant Hire** > **Setup** > **Review / Edit Processed Hire Charges**.



On this screen you can review the plant hire charges processed for the period nominated. Each line displays:

- Equipment
- From and To dates
- Financial Period Month / Year
- Overridden: indicates if you have manually overridden the project monthly charge
- Usage: usage of the equipment in that month
- Total: total value of plant hire charge for the month
- Rate
- Project charged for hire
- % of invoice split between projects (may have worked two locations in month)

Double-click any of the lines of transactions to view the detail and the underlying charges for each expense element. Delete the transaction by clicking **Delete**. Re-process the transaction from the project's base plant hire rate by selecting **Re-Process Transaction** at the bottom right of the screen and clicking **Perform**.



Once the invoice details screen is open you can manually override the:

- Start and End Usage
- Revenue per expense element

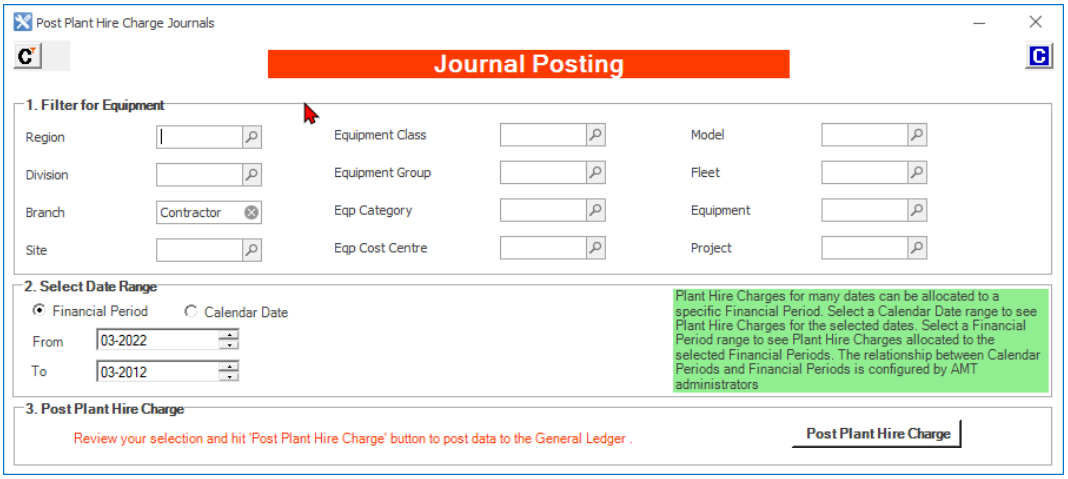
If you make manual adjustments, the reasons for the adjustment should be noted in the **Comments** section.

- If the usage for a month is less than the minimum usage required for depreciation, the usage charged would be the minimum hours, provided there were no funds in credit from the usage being higher than the minimum usage in previous months.
- AMT calculates the minimum usage requirements on a cumulative basis and compares the actual usage to the current period. AMT makes adjustments to the usage charged for depreciation each month based on this method.

Post Plant Hire Rates

This step is performed to export the plant hire invoice transactions to an ERP system where the journal entries are posted. This process creates an export interface file.

1. Navigate to **Plant Hire** > **Setup** > **Post Plant Hire Rate**.
2. Filter for the **Equipment** and **Period**.
3. Click **Post Plant Hire Charge**.



After the invoices are posted, they are flagged in AMT as posted. If these invoices are edited and saved in AMT, a reversal of the journal is sent and a new journal is sent.

Posting invoices can also be performed from the footer of the Review & Edit Processed Hire Charges screen.

Project Allocations Gantt

The Projects Allocation Gantt chart provides functionality to manage the future allocations of Equipment to Projects. Navigate to **Plant Hire** > **Setup** > **Projects Allocation Gantt**.

Functionality includes:

- displaying a line for each equipment or project rates schedule
- assigning a probability to future assignments - based on the probability AMT assigns it a **Hire Status** of
  - 100% = Committed
  - 50-100% = Probable
  - 0-100% = Potential

  Current projects are always treated as 100%, no matter what percentage is entered as a probability.

- the colour of the bar aligns to the **Hire Status**
- in the gird you can edit the **Project**, **Start Date**, **End Date** and **Probability %**

  To create a new **Projects Rate Schedule** navigate to **Equipment Plan** > **Plant Hire**.

(Rpt_565) Plant Hire Rates Schedules

Outputs Plant Hire Rate Schedules into a format which can be easily reviewed.

Tips

This report is useful to validate consistency of rates across equipment and rate periods set up in AMT are correct.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|LogoLocation|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Centre, Expense Element, UOM, Currency

Centrelines

The characteristics of centreline equipment include:

- It represents a standard template or baseline, and is the default maintenance plan for a particular model of equipment.
- Centrelines can be created as new, or developed from existing equipment or centrelines.
- There can be many centrelines for a particular model. For example, there could be a centreline for different applications.
- Costing is at the component or task level, which may be linked to standard jobs.
- Different cost categories can be used, for example parts, labour, misc.
- Different maintenance activities performed on the same component can be budgeted in the centreline for example, remove & install, rebuild, replace with new.
- Component life can be based on specific units of measure for example, engine hrs, fuel burn, tramming.
- Centrelines can contain all work that would be done on a machine, not just the work that is the Dealer's responsibility, for example, Ground Engaging Tools (GET), undercarriage, tyres.

Reports

Only reports that have the suffix " – Centreline" can be run for centreline equipment. Other reports will not function for this type of equipment.

Administration

Branch Administration

Branches are the levels below the Company (this is the highest level in the hierarchy in AMT). Use the Branch Administration function to add, edit and delete branches in AMT.

Field	Description
Branch Description	Name of the branch.
Branch Code	Identifies where costs for assets are captured in the accounting system.
WO Suffix	Work Order Settlement code suffix is used when AMT opens internal work order settlements automatically for an equipment under this branch.

Site Administration

Sites are the levels below branches. This system table is used to set up sites and related details, including setting up the Shift Start Time and enabling Employee Level Resourcing.

Field	Description
Site	Name of the site.
Branch	Branch to which the site belongs.
Site Code, Address, Suburb, State	Physical location information for the site.
Resource Scheduling	Select either Work Group Scheduling or Labour Activity and Work Group Scheduling.
Enable Employee Level Resourcing	Labour resource levelling is defaulted by site to be at the work group level. If a site wants to plan labour at the employee level, this check box must be selected.
Shift Start Time	The time when the first shift of the standard working day commences.  For maintenance contract sites, the shift start time must align to the dispatch system's shift start time.
Week Start Day	The first day of the week.
Currency	The financial currency used by the site.

Fleet Administration

Fleets are the levels below Site in the asset hierarchy. Navigate to **Setup & Admin** > **Equipment** to use the **Fleet Administration** screen to add, edit and delete fleets in AMT.

Field	Description
Fleet	Name of the fleet.
Site	Site to which the fleet belongs.
Pricing Group	Parts can be grouped by price. The pricing group is usually the default pricing group. However, if the site's parts are priced at a special rate, that is, include a discount, then a specific pricing group can be selected.  Pricing groups are set up in the System Administration area of AMT, and require special parts pricing to be imported or entered into AMT.
Default	Select for the fleet to be the default fleet.
Service Interval	Standard interval between services, for example 250 hours. The service interval is

Field	Description
	used in the Service Accuracy report.
Tolerance%:	A KPI benchmark used in the Service Accuracy report to determine which services were performed within an acceptable tolerance of the standard service interval, for example 10%.
Availability Formula	Currently there are two options to work out asset availability. • Physical availability (based on scheduled/calendar hours). • Mechanical availability (based on operating hours).
Set Availability Calculation Parameters	Opens a special function designed to enable Availability Forecasting for the fleet.
Standard Job Parts Pricing Adjustment Factors	Use this function to manually adjust the pricing of parts for specific part types for this fleet. For example, if fluid was a specific part type, then you could increase the price of all fluid parts for this fleet by 10% (enter into AMT 0.1) using this function.

Equipment Administration

You can view existing equipment (real, modelling and centrelines), as well as add and register new equipment in the system using the Equipment Administration function.

The equipment registration process consists of a number of steps that support the ongoing management and control of an operational asset. This process begins with defining a method to easily and uniquely identify an asset (depending on the asset type, this can be a difficult task). After the identification method is decided, you need to gather additional information about the asset to help determine classification and data collection requirements.

Before registering the asset in AMT, a number of questions need to be answered. For example:

- At which branch, site and fleet will the asset be located?
- Is this a complex asset requiring:
  - Sophisticated maintenance planning?
  - The ability to track a component's life using different units of measure (for example, machine hours, engine hours, fuel burn, travel time)?

  If so, a Usage Profile will have to be created.
- Is this a simple, low value asset requiring basic routine maintenance?
- What is the forecasted life cycle for the asset (for example 90,000 hours)?
- Will there be a rotable component strategy for certain components on the asset?
- What is the applicable cost centre or project that the asset will be assigned to?
- Is the asset is under a maintenance contract?
  - What are the contract details? (Contract number, contract name, contract account number, customer name and contract type).

- If the asset is under contract, the contract needs to be set up in AMT using Contract Administration.

This kind of information determines how the asset will be registered in AMT.

To register equipment in AMT, either click **Add** in the Equipment Administration screen (which opens the Equipment Plan Editor screen), or run the Equipment Plan Editor function.

Multi-Editor Asset Details

Use the Multi-Editor Asset Details function to change Classification and/or Life Cycle Term parameters across many equipment.

Equipment Editor

The Equipment Editor function is used to create new equipment in AMT.

The nature of the information required when registering an equipment depends on the type of equipment being registered. Valid equipment types are Real, Centreline, Modelling. Adding new equipment to AMT requires the completion of the following.

Base Details

- Equipment Type, Model, Serial Number and Registration Counter are important fields in this section.
- Check with your System Administrator before entering the serial number. External systems, which may be linked to AMT via interfaces, may require the serial number to be entered in a specific format. For example, some systems require the serial number to actually be “MODEL:SERIAL_NUMBER” because serial numbers are not unique unless prefixed by the Model.
  - **System Configuration - Utilisation**
    - No utilisation: select when only date based scheduling and reporting is required.
    - Simple: select when only a single unit of measure (for example, engine hours) is required for scheduling and reporting.
    - Advanced: select when multiple units of measure are required for scheduling and reporting (this requires a usage profile).
  - **System Configuration - Planning**
    - None: select when AMT will not be required to do daily planning for this asset.
    - Standard: refer to Standard vs. Advanced Planning.
    - Advanced: refer to Standard vs. Advanced.
  - **System Configuration - Enabled**

    Some of the special setting options may be disabled if certain modules of AMT have not been purchased.

    - Asset Register: select this option to use AMT’s asset register.
    - Other Operating Costs: select this option to use AMT’s operating costs for the asset .
    - Plant Hire: select this option if the asset is Plant Hire.
    - MARC: select this option if the asset is under a maintenance contract
    - Multi-projection: select this option if more than one projection is required for the asset (maintenance contract equipment require this setting to be enabled)

    - **Forecast Calculations**
        - The unassigned ratio is the ratio of the final hour/months unassigned cost to the first hour/month. The logic is that the average unassigned costs are still based on the value entered.
    - **Equipment Interfaces**
        - Allow Equipment Detail Updates: select this option if the equipment import interface should be allowed to update asset details.
    - **Quick Watch**

Term

- **Utilisation**: options presented depend on the type of utilisation selected in Configuration.
    - Usage Profile can only be selected with Advanced utilisation.
- **Start Date**: date when the asset began operation.
- **Original Commission Date**: date when the asset was commissioned.
- **Start**: for each usage listed, enter the starting usage. Typically it will be set to zero for new assets, but “second-hand” asset’s will have a usage greater than zero.
- **End**: set either an End Usage or End Date for the asset. This then becomes the termination point for forecasting.

Tasks

On this tab you can view, add, edit and delete tasks in selected projections. Refer to About Strategy Tasks.

Quick Editors

On this tab you are able to make multiple changes to strategy tasks for this specific equipment. This feature incorporates many of the Multi Editor functions which already exist in AMT, but in this case, the changes are limited to the specific equipment being edited.

- **Multi-Task Cost Adjustor**

  Apply a Parts/Labour/Misc cost adjustments, or Labour Hours or Duration (hours) adjustments across multiple tasks (for example, for selected tasks, increase Parts cost by 5%).

- **Labour Rates**

  Apply new Labour Rates for Shop and Field for selected projected tasks.

- **Travel**

  Ability to change Travel costs (labour and misc) for selected projected tasks.

Classifications

- Refer to Terminology for a description of each of the classification options
  Cost Details.
- **Cost Bearer**: select the default cost bearer. If unsure, contract the AMT Administrator.

Configuration & Application

The Registration and Configuration Details available for selection depend on which options were enabled for the Model of equipment you are registering.

- **Registration Details**

  Information only fields documenting the registration information of an equipment.

- **Configuration Details**

  Information only fields capturing equipment configuration.

- **Application Details**

  Information only fields capturing details of the application in which the equipment is currently operating.

- **Other Details**

  Documents can be attached to the equipment. The process will copy the document from the specified location to the AMT in order that it can be accessed by AMT users.

- **Notes:**

  Add any notes required.

Create blank equipment - Centreline

Centrelines are generic projections for equipment, for example, templates or baselines. You should try to maintain one for each equipment model. Typically, after the factory releases a new model, you document the components that make up the model, how long each component lasts and the cost of replacing it. All of this base information is captured in a centreline.

After you have created the centreline, you can copy it and customise it for a specific tender.

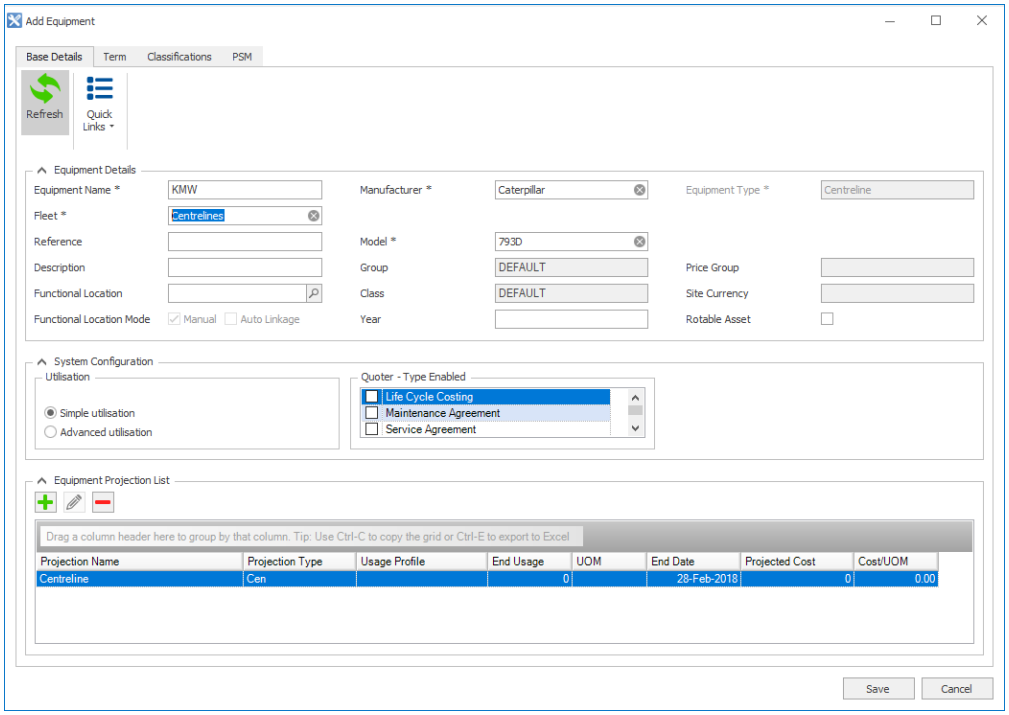
There can be many centrelines for a model of equipment, such as high altitude and low altitude. The standard jobs, imported from the **Builder Import** screen, are used to populate the centreline.

Before creating a centreline you need to use **Branch Administration**, **Site Administration** and **Fleet Administration** to create a suitable hierarchy. For example:

- Branch = Centrelines.
  - Site = C-Off Highway Trucks.
    - Fleet = C-OHT-793.
      - Centreline Equipment = C-OHT-793C.
      - Centreline Equipment = C-OHT-793D.

To create centreline equipment, navigate to **Centrelines** > **Administration** > **Create Blank Equipment - Centreline**.

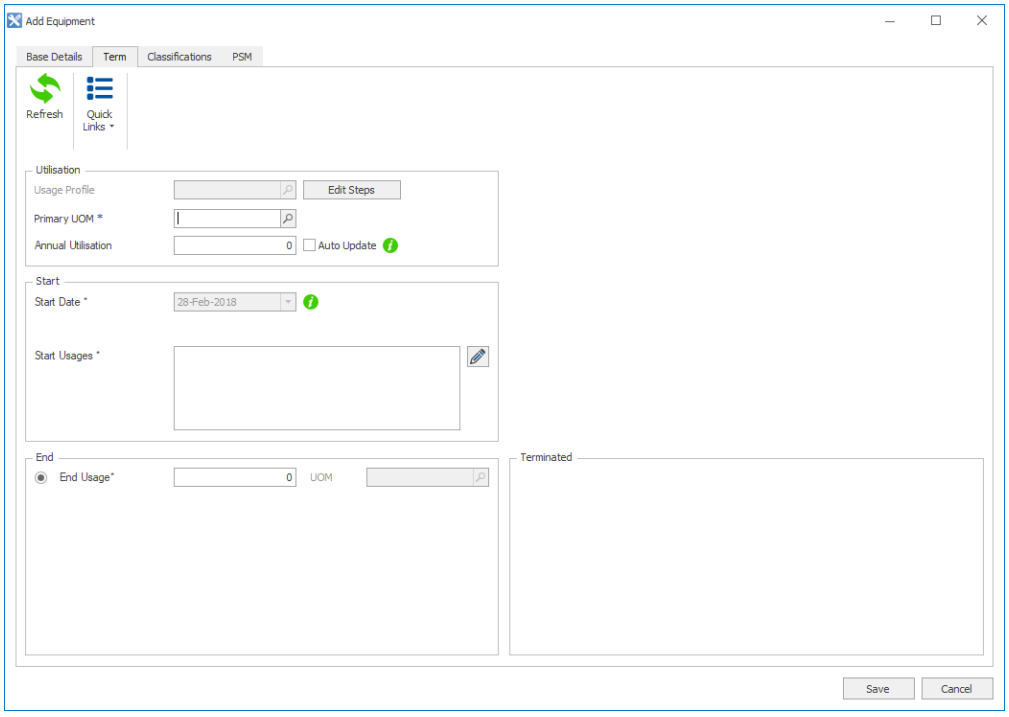
Base Details tab



On the **Base Details** tab fill out at least the following fields. The **Equipment Type** field defaults to **Centreline**.

- Equipment Name. Enter the name for the centreline which will be easy to find.
- Manufacturer
- Model

Term tab



On the **Term** tab fill out at least the following fields.

- Primary UOM, for example Eng - Hrs.
- Annual Utilisation
- End Usage (the end of the life cycle point)

Classification tab

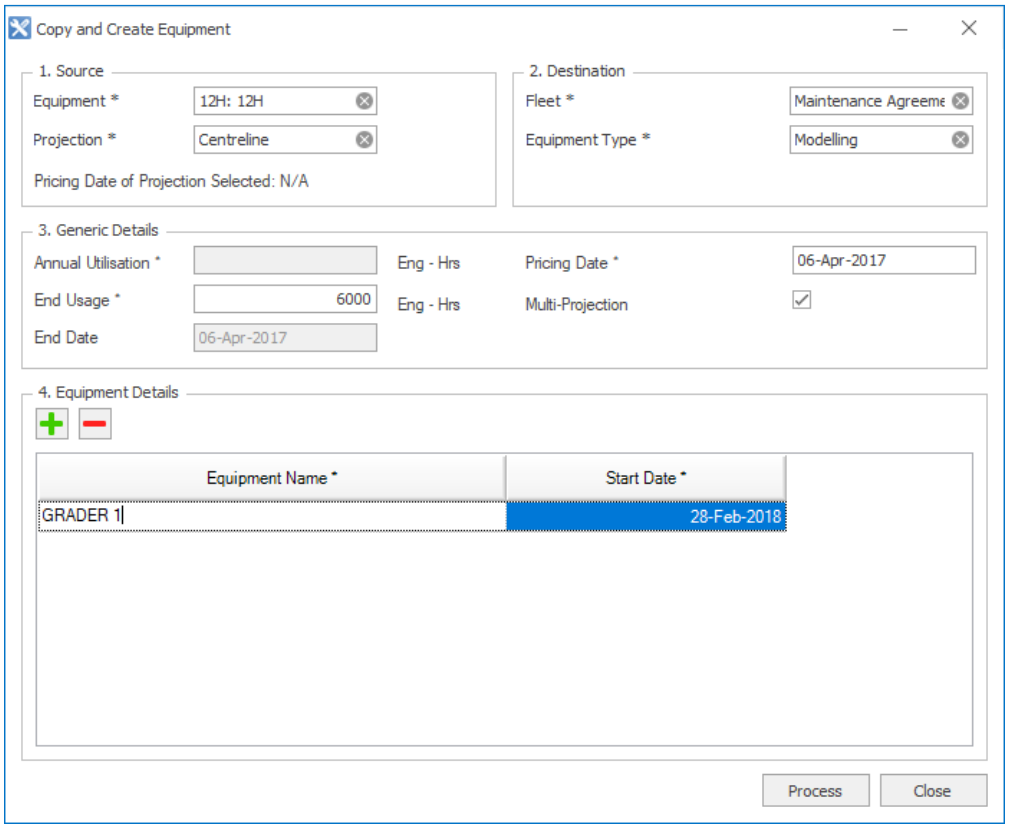
The options available on this tab come from the AMT system tables. If you want to add more options into these fields contact, your AMT Administrator. The only field you need to select on this tab is the **Default Cost Bearer** field.

Click **Save** to register the new centreline.

Register modelling equipment

On the Home Screen right panel, ensure the Centreline icon is selected.

Navigate to the Centreline, right-click it and select **Copy and Create Equipment**.



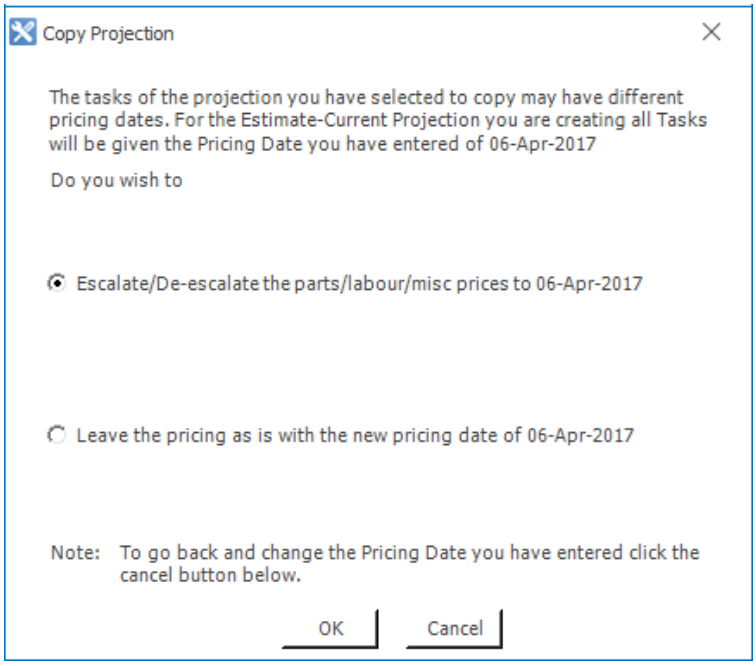
Enter the relevant information in the fields, and select the **Multi-Projection** check box.

- Source Projection = Centreline.
- Destination Equipment Type = Modelling.
- Pricing Date = the date that the pricing in the centreline is valid from.

Click **Add** under **Equipment Details** to create multiple assets from the same centreline. Creating multiple assets to represent a fleet is necessary if you need to accurately forecast labour requirements. As more assets are delivered, more labour is needed; staggering asset delivery using the start date ensures AMT properly forecasts this increase in resources as well as the costs, revenue and cash flow.

Click **Process** to create the asset.

On the **Copy Projection** dialog box, select the appropriate escalation method, then click **OK**.



Copy and Create Equipment

Create and copy new assets

The term asset is used interchangeably with equipment.

You can create a new blank asset in the following ways:

- **Setup & Admin** > **Equipment** > **Create Blank Equipment - Real**.
- **Centrelines** > **Administration** > **Create Blank Equipment - Centreline**.
- **Modelling** > **Administration** > **Equipment Hierarchy** > **Create Blank Equipment - Modelling**.

You can also create an asset by copying an existing asset. There are two ways to do this.

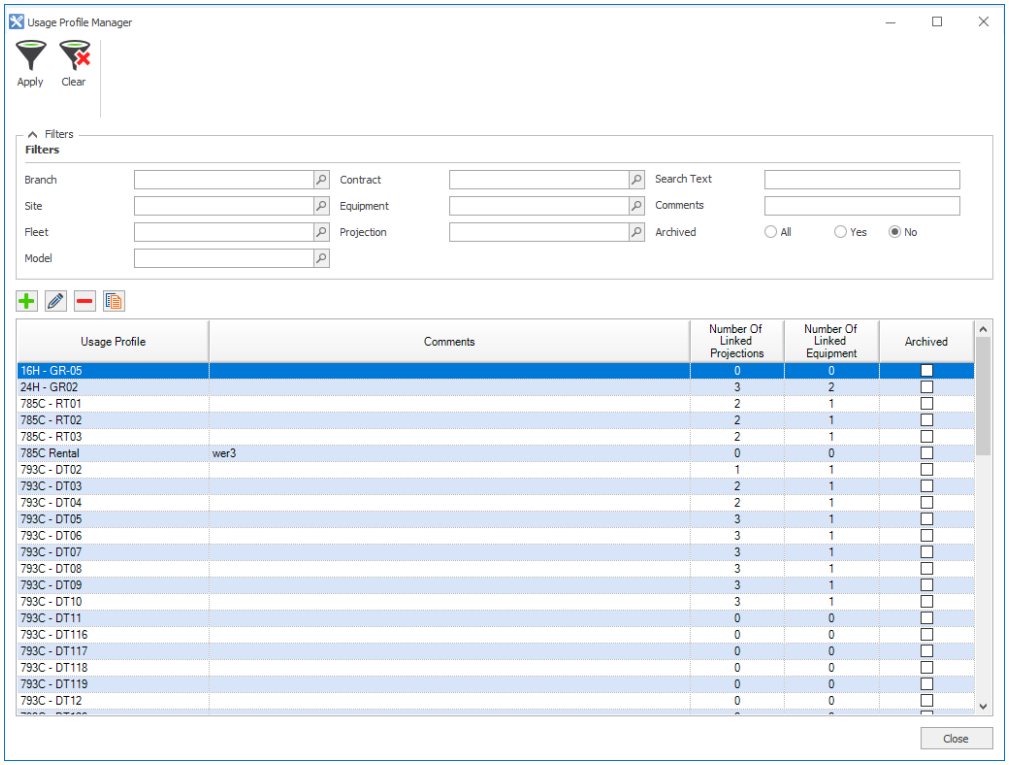
- **Modelling** > **Administration** > **Equipment Hierarchy** > **Create Real Equipment From Modelling**.
- **Setup & Admin** > **Equipment** > **Copy and Create Equipment**:

You can also copy and create equipment under the **Centreline** and **Modelling** menus.

Refer to Entering Asset Details for information on the fields on the form.

Usage Profile Manager

You can open the Usage Profile Manager by navigating to **Setup & Admin** > **Usage Profiles** > **Usage Profile Manager** or **Centrelines** > **Administration** > **Usage Profile Manager**.



Use the Usage Profile Manager function in AMT to add, edit and delete usage profiles.

To copy a usage profile click  .

Usage is the consumption of a resource by an asset. This consumption can vary over time due to changes in the working application and in working patterns. For example, consumption may be altered by changes in weather patterns; high rainfall may slow production, dry fine weather may increase production, or by physical changes in loads, haulage road conditions, holidays, and so on.

The usage profile maps the changes in consumption over time and allows the consumption to be measured by different units of measure. Usage profiles can have multiple steps and multiple branches in the consumption tree.

All usages are expressed in qualified units of measure (UOM) such as engine hours or litres of fuel. The most common unit of measure is engine hours. A usage profile is a named pattern for the rate of consumption of UOMs by an asset. Usage profiles can be linked to a single asset in a single projection or a range of assets of the same model across a fleet or site.

Notes

- A range of assets with different models can use the same usage profile, and different projections can use different usage profiles to model application, utilisation, or other what-if scenarios.
- The final usage step within a usage profile cannot have a usage value of zero.

(Rpt_311) Usage Profile Details - Centreline

Displays the different usage profiles set up with both actual and cumulative values showing. The report can be generated for a specific month period, but defaults are set to the last twelve months.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,4,|UserID|

Report parameters

Default.

Additional filters

Branch, Site, Fleet, Equipment

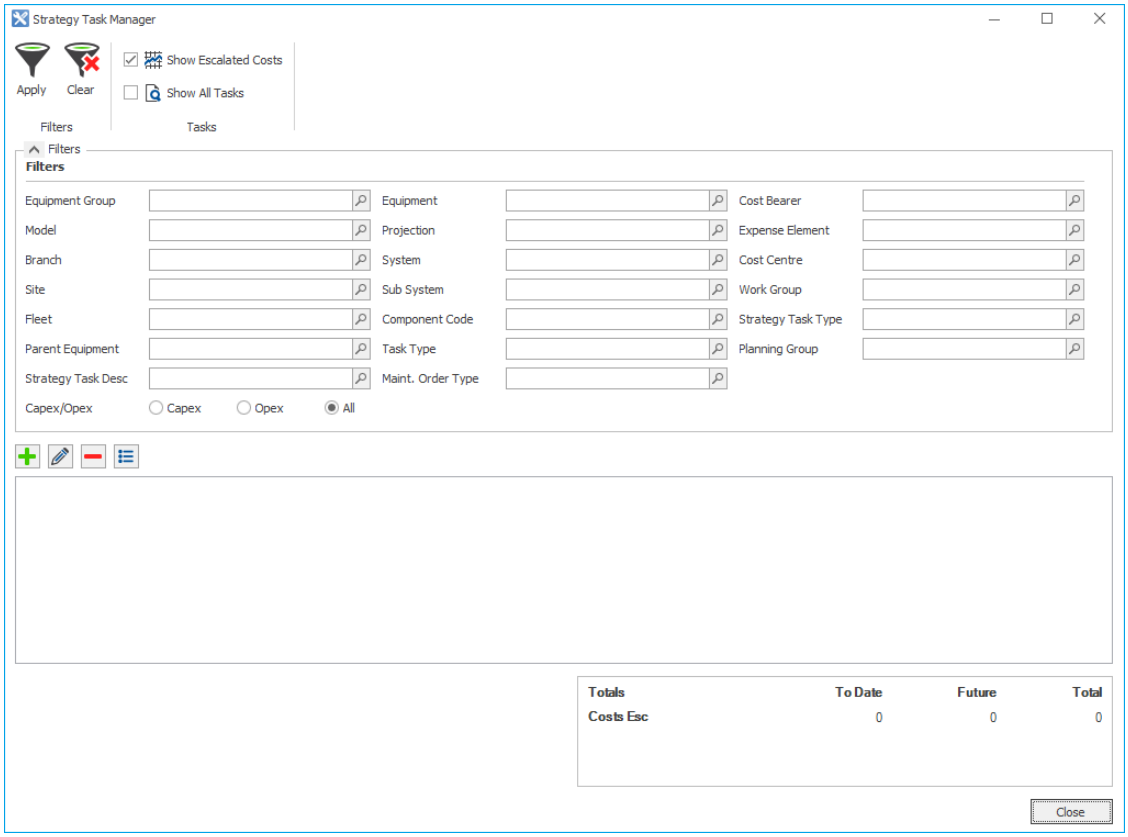
Define application severity

Application severity covers the physical environment and operating practices. In AMT, application severity is represented by Usage Profiles.

Strategy Task Manager

You can navigate to strategy task manager in the following ways:

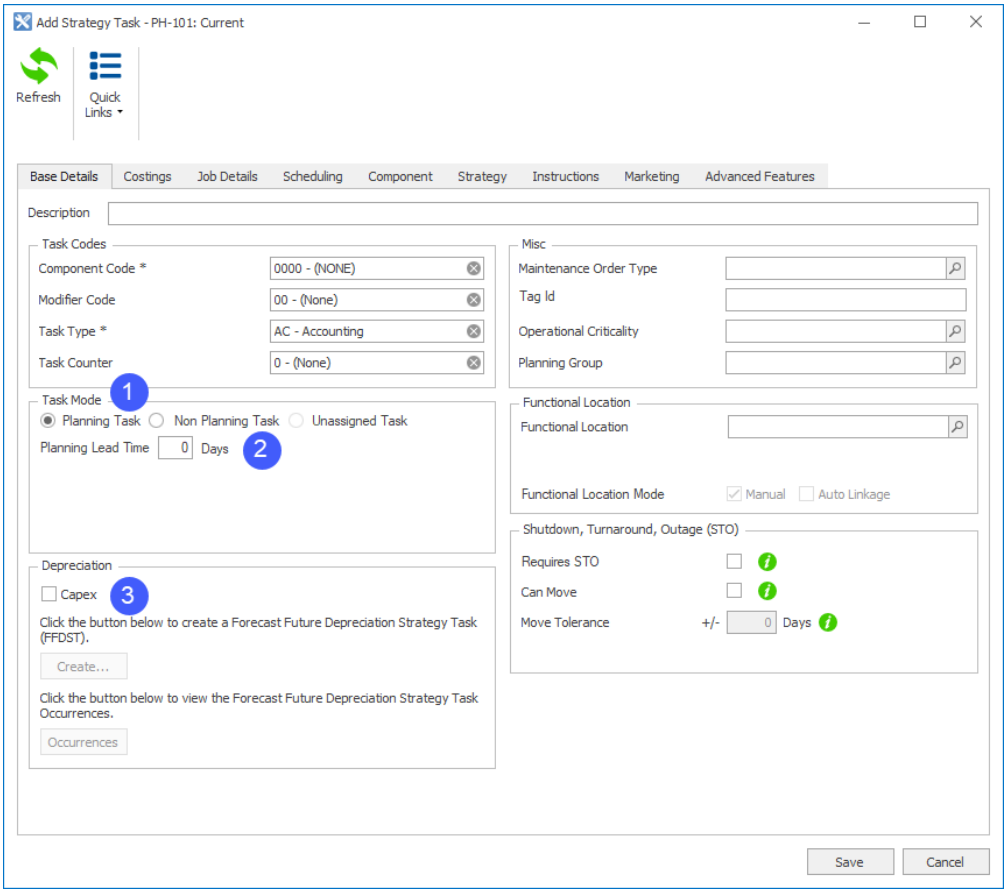
- Navigate to **Setup & Admin** > **Projections** > **Strategy Task Manager**.
- Navigate to **Centrelines** > **Administration** > **Strategy Task Manager**.
- On the right panel of the **Home** screen, select **Real Equipment**  . Use the filters to select the relevant equipment, right-click the equipment and select **Current Projection**.



Double-click a strategy task to open the **Edit Strategy Task** dialog box.

Key fields on Edit Strategy Task

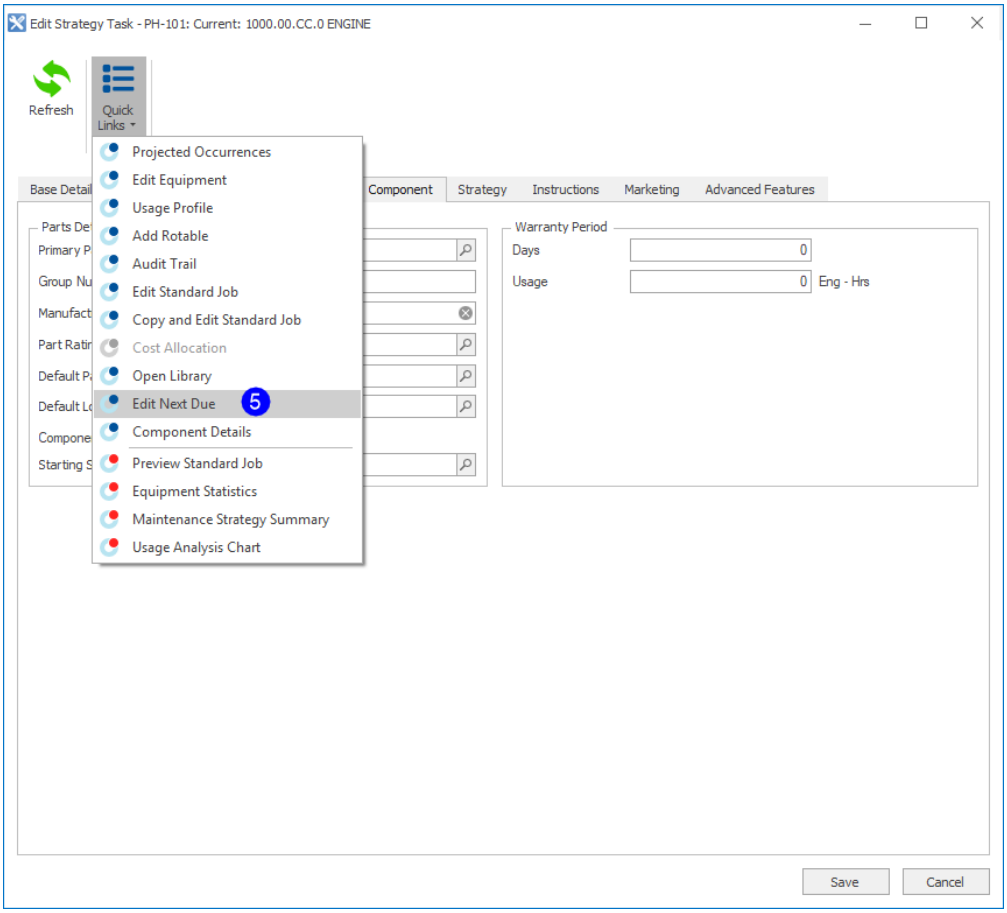
1. The Task Mode must be **Planning Task** for the component to display in the long term planning forms.
2. The **Planning Lead Time** defines when the strategy task will display in the long term planning forms.



3. The **Capex** check box, when selected, identifies the strategy task as a Capital expense. If the check box is not selected, the strategy task is an Operating expense (Opex). If a linked FFDST exists, the Capex check box is selected and cannot be changed. If a linked FFDST does not exist, the Capex check box is not selected by default, but is selectable.
4. For Dealers, **Auto generate Workorder History** should be selected for non-contract equipment. For contract equipment, it should be cleared as the dealer will be creating contract work orders for the work. These fields are on the **Advanced Features** tab.



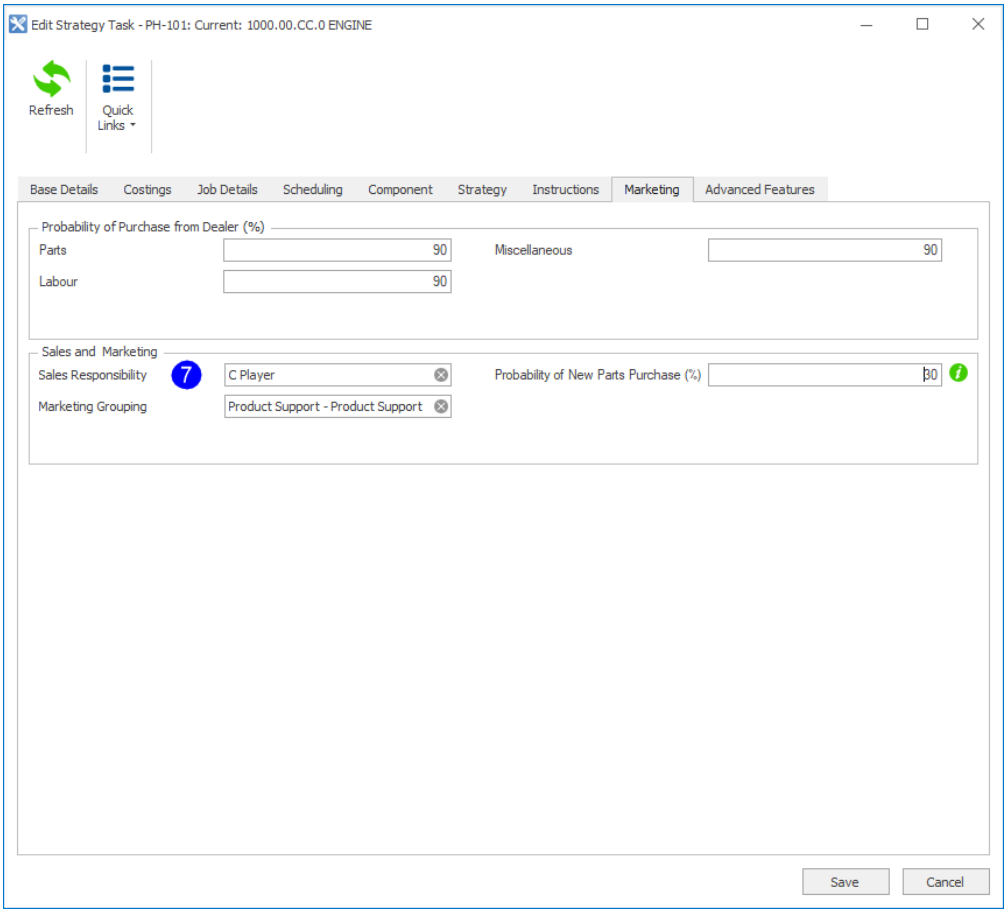
5. The **Parts Details** section displays the default settings for that strategy task. Click **Quick Links** at the top of the screen and then click **Edit Next Due**.



6. The three **Component Capacity Management** fields initially default to the settings in the **Base Details** tab, but you can edit the values.

   The defaults are only applied when a changeout takes place, so you have to set them manually initially.

7. Sales responsibility should be the sales rep, prime product sales person or account manager. This field is on the **Marketing** tab.



STO business process rules

Can Move check box

The following rules apply, depending on whether or not you have selected the **STO** and **Can Move** check boxes.

1. **STO Required is selected (that is, Y), Can Move is not selected (that is, N)**

   This means that the occurrence needs to be completed in an STO and the dates of the occurrence cannot move (unless there is manual intervention).

2. **STO Required is selected (that is, Y), Can Move is selected (that is, Y) and Tolerance = 0 days**

   This means there is no limit to the number of days that an occurrence can move, but the date movement is constrained by the STO dates (for example, must be within the STO dates).

3. **STO Required is selected (that is, Y), Can Move is selected (that is, Y) and Tolerance => 0 days (1, 2 ...)**

   This means that the occurrence move is limited by the number of days in the tolerance and constrained by the linked STO dates (for example, must be within the STO dates).

Editable Strategy Task Manager

This editor is specially designed to look and feel like a spreadsheet, where you can make changes to multiple strategy tasks across multiple equipment.

Each line in the grid shows a Strategy Task (combination of **CC**, **MC**, **TT**, **TC** and **UOM**). The count of equipment that have this Strategy Task is shown in the **Count** column. One line in the grid could represent the same Strategy Task for many equipment, therefore, when that line is updated, many Strategy Tasks are being updated at the same time. Strategy Task Job parameters can be updated in Strategy Tasks that have exactly one Projected Job. Linked Standard Jobs or Cost Allocations also restrict the Job Parameters that can be updated.

Cells which are shaded **grey** cannot be edited.

Click **Undo** to reverse changes up to the point at which the data on the form was last saved or displayed.

(Rpt_299) Projection Details - Centreline

Shows the base details of the projection, including pricing, and highlights which tasks have associated documents in the library, or has a standard job attached.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Bearer, Escalation

(Rpt_442) Projection and Interval Analysis - Centrelines

Tips

Timing of expenditure is based on the scheduling parameters of each strategy task.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Costs, Escalation

(Rpt_301) Projection Summary - Centreline

Gives an overview of the costs for the equipment by projected task and component sub system. Provides details of each projected task.

Tips

All costs are shown as cost at sell.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|EqpPlanID|,|EqpPlan|,
|ProjHeaderID|,|ProjHeader|,|ProjHeader|,|IsEscalated|,|ProjectionTypes|,
|LogoLocation|,|UserID|, |Currency|,|CurrencyID|

Report parameters

Default.

Additional filters

Cost Bearer, Escalation, Currency

(Rpt_484) Projection Summary - Customer - Centreline

Delete this text and replace it with your own content. Gives an overview of the costs for the equipment by projected task and component sub system. Provides details of each projected task. Costs are adjusted by customer risk premium and discounts.

Tips

All costs are shown as cost at sell, but adjusted by customer risk premium and discounts.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,
|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Cost Bearer, Escalation

(Rpt_427) Part Number Reference - Centreline

Report which lists the Primary Part Number, Group Number and Component Serial Number per strategy task for the selected equipment.

Tips

Component Serial Number will always be blank for Centreline equipment

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Report Type, Equipment Hierarchy

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Modifier Code, Strategy Task, Region, Division, Cost Responsibility, Eqp Location

Copy Strategy Tasks

Copy Strategy Tasks lets you copy and overwrite strategy tasks from one projection to another.

Multi-Task % Cost Adjuster

The Multi-Task Percent Cost Adjuster makes direct adjustments to the amounts in the strategy task.



It does not leave an audit trail.

It will only adjust strategy task jobs in Manual mode, therefore strategy task jobs linked to standard jobs will not be adjusted.

Multi-Task Editor

This is the most flexible multi-editor, and enables you to edit both strategy tasks and strategy task job fields. The Job Filters section applies to strategy task job fields.



There can be many strategy task jobs in one strategy task.

Multi-Task Labour Editor

This applies the labour rates entered to the labour cost of the strategy task job. It multiplies the labour hours in the strategy task job by the rates entered to obtain the labour cost.

It will only adjust strategy task jobs in Manual mode, therefore strategy task jobs linked to standard jobs will not be adjusted.

Multi-Task Scope Editor

The Multi-Task Scope Editor lets you edit the cost adjustment factors across multiple assets in the same fleet. In the strategy task these factors can be viewed by going to the Audit Trail form **Edit Strategy Task** > **Quick Links** > **Audit Trail** .

Add/Copy/Delete projection

You can create, copy, delete or import projections of the following types:

- Budget
- Current
- Alternate. Alternate projections are useful for What-If analysis.
- Archive. Archive projections are useful for annual budgets.

Centreline Analyses

(Rpt_286) Cost Analysis - "Life Cycle" - Centreline

Provides users the ability to analyse the overhead costs of a project against a projection.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|CostBearerID|,|ProjectionTypes|,|EquipmentTypes|,
|ProjHeaderID|,|ProjHeader|,|UserID|,|AutoLaunch|,|BranchID|,
|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,
|Fleet|,|EqpPlan|,|AnalyseBy|,|CurrencyID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Task Mode, Costs, Parts/Labour/Misc, Currency

(Rpt_278) Cost per Interval - Centerline

Lets you nominate a period of hours to analyse what equipment costs will fall in that timeframe. Can be analysed using either the Budget or Current projection and by numerous other filters.

Tips

This report can assist management in understanding what costs have occurred during what period of time in the equipment life cycle. It can also assist in estimating other machine costs of a similar nature.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modify Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Item, Item

(Rpt_282) Cost per UOM - by Usage - Centreline

Depicts the costs at intervals of the equipment usage (engine hours) throughout the term of its use.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_284) Cost per UOM - Comparison - Centreline

Compares visually the expected hourly costs by machine. It shows the cumulative cost per UOM for the period of time selected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_280) Cost per UOM - by Date - Centreline

Visually compares the expected hourly costs by machine. It shows the cumulative cost per UOM for the period of time selected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_431) Cost per UOM per Interval - Centrelines

Report which lists the Primary Part Number, Group Number and Component Serial Number per strategy task for the selected equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Labour Activity, Job Code, Costs

(Rpt_290) High Risk - Centreline

Identifies the risk components and calculates the life buffer available to prevent an additional component changeout prior to the equipment termination.

Tips

Shows the next projected occurrence after the equipment termination, the component life to avoid and the cost risk.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|MaxCompLife|,
|MinCompCost|,|LogoLocation|,|UserID|,|AutoLaunch|,|BranchID|,
|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|,
|AnalyseBy|,|ProjectionHeaderID|,|ProjectionHeader|,|Currency|

Report parameters

Default.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, Min. Comp. Cost, Max Life Left %, Min. Comp.Cur.

(Rpt_288) High Opportunity - Centreline

Identifies components due for change just short of the contract end, and calculates the new frequency required to ensure the opportunity can be realised.

Tips

Shows the last occurrence prior to equipment termination, the component life required and the cost benefit.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|MaxCompLife|,
|MinCompCost|,|LogoLocation|,|UserID|,|AutoLaunch|,|BranchID|,
|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,
|Fleet|,|EqpPlan|,|AnalyseBy|,|ProjectionHeaderID|,|ProjectionHeader|,|Currency

Report parameters

Default.

Additional filters

Task Type, System, Component Code, Max Life Used %, Cost Type, Sub System, Min.Comp.Cost, Min. Comp. Cur.

(Rpt_303) RVIC - Chart - Centreline

Displays the movement of Residual Life over the period of the chart dates.

Tips

The chart does not display consolidated Totals for Fleet, only individual Equipment. This means that selections should consider the amount of data to be displayed in the chart. It is recommended that no more than 6-10 items of equipment be displayed.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Sub System, Task Type, System, Component Code, Cost Bearer

(Rpt_305) RVIC Analysis - Centerline

Shows defined periods and components can be used as a basis for contract variation or termination, and can also be used for assessing contingent liabilities.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Component Code, Task Type, Cost Bearer

(Rpt_292) Labour Forecast - Hours - Centreline

Lets you analyse maintenance work by hours required. The report can be grouped by a maximum of five categories for more specific analysis.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity

(Rpt_294) Labour Forecast - Persons - Centreline

Lets you analyse maintenance work by persons required. The report can be grouped by a maximum of five categories for more specific analysis.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity, Productive Hours

+ indicates Average Productive Hours per Person per Month.

(Rpt_435) Labour Hours per Interval - Centrelines

Shows labour hour requirements per interval based on the labour hour estimates within strategy tasks.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Item, Item, Labour Type

(Rpt_296) Maintenance Ratio - Centreline

High level measure of efficiency of the direct Labour hours. The report only looks at productive labour hour, so needs to be viewed with the Labour Productivity report. It is calculated as Direct Labour Hours/Engine Hours.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|CostBearerID|,|AutoLaunch|,
|EqpPlanID|,|EqpPlan|,|DateChecked|,|StartDate|,|EndDate|,|LabourType|,
|ActualProjected|,|AnalyseByProjectionTypes|,|UserID|,|BranchID|,|SiteID|,
|FleetID|,|Branch|,|Site|,|Fleet|,|ProjectionHeaderID|,|ProjectionHeader|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Cost Bearer, Task Type, Strategy Task, Job Code, Labour, Show, Shop Job Codes

(Rpt_296) Maintenance Ratio - Centreline

High level measure of efficiency of the direct Labour hours. The report only looks at productive labour hour, so needs to be viewed with the Labour Productivity report. It is calculated as Direct Labour Hours/Engine Hours.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|CostBearerID|,|AutoLaunch|,
|EqpPlanID|,|EqpPlan|,|DateChecked|,|StartDate|,|EndDate|,|LabourType|,
|ActualProjected|,|AnalyseByProjectionTypes|,|UserID|,|BranchID|,|SiteID|,
|FleetID|,|Branch|,|Site|,|Fleet|,|ProjectionHeaderID|,|ProjectionHeader|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Cost Bearer, Task Type, Strategy Task, Job Code, Labour, Show, Shop Job Codes

(Rpt_438) Duration Per Interval - Centrelines

Shows duration estimates per interval based on the duration estimates within strategy tasks.

Tips

This report is used to assist in calculating availability forecasetss.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Labour Activity, Job Code, Labour Type

(Rpt_445) Consumables Forecast - Centrelines

Shows Forecast Consumable Usage.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Part Type, Job Code, Show

(Rpt_531) Strategy Task - Standard Job Details - Centreline

Lists, for strategy tasks linked to standard jobs, the details of the parts from the standard job.

Tips

Labour and Misc data from the standard jobs are excluded from this report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

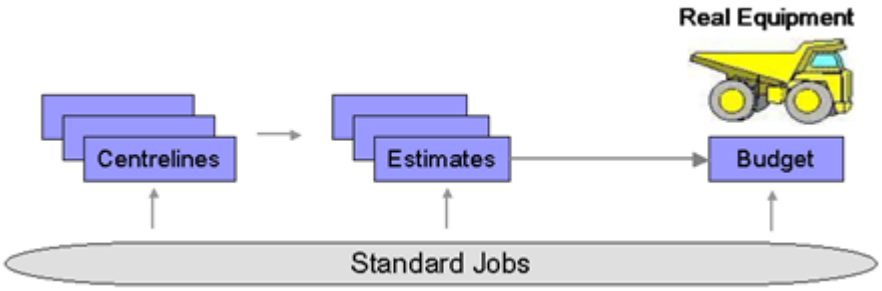
System, Sub System, Component Code, Task Type, Cost Bearer

Modelling

AMT's Modelling module allows you to:

- Create centrelines.
- Create tenders for maintenance contracts.
- Maintain audit trails of tender development.
- Model different fleet scenarios – different type of equipment with different start dates, utilisation, productivity and so on.
- Compare costs and productivity (for example, cost per tonne).

The Modelling functionality in AMT is based on the following high-level process.



- A centreline represents a generic template or baseline for a particular model.
- An estimate represents a tender, which can be for contract equipment (MARC) or non-contract equipment (Standard), and can be based on a centreline.
- If the tender is successful, the estimate becomes the budget for real equipment.

You can link standard jobs if required

Centrelines and modelling are different types of equipment in AMT. These special equipment types share the same basic configuration options and structure as real equipment. For example, each requires a fleet, projection structure is common, and the same projection editors can be used.

However, there are however a number of subtle differences. These are outlined in the related topics links below.

Centrelines Overview

Modelling

Administration

Equipment Hierarchy

Branch Administration

Branches are the levels below the Company (this is the highest level in the hierarchy in AMT). Use the Branch Administration function to add, edit and delete branches in AMT.

Field	Description
Branch Description	Name of the branch.
Branch Code	Identifies where costs for assets are captured in the accounting system.
WO Suffix	Work Order Settlement code suffix is used when AMT opens internal work order settlements automatically for an equipment under this branch.

Site Administration

Sites are the levels below branches. This system table is used to set up sites and related details, including setting up the Shift Start Time and enabling Employee Level Resourcing.

Field	Description
Site	Name of the site.
Branch	Branch to which the site belongs.
Site Code, Address, Suburb, State	Physical location information for the site.
Resource Scheduling	Select either Work Group Scheduling or Labour Activity and Work Group Scheduling.
Enable Employee Level Resourcing	Labour resource levelling is defaulted by site to be at the work group level. If a site wants to plan labour at the employee level, this check box must be selected.
Shift Start Time	The time when the first shift of the standard working day commences.  For maintenance contract sites, the shift start time must align to the dispatch system's shift start time.
Week Start Day	The first day of the week.
Currency	The financial currency used by the site.

Fleet Administration

Fleets are the levels below Site in the asset hierarchy. Navigate to **Setup & Admin** > **Equipment** to use the **Fleet Administration** screen to add, edit and delete fleets in AMT.

Field	Description
Fleet	Name of the fleet.
Site	Site to which the fleet belongs.
Pricing Group	Parts can be grouped by price. The pricing group is usually the default pricing group. However, if the site's parts are priced at a special rate, that is, include a discount, then a specific pricing group can be selected. Pricing groups are set up in the System Administration area of AMT, and require special parts pricing to be imported or entered into AMT.
Default	Select for the fleet to be the default fleet.
Service Interval	Standard interval between services, for example 250 hours. The service interval is used in the Service Accuracy report.
Tolerance%:	A KPI benchmark used in the Service Accuracy report to determine which services were performed within an acceptable tolerance of the standard service interval, for example 10%.
Availability Formula	Currently there are two options to work out asset availability. • Physical availability (based on scheduled/calendar hours). • Mechanical availability (based on operating hours).
Set Availability Calculation Parameters	Opens a special function designed to enable Availability Forecasting for the fleet.
Standard Job Parts Pricing Adjustment Factors	Use this function to manually adjust the pricing of parts for specific part types for this fleet. For example, if fluid was a specific part type, then you could increase the price of all fluid parts for this fleet by 10% (enter into AMT 0.1) using this function.

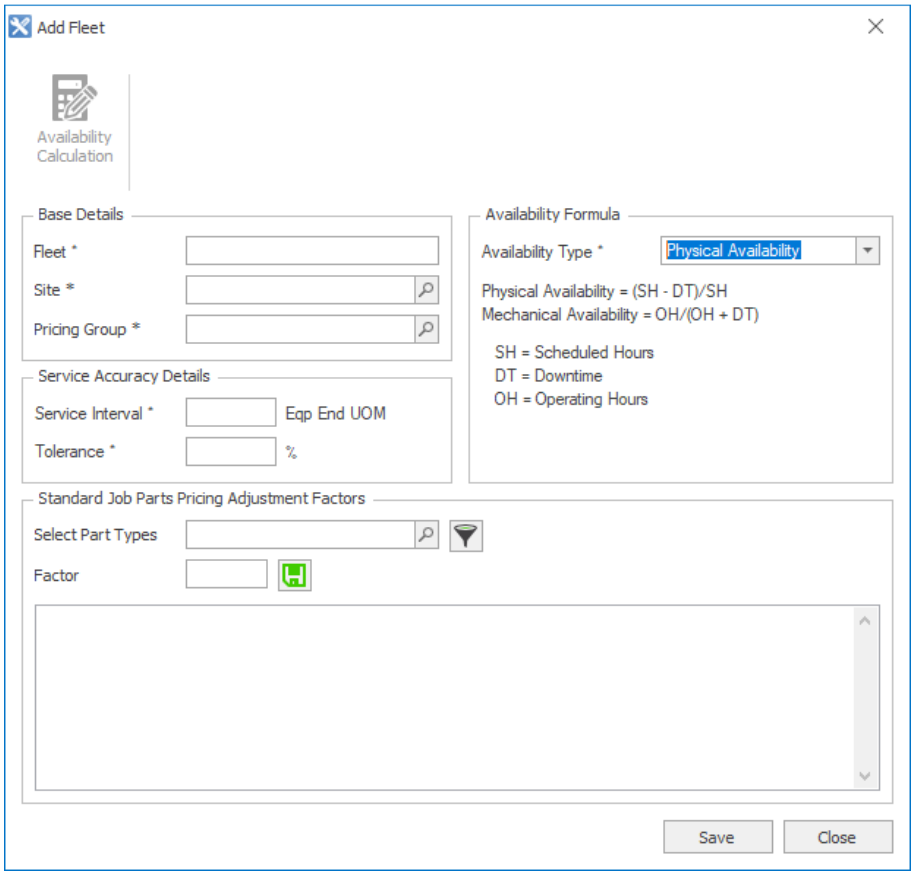
Set up Tendering (Modelling) fleet

One or more equipment can be part of a contract tender, so they need to be set up in a fleet. For modelling fleets, the fleet can be used to refer to the fleet of equipment being tendered.

For example:

- Branch = Modelling.
- Site = Customer Name + Tender ID.
- Fleet = Customer Name + Tender ID + Model.

Use the **Fleet Administration** form to set up fleets. Navigate to **Modelling** > **Administration** > **Equipment Hierarchy** > **Fleet Administration**, and click  .

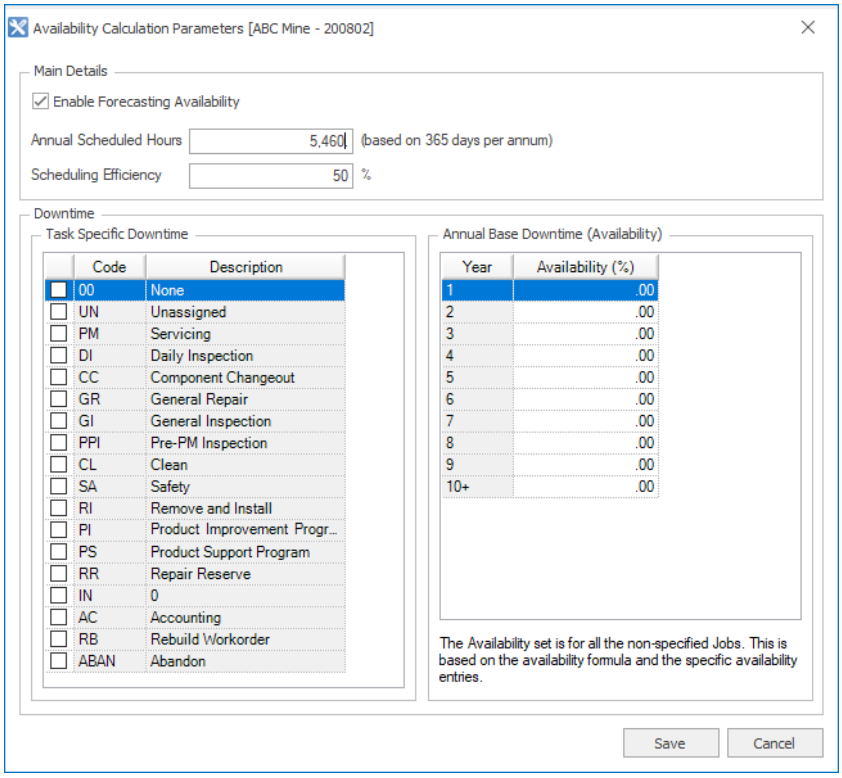


Complete the following fields:

- **Fleet**: Enter the fleet and link it to the tender at the site level.
- **Availability Type**: Select the fleet availability.
- Set the **Service Interval** and **Tolerance** for KPI reports.
- If the tender is linked to standard jobs, you can adjust the prices of specific **Part Types**.

After you save the fleet, the **Availability Calculation** button is enabled at the top left of the dialog box.

Set availability parameters



For a contract tender, you are able to forecast availability - which is typically what a customer wants to see. When you select the **Forecasting Availability** check box, the following four factors are taken into account:

1. **Annual Average Utilisation**

   The assumption of how many hours each asset in the fleet will operate per annum.

2. **Scheduling Efficiency %**

   The assumption of how efficient maintenance planning will be performed. That is, will the planner group all major tasks that are forecast to occur in the same month, to be executed at the same time (the absolute minimum stoppages ~ 100%), or will each task be performed at an independent stoppage (maximum possible stoppages ~ 0%). Usually the stoppages are set to 50%.

3. **Annual Base Downtime (Availability)**

   The baseline availability expected per year if only minor breakdown-type repairs were to be performed. Usually the rate is high, for example 95%.

4. **Task Specific Downtime**

   You can decide the task types that should be factored into the availability forecast. For example, if 95% is set as the annual base downtime rate, and you select component changeouts and PM services to be included (because they will be performed under the contract), the availability rate will drop. This is because the downtime for changeouts and PM services will be factored into the availability forecast.

Equipment Administration

You can view existing equipment (real, modelling and centrelines), as well as add and register new equipment in the system using the Equipment Administration function.

The equipment registration process consists of a number of steps that support the ongoing management and control of an operational asset. This process begins with defining a method to easily and uniquely identify an asset (depending on the asset type, this can be a difficult task). After the identification method is decided, you need to gather additional information about the asset to help determine classification and data collection requirements.

Before registering the asset in AMT, a number of questions need to be answered. For example:

- At which branch, site and fleet will the asset be located?
- Is this a complex asset requiring:
    - Sophisticated maintenance planning?
    - The ability to track a component's life using different units of measure (for example, machine hours, engine hours, fuel burn, travel time)?

  If so, a Usage Profile will have to be created.
- Is this a simple, low value asset requiring basic routine maintenance?
- What is the forecasted life cycle for the asset (for example 90,000 hours)?
- Will there be a rotable component strategy for certain components on the asset?
- What is the applicable cost centre or project that the asset will be assigned to?
- Is the asset is under a maintenance contract?
    - What are the contract details? (Contract number, contract name, contract account number, customer name and contract type).
    - If the asset is under contract, the contract needs to be set up in AMT using Contract Administration.

This kind of information determines how the asset will be registered in AMT.

To register equipment in AMT, either click **Add** in the Equipment Administration screen (which opens the Equipment Plan Editor screen), or run the Equipment Plan Editor function.

Create Blank Equipment - Modelling

Complete documentation for this topic will be available in a future release.

Multi-Editor Asset Details

Use the Multi-Editor Asset Details function to change Classification and/or Life Cycle Term parameters across many equipment.

Equipment Editor

The Equipment Editor function is used to create new equipment in AMT.

The nature of the information required when registering an equipment depends on the type of equipment being registered. Valid equipment types are Real, Centreline, Modelling. Adding new equipment to AMT requires the completion of the following.

Base Details

- Equipment Type, Model, Serial Number and Registration Counter are important fields in this section.
- Check with your System Administrator before entering the serial number. External systems, which may be linked to AMT via interfaces, may require the serial number to be entered in a specific format. For

example, some systems require the serial number to actually be “MODEL:SERIAL_NUMBER” because serial numbers are not unique unless prefixed by the Model.

- **System Configuration - Utilisation**
  - No utilisation: select when only date based scheduling and reporting is required.
  - Simple: select when only a single unit of measure (for example, engine hours) is required for scheduling and reporting.
  - Advanced: select when multiple units of measure are required for scheduling and reporting (this requires a usage profile).
- **System Configuration - Planning**
  - None: select when AMT will not be required to do daily planning for this asset.
  - Standard: refer to Standard vs. Advanced Planning.
  - Advanced: refer to Standard vs. Advanced.
- **System Configuration - Enabled**

  Some of the special setting options may be disabled if certain modules of AMT have not been purchased.

  - Asset Register: select this option to use AMT’s asset register.
  - Other Operating Costs: select this option to use AMT’s operating costs for the asset .
  - Plant Hire: select this option if the asset is Plant Hire.
  - MARC: select this option if the asset is under a maintenance contract
  - Multi-projection: select this option if more than one projection is required for the asset (maintenance contract equipment require this setting to be enabled)
- **Forecast Calculations**
  - The unassigned ratio is the ratio of the final hour/months unassigned cost to the first hour/month. The logic is that the average unassigned costs are still based on the value entered.
- **Equipment Interfaces**
  - Allow Equipment Detail Updates: select this option if the equipment import interface should be allowed to update asset details.
- **Quick Watch**

Term

- **Utilisation**: options presented depend on the type of utilisation selected in Configuration.
  - Usage Profile can only be selected with Advanced utilisation.
- **Start Date**: date when the asset began operation.
- **Original Commission Date**: date when the asset was commissioned.
- **Start**: for each usage listed, enter the starting usage. Typically it will be set to zero for new assets, but “second-hand” asset’s will have a usage greater than zero.
- **End**: set either an End Usage or End Date for the asset. This then becomes the termination point for forecasting.

Tasks

On this tab you can view, add, edit and delete tasks in selected projections. Refer to About Strategy Tasks.

Quick Editors

On this tab you are able to make multiple changes to strategy tasks for this specific equipment. This feature incorporates many of the Multi Editor functions which already exist in AMT, but in this case, the changes are limited to the specific equipment being edited.

- **Multi-Task Cost Adjustor**

  Apply a Parts/Labour/Misc cost adjustments, or Labour Hours or Duration (hours) adjustments across multiple tasks (for example, for selected tasks, increase Parts cost by 5%).

- **Labour Rates**

  Apply new Labour Rates for Shop and Field for selected projected tasks.

- **Travel**

  Ability to change Travel costs (labour and misc) for selected projected tasks.

Classifications

- Refer to Terminology for a description of each of the classification options
  Cost Details.
- **Cost Bearer**: select the default cost bearer. If unsure, contract the AMT Administrator.

Configuration & Application

The Registration and Configuration Details available for selection depend on which options were enabled for the Model of equipment you are registering.

- **Registration Details**

  Information only fields documenting the registration information of an equipment.

- **Configuration Details**

  Information only fields capturing equipment configuration.

- **Application Details**

  Information only fields capturing details of the application in which the equipment is currently operating.

- **Other Details**

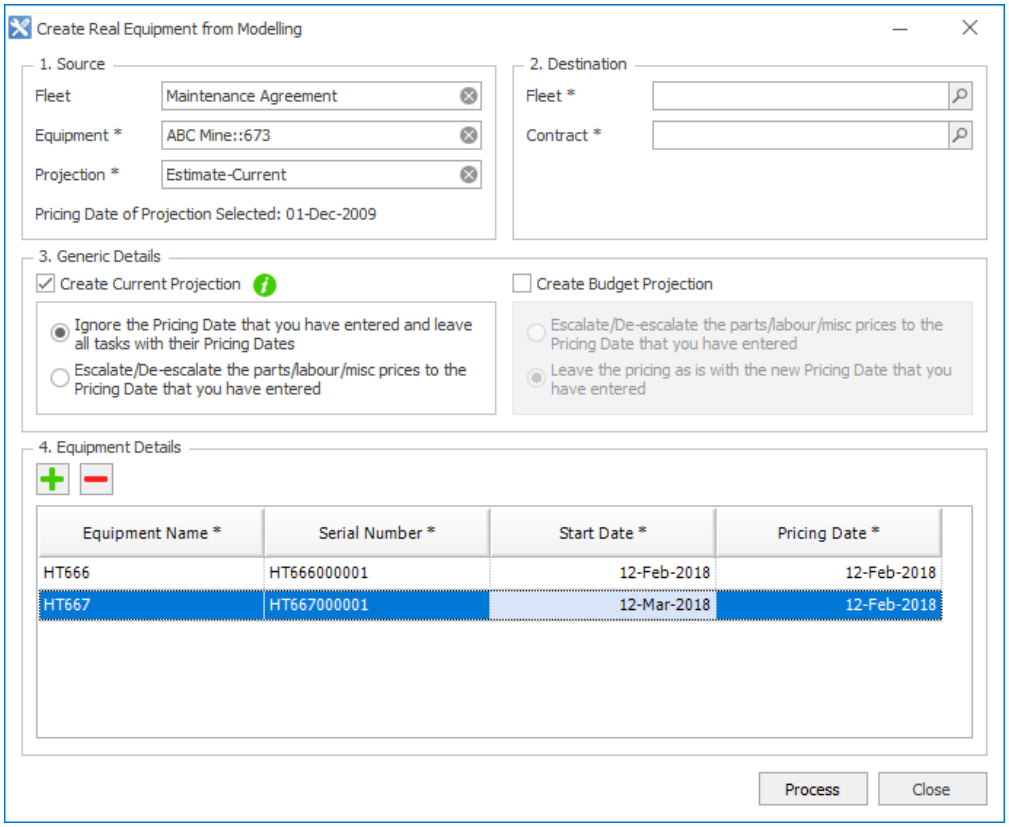
  Documents can be attached to the equipment. The process will copy the document from the specified location to the AMT in order that it can be accessed by AMT users.

- **Notes:**

  Add any notes required.

Create Real Equipment from Modelling

Modelling > **Administration** > **Equipment Hierarchy** > **Create Real Equipment From Modelling**.



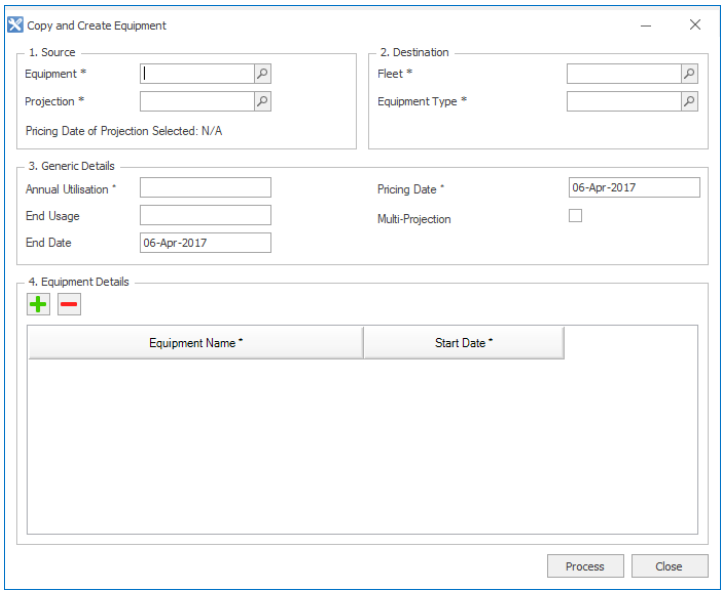
Field	Description
Source	
Fleet	Select the required source asset group.
Equipment	Select the specific equipment (asset).
Projection	Select the projection type. This may be automatically populated depending on the asset you selected in the **Equipment** field.
Destination	
Fleet	Select the destination asset group.
Contract	Select the type of contract.
Generic Details	
Create Current Projection	The current projection reflects actual costs incurred to date, plus occurrences and costs from today (system date) to the end date of the projection. All “live” forecasts (that is, component, financial, parts, and rotables) are calculated using the current projection.

Field	Description
	Therefore, changes to the current projection immediately change the live forecasts. Select the relevant option. If you do not select **Create Current Projection**, a blank projection is created.
Create Budget Projection	The budget projection establishes a baseline for future comparison with another AMT projection (usually the current projection). For assets under a maintenance contract, the budget projection usually reflects the commercial assumptions made to determine the contractual billing rates. Select the relevant option.
Equipment Details	Click **Add** to enter the details of the equipment.

If you copy an Asset that is configured as a contract (MARC) asset (see Base Details tab in the **Asset** form), the Assets being created will have the same configuration. In this case, **Contract** will be enabled under **Fleet** in destination. All other configurations and classifications will be copied from the source modelling asset. This includes the utilisation configuration and rates.

Create a Centreline or Modelling asset

Setup & Admin > **Equipment** > **Copy and Create Equipment**.



Panel	Description
Source	
Equipment	Select the required source asset group.
Projection	Select the projection type. This may be automatically populated depending on the asset you selected in the **Equipment** field.
Destination	
Fleet	Select the required destination asset group.
Equipment Type	Select the required equipment type, either centreline or modelling.
Generic Details	
Annual Utilisation	Enter the ratio of productive hours to available hours.
End Usage	Enter the projected last unit of measure reading of a projection.
End Date	Enter the end date for the asset.
Pricing Date	Enter the date at which the pricing entered for parts, labour and miscellaneous was relevant. This may differ from the budget pricing date.
Multi-Projection	Select this check box if more than one projection is required for the asset (maintenance contract equipment require this setting to be enabled).
Equipment Details	Click **Add** to enter the details of the equipment.

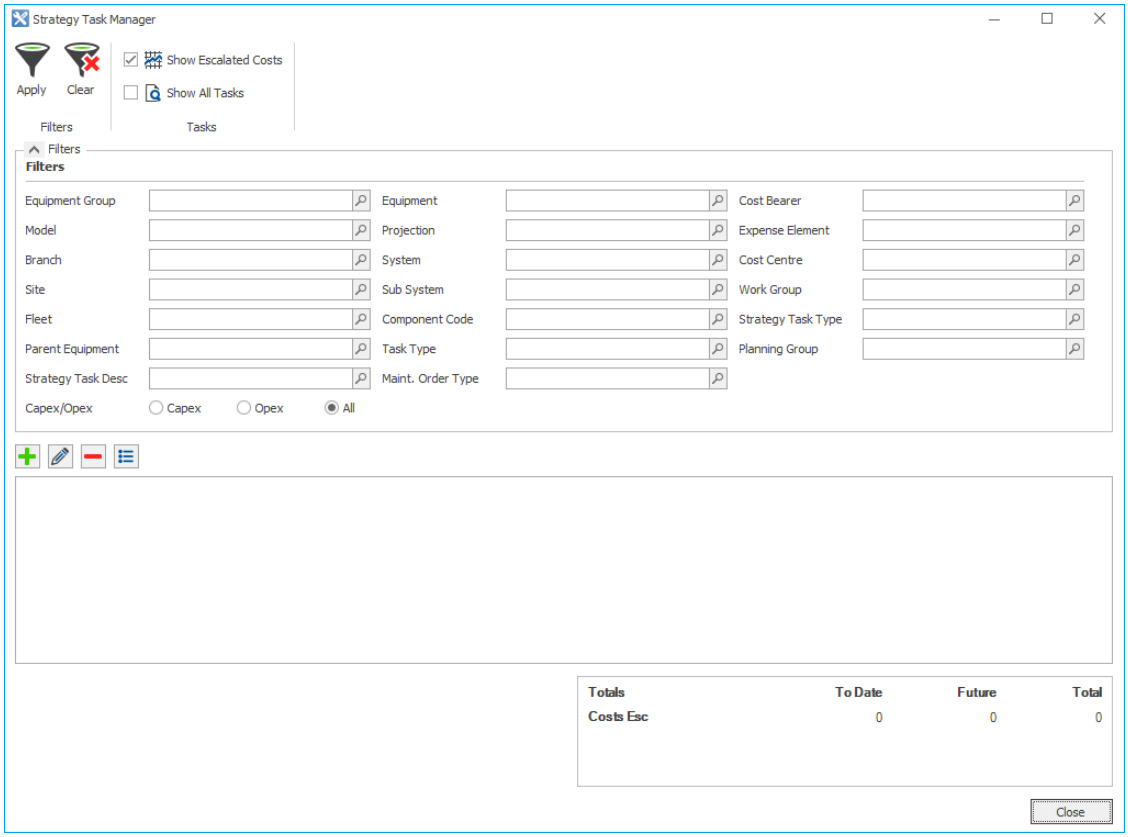
If you copy an asset that is linked to a usage profile, the assets created will be linked to the same usage profile. In this situation, the **Annual Utilisation** field will be disabled. Either **End Date** or **End Usage** will be disabled. The enabled option will default to the same as the source asset being copied.

Projections

Strategy Task Manager

You can navigate to strategy task manager in the following ways:

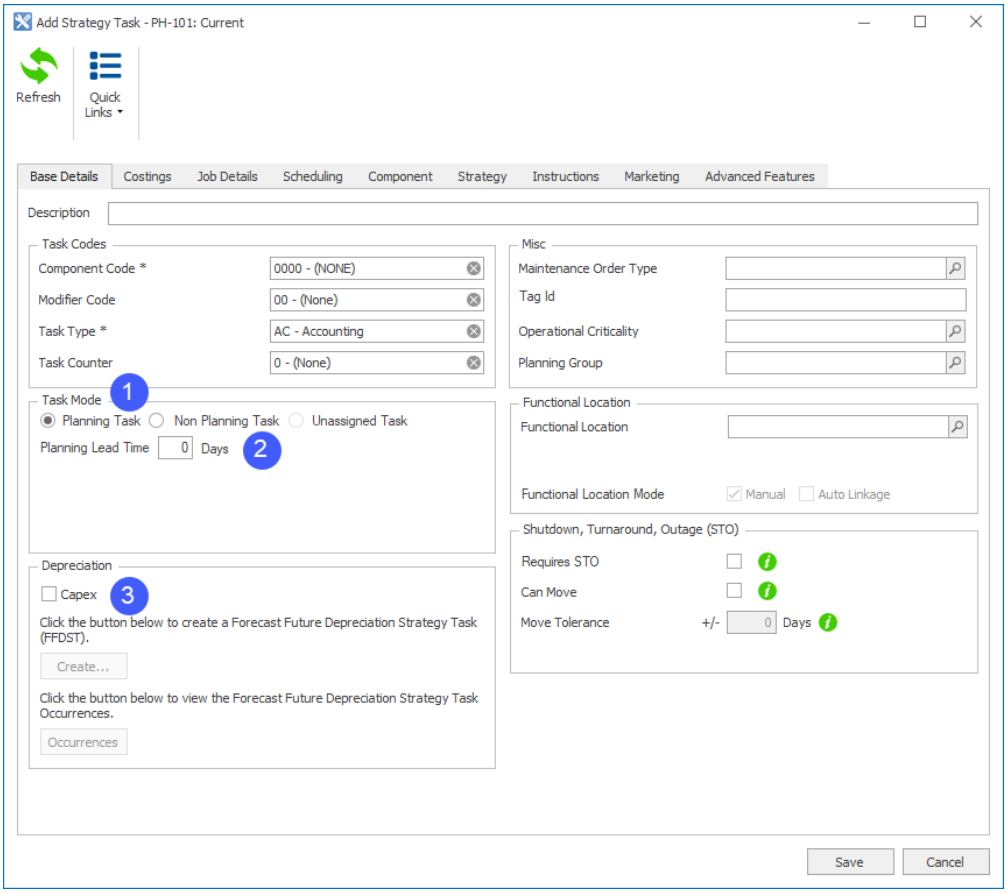
- Navigate to **Setup & Admin** > **Projections** > **Strategy Task Manager**.
- Navigate to **Centrelines** > **Administration** > **Strategy Task Manager**.
- On the right panel of the **Home** screen, select **Real Equipment**  . Use the filters to select the relevant equipment, right-click the equipment and select **Current Projection**.



Double-click a strategy task to open the **Edit Strategy Task** dialog box.

Key fields on Edit Strategy Task

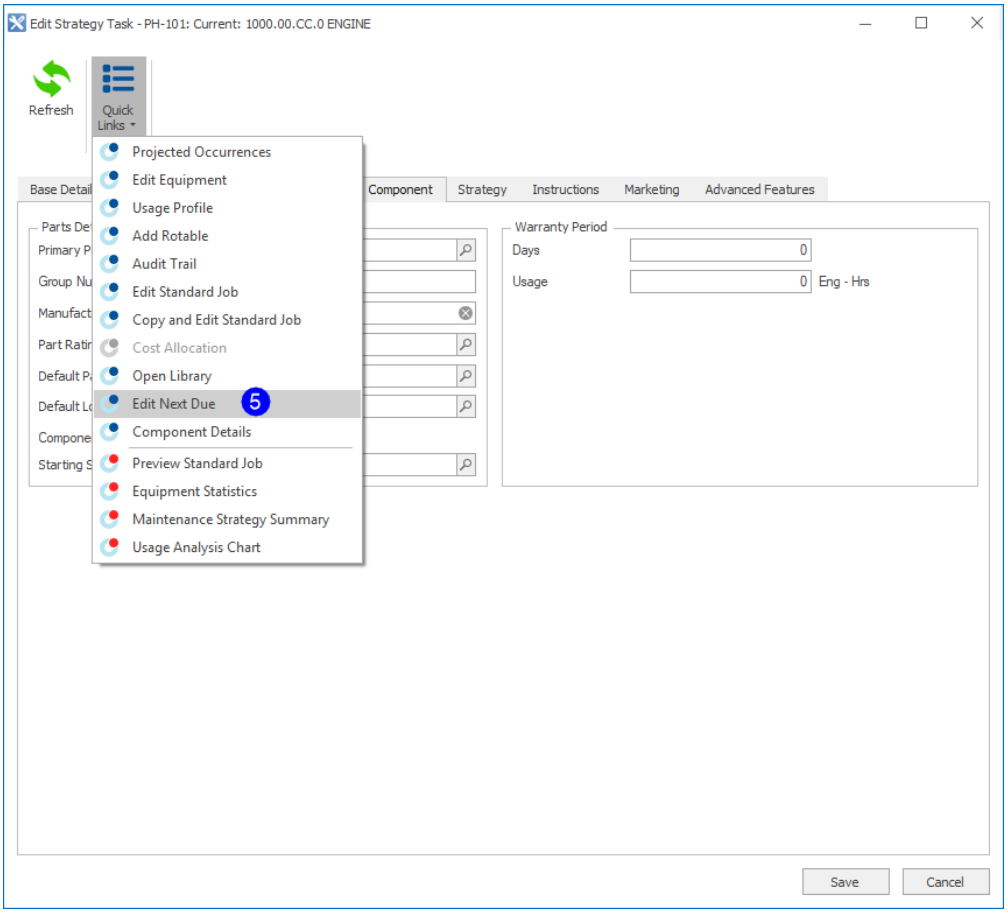
1. The Task Mode must be **Planning Task** for the component to display in the long term planning forms.
2. The **Planning Lead Time** defines when the strategy task will display in the long term planning forms.



3. The **Capex** check box, when selected, identifies the strategy task as a Capital expense. If the check box is not selected, the strategy task is an Operating expense (Opex). If a linked FFDST exists, the Capex check box is selected and cannot be changed. If a linked FFDST does not exist, the Capex check box is not selected by default, but is selectable.
4. For Dealers, **Auto generate Workorder History** should be selected for non-contract equipment. For contract equipment, it should be cleared as the dealer will be creating contract work orders for the work. These fields are on the **Advanced Features** tab.



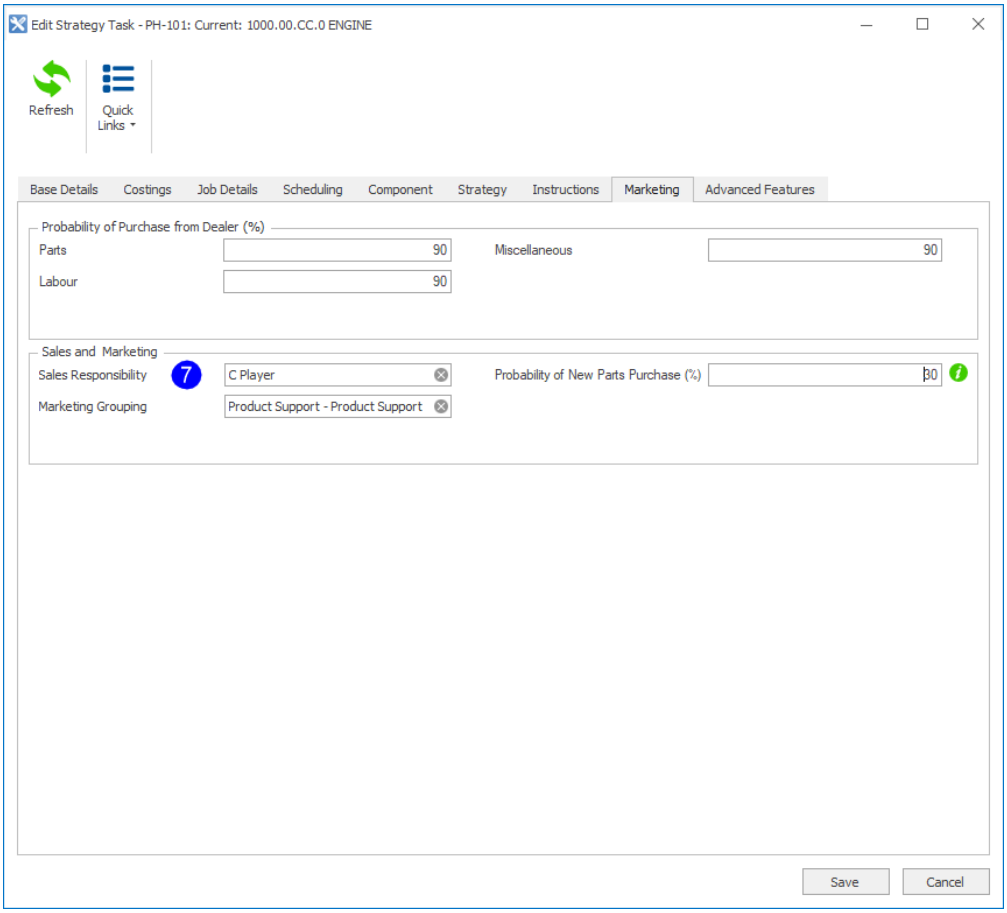
5. The **Parts Details** section displays the default settings for that strategy task. Click **Quick Links** at the top of the screen and then click **Edit Next Due**.



6. The three **Component Capacity Management** fields initially default to the settings in the **Base Details** tab, but you can edit the values.

   The defaults are only applied when a changeout takes place, so you have to set them manually initially.

7. Sales responsibility should be the sales rep, prime product sales person or account manager. This field is on the **Marketing** tab.



STO business process rules

Can Move check box

The following rules apply, depending on whether or not you have selected the **STO** and **Can Move** check boxes.

1. **STO Required is selected (that is, Y), Can Move is not selected (that is, N)**

   This means that the occurrence needs to be completed in an STO and the dates of the occurrence cannot move (unless there is manual intervention).

2. **STO Required is selected (that is, Y), Can Move is selected (that is, Y) and Tolerance = 0 days**

   This means there is no limit to the number of days that an occurrence can move, but the date movement is constrained by the STO dates (for example, must be within the STO dates).

3. **STO Required is selected (that is, Y), Can Move is selected (that is, Y) and Tolerance => 0 days (1, 2 ...)**

   This means that the occurrence move is limited by the number of days in the tolerance and constrained by the linked STO dates (for example, must be within the STO dates).

Editable Strategy Task Manager

This editor is specially designed to look and feel like a spreadsheet, where you can make changes to multiple strategy tasks across multiple equipment.

Each line in the grid shows a Strategy Task (combination of **CC**, **MC**, **TT**, **TC** and **UOM**). The count of equipment that have this Strategy Task is shown in the **Count** column. One line in the grid could represent the same Strategy Task for many equipment, therefore, when that line is updated, many Strategy Tasks are being updated at the same time. Strategy Task Job parameters can be updated in Strategy Tasks that have exactly one Projected Job. Linked Standard Jobs or Cost Allocations also restrict the Job Parameters that can be updated.

Cells which are shaded **grey** cannot be edited.

Click **Undo** to reverse changes up to the point at which the data on the form was last saved or displayed.

(Rpt_302) Projection Summary - Modelling

Gives an overview of the costs for the equipment by projected task and component sub system. Provides details of each projected task.

Tips

All costs are shown as cost at sell.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|AutoLaunch|,|EqpPlanID|,|EqpPlan|,
|ProjHeaderID|,|ProjHeader|,|ProjHeader|,|IsEscalated|,|ProjectionTypes|,
|LogoLocation|,|UserID|, |Currency|,|CurrencyID|

Report parameters

Default.

Additional filters

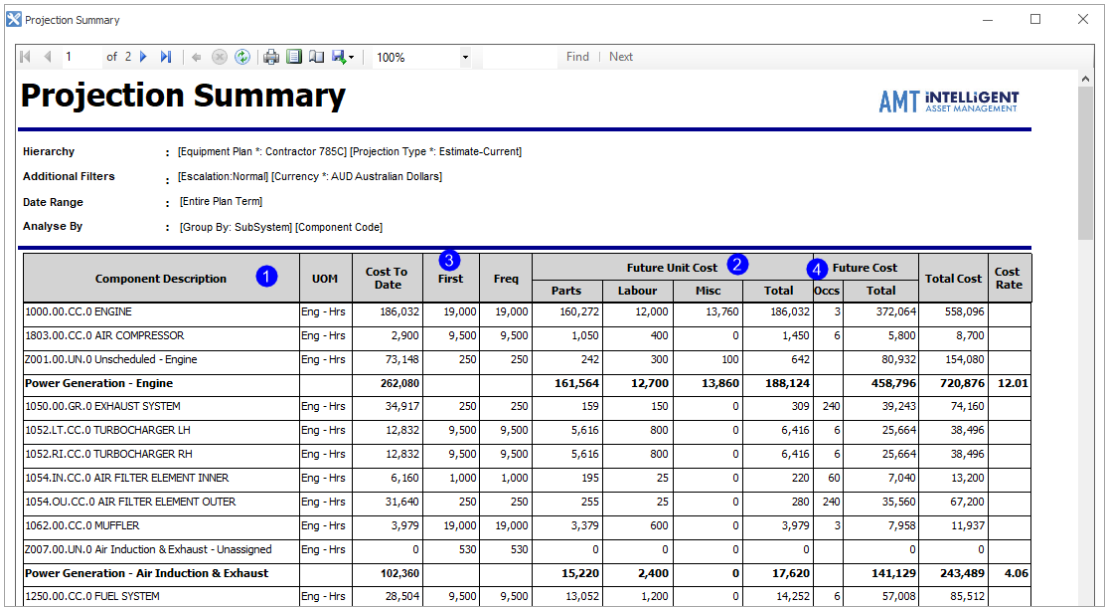
Cost Bearer, Escalation, Currency

Validate contract

To validate the contract, use the Modelling module's reporting functionality.

Use the Projection Summary and Projection Details reports to review the inclusions of the contract, and validate the contract term.

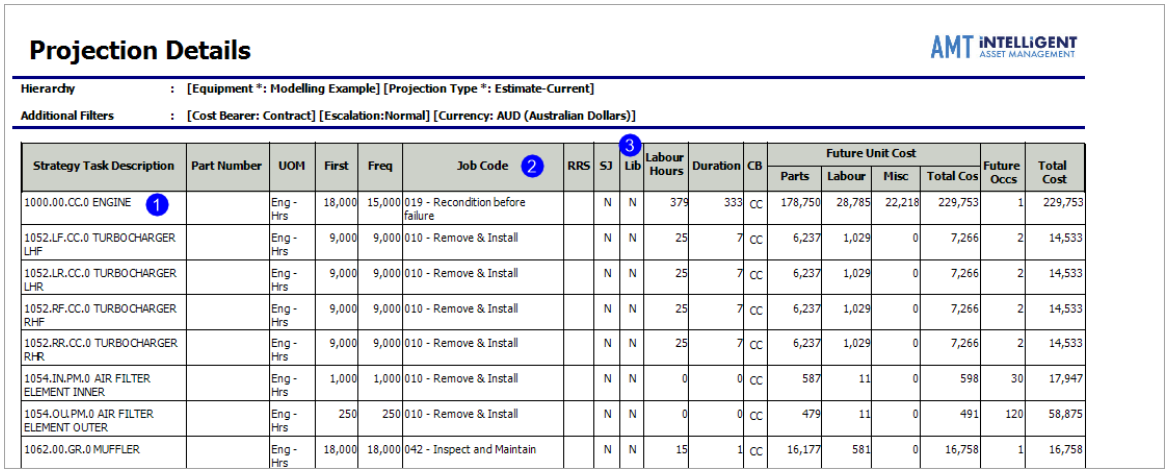
Navigate to **Modelling** > **Administration** > **Projections** > Projection Summary - Modelling.



Component Description 1	UOM	Cost To Date	First 3	Freq	Future Unit Cost 2				Future Cost 4		Total Cost	Cost Rate
					Parts	Labour	Misc	Total	Occs	Total		
1000.00.CC.0 ENGINE	Eng - Hrs	186,032	19,000	19,000	160,272	12,000	13,760	186,032	3	372,064	558,096	
1803.00.CC.0 AIR COMPRESSOR	Eng - Hrs	2,900	9,500	9,500	1,050	400	0	1,450	6	5,800	8,700	
Z001.00.UN.0 Unscheduled - Engine	Eng - Hrs	73,148	250	250	242	300	100	642		80,932	154,080	
Power Generation - Engine		**262,080**			**161,564**	**12,700**	**13,860**	**188,124**		**458,796**	**720,876**	**12.01**
1050.00.GR.0 EXHAUST SYSTEM	Eng - Hrs	34,917	250	250	159	150	0	309	240	39,243	74,160	
1052.LT.CC.0 TURBOCHARGER LH	Eng - Hrs	12,832	9,500	9,500	5,616	800	0	6,416	6	25,664	38,496	
1052.RI.CC.0 TURBOCHARGER RH	Eng - Hrs	12,832	9,500	9,500	5,616	800	0	6,416	6	25,664	38,496	
1054.IN.CC.0 AIR FILTER ELEMENT INNER	Eng - Hrs	6,160	1,000	1,000	195	25	0	220	60	7,040	13,200	
1054.OU.CC.0 AIR FILTER ELEMENT OUTER	Eng - Hrs	31,640	250	250	255	25	0	280	240	35,560	67,200	
1062.00.CC.0 MUFFLER	Eng - Hrs	3,979	19,000	19,000	3,379	600	0	3,979	3	7,958	11,937	
Z007.00.UN.0 Air Induction & Exhaust - Unassigned	Eng - Hrs	0	530	530	0	0	0	0		0	0	
Power Generation - Air Induction & Exhaust		**102,360**			**15,220**	**2,400**	**0**	**17,620**		**141,129**	**243,489**	**4.06**
1250.00.CC.0 FUEL SYSTEM	Eng - Hrs	28,504	9,500	9,500	13,052	1,200	0	14,252	6	57,008	85,512	

Number	Description
1	The **Component Description** column displays the strategy tasks (components, PMs) included in the contract.
2	The **Future Unit Cost** column displays the estimated cost per occurrence of the task.
3	The **First** column displays the component life estimates.
4	The **Occs** column displays the Projected occurrences for the term of the contract.

Navigate to **Modelling** > **Administration** > **Projections** > **Projection Details - Modelling**.



Strategy Task Description 1	Part Number	UOM	First	Freq	Job Code 2	RRS	SJ	Lib 3	Labour Hours	Duration	CB	Future Unit Cost				Future Occs	Total Cost
												Parts	Labour	Misc	Total Cos		
1000.00.CC.0 ENGINE		Eng - Hrs	18,000	15,000	019 - Recondition before failure		N	N	379	333	CC	178,750	28,785	22,218	229,753	1	229,753
1052.LF.CC.0 TURBOCHARGER LHF		Eng - Hrs	9,000	9,000	010 - Remove & Install		N	N	25	7	CC	6,237	1,029	0	7,266	2	14,533
1052.LR.CC.0 TURBOCHARGER LHR		Eng - Hrs	9,000	9,000	010 - Remove & Install		N	N	25	7	CC	6,237	1,029	0	7,266	2	14,533
1052.RF.CC.0 TURBOCHARGER RHF		Eng - Hrs	9,000	9,000	010 - Remove & Install		N	N	25	7	CC	6,237	1,029	0	7,266	2	14,533
1052.RR.CC.0 TURBOCHARGER RHR		Eng - Hrs	9,000	9,000	010 - Remove & Install		N	N	25	7	CC	6,237	1,029	0	7,266	2	14,533
1054.IN.PM.0 AIR FILTER ELEMENT INNER		Eng - Hrs	1,000	1,000	010 - Remove & Install		N	N	0	0	CC	587	11	0	598	30	17,947
1054.OU.PM.0 AIR FILTER ELEMENT OUTER		Eng - Hrs	250	250	010 - Remove & Install		N	N	0	0	CC	479	11	0	491	120	58,875
1062.00.GR.0 MUFFLER		Eng - Hrs	18,000	18,000	042 - Inspect and Maintain		N	N	15	1	CC	16,177	581	0	16,758	1	16,758

Number	Description
1	The **Strategy Task Description** column displays the strategy tasks included in the contract.
2	The **Job Code** column displays the jobs to be performed as part of each task.
3	The **Lib** column displays whether the tasks are linked to standard jobs.

(Rpt_300) Projection Details - Modelling

Shows the base details of the projection, including pricing, and highlights which tasks have associated documents in the library, or has a standard job attached.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Bearer, Escalation

(Rpt_441) Projection and Interval Analysis - Modelling

This report shows a strategy task's estimated cost, and then how much is Forecasted to be spent per interval.

Tips

Timing of expenditure is based on the scheduling parameters of each strategy task.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Costs, Escalation

(Rpt_426) Part Number Reference - Modelling

Report which lists the Primary Part Number, Group Number and Component Serial Number per strategy task for the selected equipment.

Tips

Component Serial Number will always be blank for Modeling equipment

Initial parameters

|ControlType|,|IDlist|,|TextList|,|EquipmentTypes|,|UserID|

Report parameters

Report Type, Equipment Hierarchy

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Modifier Code, Strategy Task, Region, Division, Cost Responsibility, Eqp Location

Add/Copy/Delete projection

You can create, copy, delete or import projections of the following types:

- Budget
- Current
- Alternate. Alternate projections are useful for What-If analysis.
- Archive. Archive projections are useful for annual budgets.

Copy Strategy Tasks

Copy Strategy Tasks lets you copy and overwrite strategy tasks from one projection to another.

Multi-Equipment Delete Task

Enables you to delete multiple strategy tasks.

Multi-Task % Cost Adjuster

The Multi-Task Percent Cost Adjuster makes direct adjustments to the amounts in the strategy task.



It does not leave an audit trail.

It will only adjust strategy task jobs in Manual mode, therefore strategy task jobs linked to standard jobs will not be adjusted.

Multi-Task Editor

This is the most flexible multi-editor, and enables you to edit both strategy tasks and strategy task job fields. The Job Filters section applies to strategy task job fields.



There can be many strategy task jobs in one strategy task.

Multi-Task Labour Editor

This applies the labour rates entered to the labour cost of the strategy task job. It multiplies the labour hours in the strategy task job by the rates entered to obtain the labour cost.

It will only adjust strategy task jobs in Manual mode, therefore strategy task jobs linked to standard jobs will not be adjusted.

Multi-Task Scope Editor

The Multi-Task Scope Editor lets you edit the cost adjustment factors across multiple assets in the same fleet. In the strategy task these factors can be viewed by going to the Audit Trail form **Edit Strategy Task** > **Quick Links** > **Audit Trail** .

Refresh Standard Job Pricing

This function is designed to refresh the pricing of strategy tasks, within Modelling only, based on changes to standard jobs to which the strategy tasks are linked.

(Rpt_556) Projection Summary and Cost per Interval - Modelling

Provides a summary of Equipment life cycle costs along with an analysis of costs per interval.

Tips

The interval is provided by the user when the report is run.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Costs, Escalation, Comp-Job-Mod Desc

Usage Profile

Usage profile

Unit of measure (UOM) refers to any cumulative usage related to an asset, such as machine hours, fuel litres, kilowatt hours and tonnes. They are also referred to as **Usages** in AMT. Usages are used to:

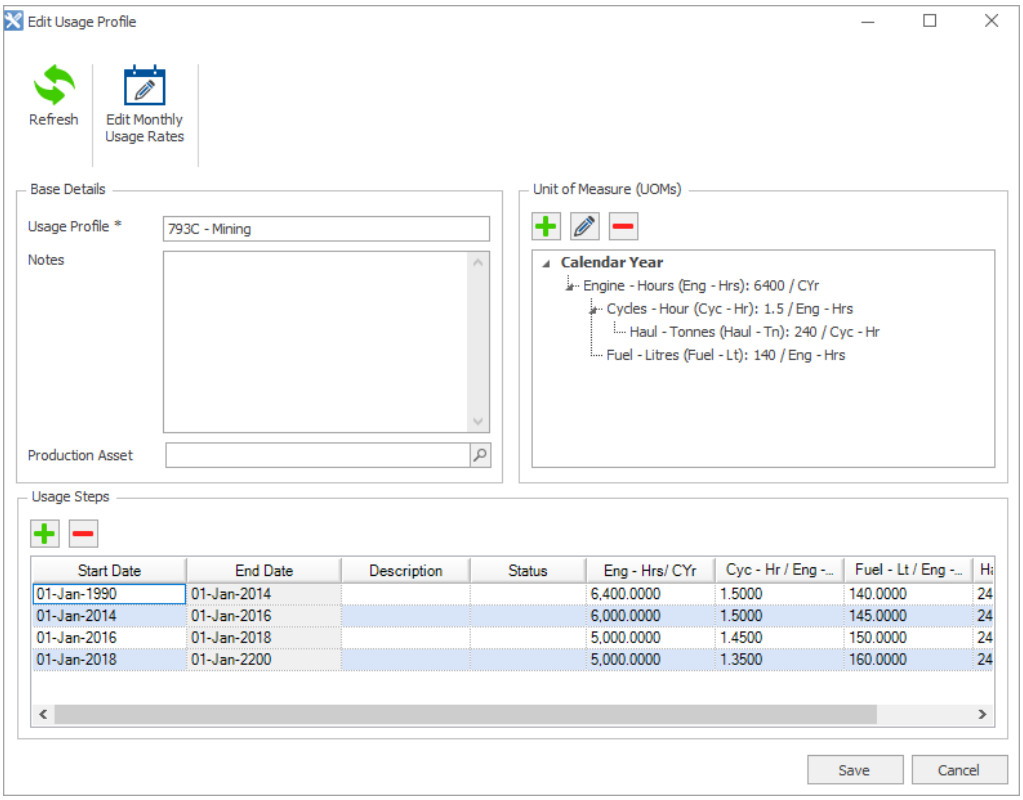
1. **Drive maintenance schedules**
   For example, an engine is scheduled based on fuel litres. In the strategy task you can select the UOM appropriate for that asset.

2. **Forecast productivity**
   For example, tonnes moved can be set up as a UOM. Tonnes moved can then be used to forecast production.

Usage Profiles enable you to:

- Set up a profile for the usage rates or utilisation for groups of assets.
- Set up relationships between the usages (for example, tonnes is a dependency of engine hours).
- Vary the usage rates over time to reflect changing operating conditions (for example, deeper pit, rainy season, winter and other conditions).

  The example screenshot below shows a usage profile with a hierarchy of usages.

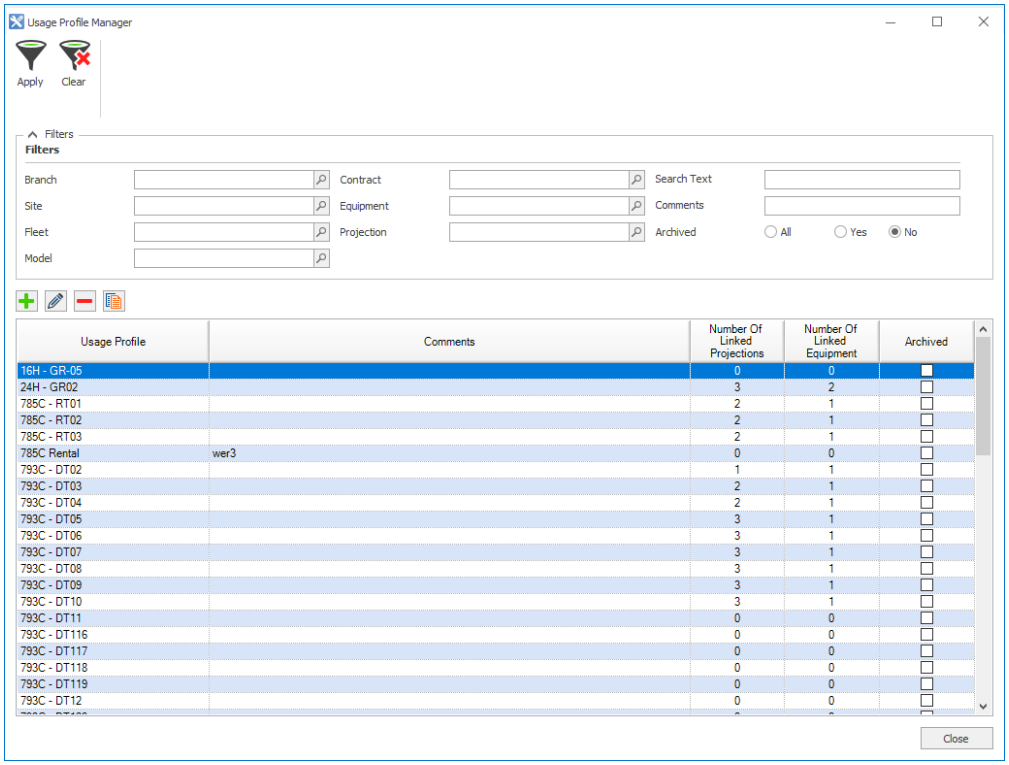


- Usage profiles allow you to set up the severity of the situation. For example, if the pit in a mine becomes deeper, the fuel burn rate increases and the engine life in hours will decrease.
- Actual usages can be captured in AMT and plotted against the usage profile.
- Usage profiles are optional for assets. You can elect to either have no usage measure for an asset (all strategy tasks are then date-based) or apply a simple utilisation where one UOM is selected and one utilisation rate set (for example, 4,000 engine hours per annum).

The **Status** field within a usage profile step is only applicable for the tendering and forecasting of a specific type of equipment which is configured for this functionality, and therefore, in most cases, does not impact the usage forecast by AMT.

Usage Profile Manager

You can open the Usage Profile Manager by navigating to **Setup & Admin** > **Usage Profiles** > **Usage Profile Manager** or **Centrelines** > **Administration** > **Usage Profile Manager**.



Use the Usage Profile Manager function in AMT to add, edit and delete usage profiles.

To copy a usage profile click  .

Usage is the consumption of a resource by an asset. This consumption can vary over time due to changes in the working application and in working patterns. For example, consumption may be altered by changes in weather patterns; high rainfall may slow production, dry fine weather may increase production, or by physical changes in loads, haulage road conditions, holidays, and so on.

The usage profile maps the changes in consumption over time and allows the consumption to be measured by different units of measure. Usage profiles can have multiple steps and multiple branches in the consumption tree.

All usages are expressed in qualified units of measure (UOM) such as engine hours or litres of fuel. The most common unit of measure is engine hours. A usage profile is a named pattern for the rate of consumption of UOMs by an asset. Usage profiles can be linked to a single asset in a single projection or a range of assets of the same model across a fleet or site.

Notes

- A range of assets with different models can use the same usage profile, and different projections can use different usage profiles to model application, utilisation, or other what-if scenarios.
- The final usage step within a usage profile cannot have a usage value of zero.

(Rpt_312) Usage Profile Details - Modelling

Displays the different usage profiles set up with both actual and cumulative values showing. The report can be generated for a specific month period, but defaults are set to the last twelve months.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,5,|UserID|

Report parameters

Default.

Additional filters

Branch, Site, Fleet, Equipment

Asset Ledger

Capital Details

You can open the Capital Details screen in the following ways:

- **Setup & Admin** > **Asset Ledger**. Select an equipment and click OK. The **Capital Details** tab displays.
- **Modelling** > **Administration** > **Asset Ledger** > **Capital Details** tab. Select an equipment and click OK. The **Capital Details** tab displays.
- On the right of the home screen, in the **Equipment Navigator** on the right of the screen, drill-down into **Contractor** > **Plant Hire** > **Project** and double-click an equipment. Click **Show Details** to display the input fields.

The capital detail function in AMT has a number of input fields. First nominate if the equipment is owned or leased. Select the check box to indicate if the equipment will be replaced. If this check box is selected the estimated replacement cost field can be updated with the cost. You can also enter the estimated disposal price at this point.

Other fields to complete for depreciation are:

- Component code.
- Cost bearer, for example project or plant group.
- Expense element.
- Journal applicable. Select this box if transactions will be posted to the general ledger using an interface.
- Entry distribution code (general ledger codes for posting via an interface).

The current depreciation position is displayed on the screen to indicate the gross cost, accumulated depreciation, book value and CBD provision at the end of the month. The depreciation rate is also shown here, whether on a fixed basis or UOM. On the lower half of the screen you can set up capital schedules and add or adjust the capital details.

After the capital details are set up, you must save the records before the acquisition details or schedules can be set up.

Acquisition/Disposal Details

- Enter the purchase price and any accumulated depreciation.
- These details set the book value to be depreciated in the schedules. The purchase order, delivery and warranty details of the equipment are also stored on this screen for reference by AMT.

Technical Details

- Specifications and general information for an item of equipment.
- Record information such as registration, chassis ID, wheelbase and tare weights. The current driver or operator of a vehicle can also tracked in this screen.

Capital Schedule

- After the base details are complete, you can create a capital schedule.
- The capital schedule is added from the capital detail screen by clicking **Add**.
- The screen displays reference data and contains fields to nominate various parameters for the capital schedule.
- The starting period defaults to the current month but should be changed to the equipment start month if no processing has been performed previously.
    - The capital adjustment/addition section allows any purchases or adjustments to the book value to be modified. There may be an additional capital item purchased for the equipment throughout its life and this adjusts the value to be depreciated.
    - The depreciation schedule can be given a nominated residual value to remain at the end of the term. The schedule has the option of monthly or by a UOM. The end point will change depending on whether the schedule is by date or UOM.
- The capital schedule grid displays a summary section to indicate the current position.



It is important for you to refresh the screens after changes are made so that the summary displays the modified information.

Plant Hire Administration

Set/Edit Plant Hire Rate

(Rpt_298) Plant Rate Analysis - Modelling

A breakdown of the expense elements and the associated costs. The report displays three different scenarios based on the simulated hours you have entered into the report parameters.

Tips

This report allows analysis between the usage per annum and the plat rate by expense element.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Today|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Simulation Usage 2, Simulation Usage 3, Currency

Simulation usage is the annual usage using the primary UOM.

MARC Administration

Equipment Contract Details

This function is only relevant to equipment configured as MARC, which must be selected from the Equipment Selector.

Upon running this function, you are presented with the MARC tab of the equipment. The screen is split into a number of sections, which are explained below.

Contract Parameters

- **Contract** is a look-up field which references a Contract using the Contract Administration function.
- **Record Test Time** should be selected if actual usages consumed during a downtime event are to be excluded from the equipment's billing forecast. Selecting this check box will result in a **Test Time** box appearing in every AMT Event for this equipment, where you can enter the number of hours to be dedicated from the equipment's expected contract billing for the month in which the Event is performed.
- **Billings Split %** is only used in select MARC Intercompany Analysis report.
- **Budget Margins** are used to calculate the parts, labour and miscellaneous Cost@Cost values for the linked budget projection.

Billings and Cost Summary

This section contains billing projections, and at least one linked billing schedule. A billing projection is a summary of all linked billing schedules. If a billing schedule's Start or End date do not align to the equipment's Start date/usage or End date/usage, the misaligned value will be displayed in **red** text.

The Cost@Sell and Cost@Cost values in the grid, for the Budget projection, are not escalated. Cost values for all other projection types are escalated.

The Re-link Actual Billings button triggers AMT to link existing actual billing records, which potentially are not linked to the equipment. Triggering this process is only required if chances to equipment model, serial number or registration counter were made, which may cause previously linked actual billings to be unlinked.

To create or edit a billing schedule, refer to Create an equipment billing projection.

Scope

Provides additional information related to the scope of the contract. The attributes listed are sourced from the Tech Fields defined against the Equipment Group to which the equipment is linked using the equipment's Model. The selections made are for information only.

Overheads

Overhead Navigator

Some maintenance contracts include overheads costs as part of the direct equipment maintenance costs and billings calculations. This is captured in the estimate-current projection's costs and billing schedule.

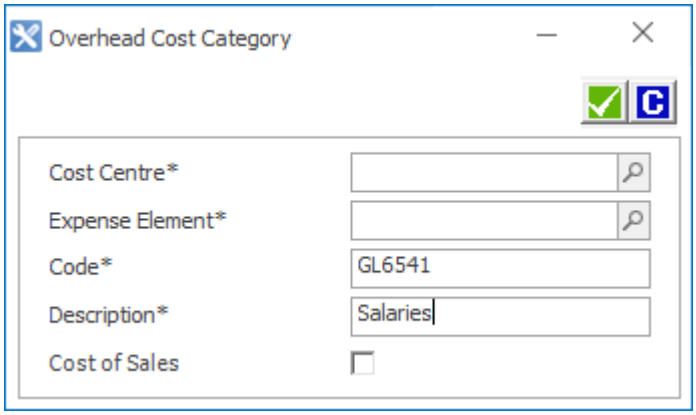
Other types of maintenance contracts classify contract overhead costs and revenue separately. The overhead costs are factored into the direct equipment maintenance costs, but the revenue is captured separately or vice-versa. The **Overheads** module of AMT handles this case; this module is a separate area where overhead costs and revenue can be estimated and managed for contracts

The Overheads module can also be used for non-contract overheads.

Before setting up overheads, it is important to validate:

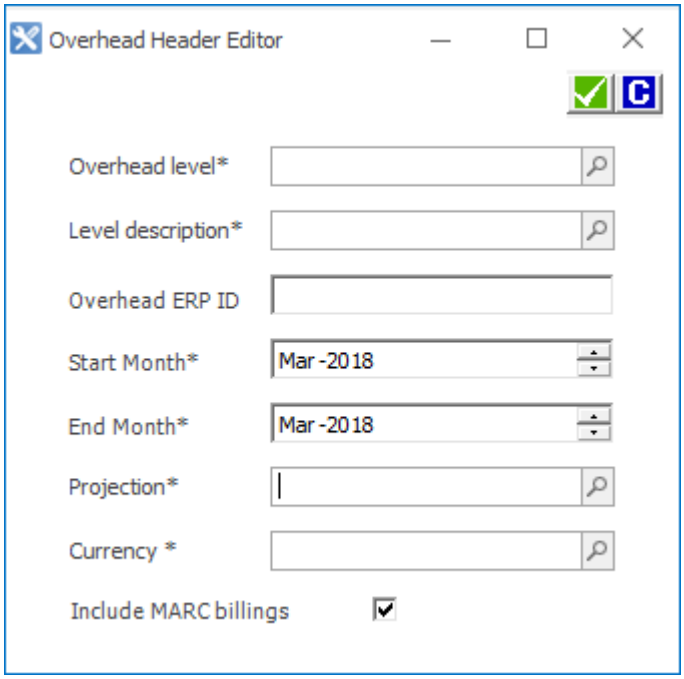
- How are overheads costed in the contract; are they part of the direct maintenance costs or separate?
- How will overhead costs will be recovered - as a separate fixed monthly fee or calculated into the utilisation billing rate?
- How are these costs and revenue being captured in the ERP system?
- How are these costs and revenue captured if AMT manages the equipment or contract when they become live?

Set up overheads by navigating to **System Administration** > **System Tables** > **Financial** > **Overhead Cost Category**, then clicking . The Overhead Cost Category dialog box displays.



The **Code** field is the overhead cost category definition. Select the **Cost of Sales** check box means to report overhead costs as cost of sales rather than expenses in some profit & loss reports.

To set up overhead costs and billings for a contract tender, use the **Overhead Header Editor** function. Navigate to **Setup & Admin** > **Overheads** > **Overhead Navigator**, then click .



Field	Description
Overhead level	The overhead level at which the overheads are costed and the overhead billing (management fee) is charged. For example, does the management fee cover all equipment under contract? Does it only apply to specific fleets or a specific branch or site?
Level description	Describes the overhead level chosen.
Overhead ERP ID	Not relevant for modelling overheads (only real overheads).
Start and End Month	Set to the period of time during which the overheads will be incurred.
Projection	Select the projection type.
Currency	Select the currency of the overheads.
Include MARC Billings	If a non-utilisation fee is charged to the customer to recover overhead costs, select this check box. If the check box is not selected, you can set up a schedule of overhead costs, but not management fees.

Populate the overhead's schedule with costs and billings on the **Overhead Header Editor**. You can either enter values directly on the grid, or copy and paste them from Excel.

Equipment Evaluation

(Rpt_307) Summary Analysis - Modelling

Provides summary details of the cost per UOM to run the equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, Cost Type, Contract, Costs, Cost Bearer, Discount Factor %, Revenue

 **+** indicates that the UOM selected is used as a basis for the equipment evaluation.

(Rpt_281) Cost per UOM - by Date - Modelling

Visually compares the expected hourly costs by machine. It shows the cumulative cost per UOM for the period of time selected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_283) Cost per UOM - by Usage - Modelling

Depicts the costs at intervals of the equipment usage (engine hours) throughout the term of its use.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_285) Cost per UOM - Comparison - Modelling

Visually compares the expected hourly costs by machine. It shows the cumulative cost per UOM for the period of time selected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_430) Cost per UOM per Interval - Modelling

Similar to the Cost Per Interval report except that this report shows costs as a cost per UOM (e.g. engine hours) rather than a total cost.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Labour Activity, Job Code, Costs

(Rpt_289) High Opportunity - Modelling

Identifies components due for change just short of the contract end, and calculates the new frequency required to ensure the opportunity can be realised.

Tips

Shows the last occurrence prior to equipment termination, the component life required and the cost benefit.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|MaxCompLife|,
|MinCompCost|,|LogoLocation|,|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,
|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,|EqpPlan|,|AnalyseBy|,|
ProjectionHeaderID|,|ProjectionHeader|,|Currency|

Report parameters

Default.

Additional filters

Task Type, System, Component Code, Max Life Used %, Cost Type, Sub System, Min.Comp.Cost, Min. Comp. Cur.

(Rpt_291) High Risk - Modelling

Identifies the risk components and calculates the life buffer available to prevent an additional component changeout prior to the equipment termination.

Tips

Shows the next projected occurrence after the equipment termination, the component life to avoid and the cost risk.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|MaxCompLife|,
|MinCompCost|,|LogoLocation|,|UserID|,|AutoLaunch|,|BranchID|,
|SiteID|,|FleetID|,|EqpPlanID|,|Branch|,|Site|,|Fleet|,
|EqpPlan|,|AnalyseBy|,|ProjectionHeaderID|,|ProjectionHeader|,|Currency|

Report parameters

Default.

Additional filters

Task Type, Cost Type, System, Sub System, Component Code, Min. Comp. Cost, Max Life Left %, Min. Comp.Cur.

(Rpt_306) RVIC Analysis - Modelling

Shows defined periods and components can be used as a basis for contract variation or termination, and can also be used for assessing contingent liabilities.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

System, Component Code, Task Type, Cost Bearer

(Rpt_304) RVIC - Chart - Modelling

Displays the movement of Residual Life over the period of the chart dates.

Tips

The chart does not display consolidated Totals for Fleet, only individual Equipment. This means that selections should consider the amount of data to be displayed in the chart. It is recommended that no more than 6-10 items of equipment be displayed.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, Sub System, Task Type, System, Component Code, Cost Bearer

Cost Analysis

(Rpt_277) Cost Analysis - "Life Cycle" - Modelling

Shows graphically, or in text, the costs for the entire plan term or to the month to date.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,|CostBearer|,|CostBearerID|,
|ProjectionTypes|,|EquipmentTypes|,|ProjHeaderID|,|ProjHeader|,
|UserID|,|AutoLaunch|,|BranchID|,|SiteID|,|FleetID|,|EqpPlanID|,
|Branch|,|Site|,|Fleet|,|EqpPlan|,|AnalyseBy|,|CurrencyID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance

Type, Task Mode, Costs, Parts/Labour/Misc, Currency

(Rpt_279) Cost per Interval - Modelling

Lets you nominate a period of hours to analyse what equipment costs will fall in that timeframe. Can be analysed using either the Budget or Current projection and by numerous other filters.

Tips

This report can assist management in understanding what costs have occurred during what period of time in the equipment life cycle. It can also assist in estimating other machine costs of a similar nature.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Item, Item

(Rpt_797) Asset Cost Worksheet - Modelling

Flexible cost analysis report that combines the Strategy Tasks costs set up in a Projection with the Other Operating Costs set up for that equipment.

Tips

Lets you evaluate and forecast all equipment-related costs - repair and maintenance costs, insurance, fuel, depreciation and so on.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Project, Region, Division, industry Group, Cost Responsibility, Component Cable, Equipment Class, Equipment Group, Eqp Category, Cost Bearer, Cost Centre, Expense Element, System, Sub System, Cost Type, Eqp Criticality, Manufacturer

(Rpt_342) Total Cost Worksheet - Modelling

Lets you analyse costs by several different categories within the budget period. You are able to group the report by three categories for viewing particular cost information.

Tips

This report will return results for both real and modelling, if there is a mixture of equipment in the fleet selected.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

CC Activity, CC Location, CC Responsibility, Cost Centre, Cost Bearer, Expense Element, Cost Category, Include, Costs

(Rpt_383) Financial Summary - Modelling

Shows the base financial details of the equipment.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Branch, Site, Fleet, Equipment, Business Group, Contract Type, Contract, Model

(Rpt_388) Forecast Worksheet - Modelling

Used for forecasting costs by expense group. It shows future and historical costs at sell, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|EquipmentTypes|,
|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|,|Feature_32|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

(Rpt_817) Forecast Worksheet - @Cost - Modelling

Used for Forecasting costs by expense group. Shows future and historical costs at cost, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|EquipmentTypes|,|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|,|Feature_32|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Workorder, Currency

(Rpt_821) Forecast Worksheet (Dealer Adjusted)

- Modelling

Used for forecasting costs by expense group and optional dealer market share. Shows future and historical costs at sell, categorised by up to five different areas.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|,|LanguageKey|,|ReportStyleKey|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Strategy Task Desc, Task Type, Component Code, Modifier Code, Cost Bearer, Task Counter, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Costs, Parts/Labour/Misc, Planned, Workorder, Currency

(Rpt_496) Budget Compliance - Modelling

Compares actual costs for the current month, financial year to date and projected for the current financial year to a budget.

Tips

Applicable to Modelling equipment only. Only budgets created in AMT's Budgeting module can be used in this report for comparison.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|EquipmentTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Task Counter, Work Group

(Rpt_497) Cost Trending - Modelling

Trends actual and projected costs of an equipment over time.

Tips

Applicable to modelling equipment only.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,
|EquipmentTypes|,|ProjHeaderID|,|ProjHeader|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Work Group, Cost Centre, Cost Category, Expense Element, Maintenance Type, Parts/Labour/Misc, Display, Currency

Availability

(Rpt_ 287) Forecast Availability - Modelling

This report relates to the Budgeting Module, and lets you define and set or edit availability parameters for Forecast the availability of the fleet over a period of time. It reports on the base downtime, monthly downtime and availability.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

None.

(Rpt_437) Duration Per Interval - Modelling

Shows duration estimates per interval based on the duration estimates within strategy tasks.

Tips

This report is used to assist in calculating availability Forecasts.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Labour Activity, Job Code, Labour Types

Labour Analysis

(Rpt_293) Labour Forecast - Hours - Modelling

Lets you analyse maintenance work by hours required. The report can be grouped by a maximum of five categories for more specific analysis.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity

(Rpt_295) Labour Forecast - Persons - Modelling

Lets you analyse maintenance work by persons required. The report can be grouped by a maximum of five categories for more specific analysis.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Task Type, Strategy Task, Maintenance Type, Eqp Cost Centre, Responsibility, Expense Element, Labour Activity, Productive Hours



+ indicates Average Productive Hours per Person per Month.

(Rpt_434) Labour Hours per Interval - Modelling

Shows labour hour requirements per interval based on the labour hour estimates within strategy tasks.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Eqp Cost Centre, Expense Element, Maintenance Type, Item, Item, Labour Type

(Rpt_297) Maintenance Ratio - Modelling

High level measure of efficiency of the direct Labour hours. The report only looks at productive labour hour, so needs to be viewed with the Labour Productivity report. It is calculated as Direct Labour Hours/Engine Hours.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|CostBearerID|,|AutoLaunch|,
|EqpPlanID|,|EqpPlan|, |DateChecked|,|StartDate|,|EndDate|,
|LabourType|,|ActualProjected|,|AnalyseByProjectionTypes|,|UserID|,|BranchID|,
|SiteID|,|FleetID|,|Branch|,|Site|,|Fleet|,|ProjectionHeaderID|,|ProjectionHeader|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Cost Bearer, Task Type, Strategy Task, Job Code, Labour, Show, Shop Job Codes

What-if

(Rpt_316) What-if - Life Cycle Analysis - Modelling

Allows analysis of costs between a budget or current projection and one created within the scenario manager.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Cost Selector, Component Code, Task Type, Strategy Task

(Rpt_317) What-if - Period Analysis - Modelling

Compares the Cost between two projections for a nominated time period.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Strategy Task, Task Type, Maintenance Type, Cost Centre, Responsibility, Expense Element, Labour Activity, Parts/Labour/Misc, Currency

(Rpt_314) What-if - Cost Analysis - Modelling

Compares the cost between two projections for a nominated time period.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Strategy Task, Task Type, Maintenance Type, Cost Centre, Responsibility, Expense Element, Labour Activity, Parts/Labour/Misc, Currency

(Rpt_315) What-if - Cost per UOM - Modelling

Compares the Cost per UOM between two projections selected by the user.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|QUOM|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

UOM, System, Sub System, Modifier Code, Escalation, Fleet, Equipment, Cost Type, Cost Bearer, Component Code, Strategy Task, Cost Selector

(Rpt_313) Variance Analysis Worksheet - Modelling

Lets you analyse between projections and group by a maximum of five categories to show variances in a period of time - default date is last month but this can be modified as required.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Component Code, Strategy Task, Task Type, Maintenance Type, Cost Bearer, Cost Centre, Responsibility, Expense Element, Labour Activity, Parts/Labour/Misc, Currency

Components

(Rpt_386) Changeout Schedule - Modelling

Provides details of next changeout scheduled to allow for parts supply, labour resources and production schedules to be managed effectively.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Task Type, System, Component Code, Cost Type, Sub System, Task Counter, Strategy Task, Part Classification, Part Rating, Primary Part No, Rebuild Location, Review Status, Sales Status

(Rpt_385) Changeout Analysis - Modelling

Provides a graphical view of the number of components to be changed in the time period required.

Tips

Date selector defaults to entire plan term, but can be modified to suit user requirements.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Feature_32|

Report parameters

Default.

Additional filters

System, Strategy Task, Modifier Code, Occurrence Type, Cost Type, Sub System, Task Type, Component Code, Part Classification, Part Rating, Primary Part No, Rebuild Location, Review Status, Sales Status.

(Rpt_387) Changeout Analysis - Detailed - Modelling

Provides a graphical view of the number of components to be changed in the time period required.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|UserID|,
|Feature_32|

Report parameters

Default.

Additional filters

System, Strategy Task, Strategy Task Desc, Modifier Code, Occurrence Type, Cost Type, Sub System, Task Type, Component Type, Region, Division, Cost Responsibility, Equipment Location, Part Classification, Part Rating, Primary Part No, Rebuild Location, Review Status, Sales Status

Marketing and Customer Support Editor

To make changes to any missing classifications:

- Navigate to **Product Support Marketing** > **PSM Administration** > **Marketing and Customer Support Editor**, filter for the required equipment and select the **Display Fields** filter.
- Double-click **Sales Responsibility**, **% of New Parts Purchase**, **Parts % of Purchase from Dealer**, **Labour % of Purchase from Dealer**, **Misc % of Purchase from Dealer**, **Marketing Group** and click **OK**.
- Once you have made all the selections, click **Apply Filters**, make the changes and click **Apply Changes**.

(Rpt_454) Parts Demand Analysis - Modelling

Forecasts expected parts demand based on probability of sale.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|EquipmentTypes|,|ProjHeaderID|, |ProjHeader|,|LogoLocation|,|UserID|,|Feature_32|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Cod, Cost Bearer, Part Type, Part Manufacturer, Primary Part No, Sales Resp., Part Classification, Part Rating, Rebuild Location, Review Status, Sales Status, Task Mode

Consumables

(Rpt_444) Consumables Forecast - Modelling

Shows Forecast Consumable Usage.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Part Type, Job Code, Show

MARC Analyses

MARC Summary

(Rpt_332) MARC Financial Summary - Modelling

Shows the base financial details of the equipment, including percentage of contract completion, historical and projected costs and billings.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|EquipmentTypes|,
|UserID|,|Currency

Report parameters

Default.

Additional filters

Overheads/Exp, Currency

(Rpt_181) Internal Summary

List of reports that have been attached to a particular agreement type that need to be printed out with an estimation generated from the Sales Tool module.

Tips

Internal

Initial parameters

SalesToolReport,|UserID|

Report parameters

Equipment

Additional filters

Equipment Class, Equipment Group, Manufacturer, Model, Fleet, Equipment Year, Serial Number, Eqp type, Eqp Criticality, Eqp Location, Eqp Category, Contract, Region, Division, Branch, Site

MARC Cash Flow

(Rpt_333) Cash Flow Chart - Modelling

Bar chart displaying Cash Inflow and Outflows with the resulting Period (Net) Cash Flows.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,
|AutoLaunch|,|FleetID|,|EqpPlanID|,|Fleet|,|EqpPlan|,
|AnalyseBy|,|ProjectionTypeID|,|ProjectionType|

Report parameters

Default.

Additional information

None.

(Rpt_334) Cash Flow Statement - Modelling

Provides the ability to project forward future cash flows to highlight any potential problems so that management can start planning for issues now. Cash in and cash out are significantly different for contracts.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|
AutoLaunch|,|FleetID|,|EqpPlanID|,|Fleet|,|EqpPlan|,|AnalyseBy|,
|ProjectionTypeID|,|ProjectionType|,|GroupBy|

Report parameters

Default.

Additional filters

Business Group, Contract, Equipment, Projection Type, Contract Type, Fleet, Model

(Rpt_335) Equipment Cash Flow Chart - Modelling

Graphical display of cash flow. Refer to the Cash Flow Statement report.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Business Group, Contract, Equipment, Contract Type, Fleet, Projection Type

MARC Cost at Cost

(Rpt_336) Costs at Cost - Modelling

Provides the ability for sharing historic and forecast costs using a variety of report filters.

Tips

Initially analyse costs by System, then drill down into sub systems and components to identify the major contributors to high costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,|CostBearer|,
|ProjectionTypes|,|EquipmentTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Region, Eqp Criticality, Equipment Group, Eqp Category, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Modifier Code, Divison, Equipment Class, Equipment Location, Parts/Labour/Misc, Currency

MARC Customer Reports

(Rpt_121) Agreement Summary

This report provides a summary of the equipment life cycle costs updated from the details entered into the scoping area of the Sales Tool and Agreement Manager.

Initial parameters

SalesToolReport,|UserID|

Report parameters

Default.

Additional filters

Equipment Class, Eqp Type, Region, Equipment Group, Eqp Criticality, Division, Manufacturer, Eqp Location, Branch, Model, Eqp Category, Site, Fleet, Contract, Equipment Year, Serial Number

(Rpt_450) Agreement Summary 2 - Modelling

Provides a summary of the equipment life cycle costs updated from the details entered into the scoping area of the Sales Tool/Agreement Manager.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Cost Bearer, Escalation, Availability Table, Labour Hours Table, Labour Rate Table, Lube Cost Table, Lube Quantity Table, Projection Details, Costs Equal Bilings, Comp-Job-Mod Desc, Cost Bearer for Lube only

(Rpt_179) Inclusions and Exclusions

Provides a listing of included and excluded tasks in any given agreement - relates to the Sales Tool module.

Initial parameters

SalesToolReport,|UserID|

Report parameters

Equipment

Additional filters

Equipment Class, Equipment Group, Manufacturer, Model, Fleet, Equipment Year, Serial Number, Eqp Type, Eqp Criticality, Eqp Location, Eqp Category, Contact, Region, Division, Branch, Site

(Rpt_220) Plant Rate Analysis

A breakdown of the expense elements and the associated costs. Displays three different scenarios based on the simulated hours you have entered into the report parameters.

Tips

This report allows analysis between the usage per annum and the plant rate by expense element.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Today|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Simulation Usage 2, Simulation Usage 3, Currency



Simulation usage is the annual usage using the primary UOM.

(Rpt_483) Projection Summary - Customer - Modelling

Gives an overview of the costs for the equipment by projected task and component sub system. Provides details of each projected task. Costs are adjusted by customer risk premium and discounts.

Tips

All costs are shown as cost at sell, but adjusted by customer risk premium and discounts.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|CostBearer|,|ProjectionTypes|,
|LogoLocation|,|UserID|

Report parameters

Default.

Additional filters

Cost Bearer, Escalation

(Rpt_486) Contract Header

This report generates the Contract Header for a Modelling equipment, where the Modelling equipment, where the Modelling equipment represents a contract developed in AMT by an OEM or Dealer.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|LogoLocation|,|UserID|

Report parameters

Default.

.

(Rpt_535) Cost Per Interval - Customer

This is the standard Cost Per Interval report, but in addition it contains logic which enables it to balance to provisions made to an estimate created using the Sales Tool.

Tips

This report is only relevant where the Sales Tool module in AMT is used.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|Enable_Cost_At_Cost|,
|CostBearer|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Cost Type, System, Sub System, Strategy Task, Task Type, Component Code, Modifier Code, Cost Bearer, Cost Responsibility, Cost Centre, Expense Element, Maintenance Type, Labour Activity, Job Code

Quoter

The Quoter module provides you with facilities to generate estimates or customer support agreements based on a general inquiry. Being a web-based tool, Quoter allows you to generate the estimates on the spot, through the use of a connected hand-held device. It can also be configured to request the most recent price agreement for any given customer from a bolted-on ERP, guaranteeing that all generated quotes are based on updated prices.

Configurations

Configuration	Integration	Description
Standard job repricing web service	Yes	Needs to be enabled if the ERP Standard Job repricing process is to work.
AMT source	Yes	Needs to be enabled if the ERP Standard Job repricing process is to work.

Launching Quoter

You can launch Quoter in two ways.

- Navigate to **Modelling** > **MARC Analyses** > **Quoter** > **Quoter Manager**.
- In your web browser enter the following URL: **https://<AMT Server Location>/Amt.Mobile/**

Quoter setup

AMT Mobile Quoter is a tool that gives the Sales Representative or Contract Manager the power to generate quotes for customers on the spot on a mobile device. Quoter still uses the data in AMT client to calculate the quotes, therefore, in order to ensure the quoted price is accurate and up to date, it is essential to maintain a healthy AMT database.

To generate a quote, two key pieces of information are used:

- **Equipment centrelines**: This determines all the maintenance work needed for each asset (AMT strategy task), what job needs to be done and when to do them.
- **Job prices**: This determines the cost of each job. By default, the standard job linked to each strategy task will have a default price. It is important to keep the centreline job default price up-to-date. Often an equipment dealer might have different price groups and different pricing for different customers. All price group and discount information needs to be checked regularly to ensure the quote Quoter produces is accurate and up-to-date.

Quoter Manager

Quoter Manager - Add Quote Jobs

Jobs are the core information that the quote is used for to display to your customer. The pricing and instructions for the quoted jobs are contained in this area. Items on the quote will report based on the job. This is the same type of information that is contained within a strategy task, a standard job or a work order.

They are several ways to add a job to a quote and they can be used in different situations to give you the flexibility required to create the quote you need.

The methods of adding jobs to a quote are by adding:

- A standard job.
- An occurrence from long term planning.
- Tasks from a project.
- Defect work orders from the backlog.

- Notifications.
- Existing quote jobs.
- A new job created manually.

Quoter Manager- Approve and Authorise Quote

Approving the quote indicates that you, or other users, have reviewed the content of the quote and give approval that it is correct and ready for authorisation. This can be performed by many users, or none. It is not mandatory, but can be incorporated into a quote management process.

Authorising the quote can only be performed by users with **Quote Authorisation** enabled and with a maximum authority greater than the value of the quote. When a quote has been authorised, the data contained within is locked so that it cannot be modified. Only an authorised quote can be used to generate the Quote Report for printing. If you are required to make a modification after the quote has been authorised, a new version must be created and then authorised again prior to being able to print the quote.

The audit trail can be used to identify who has approved and authorised a quote and the values contained within with the quote at the time of approval or authorisation.

Quoter Manager - View and Edit Quote Jobs

Viewing quote jobs lets you make any adjustments to the pricing and scope of the job to match the requirements for the Quote. The key concepts are:

- Assigning jobs to groups.
- Fixed vs variable pricing.
- In or out of scope.
- Flat rate pricing.
- Notes.
- Override pricing.
- Quantity of jobs.

Quoter Manager - Edit Quote Pricing

The quote pricing can be adjusted from the cost at sell of the jobs by applying discounts for parts, labour, or misc costs. This applies to the sum of the cost at sell. The discount does not affect the cost at cost. Editing the default gross margin will set the values of cost at cost used in the quote for when flat rate pricing is selected for a job. It follows the same formula outlined in the procedure for override pricing for a job.

Quoter Manager - Edit Quote Status

Quote Status is used to keep track of how different quotes are progressing in your process. The status allows you to more easily sort the quotes and can be applied as a filter in **Quote Manager**.

Quoter Manager - Print Quote

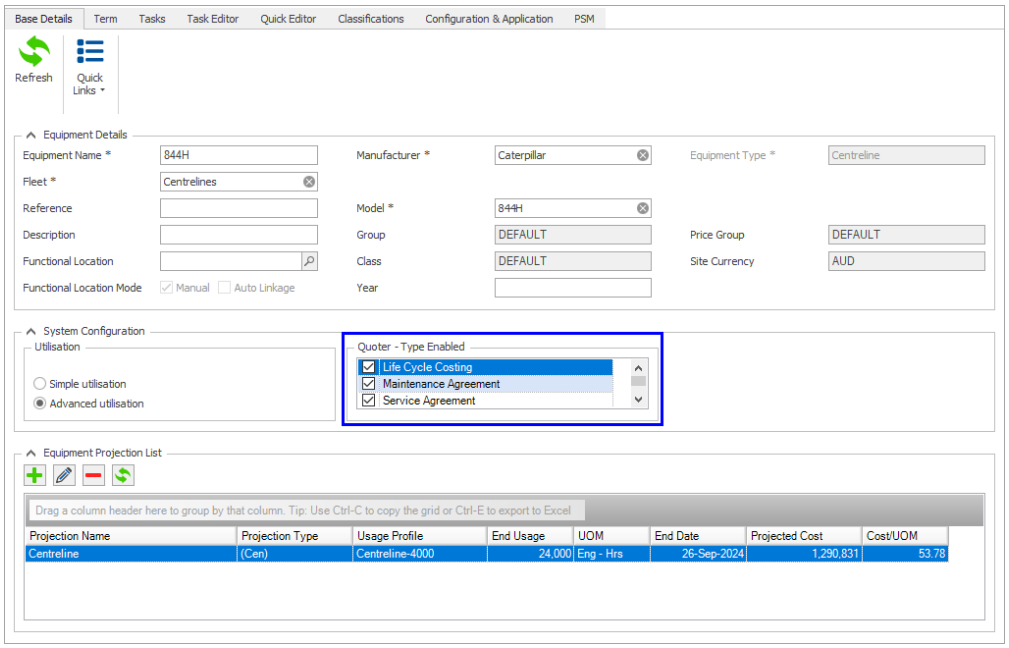
A quote can be printed once it has been authorised. This is how you can view all of the formatting and final value of the jobs in the quote once overall discounts have been applied.

Choosing to print the quote displays the **Quote Report** which you can then print, save or export.

The fields for **Printed By** and **Generating Comments** are stored in the quote header and are not printed to the quote.

Quoter centrelines

Centrelines are the starting point of every agreement. It is from a centreline that you select the basic details of a new quote, such as the model, maintenance tasks, basic additional costs, and so on. As a starting point, a single centreline can be used to create more than one agreement. If you have appropriate security access you can select which centrelines can be used to create Quoter agreements.

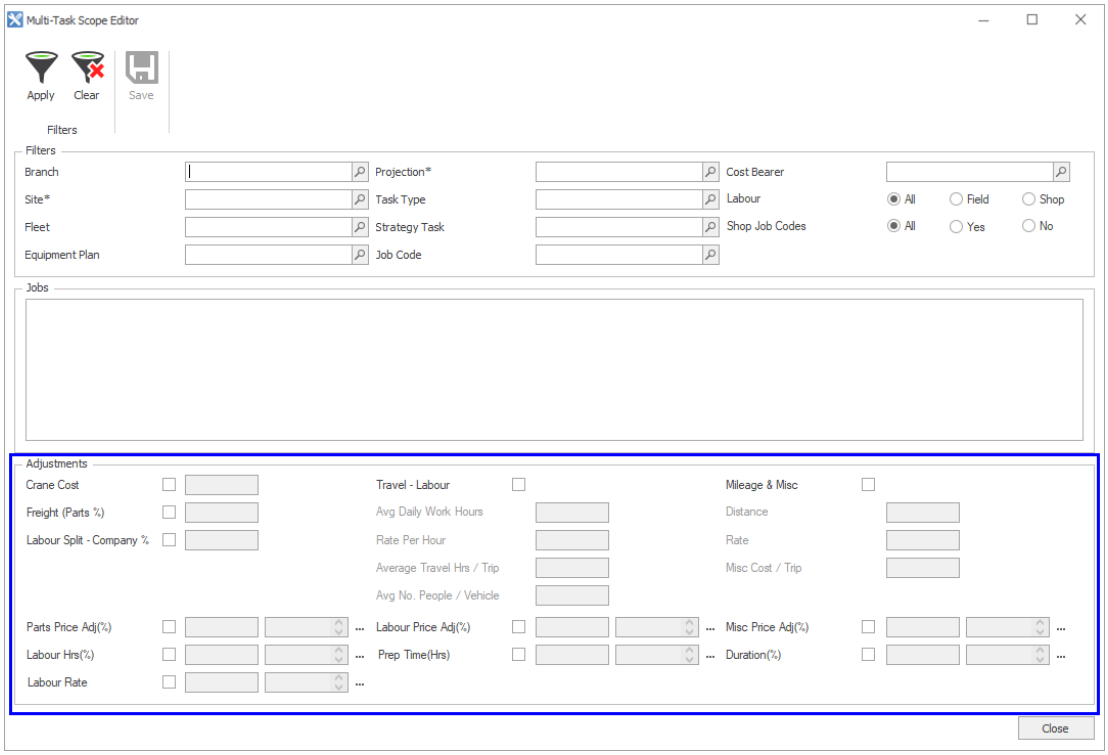


In addition to the centreline's strategy tasks, further costs details can be added to it to be used on a quote. You can update centrelines with freight, labour split and travel information (Audit Trail). The **Multi-Task Scope Editor** can then be used for bulk update of the freight, labour split and travel data.

Changes made using the Multi-Task Scope Editor are at the Strategy Task – Job level (a single strategy task can be linked to one or more jobs). Navigate to **Setup & Admin** > **Projections - Multi-editors** > **Multi-Task Scope Editor**.

Once you have applied and executed the appropriate filters, you should review the returned tasks prior to inputting the adjusted parameter on the right side of the screen. Once you have adjusted the required parameters, click **Apply Changes** . A notification displays detailing the number of records edited.

The main fields in the Multi-Task Scope Editor **Adjustment** panel are described below.



Field	Description
Crane Cost	Add the cost of craneage to miscellaneous.
Freights (% parts)	Apply freight cost as a percentage of parts cost. Parts Cost = Parts Cost + (Parts Cost x Freight).
Labour Split - Company %	Specifies what percentage of the labour hours will be performed by the dealer. This impacts labour costings for an agreement.
Travel - Labour	Enter parameters to calculate the travel labour hours and labour costs required to perform the strategy task job. This impacts labour cost.
Mileage & Misc	Enter parameters to calculate the misc. cost required to perform the strategy task job.
Parts Price Adj %	The percentage applied to adjust the strategy task job's parts cost, even if the strategy task job is linked to a standard job. Does not impact the standard job itself.
Labour Price Adj %	The percentage applied to adjust the strategy task job's labour cost, even if the strategy task job is linked to a standard job. Does not impact the standard job itself.
Misc. Price Adj %	The percentage applied to adjust the strategy task job's misc. cost, even if

Field	Description
	the strategy task job is linked to a standard job. Does not impact the standard job itself.
Labour Hrs Adj	The percentage applied to adjust the strategy task job's parts cost, even if the strategy task job is linked to a standard job. Does not impact the standard job itself.
Prep Time (Hrs)	Adds additional labour hours to the strategy task job to represent preparation time.
Duration %	Percentage applied to adjust the strategy task job's duration, even if the strategy task job is linked to a standard job.
Labour Rate	Labour rate applied to the strategy task job. If a labour rate has been entered then Labour Cost = (Labour Hours + Prep Time) x Labour Rate.

Available agreements

There are three types of agreement that can be generated from Quoter.

Maintenance Agreement Quote

The AMT Maintenance Agreement provides the ability to model different equipment or fleet scenarios made up of different types of models with different terms and annual average utilisation.

Service Agreement Quote

The Sales Tool Service Agreement is very similar to the Maintenance Agreement with the exception that:

- PMs are automatically selected in the scope section.
- The Internal Summary report points to a different report.

Life Cycle Costing Quote

Life Cycle costing is a methodology for calculating the whole cost of a system from inception to disposal. The AMT Life Cycle Cost function is designed to easily generate life cycle cost information quickly to customers based on a general inquiry. Costs include projected tasks, lubrication, tyres and component changeout activities.

Contract Tendering

Set Contract Tendering System Tables

Ensure the basic system tables are correctly set up. The Modelling system tables are listed below.

- Manufacturer (equipment manufacturers)
- Model (models of equipment made by specific manufacturers).
- Branch (use the AMT function **Branch Administration**).
- Industry (for example mining, oil & gas).
- Industry Group (for example mining - coal, mining - gold, rail, construction - heavy).
- Site (use the AMT function **Site Administration**).
- Customer (companies to which the tender will be submitted).
- Contract Type (for example, MARC, PM service, cost cap, HME).
- Parts Price Group (For example, a particular customer receives a 20% discount on the standard list price. In AMT, this is represented as a specific parts price group).

Site

Navigate to **System Administration** > **System Tables** > **Organisation** > **Site Administration**.

Set up the data for the Shift Start Time and to include The Employee Level Resourcing.

Customer

Navigate to **System Administration** > **System Tables** > **Other** > **Customer**.

A list of customers that identifies the customer who is operating the equipment.

Contract type

Navigate to **System Administration** > **System Tables** > **Other** > **Contract Type**.

Differentiate the types of maintenance contracts, for example, full maintenance contract, cost cap, parts cost guarantee.

Parts price group

Navigate to **System Administration** > **System Tables** > **Parts & Standard Jobs** > **Parts Price Group**.

Different groups of part prices. For example, a customer receives a 20% discount on the standard list price. In AMT, this arrangement is represented as a specific parts pricing group



For modelling equipment, branch, site and fleet may not refer to a geographic location in the typical sense. It could be a group of equipment types for a specific tender.

View / Edit Contract Tendering Workflow

System Administration

AMT System Administration Responsibilities

- Ensure properly trained and authorised staff are appointed as AMT administrators.
- Ensure that AMT users are properly authorised and licensed.
- Ensure that the System Administration settings of AMT are modified in a timely and appropriate manner.
- Ensure adequate security measures are implemented and maintained for AMT system settings .
- Provide an agreed process for people or staff wanting to change AMT system settings.
- Ensure changes to AMT system settings are communicated to the appropriate personnel within your organisation.
- Ensure the AMT database is correctly administered.
- Ensure licensing agreements are in place before adding new AMT users and machines to AMT.
- Maintain integrity of all AMT system tables under the **System Administration** menu list.

System Administration Procedural Guidelines

- Do not change the operation of the AMT server or SQL Server databases without the prior consent of the DBA.
- Do not connect new users of AMT to AMT unless they are covered by the License Agreement.
- Do not change system settings without authorisation from the AMT Administrator's next level of supervision.
- Consider the consequences of the proposed change, on users other than those requesting the change, should be considered before the change is implemented.
- Communicate the changes to staff as soon as possible after the change of settings, preferably before.
- Ensure procedures are kept up-to-date and changes to the procedure are communicated to all AMT users.

User Profiles

User Accounts

System Administration > User Profiles > User Accounts

Whenever a new user is added or an existing user's permission needs to be modified, the system administrator maintains groups, user names, passwords and group permissions.

This feature links a user with a **Security Group**. A user must belong to a specified security group. The permissions for each group are set in group permissions. To open this dialog box, double-click **User Accounts**. A user account requires the following settings:

- Login ID (user name).
- Full name.

Other settings include:

- Further user details.
- Restriction for which branches or sites the user can access in **Equipment Statistics**, and in the **Equipment List**. This does not restrict the branches or sites available to the user when running reports or functions in the main AMT client application.
- Work Order and Quote Authorisations.
- Budget and Budget Status Restrictions.

For Windows authentication in the **User Account** screen, enter the user name in the **Login name** field in the format domain\username, for example RPM\John.Doe. For the Web Application, it is necessary in IIS to switch off anonymous authentication for all AMT applications and switch on integrated Windows authentication.

In the User Accounts screen, click  to **Add** a new user, and  to **Edit** an existing user.

Budget cost centre restrictions and Budget status permissions

These features let you set whether a user's ability to edit or influence a budget should include cost centres.

When a budget is created by a user and includes equipment with task costs for a cost centre for which the user does not have responsibility (that is, no access), the user will be provided with the following message: *Equipment selected include cost centre costs for cost centres you can't access. These costs are excluded. Please refer to your management for budgetary access requirements.*

A user may have one or more budget statuses depending on their organisational role. When a user account is created, the default status is **No Access**. If Access is granted, then the user account would then select one or more codes for the Edit Status from the Budget Status system table.

Group Administration

This feature shows the AMT access permission for each security group. The list of available groups, and the type of access each group has to specific areas within AMT, need to be defined before setting the permissions and users. To manage user groups, search and select the function **Group Administration**.

Change Password

You can change your password by clicking **Change Password** in the login window or search and select the **Change Password** function.

There is no restriction on the length, case or type of password that can be entered.

Delete this text and replace it with your own content.

Message Group Administration

Condition Based Maintenance (CBM) module can generate internal AMT messages, like email. The messages are distributed in AMT to message groups; a message group contains specific AMT users. Use the **Message Group Administration** function to define the name of the message groups and which AMT users belong to each group. An AMT user can belong to multiple message groups.

The Budget and Forecast module can generate actual emails to AMT users within a message group. Using the same **Message Group Administration** function, define the name of the message groups and which AMT users belong to each group. An AMT user can belong to multiple message groups. Then assign a budget or forecast to the message group.

System Tables

System Administration > System Tables

The purpose of a system table and its contents are defined during the initial set up and installation of AMT. From that point forward, the AMT system administrator is usually the only person who has access to directly modify the tables.

Interfaces from external systems to AMT can indirectly modify AMT system tables. For example, if AMT receives an equipment interface file that contains a Region value (that does not already exist in AMT's Region system table), the interface process adds the new region to the system table to permit the interface to run successfully. The system tables that are exceptions to this rule are those where the system table's uniqueness is dependent on another field. For example, AMT cannot add a new Work Location via an interface because a work location has to be associated with a site. The only exception to this rule is the Component Code Hierarchy where new codes are added to the Subsystem flagged as the default.



A **Default** flag available in some system tables lets the System Administrator indicate which option is the default. In normal operation, you can make different selections if the default is not applicable.

When setting up the Asset Condition, Equipment Criticality and Operation Criticality system tables make sure you create the **Default** category first.



A **Budget** or **Budget Code** flag is used in some system tables to define which component codes can be used in the creation of a Strategy Task.

Organisation

These tables are used to define the company's organisation in AMT:

- Company
- Region
- Division
- Branch Administration
- Site Administration
- Fleet Administration
- Business Group
- Industry
- Industry Group
- Work Location
- Work Group
- Employee

Company Editor

Name of the company that has bought AMT, for example, BHP. It is the highest level of the equipment hierarchy.

Only relevant for reporting purposes.

Region

Lists the countries and regions in which equipment can be located, for example, Australasia, Eastern Europe, South-east Asia. Some financial and component forecasting reports can be run per region.

Division

Lists the areas or divisions within a company structure to which an asset can belong, for example, Mining, Civil Construction, Heavy Construction, Local and State Government, Plant Group. Some financial and component forecasting reports can be run per Division.

Branch Administration

Branches are the levels below the Company (this is the highest level in the hierarchy in AMT). Use the Branch Administration function to add, edit and delete branches in AMT.

Field	Description
Branch Description	Name of the branch.
Branch Code	Identifies where costs for assets are captured in the accounting system.
WO Suffix	Work Order Settlement code suffix is used when AMT opens internal work order settlements automatically for an equipment under this branch.

Fleet Administration

A collection of equipment on a site, typically grouped by their similarities. For example, if a mine site has trucks and excavators, all the trucks might be grouped into a fleet called Haul Trucks and the excavators into a fleet called Hydraulic Shovels.

Fleet is one element of the equipment hierarchy (Branch > Site > Fleet > Equipment). A fleet can only be linked to one site, but it can be linked to many equipment.

Site Administration

Sites are the levels below branches. This system table is used to set up sites and related details, including setting up the Shift Start Time and enabling Employee Level Resourcing.

Field	Description
Site	Name of the site.
Branch	Branch to which the site belongs.
Site Code, Address, Suburb, State	Physical location information for the site.
Resource Scheduling	Select either Work Group Scheduling or Labour Activity and Work Group Scheduling.

Field	Description
Enable Employee Level Resourcing	Labour resource levelling is defaulted by site to be at the work group level. If a site wants to plan labour at the employee level, this check box must be selected.
Shift Start Time	The time when the first shift of the standard working day commences.  For maintenance contract sites, the shift start time must align to the dispatch system's shift start time.
Week Start Day	The first day of the week.
Currency	The financial currency used by the site.

Business Group

Designates the type of business specific to the asset being registered, for example Mining, Forestry, Construction.

This should correlate with the Industry and Industry Group system tables.

Industry

Overview

Description: The type of industry in which the asset operates, for example, Mining, Oil and gas, Forestry.

Relevance: Required for identifying different industry groups.

Special Fields

None.

Industry Group

Sub-sets of an Industry, for example, if the Industry is Mining, Industry Groups could be Coal Mining, Gold Mining, and so on.

At a high-level, this classification allows you to run certain reports in order to gauge how assets perform in different applications, identified by different industries.

Work Location

Locations at which maintenance is performed, for example, Workshop Bay 1, Field, Service Bay.

By nominating where maintenance events will take place, you can run reports to see if more than one event is planned to occur in the same location. You can also inform staff where they need to go to perform the jobs.

A work location is specific to a site.

Work Group

Contains work groups hat can be assigned tasks to execute; for example, field service crew, welding crew.

From a planning perspective, you can nominate the number of available work hours per week per work group, not per employee. You can also assign certain work groups to the execution of certain tasks.

Special Fields	Description
Default	Each site must have only one default work group, otherwise AMT will not successfully create work orders in the Planning module from Long-term Planning.
Site	A work group is specific to a site in AMT. If a work group spans more than one AMT site, then the same work group can be created for each site.
Expired	If a work group is no longer required, that is, no longer available to you, then select the **Expired** check box. Historical records will remain in the database for reporting purposes.
Target Allocation %	Used to assess whether work group resources are over or under allocated in work plans generated in the Planning module. Typically this parameter is set to 80%.

Employee

Used to nominate the high-level employee groups on particular sites, both internal employees and external contractors.

On the Employee system table, click  to open the **Employee** screen.

Fields	Description
Branch	The branch where the employee is located.
AMT User Name	If applicable, an employee can be linked to an AMT User.
Status	The status chosen affects the forecast availability of the person; options are Available, Training, Annual Leave or Sick Leave.
Roster	Select this check box to indicate the employee has been created for the specific purpose of creating rosters for other employees. The person will not be selectable for work and will not show in reports. However, the roster employee's working time can be copied to other employees to create the initial roster.
Terminated	Select this check box to indicate that the person is no longer employed.
Product Support Sales Rep	Sales person responsible for parts, labour and miscellaneous sales. This is relevant to OEMs and dealers who use AMT.
Product Support	Group of product support sales representatives.

Fields	Description
Sales Group	
Account Manager	Person responsible for a sales account. This is relevant to OEMs and dealers who use AMT.
Account Manager Group	A particular group of account managers. This is relevant to OEMs and dealers who use AMT.
Prime Product Sales Person	Sales person responsible for equipment sales. This is relevant to OEMs and dealers who use AMT.
Prime Product Sales Group	A group of prime product sales persons.

On the Employee screen., click  to open the **Employee Sites** screen.

Fields	Description
Site	An Employee can be linked to more than one site, but only one Site can be the primary Site for the employee. This is because Employee – Work Order labour scheduling is determined based on the Primary Site of the Employee.
Work Group	The work group the employee belongs to, for example, workshop. The selection depends on which Site is selected.
Labour Activity	An Employee can be linked to one or more labour activities, but only one Labour Activity is the default. This is because Employee and Work Order labour scheduling is determined based on the Primary Site of the Employee.
Default Labour Activity	An employee may have different labour activities (qualifications) at different sites, for example, they may be an ELEC Class B at Site 214 and ELEC at Site 435. An employee can only perform work and review assets on sites to which they have permission and for which they are qualified.

Financial

These tables are for configuring the financial reporting parameters of AMT.

- Cost Centre
- Expense Element
- Cost Responsibility
- Currency
- Entry Distribution Code
- Escalation

- Exchange Rate
- Financial Periods
- User Defined Financial Periods
- Overhead Cost Category
- Set Margin
- Sales Status

Expense Element

The Expense Element system table is used to group costs for financial reporting. The expense elements reflect the different categories of costs, as set up in the company's accounting system. A default expense element is set for each equipment, however specific expense elements can be set for parts, labour and other miscellaneous items within each strategy task, for example tooling, emulsion, steel, fuel.

The expense element is one of the key codes used to produce and analyse accounting budgets. Your company's finance group should define these codes.

	Description
Depreciation	Indicates which expense elements are used for the parent asset, such as equipment depreciation, which is used by the Asset Ledger module of AMT.
Ownership Cost	Indicates which expense elements are used for parent asset ownership costs, for example insurance, finance and registration.
CBD	Indicates which expense elements are used for component based depreciation on child assets, such as components, which is used by the Asset Register module of AMT.
Global Code	Links to the Global Expense Element system table.
Standard Job Parts Default	Sets the expense element of a part item on a standard job to this expense element unless the part already has an expense element assigned using Parts Manager, or via a standard job imported from an external system.
Standard Job Labour Default	Sets the expense element of a labour item on a standard job to this expense element unless the labour activity already has an expense element defined (refer to the Labour Activity system table, or via standard job imported from an external system.
Standard Job Misc Default	Sets the expense element of a miscellaneous item on a standard job to this expense element unless the Miscellaneous Category already has an expense element defined, or via a standard job imported from an external system.
Inactive	If flagged as inactive, the expense element cannot be selected by a user on AMT screens. Existing records linked to an inactive expense changed.

Cost Responsibility

A high-level system table that defines the financial responsibility for equipment, for example, Site A Production. It is set for each equipment in the Equipment Plan Editor.

Often this is a higher grouping of cost centres. In AMT it is used primarily for financial and budget reporting.

Currency

Lists all the currencies that could be used by an organisation for transactions. These currencies are set up in AMT during installation and should only be changed in consultation with RPM. Enter a symbol to denote the currency, for example $, and default labour rate per hour for that currency.

AMT can process multiple currencies because some organisations incur costs and make sales in different currencies. These currencies are all converted back to the primary currency for reporting.

Special Field	Description
Default Labour Rate	Baseline labour rate which is applied to the Labour Rate system table when new branches are created in AMT.  The first currency in the currency table is the primary currency for the database. Therefore, all other currencies are converted using exchange rates for reporting purposes. Once the primary currency is set, it cannot be changed. Deleting an unused currency causes all related exchange rate data to be deleted as well.

Entry Distribution Code

Distribution codes allow the APIs to post plant hire costs to the ERP system's general ledger using a journal transaction.

Escalation

This system table allows you to enter cost escalation into AMT on a specific date for labour, miscellaneous, and parts costs. Parts escalations are listed per Manufacturer.

Escalation refers to price changes over time and is sometimes referred to as indexation. If a budget projection is priced in Year 1, for a contract spanning several years, actual costs could be higher than the budget pricing, simply as a function of inflation over time.

To harmonise comparisons between budget and current projections, escalation is applied to budget prices to adjust these values to a current levels.

 Future projections of both current and budget are made at today's escalated values.

Escalation - Equipment costs

Cost escalations are entered into the Escalation system table. Escalation percentage can be entered for any date by labour, miscellaneous and parts supplier categories. Escalations are applied to all equipment – you cannot escalate costs for one fleet differently to another. How escalations are applied in AMT depends on the projection and the analysis being performed.

As a general rule, for all reports in AMT, actual costs are never escalated. This includes all financial reports. The following reports are exceptions and escalate actual costs to today's value (these are reports that show average cost analysis and benchmarking).

- Average Work Order Costs
- Fleet Component Performance Summary
- Strategy Task Analyser
- Technical Performance Analysis
- Technical Performance Analysis Chart
- Technical Performance Details

In some cases you have the option to view actual costs as escalated (to today's values) or unescalated.

All future costs from the current projection use costs escalated to today's values. Strategy tasks have a pricing date nominated. Strategy task costs are escalated from this pricing date to today.

Example

- On 1 January 2019 you enter a cost of $1,000 for a strategy task . If there is a 5% escalation on 1 March 2019 and you run a forecast report on 1 June 2019, then the forecast will use $1,050 for the forecast cost for that strategy task.

Where a strategy task is linked to a standard job, it is assumed the pricing is always up to date and no escalation is applied. All future costs are forecast at today's values. It is incorrect to enter future escalations into AMT.

The costs entered into the budget projection have a pricing date that sets the point from which they are escalated. For the budget projection, the pricing date is set in the projection header screen and applies to all strategy tasks. The general principal is that budget costs are escalated over time. This means that the costs are escalated to the point they were budgeted to occur.

Example

For the engine on Truck 1 assume:

- Budget pricing date = 1 January 2018.
- Term = 25,000 hours.
- Engine Strategy Task Cost = $100,000 at frequency of 10,000 hours.
- Truck reached 10,000 on 27 October 2019.
- Escalation history:
  - 5% on 1 December 2018
  - 3% on 1 July 2019.
  - 4% on 1 January 2020.
- If you run the Life Cycle Cost report on 1 May 2020 it will show:
  - Budget costs to date = $108,150 (5% + 3% cumulative).
  - Budget future costs = $112, 476 (5% + 3% + 4% cumulative).

All future costs from a budget projection use costs escalated to today's values. These costs are entered into strategy tasks and the pricing date is set in the projection header. Therefore, all strategy tasks in a budget projection have the same pricing date.

A strategy task in a budget projection may be linked to a standard job, but the linked standard job will not update the cost in the budget projection's strategy tasks. This ensures the budget never changes.

All future costs are forecast at today's values. It is incorrect to enter future escalations into AMT.

Apply cost escalations

When the Contract Manager requests from management any recent updates to price increases from external suppliers (for example, different parts suppliers) and internal labour rate increases, these increases are inserted into the AMT System Table - Escalation.

Navigate to **System Administration** > **System Tables** > **Financial** > **Escalation**.

Click  to open the **Escalation Index** dialog box.



Saving your changes triggers AMT to immediately recalculate all current projections, which can take some time to complete depending on the number of strategy tasks in the AMT database that are not linked to standard jobs.

- The dates entered into this table must reflect when the cost escalations were officially applied internally within the organisation.
- Values can be positive (escalation increases), or negative (escalation decreases).

Cost Centre

A cost centre typically refers to an organisational or departmental entity of a company, and exists in the company's accounting system. For example, a branch or a site might be considered a cost centre. A default cost centre is applied to each equipment, however specific cost centres can be set for each strategy task.

This is one of the key codes used for producing and analysing accounting budgets. The finance group in the company should define these codes.

Special Fields	Description
CC Activity	Cost centre's activity code or description which is used for reporting in the Budget and Forecasting module.
CC Location	Cost centre's location code or description which is used for reporting in the Budget and Forecasting module.
CC Responsibility	Cost centre's responsibility code or description which is used for reporting in the Budget and Forecasting module.

Cost escalations

Cost escalation is an increase or decrease in the cost of parts, labour and miscellaneous costs over time.

Cost escalations are entered into the Escalation system table. Escalation indexes are created, and there is a default index for each of parts, labour and misc. These indexes can be applied to parts, labour or misc. at the strategy task level. The default parts, labour and misc. indexes are used where no index is applied.

Escalation percentage can be entered for any date for these Escalation Indexes.

Escalations are applied to all asset types. How escalations are applied in AMT depends on the projection and the analyses being performed.

Current projections - actual costs

As a general rule, for all reports in AMT, actual costs are never escalated. This includes all financial reports.

However, there are exceptions where the actual costs are escalated to today's value. This occurs when average cost analyses and benchmarking are performed. The forms and reports where this occurs are:

- Average work order costs.
- Fleet component performance summary.
- Strategy task analyser.
- Technical performance analysis.
- Technical performance analysis chart.
- Technical performance details.

In some other forms, you can view actual costs as escalated (to today's values), or unescalated.

Current projections - future costs

All future costs on the current projection use costs escalated to today's values. These costs are entered into strategy tasks and have a pricing date nominated in the strategy task. These costs are escalated from this date to today.

Example

On 1 January 2019, you enter a cost of $1,000 for a strategy task. If there is a 5% escalation on 1 March 2019, and you run a forecast report on 1 June 2019, the report uses $1,050 for the forecast cost for that strategy task.

Where a strategy task is linked to a standard job, AMT assumes the standard job pricing is always up-to-date and no escalation is applied.

All future costs are forecast at today's values. Do not enter future escalations into AMT.

Budget projections - costs to date

The costs entered into the budget projection have a pricing date that sets the point from which they are escalated. For the budget projection, the pricing date is set in the **Projection Header** form and applies to all strategy tasks.

The general principle is that budget costs are escalated over time. This means that the costs are escalated to the point that they were budgeted to occur.

Example

For the engine on truck 1 assume:

- Budget pricing date = 1 January 2016.
- Term = 25,000 hours.
- Engine strategy task: cost = $100,000 at a frequency of 10,000 hours.
- The truck reached 10,000 hours on 27 October 2017.
- Escalation history.
    - 5% on 1 December 2016.
    - 3% on 1 July 2017.
    - 4% on 1 Jan 2018.

If you run the life-cycle cost report on 1 May 2018, it will show:

- Budget costs to date: $108,15 (5% + 3% cumulative)
- Budget future costs: $112,476 (5% + 3% + 4% cumulative)

However, there are exceptions where the actual costs are escalated to today's value - when average cost analyses and benchmarking are performed. The forms and reports where this occurs are:

- Strategy Task Analyser
- Fleet Component Performance Summary

In some other forms you have the option to view actual costs as escalated (to today's values) or unescalated.

Budget projections - future costs

All future costs from the budget projection use costs escalated to today's values. These costs are entered into strategy tasks and have a pricing date set in the projection header.

See example above for Budget Project – costs to date.

A strategy task in the budget may reference a standard job, but it is not linked to the standard job. Therefore, the standard job will not update the cost in the strategy task. This ensures the budget never changes.

All future costs are forecast at today's values. Do not enter future escalations into AMT.

Alternate projection - actual costs and future costs

The same rules apply as for the current projection.

Archive projection

Archive projections are created from current, budget or alternate projections. Once created, the archive projection never changes. Therefore, it reflects the rules that apply to their source projection when it was created.

Centreline projections

Centreline projections only forecast future events and costs. There is no historical element.

All costs in the centreline projection use costs escalated to today's values. These costs are entered into strategy tasks and have a pricing date nominated in the strategy task. These costs are escalated from the pricing date to today.

Example

On 1 January 2018, you enter a cost of $1,000 for a strategy task. If there is a 5% escalation on 1 March 2018, and you run a forecast report on 1 June 2018, $1,050 is the forecast cost for that strategy task.

Where a strategy task is linked to a standard job, the program assumes the standard job pricing is always up-to-date and no escalation is applied.

All future costs are forecast at today's values. Do not enter future escalations into AMT.

Modelling Estimates (current and drafts)

The same rules apply as for budget projections.

Cost Reports

A high-level cost review of actual costs versus the annual projected costs provides an overview of whether there are sufficient strategy tasks and whether they need to be optimised. Consistent significant variances will provide triggers to investigate the causes further.

- Cost vs Budget

  Navigate to **Strategy Optimisation** > **Life Cycle Analysis** > Budget vs Current Projection Details. This report compares the actual performance to the original budget.

- Projected Cost Variance (PCV).

  Navigate to **Reporting & Analyses** > **Costs** >**Costs** > PCV Chart. This report provides the ability for analysing historic and forecast costs.

Exchange Rate

The exchange rate of a nominated currency expressed as one unit of the primary currency, for example the New Zealand Dollar to Australian Dollar exchange rate =1.13, where the Australian Dollar is the primary currency.

AMT forecasts future costs and sales, which come from projections, typically current projections. The costs and sales in projections can be set up with different currencies; however they have to be converted to be reported, generally, in the equipment's site currency. Therefore, the exchange rate tables provide the current exchange rates at which to convert the costs or sales in projections to the primary currency to report future expected costs or sales.

Financial Periods

The company's accounting periods. If the financial periods differ from calendar periods, a translation process converts financial year and month records to Calendar Year and Month, in the format YYYYMM. For example, if your organisation's financial year ends in June 2019, and an accounting record for a work order settlement signifies the invoice period is 201908, then this will translate to Feb 2019 in Calendar years (201902).

Certain financial reports require you to select the accounting period rather than the calendar period.



This system table should not be changed without consulting RPM.

Reporting Calendar

This system table is for defining custom reporting calendars.

User Defined Financial Periods

User defined financial periods in AMT will eventually allow you to define financial periods that don't align with the first and last days of a calendar month. Currently in AMT, you can only align user defined financial periods with the first and last day of the calendar month.



Any number of financial periods can be created.

A user defined financial period status can be:

- **Current**
  Identifies the period that is current for depreciation calculation purposes. The current period should generally align with the actual current month. Asset-related journals are posted to this period.
- **Failure**
  A period that will become current in the future for depreciation purposes. This period does not accept asset-related journals.
- **Closed**
  A period that is in the past from a depreciation perspective. This period does not accep asset-related journals.
- **Open**
  A prior period (that is, one that has been closed) that has been reopened. An open period accepts asset-related journals while open.



During typical operation of AMT there should only be one period with a status of current.

Overheads

Some maintenance contracts include overheads costs as part of the direct equipment maintenance costs and billings calculations. This is captured in the estimate-current projection's costs and billing schedule.

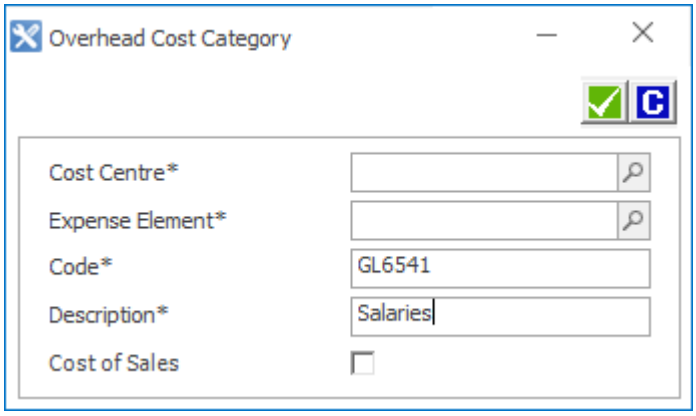
Other types of maintenance contracts classify contract overhead costs and revenue separately. The overhead costs are factored into the direct equipment maintenance costs, but the revenue is captured separately or vice-versa. The **Overheads** module of AMT handles this case; this module is a separate area where overhead costs and revenue can be estimated and managed for contracts

The Overheads module can also be used for non-contract overheads.

Before setting up overheads, it is important to validate:

- How are overheads costed in the contract; are they part of the direct maintenance costs or separate?
- How will overhead costs will be recovered - as a separate fixed monthly fee or calculated into the utilisation billing rate?
- How are these costs and revenue being captured in the ERP system?
- How are these costs and revenue captured if AMT manages the equipment or contract when they become live?

Set up overheads by navigating to **System Administration** > **System Tables** > **Financial** > **Overhead Cost Category**, then clicking . The Overhead Cost Category dialog box displays.



The **Code** field is the overhead cost category definition. Select the **Cost of Sales** check box to report overhead costs as cost of sales rather than expenses in some profit & loss reports.

When a dealer defines and calculates a contract's overhead costs, there are numerous expense categories (accounts) considered, sometimes detailed such as printer consumables. AMT is not concerned with that degree of detail. Like equipment, AMT is concerned with tracking the significant cost items (for example, wages/salaries, accommodation, vehicles, travel) with minor expenses grouped into general categories or as Other Expenses.

Overhead Cost Category

Categories for the different types of overhead costs. The category can be related to a specific combination of cost centre and expense element. For example, Mining Maintenance (Overhead Cost Category) - Hauling (Cost Centre) - Tooling (Expense Element)

The categories entered in this table reflect the type of overheads to be recorded and reported.

Special Field	Description
Cost Centre	The accounting group of the overhead cost, for example branch or site.
Expense Element	The category of the cost.
Code	The overhead cost category definition.
Description	Description of the cost category.
Cost of Sales	If an overhead cost is flagged as being COS (Cost of Sales), rather than as an operational expense, this cost will be included in the Cost of Sales section of certain profit and loss (P&L) financial reports instead of appearing in the expenses section. Overhead categories should only be flagged as COS if these costs are reported in the cost of sales section of the company's own internal profit and loss reports.

Set Margin

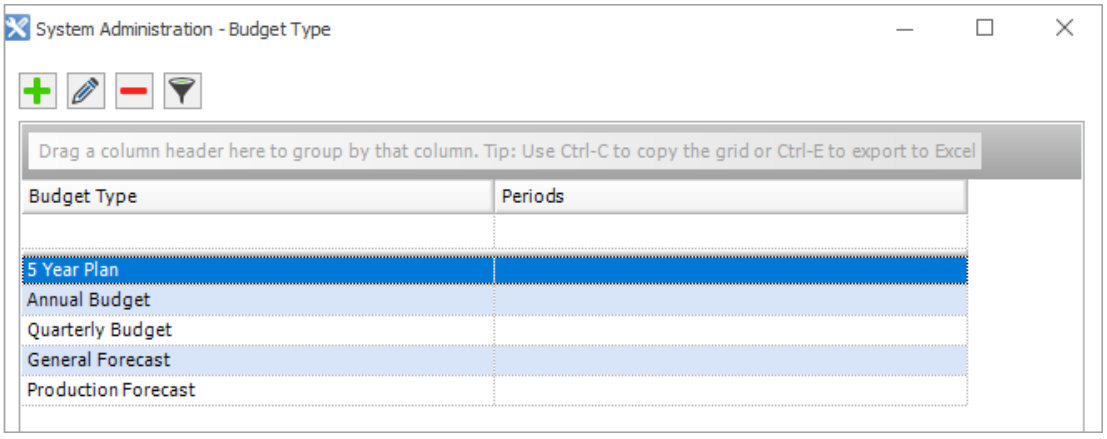
This table allows the setting of default margin percentages to enable the automatic calculation of cost from sell values for future projected costs. Parts margins can be set at individual system levels within any fleet. Labour and miscellaneous can be set at any fleet level.

Sales Status

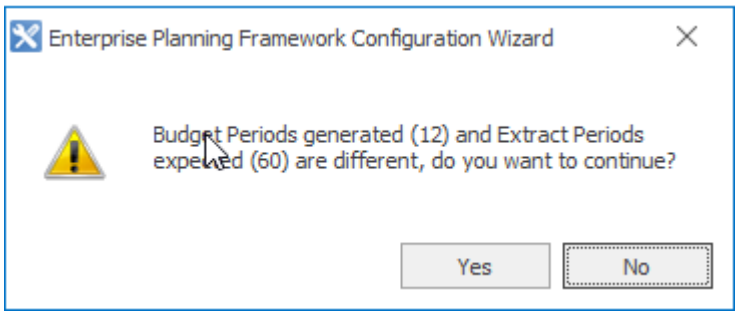
Defines the status values for sales, for example confident, sale confirmed, lost.

Budget Type

This system table gives the user (preferably a system administrator or capable user), an AMT Client method to add new, update or delete budget types. This previously required an SQL database update.



A user can associate a number of periods with a budget which is used as part of the Publish process. The number of periods associated with a budget will be compared with the number of periods of the associated Data Set Definition (DSD) during the EPF Configuration Wizard process. If the DSD Periods (what the export expects) do not equal the Budget Type periods (what AMT is generating), the user will be given a warning message.



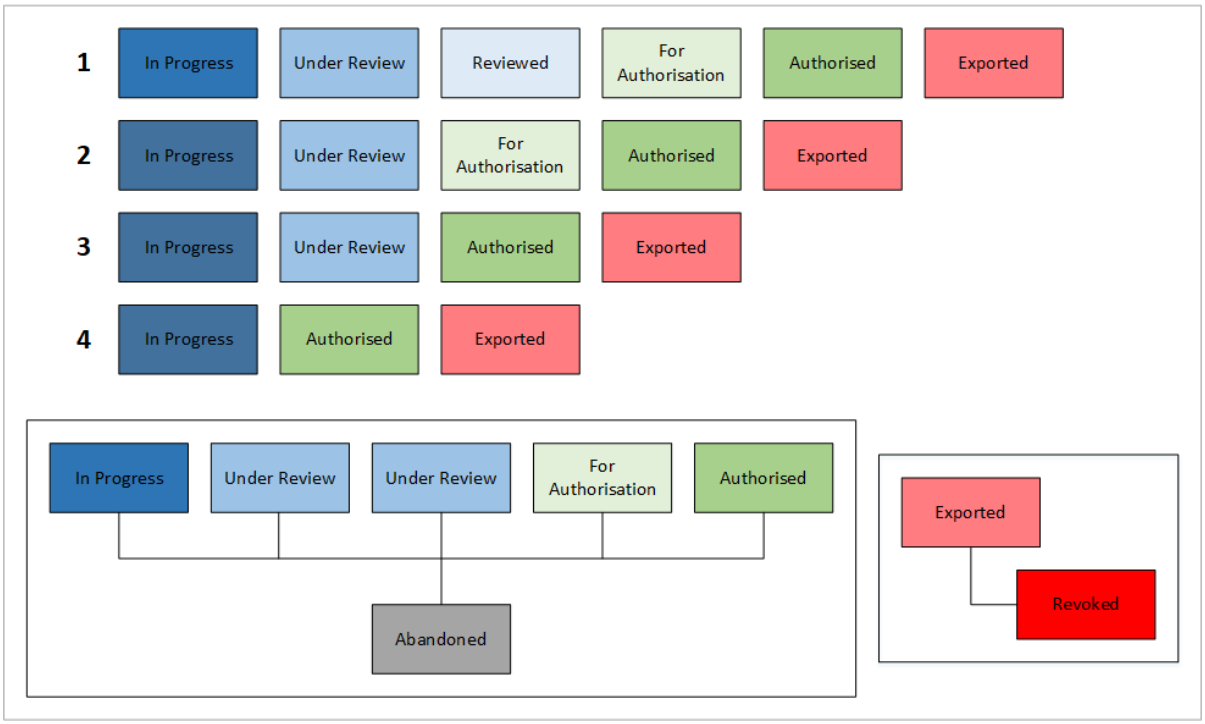
Budget Status

This system table is used to validate entries in Edit Status field in Budget header with columns for Code, Description, Default Record, Editable, Next Available Status and Read Only.

A status code cannot be both editable and locked. If the **Editable** check box is not selected, this defines the item as Locked.

Users cannot use the system table to add or delete states: The user interface can only be used to edit the **Editable** status and the **Next Available Status**.

Typical budget approval status progressions



Maintenance

These tables are for registering maintenance activity data. All costs incurred by an asset are allocated to a component code. Component codes are used to create projections and to identify on which part of an asset work was performed.

The component code hierarchy therefore establishes the relationship between different components codes, usually in terms of location on the asset. There are four levels to the component code hierarchy. The top level is cost type, then system, subsystem and component code.

- Cost Type
- System
- Subsystem
- Component Code
- Job Code
- Modifier Code
- Occurrence Type
- Task Counter
- Task Type
- Maintenance Type
- Maintenance Order Type
- Activity
- Cause
- Delay Code
- Event Prevention
- Source
- Symptom
- Task Planned Time
- Variance Cause
- Issue Type
- Work order Settlement Status
- Supplier
- Scheduling Group
- Task Misc Category
- OH&S
- "User Work Order Status " on page 878
- Planning Group
- Operational Criticality
- Notification Type
- Operation Work Status

Cost Type

Cost types are part of an equipment hierarchy, and are items such as repair and maintenance, capital, operator.

Cost type is part of the component code hierarchy. The component code hierarchy table also provides a structure for analysing data. The structure is Cost Type > System > Subsystem > Component Code.

System

Systems are part of an equipment hierarchy, for example, Repair and Maintenance > Engine (Engine is the system), or Repair and Maintenance > Drive Train (Drive Train is the system).

The component code hierarchy tables provide a structure for analysing data. The structure is Cost Type > System > Subsystem > Component Code.

Special Field	Description
Cost Type	Systems have to be associated to a cost type.

Subsystem

Subsystems are part of an equipment hierarchy, for example, Repair and Maintenance > Engine > Fuel System (Fuel System is the subsystem).

The component code hierarchy tables also provide a structure for analysing data. The structure is Cost Type > System > Subsystem > Component Code.

Special Fields	Description
Default for Unassigned	All new component codes received by AMT through an interface are allocated to this subsystem.
Expense Element	Default expense element that is assigned to the subsystem's unassigned task.
Systems	Subsystems have to be associated to a system.

Component Code Hierarchy

Component codes are the lowest level of the component code hierarchy. The structure is Cost Type > System > Subsystem > Component Code. An example of a component code is a fuel tank. The full hierarchy would be: Repair and Maintenance > Engine > Fuel System > A982 Fuel Tank.

Special Fields	Description
Budget Code	Only component codes that are budget codes can be used in a strategy task.
Global Component	Links to the Global Component Code system table.
Alert	Displays a warning when the component code description is the same as the component code, which is usually a result of AMT receiving a new code through an interface

Job Code

A list of the types of jobs that can be performed when a task is executed, for example, J01 – Remove and install, J02 – Replace with exchange.

A task may consist of one or more jobs, each of which can have a cost and an associated bill of materials. AMT usually replicates the job codes that already exist in the ERP system.

Special Fields	Description
Budget Code	Only job codes that are budget codes can be used in a strategy task.
Global Job Code	Links to the Global Job Code system table.
Flag as a Lube	Flag a job code as lube if it is to be used in a strategy task configured as a consumable. This enables the consumable strategy task to be forecast in the Consumable Forecast report.
Consumable Description	If the job code is flagged as Lube, enter the description of the consumable, for example, Engine Oil.
Consumable UOM	If the job code is flagged as Lube then enter the Unit of Measure of the consumable, for example, Litres.
Remove and Install	Flag the job code as a remove and install job code in order for it to be classified as that on the Agreement Summary 2 report.

Modifier Code

A list of codes to provide an additional identification to the component code, such as its position on the asset. For example: R – Right, L – Left, RF – Right Front, T – Top.

Special Field	Description
Budget Code	Determines which Modifier Codes can be used in Strategy Tasks.

Occurrence Type

A code and description of an occurrence of a specific failure mode that you can select when recording the nature of a maintenance task being performed. For example, AF - After Failure, BF - Before Failure.

Used by the technical analysis reports to highlight the nature of maintenance tasks performed, for example, the number of components changed out after failure vs before failure.

Task Counter

For a strategy task imported from an external system, the Task Counter is the identification number of the strategy task in the external system for the specific asset. Where the strategy task is defined in AMT, the task counter field usually remains blank or it is used to distinguish between strategy tasks that have the same combination of Component Code + Modifier Code + Task Type. For example: 1001, ZB01.

When AMT is integrated with Ellipse, the task counter equates to an MST number or standard job number. If you are a mining company setting up an equipment under a MARC, the task counter is used to distinguish between different billing steps.

Special Field	Description
Budget Code	Determines which Modifier Codes can be used in Strategy Tasks.

Task Type

A code and description combination, which identifies the general activity at an occurrence of a task. For example, CC - Component Changeout, IN - Inspection, GR - General Repair.

Task Type is a code unique to AMT. If a task type is used to identify Preventive Maintenance (PM) services, this should be flagged as a PM planning task.

Special Fields	Description
Maintenance Type	Links to the Maintenance Type system table.
PM Planning	Determines which tasks are displayed in the PM Schedule function.
PM	Task types flagged as PM are included in Service Accuracy statistics.
QW PM	Task types flagged as PM are included in Quick Watch's First Stoppage After PM statistic

Maintenance Type

A high-level maintenance code used for reporting purposes, For example: Shop, Field.

Maintenance Order Type

This table lets you set the Maintenance Type for reporting purposes. The code is not dependent of the AMT Task Type.

Activity

The downtime allocation of an event and the reason behind the cause of the downtime delay. For example, performing maintenance, washing machine, waiting parts.

In the Equipment Status screen you can select different activities in the downtime allocation section of an event which describe what happened during a certain period of time. For each activity, there must be a variance cause (see the Variance Cause table).

Cause

A list of reasons used to describe why work is being undertaken on a piece of equipment, for example, fatigue, cracked, burst, over pressure.

This system table is a mechanism supporting Reliability Centred Maintenance through which you can analyse the symptoms and causes of failure.

Special Field	Description
Catalogue Profile	An optional attribute of Assets and Notifications. When an asset or notification is associated with a Catalogue Profile, the Symptom, Cause and Object Parts available for selection are limited to those defined in the related Catalogue Profile.
Global Cause	Links to the Global Cause system table.

Delay Code

Define delay codes to identify the reasons for a work order being delayed.

Event Prevention

Describes the actions that might have prevented a breakdown downtime event, for example, oil analysis, inspection, operator training.

If an event is classified as a breakdown, you must nominate how the breakdown could have been prevented by selecting an option from this table.

Source

Possible ways in which a new task could have been identified by someone before the task was registered in AMT. For example: inspection, defect, application study, operator comments.

This table provides an audit trail back to the originator of a new task in case more information is required.

Special Fields	Description
EMI Action	The source assigned to work orders generated through the Equipment Management (EM) Issues module.
Customer Reported	The Product Incident Report Capture report requires a default source to reflect work orders raised through this process.

Symptom

The reason describing why work is being undertaken on a piece of equipment, for example, leaking oil, vibration, power failure.

This table supports Reliability Centred Maintenance, to help you analyse the symptom and causes of failure.

Special Field	Description
Strategy Default	Indicates the symptom that should be assigned to work orders raised by strategy tasks.
Catalogue	An optional attribute of Assets and Notifications. When an asset or notification is

Special Field	Description
Profile	associated with a Catalogue Profile, the Symptom, Cause and Object Parts available for selection are limited to those defined in the related Catalogue Profile.

Task Planned Time

Lists options available for when a task can be planned for in the future. For example, next operational stoppage, next planned stoppage.

This table provides guidance to a planner as to when a task should be scheduled into a work plan.

Special Fields	Description
Plan Days Offset	When a backlog work order is assigned a Task Planned Time, the work order's Target Date will be calculated as today's Date + Plan Days Offset.

Variance Cause

The reason behind the variance between the planned and actual downtime for the allocation of an event. For example, insufficient parts in stock.

This table is used by maintenance groups to identify inefficiencies in maintenance processes.

Special Field	Description
Activity	For each Activity (see the Activity system table) there must be at least one variance cause, and the description of the variance cause should be relevant to the activity. For example, if the activity is waiting parts, a variance cause might be insufficient parts in stock or parts not supplied in advance.
Default Unallocated Downtime	A record needs to be flagged the default Unallocated Variance Cause in order for downtime allocation merging logic to be applied when merging existing AMT events, or merging existing AMT events with external downtime events. If, after merging events, the sum of downtime allocation is greater than the event's downtime, then a downtime allocation record will be created in the merged event, with the Default Unallocated Downtime record, with the balancing negative number of downtime hours. As a result, the total downtime of the downtime allocations will balance to the total downtime of the event itself.

Issue Type

General classifications given to EM Issues, for example, WHS - Workplace Health and Safety, Product Improvement, Application, Operation.

Allows you to run reports by EM Issue Type because some issues may require more urgent attention than others, for example, safety issues.

Work Order Settlement Status

This table allows you to designate the status of a work order settlement, for example open, in progress, technically complete.

Scheduling Group

List of scheduling groups used for suppression and dependency logic, for example, Suppression A, Dependency X.

Strategy tasks that use Advanced Scheduling Logic can be assigned to scheduling groups. These groups of tasks can be performed either cyclically using dependency or in a parent/child relationship using suppression.

Supplier

Table for registering suppliers of parts and miscellaneous resources, such as ABC Hydraulics – ABC.

Assigning suppliers to parts and miscellaneous items on a work order is required to generate purchase orders in AMT, which can be emailed, printed or sent to external systems.

Task Misc Category

Categories for different types of miscellaneous maintenance resources, such as consumable, paint, subcontractor.

OH&S

Occupational Health and Safety (OH&S) attributes of a work order, for example, statutory, environmental, safety.

Used to describe the OH&S aspects of an AMT work order.

Special Field	Description
Auditable Record	If you select the OH&S option, which is flagged as the Auditable Record, in an AMT work order, certain fields are disabled. The specific fields disabled are Work Order Description, Component Code, Modifier Code, Task Type and Task Counter, Source, Raised By and Date Notified.

User Work Order Status

User work order status codes reflect the state of a work order from a user perspective not covered by the system status.

These codes are available to assist managing the status (or state) of a work order from a user perspective during the planning and scheduling lifecycle of a work order. For example, you may wish to know work is pending production, pending scope, awaiting parts or awaiting external labour. Such states can be defined as a user status and applied to work orders. Work orders can be searched using this status as well. The user status cannot be changed on a work order with a system status of In progress, Abandoned or Completed.

The following screens have the **User WO Status** field:

- **Daily Planning** > **Planning**
  - Planning, Work Orders, Backlogs tabs.
- **Daily Operations** > **Equipment Status**
  - Planning, Work Orders, Backlogs tabs.
- **Daily Operations** > **Daily Reports** > Workorder Details

You can filter for the user work order status on these screens. The field has a one-to-one relationship with a work order. When you add a work order and set a user work order status you have the option of adding comments about the work order.

Planning Group

A Planning Group is a person or group of people that organise maintenance activities. You can belong to zero, one or many Planning Groups. Including a Message Group reference lets email notifications to be sent to the Planning Group.

Operational Criticality



When setting up the system table make sure you create the **Default** category first.

Operational Criticality enables a risk rating or priority to be given to the maintenance activity. If the AMT Risk Feature is enabled, ranking supports further calculations of overall risk in long term planning and budgeting. Operational Criticality can be used without the Risk Feature enabled but will not support risk calculations.

Notification Type

This table is used to identify different reasons for creating a notification. Different types of notifications may be addressed differently from a work management perspective.

Operation Work Status

This table is used to identify different statuses for a work operation. Operation Work Status allows mobile users to redirect work order operations and alert supervisors and next shift work groups of a work order operation's progress.

A work order operation can have only one status at any point in time.

Regulatory Code

This table is used to identify different legislative or regulatory compliance applicable to a work order or work order operation. One, and only one, code can be held against a work order header or work order operation. The work operation code can be different from that at the header.

The Regulatory Code can be defined to suit reporting requirements. It could be a simple yes/no option or it can be more detailed and identify the different compliance reporting bodies. If identification is needed to cover multiple legislative bodies, a code will need to be defined to cover this.

Source of Supply

Used to identify the supplier or origin of a part. The **Code** indicates the code of the supplier from whom the part was purchased.

Object Part

Object Part is used to identify an asset part impacted when recording failure assessments, and is synonymous with a component code, or may be a different level of detail (greater or lesser) depending on any FEMA or FEMCA reporting being undertaken.

An object part consists of:

- A unique code
- Description

Special Field	Description
Catalogue Profile	An optional attribute of Assets and Notifications. When an asset or notification is associated with a Catalogue Profile, the Symptom, Cause and Object Parts available for selection are limited to those defined in the related Catalogue Profile.

Parts & Standard Jobs

These tables are used to set up the basic parameters for parts and standard jobs in AMT.

- Labour Activity
- Labour Rate
- Part Type
- Parts Price Group
- Parts Manager
- Part Classification
- Parts Warranty Group
- Part Rating
- Inventory Parts
- Commodity Code
- BEC Code
- PACC

Labour Activity

A description of labour activity only used in standard jobs and work orders. Each branch in AMT can have a different labour rate per labour activity. Standard jobs and work orders may require one or more different Labour Activities, each of which can have a specific Labour Rate.

If the **Inactive** check box is selected, the labour activity code will not be used when a new standard job is created.

When creating a new labour activity, it should have an **Expense Element**. If you don't select an expense element the default expense element is used.

Labour Rate

A combination of labour activity, branch and currency for which an hourly labour rate can be assigned.

Review labour activities and rates

In some cases labour will form part of the costing of a standard job. Labour costs can be defined manually in a standard job, or the standard job can link to specific labour activities (for example, welding, field service mechanic, shop mechanic).

Create and view labour activities by navigating to **System Administration** > **System Tables** > **Parts & Standard Jobs**.

Labour Rates apply to Standard Jobs or Workorders which include Labour Activities. When a Labour Rate is changed in the Labour Rate table, all priced Standard Jobs that are "Calculated from Operations" and for which there is an Operation using the changed Labour Rate are updated. There can only be one labour rate for every combination of Branch, Currency, Labour Activity, Pay Type.

Part Type

This table defines groupings of parts, such as exchange, new, reman, fluid.

Parts Manager

Parts Manager lists all the part numbers in AMT. Different attributes and assumptions can be stored against each part number . Some of the attributes are automatically populated by the different routines that import and add part numbers into the table; other attributes need to be added manually.

Navigate to **System Administration** > **System Tables** > **Parts & Standard Jobs** > **Parts Manager**.

Button	Action
	Create a new part
	Delete the highlighted standard job
	Supersede a part with one or more parts
	Save
	Close or Cancel

Grid fields (Key fields)

Field	Description
Source of Supply	The code of the supplier from whom the part was purchased.
Price Group	Parts Price Group.

Field	Description
Parts Superseded	Flag indicating whether the part has been superseded. This flag does not prevent the part from being used in a standard job.
Last Price Update	The date when the price of the part was last changed.
Used In Std Job	The number of standard jobs in which the part is used.

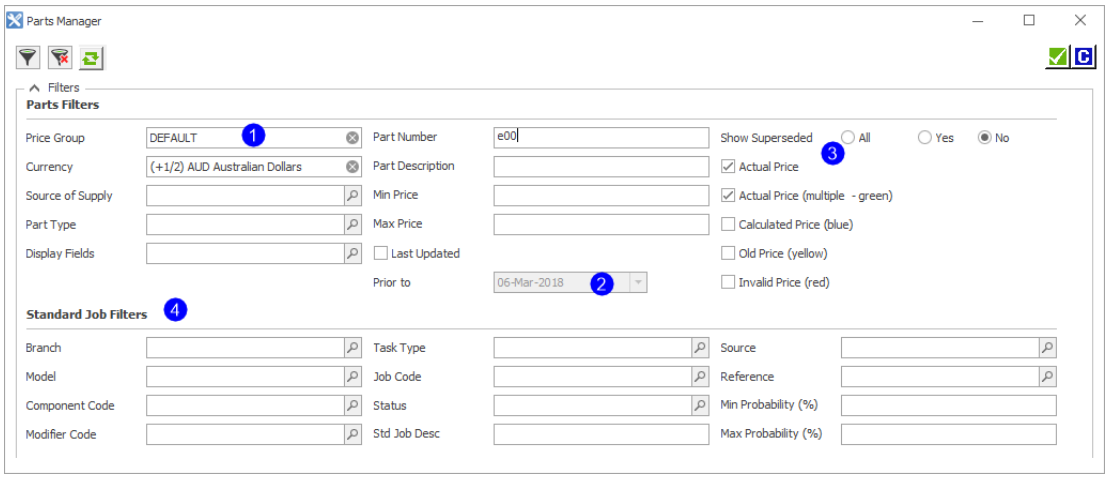
The **Budget Time Waiting For Rebuild**, **Budget Rebuild Duration**, **Budget Delivery Time to Warehouse** and **Budget Total Turn Time** fields are used in component capacity management. The **Budget Total Turn Time** field is used when forecasting the inventory level, and the other fields are used to forecast the workload through rebuild centres.

Manually add parts using Parts Manager

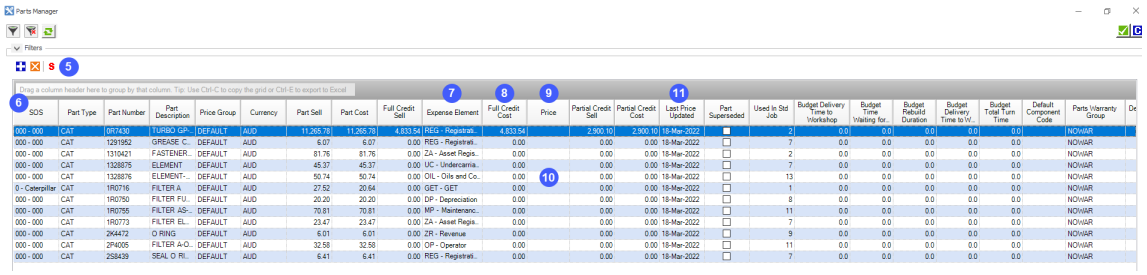
You can use Parts Manager to

- Add new parts to the AMT database.
- Edit the details of existing parts (including pricing).
- Supersede parts.
- Delete parts.
- Identify whether specific parts are used in standard jobs, and if so, how many standard jobs.
- Determine when the pricing of certain parts was last updated.

Navigate to **System Administration** > **System Tables** > **Parts & Standard Jobs** > **Parts Manager**.



Parts Manager grid



Item	Description
1	Parts Price Group Filter.
2	Show parts updated prior to a certain date.
3	Show parts which are or are not superseded.
4	These filters allow you to search for parts which are currently used in a standard job.
5	Add, Delete or Supersede the highlighted part.
6	Parts Price Group. Actual Part Price - there is exactly one part price (in the Price Group Currency) for the part. This is the normal case. Multiple Part Prices - part prices exist for more than one currency. This is also normal, but it means there are also part prices stored for other currencies. Calculated Price - there is a part price but for a different currency, so a currency conversion is performed using the AMT exchange rate. Old price that part price has not been updated for 180 days. 180 is a system configuration setting that you can change. Invalid Price - there is no part price. In this case, AMT looks up the price from the default price group. If required, AMT converts the price between currencies using the appropriate exchange rate.
7	Parts cost information for the specific parts pricing group and currency.
8	Date when past details were last modified.
9	Parts source of supply.
10	The part price validation.
11	Expense element.

Flagging a part as being superseded in the Parts Manager grids is only for information purposes. It does not trigger the automatic suppression of the nominated part by other parts, that is the purpose of the **Supersede** function.

Parts within AMT can be analysed in Excel by exporting the Parts Interrogation report. Navigate to the Parts Interrogation report by clicking **Knowledge Database** > **Parts Interrogation**.

Update part pricing

Once you have loaded all required parts into AMT, the new parts plus existing parts should have their prices updated to reflect current pricing as per the currency per parts price group.

Part pricing can be bulk-updated in AMT using the Part and Part Price Update interface process from the enterprise resource planning, or each part's price can be updated manually in Parts Manager.

Part price validation

In the Parts Manager, there is a **Price** column, which highlights which parts have invalid part prices. Part prices show as **Invalid** if there is no record of the required parts price in the Parts Manager for the selected price group.

Colour Code	Definition
	Actual Part Price - there is exactly one part price (in the Price Group Currency) for the part. This is the normal case.
	Multiple Part Prices - part prices exists for more than one currency. This is also normal, but it means there are also part prices stored for other currencies.
	Calculated Price - there is a part price but for a different currency, so a currency conversion is performed using the AMT exchange rate.
	Old Price - the part price has not been updated for 180 days (180 is a system configuration setting that can be changed)
	Invalid Price - there is no part price. In this case AMT looks up the price from the default price group. If required, AMT will convert the price between currencies using the appropriate exchange rate.

Part Classification

Defines the relevant approach for rebuilding rotable components. It is particularly important for the component capacity management functionality, as you can define which rotable component cores are retained in the system vs. those that leave the system. You can also produce forecasts by supplier or specific rebuild centre.

Parts Warranty Group

Defines the warranty period for specific groupings of parts, for example GETP - GET Parts - 90 days.

A special interface can be implemented that imports historic parts usage into AMT, along with the parts warranty group to which each used part belongs. Then, using the parts warranty group information, AMT endeavours to

determine whether the parts imported into AMT have been replaced within the warranty term of the nominated parts warranty group. They are flagged by the system as being potentially under warranty.

Part Rating

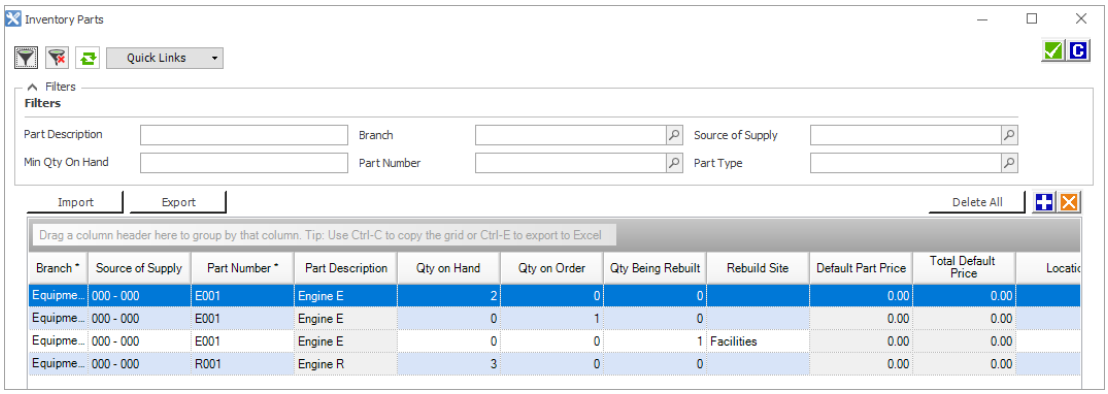
This table is designed to capture a comparison attribute of the component. For example, if the component code is Pump, the part rating could be 250kw.

Inventory Parts

This table is used to manage the status of components inventory or on order.

Import component inventory

To validate inventory capacity, you need to import or enter the current component inventory level. You do this on the Inventory Parts form, which you can open by navigating to **System Administration** > **System Tables** > **Parts & Standard Jobs** > **Inventory Parts**.



There are three *inventory classifications* - **Qty on Hand**, **Qty on Order** and **Qty being Rebuilt**.

The **Expected Delivery Date** column applies to **Qty on Order** and **Qty Being Rebuilt**, and represents the date that the component will be in stock and available for use. Because the delivery date will probably be different for components on order or being rebuilt, there can be many rows for the same part number.

You can edit fields directly on the grid, use **Ctrl-C** to copy the grid, and use **Ctrl-E** to export the grid to Excel.

Inventory capacity is managed at the **Branch** level in AMT. The forecast demand for equipment in a branch is compared with the nominated inventory levels for that branch.

Commodity Code

This table allows you to create commodity codes, for example fluids, brake parts.

BEC Code

This table allows you to create Broad Economic Category (BEC) Codes. These codes group transportable goods according to their main end use.

PACC

This is a system table for PACC.

Delivery Method

Part Rating is designed to capture a comparison attribute of the component (for example, if the component code is Pump, then Part Rating could be 250kw).

Equipment

These tables are used to set up the classification of the different types of equipment that will be registered in AMT.

- Equipment Class
- Equipment Criticality
- Equipment Group
- Manufacturer
- Model
- Equipment Category
- Equipment Location
- Asset Status

Equipment Class

High level grouping of the different equipment types, for example, mobile equipment, fixed plant, transport, infrastructure.

In financial reports, you can filter by equipment class.

Equipment Criticality



When setting up the system table make sure you create the **Default** category first.

Equipment Criticality enables the ranking of assets based on their impact to the business. If the AMT Risk Feature is enabled, ranking supports further calculations of overall risk in long term planning and budgeting. Equipment Criticality can be used without the Risk Feature enabled but will not support risk calculations.

Equipment Criticality allows a single code to be identified as the default criticality rating. Each code can have a ranking of 1 to 5 defined. Industry best practice is often 1-4 for ABCD association. Ideally, the same ranking would not be used against multiple codes, however, to support upgrading customers who have defined this field prior to the Risk Feature, and who have greater than five existing codes, AMT will support the same ranking against different codes within this table.

Equipment Group

A grouping of the model codes. Within the Asset Register, the technical details relate to the equipment group, that is, each equipment group will have its own set of technical details. Equipment group is sometimes referred to as model family.

Special Field	Description
Include Tech Fields	Where different equipment registration, configuration and application data fields can be enabled. Enabled options will display in the **Equipment Plan Editor** screen for you to populate. Options highlighted in red cannot be deselected because users have already entered data into these fields within equipment.

Manufacturer

Table for registering the manufacturer of the equipment being registered, or the manufacturer of a specific component (strategy task). Equipment and components derive escalation adjustments from specific manufacturers. Escalations per manufacturer are stored in AMT's Escalation system table.

Special Field	Description
Export Code	Short version of manufacturer code, for example TO = Toyota.

Model

Register models of equipment or assets produced by manufacturers, for example, Renault - RSC182.

Every asset registered in AMT must be linked to a model. When the manufacturer of a model is changed, AMT checks to ensure the relationship between the serial number of the asset and manufacturer are not duplicated. For example, two current assets have the same serial number, SN 01HL00003 for manufacturer XYZ, model 123 and SN 01HL00003 for manufacturer ABC, model 987. You cannot change manufacturer ABC for model 987 into manufacturer XYZ, otherwise the serial number 01HL00003 will not be unique for the manufacturer XYZ.

Special Fields	Description
Manufacturer	A model must be associated to one manufacturer.
Equipment Group	A model must be linked to one equipment group.

Equipment Category

This is a flexible system table that is used for financial reporting. It is set at the Equipment Plan.

Organisations may use different codes for different purposes. Contract your system administrator for the defined use of codes for your organisation.

Equipment Location

General categories describing locations of equipment. Locations could be production areas such as Pit A; but should not replicate regions, branches or site.

Organisations can use this flexible system table for different purposes. This code is not linked to the equipment hierarchy of Branch > Site > Fleet.

Asset Status

This table allows you to define the asset status value for Equipment.

Asset Condition



When setting up the system table make sure you create the **Default** category first.

Asset Condition is used to identify the health of an asset. Example Asset Condition Codes, Asset Condition Descriptions and Rating are below. Ratings are 1-10.

Asset Condition Code	Asset Condition Description	Rating
New	Excellent	5
Refurb	As New	4
Good	Good	4
OK	Fair	3
Poor	Poor	2
Fail	Very Poor	1

Functional Location

These tables are used to set up functional location categories, levels and attributes.

- Functional Location Category
- Functional Location Level
- Functional Location Attribute

Functional Location Category

Allows you to add, edit and delete functional location categories.

Functional Location Level

Allows you to add, edit and delete functional location levels. These are used by AMT to designate the labels for the levels of the hierarchy. You are able to change the descriptions associated with each level.

Functional Location Attribute

Allows you to add, edit and delete functional location attributes. AMT can accommodate up to 30 attributes for a functional location. This allows you to ensure that any functional locations from your ERP system are replicated by AMT.

Usages & Measurements

These tables are used to set up units of measure, measurement codes and production statistics codes.

- Units of Measure
- Measurement Code
- Production Statistics Codes

Unit of Measure

A defined unit of measure (UOM), such as engine hours, fuel litres, travel hours

This table is required to read measurements and create usage profiles.

Special Fields	Description
External UOM	Unit of measure code used by an external system that aligns to the UOM in AMT
Operating Hours	For any equipment to show data in the **Scheduled Hours** screen, its Primary UOM must be flagged as Operating Hours. It is possible to flag multiple UOMs as Operating Hours.
Revenue Rate	This field is not used.

Measurement Code

Code for specific measurements, for example, fuel quantity, oil level, engine temperature.

This code is used in consumption reports and in the Condition Based Monitoring module to define measurement translation rules and capture actual condition monitoring data.

Configure Measurement Codes

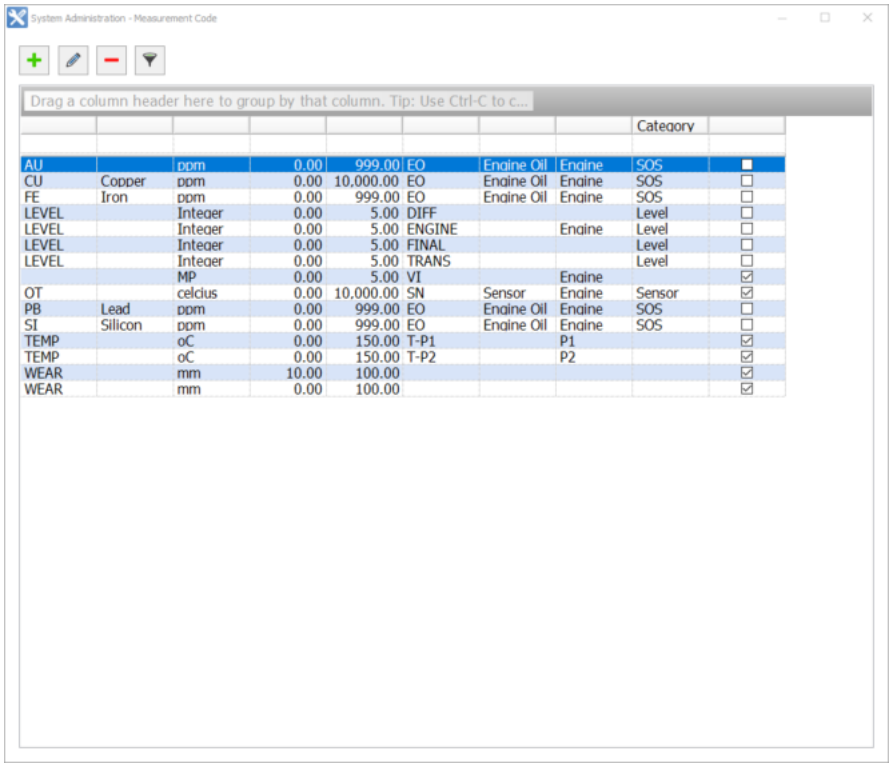
When actual measurements are received by AMT, they are classified into four ratings based on user-defined translation rules.

- 1 - OK.
- 2 - Low Alert.
- 3 - Medium Alert.
- 4 - High Alert.

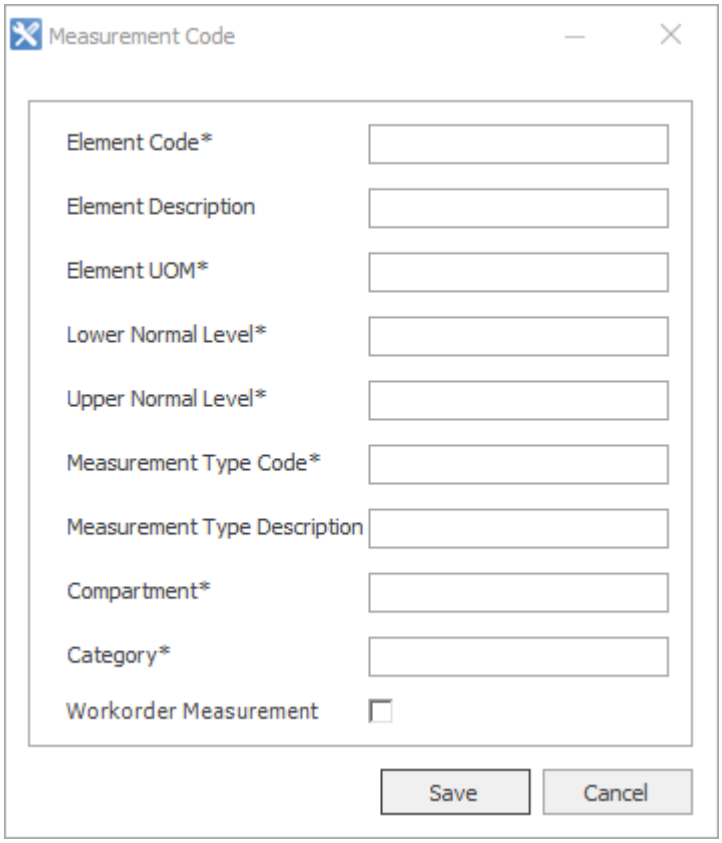
You can set AMT to auto-create a notification or work order if the measurement reading reaches a certain rating.

You must configure measurement codes before setting up Measurement Translation Rules.

To configure measurement codes, navigate to **System Administration** > **System Tables** > **Usages & Measurements** > **Measurement Code**.



When you click **Add**, the dialog box below displays



Fields and columns

Column	Description	Example
Element Code	Measurement code.	PB for Lead level.
Element Description	Description of the Measurement Code.	Lead. The lead level in Engine Oil.
Element UOM	Unit of measure for the element.	ppm, mm, Celsius, Integer.
Lower Normal Level	The normal lower limit for the measurement	0
Upper Normal Level	The normal upper limit for the measurement.	100
Measurement Type Code	The code for the measurement type.	EO, TTD.
Measurement Type Description	The description of the measurement type code.	Engine Oil, Tyre Tread Depth.
Compartment	Which part of the equipment the measurement is taken from.	Engine, Final Drive, Front left tyre.
Category	What type of measurement it is.	Inspection, Level, Telemetry.
Component Code	Which component the measurement was taken from.	"4000" for Engine.
Work Order Management	Select the check box to be able to view and edit the measurement code in the work order.	

Production Statistics Codes

Codes used to describe the different operational statuses of an equipment, for example, idle, productive, under maintenance.

These codes are only used in production statistics reports.

Special Fields	Description
Downtime	The code flagged as downtime inherits actual downtime from AMT Events.
Site	Different sites can have different production statistics codes.
Primary UOM	Indicates whether the values are the same as the equipment's primary UOM for reporting purposes.

Other

These tables are used to set up the classification of the different types of equipment that will be registered in AMT.

- Contract Type
- Customer
- Multi-Customer Editor
- Document Type
- Global Cause Code
- Global Component Code
- Global Job Code
- Global Part Type Code
- Global Expense Element
- Strategy Task Marketing Grouping
- Log List Editor
- Quote Terms

Contract Type

Use this table to differentiate different types of maintenance contracts, for example, full maintenance contract, cost cap, parts cost guarantee.

Customer

Enter your customers' details in this table.

Multi-Customer Editor

You can edit customer details in this table.

(Rpt_634) Customer Audit

Displays Customers of AMT.

Tips

Use the Blank Fields filter to identify Customers with missing classifications.

Initial parameters

|ControlType|,|IDList|,|TextList|,|EquipmentTypes|,|LogoLocation|,|UserID|

Report parameters

Report Type, Equipment Hierarchy

Additional filters

Display Columns, Blank Fields, Industry, PSS Responsibility

Document Type

Categories of documents used in AMT's library, for example, training presentations, workscopes, checksheets.

Global Cause

See Cause.

Global Component Code

See Component Code.

Global Job Code

See Job Code.

Global Part type Code

See Part Type.

Global Expense Element

See Expense Element.

Strategy Task Marketing Grouping

This table is part of product support marketing. The marketing grouping is an additional grouping for strategy tasks which may be used by the sales responsibility to simplify groupings. The market grouping can be maintained in individual strategy tasks or using the marketing and customer support editor.

Log List Editor

Use this table to edit the reason codes for logging changes. This table is used when you change the next due details for strategy tasks. If AMT is configured to record a comment when changing strategy task next due occurrences, a dialog box is displayed and you select a reason and enter a comment. The comment is recorded in the log entries. The information entered is recorded in AMT logs and you can view the log with the Log Entries function (but not the reason code).

Risk Grouping

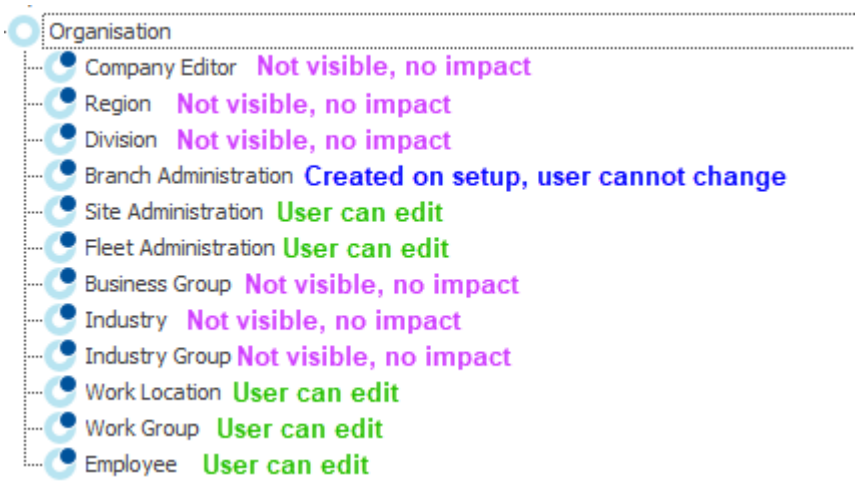
Risk Grouping will normalise AMT's Calculated Risk (based on Equipment Criticality * Asset Condition * Operational Criticality) into user defined 'bands'. Risk Grouping is not relevant without AMT's Risk Feature being enabled. It is not mandatory to Asset Condition can be used without the Risk Feature enabled but will not support risk calculations.

It is not mandatory to configure **Risk Grouping**; however, not doing so will result in a single risk group with no description. The risk bands or groups represent a percentage; therefore the highest risk value cannot be greater than 100. Risk bands can not overlap one another. A 25% banding would result in 4 items which may be represented as Low (0 to 25); Moderate (26 to 50); High (51 to 75) and Critical (76 to 100).

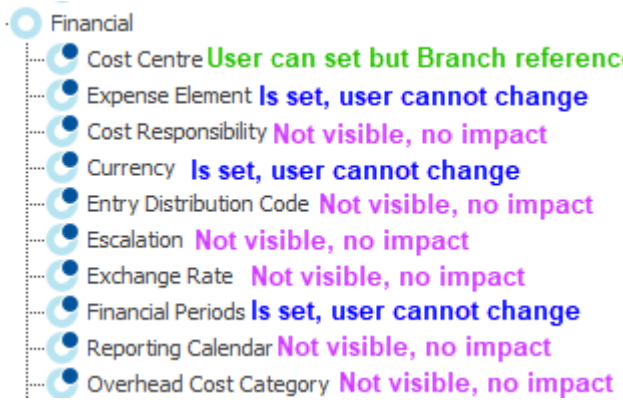
Multi tenancy system tables

The strategy for each multi tenancy system table is below. If the system table is set as **User can edit** this indicates that the table contains organisation-specific data.

Organisation

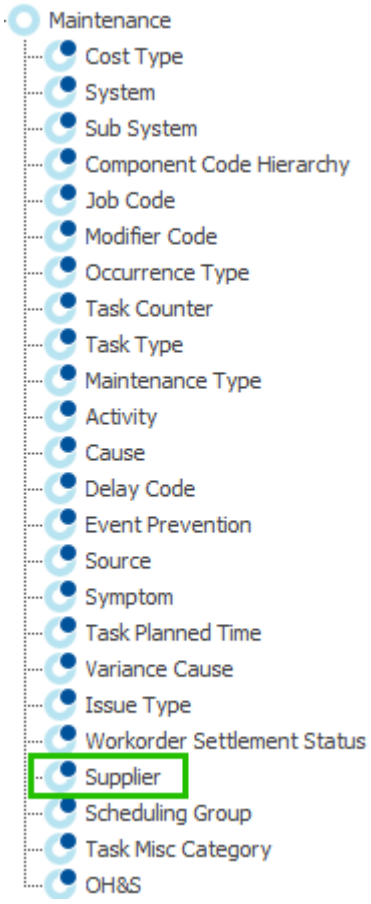


Financial

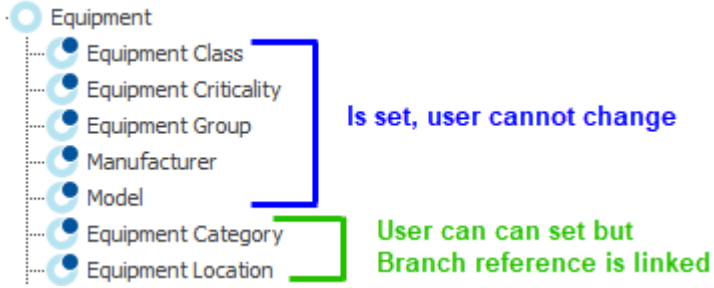


Maintenance

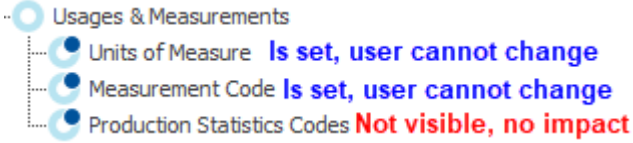
All of the Maintenance System Tables strategies are **is set, user cannot change**, except for Supplier (highlighted below) which is **User can set but Branch reference is linked**.



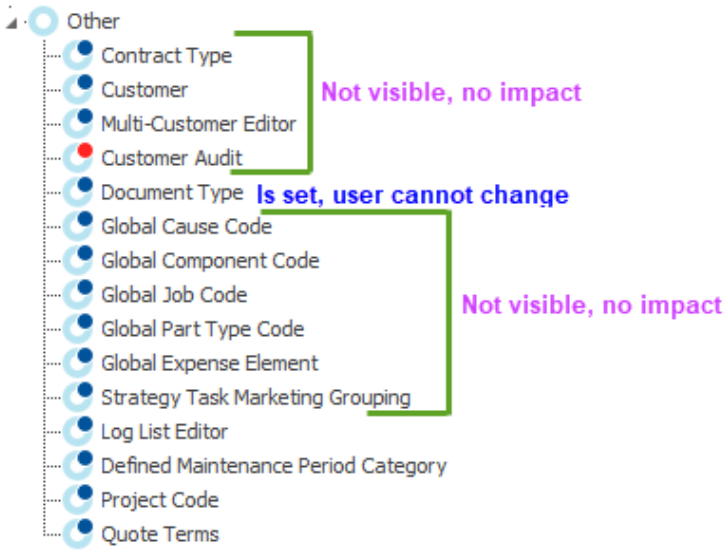
Equipment



Usages & Measurements



Other



Data Translation and Validation

Work order settlement code translation

Code translation in AMT is designed to allow work order settlement component code combinations to be automatically mapped to AMT codes during the work order settlement interface process. This facility is typically used to allow a wide variety of component code combinations within an enterprise resource planning transaction system to be expressed as a condensed version within AMT.

In work order settlement code translation, you can:

- Add sets of external transaction system codes, for example component code, modifier code, job code and link them to associated sets of internal AMT codes. They are also referred to as external to internal code mappings, so that when the transaction system codes are imported into AMT, they are translated to codes that make sense inside AMT.

  The AMT codes have an extra field, Occurrence Type.

- Remove existing sets of external to internal code mappings.
- Edit existing sets of external to internal code mappings.
- Set the order of translation where there are data conflicts.

Additional notes on code translation

In AMT, each work order settlement contains two sets of **defining** codes such as component code, modifier code, task type, task counter, job code and occurrence type. One set contains the values of the codes received from the external system. These values are Null for internal AMT work order settlements. The other set contains the **translated** set of codes. The codes received from the external system are functionally ignored, and are provided for information purposes; conversely, AMT logic requires that the value of the codes will use the translated values.

AMT cycles through the set of new or updated work order settlements and for each work order settlement, performs the following:

- If the work order settlement has a standard job as its template, the internal AMT codes of the work order settlement are set to those of the standard job.
- If the external set of codes that define the work order settlement match a set of codes specified in the list of code translation rules, the internal AMT codes of the work order settlement are set to the values specified in that code translation rule. If the codes do not match, the internal AMT codes of the work order settlement are set to the values of the external codes.

When a field in a code translation rule has a value of Null, this acts as a wild card for that field. For example, if there is just one code translation rule defined, as follows:

CC	MC	TT	TC	JC		CC	MC	TT	TC	JC	OT
1000	-	PM	-	-		1000	-	CC	0	010	B/F

The following work order settlements are imported into AMT from an external system, with codes as follows.

Work Order Settlement Number	Component Code	Modifier Code	Task Type	Task Counter	Job Code	Occurrence Type
WOA123	1000		PM	1	010	REMAN
WOB156	1000					
WOC964	1052		PM			
WOD777	1000		PM			

Work order settlements WOA and WOD would satisfy the rules, and hence have their codes translated to those specified in the rule (that is, WOA and WOD would end up with exactly the same set of codes). However, WOB and WOC would not have their codes translated - WOB does not have Task Type = PM and WOC does not have Component Code = 1000 (that is, WOB and WOC would be added to AMT with no change to their codes).

Order of priority

All code translations are assigned an order of priority, as defined in the **Order** field. This means that if two or more work order settlements contain, say four out of the five available code combinations, the translation will be determined on the Code priority setting, so that only the highest priority code translation rule is applied to any particular work order settlement.

Set GL reconciling items

Use this function to manually reconcile transactions in AMT to records imported through the General Ledger interface, where the interface was unable to link the records automatically such as auto-reconcile.

(Rpt_186) Ledger Missing Equipment

This report displays transaction records imported through the General Ledger interface that do not link to AMT equipment, and shows technical details for component changeouts. Can be extremely critical in highlighting data problems.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

GL Account, Invoice, Workorder, Serial Number

Ledger Missing Equipment

This report displays transaction records imported through the General Ledger interface that do not link to AMT equipment.

(Rpt_396) AMT to GL Reconciliation

Summarises the balance between the work order costs in AMT and the costs in the General Ledger for maintenance contract equipment. The report also reconciles contract billings.

Tips

This report does not reconcile Overheads, Plant Hire or other Operating Costs.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|,|Currency|

Report parameters

Default.

Additional filters

Contract, AMT Equipment, Currency, Cost Bearer, Parts/Labour/Misc

Click here to go to a description of the report output.

Reconciling work order settlements

A reconciliation process between AMT and the work order settlement system must be performed at least once a month, regardless of whether there is an interface between the two systems or not.

If AMT does not balance to the work order settlement system, management will not have confidence in the financial reporting produced by AMT.

The items that **must** reconcile between the two systems are the costs that have been incurred by each asset registered in AMT. The type of costs which have to be reconciled is dependent on the nature of the assets registered in AMT, for example, are the assets part of a maintenance contract, does the asset only incur direct maintenance costs and so on.



Types of costs which have to be reconciled based on the nature of the asset should be documented in a procedure prior to the installation of the software.

In certain implementations of AMT there will be a general ledger interface which provide you with the ability to perform a reconciliation within AMT and identify the items which do not balance.

For implementations which do not have this interface in place, the organisation will have to develop a procedure that ensures:

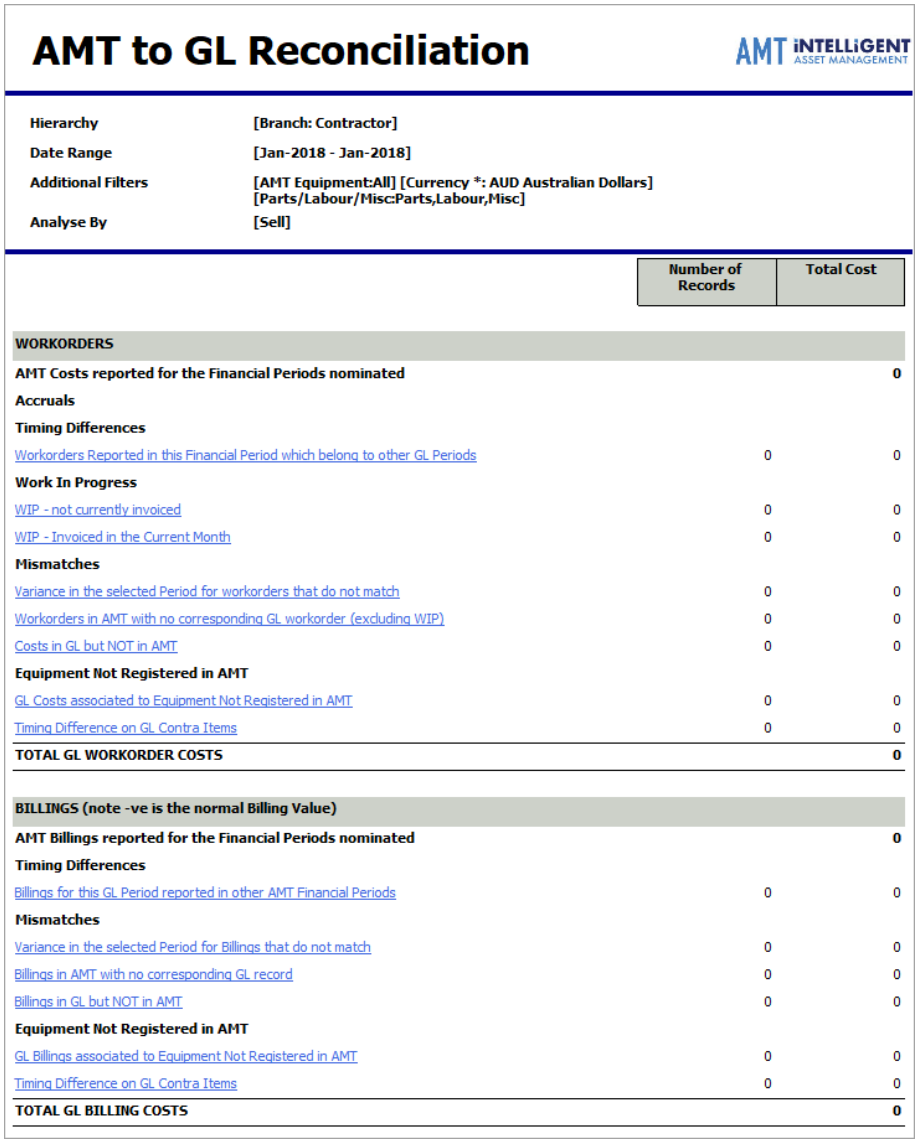
1. Reconciliations are performed on a routine basis.
2. Items which do not reconcile are investigated and addressed properly.

AMT to GL Reconciliation Report

To run this report, navigate to **System Administration** > **Data Translation and Validation** > AMT to GL Reconciliation.

An example of the report is below, followed by a description of the major sections.

There is no data in the example screenshot.



AMT to GL Reconciliation

AMT INTELLIGENT ASSET MANAGEMENT

Hierarchy	**[Branch: Contractor]**
Date Range	**[Jan-2018 - Jan-2018]**
Additional Filters	**[AMT Equipment:All] [Currency *: AUD Australian Dollars] [Parts/Labour/Misc:Parts,Labour,Misc]**
Analyse By	**[Sell]**

	Number of Records	Total Cost
WORKORDERS		
AMT Costs reported for the Financial Periods nominated		**0**
Accruals		
Timing Differences		
Workorders Reported in this Financial Period which belong to other GL Periods	0	0
Work In Progress		
WIP - not currently invoiced	0	0
WIP - Invoiced in the Current Month	0	0
Mismatches		
Variance in the selected Period for workorders that do not match	0	0
Workorders in AMT with no corresponding GL workorder (excluding WIP)	0	0
Costs in GL but NOT in AMT	0	0
Equipment Not Registered in AMT		
GL Costs associated to Equipment Not Registered in AMT	0	0
Timing Difference on GL Contra Items	0	0
TOTAL GL WORKORDER COSTS		**0**
BILLINGS (note -ve is the normal Billing Value)		
AMT Billings reported for the Financial Periods nominated		**0**
Timing Differences		
Billings for this GL Period reported in other AMT Financial Periods	0	0
Mismatches		
Variance in the selected Period for Billings that do not match	0	0
Billings in AMT with no corresponding GL record	0	0
Billings in GL but NOT in AMT	0	0
Equipment Not Registered in AMT		
GL Billings associated to Equipment Not Registered in AMT	0	0
Timing Difference on GL Contra Items	0	0
TOTAL GL BILLING COSTS		**0**

Work orders

The work order section of the report first establishes the total work order settlement cost in AMT, and then through a series of analyses, determines the total cost in the General Ledger (GL). The end result should be a balance between AMT and GL.

AMT Costs reported for the Financial Periods nominated

This is based on the work order settlement history in AMT for the financial period selected.

If AMT has been configured to use Transaction Date reporting, then the report uses the start date of children work order settlements in order to determine which work order settlement costs fall into the selected date range. However, if non-transaction date reporting is used, then the report uses the start date of the parent work order settlement (which is usually the job date, or the event actual down date for work order settlements linked to tasks which are linked to events).

If you are unsure whether Transaction Date reporting is enabled, contract your AMT Administrator.



The decision to enable this type of reporting is dependent on the nature of the work order settlement process.

Important rules for recognising and categorising work order settlement costs:

- Work order settlements and billings manually reconciled using the **Set GL Reconciling** Items function are not included in this report.
- If the work order settlement number in AMT is linked to several assets, these records will go to the match area if the costs match to GL records.
- If one or more segments of a work order settlement do not link to an asset, then the work order settlement will go to mismatch even though the work order settlement's total costs may match to GL.
- If one work order settlement in a group Parent/Child work order settlements in AMT is closed, then all work order settlements in the group are considered to be closed.
- If an open work order settlement in AMT has records in the GL, it is considered to be closed.

Timing Differences

This section of the report identifies work order settlements reported in the current financial period which belong to other GL periods.

Total Workorder Summary			Summary for the Nominated Finacial Periods		
AMT	GL	Variance	AMT	GL	Variance
193	193	0	193	0	-193
113	113	0	113	0	-113
57	57	0	57	0	-57
363	**363**	**0**	**363**	**0**	**-363**

In the example above, while the AMT work order settlement and GL totals balance (Total Work Order Summary Variance = 0), the totals do not balance for the specific financial period (the report filter's date range). The GL records fall into a different period to that of the work order settlement in AMT.

Work In Progress (WIP)

WIP - Not currently invoiced shows work order settlements which are open in AMT but the GL does not have records for this work order settlement number.

WIP - Invoiced in the current month - shows work order settlements closed in the current month that do not have corresponding records in GL data.

Mismatches

Variance in the selected Period for Work orders that do not match lists work order settlements which have an amount different to their corresponding GL record in the same financial period.

Work orders in AMT with no corresponding GL work order (excluding WIP) shows a list of work order settlements but no GL values.

Cost in GL but NOT in AMT lists all GL records for the current financial period which do not have a corresponding work order settlement in AMT (considering all financial periods).

Timing Difference on GL Contract Items

Shows manually reconciled items for the selected period.

Billings

This section of the report is only relevant to maintenance contract equipment. A negative billing value represents a normal billing amount.

AMT Billings Reported for the Financial Period

This figure is based on the billing history in AMT for the financial period selected. This value can be reconciled to the **Billing Interrogation** report.

Timing Differences

Billings recorded in this financial period reported in other AMT Financial Periods lists matches by invoice number. This sub-report includes all billings from GL which can be linked or not linked to an AMT equipment.

If there are multiple assets for an invoice, the Invoice column will show the value **Mixed**.

Mismatches

Variance in the selected period for billings that do not match shows mismatching billing amounts for the same invoices.

Billings in GL but NOT in AMT shows GL records which are linked to the real equipment but do not have matching invoice number in AMT.

Equipment Not Registered in AMT

GL Billings associated to equipment not registered in AMT show GL records which are not linked to the real equipment and do not have the matching invoice number in AMT.

Timing Difference on GL Contract Items

Shows manually reconciled items for the selected period.

Causes of reconciliation issues

If both the GL and AMT have the same work order or billing, but the value of that record is different between the two systems, the reason for the variance is typically a result of the interface between the ERP and AMT. That is, the specific scenario of that transaction has not been handled or interpreted properly by the interface before sending the transaction to AMT. Typical errors which indicate the work order or billing interface from the ERP to AMT may not be handling correctly are:

- Special codes (for example warranty splits, flat rates) may be applied to the work order in the ERP but the work order settlement records interfaced to AMT have not had these adjustments applied.
- The work order or billing may have been backed out (reversed) in the ERP system but this change has not been interfaced to AMT.
- The work order settlement or some of its segments may be flagged as a contract cost in AMT (Cost Bearer = CC) but this may not be the case in the work order in the ERP.
- The work order settlement may be showing as closed or invoiced in AMT but in the ERP the work order is still open or vice-versa. This is caused by the Closed Period in the work order interface file not being set (it does not contain a value).
- The serial number linked to the work order settlement in AMT may be different to the serial number linked to the work order in the ERP. In this case, the values of the work orders may actually balance between AMT and the GL, but it is the serial number resulting in the imbalance.

It is possible that the GL contains cost records but AMT does not. This could be a result of:

- Miscellaneous general journals posted to the contract account in the GL rather than being processed as work orders or billings.
    - This issue is addressed by stopping the practice of using general journals and replacing them with work orders and billings.
- Users of the ERP posting transactions to the contract's GL account which should have been allocated elsewhere (such as journals mistakenly debited or credited to the wrong account code).
    - Such transactions have to be identified individually and reprocessed so that they are reversed in AMT (that is, so that AMT no longer shows the incorrect amount).

      It may not be possible to delete the transaction entirely out of AMT but it should be possible to change its value through reprocessing in the ERP and interfacing the result back to AMT.

    - Users of the ERP have to be educated so this practice does not continue.
- The interface failing to recognise work order or billings transactions in the ERP which should be interfaced to AMT.
    - You should investigate the nature of the work order or billing record in the ERP system in order to identify why it was not sent to AMT through the normal work order or billing import interface.

It is possible that AMT contains work order or billings records but the GL does not. This could be a result of:

- Users manually creating work order settlements internally within AMT that do not have a corresponding record in the ERP.
    - Users have the ability to create work order settlements in AMT using **Work order Settlement Manager** or using **Equipment Status**, but costs can only be entered onto these internal AMT work orders using **Work Order Settlement Manager**.
    - This Work Order Settlement Manager function should only be available to those who will use it responsibly.

- A timing difference between when you close or invoice a billing or work order in the ERP, to when it is actually transferred to the proper account in the ERP. Batch processes in the ERP can sometimes delay posting to certain accounts.
    - Look for these records in the **WIP – Not Currently Processed** section of the AMT to GL Reconciliation report.
    - The GL interface and reconciliation should only be performed once all such processes are completed.
- A problem with the GL interface file where some records which actually exist in the contract account in the GL are not being picked up and sent to AMT for some unknown reason.

Monthly contract overheads reconciliation

Overhead billings and costs have to be reconciled manually. The AMT to GL Reconciliation report does not reconcile overhead costs or billings Therefore, to reconcile overhead costs and billings you have to run the AMT reports for overhead costs and billings for the specific month, and then compare those results to a report from an external system.

(Rpt_224) Reconciling Open Workorder Settlements

Assists users to evaluate and action any open work orders in the data validation/ledger reconciliation area of the program. Report can be run to filter for work orders at cost or at sell and defaults to current month, but can be modified to suit the users. This report only displays data if AMT is configured in WIP mode (this mode is normally used only by OEMs).

Initial parameters

|ControlType|,|IDlist|,|TextList|,|ProjectionTypes|,|UserID|

Report parameters

Default.

Additional filters

Contract

(Rpt_532) Workorder Settlement Transactions

Displays all transactions of work order settlements, including child work order settlements which are linked to a parent. The transaction details include splits between cost centres, cost bearers and work groups.

Tips

This report is useful for reconciling AMT's work orders to external systems.

Initial parameters

|ControlType|,|IDlist|,|TextList|,|UserID|

Report parameters

Default.

Additional filters

Cost Bearer, Work Group, Cost Centre, Cost Category, Expense Element, Contract, Parts/Labour/Misc, WIP

Work order settlement transactions

This report lists all work order settlement transaction information:

- Child work orders.
- Financial period.
- Every combination of work group, cost centre, expense element and cost bearer.



This report is useful when trying to perform a manual reconciliation between AMT and an external accounting system where the General Ledger interface is not used.

Monthly contract equipment reconciliation

At the end of every month, all costs and invoices are posted to the contract's General Ledger (GL) account in order for the company to report on the contract's financial performance to date. Accordingly, reports run in AMT to report the financial performance should balance to that account in the ERP. To perform the reconciliation of equipment costs and billings follow the steps below.

Additional manual functionality

The AMT function **Set GL Reconciling Items**, under **System Administration** > **Data Translation and Validation**, is used to manually reconcile work orders and billings to GL records where AMT has not been able to automatically establish the link. AMT may not be able to establish such links if equipment, work order settlement numbers or invoice numbers do not match between GL and AMT records.

Reconciling open work orders

System Administration > **Data Translation and Validation** > **Reconciling Open Workorder Statements**

This is a report that shows the AMT and GL values for work order settlements where the AMTs work order status = Open (that is, Work In Progress).

Important report filters:

- The date filters are based on calendar date.
- The dates can only be in the past.
- All data is shown in the primary currency.

Other conditions:

- The date of the work order is based on the invoice date (if there is an invoice), otherwise, it is based on the job date of the work order.
- For the work orders which are split in AMT, if one of the work orders is closed, then all work orders in the group are considered closed.
- If GL data contains several work orders with the same work order number and serial number, and one of the work orders is in the selected period, all records are shown.

Identifying missing equipment

Use the AMT report Ledger Missing Equipment, under **Systems Administration** > **Data Translation & Validation**, to identify accounting system entries where there is no matching equipment in AMT.

The dates selected are calendar dates.

- Data can only be in the past.
- The date of the work order is based on the invoice date.

This report can be extremely critical in highlighting data problems.

Configuration

These tables are for the System Administrator to configure.

- Quote Documentation
- Group Permissions
- Cost Allocation Interface Details
- External Downtime Equipment
- External Downtime Site

Quote Documentation

There are two types of report packages generated by Quoter.

- **Internal reports**: Reports designed to be shared internally within the business.
- **Customer reports**: Reports and documentation (for example, Terms and Conditions of an Agreement, Disclaimers) to be shared with customers once the Quote agreement has been approved by internal management.

Customer reports are applicable to Maintenance and Service agreement types only.

Customer and Internal report packs contain a mixture of AMT standard reports, customer IC reports and Library documents. When setting up reports, you can select Analyse by. This will customise the reports to meet the business sales needs.

The report packages are available in Quoter.

Group Permissions

This function sets the type of access the selected group has to each function and report.

The permission levels available are:

- All - the group can add, edit and delete information presented by the function.
- View - read-only access (screens only).
- Edit - the group can edit information in the EM Issues module of AMT.
- None - the group does not have access to the function or report.

Security access for User Defined Financial Periods

Whether an AMT user can access, view, edit or modify user-defined financial periods is subject to their user permissions. A new function for user defined financial periods has been added to "Group Permissions" above. The permission options are: All, Edit, View and None. These are described in the table below as they relate to user defined financial periods.

Permission Level	Access
None	No access to user-defined financial periods.
View	Review access to user defined financial periods.
Edit	Ability to change a period from: • Closed to Open.

Permission Level	Access
	• Open to Closed.
All	Full access to user defined financial periods. Ability to change a period from: • Closed to Open. • Open to Closed. Ability to change any period to Current.

There are a number of specific workflow status changes that are valid when you attempt to change the period's status.

There are number of specific workflow status changes that are valid when a user attempts to change the status.

Current Status	Updated Status	
Closed	Current	Only possible if no asset or depreciation transactions have been posted.
Closed	Open	
Future	Current	Only possible if no asset or depreciation transactions have been posted.
Current	Closed	
Open	Closed	



It is recommended that you create a Depreciation System Administration group. Give the group a single System Administrator and a Financial Accountant, with these two users having full (All) access to user defined financial periods. Give the generic System Administration group Edit access and other groups Edit or No access.

Depending on the current configuration of AMT permissions, you could create a specific Depreciation System Administration group. Add a single System Administrator and a Financial Accountant to this group. These two users have full (All) access to user-defined financial periods. Give the generic System Administration group Edit access and other groups Edit or No access, as required.

Cost Allocation Interface Details

Use the Cost Allocation Interface Details screen to nominate which elements of the cost allocation, or job cost, interface will be updated by external systems and which element can be modified manually in AMT by users.



No changes should be made unless verified first by RPM in order to minimise potential interface issues.

External Downtime Equipment

This system table is only relevant if AMT is configured to import downtime notifications from external systems such as MineStar, Jigsaw, and Modular Mining. It is used to map the equipment name used by the external system to the equipment name registered in AMT.

External Downtime Site

This system table is only relevant if AMT is configured to import downtime notifications from external systems, such as MineStar, Jigsaw, and Modular Mining. It is used to map the site name used by the external system to the site name registered in AMT.

AMT Utilities

Refresh Reference Data

System Administration > AMT Utilities > Refresh Reference Data

Double-click this function to have the system table's cache refreshed on your local AMT client. You may need to trigger this function if the system tables have changed while you are still running the AMT client. Otherwise, every time you log into AMT, AMT refreshes the reference data automatically.

User Group Function Manager

System Administration > AMT Utilities > User Group Function Manager

This feature shows a list of all functions accessible to a specific group and the way they display in the **My Functions** tab.

Functions available to the selected group are listed under **Function Name**. Create headers and sub-headers in the adjacent window by clicking **Add**, and allocate functions to those headers, to create the hierarchy.

Recalculation Manager

System Administration > AMT Utilities > Recalculation Manager

Provides you with the ability to monitor and set projections that need recalculation.

Only current projections are recalculated automatically by AMT. You need to manually trigger all other projections to recalculate them , either through using **Recalculation Manager** or when prompted by the system.

Projection calculation statuses:

- **OK**: the projection has been recalculated.
- **Required**: the projection should be recalculated because some elements of the projection have changed.
- **Queued**: the projection is waiting to be recalculated.
- **Calculating**: the projection is currently being recalculated.

Interface Administration

The Interface Administration function in AMT (**System Administration** > **AMT Utilities** > **Interface Administration**) allows you to manually load specific interfaces into AMT by browsing to the location where the interface file has been saved. This process does not permit you to disable existing automatic interfaces, nor change the timing of when existing automatic interfaces run. The available interface import options are described below.

Overhead Costs Import

Select this option to import actual overhead costs if these costs are not already loaded into AMT using work order settlements.

Overhead Billings Import

This option is only relevant to equipment enabled as MARC (refer to Asset Ledger Overview).

Select this option to import actual overhead billings.

Planned Downtime Import (Deliverability)

This option is only relevant to deliverability installations.

Select this option to import deliverability data.

Plant Hire Charges Import

This option is only relevant to equipment enabled as Plant Hire (refer to Asset Ledger Overview).

Select this option to import actual plant hire charges.

Plant Asset Journal Import

This option is only relevant to equipment enabled as Plant Hire (refer to Asset Ledger Overview).

Add/update Parts

Select this option to import actual piece part registration or pricing data.

If the data file contains parts that do not exist in AMT, you can add the new parts, where AMT is used to define standard jobs. The default option is not to add new parts.

Usage Profile Rates Import

Select this option to import projected usage rates into AMT for equipment linked to usage profiles.

Electronic Workscope Import

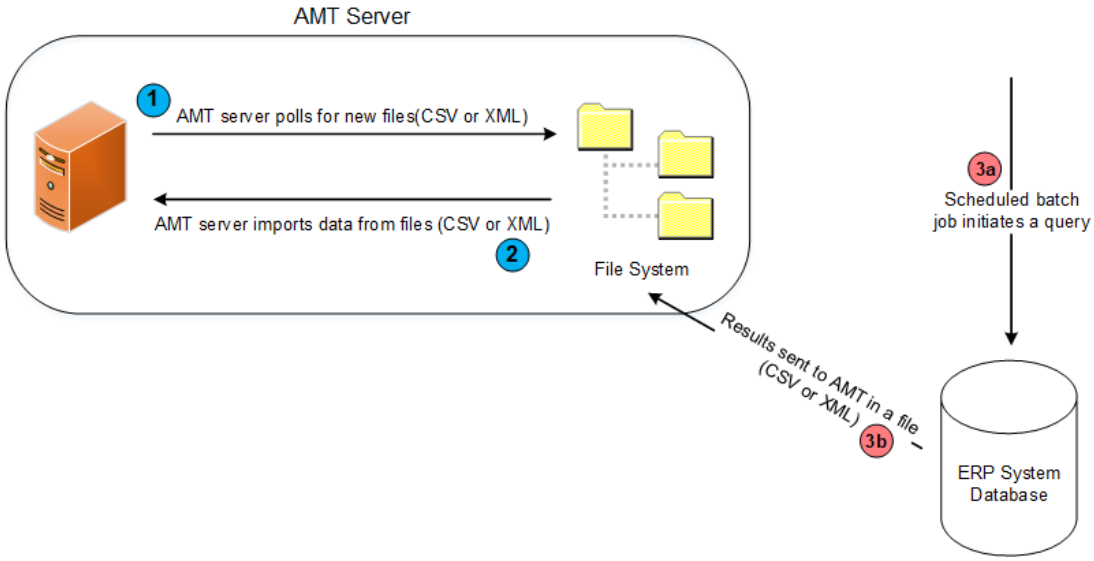
Select this option to import workscopes into AMT.

Electronic Workscope Steps Import

Select this option to import workscope steps into AMT.

Interface Data Transfer process

The illustration below shows how interface data files are passed between AMT and an ERP.



In an AMT server, the AMT Calculation and Interface Windows Service Application executes without user intervention. The service continually monitors certain file locations and imports any new files that appear in those locations. These interface files have a filename that includes an optional branch identifier and a time stamp. Once files are processed, they are moved to a Success or Failure subdirectory.

Interface file type

Interface files passed between AMT and an ERP are usually in CSV format; a file format that can be created and modified in Excel. For CSV files, the first line contains the column names, and not all columns need to be supplied. Certain system integration messages do not support the use of CSV or XML files as they are hierarchical data structures, that is, more than one entity is contained in a message.

Interface from external enterprise resource planning system

AMT receives work order settlements from an external enterprise resource planning system. Each work order settlement is identified by a work order number. The work order settlement information received by AMT from the enterprise resource planning system includes:

1. Work order settlement number.
2. Serial number and model of the equipment.
3. Codes, for example, component code, modifier code.
4. Cost (parts, cost, miscellaneous).
5. Cost Bearer, for example, contract, revenue, expenses, redo.
6. Currency and exchange rates.
7. Machine hours, for example, engine hours, and labour hours.
8. For equipment under a Maintenance and Repair Contract (MARC) AMT also receives Cost@Sell amounts.

The import routine checks if the work order settlement number exists in AMT, to determine whether to add or update a record. After the work order settlements are added or updated in AMT, the work order settlement codes are translated according to any defined code translation rules (provided the work order settlement is not reviewed).

The specific codes used for this linking are component code, modifier code, task type and task counter. The codes of the parent work order settlement are used for this linking. Therefore, it is critical that the parent work order settlement is correctly identified and if any of the work order settlements in the group match to the code translation rules, this work order settlement is set as the parent work order settlement. This maximises the change that work order settlements correctly link to strategy tasks.

You are still able to re-code the work order settlement in AMT in order to link it to a strategy task, but making the change does not update the work order settlement's codes in the enterprise resource planning system. There is no loss of the audit trail because the work order settlement number itself cannot be changed by you , therefore maintaining the "connection" between the two systems. Once you have manually set codes in AMT, you can select to mark the work order settlement as "reviewed".

Interface diagnosis

Currently there is no mechanism in AMT to generate interface errors. However, there are manual ways to check for errors. The following reports are provided through the AMT user interface:

- Generic Interface Log
- Interface Log

Using these reports, and depending on the nature of the error reported, the AMT Administrator may need to review the contents of the actual interface file. This requires access to the AMT server and the ability to copy the interface file to their local computer. When reviewing the actual file, we strongly recommend the interface specification document is at hand in order to understand the rules and data mappings used to populate the file.

Typical data issues are:

- Mandatory fields in the interface file are missing data.
- Data inserted into the file contains additional commas which can distort the separation of interface file fields. This problem is only encountered if the data in the interface file is not formatted with inverted commas; for example, oil, fuel and grease will cause a problem for CSV file processing (due to the commas), but if quote marks are used, the data is processed, for example “oil, fuel and grease”.
- Data is in the incorrect format. For example, a date field set to 31-Jan-2019 instead of 20190131.
- Data flags or elements are not set up correctly in the ERP, resulting in the wrong data being sent to AMT, data actually missing or data being sent correctly but not linking within AMT properly. For example:
    - The registration details of an asset might not be sent from the ERP to AMT if the asset is not flagged in the ERP as being required by AMT. The flag needs to be set against the asset in the ERP first.
    - Data for an asset might be sent to AMT, for example, strategy tasks, work order settlements, but as the asset is not registered in AMT, the associated data may not link to the asset properly. For example, work order settlements might link to overheads instead of linking to a proper asset.

There can be issues caused by business rule gaps not handled by the interface’s programming logic, which cause unexpected behaviour in AMT. For example, you may terminate an asset in the ERP but this may not terminate the asset in AMT. Business logic issues need to be discussed with the interface developer and potentially RPM to determine whether an interface logic change can solve the issue.



Not all business logic issues can be solved by interfaces. Where an interface cannot solve an issue, workarounds need to be defined, work procedures developed and users trained by the AMT Administrator.

Interface reprocessing

Once an interface issue has been resolved, files may need to be reprocessed into AMT. Reprocessing could be a simple case of restarting the interface between AMT and the ERP or there may be a need to reprocess a file manually.



If at all unsure about the correct approach to reprocess an interface file, contact RPM Support for assistance. Care must be taken if you manually reprocess interface files, especially when the file has been edited in Excel. Excel drops leading zeros which could cause AMT to treat a code as new and add it to a system table.

(Rpt_414) Interface Log

Contains errors found by AMT when processing certain files.

Tips

Interface Files relevant to this report are:

- ImportEquipment
- ImportProjectedTask

Initial parameters

None

Default Parameters

Report Type, Date Selector

Additional filters

File Name

Interface Log

This report displays the errors found by AMT from and to the external ERP interface for the following interface files:

- Import Equipment.
- Import Strategy Tasks.

The System Administrator can review this log to determine the root cause of why data was not inserted when the interface last ran.

(Rpt_455) Generic Interface Log

Reviews error messages related to importing data for from external systems.

Report Parameters

Report Type, Data Selector

Additional filters

Error Type, Import Reference

Generic Interface Log

This report shows errors generated by any interface file imported into AMT. The report lists the filename, error description, line of the interface that contains the error and the date when the interface file was processed.

Export Administration

It is possible to export out of AMT in the format that interfaces use to load that same data into AMT. This function provides a quick and simple way to extract, cleanse (modify) and reload data into AMT. The Export options available are described below.

Export Equipment

Equipment registration data can be exported out of AMT for specific branches, sites, fleets and equipment.

The output format is the same as the interface format used to load this data automatically into AMT.

Export Strategy Tasks

Strategy task data can be exported out of AMT for specific branches, sites, fleets and equipment.

The output format is the same as the interface format used to load this data automatically into AMT. Does not export cost allocations or standard jobs linked to strategy tasks.

Export Utilisation

Actual usages data, for example actual service metre unit readings, actual km readings, can be exported out of AMT for specific branches, sites, fleets and equipment.

The output format is the same as the interface format used to load this data automatically into AMT.Export Work Order Settlements

Actual work order settlement cost data can be exported out of AMT for specific branches, sites, fleets and equipment.

The output format is the same as the interface format used to load this data automatically into AMT. However, Export Option fields, apart from **Convert to Primary Currency**, need to be selected if you want to reload this data.

Planned Downtime Export

Export Planned Downtime.

All of the above exports can be scheduled to occur automatically by completing the automatic export details for the specific data extraction process created.

Language Type

In AMT you can define languages, but a language does not have to be a standard spoken language. It could be anything, for example, British Slang could be a language.

You can create a language using the Language Type function and then assign it to an AMT user. In AMT it is possible for one user to use German and another user to use British Slang.

Once you have created a language type, translations for each caption and label in AMT can be edited using **Language Editor**. The translation is seen on screens, dialog boxes and in reports. The list of translatable captions/labels displayed in the Editor grows as users enter new translations.

Notes

- AMT does not translate data entered into AMT by users, only captions and labels. For example, the field label Component Code can be translated, but not the actual component codes entered into the component code system table.
- RPMGlobal does not maintain or update translations created by System Administrators.
- The default AMT language is English. The maximum length of all captions and labels in AMT is based on the English word.
- AMT only supports the Gregorian calendar.
- AMT only supports left-to-right display of text.

Language Editor

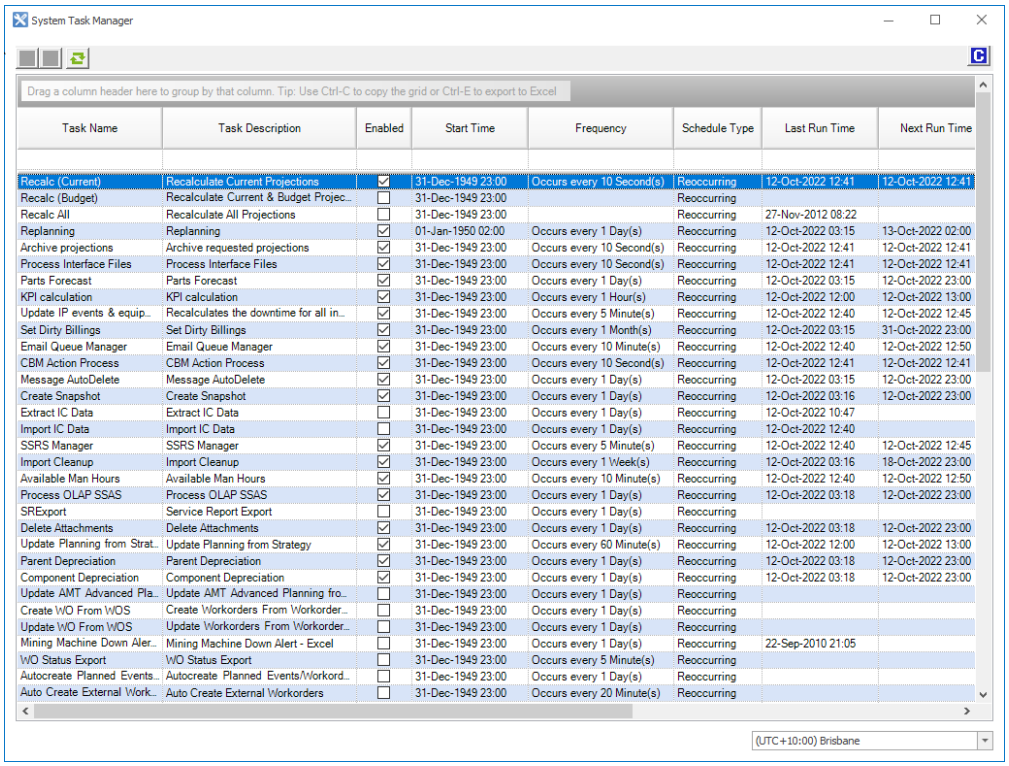
This function is used to customise and translate the labels in AMT for the Language.

System Task Manager

System Task Manager allows a System Administrator to review when internal AMT database calculation processes were last performed, and to manually start a process to execute immediately. The System Administrator can queue and run system tasks on demand by right-clicking a task and selecting **Run Now**, and view when a system task was last run. The screen displays details such as last run date and time, next scheduled run date and time.

The System Administrator should be careful not to trigger a process that may consume valuable AMT Server resources, and consequently slow down AMT performance.

To access the System Task Manager, click **System Administration** > **AMT Utilities** > **System Task Manager**. An example screenshot displays below.



Create Component Changeout Journal

This function is only relevant to Plant Hire installations of AMT.

Use this function to generate plant asset journals for component changeout work order settlements.

Create Work Order Settlement History

This function allows you to generate the work order settlement history for strategy tasks flagged **Auto generate Work Order History**.

AMT prevents you from triggering the creation of work order settlement history if the software is busy creating work order settlements from the last time you triggered the process. This prevents duplicate settlements being created.

Log Entries

Displays the entries recorded in AMT logs. See System Table - Log List Editor.

Process Work Order Settlements

This function generates work order settlements for a strategy task where the strategy task is flagged to **Auto generate work order history**.

If the strategy task is a planning task, a work order settlement will only be created when the strategy task is overdue. The work order settlement amount will match the strategy task amount at the time the Process Work Order Settlement process is triggered.

For non-planning and unassigned tasks, a work order settlement will be created per month based on strategy task assumptions. The work order settlement amount will match the strategy task amount at the time the Process Work Order Settlement is triggered.

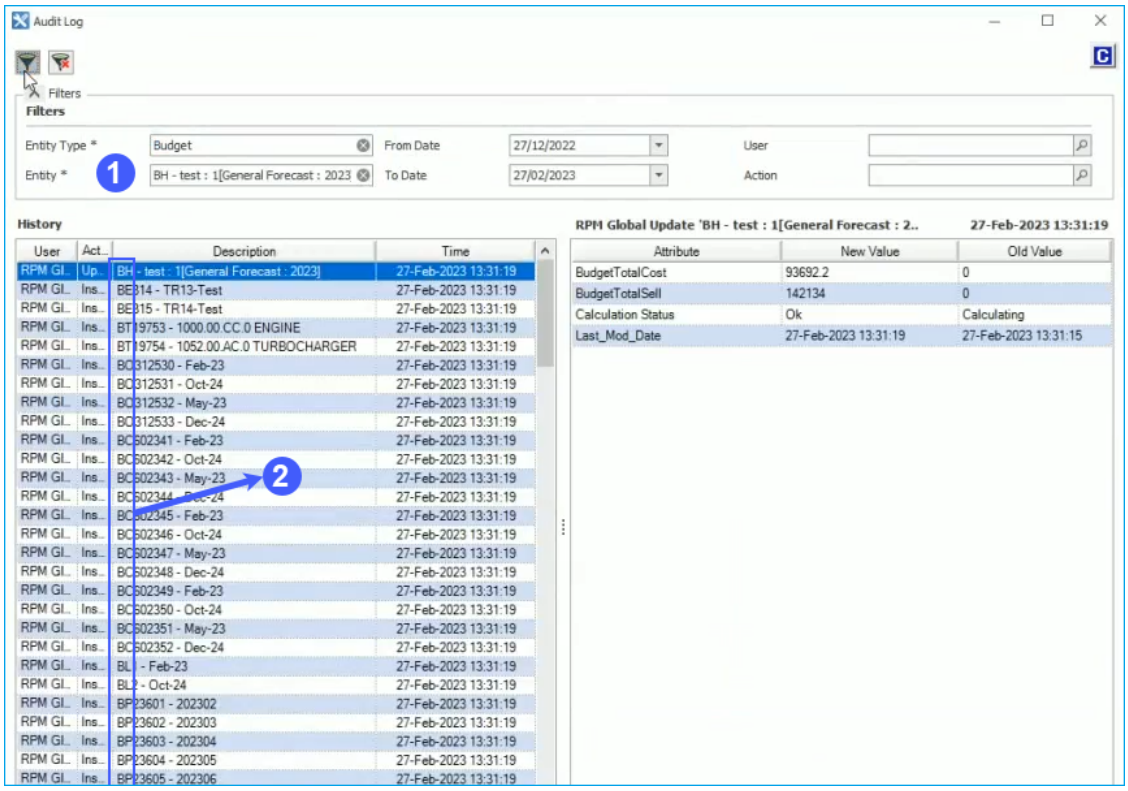
For non-planning and unassigned tasks, a work order settlement will be created per month based on strategy task assumptions. For example, for a strategy task with a cost of $200 and a frequency of 100 hours, if the equipment utilised 322 hours in the month, the cost of the work order settlement created will be $644 (that is, $200 divided by 100, then multiplied by 322).

The resultant work order settlements created by this process can be viewed using Work Order Settlement Manager.

Audit Log

This screen displays the history of changes to selected data between the From and To dates, including before and after values.

When you select the **Entity Type**, **Entity** becomes **mandatory**and you must select an Entity (1 below).



The alphabetic prefixes of the Description (2 above) are as follows:

- BH - Budget Header
- BE - Budget Equipment
- BT - Budget Task

- BO - Budget Occurrence
- BC - Budget Cost
- BL - Budget Labour Hours
- BP - Budget Period
- BU - Budget Period Usage
- BR - Budget RVIC

AMT Audit Log Report

This report provides a history of any AMT database tables enabled, and configured, for logging. The report groups by Entity (for example, Asset Condition), who has changed what and when, giving an auditable record of new and old values. Audit results are presented in reverse chronological order (latest to oldest).

When you select the **Entity Type**, **Entity** becomes **mandatory**and you must select an Entity.

The **Date Selector** defaults to the last two months. Clear the Date Selector check box to choose a different date range.



The date is the **budgeting logging period**, not the budget period.

(Rpt_842) AMT Audit Log Report

Initial parameters

|ControlType|,|Idlist|,|TextList|,|LogoLocation|,|UserID|,|LanguageKey|,
|ReportStyleKey|

Tips

The report can be run for all changes across all entities (equipment, functional location, strategy task or standard job), or by selecting one or more entities. Changes can be reported "from a date" to the current date and time by leaving the To Date and Time blank.

Report parameters

Default.

Additional filters

Entity Type, Entity, User, Action

Generic Value Pairs

Generic Value Pairs allow you to manage the generic linking of data items. You can create, modify and delete Value Pair records. Value Pair records have a **Type** used to group like records. Record names must be unique within its type.

You cannot delete a Value Pair type if records exist that reference the type.

Terminology

A

Abandoned Work Order	A work order that is neither physically completed nor closed out in any system.
Absolute Cost per Hour	A method for calculating the maintenance cost per hour for an asset. It is calculated by dividing a strategy task's cost by its frequency. Unlike traditional cost-per-hour calculation methods, this method accounts for every hour that an asset is in use. The relative cost incurred by using components must be accounted for; not just when the components need to be replaced.
Account Manager	Person responsible for a sales account. This is relevant to OEMs and Dealers who use AMT.
Account Manager Group	A particular group of account managers. This is relevant to OEMs and Dealers who use AMT.
Action Required	In the lead time prior to when a strategy task is due to be performed (sometimes referred to as the "planning window"), you can assess if the strategy task should be performed on its due date/usage or on another date/usage. The due date/usage is calculated by AMT automatically. During lead time, when the strategy task is due to be performed on the automatic due date/usage (referred to as the Strategy Date / Usage), it's in a state of action required until the due date/usage is updated by the user. You don't have to change the due date/usage, but you can review it and change it. It is a business process issue whether you change or confirm the strategy task due date/usage during the lead time.
Activity	An activity is a maintenance activity that is performed during a downtime event. Examples of maintenance activities include performing maintenance, waiting parts, waiting labour. There is a relationship between activities and variance cause.
Actual Cost Variance (ACV)	The difference between budgeted costs to date (from a budget projection) and actual costs to date (from a current projection).
Actual PSS	Actual parts and service sales. These are actual parts and service (labour) sales that AMT accepts verbatim from the

	external system. This term is used in the Product Support Marketing module.
Actual Usages	In AMT, a usage is a cumulative reading, such as 577 engine hours or 2,1000 km. An actual usage is the true usage reading at a particular point in time.
Admin Alert	A warning in the work order section of the Equipment Status module that indicates either (i) the strategy work order is not linked to a work order settlement or (ii) the status of the work order does not align to the status of the event to which the work order is linked.
Advanced Planning	If an asset is configured with Advanced Planning mode, you can group work orders into maintenance events. This mode focuses on mobile equipment and productive unit scenarios, and is suited for capturing downtime information, as it integrates with the Equipment Status module.
Advanced Utilisation	If an asset is configured with Advanced Utilisation mode, you can track multiple usages relating to its life, or the life of the asset's components. You can also vary the usage forecast of the asset over time. Assets configured for advanced utilisation require a usage profile.
Alternate Projection	An alternate projection is a different, what-if scenario that you use to compare to your current, or any other projection. To create an alternate projection, copy any projection, and change some of the parameters to test them. An alternate projection continues to reflect the actual costs incurred (similar to the current projection), but live forecasts used by AMT are calculated in the current projection.
Application Details	You can add application details to an asset when you create it in AMT. You configure what the application details are on the Equipment Group system table. Application details typically relate to the physical area of the site, such as high altitude or hard rock.
Archive Projection	A projection can be archived. When you archive a projection, you lock the parameters of the projection at that point in time. If you open the archive and report from it, the archive's parameters do not change over time. Often, you have a current projection and you create an alternate projection to consider some what-if scenarios. To keep a record of the maintenance strategy in the current projection, you archive it, and copy the alternate projection to override the current projection. Only projection parameters are copied, not work orders.
Asset	An asset is plant, machinery, property, buildings, vehicles and other items that have a distinct value to the organisation. AMT covers machinery assets; an asset is usually called an equipment. These two terms are interchangeable.
Asset Condition	Asset Condition is used to identify the overall health of an asset, for example, As New, Fair, Poor, Very Poor.
Asset Register	Asset Register is an equipment configuration, which enables you to record capital details

	of an equipment. For example, the asset's estimated disposal price, whether the asset is owned or leased, create a depreciation schedule, view the asset's written down value.
Auto Accrue Billings	In AMT, you set up a billings schedule for maintenance; this schedule defines how and when you invoice the customer. When auto accrue billings is turned on, AMT assumes that you follow the billings schedule, regardless of when you actually send invoices. You may have only sent invoices to a value of $90,000 by 20 September, 2021; if you had followed the billings schedule exactly, you should have sent invoices to a value of $100,000. In this scenario, AMT accrues $100,000, not the actual value of $90,000. Auto accrue billings are specific to the company's billings process. This functionality is designed for maintenance contracts with an external customer.
Auto Update	Auto update keeps the expected annual usage (utilisation) in synchronisation with the actual usage history, for assets configured in Standard Utilisation mode. For example, if the average actual usage for the last 6 months varies more than 5% from the annual usage, AMT updates the annual usage to the average for the last 6 months.
Availability	The availability of the asset to be used in production. The availability method is the formula used to calculate availability. Availability is defined from either a physical or a mechanical perspective. Each asset is assigned a default availability method based on the fleet to which the asset is linked. Nevertheless, even though an asset may be assigned a default availability method, you can still report availability on the asset using another method.
Availability – Mechanical	Mechanical availability is calculated as operating hours divided by (operating hours plus downtime).
Availability – Physical	Physical availability is calculated as (scheduled hours minus downtime) divided by scheduled hours. Scheduled hours are defined using the Scheduled Hours function.
Availability Target	An agreed level of productivity, for example 95%, for an asset or fleet. For assets under contract, the availability target is defined in the contract document.

Go to top

B

Backlog	In AMT, a backlog is a work order or defect that needs to be performed on an asset. The backlog list in AMT includes work orders for strategy tasks not yet performed (including current and overdue work orders), and any work orders to be performed in the future.
Base Potential	The theoretical potential parts, labour and miscellaneous sales a dealer could have made from the assets.

	Systems definition: Base potential is a summary of current projections for assets flagged for sales forecasting.
Bill of Materials (BOM)	The bill of materials, or spare parts list, contains all of the parts and consumables required to complete a specific job.
Billing Rate	The amount per unit of measure to be charged to the customer according to the contract billing schedule. For example, dollars or roubles per hour.
Billing Schedule	The billing schedule is outlined in the maintenance contract for the asset. The billing schedule includes the rates at which the customer is charged for the contracted repair and maintenance activities. An asset can have more than one billing schedule at one time. For example, a customer has one rate for most tasks, except a particular task that has a lower rate.
Billing Step	A billing step is a period of time during which a specific billing rate applies. The period could be a defined period of asset usage or a specific date range.
Billings	Billings are the amounts that have been billed or invoiced to the customer, and will be billed or invoiced to the customer in future. Billings are calculated at the asset level, following the billing schedule. In AMT, billings relate specifically to maintenance contract assets.
Branch	A branch is a centre that is responsible for managing a group of sites. A branch is the second level in the equipment hierarchy - Organisation > branch > site > fleet.
Branch Code	The branch code typically identifies where costs for assets are captured in the accounting system. For example, for dealers, the branch code represents a physical branch location. For other AMT users, the branch code might represent location or area of responsibility. The individual deployments decide if and how branches are used.
Breakdown	A breakdown occurs when assets become unexpectedly unavailable for production. A breakdown is one classification of an event. A work order for an asset configured in Standard Planning mode can be flagged as a breakdown work order.
Budget Projection	The budget projection establishes a baseline for future comparison with another AMT projection, usually the current projection. For assets under a maintenance contract, the budget projection usually reflects the commercial assumptions made to determine the contractual billing rates. For assets not under a maintenance contract, the budget projection typically represents the assumptions about the future maintenance life cycle that you had when you first bought the equipment.

Budget Pricing Date	The date when the budget projection pricing was set. This date is an index date for future escalations (changes in pricing) of all budget projection strategy tasks. The budget pricing date must be earlier than or equal to the asset plan's start date. For assets under a maintenance contract, the pricing date should reflect the date in the original contract. For example, the prices in truck contract A were relevant at the time of purchase, 1-July-2019. For billings, you need to know from which date to index the pricing. Budget pricing dates are relevant for OEMs and contractors.
Budget Contract Variance (budget variance in the system)	For an asset's total life cycle, the budget variance is the difference between total budget costs at sell (unescalated) and total budget billings. The variance should be zero, unless there is a good reason for it to be more or less than zero. Only assets under a maintenance contract have their budget costs and billings tracked. Budget contract variance is called budget variance in the AMT product.
Business Economic Code (BEC)	A reporting code used in AMT's Product Support Marketing module by OEMs and Dealers.
Business Group	The type of business specific to the asset being registered, for example, mining group, plant group, rental group. This term is used in Contractor and OEM reporting.

Go to top

C

Calculated Current Usage	The calculated current usage of an asset or component's formula is: Last Actual Reading Usage + (Projected Daily Usage * Days since Last Usage Actual Reading).
Calendar Period	A calendar month, defined in the format YYYYMM.
Calendar Year	A period used for defining consumption. In AMT, a calendar year is averaged out to 365.25 days every year. AMT has considered the extra day that occurs in a leap year, and included it in yearly calculations. For example, 5,000 engine hours per calendar year = 13.689 hours per day.
Cause	Cause is the reason for an asset failure. For example, engine oil leaking.
Centreline	Centrelines are generic projections for assets; that is, they are templates or baselines. Generally, you maintain one centreline for each asset model but you can create as many centrelines as desired. Typically, when a new model is released,

	documentation is created listing the model's components, expected life of the components, cost of replacing the components, and so on. All of this base information is included in a centreline.
Charge Hours	The number of maintenance hours an asset has consumed in a given period, based on which the customer is billed a certain amount using the billing step.
Close Period	The financial period when a work order settlement is closed in either the ERP system or in AMT, if AMT is automatically creating work orders.
Commission Date	The date when the asset started to be used in production – it arrived on site, was tested, and was approved for use. For new assets, this date is the same as the asset's start date. For second-hand assets or rolled over assets, the date is different. AMT uses this date to record the age of the asset for availability forecasting.
Commodity Code	A reporting code used in AMT's Product Support Marketing module by OEMs and Dealers.
Compartment	The section of a component in which fluids are stored, for example, transmission oil, engine oil, hydraulic oil.
Component	A component can be physical or non-physical. If the component is physical, it tends to be a maintainable item. There is some variation depending on AMT setup and business requirements. In some AMT reports and screens, component is used instead of strategy task. This is contextual.
Component Code	Component codes are part of a component code hierarchy. A component of an asset is an assembly, sub-assembly, or part of an asset. The component code hierarchy, with examples, is cost types (repair and maintenance) > system (engine) > sub-system (fuel system) > and component code (code C0723 and description fuel tank). Sometimes a component code will not reflect a maintainable part of an asset. A component code may be used to represent the type of work being performed (such as a 250 Hour Service) or a type of non-maintenance asset cost (for example, Depreciation, Finance, Insurance).
Component Structure	Component structures are used in AMT to identify the failure modes of a specific component of a specific model of asset. For each failure mode identified, you can quantify how often the failure mode occurs, cost of repair, labour hours required, and downtime incurred. You can document the preventative maintenance required to prevent the failure mode, and how to predict the failure will occur. This term is used in the Asset Health module of AMT.

Component Based Depreciation (CBD)	A method of depreciation, based on components, where the cost of replacing a component is accumulated over the life of the component. This method of depreciation is in addition to a standard method of depreciation of the entire asset over its useful life.
Condition Based Maintenance (CBM)	Condition-based maintenance is the process of interpreting condition monitoring data against reliability assumptions.
Condition Monitoring Intervention	A work order that was created based on a condition monitoring translation rule alert level or follow-up action.
Configuration Details	The specific configuration attributes of an equipment group that can be included or excluded when you create a new asset in AMT. For example, transmission type, engine type, bucket size, drill diameter.
Contract	In AMT, a contract is between two parties. A contract can refer to different maintenance, repair and/or services contracts usually provided by OEMs or contractors.
Contract Term	For assets that are on a usage-based maintenance contract, the length of the contract is expressed as a UOM value (for example, 30,000 engine hours) or a date.
Contract Type	The contract type is a grouping of similar contracts, for example, MARC, service agreement, rental.
Contract Variance	A change in the actual documented terms of a maintenance or rental contract.
Contribution	The contribution is the billings minus the cost of sales and minus the expenses.
Contribution Recognition	Contribution recognition is the contribution amount taken to an organisation's profit and loss, per contract.
Cost at Cost	The purchase cost of parts, labour and miscellaneous expenses. Historic actual cost comes from work order settlements. AMT calculates the future cost at cost using margins and escalation.
Cost at Sell	Original equipment manufacturers carry two costs, true cost (cost at cost) and the cost at which they sold something to someone else. The latter is called cost at sell, or sell price. The cost at sell is the calculated value of parts, labour and miscellaneous expenses, including a margin, added to costs (cost at cost) to allow for parts and service departmental overhead recovery and a profit margin.
Cost Bearer	The cost bearer is used to distinguish between maintenance and other costs.

Cost Cap Agreement	A cost cap agreement is a type of contract provided by OEMs that has components costed at a fixed price. Cost cap agreements do not contain billings schedules.
Cost Centre	A cost centre is typically an accounting grouping that represents an organisational, departmental or process entity of a company, and exists in the company's accounting system. For example, a branch or a site could be a cost centre. A default cost centre is assigned for each asset; specific cost centres can be assigned for each strategy task.
Cost Responsibility	The cost responsibility is the ability to group cost centres. It is primarily used for financial and budgetary reporting.
Cost To Date	Actual costs incurred to date; the costs are typically sourced from the ERP system.
Cost Type	Cost types are part of a component code hierarchy. Examples of cost types include repair and maintenance (R&M), capital, tyres.
Cumulative Usage	The last actual usage reading, plus the offset reading.
Current Projection	The current projection reflects actual costs incurred to date, plus future occurrences and costs from today (system date) to the end date or end usage of the asset. The current projection usually reflects all the current strategy task assumptions. All "live" forecasts (that is, availability, component, financial, parts, and labour) are calculated using the current projection. Therefore, changes to the current projection immediately change the live forecasts. Only current and alternate projections contains actual costs.
Customer	Customers are the owners of registered assets, or the owning party in agreements.

Go to top

D

Date Notified	The initial date when a new backlog work order was identified. The identification date can be earlier than the creation date, as an issue can be identified on one day, but the backlog work order for it created in AMT on the following day.
Defect	A defect is a repair required on an asset. Defects are generally identified by daily inspections, operator reports, and so on. A defect work order is a one-off work order that is not regularly scheduled to be performed as part of the maintenance strategy. Defect work orders become part of the AMT backlog.
Dependency	Dependency occurs when strategy tasks require prior strategy tasks to be completed in a particular order (ABCD, ABCD). The dependency order is indicated by the strategy task priority – the lowest priority is scheduled first, followed by the second-lowest

	number (1, then 2, then 3, an so on). When the cycle of strategy tasks is complete, the sequence starts again from the lowest number. For example, if you have strategy tasks that are always executed in a sequence, such as PM service 1 followed by PM service 2 followed by PM service 3, and then start again at PM service 1, dependency logic is applied.
Document Type	Documents are grouped together by type, to make it easier to search for a specific document. AMT has straightforward library capability and can group documents as procedures, flow charts, and so on.
Down Delay	The down delay is the reason why an asset is not available for production, for example, inclement weather or waiting for tyre handler.
Downtime	The downtime is the time that the asset is not available for production.
Downtime Allocation	Downtime is allocated between different codes that represent different responsibilities or different activities. Different departments (usually maintenance and production, but not always) are responsible for the activities during downtime, for example, waiting for parts, waiting for labour or washing the machine.
Draft Estimate	Modelling equipment can have many estimate-draft projections. Estimate-draft projections are the previously saved versions of the estimate-current projection.
Duration vs. Downtime	Duration is the difference between the start and end times for the work order. Downtime is the difference between the start and end times for the event. For example, an asset has 7 hours of downtime, during which 3 work orders were performed. The sum of the duration for the work orders is 14 hours, because some of the work orders were performed concurrently.

Go to top

E

Electronic Workscope	A workscope created in AMT that is used by the AMT Mobile Inspections module.
Element	An element in the same sense as the elements of the periodic table. An element can occur within a measurement; for example, the iron from an oil sample.
Employee	The Employee system table contains groups for particular sites, including both internal employees and external contractors.
Employee Number	A unique employee identification number is stored for each employee in the Employee system table. The employee number is how an employee is referenced in AMT. The employee number maps to the employee identification number in the ERP system.

End Of Life	When a component is at its end of life, it no longer has a use.
End Usage	The projected last reading of a projection or asset.
Entry Distribution Codes	In AMT, entry distribution codes allow accounting information to be recorded to drive journal processing. These codes are used by plant or equipment hire and asset capitalisation. Entry distribution codes are related to capital expenditure.
Equipment	In AMT, an equipment is an asset or piece of machinery, an item to which you can associate a maintenance strategy. In a project, each equipment is referenced as a specific serial numbered item. The last level of the AMT default equipment hierarchy, that is, organisation > branch > site > fleet. For example, mobile equipment, process plant, productive unit, rail.
Equipment Category	The Equipment Category system table is used for reporting. Organisations use this code for different purposes. For example, mobile mining, infrastructure, fixed plant. It is set at the asset level. For an OEM, example categories are contract equipment and non-contract equipment.
Equipment Class	Equipment Class is a system table. An equipment class is a group of equipment groups.
Equipment Group	An equipment group is a group of equipment models, for example, off-highway trucks (this is an OEM example). Each equipment group has its own technical details. An equipment group in AMT must belong to an equipment class.
Equipment Hierarchy (in screens on AMT)	In AMT, the equipment (or asset) hierarchy consists of these entities: Organisation > branch > site > fleet.
Equipment Management Issue (EM Issue)	An equipment management issue is any issue relating to an asset, such as application, operation, engineering, product problems, and so on. Equipment management issues have a significant impact on downtime or costs. An EM issue also has a resolution or continuous improvement process associated with it.
Equipment Start Date	The start date of the equipment's current life cycle. For a brand new equipment, this is the commissioning date. For equipment under a maintenance contract, it's the start date under the contract. The commencement of an asset or maintenance contract.
Equipment Serial	A unique alphanumeric combination that identifies a single, real asset. The serial

Number	number need not be an actual serial number. This number is one of three fields used to uniquely identify equipment.
ERP Maintenance Plan	The unique identifier of an ERP maintenance plan to which the AMT strategy task is linked.
ERP Scheduling Task	Refer to Scheduling Task.
Escalation	An escalation is the price changes over time; it is also referred to as indexation. If a budget projection is priced in year 1 for an asset spanning several years, actual costs could be higher than the budget pricing due to inflation over time. To align comparisons between budget and current projections, escalation is applied to budget prices to adjust values to current levels. To align comparisons between historic actual costs and today's cost, escalation is applied to actual costs to enable meaningful analysis. Future projections of both current and budget are made at today's escalated values.
Estimate Projection	Estimate projections are relevant to modelling equipment, and represent the life cycle cost of that equipment. Estimates are used to model a specific scenario, for example, to purchase new trucks and a loader. Estimates are specific to an environment (hard rock, high altitude) and point in time. You can also create multiple versions of estimates. For asset suppliers, estimates are prepared in maintenance contracts tendering situations. For others, estimates are used to evaluate tenders from different suppliers.
Event	An event is an occurrence of downtime. A planned event is a planned stoppage of an asset for maintenance. For example, if an asset is down for maintenance for 9 hours, that is called an event in AMT. During this period, the service meter unit (SMU) hours would normally be the same as at the beginning of the downtime. In some cases, the SMU hours increase due to asset testing as part of the downtime event.
Event Prevention	Event prevention are actions that may prevent a future breakdown event. Event prevention actions are the answers to the question "How could we have stopped this?"
Eventscope	An event scope provides an overview to the planned event for the asset, including a brief description of what will be performed, to which asset, when, and the budgeted labour hours and duration. The event scope also includes the list of work orders to be executed during the event.
Exceeds Projected Usage	The exceeds projected usage error flag indicates that an actual usage reading exceeds the estimated usage calculated by AMT, based on the asset's current projection usage profile or annual utilisation (usage) assumption.

Expense Element	Typically, expense elements reflect the different categories of costs and exist in the company's accounting system. Expense elements can be set for parts, labour, and miscellaneous items within each strategy task and work order.
Expenses	Costs, typically associated with a contract, that are not itemised in an asset. Expenses are overhead costs.

Go to top

F

Failure Mode	A particular way that a component fails, in terms of symptoms, behaviours, or internal state changes. For example, ffailure modes of an engine include overheating and vibration.
Financial Period	Financial periods are the organisation's accounting periods. If the financial period is different to the calendar period, a translation process converts financial year and month records to calendar year and month, in the format YYYYMM. For example, if the organisation's financial year ends in June 2023, and an accounting record for a work order settlement has an invoice period of 202308, this translates to February 2023 in calendar years (202302).
First	The UOM at which the first occurrence of a strategy task is expected to be performed. For example, an engine service at 15,000 hours.
Fleet	A fleet typically represents a group of like assets.
Frequency	The UOM or time between projected occurrences, for example, 15,000 engine hours between each engine replacement.
Frequency Multiplier	Frequency multiplier automatically recalculates the next occurrence of overdue non-planning strategy tasks by multiplying the strategy task's frequency by the frequency multiplier value. The frequency multiplier factor is a global setting and is configured when AMT is installed.
Future Cost	The forecasted cost from today to the end of the asset's life cycle, calculated using the asset's projections.

Go to top

G

Global Cause	A code for the possible global causes of the occurrence of a work order (for example,

Code	fatigue, cracked, over pressure). Global cause codes are mapped to local cause codes to translate imported data, such as standard jobs. Global cause codes are specific to the Intelligence Centre module's reporting and data consolidation functionality.
Global Component Code	A list of the possible global components on an asset. Global component codes map to local component codes, so that data that is imported between different site or branch databases is matched. A local component code relates to a specific database that is used by specific sites or branches. The local code is converted to a global code when the data is transferred to head office's database. For example, local code A982 Fuel Tank is converted to global code 0033 Fuel Tank. Global component codes are specific to the Intelligence Centre module's reporting and data consolidation functionality.
Global Job Code	A list of the types of global jobs that can be performed when a work order is executed, for example, remove and install, replace with exchange, perform repair, test and adjust. Global job codes are mapped to local job codes to translate imported data, such as standard jobs. Global job codes are specific to the Intelligence Centre module's reporting and data consolidation functionality.
Global Part Type Code	The code for a part type that is available throughout all branches. Some example part types are service exchange, new, re-manufactured, fluid. Global part type codes are mapped to local part type codes to translate imported data, such as standard jobs. Global part type codes are specific to the Intelligence Centre module's reporting and data consolidation functionality.
Gross Margin	The gross margin is calculated as billings less the cost of sales. Gross Margin % = (1 - (Cost of Sales)/Billings) * 100
Group Number	Number that represents a group of parts. Sometimes referred to as an assembly number.

Go to top

I

Industry	Customers are grouped into the industries in which they operate, for example, mining, oil and gas. Equipment is assigned to an industry. See also Industry Group.

Industry Group	An industry can be divided into groups to further define the specific activities of a customer. For example, mining can have the industry groups coal mining, gold mining, and so on.
Inspections	An inspection is when you view and assess an equipment, checking for potential repairs and other maintenance work.
Inspection Logic	Inspection logic is a type of scheduling logic used typically when you schedule a preventive maintenance (PM) service. You schedule an inspection to occur a defined number of hours or days prior to the PM service, so that you can identify any other issues that need to be corrected during the service.
Interface	An interface is a program that connects two software programs, so that they can send and receive data between one another, such as an application program interface (API). The interface ensures that the data exchanged between the two programs is translated so that each system receives data in a format that it understands. This relates to a series of system administration functions in AMT.
Interval	An interval is the same as a frequency in AMT. An interval is the UOM or time between projected occurrences, for example, 15,000 engine hours between each engine replacement.
Invalid Usage	A usage reading that does not comply with the data validation rules. An invalid usage is either lower than the previous usage, or is greater than the usage tolerance percentage. Invalid usage tolerance is defined in the configuration of the database; it is not an element that you can see in AMT.
Invoice number	The invoice number of the billing charged to the customer. Invoice numbers are related to equipment that is under maintenance contracts.
Issue Type	General classifications given to equipment management (EM) issues, for example, product improvement, application, operation. Issue types are specific to the Equipment Management Issues module in AMT.

Go to top

J

Job Code	The job code identifies a specific job within a strategy task. For example, J09 – Remove and Install, J10 – Replace with Exchange, J11 – Weld, J12 – Service. A task type is a high level description of the work you are performing; a job code provides more specific detail of the work. For example, tightening, welding, painting.
Job Date	The date when the work order was executed. If the work order hasn't been executed yet, the job date is the plan date.

Job Folder	A job folder is a physical folder that contains all required information for the maintenance worker to execute a job. The information in the folder includes the event scope, work scope, work order instructions, and linked documents (for example, permits, timesheets, bill of materials). Historically, the job folder was a physical folder or pack. In AMT, the job folder generates the printed documents needed to execute the job.

Go to top

L

Labour	Labour is the time or cost of the labour force working on equipment, usually employees.
Labour Activity	In standard jobs, the labour activity describes the type of labour to perform the job, for example, electrician, mechanic.
Labour Hours	The number of productive labour hours required or consumed when performing work. For example, if a work order was executed in 2 hours, and 3 people worked on it, then the total labour hours is 6.
Labour Rate	The labour rate is the cost per time period (hour, day, and so on) for a particular combination of labour activity, branch and currency.
Last Occurrence Date	The date on which the last occurrence of a strategy task happened.
Last Occurrence Usage	The usage of the last occurrence from work order settlement records.
Last Offset	The offset is the difference between a usage reading and the cumulative usage reading. Often an hour meter is replaced, which requires an offset to be applied to its reading to make the future readings accurate. The last offset is the most recent offset usage reading.
Last Performed UOM/Date	The usage or date when the strategy task was last executed. This date is used in certain scheduling setups to forecast the next occurrence of a strategy task.
Last Reading	The most recent actual usage entry for an asset.
Last Scheduled UOM/Date	The usage (UOM) or date when a strategy task was last scheduled to occur. The scheduled date can be different to the actual execution date. The next forecasted strategy task can be scheduled based on the last scheduled usage or date. If the performed usage or date is different to the scheduled usage or date, the forecast strategy task's schedule does not change. If you change the next occurrence usage or date of a strategy task manually, for

	example in the Long-term Planning screen, AMT will log this updated usage/date as the Last Scheduled point.
Lead Time (days)	Typically, lead time is the number of days required to order and receive the parts needed for the strategy task. For example, if it takes six months to order and receive an engine, the lead time is 180 days. If AMT's planning module is deployed, and depending on system configuration, AMT automatically creates a work order when today's date = the strategy task next occurrence date minus lead time days.
Life Confirmed	By default, an AMT work order is flagged as life confirmed, that is, the work order should be considered an actual occurrence of a strategy task and included in AMT reports that analyse average strategy task cost and frequency. If this is not the case, the work order should not be flagged as life confirmed and excluded from such analysis. Typically, life confirmed only relates to component change out strategy tasks and work orders.
Life Cycle Costing	Life cycle costing forecasts all maintenance activities for an equipment to the end of its useful life. Actual costs to date are included in the costing.

Go to top

M

Maintenance Agreement	A maintenance agreement is a contract. In AMT, a contract is between two parties. A contract can refer to different maintenance, repair and/or services contracts usually provided by OEMs or contractors.
Maintenance Type	High-level maintenance code used for reporting purposes. The maintenance type links to task type. Maintenance type is used in some ERP systems.
Manufacturer	The manufacturer of the asset that is registered in AMT.
MARC	In AMT, MARC refers to any type of maintenance contract. MARC is an industry term for maintenance and repair contracts between a dealer and a customer. MARC is specific to the OEM and Contractor modules.
Market Share	The proportion of potential sales within a defined market (region, equipment type, and so on) that a dealer has made. Market share only applies to original equipment manufacturers, and can be shown as an amount or a percentage.
Marketing Group	Used for reporting by AMT's Product Support Marketing module.
Measurement	A measurement is a reading or statistic, for example, air temperature, iron level, or fuel usage at a particular point in time. A measurement is not necessarily cumulative

	and it does not drive the life cycle calculation of a strategy task. Measurement is related to the Equipment Health module
Miscellaneous (Misc.)	Miscellaneous costs that are not designated as parts or labour.
Model	A specific series of asset, with similar construction, features and operation (generally a manufacturer model). A model is usually represented as a particular name or code.
Modelling Equipment	Modelling equipment is used to model a scenario, for example, purchasing new trucks and a loader, to help you decide if you want to purchase that equipment. Modelling equipment is specific to an application and a point in time. OEMs use modelling equipment when they work on tenders and contracts. Companies use modelling equipment to evaluate tenders from different suppliers. Modelling equipment can also be used to model life extensions of existing equipment in operation. OEMs may also use modelling to model contract extensions.
Modifier Code	The modifier code provides an additional reference to the component code, typically the component's location on the asset. For example, RF - Right Front, TOP – Top.
MTBS	Mean time between stoppages.
MTBUS	Mean time between stoppages which are not classified in AMT as a planned event.
MTTR	Mean time to repair.

Go to top

N

Next Usage	The usage at which the next projected occurrence of a strategy task occurs, for example, the next PM Service is at 7,500 engine hours.
Non-Planning (strategy) Task	Planning mode assigned to a strategy task. If a non-planning strategy task becomes overdue, based on system configuration, AMT automatically reschedules it using the frequency multiplier. Unlike a planning strategy task, a non-planning strategy task will not automatically create an AMT work order.
Not After	Sets a usage or date limit, after which no projected occurrence of a strategy task will be scheduled.

Go to top

O

Occurrence	An instance of a strategy task. The occurrence can be in the future or in the past.
Occurrence Type	Typically, occurrence type represents failure modes, which are tracked for condition-based maintenance and component failure analysis. In AMT, an occurrence type has a code and a description.
Offset	If an asset's meter is damaged and unable to be used, the last reading of that meter is recorded as the offset. Readings from the new meter usually start at a figure less than the offset reading. The cumulative meter reading for the asset is the offset reading plus the last recorded usage reading.
OH&S	Occupational health and safety. OH&S is typically used to classify and monitor statutory strategy tasks in AMT.
Operation	Operations are the steps to perform a specific activity on a work scope, work order, or standard job. Work scopes, work orders and standard jobs have at least one operation.
Original Commission Date	Used by AMT's Availability Forecast report. This is the date the asset was originally commissioned. The commissioning date must be earlier than or the same as the asset's start date.
Opportunity	Generally opportunity refers to component opportunity – the opportunity to reduce costs over the life cycle of the equipment. For AMT's Product Support Marketing (PSM) module, used by OEMs and dealers, opportunity has a very specific meaning. - the difference between the potential sales that a dealer could be making and the sales budgeted.
Other Operating Cost	In AMT, other operating costs is a general term used to represent ownership costs (for example, finance or registration).
Overdue Strategy Tasks	Overdue strategy tasks are occurrences of strategy tasks that have been projected, but the projected usage or date for the occurrence has passed.
Overheads	Overheads are costs (and/or revenues) that are relevant to the asset being managed in AMT, but are not directly associated to an asset using a specific work order or billing invoice. For example, salaries for planners, workshop overhead costs.
Overhead Billing	Overhead billing is typically a monthly, fixed invoice amount charged by dealers to their customers to pay the overhead costs of supporting maintenance contracts. Overhead billing is also known as the MARC overhead billing or management fee.
Overhead Cost Category	Categories for the different types of overhead costs, for example, vehicles, capital, payroll.

Overhead ERP ID	The overhead ERP ID links the actual overhead billing records to a specific, current overhead projection. When AMT receives management fees for different contracts, this ID tells AMT that the invoice belongs to a specific contract. The overhead ERP ID relates to overhead billings, not overhead costs.
Owner	An owner is the owner of an asset, for example, a mining company, a contractor.

Go to top

P

Part	Parts are used when maintaining or repairing an asset and have a part number and a description. A part is uniquely identified in AMT by the combination of part number and source of supply.
Part Classification	The part classification is used to define the type of part, particularly for rotable components. Parts are classified according to their ownership and ability to be repaired. Used for a specific purpose to drive logic in forecasting.
Part Price	The cost of a spare part.
Part Rating	The part rating defines the rating or size of a part. For example, an engine's part rating might be its capacity, 3600hp.
Part Type	Groups of parts, for example, exchange, new, remanufacture consumable. Discounts can be applied to part types per fleet.
Parts Price Group	Parts can be grouped by price. For example, a particular customer receives a 20% discount on the standard list price. In AMT, this is represented as a specific parts price group.
Partial Out	If there is a partial out, a component has been removed from an asset before it has reached its end of life. The component has minor repairs before it is installed into another asset.
Pay type	The pay type will be standard time versus over time (such as working after your shift or on a day you wouldn't normally work).
Perform By Date	The indicative date by when the work order should be executed.
Plan Name	A unique name for an equipment.

Planned	A work order is planned if it exists in an agreed maintenance plan before it is executed. The exact timeframe to be regarded as planned is defined by each site. For example, at one site, a work order is planned if it is on the weekly maintenance plan (that is, the work order was scheduled the week before). Another site may set 48 hours in advance as the timeframe for a work order to be as planned.
Planned Event	A planned event is a downtime stoppage on an equipment that is planned before it is executed. Each site can have its own definition of the time period that is required to classify an event as planned. A planned event only applies to equipment that is configured in advanced planning mode.
Planning (strategy) Task	A planning task is a strategy task that requires planning, or the execution of the strategy task requires planning in advance, for example, ordering parts. A task is created in the Planning module of AMT after the lead time has been reached. If the strategy task only captures costs, or forecasts a provision for possible maintenance, and no forward planning is required, the Planning Task flag on the strategy task is off.
Planning Window	The period of time that the planner has to prepare to execute a strategy task. The planning window is defined by the strategy task's lead time. When a strategy task enters its planning window, AMT flags the strategy task as being "action required".
Preventable	In AMT, every breakdown event is associated with information about what area of the operation could have prevented the breakdown. For example, if the cause of the breakdown could have been identified by a daily inspection before the breakdown, then why the inspection process did not pick up the issue needs to be investigated. Preventable is only associated with a breakdown event.
Preventive Maintenance (PM)	In AMT, preventive maintenance typically relates to the servicing of equipment.
Primary Part Number	Part number that represents the primary part that is installed or replaced. For example, for a component change out, the primary part number is the part number of the component being installed. For a s ice, the primary part number could be the service kit number.
Prime Product Sales Person	Sales person responsible for equipment sales. This is relevant to OEMs and dealers that use AMT.
Prime Product Sales Group	A group of prime product sales persons.

Planning Group	A person or group of people responsible for planning maintenance activities.
Plant Hire	A module of AMT. Referred to as Rental for OEM and dealer users of AMT.
Plant Hire Rate	The plant hire rate is an internal rate charged by the central asset management group to a project for the use of an asset.
Priced Job	The costs associated with a standard job can vary from branch to branch, for example, shipping or labour costs are higher in branch A than branch B. A priced job identifies the version of the standard job to be used in a specific branch.
Pricing Date	The date at which the pricing entered for parts, labour and miscellaneous was relevant. This may differ from the budget pricing date.
Pricing Group	See parts price group.
Primary Currency	The main currency in which financial data is expressed.
Primary Part Number	The part number that identifies the component, or in some circumstances, a kit number (for example, a PM service kit number). In AMT, primary part numbers are linked to strategy tasks.
Probability of New Parts Purchase	Probability that the strategy task's parts will be sourced by the dealer from the OEM and not the dealer's own stock. This factor is only relevant to OEMs and dealers who use AMT.
Probability of Purchase from Dealer	Probability that the strategy task's parts will be purchased from a dealer versus a dealer's competitor. This factor is only relevant to OEMs and dealers who use AMT.
Product Support Sales Group	A group of product support sales representatives.
Product Support Sales Rep	Sales person responsible for parts, labour and miscellaneous sales. This is relevant to OEMs and dealers who use AMT.
Production Statistics	Measures the productive or non-productive utilisation of assets.
Production Support Marketing	The Product Support Marketing (PSM) module shows the overall sales potential for a specific dealer in its territory. PSM also reflects the market share that the dealer has within the industry based on its sales records, and highlights where to target future sales.
Projected Cost	The total cost of a projection (that is, costs to date + future costs).
Projected Cost	Projected cost variance is calculated as the difference between the total escalated

Variance (PCV)	budget projection cost and the total current projection cost for the life cycle of the asset.
Projected Daily Usage	AMT's calculation of the estimated daily usage for an asset. The calculation is based on the asset's average annual utilisation divided by the number of days in an average calendar year.
Projected Occurrences	The future occurrences for a specific strategy task.
Projected Task	Projected task is a general term that encompasses strategy tasks and unassigned tasks.
Projection	A projection in AMT represents the life cycle costing for an asset, which can be broken down to the component level. An asset can have a number of projections, including current, budget, alternate, achieve, centreline and estimate-draft. The budget projection is the initial expectation of the life, cost and the billing of an asset (this billing is only applicable to OEMs and contractors). The current projection is the current expectation of the life, cost and billing of an asset. Alternate projections are a different possible projection to the current projection. Archived projections are a snapshot of a specific projection taken at a point in time.
Projection Assumptions	Projection assumptions are the assumptions about frequency, costs, duration and labour hours that AMT uses to project future costs, duration and labour hours. It is important that the projection assumptions always reflect today's best estimates. Historical budget assumptions very quickly become out of date. Keeping assumptions up-to-date keeps future projections up-to-date, which drives future forecasts, identifies key issues, and so on.

Q

Quick Watch	Quick watch is a daily control mechanism that provides management with instant feedback about the health of their operations on-site.

Go to top

R

Reading	A reading from a meter on an asset. This value is usually expressed as a cumulative amount, for example, the meter reading on an asset was 3,560 engine hours on day 1. On day 2, the reading was 3,572 engine hours, and so on.

Reading Date	The date on which the reading was taken.
Reading Time	The time at which the reading was taken.
Real Equipment	Real equipment in AMT is an operational asset (unlike centreline or modelling equipment, which are theoretical assets). A real equipment can incur actual costs and revenue, incur actual downtime, and record actual usage readings.
Rebuild	A rotable item that is being rebuilt at a rebuild location.
Recorded Usage	The actual usage recorded from the asset at a specific date and time. This reading does not include the offset.
Redo	A work order that was performed again because it was not correctly executed the first time. Also known as rework.
Registration Counter	The registration counter, along with the model and serial number, uniquely identifies real assets. The registration counter counts the number of registrations for that equipment in AMT. Sometimes an asset is registered in AMT more than once. For example, the asset was refurbished, relocated to another project, or had its maintenance contract extended. To separate the previous history of the asset from its new life, AMT uses the registration counter in combination with the model and serial number of the asset, to achieve this distinction. Certain interfaces include the registration counter as an optional field. If the interface file sent to AMT does not include the registration counter, AMT links the interface record to the current version of the asset.
Registration Details	Specific attributes of an equipment group that can be included when you create a new asset in AMT. Examples include license number, registration number, policy number.
Reman	Abbreviation for remanufacture.
Repair Code	See job code.
Repair Reserve	A repair reserve task is a type of strategy task specifically for factoring contingency costs. AMT can include different failures of components in the overall cost forecast for an equipment. To manage this, you create a repair reserve task. Repair reserve tasks are relevant to the Condition Based Maintenance module.
Responsibility	The party that is responsible for the down time when maintenance is performed on an asset.
RVIC	RVIC is residual value in components, that is, the life remaining in a component at the end of the life cycle or at a future point in time.

Go to top

S

Safety Stock Level	The minimum number of a particular rotable part that it is considered safe to have in stock. For example, the minimum level of spare haul truck engines is three for a particular site or branch. AMT doesn't manage inventory – the inventory system manages stock levels.
Sales Responsibility	The person responsible for the sale of a new equipment, parts, labour and/or services. Sales Responsibility is only relevant to OEMs and dealers who use AMT.
Scheduled Hours	The scheduled hours are the calendar hours per month that a particular machine will operate. Scheduled hours are used to calculate or understand physical availability.
Scheduling Efficiency	Scheduling efficiency represents how efficient a planner is in optimising the schedule for availability forecasting. At 100% efficiency, all strategy tasks in the same month are performed at the same time as the strategy task with the greatest single duration. 0% efficiency means that the strategy tasks are performed on their current due date (optimisation is not considered). Scheduling efficiency only applies to strategy tasks, not to work orders, and it is only applied to AMT's Forecast Availability report.
Scheduling Group	A group of strategy tasks that belong to a specific scheduling logic. You can group strategy tasks together so that they have the same scheduling logic applied to them, such as suppression (when one strategy task suppresses another from occurring) or dependency (when one strategy task is a prerequisite for another task.
Scheduling Task	In AMT, a scheduling task links a strategy task to a specific maintenance task in an external ERP system. Typically, scheduling tasks are only relevant if the ERP system performs daily planning, and AMT performs the long-term planning function. There is an identifier for the scheduling task.
Scheduling Tolerance	Scheduling tolerance is used to group preventive maintenance services (that is, strategy tasks with task type PM) that are scheduled to occur around the same time. For example, when an asset is due for its 3,000 hour service, and this service is a combination of the 1,000 hour service and the 2,000 hour service. Both tasks would be forecast to start at around the same time. To keep the tasks together, set the tolerance to 24 hours. With this scheduling tolerance setting, any service forecast to occur within 24 engine hours of another service will be forecast to occur at the same time. Scheduling tolerance is linked to parent suppression.
Sell (Cost at Sell)	The calculated value of parts, labour and miscellaneous items, including a margin, added to costs to allow for parts and service departmental overhead recovery and a profit margin.
Serial Number	A unique alphanumeric identifier for a single, real asset.

Service Accuracy – Service Interval	The service interval is the time or usage between services. For example, if the preventive maintenance strategy is to have the PM services executed in the following sequence: 250PM, 500PM, 250PM, 1000PM, 250PM, 500PM, the service interval is 250 hours because a service occurs every 250 hours.
Service Accuracy – Tolerance %	An acceptable level of variance to the service interval is defined for the tolerance percentage, for example ±10%. Services performed outside the tolerance level for execution, that is, much earlier or later than planned by the standard interval, are reported as being outside the tolerance. The tolerance on service accuracy is reported in KPI reporting.
Service Agreement	An agreement between two parties that covers performing preventive maintenance services.
Service Code	A set of codes in AMT used to align with the company's internal service codes.
Shift Start Time	The starting time for the 24 hour shift on site. 24 hour downtime can be reported from the shift start time, rather than from midnight to midnight. A number of daily downtime reports in AMT use the shift start time to report actual downtime.
Shiftlog	In the shift log, you can record downtime for Advanced Planning mode assets. The shift log is formatted similarly to paper-based downtime logbooks maintained on site.
Shutdown	A shutdown is a period of time when assets are not working, and can be maintained. When planning the maintenance to be performed during a shutdown, work orders (standard planning) or events (advanced planning) are grouped together for the same or different assets. A shutdown is a group of work orders to reflect a shutdown of fixed plant assets. A shutdown can be for many assets.
Simple Utilisation	A simple utilisation has one UOM and one utilisation rate that represents the life of the asset and its components. Dates can also be used to represent the life of assets and their components.
Site	The location of a fleet.
Skipped Work Order	A work order that is not started, and is ignored. Even though the work order was not started, it is completed in AMT, and the next occurrence is scheduled.
SMU Hours	The engine hours recorded on the asset's service meter unit (SMU).
Source	The source is who or what identified the need to perform a work order.
Source of Supply	The supplier of different parts.

Standard Job	A standard job is a repeatable template of a work activity that can be performed on a model or asset (for example, Repair Engine Before Failure). A standard job can be made up of one of more operations. An operation typically represents a step within the work activity. An operation can have many parts, labour, or miscellaneous items associated with it.
Standard Planning	Standard planning is a simple method of planning individual work orders. In contrast to advanced planning, which captures downtime using events, standard planning does not capture downtime and therefore does not link to an event.
Start Date	The date from which the asset is managed in AMT.
Start Usage	The usage reading on the asset as of the start date. A start usage is greater than or equal to zero.
Starting Serial Number	The serial number of the component that was installed in the asset when it started being managed in AMT.
Stock Turn	Stock turn is the number of turns divided by total stock for the last 12 months. A turn is the time between when a rotable item is installed and when it is removed.
Strategy Task	Strategy tasks are tasks that are contained in the current projection, including unassigned tasks. A strategy task can contain many jobs, for example, remove and install, and recondition. A strategy task is uniquely identified within a projection by component code, modifier code, task type and task counter.
Strategy Date	The projected date of the next occurrence of a strategy task.
Strategy Usage	The projected usage of the next occurrence of a strategy task.
Sub-System	Sub-systems are the third level of the component code hierarchy. For example, the system is an engine, the sub-system is a fuel system, and the component code is C0723 fuel tank.
Supplier	Vendor that supplies parts and/or services to the maintenance operation.
Suppression - Sibling	Sibling suppression logic is where scheduled occurrences of future strategy tasks are set so that they occur at a fixed interval (or multiples thereof) after the most recent last occurrence of any strategy task in the suppression grouping. This is used primarily for preventive maintenance scheduling where there is a requirement for preventive maintenance services to occur at a set scheduling interval. For example, if a 250 hour PM service occurs at 260 hours, with sibling suppression, the next service will occur at 510 hours, not at 500 hours.
Suppression – Parent	Parent suppression logic is based on performing the parent strategy task. The parent strategy task suppresses any child tasks if they fall within a certain tolerance of the

	parent strategy tasks. Performing the parent strategy task implies that the child strategy task has been performed as well, and therefore the child strategy task is reset along with the parent. For example, strategy tasks are grouped in a hierarchy so that A1 is the parent of A2, which is the parent of A3, which is the parent of A4, and so on. When A1 occurs, the component life for all child strategy tasks are reset.
Swing Component	A component that is installed into an asset on a temporary basis while the original component undergoes a minor repair.
Swing In	Refers to the swing component that has been installed temporarily while the component taken out undergoes a minor repair.
Swing Out	The swing component is removed and the original component (that was temporarily removed) has been re-installed.
Symptom	A symptom of failure. For example, black smoke is a symptom of engine oil leaking.
System	Systems are the second level of the component code hierarchy. For example, the cost type is repair and maintenance, the system is an engine, and the sub-system is a fuel system.

Go to top

T

Target Budget Turn Time	This is the conservative target of the number of days between when a rotable part is removed from one machine, until it is available, in inventory, for use in another machine. Each rotable part has a target budget turn time.
Target R&I Time	This is the conservative target of the number of days to remove and install a rotable part on an equipment. Each rotable part has a target remove and install (R&I) time.
Target Rebuild Time	This is the conservative target of the number of days to rebuild a rotable part. Each rotable part has a target rebuild time.
Task	In AMT 8.3, the term work order replaced what was previously called a task. There is a language option in later versions of AMT to keep this earlier terminology.
Task Counter	The task counter can be used to distinguish between strategy tasks with the same component code, modifier code and task type. In certain AMT configurations, task counter is a unique ID that maps to the ERP.
Task Planned Time	A configurable list of time periods to set a default target date for a non- strategy task work order. For example, one time period is Next Planned Stoppage, which can be

	defined as five days.
Task Type	A configurable list of high-level, generalised work activities, each of which generally describes the type of work to be performed on a component and modifier code. Detailed work types are defined in AMT as job codes. For example, component change out, inspection, general repair, preventative maintenance service.
Terminated	The status of an asset when it is no longer operational. As a result, AMT will not forecast strategy tasks into the future. The status of an employee who is no longer available to be assigned to work orders. The status of a user account that is disabled.
Test Hours (Test Time)	The UOM consumed on an asset while it is down or stopped for maintenance. For example, sometimes engine hours are consumed during downtime because technicians have to perform a test on the asset before releasing it back into operation. If the asset is under a maintenance contract, test hours can be subtracted from the billable hours, for which the customer will be charged.
Transaction Date	The date when an actual cost or billing (revenue invoice) was posted to the general ledger. This is the ERP transaction date.
Travel	Travel generally represents the costs involved in travelling from a central location to a remote location in order to perform maintenance. AMT uses the following formula to calculate total travel cost: Travel labour cost = No. trips x Average travel hours per trip x Average no. people per vehicle x Travel labour rate. [No. trips = Hours required /(Average no. people * Average work time)] Mileage (Misc.) cost per trip = (distance x rate) + Misc. cost per trip Travel is used by dealers to quote and track travel costs in contracts.
Turn	A turn is another way to say life. It is the time between when a rotable item is installed and when it is removed. Rotable items can have many turns in different machines.

Go to top

U

Unassigned	Unassigned is a mechanism for budgeting for general repair costs, at the sub-system level, in order to complete a full life cycle costing model. Unassigned is also used to group actual costs for non-strategy tasks into sub-systems for high-level cost reporting.

Unit of Measure (UOM)	A defined unit of measure, for example, engine hours, litres, tonnes, kilometres, miles.
Unplanned Strategy Task	An unplanned strategy task is a strategy task that has not yet entered its planning window, and consequently AMT has not created a backlog work order for it.
Usage	A usage is a specific cumulative reading that represents how the life cycle of an asset or a component is measured (for example, machine hours, tonnes, fuel burn). Usages can also be used for forecasting maintenance activities (that is, strategy and unassigned tasks). The usage value is used to determine asset projections.
Usage Based	Denotes that the strategy task scheduling is based on usage, rather than date.
Usage Profile	The usage profile only applies to assets configured for advanced utilisation. A usage profile can contain multiple usage rates, for example, machine hours, travel hours, fuel burn. The usage rates can vary over time to reflect changing operating conditions, for example, deeper pit, rainy season, winter, and relationships between the usages can be created, for example, tonnes is a dependency of engine hours.
User Account	The user account contains details of an individual user such as login, password, level of access, and so on.
User Group	In AMT, an individual user is associated with a user group. The user group specifies the functions and reports that the user has access to, and the level of access permitted. Levels of access include View, Edit, All, None.
Utilisation	The ratio of productive hours to available hours, usually expressed as a percentage.

Go to top

V

Variance Cause	The reason why a particular activity within a downtime allocation took more or less time to complete than expected.

Go to top

W

Warranty Period - Days	The amount of time in days that the warranty policy for a certain strategy task is valid.

Warranty Period - Usage	The warranty period represented as a usage reading, for example, 100,000km instead of two years.
WBS	WBS is work breakdown structure; a term used in SAP to classify work order settlement relationships.
Work Group	The group of people who perform the work activity, for example, PM crew, welding crew.
Work Location	Location where a downtime event will take place, for example, workshop, field.
Work Order	A work order is a specific activity performed on an asset, that can be planned or executed. A work order settlement can link to a work order to provide the actual cost of performing the work order.
Work Order Settlement	In an integrated solution, a work order settlement represents the ERP's actual transactional costs. In a non-integrated solution, actual costs can be entered manually.
Work Order Settlement Status	The status of a work order settlement, for example, closed, open.
Work Scope	A work scope is a hard copy or electronic record of all work information for a work order. Some systems call a work scope a job card.

Go to top

AMT Mobile overview

AMT Mobile is a practical tool for mechanics, supervisors and ground teams which will improve the efficiency and accuracy of maintenance-related tasks and the workforce. It operates disconnected from the network and provides an overview of fleet performance and access to information across all sites

AMT Mobile is compatible with most smartphones, tablets and laptops running modern browsers including Internet Explorer 11+, Edge, Chrome, Firefox 10+ and Safari; access to the internet is a requirement as it does not work offline.

The AMT Mobile home screen displays a suite of apps which are highly configurable. You can navigate to the relevant app by tapping the relevant icon. Most apps will prompt you for further information such as a Site or Login.

AMT Mobile - Login

To navigate to AMT Mobile, open the relevant web browser and enter the AMT Mobile link, provided by RPMGlobal.

Enter your user name and password and touch **Login**. You also have the option of selecting **Keep me logged in** .

Navigation - Mobile module

The AMT Mobile home screen displays a suite of apps which are highly configurable. You can navigate to the relevant app by clicking the relevant block. You will enter the relevant app with its unique functionality. Most apps will prompt you for further information such as a Site or Login.



Customers app

This app is designed to be used as a PSSR Tool for the sales team and account managers in the field. It:

- Identifies component and service sales opportunities per customer per equipment.
- Shows asset details and health.
- Shows historic sales and market share.

On the **Home Screen**, click the **Customers** app. When you open the Customers app a list of customers displays. This list can be narrowed down by the respective account manager. The number of non-terminated assets associated to that customer is listed on the right of the screen, and the monetary value of the sales forecast is listed below the customer name.

Clicking the **Refine** button at the top right of the screen lets you **filter** by account manager.

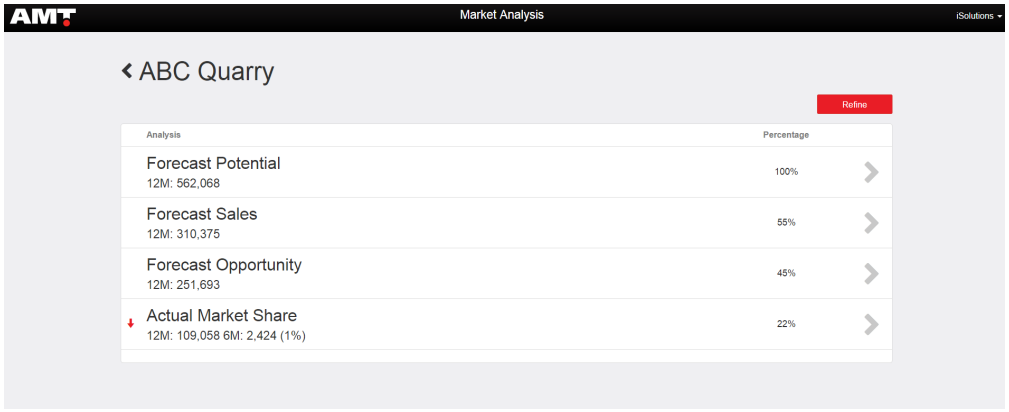
Once you have selected a particular customer, you then have the option to view four different analysis categories:

- Market Analysis.
  Displays forecast and actual figures for the selected customer.
- Alert Opportunities.
  Displays the top 20 next component sales opportunities.
- Component Forecast.
  Displays a full component forecast.

- Overdue Opportunities.
  Displays components which should have been replaced by now but have not, which if not followed up quickly could result in a lost sale.

View Market Analysis

When you select a **Customer**, then select **Market Analysis**, the Forecast and Actual figures for the selected customer are displayed.



Definitions

- **Forecast Potential** is the total number of opportunities in the next 12 months available to the dealer to make a sale.
- **Forecast Sales** is the percentage of **Forecast Potential** the dealer believes it will make in the next 12 months.
- **Forecast Opportunity** is the percentage of the **Forecast Potential** that the dealer could potentially make.
- Forecast Potential = Forecast Sales + Forecast Opportunity.

Items on the screen

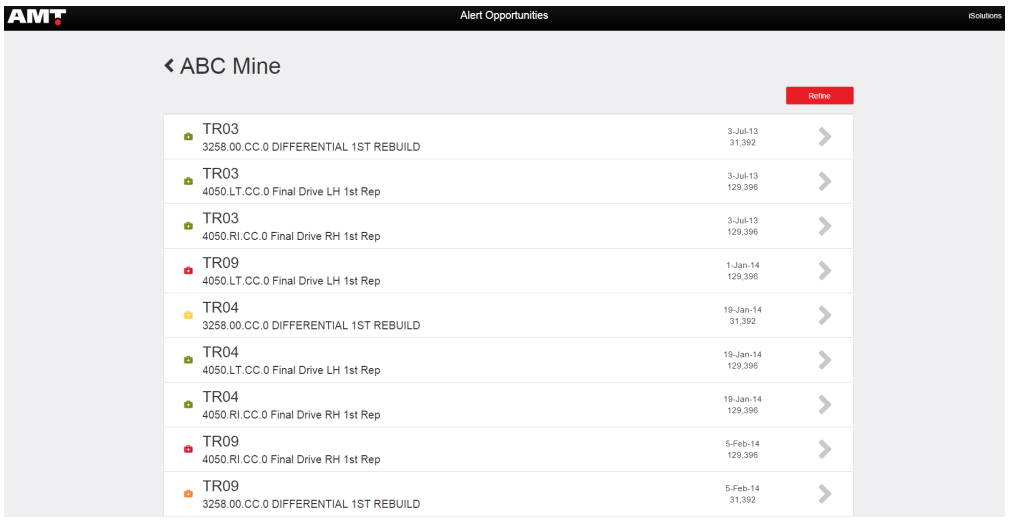
- **Refine** (Filter) by Major Commodity Code, Parts, Labour, Misc.
- Forecast Potential:
  - Potential = total sales potential (100% market share).
  - 12M = value of forecasted potential for the next 12 months.
- Forecast Sales:
  - Sales = market share targeted.
  - 12M = value of market share targeted for the next 12 months.
  - % = Forecast Sales value divided by Forecast Potential value.
- Forecast Opportunity:
  - Opportunity = the difference between Potential and Sales, that is, the value of additional sales opportunity in addition to what you are currently targeting.
  - 12M = Forecast Potential less Forecast Sales for the next 12 months.
  - % = Forecast Opportunity divided by Forecast Potential.
- Actual Market Share:

- Figures are based on actual history of pars and service sales.
- Trent of the last month over the last six months.

By selecting one of the above categories, you then get a breakdown per Major Commodity Code.

View and update Alert Opportunities

When you select a **Customer**, then select **Alert Opportunities**, a list of the next 20 component sales opportunities, based on Due Date and Sales Status displays.



You have the option to **Refine** (top right of the screen) the list by Equipment, Health Status and Component Code.

Each line item displays:

- Equipment, for example TR09.
- Component Opportunity, for example Final Drive Left Replace.
- Health of the Component (coloured icon to the left). The Health Status originates from the Condition Based Maintenance module and the Sales Status is specified in the work order or strategy task within AMT.
    - Blank - Condition monitoring not tracked in AMT for this component.
    - Green - OK.
    - Yellow - Low Level Alert.
    - Orange - Medium Level Alert.
    - Red - High Level Alert.
- Due Date.
- Value of the Opportunity escalated to today's value.

You can update the forecast and sale status of the component by clicking or tapping the arrow **>**.

View the Component Forecast

Displays a list of the next 20 sales opportunities (including overdue opportunities), filtered by Date and Equipment.

You have the option to Refine the list by Equipment, Health Status, Sales Status, or Component Code.

Each line item displays:

- Equipment, for example TR09.
- Component Opportunity, for example Final Drive Left Replace.
- Health of the Component (coloured icon to the left). The Health Status originates from the Condition Based Maintenance module and the Sales Status is specified in the Work Order or Strategy Task within AMT.
    - Blank - Condition monitoring not tracked in AMT for this component.
    - Green - OK.
    - Yellow - Low Level Alert.
    - Orange - Medium Level Alert.
    - Red - High Level Alert.
- Sales Status:
    - Unknown.
    - Unlikely.
    - Confident.
    - Sale Confirmed.
    - Lost.
- Due Date.
- Value of the Opportunity escalated to today's value.

 You can update the forecast and sale status of the component by clicking or tapping the arrow **>**.

Update Component Forecast

 You must have AMT v8.10 Update 3 or later to update the forecast on the app.

It is critical when a Sales Representative is talking to a customer about a component sales opportunity, that the Sales Representative can update information live about the opportunity based on the customer's feedback.

Using the Customer app, the Sales Representative can make the updates. Updates can be made from the following menus:

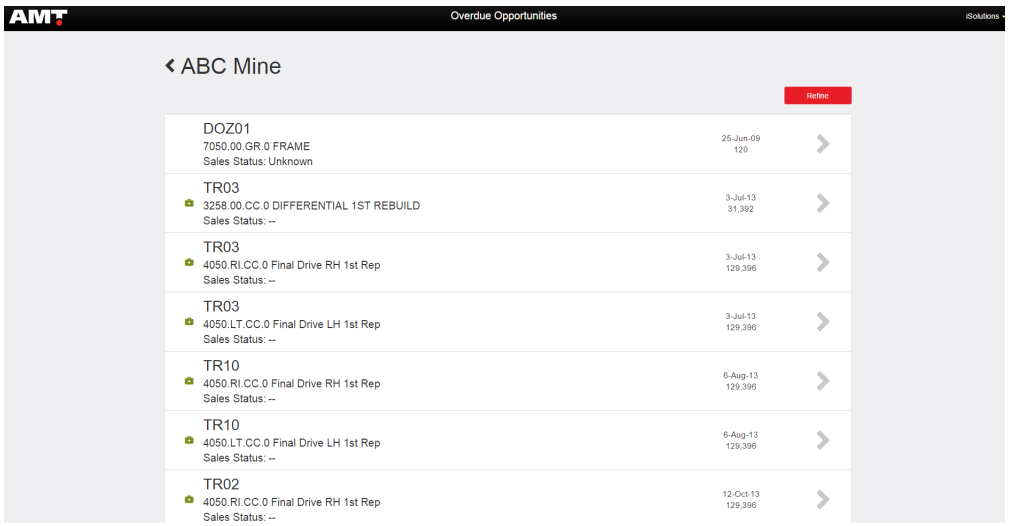
- Alert Opportunities.
- Component Forecast.
- Overdue Opportunities.

When updating the forecast, you can make three updates:

- Reschedule a component sales opportunity.
- Record the changeout of a component (the opportunity has in fact occurred).
- Update the Sales Status of a component sales opportunity.

View and update Overdue Opportunities

When you select a **Customer**, then select **Overdue Opportunities**, a list of the next 20 overdue component opportunities, filtered by Target Date and Equipment displays.



You have the option to **Refine** (top right of the screen) the list by Equipment, Health Status, Sales Status or Component Code.

Each line item displays:

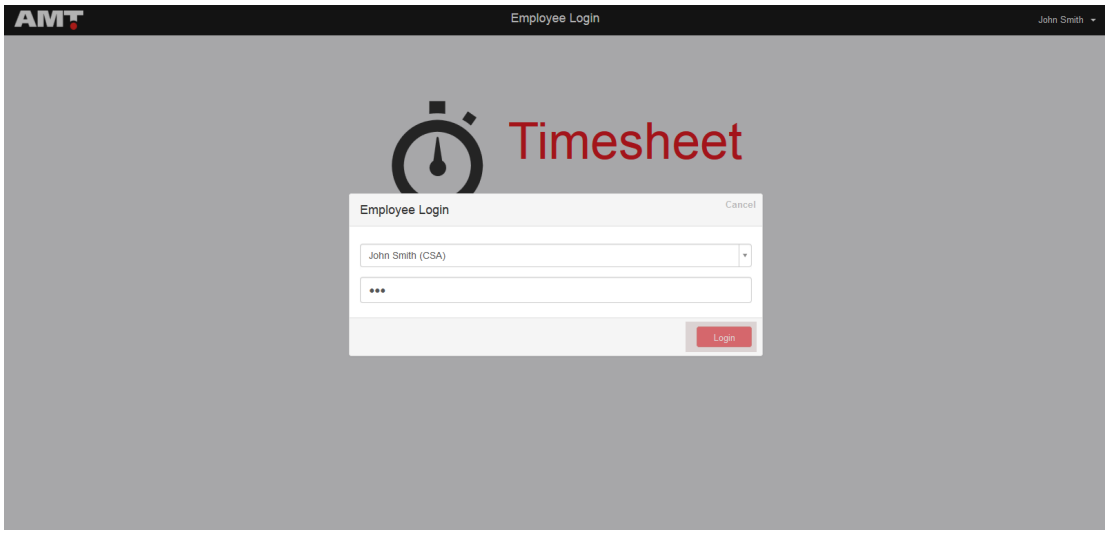
- Equipment, for example TR09.
- Component Opportunity, for example Final Drive Left Replace.
- Health of the Component (coloured icon to the left). The Health Status originates from the Condition Based Maintenance module and the Sales Status is specified in the Work order or Strategy Task within AMT.
    - Blank - Condition monitoring not tracked in AMT for this component.
    - Green - OK.
    - Yellow - Low Level Alert.
    - Orange - Medium Level Alert.
    - Red - High Level Alert.
- Sales Status:
    - Unknown.
    - Unlikely.
    - Confident.
    - Sale Confirmed.
    - Lost.
- Due Date.
- Value of the Opportunity escalated to today's value.

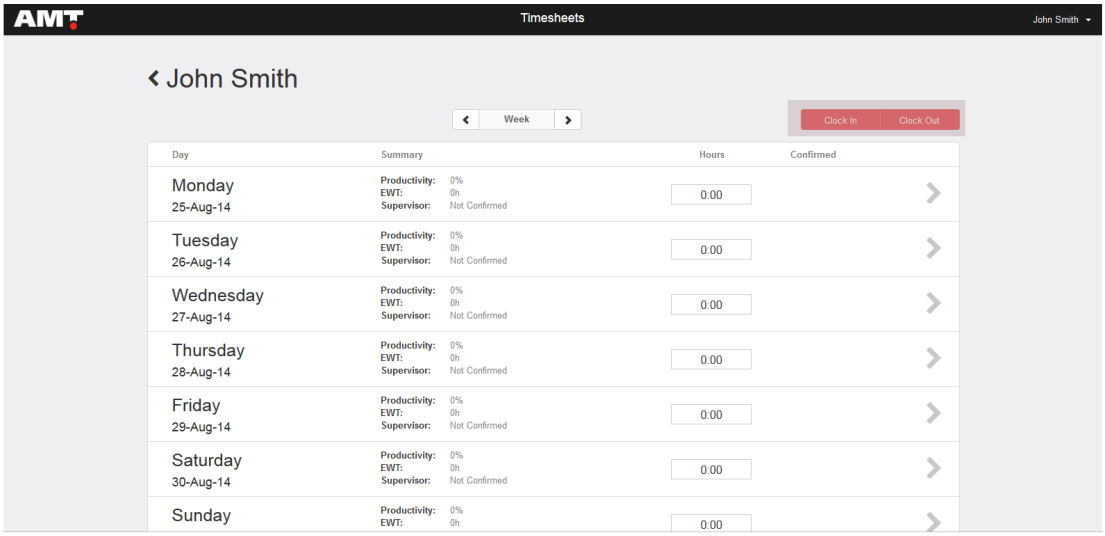
You can update the forecast and sale status of the component by clicking or tapping the arrow **>**.

Timesheet App Login

This app is designed to give you the ability to enter and review daily timesheets from anywhere.

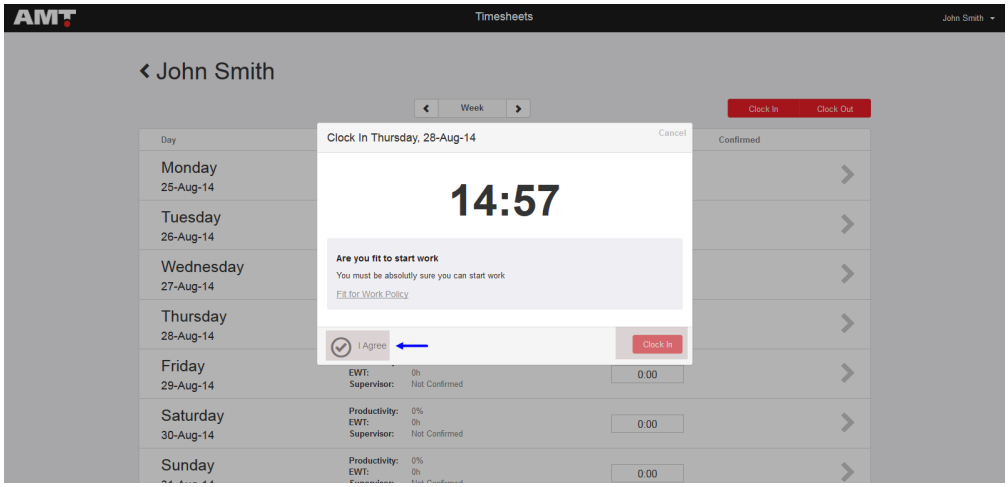
On the **Home Screen**, click the **Timesheet** app. The Employee Login screen displays. Each employee will have a login and password for the Timesheet app, and they must be entered prior to entering any time information. If you are unsure of this information, contact your AMT Administrator.





When an employee commences a shift it is a requirement that they **Clock In** and confirm they are fit for work. They do this by clicking **Clock In** at the top right of the screen.

When they click Clock In, another screen displays. showing the Clock In time and a configurable message, which specifies any relevant company policies employees are required to adhere to. For example, in the example screenshot below the employee must click **I agree** to confirm they are fit for work and then click **Clock In**.

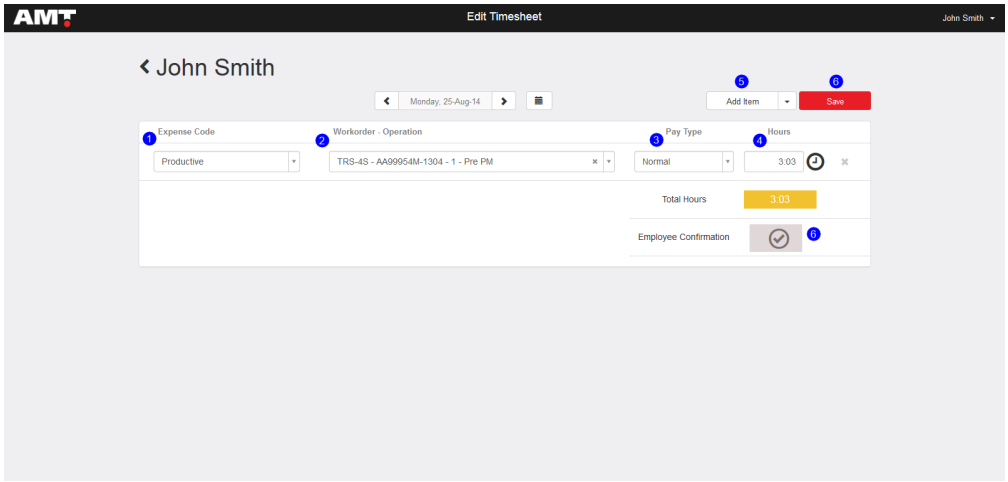


The configurable message is set up by the AMT Administrator.

Time allocation

You can allocate time to specific work orders or non-productive time by selecting the relevant day of the week and clicking **Add**. Each work order that the employee worked on that day should be listed on its own line.

If you select a non-productive Expense Code you will be able to capture non-work order time.



1. The **Expense Code** is defined by the AMT Administrator.
2. All work orders in the AMT database will be accessible to the employee. A specific work order can be found by typing in keywords in the **work order** bar.
3. **Pay Type** is defined by the AMT Administrator.
4. **Hours** is the total number of hours that the employee spent working on a particular work order.
5. To add another work order performed on the same day, click **Add Item**. This adds another line.
6. **Confirm** the total time and **Save** the time.

Clock out

On the Timesheet screen, click **Clock Out**. A new screen displays similar to the Clock In screen, showing the Clock Out time and a configurable message which specifies any relevant company policies employees are required to adhere to. For example, "Are you fit to leave work?"

Select **I agree**, and click **Clock Out**.

The configurable message is set up by the AMT Administrator.

Supervisor App

This app is designed to give Supervisors the ability to view, review, edit and approve timesheets for a specific team and specific employees.

The first screen on this app asks you to select a site, then the Supervisor Login dialog box displays.

Enter your **Name**, **PIN** and **Supervisor PIN**.

If you are unsure of your PIN, contact your AMT Administrator.

Once you have logged in you will be able to select the following:

Measurements

Defects

Usages

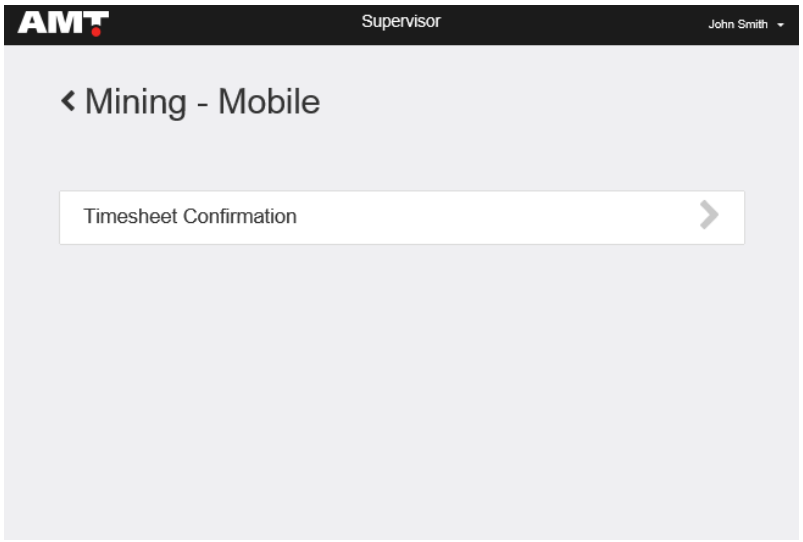
Inspections

Authorise Workorders

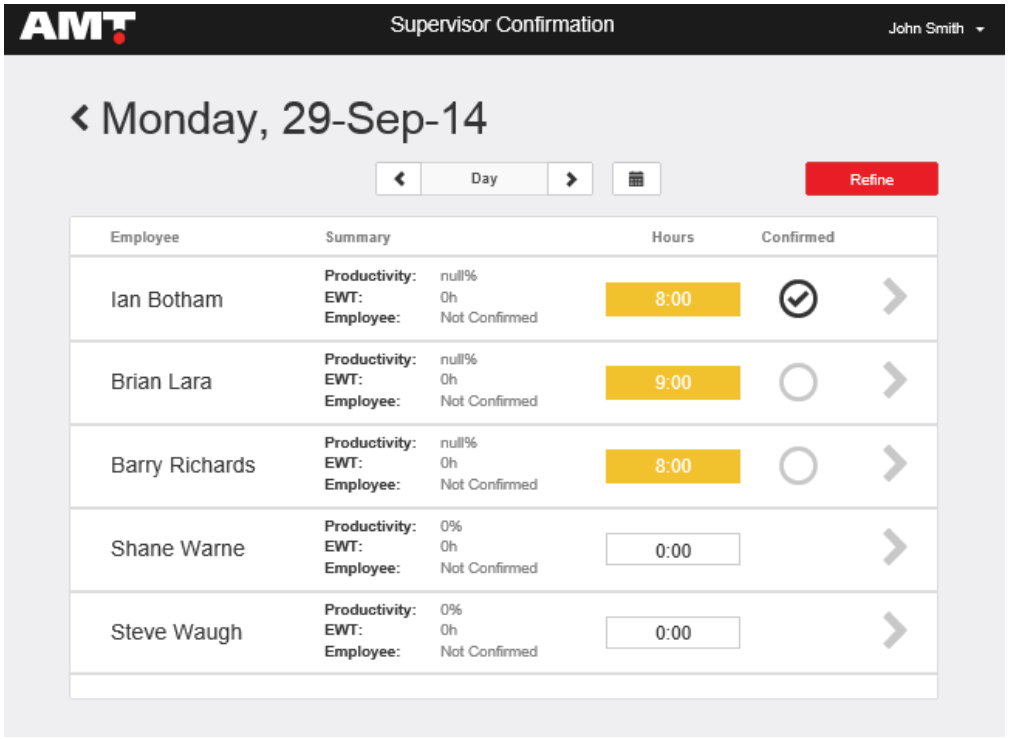
Timesheet Confirmation

Timesheet confirmation - Supervisor

Once you have **Selected a Site** to supervise, select **Timesheet Confirmation**.



The following screen displays



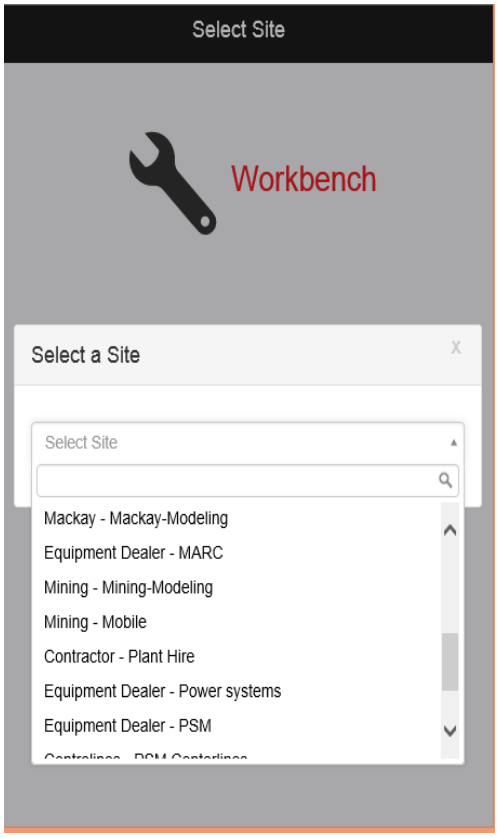
This screen lists the employee's time allocations by day for a particular site. This enables you to quickly scroll through the days and confirm employee timesheets by selecting the **Confirmed** circle.

You can search for specific employees by clicking **Refine** at the top right of the screen.

Workbench app

This app is designed to capture new defects, record actual usages and measurements.

On the **Home Screen**, click the **Workbench** app. The Workbench screen displays and you are prompted to select a site.



Once you have selected a site you will need to log in and enter a PIN. The Workbench home screen displays with the following functionality options:

- Inspections
  You can perform Inspections from within the Workbench app or by using the Inspections app.
- Statutory Compliance
- Work orders

Select an Asset and start an Inspection

Once you have logged into do the following:

- Select the **Asset** the inspection is being performed on.
- Select an **Inspection Type**. This narrows down the list of Inspections that you can select.
- Select an **Inspection**.

  If you had previously started an inspection but not completed it, the inspection list will include this. These show in the list with the previous date and time in brackets after the inspection name. You can select it to resume and complete the inspection. Inspections are created and linked to the **Inspection Type** in the **Electronic Workscope Manager**.
- Enter a **Start Date** and **Time**.

  The Start Date and Time can be modified and confirmed before commencing the Inspection. If you had selected an Inspection that was in progress, the Start Date and Time are not editable.

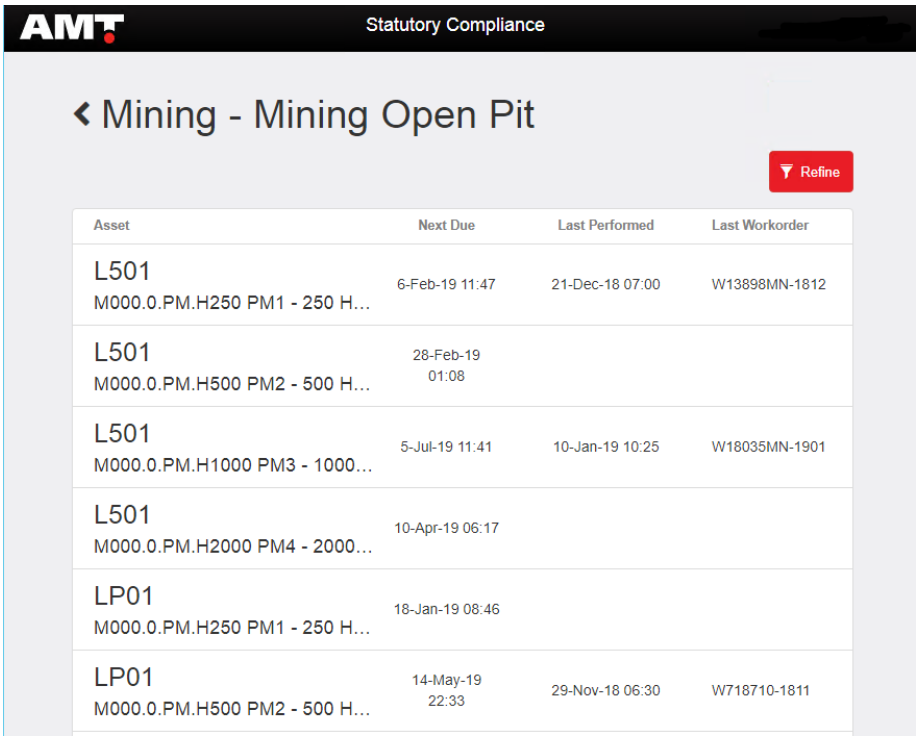
- Record **Usages**. All of the asset's nominated UOM are displayed and by selecting the relevant one, you will see what the latest valid reading is. You also have the option to update the readings. the readings are subjected to the same validation rules as any usage readings entered into AMT. If a reading is invalid it will not be visible on the AMT Mobile device.

You can now enter any Defects found during Inspection.

Statutory Compliance - Mobile

This screen gives you the ability to review all work orders which have been nominated as statutory jobs. You are able to see the name of the asset, the relevant strategy task, when it is next due, when it was last performed and what the most recent work order number was.

To narrow down the list, click **Refine**.



Inspections App Login

This app is designed to record defects, complete inspections and enter measurements directly in the field.

- Inspections are based on check lists which are easily created and modified within AMT.
- The Inspection Configuration defines the “nodes” that make up an Inspection.
- There are four different modes for displaying inspection steps.

On the **Home** screen, click the **Workbench** app. Select a **Site**, and the **Employee Login** screen displays. You are required to enter a PIN to log in.

You should be allocated to the Labour Activity Inspector in order to see and perform inspections. This can be set up in the employee system table in the AMT Mobile device and the AMT Client (**System Administration > System Tables > Organisation > Employee**.

It is assumed that at this point, the Inspections app has been configured by the AMT Administrator.

Select an Asset and start an Inspection

Once you have logged into do the following:

- Select the **Asset** the inspection is being performed on.
- Select an **Inspection Type**. This narrows down the list of Inspections that you can select.
- Select an **Inspection**.

  If you had previously started an inspection but not completed it, the inspection list will include this. These show in the list with the previous date and time in brackets after the inspection name. You can select it to resume and complete the inspection. Inspections are created and linked to the **Inspection Type** in the **Electronic Workscope Manager**.

- Enter a **Start Date** and **Time**.

  The Start Date and Time can be modified and confirmed before commencing the Inspection. If you had selected an Inspection that was in progress, the Start Date and Time are not editable.

- Record **Usages**. All of the asset's nominated UOM are displayed and by selecting the relevant one, you will see what the latest valid reading is. You also have the option to update the readings. the readings are subjected to the same validation rules as any usage readings entered into AMT. If a reading is invalid it will not be visible on the AMT Mobile device.

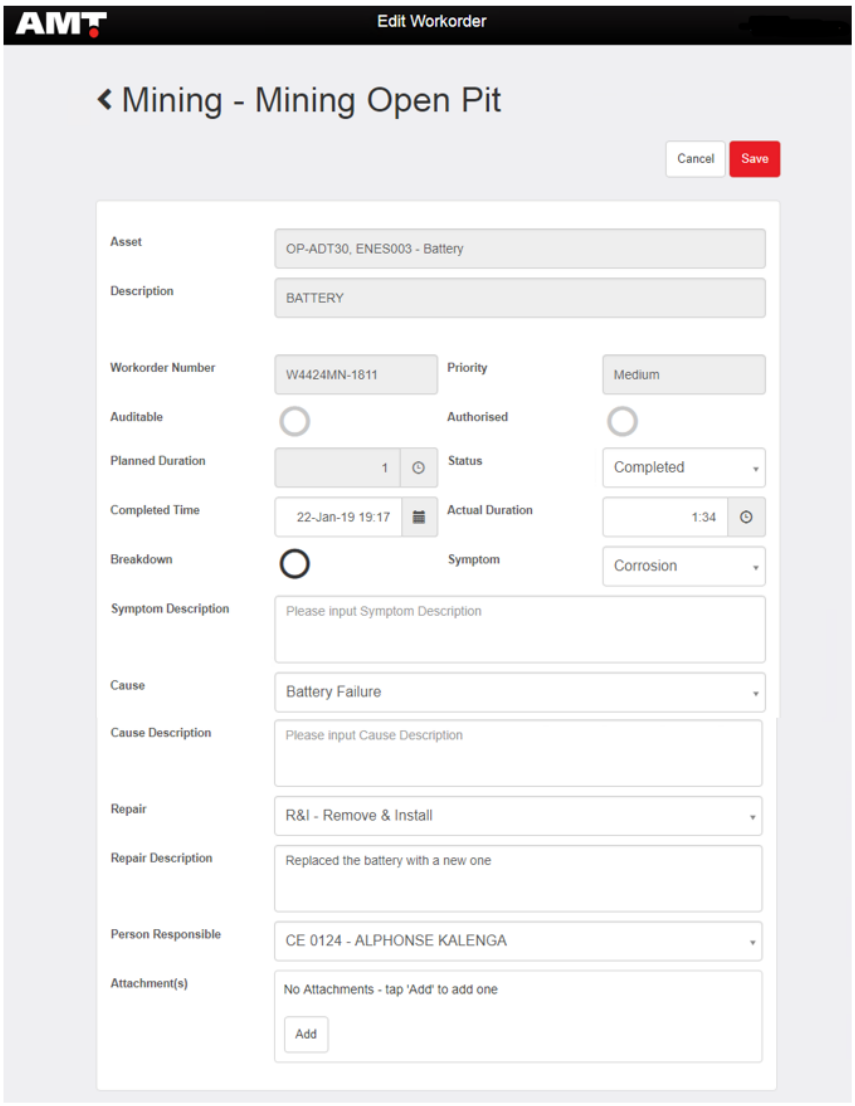
You can now enter any Defects found during Inspection.

Work Orders - Mobile

On this screen you can view, edit and perform and change the status of the work orders that are still to be performed, against all assets like to the sites you have permission to see. This function does not include the option to add a new work order.

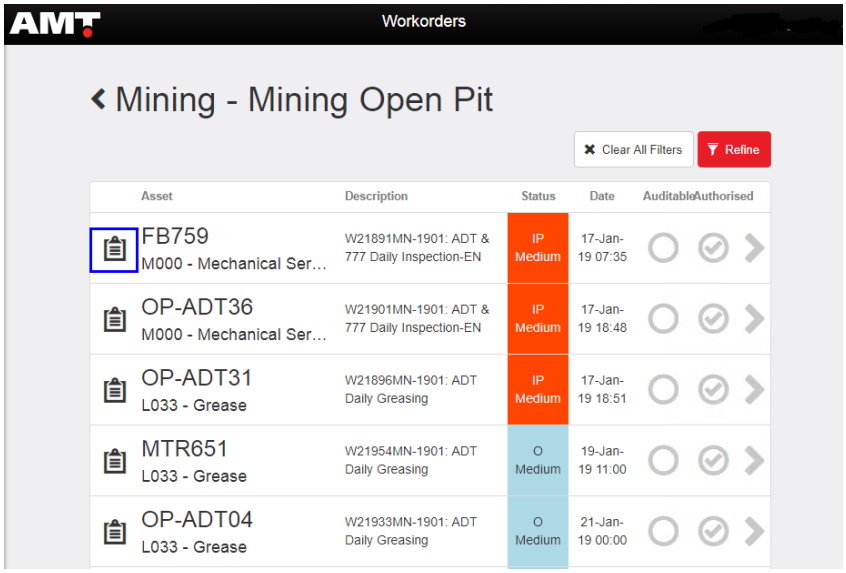
To narrow down the list, click **Refine**.

Once you have selected a work order you can update the relevant fields, and attach any photos or documentation as required.



Perform an Inspection

If an Inspection strategy task is linked to an electronic workscope, you can perform the inspection from the work order. You can identify these work orders by the clip-board icon to the left of the work order.



Select the work order, then click **Go to Inspection**. More information on performing an inspection click here, and review from Select an Inspection.

Quoter

The Quoter module provides you with facilities to generate estimates or customer support agreements based on a general inquiry. Being a web-based tool, Quoter allows you to generate the estimates on the spot, through the use of a connected hand-held device. It can also be configured to request the most recent price agreement for any given customer from a bolted-on ERP, guaranteeing that all generated quotes are based on updated prices.

Configurations

Configuration	Integration	Description
Standard job repricing web service	Yes	Needs to be enabled if the ERP Standard Job repricing process is to work.
AMT source	Yes	Needs to be enabled if the ERP Standard Job repricing process is to work.

Quoter setup

AMT Mobile Quoter is a tool that gives the Sales Representative or Contract Manager the power to generate quotes for customers on the spot on a mobile device. Quoter still uses the data in AMT client to calculate the quotes, therefore, in order to ensure the quoted price is accurate and up to date, it is essential to maintain a healthy AMT database.

To generate a quote, two key pieces of information are used:

- **Equipment centrelines**: This determines all the maintenance work needed for each asset (AMT strategy task), what job needs to be done and when to do them.

- **Job prices**: This determines the cost of each job. By default, the standard job linked to each strategy task will have a default price. It is important to keep the centreline job default price up-to-date. Often an equipment dealer might have different price groups and different pricing for different customers. All price group and discount information needs to be checked regularly to ensure the quote Quoter produces is accurate and up-to-date.

Quote Report structure

The Quote Report is the final output of the Quote Manager - Job Quoting function (**Quoting** > **Quote Manager - Job Quoting**). This is the document that is designed to be passed on to your customer for approval.

This Report Structure topic has several related topics. Click the link below to see them.

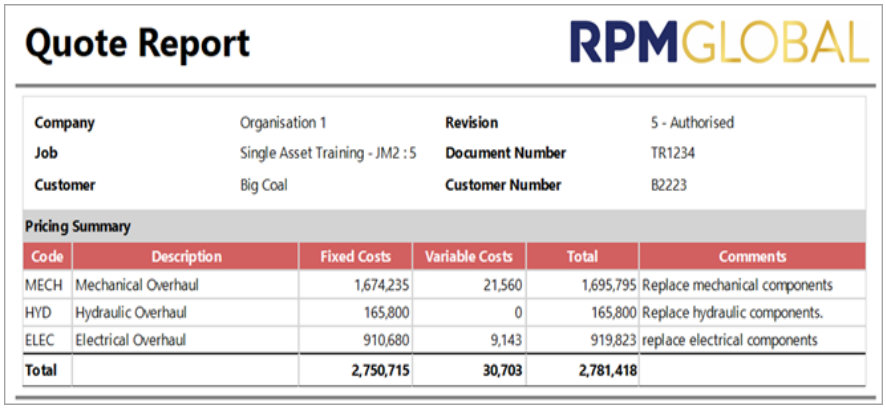
Quote Report Header and Footer

Quote Report Pricing Summary

Quote Report Pricing Details

Quote Report Terms

Quote Report - Header and Footer



Quote Report RPMGLOBAL

Company	Organisation 1	**Revision**	5 - Authorised
Job	Single Asset Training - JM2 : 5	**Document Number**	TR1234
Customer	Big Coal	**Customer Number**	B2223

Pricing Summary

Code	Description	Fixed Costs	Variable Costs	Total	Comments
MECH	Mechanical Overhaul	1,674,235	21,560	1,695,795	Replace mechanical components
HYD	Hydraulic Overhaul	165,800	0	165,800	Replace hydraulic components.
ELEC	Electrical Overhaul	910,680	9,143	919,823	replace electrical components
Total		**2,750,715**	**30,703**	**2,781,418**	

Each page of the Quote Report has a header comprising the following.

Field	Description
Company	The **Organisation** from the AMT system table.
Job	The **Title** of the quote header and version number.
Customer	The Customer assigned to the asset for a single asset quote, or the customer chosen for a multi-asset quote. The customer must be in the AMT Customers table.
Revision	The Revision Number of the quote and the Authorisation Status.
Document Number	The Document Number of the quote.

Field	Description
Customer Number	The Customer Number for the customer.
Logo (Top Right)	The company logo loaded in to the AMT database.

You can create a quote header for a single asset, or for multiple assets in a system. When you create a new quote, you are creating a new, blank, quote header. You can create a quote using Real or Modelling assets. The quote header contains **Internal Comments** and **Comments for Customer** that can be used for a general reference or place to store information.

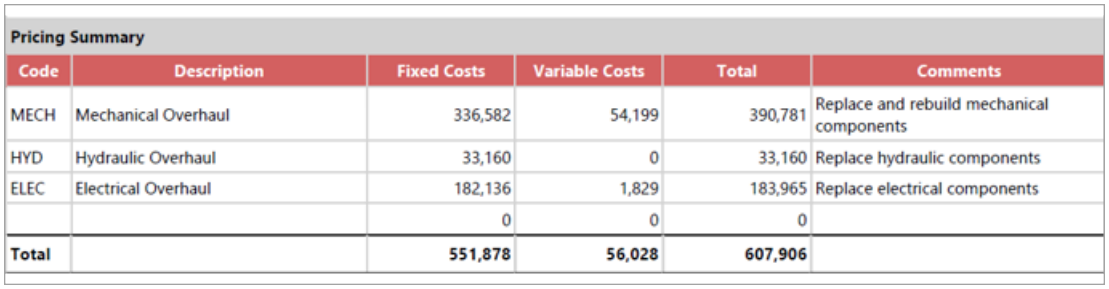
The Quote Header contains a grouping structure which is used to layout the quote in a logical sequence, and to group costs. This structure works with a parent and child relationship between the levels. For example, you could have three level 1 “parents”, each of which has 0 or more different “children”.

The codes are used for identifying sections of the quote. For example a parent code of SH and a child code of 01 will result in a line item of SH.01. Within each group, the jobs will be sorted by the component code, modifier code, task type code, task counter and job code.

Each page in the Quote Report has a footer showing details of when the report was run and the **Company**.

Quote Report - Pricing Summary

The Quote Report comprises three main sections. The first page is always a **Pricing Summary** page. This contains the Level 1 groups and a summary of the in-scope costs separated into fixed and variable costs.



Pricing Summary

Code	Description	Fixed Costs	Variable Costs	Total	Comments
MECH	Mechanical Overhaul	336,582	54,199	390,781	Replace and rebuild mechanical components
HYD	Hydraulic Overhaul	33,160	0	33,160	Replace hydraulic components
ELEC	Electrical Overhaul	182,136	1,829	183,965	Replace electrical components
		0	0	0	
Total		**551,878**	**56,028**	**607,906**	

The table below describes the fields in the grid.

Field	Description
Code	Level 1 Group Code.
Description	Level 1 Group Description.
Fixed Costs	For each line this is the sum of the **In-scope**, **Fixed** costs.
Variable Costs	For each line this is the sum of the **In-scope**, **Variable** costs.

Field	Description
Total	The Total column is the total of the **Fixed** and **Variable** costs for each Level 1 Group. The Total row is the sum of all **In-scope Fixed** and **Variable** costs for all for the Level 1 groups.
Comments	Level 1 Group Comments.

Quote Report - Pricing Details

The second page of the Quote Report is the **Pricing Details** section. This section contains the details of the individual jobs contained within the quote and is where most of the information is found.

	Description	Fixed Costs	Variable Costs	Total	Comment
MECH	**Mechanical Overhaul**				**Replace mechanical components**
3000.INB.CC.0-540	3000.INB.CC.0 Replace Cutter Gearcase Inbye	166,944	0	166,944	
3000.OUTB.CC.0-540	3000.OUTB.CC.0 Replace Cutter Gearcase Outbye	166,908	0	166,908	
7000.00.GR.0-571	**Photograph shearer before and after cleaning.**				
7000.00.GR.0-571	7000.00.GR.0 Clean shearer	440	665	1,105	Labour will be charge per hour consumed.
7000.00.GR.0-571	*Clean shearer prior to stripping any components*				
7050.INB.GI.0-040	7050.INB.GI.0 Inspect Ranging arm casing crack	0	0	0	Determine if repair is appropriate or if replacement is required.
MECH.01	Structural				Structural Jobs
7050.INB.GI.0-040	7050.INB.GI.0 Inspect Ranging arm casing crack	555	3,647	4,202	Determine if repair is appropriate or if replacement is required.
Total	Mechanical Overhaul	334,847	4,312	339,159	

The page is split into one or more tables each containing the header row and one Level 1 group. If the quote has more than one Level 1 group, there will be more than one table in this section. The tables display in the order that they appear in the **Quote Groups** window.

The second part of the page contains the job details for Level 1, and Level 2 jobs. It is structured such that each Level 1 group has a table containing the Level 2 groups and the jobs. The order of the Level 1 and Level 2 groups will be as per the **Quote Groups** in **Quote Manager – Job Quoting**. The order of the jobs is sorted by the component code, modifier code, task type code and task counter code and job code.

The formatting and location of the data in this second page of the document is determined by different fields in the **Quote Manager**.

The fields on the Pricing Details page are describe in the table below.

Field	Description
(Left most column)	• Level 1 Group Code - in **bold**. • Level 1.Level 2 Group Code - in grey row. • Job coding - CC.MC.TT.TC-Job Code.

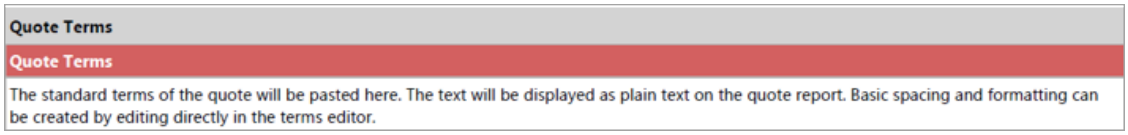
Field	Description
Description	• Level 1 Group Description - in **bold**. • Level 2 Group Description - in grey row. • Header Notes - in **bold red**. Job Description - Shown as CC.MC.TT.TC "Description". For example *7000.00GR.0 Clean Shearer*. The "Description" is from the Job Details window. • Footer notes - in *grey italics*.
Fixed Costs	For each line, this is the sum of the **In-scope**, **Fixed** costs.
Variable Costs	For each line, this is the sum of the **In-scope**, **Variable** costs.
Total	• The Total column is the total of the **Fixed** and **Variable** costs for each Level 1 group. • The Total row is the sum of all In-scope, **Fixed** and **Variable** costs for all of the Level 1 groups.
Comment	• Level 1 Group Comment - in **bold**. • Level 2 Group Comment - in grey row. • Inline Comments - in regular formatted text.

Any jobs that are set to out of scope will display in strikethrough text.

~~5050.00.GR.0 540~~	~~5050.00.GR.0 Hydraulic System Repair~~	~~0~~	~~0~~	~~0~~	

Quote Report - Terms

The third page of the Quote Report is a Terms page. This page displays the terms entered into the quote header, either manually or using stored quote text. The text displays as plain text.



If you require a specific quote terms page or other customised version of the Quote Report, it is best to speak to your RPMGlobal account manager to have a custom report written.

Quoter user security

To access Quoter, System Administrators must ensure user profiles are set up accordingly using the User Accounts function (**System Administration** > **User Profiles** > **User Accounts**). Within the **Quote Authorisation** section, you can define the following for each user:

- Enable Authorisation.
- Maximum Authority (Financial Amount).

- Maximum Discount.

The above parameters span over all Quoter agreement types.

Available agreements

There are three types of agreement that can be generated from Quoter.

Maintenance Agreement Quote

The AMT Maintenance Agreement provides the ability to model different equipment or fleet scenarios made up of different types of models with different terms and annual average utilisation.

Service Agreement Quote

The Sales Tool Service Agreement is very similar to the Maintenance Agreement with the exception that:

- PMs are automatically selected in the scope section.
- The Internal Summary report points to a different report.

Life Cycle Costing Quote

Life Cycle costing is a methodology for calculating the whole cost of a system from inception to disposal. The AMT Life Cycle Cost function is designed to easily generate life cycle cost information quickly to customers based on a general inquiry. Costs include projected tasks, lubrication, tyres and component changeout activities.

Service Agreement Quote

When creating a Service Agreement Quote, select a **Model** and then select the **Customer** and **Pricing Rule**. The other fields are optional. Enter the **Start Usage** and **Date**, **Annual Utilisation**, and either the **End Usage** or **End Date**.

Creating a Service Agreement Quote follows a similar process to creating a Maintenance Agreement Quote. The key differences are:

- PMs are automatically selected in the **Scope** section indicated by a ✓ in the **Included** column.
- The **Internal Summary** report points to additional reports.

Review the scope and continue to step 5.

Field descriptions for steps 5-7 are described in separate topics listed below. Tap a link to go to the step. The step number relates to the step number in the process (displayed at the top of the AMT Mobile screen).

Costing Parameters (Step 5)

Billings Method (Step 6)

Update Discount (Step 6)

Update Status (Step 7)

View Reports (Step 7)

Customer Agreement (Step 7)

Dealer Quote (Step 7)

Costing Parameters (Step 5)

The table below describes the fields on the Costing Parameters screen.

Field	Description
Field/Shop Labour Rates	• You can specify new labour rates, as well as define a percentage of expected overtime and a rate for these overtime hours. • The new rates will override the labour rates defaulted from the centreline where the strategy task in the centreline contains labour hours.
Dealer Labour	The percentage of field labour supplied by the Dealer. • The Labour Split is applied to all strategy task jobs where Labour Type = Field (regardless of cost bearer). • Labour Split determines the number of labour hours used to calculate labour cost if a labour rate is specified.
Travel	• **Total Travel Costed** shows travel costs already contained in the centreline and updated by using the Multi-task Scope Editor. • **Apply Travel Cost**. Selecting this overrides the costs displayed on the **Total Travel Costed** field, and enables both **Hours** and **Distance (roundtrip)** fields, so new costs can be calculated for this agreement specifically. • **Hours** and **Distance** are used in conjunction with values entered using the Multi-task Scope Editor. **If used**, will add travel costs to the billings (labour and misc) which will override any travel data already contained in the centreline.
Freight %	• Freight is calculated as a percentage of the cost of parts on each strategy task job, where the strategy tasks jobs which have Cost Bearer = Contact.  The selection you make is applied to all strategy task jobs which have Cost Bearer = Contract and Labour Type = Field.



Lubrication Costs are no longer stored in the Maintenance Agreement, it is calculated in reports only. Lubrication cost is calculated from included strategy task jobs which have
- the default Cost Bearer.
- Job Code flagged as Lube.

Billings (Step 6)

Tap beside **Billings Method** to change. You can also tap **Actions** > **Change Billings Method**.

- Hourly Rate.
- Lump Sum.
- When Services are Performed.



The Billings Method selection is for information purposes only and does not impact the contract's revenue value. The billings created on the modelling machine is set as Usage Rate by Usage Steps for any of the choices.

Update Discount (Step 6)

Tap **Actions** > **Update Discount**.

- Discount Percentage on parts, labour and misc.
- Risk Premium Percentage.



While the discount percentage is applied individually at one of the costs groups (that is, parts, labour or misc), the risk premium percentage (for example, a percentage applied to a certain contract to account for some extra financial risk) represents a percentage over the total projected cost at sell.

Update Status (Step 7)

Tap **Actions** > **Update Status**.

You can select between **Abandoned**, **Costings Completed**, **Deal Lost**, **Deal Won**, **Draft Workings** and **Proposal Sent**. These are for informational purposes only.

View Reports (Step 7)

Select **Actions** > **View Reports**.

A list of available reports displays. Select a report to view. You can click the drop-down list at the top right of the screen and choose to Export the report to PDF, Excel or Word.

Customer Agreement (Step 7)

Select **Actions** > **Customer Agreement**.

Once the Agreement has been authorised, the Customer Agreement option becomes available. This provides you with Customer-specific reports.

Delete Quote (Step 7)

Select **Actions** > **Delete Quote**.

If you have the right access you are able to delete a quote at any time. If you do not have access, see your AMT Administrator.

Maintenance Agreement Quote

Field descriptions for creating a Maintenance Agreement Quote are described below. Click on a link to expand it. The step number relates to the step number in the process (displayed at the top of the mobile screen).

Costing Parameters (Step 5)

The table below describes the fields on the Costing Parameters screen.

Field	Description
Field/Shop Labour Rates	• You can specify new labour rates, as well as define a percentage of expected overtime and a rate for these overtime hours. • The new rates will override the labour rates defaulted from the centreline where the strategy task in the centreline contains labour hours.
Dealer Labour	The percentage of field labour supplied by the Dealer. • The Labour Split is applied to all strategy task jobs where Labour Type = Field (regardless of cost bearer). • Labour Split determines the number of labour hours used to calculate labour cost if a labour rate is specified.
Travel	• **Total Travel Costed** shows travel costs already contained in the centreline and updated by using the Multi-task Scope Editor. • **Apply Travel Cost**. Selecting this overrides the costs displayed on the **Total Travel Costed** field, and enables both **Hours** and **Distance (roundtrip)** fields, so new costs can be calculated for this agreement specifically. • **Hours** and **Distance** are used in conjunction with values entered using the Multi-task Scope Editor. **If used**, will add travel costs to the billings (labour and misc) which will override any travel data already contained in the centreline.
Freight %	• Freight is calculated as a percentage of the cost of parts on each strategy task job, where the strategy tasks jobs which have Cost Bearer = Contact.  The selection you make is applied to all strategy task jobs which have Cost Bearer = Contract and Labour Type = Field.

Lubrication Costs are no longer stored in the Maintenance Agreement, it is calculated in reports only. Lubrication cost is calculated from included strategy task jobs which have
- the default Cost Bearer.
- Job Code flagged as Lube.

Billings Method (Step 6)

Select **Actions** > **Change Billings Method**.

- Hourly Rate.
- Lump Sum.
- When Services are Performed.

The Billings Method selection is for information purposes only and does not impact the contract's revenue value. The billings created on the modelling machine is set as Usage Rate by Usage Steps for any of the choices.

Update Discount (Step 6)

Select **Actions** > **Update Discount**.

- Discount Percentage on parts, labour and misc.
- Risk Premium Percentage.

While the discount percentage is applied individually at one of the costs groups (that is, parts, labour or misc), the risk premium percentage (for example, a percentage applied to a certain contract to account for some extra financial risk) represents a percentage over the total projected cost at sell.

Update Status (Step 7)

Select **Actions** > **Update Status**.

You can select between Abandoned, Costings Completed, Deal Lost, Deal Won, Draft Workings and Proposal Sent. These are for informational purposes only.

Request Authorisation (Step 7)

Select **Actions** > **Request Authorisation**.

An email can also be sent to the relevant users requesting that they log into AMT and authorise it. It will contain the Agreement Number and the user can then search for it in AMT and approve it.

Several users can be chosen to receive the email at the same time and that the name of the generated quote is provided on the email.

Confirm Authorisation (Step 7)

Select **Actions** > **Confirm Authorisation**.

If you have been enabled to authorise agreements you can authorise it directly on this step.

View Reports (Step 7)

Select **Actions** > **View Reports**.

A list of available reports displays. Select a report to view. You can click the drop-down list at the top right of the screen and choose to Export the report to PDF, Excel or Word.

Customer Agreement (Step 7)

Select **Actions** > **Customer Agreement**.

Once the Agreement has been authorised, the Customer Agreement option becomes available. This provides you with Customer-specific reports.

Delete Quote (Step 7)

Select **Actions** > **Delete Quote**.

If you have the right access you are able to delete a quote at any time. If you do not have access, see your AMT Administrator.

Life Cycle Costing Quote

Creating a Life Cycle Costing Quote follows a similar process to creating a Maintenance Agreement Quote. The key differences are:

- The Life Cycle Costing Quote is **not** an agreement.
  - It is purely a costing to give to a customer and is an estimate on the cost of an asset over its life.
- It does not have the following:
  - Step 3, Term.
  - Step 6, Billings.
  - In Step 7, Summary Page, there is no Authorise option.

Quoter Manager - Add Quote Jobs

Jobs are the core information that the quote is used for to display to your customer. The pricing and instructions for the quoted jobs are contained in this area. Items on the quote will report based on the job. This is the same type of information that is contained within a strategy task, a standard job or a work order.

They are several ways to add a job to a quote and they can be used in different situations to give you the flexibility required to create the quote you need.

The methods of adding jobs to a quote are by adding:

- A standard job.
- An occurrence from long term planning.
- Tasks from a project.
- Defect work orders from the backlog.
- Notifications.
- Existing quote jobs.
- A new job created manually.

Quoter Manager- Approve and Authorise Quote

Approving the quote indicates that you, or other users, have reviewed the content of the quote and give approval that it is correct and ready for authorisation. This can be performed by many users, or none. It is not mandatory, but can be incorporated into a quote management process.

Authorising the quote can only be performed by users with **Quote Authorisation** enabled and with a maximum authority greater than the value of the quote. When a quote has been authorised, the data contained within is locked so that it cannot be modified. Only an authorised quote can be used to generate the Quote Report for printing. If you are required to make a modification after the quote has been authorised, a new version must be created and then authorised again prior to being able to print the quote.

The audit trail can be used to identify who has approved and authorised a quote and the values contained within with the quote at the time of approval or authorisation.

Quoter Manager - View and Edit Quote Jobs

Viewing quote jobs lets you make any adjustments to the pricing and scope of the job to match the requirements for the Quote. The key concepts are:

- Assigning jobs to groups.
- Fixed vs variable pricing.
- In or out of scope.
- Flat rate pricing.
- Notes.
- Override pricing.
- Quantity of jobs.

Quoter Manager - Edit Quote Pricing

The quote pricing can be adjusted from the cost at sell of the jobs by applying discounts for parts, labour, or misc costs. This applies to the sum of the cost at sell. The discount does not affect the cost at cost. Editing the default gross margin will set the values of cost at cost used in the quote for when flat rate pricing is selected for a job. It follows the same formula outlined in the procedure for override pricing for a job.

Quoter Manager - Edit Quote Status

Quote Status is used to keep track of how different quotes are progressing in your process. The status allows you to more easily sort the quotes and can be applied as a filter in **Quote Manager**.

Quoter Manager - Print Quote

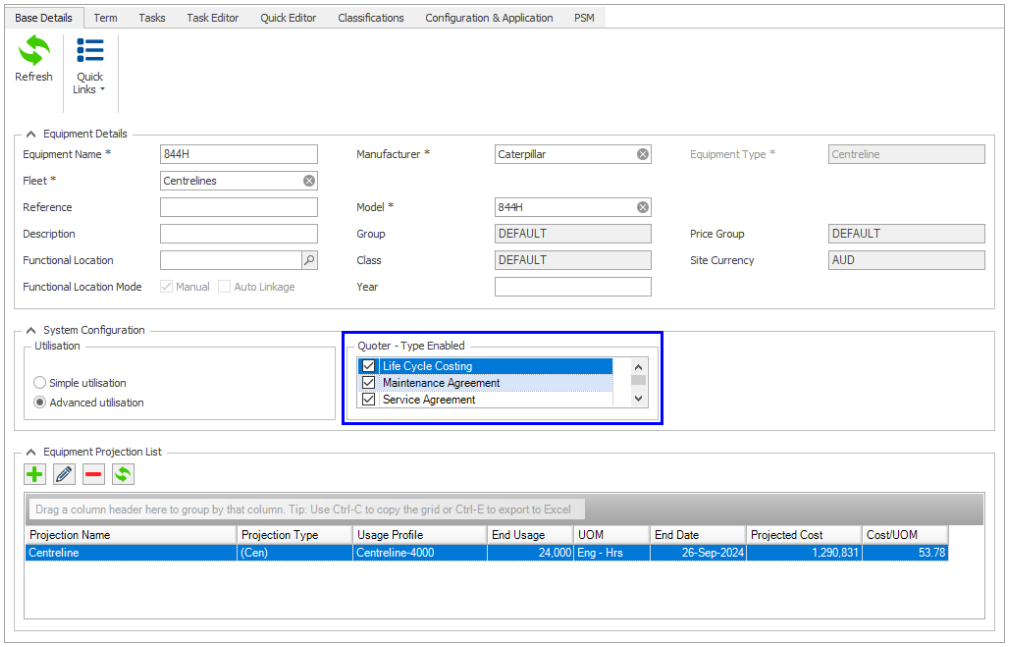
A quote can be printed once it has been authorised. This is how you can view all of the formatting and final value of the jobs in the quote once overall discounts have been applied.

Choosing to print the quote displays the **Quote Report** which you can then print, save or export.

The fields for **Printed By** and **Generating Comments** are stored in the quote header and are not printed to the quote.

Quoter centrelines

Centrelines are the starting point of every agreement. It is from a centreline that you select the basic details of a new quote, such as the model, maintenance tasks, basic additional costs, and so on. As a starting point, a single centreline can be used to create more than one agreement. If you have appropriate security access you can select which centrelines can be used to create Quoter agreements.

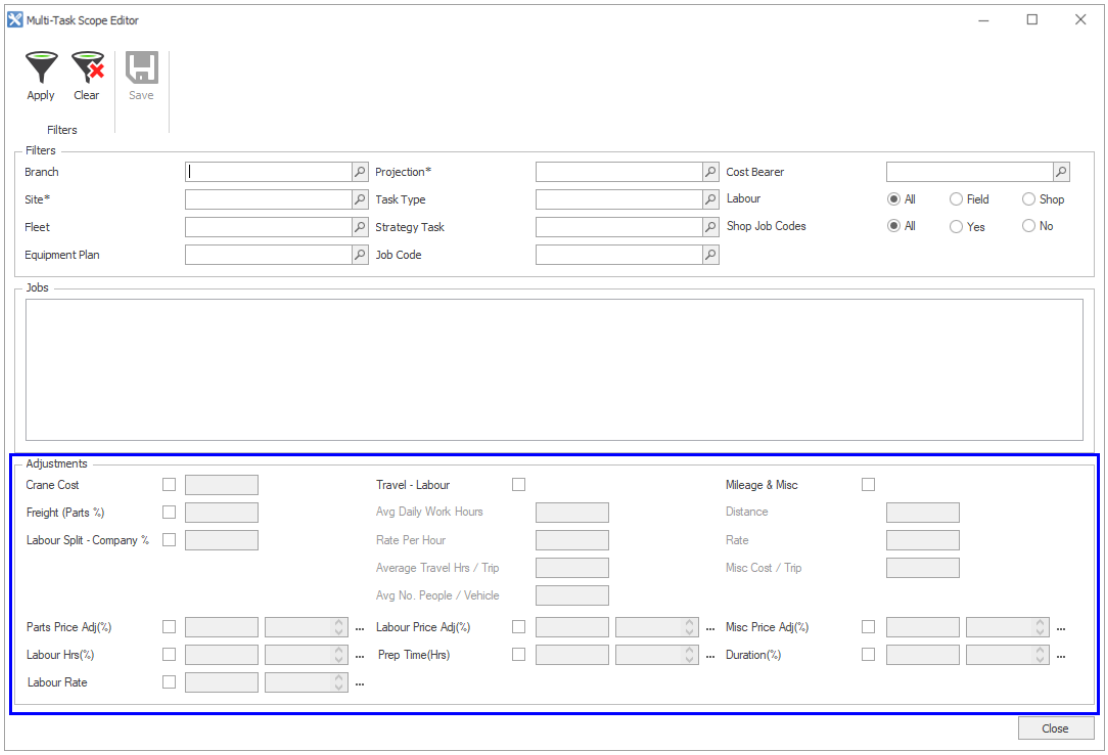


In addition to the centreline's strategy tasks, further costs details can be added to it to be used on a quote. You can update centrelines with freight, labour split and travel information (Audit Trail). The **Multi-Task Scope Editor** can then be used for bulk update of the freight, labour split and travel data.

Changes made using the Multi-Task Scope Editor are at the Strategy Task – Job level (a single strategy task can be linked to one or more jobs). Navigate to **Setup & Admin** > **Projections - Multi-editors** > **Multi-Task Scope Editor**.

Once you have applied and executed the appropriate filters, you should review the returned tasks prior to inputting the adjusted parameter on the right side of the screen. Once you have adjusted the required parameters, click **Apply Changes** . A notification displays detailing the number of records edited.

The main fields in the Multi-Task Scope Editor **Adjustment** panel are described below.



Field	Description
Crane Cost	Add the cost of craneage to miscellaneous.
Freights (% parts)	Apply freight cost as a percentage of parts cost. Parts Cost = Parts Cost + (Parts Cost x Freight).
Labour Split - Company %	Specifies what percentage of the labour hours will be performed by the dealer. This impacts labour costings for an agreement.
Travel - Labour	Enter parameters to calculate the travel labour hours and labour costs required to perform the strategy task job. This impacts labour cost.
Mileage & Misc	Enter parameters to calculate the misc. cost required to perform the strategy task job.
Parts Price Adj %	The percentage applied to adjust the strategy task job's parts cost, even if the strategy task job is linked to a standard job. Does not impact the standard job itself.
Labour Price Adj %	The percentage applied to adjust the strategy task job's labour cost, even if the strategy task job is linked to a standard job. Does not impact the standard job itself.
Misc. Price Adj %	The percentage applied to adjust the strategy task job's misc. cost, even if

Field	Description
	the strategy task job is linked to a standard job. Does not impact the standard job itself.
Labour Hrs Adj	The percentage applied to adjust the strategy task job's parts cost, even if the strategy task job is linked to a standard job. Does not impact the standard job itself.
Prep Time (Hrs)	Adds additional labour hours to the strategy task job to represent preparation time.
Duration %	Percentage applied to adjust the strategy task job's duration, even if the strategy task job is linked to a standard job.
Labour Rate	Labour rate applied to the strategy task job. If a labour rate has been entered then Labour Cost = (Labour Hours + Prep Time) x Labour Rate.

Administrator App

This app allows you to add, view and delete the following **Timesheet System Tables**:

- Pay Types
- Shift Codes
- Labour Expense Codes
- Labour Groups.

You can also add and edit **Employee Information**.

Maintenance App

This app is designed to record measurements and usages, capture new defects, perform ad hoc inspections, review statutory compliance jobs and review and perform work orders.

On the **Home** screen, click the **Maintenance** app. Select a **Site**, and the **Employee Login** screen displays. You are required to enter a PIN to log in.

The Maintenance home screen displays with the following functionality options:

- Measurements
- Defects
- Usages
- Inspections
- Statutory Compliance
- Work orders

You can also perform Inspections, Statutory Compliance and Work Orders from within the Workbench app or by using the Inspections app.

Measurements

Tap **Measurements**. This screen displays the top 20 measurements for the nominated site. To narrow down the list tap **Refine**, and select Asset, Fleet or Measurements. The Measurement categories are displayed for all assets, and by selecting the relevant one, you will see what the latest valid reading is on the right. The measurements are validated and colour-coded as shown below.

Colour	Description
	OK
	Low Level Alert
	Medium Alert
	High Alert

Tap **Record Measurements** to update the measurements directly in the Measurements screen. Once you have updated them tap **Save**. The readings are subjected to the same validation rules as any measurements entered into AMT. If a measurement is invalid it will not be visible, and AMT will revert to the previous measurement.

Defects

You can log Defects identified during an Inspection from the Workbench app. Tap **Defects**.

This displays the top 20 Defects for the selected site. To narrow down the list you can you can click the **Refine** button and clicking an Asset, Fleet or both.

To add a Defect, tap **Add New Defect**.

To edit a Defect, tap the relevant one, update it and click **Save**.

When you enter the details of the Defect you are able to also attach photos or documents.

Once you have entered all Defects, tap **Next**. A list of all steps displays.

Select relevant steps and enter the results of the inspection. Steps with bad results and those that require a defect to be created are marked in red on the left, all OK results are marked in green.

AMT Mobile Usages

This screen displays the top 20 usage readings for the nominated site. For every asset, each UOM set up in the Usage Profile is displayed. By selecting the relevant one you will see what the latest valid reading is. To narrow down the list you can tap **Refine** and select Asset, Fleet or UOM.

You can update the usages directly in the Usages screen. Once you have updated them tap **Save**. The readings are subjected to the same validation rules as any usage readings entered into AMT. If a reading is invalid it will not be visible and AMT will revert to the previous reading.

Statutory Compliance

This screen gives you the ability to review all work orders which have been nominated as statutory jobs. You are able to see the name of the asset, the relevant strategy task, when it is next due, when it was last performed and what the most recent work order number was. Navigate to **Daily Planning** > **Statutory Compliance**. The AMT Mobile screen opens in your web browser.

For each strategy task, the Statutory Compliance screen shows:

1. When the task was last performed.
2. The last work order number.
3. The next due date.

Colour coding is used to highlight the status:

Colour Coding	Description
	Not yet within 10% of frequency (system configuration variable).
	Within 10% of frequency (system configuration variable).
	Past compliance.

Offline Options

These are options for when you are not connected to the Internet.

Offline

Allows you to select the Site, Fleet and Asset to take offline.

Download Data

Downloads selected data for you to use when offline.

Discard Data

Deletes all offline data, including any Inspections performed offline.

Sync Now

Syncs all pending items.

Content feedback

If you would like to send us feedback on the AMT help content, please email contentfeedback@rpmglobal.com.

RPMGLOBAL

About

Version 8.26.